Mathematical Physics
Collected Notes and Worked Problems

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12 October 2026

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Part I
Linear Algebra Review

cf. *Change of Basis*, of Appendix B of John Lee (2012) [76].

Exercise B.22. Suppose V, W, X finite-dim. vector spaces
$S : V \to W, \quad T : W \to X$

(a) $\text{rank}S \leq \dim V$ with $\text{rank}S = \dim V$ iff S injective

(b) $\text{rank}S \leq \dim W$ with $\text{rank}S = \dim W$ iff S surjective

(c) if $\dim V = \dim W$ and S either injective or surjective, then S isomorphism

(d) $\text{rank}TS \leq \text{rank}S$ $\text{rank}TS = \text{rank}S$ iff $\text{im}S \bigcap \ker T = 0$

(e) $\text{rank}TS \leq \text{rank}T$ $\text{rank}TS = \text{rank}T$ iff $\text{im}S + \ker T = W$

(f) if S isomorphism, then $\text{rank}TS = \text{rank}T$

(g) if T isomorphism, then $\text{rank}TS = \text{rank}S$

EY : Exercise B.22(d) is useful for showing the chart and atlas of a Grassmannian manifold, found in the More examples, for smooth manifolds.

Proof. (a) Recall the **rank-nullity theorem**:

Theorem 1 (Rank-Nullity Theorem)**.**

$$dim(im(S)) + dim(ker(S)) = dim V \tag{1}$$

Now

$$\text{rank}(S) + \dim(\ker(S)) \equiv \dim(\text{im}(S)) + \dim(\ker(S)) = \dim V$$
$$\implies \boxed{\text{rank}(S) \leq \dim V}$$

If $\text{rank}(S) = \dim V$,
then by rank-nullity theorem, $\dim(\ker(S)) = 0$, implying that $\ker S = \{0\}$.

Suppose $v_1, v_2 \in V$ and that $S(v_1) = S(v_2)$. By linearity of S, $S(v_1) - S(v_2) = S(v_1 - v_2) = 0$, which implies, since $\ker S = \{0\}$, that $v_1 - v_2 = 0$.
$\implies v_1 = v_2$. Then by definition of injectivity, S injective.

If S injective, then $S(v) = 0$ implies $v = 0$. Then $\ker S = \{0\}$. Then by rank-nullity theorem, $\text{rank}(S) = \dim V$.

(b) $\forall\, w \in \text{im}(S)$, $w \in W$. Clearly $\text{rank} S \leq \dim W$.

If S surjective, $\text{im}(S) = W$. Then $\dim(\text{im}(S)) = \text{rank} S = \dim W$.

If $\text{rank} S = \dim W = m$, then $\text{im}(S)$ has basis $\{y_i\}_{i=1}^m$, $y_i \in \text{im}(S)$, so $\exists\, x_i \in V$, $i = 1 \dots m$ s.t. $S(x_i) = y_i$, with $\{S(x_i)\}_{i=1}^m$ linearly independent.

Since $\{S(x_i)\}_{i=1}^m$ linearly independent and $\dim W = m$, $\{S(x_i)\}_{i=1}^m$ basis for W.
$\forall\, w \in W$, $w = \sum_{i=1}^m w^i S(x_i) = S(\sum_{i=1}^m w^i x_i)$. $\sum_{i=1}^m w^i x_i \in V$. S surjective.

(c)

(d) Now

$$\dim V = \text{rank} TS + \text{nullity} TS$$
$$\dim V = \text{rank} S + \text{nullity} S$$

$\ker S \subseteq \ker TS$, clearly, so $\text{nullity} S \leq \text{nullity} TS$

$$\implies \boxed{\text{rank} TS \leq \text{rank} S}$$

If $\text{rank} TS = \text{rank} S$,
then $\text{nullity} S = \text{nullity} TS$
Suppose $w \in \text{Im} S \bigcap \ker T$, $w \neq 0$
Then $\exists\, v \in S$, s.t. $w = S(v)$ and $T(w) = 0$
Then $T(w) = TS(v) = 0$. So $v \in \ker TS$
$v \notin \ker S$ since $w = S(v) \neq 0$
This implies $\text{nullity} TS > \text{nullity} S$. Contradiction.
$\implies \text{Im} S \bigcap \ker T = 0$

If $\text{Im} S \bigcap \ker T = 0$,
Consider $v \in \ker TS$. Then $TS(v) = 0$.
. Then $S(v) \in \ker T$
$S(v) = 0$; otherwise, $S(v) \in \text{Im} S$, contradicting given $\text{Im} S \bigcap \ker T = 0$
$v \in \ker S$

$\ker TS \subseteq \ker S$
$\implies \ker TS = \ker S$
So $\text{nullity} TS = \text{nullity} S$
$\implies \text{rank} TS = \text{rank} S$

(e)

(f)

(g)

□

Part II
Algebra; Groups, Rings, R-Modules, Categories

We should know some algebra. I will follow mostly Rotman (2010) [57].

1 Prime numbers, GCD (greatest common denominator), integers, Euler's totient, Chinese Remainder Theorem, integer divison, modulus, remainders; Euclid's Lemma

Definition 1 (natural numbers $\mathbb{N}$)**.** natural numbers $\mathbb{N}$

$$\mathbb{N} = \{ \text{ integers } n | n \geq 0 \} \tag{2}$$

i.e. $\mathbb{N}$ is set of all nonnegative integers.

Definition 2 (prime)**.** natural number p is **prime** if $p \geq 2$, and $\nexists$ *factorization* $p = ab$, where $a < p$, $b < p$ are natural numbers.

Definition 3. $a, b \in \mathbb{Z}$ **relatively prime** if $\gcd(a, b) = 1$

Axiom 1. ***Least Integer Axiom*** $\exists$ *smallest integer in every* $C \subset \mathbb{N}$, $C \neq \emptyset$

cf. pp. 1, Ch. 1 Things Past of Rotman (2010) [57]

Theorem 2 (**Division Algorithm**)**.** $\forall\, a, b \in \mathbb{Z}$, $a \neq 0$, $\exists !\, q, r \in \mathbb{Z}$ *s.t.*

$$b = qa + r \text{ and } 0 \leq r < |a|$$

Proof. Consider $n \in \mathbb{Z}$, $b - na \in \mathbb{Z}$
Let $C = \{b - na | n \in \mathbb{Z}\} \bigcap \mathbb{N}$.
$C \neq \emptyset$ (otherwise, consider $b - na < 0$, $b < na$, then contradiction)
By Least Integer Axiom, $\exists$ smallest $r \in C$, $r = b - na$.
define $q = n$ when $r = b - na$.
Suppose

$$\begin{aligned} qa + r &= q'a + r' \\ (q - q')a &= r' - r \quad , \\ |(q - q')a| &= |r' - r| \end{aligned}$$

$0 \leq r' < |a|$. Now $0 \leq |r' - r| < |a|$
if $|q - q'| \neq 0$, $|(q - q')a| \geq |a|$

$$\implies q = q', r = r'$$

Conclude both sides are 0 (by contradiction) □

cf. pp. 2, Thm. 1.4, Ch. 1 Things Past of Rotman (2010) [57]

Definition 4 (divisor)**.** $a, b \in \mathbb{Z}$, a **divisor** of b if $\exists\, d \in \mathbb{Z}$ s.t. $b = ad$.
a **divides** b or b multiple of a, denote

$$a | b$$

$a|b$ iff b has remainder $r = 0$ after dividing by a

cf. pp. 3, Ch. 1 Things Past of Rotman (2010) [57]

1.1 Greatest Common Denominator (GCD); Euclid's Lemma

Definition 5 (common divisor)**. common divisor** of integers a and b, is integer c, s.t. $c|a$ *and* $c|b$.
greatest common divisor or **gcd** of a and b, denoted $(a, b) \equiv \gcd(a, b)$ defined by

$$(a, b) \equiv \gcd(a, b) = \begin{cases} 0 & \text{if } a = 0 = b \\ \text{the largest common divisor of } a \text{ and } b & \text{otherwise} \end{cases}$$

cf. pp. 3, Ch. 1 Things Past of Rotman (2010) [57]

Definition 6 (Greatest Common Divisor (GCD), practical definition)**.** Say $\gcd(a,b) = d$. Thus, we have that $\exists\, a = dr$ and $b = ds$, $a, b \in \mathbb{Z}$, where $\gcd(r,s) = 1$

cf. Proof of greatest common divisor, math stackexchange

Theorem 3. *If $a, b \in \mathbb{Z}$, then $gcd(a,b) \equiv (a,b) = d$ is linear combination of a and b, i.e. $\exists\, s, t \in \mathbb{Z}$ s.t.*

$$d = sa + tb \tag{3}$$

cf. pp.4, Thm. 1.7, Ch. 1 Things Past of Rotman (2010) [57]

Proof. Let $I :=$

$$I := \{sa + tb | s, t \in \mathbb{Z}\}$$

If $I \neq \{0\}$, let d be smallest positive integer in I.
$d \in I$, so $d = sa + tb$ for some $s, t \in \mathbb{Z}$.
Claim: $I = (d) \equiv \{kd | k \in \mathbb{Z}\}$ = set of all multiples of d.
Clearly $(d) \subseteq I$, since $kd = k(sa + tb) = (ks)a + (kt)b \in I$.
Let $c \in I$.
By division algorithm Thm. 2, $c = qd + r$, $0 \leq r < d$

$$r = c - qd = s'a + t'b - qsa - qtb = (s' - sq)a + (t' - qt)b \in I$$

If $r \in I$, but $r < d$, contradiction that $\min_{\substack{i \in I \\ i > 0}} i = d$.
So $r = 0$, and $d|c = c/d$.

$$c \in (d), \text{ so } I \subseteq (d) \implies I = (d)$$

□

Theorem 4 (**Euclid's Lemma**; 1.10 of Rotman (2010) [57])**.** *If p prime and $p|ab$, then $p|a$ or $p|b$.*
More generally,
if prime p divides product $a_1 a_2 \dots a_n$,
then it must divide at least 1 of the factors a_i.
i.e. (notation),
If prime p, and $ab/p \in \mathbb{Z}$,
then $a/p \in \mathbb{Z}$ or $b/p \in \mathbb{Z}$.
More generally,
if prime p, s.t. $a_1 a_2 \dots a_n / p \in \mathbb{Z}$,
then $\exists$ 1 a_i s.t. $a_i/p \in \mathbb{Z}$

Proof. If $p \nmid a$, i.e. $a/p \notin \mathbb{Z}$, then $\gcd(p,a) \equiv (p,a) = 1$.
From Thm. 3,

$$1 = sp + ta$$

$$\implies b = spb + tab = p(sb + td)$$

ab/p and so $ab = pd$, so $b = spb + tdp$, i.e. b is a multiple of p ($b/p \in \mathbb{Z} \equiv p|b$).

□

Corollary 1 (1.11 of Rotman (2010) [57])**.** *Let $a, b, c \in \mathbb{Z}$.*
If c, a relatively prime, i.e. $gcd(c,a) = 1$, and if $c|ab \equiv ab/c \in \mathbb{Z}$, then $c|b \equiv b/c \in \mathbb{Z}$

Proof.

$$\gcd(c,a) = 1 = sc + ta \implies b = sbc + tab = sbc + t(qc) = c(sb + tq) \implies b/c = sb + tq$$

□

Theorem 5 (Euclidean Algorithm)**.** *Let $a, b \in \mathbb{Z}^+$.*
$\exists$ algorithm that finds $d = \gcd a, b$

cf. pp. 5, Thm. 1.14 (Euclidean Algorithm), Ch. 1 Things Past of Rotman (2010) [57].

Proof. □

Definition 7. Let fixed $m \geq 0$. Then $a, b \in \mathbb{Z}$ are **congruent modulo** m, denoted by

$$a \equiv b \bmod m$$

if $m|(a-b)$, i.e. $(a-b)/m \in \mathbb{Z}$, i.e. if $(a-b)/m \in \mathbb{Z}$, i.e. $(a-b)$ integer multiple of m

Proposition 1. *If $m \geq 0$ is fixed, $m \in \mathbb{Z}$, then $\forall\, a, b, c \in \mathbb{Z}$*

1. $a \equiv a \bmod m$
2. *if* $a \equiv b \bmod m$, *then* $b \equiv a \bmod m$
3. *if* $a \equiv b \bmod m$, *and* $b \equiv c \bmod m$, *then* $a \equiv c \bmod m$

cf. Prop. 1.18 of Rotman (2010) [57]

Proof. 1. $(a-a)/m = 0/m = 0$

2. $(b-a)/m = (-1)(a-b)/m \in \mathbb{Z}$

3. $(a-c)/m = (a-b+b-c)/m = (a-b)/m + (b-c)/m \in \mathbb{Z}$

□

EY : 20171225 to recap,

$$\boxed{\begin{array}{c} a \equiv b \bmod n \\ \text{meaning} \\ \dfrac{a-b}{n} \in \mathbb{Z} \text{ or } a - b = kn,\ k \in \mathbb{Z} \text{ or } a = b + kN \text{ but rather} \\ a = pn + r \\ b = qn + r \end{array}} \tag{4}$$

for $a = b + kn$, but b need not be a remainder of division of a by n. More precisely, $a = b \bmod n$ asserts that a, b have the same remainder when divided by n, i.e.

$$a = pn + r$$
$$b = qn + r$$

So $a \sim b$ or $[a] = [b]$ is an equivalence relation since
$a \sim a$ since $a \equiv a \bmod N$, since $a = a + 0N$,
if $a \sim b$, then $b \sim a$, since $a - b = kN$, then $b = a - kN$

if $a \sim b$, $b \sim c$, then $a \sim c$, since $a - b = kN$, then $a - c = (k+l)N$.

$$b - c = lN$$

cf. Prop. 1.19 of Rotman (2010) [57]

Proposition 2. *Let $m \geq 0$ be fixed*

1. *If* $a = qm + r$, *then* $a \equiv r \bmod m$
2. *If* $0 \leq r' < r < m$, *then* $r \not\equiv \bmod m$ *i.e.* r *and* r' *aren't congruent* $\bmod m$
3. $a \equiv b \bmod m$ *iff* a, b *leave same remainder after dividing by* m
4. *If* $m \geq 2$, $\forall\, a \in \mathbb{Z}$, $a \equiv b \bmod m$ *for some* $b \in 0, 1, \ldots m-1$

Proof. 1. If $a = qm + r$, then $a \equiv r \bmod m$

$$\frac{a-r}{m} = q \in \mathbb{Z}$$

2. *Want*: If $0 \leq r' < r < m$, then $r \not\equiv \mod m$.
   Suppose $\frac{r-r'}{m} = k \in \mathbb{Z}$. Then $r - r' = km$ or $r = r' + km$.

$$m > r > r' \leq 0$$
$$m > r' + km > r' \leq 0$$
$$m - r' > km > 0$$

   But $k > 0$ (since $m > 0$ and $r - r' = km > 0$) and $m - r' > km > 0$ is a contradiction.

3. *Want*: $a \equiv b \bmod m$ iff a, b leave same remainder after dividing by m. By
   By Division Algorithm, this is true:

$$a = q_a m + r_a$$
$$b = q_b m + r_b$$
$$\frac{a-b}{m} = q_a + \frac{r_a}{m} - q_b - \frac{r_b}{m} = k = q_a - q_b + \frac{r_a - r_b}{m} \in \mathbb{Z}$$

   Now

$$|m| > r_a \leq 0$$
$$|m| > r_b \leq 0$$

   $2|m| > r_a + r_b$.
   And if $r_a > r_b$, $|m| > r_a > r_a - r_b > 0$.
   In both cases, $r_a = r_b$ since $q_a - q_b + \frac{r_a - r_b}{m} \in \mathbb{Z}$ needs to be enforced.

4. *Want*: If $m \geq 2$, $\forall\, a \in \mathbb{Z}$, $a \equiv b \bmod m$ for some $b \in 0, 1, \ldots m-1$.
   By Division Algorithm Thm. 2, $a = q_a m + r_a$, $\quad 0 \leq r_a < |m|$. $\frac{a-r_a}{m} = q_a \in \mathbb{Z}$ so let $b = r_a$.

□

Theorem 6 (1.26 of Rotman (2010) [57]). *If $gcd(a, m) \equiv (a, m) = 1$, then $\forall\, b \in \mathbb{Z}$, $\exists\, x$ s.t.*

$$ax \equiv b \bmod m$$

In fact, $x = sb$, where $sa \equiv 1 \bmod m$ is 1 solution. Moreover, any 2 solutions are congruent mod$m$.
i.e.
If $\gcd a, b = 1$, *then* $\forall\, y \in \mathbb{Z}$, $\exists\, x$ *s.t.* $ax \equiv y \bmod b$, $x = sy$, *where* $sa \equiv 1 \bmod b$ *is 1 solution. Moreover, any 2 solutions are congruent* mod$m$. *This implies that*

$ax \equiv y \bmod b$ *or* $\frac{Ax-y}{b} \in \mathbb{Z}$, *and* $\frac{(as-1)y}{b} \in \mathbb{Z}$.
$sa \equiv 1 \bmod b$ *or* $\frac{sa-1}{b} \in \mathbb{Z}$, *which implies that* $sa - 1 = b(-t)$ *or*

$$sa + tb = 1$$

for some $s, t \in \mathbb{Z}$.

Proof. $\gcd(a, m) = 1 = sa + tm$, by Thm. 3
Then $b = b \cdot 1 = b(sa + tm) = sab + tmb$ or $b = tbm + sab$ or $a(sb) = -tbm + b$.
So $a(sb) \bmod m \equiv b$.
Let $x := sb$ and so $ax \bmod m = b$.
Now suppose $x \neq sb$ s.t. $ax \bmod m = b$. Then $ax = qm + b$. From $a(sb) \bmod m = b$, we also get $a(sb) = q'm + b$. Then $a(x - sb) \bmod m = 0$, so $m | a(x - sb) \equiv a(x - sb)/m \in \mathbb{Z}$.
By Corollary 1 (which says, if $\gcd(c, a) = 1$ and if $ab/c \in \mathbb{Z}$, then $b/c \in \mathbb{Z}$), since $\gcd(m, a) = (m, a) = 1$, and since $a(x - sb)/m \in \mathbb{Z}$, then $(x - sb)/m \in \mathbb{Z}$. So $(x - sb) = qm$ or $(sb) \bmod m = x$.

□

Proposition 3 (3.1 of Scheinerman (2006) [59]). *Let $a, b \in \mathbb{Z}$, let $c = a \bmod b$, i.e. $a = qb + c$ s.t. $0 \leq c < b$.*
Then

$$gcd(a, b) = gcd(b, c) \tag{5}$$

cf. Sec. 3.3 Euclid's method of Scheinerman (2006) [59]

Proof. If d common divisor of a, b, i.e. $a/d, b/d \in \mathbb{Z} \equiv d|a, d|b$.
$c/d \in \mathbb{Z} \equiv d|c$ since $c = a - qb$.

If d is common divisor of b, c, i.e. $d|b, d|c \equiv c/d, b/d \in \mathbb{Z}$,
then $d|a \equiv a/d \in \mathbb{Z}$ since $a = qb + c$. So set of common divisors of a, b same as set of common divisors of b and c.

Then $\gcd(a, b) = \gcd(b, c)$. □

1.2 Euler's totient; relatively prime

cf. Ch. 5 Arrays, Sec. 5.1 Euler's totient of Scheinerman (2006) [59]

For

$$\varphi : \mathbb{Z}^+ \to \mathbb{Z}^+$$
$$\varphi : n \mapsto \varphi(n) := \text{ number of elements of } \{1, 2, \dots n\}$$

that are relative prime to

$$n = |\{i | i \in \{1, 2, \dots n\}, (n, i) = 1 \text{ or equivalently } n \propto i\}|$$

e.g. $\varphi(10) = 4$ since $\varphi(10) = |\{1, 3, 7, 9\}|$.
we want $|(a, b)|1 \leq a, b, \leq n, \gcd(a, b) \equiv (a, b) = 1|$.

$$p_n = \frac{1}{n^2}\left[-1 + 2\sum_{i=1}^{n} \varphi(k)\right] =$$

$=$ probability that 2 integers, chosen uniformly and independently from $\{1, 2, \dots n\}$ are relatively prime

If p is prime, $\forall\, i \in \{1, 2, \dots p\}$, $(p, i) \equiv \gcd(p, i) = 1$, i.e. relatively prime to p, except 1 $i \in \{1, 2, \dots p\}$.
Therefore

$$\varphi(p) = p - 1$$

Consider $\varphi(p^2)$.

$\{1, 2, \dots p^2\}$, only numbers *not* relatively prime to p^2 are multiples of p since
$p, 2p, 3p, \dots p^2$ all divide p^2, i.e. $p|p^2, 2p|p^2 \dots (p-1)p|p^2 \equiv p^2/p, p^2/2p, \dots p^2/p(1-p)$.

Assume $\varphi(p^n) = p^2 - p^{n-1} = p^{n-1}(p - 1)$.

$$\varphi(p^{n+1}) = \varphi(pp^n) = p^n\varphi(p) = p^n(p - 1)$$

Therefore,

Proposition 4 (5.1)**.** *Let p prime, $n \in \mathbb{Z}^+$*

e.g. $\varphi(77)$.
$\forall\, n$ s.t. $1 \leq n \leq 77$.

$$\gcd(n, 77) = 1$$
$$\gcd(n, 7) = 1$$
$$\gcd(n, 11) = 1$$

By Prop. 3,

$$\gcd(n, 7) = \gcd(7, n \mod 7)$$
$$\gcd(n, 11) = \gcd(11, n \mod 11)$$

cf. Example (10) of Dummit and Foote [25].
To recap,

Definition 8 (Euler φ-function)**.** $\forall\, n \in \mathbb{Z}^+$,
let $\varphi(n) :=$ number of positive integers $a \leq n$ with a relatively prime to n, i.e. $\gcd(a, n) = 1 \equiv (a, n)$

e.g. $\varphi(12) = 4$, since $1, 5, 7, 11$ are only positive integers less than or equal to 12.

If p prime, $\varphi(p) = p - 1$.

More generally,

$\forall\, a \geq 1$,

$$\boxed{\varphi(p^a) = p^a - p^{a-1} = p^{a-1}(p-1)} \tag{6}$$

φ is multiplicative in the sense that

$$\varphi(ab) = \varphi(a)\varphi(b) \text{ if } \gcd(a,b) = 1 \tag{7}$$

$\Longrightarrow$ general formula.

If $n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_s^{\alpha_s}$ (Fumdanetal Thm. of Arithmetic, $\forall\, n \in \mathbb{Z}, n > 1$), then

$$\boxed{\begin{aligned}\varphi(n) &= \varphi(p_1^{\alpha_1})\varphi(p_2^{\alpha_2}) \dots \varphi(p_s^{\alpha_s}) \\ &\quad p_1^{\alpha_1 - 1}(p_1 - 1) p_2^{\alpha_2 - 1}(p_2 - 1) \dots p_s^{\alpha_s - 1}(p_s - 1)\end{aligned}} \tag{8}$$

cf. pp. 69 Thm. 5.4 (Chinese Remainder) of Scheinerman (2006) [59].

Theorem 7. *Let $n \in \mathbb{Z}^+$,*
let $p_1, p_2, \dots p_t$ be distinct prime divisors of n (i.e. $\forall\, p_i$, $\frac{n}{p_i^{k_i}} \in \mathbb{Z}$ for some $k_i \geq 1$)
Then

$$\varphi(n) = n\left(1 - \frac{1}{p_1}\right)\left(1 - \frac{1}{p_2}\right) \dots \left(1 - \frac{1}{p_t}\right) \tag{9}$$

Proof. By Fundamental Thm. of Arithmetic,

$$n = p_1^{e_1} p_2^{e_2} \dots p_t^{e_t}$$

where p_j are distinct primes, and e_j are positive integers.

From Eqns. 6, 7, i.e. where

$$\varphi(p^a) = p^a - p^{a-1} = p^{a-1}(p-1)$$
$$\varphi(ab) = \varphi(a)\varphi(b) \text{ if } \gcd(a,b) = 1$$
$$\varphi(n) = \varphi(p_1^{e_1} p_2^{e_2} \dots p_t^{e_t}) = \varphi(p_1^{e_1})\varphi(p_2^{e_2}) \dots \varphi(p_t^{e_t}) =$$
$$= p_1^{e_1}(1 - \frac{1}{p_1}) p_2^{e_2}(1 - \frac{1}{p_2}) \dots p_t^{e_t}(1 - \frac{1}{p_t}) = n(1 - \frac{1}{p_1})(1 - \frac{1}{p_2}) \dots (1 - \frac{1}{p_t})$$

□

Exercise 10. cf. pp. 7 Exercise 10 Dummit and Foote [25].

Prove: $\forall$ given $N \in \mathbb{Z}^+$ (positive number),
$\exists$ only finite many integers n with $\varphi(n) = N$, where φ denotes Euler's φ-function.

EY, Indeed, by definition,

$$\varphi(n) = N$$
$$a_1, a_2 \dots a_N \text{ s.t. } a_i \leq n$$
$$\gcd(a_i, n) = 1 \text{ i.e. } 1 = s_i a_i + t_i n$$

Given $N \in \mathbb{Z}^+$, let $n \in \mathbb{Z}$, s.t. $\varphi(n) = N$ (given hypothesis).

Let $p =$ least (i.e. smallest) prime s.t. $p > N + 1$.

If $q \geq p$ is a prime divisor of n, i.e.

$$n = q^k m$$

for some $k \geq 1$, and m with q not dividing m.

Then

$$\varphi(n) = \varphi(q^k)\varphi(m) = q^{k-1}(q-1)\varphi(m) \geq q - 1 \geq p - 1 > N$$

Contradiction.

Thus, $\nexists$ prime divisor of n greater than $N + 1$.

Particularly, distinct prime divisors of n belong to a finite set, say these primes are $p_1, p_2 \dots p_m$.

Definition 9. prime divisor q of n if q is prime and

$$\frac{n}{q} \in \mathbb{Z} \text{ i.e. } n = q^k m \text{ for some } k \geq 1 \text{ and } \frac{m}{q} \notin \mathbb{Z}^+ \tag{10}$$

Now

$$n = p_1^{a_1} p_2^{a_2} \dots p_m^{a_m}$$

for some $0 < a_i$, so

$$\varphi(n) = \varphi(p_1^{a_1})\varphi(p_2^{a_2}) \dots \varphi(p_m^{a_m}), \text{ so } \varphi(n) = \prod_{i=1}^{m} p_i^{a_i - 1}(p_i - 1)$$

Note, $\forall$ prime p_i, $\varphi(n) \geq p_i^{a_i-1}(p_i - 1) \geq p_i - 1 > N$ for sufficiently large a_i.

Thus, $\forall p_i$, $\exists$ only finitely many permissible choices for exponents a_i.
So set of all n with $\varphi(n) = N$ is subset of finite set, hence finite.

$\forall N \in \mathbb{Z}^+$, $\exists$ largest integer n with $\varphi(n) = N$.
Thus, as $n \to \infty$, $\varphi(n) \to \infty$.

Scheinerman (2006) [59]

cf. Ex. 1.19, pp. 13, Sec. 1.1 Some Number Theory of Rotman (2010) [57] **Exercise 1.19.** If a and b are relatively prime and if each divides an integer n, then their product ab also divides n, i.e.

Theorem 8. *If* $\gcd a, b = 1$, *and if* $n/a \in \mathbb{Z} \equiv a|n$, *and* $n/b \in \mathbb{Z} \equiv b|n$, *then* $n/ab \in \mathbb{Z} \equiv ab|n$.

Proof. $\gcd a, b = 1$, so $sa + tb = 1$ for some $s, t \in \mathbb{Z}$ (Thm. 6).
$\frac{n}{a}, \frac{n}{b} \in \mathbb{Z}$, so $n = au$, $n = bv$
$n = n \cdot 1 = n(sa + tb) = bvsa + autb = ab(vs + ut)$, so $\frac{n}{ab} = vs + ut \in \mathbb{Z}$. □

1.2.1 Chinese Remainder Theorem

Theorem 9. *If* m, m' *relatively prime (i.e.* $\gcd(m, m') = 1$*), then for*

$$x \equiv b \bmod m$$
$$x \equiv b' \bmod m'$$

i.e. given $b, b' m, m'$, *and wanting to find* x, $\exists x$ *and* $\forall 2x$*'s,* $x = x' \mod mm'$, *i.e.*

Let m, n *relatively prime positive integers (i.e.* $\gcd m, n = 1$*),*
$\forall a, b \in \mathbb{Z}$,
then pair of congruences
$x \equiv a \bmod m$
$x \equiv b \bmod n$
has a solution (x)*, and this solution* x *is uniquely determined, modulo* mn.

Proof. cf. The Chinese Remainder Theorem by Keith Conrad
Suppose
$(x - a)/m \in \mathbb{Z}$ or $x - a = my$
$(x - b)/n \in \mathbb{Z}$ or $x - b = nz$ or $a + my - b = nz$
$\gcd m, n = 1$, so then $\forall b \in \mathbb{Z}$, $\exists w$ s.t. $mw \equiv b \bmod n$ i.e. $\frac{mw-b}{n} \in \mathbb{Z}$, in fact, $w = sb$, where $sm \equiv 1 \bmod n$, or $\frac{sm-1}{n} \in \mathbb{Z}$, is 1 solution (Thm. 6).

$$my = b - a + nz$$
$$smy = sb - sa + snz = (1 + nv)y = s(b - a) + snz \text{ or } y = s(b - a) + n(sz - vy)$$
$$\text{or } y \equiv s(b - a) \bmod n$$

$$x = a + my = a + m(s(b - a) + n(sz - vy)) = a + ms(b - a) + mn(sz - vy) \equiv a + ms(b - a) + mnu$$
$$x - a = m(s(b - a) + nu) \implies x = a \bmod m$$
$$x - b = a + ms(b - a) + mnu - b = a + (1 + m)(b - a) + mnu - b = m(b - a) + mnu \implies x \equiv b \bmod n$$

Uniqueness: Suppose $x, y \in \mathbb{Z}$ s.t.

$$x \equiv a \bmod m \qquad y \equiv a \bmod m$$
$$x \equiv b \bmod n \qquad y \equiv b \bmod n$$

Given $\gcd m, n = 1$, $sm + tn = 1$.
Since $\frac{x-a}{m}, \frac{y-a}{m} \in \mathbb{Z}$, $\frac{x-y}{m} \in \mathbb{Z}$, likewise, $\frac{x-a}{n}, \frac{y-a}{n} \in \mathbb{Z}$, $\frac{x-y}{n} \in \mathbb{Z}$
Since $\frac{x-y}{m}, \frac{x-y}{n} \in \mathbb{Z}$, $\frac{x-y}{mn} \in \mathbb{Z}$ by Thm. 8.
Thus, $x - y = mnk$ for some $k \in \mathbb{Z}$. For instance, $k = 0$, $x = y$.
This shows any 2 solutions are the same, modulo mn. □

cf. Ch. 1 Things Past, Thm. 1.28 of Rotman (2010) [57], pp. 68 Thm. 5.2 (Chinese Remainder) of Scheinerman (2006) [59].

2 Groups

cf. pp. 16 Chapter 1 Introduction to Groups. Dummit and Foote (2004) [25]

Definition 10 (binary operation)**.** 1. binary operation $*$ on set G is a function $* : G \times G \to G$. $\forall\, a, b \in G$, $a * b \equiv *(a, b)$

2. binary operation $*$ on set G is *associative*: if $\forall\, a, b, c \in G$, $a * (b * c) = (a * b) * c$

3. If $*$ is binary operation on set G, a, b of G commute if $a * b = b * a$.
$*$ (or G) is **commutative** if $\forall\, a, b \in G$ $a * b = b * a$.

cf. pp. 16. Sec. 1.1. Basic Axioms and Examples, Dummit and Foote (2004) [25]

Definition 11 (Group)**.** 1. Group is an ordered pair $(G, *)$ where G is a set, $*$ is a binary operation on G s.t.

   (a) $(a * b) * c = a * (b * c)$, $\forall\, a, b, c \in G$, i.e. $*$ *associative*

   (b) $\exists\, e \in G$, s.t. $\forall\, a \in G$, $a * e = e * a = a$ ($\exists$ *identity* e)

   (c) $\forall\, a \in G$, $\exists\, a^{-1} \in G$, called an *inverse* of a, s.t. $a * a^{-1} = a^{-1} * a = e$

2. (optional; abelian or *commutative*) $(G, *)$ abelian (or commutative) if $a * b = b * a$, $\forall\, a, b \in G$.

e.g.

1. $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ are groups under $+$ with $e = 0$ and $a^{-1} = -a$, $\forall\, a$.

2. $\mathbb{Q} - \{0\}$, $\mathbb{R} - \{0\}$, $\mathbb{C} - \{0\}, \mathbb{Q}^+, \mathbb{R}^+$ groups under $\times$ with $e = 1$, $a^{-1} = \frac{1}{a}$

3. (**direct product of groups**)If $(A, *), (B, \circ)$ are groups, we can form new group $A \times B$ called **direct product** s.t.
$$A \times B = \{(a, b) | a \in A, b \in B\}$$
and $(a_1, b_1)(a_2, b_2) = (a_1 * a_2, b_1 \circ b_2)$ cf. Example 6, Sec. 1.1 Dummit and Foote (2004) [25]

Proposition 5. *If G group under operation $*$, then*

1. *identity of G is unique*

2. *$\forall\, a \in G$, a^{-1} uniquely determined.*

3. *$(a^{-1})^{-1} = a \quad \forall\, a \in G$*

4. *$(a * b)^{-1} = (b^{-1}) * (a^{-1})$*

5. *$\forall\, a_1, a_2, \ldots a_n \in G$, $a_1, a_2 \ldots a_n$ independent of how expression is bracketed (generalized associative law)*

cf. Prop. 1, Sec. 1.1 Dummit and Foote (2004)[25]

2.1 $\mathbb{Z}/n\mathbb{Z}$: The Integers Modulo n

cf. Dummit and Foote (2004) [25], Sec. 0.3, $\mathbb{Z}/n\mathbb{Z}$: The Integers Modulo n.

Recall Sec. 0.2 Properties of the Integers, pp. 4, Dummit and Foote (2004) [25]. $\forall\, a, b \in \mathbb{Z}$, $a|b$ if $\exists\, c \in \mathbb{Z}$ s.t. $b = ac$.

Definition 12 (modulo equivalence relation)**.** Let $n \in \mathbb{Z}^+$ (positive integer). Define

$$a \sim b \text{ iff } n|(b-a), \text{ i.e.} \tag{11}$$

$\exists\, c \in \mathbb{Z}$ s.t. $b - a = nc$ or $a + nc = b$. Rewrite this as $a = b - nc \equiv b + Nk$ or even $a = pn + r$.
$b = qn + r$

Proposition 6 (modulo is an equivalence relation)**.** *modulo n is an equivalence relation*

Proof. Clearly $a \sim a$ (choose $k = 0$ in $a = b + Nk$), $a \sim b$ implies $b \sim a$ (choose $k \mapsto -k$) $\forall\, a, b \in \mathbb{Z}$, so $\sim$ is reflexive and symmetric.

If $a \sim b$, and $b \sim c$, $a = b + Nk$, $b = c + Nl$, so $a = c + N(k + l)$, so $a \sim c$, and $\sim$ is transitive.

$\Longrightarrow \sim$ is equivalence relation.

□

$a \equiv b \bmod n$ or $a \equiv b \bmod n$, i.e. a is congruent to $b \bmod n$ if $a \sim b$.

$\forall\, k \in \mathbb{Z}$ denote equivalence class of $a \equiv \overline{a}$, called *congruence class* or **residue class** of $a \bmod n$.

$$\overline{a} = \{a + kn | k \in \mathbb{Z}\} = \{a, a \pm n, a \pm 2n, a \pm 3n, \dots\} \tag{12}$$

If $\overline{a} = pn + r$, $\frac{\overline{a}}{n} = p + \frac{r}{n}$, $0 \le r < n$, so remainder r determines equivalence class $\overline{a}$.

These residue classes partition $\mathbb{Z}$.

Definition 13 ($\mathbb{Z}/n\mathbb{Z}$)**.** $\mathbb{Z}/n\mathbb{Z} \equiv$ set of equivalence classes under this equivalence relation $\equiv$ called integers modulo n (or integers mod n)

Definition 14 (modular arithmetic)**.** $\forall\, \overline{a}, \overline{b} \in \mathbb{Z}/n\mathbb{Z}$, define sum and product by

$$\overline{a} + \overline{b} = \overline{a + b},\ \overline{a} \cdot \overline{b} = \overline{ab} \tag{13}$$

i.e. $\forall\, \overline{a}, \overline{b}$ in $\mathbb{Z}/n\mathbb{Z}$, take *any representive* integer a in class $\overline{a}$ and *any representative* integer b in class $\overline{b}$ and add (respectively, multiply), as usual in $\mathbb{Z}$, and take equivalence class of result.

Theorem 10 (Theorem 3 of Dummit and Foote (2004) [25], Sec. 0.3)**.** *$\overline{a} + \overline{b} = \overline{a+b}$, $\overline{a} \cdot \overline{b} = \overline{ab}$ on $\mathbb{Z}/n\mathbb{Z}$ well-defined, and don't depend on representation choice, i.e. $a_1, a_2 \in \mathbb{Z}$, $b_1, b_2 \in \mathbb{Z}$, with $\overline{a}_1 = \overline{b}_1$, $\overline{a}_2 = \overline{b}_2$, then $\overline{a_1 + a_2} = \overline{b_1 + b_2}$, and $\overline{a_1 a_2} = \overline{b_1 b_2}$*

Proof. Suppose $a_1 = b_1 \bmod n$, so $a_1 = b_1 + sn$, likewise $a_2 = b_2 \bmod n$ means $a_2 = b_2 + tn$, so $a_1 + a_2 = b_1 + b_2 + (s + t)n$, so $\overline{a_1 + a_2} = \overline{b_1 + b_2}$ from $a_1 + a_2 = (b_1 + b_2) \bmod n$

This shows that sum of residue classes is independent of the representatives chosen.

Similarly,

$$a_1 a_2 = (b_1 + sn)(b_2 + tn) = b_1 b_2 + (b_1 t + b_2 s + stn)n \text{ so } a_1 a_2 = b_1 b_2 (\bmod n)$$

□

2.1.1 Elementary number theory application: Last 2 digits of 2^{1000}

What is the last 2 digits of 2^{1000}?

Now $2^{1000}/100 =$ last 2 digits of 2^{1000}. So $2^{1000} \bmod 100 =$?

Observe $2^{10} = 1024$ $\quad$ $2^{10} \bmod 100 = 24$

$$2^{20} \bmod 100 = (2^{10})^2 \bmod 100 = (24)^2 \bmod 100 = 576 \bmod 100 = 76 \bmod 100$$
$$2^{80} \bmod 100 = 2^{160} \bmod 100 = 2^{320} \bmod 100 = 2^{640} \bmod 100 = 76 \bmod 100$$

where $(2^{20})^4 \bmod 100 = 76^4 \bmod 100 = 76 \bmod 100$,
$(76)^2 \bmod 100 = 76 \bmod 100$

So $2^{1000} \bmod 100 = 2^{640}2^{320}2^{40} \bmod 100 = 76 \cdot 76 \cdot 76 \bmod 100 = 76 \bmod 100$ where $(76)^3 \bmod 100 = 76 \bmod 100$.

Note that we used $\overline{a_1 a_2} = \overline{a}_1 \cdot \overline{a}_2$ definition extensively.

$2^{1000} \bmod 100 = 76 \bmod 100$

Collection of residue classes having multiplicative inverse in $\mathbb{Z}/n\mathbb{Z}$ is an important subset of $\mathbb{Z}/n\mathbb{Z}$, $\equiv (\mathbb{Z}/n\mathbb{Z})^\times$

Definition 15 (cf. pp. 10, Dummit and Foote (2003)[25])**.**

$$(\mathbb{Z}/n\mathbb{Z})^\times = \{\overline{a} \in \mathbb{Z}/n\mathbb{Z} | \exists\, \overline{c} \in \mathbb{Z}/n\mathbb{Z} \text{ with } \overline{a} \cdot \overline{c} = 1\} \tag{14}$$

Proposition 7 (Prop. 4, pp. 10, Dummit and Foote (2003) [25])**.**

$$(\mathbb{Z}/n\mathbb{Z})^\times = \{\overline{a} \in \mathbb{Z}/n\mathbb{Z} | (a, n) = 1\}$$

, representatives of residue classes $\overline{a}$ are relatively prime to n.

$(a, n) \equiv \gcd(a, n)$, greatest common divisor of a and n.

Proof. Since $(a, n) = 1$, $1 = sa + tn$ so $as = 1 - tn$, so $as = 1 \bmod n \equiv \overline{1}$, so $\overline{s} = s \bmod n$ is a multiplicative inverse of $\overline{a}$.

$\implies \overline{a} \in (\mathbb{Z}/n\mathbb{Z})^\times$. If $\overline{a} \cdot \overline{c} = \overline{1}$, then $ac = 1 + kn$ or $ac + k'n = 1$

□

Exercise 2. cf. Ex. 2, pp. 11, Dummit and Foote (2003) [25], Sec. 0.3. $\mathbb{Z}/n\mathbb{Z} \ni \overline{a}$ if $\overline{a} = a + kn$.
By Division Algorithm, Thm. 2, $\forall\, a, N \in \mathbb{Z}$, $n \neq 0$, $a = qn + r$, $0 \leq r < |n| = n$.

Comparing expressions $a = r + qn$ and $\overline{a} = a + kn$, then equivalence classes of $\mathbb{Z}/n\mathbb{Z}$ can only be of r s.t. $0 \leq r < |n|$.

Exercise 3. cf. Ex. 3, pp. 11, Dummit and Foote (2003) [25], Sec. 0.3. If $a = \sum_{j=0}^{n-1} a_j b^j$, $b = 10$, $a \in \mathbb{Z}^+$ (positive integer), then prove $a \equiv a_n + a_{n-1} + \cdots + a_1 + a_0 (\text{mod} 9)$.

Observe that $10 \equiv 1 (\text{mod} 9)$. So,

Use the fact that $b \equiv 1 \bmod (b-1)$.

$$a_j b^j \bmod (b-1) = a_j \bmod (b-1) \cdot b^j \bmod (b-1) = a^j \bmod (b-1) \cdot (b \bmod (b-1))^j = a_j \bmod (b-1) 1^j \bmod (b-1) = a^j \bmod (b-1)$$

$$a \bmod (b-1) = \sum_{j=0}^{n-1} a^j b^j \bmod (b-1) = \sum_{j=0}^{n-1} a^j \bmod (b-1) = \left(\sum_{j=0}^{n-1} a_j \right) \bmod (b-1)$$

if $b = 0$, this says that remainder after division by 9 is same as sum of the decimal digits mod 9 - particularly, an integer is divisible by 9 iff sum of digits is divisible by 9.

If $b = 2$, this says a odd (even) if sum of bits is odd (even).

2.1.2 As an abelian group

cf. Dummit and Foote, pp. 17, Ch. 1, Sec. 1.1, Examples, Example 4.

$\mathbb{Z}/n\mathbb{Z}$ is an abelian group under operation $+$ of addition of residue classes, as described in Ch. 0. In Ch. 3 of Dummit and Foote (2003) [25], prove binary operation $+$ is well-defined and associative.

$(\overline{a})^{-1} = \overline{-a} \quad \forall\, a \in \mathbb{Z}/n\mathbb{Z}$, identity of $\angle \mathbb{Z}/n\mathbb{Z}, + \rangle = \overline{0}$.

Exercise 3. cf. Exercise 3 in pp. 21, Exercises, of Sec. 1.1, Dummit and Foote (2003) [25].

$$\overline{a} + (\overline{b} + \overline{c}) = (\overline{a} + \overline{b}) + \overline{c}$$

is associative, since,

$$\overline{a} + (\overline{b} + \overline{c}) = \overline{a} + (\overline{b + c}) = \overline{a} + (\overline{b + tn + c + un}) = \overline{a} + \overline{b + c} = \overline{a + b + c}$$
$$(\overline{a} + \overline{b}) + \overline{c} = (\overline{a + b}) + \overline{c} = (\overline{a + sn + b + tn}) + \overline{c} = \overline{a + b} + \overline{c} = \overline{a + b + c}$$

2.2 Cyclic groups and cyclic subgroups

cf. Sec. 2.3, pp. 54, Dummit and Foote (2003), [25]

Definition 16 (Cyclic group)**.** group H is **cyclic** if H can be generated by a single element, i.e. $\exists x \in H$ s.t. $H = \{x^n | n \in \mathbb{Z}\}$ ("multiplication" operation notation) $\equiv H = \{nx | n \in \mathbb{Z}\}$ ("additive" operation notation).

In both cases, write $H = \langle x \rangle$ and say H *generated* by x (i.e. x is *generator* of H)

cyclic group may have more than 1 generator; e.g. $H = \langle x \rangle$ and $H = \langle x^{-1} \rangle$ because $(x^{-1})^n = x^{-n}$ and $-n \in \mathbb{Z}$.

Exercise 19. cf. Exercise 19, Sec. 1.1, pp. 22, Dummit and Foote (2003) [25].
Let $x \in G, a, b \in \mathbb{Z}^+$
In the following steps, I'll use induction a lot.

(a) $x^{a+b} = x^a x^b$ and $(x^a)^b = x^{ab}$, prove.

$$x^a x^b = x^a \cdot x \cdot x^{b-1} = x^{a+1} x^{b-1} = \cdots = x^{a+b-1} \cdot x = x^{a+b} \quad \text{(used associativity of group operation)}$$

$$(x^a)^b = \underbrace{x^a \cdot x^a \cdots x^a}_{b \text{ times}} = x^{2a} \cdot \underbrace{x^a \ldots x^a}_{b-2 \text{ times}} = x^{3a} \cdot \underbrace{x^a \ldots x^a}_{b-3 \text{ times}} = \cdots = x^{(b-1)a} \cdot x^a = x^{(b-1)a+a} = x^{ba}$$

(b)

$$x^a \cdot x^{-a} = x^a \cdot (x^{-1})^a = x^{a-1} \cdot 1 \cdot (x^{-1})^{a-1} = \cdots = x \cdot x^{-1} = 1 \implies (x^a)^{-1} = x^{-a}$$

The steps previous depended upon defining $x^{-a} = (x^{-1})^a$

(c) if $b = 0$, $x^{a+0} = x^a = x^a x^b \implies x^0 = 1$. $x^{ab} = x^0 = (x^a)^0 = 1$.

$x^{a-b} = x^a \cdot x^{-b}$ since $x^{-b} = (x^{-1})^b$.

$(x^a)^{-b} = ((x^a)^{-1})^b = (x^{-a})^b = x^{-ab}$

So $x^{a+b} = x^a x^b$, $(x^a)^{-1} = x^{-a} \ \forall a, b \in \mathbb{Z}$

Theorem 11 (Cyclic groups are abelian)**.** *Cyclic groups are abelian since* $x^a x^b = x^{a+b} = x^{b+a} = x^b x^a$

Proposition 8. *cf. Dummit and Foote (2003) [25], Prop. 3, pp. 55, Sec. 2.3*
Let arbitrary group G, $x \in G$, *let* $m, n \in \mathbb{Z}$.

If $x^n = 1$, $x^m = 1$, *then* $x^d = 1$, *where* $d = (m, n) \equiv \gcd(m, n)$
If $x^m = 1$, *for some* $m \in \mathbb{Z}$, *then* $|x|$ *divides* m *(recall* $|x| \equiv$ *order of* $x :=$ *smallest positive integer* n *s.t.* $x^n = 1$, *Def. on pp. 20, Sec. 1.1 of Dummit and Foote (2003) [25]).*

Proof. By Euclidean Algorithm (Sec. 0.2, (6) of Dummit and Foote (2003) [25]), $\exists r, s \in \mathbb{Z}$ s.t. since $d = \gcd(m, n)$, $d|m$ and $d|n$, i.e. $m/d, n/d \in \mathbb{Z}$, i.e. $m = pd + r$, $n = qd + s$, with $r = s = 0$.

Consider $mr + ns = pdr + qds = (pr + qs)d$. Choose r, s s.t. $pr + qs = 1$ (TODO: need to prove Euclidean Algorithm)

$x^d = x^{mr+ns} = (x^m)^r (x^n)^s = 1^r 1^s = 1$.

If $x^m = 1$, let $n = |x|$. If $m = 0$, $n|m$, i.e. $m/n = 0/n$, so assume $m \neq 0$.

Since some nonzero power of x, m, is identity $n < \infty$.

Let $d = (m, n) = \gcd(m, n)$, so $x^d = 1$ by preceding result for Prop. 8.

Since $0 < d \leq n$, and n is smallest positive power of x s.t. $x^n = 1$ (def. of order of a group), $d = n$, so $n|m \equiv m/n = m/|x|$. □

Theorem 12 (Dummit and Foote, Thm. 4 (2003) [25])**.** *Any 2 cyclic groups of the same order are isomorphic; more specifically,*

(1) if $n \in \mathbb{Z}^+$, $\langle x\rangle, \langle y\rangle$ are cyclic groups of order n, then

$$\begin{aligned} \varphi : &\langle x\rangle \to \langle y\rangle \\ &x^k \mapsto y^k \end{aligned} \tag{15}$$

is well-defined and an isomorphism.

(2) If $\langle x\rangle$ is infinite cyclic group, map φ

$$\begin{aligned} \varphi : &\mathbb{Z} \to \langle x\rangle \\ &k \mapsto x^k \end{aligned} \tag{16}$$

is well-defined and is an isomorphism.

Proof. Suppose $\langle x\rangle, \langle y\rangle$ are cyclic groups of order n. Let $\varphi : \langle x\rangle \to \langle y\rangle$ s.t. $\varphi(x^k) = y^k$.

φ **well-defined**? *Prove*:
if $x^r = x^s$, then $\varphi(x^r) = \varphi(x^s)$.

Since $x^{r-s} = 1$, by Prop. 8, $|x||r-s$, i.e. $r - s/|x|$, so $r - s = |x|$ or $r = tn + s$, so

$$\varphi(x^r) = \varphi(x^{tn+s}) = y^{tn+s} = (y^n)^t y^s = 1^t y^s = y^s = \varphi(x^s)$$

φ well-defined.

$\varphi(x^a x^b) = \varphi(x^{a+b}) = y^{a+b} = y^a y^b = \varphi(x^a)\varphi(x^b)$, so φ homomorphism.

Since $y^k = \varphi(x^k)$, φ surjective.

Since both groups have same finite order, any surjection from 1 to the other is bijection, so φ isomorphism.

Alternatively, $\varphi^{-1} : \langle y\rangle \to \langle x\rangle \quad \varphi^{-1}(y^k) = x^k \qquad$ and $\qquad \begin{matrix} \varphi \circ \varphi^{-1}(y^k) = y^k \\ \varphi^{-1} \circ \varphi(x^k) = x^k \end{matrix}$.

If $\langle x\rangle$ infinite cyclic group, let $\varphi : \mathbb{Z} \to \langle x\rangle$ defined by $\varphi(k) = x^k$. φ well-defined since if $k = l$ for $x, y \in \mathbb{Z}$,

$$x^k = x^l = \varphi(k) = \varphi(l)$$

By Prop. 2 of Dummit and Foote (2003) [25], pp. 55, Sec. 2.3, (if $H = \langle x\rangle$, $|H| = |x|$), if $x^a \neq x^b$, $\forall\, a, b \in \mathbb{Z}$, $a \neq b$, φ injective. $\forall\, x^k \in \langle x\rangle$, $k \in \mathbb{Z}$, so $\varphi(k) = x^k$ is surjective.
Law of exponents ensure φ homomorphism, (as above), hence φ isomorphism.

□

Theorem 13 (Cyclic Group and Integer Modulo N are isomorphic)**.** *Up to isomorphism, Z_n is the unique cyclic group of order n and $Z_n \simeq \mathbb{Z}/n\mathbb{Z}$*

2.3 Binary Representation, fixed-width integers as Abelian group

For $x \in \mathbb{Z}$, $B2U(x) = \sum_{i=0}^{w-1} x_i \cdot 2^i \equiv \sum_{i=0}^{w-1} x_i b^i$, where $b = 2$, $w =$ total number of bits to represent x.

Consider $n = 2^w$.

Observe that if $\sum_{i=0}^{w-1} 1 \cdot 2^i = S$, $2S = \sum_{i=1}^{w-1} 2^i + 2^w = S - 1 + 2^w$. So $S = 2^w - 1$, and so consider integers modulo $n = 2^w$.

$$\mathbb{Z}/2^w\mathbb{Z} = \{\overline{a} = \{a + kn | k \in \mathbb{Z}\}\} = \{\overline{a} = \{a + k2^w | k \in \mathbb{Z}\}\}$$

and was shown that $0 \leq a < 2^w$.

Observe that if $x = 2^w + \sum_{i=0}^{w-1} x_i \cdot 2^i$,

since $x \leq 2^w - 1, \quad \forall\, x, y \in \mathbb{Z}/2^w\mathbb{Z},\ x + y \leq 2 \cdot 2^w - 2 = 2^{w+1} - 2 = \sum_{i=0}^{w} 2^i + 1 - 2 = \sum_{i=0}^{w} 2^i - 1 \sim \left(\sum_{i=0}^{w-1} 2^i -\right) \bmod 2^w$

Consider $2^w - 1 = \sum_{i=0}^{w-1} 1 \cdot 2^i$. Then $2^w = \sum_{i=0}^{w-1} 1 \cdot 2^i + 1 = \sum_{i=0}^{w} x_i \cdot 2^i$ where $x_i = 0$ unless $i = w$, so $x_w = 1$. Define one's complement addition s.t. $2^w \bmod 2^w = 0 \bmod 2^w$

Then $B2U(x) \bmod 2^w$ is a cyclic group s.t. $(\overline{1})^{2^w} \equiv \sum_{i=1}^{2^w} 1 = \sum_{i=0}^{2^w-1} 1 = 2^w \bmod 2^w = 0 \bmod w^2$, $\forall\, w \in \mathbb{Z}^+$.

3 Groups; normal subgroups

Definition 17 (normal subgroup $K \lhd G$)**.**
normal subgroup K of $G \equiv K \lhd G$ -
subgroup $K \subset G$, if $\forall\, k \in K$, $\forall\, g \in G$,

$$gkg^{-1} \in K$$

Definition 18 (quotient group)**.**
quotient group $G \mod K \equiv G/K$ -
if G/K = family of all left cosets of subgroups $K \subset G =$

$$= \{gK | g \in G, gK = \{gk | k \in K\}$$

and
K = normal subgroup of G, i.e. $K \lhd G$, and so

$$aKbK = abK \qquad \forall\, a, b \in G,$$

so G/K group.

Definition 19 (exact sequence of groups)**. exact sequence** if $\mathrm{im} f_{n+1} = \ker f_n \qquad \forall\, n$ for sequence of group homomorphisms and groups

$$G_{n+1} \xrightarrow{f_{n+1}} G_n \xrightarrow{f_n} G_{n-1} \tag{17}$$

Theorem 14. *1.*

$$1 \qquad A \xrightarrow{f} B$$

2.

$$B \xrightarrow{g} C \qquad 1$$

3.

$$1 \qquad A \xrightarrow{h} B \qquad 1$$

Proof. 1. $\mathrm{im}(1 \to A) = 1$, since $1 \to A$ is a group homomorphism ($(1 \to A)(1) = 1_A$).
if $1 \to A \overset{f}{\mapsto} B$ exact, $\ker f = \mathrm{im}(1 \to A) = 1$, so if $f(x) = 1$, $x = 1$, f injective.
If f injective, $\ker f = 1$. $1 = \mathrm{im}(1 \to A)$. $1 \to A \overset{f}{\mapsto} B$, exact.

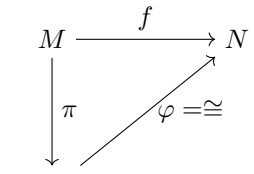
2. $\ker(C \to 1) = C$, by def. of $C \to 1$
if $B \overset{g}{\mapsto} C \to 1$ exact, $\mathrm{im} g = g(B) = \ker(C \to 1) = C$. $g(B) = C$ implies g surjective.
If g surjective, $g(B) = C = \ker(C \to 1)$. $B \overset{g}{\mapsto} C \to 1$ exact.

3. From (i), $1 \to A \overset{h}{\mapsto} B$ exact iff h injective. From (ii), $A \overset{h}{\mapsto} B \to 1$, exact iff h surjective.
h isomorphism.

□

3.1 1st, 2nd, 3rd Isomorphism Theorems

Theorem 15 (1st Isomorphism Theorem (Modules) Thm. 7.8 of Rotman (2010) [57]). *If $f : M \to N$ is R-map of modules, then $\exists$ R-isomorphism s.t.*



$$\begin{aligned} &\varphi : M/kerf \to imf \\ &\varphi : m + kerf \mapsto f(m) \end{aligned} \tag{18}$$

Proof. View M, N as abelian groups.

Recall natural map $\pi : M \to M/N$

$$m \mapsto m + N$$

Define φ s.t. $\varphi\pi = f$.

(φ well-defined). Let $m + \ker f = m' + \ker f$, $m, m' \in M$, then $\exists\, n \in \ker f$ s.t. $m = m' + n$.

$$\varphi(m + \ker f) = \varphi\pi(m) = f(m) = f(m' + n) = f(m') + f(n) = \varphi\pi(m') + 0 = \varphi(m' + \ker f)$$

$\implies \varphi$ well-defined.

(φ surjective). Clearly, $\text{im}\varphi \subseteq \text{im} f$.
Let $y \in \text{im} f$. So $\exists\, m \in M$ s.t. $y = f(m)$. $f(m) = \varphi\pi(m) = \varphi(m + \ker f) = y$. So $y \in \text{im}\varphi$. $\text{im} f \subseteq \text{im}\varphi$.
$\implies \varphi$ surjective.

(φ injective) If $\varphi(a + \ker f) = \varphi(b + \ker f)$, then

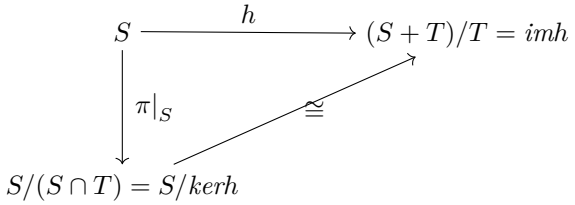
$\varphi\pi(a) = \varphi\pi(b)$ or $f(a) = f(b)$ or $0 = f(a) - f(b) = f(a - b)$ so $a - b \in \ker f (a - b) + \ker f = \ker f$ so $a + \ker f = b + \ker f$

φ isomorphism.

φ R-map. $\varphi(r(m + N)) = \varphi(rm + N) = f(rm)$.
Since f R-map, $f(rm) = rf(m) = r\varphi(m + N)$. φ is R-map indeed.

□

Theorem 16 (2nd Isomorphism Theorem (Modules) Thm. 7.9 of Rotman (2011) [57]). *If S, T are submodules of module M, i.e. $S, T \in M$, then $\exists$ $R-$isomorphism*



$$S/(S \cap T) \to (S + T)/T \tag{19}$$

Proof. Let natural map $\pi : M \to M/T$.

So $\ker\pi = T$.

Define $h := \pi|_S$, so $h : S \to M/T$, so $\ker h = S \cap T$,

$$(S + T)/T = \{(s + t) + T | a \in S + T, s \in S, t \in T\}$$

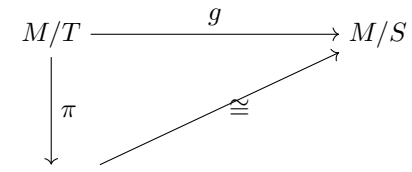
i.e. $(S + T)/T$ consists of all those cosets in M/T having a representation in S.

By 1st. isomorphism theorem,

$$S/S \cap T \xrightarrow{\cong} (S + T)/T$$

□

Theorem 17 (3rd Isomorphism Theorem (Modules) Thm. 7.10 of Rotman (2011) [57])**.** *If $T \subseteq S \subseteq M$ is a tower of submodules, then $\exists$ R-isomorphism*



$$(M/T)/(S/T) \to M/S \tag{20}$$

Proof. Define $g : M/T \to M/S$ to be **coset enlargement**, i.e.

$$g : M + T \mapsto m + S \tag{21}$$

g well-defined: if $m + T = m' + T$, then $m - m' \in T \subseteq S$, and $m + S = m' + S \implies g(m + T) = g(m' + T)$

$\ker g = S/T$ since

$$g(s + T) = s + S = S \qquad (S/T \subseteq \ker g)$$
$$g(m + T) = m + S = 0 = S = s + S, \text{ so } m = s \implies \ker g \subseteq S/T$$

$\text{im} g = M/S$ since

$$g(m + T) = m + S \implies \text{im} g \subseteq M/S$$
$$m + S = g(m + T)$$

Then by 1st isomorphism, and commutative diagram, done. □

4 Rings

cf. Ch. 7 "Introduction to Rings" pp. 223, Dummit and Foote (2014)[25]

Definition 20 (Ring)**.** ring R is a set, together with 2 binary operations $+, \times$ (addition and multiplication, $\times \equiv \cdot$) s.t.

1. (a) $(R, +)$ abelian group
   (b) $\times$ associative: $a(bc) = (ab)c \quad \forall\, a, b, c \in R$
   (c) distributivity in R: $\forall\, a, b, c \in R$

$$(a + b)c = ac + bc \text{ and } a(b + c) = ab + ac$$

2. R *commutative* if multiplication commutative
3. R has an identity 1 if $\exists 1 \in R$ s.t.

$$1a = a1 = a \quad \forall\, a \in R$$

Definition 21 (division ring)**.** ring R with identity 1, where $1 \neq 0$ is a **division ring** (or skew field) if $\forall\, a \in R$, $a \neq 0$, $\exists$ multiplicative inverse $1/a$, i.e. $\exists\, b \in R$ s.t. $ab = ba = 1$

e.g.

1. rational numbers $\mathbb{Q}$
   real numbers $\mathbb{R}$
   complex numbers $\mathbb{C}$
   are commutative rings with identity (in fact, they're fields)
   Ring axioms for each follow ultimately from ring axioms for $\mathbb{Z}$
   (verified when $\mathbb{Z}$ constructed from $\mathbb{Z}$ (Sec. 7.5)), $\mathbb{C}$ from $\mathbb{R}$ (Example 1, Sec. 13.1).
   Construction of $\mathbb{R}$ from $\mathbb{Z}$ carried out in basic analysis texts

2. **quotient group** $\mathbb{Z}/n\mathbb{Z}$ is a commutative ring with identity (element 1) under operations of addition and multiplication of residue classes (frequently referred to as "modular arithmetic").
We saw additive abelian groups axioms followed from general principles of theory of quotient groups ($\mathbb{Z}/n\mathbb{Z}$) was prototypical quotient group. cf. Example 4, pp. 224, Dummit and Foote (2014)[25]

3. **the (real) Hamiltonian Quaternions**.

Definition 22 ((real) Hamiltonian Quaternions)**.** Let $\mathbb{H} = \{a + bi + cj + dk | a, b, c, d \in \mathbb{R}\}$ s.t. "componentwise" addition is defined as

$$(a + bi + cj + dk) + (a' + b'i + c'j + d'k) = (a + a') + (b + b')i + (c + c')j + (d + d')k \tag{22}$$

and multiplication defined by expanding using distributive laws

$$(a + bi + cj + dk)(a' + b'i + c'j + d'k)$$

using

$$\begin{aligned} i^2 = j^2 = k^2 = -1 \\ ij = -ji = k \\ jk = -kj = i \\ ki = -ik = j \end{aligned} \tag{23}$$

Working out the multiplication

$$\begin{aligned} (a + bi + cj + dk)(a' + b'i + c'j + d'k) = \\ = \begin{array}{l} aa' + ab'i + ac'j + ad'k + ba'i - bb' + bc'k - bd'j + \\ ca'j - cb'k - cc' + cd'i + da'k + db'j - dc'i - dd' = \end{array} \\ = aa' - bb' - cc' - dd' + (ab' + ba' + cd' - dc')i + (ac' - bd' + ca' + db')j + (ad' + bc' - cb' + da')k \end{aligned}$$

Hamiltonian Quaternions are noncommutative ring with identity ($1 = 1 + 0i + 0j + 0k$).

Similarly define *rational* Hamiltonian Quaternions ring by taking $a, b, c, d \in \mathbb{Q}$.

real and rational Hamiltonian Quaternions both are divison rings, where inverse of nonzero element defined as

$$(a + bi + cj + dk)^{-1} = \frac{a - bi - cj - dk}{a^2 + b^2 + c^2 + d^2} \tag{24}$$

cf. Example 5, pp. 224, Dummit and Foote (2014)[25]

4. **rings of functions** (important class)
Let X be any nonempty set.
Let A be any ring.

Definition 23 (function ring)**.** collection $R = \{f : X \to A\}$ is a ring under pointwise addition and multiplication of functions s.t.

$$\begin{aligned} (f + g)(x) = f(x) + g(x) \\ (fg)(x) = f(x)g(x) \end{aligned} \tag{25}$$

cf. Example 6, pp. 225, Dummit and Foote (2014)[25]

5. cf. Example 7 of p. 225, Dummit and Foote (2014)[25] *ring which doesn't have an identity*:

- ring $2\mathbb{Z}$ of even integers (sum and product of even integers is an even integer)

- function $f : \mathbb{R} \to \mathbb{R}$ has compact support if $\exists\, a, b \in \mathbb{R}$, (depending on f) s.t. $f(x) = 0, \quad \forall\, x \notin [a, b]$
$\{f : \mathbb{R} \to \mathbb{R} | f$ has compact support $\}$ is a commutative ring without identity (since an identity couldn't have compact support),
Similarly,
$\{$ cont. $f : \mathbb{R} \to \mathbb{R} | f$ has compact support $\}$ is a commutative ring without identity.

Proposition 9 (Dummit and Foote (2014)[25], Prop. 1, pp. 226)**.** *Let R be a ring. Then*

1. $0a = a0 = 0 \quad \forall\, a \in R$
2. $(-a)0 = a(-b) = -(ab) \quad \forall\, a, b \in R$ *(recall $-a$ is additive inverse of a)*
3. $(-a)(-b) = ab \qquad \forall\, a, b \in R$
4. *if R has identity 1, identity unique and $-a = (-1)a$*

Proof. Use distributivity and additive existence inverse for abelian group $(R, +)$.

1. $$0a = (0+0)a = 0a + 0a \to 0a = 0a0 = a(0+0) = a0 + a0 \text{ so } a0 = 0$$
2. $$ab + (-a)b = (a + (-a))b = 0b = 0 \implies (-a)b = -(ab)$$
   $$ab + a(-b) = a(b + (-b)) = a0 = 0 \implies a(-b) = -(ab)$$
3. $$(-a)(b + (-b)) = (-a)b + (-a)(-b) = -(ab) + (-a)(-b) = 0 \implies (-a)(-b) = ab$$
4. $$(-1)a + a = (-1)a + 1a = (-1+1)a = 0a = 0 \text{ so } \; -a = (-1)a$$
   Suppose $\exists\, 1'$ s.t. $1'a = a1' = a$ and $1' \neq 1$
   $$-a + a = 0 = (-1)a + 1'a = (-1+1')a = 0a \implies 0 = -1 + 1'$$
   Then $1' = 1$. Contradiction.

□

Definition 24. Let R be a ring.

1. **zero divisor** a, $a \neq 0$, $a \in R$ if $\exists\, b \in R$, $b \neq 0$ so either $ab = 0$ or $ba = 0$.
2. Assume R has identity $1 \neq 0$.
   unit in R, $u \in R$ if $\exists$ some $v \in R$ s.t. $uv = vu = 1$.
   $R^x \equiv$ set of units in R.
   $$\forall\, u, v \in R^x,\ (uv)(v^{-1}u^{-1}) = u(1)u^{-1} = 1$$
   $1 \in R^x$ so, $1u = u1 = u$
   if $u \in R^x$, $u^{-1} \in R^x$ since $uu^{-1} = u^{-1}u = 1$ R^x is a **group of units**

Thus a *field* = commutative ring F with identity $1 \neq 0$ s.t. $\forall\, a \in$ field, $a \neq 0$ is a unit, i.e. $F^x = F - \{0\}$.

Proposition 10. *zero divisor can never be a unit.*

Proof. Suppose a unit in R, and $ab = 0$ for some $b \neq 0$, $b \neq R$.
Then $va = 1$ s.t. some $v \in R$ and
$$b = 1b = vab = v(ab) = v(0) = 0$$
Contradiction. □

5 Commutative Rings

cf. Ch. 3 "Commutative Rings I" of Rotman (2010) [57]

Definition 25. commutative ring R is a set with 2 binary operations, addition and multiplication, s.t.

(i) R abelian group under addition

(ii) (commutativity) $ab = ba \quad \forall\, a, b \in R$ (this isn't there for noncommutativity)

(iii) (associativity) $a(bc) = (ab)c \qquad \forall\, a, b, c \in R$

(iv) $\exists\, 1 \in R$ s.t. $1a = a \quad \forall\, a \in R \qquad$ (many names used: one, unit, identity)

(v) (distributivity) $a(b+c) = ab+ac \qquad a, b, c \in R$ (this splits up into 2 distributivity laws for noncommutativity)

To reiterate, abelian group under addition R (is defined as)

1. associative $\forall\, x, y, z \in R$, $x + (y + z) = (x + y) + z$
2. $\exists\, 0 \in R$, $0 + x = x + 0, \quad \forall\, x \in R$
3. $\forall\, x \in R$, $\exists\, (-x) \in R$ s.t. $x + (-x) = 0 = (-x) + x$

abelian, if commutativity: $x + y = y + x$.

5.1 Linear Algebra; Linear Algebra with commutative rings as fields

5.1.1 Linear Algebra

Definition 26 (subspace)**.** If V vector space over field k,
then **subspace** of V is subset U of V s.t.

1. $0 \in U$
2. $u, u' \in U$ imply $u + u' \in U$
3. $u \in U$, and $a \in k$ imply $au \in U$

proper subspace of $V \equiv U \subsetneq V$ is subspace $U \subseteq V$ with $U \neq V$.

$U = V$, $U = \{0\}$ are always subspaces of a vector space V.
Examples (Example 3.70 Rotman (2010) [57])

(ii) If $V = (a_1, \dots a_n)$, $v \neq 0$, $v \in \mathbb{R}^n$,
line through origin $l = \{av | a \in \mathbb{R}\}$ is a subspace of $\mathbb{R}^n$.
plane through origin $= \{av_1 + bv_2 | v_1, v_2$ fixed pair of noncollinear vectors, $a, b \in \mathbb{R}\}$ are subspaces of $\mathbb{R}^n$

(iii) If $m \leq n$, $\mathbb{R}^m$ regarded as set of all vectors in $\mathbb{R}^n$ s.t. last $n - m$ coordinates are 0, then $\mathbb{R}^m$ subspace of $\mathbb{R}^n$. e.g. $\mathbb{R}^2 = \{(x, y, 0) \in \mathbb{R}^3\} \subsetneq \mathbb{R}^3$

(iv) If k field, **homogeneous linear system over** k of m equations in n unknowns is a set of equations

$$\begin{aligned} a_{11}x_1 + \cdots + a_{1n}x_n &= 0 \\ a_{21}x_1 + \cdots + a_{2n}x_n &= 0 \\ &\vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n &= 0 \end{aligned}$$

where $a_{ji} \in k$.

solution of this system is vector $(c_1 \dots c_n) \in k^n$ s.t. $\sum_i a_{ji}c_i = 0$, $\forall\, j$.
solution $(c_1 \dots c_n)$ **nontrivial** if $\exists$ some $c_i \neq 0$.
solution space (or null space) of system = set of all solutions.
solution space also a subspace of k^n

e.g. $k = \mathbb{I}_p$,

$$3x - 2y + z \equiv 1 \bmod 7$$
$$x + y - 2z \equiv 0 \bmod 7$$
$$-x + 2y + z \equiv 4 \bmod 7$$

Definition 27 (list)**.** list := vector space V is ordered set $v_1 \dots v_n$ of vectors in V, i.e. $\exists$ some $n \geq 1$, $\exists$ some function φ

$$\varphi : \{1, 2 \dots n\} \to V$$

with $\varphi(i) = v_i \quad \forall\, i$

Thus, $X = \text{im}\varphi$.
X ordered, φ need not be injective.

Definition 28 (k-linear combination)**.** k-linear combination of list $v_1 \dots v_n$ in V, $V \equiv$ vector space over field k, is vector v of form

$$v = a_1 v_1 + \cdots + a_n v_n = \sum_{i=1} a_i v_i \quad \forall\, a_i \in k, \quad \forall\, i$$

Definition 29 (list)**.** If list $X = v_1 \dots v_m$ in vector space V, then
subspace spanned by X, $\langle v_1 \dots v_m \rangle :=$ set of all k-linear combinations of $v_1 \dots v_m$. Also, say $v_1 \dots v_m$ spans $\langle v_1 \dots v_m \rangle$.

Lemma 1 ($\langle v_1 \dots v_m \rangle$ is smallest subspace of V containing $v_1 \dots v_m$)**.** *(i) Every intersection of subspaces of V is itself a subspace.*

(ii) If $X = v_1 \dots v_m$ list in V, then intersection of all subspaces of V containing X is $\langle v_1 \dots v_m \rangle$, subspace spanned by $v_1 \dots v_m$, so $\langle v_1 \dots v_m \rangle$ is smallest subspace of V containing X.

cf. (Lemma 3.71 Rotman (2010) [57])

Proof. (i) Consider $\bigcap_{\alpha \in I} V_\alpha$, $\forall\, \alpha \in I$, V_α subspace of V

(i) $0 \in V_\alpha$, $\forall\, \alpha \in I$, so $0 \in \bigcap_{\alpha \in I} V_\alpha$,

(ii) Let $u, u' \in \bigcap_{\alpha \in I} V_\alpha$. Then $u, u' \in V_\alpha$, $\forall\, \alpha \in I$. Consider $\beta \in I$. $u, u' \in V_\beta$, so $u + u' \in V_\beta$. Without loss of generality, $u + u' \in V_\alpha$, $\forall\, \alpha \in I$. Then $u + u' \in \bigcap_{\alpha \in I} V_\alpha$

(iii) Let $u \in \bigcap_{\alpha \in I} V_\alpha$. Consider $\alpha \in k$. Since $u \in V_\alpha$, $\forall\, \alpha \in I$, $au \in V_\alpha$, $\forall\, \alpha \in I$.
Then $au \in \bigcap_{\alpha \in I} V_\alpha$

(ii) Let $X = \{v_1 \dots v_m\}$, let $\mathcal{S} \equiv$ family of all subspaces of V containing X.

$\bigcap_{S \in \mathcal{S}} S \subseteq \langle v_1 \dots v_m \rangle$ because $\langle v_1 \dots v_m \rangle \in \mathcal{S}$, since,
$\langle v_1 \dots v_m \rangle$ is *a* subspace of V containing X.

If $S \in \mathcal{S}$, then $S \ni v_1 \dots v_m$. As shown above, $\forall\, v \in \langle v_1 \dots v_m \rangle$, $v \in S$, and thus $v \in \bigcap_{S \in \mathcal{S}} S$. $\langle v_1 \dots v_m \rangle \subseteq \bigcap_{S \in \mathcal{S}} S$.

□

Were all terminology in algebra consistent,
$\langle v_1 \dots v_m \rangle \equiv$ subspace *generated* by X.
Reason for different terms is that group theory, rings, vector spaces developed independently of each other.
Example 3.72 of Rotman (2010) [57]

(i)

(ii)

(iii) **polynomial vector space; polynomials as a vector space**.
Vector space need not be spanned by finite list.
e.g. $V = k[x]$,
Suppose $X = f_1(x) \dots f_m(x)$ finite list in V.
If $d =$ largest degree of any of $f_i(x)$,
then every (nonzero) k-linear combination of $f_1(x), \dots f_m(x)$ has degree at most d.
Thus $x^{d+1} \notin \langle f_1(x) \dots f_m(x) \rangle$, so X doesn't span $k[x]$

Definition 30 (finite-dimensional vector space; infinite-dimensional vector space)**.** Vector space V is **finite-dimensional** if it's spanned by a finite list; otherwise V is **infinite-dimensional**.

Proposition 11 (linear dependent span properties)**.** *If vector space V, list $X = v_! \dots v_m$ spanning V, following are equivalent:*

(i) X isn't shortest spanning list

(ii) some v_i is in subspace spanned by others, i.e. $v_i \in \langle v_i \dots \widehat{v_i} \dots v_m \rangle$,

(iii) $\exists\, a_1 \dots a_m$ not all 0 s.t. $\sum_{l=1}^m a_l v_l = 0$

Proof. (i) $\implies$ (ii). If X isn't shortest spanning list, then 1 of vectors in X can be thrown out, and shorter list still spans, i.e. cf. Lemma 1(Lemma 3.71, Rotman (2010) [57]); let $\mathcal{S} \equiv$ family of all subspaces of V containing X.

EY: 20180610 Let $\bigcap_{S\in\mathcal{S}} S$. $\bigcap_{S\in\mathcal{S}} S \neq \langle v_1 \dots v_m \rangle$, $\bigcap_{S\in\mathcal{S}} S \subset \langle v_1 \dots v_m \rangle$
$\exists\, v \in \langle v_1 \dots v_m \rangle$, say$v = \sum_{i=1}^m a_i v_i$ s.t. $\exists\, S \in \mathcal{S}$, s.t. $v \notin S$.

(ii) $\implies$ (iii) If $v_i = \sum_{j\neq i} c_j v_j$, define $a_i = -1 \neq 0$, $a_j = c_j$, $\forall\, j \neq i$. Then $\sum_{l=1}^m a_l v_l = -v_i + \sum_{j\neq i} c_j v_j = 0$

(iii) $\implies$ (i) Suppose for $i \in 1 \dots m$, $a_i \neq 0$. $v_i = -\sum_{j\neq i} \frac{a_j}{a_i} v_j$. $\langle v_1 \dots \widehat{v_i} \dots v_m \rangle$ still spans V (i.e. deleting v_i gives a shorter list, which still spans).

For instance, if $v \in \langle v_1 \dots v_m \rangle$, $v = \sum_{l=1}$

□

Exercise 3.67. Suppose $\dim V > 1$. Then $\exists$ at least 2 elements in a basis of V, say e_1, e_2. (Thm. 3.78 of Rotman (2010) [57], "Every finite-dim. vector space V has a basis; Def. of dim, "number of elements in a basis of V").

Consider subspaces $\langle e_1 \rangle$, $\langle e_2 \rangle$, subspaces spanned by e_1, e_2, respectively. Whether $V = \langle e_1, e_2 \rangle$ or $V = \langle e_1, e_2 \rangle$, $\langle e_1 \rangle, \langle e_2 \rangle \neq \{0\}$ nor V. Contradiction of hypothesis.

Thus, "If only subspaces of a vector space V are $\{0\}$ and V itself, $\dim(V) \leq 1$."

Proposition 12 (Matrix representation of linear transformation; 3.94 of Rotman (2010) [57])**.** *If linear transformation $T : k^n \to k^m$, then $\exists\, A \in Mat_k(m, n)$ s.t.*

$$T(y) = Ay, \qquad \forall\, y \in k^n$$

Proof. Let $\begin{matrix}(e_1 \dots e_n) & \text{standard basis of } k^n \\ (e'_1 \dots e'_m) & \text{standard basis of } k^m\end{matrix}$
Define $A = [a_{ij}]$, s.t. $T(e_j) = A_{*j} = A_{ij} e'_i$ (jth column),
If $\begin{matrix}S : k^n \to k^m \\ S(y) = A(y)\end{matrix}$, then

$$T(e_j) = a_{ij} e'_i = Ae_j$$

and so $\forall\, y = y_j e_j \in k^n$,

$$T(y) = T(y_j e_j) = y_j T(e_j) = y_j A_{ij} e'_i = Ay$$

□

6 Modules

6.1 R-modules

cf. Sec. 7.1 Modules of Rotman (2010) [57]

Definition 31 (R-module)**.** R-module is (additive) abelian group M,

equipped with scalar multiplication $R \times M \to M$
$(r, m) \mapsto rm$
s.t. $\forall\, m, m' \in M$, $\forall\, r, r', 1 \in R$

(i) $r(m + m') = rm + rm'$

(ii) $(r+r')m = rm + r'm$

(iii) $(rr')m = r(r'm)$

(iv) $1m = m$

Example 7.1

(i) $\forall$ *vector space* over field k is a k-module. (by inspection of the axioms for a vector space, associativity, distributivity!)

(ii) $\forall$ abelian group is a $\mathbb{Z}$-module, by laws of exponents (Prop. 2.23)

Indeed, for

$$\begin{aligned} \mathbb{Z} \times M &\to M \\ (r, m) &\mapsto rm \equiv m^r \end{aligned}$$

and so

$$r(m \cdot m') \equiv (m \cdot m')^r = m^r (m')^r = rm + rm'$$

(since M abelian)

(iii) For commutative ring, scalar multiplication, defined to be given multiplication of elements of R

$$\begin{aligned} R \times R &\to R \\ (a, b) &\mapsto ab \end{aligned}$$

For reference, recall some of the properties of a commutative ring:

$$\begin{aligned} &ab = ba \\ &a(bc) = (ab)c \\ &1a = a \\ &a(b+c) = ab + ac \end{aligned}$$

$\forall$ ideal I in R is an R-module,

for if $i \in I$, then $ri \in I$.
$r \in R$

$0 \in I$
$\forall\, a, b \in I$, $a + b \in I$
If $a \in I$, $r \in R$, then $ra \in I$.

(iv)

(v) Let linear $T : V \to V$, V finite-dim. vector space over field k.
Recall $k[x] \equiv$ set of polynomials with coefficients in k.

Define $\begin{aligned} k[x] \times V &\to V \\ f(x)v = \left(\sum_{i=0}^{m} c_i x^i \right) v &= \sum_{i=0}^{m} c_i T^i(v) \end{aligned}$ $\qquad \forall\, f(x) = \sum_{i=0}^{m} c_i x^i \in k[x]$

$\implies$ denote $k[x]$-module V^T.

Special case: Let $A \in \mathrm{Mat}_k(n, n)$, let linear $T : k^n \to k^n$.

$$T(w) = Aw$$

vector space k^n is $k[x]$-module if we define scalar multiplication

$$k[x] \times k^n \to k^n$$
$$f(x)w = \left(\sum_{i=0}^{m} c_i x^i \right) w = \sum_{i=0}^{m} c_i A^i w$$

$\forall\, f(x) = \sum_{i=0}^{m} c_i x^i \in k[x]$
In $(k^n)^T$, $xw = T(w)$
In $(k^n)^A$, $xw = Ax$
$T(w) = Ax$ and so $(k^n)^T = (k^n)^A$ (EY : 20151015 because of induction?)

Definition 32 (R-homomorphism (or R-map))**.** If ring R, R-modules M, N, then function $f : M \to N$,
if $\forall\, m, m' \in M$, $\forall\, r \in R$,

$$f(m + m') = f(m) + f(m')$$
$$f(rm) = rf(m)$$

Example 7.2. of Rotman (2011) on pp. 425 [57]]

(i) If R field, then R-modules are vector spaces and R-maps are linear transformations. Isomorphisms are then nonsingular linear transformations.

(ii)

(iii)

(iv)

(v) Let linear $T : V \to V$, let $v_1 \dots v_n$ be basis of V, let A be matrix of T relative to this basis.
Let $e_1 \dots e_n$ be standard basis of k^n.

Define $\varphi : V \to k^n$
$\varphi(v_i) = e_i$

$$\varphi(xv_i) = \varphi(T(v_i)) = \varphi(v_j a_{ji}) = a_{ji}\varphi(v_j) = a_{ji}e_j$$
$$x\varphi(v_i) = A\varphi(v_i) = Ae_i$$

$\implies \varphi(xv) = x\varphi(v) \quad \forall\, v \in V$
By induction on $\deg f$, $\varphi(f(x)v) = f(x)\varphi(v) \qquad \forall\, f(x) \in k[x] \qquad \forall\, v \in V$
$\implies \varphi$ is $k[x]$-map
$\implies \varphi$ is $k[x]$-isomorphism of V^T and $(k^n)^A$.

Proposition 13 (7.3 of Rotman (2011) [57])**.** *Let vector space over field k, V, let linear $T, S : V \to V$ Then $k[x]$-modules V^T, V^S are $k[x]$-isomorphic iff $\exists$ vector space isomorphism $\varphi : V \to V$ s.t. $S = \varphi T \varphi^{-1}$.*

Proof. If $\varphi : V^T \to V^S$is a $k[x]$-isomorphism,

$$\varphi(f(x)v) = f(x)\varphi(v) \quad \forall\, v \in V, \forall\, f(x) \in k[x]$$

if $f(x) = x$, then $\varphi(xv) = x\varphi(v)$

$$xv = T(v)$$
$$x\varphi(v) = S(\varphi(v))$$
$$\implies \varphi \circ T(v) = S \circ \varphi(v) \implies \varphi \circ T = S \circ \varphi$$

φ isomorphism, so $S = \varphi \circ T \circ \varphi^{-1}$

Conversely, if given isomorphism $\varphi : V \to V$ s.t. $S = \varphi T \varphi^{-1}$, then $S\varphi = \varphi T$.

$$S\varphi(v) = \varphi T(v) = \varphi(xv) = x\varphi(v)$$

Then by induction, $\varphi(x^n v) = x^n \varphi(v)$ (for $S^n \varphi(v) = x^n \varphi(v) = (\varphi T \varphi^{-1})^n \varphi(v) = \varphi T^n v = \varphi(x^n v)$).
By induction on $\deg(f)$, $\varphi(f(x)v) = f(x)\varphi(v)$.

□

Corollary 2 (7.4 of Rotman (2011) [57]). *Let k be a field,*
Let $A, B \in Mat_k(n,n)$.
Then $k[x]$-modules $(k^n)^A$, $(k^n)^B$ are $k[x]$-isomorphic.
(recall, $k[x] \equiv$ set of polynomials with coefficients in $k = \{\sum_{i=0}^m c_i x^i | c_i \in k\}$, and define scalar multiplication

$$k[x] \times k^n \to k^n$$
$$f(x)w = \left(\sum_{i=0}^m c_i x^i\right) w = \sum_{i=0}^m c_i A^i w, \qquad \forall\, f(x) = \sum_{i=0}^m c_i x^i \in k[x]$$

)
iff $\exists$ nonsingular P with

$$B = PAP^{-1}$$

Proof. Define
$T : k^n \to k^n$
$T(y) = A(y)$ where $y \in k^n$ is a column.
By Example 7.1 (v) of Rotman (2011) [57], recall,
and so for $k[x]$-module, $(k^n)^T = (k^n)^A$.
Similarly, define

$$S : k^n \to k^n$$
$$S(y) = B(y)$$

Denote corresponding $k[x]$-module by $(k^n)^B$.
Given $(k^n)^A \cong (k^n)^B$ (isomorphic), by Prop. 13,
$\exists$ isomorphism $\varphi : k^n \to k^n$ s.t. $B = \varphi A \varphi^{-1}$.
By Prop. 12, i.e. Prop. 3.94 of Rotman (2011) [57], in that every linear transformation has a matrix representation (even in the standard "Euclidean" basis), $\exists\, P \in \mathrm{Mat}_k(n,n)$, s.t.

$$\varphi(y) = Py \qquad y \in k^n$$

(P nonsingular because φ isomorphism)
Thus,

$$B\varphi(y) = \varphi A(y)$$
$$BPy = P(Ay) \qquad \forall\, y \in k^n$$
$$\Longrightarrow PA = BP \text{or } B = PAP^{-1}$$

Conversely, given $B = PAP^{-1}$, P nonsingular matrix,
define isomorphism

$$\varphi : k^n \to k^n$$
$$\varphi(y) = Py \qquad \forall\, y \in k^n$$

By Prop. 13,
$(k^n)^B$, $(k^n)^A$ are $k[x]$-isomorphic.
i.e. $\varphi : (k^n)^A \to (k^n)^B$ is a $k[x]$-module isomorphism.

□

Definition 33 ($\mathrm{Hom}_R(M,N)$).

$$\mathrm{Hom}_R(M,N) = \{ \text{ all } R\text{-homomorphisms } M \to N \} = \{ f | f : M \to N, \text{ s.t. } \forall\, m, m' \in M, \forall\, r \in R, \begin{matrix} f(m+m') = f(m) + f(m') \\ f(rm) = rf(m) \end{matrix} \} \tag{26}$$

If $f, g \in \mathrm{Hom}_R(M, N)$,
define

$$\begin{aligned} f+g &: M \to N \\ f+g &: m \mapsto f(m)+g(m) \end{aligned} \tag{27}$$

Proposition 14 ($\mathrm{Hom}_R(M,N)$ R-module, 7.5 of Rotman (2011) [57]). *If M, N R-modules, where R commutative ring,*
then $Hom_R(M,N)$ R-module,
with addition

$$\begin{aligned} f+g &: M \to N \qquad \forall\, f, g \in Hom_R(M,N) \\ f+g &: m \mapsto f(m)+g(m) \end{aligned}$$

and scalar multiplication

$$rf : m \mapsto f(rm)$$

Moreover, distributive laws:
If $p : M' \to M$, $q : N \to N'$, then

$$(f+g)p = fp + gp \text{ and } q(f+g) = qf + qg$$

$\forall\, f, g \in Hom_R(M,N)$

Proof. $\forall\, f, g \in \mathrm{Hom}_R(M,N)$, $\forall\, r, r', 1 \in R$,

(i)
$$r(f+g)(m) = (f+g)(rm) = f(rm) + g(rm) = rf(m) + rg(m) = (rf + rg)(m)$$

(ii)
$$(r+r')f(m) = f((r+r')m) = f(rm + r'm) = f(rm) + f(r'm) = (rf + r'f)(m)$$

(iii)
$$(rr')f(m) = f(rr'm) = rf(r'm) = f(r'rm) = f(rr'm) \implies (rr')f = r(r'f)$$

(iv)
$$1f(m) = f(1m) = f(m) \implies 1f = f$$

□

Definition 34. if R-module M, the submodule N of M, denoted $N \subseteq M$, is additive subgroup N of M, closed under scalar multiplication $rn \in N$ whenever $n \in N$, $r \in R$

Definition 35 (quotient module M/N).
quotient module M/N -
For submodule N of R-module M, then,
remember M abelian group, N subgroup,
quotient group M/N equipped with scalar multiplication

$$\begin{aligned} r(m+N) &= rm + N \\ M/N &= \{m+N | m \in M\} \end{aligned}$$

natural map

$$\begin{aligned} \pi &: M \to M/N \\ m &\mapsto m+N \end{aligned} \tag{28}$$

easily seen to be R-map.
Scalar multiplication in quotient module well-defined:
If $m+N = m'+N$, $m-m' \in N$, so $r(m-m') \in N$ (because N submodule), so

$$rm - rm' \in N \text{ and } rm + N = rm' + N$$

Proposition 15 (7.15 of Rotman (2010) [57]). *(i)* $S \bigsqcup T \simeq M$

(ii) $\exists$ *injective R-maps* $i : S \to M$ *, s.t.*
$j : T \to M$

$$M = im(i) + im(j) \text{ and } im(i) \bigcap im(j) = \{0\} \tag{29}$$

(iii) $\exists$ *R-maps*

$$i : S \to M$$
$$j : T \to M$$

s.t. $\forall\, m \in M$, $\exists!$

$$s \in S$$
$$t \in T$$

with $m = is + jt$.

(iv) $\exists$ *R-maps*

$$i : S \to M \qquad p : M \to S$$
$$j : T \to M \qquad q : M \to T$$

s.t.

$$pi = 1_S \qquad pj = 0$$
$$qj = 1_T \qquad qi = 0 \qquad ip + jq = 1_M$$

Proof. • (i)→ (ii) Given $S \bigsqcup T \simeq M$,
let $\varphi : S \bigsqcup T \to M$ be this isomorphism.

Define

$$i := \varphi\lambda_S \qquad (\lambda_S : s \mapsto (s, 0)) \qquad i : S \to M$$
$$j := \varphi\lambda_T \qquad (\lambda_T : t \mapsto (0, t)) \qquad j : T \to M$$

i, j are injections, being composites of injections.

If $m \in M$, $\exists!\,(s, t) \in S \bigsqcup T$, s.t. $\varphi(s, t) = m$.

Then

$$m = \varphi(s, t) = \varphi((s, 0) + (0, t)) = \varphi\lambda_S(s)\varphi\lambda_T(t) = is + jt \in \mathrm{im}(i) + \mathrm{im}(j)$$

Let $c \in \mathrm{im}(i) + \mathrm{im}(j)$. Since $i : S \to M$, $c \in M$.
$j : T \to M$

$\implies M = \mathrm{im}(i) + \mathrm{im}(j)$.

If $x \in \mathrm{im}(i) \bigcap \mathrm{im}(j)$,

$$x = i(s) \text{ for some } s \in S$$
$$x = j(t) \text{ for some } t \in T$$

$$is = jt = \varphi\lambda_S(s) = \varphi\lambda_T(t) = \varphi(s, 0) = \varphi(0, t)$$

φ isomorphism, so $\exists\, \varphi^{-1} \implies (s, 0) = (0, t)$, so $s = t = 0$. $x = 0$

- (ii)→ (iii) Given $i : S \to M$, s.t. $M = \mathrm{im}(i) + \mathrm{im}(j)$, so
$j : T \to M$

$\forall\, m \in M$, $m = i(s) + j(t)$ for some $s \in S, t \in T$.

Suppose $s' \in S$, s.t. $m = i(s'_+ j(t')$.
$t' \in T$

$$i(s - s') = j(t - t') \in \text{im}(i) \bigcap \text{im}(j) = \{0\}$$

So $s = s', t = t'$, since i, j injective.

- (iii)→ (iv)
  Given $\forall\, m \in M$, $\exists !\, s \in S, t \in T$ s.t.

$$m = i(s) + j(t)$$

Define

$$p : M \to S \qquad q : M \to T$$
$$p(m) := s \qquad q(m) := t$$

$$pi(s) = s \qquad pj(t) = 0$$
$$qj(t) = t \qquad qi(s) = 0$$

$$(ip + jq)(m) = ip(m) + jq(m) = i(s) + j(t) = m$$

□

6.2 Vector Spaces as a Module

Lang made the key insight on vector spaces as a whole in Sec 5. "Vector Spaces" in pp. 139-140 of Lang (2005) [58]:

Theorem 18 (Existence of a basis for vector spaces, Thm. 5.1 Lang (2005) [58])**.** *Let V be a vector space over a field K, assume $V \neq \{0\}$.*
Let Γ be a set of generators of V over K and let S be a subset of Γ which is linearly independent.
Then $\exists$ basis $\mathcal{B}$ of V s.t. $S \subset \mathcal{B} \subset \Gamma$.

Indeed, while this wikipedia article [1] on Vector space does a good job generalizing the properties defining a vector in a vector space, a vector's properties is separate from what *characterizes* a vector space. Here, we can *specify* a vector space by its generators, and furthermore, from Thm. 18, it has a basis that characterizes a vector space. This can be useful for implementation in C++.

7 Vector Spaces

cf. Ch. 11, "Vector Spaces" pp. 408, Dummit and Foote [25]

Terminology for R any Ring	**Terminology for R a Field**
M is an R-module	M is a vector space over R
m is an element of M	m is a vector in M
α is a ring element	α is a scalar
N is a submodule of M	N is a subspace of M
M/N is a quotient module	M/N is a quotient space
M is a free module of rank n	M is a vector space of dimension n
M is a finitely generated module	M is a finite dimensional vector space
M is a nonzero cyclic module	M is a 1-dimensional vector space
$\varphi : M \to N$ is a R-module homomorphism	$\varphi : M \to N$ is a linear transformation
M and N are isomorphic as R-modules	M and N are isomorphic vector spaces
the subset A of M generates M	the subset A of M spans M
$M = RA$	each element of M is a linear combination of elements of A i.e., $M = \text{Span}(A)$

[1] https://en.wikipedia.org/wiki/Vector_space

cf. pp. 408, 11.1 "Definitions and Basic Theory", Dummit and Foote [25]
Vector spaces are free F-modules, i.e. vector spaces have bases.

Examples:

1. Space $V = F[x]$ of polynomials in the variable x with coefficients from field F is a vector space over F. elements $1, x, x^2, \dots$ are linearly independent by def. (i.e. polynomial is 0 iff all its coefficients are 0). Since these elements also span V, by def., they're a basis for V (basis of vector space V is ordered set of linearly independent vectors which span V)

2. Collection of solutions of a linear, homogeneous, constant coefficient differential equation (e.g. $y'' - 3y' + 2y = 0$) over $\mathbb{C}$ form a vector space over $\mathbb{C}$, since differentiation is a linear operator.
e.g. e^t, e^{2t} are linearly independent ($ae^t + be^{2t} = 0$, then so $a + b = 0\,(t = 0)$, $ae + be^2 = 0\,(t = 1) \implies a = b = 0$)
By Thm. in differential eqns. that these elements span set of solutions of this eqn., hence form a basis.

Proposition 16 (1, Dummit and Foote [25])**.** *Assume set $\mathcal{A} = \{v_1, v_2, \dots v_n\}$ spans vector space V but no proper subset of $\mathcal{A}$ spans V. Then $\mathcal{A}$ basis of V. In particular, any finitely generated (finitely spanned) vector space over F is a free F-module.*

Proof. *Want*: $v_1, v_2, \dots v_n$ linearly independent.

Suppose $\sum_{i=1}^{n} \alpha_i v_i = 0$ s.t. not all $\alpha_i = 0$.

By reordering, assume $\alpha_1 \neq 0$ and then $v_1 = \frac{-1}{\alpha_1}(\alpha_2 v_2 + \dots + \alpha_n v_n)$.
Then $\{v_2, v_3, \dots v_n\}$ also spans V since any linear combination of $v_1, v_2, \dots v_n$ can be written as a linear combination of $v_2, v_3, \dots v_n$ using $v_1 = \frac{-1}{\alpha_1}(\alpha_2 v_2 + \dots + \alpha_n v_n)$. Contradiction.

□

Example.
Let F be a field, consider $F[x]/(f(x))$ where $f(x) = x^n + \sum_{j=0}^{n-1} a_j x^j$
ideal $(f(x))$ is a subspace of vector space $F[x]$, quotient $F[x]/(f(x))$ is a vector space over F.

By Euclidean Algorithm, $\forall$ polynomial $a(x) \in F[x]$ can be written uniquely in form

$$a(x) = q(x)f(x) + r(x)$$

where $r(x) \in F[x]$, and $0 \leq \deg r(x) \leq n - 1$
Since $q(x)f(x) \in (f(x))$, it follows $\forall$ element of quotient is represented by polynomial $r(x)$ of degree $\leq n - 1$
2 distinct such polynomials cannot be the same in the quotient since this would say their difference (which is a polynomial of degree at most $n - 1$) would be divisible by $f(x)$ (which is of degree n), i.e.

$$f(x) = x^n + \sum_{j=1}^{n-1} a_j x^j$$

$\forall\, a(x) \in F[x]$, $a(x) = q(x)f(x) + r(x)$, $q(x)f(x) \in (f(x))$ (ideal)

Consider $r_1(x)$, $\deg(r_1(x)) = n_1 \leq n - 1$, $\deg(r_2(x)) = n_2 \leq n - 1$.

If $a_1(x), a_2(x) \in F[x]$, $a_1(x) = q_1(x)f(x) + r_1(x)$ $a_1(x) - a_2(x) = (q_1(x) - q_2(x))f(x) + (r_1(x) - r_2(x))$.
$a_2(x) = q_2(x)f(x) + r_2(x)$
If $r_1(x) = r_2(x)$, then $a_1(x) - a_2(x) = (q_1(x) - q_2(x))f(x)$, so $a_1(x) - a_2(x)$ divisible by $f(x)$.
$\implies \overline{1}, \overline{x}, \overline{x^2}, \dots \overline{x^{n-1}}$ (bar denotes image of these elements in the quotient) span $F[x]/f(x)$, as vector space over F.
No proper subset of these elements also spans, hence elements give basis for $F[x]/(f(x))$.

8 Categories; Category Theory

8.1 Categories

cf. 7.2 Categories of Rotman (2010) [57]

8.1.1 Russell paradox, Russell set

Definition 36 (Russell set)**.** Russell set - set S that's not a member of itself, i.e. $S \notin R$

If R is family of all Russell sets,
Let $X \in R$. Then $X \notin X$. But $X \in R$. $X \notin R$.
Let $R \notin R$. Then R in family of Russell Sets. $R \in R$. Contradiction.
Then consider *class* as primitive term, instead of set.

Definition 37 (Category)**.** Category $\mathcal{C}$ (Rotman's notation) $\equiv \mathbf{C}$ (my notation), consists of class obj($\mathcal{C}$) (Rotman's notation) $\equiv$ Obj($\mathbf{C}$) $\equiv$ Obj$\mathbf{C}$ (my notation) of objects, set of morphisms $\mathrm{Hom}(A, B)$ $\forall\,(A, B)$ of ordered tuples of objects, composition

$$\mathrm{Hom}(A, B) \times \mathrm{Hom}(B, C) \to \mathrm{Hom}(A, C)$$
$$(f, g) \mapsto gf$$

, s.t.

1. $\exists\,\mathbf{1}, \forall\, f : A \to B, \exists\ 1_A : A \to A$, s.t. $1_B \cdot f = f = f \cdot 1_A$, and
   $1_B : B \to B$

2. associativity, $\forall \begin{matrix} f : A \to B \\ g : B \to C \\ h : C \to D \end{matrix}$, then $h \circ (g \circ f) = (h \circ g) \circ f$

In summary,

$$\mathbf{C} := (\mathrm{Obj}(\mathbf{C}), \mathrm{Mor}\mathbf{C}, \circ, \mathbf{1}) \equiv (\mathrm{Obj}\mathbf{C}, \mathrm{Mor}\mathbf{C}, \circ_{\mathbf{C}}, \mathbf{1}_{\mathbf{C}}) \tag{30}$$

s.t.

$$\mathrm{Mor}\mathbf{C} = \bigcup_{A,B \in \mathrm{Obj}\mathbf{C}} \mathrm{Hom}(A, B)$$

Examples (7.25 of Rotman (2010)[57]):

(i) $\mathbf{C} = \mathrm{Sets}$

(ii) $\mathbf{C} = \mathrm{Groups} = \mathrm{Grps}$

(iii) $\mathbf{C} = \mathrm{CommRings}$

(iv) $\mathbf{C} = {}_R\mathbf{Mod}$, if $R = \mathbb{Z}$, ${}_{\mathbb{Z}}\mathbf{Mod} = \mathbf{Ab}$, i.e. $\mathbb{Z}-$modules are just abelian groups.

(v) $\mathbf{C} = \mathbf{PO}(X)$, If partially ordered set X, regard X as category, s.t. $\mathbf{Obj}, \mathbf{PO}(X) = \{x | x \in X\}$, $\forall\, \mathrm{Hom}(x, y) \in \mathbf{Mor}_{\mathbf{PO}}(X)$, $\mathrm{Hom}(x, y) = \begin{cases} \emptyset & \text{if } x \not\preceq y \\ \kappa_y^x & \text{if } x \preceq y \end{cases}$ where $\kappa_y^x \equiv$ unique element in Hom set when $x \preceq y$) s.t.

$$\kappa_z^y \kappa_y^x = \kappa_z^x$$

Also, notice that

$$1_x = \kappa_x^x$$

Definition 38 (isormorphisms or equivalences)**.** $f : A \to B$, $f \in \mathrm{Hom}(A, B)$, if $\exists$ **inverse** $g : B \to A$, $g \in \mathrm{Hom}(B, A)$, s.t.

$$gf = 1_A$$
$$fg = 1_B$$

and if $\mathbf{C} = \mathbf{Top}$, equivalences (isomorphisms) are homeomorphisms.

Feature of category ${}_R\mathbf{Mod}$ not shared by more general categories: *Homomorphisms can be added.*

Definition 39 (pre-additive Category)**.** category $\mathbf{C}$

We can force 2 overlapping subsets A, B to be disjoint by "disjointifying" them: e.g. consider $(A \cup B) \times \{1, 2\}$,

consider $A' = A \times \{1\}$.
$B' = B \times \{2\}$

$$\implies A' \cap B' = \emptyset$$

since $(a, 1) \neq (b, 2) \quad \forall\, a \in A, \forall\, b \in B$.

Let bijections $\alpha : A \to A'$, $\quad \alpha : a \mapsto (a, 1)$, denote $A' \bigcup B' \equiv A \coprod B$.
$\beta : B \to B'$ $\quad \beta : b \mapsto (b, 2)$
From Rotman (2010) [57], pp. 447,

Definition 40. coproduct $A \coprod B \equiv C \in \mathrm{Obj}(\mathcal{C})$

In my notation,
coproduct

$$\begin{aligned} &(\mu_1, A_1 \coprod A_2) \\ &(\mu_2, A_1 \coprod A_2) \end{aligned} \tag{31}$$

where injection (morphisms)

$$\begin{aligned} &\mu_1 : A_1 \to A_1 \coprod A_2 \\ &\mu_2 : A_1 \to A_1 \coprod A_2 \end{aligned} \tag{32}$$

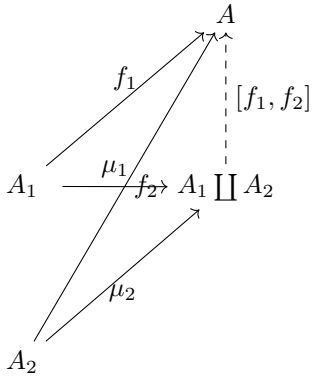
s.t.

$$\begin{aligned} \forall\, A \in \mathrm{Obj}\mathbf{A}, \forall\, f_1, f_2 \in \mathrm{Mor}\mathbf{A} \text{ s.t. } & f_1 : A_1 \to A \\ & f_2 : A_2 \to A \end{aligned}$$

then

$$\begin{aligned} \exists\, ![f_i] \equiv [f_1, f_2] \in \mathrm{Mor}\mathbf{A},\ & [f_1, f_2] : A_1 \coprod A_2 \to A \text{ s.t.} \\ & [f_1, f_2]\mu_1 = f_1 \\ & [f_1, f_2]\mu_2 = f_2 \end{aligned} \tag{33}$$

i.e.



(34)

So to generalized, for $i \in I$, (finite set I?)

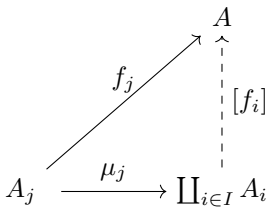
coproduct $(\mu_j, \coprod_{i\in I} A_i)_{j\in I}$, where
(family of) injection (morphisms) $\mu_j : A_j \to \coprod_{i\in I} A_i$
s.t.

$$\forall\, A \in \mathrm{Obj}\mathbf{A},\ \forall\, f_i \in \mathrm{Mor}\mathbf{A},\ i \in I,\ f_i : A_i \to A$$

then

$$\begin{gathered}\exists\,!\,[f_i] \equiv [f_i]_{i\in I} \in \mathrm{Mor}\mathbf{A},\ [f_i] : \coprod_{i\in I} A_i \to A \text{ s.t.} \\ [f_i]\mu_j = f_j \qquad \forall\, j \in I\end{gathered} \tag{35}$$

i.e.



(36)

For notation purposes only, recall that it's denoted the sets $\mathrm{Hom}(A, B)$ in ${}_R\mathbf{Mod}$ by

$$\mathrm{Hom}_R(A, B)$$

i.e., in my notation, for $A, B \in \mathrm{Obj}_R\mathbf{Mod}$, $\mathrm{Hom}(A, B) \subset \mathrm{Mor}({}_R\mathbf{Mod})$, $\mathrm{Hom}(A, B) \equiv \mathrm{Hom}_R(A, B)$

Definition 41 (pre-additive category)**.** category $\mathbf{C}$ is **pre-additive** if $\forall\, \mathrm{Hom}(A, B)$, $\mathrm{Hom}(A, B)$ equipped with binary operation $+$ s.t. $\forall\, f, g \in \mathrm{Hom}(A, B)$,

1. if $p : B \to B'$, then
$$p(f + g) = pf + pg \in \mathrm{Hom}(A, B')$$
2. if $q : A' \to A$, then
$$(f + g)q = fq + gq \in \mathrm{Hom}(A', B)$$

and

$$f + g = g + f \qquad \text{(additive abelian)}$$

8.1.2 Examples of extra assumptions on sets, ${}_R$Mod we take for granted

In Prop. 7.15(iii) Rotman (2010) [57],

$$\begin{gathered}p : M \to A \qquad pi = 1_A \\ \text{direct sum } M = A \oplus B \text{ if } \exists \text{ homomorphisms } q : M \to B \text{ s.t. } qj = 1_B, \\ i : A \to M \qquad pj = 0 \\ j : B \to M \qquad qi = 0 \\ ip + jq = 1_M\end{gathered}$$

direct sum $M = A \oplus B$ uses property that morphisms can be added ${}_R\mathbf{Mod}$ has this property. **Sets** don't.

In Corollary 7.17,
direct sum in terms of arrows,
$\exists$ map $\rho : M \to S$ s.t. $\rho(s) = s$. Moreover $\ker\rho = \mathrm{im}j$, $\mathrm{im}\rho = \mathrm{im}i$ and $\rho(s) = s$, $\forall\, s \in \mathrm{im}\rho$.

$$S \xrightarrow{\ i\ } M \xleftarrow{\ j\ } T \qquad \text{and } M \simeq S \coprod T,$$

where $i : s \mapsto s$ (i.e. inclusions)

$j : t \mapsto t$

This makes sense in **Sets**, but doesn't make sense in arbitrary categories because image of morphism may fail, e.g. $\mathrm{Mor}(\mathcal{C}(G))$ are elements in $\mathrm{Hom}(*,*) = G$, not functions.

Categorically, object S is (equivalent to) retract of object M, $S, M \in \mathrm{Obj}\mathbf{C}$, if $\exists$ morphisms $i, p \in \mathrm{Mor}(\mathbf{C})$, s.t.

$$i : S \to M$$
$$p : M \to S$$

s.t. $pi = 1_S$, $(ip)^2 = ip$ (for modules, define $\rho = ip$)

Definition 42 (free products)**.** **free products** are coproducts in groups

Prop. 7.26, Rotman (2010) [57]

Proposition 17 (7.26, Rotman)**.** *If A, B are R-modules, then their coproducts in ${}_R\mathbf{Mod}$ exists, and it's the direct sum $C = A \coprod B$.*

Proof. Define

$$\mu : A \to C \qquad \nu : B \to C$$
$$\mu : a \mapsto (a, c) \qquad \nu : b \mapsto (0, b)$$

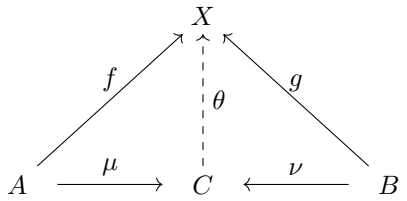
(Rotman's notation) $\alpha : A \to C$, $\beta : B \to C$

Let X be a module, $f : A \to X$, $g : B \to X$ homomorphisms □

Define

$$\theta : C \to X$$
$$\theta : (a, b) \mapsto f(a) + g(b)$$
$$\theta\mu(a) = \theta(a, 0) = f(a)$$
$$\theta\nu(b) = \theta(0, b) = g(b)$$

so diagram commutes, i.e.



If $\psi : C \to X$ makes diagram commute,

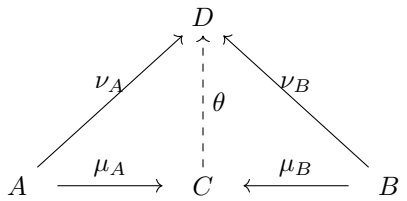
$$\psi((a, 0)) = f(a) \qquad \forall\, a \in A$$
$$\psi((0, b)) = g(b) \qquad \forall\, b \in B$$

and since ψ is a homomorphism, $\psi((a, b)) = \psi((a, 0)) + \psi((0, b)) = f(a) + g(b) = \theta((a, b))$. $\psi = \theta$.

Prop. 7.27, Rotman (2010) [57]

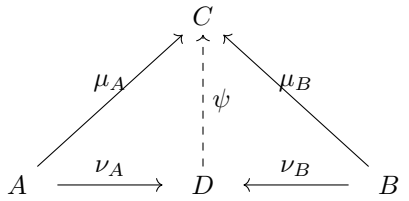
Proposition 18 (7.27, Rotman)**.** *If category $\mathcal{C} = \mathbf{C}$, and if $A, B \in Obj\mathbf{C}$, then $\forall$ 2 coproducts of A, B, if they $\exists$, are equivalent.*

Proof. Suppose C, D coproducts of A, B. Suppose coproducts $\mu_A : A \to C$, $\nu_A : A \to D$
$\mu_B : B \to C$, $\nu_B : B \to D$



Just substitute $X = D$ in diagram above.

Then substitute again:



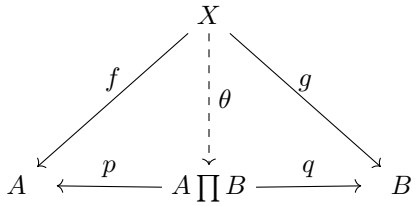
Then combine the 2 diagrams: $\psi\theta = 1_C$. Likewise by label symmetry of C, D, $\theta\psi = 1_D$.

Then C, D are equivalent. □

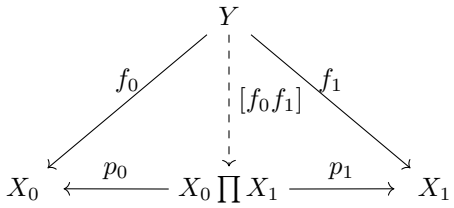
Exer. 7.29 on pp. 459 of Rotman (2010) [57]

Definition 43. If $A, B \in \text{Obj}\mathbf{C}$, then their **product**; $A \prod B = P \in \text{Obj}\mathbf{C}$, and morphisms $p : P \to A$ s.t. $q : P \to B$

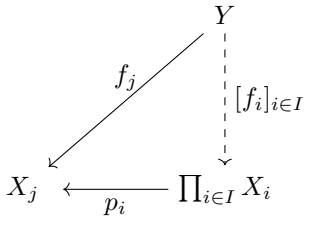
$\forall\, X \in \text{Obj}\mathbf{C}, \quad \forall\, f : X \to A \in \text{Mor}\mathbf{C},$
$g : X \to B \in \text{Mor}\mathbf{C}$
$\exists !\, \theta : X \to P$, s.t.



In the notation of Kashiwara and Schapira (2006) [24],



In general



product of X_i's,

$$\prod_i X_i \equiv \prod_{i \in I} X_i$$

given by

$$\prod_i X_i := \varprojlim \alpha \tag{37}$$

When $X_i = X$, $\forall\, i \in I$, denote product by $X^{\prod I} \equiv X^I$.

e.g. Cartesian product $P = A \times B$ of 2 sets A, B, $A, B \in \text{Obj}\mathbf{Sets}$.
Define

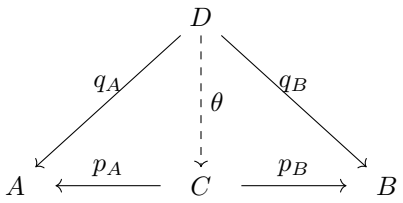
$$p : A \times B \to A \qquad q : A \times B \to B$$
$$p(a,b) \mapsto a \qquad q(a,b) \mapsto b$$

If $X \in \text{Obj}\mathbf{Sets}$,

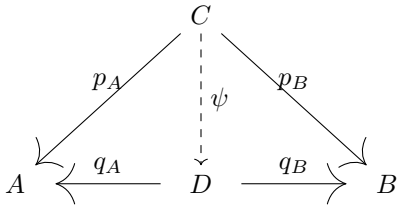
if $f : X \to A$, then $\theta : X \to A \times B$
$g : X \to B \qquad \theta : x \mapsto (f(x), g(x)) \in A \times B$

Proposition 19 (7.28 Rotman (2010); equivalence of products, if it exists)**.** *If $A, B \in Obj\mathbf{C}$, then $\forall$ 2 products of A and B, should they exist, are equivalent.*

Proof. Suppose C, D products of A, B. Suppose products $p_A : C \to A, \quad q_A : D \to A$
$p_B : C \to B, \quad q_B : D \to B$



Just substitute $X = D$ in diagram above.
Then substitute again:



Then combine the 2 diagrams: $\psi\theta = 1_C$. Likewise by label symmetry of C, D, $\theta\psi = 1_D$.
Then C, D are equivalent.

□

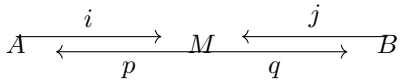
8.1.3 Products of Modules and Sets

Proposition 20 (7.29 Rotman (2010); products of R-modules are equivalent)**.** *If commutative ring R, R-modules A, B,*
then $\exists$ their (categorical) product $A \sqcup B$, in fact

$$A \sqcap B \cong A \sqcup B \tag{38}$$

Proof. If $A \sqcup B \cong M$, then $\exists$ R-maps, $i : S \to M$, $\quad p : M \to S$ s.t. $pi = 1_A$ and $pj = 0$, and $ip + jq = 1_M$,
$j : T \to M \quad q : M \to T \quad qj = 1_B \quad qi = 0$

i.e.

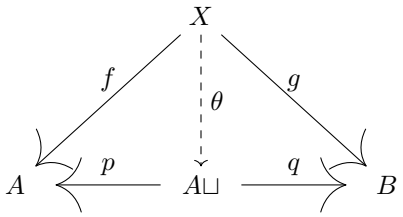


If module X, since $f : X \to A$ are homomorphisms,

$$g : X \to B$$

define $\theta : X \to A \sqcup B$ so that

$$\theta(x) = if(x) + jg(x)$$



since, $\forall\, x \in X$,

$$p\theta(x) = pif(x) + pjg(x) = pif(x) + 0 = f(x)$$

since $ip + jq = 1_{A \sqcup B}$

$$\psi = ip\psi + jq\psi = if + jf = \theta$$

so product is unique. □

Definition 44. Let R be commutative ring,
let $\{A_i : i \in I\}$ be indexed family of R-modules.

direct product $\prod_{i \in I} A_i$ is cartesian product (i.e. set of all I-tuples (a_i) whose ith coordinate a_i lies in $A_i \quad \forall\, i$) with coordinate wise addition and scalar multiplication:

$$(a_i) + (b_i) = (a_i + b_i)$$
$$r(a_i) = (ra_i)$$

where $r \in R$, $a_i, b_i \in A_i, \quad \forall\, i$

cf. Thm. 7.32 of Rotman (2010) [57]

Theorem 19 (7.32, Rotman)**.** *Let commutative ring R.*
$\forall$ R-module A, $\forall$ family $\{B_i | i \in I\}$ of R-modules,

$$Hom_R(A, \coprod_{i \in I} B_i) \simeq \coprod_{i \in I} Hom_R(A, B_i) \tag{39}$$

via R-isomorphism

$$\varphi : f \mapsto (p_i f)$$

where p_i are projections of product $\coprod_{i \in I} B_i$

Proof. Let $a \in A$, $f, g \in \mathrm{Hom}_R(A, \coprod_{i \in I} B_i)$.

$$\varphi(f + g)(a) = (p_i(f + g))(a) = (p_i(f(a) + g(a))) = (p_i f + p_i g)(a)$$

φ additive.

$\forall\, i$, $\forall\, r \in R$, $p_i r f = r p_i f$ (since product of R-modules, $\coprod_{i \in I} B_i$ is also an R-module of $\mathrm{Obj}_R\mathbf{Mod}$, by def. of product).

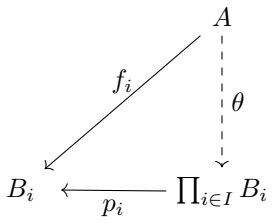
$$\varphi r f \mapsto (p_i r f) = (r p_i f) = r(p_i f) = r\varphi(f)$$

So φ is R-map.

If $(f_i) \in \coprod_i \mathrm{Hom}_R(A, B_i)$, then $f_i : A \to B_i \quad \forall\, i$

By Rotman's Prop. 7.31 (If family of R-modules $\{A_i | i \in I\}$, then direct product $C = \coprod_{i \in I} A_i$ is their product in ${}_R\mathbf{Mod}$),

By def. or product, $\exists\,!$ R-map , $\quad \theta : A \to \coprod_{i \in I} B_i$ s.t. $p_i \theta = f_i \quad \forall\, i$

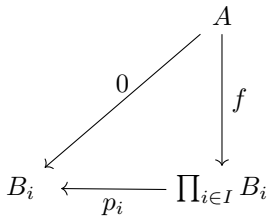


Then

$$f_i = (p_i\theta) = \varphi(\theta)$$

, and so φ *surjective*.

Suppose $f \in \ker\varphi$, so $\theta = \varphi(f) = (p_i f)$. Thus $p_i f = 0 \quad \forall\, i$



But 0-homomorphism also makes this diagram commute, so uniqueness of homomorphism $A \to \prod B_i$ gives $f = 0$.

□

9 Applications of Category Theory: Finite State Machines (FSM)

Definition 45 (Finite State Machines ≡ Finite State Automaton)**.** A deterministic finite state machine or acceptor deterministic finite state machine is a quintuple $(\Sigma, S, s_0, \delta F)$ where
$\Sigma \equiv$ input alphabet (finite, non-empty set of symbols)
$S \equiv$ finite, non-empty set of states
$s_0 \equiv$ initial state, $s_0 \in S$
$\delta \equiv$ state-transition function; $\delta : S\times\Sigma \to S$ (in a nondeterministic finite automaton, it would be $\delta : S\times\Sigma \to \mathcal{P}(S)$), i.e. δ would return a set of states; $\mathcal{P}(S) \equiv$ set of all subsets of S, including $\emptyset$ and $S \equiv$ power set.
$F \equiv$ set of final states, (possibly empty subset of S; $F \subseteq S$ or $F \subseteq S \cup \{\emptyset\}$)

Finite State Machine (FSM) is also known as a Finite State Automaton.

cf. Black, Paul E (12 May 2008). "Finite State Machine". *Dictionary of Algorithms and Data Structres*. U.S. National Institute of Standards and Technology (NIST).

For both deterministic and non-deterministic FSMs, it's conventional to allow δ to be a partial function, i.e. $\delta(q, x)$ doesn't have to be defined for every combination of $q \in S$, $x \in \Sigma$

If FSM M is in state q; the next symbol (input?) is x and $\delta(q, x)$ not defined; then M can announce an error (i.e. reject the input (???)).

Definition 46 (Alphabet)**.** alphabet := nonempty set of symbols $\equiv \Sigma$
string := finite sequence of members (i.e. symbols) of an underlying base set (i.e. **alphabet**)
$\Sigma^n \equiv$ set of all strings of length n.

In Curino and Spivak (2011)[41], on pp. 6, Sec. 4.2 "A new concept of state transformations", they say that "Legend has it that Eilenberg and MacLane spent the effort to invent category theory because they needed to formalize the concept now known as natural transformation."

It goes and says "The definition of natural transformations between states naturally captures much of the semantics for what remains unchanged when performing an update."

Part III
Set Theory

10 Axioms of Set Theory

Axioms of set Theory - Lec 02 - Frederic Schuller, Schuller (2016) [44]

10.1 1.4 The ϵ-Relation

Set theory built on the postulate that there is a fundamental relation (i.e., a predicate of two variables) called ϵ.
There will be no *definition* of what ϵ is, or of what a set is.
Instead: 9 axioms that speak of ϵ and sets.

Overview of axioms

EE PURP IC F

$$\left.\begin{matrix} E \\ E \end{matrix}\right\} \text{ basic existence axioms}$$

$$\left.\begin{matrix} P \\ U \\ R \\ P \end{matrix}\right\} \text{ construction axioms}$$

$$\left.\begin{matrix} I \\ C \end{matrix}\right\} \text{ further existence / construction}$$

$$\left.\begin{matrix} F \end{matrix}\right\} \text{ non-existence}$$

Using the ϵ-relation, we can immediately define

$$\notin (x, y) =: x \notin y$$

Part IV
Probability

11 Measure Theory

11.1 Measure Theory -Lect05- Frederic Schuller

Measure Theory -Lec05- Frederic Schuller, Schuller (2016) [44].
Lecture 5: **Measure Theory**
Short recap of basic notions of measure theory - without proofs - for two reasons:
EY: Without proofs because it would take too much time and they're relatively straightforward.

1. spectral theory (Schuller: for self-adjoint operators) requires the (projection-valued) measures.
2. **the** most commonly emerging ∞-dim. separable Hilbert space in quantum mechanics
   $L^2(\mathbb{R}^d)$ (Schuller: d typically $= 3$; you have to take the equivalence classes of square-integrable functions, almost everywhere; for physicists, theorems for exchanging integration and limits, for Riemann integral, you need uniform convergence, but for Lebesque integral you need not strong conditions).

11.1.1 1. General measure spaces

The basic definition is:

Definition 47. Let M be a non-empty set.
Then a collection of subsets

$$\sigma \subseteq \mathcal{P}(M)$$

is called
a **σ-algebra for M**
if

1. $M \in \sigma$
2. $A \in \sigma \Longrightarrow M \setminus A \in \sigma$
3. $A_1, A_2, \dots \in \sigma$ and $A_i \cap A_j = \emptyset$, $i \neq j$
$$\Longrightarrow \bigcup_{n\geq 1} A_n \in \sigma$$

Terminology $A \in \sigma$ called a **measurable set** in M.
(M, σ) is called a **measurable** space.

Schuller: Volume will be called measure soon. Only sets in σ-algebra can be assigned a volume, i.e. measure. σ-algebra could be called set of all measurable subsets.

Remark: Clearly, taking $\sigma = \mathcal{P}(M)$.
In general, however, it does not provide a useful σ-algebra if M is non-countable. (Schuller: e.g. discrete topology doesn't provide a useful topology)

($\Longrightarrow$ problem sheet, Banach-Taichi paradox) (EY: Banach-Tarski paradox)

Definition 48. A measure $\mu : \sigma \to \mathbb{R}_0^+$ on a measurable space (M, σ) is a map satisfying:

(i) $\mu(\emptyset) = 0$

(ii) $A_1, A_2, \dots \in \sigma$ pairwise disjoint: $\forall i \neq j, \quad A_i \cap A_j = \emptyset$
$\Longrightarrow \mu(\bigcup_{n\geq 1} A_n) = \sum_{n\geq 1} \mu(A_n)$ (Schuller: this is often called "sigma additivity")

Terminology: $\underbrace{(\underbrace{M, \sigma}_{\text{measurable space}}, \mu)}_{\text{measure space}}$

Properties of a measure.

(i) **monotony**: $\underbrace{A_1}_{\in\sigma} \subseteq \underbrace{A_2}_{\in\sigma} \Longrightarrow \mu(A_1) \leq \mu(A_2)$ (Schuller, EY: Equality could happen if sets 1,2 differ by a point or in general a set of measure 0)

(ii) Schuller: What if you dropped pairwise-disjoint property in the definition Def. 48?
sub-additivity $\{A_1, A_2, \dots\} \subseteq \sigma : \mu\left(\bigcup_{n\geq 1} A_n\right) \leq \sum_{n\geq 1} \mu(A_n)$

(iii) **"continuity" from below**: *Increasing exhaustive* sequence of measurable sets

$$A_q \subseteq A_2 \subseteq \dots \qquad \text{(increasing)}$$

$$\bigcup_{n\geq 1} A_n = A \qquad \text{(exhastive)}$$

Then $\lim_{n\to\infty} \mu(A_n) = \mu(A)$

(iv) **"continuity from above"**:
decreasing sequence enveloping:

$$\underbrace{A_1}_{\in\sigma} \supseteq \underbrace{A_2}_{\in\sigma} \subseteq \dots \text{ and } A = \bigcap_{n\geq 1} A_n \ (\in \sigma \text{ due to } \textbf{de Morgan} \text{ rule })$$

(Schuller: if these are measurable, $A_1, A_2, \dots$, are A measurable?)
deMorgan $M \setminus \bigcup_{n\geq 1} A_n = \bigcap_{n\geq 1} M \setminus A_n$
and $\mu(A_1) < \infty$

$$\implies \lim_{n\to\infty} \mu(A_n) = \mu(A)$$

Proof. $\implies$ Problem Sheet. □

Definition 49. A measure μ is called finite if

$$\exists \{A_1, A_2, \dots\} \subseteq \sigma$$

with $\bigcup_{n\geq 1} A_n = M$ s.t. $\mu(A_n) < \infty, \ \ \forall\, n \in \mathbb{N}$

11.1.2 2. Borel-σ-algebras

How does one specify, in practice, a σ-algebra?

Schuller: measurability isn't a property of a set, but a property of a collection of sets. It's measurable with respect to a system.

Definition 50.

Theorem 20. *The smallest σ-algebra σ for a set M that **contains** (among other subsets of M) a collection of subsets $\mathcal{E} \subseteq \mathcal{P}(M)$ is called / defined as*

$$\begin{aligned}\sigma(\mathcal{E}) &= \{A \subseteq M | \forall\, \sigma - algebras \sigma \text{ of } M \text{ with } \mathcal{E} \subseteq \sigma : A \in \sigma\} \\ &= \bigcap_{\substack{\sigma - algebra \text{ (for } M) \ \sigma \\ \mathcal{E}\subseteq\sigma}} \sigma\end{aligned}$$

Proof. (that $\sigma(\mathcal{E})$), generating set $\mathcal{E}$ □

Theorem 21. *Every σ-algebra can be written as*

$$\sigma = \sigma(\mathcal{E})$$

some $\mathcal{E}$

This construction immediately allows to link the notions of topology and σ-algebra for a set M

Definition 51. Let $(M, \mathcal{O})$ be a topological space.
Then $\sigma(\mathcal{O})$ is called the **Borel-σ-algebra** of $(M, \mathcal{O})$

Schuller the σ generated by set of open sets. $\mathcal{O}$ is a generating set of σ algebra. There are non-open set in the σ algebra.

Instructive example.
$M = \mathbb{R}$, $\mathcal{O} = \mathcal{O}_{\substack{\mathbb{R} \\ \text{standard}}}$ "standard topology".

Clearly: $\sigma := \sigma(\mathcal{O})$

$\forall\, a, b \in \mathbb{R}$: $(a, b) \in \sigma$

$\implies$ (by (iii) of σ-algebra and deMorgan)

$$\bigcap_{n\geq 1} \left(a - \frac{1}{n}, b\right) =: A_n \in \sigma(\mathcal{O}) = [a, b)$$

Schuller: So for instance, the half-open intervals are in the Borel-σ-algebra.

11.1.3 3. Lebesgue measure on $M = \mathbb{R}^d$

Schuller: There's a Lebesgue integral from measurable spaces, but only a Lebesgue measure on $\mathbb{R}^d$, not in general for measurable spaces.

Special construction of a measure

$$M = \mathbb{R}^d \qquad \text{some } d \in \mathbb{N}^+$$
$$\sigma = \sigma\left(\mathcal{O}_{\mathbb{R}^d,\text{ standard}}\right)$$

Then the ***unique*** map $\mu_{\text{Lebesgue, }\mathbb{R}^d} \equiv \lambda : \sigma(\mathcal{O}) \to \overline{\mathbb{R}}_0^+$

$\mathbb{R}^2$:
Defined by

$$\lambda\left([a_1, b_1) \times [a_2, b_2) \times \cdots \times [a_d, b_d)\right) = (b_1 - a_1) \cdot \cdots \cdot (b_d - a_d)$$

$a_i < b_i$
is called **the Lebesgue-measure on** $\mathbb{R}^d$

Remark: $(\underbrace{\mathbb{R}, +, \cdot}_{\lambda}, |\cdot|) \rightsquigarrow \mathcal{O}_s \rightsquigarrow \sigma(\mathcal{O}_s)$ $\quad (\mathbb{R}, \sigma(\mathcal{O}), \lambda)$ where $\sigma(\mathcal{O})$ is the **Borel**-σ-algebra, and λ the Lebesgue measure

From now on: tacit agreement
that $\mathbb{R}^d$ always be equipped,
if the need arises, with
Borel σ-algebra $\sigma(\mathcal{O}_{\mathbb{R}^d,s})$ and the Lebesgue measure.

Remarks:

(i) obviously, λ is a finite measure.

(ii) an alternative, unique, characterization of λ is that it is **the** measure that is invariant under translation: $\lambda(A + v) = \lambda(A) \quad \forall A \in \sigma(\mathcal{O}), \forall v \in \mathbb{R}^d$
and which sends $\lambda([0,1]^{xd}) = +1$

11.1.4 4. Measurable maps

Standard philosophy: $(M, \sigma_M) \xrightarrow{\text{study maps}} (N, \sigma_N)$

Definition 52. A map $F : M \to N$ is called **measurable** (with respect to σ_M and σ_N) if $\forall\, A \in \sigma_N : \text{preim}_f(A) \in \sigma_M$.

Lemma 2. *Schuller: all the sets A in the generator E*
*To show that f is **measurable** it suffices to check whether*

$$\forall\, A \in \mathcal{E} : preim_f(A) \in \sigma_M$$

where $\sigma_N = \sigma(\mathcal{E})$

Corollary 3. *(i) any continuous map is measurable with respect to the **Borel**-σ-algebras.*

(ii) any monotonous map

$$f : \mathbb{R} \to \mathbb{R}$$

is measurable with respect to Borel σ-algebra.

Proof. preim_f(interval) is an interval.

Schuller: because open, half-open, half-closed, closed intervals are measurable with respect to Borel σ-algebra, it's measurable. □

Theorem 22. *The composition $f \circ g : A \to C$ of two measurable maps*
$f : B \to C$ and $g : A \to B$
is again measurable.

Proof. Problem sheet. □

Schuller: many more we can Lebesgue integrate than Riemann integrate.

11.1.5 5. Push forward of a measure

Let (M, σ, μ) be a **measure** space
and let (N, τ) be a measurable space

Schuller: measurable space, you don't know the measure for each set but you know it can be assigned a measure.

Further: Let $M \xrightarrow{f} N$ be a measurable map.

$$\begin{array}{ccc} M & \xrightarrow{\quad f \quad} & N \\ \wr\downarrow & & \wr\downarrow \\ \sigma, \mu & & \tau \end{array}$$

Schuller: could I push forward σ, μ? A new μ induced?

Then the so-called **push-forward** $f_*\mu$ is a measure on (N, τ) defined by

$$\begin{aligned} f_*\mu : &\tau \to \overline{\mathbb{R}}_0^+ \\ &A \mapsto (f_*\mu)(A) := \mu(\mathrm{preim}_f(A)) \end{aligned}$$

where $\begin{array}{l} \mu : \sigma \to \overline{\mathbb{R}}_0^+ \\ f : M \to N \end{array}$

Proof. Problem sheet. □

11.1.6 6. Sets of measure zero, and "almost everywhere"

Schuller σ you can assign a measure or volume, μ is the measure or volume.

Definition 53. Let (M, σ, μ) be a measure space. Then $A \in \sigma$ is called a **set of measure zero** (null set) if

$$\mu(A) = 0$$

Remark: There may be other null sets then $\emptyset$.

Terminology: A condition, e.g. equality or convergence or ... is said to hold **"almost everywhere"** (a.e.) if it holds $\forall\, m \in M \setminus N$ ($\leftarrow$ some N with $\mu(N) = 0$).

Examples.

(a) Let $\begin{array}{l} f : M \to N \\ g : M \to N \end{array}$ maps $\begin{array}{l} (M, \sigma, \mu) \\ N \text{ and naked set} \end{array}$

Definition 54.

$$f =_{\text{a.e.}} g :\Longleftrightarrow \exists\, A \in \sigma,\, \mu(A) = 0\, \forall\, m \in M \setminus A : f(m) = g(m)$$

(b)

$$\left.\begin{array}{l} f_n : M \to N \\ f : M \to N \end{array}\right\} \begin{array}{l} (M, \sigma, \mu) \\ (N, \mathcal{O}) \end{array}$$

$$f_n \to_{\text{a.e.}} f :\Longleftrightarrow \forall\, m \in M \setminus A : f_n(m) \to f(M)$$

$$\text{where } A \in \sigma : \mu(A) = 0$$

Proof for Thm. 11.1.1, from Ch. 1. Measure Theory of Durrett (2019) [46].

Proof. 1. **monotonicity** or i.e. **monotony**: Recall set difference $B - A = B \cap A^c$. Let $x \in B - A$, so $x \in B$ and $x \neq A$. Then $x \in B \cap A^c$. If $B - A = \emptyset$ i.e. $B \cap A = B$, then $B \cap A^c = \emptyset$. Then $B - A \subseteq B \cap A^c$.
Let $x \in B \cap A^c$, so $x \in B$ and $x \in A^c$. So $x \in B$ and $x \notin A$. $x \in B - A$. $B \cap A^c \subseteq B - A$. Thus $B \cap A^c = B - A$.

Given $A \subset B$, $B = B - A + B \cap A = B - A + A$ (where + denotes disjoint union), $\mu(B) = \mu(A) + \mu(B - A) \geq \mu(A)$

2. EY: Insight, we know from Def. 48 that if we have pairwise disjoint sets, we can say something about the measure. Thus, construct pairwise disjoint sets.

Consider $B_1 = A_1$, $B_2 = A_2 - A_2 \cap A_1$, $B_3 = A_3 - A_3 \cap B_2 - A_3 \cap B_1$.

Let

$$B_i = A_i - \sum_{j=1}^{i-1} A_i \cap B_j$$

The difference with union is as follows:

$$S - (T_1 \cup T_2) = (S - T_1)(S - T_2)$$

where we use the notation $A \cup B \equiv AB$ interchangeably and because

$$S - (T_1 \cup T_2) = S \cap (T_1 \cup T_2)^c = S \cap (T_1^c T_2^c) = S T_1^c S T_2^c = (S - T_1) \cap (S - T_2)$$

Likewise, the difference with intersection is $S - (T_1 \cap T_2) = (S - T_1) \cup (S - T_2)$ since

$$S - (T_1 \cap T_2) = S(T_1 T_2)^c = S(T_1^c \cup T_2^c) = S T_1^c \cup S T_2^c = (S - T_1) \cup (S - T_2)$$

where we used set theory' distributive theorem at the second to last equality.
Thus

$$B_{i+1} \cap B_i = (A_{i+1} - \sum_{j=1}^{i} A_{i+1} \cap B_j) \cap B_i = \prod_{j=1}^{i} (A_{i+1} - A_{i+1} \cap B_j) \cap B_i = \prod_{j=1}^{i-1} (A_{i+1} - A_{i+1} \cap B_j)(A_{i+1} - A_{i+1} \cap B_i) B_i$$

$$\text{and since } (A_{i+1} A_{i+1}^c \cup A_{i+1} B_i^c) B_i = A_{i+1} B_i^c B_i = \emptyset, \text{ then}$$

$$B_{i+1} \cap B_i = (A_{i+1} - \sum_{j=1}^{i} A_{i+1} \cap B_j) \cap B_i = \emptyset \quad \text{(they're disjoint!)}$$

We can use the same step used as above for $\forall i, j$, given $i > j$ (since they're be a l, $1 \leq l < i$ such that $l = j$), and use induction to show that B_i, B_j disjoint.
By construction, $\bigcup_{n=1} B_n = \bigcup_{n=1} A_n$. And because B_i's pairwise disjoint, use Def. 48, (ii), to say

$$\mu(\bigcup_{n \geq 1} A_n) = \mu(\bigcup_{n \geq 1} B_n) = \sum_{n \geq 1} \mu(B_n)$$

Since we had defined $B_i = A_i - \sum_{j=1}^{i-1} A_i \cap B_j$, $B_i \subseteq A_i$. By Def. 48, $\mu(B_n) \leq \mu(A_n)$. Thus

$$\mu(\bigcup_{n \geq 1} A_n) \leq \sum_{n=1} \mu(A_n)$$

□

11.2 Integration of measurable functions -Lec06- Frederic Schuller

Integration of measurable functions -Lec06- Frederic Schuller, Schuller (2016) [44].

Lecture 6: **Integration of measurable functions**

Key definition: Lebesgue integral.

key application: definition: L^p spaces. special: $p = 2$

11.2.1 1. Simple functions

As a technical tool, we introduce

Schuller: for a measurable function, I need to know the σ algebra on M and σ on $\mathbb{R}_0^+$.

Definition 55. A *measurable function* $s : M \to \mathbb{R}_0^+$ is called a **simple function** if

$$s(M) = \{s_1, \dots, s_N\}$$

for some $N \in \mathbb{N}$.

Schuller: for measurable function $s : M \to \mathbb{R}_0^+$, must know σ-algebra, σ for M and usually take the Borel σ algebra $\sigma(\mathcal{O}_s|_{\mathbb{R}_0^+})$

Implication:

(1) $\text{preim}_S(\{s_i\}) \in \sigma$

s_i Borel-measurable.

(2) $s = \sum_{z \in s(M)} z \cdot \chi_{\text{preim}_s(\{z\})}$
where $s(M)$ is finite.

characteristic function

$\chi_A : M \to \mathbb{R}$

$$m \mapsto \chi_A(m) := \begin{cases} 1 & m \in A \text{ where } A \text{ is any } A \subseteq M. \\ 0 & m \notin A \end{cases}$$

Schuller: So $\sum_{z \in s(M)}$ in $s = \sum_{z \in s(M)} z \cdot \chi$ is a finite sum, and not a sequence, so there's no convergence issues.

11.2.2 2. Integration for *non-negative* measurable functions

Definition 56. By a non-negative measurable function, we mean

$$f : M \to \underbrace{\overline{\mathbb{R}}}_{\mathbb{R} \cup \{-\infty, +\infty\}}$$

such that

(i) $f(m) \geq 0$ for all $m \in M$

(ii) f measurable.

Schuller: where $\overline{\mathbb{R}}$ equipped with the σ algebra
$\overline{\sigma} = \{A \subseteq \overline{\mathbb{R}} | A \cap \mathbb{R} \in \sigma(\mathcal{O}_{\mathbb{R}})\}$

In order to define the integral of a non-neg. measurable function, we need to have a measure μ on the **domain**.

Schuller: Riemann - find area under the cover; Lebesgue - look at all the simple functions, find the supremum of the simple functions.

Definition 57. Let (M, σ, μ) be a measure space.
$[(\overline{R}, \overline{\sigma})$ is a measurable space]
Then the integral $\int f d\mu$ of a non-negative measurable function $f : M \to \overline{\mathbb{R}}$ is the number $(\in \overline{\mathbb{R}})$

$$\int f d\mu := \sup \{ \sum_{z \in s(M)} z \cdot \mu \left(\underbrace{\text{preim}_s(\{z\})}_{\in \sigma} \right) \}$$

simple functions s with $0 \leq s \leq f$

Definition 58. Let $A \in \sigma$, i.e., a measurable subset $A \subseteq M$, then we define

$$\int_A f d\mu := \int f \cdot \chi_A d\mu$$

Schuller: $f \cdot \chi_A$, these are the function proper, not the function values.

$$(f \cdot \chi_A)(m) := f(m) \underbrace{\cdot}_{\mathbb{R}} \chi_A(m)$$

Notation: It is often VERY convenient to write

$$\int f(x) \mu(dx) \text{ instead of } := \int f d\mu$$

application:
$\text{sq}_{\mathbb{R}} : \mathbb{R} \to \mathbb{R}$
$x \mapsto x^2$

$$\int \underbrace{\text{sq}_{\mathbb{R}}(x)}_{x^2} \mu(dx) \equiv \int \text{sq}_{\mathbb{R}} d\mu$$

Properties: f *non-negative* measurable

(i) Markov inequality: $z \in \mathbb{R}_0^+$

$$\int f d\mu \geq z \cdot \mu(\text{preim}_f(\{z\}))$$

(ii)

$$f \leq_{\text{almost everywhere}} g \Longrightarrow \int f d\mu \leq \int g d\mu$$

(iii)

$$f =_{\text{almost everywhere}} g \Longrightarrow \int f d\mu = \int g d\mu$$

(iv)

$$\int f d\mu = 0 \Longrightarrow f =_{\text{almost everywhere}} 0$$

(v)

$$\int f d\mu < \infty \Longrightarrow f <_{\text{almost everywhere}} \infty$$

The pivotal theorem of (Lebesgue) integration theory is **Theorem on monotone convergence**.

Let $0 \leq f_1 \leq f_2 \leq \ldots$ be a sequence of measurable functions $f_n : M \to \overline{\mathbb{R}}$

Provided $f := \sup_{n \geq 1} f_n$ **pointwise**.

Then

$$\lim_{n \to \infty} \int f_n d\mu = \int f d\mu$$

EY (20231114): Schuller says in Riemann integration, pointwise is unheard of but here we can exchange the lim with $\int$. This is a much more general, much more powerful notion of integration.

Let's collect 2 results.

Theorem 23. *Let $f, g \geq 0$ be measurable.*
Let $\alpha \in \mathbb{R}_0^+$.
Then $\int (f + \alpha g) d\mu = \int f d\mu + \alpha \int g d\mu$ (EY (20231114): Schuller says integration acts linearly as expected)

Theorem 24. *For any sequence $f_n : M \to \overline{\mathbb{R}}_0^+$ (measurable)*
We have

$$\int \left(\sum_{n=1}^{\infty} f_n \right) d\mu = \sum_{n=1}^{\infty} \int f_n d\mu$$

11.2.3 3. Integrable functions

In order for a general (not necessarily non-negative function as discussed so far) function

$$f : M \to \overline{\mathbb{R}}$$

to be integrable (having $\int f d\mu$ defined) one needs, beyond f being measurable, one very small extra technical condition. If this extra condition is satisfied, f is called **integrable**.

More precisely:

Definition 59. A function $f : M \to \overline{\mathbb{R}}$ is called **integrable** if

(i) f is **measurable**. EY (20231114): Schuller writes in thought bubble: ((M, σ) measure space, $(\overline{\mathbb{R}}, \overline{\sigma})$ measure space).

(ii) $\int |f| d\mu < \infty$

Remark (ii) $\iff \int f^+ d\mu < \infty$ *and* $\int f^- d\mu < \infty$ where $f = f^+ - f^-$ EY (20231114): Schuller has in thought bubbles $f^+ \leftarrow := \max(f, 0)$ pointwise , $f^- \leftarrow := \max(-f, 0)$ pointwise

Definition 60. The integral $\int f d\mu$ of an *integrable* f is defined as

$$\int f d\mu := \int f^+ d\mu - \int f^- d\mu$$

EY (20231114): Schuller says: restrict them, either of the 2 is ∞, $\infty - \infty$ is not good, you don't know what that is supposed to be. $0 - \infty$ is good, it's $-\infty$, $\infty - 0 = \infty$ is good.

Remark: In Quantum Mechanics, we need to deal with

$$f : M \to \mathbb{C}$$

Their integration theory follows immediately by treating $\Re(f)$ and $\Im(f)$ separately.

$$\int f d\mu := \int \Re(f) d\mu + i \int \Im(f) d\mu$$

Properties: f, g **integrable**: $M \to \mathbb{R}$

(i) $f \leq_{\text{almost everywhere}} g \implies \int f d\mu \leq \int g d\mu$

(ii) $\forall\, \alpha \in \mathbb{R}$: $\int (f + \alpha g) d\mu = \int f d\mu + \alpha \int g d\mu$

Theorem on dominant convergence

Theorem 25. *Let*

(a) $f_1, f_2, \ldots$ *sequence of measurable function.*

(b) $f_n \to_{almost\ everywhere} f$ *pointwise for some function* f*.*

(c) g *be non-negative measurable with* $\int g d\mu < \infty$ *s.t.*

$$\forall n \in \mathbb{N} : |f_n| \leq_{almost\ everywhere} g$$

Then we have

(i) f and $f_1, f_2, f_3, \ldots$ are integrable.

(ii)

$$\lim_{n\to\infty} \int |f_n - f| d\mu < \infty$$

EY (20231115) Schuller says the LHS is finite.

(iii)

$$\lim_{n\to\infty} \int f_n d\mu = \int f d\mu$$

Markov $\forall\, z \in \mathbb{R}_0^+ : \int f d\mu \geq z \cdot \mu(\mathrm{preim}_f([z, \infty)))$

11.2.4 4. The function spaces $L^p(M, \sigma, \mu)$

Fact: Quantum mechanics of a particle in 3 space-dimensions rests on the (as we will see: Hilbert) space

$$L^2(\mathbb{R}^3, \sigma(\mathcal{O}_{\mathbb{R}^3}), \lambda^3)$$

While we will need the theory for the special case $p = 2$ only, it is worthwhile to develop it for any *real* $1 \leq p < \infty$.

Definition 61. The set $\mathcal{L}^p := \{f : M \to \mathbb{C} | f \text{ measurable and } \int |f|^p d\mu < \infty\}$

Equipping $\mathcal{L}^p$ with pointwise $+$ and $\cdot$ makes $(\mathcal{L}^p(M, \sigma, \mu), +, \cdot)$ into a $\mathbb{C}$-vector space:

$$\begin{aligned} + : \mathcal{L}^p \times \mathcal{L}^p &\to \mathcal{L}^p \\ (f, g) &\mapsto (f + g) \\ (f \underbrace{+}_{\mathcal{L}^p} g)(m) := f(m) &\underbrace{+}_{\mathbb{C}} g(m) \quad \forall\, m \in M \end{aligned}$$

$$\begin{aligned} \cdot : \mathbb{C} \times \mathcal{L}^p &\to \mathcal{L}^p \\ (\alpha, f) &\mapsto \alpha \cdot_{\mathbb{C}} f \\ (\alpha \cdot f)(m) = \alpha \cdot_{\mathbb{C}} f(m) &\quad \forall\, m \in M \end{aligned}$$

Theorem 26. *The map:* $\|\cdot\|_p : \mathcal{L}^p \to \mathbb{R}$ *defined as*

$$\|f\|_p := (\int |f|^p d\mu)^{\frac{1}{\underbrace{p}_{\in\mathbb{R}}}}_{1 \leq \cdot < \infty}$$

is a ***semi-norm****:*

(i)

$$\forall\, \alpha \in \mathbb{C} : \|\alpha f\|_p = |\alpha| \cdot \|f\|_p$$

(ii)

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p$$

(iii)

$$\wr f \wr_p \geq 0$$

$$\cancel{\wr f \wr_p = 0 \Longrightarrow f = 0}$$

Remark: $\wr f \wr_p = 0 \Longrightarrow f =_{\text{almost everywhere}} 0$, and this makes

$$(\mathcal{L}^p, +, \cdot, \wr\cdot\wr) \text{ not even a normed space}$$

To remove the troublesome f which are zero almost everywhere while refusing to be zero everywhere define the following equivalence relation:

Call $f, g \in \mathcal{L}^p$ equivalent: $f \sim g :\Longleftrightarrow f =_{\text{almost everywhere}} g \qquad g =_{\text{almost everywhere}} h$

Equivalence Relation:

$$\checkmark f \sim g$$
$$\checkmark f \sim f$$
$$f \sim g \,\&\, g \sim h \Longrightarrow f \sim h$$

$$0 \leq \mu(A \bigcup B) \leq \mu(A) + \mu(B)$$
$$= 0$$
$$\mu(A) = 0$$
$$\mu(B) = 0$$

EY (20231115): Schuller says: Equivalence relations are looking at things in a more forgiving way. I choose to ignore.

For any equivalence relation $\sim$ on M (EY (20231115) Schuller says M is L^p) one has the so-called **quotient space**

$$M/\sim := \{[m] | m \in M\}$$
$$[m] := \{\overline{m} | \overline{m} \sim m\}$$

EY (20231115): Schuller says: You have the set of the boys and the girls. You ignore the tall boy, or the short boy. You say they're equivalent is by definition of gender.

$m_1 \sim m_2 :\Longleftrightarrow g(m_1) = g(m_2)$

EY (20231115): Schuller says: equivalence class is when you forget the elements inside the set. Here, we're identifying the functions that are different on set of measure zero.

For our $f \sim g :\Longleftrightarrow f =_{\text{almost everywhere}} g$ we have

$$L^p := \mathcal{L}^p/\sim = \{[f]_\sim | f \in \mathcal{L}^p\}$$

Check:

$$[f]_\sim +_{L^p} [g]_\sim := [f +_{\mathcal{L}^p} g]_\sim$$
$$\alpha \cdot_{L^p} [f]_\sim := [\alpha \cdot_{L^p} f]$$

are well-defined.

Show: $\underbrace{[0]}_{\mathcal{L}^p} + [f] = [f]$

Inherit

$$\|\cdot\| : L^p \to \mathbb{R} \quad \underbrace{[f]}_{\in \mathcal{L}^p} \mapsto \|[f]\| := \wr f \wr \qquad \textit{Well-defined}$$

Theorem 27.

$$\|\cdot\|_p : L^p \to \mathbb{R}$$

is a ***norm*** *on* L^p*.*

$$(\Longrightarrow (L^p(M, \sigma, \mu), +, \cdot, \|\cdot\|_p) \textit{ is a normed space })$$

Theorem 28. $(L^p, \dots)$ *is a* ***Banach space****.*

Fact: Hölder inequality: $1 \leq p, q < \infty$ and $\frac{1}{p} + \frac{1}{q} = 1$, then

$$\| \int \overline{f} \cdot g d\mu \| \leq \left(\int |f|^p d\mu \right)^{\frac{1}{p}} + \left(\int |g|^q d\mu \right)^{\frac{1}{q}}$$

(Hölder inequality)

Observation: Letting $\langle [f], [g] \rangle := \int \overline{f} \cdot g d\mu$ (well-defined)

$$\langle \cdot, \cdot \rangle : L^p \times L^p \to \mathbb{C}$$

Check sesquilinear inner product

$$\implies |\langle f, g \rangle| \leq \|f\|_2 + \|g\|_2$$

EY (20231115) Schuller says: Only for the case $p = 2$ do you get the Cauchy inequality.

$$\underbrace{\implies}_{p=2} |\langle f, g \rangle| \leq \|f\|_2 + \|g\|_2 \qquad \text{Cauchy inequality}$$

Theorem 29. *$(L^2(M, \sigma, \mu), +, \cdot, \langle \cdot | \cdot \rangle)$ is a Hilbert space.*

*("the (so-called) square-**integrable** [complex functions]")*

EY (20231115) Schuller uses [] to denote equivalence classes in [complex functions].

EY (20231115) Schuller says: we keep in mind that equal almost everywhere. We work with the representative of the equivalence class. We make a pick.

Part V
Combinatorics, Probability Theory

Theorem 30 (4.2. of Feller (1968) [72])**.** *Let $r_1, \ldots r_k \in \mathbb{Z}$, s.t. $r_1 + r_2 + \cdots + r_k = n$;, $r_i \geq 0$.*

Let

$$\frac{N!}{r_1! r_2! \ldots r_k!} = \tag{40}$$

number of ways in which n elemnts can be divided into k ordered parts (partitioned into k subpopulations). cf. Eq. (4.7) of Feller (1968) [72].

Note that the order of the subpopulations is essential in the sense that $(r_1 = 2, r_2 = 3)$ and $(r_1 = 3, r_2 = 2)$ represent different partitions. However, no attention is paid to the order within the groups.

Proof.

$$\binom{n}{r_1} \binom{n - r_1}{r_2} \binom{n - r_1 - r_2}{r_3} \cdots \binom{n - r_1 - \cdots - r_{k-2}}{r_{k-1}} = \frac{n!}{r_1! r_2! \ldots r_k!} \tag{41}$$

i.e. in order to effect the desired partition, we have to select r_1 elementsout of n, remaining $n - r_1$ elements select a second group of size r_2, etc. After forming the $(k-1)$st group there remains $n - r_1 - r_2 - \cdots - r_{k-1} = r_k$ elements, and these form the last group. □

cf. pp. 37 of Feller (1968) [72] Examples. (g) Bridge. 32 cards are partitioned into 4 equal groups $\to 52!/(13!)^4$.

Probability each player has an ace (?).

The 4 aces can be ordered in $4! = 24$ ways, each order presents 1 possibility of giving 1 ace to each player. Remaining 48 cards distributed $(48!)/(12!)^4$ ways.

$$\to p = 24 \frac{48!}{(12!)^4} / \frac{52!}{(13!)^4}$$

(h) A throw of 12 dice $\rightarrow 6^{12}$ different outcomes total. Event each face appears twice can occur in as many ways as 12 dice can be arranged in 6 groups of 2 each.

$$\frac{12!}{(2!)^6} / \frac{52!}{(13!)^4}$$

11.2.5 Application to Occupancy Problems; binomial coefficients

cf. Sec. 5 Application to Occupancy Problems of Feller (1968) [72].

Consider randomly placing r balls intos n cells.

Let r_k = occupancy number = number of balls in kth cell.

Every n-tuple of integers satisfying $r_1 + r_2 + \cdots + r_n = r$; $r_k \geq 0$. describes a possible configuration of occupancy numbers.

With indistinguishable balls 2 distributions are distinguishable only if the corresponding n-tuples $(r_1, \ldots r_n)$ are not identical.

(i) number of distinguishable distributions is

$$A_{r,n} = \binom{n+r-1}{r} = \binom{n+r-1}{n-1} \tag{42}$$

cf. Eq. (5.2) of Feller (1968) [72]

(ii) number of distinguishable distributions in which no cell remains empty is $\binom{r-1}{n-1}$.

Proof. Represent balls by stars, indicate n cells by n spaces between $n+1$ bars. e.g. $r = 8$ balls . $n = 6$ cells

3	10	00		4
$\|**$	$*\|$	$\|\|$	$\|$	$****\|$

Such a symbol necessarily starts and ends with a bar, but remaining $n-1$ bars and r starts appear in an arbitrary order. In this way, it becomes apparent that the number of distinguishable distributions equals the number of ways of selecting.

r places out of $n+r-1$, $\frac{(n+r-1)!}{(n-1)!r!} = \binom{n-1+r}{r}$

$$\begin{array}{ll} ||| \ldots || & n+1 \text{ bars} \\ * * * \cdots * * & r \text{ stars leave } r-1 \text{ spaces} \end{array}$$

Condition that no cell be empty imposes the restriction that no 2 bars be adjacent. r stars leave $r-1$ spaces of which $n-1$ are to be occupied by bars. Thus $\binom{r-1}{n-1}$ choices.

□

Probability to obtain given occupancy numbers $r_1, \ldots r_n = \frac{r!}{r_1! r_2! \ldots r_n!} / n^r$, with r balls given by Thm. 4.2. of n cells Feller (1968) [72], which is the Maxwell-Boltzmann distribution.

(a) Bose-Einstein and Fermi-Dirac statistics. Consider r indistinguishable particles, n cells, each particle assigned to 1 cell.

State of the system - random distribution of r particles in n cells.

If n cells distinguishable, n^r arrangements equiprobable $\rightarrow$ Maxwell-Boltzmann statistics.

Bose-Einstein statistics: only distinguishable arrangements are considered, and each assigned probability $\frac{1}{A_{r,n}}$

$$A_{r,n} = \binom{n+r-1}{r} = \binom{n-1+r}{n-1} \tag{43}$$

cf. Eq. 5.2 of Feller (1968) [72]

Fermi-Dirac statistics.

(1) impossible for 2 or more particles to be in the same cell. $\rightarrow r \leq n$.

(2) all distinguishable arrangements satisfying the first condition have equal probabilities.
$\rightarrow$ an arrangement is completely described by stating which of the n cells contain a particle
r particles $\rightarrow \binom{n}{r}$ ways r cells chosen.
Fermi-Dirac statistics, there are $\binom{n}{r}$ possible arrangements, prob. $1/\binom{n}{r}$.

pp. 39. Feller (1968) [72]. Consider cells themselves indistinguishable! Disregard order among occupancy numbers.

cf. Feller (1968) [72]

12 Distributions and Densities

cf. Ch. 5 "Important Distributions and Densities", Grinstead and Snell (1997) [73]

12.1 Discrete Uniform Distribution

$X :=$ random variable representing outcome of an experiment where all outcomes equally likely, so X *uniformly distributed*.

If sample space S is size n, where $0 < n < \infty$, then distribution function

$$m(\omega) = \frac{1}{n} \quad \forall\, \omega \in S$$

Consider $1 + \lfloor x \rfloor \in \{1 \dots n\}$,
with probability $\frac{1}{n}$ ($\lfloor x \rfloor$ denotes greatest integer not exceeding x).

If sample space is uncountable set with positive finite length, such as $[0, 1]$, then use cont. density function.

12.2 Poisson Random Variable

Sec. 5.2 "The Poisson Random Variable", Ross (2000) [74].

random variable X, $X \in \{0, 1, 2, \dots\}$ is **Poisson random variable** with parameter λ, $\lambda > 0$, if its probability mass function is

$$P\{X = i\} = e^{-\lambda}\frac{\lambda^i}{i!}, \quad i = 0, 1, \dots \tag{44}$$

Since

$$\sum_{i=0}^{\infty} p(i) = e^{-\lambda}\sum_{i=0}^{\infty}\frac{\lambda^i}{i!} = e^{-\lambda}e^{\lambda} = 1$$

then $P\{X = i\} = e^{-\lambda}\frac{\lambda^i}{i!}$ defines a probability mass function.

Moment generating function $\phi(t)$ for *Poisson random variable*

$$\phi(t) = E[e^{tX}] = \sum_{i=0}^{\infty} e^{ti}e^{-\lambda}\frac{\lambda^i}{i!} = e^{-\lambda}\sum_{i=0}^{\infty}\frac{(\lambda e^t)^i}{i!} = e^{-\lambda}e^{\lambda e^t} = \exp\left[\lambda(e^t - 1)\right] \tag{45}$$

Both *mean* and *variance* of Poisson random variable equal to parameter λ, since

$$\begin{aligned}
&\phi'(t) = \phi \cdot (\lambda e^t) = \lambda e^t \exp\left[\lambda(e^t - 1)\right] \\
&\phi''(t) = \phi'(\lambda e^t) + \phi\lambda e^t = (\lambda e^t)(\lambda e^t + 1)\phi \\
&E[X] = \phi'(0) = \lambda \\
&\mathrm{Var}(X) = \phi''(0) - (E[X])^2 = \lambda(\lambda + 1) - \lambda^2 = \lambda
\end{aligned}$$

Poisson random variable when n large, p small, may be used as an approximation for binomial random variable. Suppose X binomial random variable with parameters (n, p), $\lambda = np$

$$P\{X=i\} = \frac{n!}{(n-i)!i!}p^i(1-p)^{n-i} = \binom{n}{i}\left(\frac{\lambda}{n}\right)^i\left(1-\frac{\lambda}{n}\right)^{n-i} = \frac{n(n-1)\dots(n-i+1)}{n^i}\frac{\lambda^i}{i!}\frac{(1-\lambda/n)^n}{(1-\lambda/n)^i}$$

For n large, p small,

$$\left(1-\frac{\lambda}{n}\right)^n \approx e^{-\lambda} \qquad \frac{n(n-1)\dots(n-i+1)}{n^i} \approx 1 \qquad \left(1-\frac{\lambda}{n}\right)^i \approx 1$$

Hence, for n large, p small

$$P\{X=i\} \approx e^{-\lambda}\frac{\lambda^i}{i!} \tag{46}$$

i.e. if n independent trials, each of which results in "success" with probability $p \ll 1$, then when n large ($n \gg 1$) and p small ($p \ll 1$),
number of successes occurring is approximately Poisson random variable with mean $\lambda = np$.

Examples of n large, p small for random variables that obey to good approximation the Poisson probability law:

1. number of misprints on page (or group of pages) of book
   Suppose there's small probability p each letter typed on page will be misprinted,
   so number of misprints on a given page will be approximately Poisson with mean $\lambda = np$ where n is (presumably) the large number of letters on the page
2. number of people in community living to 100 years of age
   Suppose each person in a given community, independently, has a small probability p of reaching age 100, and so number of people that do will have approximately a Poisson distribution with mean np where n is large number of people in the community.
3. number of wrong telephone numbers that are dialed in a day
4. number of transistors that fail on first day of use
5. number of customers entering post office on given day
6. number of α-particles discharged in fixed period of time from some radioactive particle

Example 5.2a (Ross (2000) [74]), Suppose average number of accidents occurring weekly on highway equals 3. Calculate probability that there's at least 1 accident this week.
Solution Example 5.2a, Ross (2000) [74]. Let X denote number of accidents occurring on highway during this week. It's reasonable to suppose there's a large number of cars passing along stretch, each having small probability of being involved in an accident.

$$P\{X \geq 1\} = 1 - P\{X < 1\} = 1 - P\{X = 0\} = 1 - e^{-3}\frac{3^0}{0!} = 1 - e^{-3} = 0.9502$$

Example 5.2b. of Ross (2000) [74]. Suppose p = prob. item produced by certain machine is defective = 0.1. Find prob. sample of $N = 10$ items will contain at most 1 defective item.
Solution Example 5.2b, Ross (2000) [74].

$$P(X \leq 1) = \sum_{j=0}^{1}\binom{N}{j}p^j(1-p)^{N-j} = 0.7361 \text{ for } N = 10,\ p = 0.1$$

$$P(X \leq 1) \approx \sum_{j=0}^{1} e^{-\lambda}\frac{\lambda^j}{j!} = e^{-1}\left(\frac{1^0}{0!} + \frac{1^1}{1!}\right) = 2e^{-1}$$

Example 5.2c of Ross (2000) [74]. Consider experiment counting number of α-particles given off in 1-sec. interval.

If on average 3.2 α-particles given off,
What's good approximation to prob. that no more than 2 α-particles will appear?
Solution Example 5.2c, Ross (2000) [74].
$\lambda = Np = 3.2$

$$P(X \leq 2) = \sum_{j=0}^{2} e^{-\lambda} \frac{\lambda^j}{j!} = e^{-\lambda} \left(1 + \sum_{j=1}^{2} \frac{\lambda^j}{j!} \right)$$

Example 5.2d of Ross (2000) [74]. If average number of claims handled daily by insurance company is 5, what proportion of days have less than 3 claims?
What's prob. there'll be 4 claims in exactly 3 of the next 5 days?
Assume number of claims on different days is independent.
Solution Example 5.2d, Ross (2000) [74].
Suppose N = number of possible claims/clients large, so small prob. of making claim,

$$\lambda = 5$$

$$P(X < 3) = \sum_{j=0}^{2} e^{-\lambda} \frac{\lambda^j}{j!} = e^{-\lambda} \sum_{j=0}^{2} \frac{\lambda^j}{j!}$$

By law of large numbers (different days independent) $P(X < 3) = 0.125$ days have fewer than 3 claims.

$P(X = 4)$ = prob. there'll be 4 claims on a day $= e^{-\lambda} \frac{\lambda^4}{4!}$
$P(Y = 3)$ = prob. of 4 claims in exactly 3 of next 5 days $= \binom{5}{3}(P(X = 4))^3(1 - P(X = 4))^2 = \binom{5}{3}(e^{-\lambda} \frac{\lambda^4}{4!})^3(1 - e^{-\lambda} \frac{\lambda^4}{4!})^2$

Part VI
Fourier Transform

13 Derivation of Fourier Transform; Definition

We will follow Sec. 32.2 of Mauch (2004) [47].
Consider now this definition of the Fourier Transform, different from Mauch (2004) [47].

Definition 62.

$$\begin{aligned} \widehat{f}(k) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} dx f(x) e^{-ikx} \\ f(x) &= \int_{-\infty}^{\infty} dk \widehat{f}(k) e^{ikx} \quad \forall\, x \in \mathbb{R} \end{aligned} \tag{47}$$

13.1 Fourier Transform Properties

Recall "Fourier Transformation of a Derivative", Mauch (2004) [47].

$$\mathcal{F}[y'(x)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} dx y'(x) e^{-i\omega x} = \left[\frac{1}{2\pi} y(x) e^{-i\omega x} \right]_{-\infty}^{\infty} - \frac{1}{2\pi} \int_{-\infty}^{\infty} dx y(x) e^{-i\omega x} (-i\omega) =$$
$$= i\omega \mathcal{F}[y(x)]$$

Next, consider $y''(x)$:

$$\mathcal{F}[y''(x)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} dx y''(x) e^{-i\omega x} = \left[\frac{1}{2\pi} y'(x) e^{-i\omega x} \right]_{-\infty}^{\infty} + i\omega \mathcal{F}[y'(x)] =$$
$$= (i\omega)^2 \mathcal{F}[y(x)]$$

Then

$$\mathcal{F}[y^{(n)}(x)] = (i\omega)^n \mathcal{F}[y(x)] \tag{48}$$

Part VII
Laplace Transform

This is for you, you, my number one
This is for you, you, my number one

This is for you, you, my number one

Spotlights, big stage
Fifty thousand fans screamin' in a rage
Bodyguards and limousines
This is the way I see you in my dreams

- Usher, "Superstar"

Me: In which category or functor in category theory do Laplace transforms belong in?

Chat-GPT: Laplace transforms can be studied in the category of topological vector spaces. More specifically, the Laplace transform can be seen as a functor from the category of functions of exponential growth to the category of holomorphic functions defined on a half-plane in the complex plane.

In this category, the morphisms are continuous linear maps that preserve the exponential growth of functions. The Laplace transform functor takes a function in the source category and maps it to its Laplace transform in the target category.

The use of category theory provides a powerful framework for studying Laplace transforms and their properties in a more abstract and general way.

14 Laplace Transform: Definitions and Properties

14.1 Laplace Transform Definition and Examples

We will follow Mauch (2004) [47].

Definition 63.

$$\mathcal{L}[f(t)] = \int_0^\infty e^{-st} f(t) dt \equiv \widehat{f}(s) \tag{49}$$

Definition 64. A function $f(t)$ is of **exponential order** α if $\exists$ constants t_0, and M s.t.

$$|f(t)| < Me^{\alpha t} \quad \forall\, t > t_0 \tag{50}$$

If $\exists\ \int_0^{t_0} f(t)dt$ and $f(t)$ is of exponential order α, then $\exists\, \widehat{f}(s) \quad \forall\, \Re(s) > \alpha$.

Proof.

$$|\int_0^\infty e^{-st} f(t)dt| = |\int_0^\infty \exp\left(-\Re(s)t - i\Im(s)t\right) f(t)dt| \leq \int_0^\infty \exp\left(-\Re(s)t\right)|f(t)|dt <$$

$$< \int_0^\infty \exp\left(-\Re(s)t\right) M e^{\alpha t} dt = M \int_0^\infty dt \exp\left(-(\Re(s) - \alpha)t\right) = M \left(\frac{\exp\left(-(\Re(s) - \alpha)t\right)}{-(\Re(s) - \alpha)} \right) \Bigg|_0^\infty = \frac{M}{\Re(s) - \alpha}$$

if $\Re(s) > \alpha$ □

For example,
$\sin(t)$ is of exponential order 0.
te^{2t} is of exponential order α, $\forall\, \alpha > 2$.

Example 31.1 (Mauch (2004) [47]). Consider the Laplace transform of $f(t) = 1$. Since $f(t) = 1$ is of exponential order α $\forall\, \alpha > 0$, Laplace transform integral converges for $\Re(s) < 0$.

$$\widehat{f}(s) = \int_0^\infty e^{-st} dt = \left[\frac{-1}{s} e^{-st}\right]_0^\infty = \frac{1}{s}$$

Example 31.1.3. Consider Laplace transform of the Heaviside function

$$H(t-c) = \begin{cases} 0 & \text{for } t < c \\ 1 & \text{for } t > c \end{cases}$$

where $c > 0$.

$$\mathcal{L}[H(t-c)] = \int_0^\infty e^{-st} H(t-c) dt = \int_c^\infty e^{-st} dt = \left[\frac{e^{-st}}{-s}\right]_c^\infty = \frac{e^{-cs}}{s} \quad \forall\, \Re(s) > 0$$

Example 31.1.4. Consider $H(t-c)f(t-c)$

$$\mathcal{L}[H(t-c)f(t-c)] = \int_0^\infty dt e^{-st} H(t-c) f(t-c) = \int_c^\infty dt e^{-st} f(t-c) = \int_0^\infty dt e^{-s(t+c)} f(t) = e^{-cs} \widehat{f}(s)$$

14.1.1 Laplace Transform Properties and Formulae

Theorem 31.

$$\begin{aligned} \mathcal{L}\left[\frac{d}{dt} f(t)\right] &= s\widehat{f}(s) - f(0) \\ \mathcal{L}\left[\frac{d^2}{dt^2} f(t)\right] &= s^2 \widehat{f}(s) - sf(0) - f'(0) \end{aligned} \tag{51}$$

Proof. To deserve these formulas,

$$\begin{aligned} \mathcal{L}\left[\frac{d}{dt} f(t)\right] &= \int_0^\infty dt e^{-st} f'(t) = \left[e^{-st} f(t)\right]_0^\infty - \int_0^\infty dt(-se^{-st}) f(t) = -f(0) + s\widehat{f}(s) \\ \mathcal{L}\left[\frac{d^2}{dt^2} f(t)\right] &= \int_0^\infty dt e^{-st} f''(t) = \left[e^{-st} f'(t)\right]_0^\infty - \int_0^\infty dt(-se^{-st}) f'(t) = -f'(0) + s\mathcal{L}[\frac{d}{dt} f'(t)] = \\ &= -f'(0) - sf(0) + s^2 \widehat{f}(s) \end{aligned}$$

□

14.2 Inverse Laplace Transform: Definition

Definition 65. The inverse Laplace transform is denoted

$$f(t) = \mathcal{L}^{-1}[\widehat{f}(s)] \tag{52}$$

Compute the inverse Laplace transform with the Mellin inversion formula:

$$f(t) = \frac{1}{2\pi i} \int_{\alpha - i\infty}^{\alpha + i\infty} ds e^{st} \widehat{f}(s) \tag{53}$$

where α is a real constant that's to the right of the singularities of $\widehat{f}(s)$.

Consider

$$\mathcal{L}[\mathcal{L}^{-1}[\widehat{f}(s)]] = \mathcal{L}\left[\frac{1}{2\pi i} \int_{\alpha - i\infty}^{\alpha + i\infty} dz e^{zt} \widehat{f}(z)\right] = \int_0^\infty dt e^{-st} \frac{1}{2\pi i} \int_{\alpha - i\infty}^{\alpha + i\infty} dz e^{zt} \widehat{f}(z)$$

Interchange order of integration:

$$\frac{1}{2\pi i}\int_{\alpha-i\infty}^{\alpha+i\infty} dz\widehat{f}(z)\int_0^\infty dte^{-(s-z)t}$$

Since $\Re(z)=\alpha$ (because z is from $\alpha-i\infty$ to $\alpha+i\infty$, the integral in t exists for $\Re(s)>\alpha$. So

$$\int_0^\infty dte^{-(s-z)t}=\left(\frac{e^{-(s-z)t}}{-(s-z)}\right)_0^\infty=\frac{1}{s-z}\Longrightarrow\frac{1}{2\pi i}\int_{\alpha-i\infty}^{\alpha+i\infty}dz\frac{\widehat{f}(z)}{s-z}$$

Part VIII
Category Theory

15 Note on notation

From the section on "Terminology" of the Preface of Barr and Wells (1998) [26]:

> "In most scientific disciplines, notation and terminology are standardized, of- ten by an international nomenclature committee. (Would you recognize Einstein's equation if it said $p=HU^2$?) We must warn the nonmathematician reader that such is not the case in mathematics. There is no standardization body and terminology and notation are individual and often idiosyncratic."

To try to bridge the difference choice of notation and through comparison, suggest the "best" notation that's easy to remember and easy to use, I'll present all the different types of notation that I come across as much as I can. My plan of attack is the following:

1. I'll try to present different types of notation and reference the authors of the text when I can.
2. I'll try to defer to the notation used in Wikipedia, first.
3. I'll make a final decision of what notation works best (for me).

16 Category A, (definition)

Definition 66 (Category **A**)**. category A** is quadruple $\mathbf{A}=(\text{Obj}(\mathbf{A}),\text{Mor}\mathbf{A},1,\circ)$

$$\mathbf{A}=(\text{Obj}(\mathbf{A}),\text{Mor}\mathbf{A},1,\circ) \tag{54}$$

s.t.

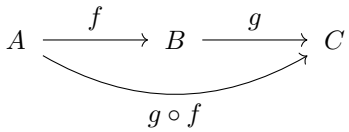
1. $\text{Obj}(\mathbf{A})$ is a *class*, whose elements, $A\in\text{Obj}(\mathbf{A})$, are called *objects*
2. Mor**A** is a *class*.
   (a) From Adámek, Herrlich, and Strecker (2004) [27], Kashiwara and Schapira (2006) [24], $\forall\, A,B\in\text{Obj}(\mathbf{A})$, $\exists\,\text{Hom}(A,B)\subseteq\text{Mor}(\mathbf{A})$. Therefore,
   $$\text{Mor}\mathbf{A}=\bigcup_{A,B\in\text{Obj}(\mathbf{A})}\text{Hom}(A,B) \tag{55}$$
   (b) $\forall\, f\in\text{Hom}(A,B)$, $f:A\to B\in\text{Hom}(A,B)$ is a *morphism*. Leinster (2014) [31] also calls them on pp. 10 **maps** or **arrow** from A to B.
   $$A\xrightarrow{\quad f\quad}B$$
3. $\forall\, A\in\text{Obj}(\mathbf{A})$, $\exists\, 1_A:A\to A$, i.e. $\exists\,\mathbf{1}_A\in\text{Hom}_{\mathbf{A}}(A,A)\equiv\text{Hom}(A,A)$,
   $$A\xrightarrow{\quad 1_A\quad}A\qquad\text{or}\qquad A\overset{\mathbf{1}_A}{\circlearrowright}A$$

4. **composition:** $\forall\, A, B, C \in \text{Obj}\mathbf{A}$, define **composition** to be a map

$$\begin{aligned} \text{Hom}_{\mathbf{A}}(A,B) \times \text{Hom}_{\mathbf{A}}(B,C) &\to \text{Hom}_{\mathbf{A}}(A,C) \\ (f,g) &\mapsto g \circ f \end{aligned} \tag{56}$$

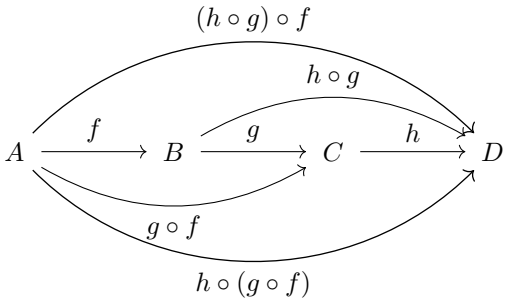
, i.e.

$\forall\, f : A \to B \in \text{Hom}(A,B)$, i.e. $f, g \in \text{Mor}\mathbf{A}$,
$g : B \to C \in \text{Hom}(B,C)$
then $g \circ f : A \to C \in \text{Hom}(A,C)$, $g \circ f \in \text{Mor}\mathbf{A}$ i.e.

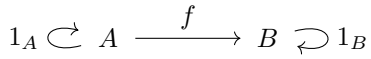


s.t.

(a) *associativity* $\forall \begin{matrix} f : A \to B \\ g : B \to C \\ h : C \to D \end{matrix}$, $h \circ (g \circ f) = (h \circ g) \circ f$ i.e.



(b) $\forall\, f : A \to B \in \text{Hom}(A,B)$, $1_B \circ f = f$ and $f \circ 1_A = f$ i.e.
$\forall\, f \in \text{Hom}_{\mathbf{A}}(A,B)$,



(c) Adámek, Herrlich, and Strecker (2004) [27] posited further that $\text{Hom}(A,B) \in \text{Mor}\mathbf{A}$ pairwise disjoint (i.e. $\text{Hom}(A,B) \bigcap \text{Hom}(C,D) \neq \emptyset$ if $C \neq A$ or $D \neq B$)

16.1 Examples of categories

- $\mathbf{Set} = (\text{Obj}(\mathbf{Set}), \text{Hom}\mathbf{Set}, \mathbf{1}, \circ)$ where
  Obj(**Set**) is the class of all sets
  Hom**Set** is the class of all functions on a set to another set

- **Vec**

  Obj**Vec** $\equiv$ all real vector spaces
  Mor**Vec** $\equiv$ all linear transformations between them (between real vector spaces)

- *preorder* cf. Ch. 1, Example 7, pp. 8, Awodey (2010) [42].
  preorder P, set P equipped with binary relation $p \leq q$, that's both
  reflexive: $a \leq a$, and
  transitive: if $a \leq b$ and $b \leq c$, then $a \leq c$.

$\forall$ preorder P can be regarded as a category:
$\text{Obj}(P) = P$
$a \to b$ iff $a \leq b \in \text{Mor}P$.
identity: $1 : a \to a$ since $a \leq a$
composition: $f : a \to b$, $g : b \to c$, $(g \circ f)a = g(b) = c$, $a \leq b$ and $b \leq c$, so $a \leq c$.

Going in the other direction, a category with at most 1 arrow between 2 objects determines a preorder.

- **Monoid**. Consider a monoid as a triple $(M, \cdot, e)$.
Every semigroup $(M, \cdot)$ (recall that a *semigroup* is a set S with binary operation $\cdot$, i.e. s.t.

$S \times S \dot{\to} S$

$\forall\, a, b, c \in S$, $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ (associativity)

(but no inverse, necessarily!)) that also has a unit e can be made into a category $\mathbf{C}$

$\implies \mathbf{C}(M, \cdot, e) = (\text{Obj}(\mathbf{C}), \text{Hom}(\mathbf{C}), \mathbf{1}, \circ)$, a category $\mathbf{C}$ with only 1 object, i.e. $\text{Obj}(\mathbf{C}) = \{M\}$, so that
$\text{Obj}(\mathbf{C}) = \{M\}$
$\text{Hom}(M, M) = M$
$\mathbf{1}_M = e$
$y \circ x = y \cdot x$

cf. pp. 10, Example 12 of Awodey (2010) [42].

16.1.1 Monoid as a category

Recall the definition of a *category* $\mathbf{A}$: category $\mathbf{A} = (\text{Obj}(\mathbf{A}), \text{Mor}\mathbf{A}, 1, \circ)$ s.t.
$A \in \text{Obj}\mathbf{A}$,
$\forall\, A, B \in \text{Obj}\mathbf{A}$, $\exists \text{Hom}(A, B) \subseteq \text{Mor}\mathbf{A}$,
$\forall\, f \in \text{Hom}(A, B)$, $f : A \to B$ is a morphism.
$\forall\, A \in \text{Obj}\mathbf{A}$, $\exists\, 1_A : A \to A$, i.e. $\exists\, 1_A \in \text{Hom}_{\mathbf{A}}(A, A)$.
$\forall\, A, B, C \in \text{Obj}\mathbf{A}$,

$$\text{Hom}_{\mathbf{A}}(A, B) \times \text{Hom}_{\mathbf{A}}(B, C) \to \text{Hom}_{\mathbf{A}}(A, C)$$
$$(f, g) \mapsto g \circ f$$

Recall the definition of a *monoid*: monoid $(M, \cdot, e)$, set M,
$\cdot : M \times M \to M$,
$e \in M$ s.t. $\forall\, x \in M$, $e \cdot x = x \cdot e = x$

$\implies \mathbf{C}(M)$ (monoid as a category)
$\text{Obj}\mathbf{C}(M) = \{M\}$
$\text{Hom}(M, M) = \text{Mor}\mathbf{C}(M)$ and $\forall\, f \in \text{Hom}(M, M)$,

$$f = m \in M$$

i.e. the morphism *is* an element in monoid M (there's a 1-to-1 correspondence).

$\forall\, n \in M$, $f : n \mapsto m \cdot n \in M$.
$1_M : M \to M$
$1_M : x \mapsto e \cdot x = x$
$m \cdot n \cdot x = m \cdot (n \cdot x) = (m \cdot n) \cdot x$ (composition)

16.1.2 Examples of monoids

Examples of monoids: $\mathbb{N}, \mathbb{Q}, \mathbb{R}$ under either addition and $e = 0$, or multiplication and $e = 1$.
$\forall$ set X, set of functions from X to X, $\mathrm{Hom}_{\mathbf{Sets}}(X, X)$ is a monoid under composition.

In general,
$\forall$ object $C \in \mathrm{Obj}(\mathbf{C})$, $\forall$ category $\mathbf{C}$, $\mathrm{Hom}_{\mathbf{C}}(C, C)$ is a monoid under composition of $\mathbf{C}$.

Since monoids are structured sets, $\exists$ category **Mon**, s.t. Obj**Mon** $\ni$ monoids, Mor**Mon** $\ni$ functions that preserve monoid structure.

In detail, homomorphism from monoid M to monoid N is function $h : M \to N$ s.t. $\forall\, m, n \in M$.

$$h(m \cdot_M n) = h(m) \cdot_N h(n)$$

and

$$h(e_M) = e_N$$

Check: Consider functor $F : M \to N$.
$F : \mathrm{Hom}_M(a, b) \to \mathrm{Hom}_N(F(a), F(b))$

$$m \cdot a = b \qquad n \cdot_N F(a) = F(b)$$
$$F(m) \cdot_N F(a) = F(b) = F(m \cdot_M a)$$

so functor F is a monoid homomorphism, and so
a monoid homomorphism from M to N is the same thing as a functor from M regarded as a category to N regarded as a category.

- **Poset** cf. Ch. 1, pp. 6, Awodey (2010) [42]. partially ordered set or poset is a set A, equipped with binary relation $a \leq_A b$ s.t. $\forall\, a, b, c \in A$,
reflexivity: $a \leq_A a$
transitivity: if $a \leq_A b$, $b \leq_A c$, then $a \leq_A c$
antisymmetry: if $a \leq_A b$, $b \leq_A a$, then $a = b$.

  An arrow from a poset A to a poset B is a function,

  $$m : A \to B$$

  that is monotone, i.e. $\forall\, a, a' \in A$,

  $$\text{if } a \leq_A a', \text{ then } m(a) \leq_B m(a')$$

  category **Pos** of posets, and monotone functions,

  $$\forall\, A \in \mathrm{Obj}(\mathbf{Pos}),\ \exists\, 1_A : A \to A, \text{ since}$$

  since if $a \leq_A a'$, then $a \leq_A a'$, so 1_A monotone.

  composition: $\forall\, f : A \to B$, $g : B \to C$, $\forall\, A, B, C \in \mathrm{Obj}(\mathbf{Pos})$, f, g monotone,
  $\forall\, a \leq_A a'$, $f(a) \leq_B f(a')$, and since
  $\forall\, b \leq_B b'$, $g(b) \leq_C g(b')$, then $g(f(a)) \leq_C g(f(a'))$, or $(g \circ f)(a) \leq_C (g \circ f)(a') \,\forall\, a \leq_A a'$, so $g \circ f : A \to C$ is monotone.
  Mor**Pos** $= \bigcup_{A,B \in \mathrm{Obj}(\mathbf{Pos})}$ monotone functions $A \to B$

  A *poset* is a *preorder*, satisfying additional condition of antisymmetry: if $a \leq b$, and $b \leq a$, $a = b$.
  e.g. $\forall\, X$, power set $P(X)$ is a poset under inclusion $U \subseteq V$, $\forall$ subsets $U, V \subseteq X$.

  - cf. Ch. 1, Example 8, pp. 9, Awodey (2010) [42]. **Pos** categories and functors: functor $F : \mathbf{D} \to \mathbf{Q}$ between poset categories $\mathbf{P}, \mathbf{Q}$.
    They are the monotone functions.

Proof. Consider

$$F : \mathrm{Hom}_{\mathbf{P}}(A, B) \to \mathrm{Hom}_{\mathbf{Q}}(F(A), F(B))$$

$f \in \mathrm{Hom}_{\mathbf{P}}(A, B)$ so if $a \leq_A a'$, $f(a) \leq_B f(a')$
Let $g \in \mathrm{Hom}_{\mathbf{Q}}(F(A), F(B))$, so if $c \leq_{F(A)} c'$, $g(c) \leq_{F(B)} g(c')$.

$F(f) \in \mathrm{Hom}_{\mathbf{Q}}(F(A), F(B))$ (by definition of a functor), so if $c \leq_{F(A)} c'$, $F(f)(c) \leq_{F(B)} F(f)(c')$ or $F(f(c)) \leq_{F(B)} F(f(c'))$.

So F itself is monotone. □

- cf. *An example from computer science*, Example 10, pp. 9 of Awodey (2010) [42]. Given a functional programming language L, $\exists$ associated category,
  Obj$\mathbf{L} \ni$ data types of L
  Mor$\mathbf{L} \ni$ computable functions of L ("processes", "procedures", "programs")
  composition (of 2 such programs) $X \xrightarrow{f} Y \xrightarrow{g} Z$ given by applying g to output of f
  (notation note: $g \circ f = f; g$)
  identity is the "do nothing" program.

  if $\mathbf{C}(L)$ is the category just defined, then
  denotational semantics of language L in category $\mathbf{D}$ of say Scott demands is simply functor

$$S : \mathbf{C}(L) \to \mathbf{D}$$

  since S assigns domains to the types of L,
  continuous functions to programs.

  This example and Example 9, pp. 9, of Awodey (2010) [42] are related to the notion of "Cartesian closed category."

  From Barr and Wells (2012) [43], a *functional programming language* L has

  1. primitive data types (built into the language)
  2. constants of each type
  3. operations, which are functions between types
  4. constructors, which are applied to data types and operations, to produce derived data types and operations of the language.

  For a $\mathbf{C}(L)$ category corresponding to functional programming language L,

  $\mathbf{C}(L)$ category corresponding to functional programming language L,
  Obj$\mathbf{C}(L) = \{$ types of $L \}$
  Mor$\mathbf{C}(L) = \{$ operations (functions; primitive and derived) of $L \}$

  identity: $\forall$ type $T \in$ Obj$\mathbf{C}(L)$, $\exists\, 1_T : T \to T$ i.e. $1_T \in \mathrm{Hom}(T, T)$.

  composition: $\forall$ types $T, U, V \in$ Obj$\mathbf{C}(L)$, defines composition

$$\begin{aligned} \mathrm{Hom}(T, U) \times \mathrm{Hom}(U, V) &\to \mathrm{Hom}(T, V) \\ (f, g) &\mapsto g \circ f \end{aligned}$$

  Assume L has a do nothing operation 1_T, and composition constructor, $\mathrm{Hom}(T, U) \times \mathrm{Hom}(U, V) \to \mathrm{Hom}(T, V)$.

  Add additional type 1 s.t. $\forall$ type $T \in$ Obj$\mathbf{C}(L)$, $\exists!$ operation (function) to $1, T \to 1$
  Interpret each constant c of type A as arrow (morphism) $c : 1 \to A$.

As a concrete example (cf. Example 2.2.5 of Barr and Wells (2012) [43]), suppose a simple language with 3 data types, `int`, `bool`, `char`.

For `int`, $\exists$ constant $0 : 1 \to \texttt{int}$, consider $\texttt{succ} : \texttt{int} \to \texttt{int} \in \text{Hom}(\texttt{int}, \texttt{int})$,
For `bool`, $\exists$ constants $\texttt{true} : 1 \to \texttt{bool}$, $\texttt{false} : 1 \to \texttt{bool}$; consider $\neg \in \text{Hom}(\texttt{bool}, \texttt{bool})$, s.t.

$$\neg \circ \texttt{true} = \texttt{false}$$
$$\neg \circ \texttt{false} = \texttt{true}$$

For `char`, $\exists$ constants $c : 1 \to \texttt{char}$, $\forall\, \texttt{char} c$.
Since $\forall T, U \in \text{Obj}(\mathbf{C}(L))$, $\exists \text{Hom}(T, U) \subseteq \text{Mor}\mathbf{C}(L)$. So consider $\texttt{ord} : \texttt{char} \to \texttt{int}$, $\texttt{chr} : \texttt{int} \to \texttt{char}$, s.t.

$$\texttt{char} \circ \texttt{ord} = 1_{\texttt{char}}$$

So $\text{Obj}\mathbf{C}(L) = \{\texttt{int}, \texttt{bool}, \texttt{char}, 1\}$,
$\text{Mor}\mathbf{C}(L)$ consists of all programs.

See 5.3.14 Record Types and 5.7.6, 14.2 for flow of control in Barr and Wells (2012) [43] to complete the programming language. For here, see section 21.2.2

- **Groups**, cf. pp. 11, Def. 1.4, Sec. 1.5 **Isomorphisms**, Awodey (2010) [42].

  A *group* G is a monoid with inverse g^{-1}, $\forall\, g \in G$.
  Thus, G is a category with 1 object, $\forall\, f \in \text{Mor}G$ is an isomorphism.

  $\forall$ set X, group $\text{Aut}(X)$ of automoprhisms (or "permutations") of X i.e. $f : X \to X$
  $\text{Aut}(X)$ is closed under composition $\circ$ because a permutation of a permutation is another permutation.

  Homomorphisms of groups $h : G \to H$ is just a homomorphism of monoids, preserving inverses.
  Given h, then $h(gg') = h(g)h(g')$.
  G, H are monoids. Then h monoid homomorphism.

Concrete categories: informally, categories which objects are sets, possibly equipped with some structure, and arrows are certain, possibly structure-preserving functions.

From pp. 10 of Mac Lane (1978) [28], Leinster (2014) [31], Examples 1.1.8 (Categories as mathematical structures), pp. 13, and Sec. 1.4 Examples of Categories, 5. Finite Categories, pp.7 of Awodey (2010) [42].

Category **1**. $\text{Obj}\mathbf{1} = \{A\}$
$\text{Mor}\mathbf{1} = \{1_A\}$; $1_A : A \to A$.

The only composition to consider is $1_A \circ 1_A : A \to A$, and $1_A \circ 1_A \circ 1_A$, so on and so for.

Category **2**. $\text{Obj}\mathbf{2} = \{A, B\}$.
$\text{Hom}(A, B) = \{f\}$, $f : A \to B$.

The only "non-trivial" composition to consider is $1_B \circ f \circ 1_A : A \to B$.

Category **3**. pp. 11, Sec. 2 Categories of Mac Lane (1978) [28], Example 5. "Finite Categories", Sec. 1.4, pp. 7 of Awodey (2010) [42].

$\text{Obj}\mathbf{3} = \{A, B, C\}$
$\text{Hom}(A, B) = \{f\}$
$\text{Hom}(A, C) = \{g\}$
$\text{Hom}(B, C) = \{h\}$
$\text{Hom}(C, \cdot) = \emptyset$
$\text{Hom}(B, A) = \emptyset$

The only "non-trivial" compositions to consider are the following: $1_C \circ h \circ f : A \to C$,
$1_C \circ g$, $1_C \circ h$ (and likewise)

16.2 Duality, opposite category

Given a category $\mathbf{A} = (\mathrm{Ob}, \hom_{\mathbf{A}}, 1, \circ)$,

Definition 67 (dual opposite category)**. dual** or **opposite** category of $\mathbf{A} = (\mathrm{Obj}(\mathbf{A}), \mathrm{Mor}\mathbf{A}, \mathbf{1}, \circ)$, denoted $\mathbf{A}^{\mathrm{op}}$, is

$$\mathbf{A}^{\mathrm{op}} = (\mathrm{Obj}(\mathbf{A}), \mathrm{Mor}\mathbf{A}^{\mathrm{op}}, \mathbf{1}, \circ^{\mathrm{op}}) \tag{57}$$

s.t.

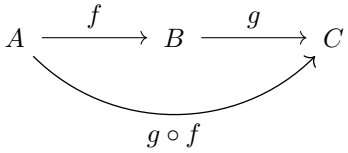
- $$\mathrm{Obj}(\mathbf{A}^{\mathrm{op}}) = \mathrm{Obj}(\mathbf{A}) \tag{58}$$

- $\forall\, A, B \in \mathrm{Obj}(\mathbf{A}^{\mathrm{op}})$, $\mathrm{Hom}_{\mathbf{A}^{\mathrm{op}}}(A, B) \subseteq \mathrm{Mor}\mathbf{A}^{\mathrm{op}}$,

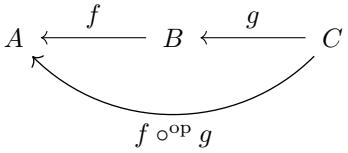
$$\mathrm{Hom}_{\mathbf{A}^{\mathrm{op}}}(A, B) = \mathrm{Hom}_{\mathbf{A}}(B, A) \subseteq \mathrm{Mor}\mathbf{A} \tag{59}$$

- *Define* the new composition

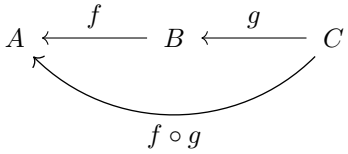
$$\begin{gathered} f \circ^{\mathrm{op}} g \text{ of } g \in \mathrm{Hom}_{\mathbf{A}^{\mathrm{op}}}(C, B) \\ f \in \mathrm{Hom}_{\mathbf{A}^{\mathrm{op}}}(B, A) \\ \text{then} \\ f \circ^{\mathrm{op}} g = g \circ f \end{gathered} \tag{60}$$



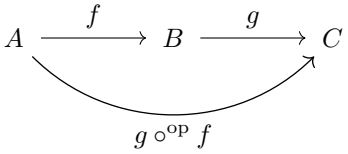
dual :



or, equivalently (notation-wise)



dual :



in that

$$\begin{gathered} g \circ^{\mathrm{op}} f \text{ of } f \in \mathrm{Hom}_{\mathbf{A}^{\mathrm{op}}}(A, B) \\ g \in \mathrm{Hom}_{\mathbf{A}^{\mathrm{op}}}(B, C) \\ \text{then} \\ g \circ^{\mathrm{op}} f = f \circ g \end{gathered}$$

i.e. (in summary) $\mathbf{C}^{\mathrm{op}}$ s.t. $\mathrm{Obj}\mathbf{C}^{\mathrm{op}} = \mathrm{Obj}\mathbf{C}$,
Denote $\forall\, \overline{C} \in \mathrm{Obj}\mathbf{C}^{\mathrm{op}}$, $\overline{C} = C \in \mathrm{Obj}\mathbf{C}$.
$\mathrm{Hom}_{\mathbf{C}^{\mathrm{op}}}(\overline{A}, \overline{B}) \ni \overline{f} : \overline{C} \to \overline{D}$, in $\mathbf{C}^{\mathrm{op}}$, for $f : D \to C$ in $\mathbf{C}$.

$$1_{\overline{C}} = 1_C$$
$$\overline{f} \circ \overline{g} \equiv \overline{f} \circ_{\mathrm{op}} \overline{g} = g \circ f$$

Diagram in $\mathbf{C}$:

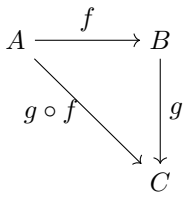
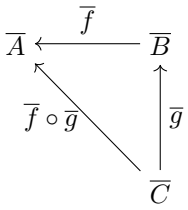


Diagram in $\mathbf{C}^{\mathrm{op}}$:



e.g. **Sets** is dual to category of complete, atomic Boolean algebras.
e.g. if $\mathbf{A} = (M, \cdot, e)$ monoid, then $\mathbf{A}^{\mathrm{op}} = (M, \widehat{\cdot}, e)$ where $a \widehat{\cdot} b = b \cdot a$

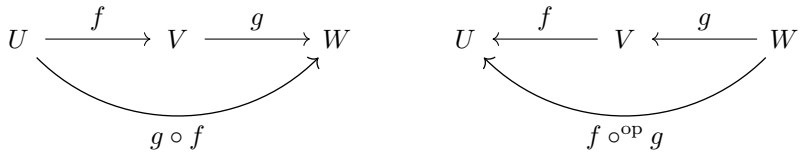
16.2.1 Example

- $\mathrm{Vec}^{\mathrm{op}}$

$$\mathbf{Vec}^{\mathrm{op}} = (\mathrm{Obj}(\mathbf{Vec}), \mathrm{Hom}_{\mathbf{Vec}^{\mathrm{op}}}, 1, \circ^{\mathrm{op}})$$

s.t.

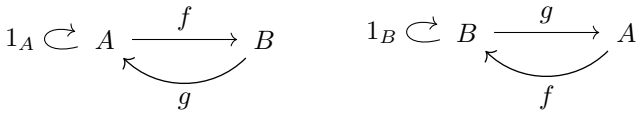
$$\mathrm{Hom}_{\mathbf{Vec}^{\mathrm{op}}}(W, V) = \mathrm{Hom}_{\mathbf{Vec}}(V, W)$$



16.3 Kinds of morphisms

cf. pp. 11, Definition 1.3 of Awodey (2010) [42]

Definition 68 (isomorphism)**. isomorphism** - $\forall$ category $\mathbf{C}$, Mor$\mathbf{C} \ni$ morphism $f : A \to B$ is an isomorphism if $\exists\, g : B \to A$ s.t. $f \circ g = 1_B$, $g \circ f = 1_A$, g unique. g called inverse of f, f^{-1}



Theorem 32 (Cayley)**.** *cf. pp. 11, Sec. 1.5.* ***Isomorphisms****, Awodey (2010) [42]*
$\forall$ *group* G,

$$G \cong \textit{ group of permutations}$$

(isomorphic $\equiv\cong$*)*

Proof. Define Cayley representation $\overline{G}$ of G to be following group of permutations:
underlying set of $\overline{G}$ is just G,
$\forall\, g \in G$, we have permutation $\overline{g}$, defined $\forall\, h \in G$,

$$\overline{g}(h) = g \cdot h$$

Let $\overline{g} = \overline{h}$.
$\forall\, g' \in G$,
$\overline{g}(g') = gg'$
$\overline{h}(g') = hg'$
$g' \in G$, so $\exists\, (g')^{-1}$ s.t. $g'(g')^{-1} = e$, so $gg'(g')^{-1} = g = h$.

Define homomorphisms $i : G \to \overline{G}$, $\quad j : \overline{G} \to G$

$i(g) = \overline{g}$ $\quad j(\overline{g}) = g$

$i(gg') = \overline{(gg')}$

$\overline{(gg')} = h = gg'h = g(g'h) = g(\overline{g}'(h)) = \overline{g}(\overline{g}'(h)) = i(g) = i(g')h$

$\implies i(gg') = i(g) \circ i(g')$

$j(\overline{g}) = j(\overline{g}')h = j(\overline{g})(g'h) = gg'h = j(\overline{gg'})h = j(\overline{g} \cdot \overline{g}')h$

$j(\overline{g}) \cdot j(\overline{g}') = j(\overline{g} \cdot \overline{g}')$

$i \circ j = 1_{\overline{G}}$, $j \circ i = 1_G$ since

$i \circ j(\overline{g}) = i \circ g = \overline{g}$

$j \circ i(g) = j(\overline{g}) = g$

i, j are isomorphisms between $G, \overline{G}$, which is in category **Groups** of groups and group homomorphisms.
permutations $\overline{g} \in \overline{G}$ are themselves isomorphisms in **Sets**.
Cayley's Thm. says that any abstract group can be represented as a "concrete" one, i.e. group of permutations of a set. □

Example 1.1.5 (Leinster (2014) [31]) isomorphisms in **Set** are exactly bijections.
$\implies$ function has 2-sided inverse iff function is injective and surjective (this is not trivial)

Example 1.1.7 Leinster (2014) [31] isomorphisms in **Top** are exactly homeomorphisms.
bijective map in **Top** is not necessarily an isomorphism:

$$[0, 1) \to \{z \in \mathbb{C} || z| = 1\}$$
$$t \mapsto e^{2\pi i t}$$

is a continuous bijection but not a homeomorphism (cont. bijection with cont. f^{-1}).

$$\ln\left(\frac{w}{2\pi i}\right) = f^{-1}(w)$$

Example 1.1.8 Leinster (2014) [31] (Categories as mathematical structures) (b) **Discrete** categories contain no maps at all apart from identities. Just a class of objects.

Exercise 1.1.13 pp. 16, Leinster (2014) [31].
Suppose for morphism $f : A \to B$, $\exists$ inverse g, s.t. $f \circ g = 1_B$, $g \circ f = 1_A$
Now $f^{-1} \circ f = 1_A$, $f \circ f^{-1} = 1_B$
$f^{-1} \circ f \circ g = f^{-1} \circ 1_B = 1_A \circ g = f^{-1} \circ 1_B \implies g = f^{1}$

Definition 69 (endomorphism)**. endomorphism** - morphism with same source and target, that is, morphism $f : A \to A$

Definition 70 (automorphism)**. automorphism** - endomorphism which is an isomorphism

Definition 71 (parallel)**. parallel** - 2 morphisms f, g are parallel if they have same source and same target:

$$f : A \to B$$
$$g : A \to B$$

Definition 72 (monomorphism)**.** monomorphism - morphism $f : A \to B$ is a monomorphism if $\forall$ pair of parallel

$g_1 : C \to A$,
$g_2 : C \to A$

$$f \circ g_1 = f \circ g_2 \text{ implies } g_1 = g_2 \tag{61}$$

i.e.

$$C \underset{f \circ g_2}{\overset{f \circ g_1}{\rightleftarrows}} B \qquad \text{implies } C \xrightarrow{g_1 = g_2} A$$

Definition 73 (epimorphism)**. epimorphism** - morphism $f : A \to B$ is an epimorphism if $f^{\text{op}} : B^{\text{op}} \to A^{\text{op}}$ is a monomorphism in $\mathbf{A}^{\text{op}}$.

Hence f epimorphism iff $\forall$ parallel morphisms $g_1 : B \to C$, $g_1 \circ f = g_2 \circ f$
$g_2 : B \to C$
implies $g_1 = g_2$

Proposition 21 (monomorphism, epimorphism iff injective)**.** *f monomorphism iff $f\circ : Hom_{\mathbf{A}}(C, A) \to Hom_{\mathbf{A}}(C, B)$ injective $\forall C \in Obj(\mathbf{A})$, i.e.*

$$\begin{gathered} Hom_{\mathbf{A}}(C, A) \xrightarrow{f\circ} Hom_{\mathbf{A}}(C, B) \\ g_1, g_2 \overset{f\circ}{\longmapsto} f \circ g_1, f \circ g_2 \\ \text{then} \\ f\circ \text{ is injective if} \\ f \circ g_1 = f \circ g_2 \Longrightarrow g_1 = g_2 \end{gathered} \tag{62}$$

f epimorphism iff map $\circ f : Hom_{\mathbf{A}}(B, C) \to Hom_{\mathbf{A}}(A, C)$ injective $\forall C \in Obj(\mathbf{A})$

$$\begin{gathered} Hom_{\mathbf{A}}(B, C) \xrightarrow{\circ f} Hom_{\mathbf{A}}(A, C) \\ g_1, g_2 \overset{\circ f}{\longmapsto} g_1 \circ f, g_2 \circ f \\ \text{then} \\ \circ f \text{ is injective if} \\ g_1 \circ f = g_2 \circ f \Longrightarrow g_1 = g_2 \end{gathered} \tag{63}$$

Definition 74 (inverses)**.** $\forall$ 2 morphisms, $f : X \to Y$, $g : Y \to X$ s.t. $f \circ g = 1_Y$,
f is called left inverse of g, g is called right inverse of f.

We also say, g is a section of f, or f is a cosection of g.
f is an epimorphism, g is a monomorphism.

16.4 More definitions with categories

Definition 75 (subcategory)**.** category $\mathbf{A}'$, $\mathbf{A}' \subset \mathbf{A}$, if $\text{Obj}(\mathbf{A}') \subset \text{Obj}(\mathbf{A})$, $\text{Hom}_{\mathbf{A}'}(A, B) \subset \text{Hom}_{\mathbf{A}}(A, B)$, $\forall A, B \in \mathbf{A}'$.

Composition in $\mathbf{A}'$ is induced by composition in $\mathbf{A}$.
identity morphisms in $\mathbf{A}'$ are identity morphisms in $\mathbf{A}$

Definition 76 (full subcategory)**.** subcategory $\mathbf{A}'$ of $\mathbf{A}$ is full if $\text{Hom}_{\mathbf{A}'}(A, B) = \text{Hom}_{\mathbf{A}}(A, B)$, $\forall A, B \in \mathbf{A}'$

Definition 77 (saturated subcategory)**.** full subcategory $\mathbf{A}'$ of $\mathbf{A}$ saturated if $A \in \mathbf{A}$ belongs to $\mathbf{A}'$ whenever A is isomorphic to object of $\mathbf{A}'$

Definition 78 (discrete category)**. discrete** - discrete category if all morphisms are identity morphisms.

Definition 79 (nonempty category)**. nonempty** - nonempty category if Obj(**A**) is nonempty

Definition 80 (groupoid)**. groupoid** - category **A** is a **groupoid** if all morphisms are isomorphisms.

Definition 81 (finite category)**.** finite - finite category if set of all morphisms in **A** (hence, in particular, set of objects) is a finite set

Definition 82 (connected)**.** connected category **A** if it's nonempty, and $\forall\, A, B \in \mathrm{Obj}\mathbf{A}$, $\exists$ finite sequence of objects $(A_0 \dots A_n)$, $A_0 = A$, $A_n = B$, s.t. at least 1 of the sets $\mathrm{Hom}_{\mathbf{A}}(A_j, A_{j+1})$ or $\mathrm{Hom}_{\mathbf{A}}(A_{j+1}, A_j)$ is nonempty $\forall\, j \in \mathbb{N}$, with $0 \leq j \leq n-1$

Definition 83 (monoid M)**. monoid** M (set endowed with internal product with associative and unital law) is nothing but a category with only 1 object (to M, associate category **M**, with single object A, and morphisms $\mathrm{Hom}_{\mathbf{M}}(A, A) = M$)

cf. Def. 1.2.5 of Kashiwara and Schapira (2006) [24].

Definition 84 (Morphisms as a category)**.** Let category $\mathcal{C} \equiv \mathbf{A}$.
Mor(**A**) is a category.
Obj(Mor(**A**)) = Mor**A** (objects of category Mor(**A**) are morphisms in **A**).

Let $f : X \to Y$, $f, g \in \mathrm{Mor}(\mathbf{A})$ (i.e. $f \in \mathrm{Hom}(X, Y)$, for $X, Y, X', Y' \in \mathrm{Obj}(\mathbf{A})$)
$g : X' \to Y'$ $g \in \mathrm{Hom}(X', Y')$
Then

$$\mathrm{Hom}_{\mathrm{Mor}(\mathbf{A})}(f, g) = \{u : X \to X', v : Y \to Y'; g \circ u = v \circ f\}$$

Composition and identity in Mor(**A**) are the obvious ones.
So

$$\mathrm{Obj}(\mathrm{Mor}(\mathbf{A})) = \mathrm{Mor}(\mathbf{A})$$

$$\mathrm{Mor}(\mathrm{Mor}(\mathbf{A})) = \bigcup_{f,g \in \mathrm{Mor}(\mathbf{A})} \mathrm{Hom}(f, g) = \bigcup_{f,g \in \mathrm{Mor}(\mathbf{A})} \{u : X \to X', v : Y \to Y'; g \circ u = v \circ f\}$$

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \downarrow{\scriptstyle u} & & \downarrow{\scriptstyle v} \\ X' & \xrightarrow{g} & Y' \end{array}$$

cf. Def. 1.2.6 of Kashiwara and Schapira (2006) [24].

Definition 85. 1. object $P \in \mathcal{C} \equiv \mathbf{A}$ is called *initial* if $\forall\, X \in \mathbf{A}$, $(\equiv \forall\, x \in \mathrm{Obj}(\mathbf{A}))$, $\mathrm{Hom}_{\mathbf{A}}(P, X) \simeq \{\mathrm{pt}\}$.
(Denote by $\emptyset_{\mathbf{A}}$ an *initial object* in **A**).
(Note that if P_1 and P_2 are initial, then $\exists!$ isomorphism $P_1 \simeq P_2$)

2. P is terminal in **A** if P is initial in $\mathbf{A}^{\mathrm{op}}$, i.e.

   $\forall\, X \in \mathbf{A}$, $\mathrm{Hom}_{\mathbf{A}}(X, P) \simeq \{\mathrm{pt}\}$.

   Denote $\mathrm{pt}_{\mathbf{A}}$ a terminal object in **A**.

3. P is zero (0) object if it's both initial and terminal.

   Such a P is denoted by 0.

   If **A** has a zero object, $\forall$ object $X, Y \in \mathbf{A} \equiv \mathrm{Obj}\mathbf{A}$, the morphism obtained as composition $X \to 0 \to Y$ is still denoted by $0 : X \to Y$.

   (Note that composition of $0 : X \to Y$, and any morphism $f : Y \to Z$ is $0 : X \to Z$)

cf. Example 1.2.7 of Kashiwara and Schapira (2006) [24]. Example

(i) In category **Set**, $\emptyset$ initial, {pt} terminal.

(ii) Zero module 0 is zero object in Mod(R).

Notation 1.2.8 of Kashiwara and Schapira (2006) [24]:

1. **Pt** ≡ category with a single object and a single morphism (the identity of this object)
2. $\emptyset$ ≡ empty category with no objects (hence, no morphisms)
3. $\bullet \to \bullet \equiv$ category which consists of 2 objects, say a, b, and 1 morphism, $a \to b$, other than $\mathrm{id}_a, \mathrm{id}_b \equiv 1_a, 1_b$. Denote this category by **Arr**.

cf. Example 1.2.9 of Kashiwara and Schapira (2006) [24].
Let R be a ring. Let $N \in \mathrm{Mod}(R^{\mathrm{op}})$, $M \in \mathrm{Mod}(R)$.
Category **C**:
Obj**C** $\ni (f, L)$, where $L \in \mathrm{Mod}(\mathbb{Z})$, f bilinear map $f : N \times M \to L$ (i.e. it's $\mathbb{Z}$-bilinear and satisfies

$$f(na, m) = f(n, am), \quad \forall\, a \in R$$

)
Morphism from $f : N \times M \to L$ to $g : N \times M$ is a linear map $h : L \to K$ s.t. $h \circ f = g$.
Since any bilinear map $f : N \times M \to L$ (i.e. any object of **C**) factorizes uniquely through

$$u : N \times M \to N \otimes_R M$$

object $(u, N \otimes_R M)$ is initial in **C**

17 Functors

cf. Def. 1.2.10 of Kashiwara and Schapira (2006) [24], with terminology from pp. 13, Sec. 3. "Functors" from Mac Lane (1978) [28]

Definition 86 ((covariant) Functor)**.** 1. (covariant functor) Let categories **C**, **D**.
(covariant) **functor** $F : \mathbf{C} \to \mathbf{D}$ consists of (Mac Lane (1978) [28] says 2 suitably related functions)

- map $F : \mathrm{Obj}(\mathbf{C}) \to \mathrm{Obj}(\mathbf{D})$ (i.e. $\forall\, C \in \mathrm{Obj}(\mathbf{C})$, $F(C) \in \mathrm{Obj}(\mathbf{D})$),
(Mac Lane (1978) [28] calls this the *object function* T or F in our notation; we'll call it the *object map*)
and
- maps $F : \mathrm{Hom}_{\mathbf{C}}(X, Y) \to \mathrm{Hom}_{\mathbf{D}}(F(X), F(Y))$, so that

$$F(f) : F(X) \to F(Y) \text{ or } F(f)(F(X)) = F(f(X))$$

$\forall\, X, Y \in \mathrm{Obj}(\mathbf{C})$ s.t.

$$F(1_X) = 1_{F(X)} \qquad \forall\, X \in \mathbf{C}$$

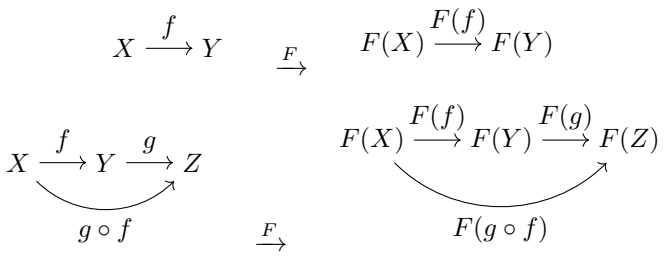
$$F(g \circ f) = F(g) \circ F(f) \qquad \forall\, f : X \to Y, \quad g : Y \to Z \qquad X, Y, Z \in \mathrm{Obj}(\mathbf{C}) \tag{64}$$

Mac Lane (1978) [28] calls this the *arrow function* (also written as T for Mac Lane's notation; F for our notation); we'll call F to be the *morphism map*.

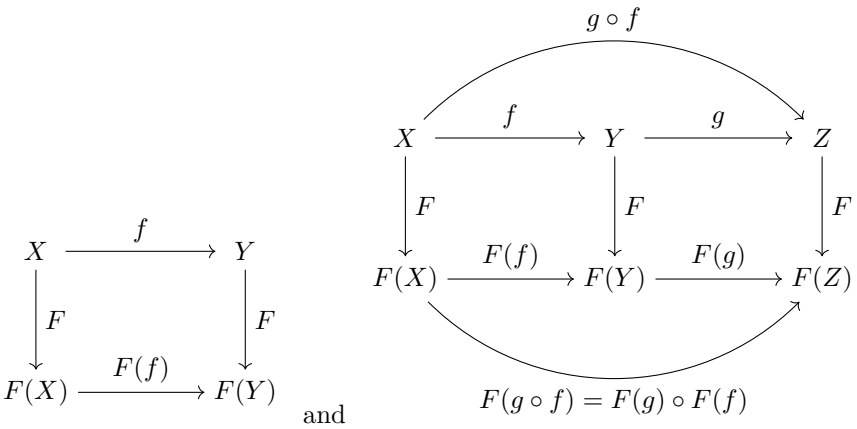
2. (composition law for functors)
For categories $\mathbf{A}, \mathbf{B}, \mathbf{C}$, functors $F : \mathbf{A} \to \mathbf{B}$, $G : \mathbf{B} \to \mathbf{C}$,
Composition $G \circ F : \mathbf{A} \to \mathbf{C}$, is a functor defined by

$$\begin{aligned} &(G \circ F)(X) = G(F(X)) \qquad \forall\, X \in \mathrm{Obj}\mathbf{A}, \text{ and} \\ &(G \circ F)(f) = G(F(f)), \qquad \forall \text{ morphism } f \in \mathrm{Mor}(\mathbf{C}) \end{aligned} \tag{65}$$

Diagrammatically,



i.e.



and

17.1 Examples of Functors

cf. pp. 13, 3. Functors, Maclane (1978) [28]

Power Set Functor.

$$P : \mathbf{Set} \to \mathbf{Set}$$
$$P : \mathrm{Obj}\mathbf{Set} \to \mathrm{Obj}(\mathbf{Set})$$
$$P : X \in \mathrm{Obj}\mathbf{Set} \mapsto 2^X \in \mathrm{Obj}\mathbf{Set}, \text{ i.e.}$$
$$P : X \mapsto 2^X = \{S | S \subseteq X\}$$
$$P : \mathrm{Hom}_{\mathbf{Set}}(X, Y) \to \mathrm{Hom}_{\mathbf{Set}}(P(X), P(Y))$$
$$P(f) : P(X) \to P(Y) \text{ or } P(f)(P(X)) = P(f(X))$$
$$P(f) : S \in 2^X \mapsto f(S) \in 2^Y \text{ or } f(S) \subseteq Y$$
$$P(1_X) = 1_{PX} \text{ and } P(g \circ f) = P(g)P(f), \text{ so } P \text{ defines a functor}$$

Algebraic topology; singular homology in a given dim. n ($n \in \mathbb{N}$) Maclane (1978) [28]

$\forall$ topological space X, assign $X \mapsto$ abelian group $H_n(X) \equiv n$th homology group of X, also $\forall$ cont. $f : X \to Y$ of spaces corresponding group homomorphism.

$$H_n(f) : H_n(X) \to H_n(Y)$$
$$H_n : \mathrm{Obj}(\mathbf{Top}) \to \mathrm{Obj}(\mathbf{Ab})$$
$$H_n : X \mapsto H_n(X)$$
$$H_n(f)(H_n(X)) = H_n(f(X))$$

$\implies H_n : \mathbf{Top} \to \mathbf{Ab}$.

e.g. if $X = Y = S^1$, $H_1(S^1) = \mathbb{Z}$

So group homomorphism $H_1(f) : \mathbb{Z} \to \mathbb{Z}$ determined by integer $d \in \mathbb{Z}$ (image of 1)

d is usual "degree" of cont. map $f : S^1 \to S^1$.

In this case, and in general, homotopic maps $f, g : X \to Y$

Then $\exists$ homotopy $H : X \times [0, 1] \to Y$ s.t. $H(x, 0) = f(x)$; $H(x, 1) = g(x) \, \forall \, x \in X$

Example 1.2.3 (forgetful functors) Leinster (2014) [31], pp. 18

(a) $U : \mathbf{Grp} \to \mathbf{Set}$
if group G then $U(G)$ is underlying set of G.
If group homomorphism $f : G \to H$,
$U(f)$ is function f itself.
So U forgets group structure of groups and forgets group homomorphisms are homomorphisms.

(c) $\mathbf{Ab} \equiv$ category of abelian groups
$\mathbf{Ring} \to \mathbf{Ab}$ forgets multiplicative structure.
$U : \mathbf{Ring} \to \mathbf{Mon}$ forgets additive structure; $\mathbf{Mon} \equiv$ category of monoids

- (monoid) homomorphism $h : M \to N$, $\forall$ monoid M, N s.t. $\forall\, m, n \in M$,
$h(m \cdot n) \equiv h(m \cdot_M n) = h(m) \cdot h(n) \equiv h(m) \cdot_N h(n)$ and
$h(e_M) = e_N$.

  $h : m \mapsto h(m)$ so $h : \mathrm{Obj}\mathbf{C}(M) \to \mathrm{Obj}\mathbf{C}(N)$.
  Let $m \cdot n \equiv f_m(n)$, $f_m \in \mathrm{Hom}\mathbf{C}(M)$

$$h(m) \cdot h(n) \equiv f_{h(m)}(h(n)),\ f_{h(m)} \in \mathrm{Hom}\mathbf{C}(N)$$
$$h(f_m) = h(m)$$
$$h(f_m)(h(n)) = h(m) \cdot h(n) = h(m \cdot n) = h(f_m \cdot n)$$

  Compare the last statement with $F(f)(F(X)) = F(f(X))$.

- poset P, Q (set P, equipped with $\leq$), poset as category $\mathbf{P}, \mathbf{Q}$, $\mathrm{Obj}\mathbf{P} = P$, $\mathrm{Obj}\mathbf{Q} = Q$
$\mathrm{Hom}\mathbf{P} \ni m$ s.t. $m : x \mapsto y$ iff $x \leq y$.

  Let functor $F : \mathbf{P} \to \mathbf{Q}$ s.t.

$$F : \mathrm{Obj}\mathbf{P} \to \mathrm{Obj}\mathbf{Q}$$
$$F : x \mapsto F(x)$$
$$F : \mathrm{Hom}\mathbf{P} \to \mathrm{Hom}\mathbf{Q}$$
$$F : m \mapsto F(m)$$

  s.t. for $m(x) = y$ and $x \leq y$,
  $F(m) : F(x) \mapsto F(y)$ and $F(x) \leq F(y)$ (so that $F(m)$ is also monotone), i.e.

$$F(m)(F(X)) = F(m(X)) = F(y)$$

Kleene closure itself is a functor from **Set** to **Set**, from A to A^*, from f to f^*.
It's the composition of $U \circ F$ of underlying functor $U : \mathbf{Mon} \to \mathbf{Set}$ and free functor $F : \mathbf{Set} \to \mathbf{Mon}$.

- *arrow category* $\mathbf{C}^{\to}$ of category $\mathbf{C}$:

$$\mathrm{Obj}\mathbf{C}^{\to} = \mathrm{Mor}\mathbf{C}$$

  Given $f : A \to B$, $f' : A' \to B'$, morphism (arrow) $g : f \mapsto f'$ is a "commutative square", i.e. $g : \mathrm{Hom}_{\mathbf{C}}(A \to B) \to \mathrm{Hom}_{\mathbf{C}}(A' \to B')$

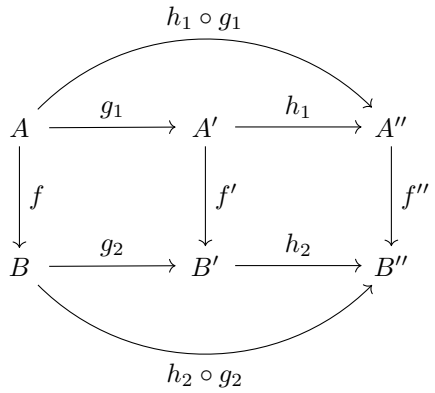
$$\begin{array}{ccc} A & \xrightarrow{g_1} & A' \\ \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f'} \\ B & \xrightarrow{g_2} & B' \end{array}$$

  where $g_1, g_2 \in \mathrm{Mor}\mathbf{C}$,
  i.e. such a morphism (arrow) is a pair of morphisms (arrows) $g = (g_1, g_2)$ in $\mathbf{C}$ s.t.

$$g_2 \circ f = f' \circ g$$

identity morphism (arrow) 1_f on object $f : A \to B$ is pair $(1_A, 1_B)$.
Composition of arrows is done componentwise:

$$(h_1, h_2) \circ (g_1, g_2) = (h_1 \circ g_1, h_2 \circ g_2)$$



Observe that there are 2 functors:

$$\mathbf{C} \xleftarrow{\mathbf{dom}} \mathbf{C}^{\to} \xrightarrow{\mathbf{cod}} \mathbf{C}$$

- *algebra* cf. pp. 14, Sec. 3, "Functors" of Mac Lane (1978) [28].
  $\forall$ commutative ring K, set of all non-singular $n \times n$ matrices with entries in K = general linear group $GL_n(K)$.
  $\forall$ homomorphism $f : K \to K'$ of rings produces
  homomorphism $GL_n f : GL_n(K) \to GL_n(K)$ of groups.

  $\forall n \in \mathbb{N}$ (natural numbers), defined functor $GL_n : \mathbf{CRng} \to \mathbf{Grp} \equiv GL_n\mathbf{CommRing} \to \mathbf{Grp}$.

  $\forall$ group G, set of all products of commutators $xyx^{-1}y^{-1}$ $(x, y \in G)$ is a normal subgroup $[G, G]$ of G called commutator subgroup.

  Since $\forall$ homomorphism $G \to H$ of groups carries commutators to commutators.
  $G \mapsto [G, G]$ defines evident functor $\mathbf{Grp} \to \mathbf{Grp}$,
  while $G \mapsto G/[G, G]$ define functor $\mathbf{Grp} \to \mathbf{Ab}$, factor-commutator functor.
  TODO - understand the previous commutators.

17.1.1 Endofunctor

Definition 87 (Endofunctor)**.** Consider **endofunctor** $T : \mathbf{C} \to \mathbf{C}$. This implies that, for the

object map $T : \mathrm{Obj}\mathbf{C} \to \mathrm{Obj}\mathbf{C}$,

$$T : X \to Y$$

object map T acts like a morphism, in $\mathrm{Mor}\mathbf{C}$ (!!!). In this case T acts like $T \in \mathrm{Hom}_{\mathbf{C}}(X, Y)$.

morphism map $T : \mathrm{Hom}_{\mathbf{C}}(X, Y) \to \mathrm{Hom}_{\mathbf{C}}(F(X), F(Y))$ s.t.

$$T(f) \equiv Tf : T(X) \to T(Y) \text{ so that } Tf \in \mathrm{Hom}_{\mathbf{C}}(T(X), T(Y))$$

$$T(1_X) = 1_{T(X)} \in \mathrm{Hom}_{\mathbf{C}}(T(X), T(Y)) \text{(implied from Kashiwara and Schapira (2006) [24] in that } 1_X \in \mathrm{Hom}(X, X))$$

$$T(g \circ f) = T(g) \circ T(f),\ \forall f : X \to Y,\ g : Y \to Z,\ X, Y, Z \in \mathrm{Obj}\mathbf{C} \tag{66}$$

So $T(f) \equiv Tf$ is indeed a morphism map.

17.1.2 Hom functors

Let $\mathbf{C}$ locally-small category (i.e. category s.t. hom-classes are actually sets and not proper classes).
$\forall\, A, B \in \mathrm{Obj}\mathbf{C}$,

Definition 88 (covariant Hom functor)**. covariant Hom functor** $\mathrm{Hom}(A,-) : \mathbf{C} \to \mathbf{Set}$

$$\begin{aligned}
&\mathrm{Hom}(A,-) : \mathrm{Obj}\mathbf{C} \to 2^{\mathrm{Mor}(\mathbf{C})} \\
&\mathrm{Hom}(A,-) : X \mapsto \mathrm{Hom}(A,X) \\
&\mathrm{Hom}(A,-) : \mathrm{Mor}\mathbf{C} \to (\mathrm{Mor}(\mathbf{C}) \to \mathrm{Mor}(\mathbf{C})) \\
&\mathrm{Hom}(A,-) : \mathrm{Hom}(X,Y) \to (\mathrm{Hom}(A,X) \to \mathrm{Hom}(A,Y)) \\
&\mathrm{Hom}(A,-) : f \mapsto \mathrm{Hom}(A,f) \text{ where} \\
&\mathrm{Hom}(A,f) : \mathrm{Hom}(A,X) \to \mathrm{Hom}(A,Y) \\
&\qquad\qquad g \mapsto f \circ g \quad \forall\, g \in \mathrm{Hom}(A,X)
\end{aligned} \tag{67}$$

Claim: $\mathrm{Hom}(A,-)$ is a functor.

Proof.

- *identity*: $\mathrm{Hom}(A,-) : 1_X \mapsto \mathrm{Hom}(A,1_X)$ where

$$\begin{aligned}
\mathrm{Hom}(A,1_X) : &\mathrm{Hom}(A,X) \to \mathrm{Hom}(A,X) \\
&g \mapsto 1_X \cdot g = g, \quad \forall\, g \in \mathrm{Hom}(A,X)
\end{aligned}$$

  Hence

$$\mathrm{Hom}(A,-)(1_X) = 1_{\mathrm{Hom}(A,X)} \in (\mathrm{Hom}(A,X) \to \mathrm{Hom}(A,X))$$

- composition: Let

$$\begin{aligned}
f &\in \mathrm{Hom}(X,Y) \\
g &\in \mathrm{Hom}(Y,Z) \\
k &\in \mathrm{Hom}(A,X)
\end{aligned}$$

$$\mathrm{Hom}(A,g) \circ \mathrm{Hom}(A,f)(k) = \mathrm{Hom}(A,g)(f \circ k) = g \circ f \circ k = (g \circ f)(k) = \mathrm{Hom}(A, g \circ f)(k)$$

□

Definition 89 (contravariant Hom functor)**. contravariant Hom functor** $\mathrm{Hom}(-,B) : \mathbf{C} \to \mathbf{Set}$

$$\begin{aligned}
&\mathrm{Hom}(-,B) : \mathrm{Obj}\mathbf{C} \to 2^{\mathrm{Mor}(\mathbf{C})} \\
&\mathrm{Hom}(-,B) : X \mapsto \mathrm{Hom}(X,A) \\
&\mathrm{Hom}(-,B) : \mathrm{Mor}\mathbf{C} \to (\mathrm{Mor}(\mathbf{C}) \to \mathrm{Mor}(\mathbf{C})) \\
&\mathrm{Hom}(-,B) : \mathrm{Hom}(X,Y) \to (\mathrm{Hom}(Y,B) \to \mathrm{Hom}(X,B)) \\
&\mathrm{Hom}(-,B) : h \mapsto \mathrm{Hom}(h,B) \text{ where} \\
&\mathrm{Hom}(h,B) : \mathrm{Hom}(Y,B) \to \mathrm{Hom}(X,B) \\
&\qquad\qquad g \mapsto g \circ h \quad \forall\, g \in \mathrm{Hom}(Y,B)
\end{aligned} \tag{68}$$

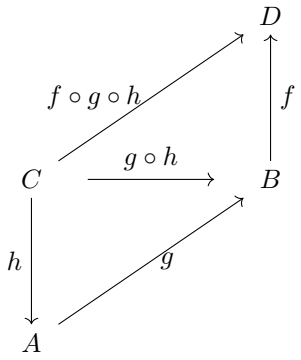
$\mathrm{Hom}(A,-)$, $\mathrm{Hom}(-,B)$ are related in a natural manner: for

$$\begin{aligned}
f : B \to B',\ f \in \mathrm{Hom}(B,B') \\
g : A \to A',\ g \in \mathrm{Hom}(A,A') \\
\mathrm{Hom}(A,f) : \mathrm{Hom}(A,B) \to \mathrm{Hom}(A,B') \\
\mathrm{Hom}(g,B) : \mathrm{Hom}(A,B) \to \mathrm{Hom}(A',B)
\end{aligned}$$

$$\begin{array}{ccc} \mathrm{Hom}(A,B) & \xrightarrow{\mathrm{Hom}(g,B)} & \mathrm{Hom}(A',B) \\ \Big\downarrow{\scriptstyle \mathrm{Hom}(A,f)} & & \Big\downarrow{\scriptstyle \mathrm{Hom}(A',f)} \\ \mathrm{Hom}(A,B') & \xrightarrow{\mathrm{Hom}(g,B')} & \mathrm{Hom}(A',B') \end{array}$$

Definition 90 (2-variable Hom functor)**.** 2-variable Hom functor $\mathrm{Hom}(-,-)$ is a bifunctor.

$$\mathrm{Hom}(-,-) : \mathbf{C}^{\mathrm{op}} \times \mathbf{C} \to \mathbf{Set}$$



$\mathrm{Hom}(-,-)$ is a functor such that:

$$\begin{gathered} \mathrm{Hom}(h,f)(g) = f \circ g \circ h \\ \mathrm{Hom}(h,f) : \mathrm{Hom}(A,B) \to \mathrm{Hom}(C,D) \\ h : C \to A,\ h \in \mathrm{Hom}(C,A) \\ f : B\dot{D},\ f \in \mathrm{Hom}(B,D) \\ (h,f) \in \mathrm{Hom}(C,A) \times \mathrm{Hom}(B,D) \mapsto \mathrm{Hom}(h,f) \in (\mathrm{Hom}(A,B) \to \mathrm{Hom}(C,D)) \\ \mathrm{Hom}(-,-) : \mathrm{Hom}(C,A) \times \mathrm{Hom}(B,D) \to (\mathrm{Hom}(A,B) \to \mathrm{Hom}(C,D)) \end{gathered}$$

and

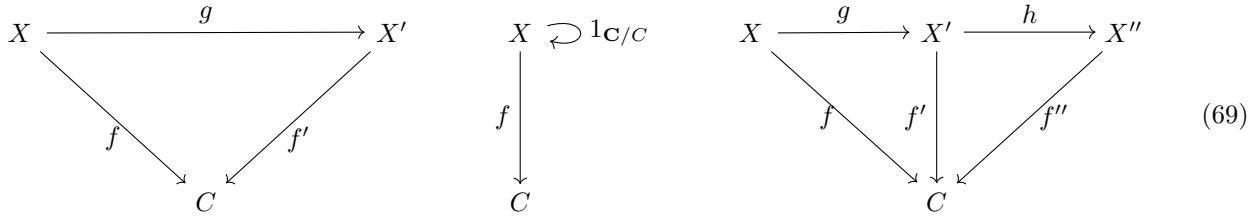
$$\begin{gathered} \mathrm{Hom}(-,-) : \mathrm{Obj}\mathbf{C}^{\mathrm{op}} \times \mathbf{C} \\ \mathrm{Hom}(-,-) : (C,D) \mapsto \mathrm{Hom}(C,D) \end{gathered}$$

18 Construction of Categories

18.1 Slice category

cf. Construction 4 of Awodey (2010) [42], pp. 15.

Slice category $\mathbf{C}/C$, of category $\mathbf{C}$ over object $C \in \mathrm{Obj}\mathbf{C}$,
$\mathrm{Obj}\mathbf{C}/C \ni f \in \mathrm{Mor}\mathbf{C}$ s.t. $\mathrm{cod}(f) = C$
$\mathrm{Mor}\mathbf{C}/C \ni g$ from $f : X \to C$ to $f' : X' \to C$ is a morphism (arrow) $g : X \to X'$ in $\mathbf{C}$ s.t. $f' \circ g = f$



(69)

If $\mathbf{C} = \mathbf{P}$ poset category, $p \in \mathbf{P}$ (i.e. $p \in \mathrm{Obj}(\mathbf{P})$), then

$$\mathbf{P}/p \simeq \downarrow (p)$$

Slice category $\mathbf{P}/p$ is just the "principal ideal" $\downarrow (p)$ of elements $q \in \mathbf{P}$ with $q \leq p$.

18.1.1 S-indexed set X and indexed functions as morphisms of slice category $\mathbf{Set}/S$

Definition 91 (typed set)**.** S-indexed set is set X together with function $\tau : X \to S$.

If $x \in X$, $\tau(x) = s$, then x is of **type** s, so X is a **typed set**.

$\{\tau^{-1}(s) | s \in S\}$, $\tau^{-1}(s) \subset X \equiv$ family of sets indexed by S.

cf. 2.6.11 of Barr and Wells (2012) [43]

Definition 92. set X typed by $S \to$ set X' typed by S that preserves typing (element of type $s \mapsto$ element of type s), is exactly an arrow (morphism) of slice category $\mathbf{Set}/S$, called **indexed function** or **typed function**.

cf. 2.6.13 Indexed functions of Barr and Wells (2012) [43]

18.1.2 Underlying functor of the slice F

Let $f : A \to C$, $f \in \mathrm{Hom}(A, C) \subset \mathrm{Mor}\mathbf{C}$ and $f \in \mathrm{Obj}\mathbf{C}/C$.

Let $f, g \in \mathrm{Obj}\mathbf{C}/C$, s.t. $f \in \mathrm{Hom}(A, C)$, $g \in \mathrm{Hom}(B, C)$
$h \in \mathrm{Hom}(f, g) \subseteq \mathrm{Mor}\mathbf{C}/C$ in that
$h : f \to g$ s.t. $h : A \to B$, so that $g \circ h = f$.

underlying functor of the slice F (notation U in Barr and Wells (2012) [43]):

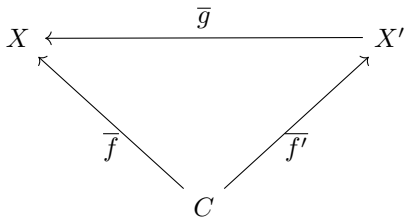
$$\begin{aligned} &F : \mathbf{C}/C \to \mathbf{C} \\ &F : \mathrm{Obj}\mathbf{C}/C \to \mathrm{Obj}\mathbf{C} \\ &F : f \mapsto F(f) = A \\ &F : \mathrm{Mor}\mathbf{C}/C \to \mathrm{Mor}\mathbf{C} \\ &F : h \mapsto h \end{aligned} \tag{70}$$

Special case: $\mathbf{C} = \mathbf{Set}$, $C = S \ni \mathrm{Obj}\mathbf{Set}$, i.e. S is a set.
For $f \in \mathrm{Obj}\mathbf{Set}/S$, $f : T \to S$, where T is a set, object f is an S-indexed set. (e.g. $x(i) \in T \subset \mathbb{R}$, where for some $y \in T \subset \mathbb{R}$, $\exists\, i \in \mathbb{Z}$ s.t. $y \mapsto i$).

$$\begin{aligned} &F : \mathbf{Set}/S \to \mathbf{Set} \\ &F : f \mapsto T \end{aligned}$$

The underlying functor F forgets the S-indexing.

18.1.3 Coslice category



Coslice category $C/\mathbf{C}$ of category $\mathbf{C}$ under object $C \in \text{Obj}\mathbf{C}$:
$\text{Obj}(C/\mathbf{C}) = \overline{f} \in \text{Mor}\mathbf{C}$ s.t. $\text{dom}\overline{f} = \overline{C} = C$,

$\text{Mor}(C/\mathbf{C}) \ni \overline{g} : \overline{X}' \to \overline{X}$s.t.

$$\overline{g} \circ \overline{f'} = \overline{f}$$

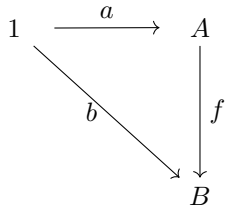
cf. Example 1.8 of Awodey (2010) [42], pp. 15. *pointed sets*

$\mathbf{Sets}_*$ of pointed sets consisting of sets A and distinguished element $a \in A$, and, i.e.
$\text{Obj}\mathbf{Set}_* \ni$ set A with distinguished element $a \in A$,
$\text{Mor}\mathbf{Set}_* \ni f : (A, a) \to (B, b)$, function $f : A \to B$ preserves the "points" $f(a) = b$.

$$\mathbf{Sets}_* = \mathbf{Set}_* \simeq 1/\mathbf{Sets}$$

i.e. $\mathbf{Sets}_*$ isomorphic to coslice category of sets "under" any singleton $1 = \{*\}$

functions $a : 1 \to A$ correspond uniquely to elements
$a(*) = a \in A$
morphisms (arrows) $f : (A, a) \to (B, b)$ correspond exactly to



18.2 Free monoid

Given set S, e.g. $S =$ "alphabet" A,
free monoid S^* is set S^*.

$S^* =$ set of all lists (finite sequences) of elements of S,
e.g. set of all *words* (finite sequence of letters) *over* A $= A^*$.

Concatenation: $* : \forall\, w, w^* \in A^*$, $w * w' = ww'$
empty list (e.g. empty word "-"), (), $*$, or ϵ is a unit. Or we'll use this notation: ""

$$\implies (S^*, *, \texttt{""})\text{is a monoid.}$$

monoid M is **freely generated** by subset A of M, if

1. "no junk": $\forall\, m \in M$, m can be written as product of elements of A,
$$m = a_1 \dots a_n,\ a_i \in A$$
2. "no noise": no "nontrivial" relations hold in M, i.e. if $a_1 \dots a_j = a'_1 \dots a'_k$, then this is required by the axioms for monoid.

$A^* =$ Kleene closure.

18.2.1 Free monoid functor: underlying functor

$$F : \mathbf{Set} \to \mathbf{Mon}$$
$$F : A \in \text{Obj}\mathbf{Set} \mapsto \text{ free monoid } F(A) = \text{ Kleene closure } A^*$$

For $f : A \to B$, $f \in \text{Hom}_{\mathbf{Set}}(A, B)$,

$$F : f \mapsto F(f) \text{ s.t.}$$
$$F(f)(w) = F(f)(a_1 \dots a_m) = (f(a_1), \dots f(a_m)) = w' \in B^*, \text{ i.e.}$$
$$F(f) = f^* : F(A) \to F(B)$$

19 Universal mapping property

19.1 Examples of Universal mapping property

cf. Ch. 0 "Introduction" of Leinster (2014) [31].

Example 0.1. Let $\{1\}$ denote set with 1 element.
Then $\forall$ sets $X \in \mathbf{Set}$, $\exists!$ map $X \to \{1\}$

$$X \xrightarrow{f} \{1\}$$

Proof. $\exists\, X \to \{1\}$, $\forall X$ because define $f : X \to \{1\}$ s.t. $f(x) = 1$, $\forall\, x \in X$.
(!) if for $X \to \{1\}$, then $x \mapsto 1$, so map is equal to f. □

Example 0.2. $\forall$ rings R with multiplicative identity 1, $\exists!$ homomorphism $\mathbb{Z} \to R$.
Define $\phi : \mathbb{Z} \to R$ by

$$\phi(n) = \begin{cases} \sum_{i=1}^{n} 1 & \text{if } n > 0 \\ 0 & \text{if } n = 0 \\ -\phi(-n) & \text{if } n < 0 \end{cases}$$

Check that ϕ homomorphism. (TODO)

(!) $\psi(1) = 1$ (homomorphisms preserve multiplicative identities)
homomorphisms preserve addition:

$$\psi(n) = \psi(\sum_{i=1}^{n} 1) = \sum_{i=1}^{n} \psi(1) = \sum_{i=1}^{n} 1 = \phi(n)$$

homomorphisms preserve zero: $\psi(0) = 0 = \phi(0)$
homomorphisms preserve negatives: $\psi(n) = -\psi(-n) = -\sum_{i=1}^{-n} 1 = -\phi(-n) = \phi(n)$.

Lemma 3 (0.3 Leinster (2014) [31])**.** *Let ring A s.t. $\forall$ rings $R, \exists!$ homomorphism $A \to R$ (i.e. A is "initial"). Then $A \cong \mathbb{Z}$ (isomorphic).*

Proof. A initial, so $\exists!$ homomorphism $\phi : A \to \mathbb{Z}$.
$\mathbb{Z}$ initial from Ex. 19.1. $\exists!$ homomorphism $\phi' : \mathbb{Z} \to A$.

$\phi' \circ \phi : A \to A$, $1_A : A \to A$ are homomorphisms.
Since A initial, $\phi' \circ \phi = 1_A$.
$\phi \circ \phi' : \mathbb{Z} \to \mathbb{Z}$. Since $\mathbb{Z}$ initial, $1_{\mathbb{Z}} : \mathbb{Z} \to \mathbb{Z} \implies \phi \circ \phi' = 1_{\mathbb{Z}}$.

$$\implies A \cong \mathbb{Z} \text{ (isomorphic)}$$

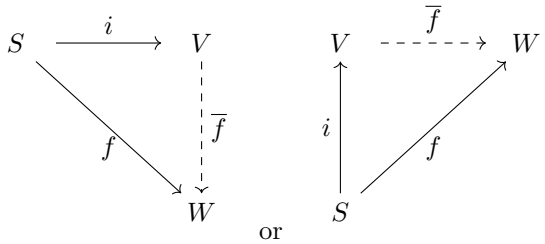
□

Example 0.4. pp. 3 Leinster (2014) [31] Let vector space V with basis $(v_s)_{s \in S}$ (.e.g if V finite-dim., e.g. $S = \{1, 2 \ldots, n\}$).
Thus $\forall$ vector space W, $\exists$ natural 1-to-1 correspondence between linear maps $\{V \to W\}$ and functions $S \to W$.

i.e. define $i : S \to V$

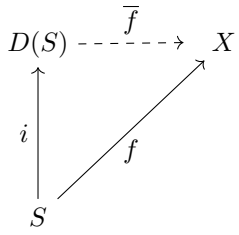
$$i(s) = v_s \quad (s \in S)$$

Then universal mapping property (given $i, f, \exists!\overline{f}$)



$$\begin{array}{c} \{S \to W\} \to \{ \text{ linear } V \to W\} \\ \overline{f} \circ i \mapsto \overline{f} \end{array} \quad \text{bijective}$$

Example 0.5. pp. 4 Leinster (2014) [31]. Given set S, topological space $D(S)$; equip $D(S)$ with discrete topology (all subsets open).
$\implies \forall$ map $S \to$ space X is cont.
i.e.



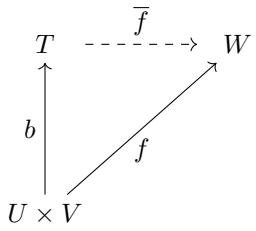
$\forall$ topological space X, $\forall f : S \to X$, $\exists!$ cont. $\overline{f} : D(S) \to X$ s.t. $\overline{f} \circ i = f$.
If $D(S)$ equipped with indiscrete topology (open $S = \{\phi, S\}$), then property is false.

Example 0.6. pp. 4 Leinster (2014) [31]. Given vector space U, V, W, bilinear map $f : U \times V \to W$ is linear.

$$f(u, v_1 + \lambda v_2) = f(u, v_1) + \lambda f(u, v_2)$$
$$f(u_1 + \lambda u_2, v) = f(u_1, v) + \lambda f(u_2, v)$$

$\forall\, u, u_1, u_2 \in U$, $v, v_1, v_2 \in V$, scalars λ.
e.g. scalar product (dot product) $\mathbf{u} \cdot \mathbf{v}$ bilniear, cross product bilinear.

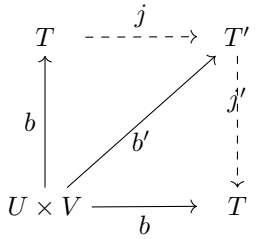
$\exists$ "universal bilinear map out of $U \times V$", i.e.
$\exists$ vector space T, $\exists$ bilinear map $b : U \times V \to T$



i.e. $\forall$ bilinear f, given b, $\exists!$ linear $\overline{f}$, s.t. $\overline{f} \circ b = f$

Lemma 4 (0.7, pp. 5 Leinster (2014) [31])**.** *Let vector spaces U, V. Suppose $b : U \times V \to T$, both universal $b' : U \times V \to T'$ bilinear maps.*
Then $T \cong T'$, i.e. $\exists!$ isomorphism $j : T \to T'$ s.t. $j \circ b = b'$

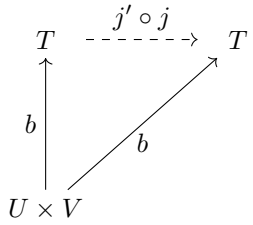
Proof. Take $b' : U \times V \to T' \implies \exists\, b$, $\exists!$ linear map $j : T \to T'$ s.t. $j \circ b = b'$.
Take $b : U \times V \to T$. $\exists\, b'$, $\exists!$ linear map $j' : T' \to T$ s.t. $j' \circ b' = b$



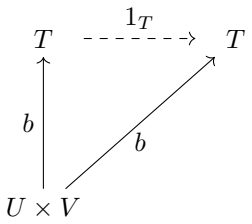
Now linear $j' \circ j : T \to T$, s.t. $(j' \circ j) \circ b = b$.
But also $1_T : T \to T$ linear and $1_T \circ b = b$

$$\implies j' \circ j = 1_T$$

i.e.



but so is



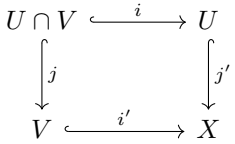
Similarly for $j \circ j' = 1_{T'}$.
So j is an isomorphism.

□

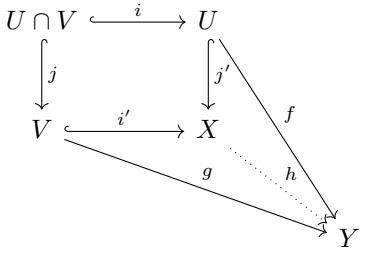
By Lemma 4, $\exists!$ tensor product $U \otimes V$, not "a" tensor product.
Example 0.8 of Leinster (2014) [31]. Let $\theta : G \to H$ be homomorphism of groups.

$$\ker(\theta) \xhookrightarrow{i} G \underset{\epsilon}{\overset{\theta}{\rightrightarrows}} H$$

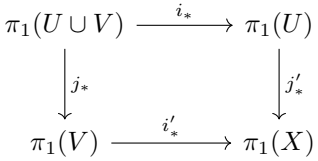
where i inclusion, ϵ trivial homomorphism, i.e.$i(x) = x \, \forall \, x \in \ker(\theta)$, $\epsilon(g) = 1 \quad \forall \, g \in G$.
map i into G s.t. $\theta \circ i = \epsilon \circ i$ and is universal (Ex. 0.11)
Example 0.9 of Leinster (2014) [31]. Let topological space $X = U \cup V$ be covered by 2 open subsets U, V.



has universal property



$\forall \, g, Y, f$ s.t. $f \circ i = g \circ j$, $\exists!$ cont. $h : X \to Y$ s.t. $h \circ j' = f$, $h \circ i' = g$.
Under favorable conditions, induced diagram of fundamental groups.

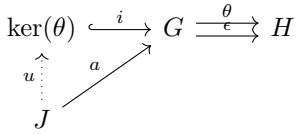


has same universal property: Van Kampen's thm.

$$\theta : G \to H$$
$$\epsilon : G \to H$$

θ, ϵ have common domain and codomain.

$i : \ker(\theta) \to G$ is s.t. $\theta \circ i = \epsilon \circ i$



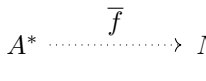
$\forall\, a : J \to G$ s.t. $\theta \circ a = \epsilon \circ a$, then $\exists !\, u : J \to \ker\theta$ s.t.

$$a = i \circ u$$

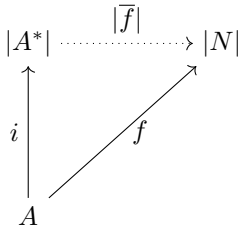
19.1.1 Free monoid A^* on set A

$\exists\, i : A \to A^*$, given any monoid N, any $f : A \to |N|$ (where $|N|$ is the underlying set of monoid N),
$\exists !$ monoid homomorphism $\overline{f} : A^* \to N$ s.t. $|\overline{f}| \circ i = f$.

Mon :



Sets :



Proposition 22 (Proposition 1.9, Awoday (2010) [42])**.** *] A^* has universal mapping property of free monoid on A.*

Proof. Givne $f : A \to |N|$, define $\overline{f} : A^* \to N$, by

$$\overline{f}(\text{""}) = e_N$$
$$\overline{f}(a_1 \dots a_n) = f(a_1) \dots f(a_n)$$

$\overline{f}$ is thus a homomorphism with $\overline{f}(a) = f(a) \qquad \forall\, a \in A$.

If $g : A^* \to N$ s.t. $g(a) = f(a) \qquad \forall\, a \in A$, then $\forall\, (a_1, \dots a_n) \in A^*$,

$$\overline{f}(a_1 \dots a_n) = \overline{f}(a_1) \dots \overline{f}(a_n) = f(a_1) \dots f(a_n) = g(a_1) \dots g(a_n) = g(a_1 \dots a_n)$$

since $\overline{f} = g$, $\overline{f}$ unique. □

Existence part of UMP is "no noise."
Unique part of UMP is "no junk."

TODO Prop. 1.10, Awodey.

Exercise 1. cf. 3.1.23 Exercises of Barr and Wells (2012) [43]

Given semigroup S, construct monoid $M = S \cup \{e\}$, $e \notin S$.
Let $e = \{S\}$.
Multiplication in M defined as

1. $xy \in S$ if $x, y \in S$
2. $xe = ex = x \quad \forall\, x \in M$

$M \equiv S^1$ in semigroup literature.

1. S^1 is a monoid (Show).
   If $a, b, c \in S \subset M$, then $(ab)c = a(bc)$ (associative holds by definition of semigroup).
   $\forall\, a \in S$, $ae = ea = a$, by how we defined multiplication.
   $ee = ee = e$ by either definition, or that multiplication in semigroup S is closed.

$$\{S\}\{S\} = \{S\}\{S\} = \{S\}$$

   If for $a, b, c \in M$, any of a, b or $c = e$, use multiplication to show equality for $(ab)c = a(bc)$; e.g.

$$(eb)c = (b) \cdot c = bc = e(bc) = bc$$
$$(ae)c = a \cdot c = a \cdot (ec) = a(ec)$$

   S^1 is a monoid.

2.

20 Actions, Finite State Machines

Definition 93. Let monoid M with identity 1, and a set S.
action of M on S, α,

$$\begin{aligned} &\alpha : M \times S \to S \text{ s.t.} \\ &\alpha(1, s) = s \quad \forall\, s \in S \\ &\alpha(mn, s) = \alpha(m, \alpha(n, s)) \quad \forall\, m, n \in M,\ s \in S \end{aligned} \tag{71}$$

Write $ms \equiv \alpha(m, s)$ then

$$1s = s$$
$$(mn)s = m(ns) \quad \forall\, m, n \in M,\ s \in S$$

Definition 94 (equivariant map)**.** Let monoid M with actions on sets S, T.
equivariant map $\phi : S \to T$ s.t. $m\phi(s) = \phi(ms)$

Suppose 2 equivariant maps ϕ_{TS}, ϕ_{UT}, sets S, T, U s.t. monoid M has actions on S, T, U:

$$\phi_{TS} : S \to T$$
$$\phi_{UT} : T \to U$$

$$m\phi_{UT} \circ \phi_{TS}(s) = m\phi_{UT}(\phi_{TS}(s)) = \phi_{UT}(m\phi_{TS}(s)) = \phi_{UT}\phi_{TS}(ms)$$

Let $\phi_{UT} \circ \phi_{TS} \equiv \phi_{US}$. ϕ_{US} equivariant.
Associativity: given equivariant maps

$$\phi_{TS} : S \to T$$
$$\phi_{UT} : T \to U$$
$$\phi_{VU} : U \to V$$

$$m\phi_{VU}\phi_{UT}\phi_{TS}(s) = m\phi_{VU}(\phi_{UT}\phi_{TS}(s)) = \phi_{VU}(m\phi_{UT}\phi_{TS}(s)) = (\phi_{VU}\phi_{UT})(\phi_{TS}(ms))$$

Thus, equivariant maps obey associativity on $\forall\, S \in \text{Obj}(M - \textbf{Act})$, (set S that M has action on) category $M - \textbf{Act}$, $\text{Obj}(M - \textbf{Act}) = \{$ sets that monoid M has action on $\}$
$\text{Mor}(M - \textbf{Act}) = \{$ equivariant maps ϕ s.t. $\forall\, m \in M$, $m\phi(s) = \phi(ms)\}$.

20.1 Actions as functors

Let α action of monoid M on set S.

Consider $\mathbf{C}(M)$ (monoid as a category; i.e. Obj$\mathbf{C}(M) = \{M\}$, Mor$\mathbf{C}(M) = M$).
action α determines function $F_\alpha : \mathbf{C}(M) \to \mathbf{Set}$,

1. $F_\alpha(*) = S \equiv F_\alpha(\{M\}) = S$
2. $F_\alpha(m) = s \mapsto \alpha(m, s) \, \forall \, m \in M, s \in S$

i.e.

$$\begin{aligned} F_\alpha &: \text{Obj}\mathbf{C}(M) \to \text{Obj}\mathbf{Set} \\ F_\alpha &: \{M\} \mapsto S \\ F_\alpha &: \text{Mor}\mathbf{C}(M) \to \text{Mor}\mathbf{Set} \\ F_\alpha &: m \mapsto (s \mapsto \alpha(m, s)) \in \text{Hom}(S, S) \end{aligned}$$

Barr and Wells (1998) [26] uses this notation: $\mathcal{M} = (A, S, s_0, \phi)$ for a "machine." Compare this to the notation used in these notes:

$$\text{FSM} = (\Sigma, S, s_0, \delta)$$

Take note that the finite set or input alphabet $A \equiv \Sigma$ is **not** a monoid.

$\forall$ string $\equiv$ finite sequence $\equiv$ "word" induces sequence of transitions in FSM starting at s_0, and ending on some final state; precisely, define

$$\begin{aligned} &\delta^* : \Sigma^* \times S \to S \\ &\delta^*(\text{""}, s) = s \quad \forall \in s \in S \\ &\delta^*((a)w, s) = \delta(a, \delta^*(w, s)) \quad \forall \, s \in S, \, w \in \Sigma^*, \, a \in A \end{aligned}$$

Recall free monoid Σ^*, $(\Sigma^*, *, \text{""})$ where $\Sigma^* \in \text{Obj}\mathbf{Set}$ is a set.

Proposition 23. *δ^* is an action of Σ^* on S.*

Proof. Assume

$$\delta^*(wv, s) = \delta^*(w, \delta^*(v, s))$$

then

$$\delta^*((a)wv, s) = \delta(a, \delta^*(wv, s)) = \delta(a, \delta^*(w, \delta^*(v, s))) = \delta^*(aw, \delta^*(v, s))$$

1st., 3rd. equalities from $\delta^*((a)w, s) = \delta(a, \delta^*(w, s))$ (definition of δ^*).
2nd. equality is from inductive hypothesis. □

recognizer - subset $L \subseteq A^*$ of strings which drive FSM from s_0 (start state) to acceptor state, is then set of strings, or language, which is **recognized** by machine FSM, this is the machine as **recognizer**.

transducer - FSM outputs string of symbols (not necessarily in same alphabet) $\forall$ state it enters, or each transition it undergoes.

20.2 Set-valued functors as actions

cf. 3.2.6 of Barr and Wells (2012) [43].

21 Products, Coproducts

21.1 Sources

It appears Adámek, Herrlich, and Strecker (2004) [27] defines *sources* to simply give a name and formalize a tuple.

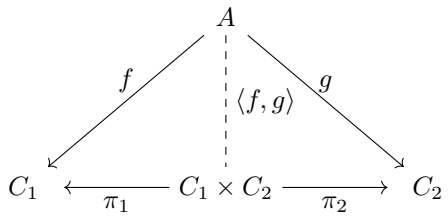
Definition 95 (source)**. source** is a tuple: $(a, (f_i)_{i \in I})$, $f_i : A \to A_i$

21.2 Products

Definition 96 (Products)**.** (in Turi's notation [30])

Given objects C_1, C_2 of category $\mathbb{C}$, **product** (if exists) consists of object $C_1 \times C_2$ of $\mathbb{C}$ and $\pi_1 : C_1 \times C_2 \to C_1$
$\pi_2 : C_1 \times C_2 \to C_2$
s.t.

$$\forall \text{ object } A \text{ of } \mathbb{C}, \quad \forall \begin{matrix} f : A \to C_1 \\ g : A \to C_2 \end{matrix} \quad \exists ! \quad \langle f, g \rangle : A \to C_1 \times C_2 \text{ s.t. } \begin{matrix} f = \pi_1 \circ \langle f, g \rangle, \text{ i.e.} \\ g = \pi_2 \circ \langle f, g \rangle \end{matrix}$$



(compare with Leinster (2014) [31])
Let category $\mathcal{A}$, $X, Y \in \mathcal{A}$, **product** of X, Y consists of object P and maps
(compare this definition with Adámek, Herrlich, and Strecker (2004) [27] and their notation)
product consisting of

$$\begin{aligned} C_1 \times C_2 \times \cdots \times C_{\mathcal{N}} &\in \text{Obj}\mathbf{C} \\ \pi_1 : C_1 \times C_2 \times \cdots \times C_{\mathcal{N}} &\to C_1 \\ \pi_2 : C_1 \times C_2 \times \cdots \times C_{\mathcal{N}} &\to C_2 \\ &\vdots \\ \pi_{\mathcal{N}} : C_1 \times C_2 \times \cdots \times C_{\mathcal{N}} &\to C_{\mathcal{N}} \end{aligned}$$

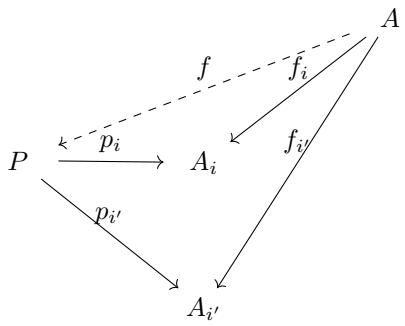
is s.t.

$$\begin{matrix} A \in \text{Obj}\mathbf{C} \\ f_1 : A \to C_1 \\ \forall \quad f_2 : A \to C_2, \\ \vdots \\ f_{\mathcal{N}} : A \to C_{\mathcal{N}} \end{matrix}$$
$$\exists ! \langle f_1, f_2, \ldots, f_{\mathcal{N}} \rangle : A \to C_1 \times C_2 \times \cdots \times C_{\mathcal{N}} \text{ s.t.}$$

$$\begin{aligned} f_1 &= \pi_1 \circ \langle f_1, f_2, \ldots f_{\mathcal{N}} \rangle \\ f_2 &= \pi_2 \circ \langle f_1, f_2, \ldots f_{\mathcal{N}} \rangle \\ &\vdots \\ f_{\mathcal{N}} &= \pi_{\mathcal{N}} \circ \langle f_1, f_2, \ldots f_{\mathcal{N}} \rangle \end{aligned}$$

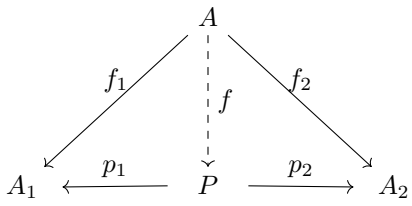
Let's use the notation of Adámek, Herrlich, and Strecker (2004) [27]. Also note that these references *assume* the *universal mapping property* in the definition of a **product**.

Definition 97 (Product (assuming univeral mapping property))**.**



(72)

For the case of only 2 objects associated with I,



(73)

product $P, \{p_i, A_i\}_{i\in I}$ s.t. $\forall\, A, \{f_i, A_i\}_{i\in I}$, $\exists!$ morphism $f : A \to P$ s.t. $f_i = p_i \circ f$, $i \in I$.

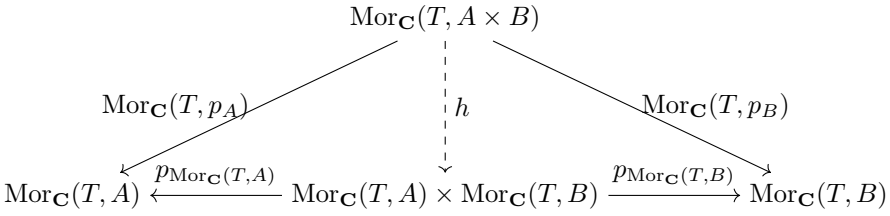
This definition is the same one used in pp. 35, Def. 2.16, of Awodey (2010) [42].
Consider the definition that doesn't use the universal mapping property from Pareigis (2004) [32].

Definition 98 (Product without universal property)**.** Given objects $A, B \in \text{Obj}\mathbf{C}$, category $\mathbf{C}$.

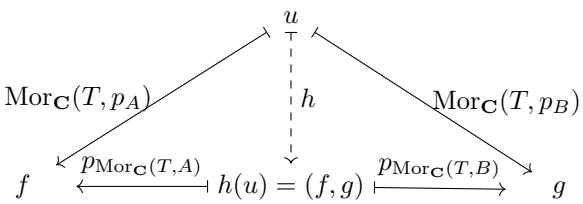
Object $A \times B$, morphisms p_A, p_B, $p_A : A \times B \to A$ is called a (not the) (categorical) product of A, B, projections p_A, p_B.
$$p_B : A \times B \to B$$
If $\forall$ object $T \in \text{Obj}\mathbf{C}$, $\exists$ isomorphism

$$\text{Mor}_{\mathbf{C}}(T, A \times B) \cong \text{Mor}_{\mathbf{C}}(T, A) \times \text{Mor}_{\mathbf{C}}(T, B) \tag{74}$$

(where $\text{Mor}_{\mathbf{C}}$ is the Cartesian product) s.t.



(75)

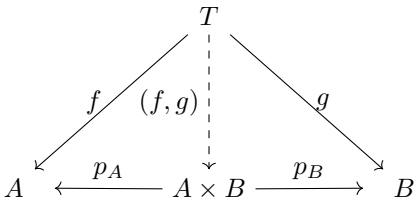


where $\mathrm{Mor}_{\mathbf{C}}(T, p_A)(u) = f = p_{\mathrm{Mor}_{\mathbf{C}}(T,A)} h(u)$
$\mathrm{Mor}_{\mathbf{C}}(T, p_B)(u) = g = p_{\mathrm{Mor}_{\mathbf{C}}(T,B)} h(u)$

Proposition 24 (Equivalent definition of product by universal mapping property)**.** *Given objects* $A, B \in Obj\mathbf{C}$*,*

category $\mathbf{C}$*, object* $A \times B$*, morphisms* $p_A : A \times B \to A$ $\quad$ *is a (categorical) product,*
$p_B : A \times B \to B$

iff $\forall$ *object* $T \in Obj\mathbf{C}$*,* $\forall\, f : T \to A$*,* $\exists !$ *morphism* $(f, g) : T \to A \times B$*, s.t.*
$g : T \to B$



(76)

cf. Proposition "Characterization of products by universal mapping property", 2.7.4 in Pareigis (2004) [32].

Proof. $\Longleftarrow$:

Let $(A \times B, p_A, p_B)$ be a "product" of $A \times B$, obeying the universal (mapping) property. Let $T \in \mathrm{Obj}\mathbf{C}$. Let $u \in \mathrm{Mor}_{\mathbf{C}}(T, A \times B)$.

Define $f := p_A \circ u : T \to A$
$g := p_b \circ u : T \to B$
Then $(f, g) \in \mathrm{Mor}_{\mathbf{C}}(T, A) \times \mathrm{Mor}_{\mathbf{C}}(T, B)$.

Define $h(u) := (f, g) \in \mathrm{Mor}_{\mathbf{C}}(T, A) \times \mathrm{Mor}_{\mathbf{C}}(T, B)$.
Then,

$$\mathrm{Mor}_{\mathbf{C}}(T, p_A)(u) = p_A \circ u = f = p_{\mathrm{Mor}_{\mathbf{C}}(T,A)} \circ h(u)$$
$$\mathrm{Mor}_{\mathbf{C}}(T, p_B)(u) = p_B \circ u = g = p_{\mathrm{Mor}_{\mathbf{C}}(T,B)} \circ h(u)$$

(so Diagram 75 commutes).

Show: h bijective.
Construct inverse map k:

$$k : \mathrm{Mor}_{\mathbf{C}}(T, A) \times \mathrm{Mor}_{\mathbf{C}}(T, B) \to \mathrm{Mor}_{\mathbf{C}}(T, A \times B)$$
$$k((f, g)) := u$$

where $(f, g) \in \mathrm{Mor}_{\mathbf{C}}(T, A) \times \mathrm{Mor}_{\mathbf{C}}(T, B)$

By universal (mapping) property, $\exists !\, u : T \to A \times B$ s.t. $p_A \circ u = f$.
$p_B \circ u = g$
Let $(f, g) \in \mathrm{Mor}_{\mathbf{C}}(T, A) \times \mathrm{Mor}_{\mathbf{C}}(T, B)$.
Then $(h \circ k)((f \circ g)) = h(u) = (p_A \circ u, p_B \circ u) = (f, g)$. Hence $h \circ k = 1$ or i.e. $h \circ k 1_{\mathrm{Mor}_{\mathbf{C}}(T,A) \times \mathrm{Mor}_{\mathbf{C}}(T,B)}$.

Let $u \in \mathrm{Mor}_{\mathbf{C}}(T, A \times B)$.
Then $(k \circ h)(u) = k((f, g)) = k((p_A \circ u, p_B \circ u)) = u'$ where $u' : T \to A \times B$ and

$$p_A \circ u' = p_A \circ k((p_A \circ u, p_B \circ u)) = p_A \circ k((f, g)) = p_A \circ u \quad (\text{since } k((f, g)) := u)$$
$$p_B \circ u' = p_B \circ k((p_A \circ u, p_B \circ u)) = p_B \circ k((f, g)) = p_B \circ u \quad (\text{since } k((f, g)) := u)$$

2nd. equality is by universal (mapping) property.
$\implies u = u'$ (by component-wise equality), and
$k \circ h = 1_{\mathrm{Mor}_{\mathbf{C}}(T, A\times B)}$.
$\implies h$ isomorphic.

$\implies$:

Given morphisms

$$f : T \to A$$
$$g : T \to B$$

then $(f, g) \in \mathrm{Mor}_{\mathbf{C}}(T, A) \times \mathrm{Mor}_{\mathbf{C}}(T, B)$.
$\forall$ object $T \in \mathrm{Obj}\mathbf{C}$, $\exists$ isomorphism h s.t. $\mathrm{Mor}_{\mathbf{C}}(T, A \times B) \cong \mathrm{Mor}_{\mathbf{C}}(T, A) \times \mathrm{Mor}_{\mathbf{C}}(T, B)$,
Then $u := h^{-1}((f, g))$ is a unique morphism, $u \in \mathrm{Mor}_{\mathbf{C}}(T, A \times B)$.
Also, by Diagram 75,

$$\mathrm{Mor}_{\mathbf{C}}(T, p_A)(u) = f = p_{\mathrm{Mor}_{\mathbf{C}}(T,A)} \circ h(u)$$
$$\mathrm{Mor}_{\mathbf{C}}(T, p_B)(u) = f = p_{\mathrm{Mor}_{\mathbf{C}}(T,B)} \circ h(u)$$

Now

$$\mathrm{Mor}_{\mathbf{C}}(T, p_A)(u) = p_A \circ u$$
$$\mathrm{Mor}_{\mathbf{C}}(T, p_B)(u) = p_B \circ u$$

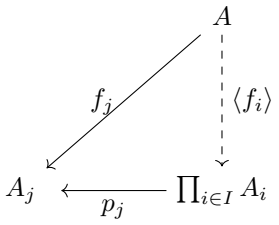
(by definition).
So

$$f = p_A \circ u$$
$$g = p_B \circ u$$

So universal mapping property is satisfied. □

Thus,

Definition 99 (Product, generalized, assuming universal property)**.** Give $A_i \in \mathrm{Obj}\mathbf{C}$, $i \in I$, category $\mathbf{C}$,
product is object $\prod_{i\in I} A_i \in \mathrm{Obj}\mathbf{C}$, morphisms $p_j : \prod_{i\in I} A_i \to A_j$, $j \in I$,
s.t. $\forall$ object $A \in\in \mathrm{Obj}\mathbf{C}$,
$\exists!$ morphism $\langle f_i\rangle : A \to \prod_{i\in I} A_i$,
s.t.



(77)

i.e. $\pi_j \circ \langle f_i\rangle = f_j \, \forall \, i, j \in I$

21.2.1 Example: Set always has products

$\forall$ sets $X, Y \in$ Obj(Set), $\exists$ product $X \times Y \in$ Obj(Set).

Let $A \in$ Obj(Set), $f_1 : A \to X$, $f_2 : A \to Y$ Define $\langle f_1, f_2\rangle : A \to X \times Y$, $\langle f_1, f_2\rangle(a) = (f_1(a), f_2(a))$

Then $\pi_1 \circ \langle f_1, f_2\rangle(a) = f_1(a)$, $\pi_2 \circ \langle f_1, f_2\rangle(a) = f_2(a)$ $\implies \pi_1 \circ \langle f_1, f_2\rangle = f_1$, $\pi_2 \circ \langle f_1, f_2\rangle = f_2$

Suppose $f' : A \to X \times Y$ s.t. $\pi_1 \circ f' = f_1$, $\pi_2 \circ f' = f_2$

Write $f'(a) = (x, y)$

$$\begin{aligned} f_1(a) &= \pi_1 \circ f'(a) = \pi_1(x, y) = x \\ f_2(a) &= \pi_2 \circ f'(a) = \pi_2(x, y) = y \end{aligned} \qquad \Longrightarrow f'(a) = (f_1(a), f_2(a)) = \langle f_1, f_2 \rangle (a)$$

$\langle f_1, f_2 \rangle$ unique.

21.2.2 Example: "Record Types", classes (in computer science, C++, Python)

cf. pp. 174, 5.3.14 "Record Types" Barr and Wells (1998) [26].

Look at Diagram 77 again.

To allow operations depending on several variables in a functional programming language L, e.g. assume $\forall$ types A_j, $\forall\, j \in I$, language L has record type $\prod_{i \in I} A_i$, and field selectors p_j

$$p_j : \prod_{i \in I} A_i \to A_j$$

Insist that any data in $\prod_{i \in I} A_i$ be determined completely by those fields A_j, $j \in I$,
then $\forall$ operations f_j, $f_j : A \to A_j$, $\quad \forall\, j \in I$,
there ought to be an unique operation $\{f_i\rangle : A \to \coprod_{i \in I} A_i$ s.t.

$$p_j \langle f_i \rangle = f_j, \quad \forall\, i, j \in I$$

This would make $\prod_{i \in I} A_i$ the product of A_i's, with selectors as product projections.

Thus, to say that one can always construct record types in a functional programming language L to say that the corresponding category $C(L)$ has finite products.

Proposition 25. *If product* $(A_1 \times \cdots \times A_{\mathcal{N}} \xrightarrow{\pi_i} A_i)_{i \in I}$, *if* $\exists\, i_0 \in I$ *s.t.* $Hom(A_{i_0}, A_i) \neq \emptyset$, $\quad \forall\, i \in I$, *then* π_{i_0} retraction

Proof. $\forall\, i \in I$, choose $f_i \in \mathrm{Hom}(A_{i_0}, A_i)$ with $f_{i_0} = 1_{A_{i_0}}$.

Then $\langle f_i \rangle : A_{i_0} \to A_1 \times \cdots \times A_{\mathcal{N}}$ is a morphism s.t.

$$\pi_{i_0} \circ \langle f_i \rangle = f_{i_0} = 1_{A_{i_0}}$$

□

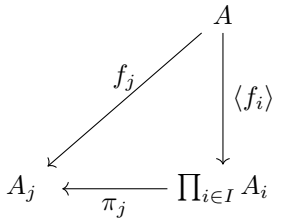
Adámek, Herrlich, and Strecker (2004) [27] and their notation) calls a **sink** what Leinster (2014) [31] calls a **cocone**.

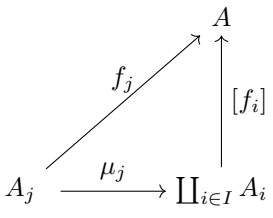
Definition 100. sink $((f_i)_{i \in I}, A) \equiv (f_i, A)_I \equiv (A_i \xrightarrow{f_i} A)_I$, object A, family of morphisms $f_i : A_i \to A$

For the *coproduct*, consider this enlightening comparision:

product $(\prod_{i \in I} A_i, \pi_j)_{j \in I}$ coproduct $(\mu_j, \coprod_{i \in I} A_i)_{j \in I}$

projection $\pi_j : \prod_{i \in I} A_i \to A_j$ injection $\mu_j : A_j \to \coprod_{i \in I} A_i$





$C \xrightarrow{\langle f,g \rangle} A \times B$ $\qquad$ $C \xleftarrow{[f,g]} A + B$

$\prod_{i \in I} f_i$, or if $i = \{1, 2\}$, $f \times g$ $\qquad$ $\coprod_{i \in I} f_i$, or if $i = \{1, 2\}$, $f + g$

21.2.3 Examples (of coproducts)

- if $(A_i)_I$ pairwise-disjoint family of sets, then $(\mu_j, \bigcup_{i\in I} A_i)_{j\in I}$ is coproduct in Set.
  If $(A_i)_I$ arbitrary set-indexed family of sets, then it can be "made disjoint" by pairing each A_i with index i, i.e. by working with $A_i \times \{i\}$ rather than A_i.

  So $\bigcup_{i\in I}(A_i \times \{i\})$ disjoint. Consider

  $$\mu_j : A_j \to \bigcup_{i\in I} A_i \times \{i\}$$
  $$\mu_i(a) = (a, j)$$

  $(\mu_j, \bigcup_{i\in I} A_i \times \{i\})_{j\in I}$ is a coproduct in Set.

  Indeed, given $f_j : A_j \to A$,
  $f_j(a) \in A$

  $$[f_i] : \coprod_{i\in I} A_i \times \{i\} \to A$$
  $$[f_i] \circ \mu_j = f_j$$

  where

  $$f_j(a) = [f_i] \circ \mu_j(a) = [f_i](a, j) = f_j(a)$$

- Top coproducts are "topological sums"; they're "concrete" coproducts (Adámek, Herrlich, and Strecker (2004) [27])

- Vec (nonconcrete) coproducts called *direct sums*

  direct sum $\bigoplus_{i\in I} A_i$ of vector spaces A_i is subspace of direct product $\prod_{i\in I} A_i$ consisting of all elements $(a_i)_{i\in I}$ with finite carrier (i.e. $\{i \in I | a_i \neq 0\}$ is finite), injections

  $$\mu_j : A_j \to \bigoplus_{i\in I} A_i$$
  $$\mu_j(a) = (a_i)_{i\in I} \text{ with } a_i = \begin{cases} a & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

- Grp has nonconcrete coproducts, "free products"

22 Naturality; Natural transformations

Definition 101 (Natural Transformation)**.** If functors F, G, categories $\mathbf{C}, \mathbf{D}$,

$$F : \mathbf{C} \to \mathbf{D}$$
$$G : \mathbf{C} \to \mathbf{D}$$

the **natural transformation** η from F to G is a family of morphisms s.t.

1. $\forall X \in \text{Obj}\mathbf{C}$, $\eta_X : F(X) \to G(X)$ s.t. $F(X), G(X) \in \text{Obj}\mathbf{D}$. morphisms $\eta_X \in \text{Hom}_{\mathbf{D}}(F(X), G(X)) \subset \text{Mor}\mathbf{D}$ is called **component** of η at X.
2. components must be s.t. $\forall$ morphism $f : X \to Y$ in $\mathbf{C}$, i.e. $f \in \text{Hom}_{\mathbf{C}}(X, Y) \in \text{Mor}\mathbf{C}$,

$$\eta_Y \circ F(f) = G(f) \circ \eta_X \tag{78}$$

i.e.

$$\begin{array}{ccc} F(X) & \xrightarrow{F(f)} & F(Y) \\ \downarrow{\scriptstyle \eta_X} & & \downarrow{\scriptstyle \eta_Y} \\ G(X) & \xrightarrow{G(f)} & G(Y) \end{array}$$

If both F, G contravariant,

$$\begin{array}{ccc} F(X) & \xleftarrow{F(f)} & F(Y) \\ \downarrow{\scriptstyle \eta_X} & & \downarrow{\scriptstyle \eta_Y} \\ G(X) & \xleftarrow[G(f)]{} & G(Y) \end{array}$$

cf. Wikipedia, "Natural transformation"

Notation: natural transformation from F to G, $\eta : F \to G$, i.e. family of morphisms $\eta_X : F(X) \to G(X)$ is natural in X.

cf. Adámek, Herrlich, and Strecker (2004) [27].

From pp. 16, Sec. 14 "Natural Transformations", Mac Lane (1978) [28],

$\forall$ arrow $f : C \to C'$ in C, i.e. notation: $\forall$ morphism $f : X \to Y$ in $\mathbf{C}$,
Mac Lane's notation:

$$\begin{array}{ccc} c & Sc & \xrightarrow{\tau c} & Tc \\ \downarrow{\scriptstyle f} & \downarrow{\scriptstyle Sf} & & \downarrow{\scriptstyle Tf} \\ c' & Sc' & \xrightarrow{\tau c'} & Tc' \end{array}$$

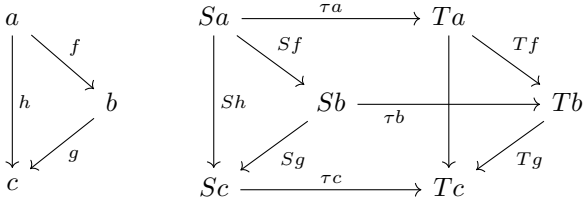
My notation:

$$\begin{array}{ccc} X & FX & \xrightarrow{\eta_X} & GX \\ \downarrow{\scriptstyle f} & \downarrow{\scriptstyle Ff} & & \downarrow{\scriptstyle Gf} \\ Y & FY & \xrightarrow{\eta_Y} & GY \end{array}$$

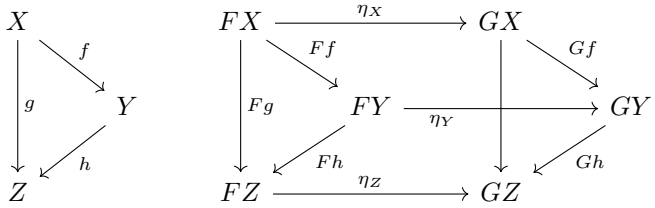
We say that $\tau_c : Sc \to Tc$ is natural in C; $\eta_X : FX \to GX$ is natural in X.

If we think of functor S as giving a picture in B of (all the objects, arrows of) C, then
natural transformation τ is set of arrow s mapping (or, translating) the picture S to picture T, with all squares (and parallelograms!) like that above commutative:

Mac Lane's notation:



My notation:



natural equivalence or i.e. natural isomorphism - natural transformation τ with $\forall$ component τc invertible in B, $\tau : S \cong T$.
notation: η with $\forall$ component η_X invertible in $\mathbf{D}$; $\eta : F \cong G$.
$\Longrightarrow$ inverses $(\tau c)^{-1}$ in B are components of natural isomorphism $\tau^{-1} : T \to S$.
notation: inverses $(\eta_X)^{-1}$ in $\mathbf{D}$ are components of natural isomorphism $\eta^{-1} : G \to F$.

22.1 Examples of Natural transformations

e.g. Hurewicz homomorphism $\pi_n(X) \to H_n(X)$ $\forall$ topological space X is a natural transformation from nth homology functor.
$\pi_n : \mathbf{Top} \to \mathbf{Grp}$ to nth homology functor $H_n : \mathbf{Top} \to \mathbf{Grp}$.

e.g. **determinant** det is a natural transformation (cf. Mac Lane (1978) [28]).
Let $\det_K M$ = determinant of $n \times n$ matrix M, with entries in commutative ring K, while
$K^* \equiv$ group of units (invertible elements) of K

Thus M non-singular when $\det_K M$ is a unit, and $\det_K$ is a morphism $GL_n K \to K^*$ of groups.
Because det is defined by same formula $\forall$ ring K, each morphism $f : K \to K'$ of commutative rings leads to

$$\begin{array}{ccc} GL_n K & \xrightarrow{\det_K} & K^* \\ \downarrow{\scriptstyle GL_n f} & & \downarrow{\scriptstyle f^*} \\ GL_n K' & \xrightarrow{\det_{K'}} & (K')^* \end{array}$$

$\Longrightarrow \det : GL_n \to ()^*$ is natural between 2 functors $\mathbf{CRng} \to \mathbf{Grp}$

TODO: Work out examples of natural transformations involving group commutators on pp. 17 of Mac Lane (1978) [28].

e.g. category $\mathbf{Finord}$ = all finite ordinal numbers n
category $\mathbf{Set}_f$ = all finite sets (in some universe U)
$\forall$ ordinal $n = \{0, 1, \ldots n-1\}$ is a finite set, so inclusion S is a functor $S : \mathbf{FinOrd} \to \mathbf{Set}_f$
$\forall$ finite set X determines ordinal number $n = \#X$, number of elements in X
$\forall X$, choose bijection $\theta_X : X \to \#X$
$\forall$ function $f : X \to Y$, X, Y finite sets,
Choose corresponding function $\#f : \#X \to \#Y$ between ordinals by $\#f = \theta_Y f \theta_X^{-1}$, so that

$$\begin{array}{ccc} X & \xrightarrow{\theta_X} & \#X \\ \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle \#f} \\ Y & \xrightarrow{\theta_Y} & \#Y \end{array} \tag{79}$$

$\Longrightarrow \#$ a functor $\# : \mathbf{Set}_f \to \mathbf{FinOrd}$
If X itself an ordinal number, take θ_X to be an identity.
$\Longrightarrow \# \circ S$ is identity functor I' of $\mathbf{FinOrd}$
$S \circ \#$ is not identity functor $I : \mathbf{Set}_f \to \mathbf{Set}_f$ because
$S \circ \#$ sends each finite set X to a special finite set - ordinal number n with same number of elements as X.
However $\theta : I \to S\#$ is a natural isomorphism by square diagram above (Eq. 79).

$$I \cong S \circ \#, \quad I' = \# \circ S$$

equivalence between categories $\mathbf{C}, \mathbf{D} =$

pair of functors $S : \mathbf{C} \to \mathbf{D}$ and
$T : \mathbf{C} \to \mathbf{D}$
natural isomorphisms $I_C \cong T \circ S$, $I_D \cong S \circ T$.

23 Limits

23.1 Pullback

Definition 102. For some category $\mathbf{A}$, and for

$$\begin{array}{ccc} & & Y \\ & & \downarrow{\scriptstyle t} \\ X & \xrightarrow[s]{} & Z \end{array}$$

$X, Y, Z \in \text{Obj}\mathbf{A}$.

$s : X \to Z$; $\quad s, t \in \text{Mor}\mathbf{A}$
$t : Y \to Z$

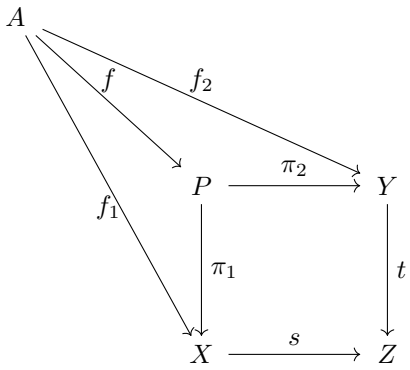
Then the **pullback** or "pullback square" consists of $P \in \text{Obj}\mathbf{A}$, $\pi_1 : P \to X$ s.t.
$\pi_2 : P \to Y$

$$\begin{array}{ccc} P & \xrightarrow{\pi_2} & Y \\ \downarrow{\scriptstyle \pi_1} & & \downarrow{\scriptstyle t} \\ X & \xrightarrow{s} & Z \end{array}$$

commutes and s.t. $\forall$ commutative square in $\mathbf{A}$

$$\begin{array}{ccc} A & \xrightarrow{f_2} & Y \\ \downarrow{\scriptstyle f_1} & & \downarrow{\scriptstyle t} \\ X & \xrightarrow{s} & Z \end{array}$$

then $\exists ! \, f : A \to P$ s.t.



Simmons (2011) [29]
cf. 3.3 *Some less simple functors*, pp. 79, of Simmons (2011) [29]

24 Monads

cf. Ch. 6, Monads and Algebras, starting on pp. 137 of Mac Lane (1971) [28].

24.1 Monads in a Category

24.1.1 Development of a Monad

Let's follow the development of a Monad according to Mac Lane (1971) [28] starting on pp. 137 in Sec. 1. "Monads in a Category."

$\forall$ endofunctor $T : X \to X$, T has composites $T^2 = T \circ T : X \to X$, $T^3 = T^2 \circ T : X \to X$ in Mac Lane's notation, X is a category. In our notation, $\mathbf{C}$ denotes the category. And so for endofunctor $T : \mathbf{C} \to \mathbf{C}$, $T^2 = T \circ T : \mathbf{C} \to \mathbf{C}$, $T^3 = T^2 \circ T : \mathbf{C} \to \mathbf{C}$.

Recall from Def. 87, that for endofunctor $T : \mathbf{C} \to \mathbf{C}$, the *object map* $T : \text{Obj}\mathbf{C} \to \text{Obj}\mathbf{C}$ acts just like a morphism in Mor$\mathbf{C}$! So $T : X \to T(X)$ for $X, T(X) \in \text{Obj}\mathbf{C}$. Also, recall that the *morphism map* $T : \text{Mor}_{\mathbf{C}} \to \text{Mor}_{\mathbf{C}}$ acts like the usual morphism map, such that $Tf : T(X) \to T(Y)$ if $f : X \to Y$, for $X, Y \in \text{Obj}\mathbf{C}$.

If $\mu : T^2 \to T$ is a *natural transformation*, then recall the definition of the natural transformation, Def. 101, and apply that onto μ.

$\forall X \in \text{Obj}\mathbf{C}$, $\mu_X : T^2(X) \to T(X)$ with $T^2(X), T(X) \in \text{Obj}\mathbf{C}$. μ_X is a morphism $\in \text{Hom}_{\mathbf{C}}(T^2(X), T(X))$ (!!), as well as being a component of μ at X.

$\forall$ morphism $f : X \to Z$ in $\mathbf{C}$, $X, Z \in \text{Obj}\mathbf{C}$ (the notation of using Z to emphasize that f can map X to *any* arbitrary object in Obj$\mathbf{C}$),

$$\mu_Z \circ T^2 f = Tf \circ \mu_X$$

i.e.

$$\begin{array}{ccc} T^2(X) & \xrightarrow{\mu_X} & T(X) \\ \downarrow{\scriptstyle T^2 f} & & \downarrow{\scriptstyle Tf} \\ T^2(Z) & \xrightarrow{\mu_Z} & T(Z) \end{array}$$

Let $T\mu : T^3 \to T^2$ *denote* a natural transformation with components $(T\mu)_X = T(\mu_X) : T^3(X) \to T^2(X)$, while $\mu T : T^3 \to T^2$ denotes another natural transformation with components $(\mu T)_X = \mu_{T(X)}$.

$T\mu$, μT are "horizontal" composites in the sense of Sec. II. 5, "The Category of All Categories". (TODO: Understand what Mac Lane (1971) [28] means when he says this on pp. 137).

Let's jump the gun and consider what natural transformation $\eta : 1_{\mathbf{C}} \to T$ means with respect to the definition of a natural transformation, Def. 101. $1_{\mathbf{C}}$ is the identity functor, mapping everything to itself.

$\forall X \in \text{Obj}\mathbf{C}$, $\eta_X : 1_{\mathbf{C}}(X) \to T(X)$ with $1_{\mathbf{C}}(X) = X, T(X) \in \text{Obj}\mathbf{C}$. η_X is a morphism $\in \text{Hom}_{\mathbf{C}}(1_{\mathbf{C}}(X), T(X)) = \text{Hom}_{\mathbf{C}}(X, T(X))$ (!!!), as well as being a component of η at X. Not only that, it applies T onto X!

$\forall$ morphism $f : X \to Z$ in $\mathbf{C}$, $X, Z \in \text{Obj}\mathbf{C}$,

$$\eta_Z \circ 1_{\mathbf{C}} f = \eta_Z \circ f = Tf \circ \eta_X$$

i.e.

$$\begin{array}{ccc} 1_{\mathbf{C}}(X) & \xrightarrow{\eta_X} & T(X) \\ \downarrow{\scriptstyle 1_{\mathbf{C}} f} & & \downarrow{\scriptstyle Tf} \\ 1_{\mathbf{C}}(Z) & \xrightarrow{\eta_Z} & T(Z) \end{array} \implies \begin{array}{ccc} X & \xrightarrow{\eta_X} & T(X) \\ \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle Tf} \\ Z & \xrightarrow{\eta_Z} & T(Z) \end{array}$$

24.1.2 Definitions and Properties of a Monad

Definition 103 (Monad)**. monad** $T = \langle T, \eta, \mu \rangle$ in category $\mathbf{C}$ consists of functor $T : \mathbf{C} \to \mathbf{C}$ and 2 natural transformations

$$\begin{aligned} \eta &: 1_{\mathbf{C}} \to T \\ \mu &: T^2 \to T \end{aligned} \tag{80}$$

s.t.

$$\begin{array}{ccc} T^3 & \xrightarrow{T\mu} & T^2 \\ \downarrow{\scriptstyle \mu T} & & \downarrow{\scriptstyle \mu} \\ T^2 & \xrightarrow{\mu} & T \end{array} \tag{81}$$

being like an "associative" diagram, and

$$\begin{array}{ccccc} 1T & \xrightarrow{\eta T} & T^2 & \xleftarrow{T\eta} & T1 \\ \| & & \downarrow{\scriptstyle \mu} & & \| \\ T & = & T & = & T \end{array} \tag{82}$$

being a diagram that could be said to express the left and right unit laws.

Formally, the monad definition is like that of a monoid M in **Sets**:
set M of elements of monad $\to$ replaced by endofunctor $T : X \to X$
Cartesian product $\times$ of 2 sets $\to$ composite of 2 functors
binary operation $\mu : M \times M \to M$ of multiplication $\to$ transformation $\mu : T^2 \to T$ and
unit (identity) element $\eta : 1 \to M$ $\to$ $\eta : 1_{\mathbf{C}} \to T$.

Thus, call
η the **unit** of monad T. The first commutative diagram, Eq. 81, is the associative law for a monad, and
μ the **multiplication** of monad T. The 2nd. and 3rd. diagrams, Eq. 82 expresses the left and right *unit laws*.

Proposition 26 (Coherence conditions)**.**

$$\begin{aligned} &\mu \circ T\mu = \mu \circ \mu T \quad (\textit{as natural transformations } T^3 \to T,\ \mu \circ T\mu \textit{ and } \mu \circ \mu T, \textit{ that is}) \\ &\mu \circ T\eta = \mu \circ \eta T = 1_T \quad (\textit{as natural transformations } T \to T,\ 1_T \equiv \textit{identity functor}) \end{aligned} \tag{83}$$

For the last statement, it is equal to commutative diagram Eq. 82, since it can be redrawn as follows:

$$\begin{array}{ccc} T & \xrightarrow{\eta T} & T^2 \\ \downarrow{\scriptstyle T\eta} & \searrow & \downarrow{\scriptstyle \mu} \\ T^2 & \xrightarrow{\mu} & T \end{array}$$

24.2 Monads and Adjoints

cf. Ch. 6, Monads and Algebras, pp. 138 of Mac Lane (1971) [28].

Proposition 27. *adjunction* $\langle F, G, \eta, \epsilon \rangle : X \to A$, $\exists$ *monad in category* X *(notation* $\langle F, G, \eta, \epsilon \rangle : \mathbf{C} \to \mathbf{D}$, $\exists$ *monad in category* $\mathbf{C}$*).*

Specifically, 2 functors $F : \mathbf{C} \to \mathbf{D}$, $G : \mathbf{D} \to \mathbf{C}$ *have composite* $T = GF$*, an endofunctor,*
unit η *of adjunction is natural transformation* $\eta : 1 \to T$*,*
counit $\epsilon : FG \to 1_A$ *of adjunction yields by horizontal composition; a natural transformation*

$$\mu = G\epsilon F : GFGF \to GF = T$$

TODO: adjunction.

24.3 Applications of Monads: Correspondence between Monads in Computer Science and Monads in Category Theory

cf. monad (in computer science), nLab

For monads in computer science,
maps type X to new type $T(X)$,
equipped with rule for composing 2 functions of form $f : X \to T(Y)$ (called Kleisli functions), $g : Y \to T(Z)$ to function

$$g \circ f : X \to T(Z) \quad \text{(their Kleisi composition)}$$

associative in evident sense, unital with respect to given unit function called $\text{pure}_X : X \to T(X)$ to be thought of as taking a value to the pure computation that simply returns that value.

e.g. when monad $T(-)$ forms product types $T(X) = X \times Q$, some fixed type Q that carries structure of monoid.
Then Kleisi function $f : X \to Y \times Q$ is a function $X \to Y$ that produces as a side effect output of type Q.

Kleisi composition of $f : X \to Y \times Q$ not only evaluates the 2 programs in sequence, but also combines their $g : Y \to Z \times Q$
Q-output using monoid operation of Q, e.g.

if $fx = (y, q)$ then $(g \circ f)(x) = (z, qq')$
$gy = (z, q')$

Let syntactic category be denoted $\mathbf{C}$.

$\text{Obj}\mathbf{C} \ni X$ $\quad \to \quad$ types of the (programming) language
$\text{Hom}_{\mathbf{C}}(X, Y)$ $\quad \to \quad$ "programs" that takes value of type X as input and returns value of type Y

endofunctor $T : \mathbf{C} \to \mathbf{C}$
object map $T : X \to T(X)$ sends each type $X \in \text{Obj}\mathbf{C}$ to another type $T(X) \in \text{Obj}\mathbf{C}$.

unit natural transformation $\epsilon : 1_{\mathbf{C}} \to T$ provides $\forall$ type X a component morphism $\text{pure}_X : X \to T(X)$ (notation, ncat uses the notation ϵ; we will use η for the "unit" natural transformation of a monad):

$$\eta : 1_{\mathbf{C}} \to T$$
$$\eta_X : X \to T(X) \equiv \text{pure}_X : X \to T(X)$$

multiplication natural transformation $\mu : T^2 \to T$ provides $\mu_X : T^2(X) \to T(X)$ which induces Klesli composition by formula

$$\begin{aligned} (g \circ f) &:= (Y \xrightarrow{g} T(Z)) \circ_{\text{Klesli}} (X \xrightarrow{f} T(Y)) \\ &:= X \xrightarrow{f} T(Y) \xrightarrow{T(g)} T(T(Z)) \xrightarrow{\mu(Z)} TZ \end{aligned} \tag{84}$$

$T(g)$ uses fact that $T(-)$ is functor morphism $\mu(Z)$ is 1 that implements "T-computation".

Now nLab says this about the "bind" operation:

"bind" $\leftrightarrow$ multiplication on monad M,

$$\begin{gathered} MA \to (MB)^A \to MB \text{ equivalent to map } MA \times MB^A \to MB \\ MA \times MB^A \xrightarrow{\text{strength}} M(A \times MB^A) \xrightarrow{M_{\text{eval}_{A,MB}}} MMB \xrightarrow{mB} MB \end{gathered} \tag{85}$$

To try to make sense of Eq. 85, consider the following:

Recall for natural transformation η, the component η_X: $\mu_X : T^2(X) \to T(X)$ and for some morphism $f \in \text{Mor}\mathbf{C}$,

$$\begin{gathered} f : X \to Z \\ Tf : T(X) \to T(Z) \end{gathered}$$

So if $f : X \to T(Y)$ instead,

$$\begin{gathered} Tf : T(X) \to T^2(Y) \\ (\mu_Y \circ Tf) : T(X) \to T(Y) \end{gathered}$$

Form a Cartesian product of $T(X)$ and $\text{Hom}_{\mathbf{C}}(X, T(Y)) \ni f$ as inputs: $T(X) \times \text{Hom}_{\mathbf{C}}(X, T(Y))$.

Then I suggest that Eq. 85 is translated as follows:

$$T(X) \times \text{Hom}_{\mathbf{C}}(X, T(Y)) \xrightarrow{\text{apply } T} T(T(X) \times \text{Hom}_{\mathbf{C}}(X, T(Y))) \xrightarrow{Tf} T^2(Y) \xrightarrow{\mu_Y} T(Y) \tag{86}$$

where apply T is on $f \in \text{Hom}_{\mathbf{C}}(X, T(Y))$ in order to form Tf.

24.3.1 More on Klesli composition

cf. Kleisli category, nLab.

Proposition 28. *Klesli composition of $g \circ f$ of $f : X \to T(Y)$ is given by*

$$g : Y \to T(Z)$$

$$X \xrightarrow{f} TY \xrightarrow{Tg} T^2 Z \xrightarrow{\mu_Z} TZ$$

i.e.

$$\boxed{g \circ_{Klesli} f = \mu_Z \circ Tg \circ f} \tag{87}$$

with $\mu_Z \circ Tg \circ f : X \to T(Z)$

Proof. Fullness of functor T:

$\forall$ morphism $g : T(X) \to T(Y)$ has antecedent the composition

$$X \xrightarrow{\eta_X} T(X) \xrightarrow{g} T(Y)$$

i.e. $g \circ \eta_X : X \to T(Y)$.

Consider $T(g \circ \eta_X) = Tg \circ T\eta_X : T(X) \to T^2(Y)$ for

$$\begin{gathered} T\eta_X : T(X) \to T^2(X) \\ Tg : T^2(X) \to T^2(Y) \\ \mu_Y : T^2(Y) \to T(Y) \end{gathered}$$

So

$$\mu_Y \circ Tg \circ T\eta_X = \mu_Y \circ T(g \circ \eta_X) : T(X) \to T(Y)$$

Because g is a morphism of algebras (TODO: understand what this means, from Kleisli category, nLab),

$$\mu_Y \circ Tg \circ T\eta_X = g \circ \mu_X \circ T\eta_X = g$$

where $\mu_Y \circ Tg = g \circ \mu_X$ uses natural transformation definition 101 for the property $\mu_Z \circ T^2 f = Tf \circ \mu_X$ (TODO: Show $g = Tf$ is of this form),
and $\mu \circ T\eta = \mu \circ \eta T = 1_T$ (coherence condition from Prop. 26).

Faithfulness:

If $\mu_Y \circ Tf = \mu_Y \circ Tg$, then precompose by η_X yields

$$\mu_Y \circ Tf \circ \eta_X = \mu_Y \circ \eta_{f(X)} \circ f = \mu_Y \circ \eta_{T(Y)} \circ f = f$$

where
$Tf \circ \eta_X = \eta_{T(Y)} \circ f$ from natural transformation def. 101 was used, and $\mu \circ \eta T = 1_T$ (coherence condition) was also used, for $f : X \to T(Y)$.

Similarly for g. $\mu_Y \circ Tf = \mu_Y \circ Tg \implies f = g$. □

Notice that the multiplication natural transformation $\mu : T^2 \to T$ was needed to induce the Klesli composition.

24.3.2 Dictionary between Monads in Computer Science and Monads in Category Theory

`join` is $\mu_X : T^2(X) \to T(X)$

`return` is $\eta_X : X \to T(X)$

`fmap` is the morphism map (arrow map) for functor F

`bind` is $\mu_Y \circ Tf : T(X) \to T(Y)$ but ...

`bind` is more nuanced in that, from the Haskell and functional programming perspective, it takes in *two* inputs.

One of them is an element x in the object X, $x \in X$. The other is a "transformation of type $X \to T(Y)$." I believe that would mean a morphism $f : X \to T(Y)$.

What Čukić in Čukić (2018) [33] calls `transform` is the *morphism map* (arrow map) for a functor F. However, note that when it comes to concrete implementation, the concrete implementation applies to *instances* of a type $X \in \text{Obj}\mathbf{C}$. For a given type X, e.g. `int` (the integer type in C or C++), there are many values that will have that type. It would be good that we work out how the definitions, theorems, and commutative diagrams apply in this case.

I don't think there's a great way to reconcile what Haskell and functional programming programmers mean by `bind` with the corresponding concept in category theory.

24.4 Examples of Monads in Computer Science

24.4.1 Reader Monad; Function Monad; Environment Monad

$[W, -] : \mathbf{C} \to \mathbf{C}$.

Given $W \in \text{Obj}\mathbf{C}$, ($W$ is a type),

$$[W, -] = (W \to (-))$$

where $[-,-] : \mathbf{C}^{\text{op}} \times \mathbf{C} \to \mathbf{C}$.

$W \mapsto$ internal hom out of W.

$\eta_X : X \to [W, -](X)$
$\eta_W : W \to [W, -](W)$
$\eta_W : w \mapsto [W, -](W)(w)$ value to constant functions with that value
$\Delta : X \in \text{Obj}\mathbf{C} \xrightarrow{(1,1)} X \times X$
$\Delta : x \mapsto (w, w)$

"read in " state of type W (an environment)
wikipedia calls this the Environment Nomad.

`reader`: $T \to E \to T = t \mapsto e \mapsto t$ corresponds to $\eta_X : X \to T(X)$ where $T(X) = \text{Hom}_{\mathbf{C}}(E, X)$ s.t. η_X sends values $x \in X$ to constant functions (on E) to value x.

Therefore, rewrite the unit component at X, i.e. `reader`, as

$$\begin{array}{ll} X \to E \to X & \eta_X : X \to \text{Hom}_{\mathbf{C}}(E, X) \\ x \mapsto e \mapsto x & \eta_X : x \mapsto (e \mapsto x) \end{array} \tag{88}$$

Thus, the *unit* component at X, called `unit` or `return` by programmers, truly maps a value $x \in X$ to a *constant function*:

$$\eta_X : x \mapsto (e \mapsto x) \in \text{Hom}_{\mathbf{C}}(E, X) \tag{89}$$

`bind`: $(E \to T) \to (T \to E \to T') \to E \to T' = r \mapsto f \mapsto e \mapsto f(re)e$ corresponds to $\mu_Y \circ Tf : T(X) \to T(Y)$ where $T(X) = \text{Hom}_{\mathbf{C}}(E, X)$, $T(Y) = \text{Hom}_{\mathbf{C}}(E, Y)$, and
$f : X \to T(Y)$ or $f : X \to \text{Hom}_{\mathbf{C}}(E, Y)$
$Tf : T(X) \to T^2(Y)$ or $Tf : \text{Hom}_{\mathbf{C}}(E, X) \to \text{Hom}_{\mathbf{C}}(E, T(Y)) = \text{Hom}_{\mathbf{C}}(E, \text{Hom}_{\mathbf{C}}(E, Y))$.

So let's rewrite `bind` in our notation:

$$\begin{array}{llll} (E \to X) \to & (X \to E \to Y) \to E \to & & Y \\ r \mapsto & & f \mapsto e \mapsto (f(r))(e)(e) & \end{array}$$

To clarify these definitions, rewrite the reader monad: the endomorphism T is defined as

$$\boxed{\begin{array}{l} T := [E, -] \\ T(X) = [E, X] \end{array}} \tag{90}$$

For any morphism $f : X \to T(Y) = [E, Y]$, then
recall the Klesli composition: $g \circ_{\text{Klesli}} f = \mu_Z \circ Tg \circ f$ with $g : Y \to T(Z) = [E, Z]$.
But if you take a look at how wikipedia defines "bind" for the reader monad, then bind *is not* the Klesli composition, but rather it's $\mu_Y \circ Tf$ (!!!):

$$\begin{array}{llll} & \mu_Y \circ Tf : T(X) \to T(Y) \text{ or } \mu_Y \circ Tf : [E, X] \to [E, Y] & & \\ r \in [E, X] & r(e) \in X \text{ for } e \in E & f(r(e)) \in T(Y) = [E, Y] & f(r(e))(e) \in Y \end{array} \tag{91}$$

bind, as defined by wikipedia for the reader monad *is not* the Klesli composition.
As an example, consider the following example:
$T(Y) \ni (E \to Y) \in \text{Hom}_{\mathbf{C}}(E, Y)$
$e =$ "dag" $\mapsto y =$ "Hello Dag!"

$T(X) \ni (E \to X) \in \text{Hom}_{\mathbf{C}}(E, X)$
$e \equiv$ `name` $\mapsto x =$ "Hi `name`!"

$f : X \to T(Y) \to f : X \to \text{Hom}(E, Y)$
$f(x) =$ replace "Hi" with "Hello" in X.

`ask`: $E \to E = 1_E$.
`ask` operation used to retrieve current context.

`local`: $(E \to E) \to (E \to T) \to E \to T = f \mapsto c \mapsto e \mapsto c(fe)$
local executes a computation in a modified subcontext. $(E \to E)$ modifies the environment, $(E \to T)$ uses the modified environment.
Rewrite `local` in our notation:

$$\begin{array}{llll} (E \to E) \to & (E \to X) \to & E \to & X \\ f \mapsto & c \mapsto & e \mapsto & c(f(e)) \end{array}$$

To make sense of this, let $f \in [E, E]$, $c \in [E, X] = T(X)$, $c \circ f \in [E, X]$ so $(c \circ f)(e) \in X$.

24.4.2 Writer Monad

$W \times (-) : \mathbf{C} \to \mathbf{C}$ where $W \in \mathrm{Obj}\mathbf{C}$. If W equipped with the structure of monoid, $W \times (-)$ canonically inherits the structure of a *monad*. Hence, the Writer *Monad*.

Recall W monoid if W is a set equipped with binary operation $\mu : M \times M \to M$ s.t. $(xy)z = x(yz)$ and $1 \in M$ s.t. $1 \cdot x = x = x \cdot 1$.

24.4.3 State Monad

cf. state monad, nLab

Input of type X, output of type Y, mutable state type W.
Consider morphism

$$X \times W \to Y \times W$$

Under (Cartesian product $\vdash$ internal hom) adjunction, this is equivalently given by its adjunt, which is a function of type

$$X \to [W, W \times Y]$$

$[W, W \times (-)]$ is the monad, induced by the above adjunction.

$X \to [W, W \times Y]$ regarded as a morphism.
$[W, W \times Y] : \mathbf{H} \to \mathbf{H}$ is called *state monad* for mutable states of type W.
TODO: Understand the above.
Writer comonad $W \times (-) : \mathbf{C} \to \mathbf{C}$,

Given by forming Cartesian product with W.

Reader Monad $\circ$ Writer Comonad is state monad.
cf. wikipedia
unit or "return":

$$\begin{array}{ll} X \to & S \to X \times S \\ x \mapsto & (s \mapsto (x, s)) \end{array} \tag{92}$$

Compare this to

$$\eta_X : X \to T(X) = [S, S \times X]$$
$$\eta_X : x \mapsto (S \to S \times X)$$

"bind" was given by wikipedia as

$$\begin{array}{llll} (S \to X \times S) \to & (X \to S \to Y \times S) \to & S \to & Y \times S \\ m \mapsto & & k \mapsto s \mapsto & k(x)(s) \end{array} \tag{93}$$

where $(x, t) = m(s)$

24.4.4 Properties of the State Monad

For the unit or "return", $\eta_X : X \to T(X) = [S, S \times X]$,

$$\begin{aligned} &\eta_X : x \mapsto (S \to S \times X) \text{ s.t.} \\ &(S \to S \times X) : s \mapsto (s, x) \end{aligned} \tag{94}$$

Therefore, for the unit component at X of a state monad,

$$\boxed{\eta_X(x)(s) = (s, x) \text{ for } s \in S} \tag{95}$$

Compare this to what wikipedia says how it defines the "`return`" operation in Eq. 92.

Consider any morphism $f : X \to T(Y) = [S, S \times Y]$,

$$\begin{aligned} &f : X \to T(Y) = [S, S \times Y] \\ &f : x \mapsto (S \to S \times Y) \text{ s.t. } (S \to S \times Y) : s \mapsto (t = f_s(s, x), y = f_x(s, x)) \end{aligned} \tag{96}$$

Allow morphism f to have "freedom" to set state to arbitrary t.

$$\begin{aligned} &Tf : T(X) \to T^2(Y) \to Tf : [S, S \times X] \to T([S, S \times Y]) = [S, S \times [S, S \times Y]] \\ &Tf : (S \to S \times X) \mapsto (S \to S \times (S \to S \times Y)) \end{aligned} \tag{97}$$

Then

$$\begin{aligned} &\mu_Y \circ Tf : T(X) \to T(Y) \text{ or } \mu_Y \circ Tf : [S, S \times X] \to [S, S \times Y] \\ &\mu_Y \circ Tf : (S \to S \times X) \mapsto (S \to S \times Y) \end{aligned} \tag{98}$$

Note that

$$\begin{aligned} &\mu_X : T^2(X) \to T(X) \\ &\mu_X : (S \to S \times (S \to S \times X)) \mapsto (S \to S \times X) \end{aligned} \tag{99}$$

$$\begin{aligned} &T^2 f : T^2(X) \to T^3(Y) \\ &T^2 f : (S \to S \times (S \to S \times X)) \mapsto (S \to S \times (S \to S \times (S \to S \times Y))) \end{aligned} \tag{100}$$

Now

$$\begin{aligned} \mu_X \circ T^2 f &: (S \to (S \times (S \to S \times X))) \mapsto (S \to S \times (S \to S \times Y)) \\ Tf \circ \mu_X &: (S \to S \times (S \to S \times X)) \mapsto (S \to S \times (S \to S \times Y)) \end{aligned}$$

and so

$$\implies \mu_X T^2 = Tf \mu_X$$

Thus, μ, as defined above, *is* a natural transformation.

Show $\eta_{T(Y)} \circ f = Tf \circ \eta_X$.

$$\begin{aligned} Tf \circ \eta_X(x) &= (S \to S \times (S \to S \times Y)) \\ \eta_{T(Y)} \circ f(x) &= \eta_{T(Y)}(S \to S \times Y) = (S \to S \times (S \to S \times Y)) \end{aligned}$$

Prove now the coherence conditions, Prop. 26. For the first one,

$$\mu \circ T\mu = \mu \circ \mu T$$

$$\begin{aligned} \mu \circ T\mu(S \to (S \to S \times (S \to S \times X))) &= \mu \circ T(S \times (S \to S \times X)) = \mu(S \to S \times (S \times (S \to S \times Y))) = \\ &= (S \times (S \to S \times Y)) \\ \mu \circ \mu T(S \to (S \to S \times (S \to S \times X))) &= \mu^2(S \to S \times (S \to (S \to S \times (S \to S \times Y)))) = \\ &= S \times (S \to S \times Y) \end{aligned}$$

$$\implies \mu \circ T\mu = \mu \circ \mu T$$

So the first coherence condition is proven.

For the second,

$$\mu \circ T\eta(S \to S \times X) = \mu \circ T(S \to S \times (S \to S \times X)) = (S \to S \times (S \to S \times X))$$
$$\mu\eta T(S \to S \times X) = \mu \circ (S \to S \times T(S \to S \times X)) = T(S \to S \times X) = (S \to S \times (S \to S \times X))$$

Thus, it's been shown that the choice of endomorphism for the state monad, $X \to [S, S \times X]$, fulfills the monad coherence conditions being of this form.

Consider what "bind" means in category theory. Given

$$\begin{aligned} &f : X \to T(Y) = [S, S \times Y] \\ &g : Y \to T(Z) = [S, S \times Z] \\ &f(x) = (S \to S \times Y) \\ &f(x)(s) = (t, y) \text{ where } s, t \in W \\ &g(y) = (S \to S \times Z) \\ &g(y)(t) = (u, z) \end{aligned} \tag{101}$$

Therefore, if we define the usual projection operations on a Cartesian product,

$$\begin{aligned} &\Pi_W : W \times X \to W \qquad \Pi_X : W \times X \to X \\ &\Pi_W(w, x) = w \qquad\qquad \Pi_X(w, x) = x \end{aligned} \tag{102}$$

$$\begin{aligned} &\mu_Z \circ Tg \circ f : X \to T(Z) = [S, S \times Z] \ni (W \to W \times Z) \\ &\mu_Z \circ Tg \circ f : x \mapsto (S \to S \times Z) \equiv \mu_Z \circ Tg \circ f(x) \end{aligned} \tag{103}$$

such that

$$\boxed{g(y)(t) = (u, z) \text{ where } y = \Pi_Y(f(x)(s)),\ t = \Pi_W(f(x)(s))} \tag{104}$$

Compare this to what wikipedia says in Eq. 93.

24.4.5 Continuation Monad

From nLab,
In a category $\mathbf{C}$ with internal homs $[-,-]$, so internal homs $[-,-]$ belong to Obj$\mathbf{C}$ (they're objects as well), given object $Y \in$ Obj$\mathbf{C}$, the *continuation monad* is endofunctor

$$\boxed{X \mapsto [[X, Y], Y]} \text{ i.e. } T(X) = [[X, Y], Y] \text{ for object } Y \in \text{Obj}\mathbf{C} \tag{105}$$

so that $T = X \mapsto [[X, Y], Y]$

From wikipedia,
unit or return:

return type Y, maps type X into functions of type $(X \to Y) \to Y$

$$\begin{aligned} &X \to \quad (X \to Y) \to Y \\ &x \mapsto \qquad\quad f \mapsto f(x) = y \end{aligned} \tag{106}$$

Compare this to

$$\begin{aligned} &T : X \to [[X, Y], Y] \\ &\eta_X : X \to T(X) = [[X, Y], Y] \end{aligned} \tag{107}$$

And so for the *unit* of the Continuation monad,

$$\boxed{\begin{aligned} &\eta_X : X \to T(X) = [[X, Y], Y] \\ &\eta_X : x \mapsto (f \to y) \\ &\eta_X(x)(f) = y = f(x) \end{aligned}} \tag{108}$$

bind:

$$\begin{array}{lll} ((X \to Y) \to Y) \to & (X \to (X' \to Y) \to Y) \to (X' \to Y) \to & Y \\ c \mapsto & f \mapsto k \mapsto & c(t \mapsto ftk) \end{array} \tag{109}$$

From nLab, if morphism $f : X \to Y$ is in Klesi category of the continuation monad, then it's a morphism in $\mathbf{C}$ of form $X \to T(Y)$ in $\mathbf{C}$. Hence morphism in original category *of the form* $X \to [[Y, S], S]$ is much like a map from $X \to Y$, only instead of "returning" output Y directly, it instead feeds it into given function $Y \to S$, which hence **continues** the computation.

So for $X \to [[Y, S], S]$, nLab gives the corresponding *computation rule* for function types of the internal hom $[X, S]$:

$$(y \mapsto a(y))(x) = a(x)$$

Hence for $X \to [[X, Y], Y]$, we have $x \mapsto f \mapsto f(x)$ because $f \in [X, Y]$ because f is treated as an object in $\mathbf{C}$.

So do the following "rewrite" or change of notation on Eq. 109:

$$\begin{array}{l} ((X \to Z) \to Z) \to (X \to (Y \to Z) \to Z) \to (Y \to Z) \to Z \\ T(X) \to (X \to T(Y)) \equiv \text{Hom}(X, T(Y)) \to \text{Hom}(Y, Z) \to Z \end{array} \tag{110}$$

Let $c \in T(X) = [[X, Z], Z]$.
Let $k \in [Y, Z]$
Let $f(x) \in [[Y, Z], Z]$ and so
$f(x)(k) \in Z$

Form $x \mapsto f(x)(k) \in [X, Z]$, so $c(x \mapsto f(x)(k)) \in Z$.

So "bind" as defined in wikipedia, Haskell/Functional Programming isn't the Klesli composition, but it's

$$\mu_Y \circ Tf : T(X) \to T(Y) = [[Y, Z], Z]$$

Write this as

$$\boxed{\begin{array}{l} (\mu_Y \circ T)(f)(c) \in T(Y) = [[Y, Z], Z] \\ (\mu_Y \circ T)(f)(c)(k) \in Z \end{array}} \tag{111}$$

call with current continuation function defined as follows:
call/cc

$$\begin{array}{lll} ((X \to (X' \to Y) \to Y) \to (X \to Y) \to Y) \to (X \to Y) & & \to Y \\ & f \mapsto k & \mapsto (f(t \mapsto x \mapsto kt)k) \end{array} \tag{112}$$

Let's look at Eq. 112, step-by-step.

$$\begin{array}{l} (X \to (X' \to Y) \to Y) \\ f \end{array} \text{ corresponds to}$$

$$\begin{array}{l} X \to [[X', Y], Y] = T(X') \\ f \equiv f_{XX'} : X \to T(X') \end{array}$$

24.4.6 `future` as Continuation Monad

Čukić (2018) [33] asks first if `std::future` in C++ is a functor. Recall the definition of a functor, Def. 86:

$$\begin{array}{l} F : \mathbf{C} \to \mathbf{D} \text{ with "object" map} \\ F : \text{Obj}\mathbf{C} \to \text{Obj}\mathbf{D} \\ F : X \to F(X) \end{array}$$

and the functor's "morphism map":

$$\begin{array}{l} Ff : F(X) \to F(Y) \\ f : X \to Y,\ f \in \text{Hom}_{\mathbf{C}}(X, Y) \end{array}$$

Suppose $X = \texttt{T1}$, $Y = \texttt{T2}$. Then in the example on pp. 220, Sec. 10.7.1 Futures as monads" example of Čukić (2018) [33],

$$F(X) = \texttt{future<T1>}$$
$$F(Y) = \texttt{future<T2>}$$

Instead, the morphisms to consider is of this form:

$$f : X \to T(Y)$$

where T is a functor but a special one: the endomorphism. And T is `std::future`.

Here are examples of morphisms f, g:

```
future<std::string> user_full_name(const std::string& login);
future<std::string> to_html(const std::string& text);
```

Now $Tf : T(X) \to T^2(Y)$, how the morphism map for T applies to morphism f, now. Recall that we can compose this with the multiplication natural transformation component at X, μ_X:

$$\mu_X \circ Tf : T(X) \to T(Y)$$
$$\mu_X : T^2(X) \to T(X)$$

In order to compose morphisms, say f, g, recall the "Klesli composition", Prop. 28

$$g \circ_{\text{Klesili}} f = \mu_Z \circ Tg \circ f \tag{113}$$

If we call $\mu_Z \circ T$ "bind", $\mu_Z \circ T : \text{Hom}(Y, T(Z)) \to \text{Hom}(T(Y), T(Z))$. The correspondence is such:

$$(\texttt{bind(to_html)})(\texttt{user_full_name}) \iff \mu_Z \circ Tg \circ f : X \to T(Z) \tag{114}$$

24.4.7 Resources for implementations of Monads

Python and C++ code.

https://github.com/Iasi-C-CPP-Developers-Meetup/presentations-code-samples, namely https://github.com/Iasi-C-CPP-Developers-Meetup/presentations-code-samples/tree/master/radugeorge

https://github.com/dbrattli/OSlash, namely https://github.com/dbrattli/OSlash/tree/master/tests

State Monad implementation in C++

25 Applications of Category Theory on Hybrid Systems

cf. Ames (2006) [34].

25.1 D-Categories

D stands for discrete.

Recall that a small category $\mathbf{C}$ is called *small* if both $\text{Obj}(\mathbf{C})$ and $\text{hom}(\mathbf{C})$ are sets, not proper classes.

Definition 104 (Axiomatic D-categories)**.** Let D-category be a *small* category $\mathbf{D}$ s.t.

1. $\forall\, D \in \text{Obj}(\mathbf{D})$,
   $\exists$ morphism $f \in \text{Mor}(\mathbf{D})$ s.t. $f \neq 1$ s.t.
   $f \in \text{Hom}(D, *)$ or $f \in \text{Hom}(*, D)$, but never both,
   i.e. $\forall$ diagram $a_0 \xrightarrow{\alpha_1} a_1 \xrightarrow{\alpha_2} \ldots \xrightarrow{\alpha_n} a_n$ in $\mathbf{D}$, all but 1 morphism must be identity (i.e. longest chain of composite non-identity morphisms is of length 1).

2. If for $D \in \text{Obj}(\mathbf{D})$, D is the domain of a non-identity morphism, i.e. $\exists\, f_1 \in \text{Mor}(\mathbf{D})$ s.t.

$$f_1 \in \text{Hom}(D, *),\ f_1 \neq 1$$

then $\exists\, f_2 \in \text{Hom}(D, *)$, $f_2 \neq f_1$, $f_2 \neq 1$ and $\forall\, f \in \text{Hom}(D, *)$ s.t. $f \neq f_1, f_2, \quad f = 1$

cf. 1.2.1 Important objects in D-categories, Ames (2006) [34].
Let

$$\text{Mor}\mathbf{A} = \bigcup_{A,B \in \text{Obj}(\mathbf{A})} \text{Hom}(A, B) \qquad \text{(my notation)}$$

$$\text{Mor}(\mathcal{D}) = \bigcup_{(a,b) \in \text{Obj}(\mathcal{D}) \times \text{Obj}(\mathcal{D})} \text{Hom}_{\mathcal{D}}(a, b) \qquad \text{(Ames' notation)}$$

Let

$$\text{Mor}_1\mathbf{A} := \{A \in \text{Mor}(\mathbf{A}) | A \neq 1\} \tag{115}$$

For a D-category, consider these subset of $\text{Obj}(\mathbf{D})$,

Definition 105 (Edge set)**.** **edge set** of $\mathbf{D}$, $E(\mathbf{D})$,

$$\begin{gathered} E(\mathbf{D}) := \{A \in \text{Obj}(\mathbf{D}) | \alpha \in \text{Hom}(A, *), \beta \in \text{Hom}(A, *), \alpha, \beta \in \text{Mor}_1(\mathbf{D}),\ \alpha \neq \beta\} \text{ i.e.} \\ E(\mathbf{D}) := \{A \in \text{Obj}(\mathbf{D}) | \alpha, \beta \in \text{Hom}(A, *),\ \alpha, \beta \neq 1,\ \alpha \neq \beta\} \end{gathered} \tag{116}$$

i.e. $\forall\, A \in E(\mathbf{D})$, $\exists\, \alpha, \beta \in \text{Mor}(\mathbf{D}), \quad \alpha, \beta \neq 1$, s.t. $\alpha, \beta \in \text{Hom}(D, *)$; denote these morphisms by s_a, t_a (this specific choice will define an **orientation**).

Conversely, given morphism $\gamma \in \text{Mor}(\mathbf{D})$, $\gamma \neq 1$, $\exists\,!\, A \in E(\mathbf{D})$ s.t. $\gamma = s_a$ or $\gamma = t_a$, i.e. $\gamma \in \text{Hom}(A, *)$.

Definition 106 (Vertex set)**.** **vertex set** of $\mathbf{D}$:

$$V(\mathbf{D}) = (E(\mathbf{D}))^c \tag{117}$$

Definition 107 (Orientation)**.** Orientation of D-category $\mathbf{D}$ is a pair of functions (s, t) between sets.

Part IX
Category Theory and Databases

26 Types

26.1 Data models for nested arrays, dictionaries, and tabular data

Definition 108 (Container types)**. Container types** - arrays, key/value pair dictionaries (or: hashes, association lists)

Definition 109 (atomic types)**.** basic atomic types (e.g. numbers, strings, Booleans)

Definition 110 (nesting)**. nesting**: containers may contain atomic values as well as other containers.

Definition 111 (flat)**.** Tabular data model (i.e. tables) is **flat**: field contains **atomic** values.

cf. Number 2, Data Models and Languages, Grust and Duta (2017) [37]

26.2 Typed Data, Untyped data, in the relational data model

Definition 112. Untyped data models - text, JSON, and tabular data models (e.g. CSV) do **not** enforce values (container or atomic) to be of specific types.

These data models are thus referred to as being **untyped**.

cf. Grust and Duta (2017) [37]
cf. Number 4, "The Relational Data Model", Grust and Duta (2017) [37]

Definition 113 (Types)**.** Let $\mathbb{T} \equiv$ set of all data types (built-in and user-defined).
$\forall$ value $v \in \mathbb{V}$ stored in a relation cell must be of type $t \in \mathbb{T}$.
e.g. When PostgreSQL starts, $\mathbb{T}$ initialized as

$$\mathbb{T} = \{\texttt{boolean}, \texttt{integer}, \texttt{text}, \texttt{bytea}, \dots\} \tag{118}$$

Consider category **Text** s.t. Obj(**Types**) $\equiv DT \in$ Obj(**Set**). Then $\mathbb{T} \equiv DT$, denoting *data type*.

Definition 114 (Values)**.** $\forall\, v \in \mathbb{V}$ stored in a relation cell, v is an element of the set of all values $\mathbb{V}$.
in the relational data model, all values $v \in \mathbb{V}$ are "atomic."

$$\mathbb{V} = \{\texttt{true}, \texttt{false}, 0, -1, 1, -2, 2, \dots\}$$

Here, I'll use the notation V to denote $\mathbb{V}$, the set of all values.

Definition 115 (Domains)**.** $\forall\, t \in DT$, its domain $\mathrm{dom}(t) :=$ set of all values of type t (i.e. $\mathrm{dom}(\cdot)$ is a function with signature

$$DT \to 2^V$$

) e.g. $\mathrm{dom}(\texttt{integer}) = \{0, -1, 1, -2, 2, \dots\}$
$\mathrm{dom}(\texttt{boolean}) = \{\texttt{true}, \texttt{false}\}$

Definition 116 (type specification)**.** type specification := function $\pi : U \to$ **DT** (Spivak's notation) $\equiv \pi : U \to DT$, $U, DT \in$ **Set**.
set $DT \equiv$ set of data types for π, set $U \equiv$ domain bundle for π.
$\forall\, t \in DT$, preimage $\pi^{-1}(t) \subset U$, $\pi^{-1}(t) \equiv$ domain of t, $x \in \pi^{-1}(t) \equiv$ object of type T.
To reconcile Grust and Duta (2017) [37]'s definition of types above, use this notation:

$$\boxed{\begin{array}{c} \textbf{type specification } \pi : V \to DT \\ \pi(v) \in DT \\ \mathrm{dom} \equiv \pi^{-1} : DT \to V, \text{ i.e.} \\ \pi^{-1}(t) \subset t \qquad \forall\, t \in DT \end{array}} \tag{119}$$

cf. Spivak (2009) [38]

Corollary 4 (type specification)**.** *If v has type T, $\pi(v) = T$, $\implies v \in \pi^{-1}(T)$*

Proposition 29 (`CREATE DOMAIN`)**.** *Consider new type t' (SQL command `CREATE DOMAIN`) so*

$$\begin{array}{c} t' \in DT \\ \pi^{-1}(t') \subseteq \pi^{-1}(t) \end{array} \tag{120}$$

27 Relational Data Model

relational data model maybe understood as a typed variant of the tabular data model.

1. $\exists$ only 1 container type: **table** (or: multisets) **of rows**
2. all rows are of same **row type** which is declared when table is created.
3. row type **consists** of sequence of **atomic types**.

In the relational data mode, data is exclusively organized in **relations**, i.e. sets of tuples of data. Data in each **attribute** (tuple component) is **atomic**, and of declared **type**.

27.1 Schemata and Relations

In the relational data model, each **attribute** of a table has a **declared type**.

If attribute has declared type t, the RDBMS will exclusively store values v in that attribute s.t.

1. $v \in \text{dom}(t)$ i.e. $v \in \pi^{-1}(t)$
2. v can successfully be casted to type t

27.2 Attributes (Columns)

Let $\mathbb{A}$ denote set of attribute names of all relations.

27.3 Attribute types

$\forall$ attribute $a \in \mathbb{A}$ has declared (attribute) type $\text{type}(a) = t \in DT$ (i.e. $\text{type}(\cdot)$ is a function with signature $\mathbb{A} \to \mathbb{T}$).

Consider Definition 2.2.3 of Spivak (2009) [38],

Definition 117 (simple schema of type π, (C, σ))**.** Let type specification $\pi : V \to DT$.

simple schema of type π consists of pair (C, σ) where C is a finite (totally) ordered set and function $\sigma : C \to DT$. $C \equiv$ **column set** or **set of attributes** for σ and π as *type specification for* σ.

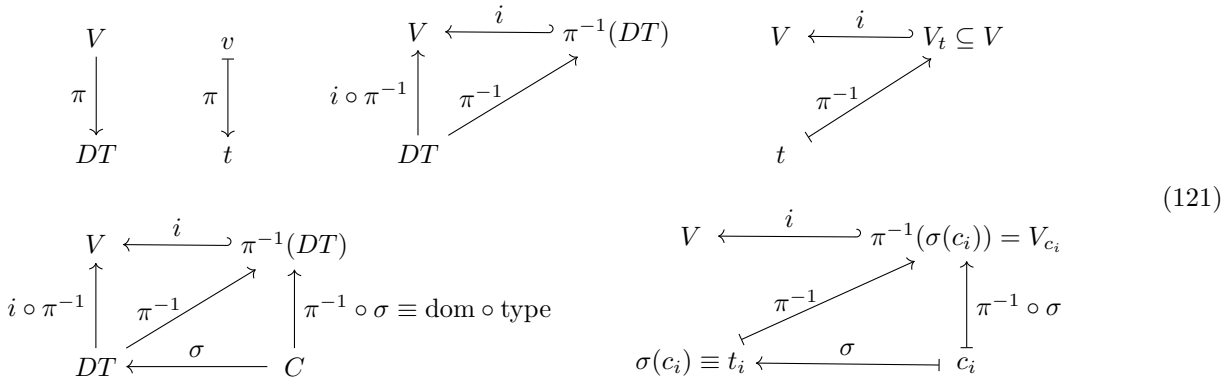
Compare the notation above. Conclude that

$$C \equiv \mathbb{A}, \quad c \equiv a$$
$$\sigma(c) \equiv \text{type}(a)$$

Here, choose the following notation:

$$\boxed{\begin{array}{c} \sigma : C \to DT \\ \pi^{-1} : DT \to V \\ \pi^{-1} \circ \sigma \equiv \text{val} \equiv \text{set of (admissable) attribute values for attribute (column) } a \ (c) \end{array}}$$

Also, summarize our definitions with these commutative diagrams:



(121)

Definition 118 (Relation Schema)**.** A relation schema associates a relation (table?) name R with its set of declared attributes (a subset of $\mathbb{A}$)

$$(R, \{a_1, \dots a_n\}) \tag{122}$$

Common notation: $R(a_1, \dots a_n)$, so that R is called a n-ary relation.

More notation: $\text{sch}(R) = \{a_1, \dots a_n\}$, and $\deg(R) = n$ (degree).

Relational database schema: a non-empty finite set of relation schemata makes a **relational database schema**

$$\{(R_1, \alpha_1), (R_2, \alpha_2), \dots\} \tag{123}$$

where $\alpha_i \subseteq \mathbb{A}_i$. In a relational database schema, the relation names R_i are unique.

Definition 119 (Tuple)**.** Given relation (i.e. table) $R(a_1, \ldots, a_n)$, a **tuple** t of R maps attributes to values, i.e. t is a function with signature $\{a_1, \ldots a_n\} \to V$ with

$$\forall\, a \in \{a_1, \ldots, a_n\} : t(a) \in \text{val}(a)$$

Common notation for $t(a)$ is $t.a$.

Recall that $\text{val}(a) := \text{dom}(\text{type}(a))$.

Take note that **tuple**, defined by Grust and Duta (2017) [37], is the *same* as **record** or **row** r, defined by Spivak (2009) [38].

Definition 120 (record or row)**. record** or **row** on (C, σ)

$$\begin{aligned} &r : C \to V_\sigma \equiv V_c \\ &r(c) \equiv v_{rc} \in \pi^{-1} \circ \sigma(c) \end{aligned} \tag{124}$$

SQL `CREATE TABLE` command prescribes an order of the attributes of a relation. This deviates from relational data model's tuple model (name-to-value mapping).

Definition 121 (Row)**.** Given SQL table $R(a_1, \ldots, a_n) \equiv R(\alpha) \equiv \tau$, a row r of τ is an ordered sequence (a_i is called the ith column)

$$r = (v_1, \ldots, v_n) \in \text{val}(a_1) \times \cdots \times \text{val}(a_n) \equiv \pi^{-1} \circ \sigma(a_1) \times \cdots \times \pi^{-1} \circ \sigma(a_n) \tag{125}$$

Thus, r is a function $\{1, \ldots, n\} \to V$ with $\forall\, i \in \{1, \ldots, n\}$, $r(i) \in \pi^{-1} \circ \sigma(a_i)$

The set of tuples (rows) stored in a relation (table) is expected to change frequently.

Definition 122 (Relation instance (state))**.** The current finite set of tuples $t_i \equiv r_i$ of relation (table) $R(a_1, \ldots a_n) \equiv (A, \sigma)$ is called the relation's **instance** (or **state**).

$$\text{inst}(R) = \{t_1, t_2, \ldots t_m\} \Longrightarrow \equiv \Gamma^\pi(\sigma) \equiv \Gamma(A, \sigma)$$

Database (instance) state - The database instance comprises instances of all its relations.

$$= \{\Gamma(A^i, \sigma^i)\}_i$$

27.4 Constraints

cf. Number 5, "Constraints" of Grust and Duta (2017) [37].

Definition 123 (Constraints)**.** An **integrity constraints** specifies conditions which table states have to satisfy at all times.

Current set of constraints, $\mathbb{C}$, is integral part of database schema:

$$(\{(R_1, \alpha_1), (R_2, \alpha_2), \ldots\}, \mathbb{C})$$

Set of constraints $\mathbb{C} \equiv$ set of morphisms of table $\tau \equiv \{(R_1, \alpha_1), (R_2, \alpha_2), \ldots\}$, $C \subset \text{Mor}\tau$.

RDBMS will refuse table state changes that violate any constraint $c \in C$.

27.5 Key constraints

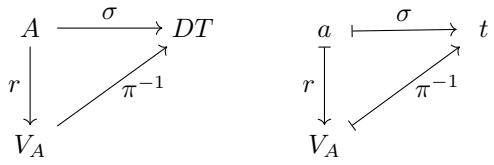
Definition 124 (Key)**.** Key of a table $R(a_1, \ldots, a_n) \equiv (\alpha, \sigma) \equiv (\alpha, \text{type}) :=$ set of columns $K \subseteq \{a_1, \ldots, a_n\}$ that unique identifies rows of R:

$$\forall\, t, u \in \text{inst}(R),\ t.K = u.K \Longrightarrow t = u$$

Read: "If 2 rows agree on the columns in K, they are indeed the same row."

I will give the following change of notation a try:

$$\forall\, r_i, r_j \in \{r_i\}_i^\tau,\ \text{ if } r_i(K) = r_j(K),\ \text{ then } r_i = r_j \tag{126}$$



(127)

Here's a "dictionary" between the definitions so far for relational databases and familiar terms for tables filled with data: cf. 08-30, Week 1 slides of Yang (2012) [40].

database - collection of relations (or tables)
relation - table
attributes - columns
tuple - row - record
relation schema - heading (heading for a table?, Grust and Duta (2017) [37])
relation contents - body (body of a table?, Grust and Duta (2017) [37])

CSV	Relational Model	SQL
	Domain	Domain
	Type	Type
	Schema	Schema
File	Relation	Table
Line	Tuple	Row
Field	Attribute	Column

Indeed, even Grust and Duta (2017) [37] remarks: "You will find that textbooks, papers, practitioners, academics, these slides, and even PostgreSQL use a mixture of terminology. Deal with it."

cf. Slides Number 7, "Referential Integrity" of Grust and Duta (2017) [37].

Definition 125 (Foreign Keys)**.** Let (S, α) and (T, β) denote 2 relational schemata (not necessarily distinct), where $K = \{b_{j_1}, \ldots, b_{j_k}\} \subseteq \beta$ is the primary key of T.
Let $F = \{a_{i_1}, \ldots, a_{i_k}\} \subseteq \alpha$ with $\text{type}(a_{i_h}) = \text{type}(b_{j_h})$, $h \in 1, \ldots, k$.

F is a **foreign key** in S **referencing** T, if

$$\forall\, s \in \text{inst}(S) : \exists\, t \in \text{inst}(T) : s.F = t.K$$

The $\forall$ and $\exists$ condition validates the assumption of K being a key in target T, i.e. that there **exists a row** in table T whose K identifier matches that of table S

K being a key in target T validates the assumption that there is **no more than one row** of in a table T with a matching key K.

In general, **a foreign key F is not a key** in source table S. 2 rows $s_1, s_2 \in \text{inst}(S)$ with $s_1.F = s_2.F$ can refer to the same row in target T.

Here is a great example of a foreign key: https://www.w3schools.com/sql/sql_foreignkey.asp

So from w3schools,

"A FOREIGN KEY is a key used to link 2 tables together.

A FOREIGN KEY is a field (or collection of fields) in 1 table that refers to the PRIMARY KEY in another table.

The table containing the foreign key is called the child table, and the table containing the candidate key is called the referenced or parent table.

"Persons" table:

PersonID	**LastName**	**FirstName**	**Age**
1	Hansen	Ola	30
2	Svendson	Tove	23
3	Pettersen	Kari	20

"Orders" table:

OrderID	**OrderNumber**	**PersonID**
1	77895	3
2	44678	3
3	22456	2
4	24562	1

Notice that the "PersonID" column in the "Orders" table points to the "PersonID" column in the "Persons" table.

The "PersonID" column in the "Persons" table is the PRIMARY KEY in the "Persons" table.

The "PersonId" column in the "Orders" table is a FOREIGN KEY in the "Orders" table.

Thus, in other words,
given 2 tables (i.e. 2 relational schemata, not necessarily distinct), (S, α), (T, β),
given primary key $PK = \{b_{j_1}, \dots, b_{j_k}\} \subseteq \beta$, primary key of of T, and
given foreign key $FK = \{a_{i_1}, \dots, a_{i_k}\} \subseteq \alpha$ s.t.

$$\text{type}(a_{i_h}) = \text{type}(b_{j_h}), \quad h \in 1, 2, \dots k$$
$$\text{(alternate notation)} \sigma(a_{i_h}) = \sigma(b_{j_h})$$

then
$\forall\, s \in \text{inst}(S)$, $\exists\, t \in \text{inst}(T)$ ((alternate notation) $s \in \Gamma(S)$, $t \in \Gamma(T)$), s.t.

$$s(FK) = t(PK)$$

Describe this categorically as

$$S \xrightarrow{FK} T$$

which is from the introductory talk by Spivak (2012)[39], "What does equivalence of paths mean?", where it states, "arrows represent foreign keys", "from table a to table b," and where "we can take any record in table a and return a record in table b.

27.6 Functional Dependency

Definition 126 (Functional Dependency (FD))**.** cf. Slides Number 10, "Functional Dependencies" of Grust and Duta (2017) [37].

Let $(R, \alpha) \equiv$ relational schema. Given $\beta \subseteq \alpha$, $c \in \alpha$,
functional dependency $\beta \to c$ holds in R if

$$\forall\, t, u \in \text{inst}(R),\ t.\beta = u.\beta \implies t.c = u.c$$

Notation: the functional dependency $\beta \to \{c_1, \dots, c_n\}$ abbreviates set of FDs $\beta \to c_1, \dots, \beta \to c_n$.

cf. "Functional dependency", wikipedia

A set of attributes $X \subseteq R$ ($\equiv X \subseteq \alpha$ for $R = (R, \alpha)$) is said to functionally determine another set of attributes $Y \subseteq R$ ($\equiv Y \subseteq \alpha$), written $X \to Y$. ($\equiv \beta \to c$)
iff $\forall\, X$ value in R is associated with precisely 1 Y value in R; R is then said to satisfy the functional dependency $X \to Y$

e.g. Cars: $\beta =$ vehicle identification number (VIN), $c =$ Engine capacity (because assume a car cannot have 2 engines).

e.g Employee department:
$\beta =$ employee ID, $c =$ employee name.
$\beta =$ employee ID, $c =$ department ID.
$\beta =$ department ID, $c =$ department name.

Recall definition of injective function, $f : X \to Y$,

$$\forall\, a, b \in X,\ f(a) = f(b) \implies a = b$$

Instead of writing the functional dependency (FD) with this notation: given $\beta, c \subseteq \alpha$,

$$\forall\, r_1, r_2 \in \text{inst}(R) \equiv \Gamma(\alpha),\ r_1(\beta) = r_2(\beta), \implies r_1(c) = r_2(c)$$

rewrite this as follows:

Given

$$\begin{gathered}\beta : \Gamma(\alpha) \to \text{val}(\beta) \equiv \text{dom}(\text{type}(\beta)) \equiv \pi^{-1} \circ \sigma(\beta), \\ c : \Gamma(\alpha) \to \text{val}(c) \equiv \text{dom}(\text{type}(c)) \equiv \pi^{-1} \circ \sigma(c), \\ \forall\, r_1, r_2 \in \text{inst}(R) \equiv \Gamma(\alpha),\ \beta(r_1), \beta(r_2) \in \text{val}(\beta) \\ \text{Then } \beta \in \text{Hom}(\text{inst}(R), \text{val}(\beta)) \equiv \text{Hom}(\Gamma(\alpha), \pi^{-1} \circ \sigma(\alpha))\end{gathered} \tag{128}$$

So define a **functional dependency** FD as

$$\begin{gathered}\text{FD} : 2^\alpha \to 2^\alpha \\ \text{FD} : \beta \mapsto c \text{ i.e. } \boxed{FD(\beta) = c}\end{gathered} \tag{129}$$

where $\beta, c \in 2^\alpha \equiv$ power set of α, i.e. set of all subsets of α, so that $\beta, c \subseteq \alpha$.

Let $FD^{-1} : 2^\alpha \to 2^\alpha$, s.t. $FD^{-1}(c) = \beta$.

So if $\forall\, r_1, r_2 \in \text{inst}(R) \equiv \Gamma(\alpha)$, for $\beta \subseteq \alpha$, $\beta(r_1), \beta(r_2) \in \text{val}(\beta)$. Then $\beta \in \text{Hom}(\text{inst}(R), \text{val}(\beta)) \equiv \text{Hom}(\Gamma(\alpha), \pi^{-1} \circ \sigma(\alpha))$.

Suppose $\beta(r_1) = \beta(r_2)$. The *key insight* is the following:

$$\beta(r_1) = \beta(r_2) = \text{FD}^{-1}(c)(r_1) = \text{FD}^{-1}(c)(r_2) = (\text{FD}^{-1} \circ c)(r_1) = (\text{FD}^{-1} \circ c)(r_2) \tag{130}$$

Then $c(r_1) = c(r_2)$ implies that FD^{-1} is *injective*.

Then FD^{-1} is a *monomorphism*, and FD^{-1} is injective, since $\forall\, r_1, r_2 \in \text{inst}(R) \equiv \Gamma(\alpha)$,

$$\beta(r_1) = \beta(r_2) = (\text{FD}^{-1} \circ c)(r_1) = (\text{FD}^{-1} \circ c)(r_2) \Longrightarrow c(r_1) = c(r_2) \tag{131}$$

$$\text{inst}(R) \xrightarrow{c} \text{val}(C) \overset{\text{FD}^{-1}}{\longrightarrow} \text{val}(\beta)$$

Then conclude that

Theorem 33. *Given a relational schema (R, α), a set of attributes α, instance of the relation (i.e. a table), $inst(R) \equiv \Gamma(\alpha)$, and $\beta, c \subseteq \alpha$,*
for morphisms β, c,

$$\begin{gathered}\beta \in Hom(\Gamma(\alpha), val(\beta)) \\ c \in Hom(\Gamma(\alpha), val(c))\end{gathered}$$

*Then for $FD : 2^\alpha \to 2^\alpha$, $FD(\beta) = c$, FD^{-1} is a **monomorphism**, i.e. $\forall\, r_1, r_2 \in \Gamma(\alpha)$,*

$$\beta(r_1) = \beta(r_2) = (FD^{-1} \circ c)(r_1) = (FD^{-1} \circ c)(r_2) \text{ i.e. } (FD^{-1} \circ c) = (FD^{-1} \circ c)$$

then $c(r_1) = c'(r_2)$ i.e. $c = c'$.

Database Queries and Constraints via Lifting Problems. David I. Spivak. https://arxiv.org/pdf/1202.2591.pdf

28 Databases and Categories

cf. Spivak (2012) [39].

$A \xrightarrow{\text{FK}} B, \quad A, B \in \text{Obj}(\mathbf{DB})$

Part X

Reading notes on Cox, Little, O'Shea's *Ideals, Varieties, and Algorithms: An Introduction to Computational Algebraic Geometry and Commutative Algebra*

29 Geometry, Algebra, and Algorithms

29.1 Polynomials and Affine Space

fields are important is that linear algebra works over *any* field

Definition 127 (2)**.** set of all polynomials in $x_1, \ldots, x_n$ with coefficients in k, denoted $k[x_1, \ldots, x_n]$

polynomial f *divides* polynomial g provided $g = fh$ for some $h \in k[x_1, \ldots, x_n]$

$k[x_1, \ldots, x_n]$ satisfies all field axioms except for existence of multiplicative inverses; commutative ring, $k[x_1, \ldots, x_n]$ *polynomial ring*

Exercises for 1

Exercise 1. $\mathbb{F}_2$ commutative ring since it's an abelian group under addition, commutative in multiplication, and multiplicative identity exists, namely 1. It is a field since for $1 \neq 0$, the multiplicative identity is 1.

Exercise 2.

(a)

(b)

(c)

29.2 Affine Varieties

29.3 Parametrizations of Affine Varieties

29.4 Ideals

29.5 Polynomials of One Variable

30 Groebner Bases

30.1 Introduction

30.2 Orderings on the Monomials in $k[x_1, \ldots, x_n]$

30.3 A Division Algorithm in $k[x_1, \ldots, x_n]$

30.4 Monomial Ideals and Dickson's Lemma

30.5 The Hilbert Basis Theorem and Groebner Bases

30.6 Properties of Groebner Bases

30.7 Buchberger's Algorithm

31 Elimination Theory

31.1 The Elimination and Extension Theorems

31.2 The Geometry of Elimination

32 The Algebra-Geometry Dictionary

32.1 Hilbert's Nullstellensatz

32.2 Radical Ideals and the Ideal-Variety Correspondence

33 Polynomial and Rational Functions on a Variety

33.1 Polynomial Mappings

34 Robotics and Automatic Geometric Theorem Proving

34.1 Geometric Description of Robots

Part XI
Reading notes on Cox, Little, O'Shea's *Using Algebraic Geometry*

Using Algebraic Geometry. David A. Cox. John Little. Donal O'Shea. Second Edition. Springer. 2005. ISBN 0-387-20706-6 QA564.C6883 2004

35 Introduction

35.1 Polynomials and Ideals

monomial

$$x_1^{\alpha_1} \dots x_n^{\alpha_n} \tag{1.1}$$ (132)

total degree of x^α is $\alpha_1 + \dots + \alpha_n \equiv |\alpha|$

field k, $k[x_1 \dots x_n]$ collection of all polynomials in $x_1 \dots x_n$ with coefficients k.

polynomials in $k[x_1 \dots x_n]$ can be added and multiplied as usual, so $k[x_1 \dots x_n]$ has structure of commutative ring (with identity)
however, only nonzero constant polynomials have multiplicative inverses in $k[x_1 \dots x_n]$, so $k[x_1 \dots x_n]$ not a field
however set of rational functions $\{f/g | f, g \in k[x_1 \dots x_n],\ g \neq 0\}$ is a field, denoted $k(x_1 \dots x_n)$

so

$$f = \sum_\alpha c_\alpha x^\alpha$$

where $c_\alpha \in k$
so

$$f \in k[x_1 \dots x_n] = \{f | f = \sum_\alpha c_\alpha x^\alpha, x^\alpha = x_1^{\alpha_1} \dots x_n^{\alpha_n}, c_\alpha \in k\}$$

f homogeneous if all monomials have same total degrees
polynomial f is homogeneous if all monomials have the *same total degree*

Given a collection of polynomials $f_1 \dots f_s \in k[x_1 \dots x_n]$, we can consider all polynomials which can be built up from these by multiplication by arbitrary polynomials and by taking sums

Definition 128 (1.3)**.** Let $f_1 \dots f_s \in k[x_1 \dots x_n]$
Let $\langle f_1 \dots f_s \rangle = \{p_1 f_1 + \dots + p_s f_s | p_i \in k[x_1 \dots x_n] \text{ for } i = 1 \dots s\}$

Exercise 1.

(a) $x^2 = x \cdot (x - y^2) + y \cdot (xy)$

(b)

$$p \cdot (x - y^2) = px - py^2$$

and for $pxy = (py)x$

(c)

$$p(y)(x - y^2) = p(y)x - p(y)y^2 \notin \langle x^2, xy \rangle$$

Exercise 2.

$$\sum_{i=1}^s p_i f_i + \sum_{j=1}^s q_j f_j = \sum_{i=1}^s (p_i + q_i) f_i, \quad p_i + q_i \in k[x_1 \dots x_n]$$

$\langle f_1 \dots f_s \rangle$ closed under sums in $k[x_1 \dots x_n]$

If $f \in \langle f_1 \dots f_s \rangle$,
$p \in k[x_1 \dots x_n]$

$$p \cdot f = p \sum_{i=1}^s q_j f_j = \sum_{i=1}^s p q_j f_j, \quad pq_j \in k[x_1 \dots x_n] \text{ so}$$
$$p \cdot f \in \langle f_1 \dots f_s \rangle$$

Done.

The 2 properties in Ex. 2 are defining properties of ideals in the ring $k[x_1 \dots x_n]$

Definition 129 (1.5)**.** Let $I \subset k[x_1 \dots x_n]$, $I \neq \emptyset$
I ideal if

(a) $f + g \in I$, $\forall f, g \in I$

(b) $pf \in I$, $\forall f \in I$, arbitrary $p \in k[x_1 \dots x_n]$

Thus $\langle f_1 \dots f_s \rangle$ is an ideal by Ex. 2.

we call it the ideal generated by $f_1 \dots f_s$.

Exercise 3. Suppose $\exists$ ideal J, $f_1 \dots f_s \in J$ s.t. $J \subset \langle f_1 \dots f_s \rangle$
if $f \in \langle f_1 \dots f_s \rangle$, $f = \sum_{i=1}^{s} p_i f_i$, $p_i \in k[x_1 \dots x_n]$

$\forall i = 1 \dots s$, $p_i f_i \in J$ and so $\sum_{i=1}^{s} p_i f_i \in J$, by def. of J as an ideal.

$$\langle f_1 \dots f_s \rangle \subseteq J \qquad \Longrightarrow J = \langle f_1 \dots f_s \rangle$$

$\Longrightarrow \langle f_1 \dots f_s \rangle$ is smallest ideal in $k[x_1 \dots x_n]$ containing $f_1 \dots f_s$

Exercise 4. For $I = \langle f_1 \dots f_s \rangle$
$J = \langle g_1 \dots g_t \rangle$

$I = J$ iff $s = t$ and $\forall f \in I$, $f = \sum_{i=1}^{t} q_i g_i$ and if $0 = \sum_{i=1}^{t} q_i g_i$, $q_i = 0$, $\forall i = 1 \dots t$, and if $0 = \sum_{i=1}^{s} p_i f_i$, $p_i = 0$, $\forall i = 1 \dots s$

Definition 130 (1.6)**.**

$$\sqrt{I} = \{g \in k[x_1 \dots x_n] | g^m \in I \text{ for some } m \geq 1\}$$

e.g. $x + y \in \sqrt{\langle x^2 + 3xy, 3xy + y^2 \rangle}$
in $\mathbb{Q}[x, y]$ since

$$(x + y)^3 = x(x^2 + 3xy) + y(3xy + y^2) \in \langle x^2 + 3xy, 3xy + y^2 \rangle$$

- (Radical Ideal Property) $\forall$ ideal $I \subset k[x_1 \dots x_n]$, $\sqrt{I}$ ideal, $\sqrt{I} \supset I$
- **(Hilbert basis Thm.)** $\forall$ ideal $I \subset k[x_1 \dots x_n]$
  $\exists$ finite generating set,
  i.e. $\exists \{f_1 \dots f_2\} \subset k[x_1 \dots x_n]$ s.t. $I = \langle f_1 \dots f_s \rangle$
- (Division Algorithm in $k[x]$) $\forall f, g \in k[x]$ (EY : in 1 variable)
  $\forall f, g \in k[x]$ (in 1 variable)
  $f = qg + r$, $\exists!$ quotient q, $\exists$ remainder r

35.2

35.3 Gröbner Bases

Definition 131 (3.1)**.** Gröbner basis for $I \equiv G = \{g_1 \dots g_k\} \subset I$ s.t. $\forall f \in I$, LT(f) divisible by LT(g_i) for some i

- (Uniqueness of Remainders) let ideal $I \subset k[x_1 \dots x_n]$
  division of $f \in k[x_1 \dots x_n]$ by Grö bner basis for I, produces $f = g + r$, $g \in I$, and no term in r divisible by any element of LT(I)

35.4 Affine Varieties

affine n-dim. space over k $\quad k^n = \{(a_1 \dots a_n) | a_1 \dots a_n \in k\}$

$\forall$ polynomial $f \in k[x_1 \dots x_n]$, $(a_1 \dots a_n) \in k^n$

$f : k^n \to k$

$f(a_1 \dots a_n)$ s.t. $x_i = a_i$ i.e.

if $f = \sum_\alpha c_\alpha x^\alpha$ for $c_\alpha \in k$, then

$f(a_1 \dots a_n) = \sum_\alpha c_\alpha a^\alpha \in k$, where $a^\alpha = a_1^{\alpha_1} \dots a_n^{\alpha_n}$

Definition 132 (4.1)**.** affine variety $\mathbf{V}(f_1 \dots f_s) = \{(a_1 \dots a_n) | (a_1 \dots a_n) \in k^n, f_1(x_1 \dots x_n) = \dots = f_s(x_1 \dots x_n) = 0\}$

subset $V \subset k^n$ is affine variety if $V = V(f_1 \dots f_s)$ for some $\{f_i\}$, polynomial $f_i \in k[x_1 \dots x_n]$

- (Equal Ideals Have Equal Varieties) If $\langle f_1 \dots f_s \rangle = \langle g_1 \dots g_t \rangle$ in $k[x_1 \dots x_n]$, then $\mathbf{V}(f_1 \dots f_s) = \mathbf{V}(g_1 \dots g_t)$

so, recap

if $\langle f_1 \dots f_s \rangle = \langle g_1 \dots g_t \rangle$ in $k[x_1 \dots x_n]$,

then $V(f_1 \dots f_s) = V(g_1 \dots g_t)$

Recall Hilbert basis Thm. $\forall$ ideal $I \subset k[x_1 \dots x_n]$

$$I = \langle f_1 \dots f_s \rangle$$

$\implies$ if $I = J$, then $V(I) = V(J)$

think of V defined by I, rather than $f_1 = \dots = f_s = 0$

Exercise 3.

Recall Def. 1.5 Let $I \subset k[x_1 \dots x_n]$

I ideal if $f + g \in I \quad \forall f, g \in I$

$pf \in I, \quad \forall f \in I$ arbitrary $p \in k[x_1 \dots x_n]$

Let $f, g \in I(V)$

$$(f+g)(a_1 \dots a_n) = f(a_1 \dots a_n) + g(a_1 \dots a_n) = 0 + 0 = 0 \qquad f + g \in I(V)$$
$$pf(a_1 \dots a_n) = p(a_1 \dots a_n) f(a_1 \dots a_n) = 0 \qquad pf \in I(V)$$

Then $I(V)$ an ideal.

$V = V(x^2)$ in $\mathbb{R}^2$

$I = \langle x^2 \rangle$ in $\mathbb{R}[x, y]$, $\quad I = \{px^2 | p \in k[x, y]\}$

$I \subset I(V)$, since $px^2 = 0$ for $x^2 = 0$, $(0, b)$, $\quad b \in \mathbb{R}$

But $p(x, y) = x \in I(V)$, as

$$I(V) = \{f \in k[x_1 \dots x_n] | f(a_1 \dots a_n) = 0, \forall (a_1 \dots a_n) \in V\}$$

$p(0, b) = x = 0$

But $x \notin I$

Exercise 4. $I \subset \sqrt{I}$

Recall Def. 1.6 $\sqrt{I} = \{g \in k[x_1 \dots x_n] | g^m \in I \text{ for some } m \geq 1\}$

$\forall f \in I$, $f = f^1$, $m = 1$, so $f \in \sqrt{I}$, $\qquad I \subset \sqrt{I}$

Hilbert basis thm., $\forall$ ideal $I \subset k[x_1 \dots x_n]$ s.t. $I = \langle f_1 \dots f_s \rangle$

$\{V(I) = \{(a_1 \dots a_n) | (a_1 \dots a_n) \in k^n, f_1(a_1 \dots a_n) = \dots = f_s(a_1 \dots a_n) = 0\}\}$

$\mathbf{I}(\mathbf{V}(I)) = \{f \in k[x_1 \dots x_n] | f(a_1 \dots a_n) = 0 \quad \forall (a_1 \dots a_n) \in V(I)\}$

Let $g \in \sqrt{I}$, $\quad g^m \in I$, $\quad g^m = g^{m-1} g$

$g^m(a_1 \dots a_n) = 0 = g^{m-1}(a_1 \dots a_n) g(a_1 \dots a_n) = 0$. Then $g(a_1 \dots a_n) = 0$ or $g^{m-1}(a_1 \dots a_m) = 0$

as $g^m \in I$, and $V(I)$ is s.t. $f_1(a_1 \dots a_n) = \dots = f_s(a_1 \dots a_n) = 0$ for $I = \langle f_1 \dots f_s \rangle$

- (Strong Nullstellensatz) if k algebraically closed (e.g. $\mathbb{C}$), I ideal in $k[x_1 \dots x_n]$, then

$$\mathbf{I}(\mathbf{V}(I) = \sqrt{I}$$

- (Ideal-variety correspondence) Let k arbitrary field

$$I \subset I(V(I))$$
$$V(I(V)) = V \quad \forall V$$

Additional Exercises for Sec.4

Exercise 6.

36 Solving Polynomial Equations

36.1

36.2 Finite-Dimensional Algebras

Gröbner basis $G = \{g_1 \dots g_t\}$ of ideal $I \subset k[x_1 \dots x_n]$,
recall def.: Gröbner basis $G = \{g_1 \dots g_t\} \subset I$ of ideal I, $\forall f \in I$, $\mathrm{LT}(f)$ divisible by $\mathrm{LT}(g_i)$ for some i
$f \in k[x_1 \dots x_n]$ divide by G produces $f = g + r$, $g \in I$, r not divisible by any $\mathrm{LT}(I)$ uniqueness of r
$f \in k[x_1 \dots x_n]$ divide by G,
Recall from Ch. 1, divide $f \in k[x_1 \dots x_n]$ by G, the division algorithm yields

$$f = h_1 g_1 + \cdots + h_t g_t + \overline{f}^G \tag{2.1}$$ (133)

where remainder $\overline{f}^G$ is a linear combination of monomials $x^\alpha \notin \langle \mathrm{LT}(I) \rangle$
since Gröbner basis, $f \in I$ iff $\overline{f}^G = 0$
$\forall f \in k[x_1 \dots x_n]$, we have coset $[f] = f + I = \{f + h | h \in I\}$ s.t. $[f] = [g]$ iff $f - g \in I$
We have a 1-to-1 correspondence

$$\text{remainders} \leftrightarrow \text{cosets}$$
$$\overline{f}^G \leftrightarrow [f]$$

algebraic

$$\overline{f}^G + \overline{g}^G \leftrightarrow [f] + [g]$$
$$\overline{\overline{f}^G \cdot \overline{g}^G} \leftrightarrow [f] \cdot [g]$$

$B = \{x^\alpha | x^\alpha \notin \langle \mathrm{LT}(I) \rangle\}$ is a basis of A, basis monomials, standard monomials
20141023 EY's take
$\forall [f] \in A = k[x_1 \dots x_n]/I$, $[f] = p_i b_i$; $b_i \in B = \{x^\alpha | x^\alpha \notin \langle \mathrm{LT}(I) \rangle\}$
For $I = \langle G \rangle$
e.g. $G = \{x^2 + \frac{3}{2}xy + \frac{1}{2}y^2 - \frac{3}{2}x - \frac{3}{2}y, xy^2 - x, y^3 - y\}$
$\langle \mathrm{LT}(I) \rangle = \langle x^2, xy^2, y^3 \rangle$
e.g. $B = \{1, x, y, xy, y^2\}$
$[f] \cdot [g] = [fg]$
e.g. $f = x$, $g = xy$, $[fg] = [x^2 y]$
now $f = h_1 g_1 + \cdots + h_t g_t + \overline{f}^G$

36.3

36.4 Solving Equations via Eigenvalues and Eigenvectors

37 Resultants

38 Computation in Local Rings

38.1 Local Rings

Definition 133 (1.1)**.**

$$k[x_1 \dots x_n]_{\langle x_1 \dots x_n \rangle} \equiv \{\frac{f}{g} | \text{ rational functions } \frac{f}{g} \text{ of } x_1 \dots x_n \text{ with } g(p) \neq 0 \text{ at } p\}$$

main properties of $k[x_1 \dots x_n]_{\langle x_1 \dots x_n \rangle}$

Proposition 30 (1.2)**.** *Let $R = k[x_1 \dots x_n]_{\langle x_1 \dots x_n \rangle}$. Then*

(a) R subring of field of rational functions $k(x_1 \dots x_n) \supset k[x_1 \dots x_n]$

(b) Let $M = \langle x_1 \dots x_n \rangle \subset R$ (ideal generated by $x_1 \dots X_n$ in R)
Then $\forall \frac{f}{g} \in R \backslash M$, $\frac{f}{g}$ unit in R (i.e. multiplicative inverse in R)

(c) M maximal ideal in R

Exercise 1. if $p = (a_1 \dots a_n) \in k^n$, $R = \{\frac{f}{g} | f, g \in k[x_1 \dots x_n],\ g(p) \neq 0\}$

(a) R subring of field of rational functions $k(x_1 \dots x_n)$

(b) Let M ideal generated by $x_1 - a_1 \dots x_n - a_n$ in R
Then $\forall \frac{f}{g} \in R \backslash M$, $\frac{f}{g}$ unit in R (i.e. multiplicative inverse in R)

(c) M maximal ideal in R

Proof. let $p = (a_1 \dots a_n) \in k^n$
let $g_1(p) \neq 0$, $g_2(p) \neq 0$

$$\frac{f_1}{g_1} + \frac{f_2}{g_2} = \frac{f_1 g_2 + f_2 g_1}{g_1 g_2} \qquad g_1(p) g_2(p) \neq 0 \text{ so } \frac{f_1}{g_1} + \frac{f_2}{g_2} \in R$$
$$\frac{f_1}{g_1} \cdot \frac{f_2}{g_2} = \frac{f_1 f_2}{g_1 g_2} \qquad g_1(p) g_2(p) \neq 0 \text{ so } \frac{f_1}{g_1} \frac{f_2}{g_2} \in R$$

$f = \frac{f}{1} \in R, \qquad \forall f \in k[x_1 \dots x_n]$, so $k[x_1 \dots x_n] \subset R$

□

EY : 20141027, to recap,
Let $V = k^n$
Let $p = (a_1 \dots a_n)$
single pt. $\{p\}$ is (an example of) a variety
$I(\{p\}) = \{x_1 - a_1 \dots x_n - a_n\rangle \subset k[x_1 \dots x_n]$

$R \equiv k[x_1 \dots x_n]_{\langle x_1 - a_1 \dots x_n - a_n \rangle}$

$$R = \{\frac{f}{g} | \text{ rational function } \frac{f}{g} \text{ of } x_1 \dots x_n,\ g(p) \neq 0,\ p = (a_1 \dots a_n)\ \}$$

Prop. 1.2. properties

(a) R subring of field of rational functions $k(x_1 \dots x_n) \qquad k(x_1 \dots x_n) \subset R$

(b) $M = \langle x_1 \dots a_1 \dots x_n - a_n \rangle \subset R$. ideal generated by $x_1 - a_1 \dots x_n - a_n$
Then $\forall \frac{f}{g} \in R \backslash M$, $\frac{f}{g}$ unit in R ($\exists$ multiplicative inverse in R)

(c) M maximal ideal in R.
in R we allow denominators that are not elements of this ideal $I(\{p\})$

Definition 134 (1.3)**.** local ring is a ring that has exactly 1 maximal ideal

Proposition 31 (1.4)**.** *ring R with proper ideal $M \subset R$ is local ring if $\forall \frac{f}{g} \in R \backslash M$ is unit in R*

localization Ex. 8, Ex. 9
parametrization

Exercise 2.

$$x = x(t) = \frac{-2t^2}{1+t^2}$$
$$y = y(t) = \frac{2t}{1+t^2}$$

$k[t]_{\langle t \rangle}$ $\frac{-2t^2}{1+t^2}$ rational function of t. $1 + t^2 \neq 0$
if $k = \mathbb{C}$ or $\mathbb{R}$

Consider set of convergent power series in n variables

$$k\{x_1 \dots x_n\} = \{ \sum_{\alpha \in \mathbb{Z}^n_{\geq 0}} c_\alpha x^\alpha | c_\alpha \in k, \text{ series converges in some open } U \ni 0 \in k^n \ \} \tag{1.5}$$ (134)

Consider set $k[[x_1 \dots x_n]]$ of formal power series

$$k[[x_1 \dots x_n]] = \{ \sum_{\alpha \in \mathbb{Z}^n_{\geq 0}} c_\alpha x^\alpha | c_\alpha \in k \} \text{ series need not converge} \tag{1.6}$$ (135)

variety V

$k[x_1 \dots x_n]/\mathbf{I}(V)$ variety V

38.2 Multiplicities and Milnor Numbers

if I ideal in $k[x_1 \dots x_n]$, then denote $Ik[x_1 \dots x_n]_{\langle x_1 \dots x_n \rangle}$ ideal generated by I in larger ring $k[x_1 \dots x_n]_{\langle x_1 \dots x_n \rangle}$

Definition 135 (2.1)**.** Let I 0-dim. ideal in $k[x_1 \dots x_n]$, so $V(I)$ consists of finitely many pts. in k^n.
Assume $(0 \dots 0) \in V(I)$
multiplicity of $(0 \dots 0) \in V(I)$ is

$$\dim_k k[x_1 \dots x_n]_{\langle x_1 \dots x_n \rangle} / Ik[x_! \dots x_n]_{\langle x_1 \dots x_n \rangle}$$

generally, if $p = (a_1 \dots a_n) \in V(I)$
multiplicity of p, $m(p) = \dim k[x_1 \dots x_n]_M / Ik[x_1 \dots x_n]_M$

$$\dim k[x_1 \dots x_n]_M / Ik[x_1 \dots x_n]_M$$

localizing $k[x_1 \dots x_n]$ at maximal ideal $M = I(\{p\}) = \langle x_1 - a_1 \dots x_n - a_n \rangle$

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41 Polytopes, Resultants, and Equations

42 Polyhedral Regions and Polynomials

42.1 Integer Programming

Prop. 1.12.

Suppose 2 customers A, B ship to same location
A: ship 400 kg pallet taking up $2\,m^3$ volume
B: ship 500 kg pallet taking up $3\,m^3$ volume

shipping firm trucks carry up to 3700 kg, up to $20\,m^3$

B's product more perishable, paying \$ 15 per pallet

A pays \$ 11 per pallet
How many pallets from A, B each in truck to maximize revenues?

$$\begin{array}{cc} & 4A + 5B \leq 37 \\ (1.1) & 2A + 3B \leq 20 \\ & A, B \in \mathbb{Z}^*_{\geq 0} \end{array} \tag{136}$$

maximize $11A + 15B$

integer programming.
max. or min. value of some linear function

$$l(A_1 \dots A_n) = \sum_{i=1}^{n} c_i A_i$$

on set $(A_1 \dots A_n) \in \mathbb{Z}^n_{\geq 0}$ s.t.
3. Finally, by introducing additional variables; rewrite linear constraint inequalities as equalities. The new variables are called "slack variables"

$$(1.4) \qquad a_{ij} A_j = b_i, \quad A_j \in \mathbb{Z}_{\geq 0} \tag{137}$$

introduce indeterminate z_i, $\forall$ equation in (1.4)

$$z_i^{a_{ij} A_j} = z_i^{b_i}$$

m constraints

$$\prod_{i=1}^{m} z_i^{a_{ij} A_j} = \prod_{i=1}^{m} z_i^{b_i} = \left(\prod_{i=1}^{m} z_i^{a_{ij}} \right)^{A_j}$$

Proposition 32 (1.6)**.** *Let k field, define $\varphi : k[w_1 \dots w_n] \to k[z_1 \dots z_m]$ by*

$$\varphi(w_j) = \prod_{i=1}^{m} z_i^{a_{ij}} \qquad \forall\, j = 1 \dots n$$

and

$$\varphi(g(w_1 \dots w_n)) = g(\varphi(w_1) \dots \varphi(w_n))$$

$\forall$ *general polynomial* $g \in k[w_1 \dots w_n]$
Then $(A_1 \dots A_n)$ *integer pt. in feasible region iff* $\varphi : w_1^{A_1} \dots w_n^{A_n} \mapsto z_1^{b_1} \dots z_m^{b_m}$

Exercise 3.
Now

$$\varphi(w_j) = \prod_{i=1}^{m} z_i^{a_{ij}}$$
$$z_i^{a_{ij}A_j} = z_i^{b_i}$$

If $(A_1 \dots A_n)$ an integer pt. in feasible region, $a_{ij}A_j = b_i$

$$z_i^{a_{ij}A_j} = z_i^{b_i} = \prod_{j=1}^{n} z_i^{a_{ij}A_j} \Longrightarrow \prod_{j=1}^{n}\prod_{i=1}^{m}(z_i^{a_{ij}})^{A_j} = \prod_{i=1}^{m} z_i^{b_i} = \prod_{j=1}^{n} \varphi(w_j)^{A_j} = \prod_{j=1}^{n} \varphi(w_j)^{A_j} = \varphi\left(\prod_{j=1}^{n} w_j^{A_j}\right) = \prod_{i=1}^{m} z_i^{b_i}$$

since $\varphi(g(w_1 \dots w_n)) = g(\varphi(w_1) \dots \varphi(w_n))$

If $\varphi : \prod_{j=1}^{n} w_j^{A_j} \mapsto \prod_{i=1}^{m} z_i^{b_i}$

$$\varphi\left(\prod_{j=1}^{n} w_j^{A_j}\right) = \prod_{j=1}^{n}(\varphi(w_j))^{A_j} = \prod_{i=1}^{m} z_i^{b_i} = \prod_{j=1}^{n}\left(\prod_{i=1}^{m} z_i^{a_{ij}}\right)^{A_j} \Longrightarrow \prod_{j=1}^{n} z_i^{a_{ij}A_j} = z_i^{b_i}$$

or $a_{ij}A_j = b_i$. So $(A_1 \dots A_n)$ integer pt.

Exercise 4.

$$\prod_{i=1}^{m} z_i^{b_i} = \prod_{i=1}^{m}\prod_{j=1}^{n} z_i^{a_{ij}A_j} = \prod_{j=1}^{n}\left(\prod_{i=1}^{m} z_i^{a_{ij}}\right)^{A_j} = \prod_{j=1}^{n} \varphi(w_j)^{A_j} = \varphi\left(\prod_{j=1}^{n} w_j^{A_j}\right)$$

So if given $(b_1 \dots b_m) \in \mathbb{Z}^m$, and for a given a_{ij}, $a_{ij}A_j = b_i$

For $m \le n$, then a_{ij} is surjective, so $\exists\, A_j$ s.t. $\prod_{i=1}^{m} z_i^{b_i} = \varphi\left(\prod_{j=1}^{n} w_j^{A_j}\right)$

Proposition 33 (1.8)**.** *Suppose* $f_1 \dots f_n \in k[z_1 \dots z_m]$ *given*
Fix monomial order in $k[z_1 \dots z_n, w_1 \dots w_n]$ *with elimination property:*
$\forall$ *monomial containing 1 of* z_i *greater than any monomial containing only* w_j

Let $\mathcal{G}$ *Gröbner basis for ideal*

$$I = \langle f_1 - w_1 \dots f_n - w_n \rangle \subset k[z_1 \dots z_m, w_1 \dots w_n]$$

$\forall\, f \in k[z_1 \dots z_m]$, *let* $\overline{f}^{\mathcal{G}}$ *be remainder on division of* f *by* $\mathcal{G}$
Then

(a) *polynomial* f *s.t.* $f \in k[f_1 \dots f_n]$ *iff* $g = \overline{f}^{\mathcal{G}} \in k[w_1 \dots w_n]$

(b) *if* $f \in k[f_1 \dots f_n]$ *as in part (a),*
$g = \overline{f}^{\mathcal{G}} \in k[w_1 \dots w_n]$

then $f = g(f_1 \dots f_n)$ *, giving an expression for* f *as polynomial in* f_j

(c) *if* $\forall\, f_i, f$ *monomials,* $f \in k[f_1 \dots f_n]$,
then g *also a monomial.*

42.2 Integer Programming and Combinatorics

43 Algebraic Coding Theory

44 The Berlekamp-Massey-Sakata Decoding Algorithm

Gröbner Bases, Martin R. Albrecht of the DTU Crypto Group

Part XII
Algebraic Geometry

45 Affine and Projective Varieties

cf. Harris (1992)[71]

For (algebraically closed) field K,
vector space K^n,
affine space $\mathbb{A}^n_K \equiv \mathbb{A}^n = K^n$, but origin plays no special role in affine space.
Affine variety $X \subset \mathbb{A}^n :=$ common zero locus of collection of polynomials $f_\alpha \in K[z_1 \dots z_n] :=$

$$X = \{Z | f_\alpha(Z) = 0 \quad \forall\, \alpha, \quad f_\alpha \in K(z_1 \dots z_n), Z = (z_1 \dots z_n)\}$$

45.1 Projective Space and Projective Varieties

Projective space over field $K =$ set of 1-dim. subspaces of vector space $K^{n+1} \equiv \mathbb{P}^n_K \equiv \mathbb{P}^n = (K^{n+1} - \{0\})/K^*$, where $(K^{n+1} - \{0\})/K^*$ is the quotient of $K^{n+1} - \{0\}$ by the action of the group K^n acting by scalar multiplication.

$\mathbb{P}(V) \equiv \mathbb{P}V \equiv$ projective space of 1-dim. subspaces of a vector space V over field K.
$P \in \mathbb{P}^n$ usually written as homogeneous vector $[Z_0 \dots Z_n]$, by which be mean line spaced by $(Z_0 \dots Z_n) \in K^{n+1}$.

For U_n s.t. $\forall\, P \in U_n \subset \mathbb{P}^n \subset V^{n+1}$, $Z_n \neq 0$. Then $[Z_0 \dots Z_n] \sim \left[\frac{Z_0}{Z_n}, \dots, \frac{Z_{n-1}}{Z_n}, 1\right] \cong \left[\frac{Z_0}{Z_n}, \dots \frac{Z_{n-1}}{Z_n}\right] \in K^n$.

$\forall\, v \neq 0$, $v \in V$, $[v] =$ corresponding pt. in $\mathbb{P}V \cong \mathbb{P}^n$

Polynomial $F \in K[Z_0 \dots Z_n]$ on vector space K^{n+1} doesn't define a function on $\mathbb{P}^n$, but
if F is homogeneous of degree d,
then since

$$F(\lambda Z_0, \dots, \lambda Z_n) = \lambda^d F(Z_0 \dots Z_n)$$

it does make sense to talk about 0 locus of polynomial F.

Definition 136 (Projective variety)**.** projective variety $X \subset \mathbb{P}^n = \{P | F_\alpha(P) = 0\ \forall\, \alpha,\ F_\alpha(\lambda P) = \lambda^d F_\alpha(P)\} =$ zero locus of a collection of homogeneous polynomials F_α.

Group $PGL_{n+1}K$ acts on space $\mathbb{P}^n$ (in Lecture 18, $PGL_{n+1}K$ are automorphisms of $\mathbb{P}^n$).
Varieties $X, Y \subset \mathbb{P}^n$ are projectively equivalent, if they're congruent, modulo this group.
Note that if $\mathbb{P}^n = \mathbb{P}V$ is projective space associated with vector space V,

- homogeneous coordinates on $\mathbb{P}V$ correspond to elements of dual space V^*
- similarly, space of homogeneous polynomials of degree d on $\mathbb{P}V$ naturally identified with vector space $\mathrm{Sym}^d(V^*)$

Meaning, set of linear coordinates on vector space V, $\dim V = n$, over field K (so $V = K^n$), $\alpha_i \equiv z_i$, $i = 1 \dots n$, is a *basis* (α_i) of V^*, since

$$\begin{array}{c} \alpha : V \to K^n \\ v \mapsto (\alpha_1(v), \dots \alpha_n(v)) \end{array} \text{ i.e. } \equiv \begin{array}{c} z : V \to K^n \\ v \mapsto (z_1(v) \dots z_n(v)) \end{array}$$

Now $\mathbb{P}(V) = (V \backslash \{0\})/K^*$ and homogeneous coordinates on $\mathbb{P}(V)$ are just linear coordinates on V up to action K^*

cf. "Correspondence between the projective space associated to a vector space and the dual space of the vector space?", stackexchange, Can dual vector spaces be thought of as linear coordinate functions? stackexchange

From $Z_i \in V^*$, $i = 0, 1 \dots n$, $Z_i : V \to K$, $Z_i : v \mapsto Z_i(v) = Z_i \in K$,
let f be a homogeneous polynomial of degree d on $\mathbb{P}V$:

$$f = \sum a_{i_0 i_1 \dots i_n} z_0^{i_0} z_1^{i_1} \dots z_n^{i_n}$$

where summation $\sum$ is over $0 \le i_0, i_1, \dots i_n \le d$ s.t. $\sum_{j=0}^n i_j = d$.

$\dim \mathrm{Sym}^d(V^*) = \binom{d+n}{n}$

$\{z_0^{i_0} z_1^{i_1} \dots z_n^{i_n}\}_{\substack{0 \le i_0, i_1 \dots i_n \le d \\ \sum_{j=0}^n i_j = d}}$ form a basis for $\mathrm{Sym}^d(V^*)$

Let $U_i \subset \mathbb{P}^n$, $U_i = \{[Z_0 \dots Z_n] | Z_i \neq 0\}$. Then $[Z_0 \dots Z_n] \sim \left[\frac{Z_0}{Z_i} \dots \frac{Z_{i-1}}{Z_i}, 1 \dots \frac{Z_n}{Z_i}\right] \equiv [z_0, \dots z_{i-1}, 1, z_i \dots z_{n-1}] \cong (z_0, z_1 \dots z_{n-1}) \in K^n$.

So there's a bijection $U_i \to K^n$.

Geometrically, this map is associating line $L \subset K^{n+1}$ not contained in hyperplane $(Z_i = 0)$, its pt. p of intersection with affine plane $(Z_i = 1) \subset K^{n+1}$.

Coordinates z_j on U_i are called affine or Euclidean coordinates on projective space or open set U_i
- open sets U_i comprise standard cover of $\mathbb{P}^n$ by affine open sets.

If $X \subset \mathbb{P}^n$ is a variety, $X_i = X \cup U_i$ is affine variety:
if X given by polynomials $F_\alpha \in K[Z_0, \dots Z_n]$, then e.g. X_0 will be zero locus of polynomials

$$f_\alpha(z_0 \dots z_n) = F_\alpha(Z_0 \dots Z_n)/Z_0^d = F_\alpha(1, z_1 \dots z_n)$$

where $d = \deg F_\alpha$.

For (projective) variety $X \subset \mathbb{P}^n$, $X = \{P | F_\alpha(P) = 0, \forall\, \alpha, F_\alpha$ homogeneous, $P = [Z_0, Z_1 \dots Z_n] \in \mathbb{P}^n\}$,
obtain affine variety $X_i = X \cup U_i$ as follows: for

$$z_j = \begin{cases} \frac{Z_{j-1}}{Z_i} & j \le i \\ \frac{Z_j}{Z_i}, & j > i \end{cases}$$

$$f_\alpha(z_1 \dots z_n) = f_\alpha\left(\frac{Z_0}{Z_i}, \dots \frac{Z_{i-1}}{Z_i}, \frac{Z_{i+1}}{Z_i} \dots \frac{Z_n}{Z_i}\right) = \frac{1}{Z_i}^{d_\alpha} F_\alpha(Z_0 \dots Z_n) = F_\alpha(z_1 \dots z_i, 1, z_{i+1}, \dots z_n)$$

If $F_\alpha(Z_0 \dots Z_n) = 0$, then $f_\alpha(z_1 \dots z_n) = 0$

$\forall$ projective variety X, X is union of affine varieties.

If affine variety $X_i \subset K^n \cong U_i \subset \mathbb{P}^n$, by def. X_i given by polynomials $\{f_\alpha\}_\alpha$

$$f_\alpha(z_1 \dots z_n) = \sum a_{i_1 \dots i_n} z_1^{i_1} \dots z_n^{i_n} = 0$$

of degree d_α (i.e. $i_1 + \dots i_n = d_\alpha$)

Then

$$F_\alpha(Z_0 \dots Z_n) = Z_i^{D_\alpha} F_\alpha\left(\frac{Z_0}{Z_i} \dots \frac{Z_n}{Z_i}\right) = Z_i^{D_\alpha} f_\alpha(z_1 \dots z_n) = \sum a_{i_1 \dots i_n} Z_i^{D_\alpha - \sum i_l} Z_0^{i_0} \dots Z_n^{i_n} =$$
$$= \sum a_{i_1 \dots i_n} Z_i^{D_\alpha - d_\alpha} Z_0^{i_0} \dots \widehat{Z}_i^{i_i} \dots Z_n^{i_n}$$

45.1.1 Example: ellipse

$$\begin{gathered} \mathbb{P}^n \to U_Z \cong K^n \\ [X, Y, Z] \mapsto (x, y) = \left(\frac{X}{Z}, \frac{Y}{Z}\right) \end{gathered} \tag{138}$$

Consider

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \text{ or } f(x, y) = \frac{x^2}{a^2} + \frac{y^2}{b^2} - 1 \tag{139}$$

For affine variety $X_Z \subset K^2$,

$$F(X, Y, Z) = \left(\frac{X^2}{Z^2 a^2} + \frac{Y^2}{Z^2 b^2} - 1\right) Z^2 = \frac{X^2}{a^2} + \frac{Y^2}{b^2} - Z^2 \tag{140}$$

46 Algebraic Curves; Conic sections

cf. Reid (2013) [70].

cf. Ch. 0 "Woffle" of Reid (2013) [70].

Given field k, $k[x_1 \dots x_n]$ colelction of all polynomials in $x_1 \dots x_n$, with coefficients in k,

$$f \in k[x_1 \dots x_n] = \{f | f = \sum_\alpha c_\alpha x^\alpha, x^\alpha = x_1^{\alpha_1} \dots x_n^{\alpha_n},\ c_\alpha \in k\}$$

Variety is (roughly) locus defined by polynomial equations

$$V = \{P \in k^n | f_i(P) = 0\} \subset k^n,\ f_i \in k[x_1 \dots x_n]$$

e.g. plane curves C: $(f(x,y) = 0) \subset \mathbb{R}^2$ or $\mathbb{C}^2$

Groups of transformations (i.e. transformation groups) are of central importance throughout geometry; properties of geometric figures must be invariant under appropriate kind of transformations before they're significant.

affine change of coordinates in $\mathbb{R}^2$ is of form

$$T(\mathbf{x}) = A\mathbf{x} + B \quad \text{(affine change of coordinates)} \tag{141}$$

where $\mathbf{x} = (x,y) \in \mathbb{R}^2$, A 2×2 invertible matrix (i.e. $A \in GL(2,\mathbb{R})$), $B \in \mathbb{R}^2$.

If A orthogonal, transformation T is *Euclidean.*

$\forall$ nondegenerate conic can be reduced to "standard form" by Euclidean transformation.

projectivity or projective transformation $\mathbb{P}^2_\mathbb{R}$ is map $T(\mathbf{X}) = M\mathbf{X}$, $M \in GL(3,\mathbb{R})$.

Understand T on affine piece $\mathbb{R}^2 \subset \mathbb{P}^2_\mathbb{R}$ is partially defined map $\mathbb{R}^2 \to \mathbb{R}^2$; it's a fractional linear transformation.

$$(x,y) \overset{\cong}{\longmapsto} [x,y,1]$$
$$(x,y) \mapsto \begin{pmatrix} A\begin{pmatrix} x \\ y \end{pmatrix} + B \\ cx + dy + e \end{pmatrix}$$

where

$$M = \left(\begin{array}{cc|c} & A & B \\ c & d & e \end{array} \right)$$

e.g. 2 different photographs of same (plane) object are obviously related by a projectivity.

For inhomogeneous quadratic polynomial q, homogeneous quadratic polynomial Q, then there exists bijection

$$q \in K[x,y] \overset{\cong}{\longmapsto} Q \in K[X,Y,Z]$$
$$q(x,y) = ax^2 + bxy + cy^2 + dx + ey + f \mapsto Q(X,Y,Z) = aX^2 + bXY + cY^2 + dXZ + eYZ + fZ^2$$

so

$$q(x,y) = Q\left(\frac{X}{Z}, \frac{Y}{Z}, 1\right) \text{ with } x = X/Z,\ y = Y/Z$$

inverse:

$$Q = Z^2 q(X/Z, Y/Z)$$

46.0.1 "Line at infinity" and asymptotic directions

cf. Ch. 1 of Reid (2013)

Points of $\mathbb{P}^2$ with $Z = 0$, $[X,Y,0]$, form *line at infinity*, a copy of $\mathbb{P}^1_\mathbb{R} = \mathbb{R} \cup \{\infty\}$ (since $[X,Y] \mapsto X/Y$) define bijection $\mathbb{P}^1_\mathbb{R} \to \mathbb{R} \cup \{\infty\}$.

Line in $\mathbb{P}^2$, L, $L := \{[X,Y,Z] | aX + bY + cZ = 0\}$.

L passes through $(X,Y,0) \iff aX + bY = 0$.

(a) hyperbola $\left(\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1\right)$. Recall that the lines of asymptotes (asymptotic lines). They are found in the following manner:

$$\frac{(bx - ay)(bx + ay)}{a^2 b^2} = 1 \text{ or } \frac{bx - ay}{a^2 b^2} = \frac{1}{bx + ay} \xrightarrow{x,y \to \infty} \frac{bx - ay}{a^2 b^2} = 0 \text{ or } y = \frac{b}{a}x$$

Now, $\left(\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1\right)$ in $\mathbb{R}^2$ corresponds in $\mathbb{P}^2_{\mathbb{R}}$ to $C : \left(\frac{X^2}{a^2} - \frac{Y^2}{b^2} = Z^2\right)$.

This meets $(Z = 0)$ in 2 pts. $(a, \pm b, 0) \in \mathbb{P}^2_{\mathbb{R}}$, corresponding to asymptotic lines of hyperbola, $y = \frac{b}{a}x$, $y = \frac{-b}{a}x$

For affine piece $U_x \subset \mathbb{P}^2_{\mathbb{R}}$, $U_x = \{p \in \mathbb{P}^2_{\mathbb{R}} | p = [X, Y, Z] \text{ s.t. } X \neq 0\}$, then
bijection $U_x \to \mathbb{R}^2$,

$$[X, Y, Z \sim [1, \frac{Y}{X}, \frac{Z}{X}] \mapsto (u, v) = \left(\frac{Y}{X}, \frac{Z}{X}\right), \text{ so}$$

$$C : X^2/a^2 - Y^2/b^2 = Z^2 \mapsto u^2 + \frac{v^2}{b^2} = \frac{1}{a^2} \text{ or } \frac{u^2}{1/a^2} + \frac{v^2}{(b/a)^2} = 1 \qquad \text{(an ellipse!)}$$

(b) $y = mx^2$ (parabola) in $\mathbb{R}^2 \mapsto C : YZ = mX^2$ in $\mathbb{P}^2_{\mathbb{R}}$.

For $Z = 0$, C meets $Z = 0$ at single pt. $[0, 1, 0] \sim [0, Y, 0]$. So in $\mathbb{P}^2$, "2 branches of parabola meet at infinity."

46.0.2 Classification of conics in $\mathbb{P}^2$

cf. 1.6. Classification of conics in $\mathbb{P}^2$, Reid (2013) [70]

Let K be any field of characteristic $\neq 2$.

Recall 2 linear algebra results for quadratic forms:

Proposition 34. *$\exists$ bijections*

{ homogeneous quadratic polynomials } = { quadratic forms $K^3 \to K$ } $\cong$ { symmetric bilinear forms on K^3 } given by

$$aX^2 + 2bXY + cY^2 + 2dXZ + 2eYZ + fZ^2 \cong \begin{pmatrix} a & b & d \\ b & c & e \\ d & e & f \end{pmatrix} \text{ since}$$

$$[X \quad Y \quad Z] \begin{pmatrix} a & b & d \\ b & c & e \\ d & e & f \end{pmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = aX^2 + 2bXY + cY^2 + 2dXZ + 2eYZ + fZ^2$$

Quadratic form nondegenerate if corresponding bilinear form nondegenerate, i.e. matrix is nonsingular.

Theorem 34. *Let V be vector space over K, quadratic form $Q : V \to K$, then $\exists$ basis of V s.t.*

$$Q = \epsilon_1 x_1^2 + \epsilon_2 x_2^2 + \cdots + \epsilon_n x_n^2 \text{ with } \epsilon_i \in K \tag{142}$$

This theorem is proved by Gram-Schmidt orthogonalization.

For $\lambda \in K \backslash \{0\}$, $x_i \mapsto \lambda x_i$ takes $\epsilon_i \mapsto \lambda^{-2}\epsilon_i$.

Corollary 5. *In a suitable coordinate system, any conic in $\mathbb{P}^2$ is one of*

(a) nondegenerate conic $C : (X^2 + Y^2 - Z^2 = 0)$

46.0.3 Parametrization of a conic

Let C be a nondegenerate, nonempty conic of $\mathbb{P}^2_{\mathbb{R}}$.

Then by Corollary 5 (cf. Corollary 1.6 (cf. Reid (2013) [70])), and taking new coordinates $[X + Z, Y, Z - X]$,

$$X^2 + Y^2 - Z^2 = 0 \mapsto (X + Z)^2 + Y^2 - (Z - X)^2 = X^2 + 2XZ + Z^2 + Y^2 - (Z^2 - 2ZX + X^2) = Y^2 + 4XZ = 0$$

$\implies C$ is projectively equivalent to curve $(Y^2 = XZ)$.

This is a curve parametrized by

$$\Phi : \mathbb{P}^1_{\mathbb{R}} \to C \subset \mathbb{P}^2_{\mathbb{R}}$$
$$[U, V] \mapsto [U^2, UV, V^2]$$

This is because

$$[X, Y, Z] \sim [X^2, XY, XZ] = [X^2, XY, Y^2]$$

and so let $U = X, V = Y$. Note that if $X \mapsto X + Z$, then $U = X + Z$.

Inverse map $\Psi = \Phi^{-1}$, $\Psi : C \to \mathbb{P}^1_{\mathbb{R}}$ given by

$$[X, Y, Z] \mapsto [X, Y] = [Y, Z]$$

$[X, Y]$ defined if $X \neq 0$, $[Y, Z]$ defined if $Z \neq 0$.

Φ, Ψ are inverse isomorphisms of varieties.

cf. Ch. 2 "Cubics and the group law" of Reid (2013) [70].

cf. Sec. 2.1 "Examples of parametrized cubics" in Ch. 2 of Reid (2013) [70].

Nodal cubic: $C : (y^2 = x^3 + x^2) \subset \mathbb{R}^2$, is image of map $\varphi : \mathbb{R}^1 \to \mathbb{R}^2$, $t \mapsto (t^2 - 1, t^3 - t)$, since

$$(t^2-1)^3 + (t^2-1)^2 = t^6 - 3t^4 + 3t^2 - 1 + t^4 - 2t^3 + 1 = t^6 - 2t^4 + t^2 = t^2(t^4 - 2t^2 + 1) = t^2(t^2-1)^2 = y^2$$

Cuspidal cubic $C : (y^2 = x^3) \subset \mathbb{R}^2$ is image of $\varphi : \mathbb{R}^1 \to \mathbb{R}^2$, $t \mapsto (t^2, t^3)$

46.0.4 Curve $y^2 = x(x-1)(x-\lambda)$ has no rational parametrization

cf. Sec. 2.2 "Curve $y^2 = x(x-1)(x-\lambda)$" in Ch. 2 of Reid (2013) [70].

$f = f(t)$ *rational function* if it's a quotient of 2 polynomials.

Lemma 5. *Let $\overline{K}$ algebraically closed field, $p, q \in \overline{K}[t]$ coprime elements (i.e. if $\exists x$ s.t. p/x, $q/x \in \overline{K}$ (i.e. $x|p$, $x|q$), then $x = 1$),*
assume 4 distinct linear combinations (i.e. $\lambda p + \mu q$ for 4 distinct ratios $(\lambda : \mu) \in \mathbb{P}^1 K$) are squares in $\overline{K}[t]$, then $p, q \in \overline{K}$

cf. Lemma 2.3 of Reid (2013) [70]

Proof. (Fermat's method of "infinite descent")

Without loss of generality,

$$p' = ap + bq$$
$$q' = cp + dq$$

$a, b, c, d \in K$, $ad - bc \neq 0$.

Hence, assume 4 given squares are

$$p, p - q, p - \lambda q, q$$

i.e. $\lambda p + \mu q$, for $\lambda = 1, \mu = 0$; $\lambda = 1, \mu = -1$; $\lambda = 1, \mu = -\lambda$; $\lambda = 0, \mu = 1$

Since a, b, c, d arbitrary linear transformation.

Then $p = u^2, q = v^2$, $u, v \in \overline{K}[t]$ are coprime, with

$$\max(\deg u, \deg v) < \max(\deg p, \deg q)$$

Suppose $\max(\deg p, \deg q) > 0$ and is minimal among all p, q satisfying lemma condition.

Then

$$p - q = u^2 - v^2 = (u - v)(u + v)$$
$$p - \lambda q = u^2 - \lambda v^2 = (u - \mu v)(u + \mu v)$$

where $\mu = \sqrt{\lambda}$, are squares in $\overline{K}[t]$.

So by u, v being coprime,

Then $u - v, u + v, u - \mu v, u + \mu v$ are squares.

This contradicts minimality of $\max(\deg p, \deg q)$

□

Theorem 35 ($y^2 = x(x-1)(x-\lambda)$ has no rational parametrization)**.** *Let K be field of characteristic $\neq 2$, let $\lambda \in K$, $\lambda \neq 0, 1$; let $f, g \in K(t)$ be rational functions s.t.*

$$f^2 = g(g-1)(g-\lambda)$$

Then $f, g \in K$.

EY (20181229). Recall, *characteristic of ring* R (e.g. field), $\text{char}(K)$, smallest number of times 1 must using ring's multiplicative identity 1 in a sum to get additive identity (0).

$\text{char}(K) = 0$ for case that $\underbrace{1 + \cdots + 1}_{n} = \sum_{i=1}^{n} 1 \neq 0 \quad \forall\, n \in \mathbb{Z}^+$.

Theorem 35 is equivalent to $\nexists$ nonconstant map $\mathbb{R}^1 \to C : (y^2 = x(x-1)(x-\lambda))$ given by rational functions.

Proof. $K[t]$ UFD; unique factorization domain (given).

EY: 20181229, recall the definitions: integral domain - nonzero commutative ring in which product of any 2 nonzero elements is nonzero.

unique factorization domain is an integral domain R s.t. $\forall\, x \in R$, $x \neq 0$, x can be written as

$$x = up_1p_2 \dots p_n, \quad n \geq 0$$

with irreducible elements p_i of R, unit u.

$$\implies \begin{matrix} f = r/s & r, s \in K[t] \text{ and coprime} \\ g = p/q & p, q \in K[t] \text{ and coprime} \end{matrix}$$

$$\implies f^2 = g(g-1)(g-\lambda) = \frac{r^2}{s^2} = \frac{p}{q}\left(\frac{p-q}{q}\right)\left(\frac{p-\lambda q}{q}\right) \implies r^2q^3 = s^2p(p-q)(p-\lambda q)$$

r, s are coprime, so RHS s^2 must divide q^3.

p, q are coprime, LHS q^3 must divide s^2

EY (20181229): observe that LHS and RHS are different and equal. How to get them into the same form? Try to divide both sides!

$$\implies s^2 | q^3 \text{and } q^3 | s^2, \text{ so } s^2 = aq^3 \text{ with } a \in K$$

Then $aq = (s/q)^2$ is square in $K[t]$

Then $r^2 = ap(p-q)(p-\lambda q)$

Consider factorization into primes $\implies$ nonzero constants $b, c, d \in K$, s.t. $bp, c(p-q), d(p-\lambda q)$ are all squares in $K[t]$.

Let algebraic closure $\overline{K}$ (algebraic extension of K s.t. $\overline{K}$ algebraically closed, i.e. $\forall$ nonconstant $f(x) \in K[x]$ has a root in K).

Then $\forall\, p, q \in \overline{K}(t)$, by lemma, $p, q \in \overline{K}$. Then $r, s \in \overline{K}$. Then $\boxed{f, g \in \overline{K}}$.

□

cf. Sec. 2.4 "Linear systems" in Ch. 2 of Reid (2013) [70].

Let $S_d \equiv \{$ forms of degree d in (X, Y, Z) $\}$; recall form is just a homogeneous polynomial.

$\forall\, F \in S_d$, $\exists$ unique form for F: $F = \sum a_{ijk} X^i Y^j Z^k$, $a_{ijk} \in K$, and $\sum \equiv \sum_{\substack{i,j,k \geq 0 \\ i+j+k=d}}$.

$\implies S_d$ is K-vector space with basis $\{Z^d, XZ^{d-1}, YZ^{d-1}, \dots X^{d-2}Y^2 \dots Y^d\}$, where

$$\boxed{\dim S_d = \binom{d+2}{2}}$$

(to see this, imagine d stars, 2 bars, and the 2 bars distinguish which are X's, Y's, or Z's).

For $P_1 \dots P_n \in \mathbb{P}^2$, let

$$S_d(P_1 \dots P_n) = \{F \in S_d | F(P_i) = 0 \quad \forall\, i = 1 \dots n\} \subset S_d$$

$\forall$ condition $F(P_i) = 0$ (e.g. $F(X_i, Y_i, Z_i) = 0$, where $P_i = (X_i, Y_i, Z_i)$) is 1 linear condition on F, so $S_d(P_1 \dots P_n)$ is a vector space of dim $\geq \binom{d+2}{2} - n$

Lemma 6 (Special case of Nullstellensatz). *(i) Let $L \subset \mathbb{P}^2_K$ be a line; if $F \equiv 0$ on L, then F divisible in $K[X, Y, Z]$ by equation of L, i.e. $F = H \cdot F'$, where H is equation of L, and $F' \in S_{d-1}$.*

(ii) Let $C \subset \mathbb{P}^2_K$ be nonempty nondegenerate conic; if $F = 0$ on C, then F divisible in $K[X, Y, Z]$, by equation of C, i.e. $F = QF'$, where Q is equation of C, and $F' \in S_{d-2}$.

cf. Lemma 2.5 of Reid (2013).

Proof. (i) By change of coordinates, assume $H = X$, Then, $\forall\, F \in S_d$, $\exists !\, F = X \cdot F'_{d-1} + G(Y, Z)$, since, just gather together all monomials involving X into 1st. summand, and what's left must be a polynomial Y, Z.

Since
$F = 0$ on L, $F(0) = 0 = 0 \cdot F'_{d-1} + G(Y, Z) \implies G(Y, Z) = 0 \quad \forall\, Y, Z.$

Otherwise, if $G(Y, Z) \neq 0$, then it has at most d zeros on $\mathbb{P}^1_K$, whereas if K is infinite, then so is $\mathbb{P}^1_K$.

(ii) By change of coordinates $Q = XZ - Y^2$,
Consider why

$$F = QF'_{d-2} + A(X, Z) + YB(X, Z)$$

where $d - 2$ in F'_{d-2} denotes the *degree* of the polynomial (to be $d - 2$).

This is because if $Y^2 = XZ - Q$, then $F(Y^2 = XZ - Q)$ has degree ≤ 1 in Y, and so would have the form

$$F(Y^2 = XZ - Q) = A(X, Z) + YB(X, Z)$$

C is a parametrized conic given by

$$X = U^2, Y = UV,\ Z = V^2$$

so that,
$F = 0$ on $C \iff A(U^2, V^2) + UVB(U^2, V^2) = 0$ on C
$\implies A(U^2, V^2) + UVB(U^2, V^2) = 0 \in K[U, V].$
$\implies A(X, Z) = B(X, Z) = 0$

Since here the last equality comes by considering separately terms of even and odd degrees in form

$$A(U^2, V^2) + UVB(U^2, V^2)$$

□

cf. Exercises to Ch. 2, Reid (2013)

Exercise 2.2. Let $\varphi : \mathbb{R}^1 \to \mathbb{R}^2$.

$$t \mapsto (t^2, t^3)$$

$\forall$ polynomial $f \in \mathbb{R}[X, Y]$, s.t. $f = 0$ on image $C = \varphi(\mathbb{R}^1)$, f divisible by $Y^2 - X^3$.

Proof. Given $\varphi(t) = (t^2, t^3) = (x, y)$, then $y^2 = x^3 \quad \forall\, t \in \mathbb{R}$, or $y^2 - x^3 = 0$.
Let $q = q(x, y) = y^2 - x^3 \in K[x, y]$.
Suppose f of degree d.
Then

$$f = qf'_{d-2} + a(x) + yb(x)$$

This is because, if $y^2 = q - x^3$, $f(y^2 = q - x^3)$ has degree ≤ 1 in y, so would have the previous form.
Now

$$f(y^2 = q - x^3) = 0 = 0 + a(x) + yb(x)$$

$f = 0$ on $C = \varphi(\mathbb{R}^1) \implies a(x) + yb(x) = 0 = a(t^2) + t^3 b(t^2) = 0.$
Suppose for $t_1 > 0$, $t_1^3 b(t_1^2) = -a(t_1^2)$.
Consider $-t_1 < 0$:

$$\implies -t_1^3 b(t_1^3) = -a(t_1^2) \implies a(t_1^2) = 0 \quad \forall\, t_1 > 0$$

Then $b(t_1^2) = 0\ \forall\, t_1 > 0$.
$\implies f = qf'_{d-2}$ where $q = y^2 - x^3$.
K needs to have "negative numbers" (i.e. additive inverses) to exist, for this proof to work. □

Part XIII
Algebraic Geometry applications

47 References for Algebraic Geometry applications

A tutorial on geometric programming Stephen Boyd, Seung-Jean Kim, Lieven Vandenberghe, Arash Hassibi
https://math.berkeley.edu/~bernd/math275old.html
https://arxiv.org/pdf/math/0103170.pdf
https://www.researchgate.net/publication/257426699_Lossless_Convexification_of_Nonconvex_Control_Bound_and_Pointing_Constraints_in_the_Soft_Landing_Optimal_Control_Problem

Part XIV
Algebraic Topology

cf. Bredon (1997) [54]

48 Simplicial Complexes

cf. pp. 245, from Sec. 21 Simplicial Complexes of Ch. 4 Homology Theory in Bredon (1997) [54]
$\mathbf{v}_0, \dots \mathbf{v}_n \in \mathbb{R}^\infty$, "affinely independent" if they span an affine n-plane, i.e.

$$\text{if } \left(\sum_{i=0}^{n} \lambda_i \mathbf{v}_i = 0, \sum_{i=0}^{n} \lambda_i = 0 \right), \text{ then } \implies \forall \lambda_i = 0$$

If not, then, e.g. $\lambda_0 \neq 0$, assume $\lambda_0 = -1$, and solve the equations to get

$$\mathbf{v}_0 = \sum_{i=1}^{n} \lambda_i \mathbf{v}_i$$
$$\sum_{i=1}^{n} \lambda_i = 1$$

i.e. $\mathbf{v}_0$ is in affine space spanned by $\mathbf{v}_1 \dots \mathbf{v}_n$.
If $\mathbf{v}_0, \dots \mathbf{v}_n$ affinely independent, then

$$\sigma = (\mathbf{v}_0, \dots \mathbf{v}_n) = \{ \sum_{i=0}^{n} \lambda_i \mathbf{v}_i | \sum_{i=0}^{n} \lambda_i = 1, \lambda_i \geq 0 \} \tag{143}$$

is "affine simplex" spanned by $\mathbf{v}_i$; also convex hull of $\mathbf{v}_i$.
$\forall k \leq n$, k-face of σ is any affine simplex of form $(\mathbf{v}_{i_1}, \dots \mathbf{v}_{i_k})$, where vertices all distinct, so are affinely independent.

Definition 137. (geometric) simplicial complex $K :=$ collection of affine simplices s.t.

1. $\sigma \in K \implies$ any face of $\sigma \in K$; and
2. $\sigma, \tau \in K \implies \sigma \bigcap \tau$ is a face of both σ and τ, or $\sigma \bigcap \tau = \emptyset$

If K simplicial complex, $|K| = \bigcup \{\sigma | \sigma \in K\} \equiv$ "polyhedron" of K

Definition 138 (Def. 21.2 of Bredon (1997) [54])**.** polyhedron := space X if $\exists$ homeomorphism $h : |K| \xrightarrow{\approx} X$ for some simplicial complex K. h, K is triangulation of X; (map h, complex K)

Let K finite simplicial complex.
Choose ordering of vertices $\mathbf{v}_0, \mathbf{v}_1 \dots$ of K.
If $\sigma = (\mathbf{v}_{\sigma_0}, \dots \mathbf{v}_{\sigma_n})$ is simplex of K, where $\sigma_0 < \cdots < \sigma_n$, then
let $f_\sigma : \Delta_n \to |K|$ be

$$f_\sigma = [\mathbf{v}_{\sigma_b}, \dots \mathbf{v}_{\sigma_n}]$$

in notation of Def. 1.2. Bredon (1997) [54].

Then this gives CW-complex structure on $|K|$ with f_σ as characteristic maps.

Part XV
Graphs, Finite Graphs

49 Graphs, Finite Graphs, Trees

Serre (1980) [56]

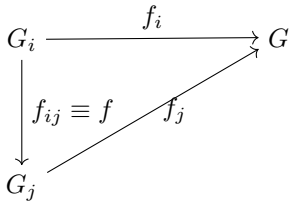
cf. Chapter I. Trees and Amalgams, Section 1 Amalgams, Subsection 1.1 Direct limits of Serre (1980) [56]

Let $(G_i)_{i \in I}$, family of groups.
$\forall$ pair (i, j), let F_{ij} = set of homomorphisms of G_i into G_j
Want: group $G = \varinjlim G_i$ and

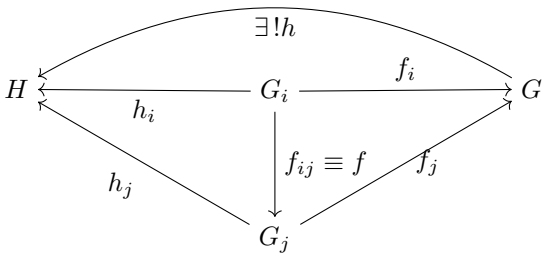
$$\{f_i | f_i : G_i \to G\} \text{s.t. } f_j \circ f = f_i \quad \forall f \in F_{ij}$$

group G and family $\{f_i\}$ universal in that
(*) if H group, if $\{h_i | h_i : G_i \to H; h_j \circ f = h_i \qquad \forall f \in F_{ij}\}$,
then $\exists ! h : G \to H$ s.t. $h_i = h \circ f_i$
i.e. $\mathrm{Hom}(G, H) \simeq \varprojlim \mathrm{Hom}(G_i, H)$, the inverse limit being taken relative to F_{ij}.
i.e. G direct limit of G_i relative to the F_{ij}.

EY : 20170918 this is my rewrite/reinterpretation:
Let $(G_i)_{i \in I}$, $\forall (i, j) \in I^2$, let $F_{ij} = \{f \equiv f_{ij} | f : G_i \to G_j, f \text{ homomorphism of } G_i \text{ into } G_j\}$.
Given group $G = \varinjlim G_i$ (for fixed i), $\{f_i | f_i : G_i \to G | f_j \circ f = f_i \quad \forall f \in F_{ij}\}$, i.e.



Then G, $\{f_i | f_i : G_i \to G | f_j \circ f = f_i \quad \forall f \in F_{ij}\}$ **universal**
if $\forall$ group H, $\forall \{h_i | h_i : G_i \to H | h_j \circ f = h_i \quad \forall f \in F_{ij}\}$,



then $\exists !$ $h : G \to H$, s.t. $h_i = h \circ f_i$ i.e.

Proposition 35. $\exists !$ *pair G, family $(f_i)_{i \in I}$, i.e. (pair consisting of $G, (f_i)_{i \in I}$, unique up to unique isomorphism.*

Proof. Define G by generators and relations.
Take generating family to be disjoint union of those for G_i.
relations - xyz^{-1} where $x, y, z \in G_i$, $z = xy \in G_i$
xy^{-1} where $x \in G_i$, $y \in G_j$, $y = f(x)$ for at least $f \in F_{ij}$.

Thus, existence of $G, \{f_i\}$.

G represents functor $H \mapsto \varprojlim \mathrm{Hom}(G_i, H)$.

Thus, uniqueness (also from universal property). □

e.g. groups A, G_1, G_2, homomorphisms $f_1 : A \to G_1$.

$$f_2 : A \to G_2$$

G obtained by amalgamating A in G_1, G_2 by $f_1, f_2 \equiv G_1 *_A G_2$.

1 can have $G = \{1\}$, even though f_1, f_2 non-trivial.

Application: (Van Kampen Thm.)

Let topological space X be covered by open U_1, U_2.

Suppose $U_1, U_2, U_{12} = U_1 \bigcap U_2$ arcwise connected.

Let basept. $x \in U_{12}$.

Then $\pi_1(X; x)$ obtained by taking 3 groups

$$\pi_1(U_1; x), \pi_1(U_2; x), \pi_1(U_{12}; x)$$

and amalagamating them according to homomorphism

$$\pi_1(U_{12}; x) \to \pi_1(U_1; x)$$
$$\pi_1(U_{12}; x) \to \pi_1(U_2; x)$$

Exercise 1. Let homomorphisms $f_1 : A \to G_1$ amalgam $G = G_1 *_A G_2$.

$$f_2 : A \to G_2$$

Define subgroups A^n, G_1^n, G_2^n, of A, G_1, G_2 recursively by

$$A^1 = \{1\}$$
$$G_1^1 = \{1\}$$
$$G_2^1 = \{1\}$$

A^n = subgroup of A generated by $f_1^{-1}(G_1^{n-1})$ and $f_2^{-1}(G_2^{n-1})$

$$G_1^n = \text{subgroup of } G_i \text{ generated by } f_i(A^n)$$

Let A^∞, G_i^∞ be unions of A^n, G_i^n resp.

Show that f_i defines injection $A/A^\infty \to G_i/G_i^\infty$.

So the amalgamation is $G \simeq G_1/G_1^\infty *_{A/A^\infty} G_2/G_2^\infty$.

Take the first induction case (for intuition about the solution).

$$A^2 = \langle f_1^{-1}(G_1^1), f_2^{-1}(G_2^1)\rangle = \langle f_1^{-1}(\{1\}), f_2^{-1}(\{1\})\rangle$$
$$G_i^2 = f_i(A^2)$$

Let $f_i(a) = f_i(b) \in G_i/G_i^\infty$; $a, b \in A/A^\infty$.

Then since $f_i(a), f_i(b) \in G_i/G_i^\infty$, $f_i(a), f_i(b) \in \{gG_i^\infty | g \in G_i\}$ (quotient is defined to be the set of all left cosets of G_i^∞, which has to be a normal subgroup for G_i/G_i^∞ to be a quotient group).

Since $a, b \in A/A^\infty$, suppose we take $a, b \in A$.

And suppose we take

$$f_i(a) = f_i(a)G_i^\infty = f_i(a)f_i(A^{n_a}) = f_i(aA^{n_a})$$
$$f_i(b) = f_i(b)G_i^\infty = f_i(b)f_i(A^{n_b}) = f_i(bA^{n_b})$$

Taking f_i^{-1} (recall for group homomorphisms, they map inverse of element of 1st. group to inverse of image of this element).

$aA^{n_a} = bA^{n_b} \in A/A^\infty$ (This is okay as we've "quotiented out A^∞; so indeed, they're equal)

cf. Subsection 1.2 Structure of amalgams of Serre (1980) [56]

Suppose given group A, family of groups $(G_i)_{i \in I}$, and, $\forall\, i \in I$, injective homomorphism $A \to G_i$.

$*_A G_i \equiv$ direct limit (cf. no. 1.1) of family (A, G_i) with respect to these homomorphisms, call it *sum* (in category theory sense, i.e. product) of G_i with A amalgamated.

e.g. $A = \{1\}$,

$*G_i \equiv$ free product of G_i.

49.0.1 reduced word

$\forall\, i \in I$, choose set S_i of right coset representations of G_i modulo A,
assume $1 \in S_i$,
$(a, s) \mapsto as$ is bijection of $A \times S_i$ onto G_i,
$A \times (S_i - \{1\}) \to G_i - A$ (onto)
Let $\mathbf{i} = (i_1 \dots i_n)$, $n \geq 0$, $i_j \in I$, s.t.

$$i_m \neq i_{m+1} \text{ for } 1 \leq m \leq n-1 \tag{144}$$

cf. (T) of Serre (1980) [56].
So *reduced word* m is defined as

$$m = (a; s_1 \dots s_n)$$

where $a \in A, s_1 \in S_{i_1} \dots s_n \in S_{i_n}$, and $s - j \neq 1\, \forall\, j$.
$f \equiv$ canonical homomorphism of A into group $G = *_A G_i$
$f_i \equiv$ canonical homomorphism of G_i into group $G = *_A G_i$
EY : 20170611 (Further explanations, basic examples, from me):
Given $A, \{G_i\}_{i \in I}$, injective (group) homomorphisms $\{f_i : A \to G_i\}_i$.
$G_i \backslash f_i(A) = \{f_i(A)g | g \in G_i\}$.
Right coset representation of $f_i(A)g \mapsto g$.

e.g. A, G_1, G_2, $f_1 : A \to G_1$.
$f_2 : A \to G_2$

$$G_1 \backslash f_1(A) = \{f_1(A)g | g \in G_1\}$$
$$G_2 \backslash f_2(A) = \{f_2(A)g | g \in G_2\}$$

$\mathbf{i} = (i_1 \dots i_n)$, $i_j \in I$, $i_m \neq i_{m+1}$ for $1 \leq m \leq n-1$.
Consider $(1212 \dots 12)$
$m = (a; f_1 g_2 f_3 g_4 \dots f_{2n-1}, g_{2n})$ where f's $\in S_1 \subset G_1$, g's $\in S_2 \subset G_2$.
and so

Definition 139 (reduced word)**. reduced word** of type $\mathbf{i}$, m,

$$m = (a; s_1 \dots s_n) \tag{145}$$

where $a \in A, s_1 \in S_{i_1}, \dots s_n \in S_{i_n}$, $s_j \neq 1 \quad \forall\, j$,
$\mathbf{i} = (i_1 \dots i_n)$, $i_j \in I$, s.t. $i_m \neq i_{m+1}$ for $1 \leq m \leq n-1$,
with $S_i = \{g | g \in f_i(A)g \in f_i(A)G_i\}$

Theorem 36 (1 of Serre (1980) [56])**.** $\forall\, g \in G$, $\exists$ *sequence* $\mathbf{i}$ *s.t.* $i_m \neq i_{m+1}$ *for* $1 \leq m \leq n-1$ *and reduced word*

$$m = (a; s_1 \dots s_n)$$

of type $\mathbf{i}$ *s.t.*

$$g = f(a) f_{i_1}(s_1) \dots f_{i_n}(s_n)$$

Furthermore, $\mathbf{i}$ and m unique.
Remark. Thm. 1 implies $f; f_i$ injective.
Then identify A and G_i with images $f(A), f_i(G_i)$ in G, and reduced decomposition (*) of $g \in G$

$$g = as_1 \dots s_n, \quad a \in A,\, s_1 \in S_{i_1} - \{1\} \dots s_n \in S_{i_n} - \{1\}$$

Likewise, $G_i \bigcap G_j = A$ if $i \neq j$.
In particular, $S_i - \{1\}$ pairwise disjoint in G.

Proof. Let $X_i \equiv$set of reduced words of type $\mathbf{i}$, $X = \coprod X_i$.
Make G act on X.
In view of universal property of G, sufficient to make $\forall\, i, G_i$ act,
check action induced on A doesn't depend on i

Suppose then that $i \in I$, and let Y_i = set of reduced words of form $(1; s_1 \dots s_n)$, with $i_1 \neq i$.
EY : 20170611
Recall that

$$S_i = \{g | g \in f_i(A)g \in f_i(A)G_i\}$$
$$A \times S_i \to G_i \text{ onto}$$
$$A \times (S_i - \{1\}) \to G_i - A \text{ onto}$$
$$(a, s) \mapsto as \text{ bijection}$$

Let Y_i = set of reduced words of form $(1; s_1 \dots s_n) = \{(1; s_1 \dots s_n) | 1 \in A; s_1 \in S_{i_1} \dots s_n \in S_{i_n}; \mathbf{i} = (i_1 \dots i_n), i_j \in I \text{ s.t. } i_m \neq i_{m+1} \text{ for } 1 \leq m \leq n-1\}$.

$$A \times Y_i \to X = \coprod_i X_i$$
$$(a, (1; s_1 \dots s_n)) \mapsto (a; s_1 \dots s_n)$$
$$A \times \{S_i - \{1\}) \times Y_i \to X$$
$$((a, s), (1; s_1 \dots s_n)) \mapsto (a; s, s_1 \dots s_n)$$

and remember that X_i = set of reduced words of type $\mathbf{i}$.

It's clear that this yields a bijection $A \times Y_i \bigcup A \times (S_i - \{1\}) \times Y_i \to X$.

Let $x \in X$. Then $x \in X_{\mathbf{i}}$ for some $\mathbf{i}$. So x is a reduced word of type $\mathbf{i}$: $x = (a; s_1 \dots s_n)$. Then clearly $x = (a; s_1 \dots s_n) \mapsto (a, (1; s_1 \dots s_n)) \in A \times Y_i$. So $\forall\, x \in X$, $\exists (a, (1; s_1 \dots s_n) \in A \times Y_i)$. So we've proven, for part of bijection, the mapping going one way.

Identify $A \bigcup A \times (S_i - \{1\})$ with G_i, with
$a \in A$, $A \times (S_i - \{1\}) \ni (a, s_i)$.

G_i act on $G_i \times Y_i$:

$$G_i \times (G_i \times Y_i) \mapsto G_i \times Y_i$$
$$g' \cdot (g, y) = (g'g, y)$$

(group $G_i, Y_i \equiv$ set of reduced words)

Transfer this to an action of G_i on X by means of θ_i:

$$G_i \times X \mapsto X$$
$$g' \cdot (a, s_1, \dots, s_n) = (g'a, s_1, \dots s_n)$$

$X \equiv$ set of reduced words of type $\mathbf{i}$.

restriction to $A \Longrightarrow a' \cdot (a; s_1, \dots s_n) = (a'a; s_1, \dots s_n)$, which doesn't depend on i.
We've now defined action of G on X.

□

cf. pp. 13, Sec. 2. Trees, 2.1 Graphs of Serre (1980) [56]

Definition 140 (1. of Serre (1980) [56])**. graph** $\Gamma = (X, Y, Y \to X \times X, Y \to Y)$, where set $X =$ vert Γ
set $Y =$ edge Γ

$$Y \to X \times X$$
$$y \mapsto (o(y), t(y))$$
$$Y \to Y$$
$$y \mapsto \overline{y}$$

s.t. $\forall\, y \in Y$, $\overline{\overline{y}} = y$, $\overline{y} \neq y$, $o(y) = t(\overline{y})$.
vertex $P \in X$ of Γ.

(oriented) edge $y \in Y$, $\overline{y} \equiv$ inverse edge.
origin of $y :=$ vertex $o(y) = t(\overline{y})$.
terminus of $y :=$ vertex $t(y) = o(\overline{y})$
extremities of $y := \{o(y), t(y)\}$
If 2 vertices **adjacent**, they're extremities of some edge.
orientation of graph $\Gamma = Y_+ \subset Y =$ edge Γ s.t. $Y = Y_+ \coprod \overline{Y}_+$. It always exists.
oriented graph defined, up to isomorphism, by giving 2 sets X, Y_+ and $Y+ \to X \times X$.
corresponding set of edges is $Y = Y_+ \coprod \overline{Y}_+$ where $\overline{Y}_+ \equiv$ copy of Y_+

49.0.2 Realization of a Graph

cf. Realization of a Graph in Serre (1980) [56].
Let graph Γ, $X = \text{vert}\Gamma$, $Y = \text{edge}\Gamma$.
topological space $T = X \coprod Y \times [0,1]$, where X, Y provided with discrete topology.
Let R be finest equivalence relation on T for which

$$\begin{aligned} &(y,t) \equiv (\overline{y}, 1-t) \\ &(y,0) \equiv o(y) \qquad\qquad \forall\, y \in Y,\ \forall\, t \in [0,1] \\ &(y,1) \equiv t(y) \end{aligned} \tag{146}$$

quotient space $\text{real}(\Gamma) = T/R$ is *realization* of graph Γ. (realization is a functor which commutes with direct limits).
Let $n \in \mathbb{Z}^+$. Consider oriented graph of $n+1$ vertices $0, 1, \dots n$,

Definition 141. path (of length n) in graph Γ is morphism c of Path_n into Γ

orientation given by n edges $[i, i+1]$, $0 \le i < n$, $o([i,i+1]) = i$
$t([i,i+1]) = i+1$
For $n \ge 1$,
$(y_1 \dots y_n)$ sequence of edges $y_i = c([i-1, i])$ s.t.

$$t(y_i) = o(y_{i+1}), \qquad 1 \le i < n \text{ determine } c$$

If $P_i = c(i)$,
c is a path from P_0 to P_n, and P_0 and P_n are *extremities of the path* c.
pair of form $(y_i, y_{i+1}) = (y_i, \overline{y}_i)$ in path is **backtracking**.
path (of length $n-2$), from P_0 to P_n given (for $n > 2$) by $(y_1 \dots y_{i-1}, y_{i+2} \dots y_n)$
If $\exists$ path from P to Q in Γ, $\exists$ one without backtracking (by induction)
direct limit $\text{Path}_\infty = \varinjlim \text{Path}_n$ provides notion of infinite path.
$\text{Path}_\infty \ni$ infinite sequence $(y_1, y_2, \dots)$ of edges s.t. $t(y_i) = o(y_{i+1}) \quad \forall\, i \ge 1$.

Definition 142 (connected graph; Def. 3 of Serre (1980) [56])**.** graph connected if $\forall$ 2 vertices, 2 vertices are extremities of at least 1 path.
maximal connected subgraphs (under relation of inclusion) are *connected components* of graph.

49.0.3 Circuits

Let $n \in \mathbb{Z}^+$, $n \ge 1$.
Consider

set of vertices $\mathbb{Z}/n\mathbb{Z}$, orientation given by n edges $[i, i+1]$, $(i \in \mathbb{Z}/n\mathbb{Z})$ with $o([i,i+1]) = i$
$t([i,i+1]) = i+1$

Definition 143 (circuit; Def. 4 of Serre (1980) [56])**.** circuit (length n) in graph is subgraph isormorphic to Circ_n.

i.e. subgraph = path $(y_1 \dots y_n)$, without backtracking, s.t. $P_i = t(y_i)$, $(1 \leq i \leq n)$ distinct, s.t. $P_n = o(y_1)$

$n = 1$ case: Circ_1, $\mathbb{Z}/\mathbb{Z} = \{0\}$, 1 edge, $[0,1]$, $0 \in \mathbb{Z}/1\mathbb{Z}$, $o([0,1]) = 0$

$$t([0,1]) = 1$$

Note Circ_1 has automorphism of order 2, which changes its orientation, i.e.
$\exists$ automorphism $\sigma \in \mathrm{Aut}(\mathrm{Circ}_1)$ s.t. $|\sigma| = 2$, i.e. $\sigma^2 = 1$.
loop := circuit of length 1; so loop $\in \overline{\mathrm{Circ}_1}$.
path (y_1), $P_1 = t(y_1) = o(y_1)$.
$n = 2$ case: Circ_2, $\mathbb{Z}/2\mathbb{Z} = \{0,1\}$, 2 edges $[0,1], [1,2]$,

path (y_1, y_2), $(1 \leq i \leq 2)$, $P_1 = t(y_1)$

$$P_2 = t(y_2) = o(y_1)$$

49.1 Combinatorial graphs

Let $(X, S) \equiv$ simplicial complex of dim. ≤ 1, with
$X \equiv$ set
$S \equiv$ set of subsets of X with 1 or 2 elements, containing all the 1-element subsets.
associates with it a graph $\Gamma = (X, \{(P,Q)\})$.
X is its set of vertices.
edges $= \{(P,Q) \in X \times X\}$ s.t. $P \neq Q$, $\{P,Q\} \in S$, with $\overline{(P,Q)} = (Q,P)$

$$o(P,Q) = P$$
$$t(P,Q) = Q$$

In this graph, 2 edges with same origin and same terminus are equal. This is equivalent to (see following Def.)

Definition 144 (combinatorial; Def. 5 of Serre (1980) [56])**.** graph is combinatorial if it has no circuit of length ≤ 2

Conversely, it's easy to see that
every combinatorial graph Γ derived (up to isomorphism) by construction above from simplicial complex (X, S), where
$X = \mathrm{vert}\Gamma$
$S =$ set of subset $\{P,Q\}$ of X s.t. P and Q either adjacent or equal.

Part XVI
Analytical Geometry

50 Applications of Vector Algebra to Analytic Geometry

cf. Ch. 13 "Applications of Vector Algebra to Analytic Geometry", Apostol (1991) [117]

50.1 Lines in n-space

cf. Sec. 13.2 "Lines in n-space", Apostol (1991) [117].

Definition 145 (Line)**.** Let given pt. P, given A nonzero vector.
Define line $L(P; A)$

$$L(P;A) := \{P + tA | t \in \mathbb{R}\} \equiv L(P;A) = \{P + tA\} \tag{147}$$

50.1.1 Some simple properties of a straight line

cf. Sec. 13.3 "Some simple properties of straight line", Apostol (1991) [117].

Theorem 37 (Thm. 13.4 of Apostol (1991) [117])**.** *2 distinct pts. determine a line, i.e. if $P \neq Q$, $\exists!$ line containing both P and Q, namely*

$$\{P + t(P - Q)\} \tag{148}$$

Proof. Let L be line parallel through P to $Q - P$, i.e. $L = L(P; Q - P) = \{P + t(Q - P)\}$.

$P, Q \in L$ for $t = 0, 1$, respectively.

Suppose L' is another line containing P and Q. Since $P \in L'$, $L' = L'(P, A)$, $Q \in L'$ so $Q = P + At$. $Q - P = tA$. $t \neq 0$ since $Q \neq P$. So $A = \frac{Q-P}{t}$, i.e. A and $Q - P$ are parallel.

By Thm. 13.1, since $L(P; Q - P), L'(P, A)$ through same pt. P and direction vectors $Q - P, A$ parallel, then $L = L'$.

□

50.2 Planes in Euclidean n-plane

cf. Sec. 13.6 "Planes in Euclidean n-space", Apostol (1991) [117].

Definition 146. Set M of pts. in V_n is called a plane if $\exists$ pt. P and 2 linearly independent vectors A and B s.t.

$$M = \{P + sA + tB | s, t \in \mathbb{R}\} \tag{149}$$

EY 20211207: Consider the intersection of 2 planes: $M \cap N$:

$$M = \{P + sA + tB | s, t \in \mathbb{R}\}$$
$$N = \{Q + uC + vD\}$$

Let $X \in M$ and $X \in N$ so $X = P + sA + tB = Q + uC + vD$.
Let $M \cap N \neq \emptyset$. Let $X_0 \in M \cap N$, $X_0 = P + s_0A + t_0B = Q + u_0C + v_0D$.

Part XVII
Manifolds

Schuller (2015) [108]

1. (strikethrough)
2.
3. transitivity
4. totality: $a \leq b$ or $b \leq a$, $\quad \forall a, b$

Definition 147 (upper bound)**.** Schuller, original, version 1:
$u \in P$ **upper bound** to a subset $T \subseteq P$ if
$\forall t \in T : t \leq u$ (EY: in $t \leq u$, $\leq$ is for partial ordered set)

version 2, possible alternative (bottomline: not right):
(of is it $u \in P$ **upper bound** to a subset $T \subseteq P$ if $\nexists t \in T$ s.t. $u \leq t$)

51 Examples

51.1 Euclidean space

51.1.1 $\mathbb{R}^2$

Consider $\mathbb{R}^2$ as a smooth manifold.

Clearly, $\mathbb{R}^2$ is covered by 1 chart, with so-called *Cartesian* coordinates (that also happen to be global coordinates, in this particular case): $(\mathbb{R}^2, \mathcal{O}_{\text{std}}, \mathcal{A} = \{= \{(\mathbb{R}^2, \varphi_{\text{cart}})\})$, with the chart (which is a homeomorphism, by definition)

$$\begin{aligned}\varphi_{\text{cart}} &: \mathbb{R}^2 \to \mathbb{R}^2\\ \varphi_{\text{cart}} &: p \equiv (x, y) \mapsto (x, y)\end{aligned}$$

Clearly, $\varphi_{\text{cart}} = 1_{\mathbb{R}^2}$ is a homeomorphism; in fact, it's a smooth diffeomorphism.

$\mathcal{O}_{\text{std}}$ denotes the so-called "standard" topology, that open sets consists of Euclidean d-balls. I will assume this topology for this discussion of $\mathbb{R}^d$'s.

Clearly, the above discussion generalizes to $\mathbb{R}^d$.

Consider the so-called *polar* coordinates

$$\begin{aligned}x &= r\cos\varphi\\ y &= r\sin\varphi\end{aligned} \tag{150}$$

where $r \geq 0$ and $\varphi \in \mathbb{R}$.

Recall the definition of a coordinate char: a *coordinate chart* (or just chart) on M is a pair (U, φ) where U is an open subset of M, homeomorphism $\varphi : U \to \widehat{U}$, where open subset $\widehat{U} = \varphi(U) \subseteq \mathbb{R}^n$.

We want to construct a homeomorphism $\varphi_{\text{pol}}^{-1}$ from an open subset $V \subseteq \mathbb{R}^2$ to $M = \mathbb{R}^2$ according to these polar coordinates:

$$\begin{aligned}\varphi_{\text{pol}}^{-1} &: V \subseteq \mathbb{R}^2 \to \varphi_{\text{pol}}^{-1}(V) \subseteq M = \mathbb{R}^2\\ \varphi_{\text{pol}}^{-1} &: (r, \varphi) \mapsto (x, y)\end{aligned}$$

Immediately, from this choice of polar coordinates, we see problems with defining a homeomorphism, which is a bijection by definition. For instance, let $(\varphi_{\text{pol}}^{-1})^{-1} \equiv \varphi_{\text{pol}}$ (defining this highly suggestive notation), and so

$$\begin{aligned}\varphi_{\text{pol}} &: \mathbb{R}^2 \to \mathbb{R}^2\\ \varphi_{\text{pol}} &: (x, y) \mapsto (r, \varphi)\end{aligned}$$

The single point, the origin of $\mathbb{R}^2$, gets mapped to many possible values! In other words,

$$\varphi_{\text{pol}}(0, 0) = \{(r, \varphi) | r = 0 \text{ and } \varphi \in \mathbb{R}\}$$

Clearly this must be the case for any kind of coordinates involving cos's, sin's, etc. (i.e. transcendental functions). So the choice of these kinds of coordinates necessitate giving up on the possibility of a single chart covering the entire manifold of the Euclidean space.

Consider

$$\begin{aligned}V &:= \{(r, \varphi) | r > 0, \varphi \in (0, 2\pi)\} \subset \mathbb{R}^2\\ [0, \infty) \times [0, 2\pi) - V &= \{(r, \varphi) | r = 0 \text{ or } \varphi = 0\}\end{aligned}$$

Now $\varphi_{\text{pol}}^{-1}([0, \infty) \times [0, 2\pi)) = \mathbb{R}^2$, i.e. $\varphi_{\text{pol}}^{-1}$ maps $[0, \infty) \times [0, 2\pi)$ *onto* $\mathbb{R}^2$. But it maps $\varphi_{\text{pol}}^{-1}(0, \varphi)$ onto $(0, 0)$ for infinitely many values of $\varphi \in [0, 2\pi)$.

Now consider if $\varphi = 0$. Then $x = r$ for $r > 0$. So $x > 0$ and $y = 0$. If $r = 0$, then $(x, y) = (0, 0)$. Thus

$$y = 0$$

$$\varphi_{\text{pol}}^{-1}([0, \infty) \times [0, 2\pi) - V) = \{(x, y) | x \geq 0 \text{ and } y = 0\}$$

(If you have trouble seeing this, think of the union of the cases when $r = 0$ or when $\varphi = 0$). Thus

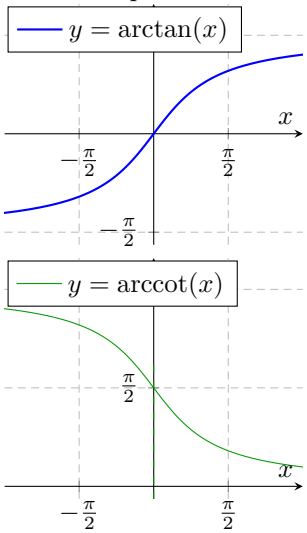
$$\varphi_{\text{pol}}^{-1}(V) = \{(x, y) | x < 0 \text{ or } y \neq 0\}$$

Thus, in summary, we can include this chart into atlas $\mathcal{A}$:

$$
\begin{aligned}
&(U, \varphi_{\text{pol}}) \in \mathcal{A} \\
&\varphi_{\text{pol}} : U \to \mathbb{R}^2 \\
U = \{(x,y) | x < 0 \text{ or } y \neq 0\} \qquad &\varphi_{\text{pol}}(x,y) = (\sqrt{x^2+y^2}, \begin{cases} \operatorname{arccot}\left(\frac{x}{y}\right) & \text{if } (x>0 \text{ and } y \geq 0) \text{ or } (x>0 \text{ and } y \leq 0) \\ \operatorname{arccot}\left(\frac{x}{y}\right) + \pi & \text{if } (x<0 \text{and } y \leq 0) \text{ or } (x \geq 0 \text{ and } y < 0) \end{cases} \\
\varphi_{\text{pol}}(U) = \{(r,\varphi) | r > 0, \varphi \in (0, 2\pi)\} \qquad &\varphi_{\text{pol}}^{-1} : \varphi_{\text{pol}}(U) \to \mathbb{R}^2 \\
&\varphi_{\text{pol}}^{-1}(r,\varphi) = (r\cos\varphi, r\sin\varphi) = (x,y)
\end{aligned} \tag{151}
$$

where $0 < \operatorname{arccot}(x) < \pi$, with limits $\lim_{x\to-\infty} \operatorname{arccot}(x) = \pi$ and $\lim_{x\to\infty} \operatorname{arccot}(x) = 0$ being well defined.

Recall the plots of arctan and arccot:



Let's try to cover $\mathbb{R}^2$ as much as possible with these "polar coordinate"-type charts. Unfortunately, we'll find out that we can't cover the origin in a well-defined (meaning with one-to-one functions) manner, so we'll have to consider $\mathbb{R}^2\backslash 0 := \mathbb{R}^2 - \{0\}$ (the Euclidean space $\mathbb{R}^2$ but without the origin).

Repeat some of the steps we did above. Consider open subset

$$U_+ = \{(x,y) | x > 0 \text{ or } y \neq 0\}$$

obtained by considering

$$
\begin{aligned}
&V_+ = \{(r,\varphi) | r > 0,\ \varphi \in (-\pi, \pi)\} \in \mathbb{R}^2 \\
&[0,\infty) \times (-\pi, \pi] - V_+ = \{(r,\varphi) | r = 0 \text{ or } \varphi = \pi\}
\end{aligned}
$$

Consider if $\varphi = \pi$. Then $x = -r \quad$ for $r > 0$, so $x < 0$ and $y = 0$. If $r = 0$, then $(x,y) = (0,0)$. So

$$
\begin{aligned}
&y = 0 \\
\varphi_{\text{pol},+}^{-1}(V_+) = \{(x,y) | x > 0 \text{ or } y \neq 0\}
\end{aligned}
$$

$$
\begin{aligned}
&(U_+, \varphi_{\text{pol},+}) \in \mathcal{A} \\
&U_+ = \{(x,y) | x > 0 \text{ or } y \neq 0\} \\
&\varphi_{\text{pol},+}(U) = \{(r,\varphi) | r > 0, \varphi \in (-\pi, \pi)\} \\
&\varphi_{\text{pol},+} : U_+ \to \mathbb{R}^2\backslash 0 \\
&\varphi_{\text{pol},+}(x,y) = (\sqrt{x^2+y^2}, \begin{cases} \arctan\left(\frac{y}{x}\right) & \text{if } (x>0 \text{ and } y \geq 0) \text{ or } (x>0 \text{ and } y \leq 0) \\ \arctan\left(\frac{y}{x}\right) + \pi & \text{if } (x \leq 0 \text{ and } y > 0)) \\ \arctan\left(\frac{y}{x}\right) - \pi & \text{if } (x \leq 0 \text{ and } y < 0) \end{cases}) \\
&\varphi_{\text{pol},+}^{-1} : \varphi_{\text{pol},+}(U) \to \mathbb{R}^2\backslash 0 \\
&\varphi_{\text{pol},+}^{-1}(r,\varphi) = (r\cos\varphi, r\sin\varphi) = (x,y)
\end{aligned} \tag{152}
$$

where $\pi/2 < \arctan(x) < \pi/2$, with limits $\lim_{x\to-\infty}\arctan(x) = -\pi/2$ and $\lim_{x\to\infty}\arctan(x) = \pi/2$ being well defined.

51.2 Spheres

cf. Example 1.4 (Spheres), Lee (2012) [76], pp. 5

$\forall\, n \geq 0$, $n \in \mathbb{Z}$, n-sphere $\mathbb{S}^n$ is Hausdorff and second-countable because it's topological subspace of $\mathbb{R}^{n+1}$.
$\forall\, i = 1, \dots n+1$, let $U_i^+ = \{(x^1, \dots x^{n+1}) \in \mathbb{R}^{n+1} : x^i > 0\}$.
Similarly, let $U_i^- = \{(x^1, \dots x^{n+1}) \in \mathbb{R}^{n+1} : x^i < 0\}$

Let $f : \mathbb{B}^n \to \mathbb{R}$ be $f(u) = \sqrt{1 - |u|^2}$.

$\forall\, i = 1, \dots n+1$, $U_i^+ \cap \mathbb{S}^n$ is graph of function

$$x^i = f(x^1 \dots \widehat{x}^i \dots x^{n+1})$$

where $\widehat{x}^i$ indicates x^i is omitted,
since

$$\sqrt{(x^1)^2 + (x^{n+1})^2} = 1^2 \text{ or } (x^i)^2 = 1 - \sum_{\substack{j\neq i \\ i=1}}^{n+1} (x^j)^2 \Longrightarrow x^i = f(x^1, \dots \widehat{x}^i, \dots x^{n+1})$$

Similarly, $U_i^i \cap \mathbb{S}^n$ is graph of

$$x^i = -f(x^1 \dots \widehat{x}^i \dots x^{n+1})$$

sicne $x^i < 0$.

Thus, $\forall$ subset $U_i^\pm \cap \mathbb{S}^n$ is locally Eucliean of dim. n nad maps

$$\varphi_i^\pm : U_i^\pm \cap \mathbb{S}^n \to \mathbb{B}^n$$
$$\varphi_i^\pm(x^1 \dots x^{n+1}) = (x^1 \dots \widehat{x}^i \dots x^{n+1})$$

are graph coordinates for $\mathbb{S}^n$.

Since $\forall$ pt. $\in \mathbb{S}^n$ is in domain of at least 1 of these $2n+2$ charts, $\mathbb{S}^n$ is a topological n-manifold.

cf. Example 1.31 (Spheres), Lee (2012) [76], pp. 20,

$\forall\, i = 1 \dots n+1$, let $(U_i^\pm, \varphi_i^\pm)$ denote graph coordinate charts, s.t.

$$U_i^\pm = \{(x^1 \dots x^{n+1}) \in \mathbb{R}^{n+1} : x^i \gtrless 0\}$$
$$\varphi_i^\pm : U_i^\pm \cap \mathbb{S}^n \to \mathbb{B}^n$$
$$\varphi_i^\pm(x^1 \dots x^{n+1}) = (x^1, \dots \widehat{x}^i \dots x^{n+1})$$

$\forall$ distinct i, j, transition map $\varphi_i^\pm \circ (\varphi_j^\pm)^{-1}$ is computed; for $i < j$,

$$\varphi_i^\pm \circ (\varphi_j^\pm)^{-1}(u^1 \dots u^n) = (u^1, \dots \widehat{u}^i \cdots \pm\sqrt{1 - |u|^2}, \dots u^n)$$

with $\sqrt{1 - |u|^2}$ in jth position; similarly for $i = j$.

Also $\varphi_i^+ \circ (\varphi_i^-)^{-1}(x^1, \dots x^n) = \varphi_i^+(x^1 \cdots - \sqrt{1 - \sum_i (x^i)^2} \dots x^n) = (x^1 \dots x^n)$.

So $\varphi_i^+ \circ (\varphi_i)^{-1} = \varphi_i^- \circ (\varphi_i^+)^{-1} = 1_{\mathbb{B}^n}$.

$\{(U_i^\pm, \varphi_i^\pm)\}$ is a smooth atlas, defines smooth structure on $\mathbb{S}^n$. This is its **standard smooth structure**.

Problem 1, Wald (1984) [107].

(a) In Wald's notation, $f_i^+ : O_i^+ \to D$; compare Lee's notation against Wald's, respectively:

$$\varphi_i^\pm : U_i^\pm \cap S^n \to \mathbb{B}^n \equiv f_i^\pm : O_i^\pm \to D^n$$

$$\varphi_i^{\pm} \circ (\varphi_j^{\pm})^{-1}(x^1 \dots x^n) = \varphi_i^{\pm}(x^1, \cdots \pm \sqrt{1 - \sum_{k=1}^{n} (x^k)^2} \dots x^n) = (x^1 \dots \widehat{x}^i \cdots \pm \sqrt{1 - \sum_{k=1}^{n} (x^k)^2}, \dots x^n)$$

Clearly, $\varphi_i^{\pm} \circ (\varphi_j^{\pm})^{-1}$ is C^∞ by taking $\frac{\partial}{\partial x^k}$ to any order, since $\mathbb{B}^n$ doesn't include its boundary.

(b) See *stereographic projection*:

north pole $N = (0, \dots 0, 1) \in S^n \subseteq \mathbb{R}^{n+1}$
south pole $S = (0, \dots 0, -1) \in S^n \subseteq \mathbb{R}^{n+1}$
stereographic projection

$$\sigma : \mathbb{S}^n \setminus \{N\} \to \mathbb{R}^n$$
$$\sigma(x^1, \dots x^{n+1}) = \frac{(x^1, \dots x^n)}{1 - x^{n+1}}$$
$$\widetilde{\sigma} : \mathbb{S}^n \setminus \{S\} \to \mathbb{R}^n$$
$$\widetilde{\sigma}(x^1, \dots x^{n+1}) = -\sigma(-x) = \frac{x^1, \dots x^n}{1 + x^{n+1}}$$

Problem 1-7, Lee (2012) [76], Stereographic projection of spheres.

Let N denote the **north pole** $(0, \dots, 0, 1) \in \mathbb{S}^n \subseteq \mathbb{R}^{n+1}$, and let S denote the **south pole** $(0, \dots, 0, -1)$. Define the **stereographic projection** $\sigma : \mathbb{S}^n \setminus \{N\} \to \mathbb{R}^n$ by

$$\sigma(x^1, \dots x^{n+1}) = \frac{(x^1, \dots x^n)}{1 - x^{n+1}}$$

Let $\widetilde{\sigma}(x) = -\sigma(-x)$ for $x \in \mathbb{S}^n \setminus \{S\}$.

(a) So given a
north pole $N = (0, \dots 0, 1) \in S^n \subseteq \mathbb{R}^{n+1}$
south pole $S = (0, \dots 0, -1) \in S^n \subseteq \mathbb{R}^{n+1}$
stereographic projection

$$\sigma : \mathbb{S}^n \setminus \{N\} \to \mathbb{R}^n$$
$$\sigma(x^1, \dots x^{n+1}) = \frac{(x^1, \dots x^n)}{1 - x^{n+1}}$$
$$\widetilde{\sigma} : \mathbb{S}^n \setminus \{S\} \to \mathbb{R}^n$$
$$\widetilde{\sigma}(x^1, \dots x^{n+1}) = -\sigma(-x) = \frac{x^1, \dots x^n}{1 + x^{n+1}}$$

Consider $\mathbf{x} \in \mathbb{S}^n \setminus \{N\} \to \mathbb{R}^n$, $\mathbf{x} = (x^1 \dots x^{n+1})$.

$$N + t(\mathbf{x} - N) = \mathbf{x}' \Longrightarrow 1 + t(x^{n+1} - 1) = 0 \text{ or } t = \frac{1}{1 - x^{n+1}}$$
$$(x')^i = \frac{x^i}{1 - x^{n+1}} \quad \forall\, i \neq n+1 \Longrightarrow \sigma(x^1 \dots x^{n+1}) = \frac{(x^1, \dots x^n)}{1 - x^{n+1}}$$

Thus we have a stereographic projection from the north pole.

$$S + t(\mathbf{x} - S) = \mathbf{x}' \Longrightarrow 1 + t(x^{n+1} + 1) = 0 \text{ or } t = \frac{1}{1 + x^{n+1}}$$
$$(x')^i = \frac{x^i}{1 + x^{n+1}} \quad \forall\, i \neq n+1$$
$$\Longrightarrow \widetilde{\sigma}(x^1 \dots x^{n+1}) = \frac{(x^1, \dots x^n)}{1 + x^{n+1}}$$

Thus we have a stereographic projection from the south pole.

b

$$\sigma^{-1}(u^1 \dots u^n) = \frac{(2u^1 \dots 2u^n, |u|^2 - 1)}{|u|^2 + 1}$$

$$\sigma \circ \sigma^{-1}(u^1 \dots u^n) = \frac{(2u^1 \dots 2u^n)}{|u|^2 + 1} / \left(1 - \frac{|u|^2 - 1}{|u|^2 + 1}\right) = (u^1, \dots u^n)$$

$$\sigma^{-1} \circ \sigma(x^1, \dots x^{n+1}) = \sigma^{-1} \frac{(x^1 \dots x^n)}{1 - x^{n+1}} = \frac{\left(\frac{2x^1}{1-x^{n+1}}, \dots \frac{2x^n}{1-x^{n+1}}, \frac{(x^1)^2 + \dots + (x^n)^2}{(1-x^{n+1})^2} - 1\right)}{\frac{(x^1)^2 + \dots + (x^n)^2}{(1-x^{n+1})^2} + 1} =$$

$$= \frac{\left(\frac{2x^1}{1-x^{n+1}} \dots \frac{2x^n}{1-x^{n+1}}, \frac{(x^1)^2 + \dots + (x^n)^2}{(1-x^{n+1})^2} - \frac{1 - 2x^{n+1} + (x^{n+1})^2}{(1-x^{n+1})^2}\right)}{\frac{2(1-x^{n+1})}{(1-x^{n+1})^2}} = \left(x^1, \dots x^n, \frac{1 - (x^{n+1})^2 - 1 + 2x^{n+1} - (x^{n+1})^2}{\frac{2}{1-x^{n+1}}(1 - x^{n+1})^2}\right) =$$

$$= (x^1, \dots x^n, \frac{2x^{n+1}(1 - x^{n+1})}{\frac{2}{1-x^{n+1}}(1 - x^{n+1})^2}) = (x^1, \dots x^{n+1})$$

(c) Compute the transition map $\widetilde{\sigma} \circ \sigma^{-1}$. (The coordinates defined by σ or $\widetilde{\sigma}$ are called **stereographic coordinates**).

$$\widetilde{\sigma} \circ \sigma^{-1}(u^1 \dots u^n) = \widetilde{\sigma} \frac{(2u^1 \dots 2u^n, |u|^2 - 1)}{|u|^2 + 1} = \frac{\frac{2u^1}{|u|^2+1}, \dots \frac{2u^n}{|u|^2+1}}{1 + \frac{|u|^2-1}{|u|^2+1}} = \frac{\frac{2u^1}{|u|^2+1}, \dots \frac{2u^n}{|u|^2+1}}{\frac{2|u|^2}{|u|^2+1}} =$$

$$= \boxed{\left(\frac{u^1}{|u|^2}, \dots, \frac{u^n}{|u|^2}\right)}$$

51.2.1 Polar coordinates

Consider

$$\mathbf{x}(\theta, \varphi) = (s\theta c\varphi, s\theta s\varphi, c\theta)$$

$$\theta = \arccos z, \quad \theta \in [0, \pi],\ z \in [0, 1]$$

$$\varphi = \begin{cases} \arctan\left(\frac{y}{x}\right) \text{ if } x > 0,\ \varphi \in (\frac{-\pi}{2}, \frac{\pi}{2}) \\ \arctan\left(\frac{y}{x}\right) + \pi if x < 0 \text{ and } y > 0,\ \varphi \in (\frac{\pi}{2}, \pi) \\ \arctan\left(\frac{y}{x}\right) - \pi if x < 0 \text{ and } y < 0,\ \varphi \in (-\pi, \frac{-\pi}{2}) \\ \frac{\pi}{2} \text{ if } x = 0, y = 1 \\ \frac{-\pi}{2} \text{ if } x = 0, y = -1 \end{cases}$$

52 Inverse Function Theorem

Shastri (2011) had a thorough and lucid and explicit explanation of the Inverse Function Theorem [78]. I will recap it here. The following is also a blend of Wienhard's Handout 4 https://web.math.princeton.edu/~wienhard/teaching/M327/handout4.pdf

Definition 148. Let (X, a) metric space.
contraction $\phi : X \to X$ if $\exists$ constant $0 < c < 1$ s.t. $\forall\, x, y \in X$

$$d(\phi(x), \phi(y)) \leq cd(x, y)$$

Theorem 38 (Contraction Mapping Principle)**.** *Let (X, d) complete metric space. Then $\forall$ contraction $\phi : X \to X$, $\exists ! y \in X$ s.t. $\phi(y) = y$, y* fixed pt.

Proof. Recall def. of complete metric space X, X metric space s.t. $\forall$ Cauchy sequence in X is convergent in X (i.e. has limit in X).

$$
\begin{aligned}
&x_1 = \phi(x_0)\\
&x_2 = \phi(x_1)\\
\forall\, x_0 \in X,\ \text{Define}\ &\vdots\\
&x_j = \phi(x_{j-1})\\
&\vdots\\
&x_n = \phi(x_{n-1})\\
&\quad d(x_{n+1}, x_n) = d(\phi(x_n), \phi(x_{n-1})) \le c d(x_n, x_{n-1}) \le \cdots \le c^n d(x_1, x_0)
\end{aligned}
$$

for some $0 < c < 1$.

$$d(x_m, x_n) \le d(x_n, x_{n-1}) + d(x_{n-1}, x_m) \le d(x_n, x_{n-1}) + d(x_{n-1}, x_{n-2}) + \cdots + d(x_{m+1}, x_m) \le \sum_{k=n-1}^{m} c^k d(x_1, x_0)$$

Thus, $\forall\, \epsilon > 0$, $\exists\, n_0 > 0$, (n_0 large enough) s.t. $\forall\, m, n \in \mathbb{N}$ s.t. $n_0 < n < m$,

$$d(x_m, x_n) \le \sum_{k=n-1}^{m} c^k d(x_1, x_0) < \epsilon d(x_1, x_0)$$

Thus, $\{x_n\}$ Cauchy sequence. Since X complete, $\exists$ limit pt. $y \in X$ of $\{x_n\}$.

$$\phi(y) = \phi(\lim_n x_n) = \lim_n \phi(x_n) = \lim_n x_{n+1} = y$$

Since by def. of y limit pt. of $\{x_n\}$, $\forall\, \epsilon > 0$, then $\{n || x_n - y| \le \epsilon,\ n \in \mathbb{N}\}$ is infinite.

Consider $\delta > \mathbb{N}$. Consider $\{n || x_n - y| \le \delta, n \in \mathbb{N}\}$

$\exists\, N_\delta \in \mathbb{N}$ s.t. $\forall\, n > N_\delta$, $|x_n - y| < \delta$; otherwise, $\forall\, N_\delta$, $\exists\, n > N_\delta$ s.t. $|x_n - y| \ge \delta$. Then $\{n || x_n - y| \le \delta, n \in \mathbb{N}\}$ finite. Contradiction.

ϕ cont. so by def. $\forall\, \epsilon > 0$, $\exists\, \delta > 0$ s.t. if $|x_n - y| < \delta$, then $|\phi(x_n) - \phi(y)| < \epsilon$.

Pick N_δ s.t. $\forall\, n > N_\delta$, $|x_n - y| < \delta$, and so $|\phi(x_n) - \phi(y)| < \epsilon$. There are infinitely many $\phi(x_n)$'s that satisfy this, and so $\phi(y)$ is a limit pt.

$$
\begin{aligned}
&\text{If } \exists\, y_1, y_2 \in X \text{ s.t. } \phi(y_1) = y_1, \text{ then}\\
&\qquad \phi(y_2) = y_2\\
&\qquad\qquad d(y_1, y_2) = d(\phi(y_1), \phi(y_2)) \le c d(y_1, y_2) \text{ with } c < 1
\end{aligned}
$$

so $c = 1$ □

Theorem 39 (Inverse Function Theorem)**.** *Suppose open $U \subset \mathbb{R}^n$, let C^1 $f : U \to \mathbb{R}^n$, $x_0 \in U$ s.t. $Df(x_0)$ invertible.*

Then $\exists$ open neighborhoods $V \ni x_0$, $W \ni f(x_0)$ s.t. $V \subseteq U$ and $W \subseteq \mathbb{R}^n$, respectively, and s.t.

(i) $f : V \to W$ *bijection*

(ii) $g = f^{-1} : V \to U$ *differentiable, i.e.* $g = f^{-1} : W \to V$ *is* C^1

(iii) $D(f^{-1})$ *cont. on* W*.*

(iv) $Dg(y) = (Df(g(y)))^{-1} \quad \forall\, y \in W$

Also, notice that $f(g(y)) = y \,\forall\, y \in W$*.*

Proof. Consider $\widetilde{f}(x) = (Df(x_0))^{-1}(f(x+x_0) - f(x_0))$. Then
$\widetilde{f}(0) = 0$ and

$$D\widetilde{f} = (Df(x_0))^{-1}(Df(x+x_0) - 0)$$
$$D\widetilde{f}(0) = (Df(x_0))^{-1}Df(x_0) = 1$$

So let $\widetilde{f} \to f$ (notation) and so assume, without loss of generality, that $U \ni 0$, $f(0) = 0$, $Df(0) = 1$

Choose $0 < \epsilon \leq \frac{1}{2}$. Let $0 < \delta < 1$ s.t. open ball $V = B_\delta(0) \subseteq U$, and $\|Df(x) - 1\| < \epsilon$. $\forall\, x \in U$, since Df cont. at 0.

Let $W = f(V)$.

$\forall\, y \in W$, define $\phi_y : V \to \mathbb{R}^n$

$$\phi_y(x) = x + (y - f(x))$$
$$D(\phi_y)(x) = 1 + -Df(x) \quad \forall\, x \in V$$
$$\|D(\phi_y)(x)\| = \|1 - Df(x)\| \leq \epsilon < 1$$

$\forall\, x_1, x_2 \in V$, by mean value Thm. (not the equality that is only valid in 1-dim., but the inequality, that's valid for $\mathbb{R}^d$,

$$\|\phi_y(x_1) - \phi_y(x_2)\| \leq \|D(\phi_y)(x')\|\|x_1 - x_2\|$$

for some $x' = cx_2 + (1-c)x_1$, $c \in [0,1]$. V only needed to be convex set.

$$\implies \|\phi_y(x_1) - \phi_y(x_2)\| \leq \epsilon\|x_1 - x_2\|$$

Then ϕ_y contraction mapping.

Suppose $f(x_1) = f(x_2) = y$, $x_1, x_2 \in V$.

$$\phi_y(x_1) = x_1$$
$$\phi_y(x_2) = x_2$$
$$\|\phi_y(x_1) - \phi_y(x_2)\| = \|x_1 - x_2\| \leq \epsilon\|x_1 - x_2\| \quad \forall\, \epsilon > 0 \implies x_1 = x_2$$

$\implies f|_U$ injective.

$W = f(V)$, so $f : V \to W$ surjective. f bijective.

Fix $y_0 \in W$, $y_0 = f(x_0)$, $x_0 \in V$.

Let $r > 0$ s.t. $B_r(x_0) \subset V$.

Consider $B_{r\epsilon}(y_0)$. If $y \in B_{r\epsilon}(y_0)$.

$$r\epsilon > \|y - y_0\| = \|y - f(x_0)\| = \|\phi_y(x_0) - x_0\| \text{ with}$$
$$\phi_y(x) = x + (y - f(x))$$

If $x \in B_r(x_0)$,

$$\|\phi_y(x) - x_0\| \leq \|\phi_y(x) - \phi_y(x_0)\| + \|\phi_y(x_0) - x_0\| \leq \epsilon\|x - x_0\| + r\epsilon < 2r\epsilon = r$$

Thus $\phi(B_r(x_0)) = B_r(x_0)$.

By contraction mapping principle, $\exists\, a \in B_r(x_0)$, s.t. $\phi_y(a) = a$. Then $\phi_y(a) = a + (y - f(a)) = a \implies f(a) = y$.

$y \in f(V) = W$.

So $B_{r\epsilon}(y_0) \subset W$. W open.

Let $\mathrm{Mat}(n,n) \equiv$ space of all $n \times n$ matrices; $\mathrm{Mat}(n,n) = \mathbb{R}^{n^2}$.

□

There is a proof of the implicit function theorem and its various forms in Shastri (2011) [78], but I found Wienhard's Handout 4 for Math 327 to be clearer.[2]

[2] https://web.math.princeton.edu/~wienhard/teaching/M327/handout4.pdf

Theorem 40 (Implicit Function Theorem)**.** *Let open* $U \subset \mathbb{R}^{m+n} \equiv \mathbb{R}^m \times \mathbb{R}^n$
C^1 $f : U \to \mathbb{R}^n$
$(a, b) \in U$ *s.t.* $f(a, b) = 0$ *and* $D_y f|_{(a,b)}$ *invertible.*
Then $\exists$ *open* $V \ni (a, b)$, $V \subset U$
$\exists$ *open neighborhood* $W \ni a$, $W \subseteq \mathbb{R}^m$
$\exists!$ C^1 $g : W \to \mathbb{R}^n$ *s.t.*

$$\{(x, y) \in V | f(x, y) = 0\} = \{(x, g(x)) | x \in W\}$$

Moreover,

$$dg_x = -\left(d_y f\right)^{-1}\big|_{(x,g(x))} d_x f|_{(x,g(x))}$$

and g *smooth if* f*.*

Proof. Define $F : U \to \mathbb{R}^{m+n}$

$$F(x, y) = (x, f(x, y))$$

Then $F(a, b) = (a, 0)$ (given), and

$$DF = \begin{bmatrix} 1 & \\ \frac{\partial f^i(x,y)}{\partial x^j} & \frac{\partial f^i(x,y)}{\partial y^j} \end{bmatrix} \equiv \begin{bmatrix} 1 & \\ D_x f & D_y f \end{bmatrix}$$

$DF(a, b)$ invertible.
By inverse function theorem, since $DF(a, b)$ invertible at pt. (a, b),

$\exists$ open neighborhoods $V \ni (a, b) \subseteq \mathbb{R}^m \times \mathbb{R}^n$ s.t. F diffeomorphism with $F^{-1} : \widetilde{W} \to V$.

$$\widetilde{W} \ni (a, 0) \subseteq \mathbb{R}^m \times \mathbb{R}^n$$

Set $W = \{x \in \mathbb{R}^m | (x, 0) \in \widetilde{W}\}$. Then $\pi_1(\widetilde{W}) = W$ open in $\mathbb{R}^m$.
Define $g : W \to \mathbb{R}^n$,

$$g(x) = \pi_2 \circ F^{-1}(x, 0) \text{ or}$$
$$F^{-1}(x, 0) = (h(x), g(x))$$

Now $FF^{-1}(x, 0) = (x, 0) = (h(x), f(h(x), g(x)))$ so $h(x) = x \,\forall\, x \in W$, $0 = f(x, g(x))$.
Then

$$\{(x, y) \in V | f(x, y) = 0\} = \{(x, y) \in V | F(x, y) = (x, 0)\} = \{(x, g(x)) | x \in W, 0 = f(x, g(x))\}$$

Since π smooth and F^{-1} is C^1, g is C^1.
To reiterate, $f(x, g(x)) = 0$ on W.
Using chain rule while differentiating $f(x, g(x)) = 0$,

$$\partial_{x^j} f(x, g(x)) = \frac{\partial f(x, g(x))}{\partial x^k} \frac{\partial x^k}{\partial x^j} + \frac{\partial f(x, g(x))}{\partial y^k} \frac{\partial g^k(x)}{\partial x^j} = D_x f|_{(x,g(x))} + (D_y f)|_{(x,g(x))} \cdot (Dg)_x = 0 \text{ or}$$
$$(Dg)_x = -\left(D_y f\right)|_{x,g(x)} D_x f|_{(x,g(x))}$$

□

53 Immersions

Definition 149 (Immersion)**.** smooth $f : M \to N$, s.t. $Df(p) : T_p M \to T_{f(p)} N$ injective. Then f **immersion** at p.

Absil, Mahony, and Sepulchre [81] pointed out that another definition for a *immersion* can utilize the theorem that rank of $Df \equiv DF = \dim T_p M$. Indeed, recall these facts from linear algebra:

for $T : V \to W$,

It's always true that rank$T \leq V$, and

$$\text{rank}T \leq W$$

rank$T = \dim V$ iff T injective.
rank$T = \dim W$ iff T surjective.

$$\begin{array}{ccc} T_xM & \xrightarrow{DF(x)} & T_{F(x)}N = T_yN \\ \uparrow & & \uparrow \\ x \in M & \xmapsto{F} & y = F(x) \in N \end{array}$$

$$M \xrightarrow{\quad F \quad} N$$

Now

$$\dim T_xM = \dim M$$
$$\dim T_{F(x)}N = \dim N$$

And

$$\text{rank}(DF(x)) \equiv \text{rank of } F$$

I know that the notation above is confusing, but this is what all Differential Geometry books apparently mean when they say "rank of F".

Now

$$\text{rank}(DF(x)) = \dim(\text{im}(DF(x))) = \dim T_xM \text{ iff } DF(x) \text{ injective}$$

If $\forall\, x \in M$, this is the case, then F an **immersion** .

Apply the rank-nullity theorem in this case:

$$\text{rank}(DF(x)) + \dim\ker(DF(x)) = \dim T_xM = \dim M$$
$$\implies \text{rank}(DF(x)) = \dim M \leq \dim T_{F(x)}N = \dim N \text{ or } \dim M \leq \dim N$$

Now

$$\text{rank}(DF(x)) = \dim T_{F(x)}N \text{ iff } DF(x) \text{ surjective}$$

If $\forall\, x \in M$, this is the case, then F an **submersion** .

$$\text{rank}(DF(x)) = \dim T_{F(x)}N = \dim N \leq \dim M$$

Shastri (2011) has this as the "Injective Form of Implicit Function Theorem", Thm. 1.4.5, pp. 23 and Guillemin and Pollack (2010) has this as the "Local Immersion Theorem" on pp. 15, Section 3 "The Inverse Function Theorem and Immersions" [77].

Theorem 41 (Local immersion Theorem i.e. Injective Form of Implicit Function Theorem)**.** *Suppose $f : M \to N$ immersion at p, $q = f(p)$.*

Then $\exists$ local coordinates around p, q, x, y, respectively s.t. $f(x_1 \dots x_m) = (x_1 \dots x_m, 0 \dots 0)$.

Proof. Choose local parametrizations

$$\begin{array}{ccc} U \subseteq M & \xrightarrow{f} & N \supseteq V \\ \downarrow \phi & & \downarrow \psi \\ \phi(U) & \xrightarrow{f} & \psi(V) \end{array} \qquad \begin{array}{l} \phi(p) = x \\ \psi(q) = y \end{array}$$

$D(\psi f \varphi^{-1}) \equiv Df$. $Df(p)$ injective (given f immersion). $Df(p) \in \mathrm{Mat}(n,m)$

By change of basis in $\mathbb{R}^n$, assume $Df(p) = \begin{pmatrix} I_m \\ 0 \end{pmatrix}$.

Now define $G : \phi(U) \times \mathbb{R}^{n-m} \to \mathbb{R}^n$

$$G(x,z) = f(x) + (0,z)$$

Thus, $DG(x,z) = 1$ and for open $\phi(U) \times U_2$, $G(\phi(U) \times U_2)$ open.
By inverse function theorem, G local diffeomorphism of $\mathbb{R}^n$, at 0.
Now $f = G \circ \mathfrak{i}$, where $\mathfrak{i}$ is canonical immersion.

$$G(x,0) = f(x)$$
$$\implies G^{-1}G(x,0) = (x,0) = G^{-1}f(x)$$

Use $\psi \circ G$ as the local parametrization of N around pt. q. Shrink U, V so that

$$\begin{array}{ccc} U \subseteq M & \xrightarrow{\quad f \quad} & N \supseteq V \\ \Big\downarrow \phi & & \Big\downarrow \psi \circ G \\ \phi(U) & \xrightarrow{\quad \mathfrak{i} \quad} & \psi \circ G(V) \end{array}$$

□

Theorem 42 (Implicit Function Thm.)**.** *Let open subset $U \subseteq \mathbb{R}^n \times \mathbb{R}^d$, $(x,y) = (x^1 \dots x^n, y^1 \dots y^k)$ on U. Suppose smooth $\Phi : U \to \mathbb{R}^k$, $(a,b) \in U$, $c = \Phi(a,b)$*

If $k \times k$ matrix $\frac{\partial \Phi^i}{\partial y^j}(a,b)$ nonsingular, then $\exists$ neighborhoods $V_0 \subseteq \mathbb{R}^n$ of a and smooth $F : V_0 \to W_0$ s.t. $W_0 \subseteq \mathbb{R}^k$ of b

$\Phi^{-1}(c) \bigcap (V_0 \times W_0)$ is graph of F, i.e.
$\Phi(x,y) = c$ for $(x,y) \in V_0 \times W_0$ iff $y = F(x)$.

54 Submersions; Rank Theorem

cf. pp. 20, Sec. 4 "Submersions", Ch. 1 of Guillemin and Pollack (2010) [77].
Consider $X, Y \in \mathbf{Man}$, s.t. $\dim X \geq \dim Y$.

Definition 150 (submersion)**.** If $f : X \to Y$,
if $Df_x \equiv df_x$ is *surjective*, $f \equiv$ **submersion** at x.

Recall that,

$$Df_x : T_x X \to T_{f(x)} Y$$
$$\dim T_x X \geq \dim T_{f(x)} Y$$

$$\mathrm{rank} Df_x \leq \dim T_{f(x)} Y, \text{ in general, while}$$
$$\mathrm{rank} Df_x = \dim T_{f(x)} Y \text{ iff } Df_x \text{ surjective}$$

Canonical submersion is standard projection:

If $\dim X = k$, $k \geq l$,
$\dim Y = l$

$$(a_1 \dots a_k) \mapsto (a_1 \dots a_l)$$

Theorem 43 (Local Submersion Theorem)**.** *Suppose $f : X \to Y$ submersion at x, and $y = f(x)$, Then $\exists$ local coordinates around x, y s.t.*

$$f(x_1 \dots x_k) = (x_1 \dots x_l)$$

i.e. f locally equivalent to canonical submersion near x

Proof. I'll have a side-by-side comparison of my notation and the 1 used in Guillemin and Pollack (2010) [77] where I can.

For charts $(U, \phi), (V, \psi)$ for X, Y, respectively, $y = f(x)$ for $x \in X$,

$$\begin{array}{ccc} U \subseteq X & \xrightarrow{f} & Y \supseteq V \\ \downarrow{\scriptstyle \phi} & & \downarrow{\scriptstyle \psi \circ G} \\ \mathbb{R}^k & \xrightarrow{\mathfrak{i}} & \mathbb{R}^l \end{array} \qquad \begin{array}{ccc} x & \xrightarrow{f} & f(x) = y \\ \downarrow{\scriptstyle \phi} & & \downarrow{\scriptstyle \psi} \\ \phi(x) = (a^1 \dots a^k) & \xrightarrow{g} & g(\phi(x)) = g(a^1 \dots a^k) = \psi(y) \end{array}$$

Dg_x surjective, so assume it's a $l \times k$ matrix $\begin{bmatrix} \mathbf{1}_l & 0 \end{bmatrix}$.

Define

$$\begin{aligned} &G : U \subset \mathbb{R}^k \to \mathbb{R}^k \\ &G(a) \equiv G(a^1 \dots a^k) := (g(a), a_{l+1}, \dots, a_k) \end{aligned} \tag{153}$$

Now

$$DG(a) = \begin{bmatrix} \mathbf{1}_l & 0 \\ & \mathbf{1}_{k-l} \end{bmatrix} = \mathbf{1}_k \tag{154}$$

so G local diffeomorphism (at 0).

So $\exists\ G^{-1}$ as local diffeomorphism of some U' of a into $U \subset \mathbb{R}^k$.

By construction,

$$g = \mathbb{P}_l \circ G \tag{155}$$

where $\mathbb{P}_l$ is the *canonical submersion*, the projection operator onto $\mathbb{R}^l$.

$$g \circ G^{-1} = \mathbb{P}_l$$

(since G diffeomorphism)

$$\begin{array}{ccc} U \subseteq X & \xrightarrow{f} & V \subseteq Y \\ {\scriptstyle \phi^{-1} \circ G^{-1}}\uparrow & & \uparrow{\scriptstyle \psi^{-1}} \\ \mathbb{R}^k & \xrightarrow{\mathbb{P}_l} & \mathbb{R}^l \end{array}$$

for

$$\begin{array}{ccc} \phi^{-1} \circ G^{-1}(a) \equiv \phi^{-1} \circ G^{-1}(a^1 \dots a^k) = x & \xrightarrow{f} & f(x) = y = \psi^{-1}(a^1 \dots a^l) \\ {\scriptstyle \phi^{-1} \circ G^{-1}}\uparrow & & \uparrow{\scriptstyle \psi^{-1}} \\ (a^1 \dots a^k) & \xrightarrow{\mathbb{P}_l} & (a^1 \dots a^l) \end{array}$$

$\Longrightarrow$

□

"An obvious corollary worth noting is that if f is a submersion at x, then it is actually a submersion in a whole neighborhood of x." Guillemin and Pollack (2010) [77]

Suppose f submersion at $x \in f^{-1}(y)$.

By local submersion theorem

$$f(x_1 \dots x_k) = (x_1 \dots x_l)$$

Choose $y = (0, \dots, 0)$.

Then, near x, $f^{-1}(y) = \{(0, \dots 0, x_{l+1} \dots x_k)\}$ i.e. let $V \ni x$ neighborhood of x, define $(x_1 \dots x_k)$ on V.

Then $f^{-1}(y) \bigcap V = \{(0 \dots 0, x_{l+1}, \dots x_k) | x_1 = 0, \dots x_l = 0\}$.

Thus $x_{l+1}, \dots x_k$ form a coordinate system on open set $f^{-1}(y) \bigcap V \subseteq f^{-1}(y)$.

Indeed,

$$\begin{array}{ccc} U \subseteq X & \xrightarrow{f} & V \subseteq Y \\ \downarrow \phi & & \downarrow \psi \\ \mathbb{R}^k & \xrightarrow{\mathbb{P}_l} & \mathbb{R}^l \end{array} \qquad \begin{array}{ccc} x & \xmapsto{f} & f(x) = y \\ \downarrow \phi & & \downarrow \psi \\ \phi(x) = (x^1 \dots x^k) & \xmapsto{\mathbb{P}_l} & (x^1 \dots x^l) \end{array}$$

and now

$$\begin{array}{ccc} f^{-1}(y) & \xleftarrow[f^{-1}]{} & y \\ \uparrow \phi^{-1} & & \downarrow \psi \\ \{(0, \dots 0, x^1 \dots x^k)\} & \xleftarrow[\mathbb{P}_l^{-1}]{} & (0 \dots 0) \end{array}$$

54.1 Rank Theorem

Lee (2012) [76] in pp. 85, Ch. 4 Submersions, Immersions, and Embeddings, combines Theorems 41, 43 (local immersion and local submersion theorems, respectively) into the "Rank Theorem" (cf. Thm 4.12 "Rank Theorem" of Lee (2012)):

Theorem 44 (Rank Theorem)**.** *Suppose smooth manifolds M, N, $dimM = m$, $dimN = n$, smooth map $F : M \to N$, F has constant rank r.*
$\forall\, p \in M$, $\exists$ smooth charts (U, φ) for M, centered at p, (V, ψ) for N, centered at $F(p)$, s.t.

$$F(U) \subseteq V$$

in which F has coordinate representation of form

$$\widehat{F}(x^1 \dots x^r, x^{r+1} \dots x^m) = (x^1 \dots x^r, 0 \dots 0) \tag{156}$$

Particularly, if F smooth submersion,

$$\widehat{F}(x^1 \dots x^n, x^{n+1} \dots x^m) = (x^1 \dots x^n)$$

and if F smooth immersion

$$\widehat{F}(x^1 \dots x^m) = (x^1 \dots x^m, 0 \dots 0)$$

$$\begin{array}{ccc} \mathbb{R}^m & \xrightarrow{\psi F \varphi^{-1}} & \mathbb{R}^n \\ \uparrow \varphi & & \uparrow \psi \\ U \subset M & \xrightarrow[F]{} & V \subset N \end{array}$$

Also remember that $DF(p) : T_pM \to T_{F(p)}N$

Proof. $DF(p)$ has rank r (given). Then $DF(p)$ is some $r \times r$ submatrix of a $n \times m$ matrix s.t. $\det DF(p)$ nonzero.

By change of basis in $\mathbb{R}^n$, or reordering coordinates, assume $DF(p)$ is upper left submatrix $\left(\frac{\partial F^i}{\partial x^j}\right)$ $\forall\, i, j = 1, \dots r$.

Relabel standard coordinate as

$$(x, y) = (x^1 \dots x^r, y^1 \dots y^{m-r}) \in \mathbb{R}^m$$
$$(v, w) = (v^1 \dots v^r, w^1 \dots w^{n-r}) \in \mathbb{R}^n$$

By initial translations of coordinates, assume without loss of generality $p = (0,0)$, $F(p) = (0,0)$

Suppose

$$F(x,y) = (Q(x,y), R(x,y))$$

for some smooth maps $Q : U \to \mathbb{R}^r$, $R : U \to \mathbb{R}^{n-r}$

Define

$$\varphi : U \to \mathbb{R}^m$$
$$\varphi(x,y) = (Q(x,y), y)$$

so

$$D\varphi(0,0) = \begin{pmatrix} \frac{\partial Q^i}{\partial x^j}(0,0) & \frac{\partial Q^i}{\partial y^j}(0,0) \\ 0 & \delta^i_j \end{pmatrix}$$

$D\varphi(0,0)$ nonsingular, since $\det \frac{\partial Q^i}{\partial x^j} \neq 0$ (by hypothesis).

By inverse function thm., $\exists$ connected neighborhoods U_0 of $(0,0)$, $\widetilde{U}_0$ of $\varphi(0,0) = (0,0)$ s.t.

$$\varphi : U_0 \to \widetilde{U}_0$$

is a diffeomorphism.

By shrinking $U_0, \widetilde{U}_0$, assume $\widetilde{U}_0$ open cube.

Write $\varphi^{-1}(x,y) = (A(x,y), B(x,y))$, for some smooth functions $A : \widetilde{U}_0 \to \mathbb{R}^r$, $B : \widetilde{U}_0 \to \mathbb{R}^{m-r}$

$$(x,y) = \varphi(A(x,y), B(x,y)) = (Q(A(x,y), B(x,y)), B(x,y))$$

$$\implies \begin{matrix} B(x,y) = y \\ \varphi^{-1}(x,y) = (A(x,y), y) \end{matrix}$$

$$\varphi\varphi^{-1} = 1 \implies x = Q(A(x,y), y)$$

Recall that we had hypotehsized that

$$F(x,y) = (Q(x,y), R(x,y))$$

Then

$$F \circ \varphi^{-1}(x,y) = F(A(x,y), y) = (Q(A(x,y), y), R(A(x,y), y)) = (x, R(A(x,y), y))$$

and so

$$F \circ \varphi^{-1}(x,y) = (x, \widetilde{R}(x,y))$$

where $\widetilde{R} : \widetilde{U}_0 \to \mathbb{R}^{n-r}$

$\widetilde{R}(x,y) = R(A(x,y), y)$

Compute

$$D(F \circ \varphi^{-1})(x,y) = \begin{pmatrix} \delta^i_j & 0 \\ \frac{\partial \widetilde{R}^i}{\partial x^j}(x,y) & \frac{\partial \widetilde{R}^i}{\partial y^j}(x,y) \end{pmatrix}$$

Since composing with a diffeomorphism doesn't change rank of map, $D(F \circ \varphi^{-1})$ has rank r everywhere in $\widetilde{U}_0$. $\begin{pmatrix} \delta^i_j \\ \frac{\partial \widetilde{R}^i}{\partial x^j}(x,y) \end{pmatrix}$ $j = 1 \ldots r$ are linearly independent, so $\frac{\partial \widetilde{R}^i}{\partial y^j}(x,y) = 0$ on $\widetilde{U}_0$, so $\widetilde{R}^i$ independent of y^j.

Let $S(x) = \widetilde{R}(x,0)$, then

$$F \circ \varphi^{-1}(x,y) = (x, S(x))$$

Let open $V_0 \subseteq V$, $(0,0) \in V$ be an open subset $V_0 = \{(v,w) \in V : (v,0) \in \widetilde{U}_0\}$.
Then V_0 is a neighborhood of $(0,0)$.

Because $\widetilde{U}_0$ is a cube, $F \circ \varphi^{-1}(x,y) = (x, S(x))$,

$$F \circ \varphi^{-1}(\widetilde{U}_0) \subseteq V_0$$

so $F(U_0) \subseteq V_0$.

Define $\psi : V_0 \to \mathbb{R}^n$

$$\psi(v,w) = (v, w - S(v))$$

Because $\psi^{-1}(s,t) = (s, t + S(s))$,
it is a diffeomorphism.

Thus (V_0, ψ) is a smooth chart.

$$\psi \circ F(\varphi^{-1}(x,y)) = \psi(x, S(x)) = (x, S(x) - S(x)) = (x, 0)$$

□

Definition 151 (regular value)**.** For smooth $f : X \to Y$, $X, Y \in$ **Man**,
$y \in Y$ is a **regular value** for f if $Df_x : T_xX \to T_yY$ surjective $\forall\, x$ s.t. $f(x) = y$.

$y \in Y$ **critical value** if y not a regular value of f.

Absil, Mahony, and Sepulchre [81] pointed out that another definition for a *regular value* can utilize the theorem that rank of $Df \equiv DF = \dim T_pN = \dim N$, iff $DF(p)$ surjective, for $p \in M$, $F : M \to N$. Then

regular value $y \in N$, of F, if rank of $F \equiv \text{rank}(DF(x)) = \dim N$, $\forall\, x \in F^{-1}(y)$, for $F : M \to N$.

Theorem 45 (Preimage theorem)**.** *If y regular value of $f : X \to Y$,*
$f^{-1}(y)$ is a submanifold of X, with $dim f^{-1}(y) = dim X - dim Y$

Proof. Given y is a regular value of $f : X \to Y$,
$\forall\, x \in f^{-1}(y)$, $Df_x : T_xX \to T_yY$ is surjective. By local submersion theorem,

$$f(x^1 \dots x^k) = (x^1 \dots x^l) = y$$

Since $x \in f^{-1}(y)$, $(x^1 \dots x^k) = (y^1 \dots y^l, x^{l+1} \dots x^k)$.

For this chart for (U, φ), $U \ni x$, consider $(U \cap f^{-1}(y), \psi)$ with $\psi(x) = (x^{l+1} \dots x^k) \quad \forall\, x \in U \cap f^{-1}(y)$.

$\forall\, f^{-1}(y)$ submanifold with $\dim f^{-1}(y) = k - l = \dim X - \dim Y$. □

Examples for emphasis

If $\dim X > \dim Y$,

if $y \in Y$, regular value of $f : X \to Y$,

f submersion, $\forall\, x \in f^{-1}(y)$

If $\dim X = \dim Y$,

f local diffeomorphism $\forall\, x \in f^{-1}(y)$

If $\dim X < \dim Y$, $\forall\, y \in f(X)$ is a critical value.

Example: $O(n)$ as a submanifold of Mat(n,n)

Given $\text{Mat}(n,n) \equiv M(n) = \{n \times n \text{ matrices }\}$ is a manifold; in fact $\text{Mat}(n,n) \cong \mathbb{R}^{n^2}$,
Consider $O(n) = \{A \in \text{Mat}(n,n) | AA^T = 1\}$.

$$AA^T \in \text{Sym}(n) \equiv S(n) = \{S \in \text{Mat}(n,n) | S^T = S\} = \{ \text{ symmetric } n \times n \text{ matrices } \} \tag{157}$$

$\text{Sym}(n)$ submanifold of $\text{Mat}(n,n)$, $\text{Sym}(n)$ diffeomorphic to $\mathbb{R}^k$ (i.e. $\text{Sym}(n) \cong \mathbb{R}^k$), $k := \frac{n(n+1)}{2}$.

$$f : \text{Mat}(n,n) \to \text{Sym}(n)$$
$$f(A) = AA^T$$

Notice f is smooth,

$$f^{-1}(1) = O(n)$$

$$Df_A(B) = \lim_{s \to 0} \frac{f(A + sB) - f(A)}{s} = \lim_{s \to 0} \frac{(A + sB)(A^T + sB^T) - AA^T}{s} = AB^T + BA^T$$

To complete the surjectivity step, let $A \in O(n)$ and let $S \in \mathrm{Sym}(n)$ be arbitrary. For this map $f(A) = AA^T$, choose $B = \frac{1}{2}SA$. The differential computed above gives

$$Df_A(\tfrac{1}{2}SA) = \tfrac{1}{2}AA^TS^T + \tfrac{1}{2}SAA^T = \tfrac{1}{2}S + \tfrac{1}{2}S = S.$$

Thus Df_A is surjective, I is a regular value, and $O(n)$ is an embedded submanifold of dimension $n^2 - n(n+1)/2 = n(n-1)/2$. Section 88.1 develops the equivalent column-inner-product map $F(A) = A^TA - I$, the determinant condition for $SO(n)$, and the tangent spaces.

Proposition 36. *If smooth $g_1 \dots g_l \in C^\infty(X)$ on X are independent $\forall\, x \in X$, s.t. $g_i(x) = 0$, $\forall\, i = 1 \dots l$, then $Z = \{x \in X | g_1(x) = \dots = g_l(x) = 0\}$ = set of "common zeros" is a* submanifold *of X s.t. $dim Z = dim X - l$.*

Take note *that $g_1 \dots g_l$ are independent at x means, really, that $D(g_1)_x \dots D(g_l)_x$ are linearly independent on T_xX.*

Proof. Suppose smooth $g_1 \dots g_l \in C^\infty(X)$ on manifold X s.t. $\dim X = k \geq l$.

Consider $g = (g_1 \dots g_l) : X \to \mathbb{R}^l$, $Z \equiv g^{-1}(0)$.

Since $\forall\, g_i$ smooth, $D(g_i)_x : T_xX \to \mathbb{R}$ linear.

Now for

$$Dg_x = (D(g_1)_x \dots D(g_l)_x) : T_xX \to \mathbb{R}^l$$

By rank-nullity theorem (linear algebra), Dg_x surjective iff $\mathrm{rank} Dg_x = l$ i.e. l functionals $D(g_1)_x \dots D(g_l)_x$ are linearly independent on T_xX.

"We express this condition by saying the l functions $g_1 \dots g_l$ are independent at x." (Guillemin and Pollack (2010) [77])

□

55 Submanifolds; immersed submanifold, embedded submanifolds, regular submanifolds

Definition 152 (Embedded Submanifold)**.**

Recall immersion:

$F : M \to N$ immersion iff DF injective, i.e. iff $\mathrm{rank} DF = \dim M$.

Consider manifolds $M \subseteq N$.

Consider inclusion map $i : M \to N$.

$$i : x \mapsto x$$

If i immersion, $Di(x) = \frac{\partial y^i}{\partial x^j} = \delta_j{}^i$ if $y^i = x^i$, $\forall\, i = 1, \dots \dim M$.

Definition 153 (immersed submanifold)**. immersed submanifold** $M \subseteq N$ if inclusion $i : M \to N$ is an immersion.

cf. 3.3 Embedded Submanifolds of Absil, Mahony, and Sepulchre [81], also Ch. 5 Submanifolds, pp. 108, **Immersed Submanifolds** of John Lee (2012) [76].

Immersed submanifolds often arise as images of immersions.

Proposition 37 (Images of Immersions as submanifolds)**.** *Suppose smooth manifold M, smooth manifold with or without boundaries N, injective, smooth immersion $F : M \to N$ (F injective itself, not just immersion)*

Let $S = F(M)$.

Then S has unique topology and smooth structure of smooth submanifolds of N s.t. $F : M \to S$ diffeomorphism.

cf. Prop. 5.18 of John Lee (2012) [76].

Proof. Define topology of S: set $U \subseteq S$ open iff $F^{-1}(U) \subseteq M$ open ($F^{-1}(U \cap V) = F^{-1}(U) \cap F^{-1}(V), F^{-1}(U \cup V) = F^{-1}(U) \cup F^{-1}(V)$).

Define smooth structure of S: $\{F(U), \varphi \circ F^{-1} | (U, \varphi) \in$ atlas for M, i.e. (U, φ) any smooth chart of $M\}$.

"smooth compatibility condition":

$$(\varphi_2 \circ F^{-1})(\varphi_i F^{-1})^{-1} = \varphi_2 \circ F^{-1} F \varphi_1^{-1} = \varphi_2 \varphi_1^{-1}$$

since $\varphi_2\varphi_1^{-1}$ diffeomorphism ($\varphi_2\varphi_1^{-1}$ bijection and it and inverse is differentiable)

F diffeomorphism onto $F(M)$.

and these are the only topology and smooth structure on S with this property:

$$S \xrightarrow{F^{-1}} M \xrightarrow{F} N \qquad = \qquad S \hookrightarrow M$$

and F^{-1} diffeomorphism, F smooth immersion, so $i : S \to M$ smooth immersion. □

56 Curves, Integral Curves, and Flows

cf. John Lee (2012) [76], Ch. 9, deals with time-dependent vector fields and I don't see other texts or references handling such an important, but overlooked, case.

56.1 Curves in Euclidean space

cf. 4.1 Jeffrey Lee (2009) [75]

If C is 1-dim. submanifold of $\mathbb{R}^n$, $p \in C$, $\exists$ chart (V, y) of C, $p \in V$ s.t. $y(V)$ is a connected open interval $I \subset \mathbb{R}$,
inverse map $y^{-1} : I \to V \subset M$ is a local parametrization.
idea is to extract information that's appropriately independent of parametrization.

If $\gamma : I \to \mathbb{R}^n$, $c : J \to \mathbb{R}^n$ curves with same image,
c is a **positive reparametrization** of γ if $\exists$ smooth $h : J \to I$ with $h' > 0$ s.t. $c = \gamma \circ h$,
in this case, γ, c have same **sense** and same **orientation**.

Assume $\gamma : I \to \mathbb{R}^n$ has $\|\gamma'\| > 0$,
Such a curve is **regular**, i.e. curve is an immersion (Recall, an immersion would be smooth $\gamma : I \to \mathbb{R}^n$, s.t. $D\gamma(p) : T_pI \to T_{\gamma(p)}\mathbb{R}^n$ injective).

Definition 154 (unit tangent field in Euclidean space, 4.3 Lee (2009) [75])**.** If $\gamma : I \to \mathbb{R}^n$ regular curve, then $\mathbf{T}(t) := \frac{\gamma'(t)}{\|\gamma'(t)\|}$ defines **unit tangent field** along γ ($\|T\| = 1$)

length of a curve defined on closed interval $\gamma : [t_1, t_2] \to \mathbb{R}^n$,

$$L = \int_{t_1}^{t_2} \|\gamma'(t)\| dt$$

Define **arc length function** for curve $\gamma : I \to \mathbb{R}^n$ by choosing $t_0 \in I$,

$$s = h(t) := \int_{t_0}^{t} \|\gamma'(t)\| d\tau$$

If curve is smooth and regular, then $h' = \|\gamma'(\tau)\| > 0$, so by inverse function theorem, $\exists$ smooth h^{-1} (since h' invertible (with $1/h'$)).

If $c(s) = \gamma h^{-1}(s)$, then $\|c'\|(s) := \|c'(s)\| = 1, \quad \forall\, s$

$$c'(s) = (\gamma(h^{-1}(s)))' = \gamma'(h^{-1}(s)) \cdot \frac{dh^{-1}}{ds}(s)$$
$$\|c'(s)\| = \|\gamma'(t)\| \|\frac{dh^{-1}}{ds}(s)\| = h' \cdot \frac{1}{h'} = 1$$

Curves parametrized by arc length are **unit speed curves**.

For a unit speed curve, $\frac{dc}{ds}(s) = \mathbf{T}(s)$.

Definition 155 (curvature vector, curvature function, principal normal in Euclidean space, 4.4 Lee (2009) [75])**.** Let $c : I \to \mathbb{R}^n$ be a unit speed curve. vector valued function

$$\kappa(s) := \frac{d\mathbf{T}}{ds}(s)$$

is called the **curvature vector**.

$$\kappa(s) := \|\kappa(s)\| = \|\frac{d\mathbf{T}}{ds}(s)\|$$

is called the **curvature function**.

If $\kappa(s) > 0$, then define principal normal

$$\mathbf{N}(s) = \|\frac{d\mathbf{T}}{ds}(s)\|^{-1}\frac{d\mathbf{T}}{ds}(s)$$

s.t. $\frac{d\mathbf{T}}{ds} = \kappa\mathbf{N}$

56.2 Integral Curves

cf. Ch. 9 of Lee (2012) [76]

integral curves - smooth curves whose velocity at each point is equal to the value of the smooth vector field there.

Let smooth manifold M with or without boundary.
If smooth curve $\gamma : J \to M$, then $\forall\, t \in J$, velocity vector $\gamma'(t)$ is a vector in $T_{\gamma(t)}M$.

Definition 156 (Integral Curve Γ)**.** Integral curve: $\mathbf{v}, \gamma$.
If vector field $\mathbf{v}$ on M,
integral curve of $\mathbf{v}$ is differentiable curve $\gamma : J \to M$, whose velocity $\forall p \in M$, $V_{\gamma(t)}$

$$\gamma'(t) = V_{\gamma(t)},\ \forall\, t \in J$$

Finding integral curves boils downs to solving a system of ordinary differential equations in a smooth chart.
Suppose smooth vector field V on M, smooth curve $\gamma : J \to M$
On smooth coordinate domain $U \subseteq M$,
Write γ in local coordinates as

$$\gamma(t) = (\gamma^1(t), \ldots, \gamma^n(t))$$
$$\gamma^i(t) = x^i \circ \gamma(t) \text{ where } x^i : U \subseteq M \to \mathbb{R}$$

Then condition for γ to be an integral curve.

$$\gamma'(t) \equiv \dot{\gamma}(t) = V_{\gamma(t)}$$

can be written as

$$\dot{\gamma}^i(t)\ \frac{\partial}{\partial x^i}\bigg|_{\gamma(t)} = V^i(\gamma(t))\ \frac{\partial}{\partial x^i}\bigg|_{\gamma(t)}$$

which reduces to an autonomous system of ordinary differential equations (ODEs)

$$\begin{aligned}
\dot{\gamma}^1(t) &= V^1(\gamma^1(t), \ldots, \gamma^n(t)) \\
&\vdots \\
\dot{\gamma}^n(t) &= V^n(\gamma^1(t), \ldots, \gamma^n(t))
\end{aligned} \tag{158}$$

20200306 EY: I use this notation which is more useful for me, e.g. taking the double derivative of the curve γ:

$$\begin{aligned}
\dot{\gamma}(t) &= v(\gamma(t)) \in T_{\gamma(t)}U \\
\dot{\gamma}(t) &= v^i(\gamma^1(t), \ldots, \gamma^n(t)) \\
\frac{d}{dt}\dot{\gamma}^i(t) = \frac{d}{dt}v^i(\gamma^1(t), \ldots, \gamma^n(t)) &= \frac{d}{dt}v^i(x^1, \ldots, x^n) \\
&= \frac{\partial v^i}{\partial x^j}(x)\dot{x}^j = \dot{\gamma}^j\frac{\partial \dot{\gamma}^i}{\partial x^j}
\end{aligned}$$

57 Vector Bundle

See Sec. 7.1 The definitions in Renteln (2014) [115].

Definition 157 (Vector Bundle)**. vector bundle** (E, M, Y, π) over manifold M is manifold E, projection map $\pi : E \to M$ s.t. (3 axioms)

1. (*All fibers are isomorphic*) $\forall$ fiber $\pi^{-1}(p)$, $p \in M$ is isomorphic to fixed vector space Y of dim. m (m is rank of vector bundle)
2. (manifold E is locally a product) $\forall p \in M$, $\exists$ neighborhood $U \ni p$ and diffeomorphism
$$\varphi_U : \pi^{-1}(U) \to U \times Y \tag{159}$$
(φ_U called **local trivialization** of E over U)
3. (local trivialization φ_U carries fibers to fibers and restriction is linear)
$$\varphi_U : \pi^{-1}(p) \to \pi_1^{-1}(p) \tag{160}$$
is linear $\forall p$. $\pi_1 : U \times Y \to U$ is a projection onto first factor $(p, g) \mapsto p$

Manifold E is called **total space** or **bundle space**, manifold M is called **base space**, π bundle projection of vector bundle (E, M, Y, π).

Suppose U, V are 2 overlapping neighborhoods of p on M. Then we get
$$\begin{aligned} &\varphi_V \circ \varphi_U^{-1} : (U \cap V) \times Y \to (U \cap V) \times Y \\ &(\varphi_V \circ \varphi_U^{-1})(p, y_U) = (p, y_V) \end{aligned} \tag{161}$$
where $y_v = g_{VU}(p) y_U$ for some nondegenerate endomorphism $g_{VU}(p) : Y \to Y$ that varies smoothly with p. Endomorphism g_{VU} is called **bundle transition** function.

TODO: Show this endomorphism g explicitly:
$$\begin{aligned} \varphi_U^{-1} : (U \cap V) \times Y &\to \pi^{-1}(U \cap V) \\ \varphi_U^{-1} : (p, y) &\mapsto e \\ \varphi_V(e) &\mapsto (p, y_V) \end{aligned}$$

Corollary 6. *These so-called* ***cocycle conditions*** *are true:*
$$\begin{aligned} &g_{UU}(p) = 1 \\ &g_{UV}(p) g_{VU}(p) = 1 \\ &g_{UV}(p) g_{VW}(p) g_{WU}(p) = 1 \end{aligned} \tag{162}$$

Proof. 1. $\varphi_U \circ \varphi_U^{-1} : U \times Y \to U \times Y$
$\varphi_U \circ \varphi_U^{-1} = 1$. So $1 : (p, y) \mapsto (p, y)$ and so $y = g_{UU}(p) y \implies g_{UU}(p) = 1$.

2. $(\varphi_V \circ \varphi_U^{-1})(p, y_U) = (p, y_V)$ s.t. $y_V = g_{VU}(p) y_U$.

$(\varphi_U \circ \varphi_V^{-1})(p, z_V) = (p, z_U)$ s.t. $z_U = g_{UV}(p) z_V$.

$(\varphi_U \circ \varphi_V^{-1})(p, y_V) = (\varphi_U \circ \varphi_V^{-1})(\varphi_V \circ \varphi_U^{-1})(p, y_U) = 1(p, y_U) = (p, g_{UV}(p) g_{VU}(p) y_U) \implies g_{UV}(p) g_{VU}(p) = 1$
Likewise,

3.
$$\begin{aligned} (\varphi_U \circ \varphi_V^{-1})(p, y_V) &= (p, g_{UV}(p) y_V) \\ (\varphi_V \circ \varphi_W^{-1})(p, y_W) &= (p, g_{VW}(p) y_W) \\ (\varphi_W \circ \varphi_U^{-1})(p, y_U) &= (p, g_{WU}(p) y_U) \end{aligned}$$
and so
$$(\varphi_U \circ \varphi_V^{-1})(\varphi_V \circ \varphi_W^{-1})(\varphi_W \circ \varphi_U^{-1})(p, y_U) = (p, g_{UV}(p) g_{VW}(p) g_{WU}(p) y_U) = 1(p, y_U)$$
so $g_{UV}(p) g_{VW}(p) g_{WU}(p) = 1$

□

Definition 158 ((vector bundle) section)**.** (smooth) section of E is smooth map

$$s : M \to E \tag{163}$$

s.t. $\pi \circ s = 1$

This definition implies that s is a vector-valued map (EY: $\pi^{-1}(p) \simeq$ vector space Y so for $p \in M$, $s(p) = (y_1, y_2, \dots y_m) = (s_1(p), s_2(p), \dots s_m(p))$.

58 Push-forward, Pull-back

58.1 Push-forward in coordinates, $f_* : T_pM \to T_{f(p)}N$

Using Nakahara (2003) [80]'s notation.

Consider a smooth mapping $f : M \to N$, $\dim M = m, \dim N = n$.

$$f_* : T_pM \to T_{f(p)}N$$

With

$$\begin{array}{lll} (U, \varphi) \subset M & \varphi : U \to \mathbb{R}^m & \varphi^i \equiv x^i : U \to \mathbb{R} \\ (V, \psi) \subset N & \psi : V \to \mathbb{R}^n & \psi^j \equiv y^j : V \to \mathbb{R} \end{array}$$

Now $X \in T_pM$, so we can write $X = X^i \dfrac{\partial}{\partial x^i}$, and $f_*X \in T_{f(p)}N$, so we can write $f_*X = Y^j \frac{\partial}{\partial y^j}$.
$Y \in T_{f(p)}N$ $\quad Y = Y^j \dfrac{\partial}{\partial y^j}$

Let $g \in \mathcal{F}(N)$
$g : N \to \mathbb{R}$
Now $g \circ \psi^{-1} : \mathbb{R}^n \to \mathbb{R}$ or $g \circ \psi^{-1} : \psi(N) \to \mathbb{R}$.

For notation, let $g \circ \psi^{-1}(y) \equiv g(y)$, in that we treat y to be a point with coordinates in $\mathbb{R}^n$ notation-wise, and not as $y^j : V \to \mathbb{R}$.

Since $g \circ f : M \to \mathbb{R}$, so $g \circ f \in \mathcal{F}(M)$.

$(f_*X)[g] \equiv X[g \circ f]$

$$(f_*X)[g] = (Y^j \frac{\partial}{\partial y^j})[g] = Y^j \frac{\partial g}{\partial y^j} = X^i \frac{\partial}{\partial x^i}[g \circ f] = X^i \frac{\partial}{\partial x^i} g(f^j(x)) = X^i \frac{\partial g}{\partial y^j}(y) \frac{\partial f^j}{\partial x^i}(x) = X^i \frac{\partial f^j}{\partial x^i}(x) \frac{\partial}{\partial y^j} g$$

$$Y^j = X^i \frac{\partial f^j}{\partial x^i} = X^i \frac{\partial y^j}{\partial x^i} = F^j{}_i(x) X^i \tag{164}$$

where $f : M \to N$, but $\psi \circ f \circ \varphi^{-1} : \varphi(U) \subset \mathbb{R}^m \to \mathbb{R}^n$, but write this notation as $f^j(x) = y^j$.

Also, notice that for Eq. 164,

$$g \circ f : M \to \mathbb{R}$$
$$g \circ \psi^{-1} \circ \psi \circ f \circ \varphi^{-1} \text{ and } g \circ f \circ \varphi^{-1} : \varphi(U) \subset \mathbb{R}^m \to \mathbb{R}$$
$$(g \circ \psi^{-1})(\psi \circ f \circ \varphi^1)(x) \equiv g(y(x))$$

58.2 Pullback in coordinates, $f^* : T^*_{f(p)}N \to T^*_pM$

For pullback $f^* : T^*_{f(p)}N \to T^*_pM$,
since $\omega \in T^*_{f(p)}N$, then write as $\omega_j dy^j$
since $f^*\omega \in T^*_pM$, so $f^*\omega = \psi_i dx^i$

$$\langle f^*\omega, X\rangle \equiv f^*\omega(X) = \langle \psi_i dx^i, X^j \frac{\partial}{\partial x^j}\rangle \equiv \psi_i dx^i (X^j \frac{\partial}{\partial x^j}) = \psi_i X^i = \langle \omega, f_* X\rangle \equiv \omega(f_* X) =$$
$$= \langle \omega_j dy^j, Y^i \frac{\partial}{\partial y^i}\rangle = \omega_j dy^j (Y^i \frac{\partial}{\partial y^i}) = \omega_j dy^j \left(X^i \frac{\partial y^k}{\partial x^i} \frac{\partial}{\partial y^k} \right) = \omega_j \frac{\partial y^i}{\partial x^i} X^i$$
$$\implies \psi_i = \omega_j \frac{\partial y^j}{\partial x^i}$$

Thus

Theorem 46 (Push-forward in coordinates)**.**

$$\begin{array}{ll} \textit{For smooth } f : M \to N, & \begin{array}{l} dimM = m \\ dimN = n \end{array} \\ X = X^i \dfrac{\partial}{\partial x^i} \in T_p M & \\ Y = Y^j \dfrac{\partial}{\partial y^j} \in T_{f(p)} N & \textit{then} \end{array}$$

$$\boxed{\begin{array}{ll} f_* : T_p M \to T_{f(p)} N & \\ f_*(X) = Y & s.t. \\ Y^j = X^i \dfrac{\partial y^j}{\partial x^i} & \end{array}} \tag{165}$$

*where f_* is the **push-forward**, and where colloquially, $y^j = y^j(x)$, but actually $y^j \equiv \psi \circ f \circ \varphi^{-1}$*

And

Theorem 47 (Pullback in coordinates)**.** *For smooth $f : M \to N$,* $\begin{array}{l} dimM = m \\ dimN = n \end{array}$

$\omega \in T^*_{f(p)} N$, $\psi = \psi_i dx^i \in T^*_p M$, then

$$\boxed{\begin{array}{ll} f^* : T^*_{f(p)} N \to T^*_p M & \\ f^*\omega = \psi_i dx^i & where \\ \psi_i = \omega_j \dfrac{\partial y^j}{\partial x^i} & \end{array}} \tag{166}$$

59 Tensors

I'll go through Ch.7 *Tensors* of Jeffrey Lee (2009) [75].

Definition 159 (7.1[75])**.** Let V, W be modules over commutative ring R, with unity.

Then, algebraic W-valued tensor on V is multilinear map.

$$\tau : V_1 \times V_2 \times \cdots \times V_m \to W \tag{167}$$

where $V_i = \{V, V^*\} \qquad \forall\, i = 1, 2, \ldots m.$

If for r, s s.t. $r + s = m$, there are r $V_i = V^*$, s $V_i = V$, tensor is r-contravariant, s-covariant; also say tensor of total type $\binom{r}{s}$.

EY : 20170404 Note that

$$(\tau^{i\alpha}_{\beta} \frac{\partial}{\partial x^i} \text{ or } \tau^{i\alpha}_{\beta} e_i)(\omega_j dx^j \text{ or } \omega_j e^j \in V^*)$$
$$(\tau^{\beta}_{i\alpha} dx^i \text{ or } \tau^{\beta}_{i\alpha} e^i)(X^j \frac{\partial}{\partial x^j} \text{ or } X^j e_j \in V)$$

$\exists$ natural map $V \to V^{**}$, $\widetilde{v} : \alpha \mapsto \alpha(v)$. If this map is an isomorphism, V is **reflexive** module, and identify $v \mapsto \widetilde{v}$
V with V^{**}.

Exercise 7.5. Given vector bundle $\pi : E \to M$, open $U \subset M$, consider sections of π on U, i.e. cont. $s : U \to E$, where $(\pi \circ s)(u) = u, \quad \forall\, u \in U$.

Consider $E^* \ni \omega = \omega_i e^i$.

$\forall\, s \in \Gamma(E)$, $\omega(s) = \omega_i (s(x))^i, \quad \forall\, x \in U \subset M$. So define $\widetilde{s} : \omega, x \mapsto \omega(s(x)), \quad \forall\, x \in U$.

If $\widetilde{s} = 0$, $\widetilde{s}(\omega, x) = \omega(s(x)) = 0 \qquad \forall\, \omega \in E^*, \forall\, x \in U$, and so $s = 0$. (Let $\omega_i = \delta_{iJ}$ for some J, and so $s^J(x) = 0$ $\forall\, J$).

$s = 0$. So $\ker(s \mapsto \widetilde{s}) = \{0\}$ (so condition for injectivity is fulfilled).

Since $\widetilde{s} : \omega, x \mapsto \omega(s(x))$, $\forall\, \omega \in E^*$, $\forall\, x \in U$, $s \mapsto \widetilde{s}$ is surjective.

$s \mapsto \widetilde{s}$ is an isomorphism so $\Gamma(E)$ is a *reflexive* module.

Proposition 38. *For R a ring (special case), $\exists$ module homomorphism:*

tensor product space $\to$ tensor, as a multilinear map, i.e. $\exists$

$$(\otimes_{i=1}^r V) \otimes \left(\otimes_{j=1}^s V^*\right) \to T^r{}_s(V; R)$$
$$u_1 \otimes \cdots \otimes u_r \otimes \beta^1 \otimes \cdots \otimes \beta^s \in (\otimes^r V) \otimes (\otimes^s V^*) \mapsto (\alpha^1 \dots \alpha^r, v_1 \dots v_s) \mapsto \alpha^1(u_1) \dots \alpha^r(u_r) \beta^1(v_1) \dots \beta^s(v_s) \tag{168}$$

Indeed, consider

$$(\alpha^1 \dots \alpha^r, v_1 \dots v_s) \in \underbrace{V^* \times \cdots \times V^*}_{r} \times \underbrace{V \times \cdots \times V}_{s} \mapsto \alpha^1(u_1) \dots \alpha^r(u_r) \beta^1(v_1) \dots \beta^s(v_s)$$

and so for

$$\begin{aligned} \alpha^i &= \alpha^i_\mu e^\mu, \quad i = 1, 2, \dots r,\ \mu = 1, 2, \dots \dim V^* & \alpha^i(u_i) &= \alpha^i_\mu u^\mu_i \\ v_i &= v^\mu_i e_\mu, \quad i = 1, 2, \dots s,\ \mu = 1, 2, \dots \dim V & \beta^i(v_i) &= \beta^i_\mu v^\mu_i \end{aligned}$$

So that

$$\begin{aligned} \alpha^1(u_1) \dots \alpha^r(u_r) \beta^1(v_1) \dots \beta^s(v_s) &= \alpha^1_{\alpha_1} u_1^{\alpha_1} \dots \alpha^r_{\alpha_r} u_r^{\alpha_r} \beta^1_{\mu_1} v_1^{\mu_1} \dots \beta^s_{\mu_s} v_s^{\mu_s} = \\ &= (u_1^{\alpha_1} \dots u_r^{\alpha_r} \beta^1_{\mu_1} \dots \beta^s_{\mu_s})(\alpha^1_{\alpha_1} \dots \alpha^r_{\alpha_r} v_1^{\mu_1} \dots v_s^{\mu_s}) \end{aligned}$$

Identify $u_1 \otimes \cdots \otimes u_r \otimes \beta^1 \otimes \cdots \otimes \beta^s$ with this multilinear map.

Proposition 39. *If V is finite-dim. vector space, or if $V = \Gamma(E)$, for vector bundle $E \to M$, map*

$$(\otimes_{i=1}^r V) \otimes \left(\otimes_{j=1}^s V^*\right) \to T^r{}_s(V; R) \tag{169}$$

is an isomorphism.

Definition 160. tensor that can be written as

$$u_1 \otimes \cdots \otimes u_r \otimes \beta^1 \otimes \cdots \otimes \beta^s \equiv u_1 \otimes \cdots \otimes \beta^s \tag{170}$$

is **simple** or **decomposable**.

Now well that not *all* tensors are simple.

Definition 161 (7.7[75], tensor product)**.** $\forall\, S \in T^{r_1}_{s_1}(V)$, $\forall\, T \in T^{r_2}_{s_2}(V)$,
define tensor product

$$\begin{gathered} S \otimes T \in T^{r_1+r_2}_{s_1+s_2}(V) \\ S \otimes T(\theta^1 \dots \theta^{r_1+r_2}, v_1 \dots v_{s_1+s_2}) := S(\theta^1 \dots \theta^{r_1}, v_1 \dots v_{s_1}) T(\theta^{r_1+1} \dots \theta^{r_1+r_2}, v_{s_1+1} \dots v_{s_1+s_2}) \end{gathered} \tag{171}$$

Proposition 40 (7.8[75])**.**

$$\tau^{i_1\dots i_r}{}_{j_1\dots j_s} e_{i_1}\otimes\cdots\otimes e_{i_r}\otimes e^{j_1}\otimes\cdots\otimes e^{j_s} = \tau(e^{i_1}\dots e^{i_r}, e_{j_1}\dots e_{j_s}) e_{i_1}\otimes\cdots\otimes e_{i_r}\otimes e^{j_1}\otimes\cdots\otimes e^{j_s} = \tau$$

So $\{e_{i_1}\otimes\cdots\otimes e_{i_r}\otimes e^{j_1}\otimes\cdots\otimes e^{j_s} | i_1\dots i_r, j_1\dots j_s \in 1\dots n\}$ spans $T^r_s(V;R)$

Exercise 7.11. Let basis for V $e_1\dots e_n$, corresponding dual basis for V^* $e^1\dots e^n$
Let basis for V $\overline{e}_1\dots\overline{e}_n$, corresponding dual basis for V^* $\overline{e}^1\dots\overline{e}^n$
s.t.

$$\overline{e}_i = C^k{}_i e_k$$
$$\overline{e}^i = (C^{-1})^i{}_k e^k$$

EY:20170404, keep in mind that

$$Ax = e_i A^i{}_k e^k (x^j e_j) = e_i A^i{}_j x^j = A^i{}_j x^j e_i$$
$$Ae_j = e_k A^k{}_i e^i(e_j) = A^k{}_j e_k = \overline{e}_j$$
$$\overline{\tau}^i{}_{jk}\overline{e}_i\otimes\overline{e}^j\otimes\overline{e}^k = \overline{\tau}^i{}_{jk} C^l{}_i e_l (C^{-1})^j{}_m e^m (C^{-1})^k{}_n e^n = \overline{\tau}^i{}_{jk} C^l{}_i (C^{-1})^j{}_m (C^{-1})^k{}_n = \tau^l{}_{mn}$$
$$\overline{\tau}^i{}_{jk} = C^c{}_k C^b{}_j (C^{-1})^i{}_a \tau^a{}_{bc}$$

On Remark 7.13 of Jeffrey Lee (2009) [75]: first, egregious typo for $L(V,V)$; it shoudl be $L(V,W)$. Onward, for $L(V,W)$,
consider $W\otimes V^*\ni w\otimes\alpha$ s.t.

$$(w\otimes\alpha)(v) = \alpha(v)w \in W,\ \forall\, v\in V,\ \text{so } w\otimes\alpha\in L(V,W)$$

Now consider (category of) left R-module,

$$_R\mathbf{Mod} \ni {}_{\mathrm{Mat}_{\mathbb{K}}(N,M)}\mathbb{K}^N \tag{172}$$

where

$$V = \mathbb{K}^N$$
$$W = \mathbb{K}^M$$

For $A\in\mathrm{Mat}_{\mathbb{K}}(N,M)$, $x\in\mathbb{K}^N$,

$$e_i A^i{}_{,\mu} e^\mu (x^\nu e_\nu) = Ax = e_i A^i{}_\mu x^\mu,\quad i=1,2,\dots M,\ \mu=1,2,\dots N$$
$$A\in\mathrm{Mat}_{\mathbb{K}}(N,M)\cong W\otimes V^*\cong L(V,W)$$

Consider

$$\alpha\in(\mathbb{K}^N)^* = V^* \qquad \alpha = \alpha_\mu e^\mu$$
$$w\in\mathbb{K}^M = W \qquad w = w^i e_i$$
$$\alpha\otimes w = w\otimes\alpha = w^i\alpha_\mu e_i\otimes e^\mu$$

(remember, isomoprhism between $\mathrm{Mat}_{\mathbb{K}}(N,M)$ and $W\otimes V^*$ guaranteed, if V,W are free R-modules, $R=\mathbb{K}$).

Let V,W be left R-modules, i.e. $V,W\in {}_R\mathbf{Mod}$.

$$V^*\in\mathbf{Mod}_R$$

For $V^*\otimes W\in\mathbf{Mod}_R\otimes {}_R\mathbf{Mod}$

$$\alpha\in V^*, w\in W$$
$$(\alpha\otimes w)(v) = \alpha(v)w,\ \text{for } v\in V\in {}_R\mathbf{Mod}$$

But $(w\otimes\alpha)(v) = w\alpha(v)$.

Note $\alpha(v)\in R$.

Let V,W be right R-modules, i.e. $V,W\in\mathbf{Mod}_R$.

$$V^*\in {}_R\mathbf{Mod}$$

For $W\otimes V^*\in\mathbf{Mod}_R\otimes {}_R\mathbf{Mod}$.

$$\alpha\in V^*,\ w\in W$$
$$(v)(w\otimes\alpha) = w\alpha(v),\ \text{with } \alpha(v)\in R,\ v\in V$$

So $W\otimes V^*\cong L(V,W)$, for $V,W\in\mathbf{Mod}_R$

Definition 162 (7.20[75], **contraction**). Let $(e_1, \dots e_n)$ basis for V, $(e^1 \dots e^n)$ dual basis.
If $\tau \in T^r_s(V)$, then for $k \leq r$, $l \leq s$, define

$$\begin{gathered} C^k_l \tau \in T^{r-1}_{s-1}(V) \\ C^k_l \tau(\theta^1 \dots \theta^{r-1}, w_1 \dots w_{s-1}) := \\ \sum_{a=1}^{n} \tau(\theta^1 \dots \underbrace{e^a}_{k\text{th position}} \dots \theta^{r-1}, w_1 \dots \underbrace{e_a}_{i\text{th position}} \dots w_{s-1}) \end{gathered} \tag{173}$$

C^k_l is called **contraction**, for some single $1 \leq k \leq r$, some single $1 \leq l \leq s$,

$$C^k_l : T^r_s(V) \to T^{r-1}_{s-1}(V)$$

s.t.

$$(C^k_l \tau)^{i_1 \dots \widehat{i_k} \dots i_r}{}_{j_1 \dots \widehat{j_l} \dots j_s} := \tau^{i_1 \dots a \dots i_r}{}_{j_1 \dots a \dots j_s}$$

Universal mapping properties can be invoked to give a basis free definition of contraction (EY : 20170405???).
IN general,

$$\forall\, v_1 \dots v_s \in V, \forall\, \alpha^1 \dots \alpha^r \in V^*$$

so that

$$\begin{aligned} v_j &= v^\mu_j e_\mu \quad j = 1 \dots s, \quad \mu = 1, \dots \dim V \\ \alpha^i &= \alpha^i_\mu e^\mu \quad i = 1 \dots r, \quad \mu = 1 \dots \dim V^* \end{aligned}$$

then $\forall \tau \in T^r_s(V)$,

$$\begin{gathered} \tau(\alpha^1 \dots \alpha^r, v_1 \dots v_s) = \tau(\alpha^1_{\mu_1} e^{\mu_1} \dots \alpha^r_{\mu_r} e^{\mu_r}, v^{\nu_1}_1 e_{\nu_1} \dots v^{\nu_s}_s e_{\nu_s}) = \\ = \alpha^1_{\mu_1} \dots \alpha^r_{\mu_r} v^{\nu_1}_1 \dots v^{\nu_s}_s \tau(e^{\mu_1} \dots e^{\mu_r}, e_{\nu_1} \dots e_{\nu_s}) = \alpha^1_{\mu_1} \dots \alpha^r_{\mu_r} v^{\nu_1}_1 \dots v^{\nu_s}_s \tau^{\mu_1 \dots \mu_r}{}_{\nu_1 \dots \nu_s} \end{gathered}$$

which is equivalent to

$$\begin{array}{ccc} \tau \in T^r_s(V) & \overset{\alpha^1 \otimes \dots \otimes \alpha^r \otimes v_1 \otimes \dots \otimes v_s \otimes}{\longmapsto} & \alpha^1 \otimes \dots \otimes \alpha^r \otimes v_1 \otimes \dots \otimes v_s \otimes \tau \\ & & \Big\downarrow \\ & C^1_{s+1} C^2_{s+2} \dots C^r_{r+s} C^r_1 C^{r+1}_2 \dots C^{r+s}_s & \\ & & \Big\downarrow \\ & & \tau(\alpha^1 \dots \alpha^r, v_1 \dots v_s) \in R \end{array}$$

where I've tried to express the right-R-module, "right action" on $\alpha^1 \otimes \dots \otimes \alpha^r \otimes v_1 \otimes \dots \otimes v_s \in V^* \otimes \dots \otimes V$.
Conlon (2008) [90]

59.1 Metric

59.1.1 Induced Metric

cf. 7.1.2. Induced metric, Nakahara (2003) [80]. Let M be m-dim. submanifold of n-dim. Riemannian manifold N with metric g_N.
If $f : M \to N$ embedding which induced the submanifold structure of M. (Sec. 5.2 of Nakahara (2003) [80]).
(Recall, smooth $f : M \to N$, $\dim M \leq N$, f immersion if $f_* : T_p M \to T_{f(p)} N$ injection, so rank $f_* = \dim M$. f embedding if f immersion and f injection. Also $f(M)$ submanifold of N).

Proposition 41. *pullback f^* induced a natural metric $g_M = f^* g_N$ on M is given by*

$$g_{M \mu n u}(x) = g_{N \alpha \beta} \frac{\partial f^\alpha}{\partial x^\mu} \frac{\partial f^\alpha}{\partial x^\nu} \tag{174}$$

Recall that

$$\begin{aligned} &f^* : T^*_{f(p)}N \to T^*_pM \qquad && \langle f^*\omega, V\rangle = \langle \omega, f_*V\rangle \\ &\omega \in T^*_{f(p)}N && \text{and} \\ &V \in T_pM && f_*V[g] = V[gf],\ g \in \mathcal{F}(N) \end{aligned}$$

Using our notation, consider

$$\begin{aligned} &(U,\varphi),\ U \subset M,\ \varphi : U \subset M \to \mathbb{R}^m \\ &(V,\psi),\ V \subset N,\ \psi : V \subset N \to \mathbb{R}^n \end{aligned}$$

Colloquially, say $y^j = y^j(x)$, but actually $y^j = \psi \circ f \circ \varphi^{-1}$, so that

$$f_*(X) = Y, \text{ where } X = X^i\frac{\partial}{\partial x^i},\ Y = Y^j\frac{\partial}{\partial y^j} = X^i\frac{\partial y^j}{\partial x^i}\frac{\partial}{\partial y^j}$$

We also observe that

$$f_*V[g] = V[g\circ f] = V^i\frac{\partial}{\partial x^i}(g\circ f) = V^i\frac{\partial g}{\partial y^i}\frac{\partial f^j}{\partial x^i} = V^i\frac{\partial f^j}{\partial x^i}\frac{\partial g}{\partial y^j}$$

where notation-wise, $X \equiv V$, $y \equiv f$ with respect to ours and Nakahara's notation.

Now $g_N : T_{f(p)}N \otimes T_{f(p)}N \to \mathbb{R}$, then if we just input one vector into g_N, it becomes a covector $g_N(U,\,) \in T^*_{f(p)}N$.

So then

$$\langle f^*g_N(U,\,), V\rangle = \langle g_N(U,\,), f_*V\rangle = \langle g_N(U,\,), V^i\frac{\partial f^j}{\partial x^i}\frac{\partial}{\partial y^j}\rangle = \langle g_N(U,\,), V^i\frac{\partial}{\partial y^j}\rangle\frac{\partial f^j}{\partial x^i}$$

$$g_N(U, V^i\frac{\partial}{\partial y^i})\frac{\partial f^j}{\partial x^i} = g_N(U^k\frac{\partial}{\partial y^k}, V^i\frac{\partial}{\partial y^j})\frac{\partial f^j}{\partial x^i} = (g_N)_{aj}V^i\frac{\partial f^j}{\partial x^i}$$

where we let $U^k = \delta^{ak}$ in the last step.

Let $V = \delta^{bi}\frac{\partial}{\partial x^i}$, so $(g_N)_{aj}\frac{\partial f^j}{\partial x^b} = \langle f^*g_N\left(\frac{\partial}{\partial y^a},\ \right), \frac{\partial}{\partial x^b}\rangle$.

By symmetry,

$$\boxed{(g_M)_{ab} = (g_N)_{ij}\frac{\partial f^i}{\partial x^a}\frac{\partial f^j}{\partial x^b}} \tag{175}$$

where $f : M \to N$.

Problem 8, pp. 28, Wald (1984) [107].

(a) The metric of flat, 3-dim. Euclidean space is

$$ds^2 = dx^2 + dy^2 + dz^2$$

Show that the metric components $g_{\mu\nu}$ in spherical polar coordinates r, θ, ϕ defined by

$$\begin{aligned} r &= (x^2+y^2+z^2)^{1/2} \\ \cos\theta &= z/r \\ \tan\phi &= y/x \end{aligned}$$

is given by

$$ds^2 = dr^2 + r^2d\theta^2 + r^2\sin^2\theta d\phi^2$$

(b) The spacetime metric of special relativity is

$$ds^2 = -dt^2 + dx^2 + dy^2 + dz^2$$

Find the components, $g_{\mu\nu}$ and $g^{\mu\nu}$, of the metric and inverse metric in "rotating coordinates," defined by

$$\begin{aligned} t' &= t \\ x' &= (x^2+y^2)^{1/2}\cos(\phi-\omega t) \\ y' &= (x^2+y^2)^{1/2}\sin(\phi-\omega t) \\ z' &= z \end{aligned}$$

where $\tan\phi = y/x$.

Solution 8, pp. 28, Wald (1984) [107].

(a) Consider $(r,\theta,\varphi) \to (x,y,z)$ s.t.

$$\begin{array}{l} x = rs\theta c\varphi \\ y = rs\theta s\varphi \text{ so consider} \\ z = rc\theta \end{array} \begin{vmatrix} s\theta c\varphi & rc\theta c\varphi & -rs\theta s\varphi \\ s\theta s\varphi & rc\theta s\varphi & rs\theta c\varphi \\ c\theta & -rs\theta & 0 \end{vmatrix}$$

Since $(g_N)_{ij} = \delta_{ij}$ in our case,

$$\begin{aligned} (g_M)_{rr} &= 1 \\ (g_M)_{\theta\theta} &= r^2 \\ (g_M)_{\varphi\varphi} &= r^2\sin^2\theta \end{aligned}$$

By summing terms across columns, $(g_M)_{ab} = 0$ if $a \neq b$ and $(g_M)_{aa}$ obtained by squaring and summing terms across a single column.

Thus, some physicists write as a (sometimes confusing) shorthand for the metric tensor

$$g_M = dr\otimes dr + r^2 d\theta\otimes d\theta + r^2\sin^2\theta d\varphi\otimes d\varphi$$

to be

$$ds^2 = dr^2 + r^2 d\theta^2 + r^2\sin^2\theta d\phi^2$$

(b) Consider $(t',x',y',z') \to (t,x,y,z)$. Let $\rho = (x^2+y^2)^{1/2}$. Consider the Jacobian:

$$\begin{vmatrix} 1 & & \\ \omega\rho s(\phi-\omega t) & \frac{x}{\rho}c(\phi-\omega t) & \frac{y}{\rho}c(\phi-\omega t) \\ -\omega\rho c(\phi-\omega t) & \frac{x}{\rho}s(\phi-\omega t) & \frac{y}{\rho}s(\phi-\omega t) \\ 0 & & 1 \end{vmatrix}$$

The terms with the same components, e.g. $g_{t't'}$, $g_{x'x'}$, ... are $(-1+\omega^2\rho^2)$, $\frac{x^2}{\rho^2}$, $\frac{y^2}{\rho^2}$, 1, and the cross terms are $\frac{xy}{\rho^2}dx\otimes dy$.

Part XVIII
Smooth Manifolds: notes and solutions for Lee, *Introduction to Smooth Manifolds*

60 Smooth Manifolds

Topological Manifolds

M topological manifold of dim$n$, or topological n-manifold

- locally Euclidean, dim$n$ - $\forall\, p \in M$, $\exists$ neighborhood $U \equiv U_p$ s.t. $U_p \approx^{\text{homeo}}$ open $V \subset \mathbb{R}^n$

Exercise (Lee 2e, Exercise 1.1, p. 3 [76])**.** Show that equivalent definitions of manifolds are obtained if instead of allowing U to be homeomorphic to *any* open subset of $\mathbb{R}^n$, we require it to be homeomorphic to an open ball in $\mathbb{R}^n$, or to $\mathbb{R}^n$ itself.

Solution. Recall, M locally Euclidean dim n $\forall\, p \in M$, $\exists$ neighborhood homeomorphic to open subset.
open subset $\mathcal{O} \subseteq \mathbb{R}^n$ homeomorphic to open ball and $\mathcal{O}$ homeomorphic is $\mathbb{R}^n$ since $\mathbb{R}^n$ homeomorphic to open ball.

To see this explicitly, that open ball $B_\epsilon(x_0) \subseteq \mathbb{R}^n$ homeomorphic to $\mathbb{R}^n$

Consider $\begin{aligned} &T : B_\epsilon(x_0) \to \mathbb{R}^n \\ &T(B_\epsilon(x_0)) = B_\epsilon(0) \\ &T(x) = x - x_0 \end{aligned}$

$T^{-1}(x) = x + x_0$. Clearly T homeomorphism.

Consider $\begin{aligned} &\lambda : \mathbb{R}^n \to \mathbb{R}^n \\ &\lambda(x) = \lambda x \end{aligned}$ for $\lambda > 0$. Clearly λ homeomorphism.

$\lambda^{-1}(x) = \dfrac{1}{\lambda} x$

Consider $B \equiv B_1(0)$.

Consider $\begin{aligned} &g : \mathbb{R}^n \to \mathbb{R}^n \\ &g(x) = \frac{x}{1 + |x|} \end{aligned}$

g cont.

Let $f : B \to \mathbb{R}^n$

$f(x) = \dfrac{x}{1 - |x|}$

How was f guessed at?

$|g(x)| = \left| \frac{x}{1+|x|} \right| = \frac{r}{1+r}$. Note $0 \leq |g(x)| < 1$

So $g(\mathbb{R}^n) = B$

For $|g(x)| = |y|$, $y \in B$, $|y|(1 + r) = r$, $\quad r = \frac{|y|}{1-|y|}$

This is well-defined, since $0 \leq |y| < 1$ and $0 < 1 - |y| \leq 1$

$$gf(x) = \frac{\frac{x}{1-|x|}}{1 + \frac{|x|}{1-|x|}} = x$$

$$fg(x) = \frac{\frac{x}{1+|x|}}{1 - \frac{|x|}{1+|x|}} = x$$

f homeomorphism between B and $\mathbb{R}^n$. B and $\mathbb{R}^n$ homeomorphic. So an open ball in $\mathbf{R}^n$ is homeomorphic to $\mathbb{R}^n$

In practice, both the Hausdorff and second countability properties are usually easy to check, especially for spaces that are built out of other manifolds, because both properties are inherited by subspaces and products (Lemmas A.5 and A.8). In particular, it follows easily that any open subset of a topological n-manifold is itself a topological n-manifold (with the subspace topology, of course).

Coordinate Charts

chart on M, (U, φ) where open $U \subset M$ and
homeomorphism $\varphi : U \to \mathbb{R}^n$, $\varphi(U)$ open.

Examples of Topological Manifolds

Example 1.3. (Graphs of Continuous Functions)
Let open $U \subset \mathbb{R}^n$
Let $F : U \to \mathbb{R}^k$ cont.
graph of F: $\Gamma(F) = \{(x, y) \in \mathbb{R}^n \times \mathbb{R}^k | x \in U, y = F(x)\}$ with subspace topology.

$\pi_1 : \mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ projection onto first factor.
$\varphi_k : \Gamma(F) \to U$ restriction of π_1 to $\Gamma(F)$
$\varphi_F(x, y) = x$, $(x, y) \in \Gamma(F)$

Example 1.4 (Spheres) $S^n = \{x \in \mathbb{R}^{n+1} || x| = 1\}$
Hausdorff and second countable because it's topological subspace of $\mathbb{R}^n$

Example 1.5 (Projective Spaces)
$U_i \subset \mathbb{R}^{n+1} - 0$ where $x^i \neq 0$
$V_i = \pi(U_i)$
Let $a \in U_i$.

$$|x - a|^2 = (x^1 - a^1)^2 + \cdots + (x^i - a^i)^2 + \cdots + (x^{n+1} - a^{n+1})^2 < \frac{(n+1)\epsilon^2}{n+1} = \epsilon^2$$

$\forall\, a^i \in \mathbb{R}$, $\exists\, x^i$, s.t. $(x^i - a^i)^2 < \frac{\epsilon^2}{n+1}$, by choice of $0 < x^i < a^i + \frac{\epsilon}{\sqrt{n+1}}$ with $0 < x^i$ for $i = i$ index.
U_i indeed open set, *saturated open set*.
open $U_i \subset \mathbb{R}^{n+1} - 0$, $x^i \neq 0$
From Lemma A.10, recall (d) restriction of π to any saturated open or closed subset of X is a quotient map.
natural map $\pi : \mathbb{R}^{n+1} - 0 \to \mathbb{R}P^n$ given quotient topology.
By Tu, Prop. 7.14, $\sim$ on $\mathbb{R}^{n+1} - 0 \to \mathbb{R}P^n$ open equivalence relation.
$\implies \pi|_{U_i}(U_i) = V_i$ open.

$$\varphi_i : V_i \to \mathbb{R}^n$$
$$\varphi_i[x^1 \dots x^{n+1}] = \left(\frac{x^1}{x^i} \dots \frac{x^{i-1}}{x^i}, \frac{x^{i+1}}{x^i} \dots \frac{x^{n+1}}{x^i}\right)$$

φ_i well-defined since

$$\varphi_i[tx^1 \dots tx^{n+1}] = \left(\frac{tx^1}{tx^i} \dots \frac{t\widehat{x^i}}{tx^i} \dots \frac{tx^{n+1}}{tx^i}\right) = \left(\frac{x^1}{x^i} \dots \frac{\widehat{x}^i}{x^i} \dots \frac{x^{n+1}}{x^i}\right) = \varphi_i[x^1 \dots x^{n+1}]$$

φ_i cont. since $\varphi_i \pi$ cont.

$$\begin{array}{ccc} U_i \subset \mathbb{R}^{n+1} - 0 & & \\ \pi \big\downarrow & \searrow^{\varphi_i \pi} & \\ V_i \subset \mathbb{R}P^n & \xrightarrow[\varphi]{} & \mathbb{R}^n \end{array}$$

$$\varphi_i : U_i \subset \mathbb{R}P^n \to \mathbb{R}^n$$
$$\varphi_i[x^1 \dots x^{n+1}] = \left(\frac{x^1}{x^i} \dots \frac{\widehat{x}^i}{x^i} \dots \frac{x^{n+1}}{x^i}\right)$$
$$\varphi_i^{-1}(u^1 \dots u^n) = [u^1 \dots u^{i-1}, 1, u^i \dots u^n]$$
$$\varphi_i^{-1}\varphi_i[x^1 \dots x^{n+1}] = \left[\frac{x^1}{x^i} \dots \frac{x^{i-1}}{x^i}, 1, \frac{x^{i+1}}{x^i} \dots \frac{x^{n+1}}{x^i}\right] = [x^1 \dots x^{i-1}, x^i, x^{i+1} \dots x^{n+1}]$$
$$\varphi_i \varphi_i^{-1}(u^1 \dots u^n) = (u^1 \dots u^{i-1}, u^i \dots u^n)$$

cont. φ_i bijective, φ_i^{-1} cont. φ_i homeomorphism.
From a previous edition:

Exercise 1.2 (Lee, 1st ed.). *Note.* Lee, first edition, Exercise 1.2. There is no second-edition exercise with this content: the argument (multiplication by t is a homeomorphism, so the quotient map π is open) is part of Exercise 1.6, p. 7 [76].

Let $\phi_t : \mathbb{R}^{n+1} \backslash \{0\} \to \mathbb{R}^{n+1} \backslash \{0\}$
$\phi_t(x) = tx$
ϕ_t invertible, $\phi_t^{-1} = \phi_{\frac{1}{t}}$

ϕ_t, ϕ_t^{-1} C^1 (and C^∞), ϕ_t homeomorphism.

Let U open in $\mathbb{R}^{n+1}\backslash\{0\}$. Then $\phi_t(U)$ open in $\mathbb{R}^{n+1}\backslash\{0\}$.
Thus $\pi^{-1}([U]) = \bigcup_{t\in\mathbb{R}} \phi_t(U)$ open in $\mathbb{R}^{n+1}\backslash\{0\}$.
Thus $[U]$ open in $\mathbb{R}P^n$. $\sim$ open.

Note $\pi : \mathbb{R}^{n+1}\backslash\{0\} \to \mathbb{R}P^n$

$$\pi(x) = \frac{x}{\|x\|}$$

$\mathbb{R}^n$ 2nd. countable, $\mathbb{R}P^n$ 2nd. countable.

Exercise (Lee 2e, Exercise 1.7, p. 7 [76])**.** Show that $\mathbb{RP}^n$ is compact. [Hint: show that the restriction of π to $\mathbb{S}^n$ is surjective.]

Note. Numbered Exercise 1.3 in an earlier edition. The work below is the hint; the full solution is under Exercise 1.7.

Solution. S^n compact.
$\pi : \mathbb{R}^{n+1}\backslash\{0\} \to \mathbb{R}P^n$
$\pi(x) = [\frac{x}{\|x\|}]$
Let $x \in \mathbb{R}P^n$

$y = \frac{x}{\|x\|} \in S^n$ and $\pi|_{S^n}(y) = [x]$

$\pi|_{S^n}$ surjective.

Exercise (Lee 2e, Exercise 1.6, p. 7 [76])**.** Show that $\mathbb{RP}^n$ is Hausdorff and second-countable, and is therefore a topological n-manifold.

Solution. First, note that $\sim$ on $\mathbb{R}^{n+1} - 0$ in the definition of $\mathbb{R}P^n$ is an open $\sim$ i.e. open equivalence relation.
This is because of the following: $\forall\, U \subset \mathbb{R}^{n+1} - 0$,
$\pi^{-1}(\pi(U)) = \bigcup_{t\in\mathbb{R}} tU$, set of all pts. equivalent to some pt. of U.
multiplication by $t \in \mathbb{R}$ homeomorphism of $\mathbb{R}^{n+1} - 0$, so tU open $\forall\, t \in \mathbb{R}$.
$\pi^{-1}(\pi(U))$ open i.e. $\pi(U)$ open (for π is cont.).
Let $X = \mathbb{R}^{n+1} - 0$.
Consider $R = \{(x, y) \in X \times X | x \sim y$ or $y = tx$ for some $t \in \mathbb{R}\}$
$y = tx$ means $y_i = tx_i \quad \forall\, i = 0 \dots n$. Then $\frac{x_i}{y_i} = \frac{x_j}{y_j} \quad \forall\, i, j = 0 \dots n$. Hence $x_i y_j - y_i x_j = 0 \quad \forall\, i, j$.

Let $f : X \times X \to \mathbb{R}$

$$f(x, y) = \sum_{i\neq j} (x_i y_j - y_i x_j)^2$$

$$\frac{\partial f}{\partial x_i} = \sum_{i\neq j} 2(x_i y_j - y_i x_j)(y_j - y_j) = 0$$

$$\frac{\partial f}{\partial y_j} = \sum_{j\neq i} 2(x_i y_j - y_i x_j)(x_i - x_i) = 0$$

Nevertheless, f is C^1 so f cont.
So $f^{-1}(0) = R$.
0 closed, so $f^{-1}(0) = R$ closed. By theorem, since $\sim$ open, $\mathbb{R}P^n = \mathbb{R}^{n+1} - 0/\sim$ Hausdorff.
cf. http://math.stackexchange.com/questions/336272/the-real-projective-space-rpn-is-second-countable
topological space is second countable if its topology has countable basis.
$\mathbb{R}^n$ second countable since $\mathcal{B} = \{B_r(q) | r, q \in \mathbb{Q}\}$ is a countable basis. $\forall\, x \in \mathbb{R}^n$

If X is second countable, with countable basis $\mathcal{B}$,

1. If $Y \subseteq X$, Y also second countable with countable basis $\{B | B \in \mathcal{B}, Y \bigcap B \neq \emptyset\}$
2. If $Z == X/\sim$, $\{\{[x] | x \in B\} | B \in \mathcal{B}\}$ is a countable basis for Z since $\mathcal{B}$ countable.

It is a basis since

$$\begin{array}{c} X = \bigcup_{B\in\mathcal{B}} B \\ \Big\downarrow \pi \\ Z = X/\sim = \pi\big(\bigcup_{B\in\mathcal{B}} B\big) = \bigcup_{B\in\mathcal{B}} \pi(B) = \bigcup_{B\in\mathcal{B}} \{[x] | x \in B\} \end{array}$$

Now let $Y = \mathbb{R}^n - 0$ and
$Z = \mathbb{R}P^n = \mathbb{R}^n - 0/\sim$

Exercise (Lee 2e, Exercise 1.7, p. 7 [76])**.** Show that $\mathbb{RP}^n$ is compact. [Hint: show that the restriction of π to $\mathbb{S}^n$ is surjective.]

Solution. S^n compact so $S^n/\{\pm\}$ compact by Theorem, as $\pi_S(S^n) = S^n/\{\pm\}$, as π_S cont. surjective ($\forall\, [x] \in S^n/\{\pm\}$, $\exists\, x \in S^n$ s.t. $\pi_S(S^n) = [x]$)
g cont. bijective as defined above so since $g(S^n/\{\pm\}) = \mathbb{R}P^n$, $\mathbb{R}P^n$ compact.

Example 1.8 (Product Manifolds)

$$M_1 = \bigcup_{\alpha\in\mathfrak{A}_1} U_\alpha^{(1)}$$
$$M_i = \bigcup_{\alpha\in\mathfrak{A}_i} U_\alpha^{(i)}$$

$$M_1 \times \cdots \times M_n = \bigcup_{\substack{\alpha_1 \in \mathfrak{A}_1 \\ \vdots \\ \alpha_n \in \mathfrak{A}_n}} U_{\alpha_1} \times \cdots \times U_{\alpha_n} \qquad \text{(by def.)}$$

$\forall\, p = (p_1 \dots p_n) \in M_1 \times \dots M_n$, consider $p_i \in M_i$. Choose coordinate chart (U_{j_i}, φ_{j_i}), $\varphi_i(U_i) \subset \mathbb{R}^{n_i}$. Then,

Consider $\varphi : U_1 \times \cdots \times U_n \to \mathbb{R}^{m_1} \times \cdots \times \mathbb{R}^{m_n} = \mathbb{R}^{m_1+\cdots+m_n}$, $(U_1 \times \cdots \times U_n, \varphi_1 \times \cdots \times \varphi_n)$
$\varphi = \varphi_1 \times \cdots \times \varphi_n$
φ also a homeomorphism. $\varphi\psi^{-1}$
$(\varphi_1 \times \cdots \times \varphi_n) \circ (\psi_1 \times \cdots \times \psi_n)^{-1}$ also a diffeomorphism, cont. bijective and C^∞
$\{(U, \varphi) = (U_1 \times \cdots \times U_n, \varphi_1 \times \cdots \times \varphi_n) | (U_i, \varphi_i) \in \{(U_i, \varphi_i) | U_i \in M_i\}\}$ also an atlas.

Topological Properties of Manifolds

Lemma 7 (1.10)**.** $\forall$ *topological M, M has countable basis of precompact coordinate balls*

Proof. First consider M can be covered by single chart.
Suppose $\varphi : M \to \widehat{U} \subseteq \mathbb{R}^n$ global coordinate map.
Let $\mathcal{B} = \{B_r(x) |$ open $B_r(x) \subseteq \mathbb{R}^n$ s.t. $r \in \mathbb{Q}$, $x \in \mathbb{Q}$, i.e. x rational coordinates , $B_{r'}(x) \subseteq \widehat{U}$, for some $r' > r\}$
Clearly, $\forall\, B_r(x)$ precompact in $\widehat{U}$
$\mathcal{B}$ countable basis for topology of $\widehat{U}$
φ homeomorphism, it follows $\{\varphi^{-1}(B) | B \in \mathcal{B}\}$ countable basis for M
Let M arbitrary,
By def., $\forall\, p \in M$, $p \in$ domain U of a chart
Prop. A.16, $\forall$ open cover of second-countable space has countable subcover.
M covered by countably many charts $\{(U_i, \varphi_i)\}$
$\forall\, U_i$, U_i has countable basis of coordinate balls precompact in U_i
union of all these coordinates bases is countable basis for M.

If $V \subseteq U_i$ one of these balls,
then $\overline{V}$ compact in U_i. M Hausdorff, so $\overline{V}$ closed.
$\overline{V}$ in M is same as $\overline{V}$ in U_i, so V precompact in M. □

Connectivity

Local Compactness and Paracompactness

Fundamental Groups of Manifolds

Smooth Structures

If open $U \subset \mathbb{R}^n$, $V \subset \mathbb{R}^m$,
$F : U \to V$ smooth (or C^∞) if $\forall$ cont. partial derivative of all orders exists.

F diffeomorphism if F *smooth*, bijective, and has a *smooth* inverse.
Diffeomorphism is a homeomorphism.

$(U, \varphi), (V, \psi)$ smoothly compatible if $UV = \emptyset$ or

$$\psi\varphi^{-1} : \varphi(UV) \to \psi(UV) \qquad \text{diffeomorphism}$$

atlas $\mathcal{A} \equiv \{(U, \varphi)\}$ s.t. $\bigcup U = M$. Smooth atlas if $\forall\, (U, \varphi), (V, \psi) \in \mathcal{A}$, $(U, \varphi), (V, \psi)$ smoothly compatible.

Smooth structure on topological n-manifold M is a maximal smooth atlas.

Smooth manifold $(M, \mathcal{A})$ where M topological manifold, $\mathcal{A}$ smooth structure on M.

Proposition 42 (1.17)**.** *Let M topological manifold.*

(a) $\forall$ smooth atlas for M is contained in ! maximal smooth atlas.

(b) 2 smooth atlases for M determine the same maximal smooth atlas iff union is smooth atlas.

Proof. Let $\mathcal{A}$ smooth atlas for M
$\overline{\mathcal{A}} \equiv$ set of all charts that are smoothly compatible with every chart in $\mathcal{A}$
Want: $\overline{\mathcal{A}}$ smooth atlas, i.e. $\forall\, (U, \varphi), (V\psi) \in \overline{\mathcal{A}}$, $\psi\varphi^{-1} : \varphi(UV) \to \psi(UV)$ smooth.
Let $x = \varphi(p) \in \varphi(UV)$
$p \in M$, so $\exists$ some chart $(W, \theta) \in \mathcal{A}$ s.t. $p \in W$.
By given, $\theta\varphi^{-1}, \psi\theta^{-1}$ smooth where they're defined.
$p \in UVW$, so $\psi\varphi^{-1} = \psi\theta^{-1}\theta\varphi^{-1}$ smooth on x.
Thus $\psi\varphi^{-1}$ smooth in a neighborhood of each pt. in $\varphi(UV)$. Thus $\overline{\mathcal{A}}$ smooth atlas.
To check maximal,

□

Local Coordinate Representations

Proposition 43 (1.19)**.** *$\forall$ smooth M has countable basis of regular coordinate balls*

Exercise (Lee 2e, Exercise 1.20, p. 15 [76])**.** Prove Proposition 1.19.

Note. Proposition 1.19: every smooth manifold has a countable basis of regular coordinate balls.

Solution. smooth manifold M has smooth structure
Suppose single smooth chart φ has entire M as domain

$$\varphi : M \to \widehat{U} \subseteq \mathbb{R}^n$$

Let $\widehat{B} = \{\widehat{B}_r(x) \subseteq \mathbb{R}^n | r \in \mathbb{Q},\, x \in \mathbb{Q},\, \widehat{B}_{r'}(x) \subseteq \widehat{U} \text{ for some } r' > r\}$

$\forall\, \widehat{B}_r(x)$ precompact in $\widehat{U}$
$\widehat{\mathcal{B}}$ countable basis for topology of $\widehat{U}$
φ homeomorphism,

Let $\varphi^{-1}(\widehat{B}_r(0)) = B$

$\varphi^{-1}(\widehat{B}_{r'}(0)) = B'$

φ homeomorphism and since $\widehat{B}$ countable basis, $\{B\}$ countable basis of regular coordinate basis.

Suppose arbitrary smooth structure.
By def., $\forall\, p \in M$, p in some chart domain
Prop. A.16., $\forall$ open cover of second countable space has countable subcover
M covered by countably many charts $\{(U_i, \varphi_i)\}$

$\forall U_i$, U_i has countable basis of coordinate balls precompact in U_i
union of all these coordinate charts is countable basis for M.
If $V \subseteq U_i$, 1 of these balls,

$\varphi(V) = B_r(0)$
$\varphi(\overline{V}) = \overline{B}_r(0)$
and $\varphi(B') = B_{r'}(0)$, $r' > r$ for countable basis for U_i
So V regular coordinate ball.

Examples of Smooth Manifolds

More Examples

Example 1.25 (Spaces of Matrices) Let $M(m \times n, \mathbb{R}) \equiv$ set of $m \times n$ matrices with real entries.

Example 1.26 (Open Submanifolds)
$\forall$ open subset $U \subseteq M$ is itself a dim$M$ manifold.
EY : $\forall$ *open subset* $U \subseteq M$ *is itself a dim*M *manifold.*

Example 1.27 (The General Linear Group)
general linear group $GL(n, \mathbb{R}) = \{A | \det A \neq 0\}$
$\det : A \to \mathbb{R}$ is cont. (by def. of $\det A = \epsilon^{i_1 \cdots i_n} a_{1 i_1} \dots a_{n i_n}$
$\det^{-1}(\mathbb{R} - 0)$ is open since $\mathbb{R} - 0$ open so $GL(n, \mathbb{R})$ open
$GL(n, \mathbb{R}) \subseteq M(n, \mathbb{R})$, $M(n, \mathbb{R})$ n^2-dim. vector space.
so $GL(n, \mathbb{R})$ smooth n^2-dim. manifold.

Example 1.28 (Matrices of Full Rank)
Suppose $m < n$

Let $M_n(m \times n, \mathbb{R}) \subseteq M(m \times n, \mathbb{R})$ with matrices of rank m
if $A \in M_m(m \times n, \mathbb{R})$,
rank$A = m$
means that A has some nonsingular $m \times m$ submatrix. (EY 20140205 ???)

Example 1.31 (Spheres)

$$\varphi_i^{\pm} : S^n \to B_1^n(0) \subset \mathbb{R}^n \quad (B_1^n(0) \text{ disk of radius 1})$$
$$\varphi_i^{\pm}(x_1 \dots x_{n+1}) = (x_1 \dots \widehat{x}_i \dots x_{n+1}) = (y_1 \dots y_n)$$

Note $x_1^2 + \cdots + x_i^2 + \cdots + x_{n+1}^2 = 1$. $\qquad x_i = \pm\sqrt{1 - (x_1^2 + \cdots + \widehat{x}_i^2 + \cdots + x_{n+1}^2)}$

$$(\varphi_i^{\pm})^{-1}(y_1 \dots y_n) = (y_1 \cdots \pm\sqrt{1 - (y_1^2 + \cdots + y_n^2)} \dots y_n) = (y_1 \dots y_{i-1}, \pm\sqrt{1 - |y|^2}, y_i \dots y_n)$$
$$\varphi_i^{\pm}(\varphi_j^{\pm})^{-1}(y_1 \dots y_n) = (y_1 \dots \widehat{y}_i \dots y_{j-1}, \pm\sqrt{1 - |y|^2}, y_j \dots y_n)$$
$$\varphi_i^{\pm}(\varphi_j^{\mp})^{-1}(y_1 \dots y_n) = (y_1 \dots \widehat{y}_i \dots y_{j-1}, \mp\sqrt{1 - |y|^2}, y_j \dots y_n)$$
$$\varphi_j^{\pm}(\varphi_i^{\mp})^{-1}\varphi_i^{\pm}(\varphi_j^{\pm})^{-1}(y_1 \dots y_n) = \varphi_j^{\pm}(y_1 \cdots \pm y_i \dots y_{j-1}, \pm\sqrt{1 - |y|^2} \dots y_n) = (y_1 \cdots \pm y_i \dots y_j \dots y_n)$$

This is symmetrical in i, j and so true if i, j reverse.

So $\varphi_i^{\pm}(\varphi_j^{\pm})^{-1}$ diff. and bijective. Likewise for $\varphi_i^{\pm}(\varphi_j^{\mp})^{-1}$.
So $\begin{matrix}\varphi_i^{\pm}(\varphi_j^{\pm})^{-1}\\ \varphi_i^{\pm}(\varphi_j^{\mp})^{-1}\end{matrix}$ diffeomorphisms.

Lemma 8 (1.35)**.** ***(Smooth Manifold Chart Lemma)*** *Let M be a set, suppose given $\{U_\alpha | U_\alpha \subset M\}$, given maps $\varphi_\alpha : U_\alpha \to \mathbb{R}^n$ s.t.*

(i) $\forall\, \alpha$, φ_α bijection between U_α and open $\varphi_\alpha(U_\alpha) \subseteq \mathbb{R}^n$

(ii) $\forall\, \alpha, \beta$, $\varphi_\alpha(U_\alpha \bigcap U_\beta)$, $\varphi_\beta(U_\alpha \bigcap U_\beta)$ open in $\mathbb{R}^n$

Example 1.36 (Grassmann Manifolds)
$G_k(V) = \{S | S \subseteq V\}$ S k-dim. linear subspace of V
$\dim V = n$, V vector space

if $V = \mathbb{R}^n$, $G_k(\mathbb{R}^n) \equiv G_{k,n} \equiv G(k, n)$ (notation) $G_1(\mathbb{R}^{n+1}) = \mathbb{R}P^n$
Let $V = P \oplus Q$, $\begin{matrix}\dim P = k\\ \dim Q = n - k\end{matrix}$
linear $X : P \to Q$

$$\Gamma(X) = \{v + Xv | v \in P\}, \quad \Gamma(X) \subseteq V, \dim \Gamma(X) = k$$

$\Gamma(X) \bigcap Q = 0$ since $\forall\, w \in \Gamma(X)$, w has a P piece, and Q complementary to P

Converse: $\forall$ subspace $S \subseteq V$, s.t. $S \bigcap Q = 0$

let $\pi_P : V \to P$ projections by direct sum decomposition $V = P \oplus Q$
$\pi_Q : V \to Q$
$\pi_P|_S : S \to P$ isomorphism
$\implies X = (\pi_Q|_S) \cdot (\pi_P|_S)^{-1}$, $X : P \to Q$

Let $v \in P$. $v + Xv = v + \pi_Q|_S\,(\pi_P|_S)^{-1} v$. Let $v \in \pi_P|_S\,(S)$ $\Gamma(X) = S$

Let $L(P; Q) = \{f |$ linear $f : P \to Q\}$, $L(P; Q)$ vector space
$U_Q \subseteq G_k(V)$, $U_Q = \{S | \dim S = k, S$ subspace , $S \bigcap Q = 0\}$
$\Gamma : L(P; Q) \to U_Q$
$X \mapsto \Gamma(X)$
Γ bijection by above
$\varphi = \Gamma^{-1} : U_Q \to L(P; Q)$

By choosing bases for P, Q, identify $L(P; Q)$ with $M((n-k)k; \mathbb{R})$ and hence with $\mathbb{R}^{k(n-k)}$

think of (U_Q, φ) as coordinate chart.
$\varphi(U_Q) = L(P; Q)$

Problems

Exercise (Lee 2e, Problem 1-7, p. 30 [76])**.** Let N denote the **north pole** $(0, \ldots, 0, 1) \in \mathbb{S}^n \subseteq \mathbb{R}^{n+1}$, and let S denote the **south pole** $(0, \ldots, 0, -1)$. Define the **stereographic projection** $\sigma \colon \mathbb{S}^n \smallsetminus \{N\} \to \mathbb{R}^n$ by

$$\sigma(x^1, \ldots, x^{n+1}) = \frac{(x^1, \ldots, x^n)}{1 - x^{n+1}}.$$

Let $\widetilde{\sigma}(x) = -\sigma(-x)$ for $x \in \mathbb{S}^n \smallsetminus \{S\}$.

(a) For any $x \in \mathbb{S}^n \smallsetminus \{N\}$, show that $\sigma(x) = u$, where $(u, 0)$ is the point where the line through N and x intersects the linear subspace where $x^{n+1} = 0$ (Fig. 1.13). Similarly, show that $\widetilde{\sigma}(x)$ is the point where the line through S and x intersects the same subspace. (For this reason, $\widetilde{\sigma}$ is called *stereographic projection from the south pole.*)

(b) Show that σ is bijective, and

$$\sigma^{-1}(u^1, \dots, u^n) = \frac{(2u^1, \dots, 2u^n, |u|^2 - 1)}{|u|^2 + 1}.$$

(c) Compute the transition map $\widetilde{\sigma} \circ \sigma^{-1}$ and verify that the atlas consisting of the two charts $(\mathbb{S}^n \smallsetminus \{N\}, \sigma)$ and $(\mathbb{S}^n \smallsetminus \{S\}, \widetilde{\sigma})$ defines a smooth structure on $\mathbb{S}^n$. (The coordinates defined by σ or $\widetilde{\sigma}$ are called *stereographic coordinates.*)

(d) Show that this smooth structure is the same as the one defined in Example 1.31.

Solution. (This was Problem 1.5 in previous editions)

$$S^n = \{(x_1 \dots x_{n+1}\} \in \mathbb{R}^{n+1} | \sum_{i=1}^{n+1} x_i^2 = 1\} \subset \mathbb{R}^{n+1}$$

Let $N = (0 \dots 0, 1)$ $\quad x \in S^n$.
$S = (0 \dots 0, -1)$

(a) Consider $t(x - N) + N = tx + (1 - t)N$ when $x_{n+1} = 0$

$$tx_{n+1} + (1 - t) = 0 \text{ or } tx_{n+1} + -1 + t = 0$$
$$\implies \frac{1}{1 - x_{n+1}} = t \quad \left(\text{or } \frac{1}{1 + x_{n+1}} \right)$$

$$\pi_1 : S^n - N \to \mathbb{R}^n$$
$$\pi_1(x_1 \dots x_{n+1}) = \left(\frac{x_1}{1 - x_{n+1}} \dots \frac{x_n}{1 - x_{n+1}}, 0 \right)$$
$$\pi_2 : S^n - S \to \mathbb{R}^n$$
$$\pi_2(x_1 \dots x_{n+1}) = \left(\frac{x_1}{1 + x_{n+1}} \dots \frac{x_n}{1 + x_{n+1}}, 0 \right)$$

Note that $-\pi_2(-x) = \pi_1$ and $\pi_1 \equiv \sigma$, $\pi_2 \equiv \widetilde{\sigma}$ in Massey's notation.

(b) Note, for $y_i = \frac{x_i}{1 - x_{n+1}}$

$$y_1^2 + \dots + y_n^2 = |y|^2 = \frac{1 - x_{n+1}^2}{(1 - x_{n+1})^2} = \frac{1 + x_{n+1}}{1 - x_{n+1}} \text{ or } x_{n+1} = \frac{|y|^2 - 1}{|y|^2 + 1}$$
$$x_i = y_i(1 - x_{n+1}) = \frac{2y_i}{1 + |y|^2}$$

$$\pi_1^{-1} : \mathbb{R}^n \to S^n - N$$
$$\pi_1^{-1}(y_1 \dots y_n) = \left(\frac{2y_1}{1 + |y|^2} \dots \frac{2y_n}{1 + |y|^2}, \frac{|y|^2 - 1}{|y|^2 + 1} \right)$$
$$\pi_2^{-1}(y_1 \dots y_n) = \left(\frac{2y_1}{1 + |y|^2} \dots \frac{2y_n}{1 + |y|^2}, \frac{1 - |y|^2}{|y|^2 + 1} \right)$$

π_1, π_2 diff., bijective, and $(S^n - N) \bigcup (S^n - S) = S^n$

(c) Computing the transition maps for the stereographic projections.
Consider $(S^n - N)(S^n - S) = S^n - N \bigcup S$

$$\pi_1\pi_2^{-1}(y_1 \dots y_n) = \left(\frac{y_1}{|y|^2} \cdots \frac{y_n}{|y|^2}, 0\right)$$
$$\pi_2\pi_1^{-1}(y_1 \dots y_n) = \left(\frac{y_1}{|y|^2} \cdots \frac{y_n}{|y|^2}, 0\right)$$

since, for example,

$$\frac{\frac{2y_i}{1+|y|^2}}{1 - \frac{1-|y|^2}{1+|y|^2}} = \frac{y_i}{|y|^2}$$

$\pi_1\pi_2^{-1}$ bijective and C^∞, $\pi_1\pi_2^{-1}$ diffeomorphism.

$\{(S^n - N, \pi_1), (S^n - S, \pi_2)\}$ C^∞ atlas or differentiable structure.

$$\partial_j \frac{y_i}{|y|^2} = \frac{-2y_iy_j}{(y_1^2 + \cdots + y_n^2)^2}$$
$$\partial_j \frac{y_j}{|y|^2} = \frac{(y_1^2 + \cdots + y_n^2) - 2y_j^2}{|y|^4} = \frac{y_1^2 + \cdots + \widehat{y_j^2} + \cdots + y_n^2 - y_j^2}{|y|^4}$$

$\det(\partial_j \pi_1\pi_2^{-1}(y)) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \partial_{\sigma_1} \frac{y_1}{|y|^2} \dots \partial_{\sigma_n} \frac{y_n}{|y|^2} = 0$ only if $y = 0$. But that's excluded

(d) Consider $\{(S^n \backslash, \pi_1), (S^n \backslash S, \pi_2)\}$
$\mathcal{A} = \{(U_i^\pm, \varphi_i^\pm)\}$

Now

$$S^n \backslash N \bigcap U_i^+ = \begin{cases} U_i^+ & \text{if } i \neq n+1 \\ U_{n+1}^+ \backslash N & \text{if } i = n+1 \end{cases}$$
$$S^n \backslash N \bigcap U_i^- = U_i^-$$

$$\pi_1(\varphi_i^\pm)^{-1}(y_1 \dots y_n)$$
$$= \pi_1(y_1 \dots y_{i-1}, \pm\sqrt{1 - |y|^2}, y_i \dots y_n)$$
$$= \left(\frac{y_1}{1-y_n} \cdots \frac{y_{i-1}}{1-y_n}, \frac{\pm\sqrt{1-|y|^2}}{1-y_n}, \frac{y_i}{1-y_n} \cdots \frac{y_{n-1}}{1-y_n}, 0\right)$$

Note $-1 < y_n < 1$ on $\varphi_i^\pm(S^n \backslash \bigcap U_i^\pm)$

$$\varphi_i^\pm \pi_1^{-1}(y_1 \dots y_n) = \varphi_i^\pm \left(\frac{2y_1}{1+|y|^2} \cdots \frac{2y_n}{1+|y|^2}, \frac{|y|^2-1}{|y|^2+1}\right) = \left(\frac{2y_1}{1+|y|^2} \cdots \frac{2\widehat{y_i}}{1+|y|^2} \cdots \frac{2y_n}{1+|y|^2}, \frac{|y|^2-1}{|y|^2+1}\right)$$

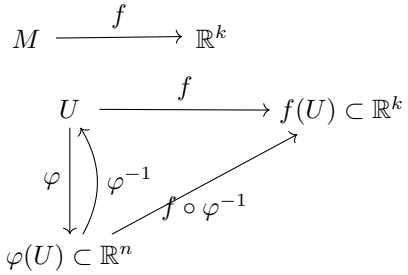
$\pi_{1,2}(\varphi_i^\pm)^{-1}, \varphi_i^\pm \pi_{1,2}^{-1}$ are diffeomorphisms (bijective and differentiable).
So $\{(S^n \backslash N, \pi_1), (S^n \backslash S, \pi_2)\} \bigcup \mathcal{A}$ also a C^∞ atlas.
So $\{(S^n \backslash N, \pi_1), (S^n \backslash S, \pi_2)\}$, $\mathcal{A}$ equivalent.

61 Smooth Maps

Smooth Functions and Smooth Maps

Definition 163. smooth function f, $f : M \to \mathbb{R}^k$, if $\forall p \in M$, $\exists$ smooth chart (U, φ), $U \ni p$, s.t. $f \circ \varphi^{-1}$ smooth on $\varphi(U) \subseteq \mathbb{R}^n$, $f \circ \varphi^{-1} : \varphi(U) \to \mathbb{R}^k$

EY : 20150717 Recall, smooth chart just means that the chart belongs to maximal smooth atlas, and smooth in that the transition maps are smooth.



Exercise (Lee 2e, Exercise 2.3, p. 33 [76])**.** Let M be a smooth manifold with or without boundary, and suppose $f\colon M \to \mathbb{R}^k$ is a smooth function. Show that $f \circ \varphi^{-1}\colon \varphi(U) \to \mathbb{R}^k$ is smooth for *every* smooth chart (U, φ) for M.

Solution. Given $f : M \to \mathbb{R}^k$ smooth,
Consider $p \in M$. M smooth manifold, so $\exists$ chart (U, φ) s.t. $p \in U$
$\varphi : U \subset M \to \mathbb{R}^m$
φ homeomorphism, $\varphi(U)$ open in $\mathbb{R}^m$ $\varphi^{-1} : \varphi(U) \to M$
Consider another smooth chart (V, ψ), $p \in V$ so that $UV \neq \emptyset$

$$f\psi^{-1} : \psi(UV) \to \mathbb{R}^k$$
$$f\psi^{-1}(y^1 \dots y^m) = (f\psi^{-1})_i(y^1 \dots y^m), \quad i = 1 \dots k$$
$$f\psi^{-1} = f\varphi^{-1}\varphi\psi^{-1} = (f\varphi^{-1})\varphi\psi^{-1}$$

$\varphi\psi^{-1}$ C^∞ (diffeomorphisms are smooth). $f\varphi^{-1}$ is smooth. $f\psi^{-1}$ also smooth.

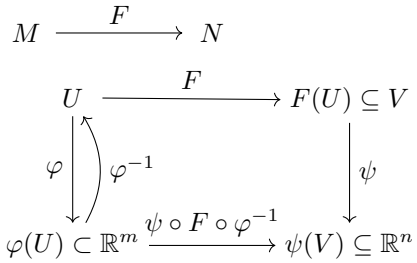
Smooth Maps Between Manifolds

Definition 164. Let M, N be smooth manifolds.

smooth map F, $F : M \to N$ if $\forall\, p \in M$, $\exists$ smooth charts (U, φ) $U \ni p$
(V, ψ) $V \ni F(p)$

s.t. $F(U) \subseteq V$ and
$\psi \circ F \circ \varphi^{-1}$ smooth from $\varphi(U)$ to $\psi(V)$.

i.e.



Exercise (Lee 2e, Exercise 2.9, p. 35 [76])**.** Suppose $F\colon M \to N$ is a smooth map between smooth manifolds with or without boundary. Show that the coordinate representation of F with respect to *every* pair of smooth charts for M and N is smooth.

Note. Numbered Exercise 2.6 in the first edition.

Solution. smooth $F : M \to N$
(U, φ), (U', φ') smooth chart for M (V, ψ), (V', ψ') smooth chart for N

$$\psi' F (\varphi')^{-1} = \psi' \psi^{-1} \psi F \varphi^{-1} \varphi (\varphi')^{-1} = (\psi' \psi^{-1})(\psi F \varphi^{-1})(\varphi (\varphi')^{-1})$$

$\psi' \psi^{-1}$, $\varphi (\varphi')^{-1}$ aresmooth (in fact diffeomorphisms).
$\psi F \varphi^{-1}$ given to be smooth.
So $\psi' F (\varphi')^{-1}$ smooth.
Example 2.5. (**Smooth Maps**)

(a) inclusion $i : S^n \hookrightarrow \mathbb{R}^{n+1}$

(b)

$\varphi_i^{\pm} : S^n \to B_1^n(0) \subset \mathbb{R}^n$ ($B_1^n(0)$ disk of radius 1; $\varphi_i^{\pm}$ is like a projection of a half hemisphere to a plane)
$\varphi_i^{\pm}(x_1 \dots x_{n+1}) = (x_1 \dots \widehat{x}_i \dots x_{n+1})$
$(\varphi_i^{\pm})^{-1}(y_1 \dots y_n) = (y_1 \dots y_{i-1}, \pm\sqrt{1 - |y|^2}, y_i \dots y_n)$ $|y|^2 = y_1^2 + \cdots + y_n^2$

$\widehat{i} : \mathbb{R}^n \hookrightarrow \mathbb{R}^{n+1}$
$\widehat{i}(u_1 \dots u_n) = i(\varphi_i^{\pm})^{-1}(u^1 \dots u^n) = (u^1 \dots u^{i-1}, \pm\sqrt{1 - |u|^2}, u_i \dots u_n)$ $|u|^2 = (u^1)^2 + \cdots + (u^n)^2$

$\widehat{i}$ is coordinate representation, and clearly, from the above formula, is smooth.

(c) $\pi : \mathbb{R}^{n+1} - 0 \to \mathbb{R}P^n$ smooth because

$$\widehat{\pi}(x^1 \dots x^{n+1}) = \varphi_i \pi(x^1 \dots x^{n+1}) = \varphi_i [x^1 \dots x^{n+1}] = \left(\frac{x^1}{x^i} \cdots \frac{\widehat{x^i}}{x^i} \cdots \frac{x^{n+1}}{x^i} \right)$$

$\varphi_i : U_i \to \mathbb{R}^n$

(d) $p : S^n \to \mathbb{R}P^n$ is restriction of $\pi : \mathbb{R}^{n+1} - 0 \to \mathbb{R}P^n$ to $S^n \subset \mathbb{R}^{n+1} - 0$. $\pi|_{S^n} = p$
$p = \pi i$ π, i smooth, so p smooth.
$p^{-1} = i^{-1}\pi^{-1}$. $|x|^2 = (x^1)^2 + \cdots + (x^{n+1})^2$
On $\mathbb{R}P^n$,

$$\begin{aligned} pp^{-1}[x^1 \dots x^{n+1}] &= pi^{-1}\pi^{-1}[x^1 \dots x^{n+1}] \\ &= \pi^{-1}\left(\frac{x^1}{|x|} \cdots \frac{x^{n+1}}{|x|} \right) \\ &= \pi i \left(\frac{x^1}{|x|} \cdots \frac{x^{n+1}}{|x|} \right) \\ &= \left[\frac{x^1}{|x|} \cdots \frac{x^{n+1}}{|x|} \right] = [x^1 \dots x^{n+1}] \end{aligned}$$

$$pp^{-1}[tx^1 \dots tx^{n+1}] = \pi^{-1}\left(\frac{tx^1}{|t||x|} \cdots \frac{tx^{n+1}}{|t||x|} \right) = [x^1 \dots x^{n+1}]$$

$$p^{-1}p(x^1 \dots x^{n+1}) = p^{-1}[x^1 \dots x^{n+1}] = (x^1 \dots x^{n+1}, \text{ as } (x^1)^2 + \cdots + (x^{n+1})^2 = 1$$

p bijective, smooth, and

$$\widehat{\pi}^{-1} = \pi^{-1}\varphi_i^{-1}(y^1 \dots y^n) = \pi^{-1}(y^1 \dots y^{i-1}, 1, y^i, y^i \dots y^n) = (y^1 \dots y^{-1}, 1, y^i \dots y^n) \text{ smooth}$$

$$\widehat{i}^{-1}(x^1 \dots x^{n+1}) = (\varphi_i^{\pm})(i^{-1}(x^1 \dots x^{n+1})) = (\varphi_i^{\pm})\left(\frac{x^1}{|x|} \cdots \frac{x^{n+1}}{|x|} \right) = \left(\frac{x^1}{|x|} \cdots \frac{\widehat{x^i}}{|x|} \cdots \frac{x^{n+1}}{|x|} \right)$$

with $|x|^2 = (x^1)^2 + \cdots + (x^{n+1})^2$
$\widehat{i}^{-1}$ smooth.
So $p^{-1} = i^{-1}\pi^{-1}$ smooth.
p diffeomorphism.

Diffeomorphisms

Lie Groups

Lie group - smooth manifold G, with $m : G \times G \to G$ $\quad$ $i : G \to G$ $\quad$ m, i smooth.

$$m(g,h) = gh \qquad i(g) = g^{-1}$$

Exercise (Lee 2e, Exercise 7.2, p. 151 [76])**.** Prove Proposition 7.1.

Note. Numbered Exercise 2.10 in the first edition. Proposition 7.1: if G is a smooth manifold with a group structure such that the map $G \times G \to G$ given by $(g,h) \mapsto gh^{-1}$ is smooth, then G is a Lie group.

Solution.

$$f : G \times G \to G$$
$$(g,h) \mapsto gh^{-1}$$
$$m(g,h) = f(g,h^{-1}) = gh \qquad (\exists\, h^{-1} \text{ since } G \text{ is a group})$$
$$f(e,g) = eg^{-1} = g^{-1} = i(g)$$

So m, i smooth as f is smooth.

Example 2.8 (Lie Group Homomorphisms)

(a)

(b)

(c)

(d)

(e)

(f) $C_g : G \to G$ conjugation.

$$C_g(h) = ghg^{-1}$$

C_g smooth because Lie group multiplication is smooth.

Suppose $hl = m$

$$C_g(hl) = ghlg^{-1} = ghg^{-1}glg^{-1} = C_g(m) = C_g(h)C_g(l)$$

Smooth Covering Maps

Partitions of Unity

Theorem 48 (2.23)**.** ***(Existence of Partitions of Unity)****. Suppose M is a smooth manifold with or without boundary, and $\chi = (X_\alpha)_{\alpha \in A}$ is any indexed open cover of M.*

Then $\exists$ smooth partition of unity subordinate to χ

Proof. □

Applications of Partitions of Unity

Lemma 9 (2.26)**.** *Suppose M smooth manifold with or without boundary.*
closed $A \subseteq M$
$f : A \to \mathbb{R}^k$ smooth function.

$\forall$ *open U, $U \supset A$,*
$\exists$ *smooth $\widetilde{f} : M \to \mathbb{R}^k$ s.t. $\widetilde{f}\big|_A = f$ and $supp\widetilde{f} \subseteq U$*

Proof. Given $A \subseteq M$
$f : A \to \mathbb{R}^k$
$\forall\, U \supseteq A$

$\forall\, p \in A$, choose neighborhood W_p of p, smooth $\widetilde{f}_p : W_p \to \mathbb{R}^k$ s.t. $\widetilde{f}_p = f$ on $W_p \bigcap A$

Replace W_p by $W_p \bigcap U$, so $W_p \subseteq U$
$\{W_p | p \in A\} \bigcup \{M \backslash A\}$ open cover of M
Let $\{\psi_p | p \in A\} \bigcup \{\psi_0\}$ smooth partition of unity subordinate to this cover, with $\text{supp}\psi_p \subseteq W_p$, $\text{supp}\psi_0 \subseteq M \backslash A$ (Thm.2.23, Existence of Partition of Unity)

$\forall\, p \in A$, $\psi_p \widetilde{f}_p$ smooth on W_p, and $\psi_p \widetilde{f}_p$ has smooth extension to all of M if $\psi_p \widetilde{f}_p = 0$ on $M \backslash \text{supp}\psi_p$ on open $W_p \backslash \text{supp}\psi_p$, they agree
define $\widetilde{f} : M \to \mathbb{R}^k$

$$\widetilde{f}(x) = \sum_{p \in A} \psi_p(x) \widetilde{f}_p(x)$$

$\{\text{supp}\psi_p\}$ locally finite, so $\sum_{p \in A} \psi_p \widetilde{f}_p(x)$ has only finite number of nonzero terms in neighborhood of $\forall\, x \in M$, so $\widetilde{f}(x)$ smooth
If $x \in A$, $\psi_0(x) = 0$, $\widetilde{f}_p(x) = f(x)$ $\quad \forall\, p$ s.t. $\psi_p(x) \neq 0$, so

$$\widetilde{f}(x) = \sum_{p \in A} \psi_p(x) f(x) = (\psi_0(x) + \sum_{p \in A} \psi_p(x)) f(x) = f(x)$$

so $\widetilde{f}$ extension of f.
Lemma 1.13(b), so $\text{supp}\widetilde{f} = \overline{\bigcup_{p \in A} \text{supp}\psi_p} = \bigcup_{p \in A} \text{supp}\psi_p \subseteq U$

□

Problems

Exercise (Lee 2e, Proposition 7.14, p. 156 [76])**.** Suppose G is a Lie group, and $W \subseteq G$ is any neighborhood of the identity.

(a) W generates an open subgroup of G.

(b) If W is connected, it generates a connected open subgroup of G.

(c) If G is connected, then W generates G.

Note. Problem 2-11 in the first edition; in the second edition it is part (c) of this proposition, proved in the text.

Solution. G connected Lie group.
$U \subset G$ neighborhood of identity e.

Let $H \leq G$ subgroup generated by U. (cf. wikipedia - Generating set of a group U of H s.t. $\forall\, h \in H$, $h =$ finite combination of u's $\in U$, u^{-1}'s)
$hU \subset H$. $\forall\, h \in H$
hU open neighborhood of h, since multiplication by h is cont. (U open, so $h^{-1}(hU)$ open)
So h open.

Let $g \in H^c$, $\left\{^{U' = \{u^{-1} | u \in U}\right\}$. $i(u) = u^{-1}$. i inversion map, cont. $i(U) = i^{-1}(U) = U'$ open.
$gU' \subset H^c$ (otherwise $g \in H$, for if $h \in gU' \bigcap H$, $g \in hU \subset H$)
H^c open, so H closed.
H open and closed, so since G connected, $H = G$. U generates G.

62 Tangent Vectors

Tangent Vectors

Geometric Tangent Vectors

Now, 1 thing that a Euclidean tangent vector provides is a means of taking "directional derivatives" of a function.

e.g. $\forall\, v_a \in \mathbb{R}^n_a$, v_a yields $D_v|_a : C^\infty\mathbb{R}^n \to \mathbb{R}$ (3.1)

$$D_v|_a\, f = D_v f(a) = \left.\frac{d}{dt}\right|_{t=0} f(a+tv)$$

which takes the directional derivative in the direction v at a.

$D_v|_a$ linear and $D_v|_a\,(fg) = f(a)\; D_v|_a\, g + g(a)\; D_v|_a\, f$ (3.1)

$$\left.\frac{d}{dt}\right|_{t=0} f(a+tv) = v^i \frac{\partial f}{\partial x^i}(a)$$

$$D_v|_a\, f = v^i \frac{\partial f}{\partial x^i}(a)$$

If $v_a = e_j|_a$,

$$D_v|_a\, f = \frac{\partial f}{\partial x^i}(a)$$

derivation at a, a linear $X : C^\infty\mathbb{R}^n \to \mathbb{R}$, $a \in \mathbb{R}^n$

$$X(fg) = f(a)Xg + g(a)Xf$$

$T_a\mathbb{R}^n$ set of all derivations of $C^\infty\mathbb{R}^n$ at a. $T_a\mathbb{R}^n$ vector space.

Lemma 10 (3.1)**.** *$X(c) = 0$, 0 const., $X(fg) = 0$ if $f(a) = g(a) = 0$*

Proposition 44 (3.2)**.** *$\forall\, a \in \mathbb{R}^n$, map $v_a \mapsto D_v|_a$ isomorphism from $\mathbb{R}^n_a$ onto $T_a\mathbb{R}^n$*

Proof. $v_a \mapsto D_v|_a$ linear.

$$D_{bv+cw}|_a\, f = D_{bv+cw} f(a) = \left.\frac{d}{dt}\right|_{t=0} f(a+t(bv+cw)) = (bv^i + cw^i)\frac{\partial f}{\partial x^j}(a) = bv^j \frac{\partial f}{\partial x^j}(a) + cw^i \frac{\partial f}{\partial x^i}(a) =$$

$$= b\left.\frac{d}{dt}\right|_{t=0} f(a+tv) + c\left.\frac{d}{dt}\right|_{t=0} f(a+tw) = b\; D_v|_a\, f + c\; D_w|_a\, f = (bD_v|_a + c\; D_w|_a) f$$

injective: $v_a \in \mathbb{R}^n_a$, write $v_a = v^i e_i\big|_a$

take f to be jth coordinate function $x^j : \mathbb{R}^n \to \mathbb{R}$, thought of as a smooth function on $\mathbb{R}^n$

$$0 = D_v|_a\,(x^j) = v^i \delta^j_i = v^j \quad \forall\, j$$

Then $v_a = 0$

surjective, let $X \in T_a\mathbb{R}^n$

define $v^i = X(x^i)$

We'll show $X = D_v|_a$, $v = v^i e_i$

Let f be any smooth function on $\mathbb{R}^n$, $f : \mathbb{R}^n \to \mathbb{R}$.

By Taylor's formula with remainder (Thm. A.58), $\exists$ smooth $g_1 \dots g_n$ on $\mathbb{R}^n$ s.t. $g_i(a) = 0$

$$f(x) = f(a) + \sum_{i=1}^n \frac{\partial f}{\partial x^i}(a)(x^i - a^i) + \sum_{i=1}^n g_i(x)(x^i - a^i)$$

Recall Lemma 3.1, and note $x^i - a^i = 0$ if $x = a$

$$Xf = X(f(a)) + \sum_{i=1}^{n} X\left(\frac{\partial f}{\partial x^i}(a)(x^i - a^i)\right) + \sum_{i=1}^{n} X(g_i(x)(x^i - a^i)) = 0 + \sum_{i=1}^{n} \frac{\partial f}{\partial x^i}(a)\left(X(x^i) - X(a^i)\right) =$$

$$= \sum_{i=1}^{n} X(x^i)\frac{\partial f}{\partial x^j}(a) = v^i \frac{\partial f}{\partial x^i}(a) = D_v|_a f$$

$$\implies X = D_v|_a$$

□

Corollary 7 (3.3). *$\forall a \in \mathbb{R}^n$, n derivatives. $\frac{\partial}{\partial x^1}\big|_a \dots \frac{\partial}{\partial x^n}\big|_a$ defined by $\frac{\partial}{\partial x^i}\big|_a f = \frac{\partial f}{\partial x^i}(a)$ form a basis for $T_a\mathbb{R}^n$, $dim T_a\mathbb{R}^n = n$*

Proof. as above, $\forall X \in T_a\mathbb{R}^n$, X derivation. $Xf = v^i \frac{\partial}{\partial x^i}\big|_a f$, so $\{\frac{\partial}{\partial x^i}\big|_a\}$ spans $T_a\mathbb{R}^n$
for linear independence, $0 = v^i \frac{\partial}{\partial x^i}\big|_a f$. Then $v^i = 0, \quad \forall i$
Note $\frac{\partial}{\partial x^i}\big|_a = D_{e_i}|_a$ with $e_i = \delta_i{}^j e_j$.

□

Tangent Vectors on a Manifold

linear $X : C^\infty M \to \mathbb{R}$ derivation at p if $X(fg) = f(p)Xg + g(p)Xf$. $\forall f, g \in C^\infty M$
tangent space T_pM = set of all derivations of $C^\infty M$

Exercise (Lee 2e, Exercise 3.5, p. 54 [76]). Prove Lemma 3.4.

Note. Numbered Exercise 3.1 in the first edition. Lemma 3.4: for $v \in T_pM$ and $f, g \in C^\infty(M)$, (a) if f is constant then $vf = 0$; (b) if $f(p) = g(p) = 0$ then $v(fg) = 0$.

Solution. Lemma 3.4

(a) f const. So let $f = 0$.

$$X(cf) = cX(f) = X(ff) = f(p)Xf + f(p)Xf = 2X(cf) \implies X(f) = 0$$

(b) if $f(p) = g(p) = 0$, $X(fg) = 0$, by definition.

Pushforwards

Let smooth $F : M \to N$
$\forall p \in M$, define $F_* : T_pM \to T_{F(p)}N$, pushforward

$$(F_*X)(f) = X(fF)$$

$f \in C^\infty N$, $fF \in C^\infty M$, so $X(fF)$ makes sense.

$$(F_*X)(fg) = X((fg)F) = X((fF)(gF)) = fF(p)X(gF) + gF(p)X(fF) = f(F(p))(F_*X)g + g(F(p))(F_*X)(f)$$

Lemma 11 (3.5). ***(Properties of Pushforwards)***

(a)

$$(F_*X)(af + bg) = X((af + bg)F) = aX(fF) + bX(gF) = aF_*X(f) + bF_*X(g)$$

F_*X *linear*

(b)

$$gf_* = g_*f_* : T_pM \to T_{gf(p)}P$$

Exercise (Lee 2e, Exercise 3.7, p. 56 [76]). Prove Proposition 3.6.

Note. Numbered Exercise 3.2 in the first edition. Proposition 3.6 (Properties of Differentials): dF_p is linear; $d(G \circ F)_p = dG_{F(p)} \circ dF_p$; $d(\mathrm{Id}_M)_p = \mathrm{Id}_{T_pM}$; if F is a diffeomorphism, dF_p is an isomorphism with $(dF_p)^{-1} = d(F^{-1})_{F(p)}$.

Solution.

(a)

(b)
$$\begin{array}{ll} f: M \to N & \text{Consider} \quad p \in M, (U,\varphi) \subset M \quad V \in T_pM \quad V = V^\alpha \dfrac{\partial}{\partial x^\alpha} \\ g: N \to P & \qquad\quad q = f(p) \in N, (V,\psi) \subset N \quad W \in T_qN \quad W = W^\beta \dfrac{\partial}{\partial y^\beta} \\ gf: M \to P & \qquad\quad r = g(q) \in P, (W,\chi) \subset P \quad X \in T_{g(q)}P \quad X = X^\gamma \dfrac{\partial}{\partial z^\gamma} \end{array}$$

$$\begin{array}{lll} f_*V \in T_{f(p)}N & g_*W \in T_{g(q)}P & gf_*V \in T_{gf(p)}P \\ f_*V[h] = V[hf] & g_*W[k] = W[kg] & gf_*V[l] = V[lgf] \\ f_*V[h\psi^{-1}(y)] = V[hf\varphi^{-1}(x)] & g_*W[k\chi^{-1}(z)] = W[kg\psi^{-1}(y)] & gf_*V[l\chi^{-1}(z)] = V[lgf\varphi^{-1}(x)] \end{array}$$

$$\Longrightarrow g_*(f_*V)[l] = f_*V[lg] = V[lgf] = gf_*V[l]$$

In coordinates,

$$\begin{aligned} g_*(f_*V)[l\chi^{-1}(z)] &= f_*V[lg\psi^{-1}(y)] \\ &= W^\alpha \frac{\partial}{\partial y^\alpha}[lg\psi^{-1}(y)] \\ = V^\mu \frac{\partial y^\alpha}{\partial x^\mu}\frac{\partial}{\partial y^\alpha}[lg\psi^{-1}(y)] &= V^\mu \frac{\partial(y^\alpha f\varphi^{-1}(x))}{\partial x^\mu}\frac{\partial}{\partial y^\alpha}[lg\psi^{-1}(y)] \\ gf_*V[l\chi^{-1}(z)] = V[lgf\varphi^{-1}(x)] &= V^\alpha \frac{\partial}{\partial x^\alpha}(lgf\varphi^{-1}(x)) \\ \Longrightarrow \frac{\partial}{\partial x^\mu}(lgf\varphi^{-1}(x)) &= \frac{\partial(y^\alpha f\varphi^{-1}(x))}{\partial x^\mu}\frac{\partial}{\partial y^\alpha}[lg\psi^{-1}(y)] \end{aligned}$$

Chain rule is reobtained.

Alternatively,

Let $F: M \to N \qquad p \in M$
$\quad G: N \to P$

$GF: M \to P \qquad . \ h \in C^\infty P$
$(GF)_*: T_pM \to T_{GF(p)}P$
Now consider

$$G_*(F_*X)(h) = F_*X(hG) = X(hGF) \Longrightarrow G_*F_* = (GF)_*$$

(c)
$$(1_M)_*X(f) = X(f1) = X(f)$$

so $(1_M)_* = 1_{T_pM}$

(d) Now
$$M \xrightarrow{F} N$$
$$T_pM \xrightarrow{F_*} T_{F(p)}N$$

cf. Tu, pp. 80, 8 Tangent Space, Corollary 8.7. If $F: M \to N$, $p \in M$, F diffeomorphism,
$F_*: T_pM \to T_{F(p)}N$ isomorphism.

Proof. To say that F is a diffeomorphism, means that it has a differentiable inverse $G : N \to M$ s.t.

$$\begin{aligned} GF &= 1_M & (GF)_* &= G_*F_* = (1_M)_* = 1_{T_pM} \\ FG &= 1_N & (FG)_* &= F_*G_* = (1_N)_* = 1_{T_{F(p)}N} \end{aligned}$$

So then F_*, G_* are isomorphisms, bijective homomorphism. □

identify T_pU with T_pM $\forall p \in U$. Since the action of a derivation on a function depends only on the values of the function in an arbitrary small neighborhood. In particular, this means that any tangent vector $X \in T_pM$ can be unambiguously applied to functions defined only in a neighborhood of p not necessarily on all of M (note partition of unity, bump functions).

Proposition 45 (3.7)**.** *open submanifold $U \subset M$, inclusion $i : U \hookrightarrow M$. $\forall p \in U$, $i_* : T_pU \to T_pM$ isomorphism.*

Exercise (Lee 2e, Proposition 3.6(d), p. 55 [76])**.** If F is a diffeomorphism, then $dF_p \colon T_pM \to T_{F(p)}N$ is an isomorphism, and $(dF_p)^{-1} = d(F^{-1})_{F(p)}$.

Note. First-edition Exercise 3.3 (local diffeomorphisms). In the second edition it follows from this part of Proposition 3.6, applied to $F|_U$, together with Proposition 3.9 (the tangent space to an open submanifold).

Solution. If $F : M \to N$ local diffeomorphism,
$\forall p \in M$, $\exists$ open $U \ni p$ s.t. $F(U)$ open in N and $F|_U : U \to F(U)$ diffeomorphism.
Consider $G : F(U) \to U$, G diff. (smooth) inverse of $F|_U$.
$F(p) \in$ open $F(U)$

$$\begin{aligned} (F|_U)_* &: T_pU \to T_{F(p)}F(U) \\ (G)_* &: T_{F(p)}F(U) \to T_pU \\ (FG)_* &= (1_{F(U)})_* = 1_{T_{F(P)}}F(U) = (F|_U)_*G_* \\ (GF)_* &= (1_U)_* = 1_{T_pU} = G_*(F|_U)_* \end{aligned}$$

Then $(F|_U)_*$, G_* are isomorphisms between $T_pU \to T_{F(p)}F(U)$. This must be true $\forall p \in M$, so $F_* : T_pM \to T_{F(p)}N$ isomorphism $\forall p \in M$

I think the idea for a local diffeomorphism is that "$F_* : T_pM \to T_{F(p)}F(M)$".

Computations in Coordinates

Change of Coordinates

The Tangent Bundle

Proposition 46 (3.18)**.** *TM has smooth structure making it $2n$-dim. smooth manifold.*
$\pi : TM \to M$ smooth

Proof. $\forall$ chart (U, φ) for M, $\varphi = (x^1 \dots x^n)$
Define $\widetilde{\varphi} : \pi^{-1}(U) \to \mathbb{R}^{2n}$

$$\widetilde{\varphi}\left(v^i \left.\frac{\partial}{\partial x^i}\right|_p\right) = (x^1(p) \dots x^n(p), v^1 \dots v^n)$$

$\widetilde{\varphi}(\pi^{-1}(U)) = \varphi(U) \times \mathbb{R}^n$, which is open
$\widetilde{\varphi}$ bijection since

$$\widetilde{\varphi}^{-1}(x^1 \dots x^n, v^1 \dots v^n) = v^i \left.\frac{\partial}{\partial x^i}\right|_{\varphi^{-1}(x)}$$

$$\widetilde{\varphi}\widetilde{\varphi}^{-1} = 1_{\mathbb{R}^{2n}},\ \widetilde{\varphi}^{-1}\widetilde{\varphi} = 1_{\pi^{-1}(U) \subset T_pM}$$

Suppose charts (U, φ) for M,
(V, ψ)

$(\pi^{-1}(U), \widetilde{\varphi})$ on TM ($\widetilde{\varphi}, \widetilde{\psi}$ homeomorphisms, cont. bijective, cont. inverse, π cont., $\pi^{-1}(U)$ open)
$(\pi^{-1}(V), \widetilde{\psi})$ $\pi^{-1}(V)$

$\widetilde{\varphi}(\pi^{-1}(U)\pi^{-1}(V)) = \varphi(UV) \times \mathbb{R}^n$ open in $\mathbb{R}^{2n}$
$\widetilde{\psi}(\pi^{-1}(U)\pi^{-1}(V)) = \psi(UV) \times \mathbb{R}^n$
$\widetilde{\psi}\widetilde{\varphi}^{-1} : \varphi(UV) \times \mathbb{R}^n \to \psi(UV) \times \mathbb{R}^n$
$\widetilde{\psi}\widetilde{\varphi}^{-1}(x^1 \dots x^n, v^1 \dots v^n) = (y^1(x) \dots y^n(x), \frac{\partial y^1}{\partial x^j}(x)v^j \dots \frac{\partial y^n}{\partial x^j}(x)v^j)$

$\widetilde{\psi}\widetilde{\varphi}^{-1}$ clearly smooth.
Choose countable cover $\{U_i\}$ of M by smooth coordinate domains.
$\{\pi^{-1}(U_i)\}$ countable cover of TM by coordinate domains.
fiber of π : $\pi^{-1}(\{p\})$ (fiber is like a preimage of a singleton set)
Consider $\widetilde{x}, \widetilde{y} \in \pi^{-1}(\{p\})$, then $\widetilde{x}, \widetilde{y} \in \widetilde{\varphi}$ (lie in 1 chart)

If $(p, X), (q, Y)$ lie in different fibers, $\exists$ disjoint smooth coordinate domains U, V for M (M Hausdorff) s.t. $p \in U$
$q \in V$
and

$\pi^{-1}(U), \pi^{-1}(V)$ disjoint, smooth coordinate neighborhoods s.t. $\pi^{-1}(U) \ni (p, X)$
$\pi^{-1}(V) \ni (q, Y)$
$\pi(x, v) = x$, so π smooth.

□

The Tangent Space to a Manifold with Boundary

define pushforward by F at $p \in M$ to be linear $F_* : T_pM \to T_{F(p)}N$ defined by $(F_*X)f = X(fF)$

Lemma 12 (3.10)**.** *If M^n with boundary, $p \in \partial M$,*
then T_pM n-dim. vector space with basis $\left(\frac{\partial}{\partial x^1}\big|_p \dots \frac{\partial}{\partial x^n}\big|_p\right)$ in any smooth chart.

Proof. T_pM vector space with basis $\left(\frac{\partial}{\partial x^i}\big|_p\right)$
$\forall$ smooth coordinate map φ, $\varphi_* : T_pM \to T_{\varphi(p)}\mathbb{H}^n$ isomorphism by the same argument as manifolds.
$\forall\, a \in \partial\mathbb{H}^n$, T_aH^n n-dim. and spanned by $\left(\frac{\partial}{\partial x^i}\big|_p\right)$.
Consider inclusion $i : \mathbb{H}^n \hookrightarrow \mathbb{R}^n$. Show $i_* : T_a\mathbb{H}^n \to T_a\mathbb{R}^n$ isomorphism.
Suppose $i_*X = 0$.
Let smooth $f \in \mathbb{R}$ on neighborhood of a in $\mathbb{H}^n$
Let $\widetilde{f}$ extension of f to smooth function on an open subset of $\mathbb{R}^n$ (by extension lemma)

$$\widetilde{f} \circ i = f$$
$$Xf = X(\widetilde{f}i) = (i_*X)\widetilde{f} = 0$$

Then $X = 0$. So i_* injective.
If arbitrary $Y \in T_a\mathbb{R}^n$, define $X \in T_a\mathbb{H}^n$, by

$$Xf = Y\widetilde{f} \qquad Y^i \left.\frac{\partial}{\partial x^i}\right|_a \widetilde{f} = Y^i \frac{\partial \widetilde{f}}{\partial x^i}(a)$$

This is well-defined because by cont. the derivatives of $\widetilde{f}$ at a are determined by those of f in $\mathbb{H}^n$

$$X(fg) = Y(\widetilde{f}\widetilde{g}) = Y^i \left.\frac{\partial}{\partial x^i}\right|_a (\widetilde{f}\widetilde{g}) = Y^i \frac{\partial \widetilde{f}(a)}{\partial x^i}\widetilde{g}(a) + Y^i \widetilde{f}(a)\frac{\partial \widetilde{g}}{\partial x^i}(a) = \widetilde{g}(a)Y(\widetilde{f}) + \widetilde{f}(a)Y(\widetilde{g}) =$$
$$= g(a)Xf + f(a)Xg$$

X derivation at a. $Y = i_*X$, so i_* surjective.

i_* isomorphism.

□

Tangent Vectors to Curves

tangent vector to γ at $t_0 \in J \subset \mathbb{R}$

$$\gamma'(t_0) = \gamma_* \left(\left.\frac{d}{dt}\right|_{t_0} \right) \in T_{\gamma(t_0)}M$$

Tangent vectors act on functions by

$$\gamma'(t_0)f = \left(\gamma_* \left.\frac{d}{dt}\right|_{t_0} \right) f = \left.\frac{d}{dt}\right|_{t_0} (f\gamma) = \frac{d(f\gamma)}{dt}(t_0)$$

$\gamma : I \to M$
$\gamma_* : T_{t_0}\mathbb{R} \to T_pM$
$\dot{\gamma} : I \to T_pM$
For (U, x^i), $p \in U$

$$\dot{\gamma}(t_0)f = \gamma_* \left(\left.\frac{d}{dt}\right|_{t_0} \right) f = \left.\frac{d}{dt}\right|_{t_0} (f\gamma) = \left.\frac{d}{dt}\right|_{t_0} (f(x^i)^{-1}x^i\gamma) = \left.\frac{d}{dt}\right|_{t_0} (f(\gamma^i)(t)) = \left.\frac{\partial f}{\partial x^i}\right|_p \left.\frac{d\gamma^i}{dt}\right|_{t_0} = \dot{\gamma}^i \left.\frac{\partial f}{\partial x^i}\right|_p$$
$$\implies \dot{\gamma}(t_0) = \dot{\gamma}^i(t_0) \left.\frac{\partial}{\partial x^i}\right|_p$$

Lemma 13 (3.11)**.** *Let $p \in M$. $\forall\, X \in T_pM$, X tangent vector is some smooth curve in M.*

Proof. Let (U, φ), $p \in U$, $X = X^i \left.\frac{\partial}{\partial x^i}\right|_p$

Define $\gamma : (-\epsilon, \epsilon) \to U$ by $\gamma(t) = (tX^1 \dots tX^n)$ i.e. $\gamma(t) = \varphi^{-1}(tX^1 \dots tX^n)$
$\gamma(0) = p$, $\gamma'(0) = X^i \left.\frac{\partial}{\partial x^i}\right|_{\gamma(0)} = X$

□

tangent vectors to curves behave well under composition with smooth maps.

Proposition 47 (3.12)**.** *(The tangent vector to a composite curve) Let smooth $F : M \to N$, smooth curve $\gamma : J \to M$*

$\forall\, t_0 \in J$, tangent vector $F\gamma : J \to N$, $t = t_0$ given by

$$(F\gamma)'(t_0) = F_*(\gamma'(t_0))$$

Proof.

$$(F\gamma)'(t_0) = (F\gamma)_* \left.\frac{d}{dt}\right|_{t_0} = F_*\gamma_* \left.\frac{d}{dt}\right|_{t_0} = F_*(\gamma'(t_0))$$

(use def. of tangent vector to a curve) □

Use it to compute pushforwards.
Suppose $F : M \to N$. $F_* = ?$
$\forall\, X \in T_pM$, choose smooth γ whose tangent vector at $t = 0$ is X,

$$F_*X = (F\gamma)'(0) \qquad (3.10)$$

Indeed, Lemma 3.11 $\gamma(0) = p$
$\dot{\gamma}(0) = X$

$$F_*(\gamma'(0)) = F_*X = (\dot{F\gamma})(0)$$

Alternative Definitions of the Tangent Space

smooth function element (f, U), open $U \subset M$, smooth $f : U \to \mathbb{R}$
$\forall\, p \in M$, $(f, U) \sim (g, V)$, if $f \equiv g$ on some neighborhood $W \ni p$

$$[(f, U)] = \text{ germ of } f \text{ at } p$$
$$\{[(f, U)]\} \text{ at } p \ = C_p^\infty$$

C_p^∞ real vector space.

$$[(f, U)] + [(g, V)] = [(f + g, UV)]$$
$$c[(f, U)] = [(cf, U)]$$
$$[(f, U)][(g, V)] = [(fg, UV)]$$

Denote $[(f, U)] = [f]_p$
$T_pM =$ set of all derivations, linear $X : C_p^\infty \to \mathbb{R}$ s.t.

$$X[fg]_p = f(p)X[g]_p + g(p)X[f]_p$$

By Prop. 3.6. $Xf = Xg$ if $f = g$ on some neighborhood W of p ($\psi \in C^\infty M$ smooth bump function with support needed).

This space is isomorphic to the tangent space as we've defined it (Prob. 3-7).

Problems

Exercise (Lee 2e, Problem 3-1, p. 75 [76])**.** Suppose M and N are smooth manifolds with or without boundary, and $F\colon M \to N$ is a smooth map. Show that $dF_p\colon T_pM \to T_{F(p)}N$ is the zero map for each $p \in M$ if and only if F is constant on each component of M.

Solution. M connected.
$X \in T_pM$

$$F_*X(fg) = X((fg)F) = X((fF)(gF)) = f(F(p))X(gF) + g(F(p))X(fF) = 0$$

Let $g = f \in C^\infty M$. $f(F(p))X(fF) = 0$ $X(fF) = 0$. fF const. by properties of tangent vector X. f arbitrary, so F const.

Exercise (Lee 2e, Theorem 2.17, p. 39 [76])**. (Diffeomorphism Invariance of Dimension).** A nonempty smooth manifold of dimension m cannot be diffeomorphic to an n-dimensional smooth manifold unless $m = n$.

Note. First-edition Problem 3-3. Second-edition Problem 3-3 is a different problem ($T(M \times N) \approx TM \times TN$).

Solution. M^m diffeomorphic to N^n by F.
T_pM isomorphic to $T_{F(p)}N$. Then $\dim(T_pM) = \dim(T_{F(p)}N)$ $m = n$

cf. Tu, Corollary 8.8. Indeed for (U, φ), $U \ni p$ $\quad (\psi F \varphi^{-1}) : \mathbb{R}^m \to \mathbb{R}^n$ is a diffeomorphism
(V, ψ), $V \ni F(p)$ $\quad (\psi F \varphi^{-1})_* : T_{\varphi(p)}\mathbb{R}^m \to T_{\psi(F(p))}\mathbb{R}^n$ is an isomorphism

cf. wj32 has some good solutions specifically for Lee (2012) http://wj32.org/wp/wp-content/uploads/2012/12/Introduction-to-Smooth-Manifolds.pdf

Exercise (Lee 2e, Problem 3-4, p. 75 [76])**.** Show that $T\mathbb{S}^1$ is diffeomorphic to $\mathbb{S}^1 \times \mathbb{R}$.

Solution. Adapted from wj32 http://wj32.org/wp/wp-content/uploads/2012/12/Introduction-to-Smooth-Manifolds.pdf

Now clearly $\forall\, p \in S^1\ \exists p \in S^1,\ \exists\,(U,\theta) \in \mathcal{A}_{S^1}$ s.t. $\theta : U \to \mathbb{R}$ $\quad U \ni p.\ T_pS^1 \ni \frac{\partial}{\partial\theta}\big|_p$.
$\theta(p) \in \mathbb{R}$

Define $F : S^1 \times \mathbb{R} \to TS^1$ s.t.

$$(p,r) \mapsto r\,\frac{\partial}{\partial\theta}\bigg|_p \text{ for } U \ni p,\ \frac{\partial}{\partial\theta}\bigg|_p \in T_pU$$

Consider smooth chart (U,θ), $U \ni p$, $\widehat{\theta} : U \times \mathbb{R} \to \mathbb{R} \times \mathbb{R}$

$$\widehat{\theta}(p,r) = (\theta(p), r)$$

The strategy is to think of the map from $\mathbb{R}^2$ to $\mathbb{R}^2$. So consider

$$F = F \circ \widehat{\theta}^{-1}\widehat{\theta}$$

Consider smooth chart $\xi_\theta : TU \to \mathbb{R}^2$

$$\xi_\theta : X \mapsto ((\theta \circ \pi)(X), (d\theta)_{\pi(X)}X)$$

$$F \circ \widehat{\theta}^{-1}(\theta, r) = r\,\frac{\partial}{\partial\theta}\bigg|_p$$

$$\xi_\theta \circ F \circ \widehat{\theta}^{-1}(\theta,r) = \left(\theta \circ \pi\left(r\frac{\partial}{\partial\theta}\right)_p, (d\theta)_{\pi(r\,\frac{\partial}{\partial\theta}|_p)}\left(r\frac{\partial}{\partial\theta}\bigg|_p\right)\right) = (\theta(p), r)$$

$F^{-1} : X_p \in T_pS^1 \mapsto (p,r)$ where $X_p = r\frac{\partial}{\partial\theta}\big|_p$

$\xi_\theta \circ F \circ \widehat{\theta}^{-1}$ clearly smooth and bijective. F diffeomorphism.

63 Submersions, Immersions, and Embeddings

rank - dim. of its image

smooth immersions - whose differentials are injective everywhere
smooth embeddings - injective smooth immersions that are also homeomorphisms onto their images

Maps of Constant Rank

Suppose smooth manifolds M, N with or without boundary.

rank of F at p - given smooth $F : M \to N$, $\quad p \in M$,
rank of linear $dF_p : T_pM \to T_{F(p)}N$, i.e. rank of Jacobian of F or dim. of Im$dF_p \subseteq T_{F(p)}N$
constant rank - if F has same rank r at any pt.

smooth $F : M \to N$ smooth submersion if F_* surjective at every pt. $\iff$ rank$F = \dim N$ $\quad (\dim M \geq \dim N)$
smooth immersion if F_* injective at every pt. $\iff$ rank$F = \dim M$ $\quad (\dim M \leq \dim N)$

EY : 20150717 I get confused between the rank of F and the rank of DF. I'm going to rewrite the above in my notation:

$$\begin{array}{ccc} M & \xrightarrow{F} & N \end{array} \qquad \begin{array}{ccc} T_pM & \xrightarrow{DF_p} & T_{F(p)}N \\ \uparrow & & \uparrow \\ p & \xmapsto{F} & F(p) \end{array}$$

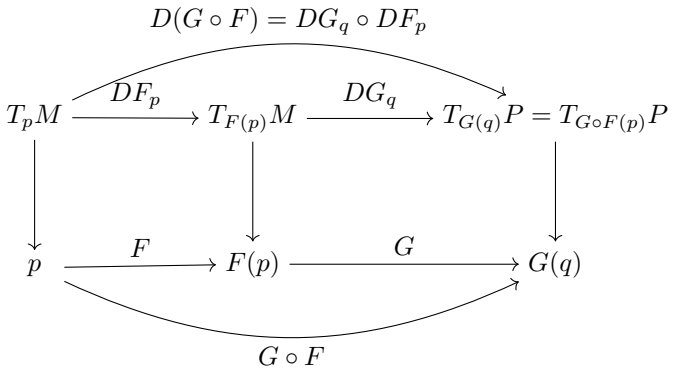
Let $\dim M = \dim T_pM = m$
$\dim N = \dim T_{F(p)}N = n$
Now $DF_p : T_pM \to \text{im}(DF_p) \subseteq T_{F(p)}N$
Let $r = \text{rank}DF_p = \text{dimim}(DF_p)$.
$r \leq n$ (clearly, since $\text{im}(DF_p) \subseteq T_{F(p)}N$)
Recall nullity-rank theorem: For linear $T : V \to T(V) = \text{im}T$,
$\text{dimim}T + \ker T = \dim V$
$\implies \text{dimim}T = \dim V - \ker T \leq \dim V$.
$\implies r \leq m$

Definition 165. For smooth $F : M \to N$
smooth submersion F if dF surjective $\iff \text{rank}DF_p = \dim T_{F(p)}N$ i.e. $r = n$
smooth immersion F if dF injective $\iff \text{rank}DF_p = \dim T_pM$ i.e. $r = m$

Exercise (Lee 2e, Exercise 4.4, p. 79 [76])**.** Show that a composition of smooth submersions is a smooth submersion, and a composition of smooth immersions is a smooth immersion. Give a counterexample to show that a composition of maps of constant rank need not have constant rank.

Solution. cf. http://www.math.ucla.edu/~iacoley/hw/diffhwfall/HW%202.pdf For $q = F(p)$,
If DF_p, DG_q surjective, $DG_q \circ DF_p = D(G \circ F)_p$ surjective. $G \circ F$ smooth submersion.
If DF_p, DG_q injective, $DG_q \circ DF_p = D(G \circ F)_p$ injective. $G \circ F$ smooth immersion.

It'd be instructive to view this as a commutative diagram. For



$F : M \to N$
$G : N \to P$
$p \in M$
$q = F(p) \in N$
$G(q) \in P$

The Inverse Function Theorem and Its Friends

Theorem 49 (7.6)**.** *(**Inverse Function Theorem**). Suppose open $U, V \subset \mathbb{R}^n$, smooth $F : U \to V$*

If $DF(p)$ nonsingular, $p \in U$, $\exists$ connected neighborhood $U_0 \subset U \ni p$
$V_0 \subset V \ni F(p)$
s.t. $F|_{U_0} : U_0 \to V_0$ diffeomorphism.

Let X metric space. $G : X \to X$ contraction if $\exists\, \lambda < 1$ s.t. $d(G(x), G(y)) \leq \lambda d(x, y)$, $\forall\, x, y \in X$.
Clearly, $\forall$ contraction is cont.

Lemma 14 (7.7)**.** *(Contraction Lemma) Let X complete metric space*
$\forall$ contraction $G : X \to X$, $\exists!$ fixed pt., i.e. $x \in X$ s.t. $G(x) = x$

Theorem 50 (7.9)**.** *(Implicit Function Theorem) Let open $U \subset \mathbb{R}^n \times \mathbb{R}^k$, $(x, y) = (x^1 \dots x^n, y^1 \dots y^k)$ coordinates on U.*

Suppose $\Phi : U \to \mathbb{R}^k$ *smooth.* $(a,b) \in U$, $c = \Phi(a,b)$
If $k \times k$ *matrix*

$$\frac{\partial \Phi^i}{\partial y^j}(a,b)$$

nonsingular,

then $\exists$ *neighborhoods* $V_0 \subset \mathbb{R}^n$,
$W_0 \subset \mathbb{R}^k$
smooth $F : V_0 \to W_0$ *s.t.*
$(\Phi^{-1}(c))V_0 \times W_0$ *is the graph of* F, *i.e.* $\Phi(x,y) = c$, $\forall\,(x,y) \in V_0 \times W_0$ *iff* $y = F(x)$

Embeddings

Definition 166. smooth embedding of M into N, F, if
smooth immersion $F : M \to N$ and
F topological embedding i.e. F homeomorphism onto its image $F(M) \subseteq N$ in subspace topology.

Exercise (Lee 2e, Exercise 4.16, p. 85 [76])**.** Show that every composition of smooth embeddings is a smooth embedding.

Solution. Let $F : M \to N$.
$G : N \to P$
F, G smooth immersions so $G \circ F$ smooth immersion (cf. Exercise 4.4, idea is composition of DF, DG is injective).
Now $(G \circ F)(M) = G(F(M))$
F, G bijective onto $F(M)$, $G(N)$. G bijective on $F(M) \subseteq N$ onto $G(F(M))$. Then $G \circ F$ bijective on M onto $G(F(M))$
F, G cont., so $G \circ F$ cont.
F^{-1}, G^{-1} cont., so $(G \circ F)^{-1} = F^{-1} \circ G^{-1}$ cont.
$G \circ F$ homeomorphism onto $G \circ F(M) \subseteq P$.
So $G \circ F$ is a smooth embedding.

Proposition 48 (4.22)**.** *Suppose smooth manifolds* M, N, *with or without boundaries, and injective smooth immersion* $F : M \to N$
If any of the following holds, then F *is a smooth embedding.*

(a) F *open or closed map*

(b) F *proper map*

(c) M *compact*

(d) M *has empty boundary and* $dim M = dim N$

64 Submanifolds

Examples of Embedded Submanifolds

Lemma 15 (8.6)**.** ***(Graphs as Submanifolds)****. If open* $U \subset \mathbb{R}^n$, *smooth* $F : U \to \mathbb{R}^k$, *then graph of* F *is an embedded* n*-dim. submanifold of* $\mathbb{R}^{n+k}$

Proof. Define $\varphi : U \times \mathbb{R}^k \to U \times \mathbb{R}^k$

$$\varphi(x,y) = (x, y - F(x))$$

φ clearly smooth.
φ diffeomorphism because its inverse can be written explicitly

$$\varphi^{-1}(u,v) = (u, v + F(u))$$

$\varphi(\Gamma(F))$ is the slice $\{(u,v) : v = 0\}$ of $U \times \mathbb{R}^k$, so graph $\Gamma(F)$ is an embedded submanifold.

□

Level Sets

Immersed Submanifolds

Definition 167. immersed submanifold of M, S, is $S \subseteq M$, with topology (not necessarily subspace topology) with respect to which it's a topological manifold (without boundary), and
smooth structure with respect to inclusion map $i : S \hookrightarrow M$ is smooth immersion (recall $Di \equiv i_*$ injective $\iff \text{rank}Di = \dim S$

$\text{codim}S = \dim M - \dim S$

smooth hypersurface is immersed submanifold of codimension 1.

65 The Cotangent Bundle

Covectors

V finite-dim. vector space
covector on V - real-valued linear functional on V, i.e. linear map $\omega : V \to \mathbb{R}$

Exercise (Lee 2e, Exercise 11.2, p. 273 [76])**.** Prove Proposition 11.1.

Note. Numbered Exercise 6.1 in the first edition; second-edition Exercise 6.1 is about sets of measure zero. Proposition 11.1: if $(E_1, \ldots, E_n)$ is a basis for V and $\varepsilon^i(E_j) = \delta^i_j$, then $(\varepsilon^1, \ldots, \varepsilon^n)$ is a basis for V^*, so $\dim V^* = \dim V$.

Solution. Suppose $\sum a_i \epsilon^i = A = 0$. Consider arbitrary $v = v^i e_i$. $A(v) = \sum a_i \epsilon^i (v^j e_j) = \sum a_i v^i = 0$
v arbitrary so let $v = \delta_k{}^j e_j$. $\forall\, i$, $a_i = 0$. linearly independent.
Consider linear map $\omega : V \to \mathbb{R}$, a covector.

$$\omega(v^i e_i) = v^i(\omega(e_i)) = k \in \mathbb{R}$$
$$\omega(e_i) = \omega_j \delta^j_i = \omega_j \epsilon^j(e_i)$$

So ω spanned by $\omega_j \epsilon^j$. Done.

Exercise (Lee 2e, Exercise 11.3, p. 273 [76])**.** Show that $A^*\omega$ is actually a linear functional on V, and that A^* is a linear map.

Note. Numbered Exercise 6.2 in the first edition.

Solution. $X \in V$

$$(A^*\omega)(aX + bY) = \omega(A(aX + bY)) = a\omega(AX) + b\omega(AY) \in \mathbb{R} \text{ since } \omega : W^* \to \mathbb{R}$$

linear map $A^*\omega : V \to \mathbb{R}$ is a functional.

$$A^*(a\omega + b\nu)(X) = (a\omega + b\nu)(AX) = a\omega(AX) + b\nu(AX) \in \mathbb{R}$$

Exercise (Lee 2e, Exercise 11.5, p. 274 [76])**.** Prove the preceding proposition.

Note. Numbered Exercise 6.3 in the first edition. The preceding proposition is Proposition 11.4: $(A \circ B)^* = B^* \circ A^*$, and $(\text{Id}_V)^* : V^* \to V^*$ is the identity map of V^*.

Solution.

(a)
$$X \xrightarrow{A} Y \xrightarrow{B} Z$$

$$X^* \xleftarrow[A^*]{} Y^* \xleftarrow[B^*]{} Z^*$$

$$(BA)^* : Z^* \to X^*$$

$$((BA)^*\zeta)(x) = \zeta(BAx) = (\zeta B)(Ax) = (B^*\zeta)(Ax) = (A^*B^*)\zeta(x)$$

(b)
$$(Id_V)^*(\nu(x)) = \nu(1x) = \nu(x)$$

Tangent Covectors on Manifolds

The Cotangent Bundle

Proposition 49 (6.5)**.** *Let M smooth manifold, $T^*M = \coprod_{p\in M} T_p^*M$*

with $\pi : T^*M \to M$
$\omega \in T_p^*M \to p$
natural vector space structure on each fiber,
$\exists!$ *smooth manifold structure making $T*M$ rank-n vector bundle over M,*
s.t. all coordinate covectors are smooth local sections

The Differential of a Function

66 Lie Groups

Basic Definitions

Lie group smooth manifold G s.t. multiplication map $m : G \times G \to G$
$$m(g,h) = gh$$

inversion map $i : G \to G$ smooth.
$$i(g) = g^{-1}$$

Proposition 50 (7.1)**.** *If $(g,h) \mapsto gh^{-1}$ smooth, G Lie group*

Exercise (Lee 2e, Exercise 7.2, p. 151 [76])**.** Prove Proposition 7.1.

Note. Proposition 7.1: if G is a smooth manifold with a group structure such that the map $G \times G \to G$ given by $(g,h) \mapsto gh^{-1}$ is smooth, then G is a Lie group.

Solution.

Proof. $\forall\, g, h \in G$, $gh^2 \in G$ since G group
$(gh^2, h) \mapsto gh$ smooth. Define $m(g,h) = (gh^2, h) \mapsto gh$. So m smooth.
$1 \in G$ since G group. $(1, g) \mapsto g^{-1}$ smooth so $i(g) = g^{-1}$, defined this way, smooth. □

Example 7.3 (Lie Groups).

(a) $A \in GL(n,\mathbb{R})$

$$(AB)_{ij} = A_{ik}B_{kj} \qquad \frac{\partial (AB)_{ij}}{\partial A_{lm}} = \delta_{il}\delta_{km}B_{kj} = \delta_{il}B_{mj}$$
$$\frac{\partial (AB)_{ij}}{\partial B_{lm}} = A_{ik}\delta_{lk}\delta_{mj} = A_{il}\delta_{mj}$$
$$(A^{-1})_{ij} = \frac{1}{\det(A)}\mathrm{adj}(A)_{ij} = \frac{1}{\det(A)}C^T_{ij} = \frac{1}{\det(A)}(-1)^{i+j}\det A_{ji}$$

AB, A^{-1} smooth functions of the entires of A_{ij}, B_{kl}, A_{ij} respectively.

(b)

(c)

Lie Group Homomorphisms

Example 7.4 (Lie Group Homomorphisms)

(a)

(b)

(c)

(d)

(e)

(f) **conjugation** by g

$$C_g : G \to G$$
$$C_g(h) = ghg^{-1}$$

$H \subseteq G$ **normal** if $C_g(H) = H, \quad \forall\, g \in G$

Theorem 51 (7.5)**.** *Every Lie group homomorphism has constant rank.*

67 Vector Fields

Exercise 4.1 (Lee, 1st ed.). *Note.* Lee, first edition, Exercise 4.1 ($T\mathbb{R}^n \approx \mathbb{R}^{2n}$). There is no second-edition exercise with this content; take $(U,\varphi) = (\mathbb{R}^n, \mathrm{Id})$ in the charts (3.13) of Proposition 3.18, p. 66 [76].

Consider $1 : \mathbb{R}^n \to \mathbb{R}^n$, smooth structure on $\mathbb{R}^n$, that's open.
$x^i(x) = x^i$

Consider $F : T\mathbb{R}^n \to \mathbb{R}^{2n}$
$F(x^1 \dots x^n, v^1 \dots v^n) = (x^1 \dots x^n, v^1 \dots v^n)$

$F = F^{-1}$, so clearly $F = 1_{T\mathbb{R}^n}$ is cont., bijective, and it's inverse cont. and smooth. F diffeomorphism.

Exercise 4.2 (Lee, 1st ed.). *Note.* Lee, first edition, Exercise 4.2 (the pushforward in coordinates). There is no second-edition exercise with this content; it is the coordinate formula (3.9)–(3.10) for the differential, p. 62 [76].
$F : M \to N$. Consider (3.6)

$$(U,\varphi) \subset M \qquad \varphi = (x^1 \dots x^m) \qquad X = X^i\frac{\partial}{\partial x^i}$$
$$(V,\psi) \subset N \qquad \psi = (y^1 \dots y^n) \qquad Y = Y^j\frac{\partial}{\partial y^j}$$

$$(F_*X)(f) = Y^j \frac{\partial}{\partial y^j} f = X(fF) = X^i \frac{\partial}{\partial x^i} fF = X^i \frac{\partial (f\psi^{-1})}{\partial y^j} \frac{\partial}{\partial x^i} (\psi F^j \varphi^{-1})(\varphi(p)) = X^i \frac{\partial F^j}{\partial x^i}(p) \frac{\partial f}{\partial y^j}$$

where

$$fF = f\psi^{-1}\psi F\varphi^{-1}\varphi \Longrightarrow fF(p) = (f\psi^{-1})(y)(\psi F\varphi^{-1})(\varphi(p))$$

(a serious case of abuse of notation)

For F_*X,

$$Y^j = X^i \frac{\partial F^j}{\partial x^i}$$

$$F_* \left.\frac{\partial}{\partial x^i}\right|_p = F_* \frac{\partial}{\partial x^i} = \delta_i{}^k \frac{\partial F^j}{\partial x^k} \frac{\partial}{\partial y^j} = \frac{\partial F^j}{\partial x^i} \frac{\partial}{\partial y^j} = \frac{\partial F^j}{\partial x^i}(p) \left.\frac{\partial}{\partial y^j}\right|_{F(p)}$$

with $X^k = \delta_i{}^k$

$$F_* : TM \to TN$$

$$F_*(x^1 \dots x^n, v^1 \dots v^n) = (y^1(x) \dots y^n(x), v^i \frac{\partial F^1}{\partial x^i} \dots v^i \frac{\partial F^n}{\partial x^i})$$

Clearly F_* smooth since F smooth.

Lemma 16 (4.8)**.** *Suppose smooth* $F : M \to N, \quad Y \in \tau(M)$
$Z \in \tau(N)$
Y, Z, F-related iff $\forall$ smooth $\mathbb{R}$-valued f on open $V \subset N$,

$$Y(fF) = (Zf)F \qquad (4.4)$$

Proof. $\forall p \in M$, $\forall$ smooth $\mathbb{R}$-valued f, f defined near $F(p)$

$$Y(fF)(p) = Y_p(fF) = (F_*Y_p)f \qquad (F_*Y)f = Y(fF)$$
$$(Zf)F(p) = (Zf)(F(p)) = Z_{F(p)}f$$

if $(Zf)F(p) = Y(fF)(p) = Zf = (F_*Y)f$

$$(Zf)F = Y(fF) \Longleftrightarrow Z = F_*Y \text{ i.e. iff } Y, Z \quad F\text{-related}$$

□

Vector Fields on a Manifold with Boundary

Lie Brackets

Lemma 17 (4.12)**.** *Lie bracket of smooth vector fields V, W, $[V, W] : C^\infty M \to C^\infty M$ is a smooth vector fields.*
$[V, W]f = VWf - WVf$

Proof. By Prop. 4.7. (M smooth, map $\mathcal{Y} : C^\infty M \to C^\infty M$ is a derivation iff $\mathcal{Y}f = Yf$, Y some smooth vector field $Y \in \tau(M)$.
Suffices to show $[V, W]$ derivation of $C^\infty M$

$$[V, W](fg) = V(W(fg)) - W(V(fg)) = V(fWg + gWf) - W(fVg + gVf) =$$
$$= VfWg + fVWg + VgWf + gVWf - WfVg - fWVg - WgVf - gWVf =$$
$$= fVWg + gVWf - fWVg - gWVf = f[V, W]g + g[V, W]f$$

□

extremely useful coordinate formula for Lie bracket

Lemma 18 (4.13). *Let* $V = V^i \dfrac{\partial}{\partial x^i}$ $\qquad [V,W] = \left(V^i \dfrac{\partial W^j}{\partial x^i} - W^i \dfrac{\partial V^j}{\partial x^i} \right) \dfrac{\partial}{\partial x^j}$ (45)

$W = W^j \dfrac{\partial}{\partial x^j}$ $\qquad [V,W] = (VW^j - WV^j) \dfrac{\partial}{\partial x^j}$ (46)

Proof. $[V,W]$ smooth vector field already, its values are determined locally $([V,W]f)|_U = [V,W](f|_U)$
It suffices to compute in a single smooth chart.

$$
\begin{gathered}
[V,W]f \\
= V^i \frac{\partial}{\partial x^i} \left(W^j \frac{\partial f}{\partial x^j} \right) - W^j \frac{\partial}{\partial x^j} \left(V^i \frac{\partial f}{\partial x^i} \right) \\
= V^i \frac{\partial W^i}{\partial x^i} \frac{\partial f}{\partial x^j} + V^i W^j \frac{\partial^2 f}{\partial x^i \partial x^j} - W^j \frac{\partial V^i}{\partial x^j} \frac{\partial f}{\partial x^i} - W^j V^i \frac{\partial^2 f}{\partial x^j \partial x^i} = \\
= \left(V^i \frac{\partial W^j}{\partial x^i} - W^i \frac{\partial V^j}{\partial x^i} \right) \frac{\partial f}{\partial x^j}
\end{gathered}
$$

□

Exercise (Lee 2e, Exercise 8.29, p. 188 [76]). Prove part (d) of the preceding proposition.

Note. Numbered Exercise 4.6 in the first edition. The preceding proposition is Proposition 8.28 (Properties of the Lie Bracket); part (d) is $[fX, gY] = fg[X,Y] + (fXg)Y - (gYf)X$.

Solution. For Lemma 4.15 (Properties of the Lie Bracket), part (d),
the point is to use the derivative properties of the vector fields.

$$
\begin{gathered}
[fV, gW]\, h = fV(gWh) - gW(fVh) = (fVg)(Wh) + g(fV(Wh)) - (gWf)(Vh) - f(gW)(Vh) = \\
= g(fVW)h - (fgWV)h + f(Vg)Wh - g(Wf)Vh
\end{gathered}
$$

Proposition 51 (4.16). *(Naturality of the Lie Bracket) Let smooth* $F : M \to N$, $\quad V_1, V_2 \in \tau(M)$, $\quad V_i$ $W_1, W_2 \in \tau(N)$

F-related to W_i, $i = 1, 2$.
Then $[V_1, V_2]$ F-related to $[W_1, W_2]$

Proof. Use Lemma 4.8, and given V_i, F-related to W_i

$$
\begin{aligned}
V_1 V_2(fF) &= V_1(W_2 f)F = W_1 W_2 f F \\
V_2 V_1(fF) &= (W_2 W_1 f)F
\end{aligned}
\qquad \Longrightarrow [V_1, V_2](fF) = ([W_1, W_2]f)F
$$

So $[V_1, V_2]$, F-related to $[W_1, W_2]$ □

Corollary 8 (4.17). *Suppose $F : M \to N$ diffeomorphism, $V_1, V_2 \in \tau(M)$*
Then $F_*[V_1, V_2] = [F_* V_1, F_* V_2]$

Proof. F diffeomorphism. Then Lemma 4.9, $\exists$ push-forward (or alternatively, by Prop. 4.16, $W_i = F_* V_i$ i.e. F-related).

$$F_*[V_1, V_2] = [W_1, W_2] = [F_* V_1, F_* V_2]$$

□

The Lie Algebra of a Lie Group

$$L_g = m i_g$$

$$G \xrightarrow{i_g} G \times G \xrightarrow[m]{} G$$

$i_g(h) = (g, h)$, m is multiplication, follows L_g smooth.

L_g diffeomorphism of G, since $L_{g^{-1}}$ smooth inverse.

$\forall$ 2 pts. $g_1, g_2 \in G$, $\exists!\, L_{g_2 g_1^{-1}}$ s.t. $L_{g_2 g_1^{-1}} g_1 = g_2$ many important properties of Lie groups follow from $L_{g_2 g_1^{-1}}$ as diffeomorphism.

vector field X on G *left invariant* if

$$(L_g)_* X_{g'} = X_{gg'} \qquad \forall\, g, g' \in G \qquad (4.8) \tag{176}$$

L_g diffeomorphism.

$$(L_g)_*(aX + bY) = a(L_g)_* X + b(L_g)_* Y$$

set of all smooth left-invariant vector fields on G is a linear subspace $\tau(M)$, *and* closed under Lie bracket.

Lemma 19 (4.18)**.** *Let G Lie group, suppose X, Y smooth left-invariant vector fields on G Then $[X, Y]$ also left invariant.*

Proof. Given $(L_g)_* X = X$ by def. of left-invariance. □

$(L_g)_* Y = Y$

Vector Fields on Manifolds

Lemma 20 (8.6)**.** ***(Extension Lemma for Vector Fields)***
M smooth manifold with or without boundary
$A \subseteq M$ closed subset.
Suppose X smooth vector field along A.
Give open $U \supset A$, $\exists$ smooth global vector field $\widetilde{X}$ on M s.t. $\widetilde{X}\big|_A = X$ and $supp \widetilde{X} \subseteq U$

Exercise (Lee 2e, Exercise 8.9, p. 177 [76])**.** Prove Proposition 8.8.

Solution.

(a) $\forall p \in M$, $X_p = X^i(p) \dfrac{\partial}{\partial x^i}$

$Y_p = Y^i(p) \dfrac{\partial}{\partial x^i}$

$f, g \in C^\infty(M)$

$$(fX)_p = f(p) X_p = f(p) X^i(p) \frac{\partial}{\partial x^i}$$

$$(gY)_p = g(p) Y_p = g(p) Y^i(p) \frac{\partial}{\partial x^i}$$

$$(fX + gY)_p = f(p) X_p + g(p) Y_p = f(p) X^i(p) \frac{\partial}{\partial x^i} + g(p) Y^i(p) \frac{\partial}{\partial x^i} = (f(p) X^i(p) + g(p) Y^i(p)) \frac{\partial}{\partial x^i}$$

$f(p) X^i(p) + g(p) Y^i(p)$ smooth so $(fX + gY)_p$ smooth.

(b) Let $g = f$

$$(fX + fY)_p = f(p) X_p + f(p) Y_p = f(p) X^i(p) \frac{\partial}{\partial x^i} + f(p) Y^i(p) \frac{\partial}{\partial x^i} = f(p)(X^i(p) + Y^i(p)) \frac{\partial}{\partial x^i} = (f(X+Y))_p$$

Let $Y = X$ so $\forall\, p$,

$$(fX + gX)_p = f(p)X_p + g(p)X_p = f(p)X^i(p)\frac{\partial}{\partial x^i} + g(p)X^i(p)\frac{\partial}{\partial x^i} = (f(p) + g(p))X^i(p)\frac{\partial}{\partial x^i} = ((f+g)X)_p$$

$$(g(fX))_p = g(p)(fX)_p = g(p)f(p)X_p = ((gf)X)_p$$

Let $f = 1$, $g = 0$, $1X = X$

Local and Global Frames

Vector Fields as Derivations of $C^\infty(M)$

if $X \in \mathfrak{X}(M)$, smooth f defined on open $U \subseteq M$, obtain

$$Xf : U \to \mathbb{R}$$
$$(Xf)(p) = X_pF$$

From J. Lee: (Be careful not to confuse the notations fX and Xf: the former is the smooth *vector field* on U obtained by multiplying X by f, while the latter is the real-valued *function* on U obtained by applying the vector field X to the smooth function f)

Proposition 52 (8.14)**.** *$X : M \to TM$*
equivalent

(a) X smooth

(b) $\forall\, f \in C^\infty(M)$, Xf smooth on M

(c) $\forall$ open $U \subseteq M$, $\forall\, f \in C^\infty(M)$, $Xf \in C^\infty(U)$

Proof. (a) $\implies$ (b), assume X smooth,
let $f \in C^\infty(M)$
M manifold, $\forall\, p \in M$, choose smooth x^i on open $U \ni p$
Then $\forall\, x \in U$,

$$Xf(x) = \left(X^i(x)\frac{\partial}{\partial x^i}\bigg|_x \right) f = X^i(x)\frac{\partial f}{\partial x^i}(x)$$

X^i smooth on U by Prop. 8.1, Xf smooth in U □

Vector Fields and Smooth Maps

Proposition 53 (8.16)**.** *Suppose smooth $F : M \to N$,* $X \in \mathfrak{X}(M)$, $Y \in \mathfrak{X}(N)$

Then X, Y F-related iff $\forall$ smooth h, defined on open $V \subset N$

$$X(hF) = (Yh)F$$

Proof. $\forall\, p \in M$, $\forall$ smooth h defined on open $V \ni F(p)$

$$X(hF)(p) = X_p(hF) = dF_p(X_p)h$$
$$(Yh)F(p) = Yh(F(p)) = Y_{F(p)}h$$

$X(hF) = (Yh)F \quad \forall\, h \in C^\infty(N)$ iff $dF_p(X_p) = Y_{F(p)} \quad \forall\, p$

□

Proposition 54 (8.19)**.** *smooth M, N, diffeomorphism $F : M \to N$*
$\forall\, X \in \mathfrak{X}(M)$, $\exists!$ smooth vector field on N F-related to X

Proof. $\forall\, p \in M,\ F(p) = q \in N$

define Y by

$$Y_q = dF_{F^{-1}(q)}(X_{F^{-1}(q)}) = dF_p(X_p)$$
$$\Longrightarrow Y_{F(p)} = dF_p(X_p)$$

$Y : N \to TN$

$$Y = N \xrightarrow{F^{-1}} M \xrightarrow{X} TM \xrightarrow{dF} TN$$

$Y = dF \circ X \circ F^{-1}$
dF, X, F^{-1} smooth. Y smooth. □

pushforward of X by F, denote F_*X

$$(F_*X)_q = dF_{F^{-1}(q)}(X_{F^{-1}(q)}) \qquad (8.7) \tag{177}$$

$(F_*X)_q = dF_p(X_p)$

Corollary 9 (8.21)**.** *Suppose diffeomorphism $F : M \to N$, $X \in \mathfrak{X}(M)$*
$\forall\, h \in C^\infty(N)$

$$((F_*X)h) \circ F = X(h \circ F)$$

Vector Fields and Submanifolds

Lie Brackets

Proposition 55 (8.26)**.** ***(Coordinate Formula for the Lie Bracket)***

$$[X, Y] = \left(X^i \frac{\partial Y^j}{\partial x^i} - Y^i \frac{\partial X^j}{\partial x^i} \right) \frac{\partial}{\partial x^j} \qquad (8.8) \tag{178}$$

The Lie Algebra of a Lie Group

Recall that G acts smoothly and transitively on itself by left translation:

$$L_g(h) = gh$$

X on G **left-invariant** if

$$d(L_g)_{g'}(X_{g'}) = X_{gg'} \quad \forall\, g, g' \in G \qquad (8.12) \tag{179}$$

L_g diffeomorphism, so

$$(L_g)_* X = X \qquad \forall\, g \in G$$

Example 8.36 (Lie Algebras)

(a)

(b)

(c)

(d)

(e)

(f) $\forall$ vector V becomes Lie algebra if $[,] = 0$
such a Lie algebra is **abelian**

Lie$G$ Lie algebra of all smooth left-invariant vector fields on Lie Group G **Lie algebra of** G

Theorem 52 (8.37)**.** $\epsilon : Lie(G) \to T_eG$

$$\epsilon(X) = X_e$$

ϵ *vector space isomorphism*

Proof. If $\epsilon(X) = X_e = 0$ for some $X \in \text{Lie}(G)$

left invariant $d(L_g)_{g'}(X_{g'}) = X_{gg'}$

$$d(L_g)_e(X_e) = X_g = 0 \quad \forall\, g \in G, \text{ so } X = 0$$

ϵ injective

Let $V \in T_eG$ arbitrary.

define (rough) vector field v^L on G by

$$v^L\big|_g = d(L_g)_e(v) \qquad (8.13) \tag{180}$$

□

Example 8.40

(a) $L_b(x) = b + x \qquad bx = b + x \qquad y = x + b$

$d(L_g) = 1$

$X_x = X^i \frac{\partial}{\partial x^i}$

$d(L_b)_x X_x = 1X_x = X_x = \widetilde{X}^i \frac{\partial}{\partial(x+b)} = \widetilde{X}^i(x+b)\frac{\partial}{\partial x} = X^i(x)\frac{\partial}{\partial x^i}$

X^i constants

$[X, Y] = 0$ if X, Y constants.

lie algebra of $\mathbb{R}^n$ abelian (cf. Example 8.36, (f))

(b)

(c)

Proposition 56 (8.41)**.** ***(Lie Algebra of the General Linear Group)***

$$Lie(GL(n,\mathbb{R})) \to T_{1_n}GL(n,\mathbb{R}) \to \mathfrak{gl}(n,\mathbb{R}) \qquad (8.14) \tag{181}$$

is isomorphism

Proof. global coordinates $X^i{}_j$ on $GL(n,\mathbb{R})$

natural isomorphism

$$T_1GL(n,\mathbb{R}) \longleftrightarrow \mathfrak{gl}(n,\mathbb{R})$$

$$A^i{}_j \left.\frac{\partial}{\partial X^i{}_j}\right|_{1_n} \longleftrightarrow (A^i{}_j)$$

Recall

$$d(L_g)_{g'}(X_{g'}) = X_{gg'}$$

Recall Lie algebra of all smooth left invariant vector fields on G

Recall (8.13)

$$v\big|_g = d(L_g)_e(v) \qquad (8.13) \tag{182}$$

L_X is restriction to $GL(n,\mathbb{R})$ of linear map $A \mapsto XA$ on $\mathfrak{gl}(n,\mathbb{R})$

$$L_X g = Xg = X^i{}_k g^k{}_j$$

$$L_X 1 = X1 = X^i{}_k \delta^k{}_j = X^i{}_j = X$$

$X^i{}_j$ global coordinates on $GL(n,\mathbb{R})$, so

$$\left.\frac{\partial}{\partial X^i{}_j}\right|_1 = \left.\frac{\partial}{\partial X^i{}_j}\right|_X$$

$$DL_X = \frac{\partial}{\partial A^k_j}(XA)^i{}_j = \frac{\partial}{\partial A^k_l} X^i{}_m A^m_j = X^i{}_m \delta^m_k \delta^l{}_j = X^i{}_k \delta^i{}_j$$

$$(DL_X)_1(A) = ((DL_X)^i{}_j{}^l{}_k A^k_l \left.\frac{\partial}{\partial X^i{}_j}\right|_X = (X^i{}_k \delta^l{}_j A^k_l) \left.\frac{\partial}{\partial X^i{}_j}\right|_X = (X^i{}_k A^k{}_j) \left.\frac{\partial}{\partial X^i{}_j}\right|_X$$

□

Problems

Exercise (Lee 2e, Problem 8-1, p. 199 [76])**.** Prove Lemma 8.6 (the extension lemma for vector fields).

Solution. $\forall\, p \in A$, choose neighborhood W_p of p, smooth $\widetilde{X} : A \to TM$ s.t. $\widetilde{X} = X$ on $W_p \bigcap A$
Replace W_p by $W_p \bigcap U$, so $W_p \subseteq U$
$\{W_p | p \in A\} \bigcup \{M \backslash A\}$ open cover of M
Let $\{\psi_p | p \in A\} \bigcup \{\psi_0\}$ smooth partition of unity subordinate to this cover, with $\text{supp}\psi_p \subseteq W_p$, $\text{supp}\psi_0 \subseteq M \backslash A$
$\forall\, p \in A$, (U_p, x^i) smooth coordinate chart

$$X_p = X^i(p) \left.\frac{\partial}{\partial x^i}\right|_p$$
$$X^i : U_p \to \mathbb{R}$$

$\psi_p \widetilde{X}^i(p)$ smooth on W_p
$\psi_p \widetilde{X}^i(p)$ has smooth extension to all of M if $\psi_p \widetilde{X}^i(p) = 0$ on $M \backslash \text{supp}\psi_p$

on open $W_p \backslash \text{supp}\psi_p$, they agree
define $\widetilde{X}^i : M \to \mathbb{R}$

$$\widetilde{X}^i(x) = \sum_{p \in A} \psi_p(x) \widetilde{X}^i(p)$$

$\{\text{supp}\psi_p\}$ locally finite, so $\sum_{p \in A} \psi_p(x) \widetilde{X}^i(p)$ has only finite number of nonzero terms in neighborhood of $\forall\, x \in M$, so $\widetilde{X}^i(x)$ smooth
If $x \in A$, $\psi_0(x) = 0$, $\widetilde{X}^i(x) = X^i(x) \qquad \forall\, p$ s.t. $\psi_p(x) \neq 0$, so

$$\widetilde{X}^i(x) = \sum_{p \in A} \psi_p(x) X^i(x) = (\psi_0(x) + \sum_{p \in A} \psi_p(x)) X^i(x) = X^i(x)$$

so $\widetilde{X}^i$ extension of X

Exercise (Lee 2e, Problem 8-2, p. 199 [76])**.** Euler's Homogeneous Function Theorem: Let c be a real number, and let $f\colon \mathbb{R}^n \setminus \{0\} \to \mathbb{R}$ be a smooth function that is *positively homogeneous of degree c*, meaning that $f(\lambda x) = \lambda^c f(x)$ for all $\lambda > 0$ and $x \in \mathbb{R}^n \setminus \{0\}$. Prove that $Vf = cf$, where V is the Euler vector field defined in Example 8.3.

Solution. Euler's Homogeneous Function Theorem $y = \lambda x$

$$\frac{\partial f}{\partial y^i}\frac{\partial y^i}{\partial \lambda} = x^i \frac{\partial f}{\partial y^i} = x^i \frac{\partial f}{\partial(\lambda x^i)} = \frac{d}{d\lambda} f(y) = \frac{d}{d\lambda} f(\lambda x) = \frac{d}{d\lambda}(\lambda^c f(x)) = c\lambda^{c-1} f(x)$$

$\lambda = 1$

$$\boxed{x^i \frac{\partial f}{\partial x^i} = V_x f(x) = c f(x)}$$

Exercise (Lee 2e, Problem 8-29, p. 203 [76])**.** Theorem 8.46 implies that the Lie algebra of any Lie subgroup of $\mathrm{GL}(n,\mathbb{R})$ is canonically isomorphic to a subalgebra of $\mathfrak{gl}(n,\mathbb{R})$, with a similar statement for Lie subgroups of $\mathrm{GL}(n,\mathbb{C})$. Under this isomorphism, show that

$$\begin{aligned}
\mathrm{Lie}\big(\mathrm{SL}(n,\mathbb{R})\big) &\cong \mathfrak{sl}(n,\mathbb{R}),\\
\mathrm{Lie}\big(\mathrm{SO}(n)\big) &\cong \mathfrak{o}(n),\\
\mathrm{Lie}\big(\mathrm{SL}(n,\mathbb{C})\big) &\cong \mathfrak{sl}(n,\mathbb{C}),\\
\mathrm{Lie}\big(\mathrm{U}(n)\big) &\cong \mathfrak{u}(n),\\
\mathrm{Lie}\big(\mathrm{SU}(n)\big) &\cong \mathfrak{su}(n),
\end{aligned}$$

where

$$\begin{aligned}
\mathfrak{sl}(n,\mathbb{R}) &= \{A \in \mathfrak{gl}(n,\mathbb{R}) : \operatorname{tr} A = 0\},\\
\mathfrak{o}(n) &= \{A \in \mathfrak{gl}(n,\mathbb{R}) : A^T + A = 0\},\\
\mathfrak{sl}(n,\mathbb{C}) &= \{A \in \mathfrak{gl}(n,\mathbb{C}) : \operatorname{tr} A = 0\},\\
\mathfrak{u}(n) &= \{A \in \mathfrak{gl}(n,\mathbb{C}) : A^* + A = 0\},\\
\mathfrak{su}(n) &= \mathfrak{u}(n) \cap \mathfrak{sl}(n,\mathbb{C}).
\end{aligned}$$

Solution.

$$\begin{aligned}
\mathfrak{o}(n) &= \{A \in \mathfrak{gl}(n,\mathbb{R}) | A^T + A = 0\}\\
\mathfrak{o}(3) &= \{A \in \mathfrak{gl}(3,\mathbb{R}) | A^T + A = 0\}
\end{aligned}$$

$$\left\{\mathfrak{su}(n) = \{A \in \mathfrak{gl}(n,\mathbb{C}) | A^* + A = 0, \mathrm{tr} A = 0\right\}$$

$$\left\{\mathfrak{su}(2) = \{A \in \mathfrak{gl}(2,\mathbb{C}) | A^* + A = 0, \mathrm{tr} A = 0\right\}$$

$\forall\, A \in \mathfrak{su}(2)$, A is of the form, for $a, b, c \in \mathbb{R}$,

$$A = \begin{pmatrix} ia & b+ic \\ -b+ic & -ia \end{pmatrix}$$

$\forall\, B \in \mathfrak{o}(3)$, B is of the form

$$B = \begin{pmatrix} & a & b \\ -a & & c \\ -b & -c & \end{pmatrix}$$

Then the following identification, F is clearly an isomorphism, as its 1-to-1 and onto:

$$F : \mathfrak{su}(2) \to \mathfrak{o}(3)$$

$$F\begin{pmatrix} ia & b+ic \\ -b+ic & -ia \end{pmatrix} = \begin{pmatrix} & a & b \\ -a & & c \\ -b & -c & \end{pmatrix}$$

68 Integral Curves and Flows

Integral Curves

If smooth curve $c : I \to M$

$$c(t) = p$$

$\forall\, t \in I,\ c' \equiv c'(t) \in T_{c(t)}M \equiv T_pM$

If X vector field on M,
integral curve of X is differentiable $c : I \to M$ s.t.
$\forall\, t \in I,\ c'(t) \equiv \dot{c} = X_{c(t)} = X_p$

EY

$$\begin{aligned} &c : I \to M \\ &c(t) = p \\ &\varphi c(t) = \varphi(p) \Longrightarrow \begin{matrix} c^i(t) = x^i(t) \\ \dot{c}^i(t) = \dot{x}^i(t) \end{matrix} \end{aligned}$$

$$X_p = X^i(p)\frac{\partial}{\partial x^i} \equiv X^i\frac{\partial}{\partial x^i} = \dot{c}^i\frac{\partial}{\partial x^i}$$

$$f : M \to \mathbb{R}$$

$$X_pf = X_pf(p) = X_pf\varphi^{-1}\varphi(p) = X_(f\varphi^{-1})(x^j) = X^i\frac{\partial f}{\partial x^i}(x^j) \equiv X^i(p)\frac{\partial f}{\partial x^i}(x^j) = \dot{c}^i\ \frac{\partial f}{\partial x^i}\bigg|_p = \frac{d}{dt}(f \circ c)(t)$$

Example 9.1. (Integral Curves)

(a) Let $X = \frac{\partial}{\partial x}$, $(x, y) \in \mathbb{R}^2$

$$\begin{gathered} c(t) = (x(t), y(t)) = x\partial_x + y\partial_y \\ c' = \dot{x}\partial_x + \dot{y}\partial_y = \partial_y \Longrightarrow \dot{y} = 0 \quad y = b \\ \dot{x} = 1 \qquad x = a + t \\ c = (a + t, b) \end{gathered}$$

(b) $X = x\partial_x - y\partial_x$. Comparing the components of these vectors, we see that this is equivalent to

$$\begin{matrix} \dot{x} = -y \\ \dot{y} = x \end{matrix} \Longrightarrow \begin{matrix} y = a\sin t + b\cos t \\ x = a\cos t - b\sin t \end{matrix}$$

Proposition 57 (9.2)**.** *Let smooth vector field X on smooth M,*
$\forall\, p \in M,\ \exists\, \epsilon > 0,\ \exists$ smooth $c : (-\epsilon, \epsilon) \to M$ i.e. integral curve of X starting at p

Proof. existence from Thm. D.1 □

Lemma 21 (9.3)**.** *(Rescaling Lemma) $\widetilde{c}(t) = c(at)$ integral curve of aX, where $\widetilde{I} = \{t | at \in I\}$*

Proof. Let smooth f defined in neighborhood of $\widetilde{c}(t_0)$
e.g. of rescaling - $2t = 2 \cdot 1 = 2 \quad a = 2$

$$\widetilde{c}(t) = c(at) = c(\tau) = p \in M$$
$$\dot{\widetilde{c}}(t)f = \frac{d}{dt}(f \circ \widetilde{c})(t)$$
$$= \frac{d}{dt}(f\varphi^{-1})(\varphi\widetilde{c}(t))$$
$$= \frac{d}{dt}(f\varphi^{-1})(\varphi c(at)) = \frac{d}{dt}(f\varphi^{-1})(c^i(at))$$
$$= \frac{\partial f}{\partial x^i}\bigg|_p \frac{dc^i}{d\tau}(\tau)a$$
$$= a\frac{d}{d\tau}(f \circ c)(\tau) = aX_pf$$

□

Lemma 22 (9.4)**.** *(Translation Lemma)*

$$\widetilde{I} = \{t | t + a \in I\}$$
$$\widetilde{c}(t) = c(t + a)$$

Exercise (Lee 2e, Exercise 9.5, p. 208 [76])**.** Prove the translation lemma.

Note. The translation lemma is Lemma 9.4: if $\gamma\colon J \to M$ is an integral curve of V, then for any $b \in \mathbb{R}$ the curve $\widehat{\gamma}\colon \widehat{J} \to M$, $\widehat{\gamma}(t) = \gamma(t + b)$, is also an integral curve of V, where $\widehat{J} = \{t : t + b \in J\}$.

Solution.

Proof.

$$\widetilde{c}(t) = c(t + a) = c(\tau) = p \in M$$
$$\dot{\widetilde{c}}(t)f = \frac{d}{dt}(f \circ \widetilde{c})(t) = \frac{d}{dt}f(\widetilde{c}(t)) = \frac{d}{dt}f(c(t + a)) = \frac{\partial f}{\partial x^i}\bigg|_p \frac{dc^i(\tau)}{d\tau} = \dot{c}^i(\tau)\,\frac{\partial f}{\partial x^i}\bigg|_p = X_p^i\,\frac{\partial f}{\partial x^i}\bigg|_p = X_pf$$

□

Proposition 58 (9.6)**.** *(Naturality of Integral curves) Suppose smooth $F : M \to N$*

Then $X \in \mathfrak{X}(M)$ $\quad$ $Y \in \mathfrak{X}(N)$ $\quad$ *F-related iff F takes integral curves of X to integral curves of Y*

Proof. Recall

$$X, Y\ F-\text{related means}\ \ dF(X) = Y$$

Let $\gamma = Fc$

$$\dot{\gamma} = \frac{d}{dt}(F \circ c)(t) = (dF)(\dot{c}) = dF(X) = Y$$

$\implies \gamma$ integral curve of Y

if $\gamma = Fc$ integral curve of Y, $\dot{\gamma} = Y$. $\quad q = F(p)$. $p = c(t)$

$$
\begin{aligned}
Yg = Y_q g &= Y^j \left.\frac{\partial g}{\partial y^j}\right|_q \\
&= \dot\gamma^j(t) \left.\frac{\partial g}{\partial y^j}\right|_{F(p)} \\
&= \frac{d}{dt}(F\circ c) \left.\frac{\partial g}{\partial y^j}\right|_{F(p)} \\
&= \frac{\partial y^j}{\partial x^k}\dot c^k(t) \left.\frac{\partial g}{\partial y^j}\right|_q \\
&= \frac{\partial y^j}{\partial x^k} X_p^k \left.\frac{\partial g}{\partial y^j}\right|_q \\
&= (F_* X)g = dF(X)g \\
&\implies dF(X) = Y
\end{aligned}
$$

□

Flows

Let $X \in \mathfrak{X}(M)$
Suppose $\forall\, p \in M$, $\exists!$ integral curve starting at p, $\phi^{(p)} : \mathbb{R} \to M$

$\forall\, t \in \mathbb{R}$, define $\phi_t : M \to M$

$$\phi_t(p) = \phi^{(p)}(t)$$

$$\theta_0(p) = \theta^{(p)}(0) = p$$

EY: ϕ_t pushes p to $\phi^{(p)}(t)$ over time interval t

translation lemma implies $t \mapsto \phi^{(p)}(t+s)$ is integral curve of X starting at $q = \phi^{(p)}(s)$

assuming uniqueness of integral curves, $\phi^{(p)}(t) = \phi^{(p)}(t+s)$, so

$$\phi_t \circ \phi_s(p) = \phi_{t+s}(p)$$
$$\phi_0(p) = \phi^{(p)}(0) = p$$

$\implies \phi : \mathbb{R} \times M \to M$ is an action of additive group $\mathbb{R}$ on M.

define *global flow* on M (1-parameter group action) - cont. left $\mathbb{R}$-action on M, i.e. cont. $\phi : \mathbb{R} \times M \to M$ s.t. $\forall\, s, t \in \mathbb{R}$, $\forall\, pM$

$$\phi(t, \phi(s,p)) = \phi(t+s, p) \quad \phi(0,p) = p \qquad (9.2) \tag{183}$$

given global flow ϕ

$\forall\, t \in \mathbb{R}$, define cont. $\phi_t : M \to M$

$$\phi_t(p) \to \phi(t,p)$$

$$\xrightarrow{(9.2)} \phi_t \cdot \phi_s = \phi_{t+s}$$
$$\phi_0 = 1_M$$

$\phi_t : M \to M$ homeomorphism; if flow smooth, ϕ_t diffeomorphism.

$\forall\, p \in M$, define $\phi^{(p)} : \mathbb{R} \to M$

$$\phi^{(p)}(t) = \phi(t,p)$$

$\phi^{(p)}$ is orbit of p under group action.

smooth global flow $\theta : \mathbb{R} \times M \to M$

$\forall\, p \in M$, define $V_p \in T_pM$

$$V_p = (\theta^{(p)})'(0)$$

Proposition 59 (9.7)**.** *Let smooth global flow $\phi : \mathbb{R} \times M \to M$ on smooth M*

infinitesimal generator X of ϕ $\quad p \mapsto X_p$ $\quad$ *is smooth vector field on M, and $\forall\, \phi^{(p)}$, $\phi^{(p)}$ integral curve of X*

$$X_p = \dot{\phi}^{(p)}(0)$$

Proof. Show X smooth. Use Prop. 8.14,
f smooth on open $U \subseteq M$, $f : U \to \mathbb{R}$

$$Xf(p) = X_p f = \dot{\phi}^{(p)}(0)f \equiv (\dot{\phi}^{(p)}(0))[f] = \frac{d}{dt}(f \circ \phi^{(p)})\Big|_{t=0} = \frac{\partial}{\partial t}\,(f \circ \phi(t,p))|_{t=0}$$

$f \circ \phi(t,p) = f(\phi(t,p))$ smooth function of (t,p) by composition, so $\partial_t(f \circ \phi)$ smooth. So Xf smooth, so X smooth.

Let $q = \phi^{(p)}(a) = \phi_a(p)$

$$\phi^{(q)}(t) = \phi_t(q) = \phi_t(\phi_a(p)) = \phi_{t+a}(p) = \phi^{(p)}(t+a) \qquad (9.4) \tag{184}$$

$$X_q f = \dot{\phi}^{(q)}(0)f = \dot{\phi}^{(q)}(0)[f] = \frac{d}{dt}\,(f \circ \phi^{(q)}(t))\Big|_{t=0} = \frac{d}{dt}\,(f \circ \phi^{(p)}(t+a))\Big|_{t=0} = \dot{\phi}^{(p)}(a)f = X_{\phi^{(p)}(a)}f \qquad (9.5) \tag{185}$$

□

So by def., $\phi^{(p)}(t)$ integral curve of X

The Fundamental Theorem on Flows

flow domain for M is open $\mathcal{D} \subseteq \mathbb{R} \times M$ s.t. $\forall\, p \in M$, $\mathcal{D}^{(p)} = \{t \in \mathbb{R} | (t,p) \in \mathcal{D}\}$ is an open interval containing 0.

flow on M is cont. $\phi : \mathcal{D} \to M$ s.t. group laws :

$$\forall\, p \in M,\ \phi(0,p) = p \qquad (9.6) \tag{186}$$

$$\forall\, s \in \mathcal{D}^{(p)} \quad \text{s.t. } s + t \in \mathcal{D}^{(p)}, \quad \phi(t,\phi(s,p)) = \phi(t+s,p) \qquad (9.7) \tag{187}$$
$$\forall\, t \in \mathcal{D}^{(\phi(s,p))}$$

Proposition 60 (9.11)**.** *If $\phi : \mathcal{D} \to M$ smooth flow,*
then infinitesimal generator X of ϕ smooth vector field and $\forall\, \phi^{(p)}$ integral curve of X

Proof. Recall that

infinitesimal generator X of ϕ, $p \mapsto X_p$ $\quad$ now on open $\mathcal{D}$, $\mathcal{D} \subseteq \mathbb{R} \times M$ s.t. $\forall\, p \in M$, $\mathcal{D}^{(p)} = \{t \in \mathbb{R} | (t,p) \in \mathcal{D}\}$

$$X_p = \dot{\phi}(0)$$

open,

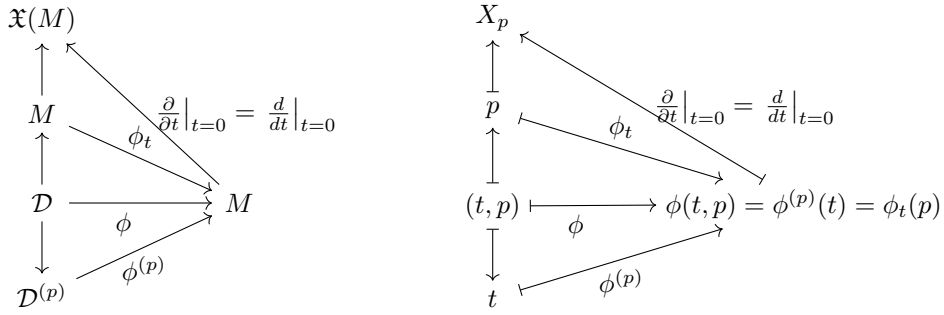
given ϕ smooth flow,
$\phi(t,q)$ defined and smooth $\forall\, (t,q)$ sufficiently close to $(0,p)$ since $\mathcal{D}$ open. With f smooth on open U in this open neighborhood of $(0,p)$,

$$\implies Xf(p) = \frac{\partial}{\partial t}(f \circ \phi(t,p))\Big|_{t=0}$$

$f\phi$ smooth (by composition), so $\partial_t f\phi$ smooth, so X smooth itself around $\forall\, p \in M$.

Suppose $t \in \mathcal{D}^{(p)}$
$\mathcal{D}^{(p)}$, $\mathcal{D}^{(\phi_t(p))} = \mathcal{D}^{(q)}$ open (by def.)
$\phi_{\Delta t}\phi_t(p) = \phi_{\Delta t + t}(p)$ by def. of flow.

□



Theorem 53 (9.12)**.** *(Fundamental Theorem on Flows) Let smooth vector field X on smooth manifold M.*

$\exists!$ *smooth maximal flow $\phi : \mathcal{D} \to M$ whose infinitesimal generator is X (recall $p \mapsto X_p$) s.t.*

$$X_p = \dot{\phi}(0)$$

(a) $\forall\, p \in M$, curve $\phi^{(p)} : \mathcal{D}^{(p)} \to M$ is unique maximal integral curve of X starting at p.

(b) If $s \in \mathcal{D}^{(p)}$, then $\mathcal{D}^{(\phi(s,p))}$ is interval $\mathcal{D}^{(p)} - s = \{t - s | t \in \mathcal{D}^{(p)}\}$

(c) $\forall\, t \in \mathbb{R}$, M_t open in M, and $\phi_t : M_t \to M_{-t}$ diffeomorphism with inverse ϕ_{-t}

Proof. From Proposition 9.2 ($\forall\, p \in M$, $\exists\, \epsilon > 0$, $\exists$ smooth $c : (-\epsilon, \epsilon) \to M$, i.e. integral curve X starting at p)

Suppose $c, \widetilde{c} : I \to M$ 2 integral curves of X, open I s.t. $c(t_0) = \widetilde{c}(t_0)$ for some $t_0 \in I$
Let $S = \{t | t \in I,\ \text{s.t. } c(t) = \widetilde{c}(t)\}$
Clearly $S \neq \emptyset$ since $c(t_0) = \widetilde{c}(t_0)$ (hypothesis)
S closed in I by continuity (of $c, \widetilde{c}$)
Suppose $t_1 \in S$
$c(t_1) = \widetilde{c}(t_1) = p$ Then in smooth coordinate neighborhood around $p = c(t_1)$, $c, \widetilde{c}$ both solutions to same ODE with same initial conditions $c(t_1) = \widetilde{c}(t_1) = p$

By uniqueness part of Thm. D.1, $c \equiv \widetilde{c}$ on interval containing t_1
$\implies S$ open in I.
Since I connected, $S = I$ (S clopen)
$c = \widetilde{c}$ $\forall\, t \in I$
Thus, $\forall\, c, \widetilde{c}$ that agrees at 1 pt. agree on common domain.

$\forall\, p \in M$, let $\mathcal{D}^{(p)} = \bigcup_\alpha I_\alpha$, open $I_\alpha \subseteq \mathbb{R}$ s.t. $0 \in I_\alpha$, and integral curve $c_\alpha : I_\alpha \to M$ starting at p is defined.

$$c_\alpha(0) = p$$

define $\phi^{(p)} : \mathcal{D}^{(p)} \to M$ where c is any integral curve s.t. $c(0) = p$ and c defined on I_α s.t. $0, t \in I_\alpha$.

$$\phi^{(p)}(t) = c(t)$$

since all integral curves agree at t by argument above, $\phi^{(p)}$ well-defined
and is obviously unique maximal integral curve starting at p.

Let $\mathcal{D} = \{(t,p) \in \mathbb{R} \times M | t \in \mathcal{D}^{(p)}\}$

define $\phi : \mathcal{D} \to M$ (notation for last statement)

$\phi(t,p) = \phi^{(p)}(t) \equiv \phi_t(p)$

By def. ϕ satisfies (a): $\forall\, p \in M$, $\exists!$ maximal integral curve of X, $\phi^{(p)}$, starting at p.

Fix $p \in M$, $s \in \mathcal{D}^{(p)}$
write $q = \phi(s,p) = \phi^{(p)}(s)$

define $\widetilde{c} : \mathcal{D}^{(p)} - s \to M$

$\widetilde{c}(t) = \phi^{(p)}(t+s)$ s.t. $\widetilde{c}(0) = \phi^{(p)}(s) = q$

By translation lemma (9.4), $\widetilde{c}(t) = c(t+s)$ e.g. $I = (-2,6),\ s = 1$ $\widetilde{c}(t)$ also integral curve of X.

$\widetilde{I} = \{t | t+s \in I\}$ $\widetilde{I} = (-3,5)$

By uniqueness of ODE solutions,
$\widetilde{c}$ agrees with $\phi^{(q)}$ on their common domain,
equivalent to second group law (9.7)

$$\widetilde{c}(t) = \phi^{(p)}(t+s) = \phi(t+s,p) = \phi^{(q)}(t) = \phi(t,q) = \phi(t,\phi(s,p))$$

□

Lemma 23 (9.19). ***(Escape Lemma)*** *Suppose smooth M, $V \in \mathfrak{X}(M)$.*
If $\gamma : J \to M$ maximal integral curve of V s.t. domain J has finite least upper bound b,
then $\forall\, t_0 \in J$, $\gamma([t_0, b))$ not contained in any compact subset of M

Proof. See Problem 9-6. Solution there. □

Flowouts

Suppose smooth M, $S \subseteq M$ embedded k-dim. submanifold.
smooth $V \in \mathfrak{X}(M)$ s.t. V nowhere tangent to S.
Let $\theta : \mathcal{D} \to M$ be flow of V
Let $\mathcal{O} = (\mathbb{R} \times S) \bigcap \mathcal{D}$
$\Phi = \theta|_{\mathcal{O}}$

(a) $\Phi : \mathcal{O} \to M$ immersion

(b) $\frac{\partial}{\partial t} \in \mathfrak{X}(\mathcal{O})$ is Φ-related to V

(c) $\exists$ smooth $\delta > 0$, $\delta : S \to \mathbb{R}$ s.t.
$\Phi|_{\mathcal{O}_\delta}$ injective, where $\mathcal{O}_\delta \subseteq \mathcal{O}$ flow domain.

$$\mathcal{O}_\delta = \{(t,p) \in \mathcal{O} | |t| < \delta(p)\} \qquad (9.9) \tag{188}$$

Thus $\Phi(\mathcal{O}_\delta)$ immersed submanifold of M containing S. V tangent to $\Phi(\mathcal{O}_\delta)$

(d) If S codim. 1, $\Phi|_{\mathcal{O}_\delta}$ diffeomorphism onto open submanifold of M

Flows and Flowouts on Manifolds with Boundary

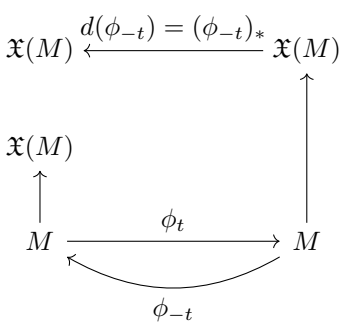
Lie Derivatives

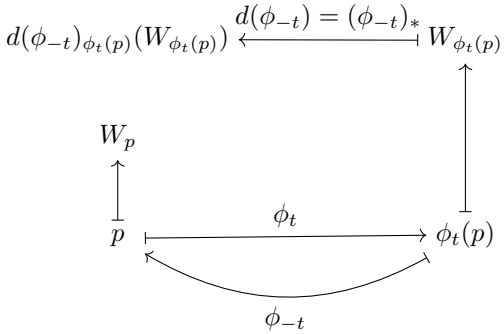
$$D_v W(p) = \left.\frac{d}{dt}\right|_{t=0} W_{p+tv} = \lim_{t\to 0} \frac{W_{p+tv} - W_p}{t} \qquad (9.15) \qquad (189)$$

$$D_v W(p) = D_v W^i(p) \left.\frac{\partial}{\partial x^i}\right|_p$$

Lie derivative of W with respect to V

$$(\mathcal{L}_V W)_p = \left.\frac{d}{dt}\right|_{t=0} d(\theta_{-t})_{\theta_t(p)}(W_{\theta_t(p)}) = \lim_{t\to 0} \frac{d(\theta_{-t})_{\theta_t(p)}(W_{\theta_t(p)}) - W_p}{t} \qquad (9.16) \qquad (190)$$





Lemma 24 (9.36)**.** *Suppose smooth M with or without ∂, $V, W \in \mathfrak{X}(M)$*
If $\partial M \neq \emptyset$, assume v trangent to ∂M
Then $\exists$ smooth $(\mathcal{L}_V W)_p \quad \forall\, p \in M$

Proof. $\forall\,(t,x) \in J_0 \times U_0$,
matrix $d(\theta_{-t})_{\theta_t(x)} : T_{\theta_t(x)}M \to T_x M$

$$\left(\frac{\partial \theta^i}{\partial x^j}(-t, \theta(t,x))\right)$$

Therefore,

$$d(\theta_{-t})_{\theta_t(x)}(W_{\theta_t(x)}) = \frac{\partial \theta^i}{\partial x^j}(-t,\theta(t,x)) W^j(\theta(t,x)) \left.\frac{\partial}{\partial x^i}\right|_x$$

□

Exercise (Lee 2e, Exercise 9.37, p. 229 [76])**.** Suppose $v \in \mathbb{R}^n$ and W is a smooth vector field on an open subset of $\mathbb{R}^n$. Show that the directional derivative $D_v W(p)$ defined by (9.15) is equal to $(\mathcal{L}_V W)_p$, where V is the vector field $V = v^i\, \partial/\partial x^i$ with constant coefficients in standard coordinates.

Solution. Given $V = v^i \frac{\partial}{\partial x^i}$ with constant coefficients (i.e. v^i constant)

$$\begin{aligned} &\dot\theta^{(x)}(t) = V \\ &\dot\theta^i(t) = v^i \qquad\qquad \Longrightarrow \quad \frac{\partial \theta^i}{\partial x^j} = \delta^i{}_j \\ &\theta^i(t) = v^i t + x^i \end{aligned}$$

$$\frac{d}{dt} W^i(\theta(t,x)) = \frac{\partial W^i}{\partial y^j}(v^j)$$

From the proof of Lemma 9.36,

$$d(\theta_{-t})_{\theta_t(x)}(W_{\theta_t(x)}) = \frac{\partial \theta^i}{\partial x^j}(-t,\theta(t,x)) W^j(\theta(t,x)) \left.\frac{\partial}{\partial x^i}\right|_x =$$

$$= \delta^i{}_j W^j(\theta(t,x)) \left.\frac{\partial}{\partial x^i}\right|_x = W^i(\theta(t,x)) \left.\frac{\partial}{\partial x^i}\right|_x$$

From (9.16)

$$(\mathcal{L}_V W)_p = \left.\frac{d}{dt}\right|_{t=0} d(\theta_{-t})_{\theta_t(p)}(W_{\theta_t(p)}) = v^j \frac{\partial W^i}{\partial x^j} \left.\frac{\partial}{\partial x^i}\right|_p = D_V W^i(p) \left.\frac{\partial}{\partial x^i}\right|_p = D_V W(p)$$

Theorem 54 (9.38)**.** *If smooth M, and $V, W \in \mathfrak{X}(M)$,*

$$\mathcal{L}_V W = [V, W]$$

Corollary 10 (9.39)**.** *(a)*

(b)

(c)

(d)

(e)

Exercise (Lee 2e, Exercise 9.40, p. 230 [76])**.** Prove this corollary.

Note. The corollary is Corollary 9.39: (a) $\mathcal{L}_V W = -\mathcal{L}_W V$; (b) $\mathcal{L}_V[W, X] = [\mathcal{L}_V W, X] + [W, \mathcal{L}_V X]$; (c) $\mathcal{L}_{[V,W]} X = \mathcal{L}_V \mathcal{L}_W X - \mathcal{L}_W \mathcal{L}_V X$; (d) $\mathcal{L}_V(gW) = (Vg)W + g\mathcal{L}_V W$ for $g \in C^\infty(M)$; (e) $F_*(\mathcal{L}_V X) = \mathcal{L}_{F_* V} F_* X$ for a diffeomorphism F.

Solution.

(a)

$$\mathcal{L}_V W = [V, W] = -[W, V] = \mathcal{L}_W V$$

(b)

(c)

(d)

(e)

Prop. 9.41 is about derivative of $d(\theta_{-t})_{\theta_t(p)}(W_{\theta_t(p)})$ at other times

Proposition 61 (9.41)**.** *Suppose smooth M with or without ∂ and $V, W \in \mathfrak{X}(M)$*
If $\partial M \neq \emptyset$, assume V tangent to ∂M
Let θ flow of V.

$\forall\, (t_0, p)$ in domain of θ

$$\left.\frac{d}{dt}\right|_{t=t_0} d(\theta_{-t})_{\theta_t(p)}(W_{\theta_t(p)}) = d(\theta_{-t_0})((\mathcal{L}_V W)_{\theta_{t_0}(p)})$$

Commuting Vector Fields

Time-Dependent Vector Fields

Let smooth manifold M

time-dependent vector field on M, V
cont. $V : J \times M \to TM$, interval $J \subseteq \mathbb{R}$
s.t.

$$V(t, p) \in T_p M \quad \forall\, (t, p) \in J \times M$$

i.e. $\forall\, t \in J$,
$V_t : M \to TM$
$V_t(p) = V(t,p)$ is a vector field on M

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time-dependent vector field on M, V is

$$\text{cont. } V : J \times M \to TM \qquad \text{interval } J \subseteq \mathbb{R}$$
$$V(t,p) \in T_pM \quad \forall\, (t,p) \in J \times M$$

i.e. $\forall\, t \in J$

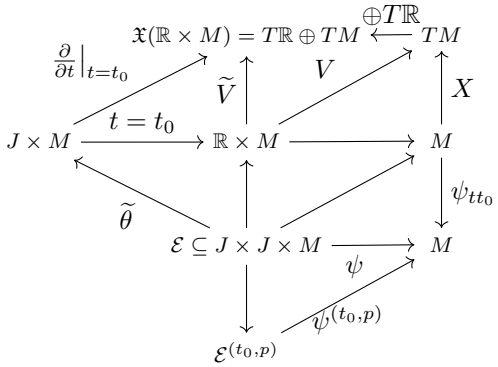
$V_t : M \to TM$
$V_t(p) = V(t,p) \in \mathfrak{X}(M)$

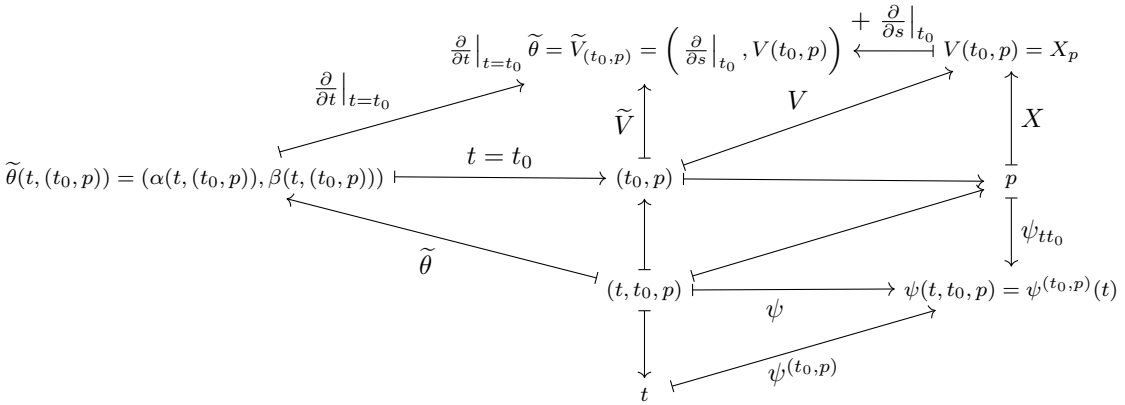
integral curve of V is diff. $\gamma : J_0 \to M$ where $J_0 \subset J$ s.t.

$$\dot\gamma(t) = V(t, \gamma(t)) \quad \forall\, t \in J_0$$

$\forall\, X \in \mathfrak{X}(M)$, determines time dependent vector field $V : \mathbb{R} \times M \to TM$ by

$$V(t,p) = X_p$$

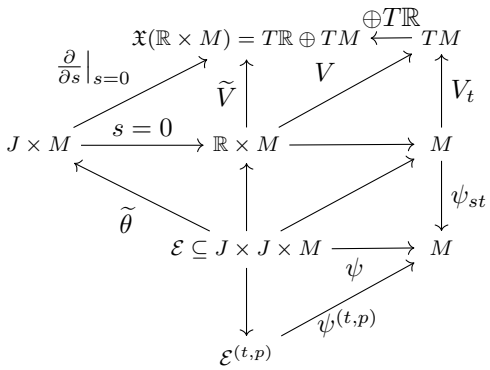


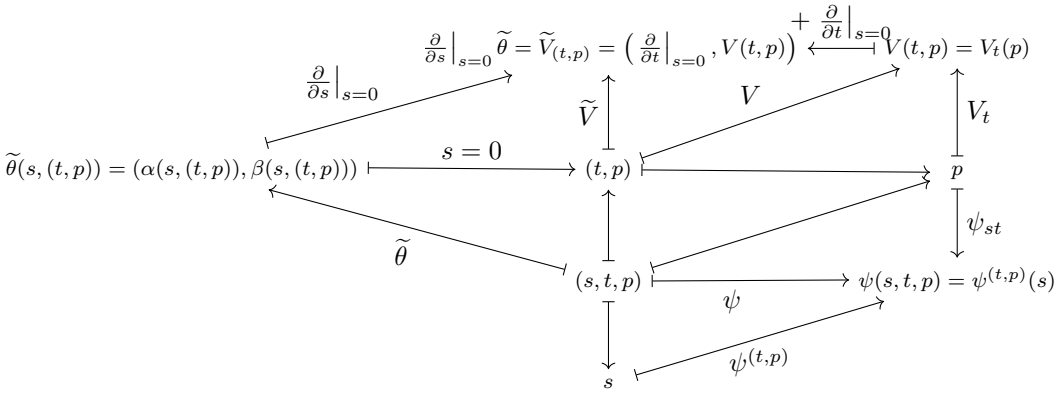


with

$$\left.\frac{\partial}{\partial t}\right|_{t=t_0} \widetilde{\theta}(t,(t_0,p)) = \left(\frac{\partial \alpha}{\partial t}(t,(t_0,p)), \frac{\partial \beta}{\partial t}(t,(t_0,p))\right)\Big|_{t=t_0} = (1, V(t_0,p))$$

EY : 20150725 I don't like Lee's choice of notation. Let me rewrite the above diagrams:





with

$$\frac{\partial}{\partial s}\Big|_{s=0}\widetilde{\theta}(s,(t,p)) = \left(\frac{\partial\alpha}{\partial s}(s,(t,p)),\frac{\partial\beta}{\partial s}(s,(t,p))\right)\Big|_{s=0} = (1,V(t,p))$$

Theorem 55 (9.48)**.** ***(Fundamental Theorem on Time-Dependent Flows)***
Let M smooth manifold
open $J\subseteq\mathbb{R}$
$V: J\times M\to TM$ smooth time-dependent vector field on M
$\exists$ open $\mathcal{E}\subseteq J\times J\times M$, smooth $\psi:\mathcal{E}\to M$ called time-dependent flow of V s.t.

(a) $\forall\, t_0\in J,\ \ \forall\, p\in M,$
open $\mathcal{E}^{(t_0,p)} = \{t\in J | (t,t_0,p)\in\mathcal{E}_0\}$ *s.t.* $t_0\in\mathcal{E}^{(t_0,p)}$

smooth curve $\psi^{(t_0,p)}:\mathcal{E}^{(t_0,p)}\to M$

$$\psi^{(t_0,p)}(t) = \psi(t,t_0,p)$$

is unique maximal integral curve of V with $\psi^{(t_0,p)}(t_0) = p$

(b) If $t_1\in\mathcal{E}^{(t_0,p)}$

$$q = \psi^{(t_0,p)}(t_1)$$

then $\mathcal{E}^{(t_1,q)} = \mathcal{E}^{(t_0,p)}$ *and*

$$\psi^{(t_1,q)} = \psi^{(t_0,p)}$$

(c) $\forall\,(t_1,t_0)\in J\times J$
$M_{t_1,t_0} = \{p\in M | (t_1,t_0,p)\in\mathcal{E}\}$ *open in M and*

$\psi_{t_1t_0}: M_{t_1t_0}\to M$ *is a diffeomorphism from $M_{t_1t_0}$ onto $M_{t_0t_1}$ with inverse $\psi_{t_0t_1}$*
$\psi_{t_1t_0}(p) = \psi(t_1,t_0,p)$

(d) If $p \in M_{t_1 t_0}$, $\psi_{t_1 t_0}(p) \in M_{t_0 t_1}$, then $p \in M_{t_2 t_0}$ and

$$\psi_{t_2 t_1} \psi_{t_1 t_0}(p) = \psi_{t_2 t_0}(p) \qquad (9.18) \tag{191}$$

Proof. Consider smooth vector field $\widetilde{V} \in \mathfrak{X}(J \times M)$ defined by

$$\widetilde{V}_{(s,p)} = \left(\left. \frac{\partial}{\partial s} \right|_s, V(s,p) \right)$$

identify $T_{(s,p)}(J \times M)$ with $T_s J \oplus T_p M$ (Prop. 3.14)

Let $\widetilde{\theta} : \widetilde{\mathcal{D}} \to J \times M$ flow of $\widetilde{V}$

$\widetilde{\theta}(t,(s,p)) = (\alpha(t,(s,p)), \beta(t,(s,p)))$

then $\alpha : \widetilde{D} \to J$ s.t.

$\beta : \widetilde{D} \to M$

$$\frac{\partial \alpha}{\partial t}(t,(s,p)) = 1 \qquad \alpha(0,(s,p)) = s$$
$$\frac{\partial \beta}{\partial t}(t,(s,p)) = V(\alpha(t,(s,p)), \beta(t,(s,p))) \qquad \beta(0,(s,p)) = p$$

$\Longrightarrow \alpha(t,(s,p)) = t + s$ so

$$\frac{\partial \beta}{\partial t}(t,(s,p)) = V(t+s, \beta(t,(s,p)) \qquad (9.19) \tag{192}$$

□

Let $\mathcal{E} \subseteq \mathbb{R} \times J \times M$ defined

$\mathcal{E} = \{(t,t_0,p) | (t - t_0, (t_0,p)) \in \widetilde{\mathcal{D}}\}$

$\mathcal{E}$ open because $\widetilde{\mathcal{D}}$ is.
since $\alpha : \widetilde{D} \to J$, if $(t,t_0,p) \in \mathcal{E}$, then $t = \alpha(t - t_0, (t_0,p)) \in J$, implies $\mathcal{E} \subseteq J \times J \times M$
$\mathcal{E}$ open so $M_{t_1 t_0} = \{p \in M | (t_1, t_0, p) \in \mathcal{E}\}$ open.

define $\psi : \mathcal{E} \to M$

$\psi(t,t_0,p) = \beta(t - t_0, (t_0,p))$

EY : 20150725 Remark: Out of the proof immediately above, there are a number of takeaways that really *should* be mentioned.

Let's collect the facts:

$\widetilde{\theta}(s,(t,p)) = (\alpha(s,(t,p)), \beta(s,(t,p))) := \widetilde{\theta}^{(t,p)}(s)$

$\widetilde{\theta}(0,(t,p)) = (\alpha(0,(t,p)), \beta(0,(t,p))) = (t,p)$

$\frac{\partial \alpha}{\partial s}(s,(t,p)) = 1$

$\frac{\partial \beta}{\partial s}(s,(t,p)) = V(\alpha(s,(t,p)), \beta(s,(t,p)))$ so

$\alpha(s,(t,p)) = s + t$

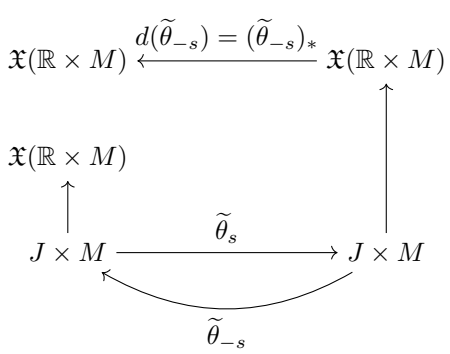
$\frac{\partial \beta}{\partial s}(s,(t,p)) = V(s+t, \beta(s,(t,p)))$

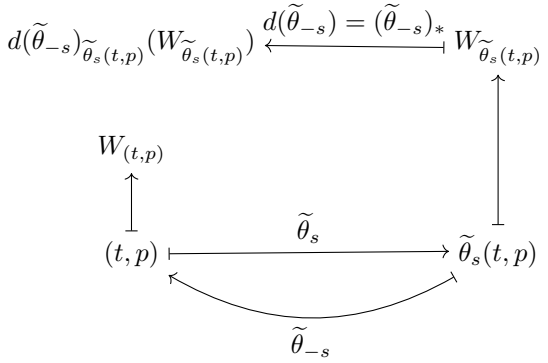
$$\left. \frac{\partial}{\partial s} \right|_{s=0} \widetilde{\theta}(s,(t,p)) = \left(\frac{\partial \alpha}{\partial s}(s,(t,p)), \frac{\partial \beta}{\partial s}(s,(t,p)) \right) \Big|_{s=0} = (1, V(t,p)) = \frac{d\widetilde{\theta}^{(t,p)}}{ds}(s=0) := \widetilde{V}_{(t,p)}$$

Also, we can write the flow $\widetilde{\theta}_s$ as

$$\widetilde{\theta}^{(t,p)}(s) = \widetilde{\theta}(s,(t,p)) = \widetilde{\theta}_s(t,p) = (\alpha(s,(t,p)), \beta(s,(t,p))) = (s+t, \beta(s,(t,p)))$$

Now consider the Lie derivative:





with $\widetilde{\theta}$ being the flow of $\widetilde{V}$. Let's define the Lie derivative:

$$\mathcal{L}_{\widetilde{V}}W=(\mathcal{L}_{\widetilde{V}}W)_{(t,p)}=\left.\frac{d}{ds}\right|_{s=0}(d\widetilde{\theta}_{-s})_{\widetilde{\theta}_s(t,p)}(W_{\widetilde{\theta}_s(t,p)})=\lim_{s\to 0}\frac{(d\widetilde{\theta}_{-s})_{\widetilde{\theta}_s(t,p)}(W_{\widetilde{\theta}_s(t,p)})-W_{\widetilde{\theta}_s(t,p)}}{s} \tag{193}$$

Use Case 1 of the proof of Lee's Theorem 9.38, for showing $\mathcal{L}_VW=[V,W]$.
Let open neighborhood $U\subseteq J\times M$, with $(t,p)\in U$. On open U, choose smooth coordinates (t,u^i) on U. By Theorem 9.22, that at a regular point $p\in M$, $\exists\,(u^i)$ coordinates s.t. $V_p=\frac{\partial}{\partial u^1}$, then consider

$$\widetilde{V}=\frac{\partial}{\partial t}+\frac{\partial}{\partial u^1}\in\mathfrak{X}(\mathbb{R}\times M)$$

with $V(t)(p)=\frac{\partial}{\partial u^1}\in\mathfrak{X}(M)$. (Remember, $V(t)$ is a vector-field that is time-dependent, but is on M. I will use this as a justification for using Thm. 9.22).

Now the flow $\widetilde{\theta}_s$ takes on these forms:

$$\widetilde{\theta}^{(t,p)}(s)=\widetilde{\theta}(s,(t,p))=\widetilde{\theta}_s(t,p)=$$
$$=(\alpha(s,(t,p)),\beta(s,(t,p)))=(s+t,\beta(s,(t,p)))$$

Given these conditions, that
$\beta(0,(t,p))=p=(u^1,u^2,\dots u^n)$ and

$$\left.\frac{\partial\beta}{\partial s}(s,(t,p))\right|_{s=0}=V(t,p)=\frac{\partial}{\partial u^1}=\left.\frac{d}{ds}\beta^{(t,p)}(s)\right|_{s=0}$$

then a β that satisfies these conditions above is

$$\beta(s,(t,p))=\beta_s(t,p)=(u^1+s,u^2\dots u^n)$$

so that we can conclude that

$$\widetilde{\theta}_s(t,p)=(t+s,u^1+s,u^2,\dots,u^n)$$

For fixed s, then

$$d(\widetilde{\theta}_{-s})_{\widetilde{\theta}_s(t,p)}=1_{T_{\widetilde{\theta}_s(t,p)}(\mathbb{R}\times M)}$$

so that

$$d(\widetilde{\theta}_{-s})_{\widetilde{\theta}_s(t,p)}(W_{\widetilde{\theta}_s(t,p)})$$
$$=d(\widetilde{\theta}_{-s})_{\widetilde{\theta}_s(t,p)}\cdot W^j(t+s,u^1+s,u^2\dots u^n)\left.\frac{\partial}{\partial u^j}\right|_{\widetilde{\theta}_s(t,p)}=W^j(t+s,u^1+s,u^2\dots u^n)\left.\frac{\partial}{\partial u^j}\right|_{(t,p)}$$
$$\implies\left.\frac{d}{ds}\right|_{s=0}W^j(t+s,u^1+s,u^2\dots u^n)\left.\frac{\partial}{\partial u^j}\right|_{(t,p)}=\left(\frac{\partial}{\partial t}W^j(t,u^1\dots u^n)+\frac{\partial}{\partial u^1}W^j(t,u^1\dots u^n)\right)\left.\frac{\partial}{\partial u^j}\right|_{(t,p)}$$

Thus, we can conclude that

$$\boxed{\mathcal{L}_{\widetilde{V}}W=\mathcal{L}_{\frac{\partial}{\partial t}+V}W=\left(\mathcal{L}_{\frac{\partial}{\partial t}V}W\right)_{(t,p)}=\left(\left(\frac{\partial}{\partial t}+V\right)W^j\right)\left.\frac{\partial}{\partial x^j}\right|_{(t,p)}} \tag{194}$$

First-Order Partial Differential Equations

Problems

Exercise (Lee 2e, Problem 9-21, p. 247 [76])**.** Let M be a smooth manifold. A *smooth isotopy of* M is a smooth map $H\colon M \times J \to M$, where $J \subseteq \mathbb{R}$ is an interval, such that for each $t \in J$, the map $H_t\colon M \to M$ defined by $H_t(p) = H(p,t)$ is a diffeomorphism. (In particular, if J is the unit interval, then H is a homotopy from H_0 to H_1 through diffeomorphisms.) This problem shows that smooth isotopies are closely related to time-dependent flows.

(a) Suppose $J \subseteq \mathbb{R}$ is an open interval and $H\colon M \times J \to M$ is a smooth isotopy. Show that the map $V\colon J \times M \to TM$ defined by

$$V(t,p) = \frac{\partial}{\partial t} H(p,t)$$

is a smooth time-dependent vector field on M, whose time-dependent flow is given by $\psi(t, t_0, p) = H_t \circ H_{t_0}^{-1}(p)$ with domain $J \times J \times M$.

(b) Conversely, suppose J is an open interval and $V\colon J \times M \to M$ is a smooth time-dependent vector field on M whose time-dependent flow is defined on $J \times J \times M$. For any $t_0 \in J$, show that the map $H\colon M \times J \to M$ defined by $H(t,p) = \psi(t, t_0, p)$ is a smooth isotopy of M.

Note. Two slips in the printed statement, reproduced above as printed: in (b) the codomain of V should be TM, and the map should read $H(p,t) = \psi(t, t_0, p)$, to match $H\colon M \times J \to M$.

Solution. Note that from wikipedia,
ambient isotopy
Let N, M manifolds,
g, h embeddings of N in M
cont. map $F : M \times [0,1] \to M$ s.t.
$F : g \mapsto h$
if $F_0 = 1$
F_t homeomorphism, $F_t : M \to M$
$F_1 : g \mapsto h$

smooth isotopy of M is smooth $H : M \times J \to M$, $J \subseteq R$ interval s.t.

$\forall\, t \in J$, $H_t : M \to M$ is a diffeomorphism.
$H_t(p) = H(p,t)$
Suppose open interval $J \subseteq \mathbb{R}$
smooth isotopy $H : M \times J \to M$

EY : 20140206
By definition

$\forall\, t$, $H_t : M \to M$
$H_t(p) = H(p,t)$
Then $DH_t = (H_t)_*$

$$DH_t : TM \to TM$$

I tried,
for some time t, consider $H(x^i(p), t) = y^j(x^i, t)$

$$\frac{\partial H(p,t)}{\partial t} = \frac{\partial y^j(x^i,t)}{\partial t}$$

Consider integral curve $x = x(t)$ s.t. $\dot{x} = X(t)$

$$\dot{y} = \frac{dy}{dt} = \frac{d}{dt} y(x(t), t) = \frac{\partial y^j}{\partial x^i} \dot{x}^i = (DH_t)X$$
$$(H_t)_* : TM \to TM$$
$$DH_t : X \mapsto \dot{y} = Y$$

$\psi(t, t_0, p) = H_t \circ H_{t_0}^{-1}(p)$ domain $J \times J \times M$??

69 Vector Bundles

Vector Bundles

(real) vector bundle of rank k over M is topological space E,
surjective cont. $\pi : E \to M$ s.t.

(i) $\forall\, p \in M$, $E_p = \pi^{-1}(p)$ endowed with structure of k-dim. real vector space

(ii) $\forall\, p \in M$,
$\exists$ neighborhood $U \ni p$
$\exists$ homeomorphism $\Phi : \pi^{-1}(U) \to U \times \mathbb{R}^k$ (local trivialization of E over U)

s.t.

- $\pi_U \circ \Phi = \pi$ (where $\pi_U : U \times \mathbb{R}^k \to U$)
- $\forall\, q \in U$, $\Phi|_{E_q}$ is vector space isomorphism from E_q to $\{q\} \times \mathbb{R}^k \cong \mathbb{R}^k$

if M, E smooth manifolds with or without boundary, π smooth, local trivializations Φ can be chosen to be diffeomorphisms,
E **smooth vector bundle**,
$\forall\, \Phi$ that's a diffeomorphism onto its image a **smooth local trivialization**

(real) line bundle - rank 1 vector bundle
complex vector bundle - $\mathbb{R}^k$ replaced by $\mathbb{C}^k$

Exercise (Lee 2e, Exercise 10.1, p. 250 [76])**.** Suppose E is a smooth vector bundle over M. Show that the projection map $\pi\colon E \to M$ is a surjective smooth submersion.

Solution. Suppose E smooth vector bundle over M.
π surjective by def.
EY : 20140204
submersion at $p \in M$, F, if for differentiable manifolds M, N, differentiable $F : M \to N$,
differential $DF_p : T_pM \to T_{F(p)}N$
is surjective

By def., $\forall\, p \in M$, $\exists\, U, \Phi$
Φ vector space isomorphism and can be chosen to be diffeomorphism
Φ diffeomorphism if Φ bijection, Φ^{-1} differentiable

$\Phi : \pi^{-1}(U) \to U \times \mathbb{R}^k$
$D\Phi : T\pi^{-1}(U) \to T(U \times \mathbb{R}^k) = TU \times T\mathbb{R}^k$

$\pi_U : U \times \mathbb{R}^k \to U$
$D\pi_U : T(U \times \mathbb{R}^k) \to TU$

$\pi : \pi^{-1}(U) \to U$

$\pi = \pi_U \circ \Phi$

So by chain rule,

$$D\pi = D\pi_U D\Phi$$
$$D\pi : T\pi^{-1}(U) \to TU$$

$D\pi_U$ clearly a surjection.

$D\Phi$ bijection.

So $D\pi$ a surjection.

---

70 The Cotangent Bundle

Covectors

Exercise (Lee 2e, Exercise 11.2, p. 273 [76])**.** Prove Proposition 11.1.

Note. Proposition 11.1: if $(E_1, \ldots, E_n)$ is a basis for V and $\varepsilon^i(E_j) = \delta^i_j$, then $(\varepsilon^1, \ldots, \varepsilon^n)$ is a basis for V^*, so $\dim V^* = \dim V$.

Solution. For linear $\omega : V \to \mathbb{R}$, $\omega \in V^*$

$$\omega(E_j) = \omega(\delta^i{}_j E_i) = \delta^i_j \omega(E_i) = \omega(E_i)\epsilon^i(E_j)$$
$$\implies \omega = \omega(E_i)\epsilon^i$$

So ω spanned by ϵ^i, $\quad i = 1 \ldots n$

Suppose $0 = \omega_i \epsilon^i$. Then $\forall\, x \in V$,

$$\omega_i \epsilon^i(x^j E_j) = \omega_i x^j \delta^i_j = \omega_i x^i = \omega(E_i)x^i = \omega(x) = 0$$

Then $\omega = 0$ since $\omega(x) = 0$, $\forall\, x \in V$. So ϵ^i form a linear basis of V^*.

By theorem, $\dim V^* = \dim V$

71 Lie Group Actions

71.1 Group Actions

action G on M cont. if $G \times M \to M$ or $M \times G \to M$ cont.

$$(g, p) \mapsto gp \qquad (p, g) \mapsto pg$$

For cont. action, θ_g homeomorphism, since $\exists$ cont. inverse $\theta_{g^{-1}}$. $\theta_g : M \to M$

orbit

$\forall\, p \in M$, orbit of $p = Gp = \{gp | g \in G\}$

action θ transitive if $\forall\, p, q \in M$, $\exists\, g$ s.t. $gp = q$, i.e. $Gp = M$

isotropy group of p, $G_p = \{g \in G | gp = p\}$

action θ is free if the only element of G that fixes any element of M is e,

i.e. if $g \cdot p = p$, $p \in M$, $g = e$.

i.e. $G_p = 1 \quad \forall\, p \in M$

Example 9.1 (Lie Group Actions)

(a) trivial action of G on M is $gp = p \quad \forall\, g \in G, \quad G_p = G$

(b) action of $GL(n,\mathbb{R})$ on $\mathbb{R}^n$, $(A,x) \mapsto Ax$; $x \in \mathbb{R}^n$ column matrix.
Ax smooth, because components of Ax depend polynomially on matrix entries of A and components of x.

Exactly only 2 orbits : $0, \mathbb{R}^n - 0$ $(\forall\, y = 0,\, y \in \mathbb{R}^n,\, Ax = y \quad)$
$x = A^{-1}y$

(c) $O(n) \times \mathbb{R}^n \to \mathbb{R}^n$
orbits : $0, S^{n-1}(R)$, $\forall\, R > 0$. To show this, complete $\frac{v}{|v|}, \frac{v'}{|v'|}$, $\forall\, v, v' \neq 0$, to orthonormal bases. Let A, A' columsn be these orthonormal bases
$A'A^{-1}(v) = v'$

(d) $O(n) \times S^{n-1} \to S^{n-1}$. $O(n)$ transitive action of $O(n)$ on S^{n-1}.
action $O(n) \times S^{n-1} \to S^{n-1}$ smooth by Corollary 8.25, S^{n-1} embedded submanifold of $\mathbb{R}^n$

Representations

If G Lie group,
(finite-dim.) representation of G is a Lie group homomorphism

$$\rho : G \to GL(V)$$

$\forall$ representation ρ yields or smooth left action of G on V,

$$g \cdot v = \rho(g)v, \qquad \forall\, g \in G,\, v \in V$$

Equivariant Maps

Proper Actions

Quotient of Manifolds by Group Actions

Suppose Lie group G acts on manifold M
$p \sim q$ if $\exists\, g \in G$ s.t. $gp = q$ equivalence classes are exactly the orbits of G in M
M/G set of orbits, with quotient topology, orbit space of the action

72 Tensors

Multilinear Algebra

$F : V_1 \times \cdots \times V_k \to W$ multilinear if $\forall\, i$, linear in each variable, $F(v_1 \dots av_1 + a'v_i' \dots v_k) = aF(v_1 \dots v_i \dots v_k) + a'F(v_1 \dots v_i' \dots v_k)$
multilinear function of 2 variables is bilinear.

$L(V_1, \dots V_k; W)$ - set of all multilinear maps from $V_1 \times \cdots \times V_k$ to W

$\{T : \underbrace{V \times \cdots \times V}_{k\text{times}} \to \mathbb{R}\} = T^k(V)$

$S \in T^k(V)$, $T \in T^l(V)$
tensor product $S \otimes T : V \times \cdots \times V \to \mathbb{R}$, covariant $(k+l)$-tensor

$$S \otimes T(x_1 \dots x_{k+l}) = S(x_1 \dots x_k)T(x_{k+1} \dots x_{k+l})$$

Exercise (Lee 2e, Exercise 12.3, p. 306 [76])**.** Show that the tensor product operation is bilinear and associative: $F \otimes G$ depends bilinearly on F and G, and $(F \otimes G) \otimes H = F \otimes (G \otimes H)$.

Solution.

$$F(v_1 \dots av_i + bw_i \dots v_k)G(v_{k+1} \dots v_{k+l}) = aF(v_1 \dots v_i \dots v_k)G + bF(v_1 \dots w_i \dots v_k)G =$$
$$= aF \otimes G(v_1 \dots v_i \dots v_k \dots v_{k+l}) + bF \otimes G(v_1 \dots w_i \dots v_k \dots v_{k+l}) = F \otimes G(v_1 \dots av_i + bw_i \dots v_k, v_{k+1} \dots v_{k+l})$$
$$F(v_1 \dots v_k)G(v_{k+1} \dots av_{k+i} + bw_{k+i} \dots v_{k+l}) =$$
$$= F(v_1 \dots v_k)(aG(v_{k+1} \dots v_{k+i}, v_{k+i+1} \dots v_{k+l}) + bG(v_{k+1} \dots w_{k+i}v_{k+i+1} \dots v_{k+l}) =$$
$$= aF \otimes G(v_{k+1} \dots v_{k+i}, v_{k+i+1} \dots v_{k+l}) + bF \otimes G(v_1 \dots w_{k+i}, v_{k+i+1} \dots v_{k+l}) =$$
$$= F \otimes G(v_1 \dots v_k, v_{k+1} \dots av_{k+i} + bw_{k+i} \dots v_{k+i+1} \dots v_{k+l})$$

$$(F \otimes G) \otimes H = (F \otimes G)(x_1 \dots x_{k+l}H(x_{k+l+1} \dots x_{k+l+m}) = F(x_1 \dots x_k)G(x_{k+1} \dots x_{k+l})H(x_{k+l+1} \dots x_{k+l+m}) =$$
$$= F(x_1 \dots x_k)(G \otimes H)(x_{k+1} \dots x_{k+l+m}) = F \otimes (G \otimes H)(x_1 \dots x_{k+l+m})$$

Proposition 62 (12.4)**.** *(A Basis for the Space of Multilinear Functions) Let V real vector space of dim. n, (E_i) any basis for V, ϵ^i dual basis.*
set of all k-tensors of form $\epsilon^{i_1} \otimes \cdots \otimes \epsilon^{i_k}$, $1 \leq i_1 \dots i_k \leq n$ basis for $T^k(V)$, dim. n^k

Proof. Let $\mathcal{B} = \{\epsilon^{i_1} \otimes \cdots \otimes \epsilon^{i_k} | 1 \leq i_1 \dots i_k \leq n\}$

Suppose arbitrary $T \in T^k(V)$
Define $T_{i_1 \dots i_k} = T(E_{i_1} \dots E_{i_k})$

$$T_{i_1 \dots i_k}\epsilon^{i_1} \otimes \cdots \otimes \epsilon^{i_k}(E_{j_1} \dots E_{j_k}) \underbrace{=}_{\text{(by definition)}} T_{i_1 \dots i_k}\epsilon^{i_1}(E_{j_1}) \dots \epsilon^{i_k}(E_{j_k})$$
$$= T_{i_1 \dots i_k}\delta^{i_1}_{j_1} \dots \delta^{i_k}_{j_k} = T_{j_1 \dots j_k} = T(E_{j_1} \dots E_{j_k})$$

T spanned by $\mathcal{B}$

□

Abstract Tensor Products of Vector Spaces

free vector space on S, $\mathbb{R}\langle S \rangle = \{\mathcal{F}\}$
finite formal linear combination - function $\mathcal{F} : S \to \mathbb{R}$ s.t. $\mathcal{F}(s) = 0$ for all but finite many $s \in S$
$\forall\, \mathcal{F} \in \mathbb{R}\langle S \rangle$, $\mathcal{F} = \sum_{i=1}^m a_i x_i$, $x_1 \dots x_m \in S$ s.t. $\mathcal{F}(x_i) \neq 0$, $a_i = \mathcal{F}(x_i)$

Exercise (Lee 2e, Exercise 12.6, p. 308 [76])**.** Prove the preceding proposition.

Note. The preceding proposition is Proposition 12.5 (Characteristic Property of the Free Vector Space): for any set S and any vector space W, every map $A\colon S \to W$ has a unique extension to a linear map $\overline{A}\colon \mathcal{F}(S) \to W$.

Solution. (Characteristic Property of Free Vector Spaces)

$$F : S \to W$$
$$x \mapsto w \in W$$

Consider $w \in W$ and $w = \sum c_\alpha w_\alpha$, $w_\alpha = F(x_\alpha)$; $x_\alpha \in S$, $w_\alpha \in W$.
Consider $\overline{F} : \mathbb{R}\langle S \rangle \to W$, $\sum_{i=1}^m c_i x_i \mapsto \sum_{I=1}^m c_i F(x_i) = \sum_{i=1}^m c_i w_i \in W$

Let $w = v$, $$w = \sum_{I=1}^m c_i w_i = \sum_{i=1}^m c_i F(x_i)$$
$$v = \sum_{i=1}^n b_i v_i = \sum_{i=1}^n b_i F(y_i)$$
$w - v = 0$ so for a vector space, this implies $w_i = v_i$, $m = n$,

$$\sum_{i=1}^m (c_i - b_i)w_i = 0, \qquad c_i = b_i$$

$\mathcal{R} \equiv$ subspace of free vector space $\mathbb{R}\langle V \times W\rangle$ spanned by

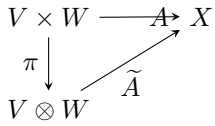
$$\begin{matrix} a(v,w) - a(v,w) \\ a(v,w) - (v, aw) \\ (v,w) + (v',w) - (v+v',w) \\ (v,w) + (v,w') - (v, w+w') \end{matrix} \quad (12.4) \tag{195}$$

tensor product of V, W

$$V \otimes W = \mathbb{R}\langle V \times W\rangle / \mathcal{R}$$

equivalence class of element (v,w) if $v \otimes w \in V \otimes W$

Proposition 63 (12.7)**.** *(Characteristic Property of the Tensor Product Space) If bilinear $A : V \times W \to X$,*



$\exists !$, $\widetilde{A} : V \otimes W \to X$, *any vector space X s.t.* *(12.6)*
$\pi(v,w) = V \otimes W$

Proof. By characteristic property of free vector space, $A : V \times W \to X$ extends uniquely to linear $\overline{A} : \mathbb{R}\langle V \times W\rangle \to X$

$$\overline{A}(v,w) = A(v,w) \text{ if } (v,w) \in V \times W \subset \mathbb{R}\langle V \times W\rangle$$

A bilinear

$$\begin{aligned} \overline{A}(av,w) = A(av,w) = aA(v,w) = a\overline{A}(v,w) = \overline{A}(a(v,w)) \\ \overline{A}(v,aw) = A(v,aw) = aA(v,w) = \overline{A}(a(v,w)) \\ \overline{A}(v+v',w) = A(v+v',w) = A(v,w) + A(v',w) = \overline{A}(v,w) = \overline{A}(a(v',w)) \end{aligned}$$

□

Likewise for (12.4)
subspace $\mathcal{R} \subset \ker \overline{A}$
$\therefore$ $\overline{A}$ descends to linear $\widetilde{A} : V \otimes W = \mathbb{R}\langle V \times W\rangle / \mathcal{R} \to X$ s.t.
$\widetilde{A} \circ \pi = \overline{A}$, $\pi : \mathbb{R}\langle V \times W\rangle \to V \otimes W$
uniqueness from $\forall\, v \otimes w \in V \otimes W$, $v \otimes w =$ linear combination of $v \otimes w$
$\widetilde{A}$ uniquely determined on $\widetilde{A}(v \otimes w) = \overline{A}(v,w) = A(v,w)$

Proposition 64 (11.4)**.** ***(Other Properties of Tensor Products)****. Let V, W, and X be finite-dimensional real vector spaces*

(a) $V^* \otimes W^*$ canonically isomorphic to $B(V,W)$, bilinear maps from $V \times W$ into $\mathbb{R}$

(b) if (E_i) basis for V , then $\{E_i \otimes F_j\}$ basis is basis for $V \otimes W$, $\therefore$ $dim(V \otimes W) = dimV\, dimW$
(F_j) basis for W

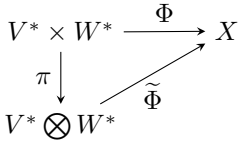
(c) $\exists !$ isomorphism $V \otimes (W \otimes X) \to (V \otimes W) \otimes X$
$v \otimes (w \otimes x) \mapsto (v \otimes w) \otimes x$

Proof. (a) canonical isomorphism (basis independence) construction between $V^* \otimes W^*$ and $B(V,W)$ (space of bilinear maps)

Define $\Phi : V^* \times W^* \to B(V,W)$
$\Phi(\omega, \eta)(v,w) = \omega(v)\eta(\omega)$

Φ bilinear (easy to check).

Prop. 11.3 $\forall$ bilinear $A : V \times W \to X$, $\exists! \widetilde{A} : V \otimes W \to X$, any vector space X s.t. $\widetilde{A}\pi = A$



i.e. descends uniquely to linear $\widetilde{\Phi} : V^* \otimes W^* \to B(V, W)$

Let (e_i) be bases for V, (ϵ^i) dual basis
(f_j) W (ϕ^j)

since $V^* \otimes W^*$ spanned by elements of the form $\omega \otimes \eta$, $\omega \in V^*$
$\eta \in W^*$
$\forall\, \tau \in V^* \otimes W^*, \quad \tau = \tau_{ij} \epsilon^i \otimes \varphi^j$

Define $\Psi : B(V, W) \to V^* \otimes W^*$

$$\Phi(b) = b(e_k, f_l)\epsilon^k \otimes \varphi^l$$

$$\Psi\widetilde{\Phi}(\tau) = \widetilde{\Phi}(\tau)(e_k, f_l)\epsilon^k \otimes \varphi^l = \tau_{ij}\widetilde{\Phi}(\epsilon^i \otimes \varphi^j)(e_k, f_l)\epsilon^k \otimes \varphi^l = \tau_{ij}\Phi(\epsilon^i, \varphi^j)(e_k, f_l)\epsilon^k \otimes \varphi^l =$$
$$= \tau_{ij}\epsilon^i(e_k)\varphi^j(f_l)\epsilon^k \otimes \varphi^l = \tau_{kl}\epsilon^k \otimes \varphi^l = \tau$$

For $v \in V$
$w \in W$

$$\widetilde{\Phi} \circ \Psi(b)(v, w) = \widetilde{\Phi} b(e_j, f_k)\epsilon^j \otimes \varphi^k(v, w) = b(e_j, f_k)\widetilde{\Phi}(\epsilon^j \otimes \varphi^k)(v, w) =$$
$$= b(e_j, f_k)\Phi(e^i, \varphi^k)(v, w) = b(e_j, f_k)e^i(v)\varphi^k(w) = b(e_j, f_k)v^i\varphi^k = b(v, w)$$

(b) Given $\{e_i | i \in I\} = \mathcal{B}_U$
$\{f_j | j \in J\} = \mathcal{B}_U$

By the bilinearity of tensor product: $a_i e_i \otimes b_j f_j = a_i b_j e_i \otimes f_j$

Consider dual basis elements $e_k^*(e_i) = \delta_{ik}$ and
$f_l^*(f_j) = \delta_{jl}$

$U \times V \to K$
$(u, v) \mapsto e_k^*(u) \cdot f_l^*(v)$

induces $U \otimes V \to K$

$$u \otimes v \mapsto e_k^*(u) \cdot f_l^*(v)$$

$$e_i \otimes f_j \mapsto \delta_{ik}\delta_{jl}$$

$$c_{ij}e_i \otimes f_j = 0 = c_{ij}\delta_{ik}\delta_{jl} = c_{kl} = 0 \qquad \forall\, k, l \text{ so } e_i \otimes f_j \text{ form a basis}$$

□

Corollary 11 (11.5)**.** *V finite-dim. real vector space, space $T^k(V)$ of covariant k-tensors on V canonically isomorphic to k-fold tensor product $V^* \otimes \cdots \otimes V^*$*

Exercise (Lee 2e, Corollary 12.12, p. 313 [76])**.** Let V be an n-dimensional real vector space. Suppose (E_i) is any basis for V and (ε^j) is the dual basis for V^*. Then the following sets constitute bases for the tensor spaces over V:

$$\begin{aligned}
&\{\varepsilon^{i_1} \otimes \cdots \otimes \varepsilon^{i_k} : 1 \le i_1, \dots, i_k \le n\} && \text{for } T^k(V^*);\\
&\{E_{i_1} \otimes \cdots \otimes E_{i_k} : 1 \le i_1, \dots, i_k \le n\} && \text{for } T^k(V);\\
&\{E_{i_1} \otimes \cdots \otimes E_{i_k} \otimes \varepsilon^{j_1} \otimes \cdots \otimes \varepsilon^{j_l} : 1 \le i_1, \dots, i_k, j_1, \dots, j_l \le n\} && \text{for } T^{(k,l)}(V).
\end{aligned}$$

Therefore, $\dim T^k(V^*) = \dim T^k(V) = n^k$ and $\dim T^{(k,l)}(V) = n^{k+l}$.

Note. First-edition Exercise 11.3, which asked for a proof of Corollary 11.5 there.

Solution. Prove Corollary 11.5.

It's enough to consider the basis (good strategy).

$T^k(V)$ basis $\mathcal{B} = \{\epsilon^{i_1} \otimes \cdots \otimes \epsilon^{i_k} | 1 \le i_1 \dots i_k \le n\}$ (Prop. 11.2) $\dim n^k$

Use Prop. 11.4(b). Surely V^* finite-dim. real vector space as well, on its own, even though it's a dual basis.

Prop.11.4(b) if (E_i) basis for V , then $\{E_i \otimes E_j\}$ basis for $V \otimes W$ and $\dim(V \otimes W) = \dim V \dim W$
(E_j) basis for W

basis for $V^* \otimes \cdots \otimes V^* = \{\epsilon^{i_1} \otimes \cdots \otimes \epsilon^{i_k} | 1 \le i_1 \dots i_k \le n\}$, $\dim(V^* \otimes \cdots \otimes V^*) = n^k$

dimensions are same. isomorphic.

Lemma 25 (11.7)**.** *Let smooth M, suppose $\sigma \in \mathcal{T}^k(M)$ $f \in C^\infty(M)$*
$\tau \in \mathcal{T}^l(M)$

Then $f\sigma$, $\sigma \otimes \tau$ also smooth tensor fields whose

$$\begin{aligned}
(f\sigma)_{i_1 \dots i_k} &= f\sigma_{i_1 \dots \sigma_k}\\
(\sigma \otimes \tau)_{i_1 \dots i_{k+l}} &= \sigma_{i_1 \dots i_k} \tau_{i_{k+1} \dots i_{k+l}}
\end{aligned}$$

Exercise (Lee 2e, Exercise 12.23, p. 318 [76])**.** Prove Proposition 12.22.

Note. First-edition Exercise 11.7 (Lemma 11.7 there). Proposition 12.22: if $A \in \mathcal{T}^k(M)$, $B \in \mathcal{T}^l(M)$ and $f \in C^\infty(M)$, then fA and $A \otimes B$ are smooth tensor fields, with components $(fA)_{i_1 \dots i_k} = fA_{i_1 \dots i_k}$ and $(A \otimes B)_{i_1 \dots i_{k+l}} = A_{i_1 \dots i_k} B_{i_{k+1} \dots i_{k+l}}$.

Solution. Prove Lemma 11.7. Note $T^0 M = T_0 M = M \times \mathbb{R}$

$$f\sigma(p, e_{i_1}^{(1)}, \dots, e_{i_k}^{(k)}) = f(p)\sigma(e_{i_1}^{(1)} \dots e_{i_k}^{(k)}) = f(p)\sigma_{i_1 \dots i_k} = (f\sigma)_{i_1 \dots i_k}$$

Suppose smooth $F : M \to N$
$\forall$ smooth covariant k-tensor field σ on N,
define k-tensor field $F^*\sigma$ on M by

$$(F^*\sigma)_p = F^*(\sigma_{F(p)})$$

explicitly, if $X_1 \dots X_k \in T_p M$, then

$$(F^*\sigma)_p(X_1 \dots X_k) = \sigma_{F(g)}(F_* X_1 \dots F_* X_k)$$

Proposition 65 (11.9)**.** *(The properties of Tensor Field Pullbacks) Suppose smooth $F : M \to N$, $\sigma \in \mathcal{T}^k(N)$,*
$G : N \to P$ $\tau \in \mathcal{T}^l(N)$

$f \in C^\infty(N)$

(a) $F^*(f\sigma) = (f \circ F)F^*\sigma$

(b) $F^*(\sigma \otimes \tau) = F^*\sigma \otimes F^*\tau$

(c) $F^*\sigma$ *smooth tensor field*

(d) $F^* : \mathcal{T}^k(N) \to \mathcal{T}^k(M)$ *linear over* $\mathbb{R}$

(e) $(GF)^* = F^*G^*$

(f) $(Id_N)^*\sigma = \sigma$

Exercise (Lee 2e, Exercise 12.26, p. 320 [76])**.** Prove Proposition 12.25.

Note. First-edition Exercise 11.9 (Proposition 11.9 there, stated above). Proposition 12.25 is Properties of Tensor Pullbacks.

Solution. Prove Prop. 11.9

Corollary 12 (11.10)**.** *Let smooth* $F : M \to N$, $\sigma \in \mathcal{T}^k(N)$
If $p \in M$, *smooth coordinates* (y^j) *for* N *on neighborhood of* $F(p)$, *then* $F^*\sigma$ *near* p

$$F^*(\sigma_{j_1\dots j_k} dy^{j_1} \otimes \cdots \otimes dy^{j_k}) = (\sigma_{j_1\dots j_k} \circ F) d(y^{j_1} \circ F) \otimes \cdots \otimes d(y^{j_k} \circ F)$$

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However, in the special case of a diffeomorphism, tensor fields of any variance can be pushed forward and pulled back at will (see Problem 11-6)

Symmetric Tensors

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Exercise (Lee 2e, Exercise 12.13, p. 314 [76])**.** Show that the following are equivalent for a covariant k-tensor α:

(a) α is symmetric.

(b) For any vectors $v_1, \dots, v_k \in V$, the value of $\alpha(v_1, \dots, v_k)$ is unchanged when $v_1, \dots, v_k$ are rearranged in any order.

(c) The components $\alpha_{i_1\dots i_k}$ of α with respect to any basis are unchanged by any permutation of the indices.

Note. First-edition Exercise 11.10.

Solution.

$$T_{i_1\dots i_k} = T(E_{i_1} \dots E_{i_k}) = T(E_{i_1} \dots E_{i_s} \dots E_{i_r} \dots E_{i_k}) = T_{i_1\dots i_s\dots i_r\dots i_k} \qquad r < s$$

set of symmetric covariant k-tensors on V by $\sum^k(V)$

define ${}^\sigma T(X_1 \dots X_k) = T(X_{\sigma(1)} \dots X_{\sigma(k)})$

define $\mathrm{Sym}T = \frac{1}{k!} \sum_{\sigma \in S_k} {}^\sigma T$

If $S \in \sum^k(V)$, define $ST = \mathrm{Sym}(S \otimes T)$

$T \in \sum^l(V)$

$$ST(X_1 \dots X_{k+l}) = \frac{1}{(k+l)!} \sum_{\sigma \in S_{k+l}} S(X_{\sigma(1)} \dots X_{\sigma(k)}) T(X_{\sigma(k+1)}, \dots, X_{\sigma(k+l)})$$

Proposition 66 (12.15)**.** *(Properties of the Symmetric Product)*

(a)

(b) if ω, η covectors, $\omega\eta = \frac{1}{2}(\omega \otimes \eta + \eta \otimes \omega)$

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Exercise (Lee 2e, Exercise 12.16, p. 315 [76])**.** Prove Proposition 12.15.

Note. Proposition 12.15 is Properties of the Symmetric Product.

Solution. Prove Proposition 12.15

(a)

(b)

$$\omega\eta(e_i, e_j) = \frac{1}{2}(\omega(e_i)\eta(e_j) + \omega(e_j)\eta(e_i)) = \frac{1}{2}(\omega(e_i)\eta(e_j) + \eta(e_i)\omega(e_j)) = \frac{1}{2}(\omega \otimes \eta + \eta \otimes \omega)(e_i, e_j)$$

direct application of definition of $ST \equiv \mathrm{Sym}(S \otimes T)$ and $S \otimes T(X_1 \dots X_{k+l}) = S(X_1 \dots X_k)T(X_{k+1} \dots X_{k+l})$ definition of tensor product.

Alternating Tensors

Lie Derivatives of Tensor Fields

Lemma 26 (12.30)**.** *smooth M, V, A, $\exists$ (12.8) $\forall p \in M$, and defines $\mathcal{L}_V A$ as smooth tensor field on M*

Exercise (Lee 2e, Exercise 12.31, p. 321 [76])**.** Prove the preceding lemma.

Note. The preceding lemma is Lemma 12.30: the derivative in (12.8) exists for every $p \in M$ and defines $\mathcal{L}_V A$ as a smooth tensor field on M.

Solution. Suppose smooth M, smooth V, smooth covariant tensor A
$I = (i_1 \dots i_k)$
$i_i = 1 \dots \dim M$

$\theta_t(p) = y$ (notation)

$$A = A(y) = A(\theta_t(p)) = A_I(y)dy^I = A_I(\theta_t(p))dy^I$$

with A_I smooth function of $\theta_t(p) = y$

$$v_1 = \delta^{j_1}_{i_1}\frac{\partial}{\partial x^{j_1}} \qquad v^{j_1}_{(1)} = \delta^{j_1}_{i_1}$$

$$d(\theta_t)^*_p(A_{\theta_t(p)})\frac{\partial}{\partial x^I} = A_{\theta_t(p)}(d(\theta_t)_p\frac{\partial}{\partial x^I})$$

$d(\theta_t)_p = \frac{\partial y^i}{\partial x^j}$

$$d(\theta_t)_p v_1 = \frac{\partial y^{i_1}}{\partial x^{j_1}} v^{j_1}_{(1)}\frac{\partial}{\partial y^{i_1}}$$

$$\implies \frac{\partial y^{k_1}}{\partial x^{j_1}}\delta^{j_1}_{i_1}\frac{\partial}{\partial y^{k_1}} = \frac{\partial y^{j_1}}{\partial x^{i_1}}\frac{\partial}{\partial y^{j_1}}$$

$\frac{\partial}{\partial x^I} = \frac{\partial}{\partial x^{i_1}} \otimes \dots \otimes \frac{\partial}{\partial x^{i_k}}$

$$d(\theta_t)_p\frac{\partial}{\partial x^I} = d(\theta_t)_p v_1 \dots d(\theta_t)_p v_k = \frac{\partial y^J}{\partial x^I}\frac{\partial}{\partial y^J}$$

$$A_{\theta_t(p)}(d(\theta_t)_p(v_1) \dots d(\theta_t)_p(v_k)) = A_J(\theta_t(p))\frac{\partial y^J}{\partial x^I}$$

$$d(\theta_t)_p^*(A_{\theta_t(p)}) = A_J^* dx^J$$
$$d(\theta_t)_p^*(A_{\theta_t(p)})\frac{\partial}{\partial x^I} = A_I^* = A_J \frac{\partial y^J}{\partial x^I}$$
$$\Longrightarrow (\mathcal{L}_V A)_p = \left.\frac{d}{dt}\right|_{t=0} (\theta_t^* A)_p = \left.\frac{d}{dt}\right|_{t=0} A_J \left.\frac{\partial \theta^J(t,x)}{\partial x^I}\right|_p dx^I$$

$A_J = A_J(\theta_t(p)) = A_J(\theta(t,x))$

θ smooth in t, x, so A_J smooth in t and x

so since $(\mathcal{L}_V A)_I|_p$ smooth $\forall\, I \in \{(i_1 \dots i_k) | i_i = 1 \dots \dim M,\, i = 1 \dots k\}$,
$(\mathcal{L}_V A)|_p$ smooth tensor field on M

Proposition 67 (12.32)**.** *(a)* $\mathcal{L}_V f = Vf$

(b) $\mathcal{L}_V(fA) = (\mathcal{L}_V f)A + f\mathcal{L}_V A$

(c) $\mathcal{L}_V(A \otimes B) = (\mathcal{L}_V A) \otimes B + A \otimes \mathcal{L}_V B$

(d) If $X_1 \dots X_k$ *smooth vector fields,* A *smooth* k*-tensor field,*

$$\mathcal{L}_V(A(X_1 \dots X_k)) = (\mathcal{L}_V A)(X_1 \dots X_k) + A(\mathcal{L}_V X_1 \dots X_k) + \cdots + A(X_1 \dots \mathcal{L}_V X_k) \qquad (12.9) \quad (196)$$

Corollary 13 (12.33)**.**

$$(\mathcal{L}_V A)(X_1 \dots X_k) = V(A(X_1 \dots X_k)) - A([V, X_1], X_2 \dots X_k) - \cdots - A(X_1 \dots X_{k-1}, [V, X_k]) \qquad (12.10) \quad (197)$$

73 Riemannian Metrics

Riemannian Manifolds

Riemannian metric on M - smooth symmetric 2-tensor field positive definite at each pt.

Riemannian manifold - pair (M, g)

If g on M, then $\forall p \in M$, g_p inner product on T_pM. Because of this, we will often use the notation $\langle X, Y\rangle_g$ to denote

$$g_p(X, Y) \in \mathbb{R} \quad \forall\, X, Y \in T_pM$$

$\forall$ smooth, local coordinates (x^i), write Riemannian metric

$$g = g_{ij} dx^i \otimes dx^j$$

where g_{ij} symmetric positive definite matrix of smooth functions. $g_{ij} = g_{ji}$

$$g = g_{ij} dx^i \otimes dx^j = \frac{1}{2}(g_{ij} dx^i \otimes dx^j + g_{ji} dx^i \otimes dx^j) = \frac{1}{2}(g_{ij} dx^i \otimes dx^j + g_{ij} dx^j \otimes dx^i) =$$
$$= g_{ij} dx^i dx^j \quad \text{(by Prop. 12.15(b))} \quad \text{Notice that } dx^i dx^j \text{ is symmetrized!}$$

Example 13.1 (The Euclidean Metric) Euclidean metric on $\mathbb{R}^n$, defined in standard coordinates

$$g = \delta_{ij} dx^i dx^j$$

It is common to use the abbreviation ω^2 for the symmetric product of a tensor ω with itself, so the Euclidean metric can also be writen

$$\overline{g} = (dx^1)^2 + \cdots + (dx^n)^2$$

Applied to $v, w \in T_p\mathbb{R}^n$

$$\overline{g}_p(v, w) = \delta_{ij} v^i w^j = \sum_{i=1}^n v^i w^i = v \cdot w$$

under coordinate change, use Corollary 11.10

Proposition 68 (13.3)**.** ***(Existence of Riemannian Metrics)***
$\forall$ *smooth manifold* M*,* M *with or without* ∂M*,* $\exists$ *Riemannian metric* g

Proof. Choose covering of M by smooth coordinate charts $(U_\alpha, \varphi_\alpha)$
$\overline{g}$ Euclidean metric
$\forall U_\alpha$, $\exists$ Riemannian metric $g_\alpha = \varphi_\alpha^* \overline{g}$

$$\varphi_\alpha : U_\alpha \to \mathbb{R}^n$$

Let $\{\psi_\alpha\}$ smooth partition of unity subordinate to cover $\{U_\alpha\}$
Define $g = \sum_\alpha \psi_\alpha g_\alpha$
s.t. $\forall\, g_\alpha$, $\psi_\alpha g_\alpha = 0$ outside suppψ_α
By local finiteness, $\exists$ only finitely many $\psi_\alpha g_\alpha \neq 0$ in neighborhood of each pt.
so $g = \sum_\alpha \psi_\alpha g_\alpha$ defines a smooth tensor field

□

defined on Riemannian manifold (M, g)

- length or norm of $X \in T_pM$ defined

$$|X|_g = \langle X, X\rangle_g^{1/2} = g_p(X, X)^{1/2}$$

- angle between $X, Y \in T_pM$, $X, Y \neq 0$ is unique $\theta \in [0, \pi]$ satisfying

$$\cos\theta = \frac{\langle X, Y\rangle_g}{|X|_g |Y|_g}$$

- $X, Y \in T_pM$ orthogonal if $\langle X, Y\rangle_g = 0$
- If $\gamma : [a, b] \to M$ piecewise smooth curve segment, length of γ is

$$L_g(\gamma) = \int_a^b |\gamma'(t)|_g dt$$

Just as we did in Chapter 8 for $\mathbb{R}^n$
define orthonormal frame for M to be local frame $(E_1 \dots E_n)$ defined on some open subset $U \subset M$ s.t.
$(E_1|_p \dots E_n|_p)$ orthonormal basis for T_pM $\quad \forall p \in U$, or equivalently s.t. $\langle E_i, E_j\rangle_g = \delta_{ij}$

Example 13.14. coordinate frame $\left(\frac{\partial}{\partial x^i}\right)$ global orthonormal frame on $\mathbb{R}^n$

Corollary 14 (13.8)**.** *(Existence of Local Orthonormal Frames). Let* (M, g) *Riemannian manifold.*
$\forall p \in M$*,* $\exists$ *smooth orthonormal frame on neighborhood of* p*.*

Observe Corollary 13.8. doesn't show that $\exists$ smooth coordinates near p for which coordinate frame is orthonormal.

Pullback Metrics

Suppose (M, g) Riemannian manifolds.
$(\widetilde{M}, \widetilde{g})$

isometry - smooth $F : M \to \widetilde{M}$ if F diffeomorphism s.t. $F^*\widetilde{g} = g$
if $\exists$ isometry F, $M, \widetilde{M}$ isometric.
F local isometry is $\forall p \in M$, $\exists$ neighborhood U s.t. $F|_U$ isometry of U onto open $\widetilde{U} \subset M$
g on M flat if $\forall p \in M$, $\exists$ neighborhood $U \subset M$ s.t. $(U, g|_U)$ isometric to open $\widetilde{U} \subset \mathbb{R}^n$ with Euclidean metric.

Prob. 11-14 shows $\exists$ only if metric flat.

Riemannian Submanifolds

$S \subset M$
define $g|_S = i^* g$, for $i : S \hookrightarrow M$

$$(g|_S)(X,Y) = i^* g(X,Y) = g(i_* X, i_* Y) = g(X,Y)$$

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in general, $S \subset M$

$F : S \to M$ e.g. $F(u^1, u^2) = (x^1, x^2, x^3)$

$$F(u^1 \dots u^s) = (x^1 \dots x^m) \text{ s.t. } s \leq m \quad \text{(for this case)}$$

Note

$$x^i = x^i(u^1 \dots u^s) \text{ or e.g. } x^2 = x^2(u^1, u^2)$$

Recall these facts about pullbacks and pushforwards.

$F^* : T_{F(p)} M \to T_p S$ (pullback!)
$F_* : T_p S \to T_{F(p)} M$ (push forward; remember we can only pushforward if F diffeomorphism, i.e. F, F^{-1} diff. and F bijective)
$F^* : \tau^2(M) \to \tau^2(S)$ (can always pullback tensors; in this case (rank 2))

Consider charts (U,u), $p \in U \subset S$, (V,x), $F(p) \in V \subset M$, $V \subset F(U)$
For $f : M \to \mathbb{R}$, i.e. $f \in \mathcal{C}^\infty(M)$
$fF : S \to \mathbb{R}$ i.e. $fF \in \mathcal{C}^\infty(S)$

$$fF = f(x^i)^{-1} x^i F(u^j)^{-1} u^j = (f(x^i)^{-1})(x^i F(u^j)^{-1}) u^j = f(x^i(u^j))$$

Consider $\overline{g}(x_i, x_j)$
$\overline{g} = \delta_{ij} dx^i dx^j$ ($\overline{g}$ as a tensor (rank 2) in its local coordinate form, with coordinates y^i. So $\overline{g}$ is (like, or is) a Euclidean metric)

$$F_* E_i(f) = E_i(fF) = \omega^k_{(i)} \frac{\partial}{\partial x^k} f = \frac{\partial}{\partial u^i} f(F(u)) = \frac{\partial f}{\partial x^k} \frac{\partial x^k}{\partial u^i} \qquad \omega^k_{(i)} = \frac{\partial x^k}{\partial u^i}$$

By definition, for
$F^* \overline{g}(x^{(i)}, x^{(j)})$ on by (notation) $F^* g(A,B)$, $A, B \in T_p S$
$F^* \overline{g}(E_i, E_j) = \overline{g}(F_* E_i, F_* E_j)$

$$F^* \overline{g}(E_i, E_j) = \overset{\circ}{g}(E_i, E_j) = \overset{\circ}{g}_{ij} = \overline{g}(F_* E_i, F_* E_j) = \overline{g}\left(\frac{\partial x^k}{\partial u^i} \frac{\partial}{\partial x^k}, \frac{\partial x^l}{\partial u^j} \frac{\partial}{\partial x^l} \right) =$$

$$= \overline{g}_{kl} \frac{\partial x^k}{\partial u^i} \frac{\partial x^l}{\partial u^j}$$

Formula for pullback of metric on M to metric on S i.e. formula for metric on S

$$\boxed{(F^* \overline{g})_{ij} = \overset{\circ}{g}_{ij} = \overline{g}_{kl} \frac{\partial x^k}{\partial u^i} \frac{\partial x^l}{\partial u^j}}$$

For $\overline{g}_{kl} = \delta_{kl}$ (Euclidean metric)

$$\overset{\circ}{g}_{ij} = \frac{\partial x^k}{\partial u^i} \frac{\partial x^k}{\partial u^j} = \left(\frac{\partial x^i}{\partial u^k} \right)^T \frac{\partial x^k}{\partial u^j} = (D_u x)^T (D_u x) \equiv (D_u x)^2$$

$\overset{\circ}{g}_{ij}$ is just the square of the Jacobian (The square of the Jacobian is the metric in S^1).
Then you could get the matrix form of the metric.

Example 13.16

$\mathring{g} = \overline{g}|_{S^n}$ $\quad S^n \hookrightarrow \mathbb{R}^{n+1}$ round metric (or standard metric) on sphere.

It's usually easiest to compute the induced metric on a Riemannian submanifold in terms of local parametrizations (see Chapter 5)

Example 13.17 **(Induced Metrics in Graph Coordinates.)**

Let open $U \subset \mathbb{R}^n$

$M \subset \mathbb{R}^{n+1}$ graph of smooth $f : U \to \mathbb{R}$

Then $X : U \to \mathbb{R}^{n+1}$

$X(u^1 \dots u^n) = (u^1 \dots u^n, f(u))$ smooth (global) parametrization of M

induced metric on M,

$\overline{g} = \delta_{ij} dy^i dy^j$ (note y^i local coordinates on $\mathbb{R}^{n+1}$)

Recall Prop.11.9. $F^*(\sigma \otimes \tau) = F^*\sigma \otimes F^*\tau$

Corollary 11.10. $F : M \to N$,

$F^*(\sigma_{j_1 \dots j_k} dy^{j_1} \otimes \cdots \otimes dy^{j_k}) = (\sigma_{j_1 \dots j_k} \circ F) d(y^j F) \otimes \cdots \otimes d(y^{j_k} F)$

$$X^*\overline{g} = (\delta_{ij} \circ X) d(y^i X) d(y^j X) = (du^1)^2 + \cdots + (du^n)^2 + (df)^2$$

$$X^*\overline{g}_p(E_i, E_j) = \overline{g}_{X(p)}(X_* E_i, X_* E_j)$$

The Normal Bundle

Suppose (M, g), Riemannian submanifold $S \subset M$

$\forall\, p \in S$, vector $N \in T_pN$ normal to S if N orthogonal to T_pS with respect to g

$N_pS \subset T_pM$, $N_pS =$ all vectors normal to S at $p = \{N | \langle N, X \rangle_g = 0,\ \forall\, X \in T_pS\}$ normal space to S at p

The Riemannian Distance Function

Exercise (Lee 2e, Exercise 13.23, p. 337 [76])**.** Suppose $\gamma\colon [a, b] \to M$ is a piecewise smooth curve segment and $a < c < b$. Show that

$$L_g(\gamma) = L_g(\gamma|_{[a,c]}) + L_g(\gamma|_{[c,b]}).$$

Solution.

$$L_g(\gamma) = \int_a^b |\gamma'(t)|_g dt = \int_a^c |\gamma'(t)|_g dt + \int_c^b |\gamma'(t)|_g dt = L_g(\gamma|_{[a,c]}) + L_g(\gamma|_{[c,b]})$$

Exercise (Lee 2e, Exercise 13.24, p. 337 [76])**.** Show that lengths of curves are local isometry invariants of Riemannian manifolds. More precisely, suppose (M, g) and $(\widetilde{M}, \widetilde{g})$ are Riemannian manifolds with or without boundary, and $F\colon M \to \widetilde{M}$ is a local isometry. Show that $L_{\widetilde{g}}(F \circ \gamma) = L_g(\gamma)$ for every piecewise smooth curve segment γ in M.

Solution. On every coordinate patch, consider on some interval $I \subset \mathbb{R}$ parametrizing curve γ on M and $\widetilde{\gamma}$ on $\widetilde{M}$ in the same way, and that $F^*\widetilde{\gamma} = \widetilde{\gamma}$

$$L_{\widetilde{g}}(F \circ \gamma) = \int_I |F\gamma|_{\widetilde{g}} ds$$

$$= \int_I (\widetilde{g}(F\dot{\gamma}(t), F\dot{\gamma}(t))^{1/2} ds$$

$$= \int_I \left(\widetilde{g}(\dot{\gamma}^i \frac{\partial y^j}{\partial x^i} \frac{\partial}{\partial y^j}, \dot{\gamma}^k \frac{\partial y^l}{\partial x^k} \frac{\partial}{\partial y^l}) \right)^{1/2} ds$$

$$= \int_I (\dot{\gamma}^i \dot{\gamma}^k)^{1/2} \left(\widetilde{g}_{jl} \frac{\partial y^j}{\partial x^i} \frac{\partial y^l}{\partial x^i} \right)^{1/2} ds =$$

$$= \int_I (F^*\widetilde{g}(\dot{\gamma}(t), \dot{\gamma}(t)))^{1/2} dt = \int_I (g(\dot{\gamma}(t), \dot{\gamma}(t)))^{1/2} = L_g(\gamma)$$

Rather, think in terms of a coordinate-free manner.

$$L_g(\gamma) = \int_a^b |\gamma(t)|_g dt = \int_a^b (g(\dot\gamma(t),\dot\gamma(t)))^{1/2} dt = \int_a^b dt(F^*\widetilde{g}(\dot\gamma(t),\dot\gamma(t)))^{1/2} = \int_a^b dt(\widetilde{g}(F_*\dot\gamma(t),F_*\dot\gamma(t)))^{1/2} =$$
$$= \int_a^b |\dot{F\gamma}(t)|_{\widetilde{g}} dt = L_{\widetilde{g}}(F\gamma)$$

Proposition 69 (13.25)**.** ***(Parameter independence of Length)***
Let (M,g), $\gamma : [a,b] \to M$ piecewise smooth curve segment
If $\widetilde{\gamma}$ any reparametrization of γ, then $L_g(\widetilde{\gamma}) = L_g(\gamma)$

Proof. Suppose γ smooth.
$\varphi : [c,d] \to [a,b]$ diffeomorphism s.t. $\widetilde{\gamma} = \gamma \circ \varphi$
φ diffeomorphism implies $\varphi' > 0$ or $\varphi' < 0$ everywhere.

(Recall diffeomorphism (cf. wikipedia) *differentiable*, bijective, inverse *differentiable*; so DF, Jacobian *matrix*, bijective, F differentiable, so it can't be 0 at any pt. (linear algebra, need $\exists$ inverse))

Assume $\varphi' > 0$

$$L_g(\widetilde{\gamma}) = \int_c^d |\widetilde{\gamma}'(t)|_g dt = \int_c^d \left|\frac{d}{dt}(\gamma\circ\varphi)\right|_g dt = \int_c^d |\gamma'(\varphi(t))\dot\varphi|_g dt = \int_a^b |\gamma'(\varphi(t))|_g \dot\varphi dt = \int_a^b |\gamma'(s)|_g =$$
$$= \int_a^b |\gamma'(s)|_g ds = L_g(\gamma)$$

where second-to-last equality follows from change of variables formula for ordinary integrals.

□

If (M,g) connected Riemannian manifold
$d_g(p,q)$ (Riemannian) distance between p,q - infinum of $L_g(\gamma)$ over all piecewise smooth curve segments γ from p to q.

The key is the following technical lemma, which shows that any Riemannian metric is locally comparable to Euclidean metric in coordinates.

Lemma 27 (13.28)**.** *Let g on open $U \subset \mathbb{R}^n$*

For compact $K \subset U$, $\exists$ constants c, C s.t. $\forall\, x \in K$
$$\forall\, v \in T_x\mathbb{R}^n$$
$$c|v|_{\overline{g}} \le |v|_g \le C|v|_{\overline{g}}$$

Theorem 56 (13.29)**.** *(Riemannian Manifolds as Metric Spaces)*
Let connected (M,g)
with $d_g(p,q)$; M metric space whose metric topology same as original manifold topology.

local orthonormal frame $(E_1 \dots E_n)$ for M on open $U \subset M$ is adapted to S if first k vectors $(E_1|_p \dots E_k|_p)$ span $T_pS \quad \forall\, p \in S$.
follows $(E_{k+1}|_p \dots E_n|_p)$ span N_pS

Prop. 11.24 proved exactly some way as counterpart for submanifolds of $\mathbb{R}^n$ (Prop. 10.17)

Proposition 70 (11.24)**.** *(Existence of Adapted Orthonormal Fames) Let $S \subset M$ embedded Riemannian submanifold*
$\forall\, p \in S$, $\exists$ smooth adapted orthonormal frame on neighborhood $U \ni p \subset M$

Recall $F : M \to N$ immersion if DF injective everywhere, F embedding if F injective (homeomorphism onto its image) and F immersion.

normal bundle to S
$$NS = \coprod_{p\in S} N_pS$$

The Tangent-Cotangent Isomorphism

EY 20140521, Below, in between the lines, are my notes off the previous edition. It's frustrating to not be able to obtain instantly the most up-to-date edition automatically, online, available freely for download. Notation had changed. It's important to me to keep up-to-date with the latest notation; it's not trivial (cf. Zee, A.; Srednicki's QFT vs. previous QFT notation)

Given (M, g) define bundle map $\widetilde{g} : TM \to T^*M$
$\forall\, p \in M$, $\forall\, X_p \in T_pM$, $\widetilde{g}(X_p) \in T_p^*M$ be covector defined $\widetilde{g}(X_p)(Y_p) = g_p(X_p, Y_p) \quad \forall\, Y_p \in T_pM$

To see this is a smooth bundle map, consider its action on smooth vector fields:

$$\widetilde{g}(X)(Y) = g(X, Y) \quad \forall\, X, Y \in \tau(M)$$

Because $\widetilde{g}(X)(Y)$ linear over $C^\infty(M)$ as a function of Y,
from Prob. 6-8, $\widetilde{g}(X)$ smooth covector field.
because $\widetilde{g}(X)$ linear over $C^\infty(M)$ as a function of X, $\widetilde{g}$ smooth bundle map by def. by Prop. 5.16.

Use same symbol: pointwise bundle map $\widetilde{g} : TM \to T^*M$
linear map on sections $\widetilde{g} : \mathcal{T}(M) \to \mathcal{T}^*(M)$

$\widetilde{g}$ injective: $\widetilde{g}(X_p) = 0$ implies $0 = \widetilde{g}(X_p)(X_p) = \langle X_p, X_p \rangle_g$ so $X_p = 0$
By dim., $\widetilde{g}$ bijective, so it's a bundle isomorphism (Prob. 5-9)

If X, Y smooth vector fields,

$$\widetilde{g}(X)(Y) = g_{ij} X^i Y^j$$
$$\widetilde{g}(X) = g_{ij} X^i dy^j$$

customary to denote $X_j = g_{ij} X^i$ so $\widetilde{g}(X) = X_j dy^j$

$\widetilde{g}^{-1} : T_p^*M \to T_pM$ is inverse of (g_{ij}) (Because (g_{ij}) matrix of the isomorphism $\widetilde{g}$, it is invertible $\forall\, p$)
let (g^{ij}) inverse of $g_{ij}(p)$ so $g^{ij} g_{jk} = g_{kj} g^{ji} = \delta^i_k$
Thus for covector field $\omega \in \mathcal{T}^*M$,

$$\widetilde{g}^{-1}(\omega) = \omega^i \frac{\partial}{\partial x^i}, \quad \omega^i = g^{ij} \omega_j$$

ω^i is a vector, which we visualize as a (sharp) arrow, while X_j covector, which we visualize by means of its (flat) level sets.

$\forall$ smooth f on (M, g), $f \in \mathbb{R}$, define vector field $\quad \mathrm{grad} f = \widetilde{g}^{-1}(df)$

$\forall\, X \in \mathcal{T}(M)$

$$\langle \mathrm{grad} f, X \rangle_g = \widetilde{g}(\mathrm{grad} f)(X) = df(X) = Xf$$

thus $\langle \mathrm{grad} f, X \rangle_g = Xf \quad \forall\, X \in \chi(M)$
or equivalently $\langle \mathrm{grad} f, \cdot \rangle_g = df$

$$\mathrm{grad} f = g^{ij} \frac{\partial f}{\partial x^i} \frac{\partial}{\partial x^j}$$

---

bundle isomorphism $\widehat{g} : TM \to T^*M$

$\forall\, p \in M$ $\quad \widehat{g}(v) \in T_p^*M$ defined by
$\forall\, v \in T_pM$ $\quad \widehat{g}(v)(w) = g_p(v, w) \quad \forall\, w \in T_pM$

$$\widehat{g}(X)(Y) = g(X, Y) \qquad \forall\, X, Y \in \mathfrak{X}(M)$$

$\widehat{g}(X)(Y)$ linear over $C^\infty(M)$ as a function of Y, Lemma 12.24 $\Longrightarrow \widehat{g}(X)$ smooth covector field
$g = g_{ij} dx^i dx^j$

$$\widehat{g}(X)(Y) = g_{ij} X^i Y^j \Longrightarrow \widehat{g}(X) = g_{ij} X^i dx^j = X_j dx^j \text{ where } X_j = g_{ij} X^i$$

$X^\flat = \widehat{g}(X)$
Now

$$\widehat{g}^{-1} : T_p^* M \to T_p M$$

$\forall$ covector field $\omega \in \mathfrak{X}^*(M)$

$$\widehat{g}^{-1}(\omega) = \omega^i \frac{\partial x^i}{,} \quad \omega^i = g^{ij} \omega_j$$

$omega^\sharp = \widehat{g}^{-1}(\omega)$
gradient of f by $\text{grad} f = (df)^\sharp = \widehat{g}^{-1}(df)$
$\forall X \in \mathfrak{X}(M)$

$$\langle \text{grad} f, X \rangle_g = \widehat{g}(\text{grad} f)(X) = df(X) = Xf$$

or
$\langle \text{grad} f, \cdot \rangle_g = df$
$\text{grad} f = g^{ij} \frac{\partial f}{\partial x^i} \frac{\partial}{\partial x^j}$ so $\text{grad} f$ is smooth

Problems

Exercise (Lee 2e, Problem 12-3, p. 324 [76])**.** Let V and W be finite-dimensional real vector spaces. Show that the tensor product space $V \otimes W$ is uniquely determined up to canonical isomorphism by its characteristic property (Proposition 12.7). More precisely, suppose $\widetilde{\pi} \colon V \times W \to Z$ is a bilinear map into a vector space Z with the following property: for any bilinear map $B \colon V \times W \to Y$, there is a unique linear map $\widetilde{B} \colon Z \to Y$ such that the following diagram commutes:

$$\begin{array}{ccc} V \times W & \xrightarrow{B} & Y \\ {\scriptstyle \widetilde{\pi}}\downarrow & \nearrow {\scriptstyle \widetilde{B}} & \\ Z & & \end{array}$$

Then there is a unique isomorphism $\Phi \colon V \otimes W \to Z$ such that $\widetilde{\pi} = \Phi \circ \pi$, where $\pi \colon V \times W \to V \otimes W$ is the canonical projection. [Remark: this shows that the details of the construction used to define the tensor product space are irrelevant, as long as the resulting space satisfies the characteristic property.]

Note. First-edition Problem 11-1. The statement runs onto p. 325.

Solution. Recall $\forall$ bilinear $A : V \times W \to Y$, $\exists!$ linear $\widetilde{A} : Z \to Y$ s.t.

$$\begin{array}{ccc} V \times W & \xrightarrow{\widetilde{\pi}} & Z \\ {\scriptstyle \otimes}\downarrow & \nearrow {\scriptstyle \exists!\, \pi} & \\ V \otimes W & & \end{array}$$

is the universal property s.t. $\pi\otimes = \widetilde{\pi}$
Suppose bilinear $\widetilde{\pi} : V \times W \to Z$ s.t.,
$\forall$ bilinear $A : V \times W \to Y$, $\exists!$ linear $\widetilde{A} : Z \to Y$ s.t.

$$\begin{array}{ccc} V \times W & \xrightarrow{A} & Y \\ {\scriptstyle \widetilde{Z}}\downarrow & \nearrow {\scriptstyle \widetilde{A}} & \\ Z & & \end{array}$$

$$\begin{array}{ccc} V \times W & \xrightarrow{\widetilde{\pi}} & Z \\ {\scriptstyle \otimes}\downarrow & \swarrow {\scriptstyle \lambda} \quad \searrow {\scriptstyle \widetilde{\Phi}} & \\ V \otimes W & & \end{array}$$

Consider

$\Phi\otimes = \widetilde{\pi}$ (by universal property)

$$\Phi(v \otimes w) = \widetilde{\pi}(v, w)$$

$\exists!$ linear $\lambda : Z \to V \otimes W$

$$\lambda \circ \widetilde{\pi} = \otimes$$
$$\lambda \otimes \widetilde{\pi}(v, w) = v \otimes w$$
$$\Phi\lambda(\widetilde{\pi}(v, w)) = \Phi(v \otimes w) = \widetilde{\pi}(v, w)$$
$$\Phi\lambda = \mathrm{id}_Z$$
$$\lambda\Phi(v \otimes w) = \lambda\widetilde{\pi}(v, w) = v \otimes w$$
$$\lambda\Phi = \mathrm{id}_{V\otimes W}$$

So Φ is an isomorphism between $Z, V \otimes W$. As λ is unique, so is Φ

Exercise (Lee 2e, Problem 12-2, p. 324 [76])**.** For any finite-dimensional real vector space V, prove that there are canonical isomorphisms $\mathbb{R} \otimes V \cong V \cong V \otimes \mathbb{R}$.

Note. First-edition Problem 11-2.

Solution. tensor product of U, V is vector space $U \otimes V$ with bilinear map $\otimes : U \times V \to U \otimes V$
$(u, v) \mapsto u \otimes v$

with universal property with any vector space W.

$K \cong k1$

By bilinearity,

$$U \otimes K \longrightarrow U \otimes 1 \longrightarrow U$$

$$u \otimes k \longmapsto ku \otimes 1 \overset{q}{\twoheadleftarrow} ku$$

$$u \otimes k \longmapsto ku \otimes 1 \overset{q}{\twoheadleftarrow} ku$$

$q : U \to U \otimes 1$
$u \mapsto u \otimes 1$

then $U \otimes K \simeq U$

Exercise (Lee 2e, Problem 12-4, p. 325 [76])**.** Let $V_1, \ldots, V_k$ and W be finite-dimensional real vector spaces. Prove that there is a canonical (basis-independent) isomorphism

$$V_1^* \otimes \cdots \otimes V_k^* \otimes W \cong \mathrm{L}(V_1, \ldots, V_k; W).$$

(In particular, this means that $V^* \otimes W$ is canonically isomorphic to the space $\mathrm{L}(V; W)$ of linear maps from V to W.)

Note. First-edition Problem 11-3. The solution below is the case $k = 1$, $U^* \otimes V \cong \mathrm{Hom}(U, V)$.

Solution. $U^* \times V \to \mathrm{Hom}(U, V)$ induces a natural homomorphism (injective), $U^* \otimes V \to \mathrm{Hom}(U, V)$
$(u^*, v) \to (u \to u^*(u), v)$

$$\begin{array}{ccc} U^* \otimes V & \longrightarrow & \mathrm{Hom}(U, V) \\ & \searrow^{\otimes} & \nearrow \end{array}$$

If $\dim U, \dim V < \infty$,

for arbitrary $v_i \in V$, $\sum_i e_i^* \otimes v_i \in U^* \otimes V$

$$e_j \to e_i^*(e_j)v_i = v_j$$

$\sum_i e_i^* \otimes v_i$ corresponds to homomorphism $U \to V$ mapping $e_i \to v_i$
$U^* \otimes V \to \text{Hom}(U, V)$ surjective.
$\dim U^* \otimes V = \dim U^* \dim V$
isomorphism by dim. reason.

74 Differential Forms

The Geometry of Volume Measurement

The Algebra of Alternating Tensors

$$\text{Alt} : T^k(V) \to \Lambda^k(V)$$
$$(\text{Alt}T)(X_1 \dots X_k) = \frac{1}{k!} \sum_{\sigma \in S_k} (\text{sgn}\sigma) T(X_{\sigma(1)} \dots X_{\sigma(k)})$$

Exercise (Lee 2e, Exercise 14.4, p. 351 [76])**.** Prove Proposition 14.3.

Note. First-edition Exercise 12.2. Proposition 14.3 (Properties of Alternation): (a) $\text{Alt}\,\alpha$ is alternating; (b) $\text{Alt}\,\alpha = \alpha$ if and only if α is alternating. The solution below is part (b).

Solution. If T alternating, by Exercise 12.1, $T(X_{\sigma(1)} \dots X_{\sigma(k)}) = (\text{sgn}\sigma) T(X_1 \dots X_k)$

$$\begin{aligned} &\frac{1}{k!} \sum_{\sigma \in S_k} (\text{sgn}\sigma) T(X_{\sigma(1)} \dots X_{\sigma(k)}) \\ = &\frac{1}{k!} \sum_{\sigma \in S_k} (\text{sgn}\sigma)(\text{sgn}\sigma) T(X_1 \dots X_k) \\ = &\frac{1}{k!} \sum_{\sigma \in S_k} T(X_1 \dots X_k) \\ = &\, T(X_1 \dots X_k) \Longrightarrow \text{Alt}T = T \end{aligned}$$

Elementary Alternating Tensors

$$I = (i_1 \dots i_k)$$
$$I_\sigma = (i_{\sigma(1)} \dots i_{\sigma(k)}) \qquad \sigma \in S_k$$
$$\delta_I{}^J = \begin{cases} \text{sgn}\sigma & \text{if } I \text{ and } J \text{ has no repeated indices} \\ & \quad \text{and } J = I_\sigma \text{ for some } \sigma \in S_k \\ 0 & \text{if } I \text{ and } J \text{ has repeated index} \\ & \quad \text{or } J \neq I_\sigma \quad \forall\, \sigma \in S_k \end{cases}$$

Let V vector space, $(\epsilon^1 \dots \epsilon^n)$ basis for V^*
define covariant k-tensor ϵ^I by

$$\epsilon^I(x_1 \dots x_k) = \det \begin{pmatrix} \epsilon^{i_1}(X_1) & \dots & \epsilon^{i_k}(X_k) \\ \vdots & & \vdots \\ \epsilon^{i_k}(X_1) & \dots & \epsilon^{i_k}(X_k) \end{pmatrix} = \det \begin{pmatrix} X_1^{i_1} & \dots & X_k^{i_k} \\ \vdots & & \vdots \\ X_1^{i_k} & \dots & X_k^{i_k} \end{pmatrix}$$

denote sum over only increasing multi-indice

$$\sum_I{}' T_I \epsilon^I = \sum_{\{I : 1 \le i_1 < \dots < i_k \le n\}} T_I \epsilon^I$$

Proposition 71 (12.5)**.** *for $k \le n$*
$\{\epsilon = \epsilon^I : I$ an increasing multi-index of length k $\}$ is a basis for $\Lambda^k(V)$

$$\implies \dim\Lambda^k(V) = \binom{n}{k}$$

Lemma 28. *12.6* $\omega \in \Lambda^n(V)$
If linear $T : V \to V$, $X_1 \dots X_n \in V$

$$\omega(TX_1 \dots TX_n) = detT\omega(X_1 \dots X_n) \qquad (12.2) \tag{198}$$

Proof. By Prop. 12.5, $\mathcal{E} = \{\epsilon^I | I$ increasing multi-index of length k $\}$ basis for $\Lambda^k(V)$

$$\omega = c\epsilon^{1\dots n} \qquad \binom{n}{n} = 1$$

By multilinearity of $\omega(TX_1 \dots TX_n)$ and $(\det T)\omega(X_1 \dots X_n)$, it suffices to verify it in special case

$$X_i = E_i, \quad i = 1 \dots n$$

□

$$\det T\omega(X_1 \dots X_n) = \det Tc\epsilon^{1\dots n}(E_1 \dots E_n) = c\det T$$
$$\omega(TE_1 \dots TE_n) = c\epsilon^{1\dots n}(T_1 \dots T_n) = c\det(\epsilon^j(T_i)) = c\det(\epsilon^j(T_i{}^k e_k)) = c\det T_i{}^j$$
where, recall $\epsilon^I(X_1 \dots X_k) = \det X_j{}^{i_k}$

The Wedge Product

Lemma 29 (14.10)**.**

$$\epsilon^I \wedge \epsilon^J = \epsilon^{IJ} \tag{199}$$

Differential Forms on Manifolds

Recall T^kT^*M bundle of covariant k-tensors on M
alternating tensors $\Lambda^kT^*M \subset T^kT^*M$

$$\Lambda^kT^*M = \coprod_{p\in M} \Lambda^k(T_p^*M)$$

Exercise (Lee 2e, Exercise 14.14, p. 359 [76])**.** Show that Λ^kT^*M is a smooth subbundle of T^kT^*M, and therefore is a smooth vector bundle of rank $\binom{n}{k}$ over M.

Solution. Λ^kT^*M smooth subbundle of T^kT^*M
EY : 20140501 EY? ???

---

denote section

$$\Omega^k(M) = \Gamma(\Lambda^kT^*M)$$

in any smooth chart, k form ω can be written locally as

$$\omega = \sum_I{}' \omega_I dx^{i_1} \wedge \dots \wedge dx^{i_k} = \sum_I{}' \omega_I dx^I$$

By Lemma 14.7(c)

$$dx^{i_1} \wedge \dots \wedge dx^{i_k}\left(\frac{\partial}{\partial x^{j_1}} \dots \frac{\partial}{\partial x^{j_k}}\right) = \delta^I{}_J$$

component functions ω^I of ω determined by

$$\omega^I = \omega\left(\frac{\partial}{\partial x^{i_1}} \dots \frac{\partial}{\partial x^{i_k}}\right)$$

$$(F^*\omega)(v_1 \dots v_k) = \omega(dF(v_1) \dots dF(v_k))$$

$$e_i \in TM$$
$$f_j \in TN$$
$$F(x) = F^j(x) = y^j(x^i)$$

$$F_*(v) = w = w^j f_j$$
$$F_*(v) = vF = v^i \frac{\partial F^j}{\partial x^i} f_j \qquad \Longrightarrow w^j = v^i \frac{\partial^j F}{\partial x^i}$$

$$(F^*\omega)(v_1 \dots v_k) = \omega(F_* v_1 \dots F * v_k) = \omega(v_1 F \dots v_k F)$$
$$(F^*\omega) = (F^*\omega)_{\underline{I}} dx^{\underline{I}}$$
$$(F^*\omega) e_I = (F^*\omega)_{\underline{I}}$$
$$\omega(dF(e_I)) = \omega(F_* e_I) = \omega_{\underline{J}} (DF)^{\underline{J}}_{I}$$

$$F_* e_I = (F_* e_I)^J f_J = (DF)^J{}_I f_J$$
$$(F^*\omega)_{\underline{I}} = \omega_{\underline{J}} (DF)^{\underline{J}}_{I}$$

Lemma 30 (14.16)**.** *(a) $F^* : \Omega^k(N) \to \Omega^k(M)$ linear over $\mathbb{R}$*

(b) $F^*(\omega \wedge \eta) = (F^*\omega) \wedge (F^*\eta)$

(c) in any smooth chart

$$F^*(\omega_{\underline{I}} dy^{\underline{I}}) = (\omega_{\underline{I}} F)(dy^{\underline{I}} F)$$

Exercise (Lee 2e, Exercise 14.17, p. 361 [76])**.** Prove this lemma.

Note. The lemma is Lemma 14.16: for smooth $F\colon M \to N$, (a) $F^*\colon \Omega^k(N) \to \Omega^k(M)$ is linear over $\mathbb{R}$; (b) $F^*(\omega \wedge \eta) = (F^*\omega) \wedge (F^*\eta)$; (c) the coordinate formula for F^*.

Solution. $\omega, \eta \in \Omega^k(N)$ $\quad F^*\omega = \alpha$, $F^*\eta = \beta$

$$(F^*\omega) = (F^*\omega)_{\underline{I}} e^{\underline{I}} = \alpha_{\underline{I}} e^{\underline{I}}$$
$$(F^*\eta) = (F^*\eta)_{\underline{J}} e^{\underline{J}} = \beta_{\underline{J}} e^{\underline{J}}$$

(a)

$$\alpha + \beta = F^*\omega + F^*\eta$$
$$= (\alpha_{\underline{I}} + \beta_{\underline{I}}) dx^{\underline{I}}$$
$$= (\omega(dF(e_{\underline{I}})) + \eta(dF(e_{\underline{I}}))) dx^{\underline{I}}$$
$$= (\omega + \eta) dF(e_I) dx^{\underline{I}}$$
$$= F^*(\omega + \eta)(e_{\underline{I}}) dx^{\underline{I}} =$$
$$= F^*(\omega + \eta)$$
$$F^* c\omega = c\omega DF = cF^*\omega$$

(b)

$$\alpha \wedge \eta = \alpha_{\underline{I}} \beta_{\underline{J}} e^{\underline{I}} \wedge e^{\underline{J}}$$
$$\alpha \wedge \eta = (\alpha \wedge \eta)_{\underline{K}} e^{\underline{K}}$$
$$(\alpha \wedge \eta)_{\underline{K}} = \alpha_{\underline{I}} \beta_{\underline{J}} \delta^{\underline{IJ}}_{\underline{K}}$$
$$\omega \wedge \eta = \omega_{\underline{I}} f^{\underline{I}} \wedge \eta_{\underline{J}} f^{\underline{J}} = \omega_{\underline{I}} \eta_{\underline{J}} f^{\underline{I}} \wedge f^{\underline{J}}$$

$$F^*(\omega \wedge \eta) e_{\underline{K}} = (\omega \wedge \eta) DF e_{\wedge K} = \omega_{\underline{I}} \eta_{\underline{J}} f^{\underline{I}} \wedge f^{\underline{J}} DF^{\underline{K}}_{\underline{K}} f_L = \omega_{\underline{I}} \eta_{\underline{J}} DF^{\underline{L}}_{\underline{K}} \delta^{\underline{IJ}}_{\underline{L}} = \omega_{\underline{I}} \eta_{\underline{J}} DF^{\underline{IJ}}_{\underline{K}}$$

In the last equality, $\underline{IJ}$ in $DF^{\underline{IJ}}_{\underline{K}}$ is some permutation of $\underline{IJ}$

$$(F^*\omega) \wedge (F^*\eta) e_{\underline{K}} = \omega_{\underline{J}} DF^{\underline{J}}_{\underline{I}} \eta_{\underline{L}} DF^{\underline{L}}_{\underline{M}} (e^{\underline{I}} \wedge e^{\underline{M}}) e_{\underline{K}} = \omega_{\underline{J}} \eta_{\underline{L}} DF^{\underline{J}}_{\underline{I}} DF^{\underline{L}}_{\underline{M}} \delta^{\underline{IM}}_{\underline{K}} = \omega_{\underline{I}} \eta_{\underline{J}} DF^{\underline{I}}_{\underline{L}} DF^{\underline{J}}_{\underline{M}} \delta^{\underline{LM}}_{\underline{K}}$$

(c)

Exterior Derivatives

$\forall$ manifold, $\exists$ differential operator

$$d : \mathcal{A}^k(M) \to \mathcal{A}^{k+1}(M)$$

s.t. $d(d\omega) = 0 \quad \forall \omega$

necessary: if smooth k-form $\omega = d\eta$, some $(k-1)$ form η, then $d\omega = 0$

in coordinates

$$d\left(\sum_J{}' \omega_J dx^J\right) = \sum_J{}' d\omega_J \wedge dx^J \qquad (12.15) \tag{200}$$

$$d\left(\sum_J{}' \omega_J dx^{j_1} \wedge \cdots \wedge dx^{j_k}\right) = \sum_J{}' \sum_i \frac{\partial \omega_J}{\partial x^i} dx^i \wedge dx^{j_1} \wedge \cdots \wedge dx^{j_k} \qquad (12.16) \tag{201}$$

Theorem 57 (12.14)**.** *(The Exterior Derivative)*

$\forall$ *smooth* M, $\exists!$ *linear* $d : \mathcal{A}^k(M) \to \mathcal{A}^{k+1}(M)$ *s.t.*

(i) if f *smooth,* $f \in \mathbb{R}$ *(0-form), then differential* df*, defined*

$$df(X) = Xf$$

(ii) if $\omega \in \mathcal{A}^k(M)$*, then* $d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^k \omega \wedge d\eta$
$\eta \in \mathcal{A}^l(M)$

(iii) $d^2 = 0$

(a) $\forall$ *smooth coordinate chart,* d *given by (12.15)*

(b) d *local; if* $\omega = \omega'$ *on open* $U \subset M$*, then* $d\omega = d\omega'$ *on* U

(c) d *commutes with restriction if* $U \subset M$ *any open set*

$$d(\omega|_U) = d(\omega)|_U \qquad (12.17) \tag{202}$$

Proof. Suppose M covered by a single chart.

define d by (12.15)

$$d\left(\sum_J{}' \omega_J dx^J\right) = \sum_J{}' d\omega_J \wedge dx^J \qquad (12.15)$$

$$df(X) = df(X^i e_i) = X^i f(e_i)$$

d linear, (i) satisfied.

Consider $\omega = f dx^I$

$\eta = g dx^J$

$$d(\omega \wedge \eta) = d((f dx^I) \wedge (g dx^J)) = d(fg dx^I \wedge dx^J) = (g df + f dg) \wedge dx^I \wedge dx^J =$$
$$= (df \wedge dx^I) \wedge (g dx^J) + (-1)^k (f dx^I) \wedge (dg \wedge dx^J) = d\omega \wedge \eta + (-1)^k \omega \wedge d\eta$$

(ii) proved.

0-form

$$d(df) = d\left(\frac{\partial f}{\partial x^j} dx^j\right) = \frac{\partial^2 f}{\partial x^i \partial x^j} dx^i \wedge dx^j = \sum_{i<j} \partial^2_{ij} f dx^i \wedge dx^j + \sum_{j<k} \partial^2_{ij} f dx^j \wedge dx^i =$$
$$= \sum_{i<j} \left(\frac{\partial^2 f}{\partial x^i \partial x^j} - \frac{\partial^2 f}{\partial x^j \partial x^i}\right) dx^i \wedge dx^j = 0$$

for k-form, use $k = 0$ case, (ii)

$$d(d\omega) = d\left(\sum_J{}' d\omega_J \wedge dx^{j_1} \wedge \cdots \wedge dx^{j_k}\right) =$$
$$= \sum_J{}' d(d\omega_J) \wedge dx^{j_1} \wedge \cdots \wedge dx^{j_k} + \sum_J{}' \sum_{i=1}^k (-1)^i d\omega_J \wedge dx^{j_1} \wedge \cdots \wedge d(dx^{j_i}) \wedge \cdots \wedge dx^{j_k} = 0$$

from (12.17), $d(\omega|_U) = (d\omega)|_U$

$$(d_U \omega)|_{UU'} = d_{UU'}\omega = (d_{U'}\omega)|_{UU'}$$

Suppose $\exists$ another operator $\widetilde{d} : \mathcal{A}^k(M) \to \mathcal{A}^{k+1}(M)$

$\eta = \omega - \omega'$, let $p \in U$.

Let $\varphi \in \mathcal{C}^\infty(M)$ smooth bump function, $\varphi = 1$ in neighborhood of p, supported in U.

Then $\varphi\eta = 0$ in M, so

$$0 = \widetilde{d}(\varphi\eta)_p = d\varphi_p \wedge \eta_p + \varphi(p)\widetilde{d}\eta_p = \widetilde{d}\eta_p$$

because $\varphi \equiv 1$ in neighborhood of p

p arbitrary, so $d\eta = 0$ on U. $d\omega = d\omega'$ (locality)

□

Antiderivation of degree $g \in \mathbb{Z}$ on $\mathbb{Z}$-graded $\mathbb{R}$-algebra $A = \bigoplus_{k\in\mathbb{Z}} A_k$ in $\mathbb{R}$-linear $D : A \to A$

$$D(A_k) = A_{k+g}$$

s.t.

$$D(a_k a_l) = (Da_k)a_l + (-1)^k a_k (Da_l) \qquad a_k \in \mathcal{A}_k,\ a_l \in A_l$$

Example 12.15.

$$\omega = P dx + Q dy + R dz$$

Recall

$$d\left(\sum_J{}' \omega_J dx^J\right) = \sum_J{}' d\omega_J \wedge dx^J$$

$$d\omega = dP \wedge dx + dQ \wedge dy + dR \wedge dz =$$
$$= \left(\frac{\partial P}{\partial x}dx + \frac{\partial P}{\partial y}dy + \frac{\partial P}{\partial z}dz\right) \wedge dx + \left(\frac{\partial Q}{\partial x}dx + \frac{\partial Q}{\partial y}dy + \frac{\partial Q}{\partial dz}dz\right) \wedge dy + \left(\frac{\partial R}{\partial x}dx + \frac{\partial R}{\partial y}dy + \frac{\partial R}{\partial z}dz\right) \wedge dz$$
$$= \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx \wedge dy + \left(\frac{\partial R}{\partial x} - \frac{\partial P}{\partial z}\right) dx \wedge dz + \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}\right) dy \wedge dz$$

An Invariant Formula for the Exterior Derivative

Proposition 72 (14.29 (Exterior Derivative of a 1-Form))**.**

$$d\omega(X,Y) = X(\omega(Y)) - Y(\omega(X)) - \omega([X,Y]) \tag{203}$$

Proof. $\forall\, \omega \in \Omega^1(M)$, $\omega = udv$ for some $u, v \in C^\infty(M)$

$$\begin{aligned}
&d\omega(X,Y) = d(udv)(X,Y) = du \wedge dv(X,Y) = du(X)dv(Y) - dv(X)du(Y) = X(u)Y(v) - X(v)Y(u)\\
&X(udv(Y)) = X(uY(v)) = X(u)Y(v) + uXY(v)\\
&Y(udv(X)) = Y(uX(v)) = Y(u)X(v) + uYX(v)\\
&udv([X,Y]) = uXYv - uYXv\\
&\implies d\omega(X,Y) = X(\omega(Y)) - Y(\omega(X)) - \omega([X,Y])
\end{aligned}$$

□

Proposition 73 (14.30)**.** *Let smooth manifold M of $dim M = n$,*
Let (E_i) smooth local frame for M, let (ϵ^i) dual coframe.
Let $d\epsilon^i = \sum_{j<k} b^i_{jk}\epsilon^j \wedge \epsilon^k$
$[E_j, E_k] = c^i_{jk}E_i$
Then $b^i_{jk} = -c^i_{jk}$

Proof is Exercise 14.31.

Exercise (Lee 2e, Exercise 14.31, p. 369 [76])**.** Use (14.28) to prove the preceding proposition.

Note. The preceding proposition is Proposition 14.30: for a smooth local frame (E_i) with dual coframe (ε^i), if $d\varepsilon^i = \sum_{j<k} b^i_{jk}\,\varepsilon^j \wedge \varepsilon^k$ and $[E_j, E_k] = c^i_{jk}E_i$, then $b^i_{jk} = -c^i_{jk}$.

Solution.

Proof. Assume $j < k$ without loss of generality.

$$\begin{aligned}
d\epsilon^i(E_j, E_k) = \sum_{j'<k'} b^i_{j'k'}(\delta^{j'}_j\delta^{k'}_k - \delta^{j'}_k\delta^{k'}_j = b^i_{jk} = E_j\delta^i_k - E_k\delta^i_j - c^i_{jk}\\
\implies b^i_{jk} = -c^i_{jk}
\end{aligned}$$

□

Lie Derivatives of Differential Forms

Proposition 74 (14.33)**.** *Suppose M smooth manifold, $V \in \mathfrak{X}(M)$, $\omega, \eta \in \Omega^*(M)$*

$$\mathcal{L}_V(\omega \wedge \eta) = (\mathcal{L}_V\omega) \wedge \eta + \omega \wedge (\mathcal{L}_V\eta)$$

Theorem 58 (14.35)**.** *(Cartan's Magic Formula)*

$$\mathcal{L}_V\omega = V \lrcorner (d\omega) + d(V \lrcorner \omega) =$$

EY

$$= i_V(d\omega) + d(i_V\omega)$$

Corollary 15 (14.36)**.** *(The Lie Derivative Commutes with d)*

$$\mathcal{L}_V(d\omega) = d(\mathcal{L}_V\omega)$$

75 Orientations

Orientations of Vector Spaces

Exercise (Lee 2e, Exercise 15.1, p. 379 [76])**.** Show that being consistently oriented is an equivalence relation on the set of all ordered bases for V, and show that there are exactly two equivalence classes.

Note. First-edition Exercise 13.1.

Solution.

$$(E_1 \dots E_n) \sim (E_1 \dots E_n)$$
$$E_i = \delta_i{}^j E_j$$
$$\det(\delta_i{}^j) = 1$$

If $E_i = B_i{}^j \widetilde{E}_j$, $\det B_i{}^j > 0$,
$\det(BB^{-1}) = \det B \det B^{-1} = 1$. $\det B^{-1} > 0$ since $\det B > 0$.
If $E_i = B_i{}^j \widetilde{E}_j$

$$\widetilde{\widetilde{E}}_i = A_i{}^j E_j$$
$$\widetilde{\widetilde{E}}_i = A_i{}^k B_k{}^j \widetilde{E}_j \qquad \det AB = \det A \det B > 0$$

So it's an equivalence relation and since $\det B \neq 0$, $\det B > 0$ or $\det B < 0$ and as above there are only 2 equivalence classes; $\det A \gtrless 0$

Orientations of Manifolds

lcaol frame (E_i) for M is (positively) oriented if $(E_1|_p \dots E_n|_p)$ positively oriented basis for T_pM. $\forall\, p \in U$ cont. if $\forall\, p \in M$, $p \in$ domain of oriented local frame.
orientation of M is cont., pointwise orientation.
M orinetable if $\exists$ orientation to M

Exercise (Lee 2e, Exercise 15.4, p. 381 [76])**.** Suppose M is an oriented smooth n-manifold with or without boundary, and $n \geq 1$. Show that every local frame with connected domain is either positively oriented or negatively oriented. Show that the connectedness assumption is necessary.

Note. First-edition Exercise 13.2.

Solution. connected domain U. Consider some $\Omega \in \Lambda^m(V)$, consider

$$U \to \mathbb{R}$$
$$p \mapsto (E_1|_p \dots E_n|_p) \mapsto \Omega(E_1|_p \dots E_n|_p) = \det(\epsilon^i(E_j)\big|_p)$$

det cont. function s.t. $\det = \begin{cases} +1 \\ -1 \end{cases}$ on U. U connected so $\forall$ cont. functions from X to $\{0,1\}$ or, the same, $\{1,-1\}$, constant.
So $\Omega(E_1|_p \dots E_n|_p) = \det(\epsilon^i(E_j)\big|_p)$ constant on U, otherwise U separated.

Proposition 75 (13.4)**.** *$dim M = m \geq 1$*
$\forall\, m$-form Ω on M, $\Omega \neq 0$ determines a unique orientation of M s.t. Ω positively oriented $\forall\, p \in M$, Conversely, if M given orientation, then $\exists$ smooth m-form Ω on M that's positively oriented $\forall\, p \in M$

$F : M \to N$ local diffeomorphism.
$\forall\, p \in M$, $\exists\, (U, \varphi)$ chart and consider $U_F \subset U$ s.t. $F(U_F)$ open, $F|_{U_F} : U_F \to F(U_F)$ diffeomorphism.
For $F(p) \in N$, $\exists\, (V, \psi)$ chart and consider $F(U_F) \cap V$
$\psi F \varphi^{-1}(x^1 \dots x^m) = F^j(x^1 \dots x^m) \qquad \det(\partial_i F^j) > 0$ suppose.

The Orientation Covering

Orientations of Hypersurfaces

interior multiplication or contraction with X

$$i_X\omega(Y_1 \dots Y_{k-1}) = \omega(X, Y_1 \dots Y_{k-1})$$

$i_X\omega$ obtained by inserting X into the first slot.

Notation $X \lrcorner \omega = i_X\omega$

Suppose M smooth manifold

$S \subset M$ submanifolds (immersed or embedded)

vector field along S is cont. $N : S \to TM$ s.t. $N_p \in T_pM \quad \forall\, p \in S$

(Note difference between vector field along S and vector field on S, s.t. $N_p \in T_pS \,\forall\, p$)

$N_p \in T_pM$, $p \in S$ transverse to S if T_pM spanned by N_p, T_pS

Similarly, vector field N along S transverse to S if N_p transverse to S, $\quad \forall\, p \in S$

Proposition 76 (15.21, 13.12 in previous version)**.** *Suppose M oriented smooth m-manifold*
S immersed hypersurface in M.
N transverse vector field along S.

Then S has unique orientation s.t. $\forall\, p \in S$, $(E_1 \dots E_{n-1})$ oriented basis for T_pS iff $(N_p, E_1 \dots E_{n-1})$ oriented basis for T_pM

If Ω orientation form for M,
then $(N \lrcorner \Omega)|_S \equiv i_N\Omega|_S$ orientation form for S with respect to this orientation.

Recall that smooth hypersurface S is $S \subseteq M$ equipped with $i : S \hookrightarrow M$, smooth immersion, i.e. Di injective.

Now orientation form $\omega = dN_p \wedge dE_1 \wedge \cdots \wedge dE_{n-1}$, then
$i_{N_p}\omega = dE_1 \wedge \cdots \wedge dE_{n-1}$

$$i_S^*(i_{N_p}\omega) = i_S^*(dE_1 \wedge \cdots \wedge dE_{n-1}) = i_S^*(dE_1) \wedge \cdots \wedge i_S^*(dE_{n-1}) = d(i_S^*E_1) \wedge \cdots \wedge d(i_S^*E_{n-1})$$

Proof. Let Ω

$\omega = (N \lrcorner \Omega)|_S$ $m-1$ form.

$(E_1 \dots E_{n-1})$ basis for T_pS

N transverse to S implies $(N_p, E_1 \dots E_{n-1})$ basis for T_pM.

Ω orientation form so Ω nonvanishing.

$$\omega_p(E_1 \dots E_{n-1}) = \Omega_p(N_p, E_1 \dots E_{n-1}) \neq 0$$

since $\omega_p(E_1 \dots E_{n-1}) > 0$ iff $\Omega_p(N_p, E_1 \dots E_n) > 0$,
orientation determined by ω is the 1 defined in the statement of the proposition.

□

76 Integration on Manifolds

Integration of Functions on Riemannian Manifolds

Proposition 77 (16.28)**.** *(M, g) oriented Riemannian manifold with or without ∂*
Suppose f compactly supported cont. on M, $f \in \mathbb{R}$, $f \geq 0$
Then $\int_M f dV_g \geq 0$
$\int_M f dV_g = 0$ iff $f = 0$

Proof. If f supported in
domain of single oriented smooth chart (U, φ)

By Prop. 15.31

$$\int_M f dV_g = \int_{\varphi(U)} f(x)\sqrt{\det(g_{ij})}dx^1 \dots dx^n \geq 0$$

□

Exercise (Lee 2e, Exercise 16.29, p. 422 [76]). Suppose (M, g) is an oriented Riemannian manifold and $f\colon M \to \mathbb{R}$ is continuous and compactly supported. Prove that $\left|\int_M f\, dV_g\right| \le \int_M |f|\, dV_g$.

Solution. given oriented Riemannian manifold (M, g)
compact supported cont. $f : M \to \mathbb{R}$

Then if f supported in
domain of single oriented smooth chart (U, φ)

$$\left|\int_M f dV_g\right| = \left|\int_{\varphi(U)} f(x)\sqrt{\det(g_{ij})}dx^1 \dots dx^n\right| \ge \int_{\varphi(U)} |f(x)|\sqrt{\det(g_{ij})}dx^1 \dots dx^n = \int_M |f| dV_g$$

where inequality above is from some thm. in calculus.

The Divergence Theorem

$$\begin{aligned} &* : C^\infty(M) \to \Omega^n(N) \\ &* f = f dV_g \end{aligned} \qquad (16.10) \tag{204}$$

smooth bundle isomorphism
EY : 20140501 smooth bundle isomorphism?
smooth bundle isomorphism

$$\begin{aligned} &\beta : \mathfrak{X}(M) \to \Omega^{n-1}(M) \\ &\beta(X) = X \lrcorner dV_g \end{aligned}$$

technical lemma

Lemma 31 (16.30). *(M, g) oriented Riemannian manifold with or without ∂*
Suppose $S \subseteq M$ immersed hypersurface with orientation by
unit normal vector field N and
$\widetilde{g}$ induced metric on S

If X any vector field along S,

$$i_S^*(\beta(X)) = \langle X, N\rangle_g dV_{\widetilde{g}} \qquad (16.12) \tag{205}$$

Proof. Define vector fields X^T, $X^\perp$ along S

$$\begin{aligned} X^\perp &= \langle X, N\rangle_g N \\ X^T &= X - X^\perp \end{aligned}$$

$$\beta(X) = X \lrcorner dV_g = X^\perp \lrcorner dV_g + X^T \lrcorner dV_g$$

pull back to S
Prop. 15.32

$$i_S^*(X^\perp \lrcorner dV_g) = \langle X, N\rangle_g i_S^*(N \lrcorner dV_g) = \langle X, N\rangle_g dV_{\widetilde{g}}$$

If $X_1 \dots X_{n-1}$ any vectors tangent to S

$$(X^T \lrcorner dV_g)(X_1 \dots X_{n-1}) = dV_g(X^T, X_1 \dots X_{n-1}) = 0$$

□

77 De Rham Cohomology

$d\omega = 0$ closed
$\omega = d\eta$ exact.

Prop. 6.24 smooth 1-form conservative iff exact.

The de Rham Cohomology Groups

closed 1-form

$$\omega = \frac{xdy - ydx}{x^2 + y^2} \qquad (15.1) \tag{206}$$

Suppose

$$\begin{aligned} x &= r\cos\theta & dx &= c_\theta dr + -rs_\theta d\theta \\ y &= rs_\theta & dy &= s_\theta dr + rc_\theta d\theta \end{aligned}$$

$$d\alpha = \omega = \frac{xdy - ydx}{x^2 + y^2} = \frac{x}{x^2 + y^2} - \frac{y}{x^2 + y^2}dx = \frac{1}{r}c_\theta(s_\theta dr + rc_\theta d\theta) - \frac{s_\theta}{r}(c_\theta dr - rs_\theta d\theta) = d\theta$$

M smooth manifold
$d : \mathcal{A}^p(M) \to \mathcal{A}^{p+1}(M)$ linear, ker, im linear subspaces.
$\mathcal{Z}^p(M) = \ker[d : \mathcal{A}^p(M) \to \mathcal{A}^{p+1}(M)] = \{$ closed p-forms on M $\}$
$\mathcal{B}^p(M) = \mathrm{im}[d : \mathcal{A}^{p-1}(M) \to \mathcal{A}^p(M)] = \{$ exact p-forms on M $\}$
By convention, $\mathcal{A}^p(M)$ zero vector space, $p < 0$ or $p > n = \dim M$
e.g.

$$\begin{aligned} \mathcal{B}^0(M) &= 0 \\ \mathcal{Z}^n(M) &= \mathcal{A}^n(M) \end{aligned}$$

$d^2 = 0$ so
$\forall$ exact form closed. $\mathcal{B}^p(M) \subset \mathcal{Z}^p(m)$

$$H^p_{dR}(M) = \frac{\mathcal{Z}^p(M)}{\mathcal{B}^p(M)}$$

Homotopy Invariance

Lemma 32 (15.4)**.** *(Existence of a Homotopy Operator)* $\forall$ *smooth manifold* M, $\exists$ *homotopy operator* $\rangle_0^*$, $\rangle_1^*$

Proof. $\forall p$, define linear $h : \mathcal{A}^p(M \times I) \to \mathcal{A}^{p-1}(M)$ s.t.

$$h(d\omega) + d(h\omega) = \rangle_1^*\omega - \rangle_0^*\omega \qquad (15.5) \tag{207}$$

define $h\omega = \int_0^1 \left(\frac{\partial}{\partial t} \lrcorner \omega\right) dt$
$h\omega$ $(p-1)$ form on M whose action on $X_1 \dots X_{p-1} \in T_qM$ is

$$(h\omega)_q(X_1 \dots X_{p-1}) = \int_0^1 \left(\frac{\partial}{\partial t} \lrcorner \omega(q,t)\right)(X_1 \dots X_{p-1})dt = \int_0^1 \omega_{(q,t)}\left(\frac{\partial}{\partial t}, X_1 \dots X_{p-1}\right) dt$$

choose smooth local coordinates (x^i) on M.

Consider separately $\omega = f(x,t)dt \wedge dx^{i_1} \wedge \dots \wedge dx^{i_{p-1}}$
$\omega = f(x,t)dx^{i_1} \wedge \dots \wedge dx^{i_p}$

□

78 Distributions and Foliations

Distributions and Involutivity

distribution on M of rank k is rank-k subbundle of TM, **smooth distribution** if it's smooth subbundle
Often rank-k distribution described by specifying $\forall\, p \in M$ linear subspace $D_p \subseteq T_pM$ of $\dim D_p = k$,
$D = \bigcup_{p\in M} D_p$

Lemma 10.32, local frame criterion for subbundles, that D smooth distribution iff $\forall\, p \in M$, $\exists$ open $U \ni p$ on which $\exists$ smooth vector fields $X_1 \dots X_k : U \to TM$ s.t. $X_1|_q \dots X_k|_q$ is basis for D_q $\forall\, q \in U$

Integral Manifolds and Involutivity

Suppose smooth distribution $D \subseteq TM$
integral manifold of D : immersed submanifold $N \neq \emptyset$, $N \subseteq M$ if $T_pN = D_p$ $\forall\, p \in N$

Example 19.1 (Distributions and Integral Manifolds)

(a)

(b)

(c)

(d)

D **involutive** if $\forall$ pair of smooth local sections of D (i.e. smooth vector fields X, Y defined on open subset of M s.t. $X_p, Y_p \in D_p \quad \forall p$)

integrable : smooth distribution D on M integrable if $\forall\, p \in M$, p in integral manifold of D, i.e.
$T_pM = D_p$

Proposition 78 (19.3)**.** $\forall$ *integrable distribution is involutive.*

Proof. Let $D \subseteq TM$ is integrable distribution.
suppose smooth local sections of D, X, Y on some open $U \subseteq M$.
$\forall\, p \in U$, let N integral manifold of D, $N \ni p$
X, Y sections of D, so X, Y tangent to N
By Corollary 8.32, $[X, Y]$ also tangent to N, so $[X, Y]_p \in D_p$ □

Involutivity and Differential Forms

Lemma 33 (19.5)**.** *(**1-form Criterion for Smooth Distributions**) Suppose smooth n-dim. manifold M, distribution $D \subseteq TM$, rank k*
D smooth iff $\forall\, p \in M$, $\exists$ neighborhood U on which $\exists$ smooth 1-forms $\omega^1 \dots \omega^{n-k}$ s.t. $\forall\, q \in U$,

$$D_q = ker\, \omega^1\big|_q \bigcap \cdots \bigcap ker\omega^{n-k}\big|_q \qquad (19.1) \tag{208}$$

Proof. By Prop. 10.15, complete forms $\omega^1 \dots \omega^{n-k}$ to smooth coframe $(\omega^1 \dots \omega^n)$ $\quad \forall\, p$
if $(E_1 \dots E_n)$ dual frame, easy to sheet that D locally spanned by $E_{n-k+1}, \dots, E_n$, so smooth by local frame criterion.

Converse, suppose D smooth.
$\forall$ open $U \ni p \in M$, $\exists$ smooth vector fields $Y_1 \dots Y_k$ spanning D.
By Prop. 16.5, complete $Y_1 \dots Y_k$ to smooth local frame $(Y_1 \dots Y_n)$ for M in open $U \ni p$
with dual coframe $(\epsilon^1 \dots \epsilon^n)$, it follows easily that D characterized locally by $D_q = \ker\, \epsilon^{k+1}\big|_q \bigcap \cdots \bigcap \ker\, \epsilon^n\big|_q$
□

if D rank-k distribution on smooth n-manifold M, any
$n-k$ linearly independent 1-forms $\omega^1 \dots \omega^{n-k}$ on open $U \subseteq M$ s.t. (19.1)

$$D_q = \ker\omega^1\big|_q \bigcap \cdots \bigcap \ker\omega^{n-k}\big|_q = \{X | X = X^i X_i,\, i = 1 \dots k,\, \omega^1(X) = 0\} \bigcap \cdots \bigcap \{X | \omega^{n-k}(X) = 0\}$$

$\forall\, q \in U$ are **local defining forms** for D

Proposition 79 (19.8)**.** ***(Local Coframe Criterion for Involutivity)*** *Let D smooth distribution of rank k on smooth n-manifold M*
let $\omega^1 \dots \omega^{n-k}$ smooth defining forms for D on open $U \subseteq M$.
The following are equivalent:

(a) D is involutive on U

(b) $d\omega^1 \dots d\omega^{n-k}$ annihilate D

(c) $\exists$ smooth 1-forms $\{\alpha^i_j | i, j = 1 \dots n-k\}$ s.t.

$$d\omega^i = \sum_{j=1}^{n-k} \omega^j \wedge \alpha^i_j \qquad \forall\, i = 1 \dots n-k$$

Exercise (Lee 2e, Exercise 19.9, p. 495 [76])**.** Prove the preceding proposition.

Note. The preceding proposition is Proposition 19.8 (Local Coframe Criterion for Involutivity).

Solution. Prove the preceding proposition, 19.8.

Proof. (a) $\implies$ (b)
On open $U \subseteq M$, $\forall\, q \in U$, ω^i smooth defining form for D, $i = 1 \dots k$, and $\omega^i(X) = 0 \quad \forall\, X \in D_q$
Then $d\omega^i$ also annihilates D on U (Thm. 19.7 1-form Criterion for Involutivity (19.3))
$d\omega^1 \dots d\omega^{n-k}$ annihilate D
(b) $\implies$ (c)
$d\omega^i \in \Omega^2_q(M)$, $\forall\, q \in U$
By Lemma 19.6, smooth p-form η on U annihilates D iff η ofform $\eta = \sum_{i=1}^{n-k} \omega^i \wedge \beta^i$, for $(p-1)$ forms $\beta^1 \dots \beta^{n-k}$ on U
$d\omega^i$ annihilates D
$\implies d\omega^i = \sum_{j=1}^{n-k} \omega^j \wedge \beta^i_j \qquad \beta^i_j$ smooth 1-forms on U, $i, j = 1 \dots n-k$
(c) $\implies$ (a)
Use Thm. 19.7 Proof

$$\omega^i([X,Y]) = X(\omega^i(Y)) - Y(\omega^i(X)) - d\omega^i(X,Y) = 0 - 0 - d\omega^i(X,Y)$$

$$d\omega^i(X,Y) - \sum_{j=1}^{n-k} \omega^j \wedge \alpha^i_j(X,Y) = \sum_{j=1}^{n-k} \omega^j(X)\alpha^i_j(Y) - \alpha^i_j\omega^j(Y) = 0 - 0 = 0$$

where I used this local formula:

$$(\alpha \wedge \beta)_p(v,w) = \alpha_p(v)\beta_p(w) - \alpha_p(w)\beta_p(v)$$

$\omega^i([X,Y]) = 0$ so $[X,Y] \in \ker\omega^i \qquad \forall\, i = 1 \dots n-k$

□

Problems

Exercise (Lee 2e, Problem 19-3, p. 512 [76])**.** Let ω be a smooth 1-form on a smooth manifold M. A smooth positive function μ on some open subset $U \subseteq M$ is called an *integrating factor* for ω if $\mu\omega$ is exact on U. Prove the following statements:

(a) If ω is nowhere-vanishing, then ω admits an integrating factor in a neighborhood of each point if and only if $d\omega \wedge \omega \equiv 0$.

(b) If $\dim M = 2$, then every nonvanishing smooth 1-form admits an integrating factor in a neighborhood of each point.

Solution. Let $\omega \in \Omega^1(M)$
integrating factor μ for $\omega \equiv \mu \in C^\infty(M)$, $\mu > 0$, and $\mu\omega$ exact on U, i.e. $\mu\omega = df$, for some $f \in \mathcal{C}^\infty(M)$

(a) If $\omega \neq 0$ on U,
Suppose ω admits an integrating factor μ.

$$d\omega \wedge \omega = d\left(\frac{df}{\mu}\right) \wedge \frac{df}{\mu} = \left(\frac{d^2 f}{\mu} + -\frac{df}{\mu^2}\frac{\partial \mu}{\partial x^i} dx^i \wedge df\right) \wedge \frac{df}{\mu} = 0$$

as $d^2 f = 0$ and $df \wedge df = 0$

If $d\omega \wedge \omega = 0$, consider $\mu \in \mathcal{C}^\infty(M)$ s.t. $\mu > 0$ (i.e. positive) on open $U \subseteq M$ (build it up with partitions of unity if need to).

Now, using the formula for exterior differentiation,

$$d(\mu\omega) = d\mu \wedge \omega + (-1)^0 \mu d\omega$$

so that

$$d(\mu\omega) \wedge \omega = d\mu \wedge \omega \wedge \omega + \mu d\omega \wedge \omega = 0 + \mu d\omega \wedge \omega = 0 + 0 = 0$$

ω nonzero, so $d(\mu\omega) = 0$. EY : 20150221 I'm not sure about this statement. Surely, locally,

$$d(\mu\omega) \wedge \omega = \frac{1}{2}(d(\mu\omega))_{ij}\omega_k dx^i \wedge dx^j \wedge dx^k = d(\mu\omega)_{\underline{I}}\omega_k dx^{\underline{I}} \wedge dx^k$$

with $\underline{I} = (i_1, i_2)$ and $i_1 < i_2$.

By considering every $k \neq \underline{I}$, then I think one can conclude, component by component, that $d(\mu\omega) = 0$.

Then, consider a compact submanifold B, $\dim B = 3$ that is a submersion of U. Then use Stoke's theorem in the following:

$$\int_B d(\mu\omega) = \int_{\partial B} \mu\omega = 0 \Longrightarrow \mu\omega = df$$

So

If $\omega \neq 0$ on U
ω admits an integrating factor μ iff $d\omega \wedge \omega = 0$

EY 20150221 : I didn't use Frobenius' theorem for the converse. Should I have?

(b) If $\dim M = 2$, $d\omega \wedge \omega = 0$ (immediately)
Then ω admits an integrating factor by the above solution.

79 The Exponential Map

79.1 One-Parameter Subgroups and the Exponential Map

79.1.1 One-Parameter Subgroups

Theorem 59 (20.1)**.** ***(Characterizations of One-Parameter Subgroups)***

Proposition 80 (20.8)**.** ***(Properties of Exponential Map)*** *Let* G *Lie group*
$\mathfrak{g}$ *Lie algebra*

(a) $\exp : \mathfrak{g} \to g$ *smooth*

(b) $\forall\, X \in \mathfrak{g}$, $s, t \in \mathbb{R}$, $\exp(s+t)X = \exp sX \exp tX$

(c) $\forall\, X \in \mathfrak{g}$, $(\exp X)^{-1} = \exp(-X)$

(d) $\forall\, X \in \mathfrak{g},\ n \in \mathbb{Z},\ (\exp X)^n = \exp(nX)$

(e) *differential* $(d\exp)_0 : T_0\mathfrak{g} \to T_eG$ *is identity, under canonical identifications of both* $T_0\mathfrak{g}$ *and* T_eG *with* $\mathfrak{g}$ *itself.*

(f) $\exp$ *restricts to diffeomorphism from some neighborhood of* 0 *in* $\mathfrak{g}$ *to neighborhood of* e *in* G.

(g) *if* $\Phi : G \to H$ *lie group homomorphism, following commutes*

$$\begin{array}{ccc} \mathfrak{g} & \xrightarrow{\Phi_*} & \mathfrak{h} \\ {\scriptstyle \exp}\downarrow & & \downarrow{\scriptstyle \exp} \\ G & \xrightarrow{\Phi} & H \end{array}$$

(h) *flow* θ *of left-invariant vector field* X, $\theta_t = R_{\exp tX}$

Proof. (a)

(b)

(c)

(d)

(e) Let $X \in \mathfrak{g}$ arbitrary, let $\begin{array}{l} \sigma : \mathbb{R} \to \mathfrak{g} \\ \sigma(t) = tX \\ \dot\sigma(0) = X \end{array}$

$$\begin{array}{ccc} \mathfrak{g} & \xrightarrow{\exp} & G \\ \downarrow & & \downarrow{\scriptstyle (d\exp)_0} \\ T_0\mathfrak{g} & \longrightarrow & T_eG \end{array}$$

$$(d\exp)_0(X) = (d\exp)_0(\dot\sigma(0)) = (\exp\circ\sigma)'(0) = (\exp(tX))'(0) = \left.\frac{d}{dt}\right|_{t=0} \exp(tX) = X$$

(f) $(d\exp)_0 = 1$ and so by inverse function thm., $\exists\, (d\exp)_0^{-1} = 1^{-1} = 1$, and so $\begin{array}{ccc} U & \ni & 0 \\ \rotatebox{90}{$\subseteq$} & & \rotatebox{90}{$\subseteq$} \\ \mathfrak{g} & & G \end{array} \overset{\simeq}{\to} \begin{array}{ccc} V & \ni & e \\ \rotatebox{90}{$\subseteq$} & & \rotatebox{90}{$\subseteq$} \\ G & & G \end{array}$

□

80 Quotient Manifolds

Problems

Exercise (Lee 2e, Problem 21-6, p. 560 [76])**.** Suppose a Lie group G acts smoothly, freely, and properly on a smooth manifold M. Show that M is the total space of a smooth fiber bundle with base M/G, model fiber G, and projection equal to the quotient map $\pi\colon M \to M/G$. [Hint: see the hint for Problem 21-5.] Conclude from Problem 10-19 that M is compact if and only if both G and M/G are compact. [Remark: any fiber bundle obtained in this way is called a *principal G-bundle*.]

Solution. Suppose Lie group G acts smoothly, freely, and properly on smooth manifold M

81 Symplectic Manifolds

81.1 Symplectic Tensors

Exercise (Lee 2e, Exercise 22.1, p. 565 [76])**.** Show that the following are equivalent for 2-covector ω on a finite-dimensional vector space V:

(a) ω is nondegenerate.

(b) For each nonzero $v \in V$, there exists $w \in V$ such that $\omega(v, w) \neq 0$.

(c) In terms of some (hence every) basis, the matrix (ω_{ij}) representing ω is nonsingular.

Solution. http://math.stackexchange.com/questions/342267/non-degenerate-bilinear-forms-and-invertible-matrices (shout out to Branimir Cacic for the answer) gave me a hint at how to approach this exercise, even though the original question was for symmetric bilinear forms.

Let $\{e_1 \dots e_n\}$ be (some) basis of V
Let $\{f^1 \dots f^n\}$ be dual basis of V s.t. $f^i(e_j) = \delta^i{}_j$

Now

$$\widehat{\omega}(e_i) = (\widehat{\omega}(e_i))_j f^j$$
$$\widehat{\omega}(e_i)e_j = (\widehat{\omega}(e_i))_j = i_{e_i}\omega(e_j) = \omega(e_i, e_j) \equiv \omega_{ij}$$

so $\omega_{ij} = (\widehat{\omega}(e_i))_j$ (i.e. matrix ω_{ij} is precisely $(\widehat{\omega}(e_i))_j$.

If ω_{ij} nonsingular, i.e. $\exists\, \omega_{ki}^{-1}$ s.t. $\omega_{ki}^{-1}\omega_{ij} = \omega_{ki}\omega_{kj}^{-1} = \delta_{kj}$ (by def.)

If $\widehat{\omega}$ invertible, $\widehat{\omega}^{-1}\widehat{\omega}(v) = v$

$$\widehat{\omega}^{-1}\widehat{\omega}(e_i) = \widehat{\omega}^{-1}(\widehat{\omega}(e_i))_j f^j = (\widehat{\omega}(e_i))_j \widehat{\omega}^{-1}(f^j) = (\widehat{\omega}(e_i))_j (\widehat{\omega}^{-1}(f^j))^k e_k = e_i$$
$$\implies (\widehat{\omega}(e_i))_j (\widehat{\omega}^{-1}(f^j))^k = \delta_i{}^k$$

So if ω_{ij} nonsingular, $(\widehat{\omega}^{-1}(f^j))^k$ exists and $(\widehat{\omega}(f^j))^k = \omega_{jk}^{-1}$

if $\widehat{\omega}$ invertible, ω_{jk}^{-1} exists and is given by $\omega_{jk}^{-1} = (\widehat{\omega}(f^j))^k$

So the (a) $\iff$(c) part of the exercise is done.

Show (a) $\iff$(b) and we're done.

If $\widehat{\omega} : V \to V^*$ linear isomorphism,

$$\ker\widehat{\omega} = 0$$

Suppose $\nexists\, w \in V$ s.t. $\omega(v, w) \neq 0$ (proof by contradiction strategy)
Then $\forall\, w \in V$, $\omega(v, w) = 0$
$\omega(v, w) = 0 = \widehat{\omega}(v)(w) \quad \forall\, w \in V$.
Then $v = 0$. Contradiction.

(b) $\implies$ (a): if $\forall\, v \neq 0$, $\exists\, w \in V$ s.t. $\omega(v, w) \neq 0$
then if $\forall\, w \in V$, $\omega(v, w) = 0$, then $v = 0$
$\omega(v, w) = \widehat{\omega}(v)(w) = 0$ implies $v = 0$, $\forall\, w \in V$.
Then $\ker\widehat{\omega} = 0$. So $\widehat{\omega}$ linear isomorphism. So ω nondegenerate.

Exercise (Lee 2e, Exercise 22.4, p. 566 [76])**.** Let (V, ω) be a symplectic vector space and $S \subseteq V$ be a linear subspace. Show that $(S^\perp)^\perp = S$.

Solution. There was a proof of this in Konstantin Athanassopoulos, **Notes on Symplectic Geometry**, Iraklion, 2013 http://www.math.uoc.gr/~athanako/symplectic.pdf

Recall that
$S^\perp = \{v \in V | \omega(v, w) = 0 \quad \forall\, w \in S\}$
$(S^\perp)^\perp = \{u \in V | \omega(u, v) = 0 \quad \forall\, v \in S^\perp\}$

Let $s \in S$. $\omega(s, v) = 0 \quad \forall\, v \in S^\perp$, by def. of $S^\perp$

$s \in (S^\perp)^\perp$
$\implies S \subseteq (S^\perp)^\perp$

Then $\dim S \leq \dim(S^\perp)^\perp$ with equality iff $S = (S^\perp)^\perp$

Now by Lemma 22.3,

$\dim S + \dim S^\perp = \dim V$, $\forall$ linear subspace $S \subseteq V$

$\dim S^\perp + \dim(S^\perp)^\perp = \dim V = \dim S + \dim S^\perp \Longrightarrow \dim S = \dim(S^\perp)^\perp$

$\dim S \leq \dim(S^\perp)^\perp$ with equality iff $S = (S^\perp)^\perp$

81.2 Symplectic Structures on Manifolds

Exercise (Lee 2e, Exercise 22.10, p. 568 [76])**.** Suppose (M, ω) is a symplectic manifold and $F\colon N \to M$ is a smooth immersion. Show that F is isotropic if and only if $F^*\omega = 0$, and F is symplectic if and only if $F^*\omega$ is a symplectic form.

Solution. $F : N \to M$ smooth immersion. Recall definition: F_* injective. Recall Appendix B, Exercise B.22 (EY: 20150512 This exercise is **very useful**; I can't emphasize that enough). F_* injective so $\text{rank}F_* = \dim N$. (implying $\ker F_* = 0$).

Recall that F isotropic if

$(F_*)_p(T_pN) \subseteq T_{F(p)}M$ isotropic, i.e.

$$(F_*)_p(T_pN) \subseteq ((F_*)_p(T_pN))^\perp$$

Consider $X, Y \in T_pN$, with X, Y nonzero. Then, as F_* injective, $Z, W \in ((F_*)_p(T_pN))$ nonzero, for

$Z = (F_*)_p X$

$W = (F_*)_p Y$

Suppose $\omega(Z, W) = 0$, $\forall\, W \in (F_*)_p(T_pN)$. $Z \in ((F_*)_p(T_pN))^\perp$.

$$\omega(Z, W) = \omega((F_*)_p X, (F_*)_p Y) = (F^*)_p \omega(X, Y) = 0$$

If F isotropic, then this is the case $\forall\, Z \in ((F_*)_p(T_pN)) \subseteq ((F_*)_p(T_pN))^\perp$.

Then since $(F^*)_p\omega(X, Y) = 0 \quad \forall\, p \in N$, $\forall\, X, Y \in T_pN$, then $F^*\omega = 0$.

If F symplectic, $(F_*)_p(T_pN) \cap ((F_*)_p(T_pN))^\perp = 0$

Likewise, for the reverse.

If F symplectic,

$$(F_*)_p(T_pN) \cap ((F_*)_p(T_pN))^\perp = 0$$

For $X, Y \in T_pN$, suppose

$$F_p^*\omega(X, Y) = \omega((F_*)_p X, (F_*)_p Y) = 0$$

This implies $(F_*)_p X \in (F_*)_p(T_pN) \cap ((F_*)_p(T_pN))^\perp$

Then

since F_* immersion, $X, Y = 0$. So $F^*\omega$ is nondegenerate and so is a symplectic form.

81.2.1 the Canonical Symplectic Form on the Cotangent Bundle

The most important symplectic manifolds are total spaces of cotangent bundles, which carry canonical symplectic structures that we now define.

81.3 The Darboux Theorem

81.4 Hamiltonian Vector Fields

Hamiltonian vector field of f

$$X_f = \widehat{\omega}^{-1}(df)$$

Hamiltonian vector field of f in Darboux coordinates:

$$X_f = \sum_{i=1}^{n} \left(\frac{\partial f}{\partial y^i} \frac{\partial}{\partial x^i} - \frac{\partial f}{\partial x^i} \frac{\partial}{\partial y^i} \right) \tag{209}$$

(22.9)

smooth $X \in \mathfrak{X}(M)$ **symplectic** if ω invariant under flow of X, i.e. $\mathcal{L}_X \omega = 0$

81.4.1 Poisson Brackets

$f \in C^\infty(M)$ **conserved quantity** if f constant on every integral curve of X_H.

smooth $V \in \mathfrak{X}(M)$ **infinitesimal symmetry** of (M, ω, H) if ω, H invariant under flow of V, i.e. EY (20150521)

$$\mathcal{L}_V \omega = 0 \quad \mathcal{L}_V H = 0$$

Proposition 81 (22.21)**.** *Let (M, ω, H) Hamiltonian system*

(a) $f \in C^\infty(M)$ conserved quantity iff $\left\{ \substack{\{f,H\} \\ =0} \right\}$

(b) infinitesimal symmetries of (M, ω, H) are precisely symplectic fields V s.t. $VH = 0$

(c) if θ flow of infinitesimal symmetry and γ trajectory of system

Proof. This is the solution to Problem 22-18.

(a) if $f \in C^\infty(M)$ conserved quantity, by def. f constant on every integral curve of X_H

$$\{f, H\} = \frac{\partial f}{\partial x^i} \frac{\partial H}{\partial y^i} - \frac{\partial f}{\partial y^i} \frac{\partial H}{\partial x^i} = X_H f = 0$$

for

$$X_H = \frac{\partial H}{\partial y^i} \frac{\partial}{\partial x^i} - \frac{\partial H}{\partial x^i} \frac{\partial}{\partial y^i}$$

likewise, if $\{f, H\} = 0$, then $X_H f = 0$, $X_H f = \mathcal{L}_{X_H} f = 0$, so f constant on flow of X_H

(b) Recall smooth $V \in \mathfrak{X}(M)$ infinitesimal symmetry of (M, ω, H) if ω, H invariant under flow of V, i.e.

$$\mathcal{L}_V \omega = 0 \quad \mathcal{L}_V H = 0$$

smooth $V \in \mathfrak{X}(M)$ symplectic if ω invariant under flow of V, i.e. $\mathcal{L}_V \omega = 0$

$$\mathcal{L}_V H = VH = 0$$

(c) EY : 20150521 I'm not sure how to go about this because what is a trajectory?

$\gamma : I \to M$

θ flow of an infinitesimal symmetry, so (collecting facts)

$$\mathcal{L}_{\dot\theta} \omega = d i_{\dot\theta} \omega + i_{\dot\theta} d\omega = d i_{\dot\theta} \omega \quad (\omega \text{ closed so } i_{\dot\theta} d\omega)$$
$$\dot\theta H = 0$$

Now $\theta_s \circ \gamma : I \to M$

$$\frac{d}{dt}(\theta_s \circ \gamma)(t) = (D\theta_s)(\gamma(t))\dot\gamma(t) = V_{s,\gamma(t)} \dot\gamma(t)$$

□

Exercise (Lee 2e, Problem 22-1, p. 590 [76])**.** Prove Proposition 22.5 (properties of symplectic, isotropic, coisotropic, and Lagrangian subspaces).

Note. Proposition 22.5: (a) S is symplectic iff $S^\perp$ is; (b) S is symplectic iff $\omega|_S$ is nondegenerate; (c) S is isotropic iff $\omega|_S = 0$; (d) S is coisotropic iff $S^\perp$ is isotropic; (e) S is Lagrangian iff $\omega|_S = 0$ and $\dim S = \frac{1}{2} \dim V$.

Solution.

Proof. (a) If S symplectic, $S \cap S^\perp = 0$. $S = (S^\perp)^\perp$ so $(S^\perp)^\perp \cap S^\perp = 0$. $S^\perp$ symplectic.
If $S^\perp$ symplectic, $S^\perp \cap (S^\perp)^\perp = 0$. $S = (S^\perp)^\perp$ so $(S^\perp)^\perp = S \cap S^\perp = 0$. S symplectic.

(b) Suppose for $s \in S \cap S^\perp$, $s \neq 0$. Then as $s \in S^\perp$, $\omega(s, w) = 0 \,\forall\, w \in S^\perp$.
Then $\omega(s, s) = 0$. But ω nondegenerate so $s = 0$. Contradiction.
Suppose S symplectic. For $\omega|_S(s, t) = \omega(s, t) = 0$, for some $s \in S$, $\forall\, t \in S$, then $S \cap S^\perp = 0$ implies that $s, t = 0$. Then $\omega|_S$ nondegenerate.

(c) If S isotropic, $S \subseteq S^\perp$ so that $\omega(s, t) = 0 \quad \forall\, t \in S$ (def. of $S^\perp$). $\omega|_S = 0$ as $\omega(s, t) = 0, \quad \forall\, s, t \in S$.
If $\omega|_S = 0$, then $\forall\, s, t \in S$, $\omega(s, t) = 0 \,\forall\, t \in S$. By def. of $S^\perp$, $S \subseteq S^\perp$.

(d) if S coisotropic, $S \supseteq S^\perp$. $S^\perp \subseteq S = (S^\perp)^\perp$. Then $S^\perp$ isotropic.
If $S^\perp$ isotropic, $S^\perp \subseteq (S^\perp)^\perp = S$, so S coisotropic.

(e) If S Lagrangian, $\forall\, s \in S$, $s \in S^\perp$, so that $\omega(s, t) = 0 \quad \forall\, t \in S$. Then $\omega|_S = 0$, (i.e. identically 0).

$\dim S + \dim S^\perp = 2\dim S = \dim V$ by Lemma 22.3, so $\dim S = \frac{1}{2}\dim V$.
If $\dim S = \frac{1}{2}\dim V$, $\dim S^\perp = \frac{1}{2}\dim V = \dim S$. $\omega|_S = 0$, so S isotropic, i.e. $S \subseteq S^\perp$. $\dim S \leq \dim S^\perp$, with equality iff $S = S^\perp$.

□

Exercise (Lee 2e, Problem 22-17, p. 593 [76])**.** Consider the 2-body problem, that is, the Hamiltonian system (T^*Q, ω, E) described in Example 22.18 in the special case $n = 2$. Suppose that the potential energy V depends only on the distance between the particles. More precisely, suppose that $V(q) = v(r(q))$ for some smooth function $v\colon (0, \infty) \to \mathbb{R}$, where

$$r(q) = |\mathbf{q}_1 - \mathbf{q}_2| = \sqrt{(q_1^1 - q_2^1)^2 + (q_1^2 - q_2^2)^2 + (q_1^3 - q_2^3)^2}.$$

(a) Let $\mathbf{u} = (u^1, u^2, u^3)$ be a unit vector in $\mathbb{R}^3$, and show that the function $P\colon T^*Q \to \mathbb{R}$ defined by

$$\begin{aligned} P(q, p) &= \mathbf{u} \cdot \mathbf{p}_1 + \mathbf{u} \cdot \mathbf{p}_2 \\ &= u^1 p_1^1 + u^2 p_1^2 + u^3 p_1^3 + u^1 p_2^1 + u^2 p_2^2 + u^3 p_2^3 \end{aligned}$$

is a conserved quantity (called the *total linear momentum in the* $\mathbf{u}$*-direction*), and that the corresponding infinitesimal symmetry generates translations in the $\mathbf{u}$-direction:

$$\begin{aligned} \theta_t(q, p) &= (\mathbf{q}_1 + t\mathbf{u}, \mathbf{q}_2 + t\mathbf{u}, \mathbf{p}_1, \mathbf{p}_2) \\ &= (q_1^1 + tu^1, q_1^2 + tu^2, q_1^3 + tu^3, q_2^1 + tu^1, q_2^2 + tu^2, \\ &\quad q_2^3 + tu^3, p_1^1, p_1^2, p_1^3, p_2^1, p_2^2, p_2^3). \end{aligned}$$

(b) Show that the function $L\colon T^*Q \to \mathbb{R}$ defined by

$$L(q, p) = q_1^1 p_1^2 - q_1^2 p_1^1 + q_2^1 p_2^2 - q_2^2 p_2^1$$

is a conserved quantity (called the *total angular momentum about the z-axis*), and find the flow of the corresponding infinitesimal symmetry. Explain what this has to do with rotational symmetry.

Solution. Given Hamiltonian system (T^*Q, ω, E).
Recall that

$$\begin{aligned} q(t) &= (q_1^1(t), q_1^2(t), q_1^3(t) \dots q_n^1(t), q_n^2(t), q_n^3(t)) = 0 \\ &= (q^1(t) \dots q^{3n}(t)) \end{aligned}$$

Now $p(t) = (p_1^1, p_1^2, p_1^3 \dots p_n^1, p_n^2, p_n^3)$ and
$p_i(t) = M_{ij}\dot{q}^j(t)$ with
M_{ij} $3n \times 3n$ diagonal matrix $(m_1, m_1, m_1, m_2, m_2, m_2 \dots m_n, m_n, m_n)$
Now $E \in C^\infty(T^*Q)$ where

$$E(q, p) = V(q) + K(p) = V(q) + \frac{1}{2} M^{ij} p_i p_j$$

(a) Let $\mathbf{u} = (u^1, u^2, u^3)$

$$\begin{aligned} &P : T^*Q \to \mathbb{R} \\ &P(q,p) = \mathbf{u}\cdot\mathbf{p}_1 + \mathbf{u}\cdot\mathbf{p}_2 \\ &\qquad = u^1p_1^1 + u^2p_1^2 + u^3p_1^3 + u^1p_2^1 + u^2p_2^2 + u^3p_2^3 \end{aligned}$$

Recall Prop. 22.21, Let (M,ω,H) Hamiltonian system

(a) $f \in C^\infty(M)$ conserved quantity iff $\{f,H\} = 0$

Now in Darboux coordinates,

$$\{f,g\} = \sum_{i=1}^{n} \frac{\partial f}{\partial x^i}\frac{\partial g}{\partial y^i} - \frac{\partial f}{\partial y^i}\frac{\partial g}{\partial x^i}$$

(22.16)

For

$$E = V(|\mathbf{q}_2 - \mathbf{q}_1|) + \frac{1}{2}M^{ij}p_ip_j$$

note that

$$\begin{aligned} &V(|\mathbf{q}_2 - \mathbf{q}_1|) = V(r) \text{ with} \\ &r = \sqrt{(q_2^1 - q_1^1)^2 + \cdots + (q_2^3 - q_1^3)^2} \\ &\frac{\partial V}{\partial q_i^j} = \frac{\partial V}{\partial r}\frac{1}{2}\frac{1}{r}(2)(q_2^j - q_1^j)(-1)^i = \frac{\partial V}{\partial r}\frac{1}{r}(q_2^j - q_1^j)(-1)^i \end{aligned}$$

then

$$\begin{aligned} \frac{\partial E}{\partial p_i^j} &= \frac{p_i^j}{m_i} \\ \frac{\partial E}{\partial q_i^j} &= \frac{\partial V}{\partial r}\frac{1}{r}(-1)^i(q_2^j - q_1^j) \end{aligned}$$

with $r = |\mathbf{q}_2 - \mathbf{q}_1| = \sqrt{(q_2^1 - q_1^1)^2 + \cdots + (q_2^3 - q_1^3)^2}$

$$\begin{aligned} P = \mathbf{u}\cdot(\mathbf{p}_1 + \mathbf{p}_2) = u^ip_1^i + u^ip_2^i \\ \frac{\partial P}{\partial p_i^j} = u^j \\ \frac{\partial P}{\partial q} = 0 \end{aligned}$$

$$\{P,E\} = 0 - u^j\frac{\partial V}{\partial r}\frac{1}{r}(-1)^i(q_2^j - q_1^j) = -\mathbf{u}\cdot(\mathbf{q}_2 - \mathbf{q}_1)\frac{\partial V}{\partial r}\frac{1}{r}(-1) + -\mathbf{u}\cdot(\mathbf{q}_2 - \mathbf{q}_1)\frac{\partial V}{\partial r}\frac{1}{r} = 0$$

(b)

$$\begin{aligned} L(q,p) = q_1^1p_1^2 - q_1^2p_1^1 + q_2^1p_2^2 - q_2^2p_2^1 \\ \frac{\partial L}{\partial q_i^j} = p_i^k\epsilon^{jk} \\ \frac{\partial L}{\partial p_i^k} = q_i^j\epsilon^{jk} \end{aligned}$$

$$\{L,E\} = p_i^k\epsilon^{jk}\frac{p_i^j}{m_i} - q_i^k\epsilon^{kj}\frac{\partial V}{\partial r}\frac{1}{r}(-1)^i(q_2^j - q_1^j) = \frac{p_i^2p_i^1}{m_i} - \frac{p_i^1p_i^2}{m_i} - q_i^k\epsilon^{kj}\frac{\partial V}{\partial r}\frac{1}{r}(-1)^i(q_2^j - q_1^j) =$$
$$= 0 - \frac{\partial V}{\partial r}\frac{1}{r}(-q_1^2(q_2^1 - q_1^1)(-1) + q_1^1(q_2^2 - q_1^2)(-1) - q_2^2(q_2^1 - q_1^1) + q_2^1(q_2^2 - q_1^2)) = 0$$

Exercise (Lee 2e, Problem 22-18, p. 593 [76])**.** Prove Proposition 22.21 (properties of conserved quantities and infinitesimal symmetries).

Solution.

(a) if $f \in C^\infty(M)$ conserved quantity, by def. f constant on every integral curve of X_H

$$\{f, H\} = \frac{\partial f}{\partial x^i}\frac{\partial H}{\partial y^i} - \frac{\partial f}{\partial y^i}\frac{\partial H}{\partial x^i} = X_H f = 0$$

for

$$X_H = \frac{\partial H}{\partial y^i}\frac{\partial}{\partial x^i} - \frac{\partial H}{\partial x^i}\frac{\partial}{\partial y^i}$$

likewise, if $\{f, H\} = 0$, then $X_H f = 0$, $X_H f = \mathcal{L}_{X_H} f = 0$, so f constant on flow of X_H

(b) Recall smooth $V \in \mathfrak{X}(M)$ infinitesimal symmetry of (M, ω, H) if ω, H invariant under flow of V, i.e.

$$\mathcal{L}_V \omega = 0 \quad \mathcal{L}_V H = 0$$

smooth $V \in \mathfrak{X}(M)$ symplectic if ω invariant under flow of V, i.e. $\mathcal{L}_V \omega = 0$

$$\mathcal{L}_V H = VH = 0$$

(c) EY : 20150521 I'm not sure how to go about this because what is a trajectory?
$\gamma : I \to M$
θ flow of an infinitesimal symmetry, so (collecting facts)

$$\mathcal{L}_{\dot\theta}\omega = di_{\dot\theta}\omega + i_{\dot\theta}d\omega = di_{\dot\theta}\omega \quad (\omega \text{ closed so } i_{\dot\theta}d\omega)$$
$$\dot\theta H = 0$$

Now $\theta_s \circ \gamma : I \to M$

$$\frac{d}{dt}(\theta_s \circ \gamma)(t) = (D\theta_s)(\gamma(t))\dot\gamma(t) = V_{s,\gamma(t)}\dot\gamma(t)$$

82 Appendix A: Review of Topology

Topological Spaces

- **neighborhood of** p open subset $\mathcal{O}$ containing p

Bases and Countability

Suppose X topological space.
basis for topology of X

$$\mathcal{B} = \{B| \text{ open } B \subseteq X, \forall \text{ open } \mathcal{O} \subset X, \mathcal{O} = \bigcup_\alpha B_\alpha\}$$

neighborhood basis at p $\mathcal{B}_p = \{\text{neighborhoods } B_p \text{ of } p \,|\forall\, \mathcal{O} \text{ of } p, B_p \subset \mathcal{O}, B_p \in \mathcal{B}_p \text{ for at least 1 }\}$
X **first countable** - if $\exists$ countable neighborhood basis $\forall\, p$
second countable if $\exists$ countable basis

Proposition 82 (A.16)**.** *Let X second countable topological space.*
$\forall$ *open cover of X has countable subcover*

Proof. X second countable. $\exists$ countable basis $\mathcal{B}$ for X
Let $\mathcal{U}$ arbitrary open cover of X.
Let $\mathcal{B}' \subseteq \mathcal{B}$ s.t. $\mathcal{B}' = \{B|B \in \mathcal{B}, B \subseteq U \text{ for some } U \in \mathcal{U}\}$

$\forall\, B \in \mathcal{B}'$, choose particular $U_B \in \mathcal{U}$ containing B

$\{U_B | B \in \mathcal{B}'\}$ countable

$\forall\, x \in X$, $\exists\, V \in \mathcal{U}$, $x \in V$
$\mathcal{B}$ basis, s.t. $\exists\, B \in \mathcal{B}$ s.t. $x \in B \subseteq V$
then $B \in \mathcal{B}'$, so $x \in B \subseteq U_B$ □

Subspaces, Products, Disjoint Unions, and Quotients

Review of Topology

Topological Spaces

Let X set
topology on X = collection $\tau = \{U | U \subseteq X\}$, U called open subsets s.t.

(i) $X, \emptyset$ open

(ii) $\bigcup_\alpha U_\alpha$ is open, $\forall\, U_\alpha$ open subset

(iii) $\bigcap_{i=1}^n U_i$ is open

(X, τ) topological space
Example A.5. (Metric Spaces) metric space = set M with metric $d : M \times M \to \mathbb{R}$ s.t. $\forall\, x, y, z \in M$

(i) Positivity $d(x, y) \geq 0$, $d(x, y) = 0$ iff $x = y$

(ii) Symmetry $d(x, y) = d(y, x)$

(iii) Triangle inequality $d(x, z) \leq d(x, y) + d(y, z)$

if M metric space, $x \in M$, $r > 0$
open ball of radius r **around** x

$$B_r(x) = \{y \in M | d(x, y) < r\}$$

closed ball of radius r

$$\underline{B}_r(x) = \{y \in M | d(x, y) \leq r\}$$

metric topology on M defined by declaring $S \subseteq M$ open iff $\forall\, x \in S$, $\exists\, r > 0$ s.t. $B_r(x) \subseteq S$

83 Appendix B: Review of Linear Algebra

83.1 Linear Maps

Exercise (Lee 2e, Exercise B.1, p. 619 [76])**.** Let V be a vector space. Prove the following statements.

(a) If $S \subseteq V$ is linearly independent, then every subset of S is linearly independent.

(b) If $S \subseteq V$ is linearly dependent or spans V, then every subset of V that properly contains S is linearly dependent.

(c) A subset $S \subseteq V$ containing more than one element is linearly dependent if and only if some element $v \in S$ can be expressed as a linear combination of elements of $S \setminus \{v\}$.

(d) If $(v_1, \ldots, v_k)$ is a linearly dependent k-tuple in V with $v_1 \neq 0$, then some v_i can be expressed as a linear combination of the *preceding* vectors $(v_1, \ldots, v_{i-1})$.

Solution.

(a)

(b)

(c)

(d) **Want**: if $(v_1 \dots v_k)$ linearly dependent k-tuple in V, $v_1 \neq 0$,
then some $v_i = \sum_{j=1}^{i-1} c^j v_j$

Proof. $(v_1 \dots v_k)$ linearly dependent, so if $\sum_{i=1}^{k} a^i v_i = 0$, a^i not all equal to 0.

Suppose for fixed i, $2 \leq i \leq k$, $a^{i+1} = \dots = a^k = 0$.
Indeed, suppose for $\sum_{i=1}^{k} a^i v_i = 0$, $a^2 = \dots = a^k = 0$. $a^1 v_1 = 0$, $v_1 \neq 0$, $a^1 = 0$. Then $(v_1 \dots v_k)$ linearly independent. Contradiction.

if $i = 2$,
$a^1 v_1 + a^2 v_2 = 0$
$\implies v_2 = \frac{-a^1}{a^2} v_1$

if $i = k$, $v_k = \frac{-\sum_{i=1}^{k-1} a^i v_i}{a^k}$

So in general, $v_i = \frac{-\sum_{j=1}^{i-1} a^j v_j}{a^i}$ □

Exercise (Lee 2e, Exercise B.9, p. 621 [76])**.** Let V be a finite-dimensional vector space. Show that every subspace $S \subseteq V$ has a complementary subspace in V. In fact, given an arbitrary basis $(E_1, \dots, E_n)$ for V, show that there is some subset $\{i_1, \dots, i_k\}$ of the integers $\{1, \dots, n\}$ such that $\text{span}(E_{i_1}, \dots, E_{i_k})$ is a complement to S. [Hint: choose a basis $(F_1, \dots, F_m)$ for S, and apply Exercise B.1(d) to the ordered $(m+n)$-tuple $(F_1, \dots, F_m, E_1, \dots, E_n)$.]

Solution. given $(E_1 \dots E_n)$ basis for V
$\exists \{i_1 \dots i_k\} \subset \{1 \dots n\}$ s.t.
$\text{span}(E_{i_1} \dots E_{i_k})$ is complement to S

Hence $\forall$ subspace $S \subseteq V$, $\exists$ complementary subspace T in V, so $V = S \oplus T$

Proof. $\forall$ subspace S is itself a vector space, closed under addition and multiplication.
Hence S has basis $(F_1 \dots F_m)$ with $\dim S = m$

Consider ordered $(m+n)$-tuple

$$(F_1 \dots F_m, E_1 \dots E_n)$$

$(F_1 \dots F_m, E_1 \dots E_n)$ linearly dependent in V, by linear algebra.
For $j_1 \in \{1 \dots n\}$, E_{j_1} linear combination of previous vectors (cf. Exercise B.1(d))
eliminate E_{j_1} : $(F_1 \dots F_m, E_1 \dots \widehat{E}_{j_1} \dots E_n)$.

Repeat, until there are $n-m$ E basis vectors left, labeled $i_1 \dots i_{n-m}$ (hence **use Exercise B.1(d)** many and enough times, m times)

$$\implies (F_1 \dots F_m, E_{i_1} \dots E_{i_{n-m}})$$

By linear algebra, $(F_1 \dots F_m, E_{i_1} \dots E_{i_{n-m}})$ a basis for V, linearly independent. The procedure wouldn't have "overshot" by a Thm. (see Apostol's **Calculus** Vol. 2, first few chapters, linear algebra part)

$\forall v \in V$, $v = a^i F_i + b^{i_j} E_{i_j}$ with $a^i F_i \in S$. Then $b^{i_j} E_{i_j} \in T$.
since $V = S \oplus T$, T complement to S

$\implies$ given fixed basis of V, $(E_1 \dots E_n)$, subspace $S \subseteq V$, S having basis $(F_1 \dots F_m)$, S has complementary subspace in V, T,

s.t. basis of T is $(E_{i_1} \dots E_{i_{n-m}})$ and $V = S \oplus T$

□

Exercise (Lee 2e, Exercise B.13, p. 623 [76])**.** Let V and W be vector spaces, and let $(E_1, \dots, E_n)$ be a basis for V. For any n elements $w_1, \dots, w_n \in W$, show that there is a unique linear map $T\colon V \to W$ satisfying $T(E_i) = w_i$ for $i = 1, \dots, n$.

Solution. Suppose $\exists$ linear $T : V \to W$ s.t. $T(E_i) = w_i, \quad i = 1 \dots n$
Suppose $\exists$ linear $T' : V \to W$ s.t. $T'(E_i) = w_i, \quad i = 1 \dots n$

Let $x \in V$, so $x = x^i E_i$ (the key idea is that with a basis, the vector space is completely determined, vectors in the vector space are spanned by the basis elements)

$$(T - T')(x) \equiv T(x) - T'(x) = x^i w_i - x^i w_i = 0$$

$T(x) = T'(x) \quad \forall x \in V$
so $T = T'$. T unique.

T exists by construction. ———————————————————————————————

affine subspace of V parallel to S, linear subspace $S \subseteq V$, $v + S = \{v + w | w \in S\}$, some fixed $v \in V$

affine map $F : V \to W$ if $F(v) = w + Tv$ for some $T : V \to W$, some fixed $w \in W$

Exercise (Lee 2e, Exercise B.16, p. 623 [76])**.** Suppose $F\colon V \to W$ is an affine map. Show that $F(V)$ is an affine subspace of W, and the sets $F^{-1}(z)$ for $z \in W$ are parallel affine subspaces of V.

Solution. Let $a, b \in \mathbb{C}$, $x, y \in F(V)$

Now
$$F(V) = \{y | y = w + Tv = F(v),\ v \in V,\ \text{fixed } w \in W,\ \text{some } T\}$$

83.1.1 Change of Basis

Exercise (Lee 2e, Exercise B.22, p. 627 [76])**.** Suppose V, W, X are finite-dimensional vector spaces, and $S\colon V \to W$ and $T\colon W \to X$ are linear maps. Prove the following statements.

(a) $\operatorname{rank} S \leq \dim V$, with equality if and only if S is injective.

(b) $\operatorname{rank} S \leq \dim W$, with equality if and only if S is surjective.

(c) If $\dim V = \dim W$ and S is either injective or surjective, then it is an isomorphism.

(d) $\operatorname{rank}(T \circ S) \leq \operatorname{rank} S$, with equality if and only if $\operatorname{Im} S \cap \operatorname{Ker} T = \{0\}$.

(e) $\operatorname{rank}(T \circ S) \leq \operatorname{rank} T$, with equality if and only if $\operatorname{Im} S + \operatorname{Ker} T = W$.

(f) If S is an isomorphism, then $\operatorname{rank}(T \circ S) = \operatorname{rank} T$.

(g) If T is an isomorphism, then $\operatorname{rank}(T \circ S) = \operatorname{rank} S$.

Solution. Suppose V, W, X finite-dim. vector spaces
$S : V \to W, \quad T : W \to X$

(a) $\operatorname{rank} S \leq \dim V$ with $\operatorname{rank} S = \dim V$ iff S injective

(b) $\operatorname{rank} S \leq \dim W$ with $\operatorname{rank} S = \dim W$ iff S surjective

(c) if $\dim V = \dim W$ and S either injective or surjective, then S isomorphism

(d) $\operatorname{rank} TS \leq \operatorname{rank} S$ $\operatorname{rank} TS = \operatorname{rank} S$ iff $\operatorname{im} S \bigcap \ker T = 0$

(e) $\text{rank}TS \leq \text{rank}T$ $\quad$ $\text{rank}TS = \text{rank}T$ iff $\text{im}S + \ker T = W$

(f) if S isomorphism, then $\text{rank}TS = \text{rank}T$

(g) if T isomorphism, then $\text{rank}TS = \text{rank}S$

EY : Exercise B.22(d) is useful for showing the chart and atlas of a Grassmannian manifold, found in the More examples, for smooth manifolds.

Proof. (a) Recall the **rank-nullity theorem**:

Theorem 60 (Rank-Nullity Theorem)**.**

$$dim(im(S)) + dim(ker(S)) = dimV \tag{210}$$

Now

$$\text{rank}(S) + \dim(\ker(S)) \equiv \dim(\text{im}(S)) + \dim(\ker(S)) = \dim V$$
$$\implies \boxed{\text{rank}(S) \leq \dim V}$$

If $\text{rank}(S) = \dim V$,
then by rank-nullity theorem, $\dim(\ker(S)) = 0$, implying that $\ker S = \{0\}$.
Suppose $v_1, v_2 \in V$ and that $S(v_1) = S(v_2)$. By linearity of S, $S(v_1) - S(v_2) = S(v_1 - v_2) = 0$, which implies, since $\ker S = \{0\}$, that $v_1 - v_2 = 0$.
$\implies v_1 = v_2$. Then by definition of injectivity, S injective.

If S injective, then $S(v) = 0$ implies $v = 0$. Then $\ker S = \{0\}$. Then by rank-nullity theorem, $\text{rank}(S) = \dim V$.

(b) $\forall\, w \in \text{im}(S)$, $w \in W$. Clearly $\text{rank}S \leq \dim W$.

If S surjective, $\text{im}(S) = W$. Then $\dim(\text{im}(S)) = \text{rank}S = \dim W$.

If $\text{rank}S = \dim W = m$, then $\text{im}(S)$ has basis $\{y_i\}_{i=1}^m$, $y_i \in \text{im}(S)$, so $\exists\, x_i \in V$, $i = 1 \dots m$ s.t. $S(x_i) = y_i$, with $\{S(x_i)\}_{i=1}^m$ linearly independent.

Since $\{S(x_i)\}_{i=1}^m$ linearly independent and $\dim W = m$, $\{S(x_i)\}_{i=1}^m$ basis for W.
$\forall\, w \in W$, $w = \sum_{i=1}^m w^i S(x_i) = S(\sum_{i=1}^m w^i x_i)$. $\sum_{i=1}^m w^i x_i \in V$. S surjective.

(c)

(d) Now

$$\dim V = \text{rank}TS + \text{nullity}TS$$
$$\dim V = \text{rank}S + \text{nullity}S$$

$\ker S \subseteq \ker TS$, clearly, so $\text{nullity}S \leq \text{nullity}TS$

$$\implies \boxed{\text{rank}TS \leq \text{rank}S}$$

If $\text{rank}TS = \text{rank}S$,
then $\text{nullity}S = \text{nullity}TS$
Suppose $w \in \text{Im}S \bigcap \ker T$, $w \neq 0$
Then $\exists\, v \in S$, s.t. $w = S(v)$ and $T(w) = 0$
Then $T(w) = TS(v) = 0$. So $v \in \ker TS$
$v \notin \ker S$ since $w = S(v) \neq 0$
This implies $\text{nullity}TS > \text{nullity}S$. Contradiction.
$\implies \text{Im}S \bigcap \ker T = 0$

If $\text{Im}S \bigcap \ker T = 0$,
Consider $v \in \ker TS$. Then $TS(v) = 0$.
. Then $S(v) \in \ker T$

$S(v) = 0$; otherwise, $S(v) \in \text{Im}S$, contradicting given $\text{Im}S \bigcap \ker T = 0$
$v \in \ker S$

$\ker TS \subseteq \ker S$
$\implies \ker TS = \ker S$
So $\text{nullity}TS = \text{nullity}S$
$\implies \text{rank}TS = \text{rank}S$

(e)

(f)

(g)

□

Inner Products and Norms

Norms

If V real vector space,
norm on V, $v \mapsto |v| \in \mathbb{R}$ s.t.

(i) Positivity $|v| \geq 0, \quad \forall\, v \in V,\ |v| = 0$ iff $v = 0$

(ii) Homogeneity $|cv| = |c||v| \qquad \forall\, c \in \mathbb{R},\ v \in V$

(iii) Triangle Inequality $|v + w| \leq |v| + |w|, \quad \forall\, v, w \in V$

2 norms $|\cdot|_1, |\cdot|_2$ on vector space V equivalent if $\exists$ constants $c, C > 0$ s.t.

$$c|v|_1 \leq |v|_2 \leq C|v|_1 \qquad \forall\, v \in V$$

Exercise (Lee 2e, Exercise B.49, p. 638 [76])**.** Show that equivalent norms determine the same topology.

Solution. $\forall\, x \in V$,
Consider $B_r(x) = \{y \in V | |y - x|_2 < r\}$
Note $y - x \in V$ as V is a vector space

$$c|y - x|_1 \leq |y - x|_2 \leq C|y - x|_1$$
$$c|y - x|_1 \leq |y - x_2| < r$$
$$|y - x|_1 < \frac{r}{c}$$

Now suppose $S \subseteq V$ open in $|\cdot|_2$
But S also open in $|\cdot|_1$ as $\exists\, \frac{r}{c'} > 0$ s.t. $B_{\frac{r}{c'}}(x) \subseteq S$ for the exact same pts. as S
so $|\cdot|_1, |\cdot|_2$ equivalent norms yield the same metric topology.
EY : 20141220

Direct Products and Direct Sums

84 Appendix C: Review of Calculus

Total and Partial Derivatives

Proposition 83 (C.3)**.** *(The Chain Rule for Total Derivatives)*
Suppose V, W, X finite-dim. vector spaces

open $U \subseteq V$
$\widetilde{U} \subseteq W$

maps $F : U \to \widetilde{U}$

$G : \widetilde{U} \to X$

if F *diff. at* $a \in U$, G *diff. at* $F(a) \in \widetilde{U}$,
then GF *diff. at* a

$$D(GF)(a) = DG(F(a)) \circ DF(a)$$

Proposition 84 (C.4)**.** *Suppose* $U \subseteq \mathbb{R}^n$, $F : U \to V$ *diffeomorphism* $V \subseteq \mathbb{R}^m$

Then $m = n$ *and*
$\forall\, a \in U$, $DF(a)$ *invertible, with*

$$DF(a)^{-1} = D(F^{-1})(F(a))$$

Proof. $F^{-1}F = 1_U$
Chain rule implies $\forall\, a \in U$,

$$1_{\mathbb{R}^n} = D(1_U)(a) = D(F^{-1}F)(a) = DF^{-1}(F(a))DF(a) \qquad (C.5) \tag{211}$$

Similarly $FF^{-1} = 1_V$ implies

$$DF(a)DF^{-1}(F(a)) = 1_{\mathbb{R}^m}$$

so thus

$$DF(a)^{-1} = D(F^{-1})(F(a))$$

□

Part XIX
Lie Groups, Lie Algebra

85 Lie Groups

Lie Groups

Groups

Ring

group algebra

Group Ring

Representation Theory

Modules

kG-modules

From Sec. 8.1 "Noncommutative Rings" of Rotman (2010) [83]:

Definition 168. ring R - additive abelian group equipped with multiplication $R \times R \to R$ s.t. $\forall\, a, b \in R$ $(a, b) \mapsto ab$

(i) $a(bc) = (ab)c$

(ii) $a(b + c) = ab + ac$, $(b + c)a = ba + ca$

(iii) $\exists 1 \in R$ s.t. $\forall a \in R$, $1a = a = a1$

Example 8.1[83]

(ii) group algebra kG, k commutative ring, G group, "its additive abelian group is free k-module having basis labeled by elements of G,
i.e. $\forall a \in kG$, $a = \sum_{g \in G} a_g g$, $a_g \in k$, $\forall g \in G$, $a_g \neq 0$ for only finitely many $g \in G$.

define (ring) multiplication $kG \times kG \to kG$ $\forall a, b \in kG$, $\begin{matrix} a = \sum_{g \in G} a_g g \\ b = \sum_{h \in G} b_h g \end{matrix}$ to be
$ab = ab$

$$\left(\sum_{g \in G} a_g g \right) \left(\sum_{h \in G} b_h h \right) = \sum_{z \in G} \left(\sum_{gh = z} a_g b_h \right) z$$

Definition 169. Given R ring, left R-module is (additive) abelian group M equipped with

scalar multiplication $R \times M \to M$ s.t. $\forall m, m' \in M$, $\forall r, r', 1 \in R$
$(r, m) \mapsto rm$

(i) $r(m + m') = rm + rm'$

(ii) $(r + r')m = rm + r'm$

(iii) $(rr')m = r(r'm)$

(iv) $1m = m$

EY : 20150922 Example : for kG-module V^σ, for $r \in kG$, so $r = \sum_{g \in G} a_g g$

$$\begin{matrix} R \times M \to M \\ (r, m) \mapsto rm \end{matrix} \implies \begin{matrix} kG \times V \to V \\ (r, v) \mapsto tv \end{matrix}$$

For some representation $\sigma : G \to GL(V)$,

$$rv = \sum_{g \in G} a_g g \cdot v = \sum_{g \in G} a_g \sigma_g(v)$$

So a kG-module needs to be associated with some chosen representation.
Note for V as an additive abelian group, $\forall u, v, w \in V$,

$$\begin{aligned} &v + w = w + v,\ (u + v) + w = u + (v + w) \\ &v + 0 = v \quad \forall v \in V \text{ for } 0 \in V \\ &v + (-v) = 0 \quad \forall v \in V \end{aligned}$$

So a vector space can be an additive abelian group.
Note that

$$r(v + w) = \left(\sum_{g \in G} a_g g \right)(v + w) = \left(\sum_{g \in G} a_g \sigma_g \right)(v + w) = \sum_{g \in G} a_g \sigma_g(v) + \sum_{g \in G} a_g \sigma_g(w) = rv + rw$$

$$(r + r')v = \left(\sum_{g \in G} a_g g + b_g g \right) v = \sum_{g \in G} (a_g \sigma_g + b_g \sigma_g) v = \sum_{g \in G} a_g \sigma_g(v) + \sum_{g \in G} b_g \sigma_g(v) = rv + r'v$$

$$(rr')v = \left(\sum_{g\in G} a_g g \sum_{h\in G} b_h h\right) v = \left(\sum_{z\in G}\sum_{gh=z} a_g b_h z\right) v = \sum_{z\in G}\sum_{gh\in z} a_g b_h \sigma_z(v) = \sum_{g\in G}\sum_{h\in G} a_g b_h \sigma_g\sigma_h(v)$$

since $\sigma(gh) = \sigma(g)\sigma(h) = \sigma_g\sigma_h = \sigma_{gh}$ (σ homomorphism)

$1v = \sigma(1)v = 1v = v$

From Sec. 8.3 "Semisimple Ring" of Rotman (2010) [83]:

Definition 170. k-representation of group G is homomorphism

$$\sigma : G \to GL(V)$$

where V is vector field over field k

Proposition 85 (8.37 Rotman (2010)[83])**.** *$\forall$ k-representation $\sigma : G \to GL(V)$ equips V with structure of left kG-module, denote module by V^σ.*
Conversely, $\forall$ left kG-module V determines k-representation $\sigma : G \to GL(V)$

Proof. Given $\sigma : G \to GL(V)$,

$\sigma_g =: \sigma(g) : V \to V$

define

$$kG \times V \to V$$

$$\left(\sum_{g\in G} a_g g\right) v = \sum_{g\in G} a_g \sigma_g(v)$$

$v, w \in V$

Let $r, r', 1 \in kG$

$r = \sum_{g\in G} a_g g$

$$r(v+w) = \left(\sum_{g\in G} a_g g\right)(v+w) = \left(\sum_{g\in G} a_g \sigma_g\right)(v+w) = \sum_{g\in G} a_g\sigma_g(v) + \sum_{g\in G} a_g\sigma_g(w) = rv + rw$$

$$(r+r')v = \left(\sum_{g\in G} a_g g + b_g g\right) v = \sum_{g\in G}(a_g\sigma_g + b_g\sigma_g)v = \sum_{g\in G} a_g\sigma_g(v) + \sum_{g\in G} b_g\sigma_g(v) = rv + r'v$$

$$(rr')v = \left(\sum_{g\in G} a_g g \sum_{h\in G} b_h h\right) v = \left(\sum_{z\in G}\sum_{gh=z} a_g b_h z\right) v = \sum_{z\in G}\sum_{gh\in z} a_g b_h \sigma_z(v) = \sum_{g\in G}\sum_{h\in G} a_g b_h \sigma_g\sigma_h(v)$$

since $\sigma(gh) = \sigma(g)\sigma(h) = \sigma_g\sigma_h = \sigma_{gh}$ (σ homomorphism)

$1v = \sigma(1)v = 1v = v$

Conversely, assume V left kG-module.

If $g \in G$, then $v \mapsto gv$ defines $T_g : V \to V$. T_g nonsingular since $\exists T_g^{-1} = T_{g^{-1}}$

Define $\sigma : G \to GL(V)$

$\sigma : g \mapsto T_g$

σ k-representation

$$\sigma(gh) = T_{gh} = T_g T_h = \sigma(g)\sigma(h)$$
$$\sigma(gh)(v) = T_{gh}v = ghv = T_g T_h v = \sigma(g)\sigma(h)v \quad \forall\, v \in V$$

□

Proposition 86. *Let group G, let $\sigma, \tau : G \to GL(V)$ be k-representations, field k.*
If V^σ, V^τ corresponding kG-modules in Prop. 85 (Prop. 8.37 in Rotman (2010) [83]), then $V^\sigma \simeq V^\tau$ as kG-modules iff $\exists$ nonsingular $\varphi : V \to V$ s.t.

$$\varphi\tau(g) = \sigma(g)\varphi \quad \forall\, g \in G$$

Proof. If $\varphi : V^\tau \to V^\sigma$ kG-isomorphism, then $\varphi : V \to V$ isomorphism s.t.

$$\varphi(\sum a_g gv) = (\sum a_g g)\varphi(v) \quad \forall\, v \in V, \forall\, g \in G$$

in V^τ, $kG \times V \to V$ $\quad gv = \tau(g)(v)$

in V^σ, $kG \times V \to V$ scalar multiplication $\quad gv = \sigma(g)(v)$

$$\implies \forall\, g \in G, v \in V, \quad \varphi(\tau(g)(v)) = \sigma(g)(\varphi(v))$$

I think

$$\varphi(gv) = \varphi(\tau(g)(v)) = g\varphi(v) = \sigma(g)\varphi(v)$$
$$\implies \varphi\tau(g) = \sigma(g)\varphi \quad \forall\, g \in G$$

Conversely, if $\exists$ nonsingular $\varphi : V \to V$ s.t. $\varphi\tau(g) = \sigma(g)\varphi \quad \forall\, g \in G$

$$\varphi\tau(g)v = \varphi(\tau(g)v) = \sigma(g)\varphi(v) \quad \forall\, g \in G, \forall\, v \in V$$

Consider scalar multiplication

$$kG \times V \to V$$
$$\sum_{g\in G} a_g g(v) = \sum_{g\in G} a_g \tau_g(v)$$

$$\varphi\left(\sum_{g\in G} a_g \tau_g(v)\right) = \varphi\left(\sum_{g\in G} a_g \tau(g)v\right) = \sum_{g\in G} a_g \sigma(g)\sigma(g)\varphi(v) = \left(\sum_{g\in G} a_g g\right)\varphi(v)$$

□

Admittedly, after this exposition from Rotman (2010) [83], I still didn't understand how kG-modules relate to representation theory and group rings. I turned to Baker (2011) [84], which we'll do right now. Note that I found a lot of links to online resources on representation theory from Khovanov's webpage http://www.math.columbia.edu/~khovanov/resource/.

Note,

Definition 171. vector subspace $W \subseteq V$ is called a
G-submodule, G-subspace, EY : 20150922 "invariant" subspace?
if $\forall\, g \in G$, for representation $\rho : G \to GL_k(V)$, $\rho_g(w) \in W$, $\quad \forall\, w \in W$, $\forall\, g \in G$ i.e. closed under "action of elements of G" with
$\rho_g =: \rho(g) : V \to V$

Given basis $\mathbf{v} = \{v_1 \dots v_n\}$ for V, $\dim_k V = n$, $\forall\, g \in G$,

$$\rho_g v_j = \rho(g)v_j = r_{kj}(g)v_k$$

for, indeed,

$$\rho_g x^j v_j = \rho(g)x^j v_j = x^j \rho(g)v_j = x^j r_{kj}(g)v_k = r_{kj}x^j v_k$$

so that

$$\rho : G \to GL_k(V)$$
$$\rho(g) = [r_{ij}(g)]$$

Example 2.1 (Baker (2011) [84]): Let $\rho : G \to GL_k(V)$ where $\dim_k V = 1$

$$\forall\, v \in V,\, v \neq 0, \forall\, g \in G,\, \lambda_g \in k \text{ s.t. } g \cdot v = \rho_g(v) = \lambda_g v$$
$$\rho(hg)v = \rho_h \rho_g v = \lambda_{hg} v = \lambda_h \lambda_g v \implies \lambda_{hg} = \lambda_h \lambda_g$$

$\implies \exists$ homomorphism $\Lambda : G \to k^\times$

$$\Lambda(g) = \lambda_g$$

From Sec. 2.2 "G-homomorphisms and irreducible representations" of Baker (2011) [84], suppose $\rho : G \to GL_k(V)$, $\sigma : G \to GL_k(W)$ are 2 representations

Many names for the same thing: G-equivalent, G-linear, G-homomorphism, EY : 20150922 kG-isomorphic?

If $\forall\, g \in G$,

$$\begin{array}{ccc} V & \xrightarrow{\varphi} & V \\ {\scriptstyle \tau_g}\downarrow & & \downarrow{\scriptstyle \sigma_g} \\ V & \xrightarrow{\varphi} & V \end{array} \iff V^\tau \overset{\varphi}{\simeq} V^\sigma$$

Indeed, define

$$\varphi : V^\tau \to V^\sigma$$
$$\varphi(v+w) = \varphi(v) + \varphi(w)$$
$$\varphi(rv) = \varphi(\sum_{g\in G} a_g g \cdot v) = \varphi(\sum_{g\in G} a_g \tau_g(v)) = \sum_{g\in G} a_g \varphi(\tau_g(v)) = \sum_{g\in G} a_g \sigma_g \cdot \varphi(v) = r\varphi(v)$$

EY : 20150922 So φ is a kG-isomorphism between left kG modules V^τ and V^σ if it's bijective and is "linear" in "scalars" $r \in kG$, i.e. $\varphi(rv) = r\varphi(v)$.

Define action of G on $\mathrm{Hom}_k(V,W)$ ($\mathrm{Hom}_k(V,W)$ is the vector space of k-linear transformations $V \to W$)

$$\begin{array}{ccc} V & \xrightarrow{f} & W \\ v & \longmapsto & f(v) \end{array}$$

$\forall\, f \in \mathrm{Hom}_k(V,W)$, $f : V \to W$

$$f(v) \in W$$

Consider

$$G \times \mathrm{Hom}_k(V,W) \to \mathrm{Hom}_k(V,W)$$
$$(g \cdot f) \mapsto (\sigma_g f) \circ \rho_{g^{-1}} \text{ i.e. } (g \cdot f)(v) = \sigma_g f(\rho_{g^{-1}} v) \quad (f \in \mathrm{Hom}_k(V,W))$$

Let $g, h \in G$,

$$(gh \cdot f)(v) = g \cdot \sigma_h f(\rho_{h^{-1}} v) = \sigma_g \sigma_h f \rho_{h^{-1}} \rho_{g^{-1}}(v) = (\sigma_{gh} f \rho_{(gh)^{-1}})(v)$$

Thus, $G \times \mathrm{Hom}_k(V,W) \to \mathrm{Hom}_k(V,W)$ is thus another G-representation of G.

$(g \cdot f) \mapsto (\sigma_g f) \circ \rho_{g^{-1}}$

For k-representation ρ, if the only G-subspaces of V are $\{0\}$, V, ρ **irreducible** or **simple**.

$$\rho_g(\{0\}) = \{0\}$$
$$\rho_g(V) = V$$

given subrepresentation $W \subseteq V$, V/W admits linear action of G, $\overline{\rho}_W : G \to GL_k(V/W)$ quotient representation

$$\overline{\rho}_W(g)(v+W) = \rho(g)(v) + W$$

if $v' - v \in W$

$$\rho(g)(v') + W = \rho(g)(v + (v'-v)) + W = (\rho(g)(v) + \rho(g)(v'-v)) + W = \rho(g)(v) + W$$

Proposition 87 (2.7 Baker (2011)[84]). *if $f : V \to W$ G-homomorphism, then*

(a) $\ker f$ is G-subspace of V

(b) $\mathrm{im}f$ is G-subspace of W

$$\begin{array}{ccc} V & \xrightarrow{f} & W \\ {\scriptstyle \rho_g}\downarrow & & \downarrow{\scriptstyle \sigma_g} \\ V & \xrightarrow{f} & W \end{array}$$

Proof. Recall

(a) Let $v \in \ker f$. Then $\forall\, g \in G$,

$$f(\rho_g v) = \sigma_g f(v) = 0$$

so $\rho_g v \in \ker f$, $\forall\, g \in G$. So $\ker f$ is G-subspace of V

(b) Let $w \in \mathrm{im} f$. So $w = f(u)$ for some $u \in V$

$$\sigma_g w = \sigma_g f(u) = f(\rho_g u) \in \mathrm{im} f$$

So $\mathrm{im} f$ is G-subspace of W

□

Theorem 61 (Schur's Lemma)**.** *Let $\rho : G \to GL_{\mathbb{C}}(V)$ be irreducible representations of G over field $k = \mathbb{C}$; let $\sigma : G \to GL_{\mathbb{C}}(W)$ $f : V \to W$ be G-linear map.*

(a) if $f \neq 0$, f isomorphism. True $\forall\, k$ field, not just $\mathbb{C}$

(b) if $V = W$, $\rho = \sigma$, then for some $\lambda \in \mathbb{C}$, f given by $f(v) = \lambda v$ $(v \in V)$ (true for algebraically closed fields)

Proof. (a) By Prop. 87, $\ker f \subseteq V$, $\mathrm{im} f \subseteq W$ are G-subspaces.
For ρ, only G-subspaces are 0 or V, so if $\ker f = V$, $f = 0$. If $\ker f = 0$, f injective.
For σ, only G-subspaces are 0 or V, so $\mathrm{im} f = 0$, $f = 0$. If $\mathrm{im} f = V$, f surjective.
$\implies f$ isomorphism.

(b) Let $\lambda \in \mathbb{C}$ be an eigenvalue of f, $f(v_0) = \lambda v_0$ eigenvector, $v_0 \neq 0$.
Let linear $f_\lambda : V \to V$ s.t.

$$f_\lambda(v) = f(v) - \lambda v \quad (v \in V)$$

$\forall\, g \in G$

$$\rho_g f_\lambda(v) = \rho_g f(v) - \rho_g \lambda v = f(\rho_g v) - \lambda \rho_g v = f_\lambda(\rho_g v)$$

So f_λ is G-linear, for

$$\begin{array}{ccc} V & \xrightarrow{f} & V \\ {\scriptstyle \rho_g}\downarrow & & \downarrow{\scriptstyle \rho_g} \\ V & \xrightarrow{f} & V \end{array}$$

Since $f_\lambda(v_0) = 0$, by Prop. 87, $\ker f_\lambda = V$, (for $\ker f_\lambda \neq 0$ and so $\ker f_\lambda = V$)
By rank-nullity theorem, $\dim V = \dim\ker f_\lambda + \dim\mathrm{im} f_\lambda$.
So $\mathrm{im} f_\lambda = 0$, and so $f_\lambda(v) = 0$ $(\forall\, v \in V) \implies f(v) = \lambda v$

□

Schur's lemma, at least the first part, implies that the left kG-modules associated with representations ρ, σ are kG-isomorphic, i.e.

$$\begin{array}{ccc} V & \xrightarrow{f} & W \\ {\scriptstyle \rho_g}\downarrow & & \downarrow{\scriptstyle \sigma_g} \\ V & \xrightarrow{f} & W \end{array} \iff V^\rho \overset{f}{\simeq} V^\sigma$$

with f being an isomorphism between V^ρ and V^σ s.t.

$$f(v+w) = f(v) + f(w) \quad \forall\, v, w \in (V^\sigma, +)$$
$$f(rv) = rf(v) \quad \forall\, r = \sum_{g \in G} a_g g \in kG$$

Kosmann-Schwarzbach's **Groups and Symmetries**[85] is a very lucid text that's mathematically rigorous enough and practical for physicists. It's really good and very clear. Let's follow its development for $SU(2), SO(3), SL(2,\mathbb{C})$ and corresponding Lie algebras $\mathfrak{su}(2)$, $\mathfrak{so}(3)$, $\mathfrak{sl}(2,\mathbb{C})$.

From Chapter 2 "Representations of Finite Groups" of Kosmann-Schwarzbach (2010) [85]

Definition 172 (2.1 Kosmann-Schwarzbach (2010)[85])**.** On $L^2(G)$, scalar product defined by

$$\langle f_1 | f_2 \rangle = \frac{1}{|G|} \sum_{g \in G} \overline{f_1(g)} f_2(g)$$

$f_1, f_2 \in \mathcal{F}(G) \equiv \mathbb{C}[G]$ vector space of functions on G taking values on $\mathbb{C}$

Definition 173 (2.3 Kosmann-Schwarzbach (2010)[85])**.** Let (E, ρ) be representation of G

$$\rho \equiv \chi_\rho : G \to \mathbb{C}$$

character of

$$\chi_\rho(g) = \mathrm{tr}(\rho(g)) = \sum_{i=1}^{n} (\rho(g))_{ii}$$

Note: equivalent representations have same character
each conjugacy class of G, function χ_p is constant

Looking at Def. 172

$$\langle \chi_{\rho_1} | \chi_{\rho_2} \rangle = \frac{1}{|G|} \sum_{g \in G} \chi_{\rho_1}(g^{-1}) \chi_{\rho_2(g)}$$

since $\overline{\chi_{\rho_1(g)}} = \chi_{\rho_1}(g^{-1})$ by unitarity of representation with respect to scalar product $\langle\, , \rangle$

Proposition 88 (2.7 Kosmann-Schwarzbach (2010)[85])**.** *Let (E_1, ρ_1) be representations of G, let linear $u : E_1 \to (E_2, \rho_2)$*

E_2.

Then $\exists$ linear T_u s.t.

$$\begin{aligned} &T_u : E_1 \to E_2 \\ &T_u = \frac{1}{|G|} \sum_{g \in G} \rho_2(g) u \rho_1(g)^{-1} \end{aligned} \tag{212}$$

so that $\rho_2(g) T_u = T_u \rho_1(g) \quad \forall\, g \in G$

Proof.

$$\rho_2(g)T_u = \frac{1}{|G|}\sum_{h\in G}\rho_2(gh)u\rho_1(h^{-1}) = \frac{1}{|G|}\sum_{k\in G}\rho_2(k)u\rho_1(k^{-1}g) = T_u\rho_1(g)$$

□

Thus, diagrammatically, we have that

$$E_1 \xrightarrow{u} E_2 \quad\Longrightarrow\quad \begin{array}{ccc} E_1 & \xrightarrow{T_u} & E_2 \\ \downarrow{\scriptstyle \rho_1(g)} & & \downarrow{\scriptstyle \rho_2(g)} \\ E_1 & \xrightarrow{T_u} & E_2 \end{array}$$

From Definition 1.12 of Kosmann-Schwarzbach [85], "representations ρ_1 and ρ_2 are called **equivalent** if there is a bijective intertwining operator for ρ_1 and ρ_2." So I will interpret this as if an intertwining operator is not bijective, then the representations ρ_1, ρ_2 are not equivalent.

Proposition 89 (2.8 Kosmann-Schwarzbach (2010)[85])**.** *Let (E_1,ρ_1) be irreducible representations of G, let (E_2,ρ_2) linear $u : E_1 \to E_2$, define T_u by $T_u = \frac{1}{|G|}\sum_{g\in G}\rho_2(g)u\rho_1(g)^{-1}$ by Eq. 212.*

(i) If ρ_1, ρ_2 inequivalent, then $T_u = 0$

(ii) If $E_1 = E_2 = E$ and $\rho_1 = \rho_2 = \rho$, then

$$T_u = \frac{tr(u)}{dimE}1_E$$

Proof. (i) if ρ_1, ρ_2 are inequivalent, by definition, T_u is not isomorphic. Then by Schur's lemma (first part), $T_u = 0$

(ii) By Schur's lemma, $T_u(v) = \lambda v \quad \forall\, v \in E = E_1 = E_2$. So $T_u = \lambda 1_E$. $\mathrm{tr}T_u = \lambda \mathrm{dim}E$ or $\lambda = \frac{\mathrm{tr}T_u}{\mathrm{dim}E}$. Thus, $T_u = \frac{\mathrm{tr}T_u}{\mathrm{dim}E}1_E$

□

Let $(e_1 \dots e_n)$ basis of E
$(f_1 \dots f_p)$ basis of F

$$\forall\, u \in \mathcal{L}(E,F),\quad \begin{array}{l} u : E \to F \\ u(x) = u(x^j e_j) = x^j u(e_j) = x^j u^i{}_j f_i \\ u = u^i{}_j e^j \otimes f_i \end{array} \quad \begin{array}{l} \text{for } x = x^j e_j \in E \\ \qquad\; y = y^i f_i \in F \end{array}$$

For
$T : E^* \otimes F \to \mathcal{L}(E,F)$
$T(\xi \otimes y) = u^i{}_j e^j \otimes f_i$ i.e. set $T(\xi\otimes y)$ to this u
$T(\xi \otimes y) = T(\xi_l e^l \otimes y^k f_k) = \xi_l y^k T(e^l \otimes f_k) = (\xi_l y^k T^{li}_{kj})e^j \otimes f_i \Longrightarrow \xi_l y^k T^{li}_{kj} = u^i{}_j$

Exercises

Exercises of Ch. 2 Representations of Finite Groups [85]

Exercise 2.6.[85] *The dual representation.*
Let (E,π) representation of group G.
$\forall\, g \in G$, $\xi \in E^*$, $x \in E$, set $\langle \pi^*(g)(\xi), x\rangle = \langle \xi, \pi(g^{-1})(x)\rangle$

(a) *dual* (or *contragredient*) of π, $\pi^* : G \to \mathrm{End}(E^*)$, π^* is a representation, since

$$\langle \pi^*(gh)(\xi), x\rangle = \langle \xi, \pi((gh)^{-1})(x)\rangle = \langle \xi, \pi(h^{-1}g^{-1})(x)\rangle = \langle \xi, \pi(h^{-1})\pi(g^{-1})(x)\rangle = \langle \xi, \pi(h^{-1})(\pi(g^{-1})(x))\rangle = \\ = \langle \pi^*(h)(\xi), \pi(g^{-1})(x)\rangle = \langle \pi^*(g)\pi^*(h)(\xi), x\rangle$$

since this is true, $\forall\, x \in E$, $\forall\, \xi \in E^*$, $\pi^*(gh) = \pi^*(g)\pi^*(h)$.

dual π^* of π is a representation.

(b) Consider $G \times \mathcal{L}(E,F) \to \mathcal{L}(E,F)$.

$$g \cdot u = \rho(g) \circ u \circ \pi(g^{-1})$$

Define

$$\sigma : G \to \mathrm{End}(\mathcal{L}(E,F))$$
$$\sigma(g) : \mathcal{L}(E,F) \to \mathcal{L}(E,F)$$
$$\sigma(g)(u) = \rho(g) \circ u \circ \pi(g^{-1})$$

Let $(e_1 \dots e_n)$ be a basis of E. Let $\xi = \xi_i e^i \in E^*$, $x = x^j e_j \in E$.

Consider the isomorphism $T : E^* \otimes F \to \mathcal{L}(E,F)$ defined as[3]

$$T : E^* \otimes F \to \mathcal{L}(E,F) = \mathrm{Hom}(E,F)$$
$$\xi \otimes y \mapsto (x \mapsto \xi(x)y)$$

Choose bases $\begin{matrix}(e_1 \dots e_n) \text{ of } E \\ (e^1 \dots e^n) \text{ of } E^* \\ (f_1 \dots f_p) \text{ of } F\end{matrix}$. Then

$$T(e^j \otimes f_i)(x) = T(e^j \otimes f_i)(x^k e_k) = \delta^j{}_k x^k f_i = x^j f_i$$
$$T(e^j \otimes f_i)(e_k) = \delta^j{}_k f_i$$

Consider

$$u \in \mathcal{L}(E,F)$$
$$u : E \to F$$
$$u(x) = u(x^j e_j) = x^j u(e_j) = x^j u^i{}_j f_i$$
$$u(e_j) = u^i{}_j f_i \text{ i.e. } u : e_j \to u^i{}_j f_i$$

Then $\forall u \in \mathcal{L}(E,F)$,

$$T(u^i{}_j e^j \otimes f_i)(e_k) = u^i{}_j \delta^j{}_k f_i = u^i{}_k f_i = u(e_k) \implies u = T(u^i{}_j e^j \otimes f_i)$$

so T is surjective.

With $T(\xi \otimes y) = T(\xi' \otimes y')$,

$$T(\xi \otimes y)(x) = T(\xi' \otimes y')(x)$$
$$\xi(x)y = \xi'(x)y' \implies \xi(x)y - \xi'(x)y' = 0$$

which implies that $\xi \otimes y = \xi' \otimes y'$. So T is injective. Or, one could consider that $T^{-1} : \mathcal{L}(E,F) \to E^* \otimes F$, $T^{-1} : u \mapsto u^i{}_j e^j \otimes f_i$, which is the inverse of T.

[3] *Mathematics stackexchage* Isomorphism between Hom and tensor product [duplicate] http://math.stackexchange.com/questions/428185/isomorphism-between-hom-and-tensor-product
http://math.stackexchange.com/questions/57189/understanding-isomorphic-equivalences-of-tensor-product

Remark 1.

$$E^* \otimes F \overset{T}{\simeq} \mathcal{L}(E,F) = \mathrm{Hom}(E,F)$$
$$(\xi, y) \mapsto (x \mapsto \xi(x)y)$$

and so $(e^j \otimes f_i) \mapsto (x \mapsto e^j(x) f_i = x^j f_i)$
So $E^* \otimes F$ is isomorphic to $\mathcal{L}(E,F) = \mathrm{Hom}(E,F)$

For representation π,

$$\pi : G \to \mathrm{End}(E)$$
$$\pi(g) : E \to E$$
$$\pi(g)(x) = \pi(g)(x^j e_j) = x^j \pi(g)(e_j) = x^j \pi(g)^i{}_j e_i = (\pi(g)^i{}_j x^j e_i$$

Consider this matrix formulation:

$$\pi^*(g)(\xi) = \pi^*(g)(\xi_i e^i) = \xi_i \pi^*(g)(e^i) = \xi_i (\pi^*(g))^i{}_j e^j$$
$$\implies \langle \pi^*(g)(\xi), x \rangle = \xi_i (\pi^*(g))^i{}_j x^j$$

and

$$\langle \xi, \pi(g^{-1})(x) \rangle = \xi_i \pi(g^{-1})^i{}_j x^j$$

so that

$$\langle \pi^*(g)(\xi), x \rangle = \langle \xi, \pi(g^{-1})(x) \rangle \implies \pi(g^{-1})^i{}_j = (\pi^*(g))^i{}_j$$

Thus, given a choice of basis for E, the *dual* of π, $\pi^*(g)^i{}_j$, and $\pi(g^{-1})^i{}_j$ are formally equal.
So for a choice of basis of E and of F,

$$(\pi^* \otimes \rho)(g)(\xi, y) = (\pi^*(g) \otimes \rho(g))(\xi, y) = \pi^*(g)\xi \otimes \rho(g)y = \xi_l \pi(g^{-1})^l{}_j e^j \otimes \rho(g)^i{}_k y^k f_i = \rho(g)^i{}_k y^k \xi_l \pi(g^{-1})^l{}_j e^j \otimes f_i$$

Applying T,

$$T(\pi^* \otimes \rho)(g)(\xi, \rho) = \rho(g)^i{}_k y^k \xi_l \pi(g^{-1})^l{}_j = \rho(g) T(\xi, y) \pi(g^{-1})$$

$$\begin{array}{ccc} E^* \otimes F & \xrightarrow{\quad T \quad} & \mathcal{L}(E,F) \\ \Big\downarrow{\scriptstyle (\pi^* \otimes \rho)(g)} & & \Big\downarrow{\scriptstyle \sigma(g)} \\ E^* \otimes F & \xrightarrow[\quad T \quad]{} & \mathcal{L}(E,F) \end{array} \qquad \begin{array}{ccc} (\xi, y) & \xmapsto{\quad T \quad} & (x \mapsto \xi(x)y) = y^i \xi_j \\ \Big\downarrow{\scriptstyle (\pi^* \otimes \rho)(g)} & & \Big\downarrow{\scriptstyle \sigma(g)} \\ \pi^*(g)(\xi) \otimes \rho(g)y & \xmapsto[\quad T \quad]{} & \rho(g) y^i \xi_j \pi(g^{-1}) = \rho(g) T(\xi, y) \pi(g^{-1}) \end{array}$$

Thus
Thus, representation $\sigma(g)$ is equivalent to representation $(\pi^* \otimes \rho)$, a tensor product of representations.

Exercise 2.15. *Representation of $GL(2,\mathbb{C})$ on the polynomials of degree 2*
Let group G, let representation ρ of G on $V = \mathbb{C}^n$, i.e. $\rho : G \to \mathrm{End}(V)$
Let $P^{(k)}(V)$ vector space of complex polynomials on V that are homogeneous of degree k.
For $f \in P^{(k)}(V)$, the general form is

$$f = \sum_{\substack{i_1 + i_2 + \cdots + i_n = k \\ 0 \le i_j \le k}} a_{i_1 i_2 \ldots i_d} x_1^{i_1} x_2^{i_2} \ldots x_n^{i_n}$$

Given

$$\binom{n+k}{k} = \binom{k-1}{k-1} + \binom{k}{k-1} + \cdots + \binom{n+k-1}{k-1} = \sum_{i=0}^{n} \binom{k-1+i}{k-1}$$

$\binom{k+n-1}{n-1}$ is number of monomials of degree k.

So $\dim P^{(k)}(V) = \binom{k+n-1}{n-1}$. This is a very lucid and elementary exposition on the basics of polynomials which I found was useful for the basic facts I forgot[4].

So we have the graded algebra

$$P(V) = \bigoplus_{k=0}^{\infty} P^{(k)}(V)$$

$$\rho^{(k)} : G \to \mathrm{End}(P^{(k)}(V))$$
$$\rho^{(k)}(g) : P^{(k)}(V) \to P^{(k)}(V)$$
$$\rho^{(k)}(g)(f) = f \circ \rho(g^{-1})$$

This is a representation of G since

(a)
$$\left.\begin{aligned}\rho^{(k)}(gh)(f) &= f \circ \rho((gh)^{-1}) = f \circ \rho(h^{-1}g^{-1}) = f \circ \rho(h^{-1}\rho(g^{-1}) \\ \rho^{(k)}(g)\rho^{(k)}(h)(f) &= \rho^{(k)}(g)(f \circ \rho(h^{-1})) = f \circ \rho(h^{-1}) \circ \rho(g^{-1})\end{aligned}\right\} \Longrightarrow \rho^{(k)}(gh) = \rho^{(k)}(g)\rho^{(k)}(h)$$

(b) Choose basis $(e_1 \dots e_n)$ of V, $x = x^j e_j \in V$, $\rho : G \to \mathrm{End}(V)$, and so $\rho(g)(x) = \rho(g)(x^j e_j) = x^j \rho(g)(e_j) = x^j (\rho(g))^i{}_j e_i$.
With $\xi(e_i) = \xi_i \Longrightarrow \langle \xi, \rho(g^{-1})x \rangle = \xi_i x^j (\rho(g^{-1}))^i{}_j$
$\forall\, \xi \in V^*$, $\xi = \xi_i e^i$,
$$\rho^*(g)(\xi) = \rho^*(g)(\xi_i e^i) = \xi_i \rho^*(g)^i{}_j e^j$$
$$\Longrightarrow \langle \rho^*(g)(\xi), x \rangle = \xi_i x^j (\rho^*(g))^i{}_j \Longrightarrow (\rho^*(g))^i{}_j = (\rho(g^{-1}))^i{}_j$$

So $\forall f \in P^{(1)}(V)$, $x \in V$, $\rho(g^{-1})x = x^j(\rho(g^{-1}))^i{}_j e_i$. So $f \circ \rho(g^{-1})(x) = \sum_{i=1}^n a_i (\rho(g^{-1}))^i{}_j x^j = \sum_{i=1}^n a_i (\rho^*(g))^i{}_j x^j$
$\Longrightarrow \rho^{(1)}(g)(f) = f \circ \rho^*(g)$

(c) Suppose $G = GL(2,\mathbb{C})$, $V = \mathbb{C}^2$, ρ fundamental representation $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $g^{-1} = \frac{1}{\det g}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ for $\det g = ad - bc$.
Let $k = 2$, $\dim P^{(2)}(\mathbb{C}^2) = \binom{2+2-1}{2-1} = \binom{3}{1} = 3$
$\forall\, f \in P^{(2)}(\mathbb{C}^2)$, $f(x,y) = Ax^2 + 2Bxy + Cy^2$
Let
$$P^{(2)}(\mathbb{C}^2) \to \mathbb{C}^3$$
$$f(x,y) = Ax^2 + 2Bxy + Cy^2 \mapsto \begin{pmatrix} A \\ B \\ C \end{pmatrix} \in \mathbb{C}^3$$

Call this transformation T, $T : P^{(2)}(\mathbb{C}^2) \to \mathbb{C}^3$.
$\forall \begin{pmatrix} A \\ B \\ C \end{pmatrix} \in \mathbb{C}^3$, $f(x,y) = Ax^2 + 2Bxy + Cy^2$ and $Tf(x,y) = \begin{pmatrix} A \\ B \\ C \end{pmatrix}$. T surjective.
Suppose $Tf(x,y) = Tf'(x,y)$,
$$\Longrightarrow Ax^2 + 2Bxy + Cy^2 = A'x^2 + 2B'xy + C'y^2$$
$$\Longrightarrow (A - A')x^2 + 2(B - B')xy + (C - C')y^2 = 0$$

Then since the monomials form a basis, and its basis elements are independent (by definition), then $A = A'$, $B = B'$, $C = C'$. T injective. So T is bijective, an isomorphism.

(This is all in `groups.sage`)

[4]Polynomials. Math 4800/6080 Project Course http://www.math.utah.edu/~bertram/4800/PolyIntroduction.pdf

```
sage: P2CC.<x,y> = PolynomialRing(CC,2) # this declares a PolynomialRing of field of
    complex numbers,
# of order 2 (i.e. only 2 variables for a polynomial, such as x, y)
sage: A = var('A')
sage: assume(A,''complex'')
sage: B = var('B')
sage: assume(B,''complex'')
sage: C = var('C')
sage: assume(C,''complex'')
sage: f(x,y) = A*x**2 +2*B*x*y + C*y**2

sage: a = var('a')
sage: assume(a,''complex'')
sage: b = var('b')
sage: assume(b,''complex'')
sage: c = var('c')
sage: assume(c,''complex'')
sage: d = var('d')
sage: assume(d,''complex'')
sage: g = Matrix([[a,b],[c,d]] )
sage: X = Matrix([[x],[y]])
sage: f( (g.inverse()*X)[0,0], (g.inverse()*X)[1,0] ).expand()
sage: f( (g.inverse()*X)[0,0], (g.inverse()*X)[1,0] ).expand().coefficient(x^2).
    full_simplify()
(C*c^2 - 2*B*c*d + A*d^2)/(b^2*c^2 - 2*a*b*c*d + a^2*d^2)
sage: f( (g.inverse()*X)[0,0], (g.inverse()*X)[1,0] ).expand().coefficient(x*y).
    full_simplify()
-2*(C*a*c + A*b*d - (b*c + a*d)*B)/(b^2*c^2 - 2*a*b*c*d + a^2*d^2)
sage: f( (g.inverse()*X)[0,0], (g.inverse()*X)[1,0] ).expand().coefficient(y^2).
    full_simplify()
(C*a^2 - 2*B*a*b + A*b^2)/(b^2*c^2 - 2*a*b*c*d + a^2*d^2)
```

So

$$\rho^{(2)}(g)(f)(x,y) = f \circ \rho(g^{-1})(x,y) =$$
$$= \frac{Cc^2 - 2Bcd + Ad^2}{(ad-bc)^2}x^2 + -2\frac{(Cac + Abd - (bc+ad)B)}{(ad-bc)^2}xy + \frac{Ca^2 - 2Bab + Ab^2}{(ad-bc)^2}y^2$$

So define $\widetilde{\rho} : G \to \mathrm{End}(\mathbb{C}^3)$. $\widetilde{\rho}$ is a representation, for

$$\forall v = \begin{pmatrix} A \\ B \\ C \end{pmatrix} \in \mathbb{C}^3, \quad \widetilde{\rho}(gh)(v) = T \circ f \circ \rho((gh)^{-1}) = T \circ f \circ \rho(h^{-1}g^{-1}) = T \circ f \circ \rho(h^{-1})\rho(g^{-1})$$

$$\text{Now } \widetilde{\rho}(h)(v) = T \circ f \circ \rho(h^{-1})$$
$$\implies \widetilde{\rho}(g)\widetilde{\rho}(h)(v) = T \circ (f \circ \rho(h^{-1})) \circ \rho(g^{-1}) = T \circ f \circ \rho(h^{-1})\rho(g^{-1}) \text{ and so}$$
$$\widetilde{\rho}(gh) = \widetilde{\rho}(g)\widetilde{\rho}(h)$$

And so

$$\widetilde{\rho}^*(g)(v) = Tf\rho(g^{-1})$$

and consider this commutation diagram, that (helped me at least and) clarifies the relationships:

$$\begin{array}{ccc} P^{(2)}(\mathbb{C}^2) & \xrightarrow{T} & \mathbb{C}^3 \\ \downarrow{\scriptstyle \rho^{(2)}(g)} & & \downarrow{\scriptstyle \widetilde{\rho}(g)} \\ P^{(2)}(\mathbb{C}^2) & \xrightarrow[T]{} & \mathbb{C}^3 \end{array} \qquad \begin{array}{ccc} f & \xrightarrow{T} & \begin{pmatrix} A \\ B \\ C \end{pmatrix} \\ \downarrow{\scriptstyle \rho^{(2)}(g)} & & \downarrow{\scriptstyle \widetilde{\rho}(g)} \\ f \circ \rho(g^{-1}) & \xrightarrow[T]{} & \begin{pmatrix} D \\ E \\ F \end{pmatrix} \end{array}$$

with

$$\begin{pmatrix} D \\ E \\ F \end{pmatrix} = \begin{pmatrix} \frac{Cc^2 - 2Bcd + Ad^2}{(ad-bc)^2} \\ -2\frac{(Cac + Abd - (bc+ad)B)}{(ad-bc)^2} \\ \frac{Ca^2 - 2Bab + Ab^2}{(ad-bc)^2} \end{pmatrix}$$

Now define the dual $\widetilde{\rho}^*$ as such:

$$\widetilde{\rho}^*(g) : (\mathbb{C}^3)^* \to (\mathbb{C}^3)^*$$
$$\widetilde{\rho}^*(g) = \widetilde{\rho}(g^{-1})$$
$$\forall \xi \in (\mathbb{C}^3)^*$$
$$\widetilde{\rho}^*(g)\xi = \xi_i(\widetilde{\rho}^*(g))^i{}_j e^j = \xi_i(\widetilde{\rho}(g^{-1}))^i{}_j e^j$$

So for $v = \begin{pmatrix} A \\ B \\ C \end{pmatrix} \in \mathbb{C}^3$, $f = T^{-1}v = Ax^2 + 2Bxy + Cy^2 \in P^2(\mathbb{C}^2)$,

$$\widetilde{\rho}(g^{-1})(v) = T \circ (f\rho(g)) = \begin{bmatrix} Aa^2 + 2Bac + Cc^2 \\ Aab + Bbc + Bad + Ccd \\ Ab^2 + 2Bbd + Cd^2 \end{bmatrix}$$

which was found using Sage Math:

```
sage: f((g*X)[0,0],(g*X)[1,0])
(a*x + b*y)^2*A + 2*(a*x + b*y)*(c*x + d*y)*B + (c*x + d*y)^2*C
sage: f((g*X)[0,0],(g*X)[1,0]).expand()
A*a^2*x^2 + 2*B*a*c*x^2 + C*c^2*x^2 + 2*A*a*b*x*y + 2*B*b*c*x*y + 2*B*a*d*x*y + 2*C*c*d*x
    *y + A*b^2*y^2 + 2*B*b*d*y^2 + C*d^2*y^2
sage: f((g*X)[0,0],(g*X)[1,0]).expand().coefficient(x^2)
A*a^2 + 2*B*a*c + C*c^2
sage: f((g*X)[0,0],(g*X)[1,0]).expand().coefficient(x*y)
2*A*a*b + 2*B*b*c + 2*B*a*d + 2*C*c*d
sage: f((g*X)[0,0],(g*X)[1,0]).expand().coefficient(y^2)
A*b^2 + 2*B*b*d + C*d^2
```

---

or

```
sage: T( f((g*X)[0,0],(g*X)[1,0]).expand() )
[A*a^2 + 2*B*a*c + C*c^2,
2*A*a*b + 2*B*b*c + 2*B*a*d + 2*C*c*d,
A*b^2 + 2*B*b*d + C*d^2]
```

So then

$$\widetilde{\rho}(g^{-1}) = \begin{bmatrix} a^2 & 2ac & c^2 \\ 2ab & 2(ad+bc) & 2cd \\ b^2 & 2bd & d^2 \end{bmatrix}$$

So then

$$\widetilde{\rho}^*(g) = \begin{bmatrix} a^2 & 2ac & c^2 \\ 2ab & 2(ad+bc) & 2cd \\ b^2 & 2bd & d^2 \end{bmatrix}$$

and operate on row vectors $\xi \in (\mathbb{C}^3)^*$ with $\widetilde{\rho}^*(g)$ from the row vector's right.

More: Let $G = SU(2)$. Then $U = e^{i\phi}\begin{bmatrix} a & b \\ -\overline{b} & \overline{a} \end{bmatrix}$

$$\widetilde{\rho} : SU(2) \to \mathrm{End}(\mathbb{C}^3)$$
$$\widetilde{\rho}(U) : \mathbb{C}^3 \to \mathbb{C}^3$$
$$\widetilde{\rho}(U)(v) = e^{-2i\varphi}\begin{bmatrix} A\overline{a}^2 + 2B\overline{a}\overline{b} + C\overline{b}^2 \\ -A\overline{a}b + B + Ca\overline{b} \\ Ab^2 - 2Bab + Ca^2 \end{bmatrix}$$
$$\implies \widetilde{\rho}(U) = e^{-2i\varphi}\begin{bmatrix} -\overline{a}^2 & 2\overline{a}\overline{b} & \overline{b}^2 \\ -\overline{a}b & 1 & a\overline{b} \\ b^2 & -2ab & a^2 \end{bmatrix}$$

cf. Ch. 5 Lie Groups of Jeffrey Lee (2009) [75]

Definition 174 (Lie Group)**. Lie Group** $G :=$ smooth manifold G is a **Lie Group** if G is a group (abstract group), s.t.

multiplication map $\mu : G \times G \to G$
$$\mu(g,h) = gh$$

inverse map $\text{inv} : G \to G$
$$\text{inv}(g) = g^{-1}$$
are C^∞ maps.

If group is abelian, use additive notation $g + h$ for group operation.

Definition 175 ($GL(n,\mathbb{R})$)**.** $GL(n,\mathbb{R}) :=$ group of all invertible real $n \times n$ matrices.

global chart on $GL(n,\mathbb{R}) = \{x^i_j\}$, n^2 functions x^i_j, where if $A \in GL(n,\mathbb{R})$, then $x^i_j(A)$ is ijth entry of A.

Claim: $GL(n,\mathbb{R})$ is a Lie group.

Proof. multiplication is clearly smooth: $(AB)_{ij} = A_{ik}B_{kj}$,

$$\frac{\partial}{\partial x^l_m}(x^i_k(A)x^k_j(B)) = \delta^i_l\delta^m_k x^k_j(B) + x^i_k(A)\delta^k_l\delta^m_j$$

inversion map; appeal to formula for A^{-1}, $A^{-1} = \text{adj}(A)/\det(A)$, $\text{adj}(A) \equiv$ adjoint matrix (whose entries are cofactors).

$\implies A^{-1}$ depends smoothly on entries of A.

Similarly, $GL(n,\mathbb{C})$, group of invertible $n \times n$ complex matrices, is a Lie group. □

Exercise 5.5. Let subgroup H of G, consider cosets gH, $g \in G$.

Recall G is disjoint union of cosets of H.

Claim: if H open, so are all its cosets. And H closed.

Proof. cf. stackexchange: Open subgroups of a topological group are closed

$gH = \{gh | h \in H\}$ is an open neighborhood of g (since $1 \in H$, and mapping $h \mapsto gh$ sends open sets to open sets, since its inverse, $gh \mapsto h$, is C^∞ (so continuous)).

$$gH \to H \qquad\qquad H \to gH$$
$$gh \xrightarrow{g^{-1}} h = \mu(g^{-1}, gh) \qquad h \xrightarrow{g} gh = \mu(g,h)$$

Then $\forall$ coset gH, gH is open.

Suppose $g' \in H^c \equiv G - H \equiv G \backslash H$.

Consider $h \in H$, if $g'h \in H$, then $g' = (g'h)h^{-1} \in H$ (recall $h^{-1} \in H$, and H is a subgroup).

Contradiction.

$\implies \forall\, g' \in H^c$, $\exists$ open neighborhood $g'H \subset H^c$, so H^c open (by definition). Then H closed.

□

cf. Thm. 5.6 in Jeffrey Lee (2009) [75].

Theorem 62. *If G connected Lie group, U neighborhood of identity element e, then U generates the group, i.e. $\forall\, g \in G$, g is a product of elements of U.*

Proof. Note $V = \text{inv}(U) \cap U$ is an open neighborhood of e. Note $\text{inv}(V) = V$. $\text{inv}(V) \equiv V^{-1} = \{V^{-1} | v \in V\}$. We say that V is *symmetric*.

Claim: V generates G.

$\forall$ open W_1, open $W_2 \subset G$,
$W_1W_2 = \{w_1w_2 | w_1 \in W_1, w_2 \in W_2\}$ is an open set being a union of open sets $\bigcup_{g \in W_1} gW_2$.

Thus, inductively defined sets
$$V^n = VV^{n-1}, \quad n = 1,2,3,\ldots$$
are open.

$$e \in V \subset V^2 \subset \dots V^n \subset \dots$$

It's easy to check that each V^n is symmetric.

$$\begin{aligned}
&\text{inv}(V) = V \\
&\text{inv}(V^2) = \text{inv}(\bigcup_{v \in V} vV) = V\text{inv}(V) = V = V^2 \\
&\text{inv}(V^{n+1}) = \text{inv}\left(\bigcup_{v \in V} vV^n\right) = V\text{inv}(V^n) = VV^n = V^{n+1}
\end{aligned}$$

so $V^\infty := \bigcup_{n=1}^\infty V^n$ is symmetric.

V^∞ closed under inversion, also multiplication. Thus V^∞ is an open subgroup.

From Exercise 5.5, Jeffrey Lee (2009) [75], i.e. Exercise 85, V^∞ also closed, since G is connected, $V^\infty = G$. (a topological space X is **connected** iff the only open and closed (clopen) sets are $\emptyset$ and X).

□

Definition 176. Identity component of G, G_0.

$G_0 :=$ connected component of Lie group G that contains identity;
G_0 is a Lie group, and is generated by any open neighborhood of the identity.

Definition 177. For Lie group G, fixed element $g \in G$,
left translation (by g) $L_g : G \to G$, $L_g x = gx, \quad \forall\, x \in G$
right translation (by g) $R_g : G \to G$, $R_g x = xg, \quad \forall\, x \in G$

L_g, R_g are diffeomorphisms with $L_g^{-1} = L_{g^{-1}}$, $R_g^{-1} = R_{g^{-1}}$.

Definition 178 (Product Lie group)**.** If G, H are Lie groups, then product manifold $G \times H$ is a Lie group, where multiplication

$$(g_1, h_1) \cdot (g_2, h_2) = (g_1 g_2, h_1 h_2)$$

Lie group $G \times H$ is called **product Lie group**

e.g. product group $S^1 \times S^1 \equiv$ 2-torus group.
Generally, higher torus groups $T^n = S^1 \times \dots \times S^1$ (n factors).

Definition 179 (Lie subgroup of G, H)**.** Let H be an abstract subgroup of Lie group G.
If H is a Lie group s.t. inclusion map $i : H \to G \equiv H \hookleftarrow G$ is an immersion, then H is a **Lie subgroup** of G.

Recall $i : H \to G$ immersion iff Di injective, i.e. iff $\text{rank} Di = \dim H$
cf. Prop. 5.9 in Jeffrey Lee (2009) [75].

Proposition 90. *If H abstract subgroup of Lie group G, that's also a regular submanifold $\equiv$ embedded submanifold, then H closed Lie subgroup.*

Recall that

embedded submanifold $\equiv$ regular submanifold

Each name is used frequently and we shouldn't be biased against one or the other; we'll have to refer to both, to emphasize they're *exactly the same.*

embedded submanifold $\equiv$ regular submanifold is an immersed submanifold s.t. inclusion map i is a topological embedding,
i.e. embedded submanifold $\equiv$ regular submanifold $S \subset M$,
immersed submanifold S if $i : S \to M \equiv S \hookrightarrow M$ is an immersion, i.e. Di injective, i.e. $\text{rank} Di \equiv \dim S$.
topological embedding := homeomorphism onto its image, i.e.
injective cont. map $f : X \to Y$, X, Y topological spaces, is a **topological embedding**
if f is a homeomorphism between X and $f(X)$.
f homeomorphism is a bijection, continuous, and f^{-1} continuous.

e.g. $\forall$ embedding $f : M \to N$, $f(M) \subset N$ naturally has the structure of an embedding submanifold $\equiv$ regular submanifold.

Useful, intrinsic definition of **embedded submanifold** $\equiv$ regular submanifold.
Let manifold M, $\dim M = n$, let $k \in \mathbb{Z}^+$, s.t. $0 \leq k \leq n$.

A k-dim. embedded submanifold $\equiv$ regular submanifold S is subset $S \subset M$ s.t. $\forall p \in S$, $\exists$ chart $(U \subset M, \varphi : U \to \mathbb{R}^n \ni 0)$, s.t. $\varphi(S \cap U)$ is the intersection of a k-dim. plane with $\varphi(U)$.
(pairs $(S \cap U, \varphi|_{S \cap U})$ form an atlas for differential structure on S).

Proof 1:

Proof. H subgroup of G, so
multiplication map $H \times H \to H$
inversion map $H \to H$
are restrictions of multiplication and inversion maps on G.

Since H regular submanifold, maps are smooth.
Recall H regular submanifold iff H immersive submanifold (i.e. $H \hookleftarrow G$ is an immersion) and H topological subspace of G, i.e. submanifold topology on H is same as subspace topology.

Claim: H closed.
Let $x_0 \in \overline{H}$
Let (U, x) be a chart adapted to H, whose domain contains e.
Let

$$\delta : G \times G \to G$$
$$\delta(g_1, g_2) = g_1^{-1} g_2$$

Choose open set V s.t. $e \in V \subset \overline{V} \subset U$.
By continuity map δ, find open neighborhood O of identity e s.t. $O \times O \subset \delta^{-1}(V)$
If $\{h_i\}$ sequence in H converging to $x_0 \in \overline{H}$, then $x_0^{-1} h_i \to e$ and $x_0^{-1} h_i \in O$ for all sufficiently large i.
Since $h_j^{-1} h_i = (x_0^{-1} h_j)^{-1} x_0^{-1} h_i$, $h_j^{-1} h_i \in V$ for sufficiently large i, j.

For any sufficiently large fixed j,

$$\lim_{i \to 0} h_j^{-1} h_i = h_j^{-1} x_0 \in \overline{V} \subset U$$

Since U is domain of a single-slice chart, $U \subset H$ closed in U.

Thus, since $\forall\, h_j^{-1} h_i \in U \cap H$, $h_j^{-1} x_0 \in U \cap H \subset H$, $\quad \forall$ sufficiently large j.
$\implies x_0 \in H$, and since x_0 arbitrary, done.

□

Proof 2:
cf. 9.2 The Closed Subgroup Theorem I of 427 Notes[5]

Proof. **Claim**: Since H is an embedded submanifold $\equiv$ regular submanifold, $\exists$ neighborhood U of 1, $1 \in G$, s.t. $U \cap H$ closed in U.

Let $x_0 \in \overline{H}$, $\overline{H} \equiv$ closure of x_0.
Then $x_0 U^{-1} \subseteq G$ is a neighborhood of x_0 in G (since $1 \in U^{-1}$, $x_0 1 = x_0 \in x_0 U^{-1}$)

$$\implies x_0 U^{-1} \cap H \neq \emptyset$$

$\forall\, x \in x_0 U^{-1} \cap H$, $x = x_0 U^{-1}$ for some $u \in U$. Thus, $x^{-1} x_0 = u \in U$.

Now
$L_{x^{-1}} : G \to G$ is a homeomorphism, so $L_{x^{-1}}(H) = H$. By continuity, $L_{x^{-1}}(\overline{H}) = \overline{H}$. Thus $x^{-1} x_0 \in \overline{H}$.

[5] https://faculty.math.illinois.edu/~lerman/519/s12/427notes.pdf

Claim: $x^{-1}x_0 \in H \cap U$.
Since $x^{-1}x_0 \in \overline{H} \cap U$, $\exists$ sequence $\{h_i\} \subset H \cap U$ s.t. $h_0 \to x^{-1}x_0$.
But recall $H \cap U$ closed in U, so $x^{-1}x_0 \in H \cap U$.

$$\implies x_0 \in xH = H, \quad \overline{H} \subseteq H$$

Thus H closed.

Claim: If H abstract subgroup of Lie group G, that's also an embedded submanifold $\equiv$ regular submanifold, then H is a Lie subgroup.

Recall that by definition, Lie group has group multiplication and inverse map to be C^∞. Then, just show group multiplication is C^∞, first.

Since G is a Lie group, then

$$\mu : G \times G \to G$$
$$\mu(x, y) = xy$$

is C^∞ (by definition).

Then $\mu : G \times G \to G$ cont.
Consider subgroup $H \subseteq G$ and $\mu : H \times H \to H$.
Since $H \times H \subseteq G \times G$, $\forall\, (x, y) \in H \times H$ (fix $(x, y) \in H \times H$), $\forall$ neighborhood V of $\mu(x, y) = xy$, $V \subset G$, $\exists$ neighborhood U of (x, y) s.t. $\mu(U) \subseteq V$ (by $\mu : G \times G \to G$ cont.).

Since H embedded submanifold $\equiv$ regular submanifold of G,
$\exists$ neighborhood $V' \subseteq V$ of $xy \in G$, coordinate map $\varphi : V' \to \mathbb{R}^n$ $(n = \dim G)$ s.t.

$$\varphi(H \cap V') = \varphi(V') \cap (\mathbb{R}^k \times \{0\})$$

where $k = \dim H$

(since H is a k-dim. embedded submanifold $\equiv$ regular submanifold, $H \subseteq G$, s.t. $\forall p \in H$, $\exists$ chart $(V \subset G, \varphi : U \to \mathbb{R}^n \ni 0)$, s.t. $\varphi(U \cap V) = \varphi(V) \cap (\mathbb{R}^k \times \{0\})$).

Now

$$\varphi \circ \mu : \mu^{-1}(V') \cap U \to \mathbb{R}^n \text{ is } C^\infty, \text{ and } \varphi \circ \mu(\mu^{-1}(V') \cap U) \subseteq \mathbb{R}^k \times \{0\}$$

Let projection $\pi : \mathbb{R}^n \to \mathbb{R}^k$ be the standard projection,

$$\pi \circ \varphi \circ \mu : \mu^{-1}(V') \cap U \to \mathbb{R}^k \text{ is } C^\infty$$

$\implies \mu$ is C^∞ □

From Chapter 4 "Lie Groups and Lie Algebras" of Kosmann-Schwarzbach (2010) [85]
While Proposition 2.6 of Kosmann-Schwarzbach (2010) [85] states that

$$\det(\exp(X)) = \exp(\operatorname{tr} X)$$

here are some other resources online that gave further discussion on the characteristic polynomial, $\det(A - \lambda 1)$ and the different terms of it, called Newton identities:

- http://scipp.ucsc.edu/~haber/ph116A/charpoly_11.pdf
- http://math.stackexchange.com/questions/1126114/how-to-find-this-lie-algebra-proof-that-mathfraksl-is-trace-zero-matrice
- http://mathoverflow.net/questions/131746/derivative-of-a-determinant-of-a-matrix-field

Theorem 63 (5.1 [85]). *Consider* $\mathfrak{g} = \{X = \gamma'(0) | \gamma : 1 \to G$ *of class* C^1 , $\gamma(0) = 1\}$
Let Lie group G

(i) $\mathfrak{g}$ *vector subspace of* $\mathfrak{gl}(n, \mathbb{R})$

(ii) $X \in \mathfrak{g}$ *iff* $\forall\, t \in \mathbb{R}$, $\exp(tX) \in G$

(iii) if $X \in \mathfrak{g}$, *if* $g \in G$, *then* $gXg^{-1} \in \mathfrak{g}$

(iv) $\mathfrak{g}$ *closed under matrix commutator, i.e. if* $X, Y \in \mathfrak{g}$, $[X, Y] \in \mathfrak{g}$

Proof. (i)

(ii) If $\exp(tX) \in G$, then $X \frac{d}{dt} \exp(tX)\big|_{t=0} \in \mathfrak{g}$ (by def.)

If $X \in \mathfrak{g}$, then by def., $X = \frac{d}{dt}\gamma(t)\big|_{t=0}$ with $\gamma(t) \in G$.

Now Taylor expand; $\forall\, k \in \mathbb{Z}^+$

$$\gamma\left(\frac{t}{k}\right) = 1 + \frac{t}{k}X + O\left(\frac{1}{k^2}\right) = \exp\left(\frac{t}{k}X + O\left(\frac{1}{k^2}\right)\right)$$

$$\implies \left(\gamma\left(\frac{t}{k}\right)\right)^k = \exp(tX)$$

$$\gamma\left(\frac{t}{k}\right) \in G \quad \forall\, k \in \mathbb{Z}^+$$

G closed subgroup, so $\lim_{k\to\infty}(\gamma\left(\frac{t}{k}\right))^k = \exp(tX) \in G$

(iii)

(iv)

□

Definition 180 (Lie algebra)**.** Lie algebra $\mathfrak{g}$, tangent space to G at 1, i.e. $\mathfrak{g} := T_1 G$ is called *Lie algebra* of Lie group G.

$$\mathfrak{g} := \{X = \gamma'(0) | \gamma : 1 \to G \text{ of class } C^1\ , \gamma(0) = 1\} = T_1 G$$

This is based on Proposition 5.3 of Kosmann-Schwarzbach (2010) [85].

For Lie group

$$U(n) = \{U \in GL(n, \mathbb{C}) | UU^\dagger = 1\}$$

If $X \in \mathfrak{u}(n)$, then $\exp(tX) \in U(n)$. Then

$$\exp(tX)\exp(tX)^\dagger = (1 + tX + O(t^2))(1 + tX^\dagger + O(t^2)) = 1 + t(X + X^\dagger) + O(t^2) = 1 \forall\, t \in \mathbb{R} \implies X + X^\dagger = 0$$

i.e. $X \in \mathfrak{u}(n)$ is an anti-Hermitian complex $n \times n$ matrix.

$\mathfrak{u}(n) = \{X \in \mathfrak{gl}(n, \mathbb{C}) | X + X^\dagger = 0\}$

Physicists: $X = iA$ and so $A - A^\dagger$. $A \in \mathfrak{u}(n)$ is a Hermitian complex $n \times n$ matrix.

$\mathfrak{u}(n) = \{A \in \mathfrak{gl}(n, \mathbb{C}) | A - A^\dagger = 0\}$

Regardless, $\dim_{\mathbb{R}} \mathfrak{u}(n) = n^2 = 2n^2 - n^2$

For Lie group

$$SU(n) = \{U \in GL(n, \mathbb{C}) | UU^\dagger = 1, \det U = 1\}$$

Then

$$\mathfrak{su}(n) = \{X \in \mathfrak{gl}(n, \mathbb{C}) | X + X^\dagger = 1,\ \mathrm{tr} X = 0\}$$

is the Lie algebra of traceless anti-Hermitian complex $n \times n$ matrices, and that

$$\dim_{\mathbb{R}} \mathfrak{su}(n) = n^2 - 1$$

In summary,

$$\begin{array}{ll} \mathfrak{u}(n) = \{X \in \mathfrak{gl}(n,\mathbb{C}) | X + X^\dagger = 0\} & \mathfrak{su}(n) = \{X \in \mathfrak{gl}(n,\mathbb{C}) | X + X^\dagger = 0, \mathrm{tr} X = 0\} \\ \Big\downarrow \exp(tX) & \Big\downarrow \exp(tX) \\ U(n) = \{U \in GL(n,\mathbb{C}) | UU^\dagger = 1\} & SU(n) = \{U \in GL(n,\mathbb{C}) | UU^\dagger = 1, \mathrm{det} U = 1\} \end{array}$$

$$\dim_{\mathbb{R}} \mathfrak{u}(n) = n^2 \qquad \dim_{\mathbb{R}} \mathfrak{su}(n) = n^2 - 1$$

From Chapter 5 "Lie Groups $SU(2)$ and $SO(3)$" of Kosmann-Schwarzbach (2010) [85],

85.0.1 Bases of $\mathfrak{su}(2)$, Subsection 1.1 of Chapter 5of Kosmann-Schwarzbach (2010) [85]

Recall that

$$\begin{array}{l} \mathfrak{su}(n) = \{X \in \mathfrak{gl}(n,\mathbb{C}) | X + X^\dagger = 0, \mathrm{tr} X = 0\} \\ \Big\downarrow \exp(tX) \\ SU(n) = \{U \in GL(n,\mathbb{C}) | UU^\dagger = 1, \mathrm{det} U = 1\} \end{array}$$

$$\dim_{\mathbb{R}} \mathfrak{su}(n) = n^2 - 1$$

and so

$$\begin{array}{l} \mathfrak{su}(2) = \{X \in \mathfrak{gl}(2,\mathbb{C}) | X + X^\dagger = 0, \mathrm{tr} X = 0\} \\ \Big\downarrow \exp(tX) \\ SU(2) = \{U \in GL(n,\mathbb{C}) | UU^\dagger = 1, \mathrm{det} U = 1\} \end{array}$$

$$\dim_{\mathbb{R}} \mathfrak{su}(2) = 3$$

Also, recall that $\mathfrak{g} \subseteq \mathfrak{gl}(n,\mathbb{C})$ is a vector subspace (63) and that
$X \in \mathfrak{g}$ iff $\forall t \in \mathbb{R}$, $\exp(tX) \in G$.
if $X \in \mathfrak{g}$, if $g \in G$, then $gXg^{-1} \in \mathfrak{g}$

$\mathfrak{g}$ closed under $\mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$

$$(X, Y) \mapsto [X, Y]$$

and so with $\mathfrak{g}$ as a vector space, we can have a choice of bases.

(a)
$$\xi_1 = \frac{i}{2} \begin{pmatrix} & 1 \\ 1 & \end{pmatrix}$$
$$\xi_2 = \frac{1}{2} \begin{pmatrix} & -1 \\ 1 & \end{pmatrix}$$
$$\xi_3 = \frac{i}{2} \begin{pmatrix} 1 & \\ & -1 \end{pmatrix}$$

satisfying

$$[\xi_k, \xi_l] = \epsilon_{klm} \xi_m$$

(b) *Physics*

$$\sigma_1 = -2i\xi_1 = \begin{pmatrix} & 1 \\ 1 & \end{pmatrix}$$

$$\sigma_2 = 2i\xi_2 = \begin{pmatrix} & -i \\ i & \end{pmatrix}$$

$$\sigma_3 = -2i\xi_3 = \begin{pmatrix} 1 & \\ & -1 \end{pmatrix}$$

satisfying

$$[\sigma_k, \sigma_l] = 2i\epsilon_{klm}\sigma_m$$

EY : 20151001 Sage Math 6.8 doesn't run on Mac OSX El Capitan: I suspect that it's because in Mac OSX El Capitan, `/usr` cannot be modified anymore, even in an Administrator account. The TUG group for MacTeX had a clear, thorough, and useful (i.e. copy UNIX commands, paste, and run examples) explanation of what was going on:

http://tug.org/mactex/elcapitan.html

So keep in mind that my code for Sage Math is for Sage Math 6.8 that doesn't run on Mac OSX El Capitan. I'll also use `sympy` in Python as an alternative and in parallel.

One can check in `sympy` the traceless anti-Hermitian (or Hermitian) property of the bases and Pauli matrices, and the commutation relations (see `groups.py`):

```
import itertools
from itertools import product, permutations

import sympy
from sympy import I, LeviCivita
from sympy import Rational as Rat

from sympy.physics.matrices import msigma # <class 'sympy.matrices.dense.MutableDenseMatrix'>

def commute(A,B):
"""
commute = commute(A,B)
commute takes the commutator of A and B
"""
return (A*B - B*A)

def xi(i):
"""
xi = xi(i)
xi is a function that returns the independent basis for
Lie algebra su(2)\equiv su(2,\mathbb{C}) of Lie group SU(2) of
traceless anti-Hermitian matrices, based on msigma of sympy
cf. http://docs.sympy.org/dev/_modules/sympy/physics/matrices.html#msigma
"""
if i not in [1,2,3]:
raise IndexError("Invalid Pauli index")
elif i==1:
return I/Rat(2)*msigma(1)
elif i==2:
return -I/Rat(2)*msigma(2)
elif i==3:
return I/Rat(2)*msigma(3)
```

```
## check anti-Hermitian property and commutation relations with xi
# xi is indeed anti-Hermitian
xi(1) == -xi(1).adjoint() # True
xi(2) == -xi(2).adjoint() # True
xi(3) == -xi(3).adjoint() # True

# xi obeys the commutation relations

for i,j in product([1,2,3],repeat=2): print i,j

for i,j in product([1,2,3],repeat=2): print i,j, "\t Commutator: ", commute(xi(i),xi(j))

## check traceless Hermitian property and commutation relations with Pauli matrices
# Pauli matrices i.e. msigam is indeed traceless Hermitian

msigma(1) == msigma(1).adjoint() # True
msigma(2) == msigma(2).adjoint() # True
```

```
msigma(3) == msigma(3).adjoint() # True

msigma(1).trace() == 0 # True
msigma(2).trace() == 0 # True
msigma(3).trace() == 0 # True

# Pauli matrices obey commutation relation
print "For Pauli matrices, the commutation relations are:\n"
for i,j in product([1,2,3],repeat=2): print i,j, "\t Commutator: ", commute(msigma(i),msigma(j))

for i,j,k in permutations([1,2,3],3): print "Commute: ", i,j,k, msigma(i), msigma(j), \ \
": and is 2*i of ", msigma(k), commute(msigma(i),msigma(j)) == 2*I*msigma(k)*LeviCivita(i,j,k)
```

And finally the traceless property of the Pauli matrices:

```
>>> msigma(1).trace()
0
>>> msigma(2).trace()
0
>>> msigma(3).trace()
0
```

85.1 Rotation Group $SO(N)$ and Lie Algebra so(3)

cf. Kambe (2009) [95], Appendix C

$g \in SO(N)$ represented by $N \times N$ orthogonal matrix (i.e. $gg^T = 1$) s.t. $\det g = 1$.

Let curve $\xi(t)$ on $SO(N)$, initially from identity 1, with tangent vector $\mathbf{a}$ at 1.

$\Longrightarrow \xi(t) = 1 + t\mathbf{a} + O(t^2)$ for an infinitesimal parameter t.

$\mathbf{a} = \xi'(0)$ is an element of tangent space $T_1 SO(N)$ i.e. the Lie algebra $\mathbf{so}(N)$, i.e. $T_1 SO(N) = \mathbf{so}(N)$ by orthogonality condition $\xi(t)\xi^T(t) = 1$, then

$$(1 + t\mathbf{a} + \dots)(1 + t\mathbf{a}^T + \dots) = 1 \Longrightarrow \mathbf{a} + \mathbf{a}^T = 0 \text{ or } \mathbf{a} = \mathbf{a}^T \tag{213}$$

$\Longrightarrow \mathbf{a}$ skew-symmetric.

Correction (skew symmetry). The alternative in Eq. (213) should read $\mathbf{a} = -\mathbf{a}^T$. The first-order term of the product is $t(\mathbf{a} + \mathbf{a}^T)$, and it must vanish. The derivation for $\mathfrak{so}(n)$ is in §88.

85.1.1 so(3)

$\dim(\mathbf{so}(3)) = 3$, vector space of Lie algebra $\mathbf{so}(3)$ represented by skew-symmetric basis (E_1, E_2, E_3)

$$E_1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}, \quad E_2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}, \quad E_3 = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$[E_i, E_j] = \epsilon_{ijk} E_k \text{ where } [E_i, E_j] = E_i E_j - E_j E_i$$

85.2 Spin

Let's follow the development by Baez and Muniain (1994) on pp. 175 of the Section II.1 "Lie Groups", the second (II) chapter on "Symmetry" [82].

Let $V = \mathbb{C}^2$, $G = SU(2)$. Then consider the graded algebra of polynomials on $V = \mathbb{C}^2 \ni (x, y)$

$$P(V) = \bigoplus_{k=0}^{\infty} P^{(k)}(V) = \bigoplus_{\substack{j=0 \\ 2j \in \mathbb{Z}}}^{\infty} P^{(2j)}(V) = \bigoplus_{\substack{j=0 \\ j \in \mathbb{Z}}}^{\infty} P^{(2j)}(V) \oplus \bigoplus_{\substack{j=1/2 \\ 2j \text{ odd}}}^{\infty} P^{(2j)}(V)$$

$$P^{(2j)}(V) \equiv \text{ vector space of complex polynomials of degree } 2j$$

and recall this representation on $P^{(2j)}(V)$

$$\rho^{(2j)} : G \to \text{End}(P^{(2j)}(V))$$
$$\rho^{(2j)} : P^{(2j)}(V) \to P^{(2j)}(V)$$
$$\rho^{(2j)}(g)(f) = f \circ \rho(g^{-1}) \text{ where } \rho \text{ is the fundamental representation of } G = SU(2)$$
$$\rho^{(2j)}(g)(f)(v) = f \circ \rho(g^{-1})(v) \quad \forall f \in P^{(2j)}(V), \forall v \in V = \mathbb{C}^2$$

Note,
$\dim P^{(2j)} = \binom{2j+2-1}{2-1} = 2j+1$

Exercise 21. [82] *spin-0* Consider the trivial representation τ:

$$\tau : G \to \text{End}(\mathbb{C})$$
$$\tau(g) : \mathbb{C} \to \mathbb{C}$$
$$\tau(g) = 1_{\mathbb{C}}$$

$$\begin{array}{ccc} \mathbb{C} & \xrightarrow{T} & P^{(0)}(V) \\ \tau(g) \downarrow & & \downarrow \rho^{(0)}(g) \\ \mathbb{C} & \xrightarrow{T} & P^{(0)}(V) \end{array}$$

Clearly, $P^{(0)}(V) = \mathbb{C}$, since $P^{(0)}(V)$ consists of polynomials of constants in $\mathbb{C}$.
Consider $c_0 \in \mathbb{C}$, $f = k_0 \in P^{(0)}(V)$
$\rho^{(0)}(g)(f) = f \circ \rho(g^{-1}) = k_0$
$\implies \rho^0(g)T(c_0) = T \circ \tau(g)c_0 = T(c_0)$. Let $T = 1_{\mathbb{C}} = 1_{P^0(V)}$
So $\rho^{(0)}(g) = \tau(g) = 1$. $T = 1$. So representations $\rho^{(0)}$ and trivial representation τ on G are equivalent.

Exercise 22. [82] *spin-$\frac{1}{2}$* For spin-$\frac{1}{2}$, $j = \frac{1}{2}$, $2j = 1$.

$\forall f \in P^{(1)}(V)$, $V = \mathbb{C}^2$. So in general form, $f(x,y) = ax + by \in P^{(1)}(V)$, $\begin{pmatrix} x \\ y \end{pmatrix} \in V = \mathbb{C}^2$

Recall the fundamental representation
$$\rho : G \to GL(2,\mathbb{C}) \equiv GL(\mathbb{C}^2)$$
$$\rho(g) : \mathbb{C}^2 \to \mathbb{C}^2$$
$$\rho(g) = g$$

So consider T such that

$$\begin{array}{ccc} \mathbb{C}^2 & \xrightarrow{T} & P^{(1)}(V) \\ \rho(g) \downarrow & & \downarrow \rho^{(1)}(g) \\ \mathbb{C}^2 & \xrightarrow{T} & P^{(1)}(V) \end{array}$$

Consider $\forall v \in \mathbb{C}^2$, $v = \begin{pmatrix} x \\ y \end{pmatrix}$, then

$$\rho(g)v = gv = \begin{bmatrix} ax + by \\ cx + dy \end{bmatrix}$$

```
sage: g*X
[a*x + b*y]
[c*x + d*y]
```

For notation, let $U \in G = SU(2)$ s.t. $UU^\dagger = 1$.
Consider $(\rho^{(2j)}(U)(f))(x) = f(U^{-1}x)$, $\forall x \in \mathbb{C}^2$.

Choose $f(x,y) = x$. So for $f(x,y) = Ax + By$, $A = 1, B = 0$. Choose $U = \begin{pmatrix} a & b \\ -\overline{b} & \overline{a} \end{pmatrix}$ so $U^{-1} = \begin{pmatrix} \overline{a} & -b \\ \overline{b} & a \end{pmatrix}$.

Then $U^{-1}x = \begin{pmatrix} \overline{a}x - by \\ \overline{b}x + ay \end{pmatrix}$

So

$$(\rho^{(1)}(U)(f))(x) = f(U^{-1}x) = \overline{a}x - by$$
$$(\rho^{(1)}(U)(f))(x) = f(U^{-1}x) = \overline{b}x + ay \text{ for } f(x,y) = y$$

Let $f(x,y) = Ax + By$

$$(\rho^{(1)}(U)(f))(x) = f(U^{-1}x) = (A\overline{a} + B\overline{b})x + (Ba - Ab)y = (\overline{a}x - by)A + (\overline{b}x + ay)B = (A\overline{a} + B\overline{b})x + (Ba - Ab)y$$

which was calculated with the assistance of Sage Math:

```
sage: U_try1 = Matrix( [[a.conjugate(),-b],[b.conjugate(),a ] ] )
sage: f1( U_try1*X).coefficient(x)
A*conjugate(a) + B*conjugate(b)
sage: f1( U_try1*X).coefficient(y)
B*a - A*b
```

Treating $P^{(1)}(\mathbb{C}^2)$ as a vector space, in its matrix formulation, then $f(x,y) = Ax + By \in P^{(1)}(\mathbb{C}^2)$ is treated as $\begin{bmatrix} A \\ B \end{bmatrix}$, then $(\rho^{(1)}(U)f)$ is

$$\implies \begin{bmatrix} \overline{a} & \overline{b} \\ -b & a \end{bmatrix} \begin{bmatrix} A \\ B \end{bmatrix} = \begin{bmatrix} A\overline{a} + B\overline{b} \\ -Ab + Ba \end{bmatrix}$$

so conclude in general that $\rho^{(1)}(U) = (U^{\dagger})^T$.

Now, as Kosmann-Schwarzbach (2010) [85] says, on pp. 13, Chapter 2 Representations of Finite Groups, "Two representations (E_1, ρ_1) and (E_2, ρ_2) are equivalent if and only if there is a basis B_1 of E_1 and a basis B_2 of E_2 such that for every $g \in G$, the matrix of $\rho_1(g)$ in the basis B_1 is equal to the matrix of $\rho_2(g)$ in the basis B_2. In particular, if the representations (E_1, ρ_1) and (E_2, ρ_2) are equivalent, then E_1 is isomorphic to E_2." So we need a change of basis between $\rho(U) = U$ and $\rho^{(1)}(U)$. What's the linear transformation T s.t.

$$T^{-1}\rho^{(1)}(U)T = U?$$

By intuition,

$$T = \sigma_x\sigma_z \equiv \sigma_1\sigma_3$$

where σ_i's are Pauli matrices.

Indeed,

```
sage: Paulimat[3] * Paulimat[1]*U_try*Paulimat[1] * Paulimat[3]
[conjugate(a) conjugate(b)]
[          -b           a]
```

Then $\rho^{(1)}(U) \circ T = TU$, so this $T = \sigma_1\sigma_3$ is an "intertwining operator" between $\rho^{(1)}(U)$ and fundamental representation $\rho(U) = U$, with $T = \begin{bmatrix} & -1 \\ 1 & \end{bmatrix}$, and $T^{-1} = \begin{bmatrix} & 1 \\ -1 & \end{bmatrix}$.

T is an isomorphism between $\mathbb{C}^2$ and $P^{(1)}(\mathbb{C}^2)$. So fundamental representation ρ of $G = SU(2)$ is equivalent to $\rho^{(1)}(U)$ on $P^{(1)}(\mathbb{C}^2)$.

Exercise 23.[82] (Also from Exercise 2.6 of Kosmann-Schwarzbach (201) [85])

Let (E, π) representation of group G.

$\forall\, g \in G$, $\xi \in E^*$, $x \in E$, set $\langle \pi^*(g)(\xi), x\rangle = \langle \xi, \pi(g^{-1})(x)\rangle$

dual (or *contragredient*) of π, $\pi^* : G \to \mathrm{End}(E^*)$, π^* is a representation, since

$$\langle \pi^*(gh)(\xi), x\rangle = \langle \xi, \pi((gh)^{-1})(x)\rangle = \langle \xi, \pi(h^{-1}g^{-1})(x)\rangle = \langle \xi, \pi(h^{-1})\pi(g^{-1})(x)\rangle = \langle \xi, \pi(h^{-1})(\pi(g^{-1})(x))\rangle =$$
$$= \langle \pi^*(h)(\xi), \pi(g^{-1})(x)\rangle = \langle \pi^*(g)\pi^*(h)(\xi), x\rangle$$

since this is true, $\forall\, x \in E$, $\forall\, \xi \in E^*$, $\pi^*(gh) = \pi^*(g)\pi^*(h)$.

dual π^* of π is a representation.

85.3 Adjoint Representation

I will first follow Sec. 7.3 The Adjoint Representation of Ch. 4 Lie Groups and Lie Algebras of Kosmann-Schwarzbach (201) [85]).

The *conjugation action* $\mathcal{C}_g : G \to G$ is defined as

$$\mathcal{C}_g : G \to G$$
$$\mathcal{C}_g : h \mapsto ghg^{-1}$$

So

$$\mathcal{C} : G \to \mathrm{Aut}(G)$$
$$\mathcal{C}g = \mathcal{C}_g$$

Now define the *adjoint action* of g as the differential or push forward of $\mathcal{C}_g$:

$$\mathrm{Ad}_g := D_1\mathcal{C}_g \equiv (\mathcal{C}_g)_{*1} \equiv (\mathcal{C}_g)_*|_{g=1} \qquad \text{(adjoint action of } g\text{)}$$

Now $\mathrm{Ad}_g : \mathfrak{g} \to \mathfrak{g}$, so $\mathrm{Ad} : G \to \mathrm{End}(\mathfrak{g})$

$$\mathrm{Ad}(g) \equiv \mathrm{Ad}_g$$

Note $\mathcal{C}_{gg'} = \mathcal{C}_g\mathcal{C}_{g'} \equiv \mathcal{C}(gg') = \mathcal{C}(g) \circ \mathcal{C}(g')$ and so

$$\xrightarrow{D_1} \mathrm{Ad}_{gg'} = \mathrm{Ad}_g \circ \mathrm{Ad}_{g'}$$

Kosmann-Schwarzbach (201) [85]) claims, because $\mathrm{Ad}_g = 1_{\mathfrak{g}}$ when $g = 1$,

$\mathrm{Ad} : G \to GL(\mathfrak{g})$ is a representation of G on $\mathfrak{g}$. (EY : 20160505 ???)

$\mathrm{Ad} : g \mapsto \mathrm{Ad}_g$

Definition 181. representation Ad of G on $V = \mathfrak{g}$ is called adjoint representation of Lie group G.

Denote adjoint representation of Lie algebra $\mathfrak{g}$, ad.

By definition, $\mathrm{Ad}_{\exp(tX)} = \exp(t\mathrm{ad}_X)$

cf. Prop. 7.8 of Kosmann-Schwarzbach (201) [85])

Proposition 91. *1. Let A invertible matrix, $A \in$ Lie group G.*
Let X matrix s.t. $X \in \mathfrak{g}$. Then

$$Ad_A(X) = AXA^{-1}$$

2. Let $X, Y \in \mathfrak{g}$. Then

$$ad_X(Y) = [X, Y]$$

3. Let $X, Y \in \mathfrak{g}$. Then

$$ad_{[X,Y]} = [ad_X, ad_Y]$$

Proof. 1. By def., $\forall\, B \in G$, $\mathcal{C}_A(B) = ABA^{-1}$, and thus

$$\mathrm{Ad}_A(X) = \frac{d}{dt} A \exp(tX) A^{-1}\Big|_{t=0} = AXA^{-1}$$

2.

$$\mathrm{ad}_X(Y) = \frac{d}{dt}\mathrm{Ad}_{\exp(tX)}(Y)\Big|_{t=0} = \frac{d}{dt}\exp(tX)Y\exp(tX)\Big|_{t=0} =$$
$$= XY - YX = [X, Y]$$

3. Use Jacobi identity:

$$[A, [B, C]] + [B, [C, A]] + [C, [A, B]] = 0 \text{ or}$$
$$[[A, B], C] = [A, [B, C]] - [B, [A, C]]$$
$$\mathrm{ad}_{[X,Y]}C = [[X, Y], C] = [X, [Y, C]] - [Y, [X, C]] = [X, \mathrm{ad}_Y C] - [Y, \mathrm{ad}_X C] \text{ and that}$$
$$\mathrm{ad}_X \mathrm{ad}_Y C = [X, [Y, C]] \Longrightarrow \mathrm{ad}_{[X,Y]}C = [\mathrm{ad}_X, \mathrm{ad}_Y]C$$

□

85.4 Invariant Vector Fields, Left-invariant Vector Field, Right-invariant Vector Field, Invariant Vector Fields

cf. pp. 26, Sec. 1.7 Lie Group and Invariant Vector Fields, Kambe (2009) [95]

Consider group G of smooth transformations (maps) of a manifold M into itself, i.e. Lie group.

Given fixed element $h \in G$, then

$$\begin{aligned} L_h &: g \mapsto hg \text{ i.e. } L_h(g) = hg \\ R_h &: g \mapsto gh \text{ i.e. } R_h(g) = gh \end{aligned} \tag{214}$$

where $L_h \equiv$ left translation of the group

$R_h \equiv$ right translation of the group

Suppose g_t is a curve on G described in terms of parameter t.
Left translation of g_t by $g_{\Delta t}$ for infinitesimal Δt given by $g_{\Delta t} \circ g_t$.

$$\dot{g}_t = \lim_{\Delta t \to 0} \frac{g_{t+\Delta t} - g_t}{\Delta t} = \lim_{\Delta t \to 0} \frac{(g_{\Delta t} - 1)g_t}{\Delta t} = X \circ g_t \text{ with } X := \lim_{\Delta t \to 0} \frac{g_{\Delta t} - 1}{\Delta t} \in T_1 G \tag{215}$$

Thus, left translation $L_{g_{\Delta t}}$ leads to **right**-invariant vector field.
Similarly, right translation $R_{g_{\Delta t}}$ leads to **left**-invariant vector field.
In summary,

$\dot{g}_t$ is said to be tangent vector at g_t.

Vector field $X^L, (X^R)$ on G is left invariant (right invariant) if it's invariant under all left-translations (right translations) respectively, i.e. $\forall\, g, h \in G$, if

$$\begin{aligned} (L_h)_* X^L_g &= X^L_{hg} \\ (R_h)_* X^R_g &= X^R_{hg} \end{aligned} \tag{216}$$

Given tangent vector X to G, at e, one may left-translate (right translate) X to every pt. $g \in G$ as

$$\begin{aligned} X^L_g &= (L_g)_* X = g \circ X = gX \\ X^R_g &= (R_g)_* X = X \circ g = Xg \end{aligned} \tag{217}$$

Since

$$(R_h)_* X^R_g = (R_h)_* (R_g)_* X = X \cdot gh = X^R_{gh}$$

Then $(R_h)_*$ operation does yield a right-invariant field.

Hence $(R_h)_*$ transformation in Eq. 217 gives a right-invariant field generated by X.

In other words, in summary

$$\begin{aligned} \text{Left translation of } g_t \text{ by } g_{\Delta t}\text{: } & L_{g_{\Delta t}} : g_t \mapsto g_{\Delta t} g_t = g_{t+\Delta t} \\ & \dot{g}_t = X g_t \text{ so } X \in T_1 G \text{ is right-invariant vector field} \\ & \text{i.e. } X^R_g = (R_g)_* X = Xg \end{aligned} \tag{218}$$

Consider curves $\xi_t : \mathbb{R} \to G$ with tangent $\dot{\xi}_0 = X$ at $t = 0$.

$\xi : t \mapsto \xi(t), \quad t \in \mathbb{R}$

Left-invariant field $X^L_s = \frac{d}{dt}(g \circ \xi_t) \quad \forall\, g_s \in G$, ($s$ a parameter)
Right-invariant field $X^R_s = \frac{d}{dt}(\xi_t \circ g_s)$

Lie group G acts as a group of linear transformations on its own Lie algebra $\mathfrak{g}$, namely
$\forall\, g \in G$, $\exists$ operator Ad_g s.t.

$$\mathrm{Ad}_g Y := (L_g)_* \circ (R_{g^{-1}})_* Y = gYg^{-1}, \quad \forall\, Y \in \mathfrak{g} \tag{219}$$

Operator Ad_g transforms $Y \in \mathfrak{g}$ into $\mathrm{Ad}_g Y \in \mathfrak{g}$, linearly.

Set of all such Ad_g, i.e. $\mathrm{Ad}(G)$ is called the adjoint representation of G, an adjoint group.

Setting g to be the inverse $\xi_t^{-1} := (\xi_t)^{-1}$,
adjoint transformation $\mathrm{Ad}_{\xi_t^{-1}} Y$ is a function of t.

Its derivative with respect to t is a linear transformation from Y to $\mathrm{ad}_X Y$:

$$\begin{aligned} \mathrm{ad}_X Y &= \frac{d}{dt} \xi_t^{-1} Y \xi_t \Big|_{t=0} := [X, Y] \\ \mathrm{ad}_X &: \mathfrak{g} \to \mathfrak{g} \end{aligned} \tag{220}$$

ad_X is a linear transformation $\mathfrak{g} \to \mathfrak{g}$.

86 Lie Groups and their Lie Algebras

Lie groups and their Lie algebras - Lec 13 - Frederic Schuller, Schuller (2015) [108]

86.1 Chapter 4. Lie Theory, Schuller (2015) [108]

86.1.1 4.1 Lie groups, Schuller (2015) [108]

Definition 182 (Lie group, Schuller (2015) [108])**.** A lie group $(G, \cdot)$ is

Part XX
Quaternions, Spin(3), and $SU(2)$

87 Conventions

Use the Hamiltonian quaternions $\mathbb{H}$ with the multiplication rules $i^2 = j^2 = k^2 = -1$, $ij = -ji = k$, $jk = -kj = i$, $ki = -ik = j$, as defined in the ring examples of the algebra part. Write the imaginary units as

$$(e_1, e_2, e_3) = (i, j, k), \qquad e_1 e_2 = e_3, \quad e_2 e_3 = e_1, \quad e_3 e_1 = e_2, \qquad e_a^2 = -1. \tag{221}$$

A quaternion is written scalar first,

$$q = q_0 + q^a e_a, \qquad q_0 \in \mathbb{R}, \quad (q^1, q^2, q^3) \in \mathbb{R}^3. \tag{222}$$

Its conjugate is $\bar{q} = q_0 - q^a e_a$ and its norm satisfies $|q|^2 = q\bar{q} = q_0^2 + q^a q^a$. A vector $\mathbf{x} \in \mathbb{R}^3$ is identified with the pure quaternion $x^a e_a$.

For attitude, the unit quaternion q carries body components to inertial components by

$$x_I^a e_a = q \, (x_B^a e_a) \, \bar{q}, \tag{223}$$

so it represents the same rotation R as in Eq. (553), where $\mathbf{x}_I = R\mathbf{x}_B$. The rotation acts on vectors (active form).

87.1 Notation of these notes and of the sources

Each book on this subject has its own symbols for the same objects. These notes fix one set, which we call *our* notation, and translate every other author into it at the first use. Our notation is the one already used in the Lie-groups part (§85), which follows Kosmann-Schwarzbach [85]:

- the Lie algebra of a matrix group G is $\mathfrak{g} = \{\gamma'(0) : \gamma \text{ a } C^1 \text{ curve in } G,\ \gamma(0) = 1\}$, and $X \in \mathfrak{g}$ if and only if $\exp(tX) \in G$ for all t (Theorem 63, which is Kosmann-Schwarzbach's Theorem 5.1);
- $\mathfrak{su}(2) = \{X \in \mathfrak{gl}(2, \mathbb{C}) : X + X^\dagger = 0,\ \mathrm{tr}\, X = 0\}$ with the basis ξ_1, ξ_2, ξ_3 (§85.0.1);
- conjugation $\mathcal{C}_g(h) = ghg^{-1}$, the adjoint action $\mathrm{Ad}_g = D_1 \mathcal{C}_g$, so that $\mathrm{Ad}_g Y = gYg^{-1}$, and its derivative $\mathrm{ad}_X Y = [X, Y]$ (§85.3).

To these we add the symbols introduced in this part: the hat map $\widehat{\boldsymbol{\omega}} = \omega^j E_j$ (Eq. (257)), the map $\Sigma(\mathbf{x}) = x^k \sigma_k$ to traceless Hermitian matrices (Eq. (268)), and the homomorphism $\Phi\colon SU(2) \to SO(3)$ (Eq. (272)). The table lists Frankel's and Kosmann-Schwarzbach's symbols for each object beside ours; page numbers for Kosmann-Schwarzbach are those of the 2010 Universitext edition, Chapter 5 unless stated. Holm's symbols are given in the text where they occur.

object	Frankel [11]	Kosmann-Schwarzbach [85]	ours
Lie algebra of G	$\mathfrak{g}$	$\mathfrak{g}$	$\mathfrak{g}$
Lie algebra of $SU(2)$	$\mathfrak{g}$	$\mathfrak{su}(2)$	$\mathfrak{su}(2)$
traceless Hermitian 2×2	$i\mathfrak{g}$	$i\,\mathfrak{su}(2)$	$i\,\mathfrak{su}(2)$
basis of $\mathfrak{su}(2)$ with $\mathfrak{so}(3)$ brackets	$\sigma_a/2i$	ξ_k (p. 71)	ξ_k
quaternion units in $\mathfrak{su}(2)$	σ_a/i	$\mathcal{I},\mathcal{J},\mathcal{K} = 2\xi_k$ (p. 72)	$e_a = -i\sigma_a$
basis of $\mathfrak{so}(3)$	E_a	η_k (p. 73)	E_k
cross product	$\times$	$\wedge$	$\times$
generator of rotations	$\mathbf{E}\bullet\boldsymbol{\omega}$	M_a, $M_a x = a\wedge x$ (p. 75)	$\widehat{\boldsymbol{\omega}}$
$\mathbb{R}^3 \to i\,\mathfrak{su}(2)$	$\mathbf{x}\mapsto \mathbf{x}_*$	—	$\mathbf{x}\mapsto\Sigma(\mathbf{x})$
$\mathbb{R}^3 \to \mathfrak{su}(2)$	—	$a\mapsto X_a = a^1\mathcal{I}+a^2\mathcal{J}+a^3\mathcal{K}$ (p. 77)	$\mathbf{x}\mapsto -i\Sigma(\mathbf{x})$
conjugation $h\mapsto ghg^{-1}$	—	$\mathcal{C}_g$ (Ch. 4, p. 64)	$\mathcal{C}_g$
action on $\mathfrak{g}$	$\mathrm{Ad}(g)$	Ad_g (Ch. 4, p. 65)	Ad_g
$SU(2)\to SO(3)$	Ad	φ (p. 77)	Φ
derivative at I	Ad_*	ad (p. 78)	$d\Phi$
rotation by θ about $\mathbf{n}$	$R(\theta)$ (19.4)	$\mathrm{Rot}(n,\theta)$ (p. 74)	$R_{\mathbf{n}}(\theta)$
"corresponds under $_*$"	$\sim$	—	an equation in Σ

Kosmann-Schwarzbach's φ is not literally our Φ: her $\mathcal{J} = -i\sigma_2$ but $\mathcal{I} = +i\sigma_1$ and $\mathcal{K} = +i\sigma_3$, and §95.1 shows $\varphi(g) = \Phi(ugu^{-1})$ for a fixed $u \in SU(2)$. Frankel uses Ad both for the action on matrices and for the resulting 3×3 matrix, and writes $\sim$ between a vector and the matrix it corresponds to. We keep the two apart: Ad_u acts on matrices, $\Phi(u)$ acts on $\mathbb{R}^3$, and Eq. (272) relates them with an equality.

88 The Lie algebra $\mathfrak{so}(n)$ and Frankel's basis of $\mathfrak{so}(3)$

Let $G \subseteq GL(n,\mathbb{R})$ be a closed matrix group. Its Lie algebra is

$$\mathfrak{g} = \{X \in \mathfrak{gl}(n,\mathbb{R}) \;:\; \exp(tX) \in G \text{ for all } t \in \mathbb{R}\}, \tag{224}$$

the characterization recorded from Kosmann-Schwarzbach in the Lie-groups part. If $g(t)$ is a differentiable curve in G with $g(0) = I$, then $\frac{dg}{dt}(0) \in \mathfrak{g}$; this is how Frankel computes the basis below.

88.1 Orthogonal groups as regular level sets

Frankel [11], Section 1.1d, pp. 9–11, proves that the orthogonal matrices form a submanifold. We give the argument for $n\times n$ matrices, using the differential notation of our manifold notes. His matrix x becomes A, an orthogonal base point x_0 becomes R, and $F_* v = w$ becomes $DF_R(H) = S$. Here $\mathrm{Sym}(n)$ denotes the symmetric $n\times n$ matrices; its argument is the matrix size. For $n = 3$ this is Frankel's six-dimensional space Sym_6.

Identify $\mathrm{Mat}(n,n)$ with $\mathbb{R}^{n^2}$ by its entries. A symmetric matrix is specified by its diagonal and entries above it, so

$$\mathrm{Sym}(n) = \{S \in \mathrm{Mat}(n,n) : S^T = S\}, \qquad \dim \mathrm{Sym}(n) = \frac{n(n+1)}{2}. \tag{225}$$

Define the smooth polynomial map

$$F : \mathrm{Mat}(n,n) \longrightarrow \mathrm{Sym}(n), \qquad F(A) = A^T A - I. \tag{226}$$

Its zero set is $O(n)$. For $n = 3$ its entries are the column inner products $(F(A))_{ij} = \sum_{k=1}^{3} A_{ki}A_{kj} - \delta_{ij}$. Since $(F(A))_{ij} = (F(A))_{ji}$, there are six distinct scalar equations. To subtract six from nine, we must also prove independence of their differentials at every orthogonal matrix.

Hold A and the ambient direction $H \in \mathrm{Mat}(n,n)$ fixed. Expanding before taking the derivative gives

$$F(A+sH) - F(A) = s(H^T A + A^T H) + s^2 H^T H,$$

$$DF_A(H) = \lim_{s\to 0} \frac{F(A+sH) - F(A)}{s} = H^T A + A^T H. \tag{227}$$

Now let $R \in O(n)$ and let $S \in \mathrm{Sym}(n)$ be arbitrary. Choose $H = \frac{1}{2}RS$. Substitute $R^T R = I$ and $S^T = S$ into Eq. (227):

$$\begin{aligned} DF_R\left(\tfrac{1}{2}RS\right) &= \tfrac{1}{2}S^T R^T R + \tfrac{1}{2}R^T RS \\ &= \tfrac{1}{2}S + \tfrac{1}{2}S = S. \end{aligned} \tag{228}$$

Thus $S \mapsto \frac{1}{2}RS$ is a right inverse of DF_R, so DF_R is surjective at every $R \in F^{-1}(0)$. Zero is a regular value. The preimage theorem in the manifold part, equivalently Lee's regular level set theorem [76], Chapter 5, makes $O(n)$ an embedded submanifold with

$$\dim O(n) = n^2 - \frac{n(n+1)}{2} = \frac{n(n-1)}{2}. \tag{229}$$

This proves differential independence on the level set, rather than inferring it merely from the number of displayed equations.

For $R \in O(n)$, $(\det R)^2 = \det(R^T R) = 1$, so $\det R \in \{+1, -1\}$. Moreover,

$$SO(n) = O(n) \cap \overset{-1}{\det}((0,\infty)).$$

The determinant is continuous, so $SO(n)$ is open in $O(n)$. It inherits the same dimension and local submanifold charts. Consequently $SO(3)$ is an embedded three-dimensional submanifold of $\mathbb{R}^9$; the determinant condition selects a sign and imposes no additional local scalar constraint. Locally, the implicit function theorem lets us solve six matrix entries smoothly in terms of three others, with the choice depending on the point.

The tangent space is the kernel of the differential:

$$\begin{aligned} T_R O(n) &= \{H : H^T R + R^T H = 0\} \\ &= \{RX : X^T = -X\}. \end{aligned} \tag{230}$$

Indeed, set $X = R^T H$, so $H = RX$ and $X^T + X = H^T R + R^T H$. At $R \in SO(n)$ the tangent space is the same, because $SO(n)$ is open in $O(n)$. At $R = I$ it consists of the skew-symmetric matrices. Along a curve $R(t) \in SO(3)$ this gives $R^T \dot{R} \in \mathfrak{so}(3)$, the angular-velocity construction in Eq. (552).

88.2 $\mathfrak{so}(n)$ is the skew-symmetric matrices

Take $G = SO(n) = \{g \in GL(n,\mathbb{R}) : g^T g = I,\ \det g = 1\}$ and a curve $g(t)$ in $SO(n)$ with $g(0) = I$ and $X = \frac{dg}{dt}(0)$. Differentiate $g(t)^T g(t) = I$ by the product rule and evaluate at $t = 0$:

$$\begin{aligned} \frac{dg}{dt}(t)^T g(t) + g(t)^T \frac{dg}{dt}(t) = 0, \\ X^T + X = 0. \end{aligned} \tag{231}$$

Conversely, let $X^T = -X$. Then X^T and X commute, so $\exp(tX)^T \exp(tX) = \exp(tX^T)\exp(tX) = \exp(-tX)\exp(tX) = I$, and $\det \exp(tX) = e^{t \operatorname{tr} X} = e^0 = 1$ because the diagonal of a skew-symmetric matrix vanishes. By Eq. (224),

$$\mathfrak{so}(n) = \{X \in \mathfrak{gl}(n,\mathbb{R}) : X^T = -X\}, \qquad \dim \mathfrak{so}(n) = \frac{n(n-1)}{2}, \tag{232}$$

the dimension being the number of entries above the diagonal.

The regular-value argument in §88.1 justifies the group dimension in Eq. (229); Eq. (232) independently gives the same dimension for its Lie algebra. Axis and angle describe rotations, but require identifications at zero angle and at angle π and are not a single global coordinate chart.

88.3 The basis E_{ab} and its brackets

Let e_a be the standard basis column vectors of $\mathbb{R}^n$ and define

$$E_{ab} = e_a e_b^T - e_b e_a^T, \qquad (E_{ab})_{cd} = \delta_{ac}\delta_{bd} - \delta_{bc}\delta_{ad}. \tag{233}$$

Then $E_{ba} = -E_{ab}$, and $\{E_{ab} : a < b\}$ is a basis of $\mathfrak{so}(n)$. Using $e_b^T e_c = \delta_{bc}$, multiply out:

$$\begin{aligned} E_{ab}E_{cd} &= \delta_{bc}\, e_a e_d^T - \delta_{bd}\, e_a e_c^T - \delta_{ac}\, e_b e_d^T + \delta_{ad}\, e_b e_c^T, \\ E_{cd}E_{ab} &= \delta_{ad}\, e_c e_b^T - \delta_{bd}\, e_c e_a^T - \delta_{ac}\, e_d e_b^T + \delta_{bc}\, e_d e_a^T. \end{aligned}$$

Subtract the second line from the first and pair terms with the same Kronecker delta:

$$\begin{aligned} [E_{ab}, E_{cd}] &= \delta_{bc}\big(e_a e_d^T - e_d e_a^T\big) - \delta_{bd}\big(e_a e_c^T - e_c e_a^T\big) \\ &\quad - \delta_{ac}\big(e_b e_d^T - e_d e_b^T\big) + \delta_{ad}\big(e_b e_c^T - e_c e_b^T\big) \\ &= \delta_{bc}E_{ad} - \delta_{bd}E_{ac} - \delta_{ac}E_{bd} + \delta_{ad}E_{bc}. \end{aligned} \tag{234}$$

Equation (234) holds for every n. A pair of generators has a nonzero bracket only when the two planes share an axis.

88.4 Three dimensions: Frankel's E_i

For $n = 3$ the dimension is 3, and each coordinate plane has a unique normal axis. Define

$$E_i = -\tfrac{1}{2}\epsilon_{ijk}E_{jk}. \tag{235}$$

By Eq. (233),

$$(E_i)_{ab} = -\tfrac{1}{2}\epsilon_{ijk}\big(\delta_{ja}\delta_{kb} - \delta_{ka}\delta_{jb}\big) = -\tfrac{1}{2}\big(\epsilon_{iab} - \epsilon_{iba}\big) = -\epsilon_{iab}. \tag{236}$$

Reading off the entries, $(E_1)_{23} = -1$, $(E_1)_{32} = 1$, and similarly for E_2 and E_3:

$$E_1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}, \qquad E_2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}, \qquad E_3 = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \tag{237}$$

These are Frankel's Eq. (19.1), p. 492, and the basis (E_1, E_2, E_3) recorded from Kambe in §85.1. In the notation of Eq. (233), $E_1 = E_{32}$, $E_2 = E_{13}$, $E_3 = E_{21}$.

Frankel obtains E_3 from the rotations about the z axis,

$$R(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{238}$$

The addition formulas give $R(\theta)R(\varphi) = R(\theta + \varphi)$: the upper-left block of the product is

$$\begin{bmatrix} \cos\theta\cos\varphi - \sin\theta\sin\varphi & -\cos\theta\sin\varphi - \sin\theta\cos\varphi \\ \sin\theta\cos\varphi + \cos\theta\sin\varphi & -\sin\theta\sin\varphi + \cos\theta\cos\varphi \end{bmatrix} = \begin{bmatrix} \cos(\theta+\varphi) & -\sin(\theta+\varphi) \\ \sin(\theta+\varphi) & \cos(\theta+\varphi) \end{bmatrix}, \tag{239}$$

and the third row and column are those of I. So R is a one-parameter subgroup, and its generator is its derivative at the identity. Differentiate entry by entry, using $\frac{d}{d\theta}\cos\theta = -\sin\theta$ and $\frac{d}{d\theta}\sin\theta = \cos\theta$, then set $\theta = 0$:

$$\frac{dR}{d\theta}(0) = \begin{bmatrix} -\sin 0 & -\cos 0 & 0 \\ \cos 0 & -\sin 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = E_3. \tag{240}$$

In the other direction, sum $\exp(\theta E_3)$. Let $P = \operatorname{diag}(1, 1, 0)$. Direct multiplication gives $E_3^2 = -P$, and $P^2 = P$, $PE_3 = E_3$. Hence for $m \geq 1$

$$E_3^{2m} = (-1)^m P, \qquad E_3^{2m+1} = (-1)^m E_3, \tag{241}$$

the second also for $m = 0$. Separate the $m = 0$ even term and split the series into even and odd powers:

$$\begin{aligned} \exp(\theta E_3) &= I + \sum_{m=1}^{\infty} \frac{(-1)^m \theta^{2m}}{(2m)!} P + \sum_{m=0}^{\infty} \frac{(-1)^m \theta^{2m+1}}{(2m+1)!} E_3 \\ &= I + (\cos\theta - 1)P + \sin\theta\, E_3. \end{aligned} \tag{242}$$

The first sum is the cosine series without its leading 1. The diagonal of Eq. (242) is $(\cos\theta, \cos\theta, 1)$ and its off-diagonal part is $\sin\theta\, E_3$, which is Eq. (238). Thus $R(\theta) = \exp(\theta E_3)$.

88.5 The brackets of $\mathfrak{so}(3)$

Frankel states $[E_1, E_2] = E_3$. Multiply the matrices of Eq. (237). Row 2 of E_1 is $-e_3^T$, so row 2 of E_1E_2 is minus row 3 of E_2; row 3 of E_1 is e_2^T, so row 3 of E_1E_2 is row 2 of E_2, which is zero. Likewise for E_2E_1:

$$E_1E_2 = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \qquad E_2E_1 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \qquad E_1E_2 - E_2E_1 = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = E_3. \tag{243}$$

All nine brackets follow at once from the action of E_i on vectors. By Eq. (236), for $\mathbf{r} \in \mathbb{R}^3$,

$$(E_i\mathbf{r})_a = -\epsilon_{iab} r^b = \epsilon_{aib} r^b = (\mathbf{e}_i \times \mathbf{r})_a, \tag{244}$$

using $(\mathbf{u}\times\mathbf{v})_a = \epsilon_{acb} u^c v^b$ with $\mathbf{u} = \mathbf{e}_i$. The Jacobi identity for the cross product, $\mathbf{a}\times(\mathbf{b}\times\mathbf{r})+\mathbf{b}\times(\mathbf{r}\times\mathbf{a})+\mathbf{r}\times(\mathbf{a}\times\mathbf{b}) = 0$, rearranges to $\mathbf{a}\times(\mathbf{b}\times\mathbf{r}) - \mathbf{b}\times(\mathbf{a}\times\mathbf{r}) = (\mathbf{a}\times\mathbf{b})\times\mathbf{r}$. With $\mathbf{a} = \mathbf{e}_i$, $\mathbf{b} = \mathbf{e}_j$,

$$\begin{aligned} [E_i, E_j]\mathbf{r} &= \mathbf{e}_i \times (\mathbf{e}_j \times \mathbf{r}) - \mathbf{e}_j \times (\mathbf{e}_i \times \mathbf{r}) \\ &= (\mathbf{e}_i \times \mathbf{e}_j) \times \mathbf{r} = \epsilon_{ijk}\, \mathbf{e}_k \times \mathbf{r} = \epsilon_{ijk} E_k \mathbf{r}. \end{aligned}$$

Since $\mathbf{r}$ is arbitrary,

$$[E_i, E_j] = \epsilon_{ijk} E_k, \qquad c_{ij}{}^k = \epsilon_{ijk}, \tag{245}$$

Frankel's Eq. (19.2) and his structure constants of $SO(3)$.

88.6 Rotation about an axis and the angular velocity vector

Frankel's Figure 19.1 (p. 492) shows a point $\mathbf{r}$ carried around the z axis, and he states $\frac{d\mathbf{r}}{dt}(0) = \boldsymbol{\omega} \times \mathbf{r}(0)$. This follows from what a rotation about an axis does to a vector, together with the definition of the angular velocity vector.

Let $\mathbf{n}$ be a unit vector and $\mathbf{r}_0 \in \mathbb{R}^3$. Split $\mathbf{r}_0$ into its parts along and perpendicular to $\mathbf{n}$:

$$\mathbf{r}_0 = (\mathbf{n}\cdot\mathbf{r}_0)\,\mathbf{n} + \mathbf{r}_\perp, \qquad \mathbf{r}_\perp = \mathbf{r}_0 - (\mathbf{n}\cdot\mathbf{r}_0)\,\mathbf{n}, \qquad \mathbf{n}\cdot\mathbf{r}_\perp = 0. \tag{246}$$

Since $\mathbf{n}\times\mathbf{n} = 0$, we have $\mathbf{n}\times\mathbf{r}_0 = \mathbf{n}\times\mathbf{r}_\perp$. This vector is perpendicular to both $\mathbf{n}$ and $\mathbf{r}_\perp$, and has length $|\mathbf{n}||\mathbf{r}_\perp|\sin\frac{\pi}{2} = |\mathbf{r}_\perp|$. So $\mathbf{r}_\perp$ and $\mathbf{n}\times\mathbf{r}_0$ are perpendicular vectors of equal length spanning the plane perpendicular to $\mathbf{n}$. A rotation by the angle θ about $\mathbf{n}$, counterclockwise when viewed from the tip of $\mathbf{n}$, leaves the component along $\mathbf{n}$ unchanged and turns $\mathbf{r}_\perp$ through θ toward $\mathbf{n}\times\mathbf{r}_0$:

$$\mathbf{r}(\theta) = (\mathbf{n}\cdot\mathbf{r}_0)\,\mathbf{n} + \cos\theta\,\mathbf{r}_\perp + \sin\theta\,\mathbf{n}\times\mathbf{r}_0. \tag{247}$$

For $\mathbf{n} = \mathbf{e}_3$ and $\mathbf{r}_0 = (x, y, z)$, the perpendicular part is $\mathbf{r}_\perp = (x, y, 0)$ and $\mathbf{e}_3 \times \mathbf{r}_0 = (-y, x, 0)$. Equation (247) then gives $(x\cos\theta - y\sin\theta,\ x\sin\theta + y\cos\theta,\ z)$, which is $R(\theta)\mathbf{r}_0$ for the matrix of Eq. (238). So "counterclockwise about $\mathbf{n}$" is the same sense as Frankel's $R(\theta)$.

Now let the angle change in time, $\theta = \theta(t)$ with $\theta(0) = 0$, about the fixed axis $\mathbf{n}$. The *angular velocity vector* is defined as

$$\boldsymbol{\omega} = \frac{d\theta}{dt}\,\mathbf{n}, \tag{248}$$

the rate of turning times the axis, oriented by the right-hand rule. Differentiate Eq. (247) with $\theta = \theta(t)$ by the chain rule. The first term is constant:

$$\frac{d\mathbf{r}}{dt} = \frac{d\theta}{dt}\Big(-\sin\theta\,\mathbf{r}_\perp + \cos\theta\,\mathbf{n}\times\mathbf{r}_0\Big). \tag{249}$$

At $t = 0$, $\theta = 0$, so $\sin\theta = 0$ and $\cos\theta = 1$:

$$\frac{d\mathbf{r}}{dt}(0) = \frac{d\theta}{dt}(0)\,\mathbf{n}\times\mathbf{r}_0 = \boldsymbol{\omega}\times\mathbf{r}(0). \tag{250}$$

This is Frankel's statement. It is the definition (248) applied to the geometry of Eq. (247), not a separate fact.

88.7 One-parameter subgroups are exponentials

Frankel next says that the rotations at constant angular velocity form a one-parameter subgroup, "of the form $R(t) = e^{tS}$ for some skew symmetric matrix S". Two facts are used: these rotations form a one-parameter subgroup, and every one-parameter subgroup is an exponential.

The rotations about a fixed axis form a one-parameter subgroup. Let $R_{\mathbf{n}}(\theta)$ denote the linear map (247). Choose a unit vector $\mathbf{u} \perp \mathbf{n}$ and put $\mathbf{v} = \mathbf{n} \times \mathbf{u}$. Then $(\mathbf{u}, \mathbf{v}, \mathbf{n})$ is a right-handed orthonormal basis, since $\mathbf{u}\times\mathbf{v} = \mathbf{u}\times(\mathbf{n}\times\mathbf{u}) = (\mathbf{u}\cdot\mathbf{u})\mathbf{n} - (\mathbf{u}\cdot\mathbf{n})\mathbf{u} = \mathbf{n}$. Applying Eq. (247) to $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{n}$, and using $\mathbf{n}\times\mathbf{v} = \mathbf{n}\times(\mathbf{n}\times\mathbf{u}) = -\mathbf{u}$,

$$R_{\mathbf{n}}(\theta)\mathbf{u} = \cos\theta\,\mathbf{u} + \sin\theta\,\mathbf{v}, \quad R_{\mathbf{n}}(\theta)\mathbf{v} = -\sin\theta\,\mathbf{u} + \cos\theta\,\mathbf{v}, \quad R_{\mathbf{n}}(\theta)\mathbf{n} = \mathbf{n}. \tag{251}$$

In the basis $(\mathbf{u}, \mathbf{v}, \mathbf{n})$ the matrix of $R_{\mathbf{n}}(\theta)$ is therefore the matrix of Eq. (238), and the addition formulas shown there give $R_{\mathbf{n}}(\theta)R_{\mathbf{n}}(\varphi) = R_{\mathbf{n}}(\theta+\varphi)$. At constant angular velocity, $\theta(t) = \frac{d\theta}{dt}t$, so $R(t) = R_{\mathbf{n}}(\frac{d\theta}{dt}t)$ satisfies $R(t)R(s) = R(t+s)$ and $R(0) = I$.

Every differentiable one-parameter subgroup is an exponential. Let $R\colon \mathbb{R} \to GL(n,\mathbb{R})$ be differentiable with $R(0) = I$ and $R(t+s) = R(t)R(s)$ for all s, t, and put $S = \frac{dR}{dt}(0)$. Differentiate $R(t+s) = R(t)R(s)$ with respect to s, holding t fixed, and set $s = 0$:

$$\frac{dR}{dt}(t) = R(t)S. \tag{252}$$

Let $M(t) = R(t)\exp(-tS)$. Since $\frac{d}{dt}\exp(-tS) = -S\exp(-tS)$, the product rule and Eq. (252) give

$$\frac{dM}{dt}(t) = R(t)S\exp(-tS) - R(t)S\exp(-tS) = 0. \tag{253}$$

So $M(t) = M(0) = I$, and multiplying on the right by $\exp(tS)$,

$$R(t) = \exp(tS), \qquad S = \frac{dR}{dt}(0). \tag{254}$$

This is the content of Theorem 63(ii) for one-parameter subgroups, and of Frankel's Section 15.2 (p. 398). When $R(t) \in SO(3)$, the generator S is skew-symmetric by Eq. (231).

Identifying S. With $\mathbf{r}(t) = R(t)\mathbf{r}_0 = \exp(tS)\mathbf{r}_0$, differentiation at $t = 0$ gives $\frac{d\mathbf{r}}{dt}(0) = S\mathbf{r}_0$. Comparing with Eq. (250),

$$S\mathbf{r}_0 = \boldsymbol{\omega} \times \mathbf{r}_0 \quad \text{for every } \mathbf{r}_0 \in \mathbb{R}^3. \tag{255}$$

88.8 The generators act by cross products

Frankel notes that $E_j(\mathbf{r}) = \mathbf{e}_j \times \mathbf{r}$. Check it for each j with $\mathbf{r} = (x, y, z)$. Multiplying the matrices of Eq. (237) into $\mathbf{r}$, and computing the cross products from $\mathbf{a}\times\mathbf{b} = (a^2b^3 - a^3b^2,\ a^3b^1 - a^1b^3,\ a^1b^2 - a^2b^1)$,

$$\begin{aligned}
E_1\mathbf{r} &= (0,\ -z,\ y), & \mathbf{e}_1\times\mathbf{r} &= (0\cdot z - 0\cdot y,\ 0\cdot x - 1\cdot z,\ 1\cdot y - 0\cdot x) = (0,\ -z,\ y),\\
E_2\mathbf{r} &= (z,\ 0,\ -x), & \mathbf{e}_2\times\mathbf{r} &= (1\cdot z - 0\cdot y,\ 0\cdot x - 0\cdot z,\ 0\cdot y - 1\cdot x) = (z,\ 0,\ -x),\\
E_3\mathbf{r} &= (-y,\ x,\ 0), & \mathbf{e}_3\times\mathbf{r} &= (0\cdot z - 1\cdot y,\ 1\cdot x - 0\cdot z,\ 0\cdot y - 0\cdot x) = (-y,\ x,\ 0).
\end{aligned} \tag{256}$$

In index form this is Eq. (244), used above for the brackets. The cross product is linear in its first factor, so for $\boldsymbol{\omega} = \omega^j\mathbf{e}_j$

$$\boldsymbol{\omega}\times\mathbf{r} = \omega^j\,\mathbf{e}_j\times\mathbf{r} = \omega^j E_j\mathbf{r}, \qquad \widehat{\boldsymbol{\omega}} := \omega^j E_j. \tag{257}$$

This $\widehat{\boldsymbol{\omega}}$ is the hat map of the rigid-body part, defined after Eq. (552), and it is Frankel's $\mathbf{E}\bullet\boldsymbol{\omega}$. Holm calls the same map the hat map.

Equations (255) and (257) give $S = \widehat{\boldsymbol{\omega}}$, and Eq. (254) becomes

$$R(t) = \exp\big(t\,\omega^j E_j\big) = \exp\big(t\,\widehat{\boldsymbol{\omega}}\big), \tag{258}$$

Frankel's Eq. (19.3), p. 493. At constant rate the angle is $\theta = t\frac{d\theta}{dt}$, so $t\boldsymbol{\omega} = \theta\mathbf{n}$ by Eq. (248), and

$$R_{\mathbf{n}}(\theta) = \exp\big(\theta\,n^j E_j\big), \tag{259}$$

Frankel's Eq. (19.4): the rotation through θ about the unit vector $\mathbf{n}$.

By Eq. (245), the hat map is a Lie algebra isomorphism from $(\mathbb{R}^3, \times)$ to $\mathfrak{so}(3)$:

$$[\widehat{\mathbf{u}}, \widehat{\mathbf{v}}] = u^i v^j [E_i, E_j] = u^i v^j \epsilon_{ijk} E_k = (\mathbf{u}\times\mathbf{v})^k E_k = \widehat{\mathbf{u}\times\mathbf{v}}. \tag{260}$$

89 The group $SU(2)$ and its Lie algebra $\mathfrak{su}(2)$

This section follows Frankel's Section 19.1b (pp. 493–495) in the notation of §87.1.

89.1 The group

As in the Lie-groups part,

$$SU(2) = \{U \in GL(2,\mathbb{C}) : UU^\dagger = 1,\ \det U = 1\}. \tag{261}$$

Dropping $\det U = 1$ gives the unitary group $U(2)$. For square matrices $UU^\dagger = 1$ and $U^\dagger U = 1$ are equivalent, and they say that U preserves the Hermitian inner product $\langle z, w\rangle = z^\dagger w$ on $\mathbb{C}^2$: $(Uz)^\dagger(Uw) = z^\dagger U^\dagger U w = z^\dagger w$.

Every element has a standard form. Write $U = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Since $\det U = 1$, the inverse is $U^{-1} = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$; since $UU^\dagger = 1$, the inverse is also $U^\dagger = \begin{bmatrix} \bar a & \bar c \\ \bar b & \bar d \end{bmatrix}$. Equating entries gives $d = \bar a$ and $-b = \bar c$, that is $b = -\bar c$. Then $\det U = a\bar a - (-\bar c)c = |a|^2 + |c|^2$. Writing $\alpha = a$, $\beta = c$,

$$SU(2) = \left\{ \begin{bmatrix} \alpha & -\bar\beta \\ \beta & \bar\alpha \end{bmatrix} : \alpha, \beta \in \mathbb{C},\ |\alpha|^2 + |\beta|^2 = 1 \right\}. \tag{262}$$

Conversely, every matrix of this form satisfies $UU^\dagger = 1$ and $\det U = 1$, by direct multiplication.

Dimension. $GL(2,\mathbb{C})$ has four complex entries, so real dimension 8. The matrix $U^\dagger U$ is Hermitian for every U, so $U^\dagger U = 1$ imposes only four real conditions: the two real diagonal entries and the real and imaginary parts of one off-diagonal entry. Hence $\dim U(2) = 8 - 4 = 4$. On $U(2)$, $|\det U|^2 = \det(U^\dagger)\det U = \det(U^\dagger U) = 1$, so $\det U = e^{i\phi}$ for some real ϕ, and $\det U = 1$ is one further real condition:

$$\dim SU(2) = 4 - 1 = 3. \tag{263}$$

Equation (262) agrees: $(\alpha, \beta) \in \mathbb{C}^2$ has four real parameters and $|\alpha|^2 + |\beta|^2 = 1$ is one real equation.

89.2 The Lie algebra

Let $U(t)$ be a curve in $SU(2)$ with $U(0) = 1$ and $X = \frac{dU}{dt}(0)$. Differentiating $U(t)U(t)^\dagger = 1$ by the product rule at $t = 0$ gives $X + X^\dagger = 0$. For the determinant use $\det\exp(tX) = e^{t\,\mathrm{tr}\,X}$: by Theorem 63(ii), $\exp(tX) \in SU(2)$ for all t, so $e^{t\,\mathrm{tr}\,X} = 1$ for all t; differentiating at $t = 0$ gives $\mathrm{tr}\,X = 0$. Conversely, if $X^\dagger = -X$ and $\mathrm{tr}\,X = 0$, then $\exp(tX)^\dagger \exp(tX) = \exp(-tX)\exp(tX) = 1$ and $\det\exp(tX) = e^0 = 1$. So

$$\mathfrak{su}(2) = \{X \in \mathfrak{gl}(2,\mathbb{C}) : X + X^\dagger = 0,\ \mathrm{tr}\,X = 0\}, \tag{264}$$

the definition recorded in §85.0.1. Frankel writes $\mathfrak{su}(2) = \mathfrak{g}$ and calls its elements skew-Hermitian matrices with trace 0.

Dimension. $X^\dagger = -X$ makes the diagonal entries purely imaginary (two real parameters) and fixes $X_{21} = -\overline{X_{12}}$ (two real parameters in X_{12}). Then $\mathrm{tr}\,X$ is purely imaginary, and $\mathrm{tr}\,X = 0$ is one real condition. Every $X \in \mathfrak{su}(2)$ is therefore

$$X = \begin{bmatrix} ia & w \\ -\bar w & -ia \end{bmatrix}, \qquad a \in \mathbb{R},\ w \in \mathbb{C}, \qquad \dim\mathfrak{su}(2) = 4 - 1 = 3 = \dim SU(2). \tag{265}$$

$\mathfrak{su}(2)$ is a vector space over $\mathbb{R}$, not over $\mathbb{C}$: if $X \in \mathfrak{su}(2)$ is nonzero then $(iX)^\dagger = -iX^\dagger = iX$, so iX is Hermitian and not in $\mathfrak{su}(2)$.

Hermitian generators. Multiplication by i is therefore a real linear isomorphism from $\mathfrak{su}(2)$ onto the traceless Hermitian matrices, which Frankel calls $i\mathfrak{g}$ and we write $i\,\mathfrak{su}(2)$. Physics texts usually work in $i\,\mathfrak{su}(2)$ (Hermitian generators, observables) and write group elements as $\exp(-iH)$ with H Hermitian; Frankel follows this with $e^{th/i}$.

Bases. The Pauli matrices $\sigma_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $\sigma_2 = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$, $\sigma_3 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ are Hermitian and traceless. The products

$$-i\sigma_1 = \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix}, \qquad -i\sigma_2 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \qquad -i\sigma_3 = \begin{bmatrix} -i & 0 \\ 0 & i \end{bmatrix} \tag{266}$$

give $x^k(-i\sigma_k) = \begin{bmatrix} -ix^3 & -x^2 - ix^1 \\ x^2 - ix^1 & ix^3 \end{bmatrix}$. This equals the X of Eq. (265) exactly when $x^3 = -a$, $x^2 = -\operatorname{Re} w$, $x^1 = -\operatorname{Im} w$, so every $X \in \mathfrak{su}(2)$ is $x^k(-i\sigma_k)$ for unique real x^k. Thus

$$\{-i\sigma_k\} = \{\sigma_k/i\} \text{ is a basis of } \mathfrak{su}(2), \qquad \{\sigma_k\} \text{ is a basis of } i\,\mathfrak{su}(2). \tag{267}$$

These are Frankel's bases. Holm uses the same basis $-i\sigma_k$ of $\mathfrak{su}(2)$ (Section 3.2.3, p. 102). Kosmann-Schwarzbach's basis ξ_k of §85.0.1 is related to them by $\sigma_1 = -2i\xi_1$, $\sigma_2 = 2i\xi_2$, $\sigma_3 = -2i\xi_3$ ([85], Ch. 5 §1.1, p. 71), so $\xi_k = \pm\frac{1}{2}(-i\sigma_k)$ with the signs $(-,+,-)$; the meaning of the signs is Eq. (316).

89.3 Vectors as traceless Hermitian matrices

Define the real linear map

$$\Sigma\colon \mathbb{R}^3 \to i\,\mathfrak{su}(2), \qquad \Sigma(\mathbf{x}) = x^k\sigma_k = \begin{bmatrix} z & x - iy \\ x + iy & -z \end{bmatrix}, \tag{268}$$

for $\mathbf{x} = (x, y, z)$; the matrix is $x\sigma_1 + y\sigma_2 + z\sigma_3$ added entry by entry, and $\Sigma(\mathbf{e}_k) = \sigma_k$. This is Frankel's Eq. (19.7), where $\Sigma(\mathbf{x})$ is written $\mathbf{x}_* = \mathbf{x} \bullet \boldsymbol{\sigma}$. Holm writes the same matrix as $\mathbf{r} \cdot \boldsymbol{\sigma}$ (Eq. (3.2.13), p. 101). Multiplying by $-i$ lands in $\mathfrak{su}(2)$ instead: Holm's tilde map (Eq. (3.2.14), p. 102) is $\widetilde{\mathbf{q}} = -i\,\mathbf{q} \cdot \boldsymbol{\sigma} = -i\Sigma(\mathbf{q})$, which is our pure quaternion $q^a e_a$ under Eq. (328). Kosmann-Schwarzbach identifies $\mathbb{R}^3$ with $\mathfrak{su}(2)$ by $a \mapsto X_a = a^1\mathcal{I} + a^2\mathcal{J} + a^3\mathcal{K} = 2a^k\xi_k$ (Ch. 5 §2.2, p. 77), which differs from $-i\Sigma$ in two signs; see §95.1.

To invert Σ, use Frankel's anticommutator $\sigma_j\sigma_k + \sigma_k\sigma_j = 2\delta_{jk}I$ (Eq. (19.18)). Taking the trace, $2\operatorname{tr}(\sigma_j\sigma_k) = 2\delta_{jk}\operatorname{tr} I = 4\delta_{jk}$, so $\operatorname{tr}(\sigma_j\sigma_k) = 2\delta_{jk}$. Then

$$\operatorname{tr}\big(\Sigma(\mathbf{x})\sigma_j\big) = x^k \operatorname{tr}(\sigma_k\sigma_j) = 2x^j, \qquad x^j = \tfrac{1}{2}\operatorname{tr}\big(\Sigma(\mathbf{x})\sigma_j\big), \tag{269}$$

Frankel's Eq. (19.8). With his real scalar product $\langle h, h'\rangle = \operatorname{tr}(hh')$ on $i\,\mathfrak{su}(2)$ (Eq. (19.9)),

$$\langle \Sigma(\mathbf{x}), \Sigma(\mathbf{y})\rangle = x^j y^k \operatorname{tr}(\sigma_j\sigma_k) = 2\,\mathbf{x}\cdot\mathbf{y}, \tag{270}$$

so the Pauli matrices are orthogonal, each of length $\sqrt{2}$. The same symmetrization as in Eq. (331) gives $\Sigma(\mathbf{x})^2 = x^j x^k \sigma_j\sigma_k = |\mathbf{x}|^2 I$, and from the matrix in Eq. (268), $\det\Sigma(\mathbf{x}) = -z^2 - (x - iy)(x + iy) = -|\mathbf{x}|^2$.

89.4 The adjoint representation

Frankel writes "Recall that every Lie group G acts on its Lie algebra $\mathfrak{g}$ by the adjoint action $\mathrm{Ad}\colon G \to Gl(\mathfrak{g})$, $\mathrm{Ad}(g)(X) = gXg^{-1}$" (p. 494). Two words need definitions. An *action* of a group G on a set M is a map $G \times M \to M$, $(g, x) \mapsto g \cdot x$, with $1 \cdot x = x$ and $(gh) \cdot x = g \cdot (h \cdot x)$. When $M = E$ is a vector space and each map $x \mapsto g \cdot x$ is linear, the action is the same thing as a group homomorphism $\rho\colon G \to GL(E)$, $\rho(g)x = g \cdot x$: a *representation* (E, ρ) of G, in Kosmann-Schwarzbach's notation (§85). So the claim is that Ad is a representation of G on its own Lie algebra. It is short enough to prove.

Proposition 92 (adjoint representation)**.** *Let $G \subseteq GL(n, \mathbb{C})$ be a matrix Lie group with Lie algebra $\mathfrak{g}$, and for $g \in G$ put $\mathrm{Ad}_g Y = gYg^{-1}$. Then*

(i) $\mathrm{Ad}_g Y \in \mathfrak{g}$ *for every* $Y \in \mathfrak{g}$*;*

(ii) $\mathrm{Ad}_g\colon \mathfrak{g} \to \mathfrak{g}$ *is linear and invertible, with inverse* $\mathrm{Ad}_{g^{-1}}$*;*

(iii) $\mathrm{Ad}_1 = \mathrm{id}$ *and* $\mathrm{Ad}_{gh} = \mathrm{Ad}_g \circ \mathrm{Ad}_h$*, so* $\mathrm{Ad}\colon G \to GL(\mathfrak{g})$ *is a representation;*

(iv) $\mathrm{Ad}_g[X, Y] = [\mathrm{Ad}_g X, \mathrm{Ad}_g Y]$*, so each* Ad_g *is a Lie algebra automorphism;*

(v) Ad_g *is the derivative at* 1 *of the conjugation* $\mathcal{C}_g(h) = ghg^{-1}$*, and* $\frac{d}{dt}\mathrm{Ad}_{\exp(tX)}Y\big|_{t=0} = [X, Y] = \mathrm{ad}_X Y$*.*

Proof. (i) In $(gYg^{-1})^k$ every inner $g^{-1}g$ cancels, for example $(gYg^{-1})^2 = gYg^{-1}gYg^{-1} = gY^2g^{-1}$. Summing the exponential series,

$$\exp\big(t\,gYg^{-1}\big) = \sum_{k=0}^{\infty} \frac{t^k}{k!} gY^k g^{-1} = g\exp(tY)g^{-1} = \mathcal{C}_g\big(\exp(tY)\big). \tag{271}$$

The right side is a product of three elements of G, so it lies in G for all t, and Theorem 63(ii) gives $gYg^{-1} \in \mathfrak{g}$.

(ii) Matrix multiplication is linear in Y. Also $\mathrm{Ad}_{g^{-1}}\mathrm{Ad}_g Y = g^{-1}gYg^{-1}g = Y$, and likewise in the other order.

(iii) $\mathrm{Ad}_1 Y = Y$, and $\mathrm{Ad}_{gh}Y = ghY(gh)^{-1} = g(hYh^{-1})g^{-1} = \mathrm{Ad}_g(\mathrm{Ad}_h Y)$, using $(gh)^{-1} = h^{-1}g^{-1}$.

(iv) Insert $g^{-1}g = 1$ between the factors: $g(XY - YX)g^{-1} = gXg^{-1}gYg^{-1} - gYg^{-1}gXg^{-1} = [\mathrm{Ad}_g X, \mathrm{Ad}_g Y]$.

(v) By Eq. (271), $\mathcal{C}_g$ carries the curve $\exp(tY)$, whose velocity at $t = 0$ is Y, to the curve $\exp(t\,gYg^{-1})$, whose velocity at $t = 0$ is gYg^{-1}. So the derivative of $\mathcal{C}_g$ at 1 sends Y to $\mathrm{Ad}_g Y$. For the last statement, differentiate $\exp(tX)Y\exp(-tX)$ at $t = 0$ by the product rule: $XY + Y(-X) = [X, Y]$. □

Kosmann-Schwarzbach takes (v) as the definition, $\mathrm{Ad}_g = D_1\mathcal{C}_g$, and derives $\mathrm{Ad}_g Y = gYg^{-1}$ (§85.3); Frankel takes the formula as the definition. By (v) the two agree.

	Frankel	Kosmann-Schwarzbach	ours
conjugation	—	$\mathcal{C}_g$	$\mathcal{C}_g$
adjoint action	$\mathrm{Ad}(g)(X)$	$\mathrm{Ad}_g = D_1\mathcal{C}_g$	$\mathrm{Ad}_g X$
its target	$Gl(\mathfrak{g})$	$GL(\mathfrak{g})$	$GL(\mathfrak{g})$
representation	"by 3×3 matrices"	(E,ρ), $\rho\colon G \to GL(E)$	(E,ρ)
derivative of Ad	Ad_*	ad_X	ad_X, $d\Phi$

For $G = SU(2)$, Frankel lets $SU(2)$ act on $i\,\mathfrak{su}(2)$ instead of $\mathfrak{su}(2)$ and still calls the action Ad (his Eq. (19.10)). This is the same action: Ad_u is complex linear on 2×2 matrices, so $\mathrm{Ad}_u(iX) = i\,\mathrm{Ad}_u(X)$. Directly, for Hermitian traceless h and $u^{-1} = u^\dagger$, $(uhu^{-1})^\dagger = (uhu^\dagger)^\dagger = uh^\dagger u^\dagger = uhu^{-1}$ and $\mathrm{tr}(uhu^{-1}) = \mathrm{tr}(hu^{-1}u) = \mathrm{tr}\,h = 0$.

89.5 The homomorphism Φ

Since $u\Sigma(\mathbf{x})u^{-1}$ is traceless Hermitian, it is Σ of exactly one vector. Define $\Phi(u)\mathbf{x}$ to be that vector:

$$\Sigma\big(\Phi(u)\mathbf{x}\big) = u\,\Sigma(\mathbf{x})\,u^{-1} = \mathrm{Ad}_u\Sigma(\mathbf{x}), \qquad u \in SU(2),\ \mathbf{x} \in \mathbb{R}^3. \tag{272}$$

$\Phi(u)$ is linear in $\mathbf{x}$, and by Eq. (269) its matrix entries are

$$\Phi(u)_{jk} = \tfrac{1}{2}\,\mathrm{tr}\big(\sigma_j\, u\sigma_k u^{-1}\big). \tag{273}$$

Equation (272) says that this square commutes:

$$\begin{array}{ccc} \mathbb{R}^3 & \xrightarrow{\Phi(u)} & \mathbb{R}^3 \\ {\scriptstyle\Sigma}\downarrow & & \downarrow{\scriptstyle\Sigma} \\ i\,\mathfrak{su}(2) & \xrightarrow[\mathrm{Ad}_u]{} & i\,\mathfrak{su}(2) \end{array} \tag{274}$$

Frankel writes both $\Phi(u)$ and Ad_u as $\mathrm{Ad}(u)$, and writes $\mathrm{Ad}\,u\,\mathbf{x} \sim \mathrm{Ad}\,u\,\mathbf{x}_*$ for Eq. (272): the $\sim$ means "corresponds under Σ". Because $\Sigma(\mathbf{x})/i \in \mathfrak{su}(2)$ and $\mathrm{Ad}_u(\Sigma(\mathbf{x})/i) = \mathrm{Ad}_u(\Sigma(\mathbf{x}))/i$, $\Phi(u)$ is also the matrix of Ad_u on $\mathfrak{su}(2)$ in the basis σ_k/i. So Φ is the adjoint representation of $SU(2)$ written as 3×3 real matrices, which is why Frankel calls it Ad.

Φ is a homomorphism. Apply Eq. (272) twice:

$$\Sigma\big(\Phi(uu')\mathbf{x}\big) = uu'\Sigma(\mathbf{x})(uu')^{-1} = u\,\Sigma\big(\Phi(u')\mathbf{x}\big)u^{-1} = \Sigma\big(\Phi(u)\Phi(u')\mathbf{x}\big), \tag{275}$$

and Σ is injective, so $\Phi(uu') = \Phi(u)\Phi(u')$.

$\Phi(u)$ is orthogonal. By Eq. (270) and cyclicity of the trace,

$$\begin{aligned} 2|\Phi(u)\mathbf{x}|^2 &= \mathrm{tr}\Big(\Sigma\big(\Phi(u)\mathbf{x}\big)^2\Big) = \mathrm{tr}\big(u\Sigma(\mathbf{x})u^{-1}u\Sigma(\mathbf{x})u^{-1}\big) \\ &= \mathrm{tr}\big(u\Sigma(\mathbf{x})^2u^{-1}\big) = \mathrm{tr}\big(\Sigma(\mathbf{x})^2\big) = 2|\mathbf{x}|^2. \end{aligned} \tag{276}$$

A linear map that preserves lengths preserves inner products, since $2\mathbf{x}\cdot\mathbf{y} = |\mathbf{x}+\mathbf{y}|^2 - |\mathbf{x}|^2 - |\mathbf{y}|^2$. So $\Phi(u)^T\Phi(u) = I$ and $\det\Phi(u) = \pm 1$. That the sign is $+1$ needs the topology of $SU(2)$.

89.6 What "yields the standard representation of $SO(3)$" means

Frankel announces (p. 493): "the adjoint representation $\mathrm{Ad}(g)Y = gYg^{-1}$ of $SU(2)$ on its 3-dimensional Lie algebra $\mathfrak{su}(2)$ yields (see Theorem (19.12)) the standard representation of $SO(3)$ on $\mathbb{R}^3$." Here are the definitions involved.

The *standard* (defining) representation of a matrix group $G \subseteq GL(n,\mathbb{R})$ is the inclusion: $g \in G$ acts on $\mathbf{x} \in \mathbb{R}^n$ by matrix multiplication, $\mathbf{x} \mapsto g\mathbf{x}$. For $SO(3)$ this is the rotation of vectors in $\mathbb{R}^3$. Composing it with Φ gives a representation $(\mathbb{R}^3, \Phi)$ of $SU(2)$.

For representations (E_1, ρ_1) and (E_2, ρ_2) of a group G, an *intertwining operator* is a linear map $T\colon E_1 \to E_2$ with $\rho_2(g)\,T = T\,\rho_1(g)$ for all $g \in G$. Following Kosmann-Schwarzbach's Definition 1.12, ρ_1 and ρ_2 are *equivalent* if there is a bijective intertwining operator for them. Equivalent representations are the same representation written in two coordinate systems: T translates one into the other.

Proposition 93. *The map $\Sigma\colon \mathbb{R}^3 \to i\,\mathfrak{su}(2)$ is a bijective intertwining operator from $(\mathbb{R}^3, \Phi)$ to $(i\,\mathfrak{su}(2), \mathrm{Ad})$. Likewise $-i\Sigma\colon \mathbb{R}^3 \to \mathfrak{su}(2)$ is one from $(\mathbb{R}^3, \Phi)$ to $(\mathfrak{su}(2), \mathrm{Ad})$. Hence the adjoint representation of $SU(2)$ is equivalent to the standard representation of $SO(3)$ composed with Φ. By Theorem 64, every rotation occurs as some $\Phi(u)$.*

Proof. Eq. (272) reads $\mathrm{Ad}_u \circ \Sigma = \Sigma \circ \Phi(u)$ for every $u \in SU(2)$, which is the intertwining condition. Σ is linear, and it is bijective because $\{\sigma_k\}$ is a basis of $i\,\mathfrak{su}(2)$ (Eq. (267)). Multiplying by $-i$ commutes with Ad_u and maps $i\,\mathfrak{su}(2)$ bijectively onto $\mathfrak{su}(2)$. The last sentence is Theorem 64. □

So "Ad yields the standard representation" combines two statements. In the coordinates Σ, the operators Ad_u *are* rotation matrices, by Proposition 93 and $\det \Phi(u) = 1$. And all rotations are obtained, by Theorem 64.

What suggests this before any computation. Four facts, each already available, point to it.

1. Dimension: $\dim \mathfrak{su}(2) = 3$ (Eq. (265)), so Ad_u is a 3×3 real matrix.
2. Invariant inner product: $\langle h, h' \rangle = \mathrm{tr}(hh')$ is positive definite on $i\,\mathfrak{su}(2)$ (Eq. (270)) and Ad_u preserves it, by cyclicity of the trace. A linear map preserving a positive-definite inner product is orthogonal in an orthonormal basis, so $\mathrm{Ad}_u \in O(3)$.
3. Matching brackets: the basis $\sigma_a/(2i)$ of $\mathfrak{su}(2)$ satisfies $[\sigma_a/2i, \sigma_b/2i] = \epsilon_{abc}\,\sigma_c/2i$ (Eq. (315)), exactly the relations $[E_a, E_b] = \epsilon_{abc}E_c$ of $\mathfrak{so}(3)$ (Eq. (245)). These are Frankel's "remarks after (19.6)". The Lie algebras are isomorphic, and by Proposition 92(v) the derivative of Ad is ad, which carries $\sigma_a/2i$ to E_a (Eq. (292)).
4. Connectedness: $SU(2)$ is connected (§90), so $\det \mathrm{Ad}_u$ cannot jump from $+1$ to -1.

Frankel adds a fifth, which he does not prove here (p. 493): $SU(2)$ is simply connected, and Lie theory then lifts the Lie algebra isomorphism $\mathfrak{su}(2) \cong \mathfrak{so}(3)$ to a group homomorphism $SU(2) \to SO(3)$. In these notes Φ is constructed directly instead, by Eq. (272).

90 $SU(2)$ is the 3-sphere

This section follows Frankel's Section 19.1c (p. 495). Let

$$S^3 = \{(z_1, z_2)^T \in \mathbb{C}^2 : |z_1|^2 + |z_2|^2 = 1\} \subset \mathbb{C}^2 \cong \mathbb{R}^4. \tag{277}$$

A unitary matrix preserves $|z|^2 = z^\dagger z$, so each $u \in SU(2)$ maps S^3 to itself. This is what Frankel means by "$SU(2)\colon S^3 \to S^3$; this is the meaning of being unitary".

90.1 The action is transitive

A group G *acts transitively* on a set M if for every pair of points $x, y \in M$ some $g \in G$ has $gx = y$ (Frankel, Section 17.2a, p. 456): one can move from any point to any other. It is enough to reach every point from one fixed point x_0. If $x = gx_0$ and $y = g'x_0$, then $g'g^{-1}x = y$.

Take $x_0 = (1, 0)^T$ and any $(z_1, z_2)^T \in S^3$. The matrix

$$u = \begin{bmatrix} z_1 & -\bar{z}_2 \\ z_2 & \bar{z}_1 \end{bmatrix} \tag{278}$$

is Eq. (262) with $\alpha = z_1$, $\beta = z_2$, so $u \in SU(2)$. Also $u(1,0)^T$ is the first column of u, which is $(z_1, z_2)^T$. This is Frankel's Eq. (19.11). So $SU(2)$ acts transitively on S^3.

90.2 From the orbit to a homeomorphism

Frankel's Fundamental Principle (17.10), Section 17.2a, p. 457, states: *if G acts transitively on M, $x_0 \in M$, and $H = \{g \in G : gx_0 = x_0\}$ is the stability (isotropy) subgroup of x_0, then $gH \mapsto gx_0$ is a one-to-one correspondence between the left cosets G/H and M.* It comes from the orbit map $g \mapsto gx_0$. This map is onto by transitivity. Two elements have the same image exactly when they lie in one coset: $gx_0 = g'x_0 \iff g^{-1}g'x_0 = x_0 \iff g^{-1}g' \in H \iff gH = g'H$. Theorem (17.11), p. 458, adds that for a closed subgroup H of a Lie group G, G/H is a manifold of dimension $\dim G - \dim H$.

For $x_0 = (1,0)^T$, write u in the form (262). Then $u(1,0)^T = (\alpha,\beta)^T$, and $u(1,0)^T = (1,0)^T$ forces $\alpha = 1$, $\beta = 0$, hence

$$u = \begin{bmatrix} 1 & -\bar{0} \\ 0 & \bar{1} \end{bmatrix} = I, \qquad H = \{I\}. \tag{279}$$

The stability subgroup is trivial because Eq. (262) determines the whole matrix from its first column. Each coset uH is the single element u, so $G/H = SU(2)$, and the correspondence of (17.10) is

$$F\colon SU(2) \to S^3, \qquad F(u) = u \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} u_{11} \\ u_{21} \end{bmatrix}. \tag{280}$$

Its inverse is the matrix of Eq. (278), $F^{-1}(z_1,z_2) = \begin{bmatrix} z_1 & -\bar{z}_2 \\ z_2 & \bar{z}_1 \end{bmatrix}$. Check both compositions. $F(F^{-1}(z_1,z_2))$ is the first column $(z_1,z_2)^T$. For u in the form (262), $F(u) = (\alpha,\beta)^T$ and $F^{-1}(\alpha,\beta) = u$. In real coordinates the entries of F and F^{-1} are linear in the real and imaginary parts of their arguments, so both maps are continuous, indeed smooth. Therefore F is a homeomorphism:

$$SU(2) \approx S^3. \tag{281}$$

The dimensions agree: S^3 is cut out of $\mathbb{R}^4$ by one equation, giving $4 - 1 = 3 = \dim SU(2)$.

In quaternion form, put $\alpha = q_0 - iq_3$ and $\beta = q_2 - iq_1$. By Eq. (266),

$$q_0 I + q^a(-i\sigma_a) = \begin{bmatrix} q_0 - iq_3 & -iq_1 - q_2 \\ -iq_1 + q_2 & q_0 + iq_3 \end{bmatrix} = \begin{bmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{bmatrix}, \tag{282}$$

and $|\alpha|^2 + |\beta|^2 = q_0^2 + q_1^2 + q_2^2 + q_3^2 = |q|^2$. Under Eq. (328), $SU(2)$ is the group of unit quaternions, the unit sphere $S^3 \subset \mathbb{H} \cong \mathbb{R}^4$. Holm writes the same matrix as Q (Eq. (3.2.11), p. 100).

90.3 Connectedness, and $\det \Phi(u) = +1$

A topological space X is *connected* if it is not the union of two disjoint nonempty open subsets. Equivalently, every continuous map $f\colon X \to \{-1,+1\}$ is constant. If f took both values, then $f^{-1}(-1)$ and $f^{-1}(+1)$ would split X; if $X = U \cup V$ were split, $f = -1$ on U and $f = +1$ on V would be continuous and not constant.

X is *path-connected* if any two points are joined by a continuous path $\gamma\colon [0,1] \to X$. Path-connected implies connected: if $f\colon X \to \{\pm 1\}$ is continuous, then $f \circ \gamma\colon [0,1] \to \{\pm 1\}$ is continuous, and by the intermediate value theorem it cannot pass from -1 to $+1$, so $f(\gamma(0)) = f(\gamma(1))$.

S^3 is path-connected. Let $p, q \in S^3$ with $q \neq -p$. If $(1-t)p + tq = 0$ for some $t \in [0,1]$, then taking lengths gives $1 - t = t$, so $t = \frac{1}{2}$ and $q = -p$. Hence the segment from p to q avoids 0, and

$$\gamma(t) = \frac{(1-t)p + tq}{|(1-t)p + tq|} \tag{283}$$

is a continuous path in S^3 from p to q. If $q = -p$, choose $r \in S^3$ with $r \neq \pm p$ and join p to r and then r to $-p$. Thus S^3 is connected, and so is $SU(2)$, its image under the continuous map F^{-1}. This is Frankel's "in particular $SU(2) = S^3$ is connected".

Now apply this to $\det \Phi(u)$. By Eq. (273), the entries of $\Phi(u)$ are polynomials in the real and imaginary parts of the entries of u, using $u^{-1} = u^\dagger$. So $u \mapsto \det \Phi(u)$ is continuous on $SU(2)$, with values in $\{\pm 1\}$ because $\Phi(u)$ is orthogonal. On the connected $SU(2)$ it is constant, and at $u = I$ we have $\Phi(I) = I$, with determinant 1. Therefore

$$\Phi\colon SU(2) \to SO(3). \tag{284}$$

By contrast, $O(3)$ is not connected: $\det\colon O(3) \to \{\pm 1\}$ is continuous and takes both values, at I and at $-I$.

91 $\Phi\colon SU(2) \to SO(3)$ is onto and two-to-one

This section follows Frankel's Section 19.1d (pp. 496–497).

Theorem 64 (Frankel (19.12)). *$\Phi\colon SU(2) \to SO(3)$ is onto. For each $R \in SO(3)$ there are exactly two elements $\pm u \in SU(2)$ with $\Phi(\pm u) = R$.*

Frankel states this for Ad of his Eq. (19.10), which is our Φ. In his proof h denotes both a vector of $\mathbb{R}^3$ and the Hermitian matrix $h^j\sigma_j$. We write $\mathbf{h} \in \mathbb{R}^3$ and $\Sigma(\mathbf{h})$. His Ad_* is the derivative of Φ at I, which we write $d\Phi$.

91.1 The derivative of Φ

One-parameter subgroups of $SU(2)$. The argument of Eqs. (252)–(254) works verbatim in $GL(2,\mathbb{C})$, so every differentiable one-parameter subgroup of $SU(2)$ is $u(t) = \exp(tX)$ with $X \in \mathfrak{su}(2)$. By Eq. (267), $X = -i\Sigma(\mathbf{h})$ for a unique $\mathbf{h} \in \mathbb{R}^3$:

$$u(t) = \exp\big(-it\,\Sigma(\mathbf{h})\big) = \exp\Big(\frac{t}{i}\,h^j\sigma_j\Big). \tag{285}$$

This is Frankel's $u(t) = e^{th/i} = e^{-ith}$.

Its image is a one-parameter subgroup of $SO(3)$. Since Φ is a homomorphism, $\Phi(u(t+s)) = \Phi(u(t)u(s)) = \Phi(u(t))\Phi(u(s))$ and $\Phi(u(0)) = I$. By Eq. (254),

$$\Phi\big(u(t)\big) = \exp(tS), \qquad S = \frac{d}{dt}\Phi\big(u(t)\big)\Big|_{t=0} =: d\Phi(X). \tag{286}$$

Computing S. Apply Eq. (272) with $u = u(t)$:

$$\Sigma\big(\Phi(u(t))\mathbf{x}\big) = e^{-it\Sigma(\mathbf{h})}\,\Sigma(\mathbf{x})\,e^{it\Sigma(\mathbf{h})}. \tag{287}$$

Differentiate at $t = 0$. On the left, Σ is linear and does not depend on t, so the derivative is $\Sigma(S\mathbf{x})$. On the right, the product rule with $\frac{d}{dt}e^{\mp it\Sigma(\mathbf{h})}\big|_{t=0} = \mp i\Sigma(\mathbf{h})$ gives

$$\Sigma(S\mathbf{x}) = -i\Sigma(\mathbf{h})\Sigma(\mathbf{x}) + \Sigma(\mathbf{x})\,i\Sigma(\mathbf{h}) = -i\big[\Sigma(\mathbf{h}),\Sigma(\mathbf{x})\big]. \tag{288}$$

Expand the commutator with Frankel's Eq. (19.6), $[\sigma_j,\sigma_k] = 2i\epsilon_{jkl}\sigma_l$, and $(\mathbf{h}\times\mathbf{x})^l = \epsilon_{ljk}h^jx^k = \epsilon_{jkl}h^jx^k$:

$$\begin{aligned} -i\big[\Sigma(\mathbf{h}),\Sigma(\mathbf{x})\big] &= -ih^jx^k[\sigma_j,\sigma_k] = -ih^jx^k\,2i\,\epsilon_{jkl}\sigma_l \\ &= 2\epsilon_{jkl}h^jx^k\sigma_l = 2(\mathbf{h}\times\mathbf{x})^l\sigma_l = \Sigma(2\mathbf{h}\times\mathbf{x}). \end{aligned} \tag{289}$$

So $\Sigma(S\mathbf{x}) = \Sigma(2\mathbf{h}\times\mathbf{x})$ for every $\mathbf{x}$. Since Σ is injective, $S\mathbf{x} = 2\mathbf{h}\times\mathbf{x}$, that is $S = \widehat{2\mathbf{h}}$ by Eq. (257). The one-parameter group $\Phi(u(t))$ rotates with angular velocity $\boldsymbol{\omega} = 2\mathbf{h}$, and Eq. (258) gives

$$\Phi\Big(\exp\big(-it\,\Sigma(\mathbf{h})\big)\Big) = \exp\big(t\,(2h)^jE_j\big). \tag{290}$$

This is Frankel's Eq. (19.13), $\mathrm{Ad}\exp(\frac{\boldsymbol{\sigma}}{i}\bullet\mathbf{h}t)\mathbf{x}_* \sim R(t)\mathbf{x} = \exp(\mathbf{E}\bullet 2\mathbf{h}t)\mathbf{x}$. Each $\sim$ there is an instance of Eq. (272): the matrix on the left is Σ of the vector on the right.

$d\Phi$ is ad. The first equality of Eq. (289) reads, with $X = -i\Sigma(\mathbf{h})$,

$$\Sigma\big(d\Phi(X)\mathbf{x}\big) = \big[X,\Sigma(\mathbf{x})\big] = \mathrm{ad}_X\Sigma(\mathbf{x}). \tag{291}$$

So $d\Phi(X)$ is ad_X written in the coordinates Σ, just as $\Phi(u)$ is Ad_u written in those coordinates. The identity $\Phi(\exp tX) = \exp(t\,d\Phi(X))$ is the relation $\mathrm{Ad}_{\exp(tX)} = \exp(t\,\mathrm{ad}_X)$ of §85.3, read in $\mathbb{R}^3$.

Frankel's Eq. (19.14). Take $X = \sigma_a/(2i) = -i\Sigma(\frac{1}{2}\mathbf{e}_a)$. Then $\mathbf{h} = \frac{1}{2}\mathbf{e}_a$, $2\mathbf{h} = \mathbf{e}_a$, and $\widehat{\mathbf{e}_a} = E_a$:

$$d\Phi\Big(\frac{\sigma_a}{2i}\Big) = E_a. \tag{292}$$

Frankel calls this unsurprising given the remarks after his Eq. (19.6): the basis $\sigma_a/(2i)$ has the same structure constants as E_a (Eq. (315)). For Kosmann-Schwarzbach's basis, $d\Phi(\xi_1) = -E_1$, $d\Phi(\xi_2) = E_2$, $d\Phi(\xi_3) = -E_3$ by Eq. (316). In either case $d\Phi$ carries a basis to a basis, so $d\Phi\colon \mathfrak{su}(2) \to \mathfrak{so}(3)$ is a linear isomorphism between spaces of dimension 3.

91.2 Example: rotations about the z axis

Take $\mathbf{h} = \mathbf{e}_3$, so $\Sigma(\mathbf{h}) = \sigma_3$; Frankel writes $\mathbf{h} = \boldsymbol{\sigma}(0,0,1)^T$, that is $\mathbf{h} = \sigma_3$. Since $\sigma_3/i = -i\sigma_3 = \mathrm{diag}(-i, i)$ is diagonal, every power is diagonal and the exponential is taken entry by entry:

$$u(\theta) = \exp\Big(\theta \frac{\sigma_3}{i}\Big) = \begin{bmatrix} e^{-i\theta} & 0 \\ 0 & e^{i\theta} \end{bmatrix}. \tag{293}$$

By Eq. (290) with $t = \theta$, $\Phi(u(\theta)) = \exp(2\theta E_3)$, which is $R(2\theta)$ by Eq. (242) with θ replaced by 2θ.

The same result follows directly from Eq. (272), without derivatives. Multiplying on the left by $u(\theta)$ multiplies row 1 by $e^{-i\theta}$ and row 2 by $e^{i\theta}$. Multiplying on the right by $u(\theta)^{-1} = \mathrm{diag}(e^{i\theta}, e^{-i\theta})$ multiplies column 1 by $e^{i\theta}$ and column 2 by $e^{-i\theta}$:

$$u(\theta)\Sigma(\mathbf{x})u(\theta)^{-1} = \begin{bmatrix} z & e^{-2i\theta}(x - iy) \\ e^{2i\theta}(x + iy) & -z \end{bmatrix}. \tag{294}$$

This is $\Sigma(\mathbf{x}')$ with $z' = z$ and $x' + iy' = e^{2i\theta}(x+iy) = (x\cos 2\theta - y\sin 2\theta) + i(x\sin 2\theta + y\cos 2\theta)$. So $\Phi(u(\theta)) = R(2\theta)$.

As θ runs from 0 to 2π, $e^{-i\theta}$ goes once around the unit circle, so $u(\theta)$ is a simple closed curve in $SU(2)$, passing through $u(\pi) = -I$. Its image $R(2\theta)$ makes two full rotations, returning to I at $\theta = \pi$ and at $\theta = 2\pi$. The half $0 \le \theta \le \pi$ is a path from I to $-I$ in $SU(2)$ whose image is a closed loop in $SO(3)$; that loop is the one studied in Frankel's Section 19.2a and in his Section 21.2.

91.3 Φ is onto

Frankel uses that every rotation is a rotation about some axis. Let $R \in SO(3)$. Then

$$\begin{aligned} \det(R - I) &= \det\big(R(I - R^T)\big) = \det R \, \det\big((I - R)^T\big) \\ &= \det(I - R) = (-1)^3 \det(R - I), \end{aligned} \tag{295}$$

using $RR^T = I$, $\det R = 1$, and $\det A^T = \det A$. So $\det(R - I) = 0$, and R has a unit eigenvector $\mathbf{n}$ with $R\mathbf{n} = \mathbf{n}$. R preserves the plane perpendicular to $\mathbf{n}$: if $\mathbf{x} \cdot \mathbf{n} = 0$ then $R\mathbf{x} \cdot \mathbf{n} = R\mathbf{x} \cdot R\mathbf{n} = \mathbf{x} \cdot \mathbf{n} = 0$. In a right-handed orthonormal basis $(\mathbf{u}, \mathbf{n} \times \mathbf{u}, \mathbf{n})$, the matrix of R is block diagonal with an upper 2×2 block A and a 1 in the corner. A has orthonormal columns and $\det A = \det R = 1$. Its first column is a unit vector $(\cos\theta, \sin\theta)$ for some θ. The second is a unit vector orthogonal to it, $\pm(-\sin\theta, \cos\theta)$, and $\det A = 1$ selects the plus sign. So $R = R_{\mathbf{n}}(\theta) = \exp(\theta\, n^j E_j)$ by Eq. (259).

Now take $\mathbf{h} = \frac{1}{2}\mathbf{n}$ and $t = \theta$ in Eq. (290):

$$\Phi\Big(\exp\Big(-i\frac{\theta}{2}\Sigma(\mathbf{n})\Big)\Big) = \exp\big(\theta\, n^j E_j\big) = R. \tag{296}$$

This is Frankel's $\mathrm{Ad}\exp(\boldsymbol{\sigma}/2i \bullet \boldsymbol{\omega}\theta) = R$, with his $\boldsymbol{\omega}$ the unit axis $\mathbf{n}$. By Eq. (333), the element of $SU(2)$ in Eq. (296) is $\cos\frac{\theta}{2} I - i \sin\frac{\theta}{2}\Sigma(\mathbf{n})$, the unit quaternion of Eq. (334). So Φ is onto.

91.4 Φ in closed form: onto and two-to-one explicitly

Every $u \in SU(2)$ is $u = q_0 I - i\Sigma(\mathbf{q})$ with $q_0^2 + |\mathbf{q}|^2 = 1$ (the quaternion form in §90, since $q^a(-i\sigma_a) = -i\Sigma(\mathbf{q})$), and $u^{-1} = u^\dagger = q_0 I + i\Sigma(\mathbf{q})$ because $\Sigma(\mathbf{q})$ is Hermitian. Computing $\Phi(u)$ from Eq. (272) needs one product rule for Pauli matrices. Adding half of Frankel's anticommutator (Eq. (19.18)) and half of his commutator (Eq. (19.6)) gives $\sigma_j\sigma_k = \delta_{jk} I + i\epsilon_{jkl}\sigma_l$, so

$$\Sigma(\mathbf{a})\Sigma(\mathbf{b}) = a^j b^k \sigma_j \sigma_k = (\mathbf{a} \cdot \mathbf{b})\, I + i\, \Sigma(\mathbf{a} \times \mathbf{b}), \tag{297}$$

Frankel's Eq. (19.19), with $(\mathbf{a} \times \mathbf{b})^l = \epsilon_{jkl} a^j b^k$. Its antisymmetric part is $[\Sigma(\mathbf{a}), \Sigma(\mathbf{b})] = 2i\,\Sigma(\mathbf{a} \times \mathbf{b})$. Now expand

$$u\Sigma(\mathbf{x})u^{-1} = q_0^2\Sigma(\mathbf{x}) + iq_0\big[\Sigma(\mathbf{x}), \Sigma(\mathbf{q})\big] + \Sigma(\mathbf{q})\Sigma(\mathbf{x})\Sigma(\mathbf{q}), \tag{298}$$

and treat the last two terms separately. The middle term is $iq_0 \cdot 2i\,\Sigma(\mathbf{x} \times \mathbf{q}) = 2q_0\Sigma(\mathbf{q} \times \mathbf{x})$. For the last term apply Eq. (297) twice. The scalar $(\mathbf{q} \times \mathbf{x}) \cdot \mathbf{q}$ vanishes, and $(\mathbf{q} \times \mathbf{x}) \times \mathbf{q} = |\mathbf{q}|^2\mathbf{x} - (\mathbf{q} \cdot \mathbf{x})\mathbf{q}$ by $(\mathbf{a} \times \mathbf{b}) \times \mathbf{c} = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{b} \cdot \mathbf{c})\mathbf{a}$:

$$\Sigma(\mathbf{q})\Sigma(\mathbf{x})\Sigma(\mathbf{q}) = \big((\mathbf{q} \cdot \mathbf{x})I + i\Sigma(\mathbf{q} \times \mathbf{x})\big)\Sigma(\mathbf{q})$$

$$= (\mathbf{q}\cdot\mathbf{x})\Sigma(\mathbf{q}) - \Sigma\big((\mathbf{q}\times\mathbf{x})\times\mathbf{q}\big)$$
$$= 2(\mathbf{q}\cdot\mathbf{x})\Sigma(\mathbf{q}) - |\mathbf{q}|^2\Sigma(\mathbf{x}). \tag{299}$$

Collecting terms inside Σ and using Eq. (272),

$$\Phi(u)\mathbf{x} = \big(q_0^2 - |\mathbf{q}|^2\big)\mathbf{x} + 2q_0\,\mathbf{q}\times\mathbf{x} + 2(\mathbf{q}\cdot\mathbf{x})\,\mathbf{q}. \tag{300}$$

In matrix form this is $\Phi(u) = (q_0^2 - |\mathbf{q}|^2)I + 2q_0\widehat{\mathbf{q}} + 2\mathbf{q}\mathbf{q}^T$, the Euler–Rodrigues formula.

Two-to-one, explicitly. Every term of Eq. (300) is quadratic in $(q_0, \mathbf{q})$. Replacing u by $-u$ replaces $(q_0, \mathbf{q})$ by $(-q_0, -\mathbf{q})$ and leaves $\Phi(u)$ unchanged. The kernel can also be read off. If $\Phi(u) = I$ and $\mathbf{q} \neq 0$, take a unit $\mathbf{x} \perp \mathbf{q}$. Then Eq. (300) gives $(q_0^2 - |\mathbf{q}|^2)\mathbf{x} + 2q_0\,\mathbf{q}\times\mathbf{x} = \mathbf{x}$. The two terms are perpendicular, so $q_0^2 - |\mathbf{q}|^2 = 1$, which together with $q_0^2 + |\mathbf{q}|^2 = 1$ forces $\mathbf{q} = 0$, a contradiction. So $\mathbf{q} = 0$, $q_0 = \pm 1$, and $u = \pm I$.

Onto, explicitly. Given $R = R_{\mathbf{n}}(\theta)$, which covers every rotation by the axis argument above, put $q_0 = \cos\frac{\theta}{2}$ and $\mathbf{q} = \sin\frac{\theta}{2}\mathbf{n}$. The double-angle formulas give $q_0^2 - |\mathbf{q}|^2 = \cos\theta$, $2q_0\mathbf{q} = \sin\theta\,\mathbf{n}$, and $2\mathbf{q}\mathbf{q}^T = (1-\cos\theta)\mathbf{n}\mathbf{n}^T$. Then Eq. (300) becomes

$$\Phi(u)\mathbf{x} = \cos\theta\,\mathbf{x} + \sin\theta\,\mathbf{n}\times\mathbf{x} + (1-\cos\theta)(\mathbf{n}\cdot\mathbf{x})\mathbf{n}, \tag{301}$$

which is Eq. (247) after writing $\mathbf{x} = (\mathbf{n}\cdot\mathbf{x})\mathbf{n} + \mathbf{x}_\perp$. So the two preimages of the rotation by θ about $\mathbf{n}$ are

$$\pm u = \pm\Big(\cos\frac{\theta}{2}\,I - i\sin\frac{\theta}{2}\,\Sigma(\mathbf{n})\Big). \tag{302}$$

The minus sign is the same formula with $\theta + 2\pi$ in place of θ, since $\cos\frac{\theta+2\pi}{2} = -\cos\frac{\theta}{2}$ and $\sin\frac{\theta+2\pi}{2} = -\sin\frac{\theta}{2}$. The rotation angles θ and $\theta + 2\pi$ give the same rotation of $\mathbb{R}^3$ but different elements of $SU(2)$.

91.5 Φ is two-to-one, and $SO(3) \approx \mathbb{R}P^3$

For every u, $(-u)\Sigma(\mathbf{x})(-u)^{-1} = (-1)(-1)^{-1}u\Sigma(\mathbf{x})u^{-1} = u\Sigma(\mathbf{x})u^{-1}$, so $\Phi(-u) = \Phi(u)$. Φ is therefore at least two-to-one; in Frankel's words, the representation is not faithful.

The kernel. $\Phi(u) = I$ means $u\Sigma(\mathbf{x})u^{-1} = \Sigma(\mathbf{x})$ for all $\mathbf{x}$. Taking $\mathbf{x} = \mathbf{e}_k$, and conversely by linearity, this holds exactly when u commutes with $\sigma_1, \sigma_2, \sigma_3$. Since $\sigma_k^{-1} = \sigma_k$, commuting with σ_k is $\sigma_k u \sigma_k = u$, Frankel's $\sigma_\alpha^{-1} u \sigma_\alpha = u$. Write u in the form (262). Conjugating by $\sigma_3 = \mathrm{diag}(1, -1)$ changes the sign of the off-diagonal entries, and conjugating by σ_1 exchanges both the rows and the columns:

$$\sigma_3 u \sigma_3 = \begin{bmatrix} \alpha & \bar\beta \\ -\beta & \bar\alpha \end{bmatrix}, \qquad \sigma_1 u \sigma_1 = \begin{bmatrix} \bar\alpha & \beta \\ -\bar\beta & \alpha \end{bmatrix}. \tag{303}$$

The first equals u only if $-\bar\beta = \bar\beta$, so $\beta = 0$. The second equals u only if $\bar\alpha = \alpha$, so α is real. Then $|\alpha|^2 + |\beta|^2 = 1$ gives $\alpha = \pm 1$, and

$$\ker\Phi = \{I, -I\}. \tag{304}$$

The fibers. $\Phi(u) = \Phi(u')$ if and only if $\Phi(u^{-1}u') = \Phi(u)^{-1}\Phi(u') = I$, if and only if $u^{-1}u' \in \{\pm I\}$, if and only if $u' = \pm u$. Since $u \neq -u$, every rotation has exactly two preimages. This completes the proof of Theorem 64.

The elementary result of group theory. Frankel continues: "It is an elementary result of group theory that if $\phi\colon G \to G'$ is a homomorphism of G onto G', then G' is isomorphic to the coset space G/H, where $H = \phi^{-1}(e')$ is the kernel" (pp. 496–497). This is the first isomorphism theorem for groups. The algebra part proves the same theorem for modules (Theorem 15, from Rotman). Modules are abelian groups, where every subgroup is normal; for groups, normality of the kernel is the extra step. Dummit and Foote state the group version as Theorem 16 of their Section 3.3 [25]. Quotient groups appear in Jordan's and Hölder's work on composition series (1870–1889); the isomorphism theorems in their general form are usually credited to Emmy Noether (1927), and they reached textbooks through van der Waerden's *Moderne Algebra* (1930).

Theorem 65 (first isomorphism theorem)**.** *Let $\varphi\colon G \to G'$ be a group homomorphism and $K = \varphi^{-1}(e')$ its kernel. Then K is a normal subgroup of G, the cosets G/K form a group under $(gK)(hK) = ghK$, and*

$$\bar\varphi\colon G/K \to \varphi(G), \qquad \bar\varphi(gK) = \varphi(g), \tag{305}$$

is a well-defined group isomorphism. If φ is onto, $G/K \cong G'$.

Proof. *K is normal.* For $k \in K$ and $g \in G$, $\varphi(gkg^{-1}) = \varphi(g)\,e'\,\varphi(g)^{-1} = e'$, so $gKg^{-1} \subseteq K$.

The coset product is well defined. If $g' = gk$ and $h' = hk'$ with $k, k' \in K$, then $g'h' = gh\,(h^{-1}kh)\,k'$, and $h^{-1}kh \in K$ by normality, so $g'h'K = ghK$.

$\bar\varphi$ is well defined and injective. For $g, h \in G$,

$$gK = hK \iff g^{-1}h \in K \iff \varphi(g^{-1}h) = e' \iff \varphi(g) = \varphi(h). \tag{306}$$

Reading the chain left to right shows that $\bar\varphi(gK)$ does not depend on the representative g; reading it right to left shows that $\bar\varphi$ is injective.

$\bar\varphi$ is a homomorphism onto $\varphi(G)$. $\bar\varphi(gK\,hK) = \bar\varphi(ghK) = \varphi(gh) = \varphi(g)\varphi(h)$, and every $\varphi(g)$ equals $\bar\varphi(gK)$. □

Frankel derives the same correspondence from his Fundamental Principle (17.10). G acts on G' by $g \cdot g' = \varphi(g)g'$. The action is transitive when φ is onto: to move g'_1 to g'_2, choose g with $\varphi(g) = g'_2 g'^{-1}_1$. The stability subgroup of e' is $\ker\varphi$. That argument gives the bijection $G/\ker\varphi \leftrightarrow G'$ of sets. Theorem 65 adds that the bijection respects multiplication. For Φ, which is onto with kernel $\{\pm I\}$,

$$SU(2)/\{\pm I\} \cong SO(3) \quad \text{as groups}, \qquad \bar\Phi\big(\{u, -u\}\big) = \Phi(u). \tag{307}$$

In the coordinates of Eq. (280), $F(-u) = -F(u)$, so each fiber $\{u, -u\}$ is a pair of antipodal points of S^3. Identifying antipodal points of S^3 gives the real projective space $\mathbb{R}P^3 = S^3/\{\pm 1\}$, the space of lines through 0 in $\mathbb{R}^4$, since each such line meets S^3 in one antipodal pair:

$$\begin{array}{ccc} S^3 & \xrightarrow{\approx} & SU(2) \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle \Phi} \\ \mathbb{R}P^3 & \xrightarrow{\approx} & SO(3) \end{array} \tag{308}$$

Here $p(x) = \{x, -x\}$. The induced map $\mathbb{R}P^3 \to SO(3)$ is a continuous bijection. $\mathbb{R}P^3$ is compact, as the image of the compact S^3 under p, and $SO(3) \subset \mathbb{R}^9$ is Hausdorff, so the map is a homeomorphism: $SO(3) \approx \mathbb{R}P^3$. By Theorem (17.11), $SU(2)$ is a principal bundle over $SU(2)/\{\pm I\} = SO(3)$ with the two-point fiber $\{\pm I\}$, that is, a two-sheeted covering. Its dimension count is $\dim SO(3) = \dim SU(2) - \dim\{\pm I\} = 3 - 0$. Also, $SO(3) = \Phi(SU(2))$ is connected, as the continuous image of a connected space.

92 The smooth statements, with Lee's theorems

The preceding sections prove statements about sets, groups and continuous maps. Frankel's "≈" and "principal bundle" need the smooth versions. Lee's *Introduction to Smooth Manifolds*, second edition [76], supplies each one; below, every step names the result it uses, with its printed page.

92.1 S^3 and $SU(2)$ are smooth manifolds

Let $f(x) = |x|^2$ on $\mathbb{R}^4$. Then $df_x = 2x^T \neq 0$ on $S^3 = f^{-1}(1)$, so 1 is a regular value. By the regular level set theorem (Lee, Corollary 5.14, p. 106), S^3 is a properly embedded submanifold of codimension 1, of dimension 3. $SU(2)$ is a properly embedded $(2^2 - 1)$-dimensional Lie subgroup of $U(2)$, hence embedded in $GL(2, \mathbb{C})$ (Lee, Example 7.30, p. 168, and Problem 7-14, p. 172). Likewise $SO(3)$ is an embedded Lie subgroup of $GL(3, \mathbb{R})$ (Lee, Example 7.28, p. 167).

Two facts about restrictions are used repeatedly. If $G\colon M \to N$ is smooth and $S \subseteq M$ is an embedded submanifold, then $G|_S$ is smooth (Lee, Theorem 5.27, p. 112). If $G\colon N \to M$ is smooth with image in an embedded submanifold $S \subseteq M$, then G is smooth as a map into S (Lee, Corollary 5.30, p. 113).

92.2 $SU(2)$ is diffeomorphic to S^3

This is Lee's Problem 7-16 (p. 172). The map F of Eq. (280) is the restriction to $SU(2)$ of the linear map $M(2, \mathbb{C}) \to \mathbb{C}^2$, $u \mapsto u(1, 0)^T$, with image in S^3. Its inverse is the restriction to S^3 of the real-linear map $\mathbb{C}^2 \to M(2, \mathbb{C})$ of Eq. (278), with image in $SU(2)$. Linear maps are smooth. By the two restriction facts, F and F^{-1} are smooth, so F is a diffeomorphism:

$$SU(2) \approx S^3 \quad \text{(diffeomorphic)}. \tag{309}$$

92.3 Φ is a smooth double covering

By Eq. (273), with $u^{-1} = u^\dagger$, each entry of $\Phi(u)$ is a polynomial in the real and imaginary parts of the entries of u. So Φ is the restriction of a smooth map $M(2,\mathbb{C}) \to M(3,\mathbb{R})$, with image in $SO(3)$, and the restriction facts make $\Phi\colon SU(2) \to SO(3)$ smooth. It is a homomorphism, hence a Lie group homomorphism. Every Lie group homomorphism has constant rank (Lee, Theorem 7.5, p. 153). By Eq. (292), $d\Phi$ at I is an isomorphism, so $d\Phi_u$ has rank 3 at every u.

Lee's Theorem 7.5 is a short computation. With L_u left multiplication, $\Phi \circ L_u = L_{\Phi(u)} \circ \Phi$ because Φ is a homomorphism. The chain rule gives $d\Phi_u \circ d(L_u)_I = d(L_{\Phi(u)})_I \circ d\Phi_I$, and both $d(L_u)_I$ and $d(L_{\Phi(u)})_I$ are isomorphisms, since L_u and $L_{\Phi(u)}$ are diffeomorphisms. So $d\Phi_u$ and $d\Phi_I$ have the same rank.

Both groups are connected (§90 and §91). Lee's Theorem 21.31 (Homomorphisms with Discrete Kernels, p. 557) states that for connected Lie groups G, H and a Lie group homomorphism $F\colon G \to H$ the following are equivalent: (a) F is surjective and has discrete kernel; (b) F is a smooth covering map; (c) F is a local diffeomorphism; (d) $F_*\colon \mathrm{Lie}(G) \to \mathrm{Lie}(H)$ is an isomorphism. Here (a) is Theorem 64 with $\ker\Phi = \{\pm I\}$, and (d) is Eq. (292); either one gives (b):

$$\Phi\colon SU(2) \to SO(3) \text{ is a smooth two-sheeted covering map.} \tag{310}$$

This is the precise content of Frankel's "principal bundle with two-point fiber".

92.4 $\mathbb{R}P^3 \approx SU(2)/\{\pm I\} \cong SO(3)$

Theorem 65 identifies $SU(2)/\{\pm I\}$ with $SO(3)$ as groups. Lee's First Isomorphism Theorem for Lie Groups (Theorem 21.27, p. 556) adds the smooth structure: the kernel $\{\pm I\}$ is a closed normal Lie subgroup, and Φ descends to a Lie group isomorphism

$$\bar\Phi\colon SU(2)/\{\pm I\} \to SO(3), \tag{311}$$

that is, a group isomorphism that is also a diffeomorphism. The quotient $SU(2)/\{\pm I\}$ carries the smooth structure of Lee's Theorem 21.29 (p. 556): $\{\pm I\}$ is a discrete subgroup of the connected $SU(2)$, so the quotient map is a smooth normal covering map.

For $\mathbb{R}P^3$, the group $\{\pm 1\}$ acts on S^3 by $x \mapsto \pm x$. The action is smooth and free, and it is proper because $\{\pm 1\}$ is compact (Lee, Corollary 21.6, p. 544). Lee's Theorem 21.13 (p. 549) makes $S^3/\{\pm 1\} = \mathbb{R}P^3$ a smooth manifold with $p\colon S^3 \to \mathbb{R}P^3$ a smooth normal covering map; for $\mathbb{R}P^n$ with its standard structure this is Lee's Problem 4-10 (p. 96). Since $F(-u) = -F(u)$, the composite $S^3 \xrightarrow{F^{-1}} SU(2) \to SU(2)/\{\pm I\}$ is constant on the fibers of p. A smooth covering map is a surjective smooth submersion, so the composite descends to a smooth map $\mathbb{R}P^3 \to SU(2)/\{\pm I\}$ (Lee, Theorem 4.30, Passing Smoothly to the Quotient, p. 90). The same argument with F in place of F^{-1} gives its smooth inverse. Composing with $\bar\Phi$,

$$\mathbb{R}P^3 \;\approx\; SU(2)/\{\pm I\} \;\cong\; SO(3), \tag{312}$$

a diffeomorphism. This upgrades the homeomorphism of the previous section.

This is Lee's Problem 21-21 (p. 562), in quaternion language. There S is the group of unit quaternions, E the imaginary quaternions, and $\rho(q)$ the matrix of $v \mapsto qvq^*$ on E in the basis $(\mathbf{i},\mathbf{j},\mathbf{k})$. Under the dictionary (328), $e_a \leftrightarrow -i\sigma_a$, the imaginary quaternion $x^a e_a$ corresponds to $-i\Sigma(\mathbf{x})$, and $q^* = q^{-1}$ for a unit quaternion. Multiplying Eq. (272) by $-i$ gives $u\big(-i\Sigma(\mathbf{x})\big)u^{-1} = -i\Sigma\big(\Phi(u)\mathbf{x}\big)$. So $\Phi(u)$ is Lee's $\rho(q)$, and the problem's parts (a)–(c) are Eqs. (272), (300) and (312).

92.5 $\pi_1(SO(3)) = \mathbb{Z}/2$

S^3 is simply connected (Lee, Proposition A.68(b), p. 614), hence so is $SU(2)$ by Eq. (309). So Φ is the universal covering of $SO(3)$, and $SU(2)$ is the universal covering group of Lee's Theorem 7.7 (p. 154).

An explicit non-contractible loop. The rotations $R(2\theta)$ about the z axis, $0 \le \theta \le \pi$, form a closed loop γ in $SO(3)$ based at I. By the example of §91, its lift starting at I is $u(\theta) = \mathrm{diag}(e^{-i\theta}, e^{i\theta})$, which ends at $u(\pi) = -I$. By Lee's monodromy theorem (Proposition A.77(c), p. 616), path-homotopic paths have lifts with the same endpoint. The constant loop lifts to the constant path at I, which ends at $I \neq -I$. So γ is not null-homotopic, and $\pi_1(SO(3)) \neq 1$. The loop $\gamma * \gamma$ (two full turns) lifts to $u(\theta)$, $0 \le \theta \le 2\pi$, a closed loop in the simply connected $SU(2)$. Contract it there and push the contraction down by Φ: $\gamma * \gamma$ is null-homotopic.

The whole group. The covering automorphisms of Φ are $u \mapsto u$ and $u \mapsto -u$. A covering automorphism is determined by the image of one point (unique lifting, Lee, Proposition A.77(a)), and it must send I into $\Phi^{-1}(I) = \{\pm I\}$. Lee's Problem 21-19 (p. 562) quotes the fact that for a universal covering the automorphism group is isomorphic to the fundamental group of the base (proved in Lee's *Introduction to Topological Manifolds*, Ch. 12). Hence

$$\pi_1\big(SO(3)\big) \cong \ker \Phi = \{\pm I\} \cong \mathbb{Z}/2. \tag{313}$$

This is the topological fact behind spin $\frac{1}{2}$. A continuous family of rotations of $\mathbb{R}^3$ that starts and ends at I lifts to $SU(2)$ and ends at I or at $-I$. A 2π turn ends at $-I$; a 4π turn ends at I.

Lie algebras do not see this. $d\Phi$ identifies $\mathfrak{su}(2)$ with $\mathfrak{so}(3)$. Since $SU(2)$ is simply connected, Lee's Theorem 20.19 (p. 530) lifts every Lie algebra homomorphism out of $\mathfrak{su}(2)$ to a Lie group homomorphism out of $SU(2)$. The inverse $d\Phi^{-1}\colon \mathfrak{so}(3) \to \mathfrak{su}(2)$ has no such lift out of $SO(3)$: a lift $\Psi\colon SO(3) \to SU(2)$ with $\Psi_* = d\Phi^{-1}$ would give the homomorphism $\Phi \circ \Psi$, whose induced Lie algebra map is the identity. By naturality of exp (Lee, Proposition 20.8(g), p. 520), $\Phi \circ \Psi(\exp X) = \exp X$ for every $X \in \mathfrak{so}(3)$, so $\Phi \circ \Psi = \mathrm{Id}$ on the image of exp, which contains a neighborhood of I (Proposition 20.8(f)). A neighborhood of I generates the connected $SO(3)$ (Lee, Proposition 7.14(c), p. 156), so $\Phi \circ \Psi = \mathrm{Id}$ everywhere. Then $\Psi \circ \gamma$ would be a closed lift of γ, contradicting the monodromy argument above.

93 Compactness, and what it buys

A topological space is compact if every open cover has a finite subcover (Lee, p. 608). Compactness is a finiteness condition. A fact that holds near each point, with its own neighborhood and its own constant, holds on finitely many of those neighborhoods, so it holds everywhere with the worst of finitely many constants. The same condition says that an infinite process cannot escape. Suppose a sequence has no cluster point. Then each point has an open neighborhood containing only finitely many terms, and these neighborhoods form an open cover with no finite subcover.

Proposition 94. *$SU(2)$ and $SO(3)$ are compact.*

Proof. $SO(3) = \{g \in \mathbb{R}^{3\times 3} : g^T g = I,\ \det g = 1\}$ is the common zero set of the continuous functions $g \mapsto g^T g - I$ and $g \mapsto \det g - 1$, so it is closed in $\mathbb{R}^9$. The columns of g are unit vectors, so $\|g\|_F^2 = \mathrm{tr}(g^T g) = \mathrm{tr}\, I = 3$, and $SO(3)$ is bounded. By Heine–Borel it is compact. In the same way, $SU(2) \approx S^3$ is closed and bounded in $\mathbb{R}^4$, so it is compact. Alternatively, $SO(3) = \Phi\big(SU(2)\big)$ is the continuous image of a compact space (Lee, Proposition A.45(a), p. 608). Hall gives the same argument for $O(n)$, $SO(n)$, $U(n)$, $SU(n)$ [14, §1.3, p. 16]. □

Compactness was used once already. In §91, the continuous bijection $\mathbb{R}P^3 \to SO(3)$ is a homeomorphism because its domain is compact and its target is Hausdorff (Lee's closed map lemma, Lemma A.52, p. 610).

93.1 No global three-parameter chart

Proposition 95. *No chart of $SO(3)$ has all of $SO(3)$ as its domain.*

Proof. A chart is a homeomorphism φ from an open subset of $SO(3)$ onto an open subset $\varphi(SO(3))$ of $\mathbb{R}^3$. If its domain were all of $SO(3)$, then $\varphi(SO(3))$ would be compact by Proposition 94, hence closed in $\mathbb{R}^3$. A nonempty subset of the connected $\mathbb{R}^3$ that is open and closed is $\mathbb{R}^3$, which is not compact. □

Euler angles, rotation vectors, Rodrigues–Gibbs parameters and modified Rodrigues parameters are charts, so each of them is singular or multivalued somewhere. Gimbal lock is the boundary of the Euler-angle chart.

93.2 No continuous choice of sign

Φ is a local diffeomorphism everywhere, so four parameters with the constraint $u \in SU(2)$ describe every rotation without singularity. The price is the sign: $\Phi(-u) = \Phi(u)$.

Proposition 96. *There is no continuous map $s\colon SO(3) \to SU(2)$ with $\Phi \circ s = \mathrm{Id}_{SO(3)}$.*

Proof. Let $\gamma(\theta) = R(2\theta)$, $0 \le \theta \le \pi$, be the loop of §92, with lift $u(\theta) = \operatorname{diag}(e^{-i\theta}, e^{i\theta})$ from $u(0) = I$ to $u(\pi) = -I$. Suppose s exists. Then $s \circ \gamma$ is continuous, closed ($s \circ \gamma(0) = s \circ \gamma(\pi) = s(I)$), and satisfies $\Phi \circ (s \circ \gamma) = \gamma$, so it is a lift of γ starting at $s(I) \in \{\pm I\}$. By unique lifting (Lee, Proposition A.77(a)), it equals u if $s(I) = I$, which ends at $-I$, and $-u$ if $s(I) = -I$, which ends at I. Either way $s \circ \gamma$ is not closed, a contradiction. □

The proof uses only the restriction of s to the loop γ. So every rule that assigns one unit quaternion to each rotation is discontinuous somewhere on every full turn about an axis. For example, the rule "$\operatorname{tr} u \ge 0$" (scalar part ≥ 0) jumps from u to $-u$ at the rotations by π, where $\operatorname{tr} u(\theta) = 2\cos\theta$ changes sign. A continuous path $g(t)$ in $SO(3)$ does have a continuous lift $u(t)$, unique once $u(0)$ is chosen (Lee, Proposition A.77(a)(b)). A propagated attitude can therefore be continuous in $SU(2)$, while the sign of a quaternion built from a rotation matrix cannot be chosen continuously.

93.3 Compactness and quantization

Remark 2. For a compact group, averaging an inner product over the group, with the invariant integral in place of a finite sum, makes every finite-dimensional representation unitary, hence completely reducible (Hall, Theorem 4.28, p. 92, proved this way). By the Peter–Weyl theorem, every irreducible unitary representation of a compact Lie group is finite-dimensional, and the irreducibles form a discrete family. For $SU(2)$ they are labeled by $m = 0, 1, 2, \ldots$, of dimension $m + 1$ (Hall, Theorem 4.32, p. 96), that is, $j = m/2 \in \{0, \frac{1}{2}, 1, \ldots\}$. The integer-$j$ representations factor through $SO(3)$; the half-integer ones do not (Hall, Proposition 4.35, p. 101), because $-I \in \ker\Phi$ acts as $(-1)^m$. A connected noncompact simple Lie group, such as the Lorentz group, has no nontrivial finite-dimensional unitary representations, so its unitary representations of physical interest are infinite-dimensional.

The same mechanism discretizes spectra. The Laplacian of a compact Riemannian manifold without boundary has discrete eigenvalues of finite multiplicity, because its resolvent is a compact operator. On S^2 the eigenvalues of $-\Delta_{S^2}$ are $\ell(\ell+1)$, the spectrum of $\mathbf{L}^2/\hbar^2$. On the noncompact $\mathbb{R}^n$ the spectrum of $-\Delta$ is continuous. The contrast between Fourier series on the compact S^1 (frequencies $\mathbb{Z}$) and the Fourier transform on $\mathbb{R}$ (frequencies $\mathbb{R}$) is the abelian case. A confining potential produces a discrete spectrum through the same condition, a compact resolvent, without a compact configuration space.

94 Dimension accounting

space	ambient	conditions	dimension
$SO(3)$	9 real entries	$g^T g = I$: 6; $\det g = 1$: none	3
$\mathfrak{so}(3)$	9	$X^T = -X$: 6	3
$SU(2)$	8 real	$U^\dagger U = 1$: 4; $\det U = 1$: 1	3
$\mathfrak{su}(2)$	8	$X^\dagger = -X$: 4; $\operatorname{tr} X = 0$: 1	3
S^3	$\mathbb{R}^4$	$\lvert z_1\rvert^2 + \lvert z_2\rvert^2 = 1$: 1	3
$\ker\Phi$	—	$\{\pm I\}$, two points	0

Φ maps a 3-dimensional group onto a 3-dimensional group with a 0-dimensional kernel, and $d\Phi$ is an isomorphism of the 3-dimensional Lie algebras. The Lie algebras are isomorphic and the groups are not: $d\Phi$ only describes Φ near the identity, and the two-point kernel is global information that $\mathfrak{su}(2) \cong \mathfrak{so}(3)$ does not detect.

95 Two bases of $\mathfrak{su}(2)$: Kosmann-Schwarzbach and Frankel

§85.0.1 records Kosmann-Schwarzbach's basis

$$\xi_1 = \frac{i}{2}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \qquad \xi_2 = \frac{1}{2}\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \qquad \xi_3 = \frac{i}{2}\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \qquad [\xi_k, \xi_l] = \epsilon_{klm}\xi_m. \tag{314}$$

Frankel's basis is $F_a = \sigma_a/(2i)$. From his Eq. (19.6), $[\sigma_a, \sigma_b] = 2i\epsilon_{abc}\sigma_c$,

$$[F_a, F_b] = \frac{1}{(2i)^2}[\sigma_a, \sigma_b] = -\frac{1}{4}\,2i\,\epsilon_{abc}\sigma_c = \epsilon_{abc}\frac{\sigma_c}{2i} = \epsilon_{abc}F_c, \tag{315}$$

using $-\frac{i}{2}=\frac{1}{2i}$. Both bases therefore have the structure constants of Eq. (245), and the linear maps $\xi_k\mapsto E_k$ and $F_a\mapsto E_a$ are both Lie algebra isomorphisms $\mathfrak{su}(2)\to\mathfrak{so}(3)$. The second is Frankel's Eq. (19.14), $\mathrm{Ad}_*(\sigma_a/2i)=E_a$.

The two isomorphisms differ. Compare entries with the Pauli matrices of Eq. (326): $\xi_1=\frac{i}{2}\sigma_1$, while $F_1=\frac{1}{2i}\sigma_1=-\frac{i}{2}\sigma_1$; next $F_2=\frac{1}{2i}\begin{bmatrix}0&-i\\ i&0\end{bmatrix}=\frac{1}{2}\begin{bmatrix}0&-1\\ 1&0\end{bmatrix}$; and $\xi_3=\frac{i}{2}\sigma_3=-F_3$. Thus

$$(\xi_1,\xi_2,\xi_3)=(-F_1,\,F_2,\,-F_3). \tag{316}$$

This change of basis is conjugation by an element of $SU(2)$. Let $u=-i\sigma_2=\begin{bmatrix}0&-1\\ 1&0\end{bmatrix}$, so $u^{-1}=i\sigma_2$. Since distinct Pauli matrices anticommute (Frankel's Eq. (19.18)) and $\sigma_2^2=I$,

$$\begin{aligned}u\sigma_1u^{-1}&=\sigma_2\sigma_1\sigma_2=-\sigma_1\sigma_2\sigma_2=-\sigma_1,\\ u\sigma_2u^{-1}&=\sigma_2\sigma_2\sigma_2=\sigma_2,\\ u\sigma_3u^{-1}&=\sigma_2\sigma_3\sigma_2=-\sigma_3\sigma_2\sigma_2=-\sigma_3,\end{aligned} \tag{317}$$

where the factors $(-i)(i)=1$ cancel. Dividing by $2i$ and comparing with Eq. (316), $\xi_a=uF_au^{-1}=\mathrm{Ad}(u)F_a$. Under Eq. (328) u is the quaternion $e_2=j$, which is Eq. (334) with $\theta=\pi$ and $\mathbf{A}=\mathbf{e}_2$: the rotation by π about the second axis, $\mathrm{diag}(-1,1,-1)$, as the signs in Eq. (316) show.

This is an instance of a general fact. Two bases with the same structure constants determine the same Lie algebra, but the isomorphism between them is fixed only up to an automorphism. For $\mathfrak{so}(3)$, identified with $(\mathbb{R}^3,\times)$ by Eq. (257), the automorphisms are exactly $SO(3)$. Let $A\in GL(3,\mathbb{R})$ satisfy $A(\mathbf{u}\times\mathbf{v})=A\mathbf{u}\times A\mathbf{v}$ for all $\mathbf{u},\mathbf{v}$.

A is orthogonal. The Killing form $B(X,Y)=\mathrm{tr}(\mathrm{ad}_X\,\mathrm{ad}_Y)$ is preserved by automorphisms, since $A[X,A^{-1}Y]=[AX,Y]$ gives $\mathrm{ad}_{AX}=A\,\mathrm{ad}_X\,A^{-1}$ and the trace is invariant under conjugation. On $(\mathbb{R}^3,\times)$, $\mathrm{ad}_{\mathbf{e}_i}=E_i$ by Eq. (244), so by Eq. (236)

$$B(\mathbf{e}_i,\mathbf{e}_j)=(E_i)_{ab}(E_j)_{ba}=\epsilon_{iab}\epsilon_{jba}=-\epsilon_{iab}\epsilon_{jab}=-2\delta_{ij}. \tag{318}$$

Hence A preserves the Euclidean inner product, $A^TA=I$.

$\det A=1$. For any $A\in GL(3,\mathbb{R})$ and all $\mathbf{u},\mathbf{v},\mathbf{w}$,

$$(A\mathbf{u}\times A\mathbf{v})\cdot A\mathbf{w}=\det[A\mathbf{u},A\mathbf{v},A\mathbf{w}]=\det A\,\det[\mathbf{u},\mathbf{v},\mathbf{w}]=\det A\,(\mathbf{u}\times\mathbf{v})\cdot\mathbf{w}, \tag{319}$$

so $A^T(A\mathbf{u}\times A\mathbf{v})=\det A\,(\mathbf{u}\times\mathbf{v})$. With $A^T=A^{-1}$ and $A\mathbf{u}\times A\mathbf{v}=A(\mathbf{u}\times\mathbf{v})$ this reads $\mathbf{u}\times\mathbf{v}=\det A\,(\mathbf{u}\times\mathbf{v})$. Choosing $\mathbf{u}\times\mathbf{v}\neq0$ gives $\det A=1$. Conversely, the same identity with $A^T=A^{-1}$ and $\det A=1$ shows that every $R\in SO(3)$ preserves the cross product. Through Eq. (257) this reads $R\,\widehat{\mathbf{v}}\,R^{-1}=\widehat{R\mathbf{v}}$, so each automorphism of $\mathfrak{so}(3)$ is $\mathrm{Ad}(R)$ for a rotation R. Kosmann-Schwarzbach's and Frankel's identifications of $\mathfrak{su}(2)$ with $\mathfrak{so}(3)$ differ by the rotation $\mathrm{diag}(-1,1,-1)=\mathrm{Ad}(j)$.

95.1 Kosmann-Schwarzbach's covering map φ

Kosmann-Schwarzbach builds her covering map from the same sign choice. Her quaternion basis of $\mathfrak{su}(2)$ is [85, Ch. 5 §1.1(d), p. 72]

$$\mathcal{I}=2\xi_1=i\sigma_1,\qquad \mathcal{J}=2\xi_2=-i\sigma_2,\qquad \mathcal{K}=2\xi_3=i\sigma_3, \tag{320}$$

with $\mathcal{I}^2=\mathcal{J}^2=\mathcal{K}^2=-I$, $\mathcal{I}\mathcal{J}=\mathcal{K}$, $\mathcal{J}\mathcal{K}=\mathcal{I}$, $\mathcal{K}\mathcal{I}=\mathcal{J}$, and she identifies the quaternion units i,j,k with $\mathcal{I},\mathcal{J},\mathcal{K}$ (p. 73). Under Eq. (328) these are $(-e_1,e_2,-e_3)=(je_1j^{-1},\,je_2j^{-1},\,je_3j^{-1})$; for instance $je_1j^{-1}=-je_1j=-(-e_3)j=e_3e_2=-e_1$. So her units are ours conjugated by j, again Hamilton's order since conjugation is an algebra automorphism.

She identifies $\mathbb{R}^3$ with $\mathfrak{su}(2)$ by $a\mapsto X_a=a^1\mathcal{I}+a^2\mathcal{J}+a^3\mathcal{K}$ (§2.2, p. 77), and $\det X_a=\|a\|^2$ (her Eq. (1.1), p. 73). Put $D=\mathrm{diag}(-1,1,-1)$. By Eq. (320),

$$X_a=ia^1\sigma_1-ia^2\sigma_2+ia^3\sigma_3=-i\big((-a^1)\sigma_1+a^2\sigma_2+(-a^3)\sigma_3\big)=-i\Sigma(Da). \tag{321}$$

Her map $\varphi\colon SU(2)\to GL(3,\mathbb{R})$ sends g to the matrix of Ad_g in the basis $(\mathcal{I},\mathcal{J},\mathcal{K})$ (§2.3, pp. 77–78), that is, $\mathrm{Ad}_gX_a=X_{\varphi(g)a}$. Compute the left side with Eq. (272), $g\Sigma(\mathbf{x})g^{-1}=\Sigma(\Phi(g)\mathbf{x})$, and $D^2=I$:

$$\mathrm{Ad}_gX_a=-i\,g\Sigma(Da)g^{-1}=-i\Sigma\big(\Phi(g)Da\big)=-i\Sigma\big(D\,(D\Phi(g)D)\,a\big)=X_{D\Phi(g)Da}. \tag{322}$$

Since $a \mapsto X_a$ is a bijection, $\varphi(g) = D\Phi(g)D^{-1}$. By Eq. (317), $u = -i\sigma_2$ satisfies $u\Sigma(\mathbf{e}_1)u^{-1} = \Sigma(-\mathbf{e}_1)$, $u\Sigma(\mathbf{e}_2)u^{-1} = \Sigma(\mathbf{e}_2)$, $u\Sigma(\mathbf{e}_3)u^{-1} = \Sigma(-\mathbf{e}_3)$, so $\Phi(u) = D$. Because Φ is a homomorphism,

$$\varphi(g) = \Phi(u)\Phi(g)\Phi(u)^{-1} = \Phi(ugu^{-1}), \qquad \varphi = \Phi \circ \mathcal{C}_u, \qquad u = -i\sigma_2. \tag{323}$$

Thus φ is Φ preceded by an automorphism of $SU(2)$, and everything proved for Φ transfers. φ is onto $SO(3)$ because $\mathcal{C}_u$ is onto $SU(2)$. Also $\varphi(g) = I$ if and only if $ugu^{-1} \in \ker\Phi = \{\pm I\}$, if and only if $g = \pm I$, which is her $\operatorname{Ker}\varphi = \{I, -I\}$ (p. 78). She concludes that $SU(2)$ is the double-sheeted universal cover of $SO(3)$ (Proposition 2.6, p. 78).

Her half angle. Kosmann-Schwarzbach writes $\operatorname{Rot}(a,t)$ for the rotation through t about the unit vector a, $\operatorname{Rot}(a,t)x = x + (1-\cos t)\,a \wedge (a \wedge x) + \sin t\, a \wedge x$ (her Eq. (2.1), p. 74). This is our $R_{\mathbf{n}}(\theta)$: expanding $a \wedge (a \wedge x) = (a \cdot x)a - x$ and writing $x = (a \cdot x)a + x_\perp$ gives Eq. (247). Her Proposition 2.5(iii) (p. 78) states

$$\varphi\big(\exp(tX_a)\big) = \operatorname{Rot}(a, 2t) \qquad (\|a\| = 1). \tag{324}$$

Check it against Eq. (302). By her Proposition 2.4 (p. 77), $\exp(tX) = \cos t\, I + \sin t\, X$ when $X \in \mathfrak{su}(2)$ and $\det X = 1$. With Eq. (321), $\exp(tX_a) = \cos t\, I - i \sin t\, \Sigma(Da)$, which is Eq. (302) with $\theta = 2t$ and $\mathbf{n} = Da$. So $\Phi(\exp(tX_a)) = R_{Da}(2t)$, and by Eq. (323)

$$\varphi\big(\exp(tX_a)\big) = D\,R_{Da}(2t)\,D^{-1} = R_{D^2a}(2t) = R_a(2t), \tag{325}$$

using $gR_{\mathbf{n}}(\theta)g^{-1} = R_{g\mathbf{n}}(\theta)$ for $g \in SO(3)$ (her Eq. (2.2), p. 75; g preserves the cross product, so $gR_{\mathbf{n}}(\theta)g^{-1}$ fixes $g\mathbf{n}$ and turns the plane perpendicular to it through θ in the same sense). Her parameter t is our half angle $\theta/2$.

Why her signs. Her Proposition 2.5(i) (p. 78) says the matrix of $\operatorname{ad}_{\xi_k}$ in the basis (ξ_1, ξ_2, ξ_3) is η_k, and her η_k (§1.2, p. 73) are the matrices E_k of Eq. (236). So $d\varphi(\xi_k) = E_k$ with no signs, and her Lie algebra isomorphism $\xi_k \mapsto \eta_k$ (p. 73) is exactly $d\varphi$. For Φ the same basis gives $d\Phi(\xi_k) = (-E_1, E_2, -E_3)$ (§91); differentiating $\varphi = D\Phi D^{-1}$ gives $d\varphi(\xi_k) = D\,d\Phi(\xi_k)D^{-1}$, and $DE_1D^{-1} = \widehat{D\mathbf{e}_1} = -E_1$, $DE_2D^{-1} = E_2$, $DE_3D^{-1} = -E_3$ by $R\hat{\mathbf{v}}R^{-1} = \widehat{R\mathbf{v}}$, which cancels the signs. Frankel's choice makes $F_a \mapsto E_a$ equal to $d\Phi$; Kosmann-Schwarzbach's makes $\xi_k \mapsto \eta_k$ equal to $d\varphi$. Each author chose the $\mathbb{R}^3$ identification that makes their own basis map to E_k.

Isomorphism theorem. Kosmann-Schwarzbach states only that a kernel is a normal subgroup and that $p\colon G \to G/H$ is a morphism (Ch. 1, p. 2). The isomorphism $SU(2)/\{\pm I\} \cong SO(3)$ is Theorem 65 applied to φ or Φ.

96 The quaternion basis in Frankel and in Holm

Frankel [11] works with the Pauli matrices

$$\sigma_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \qquad \sigma_2 = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \qquad \sigma_3 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \tag{326}$$

(his Eq. (19.5), p. 493). Here the i inside σ_2 is the complex unit, not the quaternion e_1. His product relations, Eq. (19.17), p. 499, are

$$\sigma_1\sigma_2 = i\sigma_3, \qquad \sigma_2\sigma_3 = i\sigma_1, \qquad \sigma_3\sigma_1 = i\sigma_2, \qquad \sigma_a^2 = I. \tag{327}$$

Holm [12] writes a quaternion as a tetrad $q = [q_0, \mathbf{q}] = q_0 J_0 + q^a J_a$ and identifies $J_0 = \sigma_0 = I$ and $J_a = -i\sigma_a$ (Section 3.1.1, Eqs. (3.1.1)–(3.1.3), pp. 79–81). Frankel uses σ_a/i, and $1/i = -i$, so both authors use the same matrices:

$$e_a \longleftrightarrow J_a = -i\sigma_a = \frac{\sigma_a}{i}. \tag{328}$$

Check that these matrices satisfy Eq. (221). Using Eq. (327),

$$\begin{aligned} (-i\sigma_1)(-i\sigma_2) &= (-i)^2\sigma_1\sigma_2 = -\sigma_1\sigma_2 = -i\sigma_3, \\ (-i\sigma_a)^2 &= (-i)^2\sigma_a^2 = -I. \end{aligned} \tag{329}$$

The first line is $e_1e_2 = e_3$ under Eq. (328); the cyclic permutations of Eq. (327) give $e_2e_3 = e_1$ and $e_3e_1 = e_2$ in the same way. Both books therefore follow Hamilton's order, and Eq. (328) is an algebra isomorphism from $\mathbb{H}$ onto the real span of I and $-i\sigma_a$.

Two scalings differ from ours. Frankel's real basis of $\mathfrak{su}(2)$ is $\sigma_a/(2i) = \frac{1}{2}e_a$, which has the $\mathfrak{so}(3)$ brackets (p. 493). His identification of $\mathbb{R}^3$ with traceless Hermitian matrices, Eq. (19.7), p. 494, is $\mathbf{x}_* = \Sigma(\mathbf{x}) = x^a\sigma_a$ (Eq. (268)), while our pure quaternion is

$$x^a e_a = x^a(-i\sigma_a) = -i\Sigma(\mathbf{x}), \tag{330}$$

which is Holm's $\widetilde{\mathbf{x}}$.

97 The half angle in both notations

Frankel's Eq. (19.15), p. 497, sends $\exp\big(\frac{\boldsymbol{\sigma}}{2i}\cdot\mathbf{A}\theta\big)\in SU(2)$ to $\exp(\theta\,\mathbf{E}\cdot\mathbf{A})\in SO(3)$, the rotation by θ about the unit vector $\mathbf{A}$. To sum the left side, use his anticommutator, Eq. (19.18), p. 499: $\sigma_a\sigma_b+\sigma_b\sigma_a=2\delta_{ab}I$. For a unit vector $\mathbf{A}$,

$$\begin{aligned}(\boldsymbol{\sigma}\cdot\mathbf{A})^2 &= A^aA^b\sigma_a\sigma_b=\tfrac12 A^aA^b\big(\sigma_a\sigma_b+\sigma_b\sigma_a\big)\\ &= A^aA^b\delta_{ab}I=I.\end{aligned}\tag{331}$$

The symmetrization in the first line is allowed because A^aA^b is symmetric in a,b. Put $N=\boldsymbol{\sigma}\cdot\mathbf{A}/i=A^a(-i\sigma_a)$, the pure unit quaternion A^ae_a under Eq. (328). Then Eq. (331) gives $N^2=(1/i)^2I=-I$, hence

$$N^{2m}=(-1)^mI,\qquad N^{2m+1}=(-1)^mN.\tag{332}$$

Split the exponential series into even and odd powers:

$$\begin{aligned}\exp\Big(\frac{\theta}{2}N\Big) &= \sum_{m=0}^{\infty}\frac{(\theta/2)^{2m}}{(2m)!}N^{2m}+\sum_{m=0}^{\infty}\frac{(\theta/2)^{2m+1}}{(2m+1)!}N^{2m+1}\\ &= \sum_{m=0}^{\infty}\frac{(-1)^m(\theta/2)^{2m}}{(2m)!}\,I+\sum_{m=0}^{\infty}\frac{(-1)^m(\theta/2)^{2m+1}}{(2m+1)!}\,N\\ &= \cos\frac{\theta}{2}\,I+\sin\frac{\theta}{2}\,N.\end{aligned}\tag{333}$$

Since $N=-i\,\boldsymbol{\sigma}\cdot\mathbf{A}$, Eq. (333) is Frankel's $\cos(\theta/2)I-i\sin(\theta/2)\,\boldsymbol{\sigma}\cdot\mathbf{A}$ (p. 500). In the notation of Eq. (222) it is

$$q=\cos\frac{\theta}{2}+\sin\frac{\theta}{2}\,A^ae_a.\tag{334}$$

Replacing θ by $\theta+2\pi$ changes the signs of both $\cos(\theta/2)$ and $\sin(\theta/2)$, so $q\mapsto -q$ while the rotation $\exp(\theta\,\mathbf{E}\cdot\mathbf{A})$ is unchanged.

98 Reading quaternions geometrically

Frankel [11] is the spine. Printed pages refer to the third edition. Read Sections 19.1–19.2 (pp. 491–503): $SO(3)$ and $\mathfrak{so}(3)$, the Lie algebra $\mathfrak{su}(2)$, $SU(2)\cong S^3$, the homomorphism $\mathrm{Ad}\colon SU(2)\to SO(3)$ and its kernel, spinors, Hamilton's composition of rotations, and Clifford algebras. Review Section 15.3 (pp. 402–406) for the exponential map when needed. Then read Section 21.2 (pp. 567–575) on the fundamental group, covering spaces, and the universal covering group, which gives $\pi_1(SO(3))=\mathbb{Z}_2$ and $SU(2)$ as the universal cover. Read the Hopf bundle in Section 17.4c (p. 473) and the Hopf map in Section 22.4c (pp. 606–607), after the homotopy groups of Section 22.3 (pp. 596–604). Read the spinor bundle and spin connection of Section 19.5 (pp. 515–522) after the connections part of these notes.

Holm [12] then puts the same group into mechanics. Chapter 3 (pp. 77–110) covers quaternion multiplication through Pauli matrices, Cayley–Klein parameters, pure quaternions and $SU(2)$, the tilde map $\mathbb{R}^3\cong\mathfrak{su}(2)\cong\mathfrak{so}(3)$ and its dual (Sections 3.2.3–3.2.4, pp. 102–103), and Poincaré's sphere with the Hopf fibration (Section 3.2.6, p. 105). Continue with Sections 4.1–4.2 (pp. 112–123) on Cayley–Klein rigid-body dynamics and the AD, Ad, ad, Ad^* and ad^* actions, and Chapter 5 (pp. 131–140) on $SO(3)$. Compare Section 3.2.6 with Frankel's Section 22.4c.

For Clifford algebras and spin groups in general dimension, read Chapter I of Lawson and Michelsohn [13]. Its sign convention for the Clifford relation must be compared with Frankel's Section 19.2c before the two are combined. For matrix Lie groups, the exponential, $\mathfrak{su}(2)$ representations and universal covers in a single Lie-theoretic notation, use Chapters 1–5 of Hall [14].

Derive Eq. (223) from Eq. (272) under Eq. (330), and the quaternion kinematics matching Eq. (553).

Part XXI
Cohomology; Stoke's Theorem

99 Stoke's Theorem

Theorem 66 (Stoke's Theorem)**.** *Let M be oriented, smooth n-manifold with boundary, let ω be a compactly supported smooth $(n-1)$-form on M, or if $\omega \in A_c^{n-1}(M)$, Then*

$$\int_M d\omega = \int_{\partial M} \omega \tag{335}$$

If $\partial M = \emptyset$, then $\int_{\partial M} \omega = 0$
$\int_{\partial M} \omega$ interpreted as $\int_{\partial M} i_{\partial M}^* \omega = \int_{\partial M} i^* \omega$ so

$$\int_M d\omega = \int_{\partial M} i^*(\omega) \tag{336}$$

where inclusion $i : \partial M \hookrightarrow M$

Proof. Begin with very special case:
Suppose $M = \mathbb{H}^n$ (upper half space), $\partial M = \mathbb{R}^{n-1}$
ω has compact support, so $\exists\, R > 0$ s.t. supp$\omega \subseteq$ rectangle $A = [-R, R] \times \cdots \times [-R, R] \times [0, R]$.
$\forall\, \omega \in A_c^{n-1}(\mathbb{H}^n)$

$$\omega = \sum_{j=1}^{n} (-1)^{j-1} f_j dx^1 \wedge \cdots \wedge \widehat{dx^j} \wedge \cdots \wedge dx^n \equiv \sum_{i=1}^{n} \omega_i dx^1 \wedge \cdots \wedge \widehat{dx^i} \wedge \cdots \wedge dx^n \tag{337}$$

with Conlon (2008) [90] and John Lee (2012) [76]'s notation, respectively, and where f_j has compact support.

$$i^*\omega = (f_1 \circ i) dx^2 \wedge \cdots \wedge dx^n \in A_c^{n-1}(\partial \mathbb{H}^n)$$

$$d\omega = \sum_{i=1}^{n} d\omega_i \wedge dx^1 \wedge \cdots \wedge \widehat{dx^i} \wedge \cdots \wedge dx^n = \sum_{i,j=1}^{n} \frac{\partial \omega_i}{\partial x^j} dx^j \wedge dx^1 \wedge \cdots \wedge \widehat{dx^i} \wedge \cdots \wedge dx^n =$$

$$= \sum_{i=1}^{n} (-1)^{i-1} \frac{\partial \omega_i}{\partial x^i} dx^1 \wedge \cdots \wedge dx^n$$

i.e. (for another notation)

$$d\omega = \left(\sum_{j=1}^{n} \frac{\partial f_j}{\partial x^j} \right) dx^1 \wedge \cdots \wedge dx^n \in A_c^n(\mathbb{H}^n)$$

$$d\omega = \left(\sum_{j=1}^{n} \frac{\partial f_j}{\partial x^j} \right) dx^1 \wedge \cdots \wedge dx^n \in A_c^n(\mathbb{H}^n)$$

$$\int_{\mathbb{H}^n} d\omega = \sum_{i=1}^{n} (-1)^{i-1} \int_A \frac{\partial \omega_i}{\partial x^i} dx^1 \wedge \cdots \wedge dx^n = \sum_{i=1}^{n} (-1)^{i-1} \int_0^R \int_{-R}^R \cdots \int_{-R}^R dx^1 \ldots dx^n \frac{\partial \omega_i}{\partial x^i}(x)$$

We can change order of integration in each term so to do x^i integration first.
By fundamental thm. of calculus, terms for which $i \neq n$ reduce to

$$\sum_{i=1}^{n-1} (-1)^{i-1} \int_0^R \int_{-R}^R \cdots \int_{-R}^R \frac{\partial \omega_i}{\partial x^i}(x) dx^1 \ldots dx^n = \sum_{i=1}^{n-1} (-1)^{i-1} \int_0^R \int_{-R}^R \cdots \int_{-R}^R \frac{\partial \omega_i}{\partial x^i}(x) dx^i dx^1 \ldots \widehat{dx^i} \ldots dx^n =$$

$$= \sum_{i=1}^{n-1} (-1)^{i-1} \int_0^R \int_{-R}^R \cdots \int_{-R}^R [\omega_i(x)]_{x^i=-R}^{x^i=R} dx^1 \ldots \widehat{dx^i} \ldots dx^n = 0$$

because we've chosen R large enough that $\omega = 0$ when $x^i = \pm R$.

□

Part XXII
Prástaro

Prástaro (1996) [86]

99.0.1 Affine Spaces

cf. Sec. 1.2 - *Affine Spaces* of Prástaro (1996) [86]

Definition 183 (affine space)**.**

$$\begin{array}{l} \text{affine space} \qquad (M, \mathbf{M}, \alpha) \\ \text{with} \\ M \equiv \text{ set (set of pts.)} \\ \mathbf{M} \equiv \text{ vector space (space of free vectors)} \\ \alpha \equiv \mathbf{M} \times M \to M \equiv \text{ translation operator} \\ \alpha : (v, p) \mapsto p' \equiv p + v \end{array} \tag{338}$$

Note: α is a **transitive** action and without fixed pts. (free).
i.e. $\forall p \in M$,

$\forall$ pt. $O \in M$, $\alpha : (v, O) \mapsto O' \equiv O + v$, $\alpha(\cdot, O) \equiv \alpha_O \equiv \alpha(O)$. $\alpha_O(v) = O' = O + \mathbf{v}$ $\quad \forall O' \in M, \exists \mathbf{v} \in \mathbf{M}$ s.t. $O' = O + \mathbf{v}$
$\Longrightarrow M \equiv \mathbf{M}$.
$\forall (O, \{e_i\})_{1 \le i \le n}$, where $\{e_i\}$ basis of $\mathbf{M}$, $M \equiv \mathbf{M} = \mathbb{R}^n$ so isomorphism $M \simeq \mathbb{R}^n$

i.e. α is **without fixed pts.**, meaning,
Given pointed space (M, O), where base pt. $O \in M$, we can associate $\forall p \in M$, vector $\mathbf{x} \in \mathbf{M}$, by 1-to-1 mapping $M \to \mathbf{M}$.
So for

$$\begin{aligned} &\alpha : \mathbf{M} \times M \to M \\ &\alpha(\mathbf{x}, p) = p' = p + \mathbf{x} \end{aligned}$$

Consider

$$\alpha(\mathbf{x}, O) = p = \alpha_O(\mathbf{x}) = p \Longrightarrow \exists\, \alpha_O^{-1}(p) = \mathbf{x} \in \mathbf{M}$$

1. tangent space of M in $p \in M$ is vector space $T_pM \equiv (\mathbf{M}, p) \cong M$
2. If $\mathbf{M}$ Euclidean space, affine space $(M, \mathbf{M}, \alpha)$ is Euclidean
3. Call dim. of affine space $(M, \mathbf{M}, \alpha)$, dim. of $\mathbf{M} \equiv \dim \mathbf{M}$

$\{\mathbf{e}_i\}$ basis of $\mathbf{M}$

Definition 184. $(O, \{e_i\}) \equiv$ affine frame.
$\forall$ affine frame $(O, \{e_i\})$, $\exists$ coordinate system $x^\alpha : M \to \mathbb{R}$,
where $x^\alpha(p)$ is αth component, in basis $\{e_i\}$, of vector $p - O$

Proposition 97 (1.6, Prástaro (1996) [86])**.** *$\forall O \in M$, we have canonical identification $M \equiv \mathbf{M}$, since*

$$\begin{array}{ll} \alpha_O^{-1} : M \to \mathbf{M} & \alpha_O : \mathbf{M} \to M \\ \alpha_O^{-1}(p) = \mathbf{x} & \alpha_O : \mathbf{x} = \alpha(\mathbf{x}, O) = p \end{array}$$

Furthermore,
*$\forall$ **affine frame** $(O, \{\mathbf{e}_i\})_{1 \le i \le d}$, where $\{\mathbf{e}_i\}$ basis of $\mathbf{M}$,*
$\exists$ isomorphism $M \cong \mathbb{R}^d$,
Then, $\forall (O, \{\mathbf{e}_i\})_{1 \le i \le d}$,
$\exists$ coordinate system $x^\alpha : M \to \mathbb{R}$,
where $x^\alpha(p) = \alpha$th component, in basis $\{\mathbf{e}_i\}$, of vector $p - O$.

Theorem 67 (1.4 Prástaro (1996) [86]). *Let $(x^\alpha),(\overline{a}^\alpha)$ 2 coordinate systems correspond to affine frames $(O,\{e_i\})$, $(\overline{O},\{\overline{e}_i\})$, respectively.*

$$\overline{x}^\alpha = A^\alpha_\beta x^\beta + y^\alpha \tag{339}$$

where

$$y^\alpha \in \mathbb{R}^n, \qquad A^\alpha_\beta \in GL(n;\mathbb{R})$$

Definition 185 (1.10 Prástaro (1996) [86]).

$$A(n) \equiv Gl(n,\mathbb{R}) \times \mathbb{R}^n \tag{340}$$

affine group of dim. n

Theorem 68 (1.5). *symmetry group of n-dim. affine space, called affine group $A(M)$ of M. $\exists$ isomorphism,*

$$A(M) \simeq A(n), \qquad f \mapsto (f^\alpha_\beta, y^\alpha); \qquad f^\alpha \equiv x^\alpha \circ f = f^\alpha_\beta x^\beta + y^\alpha \tag{341}$$

cf. Eq. 1.4 Prástaro (1996) [86]

Definition 186 (metric). Let smooth manifold M, $\dim M = n$, $\forall p \in M$, $\exists$ vector space T_pM, and so for

$$\begin{array}{l} g_p(T_pM)^2 \to \mathbb{R} \\ g_p : (X_p, Y_p) \mapsto g_p(X_p, Y_p) \in \mathbb{R} \end{array} \tag{342}$$

with g_p being bilinear, symmetric (in X_p, Y_p), nondegenerate (i.e. if $g_p(X_p,Y_p)=0$, then X_p or $Y_p = 0$)

Note that

$$g \in \Gamma((TM \otimes TM)^*)$$

and that for $X = X^i \dfrac{\partial}{\partial x^i}$ so

$$Y = Y^i \frac{\partial}{\partial x^i}$$

$$g(X,Y) = g_{ij} X^i Y^j$$

Now for

$$\begin{array}{ll} F : M \to N & DF \equiv F_* : T_pM \to T_{F(p)}N \\ F : x \mapsto y = y(x) & DF : X_p \mapsto (DF)(X^j \dfrac{\partial}{\partial x^j}) = X^j \dfrac{\partial y^i}{\partial x^j} \dfrac{\partial}{\partial y^i} \end{array}$$

$$(F^* g')(X,Y) = (F^* g')(X^i \frac{\partial}{\partial x^i}, Y^j \frac{\partial}{\partial x^j}) = (F^* g')_{ij} X^i Y^j = g'(F_* X, F_* Y) =$$
$$=$$

Part XXIII
Connections

100 Connections of Vector Bundles

Morita (2001) [114]

Definition 187 (Connection in a vector bundle). **connection in a vector bundle** $\pi : E \to M$ over C^∞ manifold M, is a bilinear map

$$\nabla : \mathfrak{X}(M) \times \Gamma(E) \to \Gamma(E)$$

satisfying

(i) $\nabla_{fX} s = f \nabla_X s$

(ii) $\nabla_X(fs) = f\nabla_X s + (Xf)s$ where $f \in C^\infty(M)$, $X \in \mathfrak{X}(M)$, $s \in \Gamma(E)$

$\nabla_X s$ is covariant derivative of s relative to X (for Morita (2001)[114])

Alternatively, Jost (2017) [116] defines in the following way:

Let vector bundle (E, π, M), where differentiable manifold M, vector bundle E over M; a covariant derivative or (linear) connection is map

$$D : \Gamma(E) \to \Gamma(E) \otimes \Gamma(T^*M) \equiv \Gamma(E) \otimes \mathfrak{X}(M) \tag{343}$$

(we may also consider D as map $D : \mathfrak{X}(M) \otimes \Gamma(E) \to \Gamma(E)$, with $s \in \Gamma(E), X \in \mathfrak{X}(M), DX(s) \equiv D_X s$) s.t.

1. D is tensorial in $\mathfrak{X}(M)$,
$$\begin{aligned} D_{X+Y}s &= D_X s + D_Y s \quad \forall\, X, Y \in T_xM,\ s \in \Gamma(E) \\ D_{fX}s &= fD_X s \quad \forall\, f \in C^\infty(M, \mathbb{R}),\ X \in \Gamma(TM) \end{aligned} \tag{344}$$
2. D is $\mathbb{R}$-linear in s,
$$D_X(s+t) = D_X s + D_X t \quad \forall\, X \in T_xM,\ s, t \in \Gamma(E) \tag{345}$$
and product rule
$$D_X(fs) = X(f) \cdot s + fD_X(s) \quad \forall\, f \in C^\infty(M, \mathbb{R}) \tag{346}$$

Claim: *Any vector bundle admits a connection.*

e.g. product bundle $M \times \mathbb{R}^n$. Let $x_1, \dots, x_n$ be canonical coordinates in $\mathbb{R}^n$. Take frame field $(s_1, \dots, s_n)$, where $s_i(p) = \frac{\partial}{\partial x^i}$. Set $\nabla_X s_i = 0$ $(i = 1, \dots, m)$ $\quad \forall$ vector space X,
$\forall\, s = \sum_i a_i s_i$, $\forall X \in \mathfrak{X}(M)$, set

$$\nabla_X s = \sum_{i=1}^n (Xa_i)s_i$$

For this connection $\nabla_X s$ is the partial derivative in direction of X if s is considered $\mathbb{R}^n$-valued function on M. Call it **trivial connection** in product bundle.

Indeed,

$$\begin{aligned} \nabla_X s = \nabla_{X^i \frac{\partial}{\partial x^i}} \left(s^m \frac{\partial}{\partial x^m} \right) = X^i \nabla_{\frac{\partial}{\partial x^i}} \left(s^m \frac{\partial}{\partial x^m} \right) = X^i \left(\nabla_{\frac{\partial}{\partial x^i}} s^m \right) \frac{\partial}{\partial x^m} + X^i s^m \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^m} = \\ = X^i \left(\frac{\partial}{\partial x^i} s^m \right) \frac{\partial}{\partial x^m} + X^i s^m \Gamma^q{}_{mi} \frac{\partial}{\partial x^q} = X^i \frac{\partial s^m}{\partial x^i} \frac{\partial}{\partial x^m} + 0 \end{aligned}$$

if $\Gamma^q{}_{mi} = 0$ at a chosen point p.

For arbitrary vector bundle $\pi : E \to M$, take locally finite open covering $\{U_\alpha\}_{\alpha \in A}$ s.t. $\pi^{-1}(U_\alpha)$ trivial. Denote ∇^α trivial connection $\forall\, \pi^{-1}(U_\alpha)$. Let $\{f_\alpha\}$ be a partition of unity for covering U_α, define

$$\nabla_X s := \sum_\alpha f_\alpha \nabla^\alpha_X s$$

Verify this defines connection in E:

$$\begin{aligned} \nabla_X(gs) = \sum_\alpha f_\alpha \nabla^\alpha_X (gs) = \sum_\alpha f_\alpha \left[X^i \left(\frac{\partial g}{\partial x^i} \right) s + X^i g \frac{\partial s^m}{\partial x^i} \frac{\partial}{\partial x^m} \right] = \\ = g \sum_\alpha f_\alpha \nabla^\alpha_X s + \sum_\alpha f_\alpha (Xg) s \end{aligned}$$

Proposition 98 (5.18 Morita (2001)[114])**.** *Let ∇_i $(1 \le i \le k)$ be k connections in a given vector bundle. Then $\forall$ linear combination $\sum_{i=1}^k t_i \nabla_i$, where $t_1 + \cdots + t_k = 1$ is a connection.*

Proof. TODO: Ex. 5.5 □

100.1 Directional derivatives of tensor fields; Connections on a tensor product

cf. Schuller (2015) [109], Lecture 7: Connections (International Winter School on Gravity and Light 2015)).

Definition 188 (Connection on tensor fields)**.** A **connection** ∇ or smooth manifolds $(M, \mathcal{O}, \mathcal{A})$ is a map $\nabla : \mathfrak{X}(M) \times \Gamma(TM^p \otimes T^*M^q) \to \Gamma(TM^p \otimes T^*M^q)$,
where $\binom{p}{q}$ tensor-field section $\equiv \Gamma(TM^p \otimes T^*M^q)$

1. $\nabla_X f = Xf \quad \forall\, f \in C^\infty(M)$
2. $\nabla_X(T+S) = \nabla_X T + \nabla_X S$
3. $\nabla_X(T(\omega, Y)) = (\nabla_X T)(\omega, Y) + T(\nabla_X \omega, Y) + T(\omega, \nabla_X Y)$, where $T(\omega, Y) \in C^\infty(M)$

   This is written for a $(1,1)$ tensor field T, but is analogously true $\forall\,(p,q)$ tensor field.
4. $\nabla_{fX+Z} T = f\nabla_X T + \nabla_Z Y$, where $f \in C^\infty(M)$

   Consider also product rule $\nabla_X(fT) = X(f)T + f\nabla_X T \quad \forall\, f \in C^\infty(M, \mathbb{R})$

A manifold with connection is a quadruple $(M, \mathcal{O}, \mathcal{A}, \nabla)$

Consider

$$\nabla_X Y = \nabla_{X^i \frac{\partial}{\partial x^i}} \left(Y^m \frac{\partial}{\partial x^m} \right) = X^i \nabla_{\frac{\partial}{\partial x^i}} \left(Y^m \frac{\partial}{\partial x^m} \right) = X^i \left[\left(\nabla_{\frac{\partial}{\partial x^i}} Y^m \right) \frac{\partial}{\partial x^m} + Y^m \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^m} \right] =$$
$$= X^i \frac{\partial Y^m}{\partial x^i} \frac{\partial}{\partial x^m} + X^i Y^m \Gamma^q{}_{mi} \frac{\partial}{\partial x^q}$$

so then

Definition 189 (Connection coefficient functions, i.e. Christoffel Symbols)**.**

$$\begin{aligned} \Gamma^i{}_{jk} : \mathcal{U} &\to \mathbb{R} \\ p &\mapsto (dx^i \left(\nabla_{\frac{\partial}{\partial x^k}} \frac{\partial}{\partial x^j} \right))(p) \end{aligned} \tag{347}$$

Now since

$$\nabla_{\frac{\partial}{\partial x^m}} \left(dx^i \left(\frac{\partial}{\partial x^j} \right) \right) = \nabla_{\frac{\partial}{\partial x^m}} (\delta^i{}_j) = 0 \text{ so}$$
$$\nabla_{\frac{\partial}{\partial x^m}} \left(dx^i \left(\frac{\partial}{\partial x^j} \right) \right) = \left(\nabla_{\frac{\partial}{\partial x^m}} dx^i \right) \left(\frac{\partial}{\partial x^j} \right) + dx^i \left(\nabla_{\frac{\partial}{\partial x^m}} \frac{\partial}{\partial x^j} \right) = \left(\nabla_{\frac{\partial}{\partial x^m}} dx^i \right) \left(\frac{\partial}{\partial x^j} \right) + dx^i \Gamma^q{}_{jm} \frac{\partial}{\partial x^q} = 0$$
$$\implies \boxed{\left(\nabla_{\frac{\partial}{\partial x^m}} dx^i \right) \left(\frac{\partial}{\partial x^j} \right) = -\Gamma^i{}_{jm}} \text{ since}$$
$$dx^i \Gamma^q{}_{jm} \frac{\partial}{\partial x^q} = \Gamma^q{}_{jm} dx^i \left(\frac{\partial}{\partial x^q} \right) = \Gamma^q{}_{jm} \delta^i_q = \Gamma^i{}_{jm}$$

100.1.1 Change of Γ's under change of chart

cf. 1:08:50 3. Change of Γ's under change of chart, Schuller (2015) [109]
(U, x), $(V, y) \in \mathcal{A}$ and $U \cap V \neq \emptyset$

$$\Gamma^i{}_{jk}(y) := dy^i \left(\nabla_{\frac{\partial}{\partial y^k}} \frac{\partial}{\partial y^j} \right) = \frac{\partial y^i}{\partial x^q} dx^q \left(\nabla_{\frac{\partial x^p}{\partial y^k} \frac{\partial}{\partial x^p}} \frac{\partial x^s}{\partial y^j} \frac{\partial}{\partial x^s} \right)$$

because,
since y^i is a function, i.e.
$y^i = y^i(x)$, and d acts on a function, then $dy^i = dy^i(x) = \frac{\partial y^i}{\partial x^q} dx^q$, and since

$$\frac{\partial f(y)}{\partial y^k} = \frac{\partial f(x(y))}{\partial y^k} = \frac{\partial x^p}{\partial y^k} \frac{\partial f(x)}{\partial x^p} \text{ so } \frac{\partial}{\partial y^k} = \frac{\partial x^p}{\partial y^k} \frac{\partial}{\partial x^p} \text{ by chain rule}$$

Note ∇_{fX} is C^∞-linear for fX
covector dy^i is C^∞-linear in its argument

$$\implies \Gamma^i_{jk}(y) = \frac{\partial y^i}{\partial x^q} dx^q \left(\frac{\partial x^p}{\partial y^k} \left[\left(\nabla_{\frac{\partial}{\partial x^p}} \frac{\partial x^s}{\partial y^j} \right) \frac{\partial}{\partial x^s} + \frac{\partial x^s}{\partial y^j} \left(\nabla_{\frac{\partial}{\partial x^p}} \frac{\partial}{\partial x^s} \right) \right] \right) =$$
$$= \frac{\partial y^i}{\partial x^q} \underbrace{\frac{\partial x^p}{\partial y^k} \frac{\partial}{\partial x^p}}_{\frac{\partial}{\partial y^k}} \frac{\partial x^s}{\partial y^j} \delta^q_{\ s} + \frac{\partial y^i}{\partial x^q} \frac{\partial x^p}{\partial y^k} \frac{\partial x^s}{\partial y^j} \Gamma^q_{sp}(x)$$

In summary (Schuller says to commit this to memory):

$$\boxed{\Gamma^i_{jk}(y) = \frac{\partial y^i}{\partial x^q} \frac{\partial^2 x^q}{\partial y^j \partial y^k} + \frac{\partial y^i}{\partial x^q} \frac{\partial x^s}{\partial y^j} \frac{\partial x^p}{\partial y^k} \Gamma^q_{sp}(x)} \tag{348}$$

Eq. (348) is the change of connection coefficient function under the change of chart $(U \cap V, x) \to (U \cap V, y)$ Notice especially the second-order partial derivative term: for any linear transformation, this term will "not be felt" (i.e. will be zero), but otherwise for a nonlinear transformation of coordinates, you'll "feel it."

100.1.2 Normal Coordinates

Let $p \in M$ of $(M, \mathcal{O}, \mathcal{A}, \nabla)$

Then one can construct a chart (U, x) with $p \in U$ such that

$$\Gamma(x)^i_{\ (jk)}(p) = 0 \tag{349}$$

at the point p. **Not** necessarily in any neighborhood. Note the (jk) subscript, meaning the symmetric part vanishes, i.e. equals zero. So Eq. 349 says we can choose a coordinate system such that at a point the symmetric part of the Christoffel Symbol is zero.

Recall the definition of symmetric and antisymmetric parts of some tensor T. Define the symmetric and antisymmetric parts, respectively:

$$T_{(ab)} = \frac{1}{2}(T_{ab} + T_{ba})$$
$$T_{[ab]} = \frac{1}{2}(T_{ab} - T_{ba})$$

Proof. Let (V, y) be any chart (with) $p \in V$.

Thus, in general: $\Gamma(y)^i_{\ jk} \neq 0$

Then consider a new chart (U, x) to which one transits by virtue of

$$(x \circ y^{-1})^i(\alpha^1, \dots, \alpha^d) := \alpha^i - \frac{1}{2}\Gamma(y)^i_{\ (jk)}(p)\alpha^j\alpha^k$$

for $p = x^{-1}(\alpha^1, \dots, \alpha^d)$

$$\left(\frac{\partial x^i}{\partial y^j} \right)_p = \partial_j(x^i \circ y^{-1}) = \delta^i_j - \Gamma(y)^i_{\ mj}(p)\ \alpha^m|_{\alpha=0} = \delta^i_j$$
$$\frac{\partial x^i}{\partial y^k \partial y^j}(p) = -\Gamma(y)^i_{\ (kj)}(p)$$

where the partial derivative acting on the second term with the Christoffel symbol is because of the following:

$$\frac{\partial}{\partial y^j} \frac{1}{2}\Gamma(y)^i_{\ (lk)}(p)\alpha^l\alpha^k = \frac{\partial}{\partial \alpha^j} \frac{1}{2}\Gamma(y)^i_{\ (lk)}(p)\alpha^l\alpha^k = \frac{1}{2}\Gamma(y)^i_{\ (lk)}(p)\delta^l_{\ j}\alpha^k + \frac{1}{2}\Gamma(y)^i_{\ (lk)}(p)\alpha^l\delta^k_j =$$
$$= \frac{1}{2}\Gamma(y)^i_{\ (jk)}(p)\alpha^k + \frac{1}{2}\Gamma(y)^i_{\ (lj)}(p)\alpha^l = \Gamma(y)^i_{\ (jm)}(p)\alpha^m$$

where the last step is by the symmetry property of the symmetric part of $\Gamma^i_{\ jm}$, $\Gamma^i_{\ (jm)}$.

Then use Eq. (348), the change of Christoffel symbols equation we just derived:

$$\Gamma^i_{jk}(y) = \frac{\partial y^i}{\partial x^q}\frac{\partial^2 x^q}{\partial y^j \partial y^k} + \frac{\partial y^i}{\partial x^q}\frac{\partial x^s}{\partial y^j}\frac{\partial x^p}{\partial y^k}\Gamma^q_{sp}(x) = \frac{\partial y^i}{\partial x^q}\left(-\Gamma(y)^q_{(kj)}(p)\right) + \frac{\partial y^i}{\partial x^q}\Gamma^q{}_{jk}(x)$$

If $\frac{\partial y^i}{\partial x^q} = -\delta^i{}_q$ then we obtain

$$\Gamma^i{}_{jk}(y) = +\Gamma(y)^i{}_{(jk)} - \Gamma^i{}_{jk}(x) \text{ or}$$
$$\Gamma^i{}_{jk}(x) = -\Gamma^i{}_{jk}(y) + \Gamma^i{}_{(jk)}(y) = \Gamma(y)^i{}_{[kj]}$$

Compare this against what Schuller (2015) [109] obtains:

$$\implies \Gamma(x)^i_{jk}(p) = \Gamma(y)^i{}_{(jk)}(p) - \Gamma(y)^i{}_{kj}(p) =$$
$$= \Gamma(y)^i_{[jk]}(p) = T(y)^i{}_{jk}$$

Terminology: (U, x) is called a **normal coordinate chart** of ∇ at $\mathbf{p} \in \mathbf{M}$.

□

cf. pp. 164, 4.1 *Connections in Vector Bundles*, Jost (2017) [116]

Let $x_0 \in M$, let open neighborhood U of x_0 s.t. $\exists$ chart for M, $\exists$ bundle chart for E on U. Thus, $\exists$ coordinate vector fields $\frac{\partial}{\partial x^1}, \cdots \frac{\partial}{\partial x^d}$ and through identification

$$E|_U \simeq U \times \mathbb{R}^n$$

$n =$ fiber dim. of E.

For a basis of $\mathbb{R}^n$ yields basis $\mu_1 \dots \mu_n \equiv e_q \dots e_n$ of sections of $E|_U \equiv \Gamma(E|_U)$. Define Christoffel symbols Γ^k_{ij} $(j, k = 1 \dots n, i = 1 \dots d)$ (pp. 164, Jost (2017) [116])

$$\begin{aligned} D_{\frac{\partial}{\partial x^i}}\mu_j &:= \Gamma^k_{ij} e_k \quad \text{(4.1.5, Jost (2017)[116])} \\ &\equiv D_{\frac{\partial}{\partial x^j}} e_i \Gamma^k_{ij} e_k \quad \text{(Schuller's notation)} \end{aligned} \tag{350}$$

100.2 Parallel Transport

Let $\mu \equiv s \in \Gamma(E)$, ($\mu$ is the notation of Jost (2017) [116]) locally $s(y) = a^k(y)e_k(y)$, let $c(t) =$ smooth curve in U. Let $s(t) := s(c(t))$, to define a section of E along c.
Let $X(t) = \dot{c}(t) = \dot{c}^i(t)\frac{\partial}{\partial x^i}$

Then

$$D_{X(t)}s(t) = D_{\dot{c}^i \frac{\partial}{\partial x^i}} s(t) = \dot{c}^i D_{\frac{\partial}{\partial x^i}} a^k e_k = \dot{c}^i \left(\frac{\partial a^k}{\partial x^i}\right) e_k + \dot{c}^i a^k \Gamma^q_{ki} e_q =$$
$$= \left(\dot{a}^k + \dot{c}^j a^i \Gamma^k_{ij}\right) e_k$$

where we had used chain rule:

$$\dot{a}^k \equiv \frac{da^k}{dt} = \frac{\partial a^k}{\partial x^i}(x(t))\frac{dc^i}{dt}$$

So if $D_{x(t)}s(t) = 0$, we have $k = 1 \dots n$ first order linear equations, solving for $a^1, \dots a^n(t)$.

So by a fundamental theorem for ordinary differential equations (ODEs), for given initial values $s(0) \in E_{c(0)}$, $\exists!$ solution to $D_{x(t)}s(t) = 0$.

Definition 190 (Parallel transport of section of vector bundle, Def. 4.1.2, Jost (2017)[116])**.** The solution set of $D_{x(t)}s(t) = 0$ is called the parallel transport of $s(0)$ along curve c.

Given $X \in T_xM$, let curve c in M with $c(0) = x$, $\dot{c}(0) = X$, $\forall\, s \in \Gamma(E)$, then

$$D_X s := \lim_{t \to 0} \frac{P_{c,t}(s(c(t))) - s(c(0))}{t}$$

EY: This seems to be the covariant derivative.

pp. 165 Jost (2017) [116] claims that parallel transport on Riemannian manifold can obtain covariant derivative.

Proof. Let $P_{c,t} : E_{c(t)} \to E_{c(0)}$ be identification by parallel transport along c.

Select a basis of parallel sections $\mu_1(t), \dots \mu_n(t) \equiv e_1(t) \dots e_n(t)$ of E along c, i.e.

$$D_{\dot{c}(t)} e_j(t) = 0 \quad \forall\, j = 1 \dots n$$

$\forall$ arbitrary section s of E along c, then

$$s(t) = a^k(t) e_k(t)$$

and for $X = \dot{c}(0)$,

$$D_X s(t) = D_X(a^k(t) e_k(t)) = D_X(a^k(t)) e_k(t) + a^k(t) D_X e_k(t) = \dot{a}^k(t) e_k(t)$$

where $D_X e_k(t) = 0$ by parallel transport.

So

$$(D_X s)(c(0)) = \lim_{t \to 0} \frac{a^k(t) - a^k(0)}{t} e_k(0) = \lim_{t \to 0} \frac{P_{c,t}(s(t)) - s(0)}{t}$$

□

Consider tangent space at pt. $\psi \equiv s$ to the total space E of a vector bundle $T_s E$.
Inside $T_s E$, there's a distinguished subspace, namely the tangent space to the fiber E_x containing s ($x = \pi(s)$).
This space is called vertical space V_s.

However, there's no distinguished "horizontal space" H_s complementary to V_s, i.e. $T_s E = V_s \oplus H_s$.
If we have covariant derivative D, we can parallel transport s, $\forall\, X \in T_x M$ along curve $c(t)$ with

$$c(0) = x,\ \dot{c}(0) = X$$

Thus, $\forall\, X$, we obtain a curve $s(t)$ in E.
subspace of $T_s E$ spanned by all tangent vectors to E at s of the form

$$\left. \frac{d}{dt} s(t) \right|_{t=0}$$

then is the horizontal space H_s.

100.2.1 Parallelity of vector fields

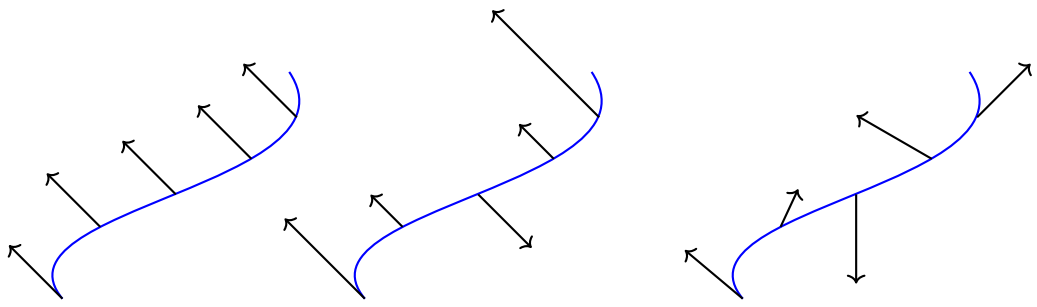
cf. Lecture 8, Schuller (2015) [109].

Assume smooth affine manifold $(M, \mathcal{O}, \mathcal{A}, \nabla)$ with connection.

Definition 191 (Parallely Transported)**.** vector field X on M is **parallely transported** along smooth curve $\gamma : \mathbb{R} \to M$ if

$$\nabla_{v_\gamma} X = 0 \tag{351}$$

Definition 192 (Parallel)**.** vector field X in M is **parallel** along curve $\gamma : \mathbb{R} \to M$ if $\nabla_{v_\gamma} = \mu \cdot X$



The left drawing is a parallely transported vector field, the middle drawing is a parallel vector field. For the middle drawing, it's important that the vector field vanishes in-between points when it points "up" vs. "down" because $\mu : \mathbb{R} \to \mathbb{R}$ is smooth. The right drawing is not even parallel.

100.2.2 Autoparallelly Transported Curves

Definition 193 (Autoparallely Transported)**.** smooth curve $\gamma : \mathbb{R} \to M$ **autoparallely transported** if

$$\nabla_{v_\gamma} v_\gamma = 0 \tag{352}$$

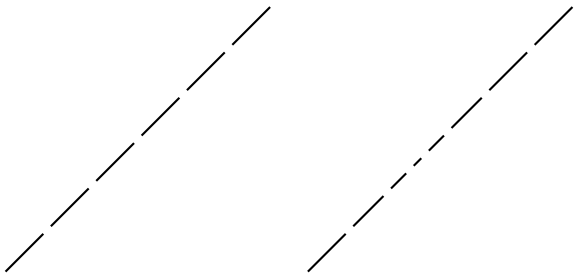
Definition 194. smooth curve $\gamma : \mathbb{R} \to M$ is autoparallel

$$\nabla_{v_\gamma} v_\gamma = \mu v_\gamma \tag{353}$$

for some smooth $\mu : \mathbb{R} \to \mathbb{R}$

For autoparallel, it means we move in the straightest line. We are not talking about the shortest line because we do not have a notion of distance.

Note that the straightest line might not actually look straight when "viewed from above." That is, if we embed the manifold into a higher dimensional one, and then just look at the curve, it might be curved. For example, on a sphere, a straight line traces out a portion of a circle around the sphere. This doesn't look straight in the (higher, Euclidean) embedding, but *on the surface* of the sphere, it's the straightest line.



Left is an autoparallely transported curve, and the right is just an autoparallel.

Consider

$$\dot{\gamma}(t) = v(\gamma(t)) = v^i(\gamma^1(t) \dots \gamma^n(t)) \frac{\partial}{\partial x^i} \in T_{\gamma(t)} U$$

Then

$$\frac{d}{dt}\dot{\gamma}(t) = \frac{d}{dt} v^i(\gamma^1(t) \dots \gamma^n(t)) \frac{\partial}{\partial x^i} = \frac{d}{dt} v^i(x^1 \dots x^n) \frac{\partial}{\partial x^i} = \frac{\partial v^i}{\partial x^j}(x) \dot{x}^j \frac{\partial}{\partial x^i} = \dot{\gamma}^j \frac{\partial \dot{\gamma}^i}{\partial x^j} \frac{\partial}{\partial x^i}$$

So

$$\nabla_{v_\gamma} v_\gamma = \nabla_{v^j \frac{\partial}{\partial x^j}} v^i \frac{\partial}{\partial x^i} = v^j \left[\frac{\partial v^i}{\partial x^j} \frac{\partial}{\partial x^i} + v^i \Gamma^k_{ij} \frac{\partial}{\partial x^k} \right] = \ddot{\gamma}(t) + \dot{\gamma}^i \dot{\gamma}^j \Gamma^k_{ij} \frac{\partial}{\partial x^k} \text{ i.e.}$$

$$\boxed{\nabla_{v_\gamma} v_\gamma = \left(\ddot{\gamma}^k(t) + \dot{\gamma}^i \dot{\gamma}^j \Gamma^k_{ij} \right) \frac{\partial}{\partial x^k}} \tag{354}$$

with $\Gamma^k_{ij} = \Gamma^k_{ij}(\gamma(t))$.

100.2.3 Torsion

cf. Lecture 8, Schuller (2015) [109]

Definition 195 (torsion)**.** The **torsion** of a connection ∇ is the $(1,2)$-**tensor field**

$$T(\omega, X, Y) := \omega(\nabla_X Y - \nabla_Y X - [X, Y]) \tag{355}$$

where $[\cdot, \cdot] : \Gamma(TM) \times \Gamma(TM) \to \Gamma(TM)$ is the commutator on $\Gamma(TM)$, that yields a vector field, $[X, Y]$ defined by

$$[X, Y] f := X(Yf) - Y(Xf) \tag{356}$$

Theorem 69. *T is C^∞-linear in each entry*

Proof. check

$$T(f\cdot\omega,X,Y)=f\cdot\omega(\dots)=fT(\omega,X,Y)$$
$$T(\omega+\psi,X,Y)=\dots=T(\omega,X,Y)+T(\psi,X,Y)$$
$$T(\omega,fX,Y)=\omega(\nabla_{fX}Y-\nabla_Y(fX)-[fX,Y])=$$
$$=\omega(f\nabla_XY-(Yf)X-f\nabla_YX-f[X,Y]+(Yf)X)=fT(\omega,X,Y)$$

since

$$[fX,Y]g=fX(Yg)-Y(fXg)=fX(Yg)-(Yf)(Xg)-fY(Xg)=$$
$$=f\cdot[X,Y]g-(Yf)Xg=(f\cdot[X,Y]-(Yf)X)g$$

$$T(\omega,X,Y)=-T(\omega,Y,X)\quad\checkmark$$

□

Definition 196. An affine manifold $(M,\mathcal{O},\mathcal{A},\nabla)$ is called torsion-free if $T=0$. This can also be written as

$$\nabla_XY-\nabla_YX=[X,Y]\quad\forall\,X,Y\in\Gamma(TM) \tag{357}$$

In a chart

$$T^i{}_{ab}:=T\left(dx^i,\frac{\partial}{\partial x^a},\frac{\partial}{\partial x^b}\right)=dx^i(\dots)$$
$$=\Gamma^i{}_{ba}-\Gamma^i{}_{ab}=-2\Gamma^i{}_{[ab]}$$

since

$$[\frac{\partial}{\partial x^a},\frac{\partial}{\partial x^b}]=0\text{ and}$$
$$\nabla_{\frac{\partial}{\partial x^a}}\frac{\partial}{\partial x^b}=\Gamma^k{}_{ba}$$
$$\nabla_{\frac{\partial}{\partial x^b}}\frac{\partial}{\partial x^a}=\Gamma^k{}_{ab}$$

So what this means is that if the torsion is 0, i.e. $T=T^i{}_{ab}=0$, then the *anti-symmetric* part of the connection coefficients, i.e. Christoffel symbols, is zero, $\Gamma^k{}_{[ij]}=0$. So the Christoffel symbols only have a symmetric part and is thus symmetric in indices i,j.

From now on, in these lectures, we only use torsion-free connections.

Suppose $X=X^i\frac{\partial}{\partial x^i}\in\mathfrak{X}(M)$, $s=s^ie_i\in\Gamma(E)$,

$$D_Xs=X^jD_{\frac{\partial}{\partial x^j}}s^ie_i=X^j\left(\frac{\partial s^i}{\partial x^j}e_i+s^iD_{\frac{\partial}{\partial x^j}}e_i\right)=X^j\left(\frac{\partial s^k}{\partial x^j}+s^i\Gamma^k_{ij}\right)e_k$$

Consider Γ^k_{ij} as an $(n\times n)$ matrix valued 1-form on $U\quad\forall\,j=1\dots d$.

$$\Gamma^k_{ij}\in\Gamma(\mathfrak{gl}(n,\mathbb{R})\otimes T^*M|_U) \tag{358}$$

cf. Eq. (4.1.12), pp. 166, Jost (2017) [116].

Abstract this to

$$D=d+A$$

where d is the exterior derivative and $A\in\Gamma(\mathfrak{gl}(n,\mathbb{R})\otimes T^*M|_U)$.

Given a^je_j

$$D(a^je_j)=e_k\otimes(da^k)+A_ja\otimes dx^j$$

where $d(a^k)(X)=(da^k)(X)=X(a^k)=X^j\frac{\partial a^k}{\partial x^j}$.

Let $(U_\alpha)_{\alpha\in A}$ be covering of M by open sets which the bundle is trivial, with transition maps

$$\varphi_{\beta\alpha}:U_\alpha\cap U_\beta\to GL(n;\mathbb{R})$$

Let section s be represented by s_α on U_α.

Thus,

$$s_\beta=\varphi_{\beta\alpha}s_\alpha\text{ on }U_\alpha\cap U_\beta$$

But then

$$\varphi_{\beta\alpha}(d+A_\alpha)s_\alpha = (d+A_\beta)s_\beta \text{ on } U_\alpha \cap U_\beta$$
$$\varphi_{\beta\alpha}ds_\alpha + \varphi_{\beta\alpha}A_\alpha s_\alpha = d\varphi_{\beta\alpha}s_\alpha + A_\beta\varphi_{\beta\alpha}s_\alpha$$
$$A_\alpha s_\alpha = -ds_\alpha + \varphi_{\beta\alpha}^{-1}d\varphi_{\beta\alpha}s_\alpha + \varphi_{\beta\alpha}^{-1}A_\beta\varphi_{\beta\alpha}s_\alpha$$

100.3 Curvature

cf. Lecture 8, Schuller (2015) [109].

Definition 197 (Riemann Curvature)**.** The **Riemann curvature** of a connection ∇ is the $(1,3)$-tensor field

$$\text{Riem}(\omega, Z, X, Y) := \omega(\nabla_X\nabla_Y Z - \nabla_Y\nabla_X Z - \nabla_{[X,Y]}Z) \tag{359}$$

Definition 198 (Ricci Curvature)**.** Let Riem be the Riemann curvature tensor of connection ∇.
Define the Ricci curvature tensor as $(0,2)$-tensor field

$$\text{Ric}(X,Y) := \text{Riem}(f^k, Y, X, e_k) \tag{360}$$

where $f^l(e_m) = \delta^l{}_m$

In terms of components, the Ricci curvature tensor is given by

$$\text{Ric}_{ab} = \text{Riem}^m{}_{amb} \tag{361}$$

Theorem 70. *Riem is C^∞-linear in each slot.*

Proof.

since $\text{Riem}(\omega, Z, X, Y) = \omega(\nabla_X\nabla_Y Z - \nabla_Y\nabla_X Z - \nabla_{[X,Y]}Z)$

$$\text{Riem}(f\omega, Z, X, Y) = f\omega(\nabla_X\nabla_Y Z - \nabla_Y\nabla_X Z - \nabla_{[X,Y]}Z) = f\text{Riem}(\omega, Z, X, Y)$$
$$\text{Riem}(\omega+\sigma, Z, X, Y) = (\omega+\sigma)(\nabla_X\nabla_Y Z - \nabla_Y\nabla_X Z - \nabla_{[X,Y]}Z) = \text{Riem}(\omega, Z, X, Y) + \text{Riem}(\sigma, Z, X, Y)$$
$$\text{Riem}(\omega, Z, fX, Y) = \omega(f\nabla_X\nabla_Y Z - Y(f)\nabla_X Z - f\nabla_Y\nabla_X Z - \nabla_{f[X,Y]-(Yf)X}Z) =$$
$$= \omega(f\nabla_X\nabla_Y Z - Y(f)\nabla_X Z - f\nabla_Y\nabla_X Z - f\nabla_{[X,Y]}Z + (Yf)\nabla_X Z) = f\text{Riem}(\omega, Z, X, Y)$$

where the second and last term cancel each other out in the second to last expression.

□

The Riemann curvature tensor Riem is antisymmetric in its last 2 entries. This is because

$$\text{Riem}(\omega, Z, Y, X) = \omega(\nabla_Y\nabla_X Z - \nabla_X\nabla_Y Z - \nabla_{[Y,X]}Z) = -\omega(\nabla_X\nabla_Y Z - \nabla_Y\nabla_X Z + \nabla_{-[X,Y]}Z) =$$
$$= -\text{Riem}(\omega, Z, X, Y)$$

since $\nabla_{-[X,Y]}Z = -\nabla_{[X,Y]}Z$

Therefore, Riem has $d^3(d-1)/2$ independent components. This is because d^4 is the total number of components (there are 4 indices, each index running from 1 to d) if the indices are completely independent of each other.

Since two entries are antisymmetric, imagine a dxd matrix with only the entries in the "top half triangle" being independent. It excludes the diagonal entries (they'd be all 0). There are $\frac{d(d-1)}{2}$ entries. For each of those antisymmetric entries, there are still d^2 ways to choose the other 2 entries of the Riem. Thus, the number of independent components are

$$d^4 = \left(\frac{d(d-1)}{2}\right)d^2 = d^4 - \frac{(d^2-d)d^2}{2} = d^4 - \frac{d^4-d^3}{2} = d^3(d-1)/2$$

Part XXIV
Holonomy

Definition 199 (Conlon, 10.1.2)**.** If $X, Y \in \mathfrak{X}(M)$, $M \subset \mathbb{R}^m$, **Levi-Civita connection** on $M \subset \mathbb{R}^m$

$$\nabla : \mathfrak{X}(M) : \mathfrak{X}(M) \to \mathfrak{X}(M)$$
$$\nabla_X Y := p(D_X Y) \tag{362}$$

with

$$D_XY := \sum_{j=1}^{m} X(Y^j)\frac{\partial}{\partial x^j} = \sum_{i,j=1}^{m} X^i \frac{\partial Y^j}{\partial x^i}\frac{\partial}{\partial x^j} \qquad \forall X = \sum_{i=1}^{m} X^i \frac{\partial}{\partial x^i}, \quad \forall Y = \sum_{i=1}^{m} Y^i \frac{\partial}{\partial x^i}$$

$$\nabla_{fX}Y = f(D_{fX}Y) = p(fD_XY) = fpD_XY = f\nabla_XY$$

$$\nabla_X fY = p(D_X fY) = p\left(\sum_{i,j=1}^{m}\left(X^i f\frac{\partial Y^j}{dx^i} + X^iY^j\frac{\partial f}{\partial x^i}\right)\frac{\partial}{\partial x^j}\right) = f\nabla_XY + p\sum_{j=1}^{m} X(f)Y^j\frac{\partial}{\partial x^j} = f\nabla_XY + X(f)p(Y)$$

Definition 200 (Conlon, 10.1.4; Christoffel symbols)**.**

$$\begin{aligned} &\nabla_{\frac{\partial}{\partial x^i}\frac{\partial}{\partial x^j}} = \Gamma_{ij}^k \frac{\partial}{\partial x^k} && \text{(Conlon's notation)} \\ &\nabla_{\frac{\partial}{\partial x^j}\frac{\partial}{\partial x^i}} = \Gamma_{ij}^k \frac{\partial}{\partial x^k} && \text{(F. Schuller's notation)} \end{aligned} \tag{363}$$

Definition 201 (torsion)**.**

$$\begin{aligned} &T : \mathfrak{X}(M) \in \mathfrak{X}(M) \to \mathfrak{X}(M) \\ &T(X,Y) = \nabla_XY - \nabla_YX - [X,Y] \end{aligned} \tag{364}$$

If $T = 0$, ∇ torsion-free or symmetric.

$$T(fX,Y) = f\nabla_XY - (f\nabla_YX + Y(f)X) - \{(fXY - (Y(f)X + fYX)\} = fT(X,Y)$$
$$T(X,fY) = f\nabla_XY + X(f)Y - f\nabla_YX - \{((X(f)Y + fXY) - fYX\} = fT(X,Y)$$

Thus, $T(X,Y)$ $C^\infty(M)$-bilinear.
$T \in \tau_1^2(M)$.
$T(v,w) \in T_xM$ defined, $\forall\, v,w \in T_xM$, $\forall\, x \in M$.
Thus, torsion is a **tensor**.

Exercise 10.1.7 Conlon (2008)[90] **..**
If $T(X,Y) = 0$,

$$T(e_i,e_j) = \Gamma_{ji}^k e_k - \Gamma_{ij}^k e_k - 0 = 0 \Longrightarrow \Gamma_{ji}^k = \Gamma_{ij}^k$$

If $\Gamma_{ij}^k = \Gamma_{ji}^k$, $T(e_i,e_j) = 0$.

Exercise 10.1.8, Conlon (2008)[90]**.**
If $M \subset \mathbb{R}^m$ smoothly embedded submanifold, $\forall \frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^i} \in T_xM$, spanning T_xM, consider $\frac{\partial}{\partial x^j} = X_j^k \frac{\partial}{\partial \widetilde{x}^k}$, $\frac{\partial}{\partial x^i} = X_i^k(\widetilde{x})\frac{\partial}{\partial \widetilde{x}^k}$

$$\nabla_{\frac{\partial}{\partial x^j}}\frac{\partial}{\partial x^i} = pD_{X_j^k\frac{\partial}{\partial \widetilde{x}^k}}X_i^l\frac{\partial}{\partial \widetilde{x}^l} = p\left(X_j^k\frac{\partial X_i^l}{\partial \widetilde{x}^k}\frac{\partial}{\partial \widetilde{x}^l}\right) = X_j^k p\left(\frac{\partial X_i^l}{\partial \widetilde{x}^k}\frac{\partial}{\partial \widetilde{x}^l}\right)$$
$$\nabla_{\frac{\partial}{\partial x^i}}\frac{\partial}{\partial x^j} = X_i^k p\left(\frac{\partial X_j^l}{\partial \widetilde{x}^k}\frac{\partial}{\partial \widetilde{x}^l}\right)$$

If $X \in \mathfrak{X}(M)$, smooth $s : [a,b] \to M$,
then $\forall\, s(t)$,

$$X'_{s(t)} = \nabla_{\dot{s}(t)}X \in T_{s(t)}M$$

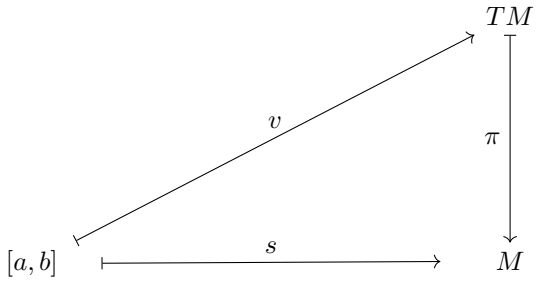
In fact, it's often natural to consider fields $X_{s(t)}$ along s, parametrized by parameter t, allowing

$$X_{s(t_1)} \neq X_{s(t_2)}$$

each of $s(t_1) = s(t_2)$.

Definition 202 (10.1.9)**.** Let smooth $s : [a, b] \to M$.

Vector field along s is smooth $v : [a, b] \to TM$ s.t.



commutes.

Note that $v \in \mathfrak{X}(s) \subset \mathfrak{X}(M)$

e.g. $(Y|s)(t) = Y_{s(t)}$, restriction of $Y \in \mathfrak{X}(M)$ to s.
e.g. $\dot{s}(t) \in \mathfrak{X}(M)$.
$\forall\, v, w \in \mathfrak{X}(s)$, $v + w \in \mathfrak{X}(s)$,

$$(fv + gv)(t) := (f(s(t)) + g(s(t)))v(t) = f(s(t))v(t) + g(s(t))v(t) = (f + g)v(t)$$

Likewise,

$$f(v + w) = fv + fw$$

$\mathfrak{X}(s)$ is a real vector space and $C^\infty[a, b]$-module.

Definition 203 (10.1.10)**.** Let conection ∇ on M.

Associated covariant derivative is operator

$$\frac{\nabla}{dt}\mathfrak{X}(s) \to \mathfrak{X}(s)$$

$\forall$ smooth s on M, s.t.

1. $\frac{\nabla}{dt}$ $\mathbb{R}$-linear
2. $\left(\frac{\nabla}{dt}\right)(fv) = \frac{df}{dt}v + f\frac{\nabla}{dt}v$, $\forall\, f \in C^\infty[a, b]$, $\forall\, v \in \mathfrak{X}(s)$
3. If $Y \in \mathfrak{X}(M)$, then
$$\frac{\nabla}{dt}(Y|s)(t) = \nabla_{\dot{s}(t)}Y \in T_{s(t)}M, \quad a \le t \le b$$

Theorem 71 (Conlon Thm. 10.1.11[90])**.** $\forall$ *connection* ∇ *on* M, $\exists!$ *associated covariant derivative* $\frac{\nabla}{dt}$

Proof. Consider arbitrary coordinate chart $(U, x^1 \dots x^n)$.

Consider smooth curve $s : [a, b] \to U$.

Let $v \in \mathfrak{X}(s)$, $v(t) = v^i(t)\frac{\partial}{\partial x^i}$; $\dot{s}(t) = s^j\frac{\partial}{\partial x^j}$.

$$\frac{\nabla v}{dt} = \frac{dv^i(t)}{dt}\frac{\partial}{\partial x^i} + v^i(t)\frac{\nabla}{dt}\frac{\partial}{\partial x^i} = \frac{dv^i}{dt}\frac{\partial}{\partial x^i} + v^i\nabla_{\dot{s}(t)}\frac{\partial}{\partial x^i} = \dot{v}^i\frac{\partial}{\partial x^i} + v^i\dot{s}^j\Gamma^k_{ij}\frac{\partial}{\partial x^k} = \left(\dot{v}^k + v^i\dot{s}^j\Gamma^k_{ij}\right)\frac{\partial}{\partial x^k}$$

This is an explicit, local formula in terms of connection, proving uniqueness.

Existence: $\forall$ coordinate chart $(U, x^1 \dots x^n)$, $\left(\dot{v}^k + v^i\dot{s}^j\Gamma^k_{ij}\right)\frac{\partial}{\partial x^k} =: \frac{\nabla v}{dt}$.

$$\frac{\nabla}{dt}(fv) = \dot{f}v^k + f\dot{v}^k + fv^i\dot{s}^j = \dot{f}v + f\frac{\nabla v}{dt}$$

If f constant, then $\frac{\nabla}{dt}$ is $\mathbb{R}$-linear.

□

Definition 204 (10.1.12 Conlon (2008)[90]). Let (M, ∇). Let $v \in \mathfrak{X}(s)$ for smooth $s : [a, b] \to M$.
If $\frac{\nabla v}{dt} \equiv 0$ on s, then v is **parallel** along s.

Theorem 72 (10.1.13). *Let (M, ∇), smooth $s : [a, b] \to M$, $c \in [a, b]$, $v_0 \in T_{s(c)}M$.*
Then $\exists !$ parallel field $v \in \mathfrak{X}(s)$ s.t. $v(c) = v_0$.
v parallel transport along s.

Proof.

$$\dot{s}(t) = \dot{s}^j(t)e_j$$
$$v(t) = v^i(t)e_i$$
$$v_0 = a^i e_i$$
$$0 = \left(\frac{dv^k}{dt}(t) + v^i(t)\dot{s}^j(t)\Gamma^k_{ij}(s(t))\right)e_k$$

or equivalently

$$\frac{dv^k}{dt} = -v^i\dot{s}^j\Gamma^k_{ij}, \qquad 1 \leq k \leq n \qquad (10.1) \tag{365}$$

with initial conditions $v^k(c) = a^k$, $1 \leq k \leq n$.
By existence and uniqueness of solutions of O.D.E.
$\exists \epsilon > 0$ s.t. $\exists !$ solutions $v^k(t)$. For $c - \epsilon < t < c + \epsilon$.
In fact, these ODEs being linear in v^k, by ODE theory (Appendix C, Thm. C.4.1).
$\nexists$ restriction on ϵ, so $\exists ! \, v^k(t) \quad \forall \, t \in [a, b]$, $1 \leq k \leq n$

□

100.4 Principal bundle, vector bundle case for parallel transport

Recall the 2 different forms or viewpoints for Lie-algebra valued 1-forms, or vector-valued 1-forms, or sections of 1-form-valued endomorphisms:

$$\omega^k_{i\mu}dx^\mu \equiv \omega^k_i \in \Omega^1(M, \mathfrak{gl}(n, \mathbb{F})) = \Gamma(\mathfrak{gl}(n, \mathbb{R} \otimes T^*M|_U))$$

for $i, k = 1 \dots n = \dim E$.
$\mu = 1 \dots d = \dim E$
Now

$$D_X\mu = X^\mu D_{\frac{\partial}{\partial x^\mu}}\mu = X^\mu\left[\left(\frac{\partial}{\partial x^\mu}\mu^k\right)e_k + \mu^i\omega^k_{i\mu}e_k\right] = \left(X(\mu^k) + \mu^i\omega^k_i(X)\right)e_k = \left(d\mu^k(X) + \mu^i\omega^k_i(X)\right)e_k$$

So then define

$$\begin{aligned} &D : \Gamma(E) \to \Gamma(E) \otimes \Gamma(T^*M) \\ &D\mu = D(\mu^i e_i) = e_k(d\mu^k + \mu^i\omega^k_i) \equiv (d + A)\mu \end{aligned} \tag{366}$$

Also, D can be defined for this case:

$$D : \Gamma(\mathrm{End}(E)) \to \Gamma(\mathrm{End}E) \otimes \Gamma(T^*M)$$

Let $\sigma = \sigma^i{}_j e_i \otimes e^j \in \Gamma(\mathrm{End}(E))$

$$\begin{aligned} D\sigma = D(\sigma^i{}_j e_i) \otimes e^j + \sigma^i{}_j e_i \otimes D^* e^j = \left(d\sigma^k{}_j + \sigma^i A^k_i\right)e_k \otimes e^j + \sigma^i{}_j e_i \otimes (A^*)^j_k e^k = \\ = (d\sigma^k{}_j + \sigma^i{}_j A^k_i)e_k \otimes e^j + \sigma^k_i e_j \otimes (-A^i{}_j)e^j = (d\sigma^k{}_j + [A, \sigma]^k{}_j)e_k \otimes e^j \end{aligned} \tag{367}$$

cf. Def. 4.1.4 of Jost (2017) [116], pp. 138.
For $\mu \in \Gamma(E)$, smooth $s : [a, b] \to M$, $X(t) = \dot{s}(t)$,

$$D_{\dot{s}(t)}\mu = \dot{s}^\mu D_{\frac{\partial}{\partial x^\mu}}\mu = \dot{s}^\mu\left[\frac{\partial \mu^k}{\partial x^\mu}e_k + \mu^i\omega^k_{i\mu}e_k\right] = \left[\dot{s}^\mu\frac{\partial \mu^k}{\partial x^\mu} + \dot{s}^\mu\mu^i\omega^k_{i\mu}\right]e_k = \frac{d}{dt}\mu(s(t)) + \mu^i\dot{s}^\mu\omega^k_{i\mu}e_k \tag{368}$$

Let $D_{\dot{s}(t)}\mu = 0$. Then,

$$\frac{d}{dt}\mu(s(t)) = -\mu^i \dot{s}^\mu \omega^k{}_{i\mu} e_k \tag{369}$$

Recall, given vector bundle $E \xrightarrow{\pi} N$, given $\varphi : M \to N$, then pullback

$$\varphi^* E \to M \tag{370}$$

i.e.

$$\begin{array}{ccc} \varphi^* E & \xleftarrow{\varphi^*} & E \\ \downarrow{\scriptstyle \psi} & & \downarrow{\scriptstyle \pi} \\ M & \xrightarrow{\varphi} & N \end{array} \qquad \begin{array}{c} (\varphi^* E)_x = E_{\varphi(x)} \\ \uparrow \\ x \in M \end{array}$$

i.e. if $s \in \Gamma(E)$,

$$\varphi^* s = s \circ \varphi \in \Gamma(\varphi^* E)$$

Thus,

$$\begin{array}{ccc} \gamma^* E & \xleftarrow{\gamma^*} & E \\ \downarrow & & \downarrow{\scriptstyle \pi} \\ [a,b] & \xrightarrow{\gamma} & M \end{array} \qquad \begin{array}{c} (\varphi^* E)_c = E_{\gamma(c)} \\ \uparrow \\ c \in [a,b] \end{array}$$

For

$$\dot{v}^k = -v^i \dot{s}^j \Gamma^k{}_{ij}$$
$$v^k(c) = v_0^k \qquad 1 \le k \le m$$
$$\dot{v} = -v^i \dot{s}^j \Gamma_{ij}$$
$$(v \dot{+} w) = -(v^i + w^i)\dot{s}^j \Gamma_{ij}(v+w)(c) = v(c) + w(c) = v_0 + w_0$$

so $v + w \in \mathfrak{X}(s)$ is parallel transport of $v_0 + w_0$.

Likewise, $\forall\, a \in \mathbb{F}$, $av \in \mathfrak{X}(s)$ is the parallel transport of av_0.

$$\dot{\mu}^k = -\mu^i \dot{s}^\mu \omega^k{}_{i\mu} = -\mu^i \omega^k{}_i(\dot{s}^\mu)$$

Suppose $\gamma^* E$ trivialized over $[a,b]$.

Closed interval is contractible, so this is always possible.

For chart (U, φ),

$$\begin{array}{ccc} \gamma^* E & \xleftarrow{\gamma^*} & E \\ \downarrow & & \downarrow{\scriptstyle \pi} \\ [a,b] & \xrightarrow{\gamma} & M \end{array} \qquad \begin{array}{ccc} E|_U & \xrightarrow{\psi} & U \times V \\ {\scriptstyle \pi^{-1}}\uparrow & \nearrow & \\ U \subset M & & \end{array}$$

Consider

$$\varphi : [a,b] \times V \to \gamma^* E$$
$$\varphi(t, \cdot) = \gamma^* \circ \psi^{-1}(\gamma(t), \cdot)$$

$\forall\, \mu \in \Gamma(E|_{x \in M})$,

$\mu = \mu^i e_i$.

$\varphi(t, e_i) = \epsilon_i$ is a basis for $\gamma^* E$.

$\forall \sigma \in \Gamma(\gamma^* E)$,

$$\sigma = \sigma^i \epsilon_i, \quad \sigma^i : [a,b] \to \mathbb{F}$$

$$\nabla_{\frac{\partial}{\partial x^\mu}} \sigma = \frac{\partial \sigma^k}{\partial x^\mu} \epsilon_k + \omega^k{}_{j\mu} \sigma^j \epsilon_k = \left(\frac{\partial \sigma^k}{\partial x^\mu} + \omega^k{}_{,j\mu} \sigma^j \right) \epsilon_k$$

$$\nabla \sigma = \epsilon_k \otimes (d\sigma^k + \omega^k{}_{j\mu} dx^\mu \sigma^j) = \epsilon_k \otimes (d\sigma^k + \omega^k{}_j \sigma^j)$$

$$\nabla_{\frac{d}{dt}} \sigma = \epsilon_k \otimes \left(\frac{d\sigma^k}{dt} + \omega^k{}_{j\mu} \dot{x}^\mu \sigma^j \right)$$

Now

$$\frac{d}{dt} = \dot{x}^\nu \frac{\partial}{\partial x^\nu}$$

Then σ parallel along γ if

$$\frac{d\sigma^k}{dt} + \omega^k{}_{j\mu} \dot{x}^\mu \sigma^j = 0$$

Definition 205 (3.1.4 [91])**.** Parallel transport along γ is

$$\begin{aligned} &P_\gamma : E_{\gamma(a)} \to E_{\gamma(b)} \\ &P_\gamma(v) \mapsto \sigma(b) \end{aligned} \tag{371}$$

where $\sigma \in \Gamma(\gamma^* E)$, σ unique and s.t. $\sigma(a) = v$.

Lemma 34 (10.1.16[90])**.** *holonomy*

$$h_s : T_x M \to T_{x_0} M$$

if ∇ around piecewise smooth loop s is a linear transformation.

Lemma 35 (10.1.18 Conlon (2008)[90])**.** *Let piecewise smooth loop $s : [a,b] \to M$ at x_0.*
Let weak reparametrization $\widetilde{s} = s \circ r : [c,d] \to M$.
If reparametrization is orientation-preserving, then $h_{\widetilde{s}} = h_s$,
If reparametrization is orientation-reversing, then $h_{\widetilde{s}} = h_s^{-1}$,

Proof. Without loss of generality, assume smooth s, r

$$\widetilde{s}(\tau) = s(r(\tau))$$
$$\widetilde{v}(\tau) = v(r(\tau))$$

$$\widetilde{u}^j(\tau) = \frac{dt}{d\tau}(\tau) u^j(r(\tau))$$
$$\frac{d\widetilde{v}^k}{d\tau}(\tau) = \frac{dr}{d\tau}(\tau) \frac{dv^k}{dt}(r(\tau))$$
$$\frac{d\widetilde{v}^k}{d\tau} = -\widetilde{v}^i \widetilde{u}^j \Gamma^k_{ij}$$

since

$$\frac{dv^k}{dt} = -v^i u^j \Gamma^k_{ij}; \qquad 1 \le k \le n$$
$$v^k(c) = a^k; \qquad 1 \le k \le a$$

$$\frac{dr}{d\tau} \frac{dv^k}{dt} = -v^i \frac{dr}{d\tau} u^j \Gamma^k{}_{ij} = \frac{d\widetilde{v}^k}{d\tau} = -\widetilde{v}^i \widetilde{u}^j \Gamma^k{}_{ij}$$

Thus, if $r(c) = a$, $r(d) = b$

$$h_{\widetilde{s}}(v_0) = \widetilde{v}(d) = v(b) = h_s(v_0)$$

If $r(c) = a$, $r(d) = b$, then

$$\widetilde{v}(c) = v(b) = h_s(v_0)$$

and

$$h_{\widetilde{s}}(h_s(v_0)) = h_{\widetilde{s}}(v(b)) = \widetilde{v}(d) = v(a) = v_0$$

At this point, I will switch to my notation because it clarified to me, at least, what was going on, in that a holonomy h_s is *invariant* under orientation-preserving reparametrization, and its inverse is well-defined.

For $\widetilde{s} = s \circ t : [c,d] \to M$,
piecewise smooth t is reparametrized, i.e.

$$t : [c,d] \to [a,b] \tag{372}$$

Now,

$$\frac{d}{d\tau}\widetilde{s}(\tau) = \frac{d}{d\tau}\widetilde{s}(t(\tau)) = \dot{s}(t)\frac{dt}{d\tau}(\tau) \equiv \dot{s}\frac{dt}{d\tau}$$
$$v^k(t) = v^k(t(\tau)) = v^k(\tau)$$
$$\frac{dv^k}{d\tau}(t(\tau)) = \frac{dv^k}{dt}\frac{dt}{d\tau} = \frac{dt}{d\tau}(-v^i(\tau)\dot{s}^j(t)\Gamma^k_{\ ij}) = -v^i(\tau)\frac{d\widetilde{s}^j}{d\tau}\Gamma^k_{\ ij}$$

Consider

$$h_s(v_0) = v(b)$$

If $t(c) = a$,
$t(d) = b$

$$h_{\widetilde{s}}(v_0) = \widetilde{v}(d) = v(t(d)) = v(b) = h_s(v_0)$$

If $t(c) = b$,
$t(d) = a$

$$h_{\widetilde{s}}(h_s(v_0)) = h_{\widetilde{s}}(v(b)) = h_{\widetilde{s}}(v(t(c))) = h_{\widetilde{s}}(\widetilde{v}(c)) =$$
$$= \widetilde{v}(d) = v(t(d)) = v(a) = v_0$$

Thus,

$$\boxed{h_{\widetilde{s}} = h_s^{-1}}$$

□

I am working through Conlon (2008) [90] , Clarke and Santoro (2012) [91], and Schreiber and Waldorf (2007)[92], concurrently, for holonomy.

101 Path Groupoid of a smooth manifold; generalization of paths

cf. Schreiber and Waldorf (2007)[92].

Definition 206 (path)**. path** is a smooth map $\gamma : [0,1] \to M$, between 2 pts. $x, y \in M$,
which has a sitting instant; i.e. number $0 < \epsilon < \frac{1}{2}$ s.t.

$$\gamma(t) = \begin{cases} x & \text{for } 0 \leq t < \epsilon \\ y & \text{for } 1 - \epsilon < t \leq 1 \end{cases} \tag{373}$$

Denote the set of such paths by PM,

$$PM \equiv \{\gamma \in \Gamma(M) | \text{ smooth } \gamma : [0,1] \to M \text{ s.t. } \exists 0 < \epsilon < \frac{1}{2} \text{ s.t. } \begin{cases} x & \text{for } 0 \leq t < \epsilon \\ y & \text{for } 1 - \epsilon < t \leq 1 \end{cases}\} \tag{374}$$

cf. Def. 2.1. of Schreiber and Waldorf (2007)[92]
Define *composition*:

Given paths γ_1, γ_2; $\gamma_1(0) = x$, $\gamma_2(0) = y$,
$\gamma_1(1) = y$ $\gamma_2(1) = z$

define composition to be path

$$\gamma_2 \circ \gamma_1$$

$$(\gamma_2 \circ \gamma_1)(t) := \begin{cases} \gamma_1(2t) & \text{for } 0 \le t \le \frac{1}{2} \\ \gamma_2(2t-1) & \text{for } \frac{1}{2} \le t \le 1 \end{cases} \tag{375}$$

$\gamma_2 \circ \gamma_1$ smooth since γ_1, γ_2 both constant near gluing pt., due to sitting instants ϵ_1, ϵ_2, respectively.
Define *inverse*:

$$\begin{aligned} \gamma^{-1} &: [0,1] \to M \\ \gamma^{-1}(t) &:= \gamma(1-t) \end{aligned} \tag{376}$$

(so that $\gamma^1(t) = \begin{cases} y & \text{for } 1-\epsilon < 1-t \le 1 \text{ or } 0 \le t < \epsilon \\ x & \text{for } 0 \le 1-t < \epsilon \text{ or } 1-\epsilon < t \le 1 \end{cases}$)

Definition 207 (thin homotopy equivalent)**.** 2 paths γ_1, γ_2 s.t. $\gamma_1(0) = \gamma_2(0) = x$, $\gamma_1(1) = \gamma_2(1) = y$ γ_1, γ_2 are thin homotopy equivalent,
if $\exists$ smooth $h : [0,1] \times [0,1] \to M$ s.t.

1. $\exists\, 0 < \epsilon < \frac{1}{2}$ with

   (a) $h(s,t) = x$ for $0 \le t < \epsilon$
   $h(s,t) = y$ for $1-\epsilon < t \le 1$

   (b)

   (c) $h(s,t) = \gamma_1(t)$ for $0 \le s < \epsilon$
   $h(s,t) = \gamma_2(t)$ for $1-\epsilon < s \le 1$

2. differential of h has at most rank 1 everywhere, i.e.

$$\operatorname{rank}(dh|_{(s,t)}) \le 1 \qquad \forall\, (s,t) \in [0,1] \times [0,1] \tag{377}$$

cf. Def. 2.2. of Schreiber and Waldorf (2007)[92]

$h(s,t) = \gamma_1(t)$ for $0 \le s < \epsilon$ is the homotopy from γ_1 to γ_2, i.e. $h(0,t) = \gamma_1(t)$
$h(s,t) = \gamma_2(t)$ for $1-\epsilon < s \le 1$ $h(1,t) = \gamma_2(t)$
and define an equivalence relation on PM.
Note that for $h : [0,1] \times [0,1] \to M$,

$$(Dh)|_{(s,t)} = \left[\frac{\partial h^i}{\partial s}, \frac{\partial h^i}{\partial t} \right]$$

$$\begin{aligned} P^1 M \equiv\ & \text{ set of thin homotopy classes of paths, i.e.} \\ P^1 M = \{[\gamma] | \gamma_1 \in PM,\ & \text{if } \exists \text{ smooth } h : [0,1] \times [0,1] \to M \text{ s.t. } h \text{ thin homotopy of } \gamma_1 \text{ and } \gamma_2, \gamma_1 \sim \gamma_2\} \end{aligned} \tag{378}$$

$\mathrm{pr} : PM \to P^1 M$ is projection to classes.

Denote thin homotopy class of path γ, $\gamma(0) = x$, by $\overline{\gamma}$, or $[\gamma]$.
$\gamma(1) = y$

101.1 Reparametrization of thin homotopies

Let $\beta : [0,1] \to [0,1]$, $\beta(0) = 0$.
$\beta(1) = 1$

Then $\forall$ path γ, $\gamma(0) = x$, $\gamma \circ \beta$ is also a path $\gamma \circ \beta(0) = x$ and
$\gamma(1) = y$ $\gamma \circ \beta(1) = y$

$$h(s,t) := \gamma(t\beta(1-s) + \beta(t)\beta(s) \tag{379}$$

defines a homotopy from γ to $\gamma \circ \beta$.

$$\gamma_1 \circ \gamma_2 \in PM \overset{\text{pr}}{\longmapsto} [\gamma_1 \circ \gamma_2] = [\gamma_1][\gamma_2] \in P^1 M$$

Composition of thin homotopy classes of paths obeys following rules:

Lemma 36. $\forall$ *path* γ, $\gamma(0) = x$
$\gamma(1) = y$

1. $\overline{\gamma} \circ \overline{id_x} = \overline{\gamma} = \overline{id_y} \circ \overline{\gamma} \equiv [\gamma]1_x = [\gamma] = 1_y[\gamma]$

2. *for paths* γ'; $\gamma'(0) = y$, $\gamma''(0) = z$
$\gamma'(1) = z$ $\gamma''(1) = w$

$$(\overline{\gamma}'' \circ \overline{\gamma}') \circ \overline{\gamma} = \overline{\gamma}'' \circ (\overline{\gamma}' \circ \overline{\gamma}) \equiv ([\gamma'']{[\gamma']})[\gamma] = [\gamma'']([\gamma']{[\gamma]}) \tag{380}$$

3. $\overline{\gamma} \circ \overline{\gamma}^{-1} = \overline{id_y}$ *and* $\overline{\gamma^{-1}} \circ \overline{\gamma} = \overline{id_x} \equiv [\gamma][\gamma^{-1}] = 1_y$ *and* $[\gamma^{-1}][\gamma] = 1_x$

cf. Lemma 2.3. of Schreiber and Waldorf (2007)[92]

Definition 208 (path groupoid)**.** $\forall$ smooth manifold M,
consider category whose set of objects is M,

whose set of morphisms is $P^1 M$, where class $[\gamma]$, $[\gamma](0) = x$ is a morphism from x to y and
$[\gamma](1) = y$
composition $[\gamma_1][\gamma_2] = [\gamma_1 \circ \gamma_2] \in P^1 M$ Lemma 36 are axioms of a category, 3rd. property says $\forall$ morphism is invertible.

Hence, we've defined a groupoid, called **path groupoid** of M, $\mathcal{P}_1(M)$.

So

$$\text{Obj}(\mathcal{P}_1(M)) = M$$
$$\text{Mor}(\mathcal{P}_1(M)) = P^1 M$$

$\forall$ smooth $f : M \to N$, denote functor f_*

$$f_* : \mathcal{P}_1(M) \to \mathcal{P}_1(N) \tag{381}$$

with

$$f_*(x) = f(x)$$
$$(f_*)([\gamma]) := [f \circ \gamma]$$

If $\gamma \sim \gamma'$, for $f \circ \gamma$, $f \circ \gamma'$,

$f \circ h(s,t)$ with $f \circ h(0,t) = f \circ \gamma(t)$,
$f \circ h(1,t) = f \circ \gamma'(t)$
so $f \circ h$ is a thin homotopy between $f \circ \gamma$, $f \circ \gamma'$ and so $[f \circ \gamma]$ *well-defined.*

Part XXV
Complex Manifolds

EY : 20170123 I don't see many good books on Complex Manifolds for physicists other than Nakahara's. I will supplement this section on Complex Manifolds with external links to the notes of other courses that I found useful to myself.

Complex Manifolds - Lecture Notes Koppensteiner (2010) [87]

Lectures on Riemannian Geometry, Part II: Complex Manifolds by Stefan Vandoren Vandoren (2008) [88]

Part XXVI
Jets, Jet bundles, h-principle, h-Prinzipien

cf. Eliashberg and Misahchev (2002) [93]

cf. Ch. 1 Jets and Holonomy, Sec. 1.1 Maps and sections of Eliashberg and Misahchev (2002) [93].

Visualize $f : \mathbb{R}^n \to \mathbb{R}^q$ as graph $\Gamma_f \subset \mathbb{R}^n \times \mathbb{R}^q$.

Consider this graph as image of $\mathbb{R}^n \to \mathbb{R}^n \times \mathbb{R}^q$, i.e.

$$x \mapsto (x, f(x))$$

$\mathbb{R}^n \to \mathbb{R}^n \times \mathbb{R}^q$ is called section (by mathematicians),

$x \mapsto (x, f(x))$
is called *field* or $\mathbb{R}^q$-valued field (by physicists).

cf. Ch. 1 Jets and Holonomy, Sec. 1.2 Coordinate definition of jets of Eliashberg and Misahchev (2002) [93].

Definition 209 (r-jet)**.** Given (smooth) $f : \mathbb{R}^n \to \mathbb{R}^q$, given $x \in \mathbb{R}^n$.

r-jet of f at x - sequence of derivatives of f, up to order r, $\equiv$

$$J^r_f(x) = (f(x), f'(x) \dots f^{(r)}(x)) \tag{382}$$

$f^{(q)}$ consists of all partial derivatives $D^\alpha f$, $\alpha = (\alpha_1 \dots \alpha_n)$, $|\alpha| = \alpha_1 + \cdots + \alpha_n = s$, ordered lexicographically.

e.g. $q = 1$,
$f : \mathbb{R}^n \to \mathbb{R}$.
1-jet of f at $x = J^1_f(x) = (f(x), f^{(1)}(x))$.

$$f^{(1)}(x) = \{D^\alpha f | \alpha = (\alpha_1 \dots \alpha_n), |\alpha| = \alpha_1 + \cdots + \alpha_n = 1\} = \left(\frac{\partial f}{\partial x^1}, \frac{\partial f}{\partial x^2}, \dots \frac{\partial f}{\partial x^n} \right)$$

Let $d_r = d(n, r) =$ number of all partial derivatives D^α of order r of function $\mathbb{R}^n \to \mathbb{R}$.

Consider r-jet $J^r_f(x)$ of map $f : \mathbb{R}^n \to \mathbb{R}^q$ as pt. of space $\mathbb{R}^q \times \mathbb{R}^{qd_1} \times \mathbb{R}^{qd_2} \times \cdots \times \mathbb{R}^{qd_r} = \mathbb{R}^{qN_r}$, where $N_r = N(n, r) = 1 + d_1 + d_2 + \cdots + d_r$, i.e.

$$J^r_f(x) = (f(x), f^{(1)}(x), \dots f^{(r)}(x)) \in \mathbb{R}^q \times \mathbb{R}^{qd_1} \times \cdots \times \mathbb{R}^{qd_r} = \mathbb{R}^{qN_r}$$

Exercise 1.

Given order r, consider n-tuple of (positive) integers $(r_1, r_2 \dots r_n)$ s.t. $r_1 + r_2 + \cdots + r_n = r$, and $r_k \geq 0$. Imagine $r_k =$ occupancy number, num ber of balls in kth cell. $(r_1 \dots r_n)$ describes a positive ocnfiguration of occupancy numbers, with indistinguishable balls; 2 distributions are distinguishable only if corresponding n-tuples $(r_1 \dots r_n)$ not identical.

Represent balls by stars, and indicate n cells by n spaces between $n + 1$ bars.

With $n + 1$ bars, r stars, 2 bars are fixed. $n - 1$ bars and r stars to arrange linearly, so a total of $n - 1 + r$ objects to arrange. r stars indistinguishable amongst themselves, so choose r out of $n - 1 + r$ to be stars.

$$\implies d_r = d(n, r) = \binom{n - 1 + r}{r} \tag{383}$$

Use *induction* (cf. Ch. 4 Binomial Coefficients).

$$N_0 = N(n,0) = \binom{n-1+0}{0} = 1$$
$$N_1 = N(n,1) = 1 + \binom{n-1+1}{1} = 1 + n = \frac{(n+1)!}{n!1!}$$

Induction step:

$$N_{r-1} = N(n,r-1) = \sum_{k=1}^{r-1} d_k + 1 = \binom{n+r-1}{r-1}$$

and so

$$N_r = N(n,r) = \sum_{k=1}^{r} d_k + 1 = \sum_{k=1}^{r} \binom{n-1+k}{k} + 1 = \sum_{k=1}^{r-1} \binom{n-1+k}{k} + \binom{n-1+r}{r} + 1 =$$
$$= \binom{n+r-1}{r-1} + \binom{n-1+r}{r} = \frac{(n+r-1)!}{(r-1)!n!} + \frac{(n-1+r)!}{r!(n-1)!} = \frac{(n+r)!}{n!r!} = \binom{n+r}{r}$$

$$\begin{array}{ccc} \mathbb{R}^{qN_r} & & \\ \uparrow{\scriptstyle J^r_f} & & \\ \mathbb{R}^n & \xrightarrow{\quad f \quad} & \mathbb{R}^q \end{array} \qquad \begin{array}{ccc} J^r_f(x) & & \\ \uparrow{\scriptstyle J^r_f} & & \\ x & \xmapsto{\quad f \quad} & f(x) \end{array}$$

Definition 210 (space of r-jets)**.** space of r-jects of maps $\mathbb{R}^n \to \mathbb{R}^q$ or space of r-jets of sections $\mathbb{R}^n \to \mathbb{R}^n \times \mathbb{R}^q \equiv$

$$J^r(\mathbb{R}^n, \mathbb{R}^q) = \mathbb{R}^n \times \mathbb{R}^{qN_r} \equiv \mathbb{R}^n \times \mathbb{R}^q \times \mathbb{R}^{qd_1} \times \mathbb{R}^{qd_2} \times \cdots \times \mathbb{R}^{qd_r} \tag{384}$$

e.g. $J^1(\mathbb{R}^n, \mathbb{R}^q) = \mathbb{R}^n \times \mathbb{R}^q \times M_{q\times n}$, where $M_{q\times n} = \mathbb{R}^{qn}$ is the space of $(q \times n)$-matrices.

Part XXVII
Morse Theory

102 Morse Theory introduction from a physicist

I needed some physical motivation to understand Morse theory, and so I looked at Hori, et. al. [89].

cf. pp. 43, Sec. 3.4 Morse Theory, from Ch. 3. Differential and Algebraic Topology of Hori, et. al. [89].

Consider smooth $f : M \to \mathbb{R}$, with non-degenerate critical points.

If no critical values of f between a and b $(a < b)$, then subspace on which f takes values less than a is deformation retract of subspace where f less than b, i.e.

$$\{x \in M | f(x) < b\} \times [0,1] \xrightarrow{F} \{x \in M | f(x) < b\}$$

$\forall\, x \in M$ s.t. $f(x) < b$,

$$F(x,0) = x$$
$$F(x,1) \in \{x \in M | f(x) < a\} \qquad \text{and } F(a',1) = a' \qquad \forall\, a' \in M \text{ s.t. } f(a') < a$$

To show this, consider $-\nabla f/|\nabla f|^2$

Morse lemma: $\forall$ critical pt. p s.t. $\exists$ choice of coordinates s.t.

$$f = -(x_1^2 + x_2^2 + \cdots + x_\mu^2) + x_{\mu+1}^2 + \cdots + x_n^2 \tag{385}$$

where $f(p) = 0$ and p is at origin of these coordinates.

- difference between
$$f^{-1}(\{x \leq -\epsilon\}),\ f^{-1}(\{x \leq +\epsilon\})$$
can be determined by local analysis and only depends on μ, $\mu \equiv$ "Morse index" = number of negative eigenvalues of Hessian of f at critical pt.

  Answer:

$f^{-1}(\{x \leq +\epsilon\})$ can be obtained from $f^{-1}(\{x \leq -\epsilon\})$ by "attaching μ-cell" along boundary $f^{-1}(0)$

- "attaching μ-cell to X mean, take μ-ball $B_\mu = \{|x| \leq 1\}$ in μ-dim. space,
identity pts. on boundary $S^{\mu-1}$ with pts. in the space X, through
cont. $f : S^{\mu-1} \to X$, i.e. take
$$X \coprod B_\mu$$
with $x \sim f(x) \quad \forall\, x \in \partial B_\mu = S^{\mu-1}$.

- find homology of M,

  f defines chain complex C^*_f, kth graded piece C^{α_k}, α_k is number of critical pts. with index k.
$$\begin{aligned} \partial &: C^k_p \to C^{k-1}_p \\ \partial x_a &= \sum_b \Delta_{a,b} x_b \end{aligned} \tag{386}$$
  where $\Delta_{a,b}$:= signed number of lines of gradient flow from x_a to x_b, b labels pts. of index $k-1$.

  Gradient flow line is path $x(t)$ s.t. $\dot{x} = \nabla(f)$, with $x(-\infty) = x_a$
  $x(+\infty) = x_b$

- To define this number ($\Delta_{a,b}$?), construct moduli space of such lines of flow (???)
by intersecting outward and inward flowing path spaces from each critical point, and then show this moduli space is oriented, 0-dim. manifold (pts. with signs)

- $\partial^2 = 0$ proof

  ∂, boundary of space of paths connecting critical points, whose index differs by 2 = union over compositions of paths between critical pts. whose index differs by 1.

  $\Longrightarrow$ coefficients of ∂^2 are sums of signs of pts. in 0-dim. space, which is boundary of 1-dim. space.

  These signs must therefore add to 0, so $\partial^2 = 0$.

Hori, et. al. [89] is good for physics, but there isn't much thorough, step-by-step explanations of the math. I will look at Hirsch (1997) [79] and Shastri (2011) [78] at the same time.

102.1 Introduction, definitions of Morse Functions, for Morse Theory

cf. Ch. 6, Morse Theory of Hirsch (1997) [79], Section 1. Morse Functions, pp. 143-

Recall for TM, $T_xM \xrightarrow{\varphi} \mathbb{R}^n$.

Cotangent bundle T^*M defined likewise:
$$T^*_x M \xrightarrow{\varphi} \text{ dual vector space } (\mathbb{R}^n)^* = L(\mathbb{R}^n, \mathbb{R})$$
i.e.
$$T^*M = \bigcup_{x \in M} (M^*_x) \qquad\qquad M^*_x = L(M_x, \mathbb{R})$$

If chart (φ, U) on M, natural chart on T^*M is

$$T^*U \to \varphi(U) \times (\mathbb{R}^n)^*$$
$$\lambda \in M_x^* \mapsto (\varphi(x), \lambda\varphi_x^{-1})$$

Projection map

$$p : T^* \to M$$
$$M_x^* \mapsto x$$

Let C^{r+1} map, $1 \leq r \leq \omega$, $f : M \to \mathbb{R}$, $\forall\, x \in M$, linear map $T_x f : M_x \to \mathbb{R}$ belongs to M_x^*

$$T_x f = Df_x \in M_x^*$$

Then

$$Df : M \to T^*M$$
$$x \mapsto Df_x = Df(x)$$

is C^r section of T^*M.

Definition 211. critical point x of f is zero of Df, i.e.

$$Df(x) = 0 \tag{387}$$

of vector space M_x^*.

Thus, set of critical pts. of f is counter-image of submanifold $Z^* \subset T^*M$ of zeros.
Note $Z^* \approx M$, codim. of Z^* is $n = \dim M$.

Definition 212. Morse function f if $\forall$ critical pts. of f are nondegenerate.

Note set of critical pts. closed discrete subset of M.
Let open $U \subset \mathbb{R}^n$, let C^2 map $g : U \to \mathbb{R}$,
critical pt. $p \in U$ nondegenerate iff

- linear $D(Dg)(p) : \mathbb{R}^n \to (\mathbb{R}^n)^*$ bijective
- identify $L(\mathbb{R}^n, (\mathbb{R}^n)^*)$ with space of bilinear maps $\mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$, $\implies$ equivalent to condition that symmetric bilinear $D^2g(p) : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ non-degenerate
- $n \times n$ *Hessian matrix*

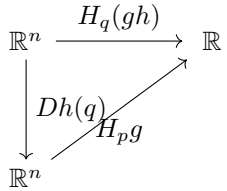
  $$\left[\frac{\partial^2 g}{\partial x^i \partial x^j}(p)\right]$$

  has rank n

Hessian of g at critical pt. p is quadratic form $H_p f$ associated to bilinear form $D^2g(p)$

$$\implies H_p f(y) = D^2g(p)(y, y) = \sum_{i,j} \frac{\partial^2 g}{\partial x^i \partial x^j}(p) y^i y^j$$

Let open $V \subset \mathbb{R}^n$, suppose C^2 diffeomorphism $h : V \to U$.
Let $q = h^{-1}(p)$, so q is critical pt. of $gh : V \to \mathbb{R}$.



(quadratic) form $(H_p f)$ invariant under diffeomorphisms.

Let C^2 $f : M \to \mathbb{R}$.
$\forall$ critical pt. x of f, define
Hessian quadratic form

$$H_x f : M_x \to \mathbb{R}$$

$$H_x f : M_x \xrightarrow{D\varphi_x} \mathbb{R}^n \xrightarrow{H_{\varphi(x)}(f\varphi^{-1})} \mathbb{R}$$

where φ is any chart at x.

Thus, critical pt. of a C^2 real-valued function nondegenerate iff associated Hessian quadratic form is nondegenerate.

Let Q nondegenerate quadratic form on vector space E.

Q negative definite on subspace $F \subset E$ if $Q(x) < 0$ whenever $x \in F$ nonzero.

Index of $Q \equiv \text{Ind}Q$, is largest possible dim. of subspace on which Q is negative definite.

cf. 1.1. Morse's Lemma of Ch. 6, pp. 145, Morse Theory of Hirsch (1997) [79]

Lemma 37 (Morse's Lemma)**.** *Let $p \in M$ be nondegenerate critical pt. of index k of C^{r+2} map $f : M \to \mathbb{R}$, $1 \leq r \leq \omega$.*

Then $\exists\, C^r$ chart (φ, U) at p s.t.

$$f\varphi^{-1}(u_1 \dots u_n) = f(p) - \sum_{i=1}^{k} u_i^2 + \sum_{i=k+1}^{n} u_i^2 \tag{388}$$

Let ${}^T Q \equiv Q^T$ denote tranpose of matrix Q.

Lemma 38. *Let $A = diag\{a_1, \dots, a_n\}$ diagonal $n \times n$ matrix, with diagonal entries ± 1.*

Then $\exists$ neighborhood N of A in vector space of symmetric $n \times n$ matrices, C^∞ map

$$P : N \to GL(n, \mathbb{R}) \tag{389}$$

s.t. $P(A) = I$, and if $P(B) = Q$, then $Q^T B Q = A$

Proof. Let $B = [b_{ij}]$ be symmetri matrix near A s.t. $b11 \neq 0$ and b_{11} has same sign as a_1.

Consider $x = Ty$ where

$$x_1 = \left[y_1 - \frac{b_{12}}{b_{11}} y_2 - \dots - \frac{b_{1n}}{b_{11}} y_n \right] / \sqrt{|b_n|}$$

$$x_k = y_k \text{ for } k = 2, \dots n$$

□

103 Lagrange multipliers

From *wikipedia:Lagrange multiplier*, https://en.wikipedia.org/wiki/Lagrange_multiplier, find local minima (maxima), pt. $a \in N$, s.t. $\exists$ neighborhood U s.t. $f(x) \geq f(a)$ $(f(x) \leq f(a))$ $\forall\, x \in U$.

For $f : U \to \mathbb{R}$, open $U \subset \mathbb{R}^n$, find $x \in U$ s.t. $D_x f \equiv Df(x) = 0$, check if Hessian $H_x f < 0$.

Maxima may not exit since U open.

References:

Relative Extrema and Lagrange Multipliers

Other interesting links:

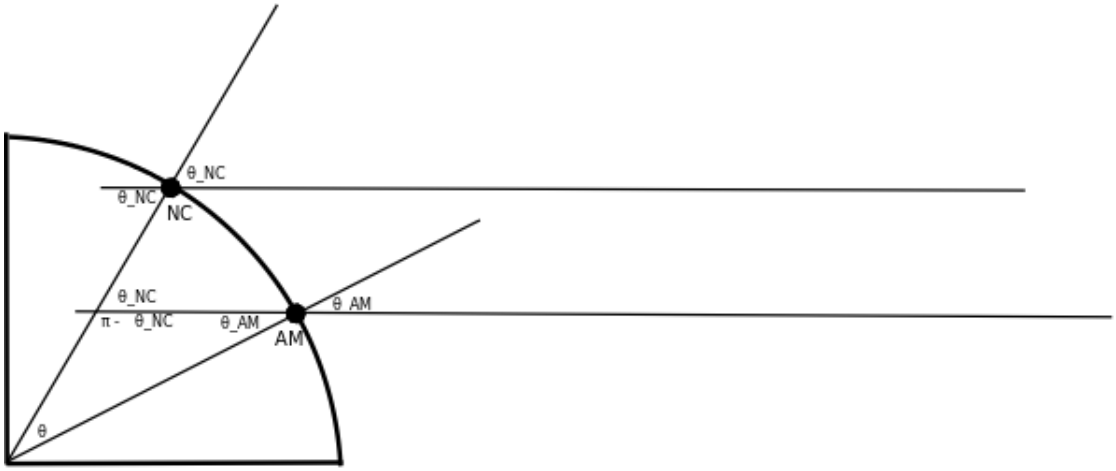
The Lagrange Multiplier Rule on Manifolds and Optimal Control of nonlinear systems

Part XXVIII
Classical Mechanics

Problem 1.4 The altitude angle of the Sun, Japan Physics Olympiad (2014). cf. Japan Physics Olympiad, (2014) [121].

Solution 1.4 The altitude angle of the Sun, Japan Physics Olympiad (2014).

$NC \equiv$ Niigata City,
$AM \equiv$ Amagi-san in Izu.



From all 3 internal angles of a triangle sum up to 180 degrees, or i.e. π radians,

$$\pi = \pi - \theta_{NC} + \theta_{AM} + \theta \Longrightarrow \boxed{\theta = \theta_{NC} - \theta_{AM}}$$

So we've shown indeed that θ is the **difference between altitude angles of the Sun** at NC and AM, respectively.

From

$$C = 2\pi R$$
$$l_M = \theta R$$
$$\frac{l_M}{l} = \frac{\theta R}{\frac{\pi}{2}R} \text{or } \frac{l_M}{l} = \frac{2\theta}{\pi}$$
$$\Longrightarrow \theta = \frac{\pi}{2}\frac{l_M}{l} = \frac{\pi}{2} \cdot \frac{334 \times 10^3\, m}{10^7 m} \cdot \frac{180^\circ}{\pi} = 3^\circ$$

Advice, remember to draw a *cross-section* to make diagram clear, rather than worry about "the other transverse dimension". Also, draw the lines literally parallel to the Sun because of how far away the Sun is and how large the Sun is.

104 Structure of Galilean Space-Time

cf. Sec. 3.1 - *Structure of Galilean Space-Time* of Prástaro (1996) [86].

Mechanics assumes a particular simple formulation if formulated with respect to some spacetime manifold.

In Galilean spacetime, it's possible to naturally recognize absolute objects, and others that depend on frames.

cf. Def. 3.1 of Prástaro (1996) [86]

Definition 213 (Galilean spacetime structure)**.** 1. Galilean spacetime structure $:= (\mathcal{G}, g)$ where
$\mathcal{G}$ is (fiber bundle space-time)

$$\mathcal{G} \equiv \{\tau : M \to T\} \tag{390}$$

where
$M =$ 4-dim. affine manifold (**space-time**); corresponding structure is $(M, \mathbf{M}, \alpha)$,

2. $T =$ 1-dim. affine space (**time**), corresponding affine structure is $(T, \mathbf{T}, \beta)$

3. $\tau =$ surjective affine mapping, of constant rank 1, s.t. $\forall p \in M$ associates its time $\tau(p) \in T$

Put $\mathbf{S} = \ker(\underline{\tau}) \equiv \ker(D\tau) \in M$,

where $\underline{\tau} \equiv D\tau$, D is symbol of derivative.

Define

$$g : M \to vS_2^0(M) \equiv M \times S_2^0(\mathbf{S})$$
$$g(p) = (p, \underline{g}) \equiv (p, Dg), \forall p \in M$$

where $\underline{g} \equiv Dg$ is a Euclidean structure on $\mathbf{S}$. g is called vertical metric field.

Thus, given $(M, \mathbf{M}, \alpha)$, $\forall (O, \{\mathbf{e}_i)_{1 \leq i \leq d}$, $\{\mathbf{e}_i\}_{i=1...d}$, is basis of $\mathbf{M}$,

$$M \cong \mathbb{R}^4, \text{and } \exists \{x^\alpha : M = \mathbb{R}^4 \to \mathbb{R}\}_{\alpha=1...4}$$

cf. 1.1.3 Principle of Relativity, pp. 9, 1 Basic Principles of Classical Mechanics, Arnold, Kozlov, Neishtadt (2006) [103].

direct product $E^3 \times \mathbb{R}\{t\}$ (spacetime) has natural structure of affine space

Galilean group is by def. group of all affine transformations of $E^3 \times \mathbb{R}$ that preserve time intervals and are isometries of space $E^3 \forall$ fixed $t \in \mathbb{R}$.

Thus if

Definition 214 (Galilean group)**. Galilean group** is group of all affine transformations of $E^3 \times \mathbb{R}$ that preserve time intervals and are isometries of space $E^3 \forall$ fixed $t \in \mathbb{R}$.

Thus, if Galilean transformation g,

$$\begin{aligned} g &: E^3 \times \mathbb{R}\{t\} \to E^3 \times \mathbb{R}\{t\} \\ g &: (s, t) \mapsto (s', t') \end{aligned} \tag{391}$$

1. $t_\alpha - t_\beta = t'_\alpha - t'_\beta$

2. if $t_\alpha = t_\beta$, then $|s_\alpha - s_\beta| = |s'_\alpha - s'_\beta|$

Consider $(s_a, t_a), (s_b, t_b)$, $t_a - t_b$, $|s_a - s_b|$.

$$\begin{aligned} g(s_a, t_a) &= (s'_a, t'_a) \\ g(s_b, t_b) &= (s'_b, t'_b) \\ t'_a - t'_b = \pi_t g(s_a, t_a) - \pi_t g(s_b, t_b) &= t_a - t_b \\ |s'_a - s'_b| = |\pi_s g(s_a, t_a) - \pi_s g(s_b, t_b)| &= |s_a - s_b| \end{aligned}$$

Galilean group g acts on $\mathbb{R}^3\{\searrow\} \times \mathbb{R}\{t\}$.

104.1 3 examples of Galilean transformations

uniform motion with constant velocity $\mathbf{v}$:

$$g_1(\mathbf{r}, t) = (\mathbf{r} + \mathbf{v}t, t) \tag{392}$$

translation of origin in spacetime

$$g_2(\mathbf{r}, t) = (\mathbf{r} + \mathbf{x}, t + \alpha) \tag{393}$$

rotation of coordinate axes

$$g_3(\mathbf{r}, t) = (G\mathbf{r}, t) \tag{394}$$

where $G : \mathbb{R}^3 \to \mathbb{R}^3$ is orthonormal transformation.

Proposition 99 (Galilean transformation decomposition)**.** $\forall$ *galilean transformation* $g : \mathbb{R}^3 \times \mathbb{R} \to \mathbb{R}^3 \times \mathbb{R}$ *can be uniquely represented as composition* $g_1 g_2 g_3$

cf. Prop. 1.1 of Arnold, Kozlov, Neishtadt (2006) [103]

Proof. Consider affine transformation of $E^4 = E^3 \times \mathbb{R}$ of general form

$$\mathbf{x} = A\mathbf{x}' + \mathbf{v}t' + \mathbf{a}$$
$$t = \langle \mathbf{l}, \mathbf{x}' \rangle + kt' + s$$

$A \in \mathrm{Mat}_{\mathbb{R}}(3,3)$, $\mathbf{v}, \mathbf{a}, \mathbf{l} \in \mathbb{R}^3$, $k, s \in \mathbb{R}$.

By galilean transformation (Def. 214), $t_2 - t_1 = t_2' - t_1'$, so

$$t_2 - t_1 = \langle \mathbf{l}, \mathbf{x}_2' \rangle + kt_2' + s - \langle \mathbf{l}, \mathbf{x}_1' \rangle - kt_1' - s = \langle \mathbf{l}, \mathbf{x}_2' - \mathbf{x}_1' \rangle + k(t_2' - t_1') = t_2' - t_1'$$

Since each of the 3 components of $\mathbf{x}_2 - \mathbf{x}_1$ are arbitrary, $\mathbf{l} = 0$, $k = 1$ clearly. Or, let $t = 2 = t_1$. Then clearly $\mathbf{l} = 0$.

By galilean transformation (Def. 214), $|\mathbf{x}_2 - \mathbf{x}_1| = |\mathbf{x}_2' - \mathbf{x}_1'|$

Let $t_2 = t_1$.

$$|\mathbf{x}_2 - \mathbf{x}_1| = |A\mathbf{x}_2' + \mathbf{v}t_2' + \mathbf{a} - A\mathbf{x}_1' - \mathbf{v}t_1' - \mathbf{a}| = |A(\mathbf{x}_2' - \mathbf{x}_1') + \mathbf{v}(t_2' - t_1')| = |A(\mathbf{x}_2' - \mathbf{x}_1')| = |\mathbf{x}_2' - \mathbf{x}_1'|$$

Now let $\mathbf{y} = \mathbf{x}_2' - \mathbf{x}_1'$.

$$|A\mathbf{y}|^2 = y^T A^T A y = y^T y$$

only if $A^T A = 1$. So A orthonormal.

Thus we've found the *general form of Galilean transformations*:

$$\boxed{\begin{array}{l} \mathbf{x} = A\mathbf{x}' + \mathbf{v}t' + \mathbf{a}, \quad A \in O(3), \mathbf{v}, \mathbf{a} \in \mathbb{R}^3, s \in \mathbb{R} \\ t = t' + s \end{array}} \tag{395}$$

Since orthogonal matrices form a 3 parameter family, $\dim A + \dim \mathbf{v} + \dim \mathbf{a} + \dim s = 3 + 3 + 3 + 1 = 10$ independent parameters.

Recall that $g_1(\mathbf{r}, t) = (\mathbf{r} + \mathbf{v}t, t)$, $g_2(\mathbf{r}, t) = (\mathbf{r} + \mathbf{a}, t + s)$, $g_3(\mathbf{r}, t) = (A\mathbf{r}, t)$, then $g = g_1 g_2 g_3$.

□

Introduce E^3 a "fixed" reference frame by fixing pt. $o \in E^3$ and choosing 3 mutually perpendicular axes.

inertial frame - frame moving uniformly and rectilinearly with respect to fixed pt. $o \in E^3$; $\forall$ element of Galilean group transforms fixed frame into inertial frame.

action of galilean group on $E^3 \times \mathbb{R}$ can be extended to action on $E^3 \times \cdots \times E^3 \times \mathbb{R}$ by the rule:
if $g : (s, t) \to (s', t')$, then
$g(s_1, \ldots s_n, t) = g(s_1', \ldots s_n', t')$

Galileo-Newton principle of relativity asserts that Newton's equations invariant under Galilean transformation group in inertial reference frame.

M = 4-dim. affine manifold (spacetime). Recall definition of an *affine manifold* from wikipedia.

Affine manifold M is real manifold with charts $\psi_i : U_i \to \mathbb{R}^n$, s.t. $\psi_i \circ \psi_j^{-1} \in \mathrm{Aff}(\mathbb{R}^n) \quad \forall i, j$ where $\mathrm{Aff}(\mathbb{R}^n)$ denotes Lie group of affine transformations;

Recall the definition of a *Lie group of affine transformations*, $\mathrm{Aff}(\mathbb{R}^n)$, the "affine structure", [6],

Definition 215 (Lie group of affine transformations, affine structure)**.**

$$\mathrm{Aff}(\mathbb{R}^n) = \{ \begin{pmatrix} A & b \\ 0 & 1 \end{pmatrix} | A \in GL(n, \mathbb{R}), b \in \mathbb{R}^n \} \tag{396}$$

[6] "Affine structures on Lie algebras" by Michel Goze and Elisabeth Remm

s.t. $\begin{pmatrix} A & b \\ 0 & 1 \end{pmatrix}$ acts on real affine space $\widetilde{\mathbb{R}^n}$ by

$$\begin{pmatrix} A & b \\ 0 & 1 \end{pmatrix} \begin{pmatrix} v \\ 1 \end{pmatrix} = \begin{pmatrix} Av + b \\ 1 \end{pmatrix} \tag{397}$$

where $A \in GL(n, \mathbb{R})$, $b \in \mathbb{R}^n$, where $(v, 1)^T \in \widetilde{\mathbb{R}^n}$

If changing notation, and keeping in mind that $n = 4$, $\psi_j \equiv y(v_{(y)}) \to \mathbb{R}^n$,

$\psi_i \equiv x : U_{(x)} \to \mathbb{R}^n$
$x(p) \in \mathbb{R}^n, x_i(p) \in \mathbb{R}$

Rewrite the Eq. 395 in matrix form:

$$\mathbf{x} = A\mathbf{x}' + \mathbf{v}t + \mathbf{b}$$
$$t = t' + s,\ A \in O(3),\ \mathbf{v}, \mathbf{b} \in \mathbb{R}^3,\ s \in \mathbb{R}$$
$$\begin{pmatrix} A & \mathbf{v} & \mathbf{b} \\ 0 & 1 & s \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \mathbf{x}' \\ t' \\ 1 \end{pmatrix} = \begin{pmatrix} \mathbf{x} \\ t \\ 1 \end{pmatrix} \text{ or } \begin{pmatrix} A & \mathbf{v} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \mathbf{x}' \\ t' \end{pmatrix} + \begin{pmatrix} \mathbf{b} \\ s \end{pmatrix} = \begin{pmatrix} \mathbf{x} \\ t \end{pmatrix}$$

Consider y being standard Euclidean coordinates in fixed frame o. x then is coordinates in any other inertial frame.

T = 1-dim. affine space since affine structure $(T, \mathbf{T}, \beta)$ is s.t. $T \in \mathbb{R}$, $\mathbf{T} \in \mathbb{R}$, $\beta : \mathbf{T} \times T \to T$
$$\beta(s, t') = t' + s = t$$

Holm (2011) [94]

105 Kinematics

105.1 Kinematics Problems from Irodov

cf. Irodov (1981) [120]
Solution 1.1. If we assume that the raft goes along downstream with no further "rowing",
$l \equiv$ distance raft travels for total time $T = 6.0$ km
$v_d \equiv$ downstream speed = raft speed
$\tau = 60\,\text{mins}$ = time before motorboat turning back
$v_m \equiv$ motorboat speed

$$\frac{l}{v_d} = \tau + \frac{\tau(v_m + v_d) - l}{v_m - v_d} = \frac{\tau(2v_m) - l}{v_m - v_d} \text{ or } \qquad (v_m - v_d)l = 2\tau v_m v_d - l v_d$$
$$v_m l = 2\tau v_m v_d$$
$$\boxed{v_d = \frac{l}{2\tau}} = \frac{6.0\,km}{2 \cdot 60\,\text{mins}} = 0.05\,\text{km / min}$$

Solution 1.2. The speed formula, recall, is $L = \overline{v}T$, or $\overline{v} = \frac{L}{T}$.

$$v_0 t_0 = \frac{L}{2} \text{ or } t_0 = \frac{L}{2v_0}$$

$$\begin{matrix} l_1 = v_1 \dfrac{T_1}{2} \\ l_2 = v_2 \dfrac{T_1}{2} \end{matrix} \text{ and } l_1 + l_2 = \frac{L}{2}, \text{ so } L = v_1 T_1 + v_2 T_1$$

$$\implies \overline{v} = \frac{L}{\frac{L}{2v_0} + \frac{L}{v_1 + v_2}} = \frac{2v_0(v_1 + v_2)}{2v_0 + v_1 + v_2}$$

Solution 1.3. Given
$w = 5.0\, m/s^2$ = constant acceleration and deceleration
$\tau = 25\,\text{sec.}$ = total time of motion
$\overline{v} = 72$ km / hr. = average velocity during that time (τ?)

Recall that

$$a = a(t) = \frac{dv}{dt} \text{ so if } a \text{ constant, then} v(t) - v_0 = at \text{ and since } v(t) = fracdxdt \text{ so}$$

$$x(t) = x_0 + v_0 t + \frac{1}{2}at^2$$

$$d = x(t) - x_0 = t(v_0 + \frac{a}{2}t) = \frac{v - v_0}{a}(v_0 + \frac{v - v_0}{2}) = \frac{v^2 - v_0^2}{2a} \text{ or } v^2 - v_0^2 = 2ad$$

Then

$$\frac{v^2}{2w} + \frac{v^2}{2w} + vT_2 = L \text{ or } \frac{v^2}{w} + vT_2 = L$$

Also

$$\tau = \frac{v}{w} + T_2 + \frac{-v}{-w} = \frac{2v}{w} + T_2 \text{ or } (\tau - T_2)\left(\frac{w}{2}\right) = v$$

Then

$$\overline{v} = \frac{\frac{v^2}{w} + vT_2}{\tau} \text{ or } \overline{v}\tau = \frac{v^2}{w} + wT_2$$

$$\implies \overline{v}\tau = v\left(\frac{v}{w} + T_2\right) = v(\frac{\tau - T_2}{2} + T_2) = \frac{w}{2}(\tau - T_2)(\frac{\tau + T_2}{2})$$

$$4\overline{v}\tau = w(\tau^2 - T_2^2) \text{ or } wT_2^2 = wT^2 - 4\overline{v}\tau$$

$$\implies \boxed{T_2 = \sqrt{\tau^2 - \frac{4\overline{v}\tau}{w}}}$$

Solution 1.5; particles collide.

Hint: *change reference frames*, in particular, into frame of references of particles 1 or 2, respectively. Consider, by label symmetry, reference frame of particle 1:

$$r_1' = r_1 - r_1 = 0 \qquad v_1' = v_1 - v_1 = 0$$
$$r_2' = r_2 - r_1 \qquad v_2' = v_2 - v_1$$

On order for a collision, in reference frame of particle 1, particle 2 will need to approach or travel towards the origin from $r_2' = r_2 - r_1$; consider
$v_2' \parallel = -r_2'$ i.e. $v_2 - v_1 \parallel -(r_2 - r_1)$, otherwise particle 2 would never reach origin where particle 1 is

$$\implies \boxed{\frac{\mathbf{v}_2 - \mathbf{v}_1}{\|\mathbf{v}_2 - \mathbf{v}_1\|} = \frac{-(\mathbf{r}_2 - \mathbf{r}_1)}{\|\mathbf{r}_2 - \mathbf{r}_1\|}}$$

Solution 1.6. $v_0 = 30\, km/hr$ = ship moves along equator to the east with speed v_0
$\varphi = 60°$ = southeastern wind blows at angle φ to equator
$v_w = 15\, km/h$ = wind speed
$v_w = v_{w,x}\mathbf{e}_x + v_{w,y}\mathbf{e}_y$
If southeastern wind means *it comes **from** the south east*, then we have

$$v_w' = v_w - v_0 = (-v_w \cos\varphi - v_0)\mathbf{e}_x + v_w s\varphi \mathbf{e}_y$$

$$\|v_w'\|^2 = v_w^2 c^2\varphi + 2v_w v_0 c\varphi + v_0^2 + v_w^2 s^2\varphi = v_w^2 + v_0^2 + 2v_w v_0 c\varphi = v_w^2 + v_0^2 + 2v_w v_0 \frac{1}{2} \approx 39.7\, km/h \approx 40\, km/h$$

$$\varphi' = \arctan\left(\frac{-v_w s\varphi}{v_w c\varphi + v_0}\right) = \arctan\left(\frac{-v_w\sqrt{3}/2}{v_w/2 + v_0}\right) \approx -19.1°$$

cf. https://physics.stackexchange.com/questions/126268/relative-wind-velocity-explanation for southeastern wind explanation.

Solution 1.7.

2 swimmers leave pt. A, pt. A lie right across point B.
stream velocity $v_0 = 2.0\,\text{km/hr}$
velocity v' of each swimmer with respect to water = 2.5 km/hr
Now $(\mathbf{v}_A + \mathbf{v}_0)T_A = \mathbf{x}_A$ and so in components, $v_{A,x}T_A = L$ and $(v')^2 = (v_{A,x})^2 + v_0^2$ or $v_{A,x} = \sqrt{(v')^2 - v_0^2}$

$T_{B,1}$ = time to swim across the river
$T_{B,2}$ = time to walk the distance he has been carried.

Since $(\mathbf{v}_B + \mathbf{v}_0)T_{B,1} = \mathbf{x}_{B,1}$ with $v'T_{B,1} = L$

$$v_0 T_{B,1} = y_B$$

$T_A = T_{B,1} + T_{B,2}$ since both swimmers reach the destination simultaneously.

$$\implies \frac{L}{v_{A,x}} = \frac{L}{v'} + \frac{v_0 L}{uv'} \text{ or } u = \frac{v_0/v'}{\frac{1}{v_{A,x}} - \frac{1}{v'}} = \boxed{\frac{v_0}{\frac{v'}{\sqrt{(v')^2 - v_0^2}} - 1}} = 3.0\,km/hr$$

Solution 1.8.

Buoy anchored at the middle of a river.
v_s = stream velocity

$$(v_A + v_s)T_{A,1} = L = (v_A - v_s)T_{A,2}$$
$$\implies (\eta + 1)v_s T_{A,1} = L = (\eta - 1)v_s T_{A,2}$$

where $T_{A,1} = \frac{L}{(\eta+1)v_s}$ is the time going away from the buoy to a distance L for person A, and $T_{A,2} = \frac{L}{(\eta-1)v_s}$ is the time to go back.

B moves away along the mutually perpendicular straight lines across the river. Then

$$(\mathbf{v}_B + \mathbf{v}_s)T_{B,1} = \mathbf{x}_B$$
$$v_{B,x}T_{B,1} = L$$
$$v_B^2 = (\eta v_s)^2 = v_{B,x}^2 + v_{B,y}^2 = v_{B,x}^2 + v_s^2$$

Now $T_{B,1} = T_{B,2}$, i.e. the time to go away from the buoy is equal to the time to go back to the buoy, and so

$$2T_{B,1} = \frac{2L}{\sqrt{(\eta v_s)^2 - v_s^2}} = \frac{2L}{v_s\sqrt{\eta^2 - 1}}$$

Thus,

$$\frac{\tau_A}{\tau_B} = \frac{T_{A,1} + T_{A,2}}{T_{B,1} + T_{B,2}} = \frac{\frac{2\eta}{\eta^2-1}}{\frac{2}{\sqrt{\eta^2-1}}} = \boxed{\frac{\eta}{\sqrt{\eta^2 - 1}}} \approx 1.81 \approx 1.8$$

Solution 1.10.

For body 1 going straight up, $y_1 = v_0 t - \frac{1}{2}gt^2$ and there's no horizontal, x_1, translation.

For body 2, the coordinates over time are the following:

$$y_2 = v_0 s\theta t - \frac{1}{2}gt^2$$
$$x_2 = v_0 c\theta t$$

$$\|\mathbf{x}_2 - \mathbf{x}_1\|^2 = (v_0 t(1 - s\theta))^2 + (v_0 c\theta t)^2 = (v_0 t)^2[2 - 2s\theta] = 2(v_0 t)^2(1 - s\theta)$$
$$\|\mathbf{x}_2 - \mathbf{x}_1\| = \boxed{v_0 t\sqrt{2}\sqrt{1 - s\theta}} = 25(1.7)\sqrt{2}\sqrt{1 - \sqrt{3}/2} \approx 22$$

Solution 1.11. 2 particles move in a uniform gravitational field with acceleration g.
At initial moment, the particles were located at 1 pt. and moved with velocities $v_1 = 3.0\,m/s$, $v_2 = 4.0\,m/s$

horizontally in opposite directions.
Typically, gravitational field is **vertical**:

$$\begin{aligned} v_{1y} &= -gt & v_{2y} &= -gt \\ v_{1x} &= -v_1 & v_{2x} &= v_2 \\ y_1 &= -\frac{1}{2}gt^2 & y_2 &= \frac{-1}{2}gt^2 \\ x_1 &= -v_1 t & x_2 &= v_2 t \end{aligned}$$

mutually perpendicular: $\mathbf{v}_1 \cdot \mathbf{v}_2 = 0 = -v_1 v_2 + g^2 t^2$ or $t = \frac{\sqrt{v_1 v_2}}{g}$

$$\|\mathbf{x}_2 - \mathbf{x}_1\|^2 = ((v_2 + v_1)t)^2 + 0 = (v_2 + v_1)^2 \frac{v_1 v_2}{g^2} \text{ or}$$

$$\boxed{\|\mathbf{x}_2 - \mathbf{x}_1\| = \frac{(v_1 + v_2)}{g}\sqrt{v_1 v_2}} = \frac{7}{9.8\, m/s^2} 2\sqrt{3} = 2.5\, m$$

Problem 1.12. Three points are located at the vertices of an equilateral triangle whose side equals a. They all start moving simultaneously with velocity v constant in modulus, with the first point heading continually for the second, the second for the third, and the third for the first. How soon will the points converge?
Solution 1.12. By symmetry at any time (point), consider component of $\mathbf{v}$ towards triangle center (EY: this is the key insight; breakup components of $\mathbf{v}$ to be along "radial" distance traveled, not just along Cartesian x and y).

By symmetry of the problem, v will always be along an equilateral triangle side, for each moving point.

Let $\theta_0 = 30°$.

Then

$$\cos\theta_0 = \frac{v_r}{v} \text{ or } v\cos\theta_0 = v_r$$

$$\cos\theta_0 = \frac{a/2}{d} \text{ or } d = \frac{a/2}{\cos\theta_0} = \frac{a/2}{\sqrt{3}/2} = \frac{a}{\sqrt{3}}$$

where d is the distance from a point to the center of the triangle.

$$v_x = \frac{ds}{dt} \Longrightarrow T = \int dt = \int \frac{ds}{v_x} = \int_{a/\sqrt{3}}^{0} \frac{ds}{-v\cos\theta_0} = \frac{-a}{\sqrt{3}} \frac{1}{v\sqrt{3}/2} = \frac{2a}{3v}$$

Problem 1.13. Point A moves uniformly with velocity v so that the vector $\mathbf{v}$ is continually "aimed" at point B which in its turn moves rectilinearly and uniformly with velocity $u < v$. At the initial moment of time $\mathbf{v} \perp \mathbf{u}$ and the points are separated by a distance l. How soon will the points converge?
Solution 1.13. Choose $\mathbf{u}$ to be in the $x+$ direction: $\mathbf{u} = u\mathbf{e}_x$.

Condition: Point A reaches point B, so total x-distance traveled during time interval T is the same:

$$\int_0^T v\cos\alpha dt = uT \tag{398}$$

Now consider the "straight line distance" between point A and point B. We're given at the initial moment of time $\mathbf{v} \perp \mathbf{u}$, that the points are separated by distance l.

Now think in terms of displacement in very short time interval Δt. Point A is traveling distance $v\Delta t$ towards point B in a straightline. But point B is moving away with component along the straight line by distance $u\cos\alpha\Delta t$.

The total straight line distance point A has to cover is distance l. The net distance traveled in Δt, $v\Delta t - u\cos\alpha\Delta t$ must sum up to l in order for point A and point B to converge. For a small time interval Δt, the $-u\cos\alpha$ term already is accounting for how much point B is moving away and v term accounts for how fast point A is traveling the distance towards point B.

So

$$\int_0^T (v - u\cos\alpha)dt = l = vT - u\left(\frac{uT}{v}\right) \text{ or } \boxed{T = \frac{lv}{v^2 - u^2}} \tag{399}$$

https://physics.stackexchange.com/questions/568531/why-is-the-following-equation-not-correct https://www.physicsforums.com/threads/a-question-of-kinematics-involving-a-changing-velocity-vector.975126/post-6210692

106 Fundamental Theorems of (Classical) Dynamics

cf. Sec. 3.4 - *Fundamental Theorems of Dynamics* of Prástaro (1996) [86].
cf. Thm. 3.20 of Prástaro (1996) [86]

Theorem 73 (Momentum Theorem)**.** *Variation of the free part of momentum of the observed motion of 1 body, in time interval* $\Delta t \equiv [0,t]$ *is equal to the corresponding impulse:*

$$I[0,t] \equiv \int_0^t F dt \equiv \left(\int_0^t F^j dt \right) \mathbf{e}_j \tag{400}$$

where $\{\mathbf{e}_j\}_{1 \leq j \leq 3}$ *is a fixed basis of* $\mathbf{S}$

$$\overline{p}_\psi(t) - \overline{p}_\psi(0) = I[0,t] \tag{401}$$

Proof.

$$\overline{p}_\psi = \mu \ddot{m}_\psi = \overline{f}_\psi \Longrightarrow \dot{\overline{p}}_\psi = \dot{p}^j \mathbf{e}_j = F^j \mathbf{e}_j \Longrightarrow \int_{[0,t]} \dot{p}^j dt = \int_{[0,t]} F^j dt$$

□

106.1 Coordinate transformation

For (U,φ), (V,ψ) charts on M, s.t. $\varphi : U \to \mathbb{R}^n$, $\quad \varphi^i \equiv x^i$, then consider
$\psi : V \to \mathbb{R}^n \quad \psi^j \equiv y^j$

$$\psi(p) = y = \psi \circ \varphi^{-1} \circ \varphi(p) = \psi \circ \varphi^{-1}(x) = y(x)$$

Consider $f : M \to M$ that is a coordinate transformation (bijective). Under this coordinate transformation, transformation of the components of the same vector $X = Y$ is the push-forward:

$$Y^j = X^i \frac{\partial y^j}{\partial x^i} \tag{402}$$

Correspondingly, basis transformation described by

$$\frac{\partial}{\partial x^j} = \frac{\partial y^k}{\partial x^j} \frac{\partial}{\partial y^k} \text{ or } \frac{\partial}{\partial y^k} = \left[\left(\frac{\partial y}{\partial x} \right)^{-1} \right]_k^j \frac{\partial}{\partial x^j} = \left(\frac{\partial x^j}{\partial y^k} \right) \frac{\partial}{\partial x^j} \tag{403}$$

Then we can show that indeed $Y = X$ under coordinate transformations:

$$Y = X = Y^k \frac{\partial}{\partial y^k} = X^j \frac{\partial}{\partial x^j} \tag{404}$$

Also, for $a_i = b_j \frac{\partial y^j}{\partial x^i}$, the pull back describing how 1-form components transform, consider the inner product:

$$b(Y) = b_i dy^i \left(Y^k \frac{\partial}{\partial y^k} \right) = b_i Y^i = b_i X^k \frac{\partial y^i}{\partial x^k} = b_i \frac{\partial y^i}{\partial x^k} X^k = a_k X^k = a(X) \tag{405}$$

$\Longrightarrow$ inner product preserved under coordinate transformation, i.e. invariance of inner product.

106.2 Transformation of reference frames

cf. Sec. 4.2.1 "Transformation of reference frames", Kambe (2009) [95].

Let us take pt. $\mathbf{y}_b$ fixed to body,
located at $\mathbf{x} = X^1\mathbf{e}_1 + X^2\mathbf{e}_2 + X^3\mathbf{e}_3 = X^i\mathbf{e}_i$ at $t = 0$,
where $\mathbf{e}_i$ $(i = 1, 2, 3)$ are orthonormal basis fixed to space $F \equiv$ nonrotating **fixed** space (inertial space)

By transformation matrix $A = (A^i{}_j) \in \mathrm{SO}(3)$, initial pt. $\mathbf{x} = (X^i)$ mapped to current pt. $\mathbf{y}_b$ at time t.

$$\mathbf{y}_b(t) = A(t)\mathbf{x} = y^i(t)\mathbf{e}_i, \quad y^i(t) = A^i{}_j(t)X^j \tag{406}$$

cf. Eq. (4.14) of Kambe (2009) [95].

i.e.

$$\mathbf{y}_b(t) = A(t)\mathbf{x} = A^i{}_k e_i \otimes e^k x^l e_l = A^i{}_k X^k e_i = y^i(t)e_i$$

EY: Observe that $A(t)$, $t \in [0, t]$ is a curve in SO(3).

In other words, we considered a point in the fixed frame F given by frame $\{\mathbf{e}_i\}$:

$$\mathbf{x} = X^i\mathbf{e}_i \text{ at } t = 0$$

Then, we considered an "active" rotation $A \in SO(3)$ of this point, so

$$\mathbf{y}(t) = A(t)\mathbf{x} = A^i{}_j(t)X^j\mathbf{e}_i = y^i(t)\mathbf{e}_i \tag{407}$$

Indeed, consider $A = \begin{bmatrix} c & -s & \\ s & c & \\ & & 1 \end{bmatrix}$. Then A would rotate points in the $x - y$ plane, such as $[1, 0, 0]$, $[0, 1, 0]$ counter-clockwise in the positive z axis direction.

On the other hand, relative to body frame F_B, which is the frame instantaneously fixed to the moving body. For the same pt., $\mathbf{y}_b(t)$, fixed to body, is

$$\mathbf{Y}(= \mathbf{y}_b) = Y^1\mathbf{b}_1 + Y^2\mathbf{b}_2 + Y^3\mathbf{b}_3 = Y^i\mathbf{b}_i$$

where $\mathbf{b}_i$ $(i = 1, 2, 3)$ are orthonormal basis fixed to body which coincided with $\mathbf{e}_i$ $(i = 1, 2, 3)$ at $t = 0$.

From property of a rigid body, it's required that $Y^i = X^i$ which don't change with t.

According to Sec. 1.5.1(b) (Kambe (2009) [95]), basis transformation written as (1.40) Kambe(2009) [95], where

$$\begin{gathered} \mathbf{b}_i = \mathbf{b}_i(t) = \mathbf{e}_j B^j{}_i(t) \\ \text{Then } \mathbf{Y} = Y^i\mathbf{b}_i = Y^i\mathbf{e}_j B^j{}_i = \mathbf{e}_j B^j{}_i Y^i \end{gathered} \tag{408}$$

(4.10) Kambe (2009) [95], where $B = (B^j_k) \in \mathrm{SO}(3)$, i.e. $BB^T = 1$.

Note that

$$\mathbf{Y} = Y^i\mathbf{b}_i = \mathbf{e}_j B^j{}_i Y^i = A^j{}_k X^k \mathbf{e}_j \tag{409}$$

implies that $B : F' \to F$ where F' rotating frame, F fixed frame, i.e. B is a coordinate transformation from coordinates in the F' rotating frame into coordinates in the F fixed frame.

Indeed, consider

$$B = \begin{bmatrix} c & -s & \\ s & c & \\ & & 1 \end{bmatrix}. \text{ Then } B \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = [c, s, 0] (\text{in } \{\mathbf{e}_j\} \text{ frame})$$

Now recall that $\mathbf{x} = X^i\mathbf{e}_i$ at $t = 0$. $\mathbf{Y} = Y^i\mathbf{b}_i$ and $\{\mathbf{b}_i\}$ frame axes had coincided with $\{\mathbf{e}_i\}$ frame axes at $t = 0$. Y^i's are constant, so $Y^i = X^i$

So for

$$\mathbf{Y} = Y^i\mathbf{b}_i = \mathbf{e}_j B^j{}_i Y^i = A^j{}_k(t)X^k\mathbf{e}_j \Longrightarrow B^j{}_i Y^i = A^j{}_k(t)X^k = A^j{}_i Y^i \text{ so } \boxed{B = A} \text{ since } X^i \text{ arbitrary} \tag{410}$$

For the basis transformation derivation, recall the push forward transformation (cf. Kambe (2009) [95]),

$$\phi : M \to N$$
$$\phi_* : T_x M \to T_y N$$
$$y = \phi(x) = F(x)$$
$$Y = \phi_* X = F_* X = \frac{d}{dt}(F(p(t)))\Big|_{t=0}$$
$$Y^k = (\phi_* X)^k = \left(\frac{\partial F^k}{\partial x^j}\right) X^j$$
$$Y = \phi_* X = \phi_* \left[X^j \frac{\partial}{\partial x^j}\right] = X^j \phi_* \left[\frac{\partial}{\partial x^j}\right] = X^j \frac{\partial y^k}{\partial x^j}\frac{\partial}{\partial y^k} = Y^k \frac{\partial}{\partial y^k}$$
$$Y^k = \frac{\partial y^k}{\partial x^j} X^j$$

106.3 Euler Angles

cf. pp. 150, 4.4 "Euler Angles", Goldstein, Poole, Safko (2001) [98]

rotation 1: about vertical axis, gives **heading** or **yaw** angle

rotation 2: around perpendicular axis fixed in vehicle and normal to figure axis; it's measured by the *pitch* or *attitude* angle

rotation 3: about figure axis of vehicle, *roll* or *bank* angle.

$\Longrightarrow$ xyz - convention, or *Tait-Bryan* angles.

Solution Exercise 24, Ch. 4 Goldstein, Poole, Safko (2001) [98].

Recall that

$$\mathbf{a}_r = \mathbf{a}_i - 2\omega \times \mathbf{v}_r - \omega \times (\omega \times \mathbf{r}) - \dot{\omega} \times \mathbf{r}$$

Let $\dot{\omega} = 0$.

Then consider, multiplying by m,

$$m\mathbf{a}_r = m\mathbf{a}_i - 2m\omega \times \mathbf{v}_r - m\omega \times (\omega \times \mathbf{r}) \text{ or}$$
$$\mathbf{F}_r = \mathbf{F}_i - 2m\omega \times \mathbf{v}_r - m\omega \times (\omega \times \mathbf{r})$$

where $\mathbf{F}_r$ is the apparent force experienced by the bug in the rotating frame, and $\mathbf{F}_i$ are the forces in the inertial reference frame.

By the geometry of the wheel and the problem, and taking the instantaneous coordinate axis $\mathbf{e}_r$ to be same for the rotating and inertial frames,

$$-\omega \times (\omega \times \mathbf{r}) = \omega^2 R \mathbf{e}_r$$
$$-2\omega \times \mathbf{v}_r = 2\omega v_r(- + \mathbf{e}_y)$$

where the direction of the last expression for the Coriolis force is negative when the apparent velocity is in the positive radial direction and vice versa.

Suppose that the frictional force acts in the opposite direction of the force. In the rotating frame, consider the sum of the 2 fictitious forces, the Coriolis force and centripetal force, which happen to be orthogonal to each other:

$$\|\mathbf{F}_{\text{fic}}\| = \|(-2\omega \times \mathbf{v}_r + \omega \times (\omega \times \mathbf{r}))m\| = m\sqrt{(2\omega v_r)^2 + (\omega^2 R)^2} = m\omega\sqrt{(2v_r)^2 + (\omega R)^2}$$

When $F_{\text{fric}} \geq F_{\text{fic}}$, then

$$(\mu g)^2 \geq \omega^2(4v_r^2 + \omega^2 R^2) \text{ or } \left(\frac{\mu g}{\omega}\right)^2 \geq 4v_r^2 + \omega^2 R^2 \text{ or } \omega^2 R^2 \leq \left(\frac{\mu g}{\omega}\right)^2 - 4v_r^2$$

With $v_r = 0.5\,\text{cm}/s^2$ while $g = 9.8\,m/s^2$, the Coriolis force is negligible compared to max. frictional force.

106.4 Right Invariance, Left Invariance, Angular Velocity

For an infinitesimal time increment δt, motion of the body from position g_t is described by the left translation $g_{t+\delta t} = g_{\delta t} g_t$.

This is interpreted as an infinitesimal rotation $\delta t \overline{\Omega}$ at g_t, matrix $\overline{\Omega}$ defined by

$$\begin{aligned} A(t+\delta t) - A(t) = A(\delta t)A(t) - A(t) = (A(\delta t) - 1)A(t) = \delta t \overline{\Omega} A(t) \\ A(t+\delta t)^i{}_k = (A(\delta t)A(t))^i{}_k = A(\delta t)^i{}_l A(t)^l{}_k \end{aligned} \tag{411}$$

cf. 4.16 Kambe (2009) [95]

$\overline{\Omega} = \frac{A(\delta t) - 1}{\delta t}$ is the tangent vector at 1.

Recall that, in general, mathematically,

$$\dot{g}_t = X \circ g_t, \text{ where } X := \lim_{\Delta t \to 0} \frac{(g_{\Delta t} - 1)}{\Delta t}$$

$$\dot{g}_t := \frac{dg_t}{dt} = \frac{dA}{dt} = \overline{\Omega} g_t \tag{412}$$

so

$$\overline{\Omega} = \dot{g}_t \circ (g_t)^{-1} \tag{413}$$

In my notation, for $O \in SO(N)$,

$$\widehat{\omega} = \dot{O} O^T \tag{414}$$

106.4.1 2 Physical Interpretations of rotations: frame transformation and "active" rotation

Consider $O \in SO(N)$, $N = 3$.

$$O : \mathbb{R}^N \to \mathbb{R}^N$$

O can be interpreted as either

O : rotating body frame $\to$ lab frame, i.e.

$O : K \to K'$ (notation used in Hand and Finch (1998) [97] where K is the body frame, K' is the fixed frame)

$O : \diagdown \mapsto \mathbf{r}'$

or O interpreted as "active" rotation of vector in $\mathbb{R}^N$.

Consider this example:

$$\begin{aligned} O = \begin{bmatrix} \cos(\omega t) & -\sin(\omega t) & 0 \\ \sin(\omega t) & \cos(\omega t) & \\ & & 1 \end{bmatrix} \qquad \dot{O} = \omega \begin{bmatrix} -s & -c & \\ c & -s & \\ & & \end{bmatrix} \\ \dot{O} O^T = \omega \begin{bmatrix} -s & -c & \\ c & -s & \\ & & \end{bmatrix} \begin{bmatrix} c & s & \\ -s & c & \\ & & 1 \end{bmatrix} = \omega \begin{bmatrix} & -1 & \\ 1 & & \\ & & \end{bmatrix} = \widehat{\omega} \end{aligned} \tag{415}$$

s.t. $\omega = \omega \mathbf{e}_z$.

$\dot{O} O^T$ is the angular velocity of the rotating frame, or angular velocity of how much the vector in $\mathbb{R}^N$ is rotated by.

106.4.2 Instantaneous Angular Velocity

Given $\mathbf{r} = (r^1, r^2, r^3) \in M$, an infinitesimal transformation of $\mathbf{r}$ by $\xi(t) = 1 + t\mathbf{a} + O(t^2)$.

$$\mathbf{r} \mapsto \mathbf{r}' = \xi(t)\mathbf{r} = \mathbf{r} + t\mathbf{a}\mathbf{r}$$

$$\mathbf{a}\mathbf{r} = \begin{bmatrix} & -a^3 & a^2 \\ a^3 & & -a^1 \\ -a^2 & a^1 & \end{bmatrix} \begin{bmatrix} r^1 \\ r^2 \\ r^3 \end{bmatrix} = \begin{bmatrix} -a^3 r^2 + a^2 r^3 \\ a^3 r^1 - a^1 r^3 \\ -a^2 r^1 + a^1 r^2 \end{bmatrix} = (\widehat{\mathbf{a}} \times \mathbf{r}) = \epsilon_{ijk} a^j r^k \mathbf{e}_i$$

So multiplication by a skew symmetric matrix is equivalent to taking the cross product with the axial vector representation.

Now $\overline{\Omega}$ shown to be skew-symmetric (from a Taylor series expansion). See either Appendix C.3 of Kambe (2009) [95] or Eq. 213.

Make the following notation change from Kambe's (2009) [95]:

$$\widehat{\Omega} \equiv \omega \tag{416}$$

where ω is an axial vector of the angular velocity of body rotation relative to the *fixed* reference frame F (because in body frame, F_B, there's no rotation!).

Solution Question 9: U Matrix, pp. 265, Ch. 7 "Rotating Coordinate Systems", Hand and Finch (1998) [97].

Find the form of $\mathbf{U}$ for a fixed rotation θ about the X axis:

$$O = \begin{bmatrix} c & -s \\ s & c \end{bmatrix}$$

Then

$$\dot{O} = \omega \begin{bmatrix} -s & -c \\ c & -s \end{bmatrix}$$

$$\dot{O}O^T = \omega \begin{bmatrix} -s & -c \\ c & -s \end{bmatrix} \begin{bmatrix} c & s \\ -s & c \end{bmatrix} = \omega \begin{bmatrix} & -1 \\ 1 & \end{bmatrix}$$

Thus

$$\dot{O}O^T = \widehat{\omega} \mapsto \omega = \omega \mathbf{e}_x$$

So we get the form for ω. Now consider

$$\widehat{\omega}_b = O^T \widehat{\omega} O = \begin{bmatrix} c & s \\ -s & c \end{bmatrix} \omega \begin{bmatrix} & -1 \\ 1 & \end{bmatrix} O = \omega \begin{bmatrix} s & -c \\ c & s \end{bmatrix} \begin{bmatrix} c & s \\ s & c \end{bmatrix} = \omega \begin{bmatrix} & -1 \\ 1 & \end{bmatrix}$$

$\widehat{\omega}_b$ is $\widehat{\omega}$ expressed in body coordinates. $\widehat{\omega} = \widehat{\omega}_b$ in this particular case because $\widehat{\omega}_b$ is the angular velocity measured in the inertial frame, but expressed in the body frame coordinates.

See https://physics.stackexchange.com/questions/88398/why-is-body-frame-angular-velocity-nonzero/88401#88401

For g_t operating on $\mathbf{x}$, let's follow Kambe's development in pp. 133-134, Eqns. (4.17)-(4.19) (Kambe (2009) [95]). So then

$$g_t \mathbf{x} = \mathbf{y}(t), \quad g_{t+\delta t}\mathbf{x} = \mathbf{y}(t+\delta t)$$

Then

$$\mathbf{y}(t+\delta t) = \mathbf{y}(t) + \frac{d\mathbf{y}}{dt}(t)\delta t + O((\delta t)^2) = \mathbf{y}(t) + (\dot{g}_t \mathbf{x})\delta t + \cdots = \mathbf{y}(t) + (\overline{\Omega} g_t \mathbf{x})\delta t + \cdots =$$
$$= g_t \mathbf{x} + (\omega \times (g_t \mathbf{x}))\delta t$$

Let

$$\mathbf{v}_y = \frac{d\mathbf{y}}{dt} = \lim_{\delta t \to 0} \frac{\mathbf{y}(t+\delta t) - \mathbf{y}(t)}{\delta t} = \omega \times (g_t \mathbf{x}) = \omega \times \mathbf{y}$$

This means that point $\mathbf{y}(t)$ is moving with velocity $\mathbf{v}_\mathbf{y} = \omega \times \mathbf{y}$, i.e. ω *angular velocity* in fixed space.

But typically, $\mathbf{x} = \mathbf{x}(t)$, i.e. the position on the body is time dependent, it just doesn't "stay on the body".

Recall that $\mathbf{x} = \mathbf{x}(t)$ are coordinates in body frame F_B. $\mathbf{y}(t)$ are coordinates in fixed reference frame F.

$$\mathbf{y}(t) = g_t \mathbf{x}(t)$$

Compare this notation to the one in Hand and Finch (1998) [97] in Eq. (7.18):

$$\mathbf{r}' = \widetilde{U}\mathbf{r} \equiv U\mathbf{r}$$

where the rotated or body frame is K, and fixed, inertial reference frame K', and I modify the notation for U to be an element in $SO(3)$.

So instead of how Kambe assumed above that the position is fixed in the body frame, assume that it's also time-dependent in the rotating frame. So use chain rule:

$$\frac{d\mathbf{y}}{dt}(t) = \frac{d}{dt}(g_t\mathbf{x}) = \dot{g}_t\mathbf{x} + g_t\dot{\mathbf{x}}(t) = \dot{g}_t\mathbf{x} + g_t\mathbf{v}_\mathbf{x} = \overline{\Omega}g_t\mathbf{x} + g_t\mathbf{v}_\mathbf{x}$$

$$\implies \mathbf{v}_y = \omega \times \mathbf{y} + g_t\mathbf{v}_\mathbf{x} \tag{417}$$

Follow the development in Hand and Finch (1998) [97]:

$$\begin{aligned}\mathbf{v}|'_{\text{space}} &= \dot{U}\mathbf{r} + U\dot{\mathbf{r}}|_{\text{body}} = \dot{U}\mathbf{r} + U\mathbf{v}|_{\text{body}} = \dot{U}U^{-1}U\mathbf{r} + U\mathbf{v}|_{\text{body}} = \dot{U}U^{-1}\mathbf{r}' + U\ \mathbf{v}|_{\text{body}} = \\ &= \omega \times \mathbf{r} + \mathbf{v}|_{\text{body}}\end{aligned} \tag{418}$$

What we wrote is essentially a summary of Eqns. (7.28), (7.29) Hand and Finch (1998) [97].

Compare this, namely Eq. 417, with wikipedia ("Rotating reference frame"), where the velocities in the 2 reference frames are related by:

$$\mathbf{v}_i = \mathbf{v}_r + \boldsymbol{\Omega} \times \mathbf{r} \tag{419}$$

where $i \equiv$ inertial frame of reference, $r \equiv$ rotating reference frame.

Now

$$\begin{aligned}\mathbf{v}_y &= \omega \times \mathbf{y} + g_t\mathbf{v}_\mathbf{x} \\ \mathbf{a}_y &= \frac{d}{dt}\mathbf{v}_\mathbf{y} \\ \frac{d}{dt}\omega \times \mathbf{y} &= \frac{d}{dt}\epsilon_{ijk}\omega_j y_k = \epsilon_{ijk}\dot{\omega}_j y_k + \epsilon_{ijk}\omega_j \dot{y}_k \\ \text{For } \dot{y}_k, \text{ consider } \frac{d\mathbf{y}}{dt} &= \omega \times \mathbf{y} + g_t\mathbf{v}_\mathbf{x} \\ \text{so } \omega \times \frac{d\mathbf{y}}{dt} &= \omega \times (\omega \times \mathbf{y}) + \omega \times g_t\mathbf{v}_\mathbf{x} \\ \implies \frac{d}{dt}(\omega \times \mathbf{y}) &= \dot{\omega} \times \mathbf{y} + \omega \times (\omega \times \mathbf{y}) + \omega \times g_t\mathbf{v}_\mathbf{x} \\ \frac{d}{dt}(g_t\mathbf{v}_\mathbf{x}) &= \dot{g}_t\mathbf{v}_\mathbf{x} + g_t\mathbf{a}_\mathbf{x} = \omega \times g_t\mathbf{v}_\mathbf{x} + g_t\mathbf{a}_\mathbf{x} \\ \implies \mathbf{a}_y &= \omega \times (\omega \times \mathbf{y}) + 2\omega \times g_t\mathbf{v}_\mathbf{x} + \dot{\omega} \times \mathbf{y} + g_t\mathbf{a}_\mathbf{x}\end{aligned} \tag{420}$$

Compare this to Hand and Finch (1998) [97], Eq. 7.32:

$$\mathbf{a}|_{\text{space}} = \omega \times (\omega \times \mathbf{r}) + 2\omega \times \mathbf{v}|_{\text{body}} + \dot{\omega} \times \mathbf{r} + \mathbf{a}|_{\text{body}} \tag{421}$$

cf. pp. 134 Kambe (2009) [95]

Now compare this, namely Eq. 420, with wikipedia ("Rotating reference frame"), where the accelerations in the 2 reference frames are related by:

$$\mathbf{a}_r = \mathbf{a}_i - 2\boldsymbol{\Omega} \times \mathbf{v}_r - \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}) - \frac{d\boldsymbol{\Omega}}{dt} \times \mathbf{r} \tag{422}$$

where $\mathbf{a}_r := \left(\frac{d^2\mathbf{r}}{dt^2}\right)_r$ is the apparent acceleration in the rotating reference frame.

Solution Ch. 7, Angular Velocity, Problem 1, (Locomotive) Hand and Finch (1998) [97].

(a)

(b)

(c)

$$\widehat{\omega} = \dot{O}O^T$$
$$O = O_2O_1 \text{ and } O^T = O_1^T O_2^T$$
$$\dot{O} = \dot{O}_2O_1 + O_2\dot{O}_1$$
$$\dot{O}O^T = \dot{O}_2O_2^T + O_2\dot{O}_1O_1^TO_2^T = \widehat{\omega}_2 + O_2\widehat{\omega}_1O_2^T = \widehat{\omega}$$

Corresponding to right-invariant $\dot{g}_t = \omega g_t$ of the tangent vector, the angular momentum is written in the right-invariant way as

$$M_t = J\dot{g}_t = \overline{M}g_t, \quad \overline{M} = J\omega \tag{423}$$

where J is an inertia operator and time dependent in fixed space. cf. Eqn. (4.20), pp. 134, Kambe (2009) [95].

Make the following notation change:

$$L_t = I\dot{g}_t = \overline{L}g_t, \quad \overline{L} = I\widehat{\omega} \tag{424}$$

where I is an inertia operator.

In mechanics, angular momentum is defined by

$$M_t = \int (\mathbf{y} \times \mathbf{v}_\mathbf{y})\rho d^3\mathbf{y} = \int (g_t\mathbf{x} \times (\widehat{\omega}g_t\mathbf{x}))g_t(\rho d^3\mathbf{x})$$

and so $M_t := \overline{M}g_t$.

Relative to body frame F_B, the same velocity (relative to fixed space) is represented as $\frac{d\mathbf{y}_b}{dt} = v_b^i\mathbf{b}_i$, where $\mathbf{v}_b = \widehat{\Omega}_b \times Y$.

$\widehat{\Omega}_b$ is angular velocity relative to body frame. Now $\overline{\Omega}_b$ derived by *left translation* of tangent vector.

$$\overline{\Omega}_b = g_t^{-1} \circ \dot{g}_t = L_{g_t^{-1}}\dot{g}_t = g_t^{-1}\overline{\Omega}g_t \tag{425}$$

as a change of notation, I will also write

$$\omega_b = g_t^{-1}\omega g_t$$

An important point is that tangent vector $\dot{g}_t$ is represented by **left-invariant** form

$$\widehat{\omega}_b = g_t^{-1}\dot{g}_t \xrightarrow{L_{g_t}} L_{g_t}\widehat{\omega}_b = g_t\widehat{\omega}_b = \dot{g}_t \text{ or } \dot{g}_t = g_t\widehat{\omega}_b$$

by ω_b at 1, whereas it's also represented by right-invariant form $\dot{g}_t = \widehat{\omega}g_t$ with $\widehat{\omega}$ at 1 in Eqn. (4.17) of Kambe (2009) [95], where, recall $\dot{g}_t = \frac{dg}{dt} = \frac{dA}{dt}$.

Indeed, note that for

$$\mathbf{b}_i(t) = \mathbf{e}_jB^j{}_i(t) \text{ then}$$
$$\mathbf{b}_i(t+\delta t) = \mathbf{e}_jB^j{}_i(t+\delta t) = \mathbf{e}_jB^j{}_k(t)B^k{}_i(\delta t)$$

where the last step is because since $\mathbf{e}_j$ on the left forces an infinitesimal change to be from the right.

Then

$$B(t+\delta t) - B(t) = B(t)B(\delta t) - B(t) = B(t)(B(\delta t) - 1) \text{ so } \dot{B}(t) = B(t)Y_1$$

where $Y_1 \in \mathfrak{so}(N)$ and $Y_1 := \lim_{\delta t \to 0}\frac{B(\delta t)-1}{\delta t}$.

Also, recall that

$$\mathbf{Y} = Y^i\mathbf{b}_i = \mathbf{e}_jB^j{}_iY^i = X^j\mathbf{e}_j \text{ implies that } B : F' \to F$$

where F' is a rotating frame, F is a fixed frame, i.e. B is a coordinate frame transformation.

Due to right-translation property of $B(t)$, then $\frac{dB}{dt} = B(t)\widehat{\Omega}_b$. Thus, we obtain a left-invariant vector field.

From Eq. 410, $A = B$ but

$$\begin{aligned} \frac{dA}{dt} &= \widehat{\Omega}A \\ \frac{dA}{dt} = \frac{dB}{dt} &= B(t)\widehat{\Omega}_b = \widehat{\Omega}A(t) \text{ or } \widehat{\Omega}_b = A^{-1}\widehat{\Omega}A \end{aligned} \tag{426}$$

107 Rigid Body Dynamics, Rigid Body Motion

107.1 Moment of Inertia

Solution Hand and Finch (1998) [97], Problem 3, pp. 327. Hand and Finch (1998) [97], Problem 3: (Moment of inertia tensor) from "Moment of Inertia Tensor", pp. 327. Which of the symmetric 3x3 matrices below could represent a physical moment of inertia tensor?

$$I_1 = \begin{pmatrix} 1 & 2 & 1 \\ \dots & 0 & 2 \\ \dots & \dots & 1 \end{pmatrix}, \quad I_2 = \begin{pmatrix} 1.94791 & .0347273 & -.394509 \\ \dots & 2.42924 & -.823746 \\ \dots & \dots & 1.62285 \end{pmatrix} \tag{427}$$

See Hand and Finch (1998) [97], (8.131)

Recall that

$$I_{ij} = \sum m(r_l^2 \delta_{ij} - x_i x_j)$$
$$I_{ii} = \sum m(r_j^2 + r_i^2)$$

I_1 is not physical since

$$\sum m(x^2 + z^2) = 0$$
$$\implies \sum mx^2 = -\sum mz^2$$

but both sums on the left and right-hand sides are positive.

107.1.1 Moments of Inertia of Some Simple Objects

(a) **Uniform thin ring of mass M and radius R, around the axis of symmetry of the ring**

See Kleppner and Kolenkow (2014) [96], pp. 247, "Angular Momentum and Fixed Axis Rotation, Example 7.3 Moments of Inertia of Some Simple Objects; Hand and Finch (1998) [97] for Problem 4: (I for a circular hoop), pp. 327, Ch. 8 "The Dynamics of Rigid Bodies".

Get the mass density λ of a thin ring or i.e. circular hoop.

$$M = \int \rho(\mathbf{x}) d\mathbf{x} \equiv \int \rho(\mathbf{s}) d\mathbf{s} = \int_0^{2\pi} \rho(\theta) R d\theta = R\lambda 2\pi \implies \boxed{\lambda = \frac{M}{2\pi R} = \rho(s)}$$

where $\lambda =$ mass density of a thin ring of total mass M, radius R and where we can say λ is constant because it's a *uniform* thin ring.

$$I = \int r^2 dm = \int_0^{2\pi} R^2 \lambda R d\theta = MR^2 \tag{428}$$

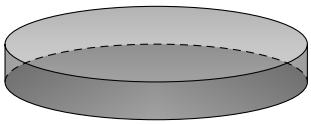
Now from Hand and Finch (1998) [97], Problem 4, pp. 327, for the principal axes and moments, recall that

$$I_3 = \int (x^2 + y^2) dm$$
$$I_2 = \int (x^2 + z^2) dm = \int x^2 dm$$
$$I_1 = \int (y^2 + z^2) dm = \int y^2 dm$$

and that

$$I_3 = \int R^2 dm = M^2 \qquad \implies I_3 = I_1 + I_2 \text{ so that by symmetry } I_1 = I_2 = \frac{MR^2}{2}$$

(b) **Uniform thin disk of mass M and radius R, around the axis of symmetry of the disk**



Using the fact that the disk is uniform, so mass density σ is uniform

$$M = \int \rho(\mathbf{x}) d\mathbf{x} = \int \rho(\mathbf{s}) d\mathbf{s} = \sigma \int r d\theta dr = \sigma \frac{1}{2} R^2 2\pi = \sigma \pi R^2$$

$$\sigma = \frac{M}{\pi R^2} = \text{ mass per unit area}$$

$$I = \int r^2 dm = \int r^2 \sigma r d\theta dr = \sigma \frac{1}{4} r^4 2\pi = \boxed{\frac{MR^2}{2}} \tag{429}$$

107.2 Euler's equations of motion, Euler's rotation equations

In the body frame, inertia tensor $J \equiv I$ is time-independent.

$$\frac{d}{dt}\widehat{\Omega} - \mathrm{ad}^*_{\widehat{\Omega}}\widehat{\Omega} = 0$$

Use (Eq. (4.36) Kambe (2009) [95]

108 Waves

108.1 Signals, Fourier Analysis

See Ch. 10 "Signals and Fourier Anaylsis" of Georgi (1992) [105].

108.1.1 Example: A pulse on a String

See 10.1.1 "A Pulse on a String", Sec. 10.1 "Signals in Forced Oscillation" of Georgi (1992) [105].

Consider transvere oscillation of a semiinfinite string stretched from $x = 0$ to $x = \infty$, driven at $x = 0$ with some arbitrary transverse signal $f(t)$, with a boundary condition at ∞ that there's no incoming traveling waves.

If given dispersion relation (10.1) of Georgi (1992) [105]

$$\omega^2 = v^2 k^2 \tag{430}$$

This implies system that satisfies wave equation

$$\frac{\partial^2}{\partial t^2}\psi(x,t) = v^2 \frac{\partial^2}{\partial x^2}\psi(x,t) \tag{431}$$

((10.2) of Georgi (1992) [105])

Its a mathematical fact that the general solution to Eq. 431

$$\psi(x,t) = g(x - vt) + h(x + vt) \tag{432}$$

where g, h arbitrary. This fact is because of this straightforward derivation:

$$\frac{\partial^2}{\partial t^2}\psi(x,t) = v^2(g''(x - vt) + h''(x + vt))$$
$$\frac{\partial^2}{\partial x^2}\psi(x,t) = g''(x - vt) + h''(x + vt)$$

Boundary condition at ∞ implies $h = 0$, because h function describes wave moving in $-x$ direction. EY: by going to infinity, by physical intuition, we don't expect a reflection back from infinity.

Boundary condition at $x = 0$ (substitute $x = 0$ into g and thus for ψ):

$$g(-vt) = \psi(0, t) = f(t) \tag{433}$$

((10.6) of Georgi (1992) [105])

EY: Consider $\tau = t - \frac{x}{v}$:

$$g(-v\tau) = g(x - vt)$$
$$f(\tau) = f(t - \frac{x}{v})$$

Thus we guess that

$$\psi(x, t) = f(t - \frac{x}{v}) \tag{434}$$

((10.7) of Georgi (1992) [105])

e.g.

$$f(t) = \begin{cases} 1 - |t| & \text{for } |t| \leq 1 \\ 0 & \text{for } |t| > 1 \end{cases} \tag{435}$$

((10.8) of Georgi (1992) [105])

108.1.2 Introduction to Fourier integrals

The idea is to use linearity in a clever way.
We can take $f(t)$ apart into its component angular frequencies.
We already know how to solve forced oscillatio nproblem for each angular frequency.

- Take individual solutions and add them back up.
- The advantage of this is that it works for any dispersion relation, not just for Eq. 430.

$$f(t) = \int_{-\infty}^{\infty} d\omega C(\omega) e^{-i\omega t} \tag{436}$$

((10.9) of Georgi (1992) [105])

Theorem 74 (Properties of Fourier integrals, coefficients)**.** *In Eq. 436*

$$C(\omega) = C(-\omega)^* \tag{437}$$

Proof. If $f(t) \in \mathbb{R}$, $f(t) = f^*(t)$ and so

$$f(t) = \int_{-\infty}^{\infty} d\omega C(\omega) e^{-i\omega t} = f(t)^* = \int_{-\infty}^{\infty} d(\omega) C^*(\omega) e^{i\omega t} = \int_{\infty}^{-\infty} d(-\omega) C(-\omega)^* e^{-i\omega t} = \int_{-\infty}^{\infty} d(\omega) C(-\omega)^* e^{-i\omega t}$$
$$\implies C(\omega) = C(-\omega)^*$$

□

Since each frequency component of the force produces a wave traveling in $+x$ direction, we use linear superposition, and so we have

$$\implies \psi(x, t) = \int_{-\infty}^{\infty} d\omega C(\omega) e^{-i\omega t + ikx}$$

108.1.3 Dispersive Media, Group Velocity

See 10.2.1 "Group Velocity", Sec. 10.2 "Dispersive Media and Group Velocity" of Georgi (1992) [105].

108.1.4 Group Velocity

You can send signals in a dispersive medium; the trick is to send the signal not directly as the function, $f(t)$, but as modulation of harmonic signal,

$$f(t) = f_s(t)\cos\omega_0 t \tag{438}$$

((10.17) of Georgi (1992) [105]), where $f_s(t)$ is the signal. Very often, you want to do this anyway because the important frequencies in your signal,
$f_s(t)$,
may not match frequencies of the waves with which you want to send the signal, ω_0 of $\cos\omega_0 t$.

Example: AM radio transmission, in which signal is derived from sound with a typical frequency of a few hundred cycles per second (Hz), but is carried as a modulation of amplitude of an electromagnetic radio wave, with frequency of a few million cycles per second (ω_0).

Consider

$$\cos(k_+x - \omega_+t) + \cos(k_-x - \omega_-t) \equiv c(k_+x - \omega_+t) + c(k_-x - \omega_-t)$$

Let $k_\pm = k_0 \pm k_s$, $\omega_\pm = \omega_0 \pm \omega_s$ for $k_s \ll k_0$, $\omega_s \ll \omega_0$.

Then

$$c(k_\pm x - \omega_\pm t) = c(k_\pm x)c(\omega_\pm t) + s(k_\pm x)s(\omega_\pm t) \tag{439}$$

and so for each term,

$$c(k_\pm x)c(\omega_\pm t) = c((k_0 \pm k_s)x)c((\omega_0 \pm \omega_s)t) = (c(k_0x)c(k_sx) \mp s(k_0x)s(k_sx))(c(\omega_0t)c(\omega_st) \mp s(\omega_0t)s(\omega_0t)) =$$
$$= c(k_sx)c(\omega_st)c(k_0x)c(\omega_0t) \mp c(k_sx)s(\omega_st)c(k_0x)s(\omega_0t) \mp s(k_sx)c(\omega_st)s(k_0x)c(\omega_0t) + s(k_sx)s(\omega_st)s(k_0x)s(\omega_0t)$$

$$s(k_\pm x)s(\omega_\pm t) = s(k_0 \pm k_s)xs(\omega_0 \pm \omega_s)t = (s(k_0x)c(k_sx) \pm c(k_0x)s(k_sx))(s(\omega_0t)c(\omega_st) \pm c(\omega_0t)s(\omega_st)) =$$
$$= c(k_sx)c(\omega_st)s(k_0x)s(\omega_0t) \pm c(k_sx)s(\omega_st)s(k_0x)c(\omega_0t) \pm s(k_sx)s(\omega_0t)c(k_0x)s(\omega_0t) + s(k_sx)s(\omega_st)c(k_0x)c(\omega_0t)$$

Terms with $\pm$ or $\mp$ in front cancel for when considering the total expression summing 2 terms in Eq. 439, and so

$$\implies 2\cdot[c(k_sx)c(\omega_st)(c(k_0x - \omega_0t)) + s(k_sx)s(\omega_st)c(k_0x - \omega_0t)] = 2c(k_sx - \omega_st)c(k_0x - \omega_0t) \tag{440}$$

((10.21) of Georgi (1992) [105])

Since $k_s \ll k_0$, and $\omega_s \ll \omega_0$, so $c(k_sx - \omega_st)$ varies slowly in x and t compared to second factor. You should think of this first factor as the signal in Eq. 440. The second factor is called the **carrier wave**.

Then Eq. 440 describes a signal that moves with velocity

$$v_s = \frac{\omega_s}{k_s} = \frac{\omega_+ - \omega_-}{k_+ - k_-} \tag{441}$$

((10.22) of Georgi (1992) [105]) while smaller waves associated with second factor, carrier wave, move with velocity

$$v_0 = \frac{\omega_0}{k_0} \tag{442}$$

((10.23) of Georgi (1992) [105])

In limit as $k_+ - k_- = 2k_s$ becomes small, $v_s \equiv \frac{\omega_+ - \omega_-}{k_+ - k_-} \mapsto \left.\frac{\partial\omega}{\partial k}\right|_{k=k_0}$.

$v_g = \left.\frac{\partial\omega}{\partial k}\right|_{k=k_0}$ is called the "group velocity"; measures the speed the signal can actually be sent.

Suppose dispersion relation is slowly varying for some range of frequencies near some frequency ω_0,

$$\omega = \omega(k) = \omega_0 + (k - k_0)\left.\frac{\partial\omega}{\partial k}\right|_{k=k_0} + \dots$$
$$\omega_0 \equiv \omega(k_0)$$

Higher order terms negligible only if $\Delta\omega \ll \omega_0$. If so let

$$\omega = vk + a \qquad \text{since} \qquad a = \omega_0 - k_v$$
$$k = \frac{\omega}{v} + b \qquad\qquad b = -\frac{a}{v} = k_0 - \frac{\omega_0}{v}$$

v is group velocity, $v = \frac{\partial\omega}{\partial k}\big|_{k=k_0}$

Send signal of form $f(t)e^{-i\omega_0 t}$ (modulate the amplitude by $f(t)$ that's non-zero for some range of frequencies, of carrier wave $e^{-i\omega_0 t}$):

$$\psi(0,t) = f(t)e^{-i\omega_0 t} = \int_{-\infty}^{\infty} d\omega C(\omega)e^{-i\omega t}e^{-i\omega_0 t} = \int_{-\infty}^{\infty} d\omega C(\omega)e^{-i(\omega+\omega_0)t} = \int_{-\infty}^{\infty} d\omega C(\omega-\omega_0)e^{-i\omega t}$$

where we used linearity or i.e. linear superposition or i.e. adding up frequency components for second equality, and $u = \omega + \omega_0$ substitution for 3rd equality.

Thus

$$\psi(x,t) = \int_{-\infty}^{\infty} d\omega C(\omega-\omega_0)e^{-i\omega t}e^{ikx}$$

For approximation $k = \frac{\omega}{v} + b$,

$$\psi(x,t) = \int_{-\infty}^{\infty} d\omega C(\omega-\omega_0)e^{-i\omega t}e^{i\left(\frac{\omega x}{v}+bx\right)} = \int_{-\infty}^{\infty} d\omega C(\omega-\omega_0)e^{-i\omega\left(t-\frac{x}{v}\right)}e^{ibx} =$$
$$= \int_{-\infty}^{\infty} d\omega C(\omega)e^{-i(\omega+\omega_0)\left(t-\frac{x}{v}\right)}e^{ibx} = f(t-\frac{x}{v})e^{-i\omega_0(t-\frac{x}{v})+ibx}$$

Thus, for signal $f(t)$ s.t. $C(\omega) \approx 0$ for $|\omega - \omega_0| > \Delta\omega$, where $\Delta\omega \ll \omega_0$,
modulation $f(t)$ travels without change of shape at group velocity $v = \frac{\partial\omega}{\partial k}\big|_{k=k_0}$.
Phase velocity $v_\phi = \frac{\omega}{k}$ has nothing to do with transmission of information, but notice because of $e^{ibx} = \exp\left(i\left(k_0 - \frac{\omega_0}{v}\right)\right)$ carrier wave travels at phase velocity

$$\frac{\omega_0}{b+\frac{\omega_0}{v}} = \frac{\omega_0}{k_0 - \frac{\omega_0}{v} + \frac{\omega_0}{v}} = \frac{\omega_0}{k_0} = v_\phi$$

"You can see the difference between phase velocity and group velocity in your pool or bathtub by making a wave packet consisting of severla shorter waves." pp. 232, Sec. 10.2 of Georgi (1992) [105]).

108.1.5 Bandwidth, Fidelity, Uncertainty

See Sec. 10.3 "Bandwidth, Fidelity, and Uncertainty" of Georgi (1992) [105].

108.1.6 Inverse Fourier Transform

Now suppose we have the inverse Fourier transform to be of this form:

$$C(\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty} dt f(t)e^{i\omega t} \tag{443}$$

Recall $f(t) = \int_{-\infty}^{\infty} d\omega C(\omega)e^{-i\omega t}$ of Eq. 436.
Given $\frac{1}{2\pi}\int_{-\infty}^{\infty} dt f(t)e^{i\omega t}$ of Eq. 443, insert Eq. 436 for $f(t)$:

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} dt \int_{-\infty}^{\infty} d\omega' C(\omega')e^{i(\omega-\omega')t} = \frac{1}{2\pi}\int_{-\infty}^{\infty} d\omega' C(\omega')\int_{-\infty}^{\infty} dt e^{i(\omega-\omega')t}$$

$\int_{-\infty}^{\infty} e^{i(\omega-\omega')t}$ averages to zero unless $\omega' = \omega$.
Therefore $\frac{1}{2\pi}\int_{-\infty}^{\infty} dt f(t)e^{i\omega t}$ is proportional to $C(\omega)$.
Now consider $f(t) = e^{-\Gamma|t|}$.

$$\int_{-\infty}^{\infty} dt e^{-\Gamma|t|}e^{i\omega t} = \int_0^{\infty} dt e^{t(i\omega-\Gamma)} + \int_{-\infty}^{0} dt e^{i(i\omega+\Gamma)} = \frac{e^{t(i\omega-\Gamma)}}{i\omega-\Gamma}\Big|_0^{\infty} + \frac{e^{t(i\omega+\Gamma)}}{i\omega+\Gamma}\Big|_{-\infty}^{0} = \frac{i\omega+\Gamma}{\omega^2+\Gamma^2} + \frac{-i\omega+\Gamma}{\omega^2+\Gamma^2} = \frac{2\Gamma}{\omega^2+\Gamma^2}$$

Note that for $\frac{1}{i\omega-\Gamma}$, $\frac{1}{i\omega+\Gamma}$, if $\omega = \mp i\Gamma$, there's a pole; otherwise $\forall\,\omega \in \mathbb{R}$, integral is well-behaved.
Then for this particular case $C(\omega) = \frac{1}{2\pi}\frac{2\Gamma}{\omega^2+\Gamma^2}$.

For $f(t) = e^{-\Gamma|t|}$, $f(0) = 1$. Then

$$\int_{-\infty}^{\infty} d\omega C(\omega) e^{-i\omega t} = \int_{-\infty}^{\infty} d\omega \frac{1}{\pi} \frac{\Gamma e^{-i\omega t}}{\omega^2 + \Gamma^2} = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{d\omega \Gamma}{\omega^2 + \Gamma^2} = \frac{1}{\pi} \int_{-\pi/2}^{\pi/2} \frac{\Gamma \sec^2 \theta d\theta \Gamma}{\Gamma^2(\tan^2 \theta + 1)} = 1$$

So $\frac{1}{2\pi}$ factor is obtained.

In another way to get $C(\omega)$, the inverse Fourier transform, as the limit of a Fourier waves, consider $\psi(-\pi l, t) = \psi(\pi l, t)$, periodic boundary condition (notice the period is $2\pi l = T$),

$$\xrightarrow{\text{in general}} \psi(x) = \sum_{j=-\infty}^{\infty} c_j e^{-ijx/l}$$

Notice that changing $c_j e^{-ijx/l}$ by $2\pi l$ just changes the phase of the exponential by some multiple of 2π and so each factor indeed obeys the periodic boundary condition.

Likewise, for $f(t)$, s.t. $f(-\pi T) = f(\pi T)$, T large

$$f(t) = \sum_{j=-\infty}^{\infty} c_j e^{-ijt/T} \qquad \text{(Fourier transform in time)}$$
$$c_j = \frac{1}{2\pi T} \int_{-\pi T}^{\pi T} dt e^{ijt/T} f(t)$$

$\lim_{t\to\infty} f(t) = 0$, then $c_j \to 0$ as $T \to \infty$. $\implies \begin{matrix} c_j T \to C(\omega) \\ j/T \to \omega \end{matrix}$, so then

$$f(t) = \sum c_j e^{-ijt/T} = \sum (c_j T) e^{-ijt/T} \frac{1}{T} \to \int_{-\infty}^{\infty} d\omega C(\omega) e^{-i\omega t}$$

So we have these important results:

$$\boxed{f(t) = \int_{-\infty}^{\infty} C(\omega) e^{-i\omega t}} \tag{444}$$

$$\boxed{C(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} dt f(t) e^{i\omega t}} \tag{445}$$

109 Fluid Mechanics, Fluid Flow

109.0.1 Courses to consider

MIT 16.01, 16.02

109.0.2 Videos to watch

Engineering MAE 130A. Intro to Fluid Mechanics. Lecture 01. UCI. Follow the rest of the videos.

109.1 Buoyancy

From Problem 1.3 of the Japan Physics Olympiad (2014) [121], Ch. 1, General Physics, Elementary Problems.
Problem 1.3. The part of the iceberg above the sea.

$$\rho_{sw} = 1024\,\text{kg}/m^3$$
$$\rho_{ice} = 917\,\text{kg}/m^3$$

with $sw \equiv$ sea water.

Find the ratio of the volume of the part of the iceberg above the sea to whole volume of the iceberg.
Solution 1.3.

The buoyancy force exerted on the iceberg is equal to the weight of the seawater displaced by the iceberg.

$$\rho_{sw} g V_{is} = \rho_{ice} g V_{ice} \Longrightarrow \frac{\rho_{sw}}{\rho_{ice}} = \frac{V_{ice}}{V_{is}}$$

with iceberg in sea $\equiv$ is.

iceberg above sea $\equiv$ as

$$V_{as}/V_{ice} = \frac{V_{ice} - V_{is}}{V_{ice}} = 1 - \frac{V_{is}}{V_{ice}} = 1 - \frac{\rho_{ice}}{\rho_{sw}} = 1 - \frac{917\,\mathrm{kg}/m^3}{1024\,\mathrm{kg}/m^3} = \frac{1024 - 917}{1024} = \frac{107}{1024}$$

The buoyancy on a body equals the resultant force due to the pressure exerted by the surrounding fluid, i.e. **The buoyancy force exerted on a body (iceberg) is equal to the resultant force due to pressure exerted by the surrounding fluid (seawater)**.

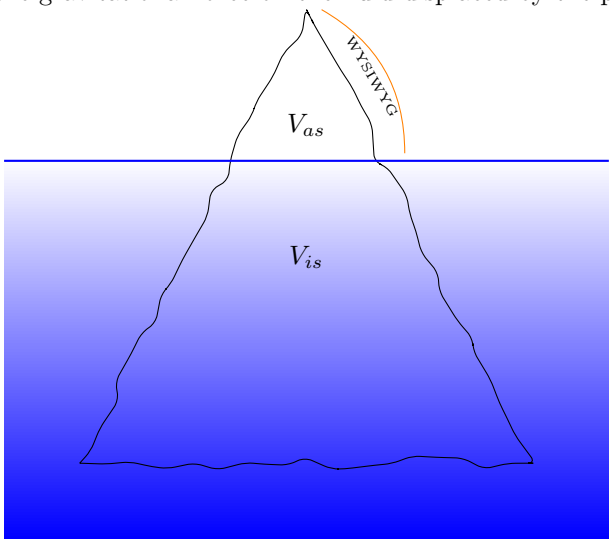
Pressure on body of volume V due to its surrounding fluid (whose density is ρ) acts perpendicularly to the boundary surface between the body and the fluid.

Since fluid pressure at a deep location is greater than that at shallow location, resultant force due to pressure on boundary surface points upward. This resultant force is the buoyancy, F, acting on body.

Consider region of fluid with same volume V as body.

buoyancy $F \equiv$ force exerted vertically on body by its surrounding fluid $\underbrace{=}_{\text{balance gravitational force acting on} V}$ $\rho g V$

For a body floating in a fluid, the magnitude of the buoyancy force acting on the body is equal to the magnitude of the gravitational force on the fluid displaced by the part of the body submerged in the fluid.



109.2 Fluids as a State of Matter

cf. Pert (2013) [99], Ch. 1 Introduction, 1.1 Fluids as a State of Matter.

fluid - isotropic, (no directional preference; identical values of a property in all directions),
locally homogeneous, macroscopic material,
whose particles are free to move within constraints established by the dynamical laws of continuum physics.
requirement that fluid be a continuum - if volume of fluid successively subdivided into smaller elements, each element will remain structurally similar to its parent, and this subdivision can be carried out down to infinitesimal volumes.

fluid particle - fictitious particle fixed within fluid continuum and moving with flow velocity;
represents average over large number of microscopic particles.

fluid point - fixed in fluid moving with flow velocity;
- fluid particle always situated at same fluid point.

infinitesimal volume - within continuum of fluid, large compared with microscopic scales, but small compared with macroscopic ones.

fluid is made up of discrete microscopic particles, namely molecules, distributed randomly, with distribution of velocities characteristic of fluid in termal equilibrium, typically Maxwell-Boltzmann distribution in a gas.

At typical experimental densities,
intermolecular separation extremely small and very much less than lab scale, i.e. $l_s \ll L_0$
possible to average over small volumes which contain very large number of particles, yet very small on lab scale, $N - V$ in V_0, $V_0^{1/3} \ll L_0$
$\implies$ allow us to recover continuum approximation
obtain terms which characterize fluid as bulk material

Typical average quantities:
Density, ρ: number of mass particles per unit volume.
temperature, τ: average energy of random motion per particle in thermal equilibrium
pressure p average momentum flow associated with random motion per unit area.
flow velocity u - mean velocity of molecules averaging out random motion.

role of collisions amongst particles important in defining irreversibility through loss of correlation between particles.

mean free path l_0 - particles collide on average after distance equal to l_0 and time after collision interval

since fluid mechanics assumes fluid particles in thermal equilibrium and randomly distributed, this condition requires that spatial and temporal averages be taken to include large number of collisions; i.e. lab scale length L_0 large compared with l_0, and time to collision interval

effects of collisions on fluid transport (momentum and energy) are averaged over thermal distribution to yield bulk properties of material, namely viscosity and thermal conduction, respectively.

theory: flow of basic fluid may be calculated using Newtonian mechanics, classical thermodynamics, viscosity and thermal conductivity values.
Conditions under which this theory may be applied:

- lab-scale lengths L_0 must be large compared with intermolecule separation and mean free path l_0
- characteristic lab times T_0 must be large compared with collision interval
- fluid locally in thermal equilibrium

cf. 1.2 The Fundamental Equations for Flows of a Dissipationless Fluid. Pert (2013) [99]
inviscid limit - viscosity and thermal conduction are neglected
adiabatic flow - specific entropy of fluid particle constant in time
isentropic flow - specific entropy of each fluid particle has same initial value

many flows are both isentropic and adiabatic, e.g. ideal steady fluid flow, whose specific entropy an entry is everywhere constant.
Eulerian frame - lab frame, coordinates fixed in space and time, derivatives are usual partial derivatives

$$\left.\frac{\partial}{\partial t}\right|_{\mathbf{r}}, \ \nabla|_t$$

Consider coordinate system (t, x^1, x^2, x^3) and so each vector field $\mathbf{e}_j$ forming basis of tangent space at each pt. expressed as linear combination of coordinate vector fields

$$\left.\frac{\partial}{\partial t}\right|_{\mathbf{r}}, \left.\frac{\partial}{\partial x^1}\right|_t, \left.\frac{\partial}{\partial x^2}\right|_t, \left.\frac{\partial}{\partial x^3}\right|_t,$$

in Lagrangian frame, spatial variation seen by particle due to its motion is absorbed into time derivative:

$$\frac{d}{dt} = \frac{\partial}{\partial t} + \mathbf{v} \cdot \nabla$$

Rather, consider total time derivative of integral of n-form:

$$\frac{d}{dt}\int_{V_0} \omega = \int_{V_0} \mathcal{L}_{\frac{\partial}{\partial t}+\mathbf{v}}\omega$$

Volume form $\Omega^n(M) = \bigwedge^n(T^*M)$

$$\text{vol}^n \in \Gamma(\Omega^n(M)) = \Gamma(\bigwedge^n T^*M)$$

i.e. vol^n is n-form, a section of the line bundle, $\Omega^n(M) = \bigwedge^n T^*M$

Now

$$\text{vol}^n = \frac{\sqrt{g}}{n!}\epsilon_{i_1,\dots i_n} dx^{i_1} \wedge \cdots \wedge dx^{i_n}$$

Consider

$$\text{vol}^{n+1} = \frac{\sqrt{g}}{(n+1)!}\epsilon_{i_0 i_1,\dots i_n} dx^{i_0} \wedge dx^{i_1} \wedge \cdots \wedge dx^{i_n}$$

109.3 Lagrangian frame

cf. Pert (2013) [99] 1.3 Lagrangian Frame.

Lagrangian frame - frame of the moving particle;
Lagrangian reference frame considers fluid from point of view of an observer on a fluid particle.

fluid particle may be conveniently identified by coordinate set, which is fixed on the particle,
$\mathbf{\Lambda} = (\lambda, \mu, \nu)$ i.e. triad of numbers.
e.g. $\mathbf{\Lambda}$ maybe initial position of particle $\mathbf{r}_0 = (x_0, y_0, z_0)$, i.e. fluid particle labeled by some (time-independent) vector field $\mathbf{x}_0 \equiv \mathbf{\Lambda}$.

$\implies$ position, velocity, thermodynamic state of particles are therefore functions of time alone.
$\implies$ leads to simple set of kinematic and dynamic relations governing motion of particle

$$\mathbf{r} = \mathbf{r}(t) = (x(t), y(t), z(t)) = (x^1(t), x^2(t), x^3(t))$$
$$\dot{\mathbf{r}} \equiv \dot{\mathbf{r}}(t) \equiv \frac{d\mathbf{r}}{dt} \equiv \frac{d\mathbf{r}}{dt}(t) = \mathbf{v} \equiv \mathbf{v}(t)$$

or

$$\begin{aligned} \frac{d\mathbf{r}}{dt} &= \mathbf{v} \\ \frac{d\mathbf{v}}{dt} &= \frac{\mathbf{F}}{m} \end{aligned} \tag{446}$$

cf. Eq. (1.2) of Pert (2013) [99], i.e.

flow described by function $\mathbf{X}(\mathbf{x}_0, t)$ giving position of particle labeled at $\mathbf{x}_0$ at time t.

109.4 Momentum conservation in Lagrangian Frame-Euler's Equation

Recall that a connection ∇ on smooth manifold $(M, \mathcal{O}, \mathcal{A})$ is a map that takes a pair consisting of a vector (field) X and (p,q)-tensor field T and sends them to a $(p,q)-$ tensor (field) $\nabla_X T$, i.e.

$$\nabla : \mathfrak{X}(M) \times (\otimes^p TM \otimes^q T^*M) \to \otimes^p TM \otimes^q T^*M$$

s.t.

$$\begin{aligned}
&\nabla_X f = Xf \quad \forall f \in C^\infty(M) \\
&\nabla_X (T+S) = \nabla_X T + \nabla_X S \\
&\nabla_X (T(\omega, Y)) = (\nabla_X T)(\omega, T) + T(\nabla_X \omega, Y) + T(\omega, \nabla_X Y) \\
&\nabla_{fX+Z} T = f\nabla_X T + \nabla_Z T, \quad f \in C^\infty(M) \quad \text{"Leibnitz" rule}
\end{aligned}$$

cf. Schuller, International Winter School on Gravity and Light held central lectures.

Remember that in the Lagrangian frame, we are an observer following an individual fluid parcel or i.e. "fluid particle" as it moves through space and time. This fluid particle will obey Newton's 2nd. law: $\nabla_u u = \frac{F}{m} \iff m \cdot \mathfrak{a} = F$.

$$\begin{aligned}
\nabla_u u = \nabla_{u^\mu \frac{\partial}{\partial x^\mu}} (u^\nu \frac{\partial}{\partial x^\nu}) = u^\mu \nabla_{\frac{\partial}{\partial x^\mu}} (u^\nu \frac{\partial}{\partial x^\nu}) = u^\mu \left(\nabla_{\frac{\partial}{\partial x^\mu}} u^\nu \right) \frac{\partial}{\partial x^\nu} + u^\mu u^\nu \nabla_{\frac{\partial}{\partial x^\mu}} \frac{\partial}{\partial x^\nu} = \\
= u^\mu \frac{\partial u^\nu}{\partial x^\mu} \frac{\partial}{\partial x^\nu} + u^\mu u^\nu \Gamma^\gamma_{\mu\nu} \frac{\partial}{\partial x^\gamma}
\end{aligned}$$

Let $u^0 = 1$.

$$\nabla_u u = \frac{\partial u^a}{\partial t} \frac{\partial}{\partial x^a} + u^b \frac{\partial u^a}{\partial x^b} \frac{\partial}{\partial x^a} + u^\mu u^\nu \Gamma^a_{\mu\nu} \frac{\partial}{\partial x^a}$$

where $\Gamma^0_{ba} = 0$ since $\nabla dt = 0$ and we choose a stratified atlas $\mathcal{A}_{\text{stratified}}$.

Then let

$$\int \rho \text{vol}^n \nabla_u u = F_{\text{tot}} = \int f^i \text{vol}^n \otimes e_i$$

From Sec. 2.4 Equations of Motion Neglecting Viscosity, pp. 38, Sabersky, et. al. (1998) [100], the principal forces are those due to pressures acting on surfaces and to forces acting directly on a mass of the particle:

$$\begin{aligned}
\int_{\partial V} p dS^i \otimes e_i = \int_V \mathbf{d}(p dS^i \otimes e_i) = \int_V \frac{\partial p}{\partial x^j} dx^j \wedge \left(\frac{\sqrt{g}}{(n-1)!} \epsilon^i_{i_2, \dots i_n} dx^{i_2} \wedge \cdots \wedge dx^{i_n} \right) \otimes e_i = \\
\int_V \frac{\partial p}{\partial x^j} g^{ij} \text{vol}^n \otimes e_i = \int_V (\text{grad} p)^i \text{vol}^n \otimes e_i \\
\implies \int \rho \text{vol}^n \nabla_u u = -\int_V (\text{grad} p)^i \text{vol}^n \otimes e_i + \int_V f^i \text{vol}^n \otimes e_i \\
\implies \rho (\nabla_u u)^i = -(\text{grad} p)^i + f^i \\
(\nabla_u u)^i = -\frac{1}{\rho} (\text{grad} p)^i + f^i
\end{aligned} \tag{447}$$

describes the motion of a perfect fluid.

Assuming a non-rotating, inertial frame (so that we can set the Christoffel symbols to 0),

$$\frac{\partial}{\partial t} u^i + u^b \frac{\partial u^i}{\partial x^b} = -\frac{1}{\rho} (\text{grad} p)^i + f^i \tag{448}$$

which describes the motion of the perfect fluid in an inertial ref. frame. Compare this to Eqns. (2.10a- 2.10c) of Sabersky (1998) [100]. Sabersky says that no assumptions were made about density so these results apply to compressible fluid as well as incompressible ones. Also Eqns. 447, 448 are called either Euler's Equation by Pert (2013) [99] or Eulerian equation by Sabersky, et. al. (1998) [100]

109.5 Mass Conservation for Fluid Flow, Continuum media

The mass of fluid in some volume $V_0 \subset N$ is $\int_{V^0} \rho \text{vol}^n$, where ρ is fluid density, $\rho \in C^\infty(N)$.

The total mass of fluid flowing out of volume V_0 is

$$\frac{d}{dt}\int_{V_0} \rho \text{vol}^n = \int_{V_0} \mathcal{L}_{\frac{\partial}{\partial t}+\mathbf{u}}(\rho \text{vol}^n) = \int_{V_0} \frac{\partial}{\partial t}\rho \text{vol}^n + \int_{V_0} \mathcal{L}_u \rho \text{vol}^n$$

$$\int_{V_0} \mathcal{L}_u \rho \text{vol}^n = \int_{V_0} d i_{\mathbf{u}} \rho \text{vol}^n + i_{\mathbf{u}} d\rho \text{vol}^n = \int_{V_0} d i_{\mathbf{u}} \rho \text{vol}^n + 0 = \int_{V_0} d i_{\mathbf{u}} \rho \text{vol}^n = \int_{\partial V_0} i_{\mathbf{u}} \rho \text{vol}^n$$

Now

$$i_u \text{vol}^n = i_u \frac{\sqrt{g}}{n!} \epsilon_{i_1 \dots i_n} dx^{i_1} \wedge \dots \wedge dx^{i_n}$$

$$i_u dx^{i_1} \wedge \dots \wedge dx^{i_n} = u^{i_1} dx^{i_2} \wedge \dots \wedge dx^{i_n} - dx^{i_1} \wedge u^{i_2} dx^{i_3} \wedge \dots \wedge dx^{i_n} + \dots + (-1)^{n+1} dx^{i_1} \wedge \dots \wedge dx^{i_{n-1}} u^{i_n} = \epsilon^{i_1 \dots i_n}_{j_1 \dots j_n} u^{j_1} dx^{j_2} \wedge$$

$$\implies i_u \text{vol}^n = \frac{\sqrt{g}}{(n-1)!} \epsilon_{j_1 \dots j_n} u^{j_1} dx^{j_2} \wedge \dots \wedge dx^{j_n} \tag{449}$$

We can also rewrite Eq. 449 to be a "surface differential":

$$\begin{gathered} \int_{V_0} \mathbf{d} i_{\mathbf{u}} \rho \text{vol}^n = \int_{\partial V_0} i_{\mathbf{u}} \rho \text{vol}^n = \\ \int_{\partial V_0} \rho \frac{\sqrt{g}}{(n-1)!} u^{j_1} \epsilon_{j_1 j_2 \dots j_n} dx^{j_2} \wedge \dots \wedge dx^{j_n} \equiv \int_{\partial V_0} \rho \mathbf{u} \cdot d\mathbf{S} \equiv \int_{\partial V_0} \rho \langle \mathbf{u}, d\mathbf{S} \rangle \end{gathered} \tag{450}$$

If $\sqrt{g} = 1$, $n = 2$,

$$i_u \text{vol}^2 = (u^1 dx^2 - u^2 dx^1) = u \cdot n_1 dx^2 + u \cdot n_2 dx^1 = u \cdot n dS$$

with $n_1 = e_1$ and $n_2 = -e_2$.

Now

$$d i_u \rho \text{vol}^n =$$

$$= \frac{\partial(\sqrt{g}\rho u^{j_1})}{\partial x^k} \frac{\epsilon_{j_1 \dots j_n}}{(n-1)!} dx^k \wedge dx^{j_2} \wedge \dots \wedge dx^{j_n} = \frac{\partial(\sqrt{g}\rho u^k)}{\partial x^k} \frac{\epsilon_{j_1 \dots j_n}}{n!} dx^{j_1} \wedge \dots \wedge dx^{j_n} = \frac{1}{\sqrt{g}} \frac{\partial(\sqrt{g}\rho u^k)}{\partial x^k} \text{vol}^n =$$

$$= \frac{\partial(\rho u^k)}{\partial x^k} \text{vol}^n + \rho u^k \frac{\partial \ln \sqrt{g}}{\partial x^k} \text{vol}^n = \text{div}(\rho u) \text{vol}^n + \rho u^k \frac{\partial \ln \sqrt{g}}{\partial x^k} \text{vol}^n$$

Now if $\sqrt{g} = 1$, then

$$\frac{d}{dt}\int_{V_0} \rho \text{vol}^n = \int_{V_0} \frac{\partial \rho}{\partial t} \text{vol}^n + \int_{V_0} d i_u \rho \text{vol}^n = \int_{V_0} \frac{\partial \rho}{\partial t} \text{vol}^n + \int_{V_0} \text{div}(\rho u) \text{vol}^n \implies \frac{\partial \rho}{\partial t} + \text{div}(\rho u) = 0$$

which is the so-called mass continuity equation. $j = \rho u$ is the mass flux density.

Thus,

$$\begin{gathered} \text{(mass conservation)} \\ m = m(t) := \int_{V_0} \rho \text{vol}^n,\ V_0 \subset N \\ \dot{m} \equiv \frac{d}{dt} m(t) = \int_{V_0} \left(\frac{\partial \rho}{\partial t} \text{vol}^n + \mathbf{d} i_{\mathbf{u}} \rho \text{vol}^n \right) = \boxed{\int_{V_0} \frac{\partial \rho}{\partial t} \text{vol}^n + \int_{\partial V_0} \rho \mathbf{u} \cdot d\mathbf{S}} \\ \text{if } \sqrt{g} = 1, \text{ and } \dot{m} = 0, \text{ then} \\ \frac{\partial \rho}{\partial t} + \text{div}(\rho \mathbf{u}) = 0 \end{gathered} \tag{451}$$

Sec. 1.4 Eulerian Frame, 1.4.1 Conservation of Mass-Equation of Continuity, pp. 8 of Pert (2013) [99] says that Eq. 451 is for the Eulerian Frame.

From wikipedia, recall that the *Eulerian specification* focuses on specific locations in space through which fluid flow as time passes. Visualize by sitting on bank of river and watch water pass fixed location. In the Eulerian specification of field, field represented as a function of position $\mathbf{x}$, time t, e.g. $\mathbf{u}(\mathbf{x},t) \equiv$ flow velocity.

TODO: 20190804 Frankel (2012) [113] in pp. 138 and onwards, for Sec. 4.3. Differentiation of Integrals posed the rightful question, "How does one compute the rate of change of an integral when the domain of integration is also changing?" Revisit the derivation from a Lie derivative and 1-parameter flow point of view.

Force should not be represented by a vector but rather by a 1-form. Then,

$$f \in \Omega^1(N),\ f = f(t,\mathbf{x}) = f_j dx^j\ \ j = 1,\dots n,\ \mathbf{x} = (x^1,\dots x^n),\ \dim N = n$$

Indeed, the reason for f to be a 1-form is that we integrate differential forms, we don't integrate vectors.

$$W = \int_C f \qquad \text{(line integral)} \tag{452}$$

If f conservative, $\exists$ scalar $U = U(x)$; $x \in N$ s.t. $f = -dU$.

Let $\mathbf{u} = u^j(t,\mathbf{x})\frac{\partial}{\partial x^j} \in \mathfrak{X}(N)$ generate flow ϕ_t.

The Lie derivative along vector field $\mathbf{u}$, $\mathcal{L}_\mathbf{u}$, can be calculated in at least 2 ways: from the definition:

$$\mathcal{L}_\mathbf{u}\omega = \frac{d}{dt}\,|_{t=0}\,\phi_t^*\omega$$

or Cartan's magic formula:

$$\begin{aligned}
\mathcal{L}_\mathbf{u}\mathrm{vol}^n = (i_\mathbf{u}\mathbf{d} + \mathbf{d}i_\mathbf{u})\mathrm{vol}^n = 0 + \mathbf{d}i_\mathbf{u}\mathrm{vol}^n = \mathbf{d}i_\mathbf{u}\left(\frac{\sqrt{g}}{n!}\epsilon_{i_1\dots i_n}dx^{i_1}\wedge\cdots\wedge dx^{i_n}\right) = \\
= \mathbf{d}\left(\frac{\sqrt{g}}{(n-1)!}\epsilon_{j_1\dots j_n}u^{j_1}dx^{j_2}\wedge\cdots\wedge dx^{j_n}\right) = \frac{\partial(\sqrt{g}u^{j_1})}{\partial x^k}\frac{\epsilon_{j_1\dots j_n}}{(n-1)!}dx^k\wedge dx^{j_2}\wedge\cdots\wedge dx^{j_n} = \\
= \frac{\partial(\sqrt{g}u^{j_1})}{\partial x^k}\frac{\epsilon_{j_1\dots j_n}}{(n-1)!}\epsilon^{kj_2\dots j_n}_{k_1,\dots,k_n}dx^{k_1}\wedge\cdots\wedge dx^{k_n}\underbrace{\left(\frac{\sqrt{g}}{\sqrt{g}}\right)}_{=1} = \frac{1}{\sqrt{g}}\frac{\partial(\sqrt{g}u^k)}{\partial x^k}\mathrm{vol}^n = \\
= (\mathrm{div}\mathbf{u})\mathrm{vol}^n
\end{aligned} \tag{453}$$

cf. https://math.stackexchange.com/questions/2566381/lie-derivative-of-volume-form?rq=1

Note that

$$\mathrm{div} : \mathfrak{X}(N) \to C^\infty(N)$$
$$\mathcal{L}_\mathbf{u} : \Omega^n(N) \to \Omega^n(N)$$

Frankel (2012) [113] on pp. lvii of the "Elasticity and Stresses" section offered this analogy: "While work in particle mechanics pairs a force covector (f_i) with a contravariant tangent vector (dx^i/dt) to a curve, work done by traction in elasticity pairs the contravariant stress force 2-form $\mathcal{S}$ with the covector valued deformation 1-form $\mathcal{E}$, to yield a scalar valued 3-form.

Instead of pushing along a curve (line) particle to do work, what does it mean to do work on a moving fluid element? One could possibly try to consider the deformation of a volume element under fluid flow. Frankel (2012) [113] in Sec. 4.2. The Lie Derivative of a Form, on pp. 132, asks "If a flow deforms some attribute, say volume, how does one measure the deformation? $\mathcal{L}_\mathbf{u}\mathrm{vol}^n$ "is the n-form that reads off the rate of change of volume of a parallelopiped spanned by n vectors that are pushed forward by the flow ϕ_t." So, "in other words, $\mathcal{L}_\mathbf{u}\mathrm{vol}^n$ *measures how volumes are changing under the flow* ϕ_t *generated by* $\mathbf{X}$" on pp. 133 on Frankel (2012) [113].

Then rate of work done on a fluid element could be the following:

$$W = \int_V \mathcal{L}_\mathbf{u}(p\mathrm{vol}^n) = \int_V \mathbf{d}i_\mathbf{u}p\mathrm{vol}^n = \int_{\partial V} i_\mathbf{u}p\mathrm{vol}^n = \int_{\partial V} p\mathbf{u}\cdot d\mathbf{S}$$

TODO: work on a fluid element?

109.6 Bernoulli's equation

109.6.1 Simple cases

cf. Ch. 3 The Bernoulli Equation, Sec. 3.2 "A Simple Form of the Bernoulli Equation", pp. 90, Sabersky, et. al. (1998) [100]

Start from Eq. 447. So

$$(\nabla_u u)^i = \frac{-1}{\rho}(\text{grad}p)^i + f^i$$

Suppose $f^i = 0$ (no external forces), $\frac{\partial u^i}{\partial t} = 0$ (steady flow). Suppose $\Gamma^a_{\mu\nu} = 0$ (Cartesian coordinates? TODO: find out the conditions when this is true).

Suppose the flow is irrotational (TODO: find out the condition for irrotational flow; is it when having streamlines or ?). For the curl, recall that an external derivative $\mathbf{d}$ takes a 1-form into a 2-form. So consider

$$\mathbf{u} \in \mathfrak{X}(V), \mathbf{u} \xrightarrow{\flat} u_i dx^i = g_{ij}u^j dx^i$$

$$d\mathbf{u}^\flat \in \Omega^2(V),\ \mathbf{d}\mathbf{u}^\flat = \frac{\partial u_j}{\partial x^i} dx^i \wedge dx^j = \frac{\partial (g_{ij}u^j)}{\partial x^a} dx^a \wedge dx^i$$

If we have the condition that the flow is irrotational, i.e.

$$\mathbf{d}\mathbf{u}^\flat = 0 \tag{454}$$

then

$$\mathbf{d}\mathbf{u}^\flat(\mathbf{u}) = \frac{\partial (u_i)}{\partial x^a} u^a dx^i - \frac{\partial (u_i)}{\partial x^a} dx^a u^i = 0 \implies u^a \frac{\partial (u_i)}{\partial x^a} dx^i = u^i \frac{\partial u_i}{\partial x^a} dx^a$$

Take the exterior derivative $\mathbf{d}$ of the inner product of a vector with itself in the most general case on a metric manifold:

$$d(u_i u^i) = d(g_{ij}u^j u^i) = \frac{\partial g_{ij}}{\partial x^k} u^j u^i dx^k + g_{ij}\frac{\partial u^j}{\partial x^k} u^i dx^k + g_{ij}u^j \frac{\partial u^i}{\partial x^k} dx^k =$$
$$= 2u_j \frac{\partial u^j}{\partial x^k} dx^k + u^i u^j \frac{\partial g_{ij}}{\partial x^k} dx^k$$

Since the Christoffel symbols are the connection coefficients of a *metric-compatible* affine connection (i.e. they can be used to construct a connection ∇ s.t. $\nabla g = 0$), then [7]

$$\nabla_i g_{jk} = \frac{\partial g_{jk}}{\partial x^i} - \Gamma^s_{ji} g_{sk} - \Gamma^s_{ki} g_{js} = 0 \tag{455}$$

and so

$$u^i u^j \frac{\partial g_{ij}}{\partial x^k} = u^i u^j \left[\Gamma^s_{ik} g_{sj} + \Gamma^s_{jk} g_{is}\right] = u^i u_s \Gamma^s_{ik} + u_s u^j \Gamma^s_{jk} = 2u_s u^i \Gamma^s_{ik} \tag{456}$$

Then

$$\frac{1}{2}\mathbf{d}(u^2) = u_j \frac{\partial u^j}{\partial x^k} dx^k + u_s u^i \Gamma^s_{ik} dx^k \tag{457}$$

Eq. 457 only depended upon the fact that the connection ∇ is *metric-compatible*.

So if we suppose that the Christoffel symbols Γ^s_{ik} are 0, and have irrotational flow, as well,

$$\mathbf{d}(u^2/2) = u_j \frac{\partial u^j}{\partial x^k} dx^k + 0 = u^a \frac{\partial u_i}{\partial x^a} dx^i$$

Now consider the RHS of Eq. 447, with $f_i = 0$,

$$\frac{-1}{\rho}(\text{grad}p)^i \xrightarrow{\flat} \frac{-1}{\rho}(\text{grad}p)^i g_{ij} dx^j = \frac{-1}{\rho}\frac{\partial p}{\partial x^k} g^{ki} g_{ij} dx^j = \frac{-1}{\rho}\frac{\partial p}{\partial x^j} dx^j = \frac{-1}{\rho}\mathbf{d}p$$

So if the fluid is incompressible, then ρ is constant and so we can write

$$\mathbf{d}\left(\frac{u^2}{2} + \frac{p}{\rho}\right) = 0 \tag{458}$$

for any irrotational, incompressible, fluid flow with no (external) body forces.

[7] "Prove Christoffel Symbol Identity", Physics StackExchange

109.6.2 Venturi tube example

cf. Example 3.1 in Sabersky, et. al. (1998) [100]

Consider flow of liquid through the tube shown. Determine pressure difference between points 1 and 2 as a function of flow rate Q, density ρ. Velocity assumed constant over cross sections, and pressure will also be assumed uniform over each of these 2 areas A_1, A_2. Also assume there's no viscosity.

Solution Example 3.1 in Sabersky, et. al. (1998) [100].

Since density is constant, continuity equation requires that for Q the volume flow rate,

$$\text{flow rate} Q = u_1 A_1 = u_2 A_2$$

$$\frac{u_1^2}{2} + \frac{p_1}{\rho_1} = \frac{u_2^2}{2} + \frac{p_2}{\rho_2} \Longrightarrow \frac{Q^2}{2A_1^2} + -\frac{Q^2}{2A_2^2} = \frac{-(p_1 - p_2)}{\rho} \quad \text{or } p_1 - p_2 = \frac{+\rho Q^2}{2A_2^2}\left(1 - \frac{A_2^2}{A_1^2}\right)$$
$$\frac{Q^2}{2}\left(\frac{1}{A_1^2} - \frac{1}{A_2^2}\right) = \frac{-(p_1 - p_2)}{\rho}$$

109.6.3 Conservative body forces with Bernoulli equation

cf. Sec. 3.3 in Sabersky, et. al. (1998) [100]

Consider $U = U(x)$, force potential.

$$-\mathbf{d}U = \frac{-\partial U}{\partial x^i} dx^i = f_i dx^i$$
$$\mathbf{d}\left(\frac{u^2}{2} + \frac{p}{\rho} + U\right) = 0$$

109.6.4 Barotropic Fluids ($\rho = \rho(p)$) with Bernoulli equation

cf. Sec. 3.4 in Sabersky, et. al. (1998) [100]

Barotropic fluid: density is single-valued function of p. In the absence of heat transfer and absence of entropy change, by thermodynamics this can be shown to be barotropic.

$$\mathbf{d}\left(\frac{u^2}{2}\right) = -\frac{1}{\rho}\mathbf{d}p - \mathbf{d}U \text{ or } \mathbf{d}\left(\frac{u^2}{2} + U\right) + \frac{1}{\rho}\mathbf{d}p$$

Suppose $i = i(p)$ s.t. $di = \frac{\partial i}{\partial p} dp = \frac{1}{\rho(p)} = \frac{di}{dp} dp$. So

$$\frac{di}{dp} = \frac{1}{\rho(p)} \text{ or } i = \int \frac{1}{\rho} dp + \text{ const}$$
$$\Longrightarrow \mathbf{d}\left(\frac{u^2}{2} + U + i\right) = 0$$

In special case of isentropic flow of perfect gas, $\frac{P}{\rho^\nu} = \text{const.}$ and $i = \int \frac{dp}{p^{1/\nu}} = \frac{k p^{1-1/\nu}}{1-1/\nu} = \text{enthalpy.}$

109.6.5 Nonsteady flow $\frac{\partial u}{\partial t}$ with Bernoulli equation

First, use the musical isomorphism to transform the (1,0) tensor:

$$\frac{\partial u^i}{\partial t}\frac{\partial}{\partial x^i} \overset{\flat}{\to} g_{ij}\frac{\partial u^i}{\partial t} dx^j$$

Along a curve $\gamma = \gamma^i(t) = x^i(t)$ s.t. $\dot{\gamma} = \dot{\gamma}^i(t)\frac{\partial}{\partial x^i} = u^i(t, \gamma(t))\frac{\partial}{\partial x^i}$. Then

$$\int_\gamma g_{ij}\frac{\partial u^i}{\partial t}dx^j = \int g_{ij}\frac{\partial u^i}{\partial t}\dot{\gamma}^j(t)dt = \int g_{ij}\frac{\partial u^i}{\partial t}u^j dt = \int \frac{\partial \|u\|}{\partial t}dt$$

The last step can be made only if the metric g is time-independent.

cf. Example 3.2 in Sabersky, et. al. (1998) [100].

Consider the *efflux* of liquid of density ρ through an orifice of area A_0 at the bottom of a large tank of cross-sectional area A_t.
The depth of liquid at the given instant is y.
pressure exerted on the liquid in the tank is maintained at p_t,
pressure outside the orifice is the atmospheric pressure p_a.

The action of gravity is to be considered, but frictional effects may be neglected. Velocity of efflux u_j is to be determined for these conditions.

efflux - something given off in or as if in a stream (Webster dictionary)

Solution Example 3.2 in Sabersky, et. al. (1998) [100].

In Sec. 3.8(b) flows through orifices discussed in great detail. For this example,
assume the orifice is carefully designed so streamlines at orifice exit are parallel and thus orifice area A_0 and cross-sectional area of jet A_j are equal there.

cf. Example 3.3 in Sabersky, et. al. (1998) [100]. Example of a problem in which nonsteady terms cannot be neglected.

110 Viscous Fluids, Navier-Stokes Equation Applications

cf. Ch. 2, Viscous Fluids, Sec. 15. The equations of motions of a viscous fluid, Landau and Lifshitz (1987) [101]

Recall Navier-Stokes equation for *compressible, viscous* fluid flow:

$$\boxed{\rho\left(\frac{\partial u^i}{\partial t} + u^j\frac{\partial u^i}{\partial x^j}\right) = -\frac{\partial p}{\partial x^i} + (\lambda+\mu)\frac{\partial}{\partial x^i}\mathrm{div}u + \mu\Delta u^i} \tag{459}$$

Compare this with the expression in Eq. 15.6, Landau and Lifshitz (1987) [101]

$$\rho\left[\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v}\cdot\mathbf{grad})\mathbf{v}\right] = -\mathbf{grad}p + \eta\Delta\mathbf{v} + (\zeta+\frac{1}{3}\eta)\mathbf{grad}\mathrm{div}\mathbf{v} \tag{460}$$

$\mu \equiv \eta$, fluid isotropic, so properties described by scalar quantities.
$\mu \equiv \eta$ viscosity coefficients,
$\lambda + \mu \equiv \zeta + \frac{1}{3}\eta$, $\zeta \equiv$ second viscosity.

110.1 Flow in a pipe

111 Thermodynamics

Let Σ be a (topological) manifold.
Suppose U is a global coordinate on Σ:

(First Law of Thermodynamics (energy conservation))

$$\boxed{dU = Q - W \text{ or } Q = dU + W} \tag{461}$$

where $dU, Q, W \in \Omega^1(\Sigma)$ (i.e. dU, Q, W are 1-forms over manifold Σ).

Consider a path in Σ, γ, $\gamma : \mathbb{R} \to \Sigma$,

and using a chart $(U, S^1, \ldots, S^n)$ (e.g. $n = 1$, $S^1 = v$ for volume)

$$\gamma(t) = (U(t), S^1(t), \ldots, S^n(t))$$
$$\dot{\gamma} \in \mathfrak{X}(\Sigma),\ \dot{\gamma} = \dot{U}\frac{\partial}{\partial U} + \dot{S}^i\frac{\partial}{\partial S^i}$$

Now

$$dU(\dot{\gamma}) = \dot{\gamma}(U) = \dot{U}\frac{\partial}{\partial U}U + 0 = \dot{U}$$
$$Q(\dot{\gamma}) = Q(t)dt\left(\dot{\gamma}\frac{\partial}{\partial t}\right) = Q(t)\dot{\gamma}$$
$$\implies \dot{U} = Q(t)\dot{\gamma} - W(t)\dot{\gamma}$$

Recall that for enthalpy H, $H := U + pV$, $H = H(\sigma, p)$

TODO 20190804 Derive and check convection form of enthalpy against both Kittle and Kroemer plus thermodynamics and Sonntag, et. al.

Incomplete:

$$dH = Q + Vdp$$
$$W = pdV = pdV + Vdp - Vdp = d(pV) - Vdp =$$
$$dE = Q - W + \mu dH$$
$$dE = Q - W + \widehat{h}dN$$

$U \to E$ notation is to promote the internal energy U to include kinetic and potential energies, so that possibly, $E = U + \text{K.E.} + \text{P.E.}$, or, i.e., E includes internal energy and mechanical energy.

$G = \mu N$

$G = U - \tau\sigma + pV \xrightarrow{U \to E} G = E - \tau\sigma + pV \implies \tau\check{h}$

$$dE = Q + W + \mu dN = Q + W + (\check{h} - \tau\check{\sigma})dN$$

$$Q = \tau d\sigma = \tau d(N\check{\sigma}) = \tau N d\check{\sigma} + \tau\check{\sigma}dN$$

$\tau N d\check{\sigma}$ is the entropy change due to change in entropy per particle; i.e. **conduction term**
$\tau\check{\sigma}dN$ is entropy change due to change in number of particles, i.e. **convection term**

$$dE = Q + W + \check{h}dN - \tau\check{\sigma}dN = \tau N d\check{\sigma} + \tau\check{\sigma}dN + W + \check{h}dN - \tau\check{\sigma}dN = \tau N d\check{\sigma} + W + \check{h}dN$$

$m(t) = MN$

Assume only 1 chemical species:

$$\check{Q} := \tau N d\check{\sigma} \qquad \widehat{h} := \frac{H}{m} = \frac{H}{MN}$$
$$\implies dE = \check{Q} + W + \check{h}dN\frac{M}{M} = \check{Q} + W + \widehat{h}dm$$

The Gibbs free energy equilibrium is given by,

$$dG = \mu dN - \sigma d\tau + Vdp$$

For a throttling valve (don't all valves throttle?) the pressure drop is accounted for by the Gibbs free energy, *not* by an isenthalpy condition.

Part XXIX
Classical Mechanics applications

cf. Arnold, Kozlov, Neishtadt (2006) [103].

If known forces $\mathbf{F}_1 \dots \mathbf{F}_n$ acts on points, then

$$\sum_{i=1}^{n} \langle m_i \ddot{\mathbf{r}}_i - \mathbf{F}_i, \xi_i \rangle = 0 \tag{462}$$

cf. Eq. (1.26) of Arnold, Kozlov, Neishtadt (2006) [103], where $\xi_1, \dots \xi_n$ are arbitrary tangent vectors to M, $\xi_i, \dots \xi_n \in TM$.

$\sum_{i=1}^{n} \langle m_i \ddot{\mathbf{r}}_i - \mathbf{F}_i, \xi_i \rangle$ called "general equation of dynamics" or d'Alembert-Lagrange principle.

112 Orbital Mechanics

References: Arnold, Kozlov, Neishtadt (2006) [103] Choquet-Bruhat (2009) [110], pp. 39, 3. General relativity Bate, Mueller, White (1971) [118], Sec. 1.2, pp. 5, https://en.wikipedia.org/wiki/N-body_problem Abraham and Marsden (2008) [104], Ch. 9 Prástaro (1996) [86], Remark 3.20 many-bodies dynamics, pp. 346

112.1 Conic sections

The following is an amalgamation or summary of the development in Ch. 13, Sections 13.18-22, from pp. 497, Apostol (1991) [117]

Definition 216 (General definition of conic sections (vector-formulated definition))**.** Given line L, called the *directrix*, pt. F, the *focus*, such that $F \notin L$.,

eccentric $e > 0$,

$d =$ distance of L from F, then

the **conic section** $C = \{X\}$ (set of all points) such that

$$\|X - F\| = ed(X, L) \tag{463}$$

Now,

$e < 1$ ellipse,

$e = 1$ parabola,

$e > 1$ hyperbola.

Further, if N unit normal to L, then $\forall\, P \in L$,

$$d(X, L) = |(X - P) \cdot N| \tag{464}$$

If $(X - P) \cdot N \gtrless 0$, $X \in$ positive half plane (negative half-plane).

Theorem 75 (Further conic section development)**.** *Building upon Def. 216,*

If $F \in$ positive half plane determined by N, then $F - dN \in L$.

If $F \in$ negative half plane determined by N, then $F + dN \in L$.

Choose this convenient, specific point: $F + dN = P \in L \quad \implies \|X - F\| = e|(X - (F + dN)) \cdot N|$

If $F = 0$, $\|X\| = e|X \cdot N - d| = r = e(d - r\cos\theta)$. Then

$$\frac{ed}{r} = 1 + e\cos\theta \text{ or } \boxed{r = \frac{ed}{1 + e\cos\theta}} \text{ for } F = 0 \tag{465}$$

Consider symmetry about the origin (i.e. of $X, \exists\, -X$).
As a caveat, let $A = e(d + F \cdot N)$. Then

$$\|X - F\| = e|(X - (F + dN)) \cdot N| = ed(X, L) = e|X \cdot N - \frac{A}{e}|$$

Then

$$\|X-F\|^2 = \|X\|^2 - 2X\cdot F + \|F\|^2 = e^2((X\cdot N)^2 - 2\frac{A}{e}X\cdot N + \left(\frac{A}{e}\right)^2) \tag{466}$$

Using the symmetry about the origin, $X \to -X$*,*

$$\implies X\cdot F = AeX\cdot N \text{ or } X\cdot(F-AeN)=0$$

Then $F = AeN$*. Note that* $A = e(F\cdot N \pm d)$ *(for* F *in negative (positive) half-plane).*

Then

$$Ae = e^2(Ae+d) \text{ or } \frac{A}{e} = Ae+d \text{ or } A\left(\frac{1}{e}-e\right) = d \text{ or } A = \frac{ed}{1-e^2}$$
$$\implies \|X-AeN\| = e|(X-(AeN+dN))\cdot N| = e|X\cdot N-(Ae+d)|$$

Let $X = aN$*. Plug this into Eq. 466.*

$$\begin{aligned} a^2 - 2aAe + A^2e^2 = e^2(a^2 - \frac{2A}{e}a + \left(\frac{A}{e}\right)^2) = e^2a^2 - 2Aae + A^2 \implies a = A \\ \implies a = \frac{ed}{1-e^2} \end{aligned} \tag{467}$$

Thus, we've shown that A *is equal to the* semi-major axis a *and that*

$$\boxed{F = aeN} \tag{468}$$

Thus, the semi-major axis is obtained when $X = \pm aN$ *(vertices), so that* $\|X\|^2 + (ae)^2 = e^2(X\cdot N)^2 + a^2$ *is satisfied.*
Likewise, the semi-minor axis is obtained if $X = \pm bN_\perp$*, so that*

$$b^2 + (ae)^2 = 0 + a^2 \implies \boxed{b = a\sqrt{1-e^2}} \tag{469}$$

We can then rewrite Eq. 465, with Eq. 467 as

$$\boxed{r = \frac{a(1-e^2)}{1+e\cos\theta}}$$

With Eq. 466, and using the symmetry of X about the origin, then

$$\|X\|^2 + \|F\|^2 = e^2(X\cdot N)^2 + a^2$$

Choose $\mathbf{N} = \mathbf{e}_x$. With $\|X\|^2 = x^2+y^2$, and $\|F\| = Ae = ae$, then

$$x^2 + y^2 + a^2e^2 = e^2x^2 + a^2 \text{ or } x^2(1-e^2) + y^2 = a^2(1-e^2) \tag{470}$$

cf. 13.23 "Cartesian equations for the conic sections", Apostol (1991) [117].

112.1.1 Ellipse

If $e < 1$, then from Eq. 470,

$$\xrightarrow{\text{if } e<1} \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Observe that what Greiner (2004) [102] in pp. 249, Ch. "Planetary Motions", (a) Ellipse, denotes c is what we and Apostol denote as $\|F\| = ae$, the distance from the origin to the focus. Observe that

$$c^2 + b^2 = a^2e^2 + a^2(1-e^2) = a^2$$

112.1.2 Hyperbola

If $e > 1$, then from Eq. 470,

$$\xrightarrow{\text{if } e>1} \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

Let $b = |a|\sqrt{e^2-1}$. The foci of a hyperbola are at $(c,0)$, $(-c,0)$ where $c = |a|e = \sqrt{a^2+b^2}$.

112.1.3 Parabola

cf. pp. 507 of 13.23 "Cartesian equations for the conic sections", Apostol (1991) [117]
Go back to Eq. 463, where

$$\|X - F\| = d(X, L)$$

Let L be a vertical line at $x = \mp c$, place focus $F = (\pm c, 0)$.
If $X = (x, y)$, $X - F = (x \mp c, y)$, so

$$\|X - F\|^2 = (x \mp c)^2 + y^2 = |x \pm c|^2 = x^2 \pm 2xc + c^2 = x^2 \mp 2xc + c^2 + y^2 \implies y^2 = \pm 4xc$$

So midway between the focus and directrix, at the origin, is the vertex of the parabola.

112.2 Central Force Problem via Energy Methods

112.2.1 Reduction to the Equivalent One-Body Problem

cf. pp. 70 of Goldstein, Poole, Safko (2001) [98]
Let $\mathbf{R}$ = radius vector to the center of mass.

Let $\mathbf{r} = \mathbf{r}_2 - \mathbf{r}_1$ for the relative position of two masses. Then the Lagrangian

$$\mathcal{L} = T(\dot{\mathbf{R}}, \dot{\mathbf{r}}) - U(\mathbf{r}, \dot{\mathbf{r}}, \dots) \tag{471}$$

cf. Eq. (3.1) of Goldstein, Poole, Safko (2001) [98]
The kinetic energy T is the sum of the kinetic energy of the center of mass motion plus the kinetic energy of motion, about the center of mass T'.

$$T = \frac{1}{2}(m_1 + m_2)\dot{\mathbf{R}}^2 + T' \text{ with } T' = \frac{1}{2}m_1\dot{\mathbf{r}}_1'^2 + \frac{1}{2}m_2\dot{\mathbf{r}}_2'^2$$

$\mathbf{r}_1', \mathbf{r}_2'$ are radii vectors relative to the center of mass, since

$$\mathbf{R} = \frac{m_1\mathbf{r}_! + m_2\mathbf{r}_2}{M}$$

so

$$\begin{aligned} \mathbf{r}_1 - \mathbf{R} &= \frac{m_2(\mathbf{r}_1 - \mathbf{r}_2)}{M} = -\frac{m_2}{M}\mathbf{r} \equiv \mathbf{r}_1' \\ \mathbf{r}_2' \equiv \mathbf{r}_2 - \mathbf{R} &= \frac{m_1(\mathbf{r}_2 - \mathbf{r}_1)}{M} = \frac{m_1}{M}\mathbf{r} \end{aligned} \tag{472}$$

cf. Eqn. (3.2) of Goldstein, Poole, Safko (2001) [98].
Thus $T' = \frac{1}{2}\frac{m_1 m_2}{m_1+m_2}\dot{\mathbf{r}}^2$
So

$$\mathcal{L} = \frac{M}{2}\dot{\mathbf{R}}^2 + \frac{1}{2}\frac{m_1 m_2}{M}\dot{\mathbf{r}}^2 - U(\mathbf{r}, \dot{\mathbf{r}}, \dots) \tag{473}$$

cf. Eqn. (3.3) of Goldstein, Poole, Safko (2001) [98].
3 coordinates $\mathbf{R}$ are cyclic, so CM is at rest or moving uniformly. Drop the first term. The rest of the Lagrangian is exactly what would be expected if we had a fixed center of force with a single particle at distance $\mathbf{r}$ from it.
Let

$$\mu = \frac{m_1 m_2}{m_1 + m_2} \equiv \text{ reduced mass}$$

112.2.2 Equations of Motion from Lagrangian and First Integrals

cf. Sec. 3.2 "The Equations of Motions and First Integrals, Chapter 3 The Central Force Problem, pp. 72 of Goldstein, Poole, Safko (2001) [98].

Goldstein, Poole, Safko argues: assume potential is of form $V(r)$ and so $V(r)$ has only radial distance, so the problem has spherical symmetry. Then, Goldstein, et. al. argues $\mathbf{L} = \mathbf{r} \times \mathbf{p}$ conserved. Thus $\mathbf{r}$ and $\mathbf{p} \perp \mathbf{L}$ always.

So

$$\mathcal{L} = T - V = \frac{1}{2}\mu(\dot{r}^2 + r^2\dot{\theta}^2) - V(r) \tag{474}$$

cf. Eqn. (3.6) Goldstein, Poole, Safko (2001) [98].

Now

$$p_\theta = \frac{\partial \mathcal{L}}{\partial \dot{\theta}} = \mu r^2 \dot{\theta}$$

Since $\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = \frac{\partial \mathcal{L}}{\partial \theta}$, then

$$\dot{p}_\theta = \frac{d}{dt}(\mu r^2 \dot{\theta}) = 0 = \frac{\partial \mathcal{L}}{\partial \theta} \tag{475}$$

cf. Eqn. (3.7) Goldstein, Poole, Safko (2001) [98].

So

$$\mu r^2 \dot{\theta} = l \tag{476}$$

cf. Eqn. (3.8) Goldstein, Poole, Safko (2001) [98], where l is the constant magnitude of the angular momentum.

So then $\frac{d}{dt}\left(\frac{1}{2}r^2\dot{\theta}\right) = 0$ where $\frac{1}{2}r^2\dot{\theta}$ is the areal velocity, the area swept out by radius vector per unit time.

Goldstein argues that $\frac{1}{2}r^2\dot{\theta}$ follows from $dA = \frac{1}{2}r(rd\theta)$, so $\frac{dA}{dt} = \frac{1}{2}r^2\frac{d\theta}{dt}$. Thus, angular momentum conservation is equal to areal velocity constant. Do keep in mind that this argument is based upon the diagram in Goldstein, Poole, Safko (2001) [98] and presumably the small angle approximation (perhaps using $\sin\theta \approx \theta$ for small θ). Thus, Kepler's 2nd. law: the radius vector sweeps out equal areas in equal times.

For the r coordinate:

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{r}}\right) = \frac{\partial \mathcal{L}}{\partial r} \Longrightarrow \frac{d}{dt}(\mu\dot{r}) = \mu r\dot{\theta}^2 - \frac{\partial V}{\partial r} \text{ or } \mu\ddot{r} - \mu r\dot{\theta}^2 = -\frac{\partial V}{\partial r}$$

Denote

$$f(r) = \mu\ddot{r} - \mu r\dot{\theta}^2 \tag{477}$$

cf. Eqn. (3.11) of Goldstein, Poole, Safko (2001) [98].

Since $\mu r^2\dot{\theta} = l$, then

$$f(r) = \mu\ddot{r} - \frac{l^2}{\mu r^3} \tag{478}$$

cf. Eqn. (3.12) of Goldstein, Poole, Safko (2001) [98].

Since

$$\mu\ddot{r} = \frac{-\partial}{\partial r}\left(V + \frac{1}{2}\frac{l^2}{\mu r^2}\right) \tag{479}$$

cf. Eqn. (3.14) of Goldstein, Poole, Safko (2001) [98] and since $\mu\ddot{r}\dot{r} = \frac{d}{dt}\left(\frac{1}{2}\mu\dot{r}^2\right)$ and $\frac{d}{dt}g(r) = \frac{\partial g}{\partial r}\dot{r}$, then

$$\frac{d}{dt}\left(\frac{1}{2}\mu\dot{r}^2\right) = \frac{-d}{dt}\left(V + \frac{1}{2}\frac{l^2}{\mu r^2}\right) \text{ or } \frac{d}{dt}\left(\frac{1}{2}\mu\dot{r}^2 + \frac{1}{2}\frac{l^2}{\mu r^2} + V\right) = 0$$

and so

$$\frac{1}{2}\mu\dot{r}^2 + \frac{1}{2}\frac{l^2}{\mu r^2} + V = \text{ constant} \tag{480}$$

cf. Eqn. (3.15) of Goldstein, Poole, Safko (2001) [98], with

$$\frac{1}{2}\frac{l^2}{\mu r^2} = \frac{1}{2\mu r^2}\mu^3 r^4\dot{\theta}^2 = \frac{\mu r^2\dot{\theta}^2}{2}$$

and so

$$\frac{1}{2}\mu\dot{r}^2 + \frac{1}{2}\mu r^2\dot{\theta}^2 + V = E = T + V \tag{481}$$

cf. Eqn. (3.13) of Goldstein, Poole, Safko (2001) [98].

For $E = \frac{1}{\textcircled{a}}\mu\dot{r}^2 + \frac{l^2}{2\mu r^2} + V$, then

$$\dot{r} = \sqrt{\frac{2}{\mu}\left(E - V - \frac{l^2}{2\mu r^2}\right)} \tag{482}$$

cf. Eqn. (3.16) of Goldstein, Poole, Safko (2001) [98], or

$$dt = \frac{dr}{\sqrt{\frac{2}{\mu}\left(E - V - \frac{l^2}{2\mu r^2}\right)}} \tag{483}$$

cf. Eqn. (3.17) of Goldstein, Poole, Safko (2001) [98].

Let

$$t = \int_{r_0}^{r} \frac{dr}{\sqrt{\frac{2}{\mu}\left(E - V - \frac{l^2}{2\mu r^2}\right)}}$$

with initial value r_0.

From $l = \mu r^2\dot{\theta}$,

$$d\theta = \frac{l dt}{\mu r^2} \tag{484}$$

cf. Eqn. (3.19) of Goldstein, Poole, Safko (2001) [98] so for $\theta_0 \equiv$ initial value of θ.

$$\theta = l\int_0^t \frac{dt'}{\mu r^2(t')} + \theta_0 \tag{485}$$

cf. Eqn. (3.20) of Goldstein, Poole, Safko (2001) [98].

In Sec. 3.3, "The Equivalent 1-Dimensional Problem and Classification of Orbits" of Goldstein, Poole, Safko (2001) [98], consider $V' = V + \frac{1}{2}\frac{l^2}{\mu r^2}$, so that $F' = \frac{-\partial V'}{\partial r} = f(r) + \frac{l^2}{\mu r^3}$.

Let $V = \frac{-k}{r}$, e.g. $V = \frac{-GMm}{r}$ with $k = GMm$.

Then consider $V' = \frac{-k}{r} + \frac{l^2}{2\mu r^2}$ and its plot.

Compare this, Thm. 75, against (26.6) of Greiner (2004) [102]:

$$r = \frac{ed}{1 + e\cos\theta}, \quad e = \frac{C\widehat{L}_0^2}{GM}, \quad d = 1/C \tag{486}$$

cf. Eq. (26.32) of Greiner (2004) [102].

Investigate which physical quantities (e.g. E, L) eccentricity e depends on.

$$u = \frac{GM}{\widehat{L_0}^2} + C\cos(\theta - \theta_0) \Longrightarrow \frac{du}{d\theta} = -C\sin(\theta - \theta_0)$$

$$\frac{1}{2}mL_0^2\left[C^2\sin^2(\theta-\theta_0) + \left(\frac{GM}{L_0^2}\right)^2 + 2C\frac{GM}{L_0^2}\cos(\theta-\theta_0) + C^2\cos^2(\theta-\theta_0)\right] =$$

$$\Longrightarrow \quad = \frac{1}{2}m\widehat{L}_0^2\left[C^2 + \left(\frac{GM}{\widehat{L}_0^2}\right)^2 + 2C\frac{GM}{\widehat{L}_0^2}\cos(\theta-\theta_0)\right] =$$

$$= E - V\left(\frac{1}{u}\right) = E + GMmu = E + \frac{(GM)^2}{\widehat{L}_0^2}m + CGMm\cos(\theta-\theta_0)$$

$$\Longrightarrow \frac{1}{2}m\widehat{L}_0^2C^2 = E + \frac{1}{2}\frac{(GM)^2}{\widehat{L_0}^2}m$$

Solve for the dummy constant C:

$$C = \sqrt{\frac{2E}{m\widehat{L}_0^2} + \frac{(GM)^2}{\widehat{L}_0^4}} \tag{487}$$

cf. Eq. (26.36) of Greiner (2004) [102].

From Eq. (26.32) of Greiner (2004) [102] or Eq. 486,

$$e = \frac{C\widehat{L}_0^2}{GM} = \sqrt{\frac{2E}{m\widehat{L}_0^2} + \frac{(GM)^2}{\widehat{L}_0^4}} \left(\frac{\widehat{L}_0^2}{GM} \right)$$

So

$$\boxed{e = \sqrt{\frac{2E\widehat{L}_0^2}{m(GM)^2} + 1}} \tag{488}$$

cf. Eq. (26.37) of Greiner (2004) [102].

Hence, shape of the path depends on total energy E, angular momentum $L_0 = m\widehat{L}_0$ of moving body (recall that $\widehat{L} := \mathbf{r} \times \mathbf{v}$).

It holds that for a

$$\begin{aligned}
&\text{parabola } e = 1, && E = 0 \\
&\text{ellipse } 0 < e < 1, && E < 0, && \frac{(GM)^2 m}{-2\widehat{L}_0^2} < E < 0 \\
&\text{circle } e = 0 && E = \frac{-(GM)^2 m}{2\widehat{L}_0^2} \\
&\text{hyperbola } e = 1, && E > 0
\end{aligned}$$

The general form of initial energy E_0 is, recall, give by

$$E_0 = \frac{1}{2}mv^2 + V(r) = \frac{1}{2}mv^2 - \frac{GMm}{r_0} = m\left(\frac{v^2}{2} - \frac{GM}{r_0} \right)$$

Notice that for the case of the parabola that
$E_0 = 0$ when

$$v = \sqrt{\frac{2GM}{r_0}} \tag{489}$$

Eq. 489 is the expression for the **escape velocity**.

Knowing what we know now, let's rewrite some of the expressions from before:

$$\begin{aligned}
r(\theta) &= \frac{\widehat{L}_0^2/GM}{1 + \frac{C\widehat{L}_0^2}{GM}\cos(\theta - \theta_0)} \\
\widehat{L}_0^2 &:= r_0^2\dot{\theta}_0 \\
C &= \sqrt{\frac{2E}{m\widehat{L}_0^2} + \frac{(GM)^2}{\widehat{L}_0^4}}
\end{aligned}$$

112.3 Path parameters (i.e. orbital parameters)

cf. pp. 262, Greiner (2004) [102].

$$a = \frac{1}{2}(r(\theta = 0) + r(\theta = \pi)) = \frac{1}{2}\left[\frac{\widehat{L}_0^2/GM}{1+e} + \frac{\widehat{L}_0^2/GM}{1-e} \right] = \frac{\widehat{L}_0^2/GM}{1-e^2}$$

The addition above for a is needed because we center origin at a focus, for this expression.

$$a = \frac{\widehat{L}_0^2/GM}{\frac{-2E\widehat{L}_0^2}{m(GM)^2}} = \frac{GMm}{-2E} \tag{490}$$

cf. Eq. (26.39) of Greiner (2004) [102].

$$b = \sqrt{a^2 - f^2} = \sqrt{a^2 - e^2 a^2} = a\sqrt{1 - e^2} = a\sqrt{\frac{-2E\widehat{L}_0^2}{m(GM)^2}} = \widehat{L}_0\sqrt{\frac{-m}{2E}} \tag{491}$$

cf. Eq. (26.40) of Greiner (2004) [102].

We can also calculate the perigee (point of closest approach to mass M):

$$r(\theta = 0) = \frac{\widehat{L}_0^2/GM}{1+e} = \frac{\widehat{L}_0^2/GM}{1 + \sqrt{\frac{2E\widehat{L}_0^2}{m(GM)^2} + 1}} \tag{492}$$

Given Eq. 490 for a, plug in for E:

$$a = \frac{-GMm}{2E} = -\frac{GMm}{2\left(\frac{1}{2}mv_0^2 - \frac{GMm}{r}\right)} = \frac{-GM}{v_0^2 - \frac{2GM}{r}} = \frac{-GM}{v_0^2\left(1 - \frac{2GM}{rv^2}\right)} \tag{493}$$

Notice when the denominator is 0. It is when v_0 is the the escape velocity (cf. Eq. 489).

112.3.1 Eccentricity from r_0, $\mathbf{v}_0$

Consider being given $\mathbf{r}$, $\mathbf{v}$, with $\mathbf{r}$ in a coordinate frame with its origin at the center of mass (or, if M large enough, from the focus where M is centered at).

The first step is to use a coordinate *frame* (set of orthonormal unit vectors) with $\mathbf{r}$, $\mathbf{v}$ forming a *plane*. Note this important fact from geometry: a plane is spanned by 2 non-collinear vectors. So $\mathbf{r}, \mathbf{v}$ better not be parallel to each other.

Let

$$\mathbf{v} = v^x\mathbf{e}_x + v^y\mathbf{e}_y + v^z\mathbf{e}_z = v^r\mathbf{e}_r + v^\varphi\mathbf{e}_\varphi$$
$$\mathbf{r} = x\mathbf{e}_x + y\mathbf{e}_y + z\mathbf{e}_z = r\mathbf{e}_r$$

Now

$$(\mathbf{r}\times\mathbf{v})_i = \epsilon_{jki}r_j v_k$$

To get the vector *orthogonal* to the plane that is formed by $\mathbf{r}, \mathbf{v}$,

$$\begin{aligned}((\mathbf{r}\times\mathbf{v})\times\mathbf{r})_m &= \epsilon_{ijm}\epsilon_{jki}r_j v_k r_l = \epsilon_{ilm}\epsilon_{ijk}r_j v_k r_l = \\ &= r^2 v_m - v_k r_k r_m = r^2 v_m - v_k r_k r_m\end{aligned} \quad \text{or } ((\mathbf{r}\times\mathbf{v})\times\mathbf{r}) = r^2\mathbf{v} - (\mathbf{v}\cdot\mathbf{r})\mathbf{r}$$

Then

$$\widehat{L}_0 = r^2\dot{\theta} = r\left(\mathbf{v}\cdot\frac{((\mathbf{r}\times\mathbf{v})\times\mathbf{r})}{|(\mathbf{r}\times\mathbf{v})\times\mathbf{r}|}\right) = r\left(\frac{r^2v^2 - (\mathbf{v}\cdot\mathbf{r})^2}{|r^2\mathbf{v} - (\mathbf{v}\cdot\mathbf{r})\mathbf{r}|}\right)$$

Thus, we can find the following:

$$e^2 = 1 + \frac{2E\widehat{L}_0^2}{m(GM)^2} = 1 + \frac{2\left(\frac{1}{2}mv_0^2 - \frac{GMm}{r}\right)\widehat{L}_0^2}{m(GM)^2} = 1 + \frac{\left(v_0^2 - \frac{2GMm}{r}\right)\widehat{L}_0^2}{(GM)^2}$$

$$a = \frac{-GMm}{2E} = -\frac{GMm}{2\left(\frac{1}{2}mv_0^2 - \frac{GMm}{r}\right)} = \frac{-GM}{v_0^2 - \frac{2GM}{r}} = \frac{-GM}{v_0^2\left(1 - \frac{2GM}{rv_0^2}\right)}$$

TODO: Calculate period of revolution.

Here are some calculations (forms) that can help with further calculations:

$$\widehat{L}_0^2 = b^2\left(\frac{2E}{-m}\right) \text{ or } \frac{\widehat{L}_0^2}{GM} = b^2\left(\frac{2E}{-GMm}\right) = \frac{b^2}{a} = a(1-e^2) \tag{494}$$

using Eq. 469, for $e^2 = 1 - \left(\frac{b}{a}\right)^2$.

112.4 Path in coordinates

For $\mathbf{r}$ centered at focus F and $\mathbf{r}, \mathbf{v}$ expressed in coordinates in the plane of the conic section path, i.e. in the plane formed by $\mathbf{r}, \mathbf{v}$, with the axes expressed as $\frac{\partial}{\partial x}$, $\frac{\partial}{\partial y}$,

$$\mathbf{r} = r\cos\theta\frac{\partial}{\partial x} + r\sin\theta\frac{\partial}{\partial y} \tag{495}$$

Compare this to Eq. (2.5-1) of Sec. 2.5.1 "Expressing r and v in the Perifocal System" of Bate, Mueller, White (1971) [118].

Recall the equation for the conic section in polar coordinates, with the coordinates centered at focus F:

$$r = \frac{a(1-e^2)}{1+e\cos\theta} \tag{496}$$

Compare this to Eq. (1.5-4) of Bate, Mueller, White (1971) [118].

Now

$$\mathbf{v} = \dot{\mathbf{r}} = (\dot{r}\cos\theta - r\dot{\theta}\sin\theta)\frac{\partial}{\partial x} + (\dot{r}\sin\theta + r\dot{\theta}\cos\theta)\frac{\partial}{\partial y} \tag{497}$$

Compare this to the equation on pp. 72, of Sec. 2.5.1 "Expressing r and v in the Perifocal System" of Bate, Mueller, White (1971) [118].

Now

$$\dot{r} = \frac{a(1-e^2)}{(1+e\cos\theta)^2}e\sin\theta\dot{\theta} = \frac{r^2}{a(1-e^2)}e\sin\theta\frac{\widehat{L}_0}{r^2} = \frac{\widehat{L}_0 e\sin\theta}{a(1-e^2)} = \frac{\widehat{L}_0 e\sin\theta}{\frac{-GMm}{2E}\left(\frac{L_0^2}{\left(\frac{(GM)^2 m}{-2E}\right)}\right)} =$$

$$\implies \dot{r} = \frac{GM}{\widehat{L}_0}e\sin\theta \tag{498}$$

Compare this to Eq. (2.5-2) of Sec. 2.5.1 "Expressing r and v in the Perifocal System" of Bate, Mueller, White (1971) [118].

Then

$$\implies \frac{a(1-e^2)}{(1+e\cos\theta)^2}\dot{\theta} = \frac{GM}{\widehat{L}_0} \text{ or}$$

$$r\dot{\theta} = \frac{GM}{\widehat{L}_0}(1+e\cos\theta) \tag{499}$$

Compare this to Eq. (2.5-3) on pp. 73 of Sec. 2.5.1 "Expressing r and v in the Perifocal System" of Bate, Mueller, White (1971) [118].

Then

$$\mathbf{v} = \frac{GM}{\widehat{L}_0}\left[(e\sin\theta\cos\theta - (1+e\cos\theta)\sin\theta)\frac{\partial}{\partial x} + \left(e\sin^2\theta + (1+e\cos\theta)\cos\theta\right)\frac{\partial}{\partial y}\right] =$$

$$\mathbf{v} = \frac{GM}{\widehat{L}_0}\left[\sin\theta\frac{\partial}{\partial x} + (e+\cos\theta)\frac{\partial}{\partial y}\right] = \frac{GM}{\widehat{L}_0}\left[\frac{y}{r}\frac{\partial}{\partial x} + (e+\frac{x}{r})\frac{\partial}{\partial y}\right] \tag{500}$$

Compare this to Eq. (2.5-4) on pp. 73 of Sec. 2.5.1 "Expressing r and v in the Perifocal System" of Bate, Mueller, White (1971) [118].

Consider this ansatz from subsection 4.4.3 "The f and g expressions from Ch. 4 section on "Position and velocity - A function of time" in Bate, Mueller, White (1971) [118].

Because $\mathbf{r}, \mathbf{v}$ lie in the plane (i.e. coplanar) containing the entire conic section trajectory, then make the following ansatz:

$$\begin{aligned}\mathbf{r} &= f\mathbf{r}_0 + g\mathbf{v}_0 \\ \dot{\mathbf{r}} =: \mathbf{v} &= \dot{f}\mathbf{r}_0 + \dot{g}\mathbf{v}_0\end{aligned} \tag{501}$$

Solve for f and g by taking the appropriate cross products:

$$\begin{aligned}\mathbf{r}\times\mathbf{v}_0 &= f\mathbf{r}_0\times\mathbf{v}_0 \\ \mathbf{r}\times\mathbf{r}_0 &= g\mathbf{v}_0\times\mathbf{r}\end{aligned} \tag{502}$$

112.5 Eccentricity anomaly; Eccentricity anomaly coordinates

Let the point of interest be P. Choose the origin for the coordinates (x, y) to have *its origin* at the *center of the ellipse*.

Let point C be the point at the center of the ellipse (origin), let point A be the projection of point P onto the x-axis, point F be the point at the focus.

Recall that the length from the center C to the focus, $\overline{\text{CF}}$, is ae, where a is the semi-major axis, and eccentricity is e.

The eccentricity anomaly E (it's just an angle) is given by

$$\cos E = \frac{x}{a} \tag{503}$$

Remember E is *not* the angle formed by triangle CAP, but rather, first take the line through P and A. This line is perpendicular to the x-axis. Then intersect that line with a circle, centered at C, of radius a. Then, the line from C to that intersection forms an angle E with the x-axis.

By using the ellipse equation and trigonometry,

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \Longrightarrow \cos^2 E + \frac{y^2}{b^2} = 1 \text{ or } \frac{y^2}{b^2} = \sin^2 E \text{ or}$$

$$\sin E = \frac{y}{b} \tag{504}$$

Now

$$\overline{\text{AF}} = \overline{\text{CF}} - \overline{\text{CA}} = ae - a\cos E$$

Use the right triangle PAF and the hypotenuse being r (distance from the focus F to point P) to find

$$\overline{\text{AF}} = r\cos(\pi - \theta) = -r\cos\theta$$

So

$$\cos\theta = \frac{a(\cos E - e)}{r}$$

Then, plug into the conic section equation, Eq. 465,

$$r = \frac{ed}{1 + e\cos\theta} = \frac{a(1-e^2)}{1 + e\cos\theta} = \frac{b^2/a}{1 + e\cos\theta} = \frac{b^2/a}{1 + e\left(\frac{a(\cos E - e)}{r}\right)}$$

$$\Longrightarrow 1 = \frac{b^2/a}{r + e(a(\cos E - e))}$$

$$\Longrightarrow r = \frac{b^2}{a} - e(a(\cos E - e)) = \frac{b^2}{a} + ae^2 - ea\cos E = a(1 - e\cos E)$$

$$\boxed{r = a(1 - e\cos E)} \tag{505}$$

Compare this to Eq. (4.6-9) on pp. 215 of Sec. 4.6 in "The Kepler Problem" of Bate, Mueller, White (1971) [118].

For the coordinates with the origin at the *focus* F, it is given by

$$\begin{aligned} x &= a(\cos E - e) \\ y &= b\sin E = a\sqrt{1-e^2}\sin E \end{aligned} \tag{506}$$

Compare this to Eqns. (4.6-7), (4.6-8) on pp. 215 of Sec. 4.6 in "The Kepler Problem" of Bate, Mueller, White (1971) [118].

Indeed, for $E = 0$, $x = a - ae$ (the distance from the focus F to the vertex is $a - ae$), $E = \pi/2$, $x = -ae$ (the distance from the focus F and back to the center C) and for $E = \pi$, $x = -a - ae$ (the distance to the other vertex).

112.5.1 Path in terms of eccentric anomaly

Take the time derivative of Eq. 506:

$$\begin{aligned}\dot{x} &= -a\sin E\dot{E} \\ \dot{y} &= a\sqrt{1-e^2}\cos E\dot{E}\end{aligned} \tag{507}$$

To determine $\dot{E}$, consider both expressions for the radial distance r for Eq. 505 and Eq. 465:

$$r = \frac{a(1-e^2)}{1+e\cos\theta} = \frac{b}{1+e\cos\theta}$$

$$\implies \dot{r} = \frac{-b}{(1+e\cos\theta)^2}(-e\sin\theta)\dot{\theta} = \frac{be\sin\theta\dot{\theta}}{(1+e\cos\theta)^2} =$$

$$\dot{r} = \frac{r^2 e}{b}\sin\theta\dot{\theta}$$

$$r = a(1-e\cos E)$$

$$\implies \dot{r} = +ae\sin E\dot{E}$$

$$\implies ae\sin E\dot{E} = \frac{r^2 e}{b}\sin\theta\dot{\theta} \text{ or } a\sin E\dot{E} = \frac{r^2}{b}\sin\theta\dot{\theta} = \frac{\widehat{L}_0\sin\theta}{b} = \frac{\widehat{L}_0}{b}\frac{y}{r}$$

$$\implies \dot{E} = \frac{\widehat{L}_0}{ab}\frac{y}{r\sin E} = \frac{\widehat{L}_0}{ar} = \frac{b}{a^{3/2}r}$$

Compare the expression for $\dot{E}$ with Eq. (4.6-11) on pp. 216 of Bate, Mueller, White (1971) [118].

Armed with an expression for $\dot{E}$, plug this back into the expression for $\dot{r}$:

$$\dot{r} = ae\sin E\left(\frac{\widehat{L}_0}{ar}\right) = \frac{e\sin E\widehat{L}_0}{r} \tag{508}$$

Compare this to the equation on pp. 216, "Position and Velocity - a function of time" in Ch. 4 of Bate, Mueller, White (1971) [118].

Use the expressions above in Eqns. 502. For $\mathbf{r}, \mathbf{r}_0$ centered at focus F:

$$xv_{0,y} - yv_{0,x} = a(\cos E - e)b\cos E_0\dot{E}_0 - b\sin E(-a\sin E_0\dot{E}_0) =$$

$$= ab\dot{E}_0\left[(\cos E - e)\cos E_0 + \sin E\sin E_0\right] = \frac{b^2}{\sqrt{ar}}\left[\cos\left(E - E_0\right) - e\cos E_0\right] = \frac{\widehat{L}^2\left(\frac{-m}{2E}\right)}{\sqrt{GM}\sqrt{\frac{m}{-2E}r}}\left[\cos\left(E - E_0\right) - e\cos E_0\right] =$$

$$= \frac{\widehat{L}_0 b}{r\sqrt{GM}}\left[\cos\left(E - E_0\right) - e\cos E_0\right]$$

Comparing this against the "right-hand side (RHS)" with $f(\mathbf{r}_0 \times \mathbf{v}_0)$, the RHS has a magnitude of $f\widehat{L}_0$ in the same direction as $(\mathbf{r} \times \mathbf{v}_0)$. Thus

$$f = \frac{b}{r\sqrt{GM}}\left[\cos\left(E - E_0\right) - e\cos E_0\right] \tag{509}$$

For the expression involving g in Eqns. 502,

$$\mathbf{r} \times \mathbf{r}_0 = xy_0 - yx_0 = a(\cos E - e)b\sin E_0 - b\sin Ea(\cos E_0 - e) = -ab\left[(\cos E_0)\sin E - (\cos E - e)\sin E_0\right]$$

with

$$ab = a\widehat{L}_0\sqrt{\frac{-m}{2E}} = a\widehat{L}_0\left(\frac{a}{GM}\right)^{1/2} = \left(\frac{a^3}{GM}\right)^{1/2}\widehat{L}_0$$

For the RHS with $g\mathbf{v}_0 \times \mathbf{r}_0$, $\mathbf{v}_0 \times \mathbf{r}_0$, by the right-hand rule or the right-hand orientation, it is in the *opposite* direction as $\mathbf{r} \times \mathbf{r}_0$. Thus, apply a negative side on the RHS.

$$\implies g = \left(\frac{a^3}{GM}\right)^{1/2}\left[(\cos E_0 - e)\sin E - (\cos E - e)\sin E_0\right] \tag{510}$$

Part XXX
Geometric Mechanics and Rigid-Body Flight

113 Configuration, velocity, and degrees of freedom

Definition 217 (Rigid body; Goldstein 3e, §§1.2–1.3, pp. 11–12, and §4.1, p. 134 [98])**.** A rigid body is a system of N mass points, with masses $m_i > 0$ and inertial positions $\mathbf{r}_i(t) \in \mathbb{R}^3$, subject for all t to the holonomic constraints

$$f_{ij}(\mathbf{r}_1, \dots, \mathbf{r}_N) = (\mathbf{r}_i - \mathbf{r}_j) \cdot (\mathbf{r}_i - \mathbf{r}_j) - c_{ij}^2 = 0, \qquad 1 \le i < j \le N, \tag{511}$$

with constants $c_{ij} = c_{ji} > 0$. Its configuration space is

$$Q_{\text{rigid}} = \big\{(\mathbf{r}_1, \dots, \mathbf{r}_N) \in \mathbb{R}^{3N} : |\mathbf{r}_i - \mathbf{r}_j| = c_{ij},\ 1 \le i < j \le N\big\}. \tag{512}$$

Goldstein writes Eq. (511) as $(\mathbf{r}_i - \mathbf{r}_j)^2 - c_{ij}^2 = 0$ (p. 12) and as Eq. (4.1), $r_{ij} = c_{ij}$, where $r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|$.

A point mass in Euclidean space has configuration space $Q = \mathbb{R}^3$. A rigid body with one material point fixed has configuration space $SO(3)$, whether or not that point is the center of mass. Choose a right-handed orthonormal body frame at that point and a right-handed inertial frame. Our rotation R maps body components to inertial components:

$$\mathbf{x}_I = R\mathbf{x}_B, \qquad R^T R = I, \qquad \det R = 1. \tag{513}$$

The columns of R are the body basis vectors expressed in the inertial frame; the inverse component transformation is $\mathbf{x}_B = R^T \mathbf{x}_I$. We distinguish configurations by a fixed material frame, even if the body's shape or inertia has a symmetry. Section 88.1 proves that these rotations form a three-dimensional embedded submanifold of $\mathbb{R}^9$. Its regular-value argument justifies the count $9 - 6 = 3$. The motion $t \mapsto R(t)$ is a curve on this manifold, rather than an arbitrary curve through nine independent matrix entries. A freely moving rigid body has configuration space $\mathbb{R}^3 \times SO(3)$, identified with $SE(3)$ as a manifold. Its group law is

$$(\mathbf{r}, R)(\mathbf{s}, S) = (\mathbf{r} + R\mathbf{s}, RS). \tag{514}$$

The rotation acts on the translation in Eq. (514); this is why the rigid-motion group is a semidirect product.

Rigid motions preserve the constraints of Definition 217. Fix body components $\mathbf{s}_i$ of the mass points, satisfying $|\mathbf{s}_i - \mathbf{s}_j| = c_{ij}$, and set $\mathbf{r}_i = \mathbf{r} + R\mathbf{s}_i$. The translation cancels in the difference, and $R^T R = I$ removes the rotation:

$$\begin{aligned}(\mathbf{r}_i - \mathbf{r}_j) \cdot (\mathbf{r}_i - \mathbf{r}_j) &= \big(R(\mathbf{s}_i - \mathbf{s}_j)\big)^T R(\mathbf{s}_i - \mathbf{s}_j) \\ &= (\mathbf{s}_i - \mathbf{s}_j)^T R^T R(\mathbf{s}_i - \mathbf{s}_j) \\ &= c_{ij}^2.\end{aligned}$$

The calculation uses only $R^T R = I$, so an improper orthogonal matrix ($\det R = -1$) also satisfies Eq. (511); it gives the mirror-image configuration. A continuous motion $R(t)$ starting at $R(0) = I$ has $\det R(t)$ continuous with values ± 1, hence $\det R(t) = 1$. This is why the configuration map (513) imposes $\det R = 1$.

Degrees of freedom count independent configuration coordinates. The three translational and three rotational degrees of freedom give twelve independent position–velocity state variables for a second-order rigid-body model. A unit quaternion uses four components to represent three rotational degrees of freedom, subject to one scalar constraint. A translational three-DOF flight model omits attitude dynamics under additional modeling assumptions; it does not follow merely by invoking symmetry reduction.

113.1 Direction cosines

Let $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ be the inertial basis and $\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3$ the body basis of Eq. (513), both orthonormal and right-handed. The direction cosines are

$$R_{ij} = \mathbf{e}_i \cdot \mathbf{b}_j = \cos \angle(\mathbf{e}_i, \mathbf{b}_j), \qquad i, j = 1, 2, 3; \tag{515}$$

the second equality holds because both are unit vectors. The nine R_{ij} form a real 3×3 matrix, $R \in M_3(\mathbb{R})$. Expand each basis in the other, with coefficients $\beta_j{}^k$ and $\gamma_i{}^k$, and dot with a basis vector:

$$\mathbf{b}_j = \beta_j{}^k \mathbf{e}_k : \qquad \mathbf{e}_i \cdot \mathbf{b}_j = \beta_j{}^k\, \mathbf{e}_i \cdot \mathbf{e}_k = \beta_j{}^k \delta_{ik} = \beta_j{}^i,$$

$$\mathbf{e}_i = \gamma_i{}^k \mathbf{b}_k : \qquad \mathbf{b}_j \cdot \mathbf{e}_i = \gamma_i{}^k \, \mathbf{b}_j \cdot \mathbf{b}_k = \gamma_i{}^k \delta_{jk} = \gamma_i{}^j .$$

Both left-hand sides are R_{ij} by Eq. (515), so $\beta_j{}^i = R_{ij}$ and $\gamma_i{}^j = R_{ij}$:

$$\mathbf{b}_j = R_{ij} \mathbf{e}_i , \qquad \mathbf{e}_i = R_{ij} \mathbf{b}_j . \tag{516}$$

Let a vector be $\mathbf{a} = a_I^i \mathbf{e}_i = a_B^j \mathbf{b}_j$; a_I^i is component i of $\mathbf{a}_I$.

$$\begin{aligned} a_I^i &= \mathbf{e}_i \cdot \mathbf{a} = a_B^j \, \mathbf{e}_i \cdot \mathbf{b}_j = R_{ij} a_B^j , \\ a_B^j &= \mathbf{b}_j \cdot \mathbf{a} = a_I^i \, \mathbf{b}_j \cdot \mathbf{e}_i = R_{ij} a_I^i . \end{aligned} \tag{517}$$

The first line is Eq. (513): R maps body components to inertial components of the same vector,

$$R \colon \mathbb{R}^3 \to \mathbb{R}^3 , \qquad (a_B^1, a_B^2, a_B^3) \mapsto (a_I^1, a_I^2, a_I^3) , \quad a_I^i = R_{ij} a_B^j .$$

This is a *passive rotation*: it changes the basis in which $\mathbf{a}$ is described; $\mathbf{a}$ itself does not move. For the *active* interpretation, hold the inertial basis fixed and let $\mathcal{R}$ rotate $\mathbf{e}_j$ into $\mathbf{b}_j$:

$$\mathcal{R} \mathbf{e}_j = \mathbf{b}_j , \qquad \mathcal{R} \mathbf{a} = a_I^j \mathbf{b}_j = R_{ij} a_I^j \mathbf{e}_i . \tag{518}$$

Here the vector changes and both its input and output components use the inertial basis. Goldstein's $A = R^T$ instead converts inertial components to body components passively, or represents $\mathcal{R}^{-1}$ actively in the fixed inertial basis (§4.2, Eqs. (4.18)–(4.19), pp. 142–143 [98]).

Orthonormality of each basis constrains the nine R_{ij}. Dot the expansions (516) pairwise:

$$\begin{aligned} \delta_{jk} = \mathbf{b}_j \cdot \mathbf{b}_k &= R_{ij} R_{lk} \, \mathbf{e}_i \cdot \mathbf{e}_l = R_{ij} R_{lk} \delta_{il} = R_{ij} R_{ik} , \\ \delta_{il} = \mathbf{e}_i \cdot \mathbf{e}_l &= R_{ij} R_{lk} \, \mathbf{b}_j \cdot \mathbf{b}_k = R_{ij} R_{lk} \delta_{jk} = R_{ij} R_{lj} . \end{aligned} \tag{519}$$

The first line is unchanged under $j \leftrightarrow k$, so it gives six equations ($j \le k$) on nine entries; the regular-value proof of Section 88.1 makes the count $9 - 6 = 3$ rigorous. The nine direction cosines are therefore not independent generalized coordinates.

Right-handedness, $\mathbf{b}_1 \times \mathbf{b}_2 = \mathbf{b}_3$, fixes the determinant. In right-handed inertial components, $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w} = \epsilon_{ikl} u^i v^k w^l$, and by Eq. (516) $\mathbf{b}_j$ has components R_{ij}:

$$\det R = \epsilon_{ikl} R_{i1} R_{k2} R_{l3} = (\mathbf{b}_1 \times \mathbf{b}_2) \cdot \mathbf{b}_3 = \mathbf{b}_3 \cdot \mathbf{b}_3 = 1 . \tag{520}$$

Hence $R \in GL(3, \mathbb{R})$, the invertible real 3×3 matrices: $\det R = 1 \neq 0$, and the second line of Eqs. (517) inverts the first. Substituting $a_I^i = R_{ik} a_B^k$ and using the first line of Eqs. (519),

$$R_{ij} a_I^i = R_{ij} R_{ik} a_B^k = \delta_{jk} a_B^k = a_B^j .$$

Equations (519) and (520) place R in $SO(3) \subset O(3) \subset GL(3, \mathbb{R})$.

Engineering texts call such a matrix the direction cosine matrix (DCM). The name does not fix which matrix is meant: R^T also consists of direction cosines, and many texts use it. Read the convention from the defining equation. In the table, the space (inertial) basis is listed first and the body basis second.

source	bases	direction cosines	matrix
ours	$\mathbf{e}_i$; $\mathbf{b}_j$	$R_{ij} = \mathbf{e}_i \cdot \mathbf{b}_j$	R
Goldstein [98]	$\mathbf{i}, \mathbf{j}, \mathbf{k}$; $\mathbf{i}', \mathbf{j}', \mathbf{k}'$	$a_{ij} = \cos\theta_{ij}$	$A = R^T$
Landau [18]	X, Y, Z; x_1, x_2, x_3	n_{3x}, n_{3y}, n_{3z}	column 3 of R
Wie [20]	$\vec{a}_i$; $\vec{b}_i$	$C_{ij} = \vec{b}_i \cdot \vec{a}_j$	$C^{B/A} = R^T$
Shuster [21]	reference; body	attitude matrix	$A = R^T$
Groves [22]	i; b	$\mathbf{x}^i = C_b^i \mathbf{x}^b$	$C_b^i = R$

Goldstein (§4.1, pp. 137–138; §4.2, pp. 139–140) takes the primed axes fixed in the body, so $\mathbf{i}, \mathbf{j}, \mathbf{k}$ are $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ and $\mathbf{i}', \mathbf{j}', \mathbf{k}'$ are $\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3$. In his Eq. (4.2) the first index of θ_{ij} is primed and the second unprimed, so $\cos\theta_{ij} = \mathbf{b}_i \cdot \mathbf{e}_j = R_{ji}$, and his $a_{ij} = \cos\theta_{ij}$ (Eq. (4.11)) gives $A = R^T$. His Eqs. (4.5) and (4.12′), $x_i' = a_{ij} x_j$, are the second of Eqs. (517). His Eq. (4.9), $\sum_l \cos\theta_{lm'} \cos\theta_{lm} = \delta_{m'm}$, becomes $R_{m'l} R_{ml} = \delta_{m'm}$ on substituting $\cos\theta_{lm} = R_{ml}$;

his Eq. (4.15), $a_{ij}a_{ik}=\delta_{jk}$, becomes $R_{ji}R_{ki}=\delta_{jk}$ on substituting $a_{ij}=R_{ji}$. Renaming $(m',l,m)\to(i,j,l)$ in the first and $(j,i,k)\to(i,j,l)$ in the second gives $R_{ij}R_{lj}=\delta_{il}$, the second of Eqs. (519).

Landau (§31, p. 96) uses fixed axes X,Y,Z and body axes x_1,x_2,x_3, and fixes the orientation by three independent angles, the Eulerian angles of §35 (p. 110). He does not name the direction cosines. In §37 (p. 121) his n_{3x},n_{3y},n_{3z}, the fixed-axis components of the unit vector $\mathbf{n}_3$ along x_3, are R_{13},R_{23},R_{33}.

Wie (§5.1, pp. 323–325) writes Eq. (5.1) as $\vec{b}_i=C_{ij}\vec{a}_j$ with $C_{ij}=\vec{b}_i\cdot\vec{a}_j$, calls $C^{B/A}$ the direction cosine matrix "to B from A" (Eq. (5.2)), and transforms components by $\mathbf{H}'=C^{B/A}\mathbf{H}$ (Eq. (5.8)). With A the inertial frame, $C^{B/A}=R^T$. His elementary rotations of Eq. (5.9) are therefore transposes of active ones: $C_k(\theta)=R_k(\theta)^T$, with $R_1=R_x$, $R_2=R_y$, $R_3=R_z$. For example,

$$C_3(\theta)=\begin{pmatrix}\cos\theta & \sin\theta & 0\\ -\sin\theta & \cos\theta & 0\\ 0 & 0 & 1\end{pmatrix}=R_z(\theta)^T.$$

Transposing Eq. (532) gives the corresponding yaw–pitch–roll product, $R^T=R_x(\phi)^TR_y(\theta)^TR_z(\psi)^T=C_1(\phi)C_2(\theta)C_3(\psi)$.

Shuster's attitude matrix maps reference-frame components to body components, $\mathbf{a}_B=A\mathbf{a}_I$, so $A=R^T$. Inertial-navigation texts such as Groves write C_β^α for the matrix taking β-frame components to α-frame components, so the body-to-inertial C_b^i is R.

The components of a fixed vector turn opposite to the frame. Let the body frame turn by α about $\mathbf{e}_3$:

$$\mathbf{b}_1=\cos\alpha\,\mathbf{e}_1+\sin\alpha\,\mathbf{e}_2,\qquad \mathbf{b}_2=-\sin\alpha\,\mathbf{e}_1+\cos\alpha\,\mathbf{e}_2,\qquad \mathbf{b}_3=\mathbf{e}_3.$$

For $\mathbf{a}=\mathbf{e}_1$, $a_I^i=\delta_{i1}$, and the second of Eqs. (517) with Eq. (515) gives

$$a_B^j=R_{ij}\delta_{i1}=R_{1j}=\mathbf{e}_1\cdot\mathbf{b}_j:\qquad a_B^1=\cos\alpha,\quad a_B^2=-\sin\alpha,\quad a_B^3=0.$$

These are the components of $\mathbf{e}_1$ rotated by $-\alpha$. At $\alpha=\pi/2$, $a_B=(0,-1,0)$, which checks against $\mathbf{b}_2=-\mathbf{e}_1$.

113.2 Coordinate changes, conjugation, and the adjoint representation

Goldstein §4.3, Eqs. (4.40)–(4.41), pp. 148–149 [98], transforms both the input and output coordinates of an operator. Write the coordinate change as

$$\begin{aligned}&B\colon\mathbb{R}^3_{\text{old}}\longrightarrow\mathbb{R}^3_{\text{new}},\qquad B\in GL(3,\mathbb{R}),\\ &x\longmapsto y=Bx,\qquad y^i=B_{ij}x^j.\end{aligned}\tag{521}$$

The subscripts label coordinate copies of $\mathbb{R}^3$ describing the same vectors. For our body-to-inertial conversion, $B=R\in SO(3)$. Let $A\in\mathrm{End}(\mathbb{R}^3_{\text{old}})$ be any linear map and $F\in\mathbb{R}^3_{\text{old}}$ any input column vector; put $G=AF$. Here $\mathrm{End}(V)$ is the vector space of linear maps $V\to V$; in these coordinates A is a 3×3 matrix. Set $F'=BF$ and $G'=BG$. Then

$$\begin{aligned}&G'=BG=BAF=BAB^{-1}F'=A'F',\\ &A'=BAB^{-1},\qquad A\in\mathrm{End}(\mathbb{R}^3_{\text{old}}),\quad A'\in\mathrm{End}(\mathbb{R}^3_{\text{new}}).\end{aligned}\tag{522}$$

This is Goldstein's *similarity transformation*: the geometric operator is unchanged. The factor B^{-1} converts the input to old coordinates; A acts; B converts the output to new coordinates.

For a matrix Lie group $G\subseteq GL(n,\mathbb{R})$ with Lie algebra $\mathfrak{g}=T_IG$ and $B\in G$, conjugation and its derivative at the identity are

$$\begin{aligned}&c_B\colon G\longrightarrow G,\quad h\longmapsto BhB^{-1},\\ &\mathrm{Ad}_B=(dc_B)_I\colon\mathfrak{g}\longrightarrow\mathfrak{g},\quad X\longmapsto\frac{d}{dt}\big(B\exp(tX)B^{-1}\big)\Big|_{t=0}=BXB^{-1}.\end{aligned}\tag{523}$$

Thus each Ad_B is a similarity transformation restricted to $\mathfrak{g}$. The *adjoint representation* is the homomorphism

$$\mathrm{Ad}\colon G\longrightarrow GL(\mathfrak{g}),\qquad B\longmapsto\mathrm{Ad}_B,$$

proved in Proposition 92. For $B\in SO(3)$, the hat map (257) makes this concrete:

$$\mathrm{Ad}_B\colon\mathfrak{so}(3)\longrightarrow\mathfrak{so}(3),\qquad\widehat{\omega}\longmapsto B\widehat{\omega}B^T=\widehat{B\omega}.\tag{524}$$

Indeed, for every $v \in \mathbb{R}^3$, rotation invariance of the cross product gives

$$(B\widehat{\omega}B^T)v = B(\omega \times B^T v) = (B\omega) \times (BB^T v) = (B\omega) \times v = \widehat{B\omega}\, v.$$

With $B = R$, this identifies the coordinate laws $\omega_I = R\omega_B$ and $\widehat{\omega_I} = R\widehat{\omega_B}R^T$.

113.3 Euler angles and Tait–Bryan coordinates

Our convention remains $a_I = Ra_B$: a passive change of components. The same matrix gives the active attitude of a body initially aligned with the inertial axes. Put $c(\phi) \equiv \cos\phi$ and $s(\phi) \equiv \sin\phi$. The elementary positive right-handed active rotations are

$$\begin{aligned} R_x(\phi) &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & c(\phi) & -s(\phi) \\ 0 & s(\phi) & c(\phi) \end{pmatrix}, \\ R_y(\phi) &= \begin{pmatrix} c(\phi) & 0 & s(\phi) \\ 0 & 1 & 0 \\ -s(\phi) & 0 & c(\phi) \end{pmatrix}, \\ R_z(\phi) &= \begin{pmatrix} c(\phi) & -s(\phi) & 0 \\ s(\phi) & c(\phi) & 0 \\ 0 & 0 & 1 \end{pmatrix}. \end{aligned} \tag{525}$$

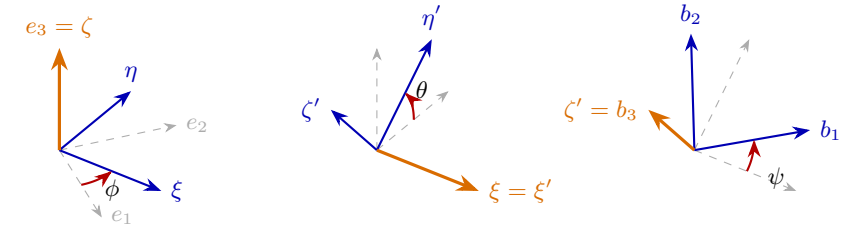
Goldstein's moving-axis ZXZ sequence. The dictionary to Goldstein §4.4, pp. 151–154 [98], is

$$(\phi_G, \theta_G, \psi_G) = (\phi, \theta, \psi), \qquad (x, x') = (a_I, a_B), \qquad A_G = R_E^T = R_E^{-1}.$$

Let Q_k have the kth frame's basis vectors as columns in inertial coordinates. His successive positive axis rotations give

$$\begin{aligned} Q_0 &= I, & Q_1 &= R_z(\phi), \\ Q_2 &= Q_1 R_x(\theta), & Q_3 &= Q_2 R_z(\psi) = R_E. \end{aligned} \tag{526}$$

The physical axes are e_3, $Q_1 e_1$ (the line of nodes), and $Q_2 e_3$.



$Q_0 \longrightarrow Q_1$ $\qquad$ $Q_1 \longrightarrow Q_2$ $\qquad$ $Q_2 \longrightarrow Q_3$

ZXZ: previous axes dashed, new axes blue, rotation axis orange. Illustrated at $(\phi, \theta, \psi) = (35°, 40°, 30°)$.

For one fixed vector $a = a_k^j Q_k e_j$, $a_0 = a_I$ and $a_3 = a_B$:

$$\begin{aligned} a_1 &= \underbrace{R_z(-\phi)}_{D_G} a_0, \\ a_2 &= \underbrace{R_x(-\theta)}_{C_G} a_1, \\ a_3 &= \underbrace{R_z(-\psi)}_{B_G} a_2 = B_G C_G D_G a_I = A_G a_I, \\ A_G &= R_z(-\psi)R_x(-\theta)R_z(-\phi). \end{aligned} \tag{527}$$

Thus, in our body-to-inertial convention,

$$R_E = R_z(\phi)R_x(\theta)R_z(\psi). \tag{528}$$

The multiplication is explicit:

$$R_z(\phi)R_x(\theta) = \begin{pmatrix} c(\phi) & -s(\phi)c(\theta) & s(\phi)s(\theta) \\ s(\phi) & c(\phi)c(\theta) & -c(\phi)s(\theta) \\ 0 & s(\theta) & c(\theta) \end{pmatrix},$$

$$R_E = \begin{pmatrix} c(\phi)c(\psi) - s(\phi)c(\theta)s(\psi) & -c(\phi)s(\psi) - s(\phi)c(\theta)c(\psi) & s(\phi)s(\theta) \\ s(\phi)c(\psi) + c(\phi)c(\theta)s(\psi) & -s(\phi)s(\psi) + c(\phi)c(\theta)c(\psi) & -c(\phi)s(\theta) \\ s(\theta)s(\psi) & s(\theta)c(\psi) & c(\theta) \end{pmatrix}. \tag{529}$$

Its transpose is Goldstein's Eq. (4.46).

Active readings: specify the axes. The positive physical rotations about those moving axes, all represented in the fixed inertial basis, are

$$U_1 = Q_1, \qquad U_2 = Q_1 R_x(\theta) Q_1^T, \qquad U_3 = Q_2 R_z(\psi) Q_2^T.$$

Their chronological product is

$$U_2 U_1 = Q_1 R_x(\theta) Q_1^T Q_1 = Q_2, \qquad U_3 U_2 U_1 = Q_2 R_z(\psi) Q_2^T Q_2 = R_E.$$

Equivalently, the same R_E actively rotates vectors about fixed axes in chronological order $z(\psi), x(\theta), z(\phi)$. Reading Goldstein's D_G, C_G, B_G actively instead gives

$$v_I \xmapsto{D_G} v_I^{(1)} \xmapsto{C_G} v_I^{(2)} \xmapsto{B_G} v_I^{(3)} = A_G v_I = R_E^{-1} v_I,$$

with all components in the inertial basis and fixed-axis rotations $z(-\phi), x(-\theta), z(-\psi)$. Goldstein defines the angles by rotating axes and uses the passive component law. For physical body motion the active R_E is equally useful; its inverse describes the opposite motion.

Goldstein's y-convention: ZYZ. Replace the second moving axis by $Q_1 e_2$:

$$R_{E,y}(\phi, \theta, \psi) = R_z(\phi) R_y(\theta) R_z(\psi), \qquad A_{G,y} = R_{E,y}^T. \tag{530}$$

The products are

$$R_z(\phi)R_y(\theta) = \begin{pmatrix} c(\phi)c(\theta) & -s(\phi) & c(\phi)s(\theta) \\ s(\phi)c(\theta) & c(\phi) & s(\phi)s(\theta) \\ -s(\theta) & 0 & c(\theta) \end{pmatrix},$$

$$R_{E,y} = \begin{pmatrix} c(\phi)c(\theta)c(\psi) - s(\phi)s(\psi) & -c(\phi)c(\theta)s(\psi) - s(\phi)c(\psi) & c(\phi)s(\theta) \\ s(\phi)c(\theta)c(\psi) + c(\phi)s(\psi) & -s(\phi)c(\theta)s(\psi) + c(\phi)c(\psi) & s(\phi)s(\theta) \\ -s(\theta)c(\psi) & s(\theta)s(\psi) & c(\theta) \end{pmatrix}. \tag{531}$$

The exact dictionary between his x- and y-conventions follows from

$$\begin{aligned} R_x(\theta) &= R_z(-\pi/2) R_y(\theta) R_z(\pi/2), \\ R_E(\phi, \theta, \psi) &= R_z(\phi - \pi/2) R_y(\theta) R_z(\psi + \pi/2) \\ &= R_{E,y}(\phi - \pi/2, \theta, \psi + \pi/2). \end{aligned}$$

The active/passive rules above apply with x replaced by y.

Standard ZYX yaw–pitch–roll convention. We use the standard aerospace 3–2–1 sequence with yaw ψ, pitch θ, and roll ϕ; our matrix direction is body to inertial. For the same angles, MathWorks's `eul2rotm` returns R, while its aerospace `angle2dcm` returns R^T [23]. The positive moving-axis sequence is $z(\psi), y'(\theta), x''(\phi)$; equivalently, the fixed-axis active order is $x(\phi), y(\theta), z(\psi)$:

$$R = R_z(\psi) R_y(\theta) R_x(\phi), \qquad a_B = R_x(-\phi) R_y(-\theta) R_z(-\psi) a_I. \tag{532}$$

The corresponding multiplication gives

$$R_y(\theta)R_x(\phi) = \begin{pmatrix} c(\theta) & s(\theta)s(\phi) & s(\theta)c(\phi) \\ 0 & c(\phi) & -s(\phi) \\ -s(\theta) & c(\theta)s(\phi) & c(\theta)c(\phi) \end{pmatrix},$$

$$R = \begin{pmatrix} c(\psi)c(\theta) & c(\psi)s(\theta)s(\phi) - s(\psi)c(\phi) & c(\psi)s(\theta)c(\phi) + s(\psi)s(\phi) \\ s(\psi)c(\theta) & s(\psi)s(\theta)s(\phi) + c(\psi)c(\phi) & s(\psi)s(\theta)c(\phi) - c(\psi)s(\phi) \\ -s(\theta) & c(\theta)s(\phi) & c(\theta)c(\phi) \end{pmatrix}. \tag{533}$$

Goldstein's "xyz-convention" means three distinct axes; here we fix its order as ZYX. The dictionary of chronological rotations is

convention	moving axes (intrinsic)	fixed axes (active)
x: ZXZ	$z(\phi), x'(\theta), z''(\psi)$	$z(\psi), x(\theta), z(\phi)$
y: ZYZ	$z(\phi), y'(\theta), z''(\psi)$	$z(\psi), y(\theta), z(\phi)$
ours: ZYX	$z(\psi), y'(\theta), x''(\phi)$	$x(\phi), y(\theta), z(\psi)$

Goldstein's names are yaw (heading), pitch (attitude), and roll (bank). For the same attitude,

$$R_E(\phi_x, \theta_x, \psi_x) = R_{E,y}(\phi_y, \theta_y, \psi_y) = R_z(\psi)R_y(\theta)R_x(\phi) = R.$$

Use the entries of this common matrix to convert between all three:

$$\begin{aligned} \phi_x &= \operatorname{atan2}(R_{13}, -R_{23}), & \theta_x &= \arccos R_{33}, & \psi_x &= \operatorname{atan2}(R_{31}, R_{32}), \\ \phi_y &= \operatorname{atan2}(R_{23}, R_{13}), & \theta_y &= \arccos R_{33}, & \psi_y &= \operatorname{atan2}(R_{32}, -R_{31}), \\ \psi &= \operatorname{atan2}(R_{21}, R_{11}), & \theta &= \arcsin(-R_{31}), & \phi &= \operatorname{atan2}(R_{32}, R_{33}). \end{aligned} \tag{534}$$

Here $0 < \theta_x, \theta_y < \pi$ and $-\pi/2 < \theta < \pi/2$; outer angles are taken modulo 2π. In particular,

$$\theta = \arcsin\bigl(-s(\theta_x)s(\psi_x)\bigr) = \arcsin\bigl(s(\theta_y)c(\psi_y)\bigr).$$

Both repeated-axis conventions are singular at $\sin\theta = 0$; for example $R_E(\phi, 0, \psi) = R_z(\phi + \psi)$. Our Tait–Bryan angles are regular near the identity and singular at $\cos\theta = 0$.

For the rate calculation use our ZYX choice (532). Let $E_i = \widehat{\mathbf{e}}_i$ be the generators of Eq. (237). Differentiating and multiplying by R^T gives

$$\begin{aligned} R^T\dot{R} = {} & E_1\dot{\phi} + R_x^T E_2 R_x \dot{\theta} \\ & + R_x^T R_y^T E_3 R_y R_x \dot{\psi}. \end{aligned}$$

Define $\widehat{\boldsymbol{\omega}}_B = R^T\dot{R}$. Using $R^T\widehat{\mathbf{a}}R = \widehat{R^T\mathbf{a}}$ gives

$$\boldsymbol{\omega}_B = \begin{pmatrix} 1 & 0 & -s(\theta) \\ 0 & c(\phi) & s(\phi)c(\theta) \\ 0 & -s(\phi) & c(\phi)c(\theta) \end{pmatrix} \begin{pmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{pmatrix}. \tag{535}$$

The determinant of the rate matrix is $c(\theta)\bigl(c(\phi)^2 + s(\phi)^2\bigr) = c(\theta)$. When $c(\theta) \neq 0$, the inverse function theorem gives a local coordinate chart after restricting the angle ranges. At $\theta = \pm\pi/2$ the angle map loses rank (gimbal lock); $SO(3)$ remains smooth. A unit quaternion avoids this angle singularity, but uses four constrained components and the two representatives $q, -q$ of one rotation.

113.4 Cayley–Klein parameters, quaternions, and angles

Use subscripts to distinguish Goldstein's complex parameters (§4.5, pp. 154–155 [98]) from the angles:

$$\begin{aligned} &\alpha_{\mathrm{CK}}, \beta_{\mathrm{CK}}, \gamma_{\mathrm{CK}}, \delta_{\mathrm{CK}} \in \mathbb{C}, \qquad \gamma_{\mathrm{CK}} = -\beta_{\mathrm{CK}}^*, \quad \delta_{\mathrm{CK}} = \alpha_{\mathrm{CK}}^*, \\ &|\alpha_{\mathrm{CK}}|^2 + |\beta_{\mathrm{CK}}|^2 = 1, \qquad K = \begin{pmatrix} \alpha_{\mathrm{CK}} & \beta_{\mathrm{CK}} \\ -\beta_{\mathrm{CK}}^* & \alpha_{\mathrm{CK}}^* \end{pmatrix}. \end{aligned} \tag{536}$$

Here * denotes complex conjugation. The constraints give

$$K^\dagger K = (|\alpha_{\mathrm{CK}}|^2 + |\beta_{\mathrm{CK}}|^2)I = I, \qquad \det K = |\alpha_{\mathrm{CK}}|^2 + |\beta_{\mathrm{CK}}|^2 = 1; \qquad K \in SU(2).$$

Dictionary to our quaternion. Goldstein's four real *Euler parameters* are

$$\begin{aligned}(e_0,e_1,e_2,e_3)_G=(q_0,q^1,q^2,q^3)\in\mathbb{R}^4,\qquad q_0^2+(q^1)^2+(q^2)^2+(q^3)^2=1,\\ \alpha_{\mathrm{CK}}=q_0+iq^3,\qquad \beta_{\mathrm{CK}}=q^2+iq^1,\qquad q=q_0+q^je_j\in S^3\subset\mathbb{H}.\end{aligned}\tag{537}$$

The e_j in q are our Hamilton units (221); the i in $\alpha_{\mathrm{CK}},\beta_{\mathrm{CK}}$ is the complex unit. Conversely,

$$q_0=\Re\alpha_{\mathrm{CK}},\quad q^3=\Im\alpha_{\mathrm{CK}},\qquad q^2=\Re\beta_{\mathrm{CK}},\quad q^1=\Im\beta_{\mathrm{CK}}.$$

Under our quaternion–Pauli dictionary (328),

$$U(q)=q_0I-iq^j\sigma_j=\begin{pmatrix}q_0-iq^3 & -q^2-iq^1\\ q^2-iq^1 & q_0+iq^3\end{pmatrix},\qquad K=U(q)^\dagger=U(\bar q).\tag{538}$$

Thus the covering map (272) gives

$$\begin{aligned}R(q)=\Phi(U(q))=(q_0^2-\mathbf{q}\cdot\mathbf{q})I+2\mathbf{q}\mathbf{q}^T+2q_0\widehat{\mathbf{q}},\\ A_G=\Phi(K)=\Phi(U(q)^{-1})=R(q)^{-1}=R(q)^T,\qquad \mathbf{q}=(q^1,q^2,q^3)^T.\end{aligned}\tag{539}$$

This is Eq. (300). For example, Goldstein's complex entry becomes

$$\begin{aligned}(A_G)_{13}&=\gamma_{\mathrm{CK}}\delta_{\mathrm{CK}}-\alpha_{\mathrm{CK}}\beta_{\mathrm{CK}}\\ &=(-q^2+iq^1)(q_0-iq^3)-(q_0+iq^3)(q^2+iq^1)\\ &=2(q^1q^3-q_0q^2)=R(q)_{31}.\end{aligned}$$

All entries are real quadratic expressions; q and $-q$ give the same $R\in SO(3)$. These are four constrained real coordinates for attitude, with no Euler-angle singularity.

Angles to quaternions and back. Put $q_a(\phi)=c(\phi/2)+e_as(\phi/2)$ for $a=1,2,3$. Hamilton multiplication follows the same order as the body-to-inertial matrices:

$$\begin{aligned}q_{ZXZ}&=q_3(\phi)q_1(\theta)q_3(\psi),\\ q_{ZYZ}&=q_3(\phi)q_2(\theta)q_3(\psi),\\ q_{ZYX}&=q_3(\psi)q_2(\theta)q_1(\phi).\end{aligned}\tag{540}$$

Writing the results scalar first, with $u=(\phi+\psi)/2$ and $v=(\phi-\psi)/2$,

$$q_{ZXZ}\longleftrightarrow\begin{pmatrix}c(\theta/2)c(u)\\ s(\theta/2)c(v)\\ s(\theta/2)s(v)\\ c(\theta/2)s(u)\end{pmatrix},\qquad q_{ZYZ}\longleftrightarrow\begin{pmatrix}c(\theta/2)c(u)\\ -s(\theta/2)s(v)\\ s(\theta/2)c(v)\\ c(\theta/2)s(u)\end{pmatrix}.\tag{541}$$

For Tait–Bryan angles,

$$q_{ZYX}\longleftrightarrow\begin{pmatrix}c(\psi/2)c(\theta/2)c(\phi/2)+s(\psi/2)s(\theta/2)s(\phi/2)\\ c(\psi/2)c(\theta/2)s(\phi/2)-s(\psi/2)s(\theta/2)c(\phi/2)\\ c(\psi/2)s(\theta/2)c(\phi/2)+s(\psi/2)c(\theta/2)s(\phi/2)\\ s(\psi/2)c(\theta/2)c(\phi/2)-c(\psi/2)s(\theta/2)s(\phi/2)\end{pmatrix}.\tag{542}$$

In particular Goldstein's ZXZ angles give directly

$$\alpha_{\mathrm{CK}}=c(\theta/2)e^{iu},\qquad \beta_{\mathrm{CK}}=s(\theta/2)\big(s(v)+ic(v)\big)=i\,s(\theta/2)e^{-iv}.$$

For the reverse conversion, compute $R(q)$ from Eq. (539) and apply Eq. (534) on its stated angle branches.

113.5 Euler's theorem: one axis and one finite angle

Goldstein §4.6, pp. 155–161, Eqs. (4.48), (4.56)–(4.61) [98]. In our notation the statement is

$$R \in SO(3) \quad \Longrightarrow \quad \exists\, n \in S^2 :\ Rn = n.$$

Indeed, putting $d = \det(R - I)$,

$$\begin{aligned} (R-I)R^T &= I - R^T, \\ d &= \det\big((R-I)R^T\big) = \det(I - R^T) = \det(I-R) = -\det(R-I) = -d, \\ d &= 0, \qquad \ker(R-I) \neq \{0\}. \end{aligned}$$

Choose $n \in \ker(R-I)$ with $|n| = 1$, $|u| = 1$, $u \perp n$, and $v = n \times u$. Then $S = (u\ v\ n) \in SO(3)$ and

$$S^T R S = \begin{pmatrix} c(\phi) & -s(\phi) & 0 \\ s(\phi) & c(\phi) & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{543}$$

The sign of n can be chosen so that $0 \le \phi \le \pi$. Consequently,

$$\begin{aligned} \operatorname{spec}_{\mathbb{C}} R &= \{1, e^{i\phi}, e^{-i\phi}\}, \qquad \operatorname{tr} R = 1 + 2c(\phi), \\ \dim\ker(R-I) &= \begin{cases} 3, & R = I, \\ 1, & R \neq I. \end{cases} \end{aligned} \tag{544}$$

The characteristic polynomial and Cayley–Hamilton identity are

$$\begin{aligned} \chi_R(\lambda) &= (\lambda - 1)(\lambda^2 - 2c(\phi)\lambda + 1), \\ \chi_R(R) &= R^3 - \tau R^2 + \tau R - I = 0, \qquad \tau = \operatorname{tr} R. \end{aligned} \tag{545}$$

The determinant argument uses odd dimension; $-I_2 \in SO(2)$ has no nonzero fixed vector.

Finite rotations and the exponential map. Goldstein §4.7, pp. 161–163, Eqs. (4.62)–(4.63), uses axis–angle coordinates for the same rotation. Put $N = \widehat{n} \in \mathfrak{so}(3)$, $Nx = n \times x$. The vector triple product gives

$$N^2 x = n \times (n \times x) = n(n \cdot x) - x, \qquad N^2 = nn^T - I, \qquad N^3 = Nnn^T - N = -N.$$

Reducing the exponential series with $N^3 = -N$,

$$\begin{aligned} \exp(\phi N) &= I + \sum_{m=0}^{\infty} \frac{(-1)^m \phi^{2m+1}}{(2m+1)!} N + \sum_{m=1}^{\infty} \frac{(-1)^{m-1}\phi^{2m}}{(2m)!} N^2 \\ &= I + s(\phi)N + (1 - c(\phi))N^2 \\ &= c(\phi) I + (1 - c(\phi)) nn^T + s(\phi)\widehat{n} =: R_{n,\phi}. \end{aligned} \tag{546}$$

Thus the exponential is surjective,

$$\exp\colon \mathfrak{so}(3) \longrightarrow SO(3), \qquad \phi\widehat{n} \longmapsto R_{n,\phi},$$

and the corresponding active vector map is precisely

$$\begin{aligned} R_{n,\phi}\colon \mathbb{R}^3 &\longrightarrow \mathbb{R}^3, \\ x &\longmapsto c(\phi)x + (1 - c(\phi))n(n \cdot x) + s(\phi) n \times x. \end{aligned} \tag{547}$$

This agrees with Eq. (247). Goldstein's Eq. (4.62) has the opposite active sense:

$$\begin{aligned} A_G &= R_{n,\phi}^T = R_{n,-\phi}, \\ A_G x &= c(\phi)x + (1 - c(\phi))n(n \cdot x) + s(\phi) x \times n. \end{aligned} \tag{548}$$

Recovering the axis and angle. Equations (544) and (546) give

$$\begin{aligned} \phi &= \arccos\!\left(\frac{\operatorname{tr} R - 1}{2}\right), \qquad R - R^T = 2s(\phi)\widehat{n}, \\ n &= \frac{1}{2s(\phi)} \begin{pmatrix} R_{32} - R_{23} \\ R_{13} - R_{31} \\ R_{21} - R_{12} \end{pmatrix}, \qquad 0 < \phi < \pi. \end{aligned} \tag{549}$$

At $R = I$, every axis is possible. At $\phi = \pi$,

$$\frac{R + I}{2} = nn^T, \qquad n \sim -n.$$

For example, choose j with maximal R_{jj} and take $n_j = \sqrt{(R_{jj}+1)/2} > 0$; then $n_i = (R_{ij} + \delta_{ij})/(2n_j)$.

One rotation, several representations. With convention subscripts on the Euler angles,

$$\begin{aligned} R &= R_z(\phi_E)R_x(\theta_E)R_z(\psi_E) \\ &= R_z(\psi_T)R_y(\theta_T)R_x(\phi_T) \\ &= \exp(\phi\widehat{n}) = \Phi(U(q)), \\ q &= \pm\big(c(\phi/2) + n^j e_j s(\phi/2)\big), \qquad K = U(q)^\dagger, \quad A_G = \Phi(K) = R^T. \end{aligned} \tag{550}$$

Use Eqs. (534), (541) and (542) for the conversions. In particular, for the principal angle $0 \le \phi \le \pi$,

$$c(\phi/2) = |q_0| = \left|c(\theta_E/2)c\big((\phi_E + \psi_E)/2\big)\right|. \tag{551}$$

Goldstein's Eq. (4.63) chooses the continuous lift near the identity; the absolute value selects the principal angle globally.

For a body-fixed vector a_B, its finite displacement between two times is

$$a_I(t_2) = R(t_2)a_B = R(t_2)R(t_1)^T a_I(t_1).$$

Euler's theorem applies to $R(t_2)R(t_1)^T$; its axis generally differs from the instantaneous angular-velocity direction.

Chasles: translation, rotation, and a screw axis. The free rigid displacement is

$$(r, R) \in SE(3), \qquad x \longmapsto r + Rx.$$

For $R = R_{n,\phi} \neq I$, put $h = n \cdot r$, $r_\perp = r - hn$, and

$$p = \frac{1}{2}\left(r_\perp + \cot(\phi/2)n \times r_\perp\right).$$

Since $R = c(\phi)I + s(\phi)N$ on $n^\perp$ and $N^2 = -I$ there,

$$\begin{aligned} (I - R)p &= \tfrac{1}{2}\big((1 - c(\phi))I - s(\phi)N\big)\big(I + \cot(\phi/2)N\big)r_\perp \\ &= \tfrac{1}{2}\Big((1 - c(\phi) + s(\phi)\cot(\phi/2))I + ((1 - c(\phi))\cot(\phi/2) - s(\phi))N\Big)r_\perp \\ &= r_\perp. \end{aligned}$$

Hence

$$r + Rx = p + R(x - p) + hn.$$

The axis is $p + \mathbb{R}n$ and the translation along it is hn. For $R = I$ the motion is pure translation.

114 Angular velocity and the rotating-frame derivative

Let $R(t)$ map body components to inertial components, with $R^T R = I$ and $\det R = 1$. Differentiating the first identity gives

$$\begin{aligned} \dot{R}^T R + R^T \dot{R} &= 0, \\ (R^T \dot{R})^T &= -R^T \dot{R}. \end{aligned} \tag{552}$$

Equation (552) makes $R^T \dot{R}$ a skew-symmetric matrix. Define $\widehat{\boldsymbol{\omega}}_B = R^T \dot{R}$ by $\widehat{\boldsymbol{\omega}}_B \mathbf{a} = \boldsymbol{\omega}_B \times \mathbf{a}$. Multiplication by R then yields

$$\dot{R} = R\widehat{\boldsymbol{\omega}}_B. \tag{553}$$

For any physical vector with components $\mathbf{a}_I = R\mathbf{a}_B$, apply the product rule and substitute Eq. (553):

$$\begin{aligned} \dot{\mathbf{a}}_I &= \dot{R}\mathbf{a}_B + R\dot{\mathbf{a}}_B \\ &= R\big(\boldsymbol{\omega}_B \times \mathbf{a}_B + \dot{\mathbf{a}}_B\big). \end{aligned} \tag{554}$$

The derivative on the left is taken in an inertial frame. The derivative of body components on the right includes the change of basis through the cross-product term.

115 Momentum balances about the center of mass

Assume a closed rigid body with constant mass $m > 0$, constant positive-definite body inertia J_B, and a body origin at its center of mass. Let $\mathbf{r}_I$ and $\mathbf{v}_I$ be its inertial position and velocity. The total external force is $\mathbf{F}_I$; the external moment about the center of mass has components $\boldsymbol{\tau}_B$. With linear momentum $\mathbf{p}_I = m\mathbf{v}_I$, Newton's balance is

$$\dot{\mathbf{r}}_I = \mathbf{v}_I, \qquad \dot{\mathbf{p}}_I = \mathbf{F}_I, \qquad m\dot{\mathbf{v}}_I = \mathbf{F}_I. \tag{555}$$

Define angular momentum $\mathbf{h}_B = J_B\boldsymbol{\omega}_B$, so that $\mathbf{h}_I = R\mathbf{h}_B$. Its inertial balance is $\dot{\mathbf{h}}_I = R\boldsymbol{\tau}_B$. Substitution into Eq. (554) gives

$$\begin{aligned} R\big(\dot{\mathbf{h}}_B + \boldsymbol{\omega}_B \times \mathbf{h}_B\big) &= R\boldsymbol{\tau}_B, \\ J_B\dot{\boldsymbol{\omega}}_B + \boldsymbol{\omega}_B \times (J_B\boldsymbol{\omega}_B) &= \boldsymbol{\tau}_B. \end{aligned} \tag{556}$$

The second line uses constant J_B and multiplication by R^T. Similarly, setting $\mathbf{v}_I = R\mathbf{v}_B$ in Eq. (555), and using Eq. (554), gives

$$m\big(\dot{\mathbf{v}}_B + \boldsymbol{\omega}_B \times \mathbf{v}_B\big) = \mathbf{F}_B, \qquad \mathbf{F}_B = R^T\mathbf{F}_I. \tag{557}$$

Equations (555), (553), and (556) form a constant-mass rigid-body six-DOF model after the force and moment laws are specified. Rocket mass loss requires open-system momentum balances; replacing m by $m(t)$ alone does not derive them.

For zero external moment, Eq. (556) implies $\dot{\mathbf{h}}_I = 0$, although $\mathbf{h}_B$ generally changes. The rotational kinetic energy satisfies

$$\begin{aligned} T_{\text{rot}} &= \tfrac{1}{2}\boldsymbol{\omega}_B^T J_B \boldsymbol{\omega}_B, \\ \dot{T}_{\text{rot}} &= \boldsymbol{\omega}_B \cdot J_B\dot{\boldsymbol{\omega}}_B \\ &= \boldsymbol{\omega}_B \cdot \boldsymbol{\tau}_B - \boldsymbol{\omega}_B \cdot \big(\boldsymbol{\omega}_B \times J_B\boldsymbol{\omega}_B\big) \\ &= \boldsymbol{\omega}_B \cdot \boldsymbol{\tau}_B. \end{aligned} \tag{558}$$

The cross-product term vanishes because it is perpendicular to $\boldsymbol{\omega}_B$. Zero moment therefore conserves both inertial angular momentum and rotational kinetic energy.

116 A free six-DOF model and its numerical realization

Take the body origin at the center of mass, SI units, constant mass $m > 0$ and constant physical inertia $J = J_B = J^T > 0$. Its principal moments satisfy $J_i \le J_j + J_k$. Write $\mathcal{R} : S^3 \to SO(3)$ for our Hamilton, scalar-first quaternion map, $q = (q_0, \mathbf{q})$, with $\mathcal{R}(q)\mathbf{a}_B = \mathbf{a}_I$. The state and evolution are

$$\mathcal{X} = \mathbb{R}^3 \times \mathbb{R}^3 \times S^3 \times \mathbb{R}^3, \qquad z = (\mathbf{r}_I, \mathbf{v}_I, q, \boldsymbol{\omega}_B),$$

$$\begin{aligned} \dot{\mathbf{r}}_I &= \mathbf{v}_I, & \dot{\mathbf{v}}_I &= m^{-1}\big(\mathcal{R}(q)\mathbf{F}_B + \mathbf{F}_{I,\mathrm{ext}}\big), \\ \dot{q} &= \tfrac{1}{2} q \otimes (0, \boldsymbol{\omega}_B), & \dot{\boldsymbol{\omega}}_B &= J^{-1}\big(\boldsymbol{\tau}_B - \boldsymbol{\omega}_B \times J\boldsymbol{\omega}_B\big). \end{aligned} \tag{559}$$

Thus $\dim \mathcal{X} = 12$ although z stores 13 real components; $q \sim -q$ gives the same physical state. The moment $\boldsymbol{\tau}_B$ is about the center of mass; a force at body offset $\boldsymbol{\rho}_B$ contributes $\boldsymbol{\rho}_B \times \mathbf{F}_B$.

For $\mathbf{F}_B = \mathbf{F}_{I,\mathrm{ext}} = \boldsymbol{\tau}_B = 0$,

$$\begin{aligned} \mathbf{r}_I(t) &= \mathbf{r}_I(0) + t\mathbf{v}_I(0), & \mathbf{p}_I &= m\mathbf{v}_I = \text{const}, \\ \mathbf{H}_I &= \mathcal{R}(q) J\boldsymbol{\omega}_B = \text{const}, & E &= \tfrac{1}{2} m\|\mathbf{v}_I\|^2 + \tfrac{1}{2}\boldsymbol{\omega}_B^T J\boldsymbol{\omega}_B = \text{const}. \end{aligned} \tag{560}$$

Here $\mathbf{H}_I$ is spin about the center of mass; total angular momentum about the inertial origin is $\mathbf{r}_I \times \mathbf{p}_I + \mathbf{H}_I$. Generally $\dot{\boldsymbol{\omega}}_B \neq 0$ even with no external wrench. For distinct principal moments, steady spin about the largest or smallest principal moment is stable; spin about the intermediate moment is unstable (Goldstein §5.6, pp. 200–208 [98]).

116.1 Midpoint momentum and a Cayley attitude step

For step size $h > 0$, let $\boldsymbol{\ell}_n = J\boldsymbol{\omega}_{B,n}$. For a body-fixed wrench held constant over the step, solve

$$\begin{aligned} \bar{\boldsymbol{\ell}} &= \tfrac{1}{2}(\boldsymbol{\ell}_n + \boldsymbol{\ell}_{n+1}), & \bar{\boldsymbol{\omega}} &= J^{-1}\bar{\boldsymbol{\ell}}, \\ \boldsymbol{\ell}_{n+1} - \boldsymbol{\ell}_n &= h\big(\boldsymbol{\tau}_B - \bar{\boldsymbol{\omega}} \times \bar{\boldsymbol{\ell}}\big), & u &= \frac{(1, \tfrac{h}{2}\bar{\boldsymbol{\omega}})}{\sqrt{1 + \tfrac{h^2}{4}\|\bar{\boldsymbol{\omega}}\|^2}}, \\ q_{n+1} &= q_n \otimes u, & R_{n+1} &= R_n \underbrace{(I - \tfrac{h}{2}\widehat{\bar{\boldsymbol{\omega}}})^{-1}(I + \tfrac{h}{2}\widehat{\bar{\boldsymbol{\omega}}})}_{\mathrm{cay}(h\widehat{\bar{\boldsymbol{\omega}}})}. \end{aligned} \tag{561}$$

The nonlinear solve uses its residual, rather than a fixed number of iterations, to decide convergence. Let $q_{1/2} = (q_n + q_{n+1})/\|q_n + q_{n+1}\|$ using these continuous quaternion lifts. Then

$$\begin{aligned} \mathbf{a}_{1/2} &= m^{-1}\big(\mathcal{R}(q_{1/2})\mathbf{F}_B + \mathbf{F}_{I,\mathrm{ext}}\big), \\ \mathbf{v}_{n+1} &= \mathbf{v}_n + h\mathbf{a}_{1/2}, \qquad \mathbf{r}_{n+1} = \mathbf{r}_n + \tfrac{h}{2}(\mathbf{v}_n + \mathbf{v}_{n+1}). \end{aligned}$$

In exact arithmetic, for zero wrench,

$$\begin{aligned} T_{\mathrm{rot},n+1} - T_{\mathrm{rot},n} &= \bar{\boldsymbol{\omega}} \cdot (\boldsymbol{\ell}_{n+1} - \boldsymbol{\ell}_n) = 0, \\ (I + \tfrac{h}{2}\widehat{\bar{\boldsymbol{\omega}}})\boldsymbol{\ell}_{n+1} = (I - \tfrac{h}{2}\widehat{\bar{\boldsymbol{\omega}}})\boldsymbol{\ell}_n \implies R_{n+1}\boldsymbol{\ell}_{n+1} &= R_n\boldsymbol{\ell}_n. \end{aligned}$$

Floating-point conservation is limited by the nonlinear residual and roundoff. This is a second-order trajectory approximation, not the exact flow: even spherical inertia gives the step angle $2\arctan(h\|\boldsymbol{\omega}\|/2)$ rather than $h\|\boldsymbol{\omega}\|$. Check both invariant drift and attitude/position convergence against an independent reference; invariant preservation alone cannot bound phase error.

116.2 What the model resolves; what to read next

Equations (559) suffice for an ideal rigid body in force-free, torque-free space. A browser clock requests elapsed time; the backend advances fixed physical steps. Applied controls are explicit external wrenches; releasing them returns to free coast.

For dynamics, read Goldstein §§5.1, 5.3–5.6 (pp. 184–208) [98]; for an engineering connection, Wie §5.5 (p. 339) and §§6.1–6.7 (pp. 349–371) [20]. Wie §§7.3–7.7 add quaternion feedback, momentum management and jet selection; Chapters 8–9 add flexibility and slosh. Spacecraft fidelity additionally requires measured mass properties and appropriate environment/actuator models. Numerical verification does not by itself validate those physical assumptions. NASA's six-DOF check cases [19] provide a later independent benchmark target.

117 Which metric makes rigid-body motion geodesic?

The submanifold proof does not by itself specify the body's dynamics. For a rigid body with a material point P fixed, assume a constant positive-definite body inertia tensor J_P about that point. The kinetic energy defines a Riemannian metric on $SO(3)$ by

$$g_R(R\widehat{\mathbf{a}}, R\widehat{\mathbf{b}}) = \mathbf{a}^T J_P \mathbf{b}, \qquad T = \tfrac{1}{2} g_R(\dot{R}, \dot{R}) = \tfrac{1}{2} \boldsymbol{\omega}_B^T J_P \boldsymbol{\omega}_B. \tag{562}$$

If P is the center of mass, J_P is the J_B used above. The metric inherited directly from the Euclidean matrix space gives

$$\operatorname{tr}\big((R\widehat{\mathbf{a}})^T R\widehat{\mathbf{b}}\big) = \operatorname{tr}(\widehat{\mathbf{a}}^T \widehat{\mathbf{b}}) = 2\mathbf{a}\cdot\mathbf{b}.$$

It is proportional to the kinetic metric only for spherical inertia $J_P = jI$. The embedding in $\mathbb{R}^9$ therefore does not determine the metric of a general rigid body.

In any local coordinates q^a on $SO(3)$, write $T = \frac{1}{2} g_{ab}(q)\dot{q}^a\dot{q}^b$. For a smooth time-independent potential $V(q)$, the Lagrangian $L = T - V$ has

$$\frac{\partial L}{\partial \dot{q}^a} = g_{ab}\dot{q}^b, \qquad \frac{\partial L}{\partial q^a} = \tfrac{1}{2}\frac{\partial g_{bc}}{\partial q^a}\dot{q}^b\dot{q}^c - \frac{\partial V}{\partial q^a}.$$

Substituting into the Euler–Lagrange equations gives

$$\frac{d}{dt}(g_{ab}\dot{q}^b) - \tfrac{1}{2}\frac{\partial g_{bc}}{\partial q^a}\dot{q}^b\dot{q}^c = -\frac{\partial V}{\partial q^a}. \tag{563}$$

For constant V, Eq. (563) is the affinely parameterized geodesic equation of g in physical time. For a general V, the right-hand side is a force term.

At fixed energy $E = T + V$, on the region $E > V$, Jacobi's principle describes the same configuration paths as geodesics of

$$g^{(E)} = 2(E - V)g. \tag{564}$$

Indeed, $ds_g/dt = \sqrt{g_R(\dot{R}, \dot{R})} = \sqrt{2(E-V)}$, so

$$ds_E = \sqrt{2(E-V)}\, ds_g, \qquad \frac{ds_E}{dt} = 2(E - V). \tag{565}$$

The reduced action $\int p_a\, dq^a$ equals $\int ds_E$ along energy-E parametrizations, because $p_a\, dq^a = g_{ab}\dot{q}^b\dot{q}^a\, dt = 2T\, dt = ds_E$. Its stationary paths are therefore geodesics for $g^{(E)}$. Frankel [11], Section 10.2c, Theorem (10.19), uses $d\rho = \sqrt{E-V}\, ds_g$, so his metric is $(E-V)g = \frac{1}{2}g^{(E)}$; the constant factor changes arc length but not the geodesic paths. The metric degenerates at $E = V$, and physical time is generally not its affine geodesic parameter.

If $E > \max_{SO(3)} V$, the Jacobi metric is positive everywhere on the compact configuration manifold. Closed-geodesic results can then be used to establish periodic motions, as in Frankel's Section 10.2d. A closed geodesic returns with the same tangent direction; returning to the same configuration alone is insufficient. Its physical period is recovered by integrating $dt = ds_E/[2(E-V)]$ around the closed geodesic.

118 Reading geometric mechanics

Holm, Schmah, and Stoica [15] provide the geometric foundation. Start with Chapter 1 (pp. 3–42), then manifolds in Chapter 2 (pp. 43–98) and geometry in Chapter 3 (pp. 99–154). Read mechanics on manifolds in Chapter 4 (pp. 155–186), Lie groups and algebras in Chapter 5 (pp. 187–208), and group actions in Chapter 6 (pp. 209–240). The first destination is Sections 7.1–7.3 (pp. 241–260): rigid-body mechanics and Euler–Poincaré reduction. Sections 7.4–7.5 (pp. 261–280) introduce the heavy top and advected parameters. Chapters 8–9 (pp. 281–324) develop momentum maps and Lie–Poisson reduction. Printed page numbers refer to the 2009 Oxford edition. The page-reconciled transcription is `resolved.md` in that book's `ocr-compare/reconciled` directory. These notes do not copy the book.

Derive Eq. (556) again from the reduced variational principle when studying Chapter 7. Compare that derivation with the momentum balance above. Keep frame conventions explicit throughout. Guidance, estimation, feedback control, aerodynamic models, and open-system propulsion balances require further material beyond these geometric-mechanics chapters.

Part XXXI
Waves

119 Forced Oscillation and Resonance, and Damped Oscillations

See Georgi (1992) [48].

119.1 Damped Oscillators

See Sec. 2.1 "Damped Oscillations", Georgi (1992) [48].

For small velocities, frictional force takes the form of

$$-m\Gamma v \tag{566}$$

See (2.1.1) of Georgi (1992) [48]. $1/\Gamma$ has the dimensions of time.

Now

$$\sum F = m\ddot{x} = -m\Gamma v - Kx$$

or

$$\ddot{x} + \Gamma\dot{x} + \frac{K}{m}x = \ddot{x} + \Gamma\dot{x} + \omega_0^2 x = 0 \tag{567}$$

where $\omega_0 = \sqrt{\frac{K}{m}}$. See (2.1.2) of Georgi (1992) [48].

Since $\ddot{x} + \Gamma\dot{x} + \omega_0^2 x = 0$ is 1. time translation invariant, i.e. if $x(t+a)$ is a solution, then $x(t)$ is a solution, and 2. linearity, then find irreducible solutions of form $z(t) = e^{\alpha t}$.

$$\implies \alpha^2 + \Gamma\alpha + \omega_0^2 = 0 \text{ or } \alpha = \frac{-\Gamma \pm \sqrt{\Gamma^2 - 4\omega_0^2}}{2}$$

119.2 Overdamped oscillations

See Sec. 2.1.1 "Overdamped Oscillations" of Georgi (1992) [48].

If $\frac{\Gamma}{2} > \omega_0$, let

$$\Gamma_\pm = \frac{\Gamma}{2} \pm \sqrt{\frac{\Gamma^2}{4} - \omega_0^2} \tag{568}$$

See (2.1.8) of Georgi (1992) [48]

$$x(t) = A_+ e^{-\Gamma_+ t} + A_- e^{-\Gamma_- t} \tag{569}$$

See (2.1.7) of Georgi (1992) [48].

119.3 Underdamped Oscillations

See Sec. 2.1.2 "Underdamped Oscillations" of Georgi (1992) [48].

If $\frac{\Gamma}{2} < \omega_0$, so

$$\alpha = \frac{-\Gamma}{2} \pm i\sqrt{\omega_0^2 - \frac{\Gamma^2}{2}}$$

so

$$\omega^2 = \omega_0^2 - \frac{\Gamma^2}{4} \tag{570}$$

$$x(t) = Ae^{\frac{-\Gamma}{2}t}\cos(\omega t - \theta) \tag{571}$$

See (2.1.10) of Georgi (1992) [48].

$$x(t) = Ae^{\frac{-\Gamma}{2}t}c(\omega t - \theta) = Ae^{\frac{-\Gamma}{2}t}(c(\omega t)c(\theta) + s(\omega t)s(\theta))$$

$$x(t) = e^{-\frac{\Gamma}{2}t}(c\cos(\omega t) + d\sin(\omega t)) \tag{572}$$

See (2.1.11) of Georgi (1992) [48], where $c = A\cos\theta$, $d = A\sin\theta$.

119.3.1 Critically Damped Oscillations

See Sec. 2.1.3 "Critically Damped Oscillations" of Georgi (1992) [48].

If $\frac{\Gamma}{2} = \omega_0$, we only get $e^{\frac{-\Gamma}{2}t}$. We know that there will be 2 solutions to the 2nd-order differential equation. One way to find the other solution is to approach this situation from underdamped case as a limit, i.e.

$$\omega \to 0 \text{ from } \omega > 0$$

$$\lim_{\omega \to 0} e^{\frac{-\Gamma}{2}(t)} c \cos(\omega t) = e^{\frac{-\Gamma}{2}t}$$
$$\lim_{\omega \to 0} e^{\frac{-\Gamma}{2}t} d \sin \omega t = 0$$

Divide the second term by ω, because ω doesn't depend on ω, $\frac{1}{\omega} e^{\frac{-\Gamma}{2}t} d \sin \omega t$ is still a solution.

$$\lim_{\omega \to 0} e^{\frac{-\Gamma}{2}t} \frac{\sin \omega t}{\omega} = te^{\frac{-\Gamma}{2}t} (\text{by L'Hopital's rule})$$

So

$$x(t) = ce^{-\Gamma t/2} + dte^{-\Gamma t/2}$$

119.4 Forced oscillations

See Sec. 2.2 "Forced Oscillations" of Georgi (1992) [48]. Equation of motion is

$$\ddot{x}(t) + \Gamma \dot{x} + \omega_0^2 x = F(t)/m \tag{573}$$

where

$$F(t) = F_0 \cos(\omega_d t) \tag{574}$$

See (2.1.5), (2.1.4) of Georgi (1992) [48], respectively.

$\frac{\omega_d}{2\pi}$ = driving frequency, driving frequency can be tuned completely independently of the other parameters of the system.

If we let

$$\ddot{z}(t) + \Gamma \dot{z}(t) + \omega_0^2 z(t) = \mathcal{F}(t)/m$$

where $\mathcal{F}(t) = F_0 e^{-i\omega_0 t}$.

For in particular numerical computation applications, we can transform Eq. 573 into a system of first order equations:

Let

$$x_1(t) = x(t)$$
$$x_2(t) = \dot{x}(t)$$

so that

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = -\Gamma x_2 - \omega_0^2 x_1 + f(t)$$

Then if we look for steady state solution of the form

$$z(t) = Ae^{-i\omega_d t}$$

One can think like the steady state solution is such that the system is driven with driving frequency ω_d.

$$(-\omega_d^2 + (-i\omega_d)\Gamma + \omega_0^2)A = \frac{F_0}{m} \tag{575}$$

See (2.20) of Georgi (1992) [48]

or

$$A = \frac{F_0/m}{\omega_0^2 - i\omega_d \Gamma - \omega_d^2} \tag{576}$$

See (2.21) of Georgi (1992) [48]

The amplitude A of the displacement is proportional to amplitude of driving force; this is as expected from linearity.

$$A = \frac{\frac{F_0}{m}(\omega_0^2 + i\omega_d\Gamma - \omega_d^2)}{(\omega_0^2 - \omega_d^2)^2 + (\omega_d\Gamma)^2} \text{ with}$$

$$\Re A = \frac{(\omega_0^2 - \omega_d^2)F_0/m}{(\omega_0^2 - \omega_d^2)^2 + (\omega_d\Gamma)^2} \qquad \Im A \equiv B = \frac{\Gamma\omega_d F_0/m}{(\omega_0^2 - \omega_d^2)^2 + (\Gamma\omega_d)^2}$$

$$\mathcal{A}e^{-i\omega_d t} = \mathcal{A}(c(\omega_d t) - is(\omega_d t)) \xrightarrow{\Re} \Re A c(\omega_d t) + \Im A s(\omega_d t)$$

where I use $\mathcal{A} \equiv A$ notation interchangeably, relying on context on whether it's real or complex, and $\cos\omega_d t \equiv c(\omega_d t)$, $\sin\omega_d t \equiv s(\omega_d t)$ as shorthand notation for sin and cos.

For

$$x(t) = \Re z(t) = (\Re A)c(\omega_d t) + Bs(\omega_d t)$$

Term proportional to B is 90° out of phase.
Real part of $\mathcal{A}$, $A = \Re A$ is the *elastic amplitude.*
Imaginary part of $\mathcal{A}$, $B = \Im A$ is called the *absorptive amplitude.*

In the case where

$$F(t) = F_0 \sin(\omega_d t)$$

we'll see that the fundamental physical characteristics don't change, but the initial conditions for the solution do change, which is important for actual modeling (but gets looked over a bit in physics that concern about high-level concepts).

If $F(t) = F_0 \sin(\omega_d t)$, while the following doesn't change:

$$z(t) = \mathcal{A}e^{-i\omega_d t}$$
$$\dot{z}(t) = (-i\omega_d)z$$
$$\ddot{z}(t) = (-\omega_d^2)z$$

but now consider

$$\mathcal{F}(t) = +iF_0 e^{-i\omega_d t} = +iF_0(c(\omega_d t) - is(\omega_d t))$$
$$\Re(\mathcal{F}(t)) = F_0 s(\omega_d t)$$

so then

$$(-\omega_d^2 + (-i\omega_d)\Gamma + \omega_0^2)\mathcal{A} = \frac{iF_0}{m} \text{ or}$$

$$\mathcal{A} = \frac{iF_0/m}{\omega_0^2 - \omega_d^2 - i\omega_d\Gamma} = \frac{\frac{iF_0}{m}(\omega_0^2 - \omega_d^2 + i\omega_d\Gamma)}{(\omega_0^2 - \omega_d^2)^2 + (\omega_d\Gamma)^2}$$

So then in this case,

$$\Re A = \frac{-\Gamma\omega_d F_0/m}{(\omega_0^2 - \omega_d^2)^2 + (\Gamma\omega_d)^2} \equiv A$$
$$\Im A = \frac{(\omega_0^2 - \omega_d^2)F_0/m}{(\omega_0^2 - \omega_d^2)^2 + (\omega_d\Gamma)^2} \equiv B \tag{577}$$

and so the general form of the steady-state solution doesn't change, but the values for the amplitudes get switched:

$$x(t) = \Re z(t) = Ac(\omega_d t) + Bs(\omega_d t)$$
$$\dot{x}(t) = \omega_d\left[-As(\omega_d t) + Bc(\omega_d t)\right]$$

119.5 Resonances

See Sec. 2.3 "Resonances" of Georgi (1992) [48].

$$\omega_d = \omega_0$$

so $\Re A = 0$, $B = \frac{F_0/m}{\Gamma\omega_d}$.

119.6 Work, Work done by forced oscillation on the system

Work is a nonlinear function of force, so we must use real force and real displacement.

$$P(t) = F(t)\frac{\partial}{\partial t}X(t) = F_0 c(\omega_d t)(\omega_d)(-\Re A s(\omega_d t) + Bc(\omega_d t))$$

The first term is proportional to $s(2\omega_d t)$ and averages to 0 over a half period.

$$\int_{t_0}^{t_0+\pi/\omega_d} s(2\omega_d t)dt = \left.\frac{-c(2\omega_d t)}{2\omega_d}\right|_{t_0}^{t_0+\pi/\omega_d} = 0$$

This is why $\Re A$ is called elastic amplitude. If A dominates, then energy fed into the system at 1 time is returned at a later time, as in elastic collision in mechanics.

$$P_{\text{average}}\text{of 2nd. term} = P_{\text{average}}$$

Consider

$$F_0\omega_d B\int_{t_0}^{t_0+\pi/\omega_d} c^2(\omega_d t)dt = F_0\omega_d B\int_{t_0}^{t_0+\pi/\omega_d}\frac{c(2\omega_d t)+1}{2}dt =$$

$$F_0\omega_d B\left(\frac{s(2\omega_d t)}{4\omega_d}+\frac{t}{2}\right)\Bigg|_{t_0}^{t_0+\frac{\pi}{\omega_d}} = F_0\omega_d B\frac{\pi}{2\omega_d} = \frac{F_0 B\pi}{2}$$

$$P_{\text{average}} = \frac{\int_{t_0}^{t_0+\pi/\omega_d}P(t)}{\Delta t}\text{ where }\Delta t = \pi/\omega_d$$

$$\Longrightarrow P_{\text{average}} = \frac{F_0 B\pi/2}{\pi/\omega_d} = \boxed{\frac{F_0\omega_d B}{2}}$$

Recall that

$$B = \frac{\Gamma\omega_d F_0/m}{(\omega_0^2-\omega_d^2)^2+(\Gamma\omega_d)^2}$$

$$P_{\text{average}} = \frac{F_0\omega_d B}{2} = \frac{(F_0\omega_d)^2\Gamma/2m}{(\omega_0^2-\omega_d^2)^2+(\Gamma\omega_d)^2}\xrightarrow{\omega_d\to\omega_0}\frac{F_0^2}{2m\Gamma}$$

The maximum average power is inversely proportional to Γ.

This analysis doesn't change when $F(t) = F_0\sin(\omega_d t)$.

$$P(t) = F(t)\dot{x}(t) = F_0 s(\omega_d t)\omega_d[-As(\omega_d t)+Bc(\omega_d t)]$$

$$\int_{t_0}^{t_0+\frac{\pi}{\omega_d}} dtc(\omega_d t)s(\omega_d t) = \left.\frac{1}{2\omega_d}s^2(\omega_d t)\right|_{t_0}^{t_0+\frac{\pi}{\omega_d}} = 0$$

$$\int_{t_0}^{t_0+\frac{\pi}{\omega_d}} dts^2(\omega_d t) = \int_{t_0}^{t_0+\frac{\pi}{\omega_d}} dt\frac{1-c(2\omega_d t)}{2} = \frac{\pi}{2\omega_d} - \left.\frac{s(2\omega_d t)}{4\omega_d}\right|_{t_0}^{t_0+\frac{\pi}{\omega_d}} = \frac{\pi}{2\omega_d}$$

$$P_{\text{average}} = \frac{\int_{t_0}^{t_0+\frac{\pi}{\omega_d}}dtP(t)}{\pi/\omega_d} = \frac{-F_0 A\pi}{2/\pi/\omega_d} = \frac{-F_0 A\omega_d}{2}$$

So B is now the elastic amplitude and A is the absorptive amplitude, but the same values are obtained, since A and B are effectively switched for $F(t) = F_0\sin\omega_d t$.

To find the width (for fixed height) of the resonance curve (curve of P_{average}, depending on ω_d), find the value $\omega_d = \omega_{1/2}$ s.t.

$$P_{\text{average}} = \frac{P_{\text{average,max}}}{2} = \frac{F_0^2}{4m\Gamma}$$

Then

$$\frac{P_{\text{average,max}}}{2} = \frac{F_0^2}{4m\Gamma} = \frac{(F_0^2\Gamma/2m)\omega_{1/2}^2}{(\omega_{1/2}^2-\omega_0^2)^2+(\Gamma\omega_{1/2})^2} \text{ or } \omega_{1/2}^4 - 2\omega_{1/2}^2\omega_0^2 + \omega_0^2 + \Gamma^2\omega_{1/2}^2 = 2\Gamma^2\omega_{1/2}^2 \text{ or}$$

$$\omega_{1/2}^4 - (2\omega_0^2+\Gamma^2)\omega_{1/2}^2 + \omega_0^4 = 0$$

$$\omega_{1/2}^2 = \frac{(2\omega_0^2+\Gamma^2) \pm \sqrt{(\Gamma^2+2\omega_0^2)^2 - 4\omega_0^4}}{2} = \omega_0^2 + \frac{\Gamma^2}{2} \pm \sqrt{\frac{\Gamma^4}{4}+\Gamma^2\omega_0^2} = \omega_0^2 + \frac{\Gamma^2}{2} \pm \Gamma\sqrt{\frac{\Gamma^2}{4}+\omega_0^2}$$

Now (EY: it seems like you would have needed to have known the answer beforehand to have tried this factorization)

$$\omega_{1/2}^2 = \left(\sqrt{\omega_0^2+\frac{\Gamma^2}{4}}\right)^2 \pm 2\frac{\Gamma}{2}\sqrt{\frac{\Gamma^2}{4}+\omega_0^2} + \left(\frac{\Gamma}{2}\right)^2 = \left(\sqrt{\omega_0^2+\frac{\Gamma^2}{4}} \pm \frac{\Gamma}{2}\right)^2$$

Then

$$\boxed{\omega_{1/2} = \sqrt{\omega_0^2+\frac{\Gamma^2}{4}} \pm \frac{\Gamma}{2}} \tag{578}$$

The "free oscillation" of the system would be the case of underdamped oscillations; recall

$$x(t) = A\exp\left(\frac{-\Gamma t}{2}\right)\cos(\omega t - \theta)$$

Notice the exponential decay behavior of the amplitude or i.e. magnitude of $x(t)$ due to the $\exp\left(\frac{-\Gamma t}{2}\right)$ factor. Let's just say the "lifetime" of free oscillation is the time for the amplitude to drop to half its value, i.e.

$$\frac{1}{2} = \exp\left(\frac{-\Gamma t_{1/2}}{2}\right) \text{ or } -\ln 2 = \frac{-\Gamma t_{1/2}}{2}, \quad \text{or } t_{1/2} = \frac{2\ln 2}{\Gamma}$$

For the time dependence of the free oscillation, i.e. $x(t) = A\exp\left(\frac{-\Gamma t}{2}\right)\cos(\omega t-\theta)$, then we say that the lifetime of the state in free oscillation is of order $1/\Gamma$, i.e.

$$t_{1/2} = \frac{2\ln 2}{\Gamma}$$

The width of the resonance peak in forced oscillation is

$$\Delta\omega_{1/2} = \Gamma$$

so width of the resonance peak in forced oscillation is inversely proportional to the lifetime of the corresponding normal mode of free oscillation.

119.7 Apply Fourier Transform to Forced Oscillations

So for the equation of motion, Eq. 573,

$$\ddot{x} + \Gamma\dot{x} + \omega_0^2 x = F(t)$$

Assume we drive the system at frequency ω_d and that system's response's frequency is also ω_d.
Then

$$(i\omega_d)^2\widehat{x} + (i\omega_d)\Gamma\widehat{x} + \omega_0^2\widehat{x} = \widehat{F}(\omega_d) \text{ or } \frac{\widehat{x}(\omega_d)}{\widehat{F}(\omega_d)} = \frac{1}{-(\omega_d^2 - i\omega_d\Gamma - \omega_0^2)}$$

So in general, $\frac{\widehat{x}(\omega_d)}{\widehat{F}(\omega_d)}$ has a real part and imaginary part.

119.8 Apply Laplace Transform to Forced Oscillations

Recall Eq. 573,

$$\ddot{x} + \Gamma\dot{x} + kx = mF(t)$$

so that if $\frac{k}{m} = \omega_0^2$, then

$$\xrightarrow{\mathcal{L}} s^2\widehat{x}(s) - sx(0) - x'(0) + \Gamma(s\widehat{x}(s) - x(0)) + \omega_0^2\widehat{x}(s) = \widehat{F}(s) \text{ or}$$
$$(s^2 + \Gamma s + \omega_0^2)\widehat{x}(s) - (\Gamma + s)x(0) - x'(0) = \widehat{F}(s)$$

If $x'(0) = 0$, and we start system at $x(0) = 0$, so that we rely upon $F(t)$ to drive system, then we get a ratio of output or "response" to input, the driving, forced oscillation, per mass,

$$\frac{\widehat{x}(s)}{\widehat{F}(s)} = \frac{1}{s^2 + \Gamma s + \omega_0^2} = \frac{1}{\left(s + \frac{\Gamma + \sqrt{\Gamma^2 - 4\omega_0^2}}{2}\right)\left(s + \frac{\Gamma - \sqrt{\Gamma^2 - 4\omega_0^2}}{2}\right)}$$

Now look at where the denominator "blows up", the singularities. We had the case that if $\Gamma^2 - 4\omega^2 \geq 0$ or $\frac{\Gamma}{2} \geq \omega_0$, we had overdamped or critically damped exponential decay. Then, the value of s where a singularity occurs would be completely real. For the underdamped case with free oscillation, the value of s where a singularity occurs would have an imaginary part.

If initial conditions can't be ignored,
then if $x'(0) = 0$,

$$(s^2 + \Gamma s + \omega_0^2)\widehat{x}(s) - x'(0) = \widehat{F}(s) \text{ or } \frac{\widehat{x}(s)}{\widehat{F}(s)} = \frac{1 + x'(0)/\widehat{F}(s)}{s^2 + \Gamma s + \omega_0^2}$$

But if $x'(0) = 0$,

$$\frac{\widehat{x}(s)}{\widehat{F}(s)} = \frac{1 + (\Gamma + s)x(0)/\widehat{F}(s)}{s^2 + \Gamma s + \omega_0^2}$$

119.9 General solution to forced oscillation

For subscript s meaning steady-state driven solution, see Eq. 576

$$x_s(t) = Ac(\omega_d t) + Bs(\omega_d t)$$
$$A = \frac{(\omega_0^2 - \omega_d^2)F_0/m}{(\omega_0^2 - \omega_d^2)^2 + (\omega_d\Gamma)^2}$$
$$B = \frac{\Gamma\omega_d F_0/m}{(\omega_0^2 - \omega_d^2)^2 + (\Gamma\omega_d)^2}$$

so then

$$x_s(0) = A$$
$$\dot{x}_s(0) = \omega_d B$$

Let x_{free} denote the response, or solution, or the resulting motion for the "free oscillation", the underdamped, exponential decaying solution, for $\omega_0 > \frac{\Gamma}{2}$. See Eq. 572,

$$x_{\text{free}}(t) = \exp\left(\frac{-\Gamma t}{2}\right)(c\cos(\omega t) + d\sin(\omega t)), \quad \omega^2 = \omega_0^2 - \frac{\Gamma^2}{2}$$
$$x_{\text{free}}(0) = c$$
$$\dot{x}_{\text{free}}(t) = \frac{-\Gamma}{2}x_{\text{free}}(t) + e^{-\frac{\Gamma}{2}t}\omega(-cs(\omega t) + dc(\omega t))$$
$$\dot{x}_{\text{free}}(0) = \frac{-\Gamma}{2}c + \omega d$$

The most general solution $x(t)$ is of the form

$$x(t) = x_{\text{free}}(t) + x_s(t)$$

with initial conditions

$$x(0) = c + A \text{ so } c = x(0) - A$$
$$\dot{x}(0) = \frac{-\Gamma}{2}c + \omega d + \omega_d B \text{ so } d = \frac{1}{\omega}\left[\dot{x}(0) + \frac{\Gamma}{2}(x(0) - A) - \omega_d B\right]$$

And so the full, general solution is

$$x(t) = e^{\frac{-\Gamma t}{2}}(c\cos(\omega t) + d\sin(\omega t)) + A\cos(\omega_d t) + B\sin(\omega_d t)$$
$$\text{where}$$
$$\begin{aligned} A &= \frac{(\omega_0^2 - \omega_d^2)F_0/m}{(\omega_0^2 - \omega_d^2)^2 + (\omega_d\Gamma)^2} \\ B &= \frac{\Gamma\omega_d F_0/m}{(\omega_0^2 - \omega_d^2)^2 + (\Gamma\omega_d)^2} \qquad\qquad \omega^2 = \omega_0^2 - \frac{\Gamma^2}{4}, \\ c &= x(0) - A, \\ d &= \frac{1}{\omega}\left[\dot{x}(0) + \frac{\Gamma}{2}(x(0) - A) - \omega_d B\right] \end{aligned} \tag{579}$$

For the case of $F(t) = F_0\sin(\omega_d t)$, i.e. the force is proportional to sin, then the exact same analysis leads to the same result as above, but for A, B values switched to be as Eq. 577.

$$\Re A = \frac{-\Gamma\omega_d F_0/m}{(\omega_0^2 - \omega_d^2)^2 + (\Gamma\omega_d)^2} \equiv A$$
$$\Im A = \frac{(\omega_0^2 - \omega_d^2)F_0/m}{(\omega_0^2 - \omega_d^2)^2 + (\omega_d\Gamma)^2} \equiv B$$

119.10 Forced Oscillation and Resonance, and Damped Oscillations

See Georgi (1992) [48].

119.10.1 Damped Oscillators

See Sec. 2.1 "Damped Oscillations", Georgi (1992) [48].

For small velocities, frictional force takes the form of

$$-m\Gamma v \tag{580}$$

See (2.1.1) of Georgi (1992) [48]. $1/\Gamma$ has the dimensions of time.

Now

$$\sum F = m\ddot{x} = -m\Gamma v - Kx$$

or

$$\ddot{x} + \Gamma\dot{x} + \frac{K}{m}x = \ddot{x} + \Gamma\dot{x} + \omega_0^2 x = 0 \tag{581}$$

where $\omega_0 = \sqrt{\frac{K}{m}}$. See (2.1.2) of Georgi (1992) [48].

Since $\ddot{x} + \Gamma\dot{x} + \omega_0^2 x = 0$ is 1. time translation invariant, i.e. if $x(t + a)$ is a solution, then $x(t)$ is a solution, and 2. linearity, then find irreducible solutions of form $z(t) = e^{\alpha t}$.

$$\implies \alpha^2 + \Gamma\alpha + \omega_0^2 = 0 \text{ or } \alpha = \frac{-\Gamma \pm \sqrt{\Gamma^2 - 4\omega_0^2}}{2}$$

119.10.2 Overdamped oscillations

See Sec. 2.1.1 "Overdamped Oscillations" of Georgi (1992) [48].

If $\frac{\Gamma}{2} > \omega_0$, let

$$\Gamma_{\pm} = \frac{\Gamma}{2} \pm \sqrt{\frac{\Gamma^2}{4} - \omega_0^2} \tag{582}$$

See (2.1.8) of Georgi (1992) [48]

$$x(t) = A_+ e^{-\Gamma_+ t} + A_- e^{-\Gamma_- t} \tag{583}$$

See (2.1.7) of Georgi (1992) [48].

119.10.3 Underdamped Oscillations

See Sec. 2.1.2 "Underdamped Oscillations" of Georgi (1992) [48].

If $\frac{\Gamma}{2} < \omega_0$, so

$$\alpha = \frac{-\Gamma}{2} \pm i\sqrt{\omega_0^2 - \frac{\Gamma^2}{2}}$$

so

$$\omega^2 = \omega_0^2 - \frac{\Gamma^2}{4} \tag{584}$$

$$x(t) = Ae^{\frac{-\Gamma}{2}t} \cos(\omega t - \theta) \tag{585}$$

See (2.1.10) of Georgi (1992) [48].

$$x(t) = Ae^{\frac{-\Gamma}{2}t} c(\omega t - \theta) = Ae^{\frac{-\Gamma}{2}t}(c(\omega t)c(\theta) + s(\omega t)s(\theta))$$

$$x(t) = e^{-\frac{\Gamma}{2}t}(c\cos(\omega t) + d\sin(\omega t)) \tag{586}$$

See (2.1.11) of Georgi (1992) [48], where $c = A\cos\theta$, $d = A\sin\theta$.

119.10.4 Critically Damped Oscillations

See Sec. 2.1.3 "Critically Damped Oscillations" of Georgi (1992) [48].

If $\frac{\Gamma}{2} = \omega_0$, we only get $e^{\frac{-\Gamma}{2}t}$. We know that there will be 2 solutions to the 2nd-order differential equation. One way to find the other solution is to approach this situation from underdamped case as a limit, i.e.

$$\omega \to 0 \text{ from } \omega > 0$$

$$\lim_{\omega\to 0} e^{\frac{-\Gamma}{2}(t)} c\cos(\omega t) = e^{\frac{-\Gamma}{2}t}$$

$$\lim_{\omega\to 0} e^{\frac{-\Gamma}{2}t} d\sin\omega t = 0$$

Divide the second term by ω, because ω doesn't depend on ω, $\frac{1}{\omega}e^{\frac{-\Gamma}{2}t} d\sin\omega t$ is still a solution.

$$\lim_{\omega\to 0} e^{\frac{-\Gamma}{2}t}\frac{\sin\omega t}{\omega} = te^{\frac{-\Gamma}{2}t} \text{(by L'Hopital's rule)}$$

So

$$x(t) = ce^{-\Gamma t/2} + dte^{-\Gamma t/2}$$

119.10.5 Forced oscillations

See Sec. 2.2 "Forced Oscillations" of Georgi (1992) [48]. Equation of motion is

$$\ddot{x}(t) + \Gamma\dot{x} + \omega_0^2 x = F(t)/m \tag{587}$$

where

$$F(t) = F_0 \cos(\omega_d t) \tag{588}$$

See (2.1.5), (2.1.4) of Georgi (1992) [48], respectively.

$\frac{\omega_d}{2\pi}$ = driving frequency, driving frequency can be tuned completely independently of the other parameters of the system.

If we let

$$\ddot{z}(t) + \Gamma\dot{z}(t) + \omega_0^2 z(t) = \mathcal{F}(t)/m$$

where $\mathcal{F}(t) = F_0 e^{-i\omega_0 t}$.

For in particular numerical computation applications, we can transform Eq. 587 into a system of first order equations:
Let

$$x_1(t) = x(t)$$
$$x_2(t) = \dot{x}(t)$$

so that

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = -\Gamma x_2 - \omega_0^2 x_1 + f(t)$$

Then if we look for steady state solution of the form

$$z(t) = Ae^{-i\omega_d t}$$

One can think like the steady state solution is such that the system is driven with driving frequency ω_d.

$$(-\omega_d^2 + (-i\omega_d)\Gamma + \omega_0^2)A = \frac{F_0}{m} \tag{589}$$

See (2.20) of Georgi (1992) [48]
or

$$A = \frac{F_0/m}{\omega_0^2 - i\omega_d\Gamma - \omega_d^2} \tag{590}$$

See (2.21) of Georgi (1992) [48]

The amplitude A of the displacement is proportional to amplitude of driving force; this is as expected from linearity.

$$A = \frac{\frac{F_0}{m}(\omega_0^2 + i\omega_d\Gamma - \omega_d^2)}{(\omega_0^2 - \omega_d^2)^2 + (\omega_d\Gamma)^2} \text{ with}$$
$$\Re A = \frac{(\omega_0^2 - \omega_d^2)F_0/m}{(\omega_0^2 - \omega_d^2)^2 + (\omega_d\Gamma)^2} \qquad \Im A \equiv B = \frac{\Gamma\omega_d F_0/m}{(\omega_0^2 - \omega_d^2)^2 + (\Gamma\omega_d)^2}$$
$$\mathcal{A}e^{-i\omega_d t} = \mathcal{A}(c(\omega_d t) - is(\omega_d t)) \xrightarrow{\Re} \Re A c(\omega_d t) + \Im A s(\omega_d t)$$

where I use $\mathcal{A} \equiv A$ notation interchangeably, relying on context on whether it's real or complex, and $\cos\omega_d t \equiv c(\omega_d t)$, $\sin\omega_d t \equiv s(\omega_d t)$ as shorthand notation for sin and cos.

For

$$x(t) = \Re z(t) = (\Re A)c(\omega_d t) + Bs(\omega_d t)$$

Term proportional to B is 90° out of phase.
Real part of $\mathcal{A}$, $A = \Re A$ is the *elastic amplitude.*
Imaginary part of $\mathcal{A}$, $B = \Im A$ is called the *absorptive amplitude.*

In the case where

$$F(t) = F_0 \sin(\omega_d t)$$

we'll see that the fundamental physical characteristics don't change, but the initial conditions for the solution do change, which is important for actual modeling (but gets looked over a bit in physics that concern about high-level concepts).

If $F(t) = F_0 \sin(\omega_d t)$, while the following doesn't change:

$$\begin{aligned} z(t) &= \mathcal{A} e^{-i\omega_d t} \\ \dot{z}(t) &= (-i\omega_d) z \\ \ddot{z}(t) &= (-\omega_d^2) z \end{aligned}$$

but now consider

$$\begin{aligned} \mathcal{F}(t) = +iF_0 e^{-i\omega_d t} &= +iF_0(c(\omega_d t) - is(\omega_d t)) \\ \Re(\mathcal{F}(t)) &= F_0 s(\omega_d t) \end{aligned}$$

so then

$$\begin{aligned} (-\omega_d^2 + (-i\omega_d)\Gamma + \omega_0^2)\mathcal{A} &= \frac{iF_0}{m} \text{ or} \\ \mathcal{A} = \frac{iF_0/m}{\omega_0^2 - \omega_d^2 - i\omega_d\Gamma} &= \frac{\frac{iF_0}{m}(\omega_0^2 - \omega_d^2 + i\omega_d\Gamma)}{(\omega_0^2 - \omega_d^2)^2 + (\omega_d\Gamma)^2} \end{aligned}$$

So then in this case,

$$\begin{aligned} \Re A &= \frac{-\Gamma\omega_d F_0/m}{(\omega_0^2 - \omega_d^2)^2 + (\Gamma\omega_d)^2} \equiv A \\ \Im A &= \frac{(\omega_0^2 - \omega_d^2)F_0/m}{(\omega_0^2 - \omega_d^2)^2 + (\omega_d\Gamma)^2} \equiv B \end{aligned} \tag{591}$$

and so the general form of the steady-state solution doesn't change, but the values for the amplitudes get switched:

$$\begin{aligned} x(t) = \Re z(t) &= Ac(\omega_d t) + Bs(\omega_d t) \\ \dot{x}(t) &= \omega_d\left[-As(\omega_d t) + Bc(\omega_d t)\right] \end{aligned}$$

119.10.6 Resonances

See Sec. 2.3 "Resonances" of Georgi (1992) [48].

$$\omega_d = \omega_0$$

so $\Re A = 0$, $B = \frac{F_0/m}{\Gamma\omega_d}$.

119.10.7 Work, Work done by forced oscillation on the system

Work is a nonlinear function of force, so we must use real force and real displacement.

$$P(t) = F(t)\frac{\partial}{\partial t}X(t) = F_0 c(\omega_d t)(\omega_d)(-\Re A s(\omega_d t) + Bc(\omega_d t))$$

The first term is proportional to $s(2\omega_d t)$ and averages to 0 over a half period.

$$\int_{t_0}^{t_0+\pi/\omega_d} s(2\omega_d t)dt = \left.\frac{-c(2\omega_d t)}{2\omega_d}\right|_{t_0}^{t_0+\pi/\omega_d} = 0$$

This is why $\Re A$ is called elastic amplitude. If A dominates, then energy fed into the system at 1 time is returned at a later time, as in elastic collision in mechanics.

$$P_{\text{average}}\text{of 2nd. term} = P_{\text{average}}$$

Consider

$$F_0\omega_d B\int_{t_0}^{t_0+\pi/\omega_d} c^2(\omega_d t)dt = F_0\omega_d B\int_{t_0}^{t_0+\pi/\omega_d}\frac{c(2\omega_d t)+1}{2}dt =$$

$$F_0\omega_d B\left(\frac{s(2\omega_d t)}{4\omega_d}+\frac{t}{2}\right)\Bigg|_{t_0}^{t_0+\frac{\pi}{\omega_d}} = F_0\omega_d B\frac{\pi}{2\omega_d} = \frac{F_0 B\pi}{2}$$

$$P_{\text{average}} = \frac{\int_{t_0}^{t_0+\pi/\omega_d}P(t)}{\Delta t}\text{ where }\Delta t = \pi/\omega_d$$

$$\Longrightarrow P_{\text{average}} = \frac{F_0 B\pi/2}{\pi/\omega_d} = \boxed{\frac{F_0\omega_d B}{2}}$$

Recall that

$$B = \frac{\Gamma\omega_d F_0/m}{(\omega_0^2-\omega_d^2)^2+(\Gamma\omega_d)^2}$$

$$P_{\text{average}} = \frac{F_0\omega_d B}{2} = \frac{(F_0\omega_d)^2\Gamma/2m}{(\omega_0^2-\omega_d^2)^2+(\Gamma\omega_d)^2}\xrightarrow{\omega_d\to\omega_0}\frac{F_0^2}{2m\Gamma}$$

The maximum average power is inversely proportional to Γ.

This analysis doesn't change when $F(t) = F_0\sin(\omega_d t)$.

$$P(t) = F(t)\dot{x}(t) = F_0 s(\omega_d t)\omega_d[-As(\omega_d t)+Bc(\omega_d t)]$$

$$\int_{t_0}^{t_0+\frac{\pi}{\omega_d}} dtc(\omega_d t)s(\omega_d t) = \frac{1}{2\omega_d}s^2(\omega_d t)\Big|_{t_0}^{t_0+\frac{\pi}{\omega_d}} = 0$$

$$\int_{t_0}^{t_0+\frac{\pi}{\omega_d}} dts^2(\omega_d t) = \int_{t_0}^{t_0+\frac{\pi}{\omega_d}} dt\frac{1-c(2\omega_d t)}{2} = \frac{\pi}{2\omega_d}-\frac{s(2\omega_d t)}{4\omega_d}\Big|_{t_0}^{t_0+\frac{\pi}{\omega_d}} = \frac{\pi}{2\omega_d}$$

$$P_{\text{average}} = \frac{\int_{t_0}^{t_0+\frac{\pi}{\omega_d}}dtP(t)}{\pi/\omega_d} = \frac{-F_0 A\pi}{2/\pi/\omega_d} = \frac{-F_0 A\omega_d}{2}$$

So B is now the elastic amplitude and A is the absorptive amplitude, but the same values are obtained, since A and B are effectively switched for $F(t) = F_0\sin\omega_d t$.

To find the width (for fixed height) of the resonance curve (curve of P_{average}, depending on ω_d), find the value $\omega_d = \omega_{1/2}$ s.t.

$$P_{\text{average}} = \frac{P_{\text{average,max}}}{2} = \frac{F_0^2}{4m\Gamma}$$

Then

$$\frac{P_{\text{average,max}}}{2} = \frac{F_0^2}{4m\Gamma} = \frac{(F_0^2\Gamma/2m)\omega_{1/2}^2}{(\omega_{1/2}^2-\omega_0^2)^2+(\Gamma\omega_{1/2})^2}\text{ or }\omega_{1/2}^4-2\omega_{1/2}^2\omega_0^2+\omega_0^2+\Gamma^2\omega_{1/2}^2 = 2\Gamma^2\omega_{1/2}^2\text{ or}$$

$$\omega_{1/2}^4-(2\omega_0^2+\Gamma^2)\omega_{1/2}^2+\omega_0^4 = 0$$

$$\omega_{1/2}^2 = \frac{(2\omega_0^2+\Gamma^2)\pm\sqrt{(\Gamma^2+2\omega_0^2)^2-4\omega_0^4}}{2} = \omega_0^2+\frac{\Gamma^2}{2}\pm\sqrt{\frac{\Gamma^4}{4}+\Gamma^2\omega_0^2} = \omega_0^2+\frac{\Gamma^2}{2}\pm\Gamma\sqrt{\frac{\Gamma^2}{4}+\omega_0^2}$$

Now (EY: it seems like you would have needed to have known the answer beforehand to have tried this factorization)

$$\omega_{1/2}^2 = \left(\sqrt{\omega_0^2+\frac{\Gamma^2}{4}}\right)^2\pm 2\frac{\Gamma}{2}\sqrt{\frac{\Gamma^2}{4}+\omega_0^2}+\left(\frac{\Gamma}{2}\right)^2 = \left(\sqrt{\omega_0^2+\frac{\Gamma^2}{4}}\pm\frac{\Gamma}{2}\right)^2$$

Then

$$\boxed{\omega_{1/2} = \sqrt{\omega_0^2 + \frac{\Gamma^2}{4}} \pm \frac{\Gamma}{2}} \tag{592}$$

The "free oscillation" of the system would be the case of underdamped oscillations; recall

$$x(t) = A \exp\left(\frac{-\Gamma t}{2}\right) \cos(\omega t - \theta)$$

Notice the exponential decay behavior of the amplitude or i.e. magnitude of $x(t)$ due to the $\exp\left(\frac{-\Gamma t}{2}\right)$ factor. Let's just say the "lifetime" of free oscillation is the time for the amplitude to drop to half its value, i.e.

$$\frac{1}{2} = \exp\left(\frac{-\Gamma t_{1/2}}{2}\right) \text{ or } -\ln 2 = \frac{-\Gamma t_{1/2}}{2}, \quad \text{or } t_{1/2} = \frac{2\ln 2}{\Gamma}$$

For the time dependence of the free oscillation, i.e. $x(t) = A \exp\left(\frac{-\Gamma t}{2}\right) \cos(\omega t - \theta)$, then we say that the lifetime of the state in free oscillation is of order $1/\Gamma$, i.e.

$$t_{1/2} = \frac{2\ln 2}{\Gamma}$$

The width of the resonance peak in forced oscillation is

$$\Delta\omega_{1/2} = \Gamma$$

so width of the resonance peak in forced oscillation is inversely proportional to the lifetime of the corresponding normal mode of free oscillation.

119.11 Apply Fourier Transform to Forced Oscillations

So for the equation of motion, Eq. 587,

$$\ddot{x} + \Gamma\dot{x} + \omega_0^2 x = F(t)$$

Assume we drive the system at frequency ω_d and that system's response's frequency is also ω_d.
Then

$$(i\omega_d)^2\widehat{x} + (i\omega_d)\Gamma\widehat{x} + \omega_0^2\widehat{x} = \widehat{F}(\omega_d) \text{ or } \frac{\widehat{x}(\omega_d)}{\widehat{F}(\omega_d)} = \frac{1}{-(\omega_d^2 - i\omega_d\Gamma - \omega_0^2)}$$

So in general, $\frac{\widehat{x}(\omega_d)}{\widehat{F}(\omega_d)}$ has a real part and imaginary part.

119.12 Apply Laplace Transform to Forced Oscillations

Recall Eq. 587,

$$\ddot{x} + \Gamma\dot{x} + kx = mF(t)$$

so that if $\frac{k}{m} = \omega_0^2$, then

$$\xrightarrow{\mathcal{L}} s^2\widehat{x}(s) - sx(0) - x'(0) + \Gamma(s\widehat{x}(s) - x(0)) + \omega_0^2\widehat{x}(s) = \widehat{F}(s) \text{ or}$$
$$(s^2 + \Gamma s + \omega_0^2)\widehat{x}(s) - (\Gamma + s)x(0) - x'(0) = \widehat{F}(s)$$

If $x'(0) = 0$, and we start system at $x(0) = 0$, so that we rely upon $F(t)$ to drive system, then we get a ratio of output or "response" to input, the driving, forced oscillation, per mass,

$$\frac{\widehat{x}(s)}{\widehat{F}(s)} = \frac{1}{s^2 + \Gamma s + \omega_0^2} = \frac{1}{\left(s + \frac{\Gamma + \sqrt{\Gamma^2 - 4\omega_0^2}}{2}\right)\left(s + \frac{\Gamma - \sqrt{\Gamma^2 - 4\omega_0^2}}{2}\right)}$$

Now look at where the denominator "blows up", the singularities. We had the case that if $\Gamma^2 - 4\omega^2 \geq 0$ or $\frac{\Gamma}{2} \geq \omega_0$, we had overdamped or critically damped exponential decay. Then, the value of s where a singularity occurs would

be completely real. For the underdamped case with free oscillation, the value of s where a singularity occurs would have an imaginary part.

If initial conditions can't be ignored,
then if $x'(0) = 0$,

$$(s^2 + \Gamma s + \omega_0^2)\widehat{x}(s) - x'(0) = \widehat{F}(s) \text{ or } \frac{\widehat{x}(s)}{\widehat{F}(s)} = \frac{1 + x'(0)/\widehat{F}(s)}{s^2 + \Gamma s + \omega_0^2}$$

But if $x'(0) = 0$,

$$\frac{\widehat{x}(s)}{\widehat{F}(s)} = \frac{1 + (\Gamma + s)x(0)/\widehat{F}(s)}{s^2 + \Gamma s + \omega_0^2}$$

119.12.1 General solution to forced oscillation

For subscript s meaning steady-state driven solution, see Eq. 576

$$\begin{aligned} x_s(t) &= Ac(\omega_d t) + Bs(\omega_d t) \\ A &= \frac{(\omega_0^2 - \omega_d^2)F_0/m}{(\omega_0^2 - \omega_d^2)^2 + (\omega_d\Gamma)^2} \\ B &= \frac{\Gamma\omega_d F_0/m}{(\omega_0^2 - \omega_d^2)^2 + (\Gamma\omega_d)^2} \end{aligned}$$

so then

$$\begin{aligned} x_s(0) &= A \\ \dot{x}_s(0) &= \omega_d B \end{aligned}$$

Let x_{free} denote the response, or solution, or the resulting motion for the "free oscillation", the underdamped, exponential decaying solution, for $\omega_0 > \frac{\Gamma}{2}$. See Eq. 586,

$$\begin{aligned} x_{\text{free}}(t) &= \exp\left(\frac{-\Gamma t}{2}\right)(c\cos(\omega t) + d\sin(\omega t)), \quad \omega^2 = \omega_0^2 - \frac{\Gamma^2}{2} \\ x_{\text{free}}(0) &= c \\ \dot{x}_{\text{free}}(t) &= \frac{-\Gamma}{2}x_{\text{free}}(t) + e^{-\frac{\Gamma}{2}t}\omega(-cs(\omega t) + dc(\omega t)) \\ \dot{x}_{\text{free}}(0) &= \frac{-\Gamma}{2}c + \omega d \end{aligned}$$

The most general solution $x(t)$ is of the form

$$x(t) = x_{\text{free}}(t) + x_s(t)$$

with initial conditions

$$\begin{aligned} x(0) &= c + A \text{ so } c = x(0) - A \\ \dot{x}(0) &= \frac{-\Gamma}{2}c + \omega d + \omega_d B \text{ so } d = \frac{1}{\omega}\left[\dot{x}(0) + \frac{\Gamma}{2}(x(0) - A) - \omega_d B\right] \end{aligned}$$

And so the full, general solution is

$$\begin{aligned} x(t) &= e^{\frac{-\Gamma t}{2}}(c\cos(\omega t) + d\sin(\omega t)) + A\cos(\omega_d t) + B\sin(\omega_d t) \\ &\text{where} \\ A &= \frac{(\omega_0^2 - \omega_d^2)F_0/m}{(\omega_0^2 - \omega_d^2)^2 + (\omega_d\Gamma)^2} \\ B &= \frac{\Gamma\omega_d F_0/m}{(\omega_0^2 - \omega_d^2)^2 + (\Gamma\omega_d)^2} \qquad \omega^2 = \omega_0^2 - \frac{\Gamma^2}{4}, \\ c &= x(0) - A, \\ d &= \frac{1}{\omega}\left[\dot{x}(0) + \frac{\Gamma}{2}(x(0) - A) - \omega_d B\right] \end{aligned} \tag{593}$$

For the case of $F(t) = F_0 \sin(\omega_d t)$, i.e. the force is proportional to sin, then the exact same analysis leads to the same result as above, but for A, B values switched to be as Eq. 591.

$$\Re A = \frac{-\Gamma\omega_d F_0/m}{(\omega_0^2 - \omega_d^2)^2 + (\Gamma\omega_d)^2} \equiv A$$
$$\Im A = \frac{(\omega_0^2 - \omega_d^2)F_0/m}{(\omega_0^2 - \omega_d^2)^2 + (\omega_d\Gamma)^2} \equiv B$$

Part XXXII
Thermodynamics

120 Scope, notation, and what was already written

This is the thermodynamics part of the mono notes. The source is one file. `master.tex` is US Letter, single column. `screen.tex` is the same text on a 1200 pt × 700 pt page in two columns, the widescreen size used in `DGDT_dump.tex` and in `Propulsion/LaTeXandpdfs/thermo.tex`. Build both with `documents/mono-notes/build.sh`. The originals below were not edited.

120.1 Where the Kittel–Kroemer notes actually are

There is no second, fuller set of reading notes on Kittel and Kroemer. The 2008 file

`mathphysics/LaTeX_and_pdfs/Kittel_Kroemer_Thermal_Physics.tex`

is a solutions manuscript with reading notes in front of some chapters. Its own abstract says the solutions are nearly complete and that notes were to be added before the problems. What is present:

- Chapter 1, States of a Model System: the heading only. Written here as §121.
- Chapter 2, Entropy and Temperature: a short derivation of thermal equilibrium, then solutions 1–4.
- Chapter 3, Boltzmann distribution and Helmholtz free energy: heat capacity, pressure, the thermodynamic identity, and the one-atom partition function, then solutions 1–11.
- Chapter 4, thermal radiation: solutions 1–15, 17, and 18. There is no solution 16 in the file.
- Chapter 5, chemical potential: notes and solutions 1–14.
- Chapter 6, ideal gas: notes and solutions. Solution 6, entropy of mixing, is a heading with no solution. The derivation is in §133. Solutions 1–4 under this heading are Fermi–Dirac and orbital-counting calculations; the next section heading is Fermi and Bose gases. They were left in the order of the manuscript.
- Chapter 7, Fermi and Bose gases: problems and solutions 1–10.
- Chapter 8, heat and work: the long geometric discussion of engines, Legendre transforms, and the Carnot cycle, then solutions 1–9.
- Chapter 9, Gibbs free energy: solutions 1–4.
- Chapter 10, phase transformations: notes and problems 1–5.
- Chapters 11–13, binary mixtures, cryogenics, and semiconductor statistics: headings only. Written here as §133, §137, and §140.
- Chapter 14, kinetic theory: the ideal-gas law, the Maxwell speed distribution, effusion, and diffusion, then the mean-speed problem.

- Chapter 15, propagation: the continuity equation written with the Hodge star, sound speed, and the start of the diffusion problems. Parts (b) and (c) of the last problem have no solution and no statement in the file, so they were not invented.

The conceptual manuscript is the 2015 file

`Propulsion/LaTeXandpdfs/thermo.tex`

already set to the same widescreen page. Its sections sit in this sequence next to the topic they extend. The part title Thermodynamics (Revisited) is not used. Overlap is kept, so nothing in either file is only in the old PDF. A preserved block begins with the year of its manuscript. Where a section title was changed so that two blocks would not share a name, the manuscript title is the italic line under the new heading.

Two neighboring files were checked and not copied. `Propulsion/thermo.py` is the SymPy scratch pad for a few numerical problems; the diesel-engine values in the 2015 notes match it, and that check is in §152.10. `Propulsion.tex` recalls Kittel–Kroemer results inside the propulsion dump. `DGDT_dump.tex` has a 2019 reminder to compare the convection form of enthalpy with Kittel–Kroemer and with Sonntag; that split is already in the 2015 manuscript, and the per-particle formula is corrected in §152.5.

The geometry books named in the 2008 bibliography are Schutz [3], Bamberg and Sternberg [4], and Frankel [5].

120.2 Notation and the energy identity

Kittel and Kroemer measure temperature and entropy in energy units. That convention is the one used in the new sections.

$$\tau = k_B T, \qquad \sigma = \ln g, \qquad S = k_B \sigma. \tag{594}$$

Here g is the multiplicity of a macrostate, T is the kelvin temperature, and k_B is Boltzmann's constant. Partial derivatives are written as fractions. The variables held fixed are the subscripts. Einstein summation is used where a coordinate index is repeated, without a separate reminder.

The energy differential adopted for the new sections, and for every correction below, is

$$dU = \tau\, d\sigma - p\, dV + \sum_j \mu_j\, dN_j. \tag{595}$$

The reversible heat 1-form is $Q = \tau\, d\sigma$. The work done on the system by a slow volume change is $-p\, dV$, so the work done by the system is $p\, dV$. The engineering line $dU = Q - W$ with that work by the system is the same statement. Equation (595) is already the identity labeled in the 2008 notes and the one used for the Helmholtz differential $dF = -\sigma\, d\tau - p\, dV$. The places where a preserved line uses the opposite work sign, or writes $(\partial F/\partial \tau)_V = \sigma$, are marked in the manuscript and collected in §152.

A correction environment in the preserved text is a later check. The surrounding sentences and formulas are the manuscript. Where they disagree, the correction is the one to use.

120.3 Order

Counting, entropy, and temperature come first, then heating and the dilute-gas law. The Boltzmann distribution is followed by the pressure derivation, the first law, and both heat capacities. Radiation, the chemical potential, and the Gibbs distribution come next, then the ideal gas and mixtures. Fermi and Bose gases are followed by engines and magnetic cooling, the catalogue of potentials, the Gibbs chapter, chemical equilibrium, and semiconductors. Phase coexistence, the Joule–Thomson expansion, kinetic theory, and hydrodynamics follow. Nonequilibrium transport is the next part, because it is the theory of fluxes and entropy production. The resolutions close the notes.

120.4 What is new

New writing is §121, §133, §137, §140, and §152. The resolutions answer the adiabatic question left open in 2015, the use of C_V on a path whose volume changes, the claim that path-dependent heat requires nontrivial topology, and the arithmetic and transcription errors listed there.

121 States of a model system

The 2008 manuscript begins with this heading and no text. The counting below is the input to the paramagnet solution in the entropy chapter. Nothing here is quoted from the book.

121.1 Two-state elements and the spin excess

Take N elements, N even, each with two states. Label the states up and down. Let $N_\uparrow$ and $N_\downarrow$ be the occupation numbers,

$$N_\uparrow + N_\downarrow = N. \tag{596}$$

The spin excess $2s$ is the difference,

$$2s = N_\uparrow - N_\downarrow, \qquad s \in \mathbb{Z}, \qquad |s| \le N/2. \tag{597}$$

Solving (596) and (597) gives

$$N_\uparrow = \frac{N}{2} + s, \qquad N_\downarrow = \frac{N}{2} - s. \tag{598}$$

Both are nonnegative integers in the range just stated. The elements are treated as distinguishable by their positions in the model, as in a lattice of fixed sites. The multiplicity of the macrostate s is the binomial count

$$g(N,s) = \frac{N!}{N_\uparrow!\, N_\downarrow!} = \frac{N!}{\left(\frac{N}{2}+s\right)!\left(\frac{N}{2}-s\right)!}. \tag{599}$$

The entropy in the units (594) is $\sigma(N,s) = \ln g(N,s)$.

An external field B is not required for the count. When the only energy is the coupling of each element to that field, take the single-element energies to be $-mB$ and $+mB$. Then

$$U(s) = -mB\, N_\uparrow + mB\, N_\downarrow = -mB\,(N_\uparrow - N_\downarrow) = -2mB\, s. \tag{600}$$

The symbol m is the magnitude of the moment of one element. Equation (600) is the energy used in the preserved paramagnet solution.

121.2 Stirling's approximation

For a positive integer n,

$$\ln n! = n \ln n - n + \tfrac{1}{2}\ln(2\pi n) + \varepsilon_n, \tag{601}$$

with $\varepsilon_n \to 0$ as $n \to \infty$. The exponent computed next uses the terms $n \ln n - n$. The square-root term is restored at the central multiplicity, and it is checked that it does not change the coefficient of s^2/N.

Substitute (601), without ε_n and without the square-root term, into (599):

$$\begin{aligned} \ln g(N,s) &= \ln N! - \ln N_\uparrow! - \ln N_\downarrow! \\ &= N \ln N - N - N_\uparrow \ln N_\uparrow + N_\uparrow - N_\downarrow \ln N_\downarrow + N_\downarrow \\ &= N \ln N - N_\uparrow \ln N_\uparrow - N_\downarrow \ln N_\downarrow. \end{aligned} \tag{602}$$

The linear terms $-N + N_\uparrow + N_\downarrow$ cancel by (596).

121.3 Gaussian peak

Set

$$x = \frac{2s}{N}, \qquad |x| \le 1, \tag{603}$$

so $N_\uparrow = (N/2)(1+x)$ and $N_\downarrow = (N/2)(1-x)$. Then

$$\begin{aligned} N_\uparrow \ln N_\uparrow &= \frac{N}{2}(1+x)\big(\ln(N/2) + \ln(1+x)\big), \\ N_\downarrow \ln N_\downarrow &= \frac{N}{2}(1-x)\big(\ln(N/2) + \ln(1-x)\big). \end{aligned} \tag{604}$$

Add (604):

$$N_\uparrow \ln N_\uparrow + N_\downarrow \ln N_\downarrow = N\ln(N/2) + \frac{N}{2}\big[\ln(1-x^2) + x\ln\big((1+x)/(1-x)\big)\big]. \tag{605}$$

The coefficient of $\ln(N/2)$ is $(N/2)(1+x)+(N/2)(1-x)=N$. Insert (605) into (602):

$$\ln g(N,s) = N\ln 2 - \frac{N}{2}\ln(1-x^2) - \frac{Nx}{2}\ln\left(\frac{1+x}{1-x}\right). \tag{606}$$

The identity $N\ln N - N\ln(N/2) = N\ln 2$ was used. Also $Nx/2 = s$.

For the peak, take $|x| \ll 1$, which is $|s| \ll N/2$. The Taylor expansions through order x^3 in the logarithms are

$$\ln(1-x^2) = -x^2 + O(x^4),$$
$$\ln\left(\frac{1+x}{1-x}\right) = 2\big(x + O(x^3)\big). \tag{607}$$

Hence

$$-\frac{N}{2}\ln(1-x^2) = \frac{N}{2}x^2 + O(Nx^4) = \frac{2s^2}{N} + O\left(\frac{s^4}{N^3}\right),$$
$$-\frac{Nx}{2}\ln\left(\frac{1+x}{1-x}\right) = -Nx^2 + O(Nx^4) = -\frac{4s^2}{N} + O\left(\frac{s^4}{N^3}\right). \tag{608}$$

The first line uses $(N/2)x^2 = (N/2)(4s^2/N^2) = 2s^2/N$. The second uses $Nx^2 = N(4s^2/N^2) = 4s^2/N$. Add them:

$$\ln g(N,s) = N\ln 2 - \frac{2s^2}{N} + O\left(\frac{s^4}{N^3}\right). \tag{609}$$

At $s=0$, the same truncation gives $\ln g(N,0) = N\ln 2$, so

$$g(N,s) \simeq g(N,0)\,\exp\left(-\frac{2s^2}{N}\right), \qquad g(N,0) \simeq 2^N, \tag{610}$$

inside the approximation that dropped both the square-root Stirling term and $O(s^4/N^3)$. Restoring $\frac{1}{2}\ln(2\pi n)$ at $s=0$ replaces the central estimate by

$$g(N,0) \simeq 2^N \sqrt{\frac{2}{\pi N}}. \tag{611}$$

The check is the standard one: $\ln N! - 2\ln((N/2)!)$ with (601) produces $N\ln 2 + \frac{1}{2}\ln(2/(\pi N))$. The s-dependent piece of those square-root terms is $-\frac{1}{2}\ln(1-x^2) = O(s^2/N^2)$, which is smaller than s^2/N by one power of N. It does not change the coefficient -2 in (609).

The peak is sharp in the fractional excess. From (609), $\sigma(s) - \sigma(0) = -2s^2/N$ at this order. At $s = \sqrt{N}$ the multiplicity has fallen by a factor e^{-2} relative to the center, while the fractional excess is $s/(N/2) = 2N^{-1/2}$. For large N almost every state in the count sits at a fractional excess of order $N^{-1/2}$ or smaller.

121.4 Temperature of the paramagnet in this approximation

Solve (600) for the excess, $s = -U/(2mB)$, and substitute into the exponent of (609):

$$\frac{2s^2}{N} = \frac{2}{N}\cdot\frac{U^2}{4m^2B^2} = \frac{U^2}{2m^2B^2N}. \tag{612}$$

Therefore

$$\sigma(N,U) = \sigma_0 - \frac{U^2}{2m^2B^2N}, \qquad \sigma_0 = \ln g(N,0), \tag{613}$$

in the same small-excess regime, and with N and B held fixed. Differentiate:

$$\frac{1}{\tau} = \left(\frac{\partial \sigma}{\partial U}\right)_N = -\frac{U}{m^2 B^2 N}. \tag{614}$$

Hence

$$U = -\frac{N m^2 B^2}{\tau}. \tag{615}$$

From (600), $U = -2mB\,s$, so the fractional excess in this regime is

$$\frac{2s}{N} = \frac{mB}{\tau}. \tag{616}$$

Equation (614) is the derivative written in the preserved paramagnet solution. That solution starts from the Gaussian multiplicity (610) and does not derive it. The canonical two-state free energy is in §137. Its small-field expansion reproduces (615). *2008 manuscript.*

122 Entropy and Temperature

Thermal Equilibrium

EY : 20150821 Based on considering the physical setup of two systems that can only exchange energy between each other, that are in thermal contact, this is a derivation of temperature.

$U = U_1 + U_2$ is constant total energy of 2 systems $1, 2$ in thermal contact
multiplicity $g(N, U)$ of combined system is

$$g(N, U) = \sum_{U_1 \leq U} g_1(N_1, U_1) g_2(N_2, U - U_1)$$

The "differential" of $g(N, U)$ is

$$dg = \left(\frac{\partial g_1}{\partial U_1}\right)_{N_1} g_2 dU + g_1 \left(\frac{\partial g_2}{\partial U_2}\right)_{N_2} dU_2 = 0$$

EY : 20150821 This step can be made mathematically sensible by considering the exterior derivative d of $g \in C^\infty(\Sigma)$, where Σ is the manifold of states of the system, with local coordinates N, U, where U happens to be a global coordinate. Then, consider a curve in Σ s.t. it has no component in $\frac{\partial}{\partial N}$, $\frac{\partial}{\partial N_1}$, and this curve is a "null curve" so that the vector field $X \in \mathfrak{X}(\Sigma)$ generated by this curve is s.t. $dg(X) = 0$.

With $-dU_1 = dU_2$,

$$\frac{1}{g_1}\left(\frac{\partial g_1}{\partial U_1}\right)_{N_1} = \frac{1}{g_2}\left(\frac{\partial g_2}{\partial U_2}\right)_{N_2} \Longrightarrow \left(\frac{\partial \ln g_1}{\partial U_1}\right)_{N_1} = \left(\frac{\partial \ln g_2}{\partial U_2}\right)_{N_2}$$

Define

$$\sigma(N, U) := \ln g(N, U)$$

Then

$$\Longrightarrow \left(\frac{\partial \sigma_1}{\partial U_1}\right)_{N_1} = \left(\frac{\partial \sigma_2}{\partial U_2}\right)_{N_2}$$

Temperature

$T_1 = T_2$ - temperatures of 2 systems in thermal equilibrium are equal.

T "must be a function of $\left(\frac{\partial \sigma}{\partial U}\right)_N$ [2].

$$\Longrightarrow \frac{1}{T} = k_B \left(\frac{\partial \sigma}{\partial U}\right)_N$$

Experimentally, $k_B = 1.381 \times 10^{-23}\, J/K = 1.381 \times 10^{-16}$ ergs$/K$.

Now

$$\frac{1}{\tau} = \left(\frac{\partial \sigma}{\partial U}\right)_N \tau = k_B T$$

Problems

Solution 1. Entropy and temperature.

(a) Recall that $\frac{1}{\tau} \equiv \left(\frac{\partial \sigma}{\partial U}\right)_{N,V}$ and $\sigma(N,U) \equiv \log g(N,U)$. Given $g(U) = CU^{3N/2}$,

$$\sigma(N,U) = \log CU^{3N/2} = \log C + \frac{3N}{2}\log U$$

$$\frac{\partial \sigma}{\partial U} = \frac{3N}{2}\frac{1}{U} = \frac{1}{\tau} \Longrightarrow \boxed{U = \frac{3N}{2}\tau}$$

(b) $\left(\frac{\partial^2 \sigma}{\partial U^2}\right)_N < 0$?

$$\left(\frac{\partial^2 \sigma}{\partial U^2}\right)_N = -\frac{3N}{2}\left(\frac{1}{U^2}\right) < 0$$

Solution 2. Paramagnetism.

$$U(s) = U_1(s_1) + U_2(s_2) = -2mB(s_1 + s_2) = -2mBs \text{ or } s = \frac{U}{-2mB}$$

i.e. potential energy $U(s) = -2s \cdot mB$.

For $|s| \ll N$, then

$$g(N,s) \simeq g(N,0)\exp\left(-2s^2/N\right) = g(N,0)\exp\left(\frac{-U^2}{2(mB)^2 N}\right)$$

$$\sigma(N,U) = \ln g(N,U) = \sigma_0 - \frac{U^2}{2m^2B^2}\frac{1}{N} \text{ where } \sigma_0 = \ln g(N,0)$$

$$\frac{1}{\tau} = \left(\frac{\partial \sigma}{\partial U}\right)_N = \frac{-U}{m^2B^2}\frac{1}{N}$$

What is the thermal equilibrium value of this N-spin system of fractional magnetization? If U denotes $\langle U \rangle$, thermal average energy, we also get the thermal average spin excess.

$$\langle U \rangle = \langle -2mBs \rangle = -2mB\langle s \rangle$$

$$\Longrightarrow \tau = \frac{m^2B^2N}{-U} = \boxed{\frac{mBN}{2\langle s \rangle}}$$

Solution 3. Quantum harmonic oscillator.

(a) Result from Ch. 1: $g(N,n) = \frac{(N+n-1)!}{n!(N-1)!}$.

Let $N-1 \to N \Longrightarrow g(N+1,n) = \frac{(N+n)!}{n!N!}$.

$$\sigma(N+1,n) \equiv \ln g(N+1,n) = \ln \frac{(N+n)!}{n!N!} = \ln(N+n)! - \ln(n!) - \ln(N!)$$

$$\approx (N+n)\ln(N+n) - N - n - n\ln n + n - N\ln N + N = \boxed{(N+n)\ln(N+n) - n\ln n - N\ln N}$$

(b) Let U denote total energy $n\hbar\omega$ of oscillators.
$U = n\hbar\omega$ or $n = \frac{U}{\hbar\omega}$

$$\sigma(N,U) = (N + \frac{U}{\omega})\ln(N + \frac{U}{\omega}) - \frac{U}{\omega}\ln\frac{U}{\omega} - N\ln N$$

At τ, $\frac{1}{\tau} = \left(\frac{\partial \sigma}{\partial U}\right)_N$.

$$\frac{1}{\tau} = \frac{1}{\omega}\ln(N + \frac{U}{\omega}) - \frac{1}{\omega}\ln\frac{U}{\omega} = \frac{1}{\omega}\ln\left(\frac{N\omega}{U} + 1\right) \text{ or } \boxed{U = \frac{N\omega}{\exp(\omega/\tau) - 1}}$$

Solution 4. The meaning of "never."

Suppose 10^{10} monkeys.

(a) Hamlet represents one specific ordering of 10^{15} with 44 possibilities for each character. The probability of hitting upon Hamlet from a given, random sequence is $\left(\frac{1}{44}\right)^{100000} = \boxed{\frac{1}{44^{100000}}}$. Given that $\log_{10} 44 = 1.64345$, then $10^{1.64345} = 44$ or $10^{-1.64345} = 44^{-1}$ so then

$$\left(\frac{1}{44}\right)^{100000} = 10^{-164345}$$

(b)

$$(\text{age of universe}) \cdot \left(\frac{10 \text{ keys}}{\text{second}}\right) = 10^{18}\, s \left(\frac{10 \text{ keys}}{\text{second}}\right) = 10^{19} \text{ keys}$$

$$10^{19} \text{ keys} \cdot 10^{10} \text{ monkeys} = 10^{29} \text{ keys typed out } \left(\frac{1 \text{ hamlet}}{10^5 \text{ characters}}\right) = 10^{24} \text{ possible "Hamlets"}$$

From part (a), the probability that a given, random sequence is Hamlet, $10^{-164345}$

$$(10^{29} \text{ characters})(10^{-164345}) = 10^{-164316}$$

Note, I think that the probability should be $(10^{29} \text{ characters}) \left(\frac{1 \text{ Hamlet}}{10^5 \text{ characters}}\right) (10^{-164345}) = 10^{-164321}$
Since we are considering the number of "Hamlet", 10^5 character sequences.

2015 manuscript.

123 Heating and the dilute gas

Manuscript section title: Background. [1]

123.1 Heating and Temperature

Heating: keep in mind 3 different types of heating for energy exchange between two systems:

1. Heating by conduction - literal contact, molecules jiggle faster from molecules jiggling faster by bouncing off them
2. Heating by radiation - em waves from hot source strike and excite target
3. heating by convection - energy transport by flow (perhaps a fluid)

This all relates to
Q

123.2 Some dilute gas relationships

Pressure according to kinetic theory

i.e. some kinetic theory
$F \equiv$ force on area A due to molecules
$\Delta p \equiv$ momentum transferred to wall per collision
$n \equiv$ number of collisions in time Δt
Thus

$$F = \frac{(\Delta p)n}{\Delta t}$$

Now $\Delta p = 2mv_x$ since $\Delta p = mv_x - (-mv_x) = 2mv_x$ (elastic collision with momentum conversation)

$v_x \Delta t A$ is a volume, inside of which gas molecules can be within distance $v_x \Delta t$ toward the wall.
$\frac{N}{V}$ number density of molecules

Assume equal distribution of velocities: Thus $\frac{1}{2}$

$$n = (v_x \Delta t A) \frac{1}{2} \frac{N}{V}$$

$$P = \frac{F}{A} = \frac{(2mv_x)(v_x \Delta t A)\frac{1}{2}\frac{N}{V}}{A\Delta t} = m v_x^2 \frac{N}{V}$$

Suppose $\langle v^2 \rangle = \langle v_x^2 \rangle + \langle v_y^2 \rangle + \langle v_z^2 \rangle = 3\langle v_x^2 \rangle = d\langle v_x^2 \rangle$.

$$\implies P = \frac{mN}{dV}\langle v^2 \rangle = \frac{2}{d}\frac{1}{2} m \langle v^2 \rangle \frac{N}{V}$$

An empirical gas law

Now

$$P = \frac{N\tau}{V} \quad \text{(empirical)}$$

$$\implies \frac{1}{2} m \langle v^2 \rangle = \frac{d}{2}\tau$$

2008 manuscript.

124 Boltzmann Distribution and Helmholtz Free Energy

cf. **Example: Energy and heat capacity of a two state system**, pp. 62 of Kittel and Kroemer [2]. Kittel and Kroemer introduces the heat capacity very early, specific to this example.

Definition 218. heat capacity C_V at constant volume is defined as

$$C_V := \tau \left(\frac{\partial \sigma}{\partial \tau} \right)_V \tag{617}$$

Recall the thermodynamic identity (which is introduced many equations later):

$$dU = \tau d\sigma - pdV \in \Omega^1(\Sigma)$$

where Σ is a manifold of states of all systems.

Consider local coordinates of Σ, (σ, V). Consider curve $c : \mathbb{R} \to \Sigma$ s.t. c generates a vector field $\dot{c} = \dot{\sigma}\frac{\partial}{\partial \sigma}$ i.e.

$$c(\tau) \in \Sigma$$

no component in the V direction. Notice the prescient choice of parameter τ.

Now for internal energy $U \in C^\infty(\Sigma)$, taking the exterior derivative d results in

$$dU = \frac{\partial U}{\partial \sigma} d\sigma + \frac{\partial U}{\partial V} dV$$

Then applying dU onto vector field $\dot{\sigma}\frac{\partial}{\partial \sigma}$,

$$dU\left(\dot{\sigma}\frac{\partial}{\partial \sigma}\right) = \frac{\partial U}{\partial \sigma}\dot{\sigma} = \dot{\sigma}\tau + 0$$

Now,

$$\left(\frac{\partial U}{\partial \sigma}\right)_V \left(\frac{\partial \sigma}{\partial \tau}\right)_V = \left(\frac{\partial U}{\partial \tau}\right)_V = \left(\frac{\partial \sigma}{\partial \tau}\right)_V \tau$$

Hence,

$$C_V := \tau \left(\frac{\partial \sigma}{\partial \tau}\right)_V = \left(\frac{\partial U}{\partial \tau}\right)_V \tag{618}$$

EY: 20150825 Why do we need differential geometry? It's because I always wondered why you could do this:

$$C_V := \tau \left(\frac{\partial \sigma}{\partial \tau}\right)_V \stackrel{?}{=} \left(\frac{\partial U}{\partial \tau}\right)_V \text{ with } \tau d\sigma = dU \Longrightarrow \tau \partial \sigma \stackrel{?}{=} \partial U$$

and talk of "differentials."

Definition: Reversible process. EY : 20150824 Mathematically, 1-forms are exact.

Pressure

Consider coordinates $(\sigma, V) \in \Sigma$ of manifold of thermodynamic states Σ.
Imagine a reversible compression of a cube system (so imagine $dV < 0$; cube's volume get smaller).
σ constant, i.e. $d\sigma = 0$ (on this curve in Σ) because as particles in cube gets squeezed, less positions particles could sit in, but they get more kinetic energy, more energetic (more momentum squared).

Now $U = U(\sigma, V) \in C^\infty(\Sigma)$.
$\Longrightarrow dU = \left(\frac{\partial U}{\partial \sigma}\right)_V d\sigma + \left(\frac{\partial U}{\partial V}\right)_\sigma dV$
Again, imagine a curve $c : \mathbb{R} \to \Sigma$, connecting 1 state $(\sigma, V) \in \Sigma$ to another state $(\sigma, V + dV) \in \Sigma$ s.t. $\dot{c} = \dot{V} \frac{\partial}{\partial V}$.

$$\Longrightarrow dU(\dot{c}) = \left(\frac{\partial U}{\partial V}\right)_\sigma \dot{V}$$

Introduce 1-form $W \in \Omega^1(\Sigma)$ of work done on the cube system from some external agent

$$W = -p dV$$

so $W > 0$ when $dV < 0$.

Then

$$W(\dot{c}) = -p\dot{V} = dU(\dot{c}) = \left(\frac{\partial U}{\partial V}\right)_\sigma \dot{V}$$

$$\Longrightarrow \boxed{p = -\left(\frac{\partial U}{\partial V}\right)_\sigma} \tag{619}$$

Consider another set of coordinates $(U, V) \in \Sigma$ for manifold Σ. Now entropy σ is a function of U, V, as $\sigma = \sigma(U, V) \in C^\infty(M)$, so that
$d\sigma = \left(\frac{\partial \sigma}{\partial U}\right)_V dU + \left(\frac{\partial \sigma}{\partial V}\right)_U dV$

Consider curve $c = (U, V) \in \Sigma$. Then $\dot{c} = \dot{U} \frac{\partial}{\partial U} + \dot{V} \frac{\partial}{\partial V}$. For this curve c, σ is constant, meaning $d\sigma(\dot{c}) = 0$ (it's a "null curve" of $d\sigma$

$$d\sigma(\dot{c}) = 0 = \left(\frac{\partial \sigma}{\partial U}\right)_V \dot{U} + \left(\frac{\partial \sigma}{\partial V}\right)_U \dot{V}$$

Now define

Definition 219.

$$\frac{1}{\tau} := \left(\frac{\partial \sigma}{\partial U}\right)_V \tag{620}$$

So then we have $\frac{1}{\tau}\dot{U} + \left(\frac{\partial \sigma}{\partial V}\right)_U \dot{V} = 0$. For the parameter of curve c, choose the parameter to be V, knowing that σ is constant on this curve, or thermodynamic process. Thus

$$\frac{1}{\tau}\left(\frac{\partial U}{\partial V}\right)_\sigma = -\left(\frac{\partial \sigma}{\partial V}\right)_U \Longrightarrow \frac{-p}{\tau} = -\left(\frac{\partial \sigma}{\partial V}\right)_U \text{ or}$$

$$\boxed{p = \tau \left(\frac{\partial \sigma}{\partial V}\right)_U} \tag{621}$$

Thermodynamic Identity

Let $\sigma = \sigma(U, V) \in C^\infty(\Sigma)$. Then
$d\sigma = \left(\frac{\partial \sigma}{\partial U}\right)_V dU + \left(\frac{\partial \sigma}{\partial V}\right)_U dV \in \Omega^1(\Sigma)$.

Now recall the quantities we've recently used: $\frac{1}{\tau} := \left(\frac{\partial \sigma}{\partial U}\right)_V$ (this is a definition) and $\frac{p}{\tau} = \left(\frac{\partial \sigma}{\partial V}\right)_U$ (it comes from the physics, of doing work on the system, by some external agent). Then the **thermodynamic identity** is obtained:

Theorem 76.

$$\boxed{\tau d\sigma = dU + pdV} \tag{622}$$

Ideal Gas: A First Look

One atom in a box

one atom of mass M in cubical box of volume $V = L^3$
$\frac{-\hbar^2}{2M}\nabla^2\psi = \epsilon\psi \qquad p = \frac{\hbar}{i}\nabla \qquad p^2\psi = \epsilon\psi$
$\implies \psi(x) = A \sin\left(\frac{n_x \pi x}{L}\right) \sin\left(\frac{n_y \pi y}{L}\right) \sin\left(\frac{n_z \pi z}{L}\right)$
$\epsilon_n = \frac{\hbar^2}{2M}\left(\frac{\pi}{L}\right)^2 (n_x^2 + n_y^2 + n_z^2)$
Then the partition function Z_1 is

$$Z_1 = \sum_n \exp\left(\frac{-\epsilon_n}{\tau}\right) = \sum_{(n_x, n_y, n_z)} \exp\left(\frac{-\hbar^2}{2M\tau}\left(\frac{\pi}{L}\right)^2 (n_x^2 + n_y^2 + n_z^2)\right)$$

Let

$$\alpha^2 = \frac{\hbar^2 \pi^2}{2ML^2\tau} \text{ or } \alpha = \frac{\hbar\pi}{(2M\tau)^{1/2} V^{2/d}}$$

Then

$$Z_1 = \int_0^\infty dn_x \int_0^\infty dn_y \int_0^\infty dn_z \exp\left[-\alpha^2(n_x^2 + n_y^2 + n_z^2)\right] = \left(\int_0^\infty dn_x \exp\left(-\alpha^2 n_x^2\right)\right)^3 = \left(\frac{1}{\alpha}\right)^3 \left(\frac{\pi^{1/2}}{2}\right)^3 = \left(\frac{\pi^{1/2}}{2\alpha}\right)^3$$

In general, $Z_1 = \left(\frac{\pi^{1/2}}{2\alpha}\right)^d$
Now

$$Z_1 = \left(\frac{\pi^{1/2} V^{1/d}}{2\frac{\hbar\pi}{(2M\tau)^{1/2}}}\right)^d = \frac{V}{\left(\frac{2\hbar\pi}{M\tau}\right)^{d/2}} = n_Q V = \frac{n_Q}{n}$$

in terms of concentration $n = 1/V$.

$n_Q := \left(\frac{M\tau}{2\pi\hbar^2}\right)^{d/2}$ is the **quantum concentration**. "*It is the concentration associated with one atom in a cube of side equal to the thermal average de Broglie wavelength.*" (pp. 73, Ch. 3: Boltzmann Distribution and Helmholtz Free Energy, Section "Ideal Gas: A First Look" of Kittel and Kroemer [2])

From Kittel and Kroemer (1980) [2], Example : N atoms in a box, Chapter 3: "Boltzmann Distribution and Helmholtz Free Energy",

state of energy $\epsilon_\alpha(1) + \epsilon_\beta(2) + \cdots + \epsilon_\xi(N)$, $\alpha, \beta, \ldots \xi$ denote orbital indces of atoms in successive boxes.
each entry occurs $N!$ times in Z_1^N (EY: 20151022) $N!$ ways to fill $\alpha, \beta \ldots \xi$ orbitals with N distinguishable particles. Thus,

$$Z_N = \frac{1}{N!} Z_1^N = \frac{1}{N!}(n_Q V)^N$$

Problems

Solution 1. Free energy of a two state system.

(a)

$$Z = 1 + e^{-\epsilon/\tau}$$
$$F = -\tau \ln Z = -\tau \ln\left(1 + e^{-\epsilon/\tau}\right)$$

(b)

$$U = -\tau^2 \frac{\partial(F/\tau)}{\partial \tau} = \frac{\epsilon e^{-\epsilon/\tau}}{1+e^{-\epsilon/\tau}}$$
$$\sigma = -\frac{\partial F}{\partial \tau} = \ln\left(1+e^{-\epsilon/\tau}\right) + \frac{\frac{\epsilon}{\tau} e^{-\epsilon/\tau}}{(1+e^{-\epsilon/\tau})}$$

Solution 2. Magnetic susceptibility

(a) Remember to calculate the **multiplicity** in the N-spin system (it's not enough to sum up $\exp(-\epsilon_s/\tau)$ factors).

$$M = 2sm \qquad U_s = -MB = -2smB \qquad N = N_+ + N_-$$
$$2s = N_+ - N_- = N_+ - (N - N_+) = 2N_+ - N$$

$$Z = \sum_{s=-N/2}^{N/2} \binom{N}{N_+} \exp\left(\frac{2smB}{\tau}\right) = \sum_{s=-N/2}^{N/2} \frac{N!}{\left(\frac{N}{2}+s\right)!\left(\frac{N}{2}-s\right)!} \exp\left(\frac{2mBs}{\tau}\right) = \sum_{s=0}^{N} \frac{N!}{s!(N-s)!} \exp\left(\frac{2mB}{\tau}\left(s-\frac{N}{2}\right)\right) =$$
$$= e^{-\frac{NmB}{\tau}}(1+e^{\frac{2mB}{\tau}})^N = 2^N \cosh^N\left(\frac{mB}{\tau}\right)$$

where it was **crucial** to use $(1+x)^N = \sum_{j=1}^{N} \binom{N}{j} x^j$. Note, in changing the sum index, since N is large, we can neglect dropping the $s=0$ term.

Correction (binomial index). The binomial theorem starts at $j=0$: $(1+x)^N = \sum_{j=0}^{N} \binom{N}{j} x^j$. The $j=0$ term is 1, and the closed form on the previous line includes it.

$$\partial_\tau Z = 2^N(N)(\cosh^{N-1}\left(\frac{mB}{\tau}\right))\sinh\left(\frac{mB}{\tau}\right)\left(\frac{-m}{\tau^2}\right)$$
$$M = -\tau^2 \frac{\partial}{\partial \tau} \ln Z = Nm \tanh\left(\frac{mB}{\tau}\right)$$
$$\chi = \frac{\partial M}{\partial B} = \frac{Nm^2}{\tau} sech^2\left(\frac{mB}{\tau}\right)$$

(b)

$$F = -\tau \ln Z = -\tau \ln\left((2\cosh\left(\frac{mB}{\tau}\right))^N\right) = -N\tau \ln\left(2\cosh\left(\frac{mB}{\tau}\right)\right)$$

For $x \equiv \frac{M}{nm} = \tanh\left(\frac{mB}{\tau}\right)$. Now $1-\tanh^2 y = sech^2 y$. $F = -N\tau \ln\left(\frac{2}{\sqrt{1-x^2}}\right) = \frac{-N\tau}{2} \ln\left(\frac{4}{1-x^2}\right)$.

(c) For $\frac{mB}{\tau} \ll 1$, $\cosh^2\left(\frac{mB}{\tau}\right) \to 1$. $\boxed{\chi = \frac{m^2 N}{\tau}}$

Solution 3. Free energy of a harmonic oscillator

(a)

$$Z = \sum_{s=0}^{\infty} \exp\left(\frac{-s\hbar\omega_0}{\tau}\right) = \frac{1}{1-e^{-\hbar\omega_0/\tau}} = (1-e^{-\hbar\omega_0/\tau})^{-1}$$
$$F = -\tau \ln Z = \tau \ln\left(1-e^{-\hbar\omega_0/\tau}\right) \simeq \tau \ln\left(\frac{\hbar\omega_0}{\tau}\right) \text{ for } 1 \gg \frac{\hbar\omega_0}{\tau}$$

(b)

$$\sigma = -\frac{\partial F}{\partial \tau} = -\{\ln\left(1-e^{-\hbar\omega_0/\tau}\right) + \frac{\tau}{1-e^{-\hbar\omega_0/\tau}}(-e^{-\hbar\omega_0/\tau})\left(\frac{\hbar\omega_0}{\tau}\right)\} = \frac{-\hbar\omega_0/\tau}{e^{\hbar\omega_0/\tau}-1} - \ln\left(1-e^{-\frac{\hbar\omega_0}{\tau}}\right)$$

Solution 4. Energy fluctuations.

$$\frac{1}{\tau} = \beta \qquad \frac{\partial}{\partial \tau} = \frac{\partial \beta}{\partial \tau}\frac{\partial}{\partial \beta} = -\frac{1}{\tau^2}\frac{\partial}{\partial \beta}$$

$$Z = \sum_s e^{-\epsilon_s \beta} \qquad \partial_\beta Z = \sum_s -\epsilon e^{-\epsilon_s \beta} \qquad \partial_\beta^2 Z = \sum_s \epsilon_s^2 e^{-\epsilon_s \beta}$$

$$U = \frac{\sum_s \epsilon_s e^{-\epsilon_s/\tau}}{Z} = -\partial_\beta \ln Z$$

$$\frac{\partial U}{\partial \tau} = -\beta^2 \frac{\partial}{\partial \beta} U = \beta^2 \left(\partial_\beta \left(\frac{\partial_\beta Z}{Z}\right)\right) = \beta^2 \left(\frac{(\partial_\beta^2 Z)Z - (\partial_\beta Z)^2}{Z^2}\right) = \beta^2 \left(\frac{\partial_\beta^2 Z}{Z} - \left(\frac{\partial_\beta Z}{Z}\right)^2\right) =$$

$$\Longrightarrow \tau^2 \frac{\partial U}{\partial \tau} = \langle \epsilon^2 \rangle - \langle \epsilon \rangle^2$$

Solution 5. Overhauser effect. System $\mathcal{S}$ in energy eigenstate $E_n = n\epsilon$.
$P(E) = (1) g_{\mathcal{R}}(E)$
Note $\Delta U_{\mathcal{R}} = (\alpha - 1)\epsilon$. $\Delta U_{\mathcal{S}} = \epsilon$. $\frac{dU_{\mathcal{S}}}{d\epsilon} + \frac{dU_{\mathcal{R}}}{d\epsilon} = 1 + (\alpha - 1) = \alpha = \frac{dU_{tot}}{d\epsilon}$ in a specific energy eigenstate; $g_{\mathcal{S}}(n\epsilon) = 1$

While $g_{\mathcal{R}}(U_{\mathcal{R}})$ = multiplicity of reservoir $\mathcal{R}$ with $U_{\mathcal{R}}$ energy.

Now

$$\frac{\partial \sigma_{\mathcal{R}}}{\partial E_s} = \frac{1}{\tau}$$

and

$$g_{\mathcal{R}}(U_{\mathcal{R}}) = \exp\left(\sigma_{\mathcal{R}}(U_{\mathcal{R}})\right)$$

If $\epsilon \frac{dU_{\mathcal{R}}}{d\epsilon} = (\alpha - 1)\epsilon = \Delta U_{\mathcal{R}}$ small compared to $U_{\mathcal{R}}$.

$$U_{\mathcal{R}}(E_{\mathcal{S}} = (n+1)\epsilon) = U_{\mathcal{R}}(n\epsilon) + \frac{dU_{\mathcal{R}}}{d\epsilon}\epsilon = U_{\mathcal{R}}(n\epsilon) + (\alpha - 1)\epsilon$$

$$\sigma_{\mathcal{R}}(U_{\mathcal{R}}((n+1)\epsilon)) \simeq \sigma_{\mathcal{R}}(U_{\mathcal{R}}(n\epsilon)) + \frac{1}{\tau}(\alpha - 1)\epsilon$$

$$\frac{P(E_{\mathcal{S}} = (n+1)\epsilon)}{P(E_{\mathcal{S}} = n\epsilon)} = \frac{\exp\left(\sigma_{\mathcal{R}}(U_{\mathcal{R}}(n\epsilon)) + \frac{1}{\tau}(\alpha - 1)\epsilon\right)}{\exp\left(\sigma_{\mathcal{R}}(U_{\mathcal{R}}(n\epsilon))\right)} = \boxed{\exp\left(-\frac{\epsilon}{\tau}(1 - \alpha)\right)}$$

Solution 6. Rotation of diatomic molecules.

(a) $\epsilon(j) = j(j+1)\epsilon_0$. $g(j) = 2j + 1$
Remember that Z is a sum over all states, not over all levels.

$$Z = \sum_{j=0}^{\infty} (2j+1) e^{-j(j+1)\epsilon_0/\tau} = \sum_{j=0}^{\infty} \frac{d}{dj}(e^{-(j^2+j)\epsilon_0/\tau}) \left(\frac{-\tau}{\epsilon_0}\right) = \frac{-\tau}{\epsilon_0} \sum_{j=0}^{\infty} \frac{d}{dj} \left(e^{-\epsilon_0/\tau}\right)^{j^2+j}$$

(b) For $1 \gg \frac{\epsilon_0}{\tau}$

$$Z_R(\tau) = -\frac{\tau}{\epsilon_0} \int_0^\infty \frac{d}{dx} \left(e^{-\epsilon_0/\tau}\right)^{x^2+x} dx = -\frac{\tau}{\epsilon_0} \left((e^{-\epsilon_0/\tau})^{x^2+x} - (e^{-\epsilon_0/\tau})^0\right) = \frac{\tau}{\epsilon_0}$$

(c) For $\frac{\tau}{\epsilon_0} \ll 1$

$$Z_R(\tau) = 1 + 3e^{-2\epsilon_0/\tau}$$

(d)

$$U = \tau^2 \frac{\partial \ln Z}{\partial \tau} \qquad \text{for } 1 \gg \frac{\epsilon_0}{\tau} \quad U = \tau^2 \partial_\tau \left(\ln \frac{\tau}{\epsilon_0}\right) = \tau$$

$$\text{for } \frac{\tau}{\epsilon_0} \ll 1,\ U = \tau^2 \left(\frac{1}{1 + 3e^{-2\epsilon_0/\tau}}\right) (3e^{-2\epsilon_0/\tau}) \left(\frac{2\epsilon_0}{\tau^2}\right) = \frac{6\epsilon_0 e^{-2\epsilon_0/\tau}}{1 + 3e^{-2\epsilon_0/\tau}}$$

$$\boxed{C_V = \left(\frac{\partial U}{\partial \tau}\right)_V = 1 \text{ when } 1 \gg \frac{\epsilon_0}{\tau}}$$

$$C_V = 6\epsilon_0 \left(\frac{e^{-2\epsilon_0/\tau}\left(\frac{2\epsilon_0}{\tau^2}\right)(1+3e^{-2\epsilon_0/\tau}) - (3e^{-2\epsilon_0/\tau})\left(\frac{2\epsilon_0}{\tau^2}\right)e^{-2\epsilon_0/\tau}}{(1+3e^{-2\epsilon_0/\tau})^2} \right) = 12\epsilon_0^2 \frac{e^{-2\epsilon_0/\tau}}{\tau^2}\left(\frac{1}{(1+3e^{-2\epsilon_0/\tau})^2}\right)$$

For very small $\frac{\tau}{\epsilon_0} \ll 1$, $\boxed{C_V \approx 12\left(\frac{e^{-2\epsilon_0/\tau}}{(\tau/\epsilon_0)^2}\right)}$

(e) See sketch.

Solution 7. Zipper problem.

(a) N links. $\epsilon_s = 0$ closed, ϵ open.

$$Z = \sum_{s=0}^{N} \exp\left(-s\epsilon/\tau\right) = \frac{1-e^{-(N+1)\epsilon/\tau}}{1-e^{-\epsilon/\tau}}$$

(b) $1 \gg \frac{\tau}{\epsilon}$.
$Z = \sum_{s=0}^{N} \exp\left(-s\epsilon B\right)$

$$\frac{-1}{\epsilon}\partial_\beta \ln Z = \frac{-1}{\epsilon}\frac{\sum_{s=0}^{N}(-s\epsilon)e^{-s\epsilon\beta}}{Z} = \langle s\rangle =$$
$$= \left(\frac{1-e^{-\epsilon/\tau}}{1-e^{-(N+1)\epsilon/\tau}}\right)\left(\frac{-e^{-(N+1)\epsilon/\tau}(-(N+1)\epsilon)(1-e^{-\epsilon/\tau}) - (-e^{-\epsilon/\tau})(-\epsilon)(1-e^{-(N+1)/\epsilon/\tau})}{(1-e^{-\epsilon/\tau})^2}\right) =$$
$$= \frac{\epsilon e^{-\epsilon/\tau}(e^{-N\epsilon\beta}(N+1)(1-e^{-\epsilon\beta}) - (1-e^{-(N+1)\epsilon/\tau}))}{(1-e^{-(N+1)\epsilon/\tau})(1-e^{-\epsilon/\tau})} \simeq \frac{\epsilon e^{-\epsilon/\tau}(e^{-N\epsilon\beta}(N+1)(1-e^{-\epsilon\beta}) - 1)}{(1-e^{-\epsilon\beta})}$$

This still does not give the desired approximation. Consider the following:

$$Z = \frac{1-e^{-(N+1)\epsilon\beta}}{1-e^{-\epsilon\beta}} = \frac{e^{\epsilon\beta}-e^{-N\epsilon\beta}}{e^{\epsilon\beta}-1}$$
$$\partial_\beta Z = \frac{\epsilon((e^{\epsilon\beta}+N\beta e^{-N\epsilon\beta})(e^{\epsilon\beta}-1) - (e^{\epsilon\beta})(e^{\epsilon\beta}-e^{-N\epsilon\beta}))}{(e^{\epsilon\beta}-1)^2} = \frac{\epsilon(e^{\epsilon\beta}(e^{\epsilon\beta}+Ne^{-N\epsilon\beta}-e^{\epsilon\beta}+e^{-N\epsilon\beta}) - e^{\epsilon\beta} - Ne^{-N\epsilon\beta})}{(e^{\epsilon\beta}-1)^2}$$

$$\frac{\partial_\beta Z}{Z} = \frac{\epsilon(e^{\epsilon\beta}(N+1)e^{-N\epsilon\beta} - e^{\epsilon\beta} - Ne^{-N\epsilon\beta})}{(e^{\epsilon\beta}-1)(e^{\epsilon\beta}-e^{-N\epsilon\beta})} \simeq \frac{\epsilon(e^{\epsilon\beta}(N+1)e^{-N\epsilon\beta} - e^{+\epsilon\beta} - Ne^{-N\epsilon\beta})}{e^{2\epsilon\beta}} =$$
$$= \frac{\epsilon(Ne^{-(N-1)\epsilon\beta} + e^{-(N-1)\epsilon\beta} - e^{\epsilon\beta} - Ne^{-N\epsilon\beta})}{e^{2\epsilon\beta}} = \frac{\epsilon((N+1)e^{-(N-1)\epsilon\beta} - (e^{\epsilon\beta}+Ne^{-N\epsilon\beta}))}{e^{2\epsilon\beta}} =$$
$$= \frac{\epsilon\left(\frac{(N+1)e^{\epsilon\beta}}{e^{N\epsilon\beta}} - \frac{n}{e^{N\epsilon\beta}} - e^{\epsilon\beta}\right)}{e^{2\epsilon\beta}} = \frac{\epsilon\left(\frac{(N+1)e^{\epsilon\beta}-N}{e^{N\epsilon\beta}} - e^{\epsilon\beta}\right)}{e^{2\epsilon\beta}} = \frac{\epsilon\left(\frac{Ne^{\epsilon\beta}+e^{\epsilon\beta}-e^{\epsilon\beta}e^{N\epsilon\beta}}{e^{N\epsilon\beta}}\right)}{e^{2\epsilon\beta}} = \epsilon\left(\frac{N-e^{N\epsilon\beta}}{e^{N\epsilon\beta}e^{\epsilon\beta}}\right) \simeq$$
$$\simeq \frac{\epsilon(-e^{N\epsilon\beta})}{e^{N\epsilon\beta}e^{\epsilon\beta}} = -\epsilon e^{-\epsilon\beta}$$

$$\Longrightarrow \boxed{\langle s\rangle = e^{-\epsilon/\tau}}$$

Solution 8. Quantum concentration. Now $\Psi(x,y,z) = A\sin\left(\frac{n_x\pi x}{L}\right)\sin\left(\frac{n_y\pi y}{L}\right)\sin\left(\frac{n_z\pi z}{L}\right)$. $p = \frac{1}{i}\nabla$, $\frac{p^2}{2m} = -\frac{1}{2m}\nabla^2$.
Ground orbital: $n_x = n_y = n_z = 1$.

$$T = \frac{3}{2m}\left(\frac{\pi}{L}\right)^2\langle\psi_0|\psi_0\rangle = \frac{3}{2m}\left(\frac{\pi}{L}\right)^2$$

where $\langle\psi_0|\psi_0\rangle = 1$, ψ normalized. It was normalized in this way:

$$\int_0^\infty \sin^2\left(\frac{n_x\pi x}{L}\right)dn_x = \int_0^L \frac{1-\cos\left(\frac{2n_x\pi x}{L}\right)}{2}dn_x = \left(\frac{n_x - \sin\left(\frac{2n_x\pi x}{L}\right)\left(\frac{L}{2n_x\pi}\right)}{2}\right)\Bigg|_0^L = \frac{L}{2}$$

$$\langle\psi|\psi\rangle = A^2\left(\frac{L}{2}\right)^3 = \frac{A^2L^3}{8} = 1 \text{ or } A^2 = \frac{8}{L^3}$$

Recall that $n_Q = \left(\frac{m\tau}{2\pi\hbar^2}\right)^{3/2}$.

Consider the condition that there will be a concentration for which the zero-point quantum kinetic energy is equal to the temperature τ:

$$\Longrightarrow \frac{3}{2m}\frac{\pi^2}{L^2} = \frac{3}{2m}\pi^2 n^{2/3} = \tau \text{ or } n^{2/3} = \frac{2m\tau}{3\pi^2\hbar^2} \text{ or } n = \left(\frac{2m\tau}{3\pi^2\hbar^2}\right)^{3/2}$$

$$\Longrightarrow \boxed{n = (\left(\frac{4}{3\pi}\right)\frac{m\tau}{2\pi\hbar^2})^{3/2} = \left(\frac{4}{3\pi}\right)^{3/2} n_Q}$$

Solution 9. Partition function for two systems.

$$Z(1+2) = \sum_{E_{1+2}} g(E_{1+2})\exp\left(\frac{-E_{1+2}}{\tau}\right) = \sum_{E_1+E_2=E_0} g(E_1+E_2)\exp\left(\frac{-E_1}{\tau}\right)\exp\left(\frac{-E_2}{\tau}\right) =$$
$$= \sum_{E_1}\sum_{E_2} g(E_1)g(E_2)\exp\left(\frac{-E_1}{\tau}\right)\exp\left(\frac{-E_2}{\tau}\right) = Z(1)Z(2)$$

since systems are independent.

Solution 10. Elasticity of polymers.

(a) Consider $2s = N_+ - N_-$; $N = N_+ + N_-$, $2s = N_+ - (N - N_+) = 2N_+ - N$. $N_+ = \frac{2s+N}{2}$.
For $2s$, consider $-2s = N_+ - (N - N_+) = 2N_+ - N$. $N_+ = \frac{-2s+N}{2}$.

$$\Longrightarrow \boxed{g(N,-s) + g(N,s) = \frac{2N!}{\left(\frac{N}{2}+s\right)!\left(\frac{N}{2}-s\right)!}}$$

(b) $|s| \ll N$

$$\sigma(l) = \ln\left(g\left(N,\frac{l}{2\rho}\right) + g\left(N,\frac{-l}{2\rho}\right)\right) = \ln\left(\frac{2(N!)}{\left(\frac{N}{2}+\frac{l}{2\rho}\right)!\left(\frac{N}{2}+\frac{-l}{2\rho}\right)!}\right) =$$
$$= \ln(2N!) = \{\left(\frac{N}{2}+\frac{L}{2\rho}\right)\ln\left(\frac{N}{2}+\frac{l}{2\rho}\right) - \left(\frac{N}{2}+\frac{l}{2\rho}\right) + \left(\frac{N}{2}-\frac{l}{2\rho}\right)\ln\left(\frac{N}{2}-\frac{l}{2\rho}\right) - \left(\frac{N}{2}-\frac{l}{2\rho}\right)\}$$

where we used $ln(x+\Delta x) \simeq \ln x + \frac{1}{x}\Delta x$.

$$\sigma(l) = \ln(2N!) - \{\left(\frac{N}{2}+\frac{l}{2\rho}\right)\left(\ln\left(\frac{N}{2}\right) + \frac{1}{N/2}\frac{l}{2\rho}\right) - \frac{N}{2} + \left(\frac{N}{2}-\frac{l}{2\rho}\right)\left(\ln\left(\frac{N}{2}\right) + \frac{1}{N/2}\left(\frac{-l}{2\rho}\right)\right)\} =$$
$$= \ln(2N!) - \{\frac{N}{2}\ln\frac{N}{2} - \frac{N}{2} + \frac{N}{2}\ln\frac{N}{2} - \frac{N}{2} + \frac{l}{2\rho}\ln\left(\frac{N}{2}\right) + \frac{l}{2\rho} + \frac{2}{N}\left(\frac{l}{2\rho}\right)^2 + \frac{-l}{2\rho} - \frac{l}{2\rho}\ln\left(\frac{N}{2}\right) + \frac{l^2}{N2\rho^2}\}$$

$$\boxed{\sigma(l) = \ln\left(\frac{2N!}{(\frac{N}{2})!\,(\frac{N}{2})!}\right) - \frac{l^2}{N\rho^2}}$$

(c)

$$\frac{\partial \sigma}{\partial l} = \frac{-2l}{N\rho^2} \quad \Longrightarrow \boxed{f = \frac{2l\tau}{N\rho^2}}$$

Solution 11. One-dimensional gas.

$$\epsilon_n = \frac{\hbar^2}{2m}\left(\frac{\pi}{L}\right)^2 n^2 \quad \text{in one dimension}$$

$$Z_1 = \sum_{n=1}^{\infty} \exp\left(\frac{-\hbar^2}{2m}\left(\frac{\pi}{L}\right)^2 n^2/\tau\right) \Longrightarrow \int_0^{\infty} dn e^{-\alpha^2 n^2} = \frac{\sqrt{\pi}}{2\alpha}$$

where $\alpha^2 = \dfrac{\hbar^2}{2m\tau}\left(\dfrac{\pi}{L}\right)^2$.

$\alpha = \dfrac{\hbar\pi/L}{\sqrt{2m}\tau^{1/2}}$

Recall that $\sigma = \left(\frac{\partial F}{\partial \tau}\right)$, $F = U - \tau\sigma$, so that

$$F = -\tau \ln Z = -\tau N \ln \frac{\sqrt{\pi}}{2\alpha} = \tau N \ln\left(\frac{2\alpha}{\sqrt{\pi}}\right) = \tau N \ln\left(\frac{2\hbar\pi/L}{\sqrt{\pi}\sqrt{2m}\tau^{1/2}}\right) =$$

$$= \tau N \ln\left(\frac{\sqrt{2\pi}\hbar}{\sqrt{m}L\tau^{1/2}}\right) = \frac{1}{2}\tau N \ln\left(\frac{2\pi\hbar^2}{mL^2\tau}\right)$$

$$\frac{\partial F}{\partial \tau} = \frac{1}{2}N \ln\left(\frac{2\pi\hbar^2}{mL^2\tau}\right) + \frac{\tau N}{2}\left(\frac{1}{\left(\frac{2\pi\hbar^2}{mL^2\tau}\right)}\right)\left(\frac{-2\pi\hbar^2}{mL^2\tau^2}\right) =$$

$$= \frac{1}{2}N \ln\left(\frac{2\pi\hbar^2}{mL^2\tau}\right) + \frac{-\tau N}{2\tau} = \boxed{\frac{N}{2}\left(\ln\left(\frac{2\pi\hbar^2}{mL^2\tau}\right) - 1\right)}$$

2015 manuscript.

125 Pressure

cf. pp. 64 of Ch. 3 "Boltzmann Distribution and Helmholtz Free Energy", Kittel and Kroemer (1980) [2].

Consider a system of quantum states of energy ϵ_s.
Assume ϵ_s is a function of volume of the system.
Volume is decreased slowly from V to $V - \Delta V$ by an applied external force.
Let volume change take place sufficiently slowly so that system remains in the same quantum state s throughout compression.

$$\epsilon_s(V - \Delta V) = \epsilon_s(V) - \frac{d\epsilon_s}{dV}\Delta V + \cdots \equiv \text{energy of states after reversible volume change}$$

cf. Eq. (21) of Kittel and Kroemer (1980) [2], pp. 64, Ch. 3.
Now

$$U(V - \Delta V) - U(V) = \Delta U = -\left(\frac{d\epsilon_s}{dV}\right)\Delta V$$

cf. Eq. (22) of Kittel and Kroemer (1980) [2], pp. 64, Ch. 3.
Let A = area of 1 face of a cube.
Then

$$\Delta V = A(\Delta x + \Delta y + \Delta z) = A\Delta x + A\Delta y + A\Delta z$$

cf. Eq. (23) of Kittel and Kroemer (1980) [2], pp. 66, Ch. 3.
(Total) work done in compression is

$$\Delta U = p_s A(\Delta x + \Delta y + \Delta z) = p_s \Delta V$$

cf. Eq. (24) of Kittel and Kroemer (1980) [2], pp. 66, Ch. 3.
Then

$$p_s = -\left(\frac{\partial \epsilon_s}{\partial V}\right)$$

cf. Eq. (25) of Kittel and Kroemer (1980) [2], pp. 66, Ch. 3.
Take the average of $p_s = -\left(\frac{\partial \epsilon_s}{\partial V}\right)$ over all states of the ensemble to obtain the average pressure p:

$$\boxed{p = -\left(\frac{\partial U}{\partial V}\right)_\sigma} \tag{623}$$

cf. Eq. (26) of Kittel and Kroemer (1980) [2], pp. 66, Ch. 3.
Entropy is held constant because the number of states in ensemble is unchanged in reversible compression.
Alternatively, consider that

$$dU = Q - pdV$$

Let $c = c(t) = (U, V) = (U(t), V(t))$ s.t. σ const. $\dot{c} = \dot{U}\frac{\partial}{\partial U} + \dot{V}\frac{\partial}{\partial V}$.

Let $t = V$. Then $\dot{c} = \left(\frac{\partial U}{\partial V}\right)_\sigma + \frac{\partial}{\partial V}$,

$$\implies dU(\dot{c}) = \left(\frac{\partial U}{\partial V}\right)_\sigma = -pdV(\dot{c}) = -p \implies \boxed{p = -\left(\frac{\partial U}{\partial V}\right)_\sigma} \tag{624}$$

We can also derive the following thermodynamic identity: for $U - U(\sigma, V)$, $dU = \tau d\sigma - pdV$, then for

$$\left(\frac{\partial U}{\partial \sigma}\right)_V = \tau$$
$$\left(\frac{\partial U}{\partial V}\right)_\sigma = -p$$

$$\implies \boxed{-\left(\frac{\partial p}{\partial \sigma}\right)_V = \left(\frac{\partial \tau}{\partial V}\right)_\sigma} \tag{625}$$

For *another expression* for pressure p, consider that entropy σ is a function of internal energy U and volume V:

$$\sigma = \sigma(U, V)$$
$$d\sigma = \left(\frac{\partial \sigma}{\partial U}\right)_V dU + \left(\frac{\partial \sigma}{\partial V}\right)_U dV$$

Let $c = (U, V)$ s.t. σ const., then

$$d\sigma(\cdot c) = 0 = \left(\frac{\partial \sigma}{\partial U}\right)_V \dot{U} + \dot{V}\left(\frac{\partial \sigma}{\partial V}\right)_U$$

Now

$$\frac{1}{\tau} := \left(\frac{\partial \sigma}{\partial U}\right)_V$$

Let $c = c(t) = (U(t), V(t))$ and $t = V$

$$\frac{1}{\tau}\left(\frac{\partial U}{\partial V}\right)_\sigma + \left(\frac{\partial \sigma}{\partial V}\right)_U = \frac{-p}{\tau} + \left(\frac{\partial \sigma}{\partial V}\right)_U = 0$$
$$p = \tau\left(\frac{\partial \sigma}{\partial V}\right)_U$$

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126 The First Law of Thermodynamics

$$dU = Q - W \text{ or } Q = dU + W$$

Consider $W = Q - dU$.

Consider path in M, γ, $\gamma : \mathbb{R} \to M = (U, V)$

$$\gamma(t) = (U(t), V(t))$$

$\dot{\gamma} \in \mathfrak{X}(M)$, $\dot{\gamma} = \dot{U}\frac{\partial}{\partial U} + \dot{V}\frac{\partial}{\partial V}$

$$W(\dot{\gamma}) = pdV(\dot{\gamma}) = p\dot{V} = Q(\dot{\gamma}) - dU(\dot{\gamma}) = Q(\dot{\gamma}) - \dot{U}$$

Suppose $Q(\dot{\gamma}) = Q(t)dt(\dot{\gamma}\frac{\partial}{\partial t}) = Q(t)\dot{\gamma}$.

$$\int p\dot{V}dt = \int Q(t)\dot{\gamma}dt - \int \dot{U}dt$$
$$\implies p\Delta V = \Delta Q - \Delta U$$

$p\Delta V$ interpreted as work done by gas. ΔQ is heat transferred to gas system. $-\Delta U$ is the drop in internal energy of gas system as it does work.

126.1 Heat capacity

$$Q = Q(\tau, V) = \left(\frac{\partial Q}{\partial \tau}\right)_V d\tau + \left(\frac{\partial Q}{\partial V}\right)_\tau dV$$

So define $C_V \equiv \left(\frac{\partial Q}{\partial \tau}\right)_V$ or interpret C_V as energy input by heating at constant volume over ensuing change in temperature.

In this case, $\left(\frac{\partial U}{\partial \tau}\right)_V$.

For the case of a monatomic gas, $U = \frac{d}{2}\tau N$, $\frac{\partial U}{\partial \tau} = \frac{d}{2}N$.

$$C_V = \frac{d}{2}N$$

$N \equiv$ number of molecules.

Now

$$\begin{array}{l} Q = \Lambda_p dp + C_p d\tau \\ Q = \Lambda_V dV + C_V d\tau \end{array} \implies \begin{array}{l} Q \wedge dp = C_p d\tau \wedge dp \\ Q \wedge dp = \Lambda_V dV \wedge dp + C_V d\tau \wedge dp \end{array}$$

Now from the thermodynamic identity, $Q = W + dU = pdV + dU$,

$$Q \wedge dp = pdV \wedge dp + dU \wedge dp$$

and from (empirical) ideal gas law, $pV = N\tau$ (which defines a hypersurface on M),

$$dpV + pdV = Nd\tau \implies pdV \wedge dp = Nd\tau \wedge dp$$

so then

$$Q \wedge dp = Nd\tau \wedge dp + dU \wedge dp$$

In the case of the monatomic gas, $U = \frac{d}{2}\tau N$, and so $dU = \frac{d}{2}Nd\tau = C_V d\tau$ and so comparing all the equations above, one recovers

$$C_p = N + C_V = \frac{2+d}{2}N$$

EY: 20151019 I'm curious to know how this all generalizes for C_V, C_P heat capacities, regardless of the type of molecule we consider.

The adiabatic relation for a classical ideal gas

Consider the adiabatic expansion (or contraction!) of a classical ideal gas.
This means that $Q = 0$; there is no heat exchange to or from the gas system.

Recall $Q = dU + W$.
If $Q = 0$, and supposing $W = pdV$, then $0 = dU + pdV$.

EY : 20151019 Either by definition, or the thermodynamic identity, $\tau d\sigma = dU + pdV$, then $C_V := \left(\frac{\partial U}{\partial \tau}\right)_V$. My question is this: for manifold of thermodynamic states M, $M = (U, V)$, i.e. U is a global coordinate and V is a "local" coordinate. One can make a Legendre transformation such that M is parametrized by (τ, V), where τ is the temperature. In general, one should say that $U = U(\tau, V) \in C^\infty(M)$, and so $dU = \frac{\partial U}{\partial \tau} d\tau + \frac{\partial U}{\partial V} dV$, $dU \in \Omega^1(M)$.

However, for this adiabatic process, we want

$$Q = 0 = dU + W = dU + pdV = C_V d\tau + pdV$$

which implies that $dU = C_V d\tau$. What happened to the $\frac{\partial U}{\partial V} dV$? Is it that in this adiabatic process, the internal energy of the gas system goes to either doing work (expansion) or increases due to work being done on it (contraction), and is characterized completely by a drop or increase in its temperature, respectively? And so $dU = C_V \tau$, and pdV completely describes what's going on with work done or work done on it?

Correction (adiabatic question). On the ideal-gas surface $(\partial U/\partial V)_\tau = 0$, so $dU = C_V\, d\tau$ on every path. The product $C_V \tau$ is not that differential. The derivation is §152.1.

Nevertheless, using the (empirical) ideal gas law, $pV = N\tau$,

$$0 = C_V d\tau + pdV = C_V d\tau + \frac{N\tau}{V} dV$$

Consider a path $\gamma : \mathbb{R} \to M$ in M, so that $\dot\gamma(t) = \dot\tau \frac{\partial}{\partial \tau} + \dot V \frac{\partial}{\partial V} \in \mathfrak{X}(M)$.
$\gamma(t) = (\tau(t), V(t))$

Thus,

$$0 = C_v \dot\tau + \frac{N\tau}{V} \dot V \text{ so } \frac{\dot\tau}{\tau} + \frac{N}{C_V} \frac{\dot V}{V}$$

$$\xrightarrow{\int dt} \ln \frac{\tau_f}{\tau_i} + \frac{N}{C_V} \ln \frac{V_f}{V_i} = 0 \text{ or } \ln \tau V^{\frac{N}{C_V}} = \text{ const.}$$

Now

$$\frac{N}{C_V} = \frac{C_P - C_V}{C_V} = \gamma - 1$$

which is true, assuming the (empirical) ideal gas law, the thermodynamic identity, and, surely, for the case of a monatomic gas.

Thus

$$\tau_f V_f^{\gamma - 1} = \tau_i V_i^{\gamma - 1}$$

Problems

Problem 4. *Adiabatic compression.*

A diesel engine doesn't have a spark plug to ignite and explode the fuel. Instead, the air in the cylinder is compressed so highly that the fuel ignites spontaneously when sprayed into the cylinder.

(a)

$$\frac{\tau_f V_f^{\gamma-1}}{\tau_i V_i^{\gamma-1}} = \frac{P_f V_f^\gamma}{P_i V_i^\gamma} = 1$$

$$\tau_f = \left(\frac{V_i}{V_f}\right)^{\gamma - 1} \tau_i$$

Run the Python script `thermo.py` to do the calculations. Here is (some of) the code from `thermo.py` for doing so (one still needs to import the necessary libraries):

```
roomtemp_K = KCconv.subs(T_C,20).rhs # room temperature in Kelvin

Prob0104ans = adia_tV.subs(gamma,1.4).subs(V_f,1).subs(V_i,15).subs(tau_i, roomtemp_K) # answer
    to Problem 4 of Chapter 1

Prob0104ans = N( Prob0104ans.lhs) # 866.016969686253 K
Prob0104ansC = solve( KCconv.subs( T_K, Prob0104ans), T_C )[0] # 592.866969686253 C
solve( FCconv.subs( T_C, Prob0104ansC ), T_F)[0] # 1099.16054543526 F
```

The final temperature is $866.01\,K$ or $592.87\,C$ or $1099.16\,F$

(b) Now from the ideal gas law, which is obeyed at all thermodynamic states

$$\frac{P_f}{P_i} = \frac{V_i}{V_f}\frac{\tau_f}{\tau_i}$$

and so $\frac{P_f}{P_i} = 44.31$

2015 manuscript.

127 Heat capacity at constant pressure

Manuscript section title: Heat Capacity.

Let Σ be a (topological) manifold of dim. 2.

Suppose U is a global coordinate on Σ. Then consider 1 chart (U, V).

Consider $\sigma = \sigma(U, V) \in C^\infty(\Sigma)$.

Recall the thermodynamic identity

$$\tau d\sigma = dU + pdV$$

Thus

$$d\sigma = \frac{1}{\tau}dU + \frac{p}{\tau}dV \Longrightarrow \begin{array}{l} \frac{p}{\tau} = \left(\frac{\partial\sigma}{\partial V}\right)_U \\ \frac{1}{\tau} = \left(\frac{\partial\sigma}{\partial U}\right)_V \end{array}$$

From Kittel and Kroemer (1980) [2], pp. 165-166, Chapter 6: Ideal Gas, "Heat Capacity",

$$Q = Q(\tau, p) = \Lambda_p dp + C_p d\tau = dU + W = dU + pdV$$

Let $c \in \Sigma$ s.t. $dp(\dot{c}) = 0$ (constant pressure). And so

$$Q(\dot{c}) = 0 + C_p d\tau(\dot{c}) = \tau d\sigma(\dot{c}) = dU(\dot{c}) + pdV(\dot{c})$$

for $c = (\tau, 0)$, $\dot{c} = \frac{\partial}{\partial\tau} \in T\Sigma$.

$$\Longrightarrow C_p = \tau\left(\frac{\partial\sigma}{\partial\tau}\right)_p = \left(\frac{\partial U}{\partial\tau}\right)_p + p\left(\frac{\partial V}{\partial\tau}\right)_p$$

for heat capacity at constant pressure is larger than C_V because additional heat must be added to perform the work needed to expand volume of gas against constant pressure.

Now recall for enthalpy

$$H = U + pV$$
$$H = H(\sigma, p)$$

Then

$$dH = dU + pdV + Vdp = Q + Vdp$$

Thus, for $c \in \Sigma$, $dp(\dot{c}) = 0$ (constant pressure). Hence

$$dH(\dot{c}) = \left(\frac{\partial H}{\partial\tau}\right)_p = Q(\dot{c}) + 0 = C_p$$

Hence,

$$C_p = \left(\frac{\partial H}{\partial\tau}\right)_p = \left(\frac{\partial Q}{\partial\tau}\right)_p$$

2008 manuscript.

128 Thermal Radiation and Planck Distribution

Problems

Solution 1. Number of thermal photons.

We consider a cavity of volume V, and of edge length L (so $V\ L^3$). So then $\omega_n = n\pi c/L$.

Now $\frac{1}{\exp\left(\frac{\hbar\omega_n}{\tau}\right)-1}$ is the thermal average number of photons in a single mode frequency ω. So then

$$\sum\langle s_n\rangle = \sum \frac{1}{\exp\left(\frac{\hbar\omega_n}{\tau}\right)-1}$$

Consider (n_x, n_y, n_z) on positive octant, and 2 independent polarization s of em field.

$$\sum_n \frac{1}{\exp\left(\frac{\hbar n\pi c}{L\tau}\right)-1} = \frac{(2)}{8}\int_0^\infty 4\pi n^2 dn \frac{1}{\exp\left(\frac{\hbar n\pi c}{L\tau}\right)-1} = \pi\int_0^\infty \frac{n^2 dn}{\exp\left(\frac{\hbar n\pi c}{L\tau}\right)-1} =$$
$$= \pi\left(\frac{L\tau}{\hbar\pi c}\right)^3\int_0^\infty\left(\frac{x^2dx}{e^x-1}\right) = \boxed{N = \frac{V}{\pi^2}\left(\frac{\tau}{\hbar c}\right)^3 (2.404)}$$

where I used the substitutions

$$x = \frac{\hbar\pi cn}{L\tau}$$
$$\frac{L\tau x}{\hbar\pi c} = n$$
$$(dx)\frac{L\tau}{\hbar\pi c} = dn$$

Now $\sigma(\tau) = (4\pi^2V/45)(\tau/\hbar c)^3$, so then

$$\frac{\sigma}{N} = \frac{1}{2.404}\left(\frac{4\pi^4}{45}\right) \simeq \boxed{3.602}$$

Now how was $\int_0^\infty dx\frac{x^2dx}{e^x-1}$ evaluated?

Solution 2. Surface temperature of a Sun. Given the **solar constant** of the Earth, the total radiant energy flux density at the Earth from the Sun normal to the incident rays, integrated over all emission wavelengths,

$$\text{solar constant } = 0.136\, J\, s^{-1}\, cm^{-2}, \qquad (49)$$

(a)

$$4\pi(1.49\times10^{11}\, m)^2\cdot 0.136\, J\, s^{-1}\, cm^{-2}\left(\frac{10^2\, cm}{1\, m}\right)^2 = (4\pi)(1.49)^2 10^{22}\cdot 0.136\times10^4 = \boxed{3.8\times10^{26}\, J\cdot s^{-1}}$$

Note that I had used $1.49\times10^{11}\, m$ as the distance of the Earth from the Sun.

(b) J_ν = energy flux density or rate of energy emission per unit area.

$\sigma_B = \frac{\pi^2k_B^4}{60\hbar^3c^2} = 5.670\times10^{-8}\, W\, m^{-2}\, K^{-4}$.

Note that $W = \frac{1\, J}{s}$. I will use $R_\circ = 6.9599\times10^{10}\, cm$ as the radius of the Sun.

$$\frac{4\times10^{26}\, J\cdot s^{-1}}{4\pi(6.9599\times10^{10}\, cm)^2} = J_\nu = \sigma_B T^4$$

$$\frac{4\times10^{26}\, J\cdot s^{-1}}{4\pi(6.9599\times10^{10}\, cm)^2}\left(\frac{1\, m^2\left(\frac{10^2\, cm}{1\, m}\right)^2}{5.670\times10^{-8}\, J/s}\right)k^4 = T^4$$

$$\boxed{T \simeq 5830\, K}$$

Solution 3. Average temperature of the interior of the Sun.

(a)

$$U = -\int_0^R \frac{G\left(\frac{4}{3}\pi\rho r^3\right)(4\pi r^2\rho dr)}{r} = -\frac{16}{3}\pi^2 G\rho^2 \int_0^R r^4 dr = \frac{-16}{15}\pi^2\rho^2 GR^5;\ M = \frac{4}{3}\pi R^3\rho$$

$$U = \frac{-3GM^2}{5R}$$

$$U = -\frac{1}{2}\int_0^R \frac{G\left(\frac{4}{3}\pi\rho r^3\right)}{r}(4\pi r^2\rho dr) = \frac{-8}{3}\pi^2 G\rho^2 \int_0^R r^4 dr = \frac{-8}{15}\pi^2\rho^2 GR^5 =$$

$$= \frac{-8}{15}\pi^2 GR^5\left(\frac{M}{\frac{4}{3}\pi R^3}\right)^2 = \frac{-3}{10}\frac{GM^2}{R} = 1.14\times 10^{41}\,J$$

(b) Using the virial theorem of mechanics, note that

$$\frac{-1}{2}U = \frac{3}{20}\frac{GM^2}{R} = \frac{3}{20}\frac{6.67\times 10^{-11}\,kg\cdot\frac{m}{s^2}\cdot\frac{m^2}{kg^2}\cdot 2\times 10^{33}\,g\left(\frac{1\,kg}{10^3\,g}\right)}{7\times 10^{10}\,cm\left(\frac{1\,m}{10^2\,cm}\right)} = 5.72\times 10^{40}\,J$$

Now $\langle s\rangle = \frac{1}{\exp(\hbar\omega/\tau)-1}$, is the Planck distribution function, giving the thermal average number of photons in a single mode frequency ω.

thermal average energy $\langle\epsilon\rangle = \langle s\rangle\hbar\omega = \frac{\hbar\omega}{\exp(\hbar\omega/\tau)-1}$ for $\tau \ll \hbar\omega$, $\langle\epsilon\rangle \simeq \tau$

So then $5.72\times 10^{40}\,J = N\langle\epsilon\rangle = N\tau$.

$$\tau = \frac{5.72\times 10^{40}\,J}{1\times 10^{57}(1.381\times 10^{-23}\,J/K)} = \boxed{4.14\times 10^6\,K}$$

Solution 4. Age of the Sun.

(a) Consider $4H \to {}^4_2He$. Then $4(1.0078) - 4.0026 = 0.0286\,amu$. Then

$$(0.0286\,amu)\left(\frac{1.6726\times 10^{-27}\,kg}{1.00727647\,u}\right)(3\times 10^8\,m/s)^2 = 4.27\times 10^{-12}\,J$$

Given $M_\odot = 2\times 10^3 3\,g$,

$$(2\times 10^{30}\,kg)(0.10)\left(\frac{1}{4\times(1.0078\,amu)}\right)\left(\frac{1.00727647\,u}{1.6726\times 10^{-27}\,kg}\right)(4.27\times 10^{-12}\,J) = 1.28\times 10^{44}J$$

So $\boxed{1.28\times 10^{44}\,J}$ energy is available.

(b)

$$\frac{1.15\times 10^{45}\,J}{4\times 10^{26}J\cdot s^{-1}}\left(\frac{1\,hr}{3600\,s}\right)\left(\frac{1\,day}{24\,hr}\right)\left(\frac{1\,yr}{365\,days}\right) = 1.02\times 10^{10}\,years$$

Solution 5. Surface temperature of the Earth. $J_S = \sigma_b T_\odot^4$ is the radiant power per unit area.
Total emitted radiation energy of the sun is $J_S 4\pi R_\odot^2$.
$\frac{4\pi R_\odot^2 J_S}{4\pi R_{ES}^2} = \frac{R_\odot^2}{R_{ES}^2}J_S =$ radiation energy hitting $1\,cm^2$ of Earth's surface in one second

Since the Earth is considered a black-body, the rate of absorption must equal the rate of emission:

$$\frac{R_\odot^2}{R_{ES}^2}\sigma_b T_\odot^4 = \sigma_b T_e^4 \text{ or } T_e^4 = T_\odot^4 R_\odot^2 \Longrightarrow T_e = 5800\,K\sqrt{\frac{7\times 10^{10}\,cm}{1.5\times 10^{13}\,cm}} = 396.2\,K = 123\,C$$

Solution 6. Pressure of thermal radiation.

(a) s_j = number of photons in that mode. Suppose modes of ω_j, $j = 0, 1, 2, \ldots$.
$\epsilon_j = s_j \hbar \omega_j$ = total energy in jth mode, s_j photons in jth mode.

$$U = \sum_j \epsilon_j = \sum_j s_j \hbar \omega_j \qquad P = -\frac{\partial U}{\partial V} = -\sum_j s_j \hbar \frac{\partial \omega_j}{\partial V}$$

(b) $\omega_j = j\pi c/L, \;\; V = L^3$. So then $\omega_j = j\pi c V^{-1/3}$.

$$\Longrightarrow \frac{d\omega_j}{dV} = \frac{-1}{3} j\pi c V^{-4/3} = \frac{-1}{3}\frac{\omega_j}{V}$$

(c) $p = \frac{U}{3V}$

(d) We want the Kinetic pressure at a concentration of $(1\,mol/cm^3)$. Recalling $P = \frac{Nk_BT}{V}$,

$$P = \left(\frac{1\,mol}{cm^3}\right)\left(\frac{6.022\times 10^{23}}{1\,mol}\right)\left(\frac{10^2\,cm}{1\,m}\right)^3\left(1.381\times 10^{-23}\,\frac{J}{K}\right)(2\times 10^7\,K) = 1.663\times 10^{14}\,\frac{N}{m^2}$$

Now for the thermal radiation pressure,

$$p = \frac{U}{3V} = \frac{1}{3}\frac{\pi^2}{15\hbar^3c^3}\tau^4 = 4.03\times 10^{13}\,\frac{N}{m^2}$$

where $t = 2\times 10^7\,K$.

For the pressures to be equal,

$$\frac{1}{3}\frac{\pi^2}{15\hbar^3c^3}k_B^4T^4 = \frac{Nk_B}{V}T \text{ or } T^3 = \frac{45(\hbar c)^3 Nk_B}{\pi^2 k_B^4 V} \text{ so that } \boxed{T = 3.2\times 10^7\,K}$$

Solution 7. Free energy of a photon gas

(a) $Z = \prod_n \frac{1}{1-e^{-\hbar\omega_n/\tau}}$ Consider $Z = \sum_{s=0}^{\infty} e^{-s\hbar\omega/\tau} = \frac{1}{1-e^{-\hbar\omega/\tau}}$ for a single mode.

(b) $F = -\tau \ln Z = \tau \sum_n \ln\left(1 - e^{-\hbar\omega_n/\tau}\right) \qquad \omega_n = \frac{n\pi c}{L}$.

$$F = \tau\sum_n \ln\left(1-e^{-\hbar\omega_n/\tau}\right) = \tau\int_0^\infty \frac{4\pi n^2 dn}{8}(2)\ln\left(1-e^{-\hbar n\pi c/\tau L}\right) = \pi\tau\int_0^\infty dn n^2 \ln\left(1-e^{-\hbar n\pi c/\tau L}\right) =$$
$$= \pi\tau\left(\left(n^3\ln\left(1-e^{-\hbar n\pi c/\tau L}\right)\right)\Big|_0^\infty - \int_0^\infty dn \frac{n^3 e^{-\hbar n\pi c/\tau L}}{1-e^{-\hbar n\pi c/\tau L}}\left(\frac{-\hbar\pi c}{\tau L}\right)\right) = -\frac{\hbar\pi^2 c}{L}\int_0^\infty dn \frac{n^3}{e^{\hbar n\pi/\tau L}-1} =$$
$$= -\left(\frac{\tau L}{\hbar\pi c}\right)^3\int_0^\infty \frac{x^3}{e^x-1} = -\frac{(\tau L)^3}{(\hbar^2c^3)}\frac{\pi^2}{45}$$

where I used $x = \frac{\hbar n\pi c}{\tau L}$.

Solution 8. Heat shields. For $J_u = \sigma_B(T_u^4 - T_l^4)$, the thermal flux without the heat shield, in the middle region. Plane m absorbs $\sigma_B T_u^4 + \sigma_B T_l^4$ and emits $J_m = \sigma_B(T_u^4 + T_l^4) = \sigma T_m^4$.

$T_m = (T_u^4 + T_l^4)^{1/4}$. The key point is that, by symmetry, plane m emits $\frac{J_m}{2}$ flux on each side.

$$J_{net} = \sigma_B T_u^4 - (\sigma_B\left(\frac{T_u^4+T_l^4}{2}\right)) = \sigma_B\left(\frac{T_u^4-T_l^4}{2}\right) = \frac{J_u}{2}$$

J_{net} is the same for the other side of the heat shield:

$$J_{net} = -\sigma_B T_l^4 + \sigma_B\left(\frac{T_u^4+T_l^4}{2}\right) = \sigma_B\left(\frac{T_u^4-T_l^4}{2}\right)$$

Solution 9. Photon gas in one dimension. $E = E_0 \sin(kx)\cos(\omega t)$ is the form of a solution with $kL = n\pi$ or $k = \frac{n\pi}{L}$, since

$$v^2 E_x x = E_t t \to v^2 k^2 = \omega^2 \text{ or } \frac{\omega}{k} = v,\ \omega_n = v\frac{n\pi}{L}$$

$$Z_j = \sum_{s=0}^{\infty} e^{-s\hbar\omega/\tau} = \frac{1}{1-e^{-\hbar\omega/\tau}}$$

where Z_j is the partition function for a particular mode frequency ω.

$$\langle s\rangle = \frac{\sum_{s=0}^{\infty} s e^{-\frac{s\hbar\omega_n}{\tau}}}{Z} = Z^{-1}\frac{d}{d(-\hbar\omega_n/\tau)}(1-\exp\left(\frac{-\hbar\omega_n}{\tau}\right))^{-1} = Z^{-1}(1-\exp\left(\frac{-\hbar\omega}{\tau}\right))^{-2}\exp\left(\frac{-\hbar\omega_n}{\tau}\right) =$$
$$= \frac{\exp(-\hbar\omega_n/\tau)}{1-\exp(-\hbar\omega_n/\tau)}$$

So

$$\langle s\rangle\hbar\omega_n = \langle\epsilon_n\rangle = \frac{\hbar\omega_n}{\exp\left(\frac{\hbar\omega_n}{\tau}\right)-1} \qquad U = \sum_n \langle\epsilon_n\rangle = \sum_n \frac{\hbar v n\pi/L}{\exp\left(\frac{\hbar v n\pi}{L\tau}\right)-1}$$

$$\frac{\partial U}{\partial\tau} = \sum_n \frac{-\hbar v n\pi/L}{\left(\exp\left(\frac{\hbar v n\pi}{L\tau}\right)-1\right)^2}\left(\exp\left(\frac{\hbar v n\pi}{L\tau}\right)\left(\frac{-\hbar v n\pi}{L\tau^2}\right)\right) = \sum_n \frac{\left(\frac{\hbar v\pi}{L\tau}\right)^2 n^2 \exp\left(\frac{\hbar v n\pi}{L\tau}\right)}{(\exp\left(\frac{\hbar n v\pi}{L\tau}\right)-1)^2} = \sum_n \frac{\left(\frac{\hbar v\pi}{L\tau}\right)^2 n^2 \exp\left(\frac{\hbar v\pi n}{L\tau}\right)}{(\exp\left(\frac{\hbar v\pi}{L\tau}n\right)-1)^2}$$

Now $\sum_n \to \int_0^\infty dn$ for one-dimensional photon. Let $\alpha = \frac{\hbar v\pi}{L\tau}$.

Letting $x = \alpha n$,

$$\frac{\partial U}{\partial\tau} = \left(\frac{nv\pi}{L\tau}\right)^2 \int dn \frac{n^2\exp(\alpha n)}{(e^{\alpha n}-1)^2} = \frac{1}{\alpha}\int dx \frac{x^2 e^x}{(e^x-1)^2} = \frac{1}{\alpha}\{\frac{-x^2}{(e^x-1)} + \int_0^\infty \frac{x\,dx}{e^x-1}\}$$

Coefficient of k term of $f(k)$ is $j = \int_0^\infty \frac{x}{e^x-1}dx$. Now $f(k) = \int_0^\infty \frac{\sin(kx)}{e^x-1}dx$, $\sin(kx) = kx - \frac{(kx)^3}{3!} + \ldots$, so that

$$\frac{\pi^2}{6} = \int_0^\infty \frac{x}{e^x-1}dx$$

$$\frac{\partial U}{\partial\tau} = \frac{L\tau}{\hbar v\pi}\left(\frac{\pi^2}{6}\right) = \boxed{\frac{L\tau\pi}{6\hbar v} = C_V}$$

Solution 10. Heat capacity of intergalactic space.

Given the density 1 atom m^{-3}, considering thermal radiation at $2.9\,K$, then $k_B T = (1.381\times10^{-23}\,J/K)(2.9\,K)$, $\hbar c = (1.05457\times10^{-34}\,J\cdot s)(3\times10^8\,\frac{m}{s})$.

Recall for radiation, that the energy per unit volume: $\frac{U}{V} = \frac{\pi^2}{15\hbar^3c^3}\tau^4$ so that $\frac{\partial U}{\partial\tau} = \frac{4\pi^2}{15\hbar^3c^3}\tau^3 V$.

Assume hydrogen atoms modeled as ideal gas: $U = \frac{3}{2}N\tau$, $\frac{dU}{d\tau} = \frac{3}{2}N$.

$$\frac{C_{V_{matter}}}{C_{V_{radiation}}} = \frac{\frac{3}{2}N}{\frac{4\pi^2}{15\hbar^3c^3}\tau^3 V} = \frac{45(\hbar c)^3(N/V)}{8\pi^2(k_BT)^3} = 2.8\times10^{-10}$$

Solution 11. Heat capacity of solids in high temperature limit. $\frac{\hbar\omega_n}{\tau} = \frac{\pi\hbar v n}{L\tau}$; $\omega_n = \frac{\pi v n}{L}$. For $\tau \ll \hbar\omega_n$, $0 \le n \le n_D$.

$$\exp\left(\frac{\hbar\omega_n}{\tau}\right) - 1 \simeq 1 + \frac{\hbar\omega_n}{\tau} + \left(\frac{\hbar\omega_n}{\tau}\right)^2\left(\frac{1}{2}\right) + \frac{1}{6}\left(\frac{\hbar\omega_n}{\tau}\right)^3 + \cdots - 1$$

By doing long division

$$\frac{\hbar\omega_n}{\exp\left(\frac{\hbar\omega_n}{\tau}\right)-1} \simeq \frac{\hbar\omega_n}{\frac{\hbar\omega_n}{\tau}\left(1+\frac{\hbar\omega_n}{2\tau}+\frac{(\hbar\omega_n)^2}{6\tau^2}\right)} = \frac{\tau}{1+\frac{\hbar\omega_n}{2\tau}+\frac{(\hbar\omega_n)^2}{6\tau^2}} = \tau + -\frac{\hbar\omega_n}{2} + \frac{(\hbar\omega_n)^2}{12\tau} + \ldots$$

For $n_p = (6N/\pi)^{1/3}$

$$U = \frac{3\pi}{2}\int_0^{n_D} dn n^2(\tau - \frac{\hbar\omega_n}{2} + \frac{(\hbar\omega_n)^2}{12\tau}) = \frac{3\pi}{2}\int_0^{n_D} dn n^2(\tau - \frac{\hbar}{2}\frac{\pi v n}{L} + \frac{\hbar^2\pi^2 v^2 n^2}{12\tau L^2}) =$$
$$= \frac{3\pi}{2}\{\frac{1}{3}n_D^3\tau - \frac{\hbar\pi v}{2L}\frac{n_D^4}{4} + \frac{\hbar^2\pi^2v^2}{12\tau L^2}\frac{1}{5}n_D^5\} = \frac{\pi}{2}\frac{6N}{\pi}\tau - \frac{3\pi^2\hbar v}{16L}\left(\frac{6N}{\pi}\right)^{4/3} + \frac{3\hbar^2\pi^3v^2(6N/\pi)^{5/3}}{120\tau L^2}$$

So

$$U = 3N\tau - \frac{3\pi^2\hbar v}{16L}\left(\frac{6N}{\pi}\right)^{4/3} + \frac{3\hbar^2\pi^3V^2(6N/\pi)^{5/3}}{120\tau L^2}$$

Now $T = \theta$, $\theta = \left(\frac{\hbar v}{k_B}\right)\frac{(6\pi^2N)^{1/3}}{L}$

So then

$$U = 3N\frac{\hbar v(6\pi^2N)^{1/3}}{L} - \frac{3\pi^2\hbar v}{16L}\frac{6^{1/3}N^{4/3}6}{\pi^{4/3}} = \frac{15}{8}\frac{(6\pi^2N)^{1/3}\hbar v}{L} = Nk_B\theta\left(\frac{15}{8}\right)$$

$$\frac{U}{\theta} = Nk_B\left(\frac{15}{8}\right) = \left(\frac{15}{8}\right)(1.381\times10^{-23}\,J/K)\left(\frac{6.022\times10^{23}\,particles}{1\,mol}\right) = 15.59$$

which is very close to experimental values.

Solution 12. Heat capacity of photons and phonons. For a photon: $U = \frac{\pi^2V}{15\hbar^3c^3}\tau^4 \qquad \partial_\tau U = \frac{4\pi^2V}{15\hbar^3c^3}\tau^3$.

phonon: $U(\tau) = \frac{3\pi^4N\tau^4}{5(k_B\theta)^3}$. $\qquad \frac{\partial U}{\partial\tau} = \frac{12\pi^4N\tau^3}{5(k_B\theta)^3}$

So then

$$C_V = \frac{12\pi^4 10^{22}}{5}\left(\frac{1}{100}\right)^3 = 2.3\times10^{18}$$

for a phonon.

For a photon,

$$\frac{4\pi^2}{15}\frac{(1.381\times10^{-23}J/K)^3}{(1.05457\times10^{-34}J\cdot s(3\times10^{10}\,cm/s))^3}\tau^3 = 220\,/K^3\tau^3$$

Temperature at which to photon contribution equals to phonon contribution:

$$(220/K^3)\tau^3 = 2.3\times10^{18} \Longrightarrow \boxed{\tau = 2.2\times10^5\,K}$$

Solution 13. Energy fluctuations in a solid at low temperatures.

$$\tau^2\left(\frac{\partial U}{\partial\tau}\right)_V = \langle(\epsilon - \langle\epsilon\rangle)^2\rangle$$

Recall that

$$\left(\frac{\partial U}{\partial\tau}\right)_V = \frac{12}{5}\frac{\pi^4N\tau^3}{(k_B\theta)^3} \text{ where } U = \frac{3\pi^4N\tau^4}{5(k_B\theta)^3}$$

$$\frac{\langle(\epsilon - \langle\epsilon\rangle)^2\rangle}{\langle\epsilon\rangle^2} = \frac{12\pi^4N\tau^5}{5(k_B\theta)^3}\left(\frac{1}{\frac{9\pi^8N^2\tau^8}{25(k_B\theta)^6}}\right) = \frac{20}{3}\left(\frac{1}{\pi^4}\right)\left(\frac{1}{N}\right)\left(\frac{1}{\tau^3}\right)(k_B\theta)^3 = \frac{0.068}{N}\left(\frac{\theta}{T}\right)^3$$

$$\mathcal{F} = \sqrt{\frac{0.070}{10^{15}}\left(\frac{200}{10^{-2}}\right)^3} = 0.02$$

Solution 14. Heat capacity of liquid ^4He at low temperatures.

(a) Given $v = 2.383\times10^4\ cm\,s^{-1}$ and accounting for only longitudinal waves (only longitudinal polarization), then the Debye temperature is

$$\theta = \left(\frac{\hbar v}{k_B}\right)\left(\frac{18\pi^2N}{V}\right)^{1/3} = \frac{(1.05457\times10^{-34}\,J\cdot s)(2.383\times10^4\,cm/s)}{1.381\times10^{-23}\,J/K}\left(\frac{18\pi^2 0.145\,g}{cm^3}\left(\frac{1.00727647\,u\,(\frac{1\,He}{4.0026u})}{1.67262\times10^{-24}\,g}\right)\right)^{1/3} =$$
$$= \boxed{28.6\,K}$$

(b) Recall the derivation for U for phonons in a solid. *Account for only longitudinal waves (only longitudinal polarization).*

$$U = \frac{\pi}{2}\int_0^{n_D} dn \frac{n^2\hbar\omega_n}{\exp(\hbar\omega_n/\tau)-1} = \frac{\pi}{2}\int_0^{n_D} n^2 dn \frac{\hbar\frac{\pi n v}{L}}{\exp(x)-1} = \frac{\hbar\pi^2 v}{2L}\int_0^{n_D}\frac{n^3}{\exp(x)-1}dn$$

With $\omega_n = \frac{\pi n}{L}v$, $x = \frac{\hbar\pi n v}{L\tau}$ or $\left(\frac{L\tau}{\hbar\pi v}\right)x = n$, then

$$U = \left(\frac{\hbar^2 v}{2L}\right)\left(\frac{L\tau}{\hbar\pi v}\right)^4\int_0^{n_D}\frac{x^3}{e^x-1}dx$$

For low temperatures, τ small so take $x_D = \left(\frac{\hbar\pi n_D v}{L\tau}\right) = \left(\frac{\hbar\pi v}{L\tau}\right)\left(\frac{18N}{\pi}\right)^{1/3} = \frac{18^{1/3}\hbar\pi^{2/3}n^{1/3}v}{\tau}$ to go to ∞.

$$U \to \left(\frac{\pi^2}{2(\hbar v)^3}\right)\frac{\tau^4}{15}V$$

Recall that $C_V = \left(\frac{\partial U}{\partial \tau}\right)_V$. Then $C_V/V = \frac{2}{15}\left(\frac{\pi^2}{(\hbar v)^3}\right)\tau^3$. Recall $\tau_B = k_B T$, and given $v = 2.383\times10^4\, cm/s$, then

$$\frac{k_B}{\hbar v} = \frac{(1.381\times10^{-23}\, J/K)}{(1.05457266\times10^{-34}\, J\cdot s)(2.383\times10^4\, cm/s)} = 5.495\times10^6\,(1/K\cdot cm)$$

So if we take C_V/V and divide by the given density $\rho = 0.145\, g/cm^3$ to get the heat capacity per gram, (and multiply by k_B, the Boltzmann constant to get the correct units; Kittel and Kroemer likes using dimensionless formulas) then

$$(C_V/V)/\rho = (k_B)\frac{2}{15}\pi^2\left(\frac{k_B}{\hbar v}\right)^3 T^3\left(\frac{cm^3}{0.145\, g}\right) = \boxed{0.0208\times T^3}$$

Solution 15. Angular distribution of radiant energy flux.

(a) Recall

$$u_\omega = \frac{\hbar}{\pi^2 c^2}\frac{\omega^3}{\exp\left(\frac{\hbar\omega}{\tau}\right)-1}$$

is the radiation energy per unit volume per unit frequency range.

cu_ω = energy per unit time, per cross sectional area per unit frequency range.

em waves emitted spherically from pt. Q.

Suppose em wave comes in at a funny angle other than directly inward.

Consider area da that's from the spherical wave from pt. Q. How much of that goes into solid angle $d\Omega$?

$$\Longrightarrow cu_\omega\cos\theta dA$$

So $cu_\omega\cos\theta$ is the energy per unit time, per cross-sectional area, per unit frequency range, that enters into some solid angle $d\Omega$.

$$\frac{r^2 d\Omega}{4\pi r^2} = \frac{d\Omega}{4\pi}$$

is the fraction of the spectral density that if arrives in solid angle $d\Omega$.

$$\Longrightarrow \boxed{cu_\omega\cos\theta\frac{d\Omega}{4\pi}}$$

is the spectral density of radiant energy flux that arrives in solid angle $d\Omega$.

(b)

$$\int \cos\theta \sin\theta d\theta d\varphi = 2\pi \int \frac{\sin 2\theta}{2} d\theta = -\pi \left(\frac{\cos 2\theta}{2}\right)\Big|_0^{\pi/2} = -\pi\left(\frac{-1-1}{2}\right) = \pi$$

$$\Longrightarrow \boxed{\frac{cu_\omega}{4}}$$

Solution 17. Entropy and occupancy.

$$Z = \sum_{s=0}^{\infty} e^{-s\hbar\omega/\tau} = \frac{1}{1-e^{-\hbar\omega/\tau}}$$

$$\partial_\tau Z = \sum_{s=0}^{\infty} e^{-s\hbar\omega/\tau}\left(\frac{s\hbar\omega}{\tau^2}\right) = -(1-e^{-\hbar\omega/\tau})^{-2}(-e^{-\hbar\omega/\tau})\left(\frac{\hbar\omega}{\tau^2}\right) = Z\frac{\hbar\omega}{\tau^2}\langle s\rangle$$

Then

$$\langle s\rangle = \frac{e^{-\hbar\omega/\tau}}{1-e^{-\hbar\omega/\tau}} \qquad \langle s+1\rangle = \frac{1}{1-e^{-\hbar\omega/\tau}}$$

$$\sigma = \partial_\tau(\tau \ln Z) = \ln Z + \tau\frac{\partial_\tau Z}{Z} = \ln\langle s+1\rangle + \tau\frac{\hbar\omega}{\tau^2}\langle s\rangle = \ln\langle s+1\rangle + \frac{\hbar\omega}{\tau}\langle s\rangle$$

Now (nyní)

$$\langle s\rangle \ln\left(\frac{\langle s+1\rangle}{\langle s\rangle}\right) = Ze^{-\hbar\omega/\tau}\ln e^{\hbar\omega/\tau} = \frac{\hbar\omega}{\tau}\langle s\rangle$$

$$\Longrightarrow \boxed{\sigma = \langle s+1\rangle \ln\langle s+1\rangle + \langle s\rangle \ln\langle s\rangle}$$

Solution 18. Isentropic expansion of photon gas.

(a) $\tau_i V_i^{1/3} = \tau_f V_f^{1/3}$ or

$$\left(\frac{V_i}{V_f}\right)^{1/3} = \frac{\tau_f}{\tau_i} = \frac{2.9\,K}{3000\,K} = 10^{-3}$$

$r = r(t) = \alpha t$ so that $\Delta r = \alpha \Delta t$ or $\frac{\Delta r}{r} = \frac{\Delta t}{t}$

$$\frac{r_f - r_i}{r_f} = 1 - \frac{r_i}{r_f} = \frac{t_f - t_i}{t_f} = 1 - \frac{t_i}{t_f}$$

Knwowing that $\frac{r_i}{r_f} = 10^{-3}$, then $\frac{t_i}{t_f} = 10^{-3}$.

(b) Now

$$\sigma = \tau V^{1/3}$$

$$\left(\frac{\sigma}{\tau}\right)^3 = V$$

For *constant entropy* expansion.

$$\frac{U}{V} = \frac{\pi^2}{15\hbar^3c^3}\tau^4 \qquad U = \frac{\pi^2}{15(\hbar c)^3}V\left(\frac{\sigma^4}{V^{4/3}}\right) = \frac{\pi^2}{15(\hbar c)^3}\frac{\sigma^4}{V^{1/3}}$$

$$\left(\frac{\partial U}{\partial V}\right)_\sigma = \frac{\pi^2}{15(\hbar c)^3}\sigma^4\left(\frac{-1}{3}V^{-4/3}\right)$$

$$W = \int pdV = \frac{\pi^2}{15(\hbar c)^3}\sigma^4\left(-V^{-1/3}\right)\Big|_{V_i}^{V_f} = \frac{\pi^2\sigma^4}{15(\hbar c)^3}\left(\frac{1}{V_i^{1/3}} - \frac{1}{V_f^{1/3}}\right) = \frac{\pi^2 V_i^{4/3}\tau_i^4}{15(\hbar c)^3}\left(\frac{1}{V_i^{1/3}} - \frac{\tau_f}{\tau_i V_i^{1/3}}\right) = \frac{\pi^2 V_i \tau_i^3}{15(\hbar c)^3}(\tau_i - \tau_f)$$

Exercise 19. Reflective heat shield and Kirchhoff's law.

For a left plane sheet at τ_u temperature, right plane sheet at τ_l temperature

$$\begin{array}{ll} +J_u = \sigma_b \tau_u^4 & -J_l = \sigma_b \tau_l^4 \\ \text{reflection } -rJ_u = -(1-a)\sigma_b \tau_u^4 & \text{reflection } rJ_l = (1-a)\sigma_b \tau_l^4 \\ \text{absorb (left) } aJ_u = a\sigma_b \tau_u^4 & \text{absorb (left) } aJ_l = a\sigma_b \tau_l^4 \end{array}$$

total absorption: $a(J_u + J_l) = a\sigma_b(\tau_u^4 + \tau_l^4)$
total emission: $a\sigma_b(\tau_u^4 + \tau_l^4)$. By symmetry, $\frac{a(J_u+J_l)}{2}$ emitted to the left, and to right.
e emissivity, where emissivity is defined so radiation flux emitted by the object is e times the flux emitted by a blackbody at the same temperature.

By Kirchhoff law, for equilibrium, $a = e$; object must emit at same rate as it absorbs.

$$J_{net} = \frac{-a(J_u+J_l)}{2} - rJ_u + J_u = \frac{-(1-r)(J_u+J_l)}{2} + (1-r)J_u = \frac{(1-r)J_u}{2} - \frac{(1-r)J_l}{2} = (1-r)\frac{(J_u-J_l)}{2} =$$
$$= (1-r)\left(\frac{\sigma_b(\tau_u^4 - \tau_l^4)}{2}\right) = \frac{a(J_u+J_l)}{2} + rJ_l - J_l = \boxed{\frac{(1-r)(J_u-J_l)}{2}}$$

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Chemical Potential and Entropy

Let $\sigma = \sigma(U, V, N)$ (so now Σ has 3 coordinates U, V, N). Then

$$d\sigma = \left(\frac{\partial \sigma}{\partial U}\right)_{V,N} dU + \left(\frac{\partial \sigma}{\partial V}\right)_{U,N} dV + \left(\frac{\partial \sigma}{\partial N}\right)_{U,V} dN$$

Consider curve $c : \mathbb{R} \to \Sigma$ s.t. $\dot{c} = \dot{U}\frac{\partial}{\partial U} + \dot{N}\frac{\partial}{\partial N} \in \mathfrak{X}(\Sigma)$ (i.e. $dV(\dot{c}) = 0$)
$c(t) \in \Sigma$

Let $\tau = \tau(U, V, N) \in C^\infty(\Sigma)$
$d\tau \in \Omega^1(\Sigma)$

Suppose $d\tau(\dot{c}) = 0$ (i.e. c is a null curve of $d\tau$ or c is an isothermal curve, or i.e. overall temperature change $d\tau$ is 0)

$$d\sigma(\dot{c}) = \left(\frac{\partial \sigma}{\partial U}\right) dU(\dot{c}) + \left(\frac{\partial \sigma}{\partial N}\right) dN(\dot{c}) \text{ or } \frac{d\sigma(\dot{c})}{dN(\dot{c})} = \left(\frac{\partial \sigma}{\partial U}\right)\frac{dU(\dot{c})}{dN(\dot{c})} + \left(\frac{\partial \sigma}{\partial N}\right)$$

Now recall that originally, considering a system with only thermal contact to a huge heat reservoir/bath (it can be the universe), then only N,U mattered in describing this system, and so $\sigma = \sigma(N, U)$, so that $\frac{1}{\tau} = \left(\frac{\partial \sigma}{\partial U}\right)_N$.

Now

$$\frac{dU(\dot{c})}{dN(\dot{c})} = \frac{\dot{c}(U)}{\dot{c}(N)} = \frac{\dot{U}}{\dot{N}} = \left(\frac{\partial U}{\partial N}\right)_{\tau,V}$$

Similarly,

$$\frac{d\sigma(\dot{c})}{dN(\dot{c})} = \frac{\dot{c}(\sigma)}{\dot{N}} = \left(\frac{\partial \sigma}{\partial N}\right)_{\tau,V}$$

So

$$\left(\frac{\partial \sigma}{\partial N}\right)_{\tau,V} = \frac{1}{\tau}\left(\frac{\partial U}{\partial N}\right)_{\tau,V} + \left(\frac{\partial \sigma}{\partial N}\right)_{U,V}$$

Consider $F \equiv U - \tau\sigma$. Now consider $dF = dU - \sigma d\tau - \tau d\sigma$. Apply dF onto $\dot{c}$, our vector field generated by our particular isothermal, and constant volume curve in Σ:

$$dF(\dot{c}) = dU(\dot{c}) - \sigma d\tau(\dot{c}) - \tau d\sigma(\dot{c}) = \dot{U} - 0 - \tau d\sigma(\dot{c})$$

The right hand side is

$$dF(\dot{c}) = \dot{U}\frac{\partial F}{\partial U} + \dot{N}\left(\frac{\partial F}{\partial N}\right)_{\tau,V} = \dot{U}(1) + \dot{N}\left(\frac{\partial F}{\partial N}\right)_{\tau,N}$$

Left hand side is

$$\dot{U} - \tau\left(\left(\frac{\partial \sigma}{\partial N}\right)_{\tau,V}\dot{N} - \frac{1}{\tau}\left(\frac{\partial U}{\partial N}\right)_{\tau,V}\dot{N} + \left(\frac{\partial \sigma}{\partial U}\right)_{V,N}\dot{U}\right)$$

Subtract $\dot{U}$ from both sides. $\dot{U}$ was arbitrary (and so is $\dot{N}$), and so

$$\Longrightarrow \mu \equiv \left(\frac{\partial F}{\partial N}\right)_{\tau,V} \equiv \left(\frac{\partial U}{\partial N}\right)_{\tau,V} - \tau\left(\frac{\partial \sigma}{\partial N}\right)_{\tau,V} \text{ or } \mu = -\tau\left(\frac{\partial \sigma}{\partial N}\right)_{U,V}$$

Thermodynamic identity

Generalize thermodynamic identity for $\sigma = \sigma(U, V, N)$

$$d\sigma = \left(\frac{\partial \sigma}{\partial U}\right)_{V,N} dU + \left(\frac{\partial \sigma}{\partial V}\right)_{U,N} dV + \left(\frac{\partial \sigma}{\partial N}\right)_{U,V} dN =$$
$$= \frac{1}{\tau}dU + \frac{p}{\tau}dV - \frac{\mu}{\tau}dN \text{ or } dU = \tau d\sigma - pdV + \mu dN$$

Solution 1. Centrifuge.

$T = \frac{1}{2}m\dot{r}^2 + \frac{1}{2}mr^2\omega^2$.

Consider $\mu_{ext} = -\frac{1}{2}Mr^2\omega^2$ (negative so for bigger r away from $r = 0$ axis, lower chemical potential μ, so to show "centrifugal force" outwards.

$\mu_{tot} = \tau \ln(n/n_Q) - \frac{1}{2}Mr^2\omega^2$.

$\mu_{tot}(r) = \mu_{tot}(0)$ *for diffusion equilibrium.*

$$\tau \ln\left(\frac{n}{n_Q}\right) - \frac{1}{2}Mr^2\omega^2 = \tau \ln\left(\frac{n(0)}{n_Q}\right) \Longrightarrow \tau \ln\left(\frac{n(r)}{n(0)}\right) = \frac{1}{2}Mr^2\omega^2$$

$$\frac{n(r)}{n(0)} = \exp\left(\frac{Mr^2\omega^2}{2\tau}\right) \text{ or } \boxed{n(r) = n(0)\exp\left(\frac{Mr^2\omega^2}{2\tau}\right)}$$

Solution 2. Molecules in the Earth's atmosphere.

Recall that for ideal gas, $F = -\tau[N \ln Z_1 - \ln N!]$; $Z_1 = n_Q V = \left(\frac{M\tau}{2\pi\hbar^2}\right)^{3/2} V$.

$$\mu = \left(\frac{\partial F}{\partial N}\right)_{\tau,V} = -\tau[\log Z_1 - \frac{d}{dN}\ln N!] = \tau \ln\left(\frac{n}{n_Q}\right)$$

where $n_Q = \left(\frac{M\tau}{2\pi\hbar^2}\right)^{3/2}$.

Now $\mu_{\int} = \tau \log\left(\frac{n}{n_Q}\right)$

$\mu_{ext} = \frac{-GM_e}{r} = -\frac{gR_m^2}{r}$ since $g = \frac{GM_e}{R_e^2}$.

$$\mu_{tot} = \tau \ln(n/n_Q) + Mgh$$

In equilibrium, this must be independent of r: $\mu_{tot}(r) = \mu_{tot}(R)$.

$$\tau \ln(n(r)/nQ) - \frac{gR^2M}{r} = \tau \ln(n(R)/n_Q) - \frac{MgR^2}{R}$$

$$\tau \ln\left(\frac{n(r)}{n(R)}\right) = Mg\left(\frac{R^2}{r} - R\right) \text{ or } \exp\left(\frac{Mg}{\tau}\left(\frac{R^2}{r} - R\right)\right) = \frac{n(r)}{n(R)}$$
$$\Longrightarrow n(r) = n(R)\exp\left(\frac{Mg}{\tau}\left(\frac{R^2}{r} - \frac{1}{R}\right)\right)$$

so that

$$\boxed{N = 4\pi n(R)\exp\frac{-MgR}{\tau}\int_R^\infty r^2 dr \exp\left(\frac{MgR^2}{\tau r}\right)}$$

Solution 3. Potential energy of gas in gravitational field.

I looked at the solution from Ed Groth [10] for Problem 3, Homework No. 4, Physics 301 for a hint, which prompted me to recall that diffusive equilibrium between 2 systems means their chemical potential μ_i's are equal. [8]

Problem 3 **Potential energy of gas in gravitational field** of Chapter 5: Chemical Potential and Gibbs Distribution from Kittel and Kroemer (1980) [2]: "Consider a column of atoms each of mass M at temperature τ in a uniform gravitational field g."

Consider an infinitesimal layer at height h and at height $h + dh$. At each height h, the layers of atoms are in *diffusive* and *thermal equilibrium*, and so

$$\tau(h + dh) = \tau(h) = \tau$$

(thermal equilibrium) and

$$\mu(h + dh) = \mu(h)$$

(diffusive equilibrium).

Now for chemical potential $\mu = \mu(\tau, V, N)$,

$$\mu = \mu_{\text{int}} + \mu_{\text{ext}} = \tau \ln\left(n\left(\frac{2\pi\hbar^2}{M\tau}\right)^{3/2}\right) + Mgh$$

Thus, to first order,

$$\tau\frac{1}{n(h)}\frac{dn(h)}{dh} = -Mg \text{ or } n(h) = n(0)\exp\left(\frac{-Mgh}{\tau}\right)$$

The thermal average potential energy per atom is

$$\langle U\rangle = \frac{\int_0^\infty dhMghn(h)}{\int_0^\infty dhn(h)} = Mg\frac{\left[\frac{h\exp\left(\frac{-Mgh}{\tau}\right)}{-Mg/\tau} + -\frac{\exp\left(\frac{-Mgh}{\tau}\right)}{(-Mg/\tau)^2}\right]\Big|_0^\infty}{\left(\frac{\exp\left(\frac{-Mgh}{\tau}\right)}{-Mg/\tau}\right)\Big|_0^\infty} = Mg\frac{\tau}{Mg} = \boxed{\tau = \langle U\rangle}$$

From Ch.3, Eq. (64) of Kittel and Kroemer (1980) [2],

$$U = \frac{\sum_{\mathbf{n}}\epsilon_{\mathbf{n}}\exp\left[-\epsilon_{\mathbf{n}}/\tau\right]}{Z_1} = \tau^2\frac{\partial \ln Z_1}{\partial\tau} = \frac{3\tau}{2}$$

So the total heat capacity per atom C is

$$C = \frac{\partial E}{\partial\tau} = \frac{\partial(\langle U\rangle + U)}{\partial\tau} = \frac{\partial}{\partial\tau}(\tau + \frac{3\tau}{2}) = \boxed{\frac{5}{2} = C}$$

Solution 4. $\mu = \mu_{\text{int}} + \mu_{\text{ext}} = \tau\ln\left(\frac{n}{n_Q}\right) + \mu_{\text{ext}}$

$$\mu_{\text{ext}} = \tau\ln\left(\frac{10^4 n}{n_Q}\right) - \tau\ln\left(\frac{n}{n_Q}\right) = \tau\ln 10^4 = 300\,K * 0.0000861733 * eV/K\ln 10^4 = 0.24\,eV$$

Note that I checked this solution afterwards with the one by John D. Naud, via Ed Groth's materials [10], for Problem 4, Homework No. 4, Physics 301 [9].

[8] John D. Naud via Ed Groth, Problem 3, Homework No. 4, Physics 301, Fall 1999, http://grothserver.princeton.edu/~groth/phys301f99/hw04sol.pdf

[9] John D. Naud via Ed Groth, Problem 4, Homework No. 4, Physics 301, Fall 1999, http://grothserver.princeton.edu/~groth/phys301f99/hw04sol.pdf

Solution 5. Magnetic concentration.

This problem pertains to Figure 5.6. Recall the example on pp. 127 of Kittel and Kroemer (1980) [2], **Example: Chemical potential of mobile magnetic particles in a magnetic field**. "Consider system of N identical particles with magnetic moment m."

Now

$$\mu_{\rm tot}(\uparrow) = \tau \ln\left(\frac{n_\uparrow}{n_Q}\right) - mB = \tau \ln n_\uparrow + \frac{3\tau}{2}\ln\left(\frac{2\pi\hbar^2}{M\tau}\right) - mB$$

$$\mu_{\rm tot}(\uparrow)(B=0) = \tau \ln\left(n_\uparrow(B=0)\right) + \frac{3\tau}{2}\ln\left(\frac{2\pi\hbar^2}{M\tau}\right)$$

$$\mu_{\rm tot}(\uparrow)(B=20\,{\rm kgauss}) = \tau \ln\left(n_\uparrow(B=20\,{\rm kgauss})\right) + \frac{3\tau}{2}\ln\left(\frac{2\pi\hbar^2}{M\tau}\right) - m(20 \quad {\rm kgauss})$$

When $n_\uparrow(B=0) = 2\times 10^7\,{\rm cm}^{-3}$,

$n_\uparrow(B=20\ \ {\rm kgauss}) = 2\times 10^9\,{\rm cm}^{-3}$

$$\mu_{\rm tot}(\uparrow)(B=0) = \mu_{\rm tot}(\uparrow)(B=20\ \ {\rm kgauss})$$

so

$$\tau \ln\left(\frac{2\times 10^9\,{\rm cm}^{-3}}{2\times 10^7\,{\rm cm}^{-3}}\right) = m(20\,{\rm kgauss})$$

Given $\tau = 300\,K$,

$$m = \frac{300\,K\cdot k_B \ln(10^2)}{20\,{\rm kgauss}} = 9.54\times 10^{-18}\,{\rm erg/gauss} = 10^3 \text{ Bohr magnetons}$$

where we take the Bohr magneton $\mu_B \equiv e\hbar/2mc$ to be 9.274×10^{-21} erg/gauss.

Solution 6. Gibbs sum for a two level system. Recall that

$$\mathcal{Z}(\mu,\tau) = \sum_{N=0}^{\infty}\sum_{s(N)} \exp\left[(N\mu - \epsilon_{s(N)})/\tau\right] = \sum_{ASN} \exp\left[(N\mu - \epsilon_{s(N)})/\tau\right]; \qquad \lambda = \exp\left(\frac{\mu}{\tau}\right)$$

(a) $\mathcal{Z} = 1 + \lambda + \lambda\exp\left(\frac{-\epsilon}{\tau}\right)$, 1 for $N=0$; for $N=1$, $\lambda + \lambda\exp\left(\frac{-\epsilon}{\tau}\right)$. λ for $\epsilon = 0$.

(b) $\langle N\rangle = \frac{0(1)+(1)(\lambda+\lambda\exp\left(\frac{-\epsilon}{\tau}\right))}{\mathcal{Z}} = \frac{\lambda(1+\exp(-\epsilon/\tau))}{\mathcal{Z}}$

(c) $\langle N(\epsilon)\rangle = \frac{0(1)+(0)(\lambda)+\lambda\exp(-\epsilon/\tau)}{\mathcal{Z}} = \frac{\lambda\exp(-\epsilon/\tau)}{\mathcal{Z}}$

(d) $\langle\epsilon\rangle = \epsilon\langle N(\epsilon)\rangle = \frac{\lambda\epsilon\exp(-\epsilon/\tau)}{\mathcal{Z}}$

(e)

$$\mathcal{Z} = 1 + \lambda + \lambda\exp(-\epsilon/\tau) + \lambda^2\exp(-\epsilon/\tau) = (1+\lambda)(1+\lambda\exp(-\epsilon/\tau)$$

where we considered the possibility the orbital at 0 and at ϵ are each occupied by one particle at the same time.

So that, for total energy being ϵ, $\exp\left[(2\mu-\epsilon)/\tau\right] = (\exp\left(\frac{\mu}{\tau}\right))^2 e^{-\epsilon/\tau} = \lambda^2 e^{-\epsilon/\tau}$.

Solution 7. States of positive and negative ionization.

$$\mathcal{Z} = e^{\frac{\delta}{2\tau}} + \lambda e^{\frac{\Delta}{2\tau}} + \lambda e^{\frac{-\Delta}{2\tau}} + \lambda^2 e^{\frac{-\delta}{2\tau}}$$

$$\partial_\lambda \mathcal{Z} = 0 + e^{\Delta/2\tau} + e^{\frac{-\Delta}{2\tau}} + 2\lambda e^{\frac{-\delta}{2\tau}}$$

$$\lambda\partial_\partial \ln\mathcal{Z} = \frac{\lambda(e^{\Delta/2\tau} + e^{-\Delta/2\tau} + 2\lambda e^{-\delta/2\tau})}{e^{\frac{\delta}{2\tau}} + \lambda e^{\frac{\Delta}{2\tau}} + \lambda e^{\frac{-\Delta}{2\tau}} + \lambda^2 e^{\frac{-\delta}{2\tau}}} = 1$$

$$\lambda e^{\Delta/2\tau} + \lambda e^{-\Delta/2\tau} + 2\lambda^2 e^{-\delta/2\tau} = e^{\delta/2\tau} = \lambda e^{\Delta/2\tau} + \lambda e^{-\Delta/2\tau} + \lambda^2 e^{-\delta/2\tau} \quad \Longrightarrow \quad \begin{array}{c}\lambda^2 e^{-\delta/2\tau} = e^{\delta/2\tau} \\ \lambda^2 = e^{\delta/\tau} \text{ or } 2\ln\lambda = \delta/\tau\end{array}$$

$$\boxed{2\tau\ln\lambda = \delta}$$

Solution 8. Carbon monoxide poisoning.

(a) $\mathcal{Z} = 1 + \lambda(O_2)e^{-\epsilon_A/\tau}$

$$P(O_2) = \frac{\lambda(O_2)e^{-\epsilon_A}\tau}{1+\lambda(O_2)e^{-\epsilon_A/\tau}} = 0.9 \text{ or } 0.9 = 0.1\lambda(O_2)\exp\left(\frac{-\epsilon_A}{\tau}\right)$$

$$\Longrightarrow \ln\left(\frac{9}{\lambda(O_2)}\right) = \frac{-\epsilon_A}{\tau};\text{ or } \tau\ln\left(\frac{\lambda(O_2)}{9}\right) = \epsilon_A$$

$$\boxed{\epsilon_A = (k_BT)\ln\left(\frac{\lambda(O_2)}{9}\right)} = (8.617\times10^{-5}\,\frac{eV}{k})(273+39)\ln\left(\frac{10^{-5}}{9}\right) = -0.3686\,eV$$

(b) $\mathcal{Z} = 1 + \lambda(O_2)e^{-\epsilon_A/\tau} + \lambda(CO)e^{-\epsilon_B/\tau}$

$$P(O_2) = \frac{\lambda(O_2)e^{-\epsilon_A/\tau}}{1+\lambda(O_2)e^{-\epsilon_A/\tau}+\lambda(CO)e^{-\epsilon_B/\tau}} = 0.1 \Longrightarrow 0.1 + 0.1\lambda(CO)e^{-\epsilon_B/\tau} = 0.9\lambda(O_2)e^{-\epsilon_A/\tau}$$

$$\ln\left(\frac{9\lambda(O_2)e^{-\epsilon_A/\tau}-1}{\lambda(CO)}\right) = \frac{-\epsilon_B}{\tau} \text{ or } \boxed{\epsilon_B = \tau\ln\left(\frac{\lambda(CO)}{9\lambda(O_2)e^{-\epsilon_A/\tau}-1}\right)} = -0.5511\,eV$$

Solution 9. Absorption of O_2 in a magnetic field. Recall that $2j+1$ = total number of spin states. $2(1)+1 = 3$.
Now $U = -\mathbf{m}\cdot\mathbf{B}$.

$$\mathcal{Z} = 1 + \lambda(O_2)e^{\frac{-\epsilon+\mu_BB}{\tau}} + \lambda(O_2)e^{-\frac{\epsilon_A}{\tau}} + \lambda(O_2)e^{\frac{-(\epsilon_A+\mu_BB)}{\tau}} =$$
$$\mathcal{Z} = 1 + \lambda(O_2)e^{-\epsilon_A/\tau}(1+2\cosh\left(\frac{\mu_BB}{\tau}\right))$$

$$0.91 = \frac{\lambda(O_2)e^{-\epsilon_A/\tau}(1+2\cosh\left(\frac{\mu_BB}{\tau}\right))}{1+\lambda(O_2)e^{-\epsilon_A/\tau}(1+2\cosh\left(\frac{\mu_BB}{\tau}\right))} \text{ or } 0.91 = 0.09\lambda(O_2)e^{-\epsilon_A/\tau}(1+2\cosh\left(\frac{\mu_BB}{\tau}\right))$$
$$\frac{1}{2}\left(\frac{91}{9}\frac{1}{\lambda(O_2)}e^{\epsilon_A/\tau}-1\right) = \cosh\left(\frac{\mu_BB}{\tau}\right) \text{ or } B = \frac{\tau}{\mu_B}arccosh\left(\frac{1}{2}\left(\frac{91}{9}\frac{1}{\lambda(O_2)}e^{\epsilon_A/\tau}-1\right)\right)$$

The Gibbs sum in the limit of zero magnetic field will differ from that of Problem 8 because there the spin multiplicity of the bound state was neglected.

$$\mathcal{Z} = 1 + 3\lambda(O_2)e^{-\epsilon_A/\tau}$$
$$P(O_2) = \frac{3\lambda(O_2)e^{-\epsilon_A/\tau}}{1+3\lambda(O_2)e^{-\epsilon_A/\tau}} = 0.9 \text{ or } 0.9 = 0.3\lambda(O_2)e^{-\epsilon_A/\tau}$$
$$\ln\frac{3}{\lambda(O_2)} = -\epsilon_A/\tau \text{ or } \epsilon_A = \tau\ln\left(\frac{\lambda(O_2)}{3}\right) = -0.6227$$

for $T = 300\,K$, so that $\tau = 0.049375\,eV$.

$$\Longrightarrow B = \frac{\tau}{\mu_B}0.59927 = \frac{0.049375\cdot0.59927}{5.7884\times10^{-11}\times10^6\,eV/T} = 511.1\,T$$

Solution 10. Concentration fluctuations.

(a) Recall that $\mathcal{Z} = \sum_{N=0}^{\infty}\sum_{s(N)}\exp\left[(N\mu-\epsilon_{s(N)})/\tau\right]$.

$$\frac{\partial^2\mathcal{Z}}{\partial\mu^2} = \sum_{ASN}\left(\frac{N}{\tau}\right)^2\exp\left[(N\mu-\epsilon_{s(N)})/\tau\right] \qquad \langle N^2\rangle = \frac{\tau^2}{\mathcal{Z}}\frac{\partial^2\mathcal{Z}}{\partial\mu^2}$$

(b)

$$\frac{\partial}{\partial\mu}\langle N\rangle = \frac{\partial}{\partial\mu}\left(\frac{\tau}{\mathcal{Z}}\left(\frac{\partial\mathcal{Z}}{\partial\mu}\right)_{\tau,V}\right) = \tau\left(\frac{-\left(\frac{\partial\mathcal{Z}}{\partial\mu}\right)^2_{\tau,V}}{\mathcal{Z}^2} + \frac{\left(\frac{\partial^2\mathcal{Z}}{\partial\mu^2}\right)_{\tau,V}}{\mathcal{Z}}\right)$$

$$\implies \tau\frac{\partial}{\partial\mu}\langle N\rangle = \langle N^2\rangle - \langle N\rangle^2 = \langle(\Delta N)^2\rangle$$

Solution 11. Equivalent definition of chemical potential.

Recall that

$$d\sigma = \left(\frac{\partial\sigma}{\partial U}\right)_{V,N}dU + \left(\frac{\partial\sigma}{\partial V}\right)_{U,N}dV + \left(\frac{\partial\sigma}{\partial N}\right)_{U,V}dN \qquad (31)$$

$$\mu = -\tau\left(\frac{\partial\sigma}{\partial N}\right)_{U,V} \qquad (35)$$

Consider when $d\sigma = dV = 0$.

$$\implies \frac{-\mu}{\tau}dN + \left(\frac{\partial\sigma}{\partial U}\right)_{V,N}dU = 0$$

$$\left(\frac{\partial\sigma}{\partial U}\right)_{V,N}\frac{dU}{dN} = \frac{\mu}{\tau}$$

Note that $\frac{dU}{dN} = \left(\frac{\partial U}{\partial N}\right)_{\sigma,V}$.

Using the definition, $\left(\frac{\partial\sigma}{\partial U}\right)_{N,V} \equiv \frac{1}{\tau}$, so then

$$\boxed{\mu = \left(\frac{\partial U}{\partial N}\right)_{\sigma,V}}$$

Now $F = U - \tau\sigma$, by definition. Consider the thermodynamic identity, $dU = \tau d\sigma - pdV + \mu dN$.

$$dF = dU - \sigma d\tau - \tau d\sigma = \tau d\sigma - pdV + \mu dN - \sigma d\tau - \tau d\sigma =$$
$$= -pdV + \mu dN - \sigma d\tau$$
$$\implies \left(\frac{\partial F}{\partial N}\right)_{\tau,V} = \mu(\tau, V, N)$$

Likewise, $\mu = \left(\frac{\partial U}{\partial N}\right)_{\sigma,V}$ is equivalent to $\mu(\tau, V, N) = \left(\frac{\partial F}{\partial N}\right)_{\tau,V}$ through the thermodynamic identity as well.

$$dU = \tau d\sigma - pdV + \mu dN$$

σ, V constant. $d\sigma, dV = 0$.

$$\implies \left(\frac{\partial U}{\partial N}\right)_{\sigma,V} = \mu$$

Solution 12. Ascent of sap in trees. Given: relative humidity $r = 0.9$. $T = 25^\circ\, C$

n_0 = concentration in saturated air that stands immediately above pool of water of water vapor in air.
rn_0 = actual concentration of water vapor in air at uppermost leaves is rn_0.

At pool, $\mu_{H_2O} = \mu_{\text{vapor}}$ for diffusion equilibrium.

Same condition at uppermost leaves, otherwise there's evaporation:
$\mu_{\text{sap}} = \mu_{\text{vapor}}(h)$.

$\mu_{\text{sap}} = \mu_{H_2O}$ (no flow going on in water). Thus, $\mu_{\text{vapor}}(h) = \mu_{\text{vapor}}(0)$ and treat water vapor as an ideal gas.

$$\tau\ln\left(\frac{n(h)}{n_Q}\right) + mgh = \tau\ln\left(\frac{n(0)}{n_Q}\right)$$

$$\tau\ln\left(\frac{rn_0}{n_0}\right) = -mgh \implies \tau\ln\left(\frac{1}{r}\right) = mgh \text{ or } h = \frac{\tau}{mg}\ln\left(\frac{1}{r}\right)$$

$$h = \frac{(1.381\times10^{-23}\,\frac{1}{K})(298\,K)(kg\cdot\frac{m^2}{s^2})\ln\left(\frac{10}{9}\right)}{(18\,amu)\left(\frac{1.67\times10^{-27}\,kg}{1\,amu}\right)(9.8\,m/s^2)} = 147.1\,m$$

Solution 13. Isentropic expansion.

(a)

(b) Recall that $F = -\tau[N\ln Z_1 - \ln N!]$ where $Z_1 = n_Q V = \left(\frac{M\tau}{2\pi\hbar^2}\right)^{3/2} V$ and $\sigma = -\left(\frac{\partial F}{\partial \tau}\right)_{V,W}$.

$$\frac{\partial F}{\partial \tau} = -[N\ln Z_1 - \ln N!] - \tau[N\frac{1}{Z_1}\left(\frac{M}{2\pi\hbar^2}\right)^{3/2}\frac{3}{2}\tau^{1/2}V] =$$
$$= -[N\ln Z_1 - \ln N!] - [N\frac{1}{Z_1}n_Q\frac{3}{2}V] = -[N\ln Z_1 - \ln N!] - \frac{3}{2}N$$

With N constant and $Z_1 = \left(\frac{M}{2\pi\hbar^2}\right)^{3/2}\tau^{3/2}V$, then for an isentropic expansion, $\tau V^{2/3}$ must remain constant.

Solution 14. Multiple binding of O_2.

(a) Be wary of the multiplicity, how you count, each of the energy states.

$$\mathcal{Z} = 1 + 4\lambda e^{-\epsilon/\tau} + \binom{4}{2}\lambda^2 e^{-2\epsilon/\tau} + \binom{4}{3}\lambda^3 e^{-3\epsilon/\tau} + \lambda^4 e^{-4\epsilon/\tau} = (1+\lambda e^{-\epsilon/\tau})^4$$

$$\boxed{P(\epsilon) = \frac{4\lambda e^{-\epsilon/\tau}}{(1+\lambda e^{-\epsilon/\tau})^4}}$$

(b)

$$P(4\epsilon) = \frac{\lambda^4 e^{-\epsilon/\tau}}{(1+\lambda e^{-\epsilon/\tau})^4} = \frac{e^{-\epsilon/\tau}}{\left(\frac{1}{\lambda+e^{-\epsilon/\tau}}\right)^4}$$

2015 manuscript.

130 Gibbs distribution

Manuscript section title: Reactions. Equilibrium in reactions is §141.

130.1 Gibbs (review)

cf. Chapter 5: Chemical Potential and Gibbs Distribution of Kittel and Kroemer (1980) [2]

Recall that

$$F = U - \tau\sigma$$
$$dF = dU - \tau d\sigma - \sigma d\tau = -pdV - \sigma d\tau \text{ or } dF = +\mu dN - pdV - \sigma d\tau$$

so $F = F(\tau, V)$.

Total entropy of system + reservoir $\sigma = \sigma_{\mathcal{R}} + \sigma_{\mathcal{S}}$

$$\sigma = \sigma_{\mathcal{R}} + \sigma_{\mathcal{S}} = \sigma_{\mathcal{R}}(U - U_{\mathcal{S}}) + \sigma_{\mathcal{S}}(U_{\mathcal{S}}) \simeq \sigma_{\mathcal{R}}(U) - U_{\mathcal{S}}\left(\frac{\partial\sigma_{\mathcal{S}}}{\partial U_{\mathcal{R}}}\right)_{V,N} + \sigma_{\mathcal{S}}(U_{\mathcal{S}})$$

Now

$$\left(\frac{\partial\sigma_{\mathcal{R}}}{\partial U_{\mathcal{R}}}\right)_{V,N} \equiv \frac{1}{\tau}$$

and so

$$\sigma = \sigma_{\mathcal{R}}(U) - \frac{1}{\tau}U_{\mathcal{S}} + \sigma_{\mathcal{S}} = \sigma_{\mathcal{R}}(U) - F_{\mathcal{S}}/\tau$$

$F_{\mathcal{S}}$ must be a minimum with respect to $U_{\mathcal{S}}$ when system in its most probable state.

EY : 20151218 Consider $\sigma = \frac{U}{\tau} - \frac{F}{\tau}$, so

$$d\sigma = \frac{dU}{\tau} - \frac{U d\tau}{\tau^2} - \frac{dF}{\tau} + \frac{F}{\tau^2} d\tau$$

Consider curve $\gamma : \mathbb{R} \to \Sigma$, s.t. $d\tau(\dot{\gamma}) = 0$, i.e. γ represents a thermodynamic process at constant temeprature. System in thermal equilibrium with reservoir.

Now for every thermodynamic process, entropy must increase, and so

$$d\sigma(\dot{\gamma}) = \frac{dU}{\tau}(\dot{\gamma}) - \frac{dF(\dot{\gamma})}{\tau} \geq 0$$

Thus, if $\int dU(\dot{\gamma}) < 0$, $\int dF(\dot{\gamma}) < 0$, and so F is a minimum.

If $d\sigma(\dot{\gamma}) = 0$, $dU(\dot{\gamma}) = dF(\dot{\gamma})$. U is minimized, so F is minimized.

Consider $\mathcal{S}_1$, $\mathcal{S}_2$ in thermal and diffusive contact, $\mathcal{S}_1$, $\mathcal{S}_2$ in thermal contact with large reservoir, with V_1,V_2 held constant,

$$F = F_1 + F_2$$
$$N = N_1 + N_2 \text{ constant}$$

$$dF = \left(\frac{\partial F_1}{\partial N_1}\right)_{\tau,V_1} dN_1 + \left(\frac{\partial F_2}{\partial N_2}\right)_{\tau,V_2} dN_2 = \left[\left(\frac{\partial F_1}{\partial N_1}\right)_{\tau,V_1} - \left(\frac{\partial F_2}{\partial N_2}\right)_{\tau,V_2}\right] dN_1 = 0$$

and so $\mu_1 = \mu_2$ for diffusive equilibrium.

Note that

$$\mu(\tau, V, N) := \left(\frac{\partial F}{\partial N}\right)_{\tau,V} \tag{626}$$

Example: Chemical potential of ideal gas. If $\mu_1 > \mu_2$, then $dF < 0$, when $dN_1 < 0$, *particles flow from high chemical potential to low chemical potential.*

Recall

$$F = -\tau[N \ln Z_1 - \ln N!]$$
$$Z_1 = n_Q V = \left(\frac{M\tau}{2\pi\hbar^2}\right)^{3/2} V$$

Then

$$\mu = -\tau(\ln Z_1 - \ln N) = \tau \ln (N/Z_1)$$
$$\mu = \tau \ln (n/n_Q)$$
$$\mu = \tau \ln \left(\frac{p}{\tau n_Q}\right)$$

Note that the *total chemical potential* μ is

$$\mu = \mu_{\text{tot}} = \mu_{\text{ext}} + \mu_{\text{int}}$$

Examples of μ:

$$\mu = \tau \ln \left(\frac{n}{n_Q}\right) + q\Delta V$$

Example: Variation of barometric pressure with altitude.

$$\mu = \tau \ln \left(\frac{n}{n_Q}\right) + Mgh$$

130.1.1 Chemical Potential and Entropy

cf. pp. 131 of Kittel and Kroemer (1980) [2], Ch.5.

$$\sigma = \sigma(U, V, N)$$
$$\tau d\sigma = dU + pdV - \mu dN$$
$$-\left(\frac{\partial \sigma}{\partial N}\right)_{U,V} = \frac{\mu}{\tau}$$

130.2 Gibbs Factor and Gibbs Sum

cf. pp. 134 of Kittel and Kroemer (1980) [2], Ch.5. Consider system $\mathcal{S}$, with (N, ϵ_S), number of particles and internal energy, respectively, and reservoir $\mathcal{R}$, with $(N_0 - N, U_0 - \epsilon_S)$, in thermal and diffusive contact.

System probability $P(N, \epsilon_S) \propto g(N_0 - N, U_0 - \epsilon_S)$

$$\frac{P(N_1, \epsilon_1)}{P(N_2, \epsilon_2)} = \frac{g(N_0 - N_1, U_0 - \epsilon_1)}{g(N_0 - N_2, U_0 - \epsilon_2)}$$

Now by definition of entropy, $g(N_0, U_0) \equiv \exp\left[\sigma(N_0, U_0)\right]$,

$$\frac{P(N_1, \epsilon_1)}{P(N_2, \epsilon_2)} = \exp\left(\Delta\sigma\right) = \exp\left(\sigma(N_0 - N_1, U_0 - \epsilon_1) - \sigma(N_0 - N_2, U_0 - \epsilon_2)\right) = \exp\left[(N_1 - N_2)\frac{\mu}{\tau} - \frac{(\epsilon_1 - \epsilon_2)}{\tau}\right]$$

so

$$\exp\left[\frac{(N\mu - \epsilon)}{\tau}\right]$$

is the Gibbs factor.

Gibbs sum or grand sum or grand partition function $\mathcal{Z}$ is

$$\mathcal{Z}(\mu, \tau) = \sum_{N=0}^{\infty} \sum_{s(N)} \exp\left[(N\mu - \epsilon_{s(N)})/\tau\right] = \sum_{ASN} \exp\left[\frac{(N\mu - \epsilon_{s(N)})}{\tau}\right]$$

where ASN = overall states of the system for all number of particles.

Problem 3 **Potential energy of gas in gravitational field** of Chapter 5: Chemical Potential and Gibbs Distribution from Kittel and Kroemer (1980) [2]: "Consider a column of atoms each of mass M at temperature τ in a uniform gravitational field g."

Consider an infinitesimal layer at height h and at height $h + dh$. At each height h, the layers of atoms are in *diffusive* and *thermal equilibrium*, and so

$$\tau(h + dh) = \tau(h) = \tau$$

(thermal equilibrium) and

$$\mu(h + dh) = \mu(h)$$

(diffusive equilibrium).

Now for chemical potential $\mu = \mu(\tau, V, N)$,

$$\mu = \mu_{\text{int}} + \mu_{\text{ext}} = \tau \ln\left(n\left(\frac{2\pi\hbar^2}{M\tau}\right)^{3/2}\right) + Mgh$$

Thus, to first order,

$$\tau\frac{1}{n(h)}\frac{dn(h)}{dh} = -Mg \text{ or } n(h) = n(0)\exp\left(\frac{-Mgh}{\tau}\right)$$

The thermal average potential energy per atom is

$$\langle U\rangle = \frac{\int_0^\infty dhMghn(h)}{\int_0^\infty dhn(h)} = Mg\frac{\left[\frac{h\exp\left(\frac{-Mgh}{\tau}\right)}{-Mg/\tau} + -\frac{\exp\left(\frac{-Mgh}{\tau}\right)}{(-Mg/\tau)^2}\right]\Big|_0^\infty}{\left(\frac{\exp\left(\frac{-Mgh}{\tau}\right)}{-Mg/\tau}\right)\Big|_0^\infty} = Mg\frac{\tau}{Mg} = \boxed{\tau = \langle U\rangle}$$

From Ch.3, Eq. (64) of Kittel and Kroemer (1980) [2],

$$U = \frac{\sum_{\mathbf{n}} \epsilon_{\mathbf{n}} \exp\left[-\epsilon_{\mathbf{n}}/\tau\right]}{Z_1} = \tau^2\frac{\partial \ln Z_1}{\partial\tau} = \frac{3\tau}{2}$$

So the total heat capacity per atom C is

$$C = \frac{\partial E}{\partial\tau} = \frac{\partial(\langle U\rangle + U)}{\partial\tau} = \frac{\partial}{\partial\tau}(\tau + \frac{3\tau}{2}) = \boxed{\frac{5}{2} = C}$$

2008 manuscript.

131 Ideal Gas

Classical Limit

Recall, Ch. 6, Sec. "Classical Limit" of Kittel and Kroemer [2], **an ideal gas is defined as a system of free noninteracting particles in the classical regime**.
$f(\epsilon) \equiv$ average occupancy of an orbital at energy ϵ
$\epsilon \equiv$ energy of orbital occupied by 1 particle; not energy of system of N particles
Fermi-Dirac and Bose-Einstein distribution $f(\epsilon) = \frac{1}{\exp[(\epsilon-\mu)/\tau]\pm 1}$
In order for $f(\epsilon) \ll 1 \ \forall$ orbitals, $\exp[(\epsilon-\mu)/\tau] \gg 1 \, \forall \, \epsilon$.

$$\implies f(\epsilon) \simeq \exp[(\mu-\epsilon)/\tau] = \lambda \exp(-\epsilon/\tau) \quad \lambda \equiv \exp\left(\frac{\mu}{\tau}\right)$$

$f(\epsilon)$, average occupancy of orbital of energy ϵ, is classical distribution function.

In Ch. 3 of Kittel and Kroemer [2], $Z_1 = n_Q V$, $n_Q = \left(\frac{M\tau}{2\pi\hbar^2}\right)^{3/2}$.
$\lambda = e^{\mu/\tau}$

$$N = \lambda Z_1 = \lambda n_Q V \quad \lambda = \frac{N}{n_Q V} = \frac{n}{n_Q}$$

$$\implies \lambda = \exp\left(\frac{\mu}{\tau}\right) = \frac{n}{n_Q}$$

Heat Capacity

Now

$$Q = Q(\tau, p) = \Lambda_p dp + C_p d\tau = dU + W = dU + pdV$$

Let $c \in \Sigma$ s.t. $dp(\dot{c}) = 0$ (constant pressure). And so

$$Q(\dot{c}) = 0 + C_p d\tau(\dot{c}) = \tau d\sigma(\dot{c}) = dU(\dot{c}) + pdV(\dot{c})$$

for $c = (\tau, 0)$, $\dot{c} = \frac{\partial}{\partial \tau} \in T\Sigma$.

$$\implies C_p = \tau \left(\frac{\partial \sigma}{\partial \tau}\right)_p = \left(\frac{\partial U}{\partial \tau}\right)_p + p\left(\frac{\partial V}{\partial \tau}\right)_p$$

for heat capacity at constant pressure is larger than C_V because additional heat must be added to perform the work needed to expand volume of gas against constant pressure.

Now recall for enthalpy

$$H = U + pV$$
$$H = H(\sigma, p)$$

Then

$$dH = dU + pdV + Vdp = Q + Vdp$$

Thus, for $c \in \Sigma$, $dp(\dot{c}) = 0$ (constant pressure). Hence

$$dH(\dot{c}) = \left(\frac{\partial H}{\partial \tau}\right)_p = Q(\dot{c}) + 0 = C_p$$

Hence,

$$C_p = \left(\frac{\partial H}{\partial \tau}\right)_p = \left(\frac{\partial Q}{\partial \tau}\right)_p$$

Now

$$\int_c dH = H_f - H_i = \int_c Q = \int_c C_p d\tau = C_p(\tau_f - \tau_i)$$

(if C_p constant over desired temperature range)

Reversible Isothermal Expansion

$Q = 0$, insulated gas, no heat flow to or from the gas ("adiabatic")
σ constant in system isolated from reservoir, if expansion reverisble (slowly)

What is the pressure after expansion?

Remember

$$\frac{C_p}{C_V} = \frac{\frac{5}{2}N}{\frac{3}{2}N} = \frac{5}{3} = \gamma$$

So let

$$\frac{1}{\gamma - 1} = \frac{3}{2}$$

$$\sigma(\tau, V) = N(\ln \tau^{\frac{1}{\gamma-1}} + \ln V + \text{ constant }) \qquad (61) \qquad (627)$$

$$\ln \tau^{\frac{1}{\gamma-1}} V = \text{ constant or } \tau^{\frac{1}{\gamma-1}} V \text{ constant} \qquad (62) \qquad (628)$$

$$\implies \tau_1^{\frac{1}{\gamma-1}} V_1 = \tau_2^{\frac{1}{\gamma-1}} V_2 \qquad (63) \qquad (629)$$

for ideal monoatomic gas.

Use $pV = N\tau$

$$\implies \frac{\tau_1^{\frac{\gamma}{\gamma-1}}}{p_1} = \frac{\tau_2^{\frac{\gamma}{\gamma-1}}}{p_2} \qquad (64) \qquad (630)$$

or $p_1^{\frac{1}{\gamma-1}} V_1^{\frac{\gamma}{\gamma-1}} = p_2^{\frac{1}{\gamma-1}} V_2^{\frac{\gamma}{\gamma-1}}$ or $p_1 V_1^\gamma = p_2 V_2^\gamma$

I will recap Problem 10 and my solution.

Isentropic relations of ideal gas.

(a) $\gamma = \frac{C_p}{C_V}$. For isentropic process, $pV^\gamma = p_i V_i^\gamma$.

Then, essentially and equivalently, take the exterior derivative:

$$V^\gamma dp + \gamma p V^{\gamma-1} dV = 0 \implies \frac{dp}{p} + \frac{\gamma dV}{V} = 0$$

For

$$\tau V^{\gamma-1} = \tau_i V_i^{\gamma-1}$$

then taking d:

$$d\tau V^{\gamma-1} + \tau(\gamma-1)V^{\gamma-2} dV = 0 \implies \frac{d\tau}{\tau} + \frac{\gamma-1}{V} dV = 0$$

For

$$\tau^{\frac{\gamma}{1-\gamma}} p = \tau_i^{\frac{\gamma}{1-\gamma}} p_i$$

then taking d:

$$dp\tau^{\frac{\gamma}{1-\gamma}} + \frac{\gamma}{1-\gamma}\tau^{\frac{2\gamma-1}{1-\gamma}} d\tau p = 0 \implies \frac{dp}{p} + \frac{\gamma}{1-\gamma}\frac{d\tau}{\tau} = 0$$

(b) isentropic bulk moduli

$$B_\sigma = -V\left(\frac{\partial p}{\partial V}\right)_\sigma = \gamma \frac{p_i V_i^\gamma}{V^\gamma} = \gamma p$$

since $p = \frac{p_i V_i^\gamma}{V^\gamma}$

$$\boxed{B_\sigma = \gamma p}$$

EY : 20150606 I think when one consider small, linear longitudinal perturbations of the gas system, with pressure being the external restoring force, then sound waves propagate (correct me if I'm wrong) and this is the way to derive B_σ.

isothermal bulk moduli $B_\tau = -V\left(\frac{\partial p}{\partial V}\right)_\tau = \frac{n\tau}{V} = p$

velocity of sound in gas is $c = \left(\frac{B_\sigma}{\rho}\right)^{1/2} = \left(\frac{\gamma p}{\rho}\right)^{1/2}$

for ideal gas of molecules of mass M, $pV = N\tau$

Manifold interpretation; I'm using Chapter 5 Applications in Physics, Section A Thermodynamics of [3]. Consider a 2-dimensional manifold of equilibrium states $\mathcal{M}$, s.t. $\dim\mathcal{M} = 2$. Local coordinates are (U, V) with U being a global coordinate.

Now $U = C_V\tau$ so use τ as a global coordinate, i.e. the local coordinates of $\mathcal{M}$ can be (τ, V).

Consider a curve in $\mathcal{M}$ parametrized by λ: $(\tau(\lambda), V(\lambda))$. The corresponding tangent vector $X = \dot\tau\frac{\partial}{\partial\tau} + \dot V\frac{\partial}{\partial V} \in$

$\mathfrak{X}(\mathcal{M})$ has components in coordinate vector basis $\dot\tau = \dot\tau(\lambda)$

$$\dot V = \dot V(\lambda)$$

Recall

$$\frac{d\tau}{\tau} + \frac{\gamma-1}{V}dV = 0$$

Consider applying this 1-form onto X:

$$\left(\frac{d\tau}{\tau} - \frac{\gamma-1}{V}dV\right)(X) = \frac{\dot\tau}{\tau} + \frac{\gamma-1}{V}\dot V = 0 \Longrightarrow \frac{d}{d\lambda}(\tau V^{\gamma-1}) = 0$$

Then the equalities of the endpoints of this curve $(\tau(\lambda), V(\lambda))$ are the equalities above. The interpretation is that the isentropic process draws out a curve in $\mathcal{M}$ and can be written as a curve or as a tangent vector field (specifically a section of $T\mathcal{M}$).

$$\tau_1 V_1^{\gamma-1} = \tau_2 V_2^{\gamma-1} \qquad (66) \tag{631}$$

$$\tau_1^{\frac{\gamma}{1-\gamma}} P_1 = \tau_2^{\frac{\gamma}{1-\gamma}} P_2 \qquad (67) \tag{632}$$

$$P_1 V_1^\gamma = P_2 V_2^\gamma \qquad (68) \tag{633}$$

for, indeed,

$$p_1 V_1^\gamma = p_2 V_2^\gamma \xrightarrow{pV=N\tau} p_1\left(\frac{N\tau_1}{p_1}\right)^\gamma = p_2\left(\frac{N\tau_2}{p_2}\right)^\gamma \text{ so}$$

$$p_1^{1-\gamma}\tau_1^\gamma = p_2^{1-\gamma}\tau_2^\gamma \text{ or } p_1\tau_1^{\frac{\gamma}{1-\gamma}} = p_2\tau_2^{\frac{\gamma}{1-\gamma}}$$

$$\Longrightarrow p_1^{\frac{1-\gamma}{\gamma}}\tau_1 = p_2^{\frac{1-\gamma}{\gamma}}\tau_2 \text{ or } \left(\frac{p_1}{p_2}\right)^{\frac{1-\gamma}{\gamma}} = \frac{\tau_2}{\tau_1} = \left(\frac{p_2}{p_1}\right)^{\frac{\gamma-1}{\gamma}}$$

$T_1 = 300\,K$, $\quad V_1/V_2 = \frac{1}{2}$,

$$T_2 = \left(\frac{1}{2}\right)^{2/3}(300\,K) = 189\,K \qquad (69) \tag{634}$$

The gas is cooled in expansion process by

$$T_1 - T_2 = 300\,K - 189\,K = 111\,K \qquad (70) \tag{635}$$

Expansion at constant entropy is important e.g. methods of refridgeration.

What is the change in energy in the expansion?

$$U_2 - U_1 = C_V(\tau_2 - \tau_1) = \frac{1}{\gamma-1}N(\tau_2 - \tau_1)$$

Problems

Solution 1. Derivative of Fermi-Dirac function. Recall $f = \frac{1}{\exp[(\epsilon-\mu)/\tau]+1} = (e^{(\epsilon-\mu)/\tau}+1)^{-1}$

$$\partial_\epsilon f = -1(e^{(\epsilon-\mu)/\tau}+1)^{-2}(e^{(\epsilon-\mu)/\tau})\left(\frac{1}{\tau}\right)$$

$$\partial_\epsilon f(\epsilon=\mu) = \frac{-1}{4\tau}$$

Solution 2. Symmetry of filled and vacant orbitals. $\epsilon = \mu + \delta$

$$f(\epsilon) = f(\mu+\delta) = \frac{1}{e^{\delta/\tau}+1} = \frac{e^{-\delta/\tau}}{1+e^{-\delta/\tau}} = 1 + \frac{-1}{e^{-\delta/\tau}+1} = 1 - f(\mu-\delta)$$

Solution 3. Distribution function for double occupancy statistics.

(a) $\eta = 1 + \lambda e^{-\epsilon/\tau} + \lambda^2 e^{-2\epsilon/\tau}$ where $\lambda = e^{\mu/\tau}$.

$$\langle N\rangle = \lambda\partial_\lambda \ln\zeta = \frac{\lambda}{\zeta}(e^{-\epsilon/\tau} + 2\lambda e^{-2\epsilon/\tau}) = \frac{\lambda e^{-\epsilon/\tau} + 2\lambda^2 e^{-2\epsilon/\tau}}{\zeta}$$

(b) $\zeta = 1 + 2\lambda e^{-\epsilon/\tau} + \lambda^2 e^{-2\epsilon/\tau}$

$$\langle N\rangle = \lambda\partial_\lambda \ln\lambda = \frac{2\lambda e^{-\epsilon/\tau} + 2\lambda^2 e^{-2\epsilon/\tau}}{\zeta} = \frac{2\lambda(e^{-\epsilon/\tau})(1+\lambda e^{-\epsilon/\tau})}{\zeta}$$

Solution 4. Energy of gas of extreme relativistic particles. For $\epsilon_p \simeq p$, $\sum_s e^{-p/tau} = Z$.
With the factor 2 for the 2 possible polarizations,

$$Z = (2)\frac{4\pi}{8}\int_0^\infty p^2 dp e^{-p/\tau} = \pi\{p^2 e^{-p/\tau}(-\tau)\Big|_0^\infty - \int_0^\infty 2pe^{-p/\tau}(-\tau)\} = 2\tau\pi\int_0^\infty pe^{-p/\tau}dp =$$
$$== 2\tau\pi\{pe^{-p/\tau}(-\tau)\Big|_0^\infty - \int_0^\infty e^{-p/\tau}(-\tau)dp\} = 2\tau^3\pi$$

$$U = \tau^2\frac{\partial \ln Z}{\partial\tau} = \tau^2\partial_\tau\{\ln(2\pi\tau^3)\} = \tau^2\partial_\tau(3\ln\tau) = \boxed{U = 3\tau}$$

Solution 5. Integration of the thermodynamic identity for an ideal gas. For constant N, recall

$$d\sigma = \frac{dU}{\tau} + \frac{pdV}{\tau} = \frac{1}{\tau}\left(\frac{\partial U}{\partial\tau}\right)_V d\tau + \frac{1}{\tau}\left(\frac{\partial U}{\partial V}\right)_\tau dV + \frac{pdV}{\tau}$$

$C_V = \left(\frac{\partial U}{\partial\tau}\right)_V$, and for an ideal gas $pV = N\tau$.

$$\int d\sigma = \sigma = C_V\ln\tau + N\ln V + \int\frac{1}{\tau}\left(\frac{\partial U}{\partial\tau}\right)_\tau dV + \sigma_1$$

Now $U = \frac{3}{2}\tau$ for an ideal gas, so $\left(\frac{\partial U}{\partial V}\right)_\tau = 0$.

$$\implies \sigma = C_V\ln\tau + N\ln V + \sigma_1$$

σ_1 independent constant of τ and V.

Solution 6. Entropy of mixing.

Correction (empty solution). The 2008 manuscript stops at this heading. The mixing entropy, the identical-gas cancellation, and the ideal-mixture chemical potential are derived in §133.

Solution 7. Relation of pressure and energy density.

(a) Recall that $U = U(\sigma, V, N)$. $p = -\left(\frac{\partial U}{\partial V}\right)_N$, and
$U = \frac{\sum_s \epsilon_s e^{-\epsilon_s/\tau}}{Z}$

$$\frac{\partial U}{\partial V} = \sum_s \left(\left(\frac{\partial \epsilon_s}{\partial V}\right)_N e^{-\epsilon_s/\tau} + \epsilon_s \left(\frac{-1}{\tau} \left(\frac{\partial \epsilon_s}{\partial V}\right)_N \right) e^{-\epsilon_s/\tau} \right) / Z - \sum_s \left(-\frac{\partial \epsilon_s}{\partial V} \right) \left(\frac{1}{\tau}\right) e^{-\epsilon_s/\tau} \sum_{s_2} \epsilon_{s_2} e^{-\epsilon_s/\tau} / Z^2$$

Now $p = -\left(\frac{\partial U}{\partial V}\right)_N$.

So if the system is in state s; then $p_s = -\left(\frac{\partial \epsilon_s}{\partial V}\right)_N$.

$$\implies \boxed{\langle p \rangle = p = \frac{\sum_s -\left(\frac{\partial \epsilon_s}{\partial V}\right)_N e^{-\epsilon_s/\tau}}{Z}}$$

(b) Now $\epsilon_s = \frac{\pi}{L^2}\frac{n^2\hbar^2}{2M} = \frac{\pi}{V^{2/3}}\frac{n^2\hbar^2}{2M}$

$$\left(\frac{\partial \epsilon_s}{\partial V}\right)_N = -\frac{2}{3} V^{-5/3} \frac{\pi n^2 \hbar^2}{2M} = \boxed{-\frac{2}{3V}\epsilon_s}$$

(c)

$$p = \frac{\sum_s \frac{-2\epsilon_s}{3V} e^{-\epsilon_s/\tau}}{Z} = \boxed{\frac{2}{3}\left(\frac{1}{V}\right) U}$$

Solution 8. Time for a large fluctuation.

(a) Recall $\sigma = N[\ln\left(\frac{n_Q}{n}\right) + \frac{5}{2}]$. $n_Q = \left(\frac{M\tau}{2\pi\hbar^2}\right)^{3/2}$.

$\sigma_f = N[\ln\left(\frac{n_{QA}2V}{N}\right) + \frac{5}{2}]$

$\sigma_i = N[\ln\left(\frac{n_{QA}V}{N}\right) + \frac{5}{2}]$

Now

$$g \sim e^{\sigma} \sim \exp\left(\ln\left(\frac{n_Q}{n}\right)^N + \frac{5N}{2}\right) = e^{\frac{5N}{2}} \left(\frac{N_Q}{n}\right)^N$$

$$n_Q = 4 \times \frac{(938\, MeV/c^2)((0.8617 \times 10^{-4}\, eV)/K)(300\, K)}{2\pi(6.582122 \times 10^{-22}\, MeV \cdot s)^2} \left(\frac{1\, MeV}{10^6\, eV}\right)\left(\frac{1\, c}{3 \times 10^8\, m/s}\right)^2 = 7.88 \times 10^{30}/m^3$$

Now $PV = N\tau$. Then

$$\frac{P}{\tau} = \frac{N}{V} = \frac{(1\, atm)(\frac{1.013\times10^5\, N/m^2}{1\, atm})\left(\frac{kg\cdot m/s^2}{1\, N}\right)}{1.381 \times 10^{-23}\, J/K(300\, K)} = 2.445 \times 10^{25}/m^3$$

Now $1\, L = 10^{-3}\, m^3$ and so for $0.1\, L$,

$$(2.445 \times 10^{25}\, \frac{1}{m^3})(10^{-4}\, m^3) = 2.445 \times 10^{21}$$

With

$$\left(\frac{7.88 \times 10^{30}/m^3}{2.445 \times 10^{25}\, /m^3}\right) = 3.22 \times 10^5$$

$$\implies g \sim e^{\frac{5}{2}(2.445\times10^{21})}(3.22 \times 10^5)^{2.445\times10^{21}}$$

(b)

(c)

Solution 9. Gas of atoms with internal degree of freedom.

For an ideal monatomic gas, assume noninteracting.

(a) $\lambda_{int} = \lambda$. $\lambda = \exp(\mu/\tau)$ ideal gas. $\mu = \tau \ln(n/n_Q)$; $\quad n_Q = \left(\frac{M\tau}{2\pi\hbar^2}\right)^{3/2}$.

$$\boxed{\lambda_{ext} = \exp(-\Delta/\tau) \text{ or } 1}$$

Z_1 is the usual canonical partition function, $Z_1 = n_Q V$ where $n_Q = \left(\frac{M\tau}{2\pi\hbar^2}\right)^{3/2}$

$$\mathcal{Z}_1 = \sum_s (\lambda \exp(-\epsilon_s/\tau)(1) + \lambda \exp\left(\frac{-\epsilon_s}{\tau}\right) \exp\left(\frac{-\Delta}{\tau}\right)) Z_1 = \lambda(1 + e^{-\Delta/\tau}) Z_1$$

$$(\mathcal{Z}_1)^N = (\lambda(1 + e^{-\Delta/\tau}) Z_1)^N$$

(b)

(c)

Solution 10. Isentropic relations of ideal gas.

(a) Isentropic process, so $pV^\gamma = p_i V_i^\gamma$.

$$V^\gamma dp + \gamma p V^{\gamma-1} dV = 0 \Longrightarrow \boxed{\frac{dp}{p} + \frac{\gamma}{V} dV = 0}$$

Dealing with an ideal gas, $pV = N\tau$ still applies.

$$\tau V^{\gamma-1} = \tau_i V_i^{\gamma-1}$$
$$d\tau V^{\gamma-1} + \tau(\gamma-1) V^{\gamma-2} dV = 0$$
$$\Longrightarrow \boxed{\frac{d\tau}{\tau} + \frac{(\gamma-1)}{V} dV = 0}$$

Using $pV = N\tau$ again, note that $p^{1-\gamma}\tau^\gamma =$ constant .

$$(1-\gamma) p^{-\gamma} dp \tau^\gamma + p^{1-\gamma} \gamma \tau^{\gamma-1} d\tau = 0 \Longrightarrow \boxed{\frac{dp}{p} + \frac{\gamma}{1-\gamma} \frac{d\tau}{\tau} = 0}$$

(b) Using $p = \frac{p_i V_i^\gamma}{V^\gamma}$,

$$\frac{\partial p}{\partial V} = \frac{-\gamma p_i V_i^\gamma}{V^{\gamma+1}}$$
$$\Longrightarrow -V \frac{\partial p}{\partial V} = \frac{\gamma p_i V_i^\gamma}{V^\gamma} = \gamma p$$

So that $B_\sigma = -V(\partial p/\partial V)_\sigma = \gamma p$, the isentropic bulk moduli.

$$B_\tau = -V \left(\frac{\partial P}{\partial V}\right)_\tau = \frac{n\tau}{V} = p$$

since

$$pV = n\tau$$
$$p = \frac{n\tau}{V}$$
$$\frac{\partial p}{\partial V} = \frac{-n\tau}{V^2}$$

2015 manuscript.

132 Ideal gas (summary)

The gist of concepts involving the ideal gas, from Chapters 3,5,6 of Kittel and Kroemer [2] are the following. Consider the partition function for 1 atom in a box, Z_1

$$Z_1 = \frac{n_Q}{n} = n_Q V \text{ where } n_Q := \left(\frac{M\tau}{2\pi\hbar^2}\right)^{d/2}$$

where d is the dimension of space, usually $d = 3$.

From Kittel and Kroemer (1980) [2], Example : N atoms in a box, Chapter 3: "Boltzmann Distribution and Helmholtz Free Energy",

state of energy $\epsilon_\alpha(1) + \epsilon_\beta(2) + \cdots + \epsilon_\xi(N)$, $\alpha, \beta, \ldots \xi$ denote orbital indces of atoms in successive boxes. each entry occurs $N!$ times in Z_1^N (EY: 20151022) $N!$ ways to fill $\alpha, \beta \ldots \xi$ orbitals with N distinguishable particles. Thus,

$$Z_N = \frac{1}{N!} Z_1^N = \frac{1}{N!}(n_Q V)^N$$

Then for N *indistinguishable particles*, the partition function of the system Z_N is

$$Z_N = \frac{1}{N!} Z_1^N = \frac{1}{N!}(n_Q V)^N \tag{636}$$

Then the Helmholtz free energy F is

$$F = -\tau \ln Z_N = -\tau \left[N \ln (n_Q V) - (N \ln N - N)\right] = \tau N \left[\ln\left(\frac{n}{n_Q}\right) - 1\right] \tag{637}$$

and so

$$\frac{\partial F}{\partial \tau} = -N \ln\left(\frac{n_Q V}{N}\right) - N - \tau N \frac{d}{2}\frac{1}{\tau} = -N\left[\ln\left(\frac{n_Q}{n}\right) + \left(\frac{d}{2} + 1\right)\right]$$

$$\sigma = \frac{-\partial F}{\partial \tau} = N\left[\ln\left(\frac{n_Q}{n}\right) + \left(\frac{d}{2} + 1\right)\right] \tag{638}$$

Also, from using

$$F := U - \tau\sigma$$

$$dF = dU - \tau d\sigma - \sigma d\tau = -pdV - \sigma d\tau$$

then

$$p = \frac{\tau N}{V} \text{ or } pV = \tau N$$

Keep in mind that

$$\mu(\tau, V, N) \equiv \left(\frac{\partial F}{\partial N}\right)_{\tau, V}$$

I shall attempt to generalize the Sackur-Tetrode equation to consider vibrational and rotational energy modes, and so we should review its derivation.

Consider the example of a molecule with translational *and* rotational degrees of freedom. Note that $\mathbf{n} = (n_x, n_y, n_z)$ are good quantum numbers discretizing the translational degrees of freedom, and $j \in \mathbb{N} = \{0, 1, \ldots\}$ is a good quantum number discretizing the rotational degrees of freedom. Then the energy (eigenvalue) for quantum state (described by particular values of) $\mathbf{n}, j$ is

$$\epsilon_{\mathbf{n},j} = \frac{\hbar^2}{2M}\left(\frac{\pi}{L}\right)^2 \mathbf{n}^2 + j(j+1)\epsilon_0 \equiv \frac{\hbar^2}{2M}\left(\frac{\pi}{L}\right)^2 (n_x^2 + n_y^2 + n_z^2) + j(j+1)\epsilon_0$$

keeping in mind the multiplicity of the energy state from the multiplicity of the rotational energy, $g(j) = 2j + 1$.

Then

$$\exp\left(-\frac{1}{\tau}\epsilon_{\mathbf{n},j}\right) = \exp\left(-\frac{\hbar^2}{2M\tau}\left(\frac{\pi}{L}\right)^2 \mathbf{n}^2\right) e^{j(j+1)\epsilon_0}$$

Then the partition function for *1* molecule in a box is

$$Z_1 = \sum_{\mathbf{n},j} g(j) \exp\left(\frac{-\hbar^2}{2\pi\tau}\left(\frac{\pi}{L}\right)^2 \mathbf{n}^2\right) e^{j(j+1)\epsilon_0} = \sum_{\mathbf{n}} \exp\left(\frac{-\hbar^2}{2\pi\tau}\left(\frac{\pi}{L}\right)^2 \mathbf{n}^2\right) \sum_j g(j) e^{j(j+1)\epsilon_0} = n_Q V \cdot \sum_j g(j) e^{j(j+1)\epsilon_0} =$$
$$= n_Q V \frac{\tau}{\epsilon_0}$$

with $n_Q := \left(\frac{M\tau}{2\pi\hbar^2}\right)^2$.

Correction (rotational factor). The Boltzmann factor of $j(j+1)\epsilon_0$ carries a minus sign and a factor $1/\tau$. In three dimensions the power on $M\tau/(2\pi\hbar^2)$ is 3/2. The integral that produces τ/ϵ_0 is §152.7.

From the indistinguishability of quantum particles, for N of these particles,

$$Z_N = \frac{(n_Q V \frac{\tau}{\epsilon_0})^N}{N!}$$

and noting that

$$\ln Z_N = N\left[\ln(n_Q V) + \ln\left(\frac{\tau}{\epsilon_0}\right)\right] - \ln N! \simeq N \ln\left(\frac{n_Q}{n}\right) + N \ln\left(\frac{\tau}{\epsilon_0}\right) + N$$

then

$$F := -\tau \ln Z_N = -\tau N\left[\ln\left(\frac{n_Q}{n}\right) + 1\right] + -\tau N \ln\left(\frac{\tau}{\epsilon_0}\right) \equiv -\tau N\left[\ln\left(\frac{n_Q}{n}\right) + 1\right] + F(\text{int})$$
$$\sigma := -\left(\frac{\partial F}{\partial \tau}\right) = N\left[\ln\left(\frac{n_Q}{n}\right) + \left(\frac{d}{2} + 1\right)\right] + N\left(\ln\left(\frac{\tau}{\epsilon_0}\right) + 1\right)$$

Examining the steps above, it's clear how to generalize to when the form of the energy eigenvalues in the vibrational and rotational degrees of freedom (internal degrees of freedom), described by quantum numbers $\mathbf{k}$, is unknown:

$$\epsilon_{\mathbf{n},\mathbf{k}} = \frac{\hbar^2}{2M}\left(\frac{\pi}{L}\right)^2 \mathbf{n}^2 + \epsilon'_{\text{int}}(\mathbf{k})$$
$$g(\mathbf{k}) \in \mathbb{Z}^+ \text{ is the multiplicity of the internal degrees of freedom}$$
$$Z_1 = n_Q V \cdot Z_{1,\text{int}} \text{ with } Z_{1,\text{int}} = \sum_{\mathbf{k}} g(k) e^{-\frac{1}{\tau}\epsilon'_{\text{int}}(\mathbf{k})}$$
$$Z_N = \frac{(n_Q V)^N}{N!}(Z_{1,\text{int}})^N \Longrightarrow \ln Z_N = N\left(\ln\left(\frac{n_Q}{n}\right) + 1\right) + N \ln Z_{1,\text{int}}$$
$$F = -\tau N\left[\ln\left(\frac{n_Q}{n}\right) + 1\right] + F(\text{int})(\tau)$$
$$\sigma = N\left[\ln\left(\frac{n_Q}{n}\right) + \left(\frac{d}{2} + 1\right)\right] + \sigma_{\text{int}}$$

with $F(\text{int})(\tau)$ emphasizing that we could surmise or conjecture that the Helmholtz free energy contribution due to internal degrees of freedom can be a function of temperature τ (cf. Eq. (42.4) of Landau and Lifshitz (1980) [7]).

133 Binary mixtures

The 2008 chapter heading was empty. The entropy-of-mixing solution under the ideal-gas heading is also empty. The 2015 manuscript derives the distinct-gas increase and the identical-gas cancellation, and its general two-volume formula repeats V_A in the second logarithm. The derivation here is the one to use. It uses the Sackur–Tetrode entropy already obtained in that manuscript,

$$\sigma = N\left[\ln\left(\frac{n_Q}{n}\right) + 1 + \frac{d}{2}\right], \qquad n = \frac{N}{V}, \qquad n_Q = \left(\frac{M\tau}{2\pi\hbar^2}\right)^{d/2}, \tag{639}$$

for a classical monatomic ideal gas of one species in d dimensions. The usual case is $d = 3$, where $1 + d/2 = 5/2$. The hypotheses are: the gas is classical, $n \ll n_Q$; the particles of one species are indistinguishable, which has already put the $1/N!$ into (639); different species are distinguishable from each other; and there is no interaction energy, so entropies of different species in the same volume add.

133.1 Two different species

Put species A in volume V_A and species B in volume V_B, at the same τ, and then let each occupy the total volume

$$V = V_A + V_B \tag{640}$$

without changing N_A, N_B, or τ. A partition removed at fixed total volume is this process. Because n_Q depends on τ and on the mass, and τ is fixed,

$$\begin{aligned} \sigma_A &= N_A\left[\ln\!\left(\frac{n_{QA}V_A}{N_A}\right) + 1 + \frac{d}{2}\right], \\ \sigma'_A &= N_A\left[\ln\!\left(\frac{n_{QA}V}{N_A}\right) + 1 + \frac{d}{2}\right]. \end{aligned} \tag{641}$$

Subtract:

$$\sigma'_A - \sigma_A = N_A \ln\!\left(\frac{V}{V_A}\right). \tag{642}$$

The same subtraction for B, with V_B in the denominator, gives the mixing entropy

$$\Delta\sigma = N_A \ln\!\left(\frac{V}{V_A}\right) + N_B \ln\!\left(\frac{V}{V_B}\right). \tag{643}$$

The 2015 general formula writes V_A in both denominators. The second term of (643) is the B contribution (642) with the labels exchanged.

If $N_A = N_B = N$ and $V_A = V_B$, then $V/V_A = 2$ and

$$\Delta\sigma = 2N \ln 2. \tag{644}$$

That is the result in the 2015 manuscript for equal amounts of two different gases.

If the initial pressures and temperature agree, the ideal-gas law $pV_i = N_i\tau$ gives $V_i/V = N_i/N$ with $N = N_A + N_B$. Then

$$\Delta\sigma = -N_A \ln x_A - N_B \ln x_B, \qquad x_i = \frac{N_i}{N}. \tag{645}$$

Each $-x \ln x$ is nonnegative for $x \in (0, 1]$, so $\Delta\sigma \geq 0$.

133.2 The same species

Now let both sides be the same species, the same n_Q, with N particles in volume V on each side. The initial entropy is the sum of two separate applications of (639):

$$\sigma_{\mathrm{i}} = 2N\left[\ln\!\left(\frac{n_Q V}{N}\right) + 1 + \frac{d}{2}\right]. \tag{646}$$

After the partition is removed there is one system of $2N$ indistinguishable particles in volume $2V$. One application of (639) gives

$$\sigma_{\mathrm{f}} = 2N\left[\ln\!\left(\frac{n_Q\,(2V)}{2N}\right) + 1 + \frac{d}{2}\right] = \sigma_{\mathrm{i}}. \tag{647}$$

The ratio $(2V)/(2N)$ equals V/N, so

$$\Delta\sigma = 0. \tag{648}$$

Counting the final state as two distinguishable species, each N in volume $2V$, would instead add $2N \ln 2$. The single factorial $(2N)!$ removes that amount. With (601) truncated at $n \ln n - n$,

$$\begin{aligned} \ln((2N)!) - 2\ln(N!) &= \big(2N\ln(2N) - 2N\big) - 2\big(N\ln N - N\big) \\ &= 2N\ln 2 + 2N\ln N - 2N\ln N \\ &= 2N\ln 2. \end{aligned} \tag{649}$$

That is the Gibbs cancellation: the extra $2N \ln 2$ from doubling the volume available to each group is the same $2N \ln 2$ that appears when two groups of N identical particles are entered as one group of $2N$.

133.3 Chemical potential in an ideal mixture

For one species, the Helmholtz free energy consistent with (639) and with $F = U - \tau\sigma$, $U = (d/2)N\tau$, is the expression in the 2015 ideal-gas summary,

$$F = N\tau\left[\ln\left(\frac{n}{n_Q}\right) - 1\right]. \tag{650}$$

Differentiate at fixed τ and V. The N inside $n = N/V$ contributes one power of N:

$$\begin{aligned}\left(\frac{\partial F}{\partial N}\right)_{\tau,V} &= \tau\left[\ln\left(\frac{n}{n_Q}\right) - 1\right] + N\tau\cdot\frac{1}{N} \\ &= \tau\ln\left(\frac{n}{n_Q}\right).\end{aligned} \tag{651}$$

So $\mu = \tau\ln(n/n_Q)$.

In a mixture of ideal gases the species share V and τ, and the free energy is the sum of (650) over species because the interaction energy was assumed to vanish. For species j,

$$\mu_j = \tau\ln\left(\frac{n_j}{n_{Qj}}\right) = \tau\ln x_j + \tau\ln\left(\frac{n}{n_{Qj}}\right), \tag{652}$$

where $n_j = N_j/V$, $n = \sum_i n_i$, and $x_j = n_j/n$. The mixture pressure is $p = n\tau$. A pure gas of species j at the same τ and at that same total pressure p would have density n, hence chemical potential $\tau\ln(n/n_{Qj})$. Call that pure-gas value $\mu_j^0(\tau, p)$. Then

$$\mu_j(\tau, p, x_j) = \mu_j^0(\tau, p) + \tau\ln x_j. \tag{653}$$

The partial pressure follows from the same ideal-gas law applied to n_j:

$$p_j = n_j\tau = x_j\,p, \qquad \sum_j p_j = p. \tag{654}$$

Equations (643), (648), (653), and (654) are the content that the empty chapter and the empty mixing solution did not contain. *2015 manuscript.*

134 Mixtures of ideal gases

134.1 Gibbs paradox

Kittel and Kroemer (1980) [2] Problem 6 **Entropy of mixing** of Chapter 6: Ideal Gas,

Prof. Steven Anlage, Introduction to Thermodynamics and Statistical Mechanics, Spring 2011, University of Maryland Physics 404, Homework 8 Solutions, Question 9, [10]

Consider adding together the entropies of ideal gas A with initial volume V, and molecular mass M_A (so it has a n_{Q_A} quantum concentration), and of ideal gas B with initial volume V, molecular mass M_B, n_{Q_B} quantum concentration:

$$\sigma_{\text{initial}} = \sigma_A + \sigma_B = N_A\ln\left(\frac{n_{Q_A}V}{N_A}\right) + N_A\left(1 + \frac{d}{2}\right) + N_B\ln\left(\frac{n_{Q_B}V}{N_B}\right) + N_B\left(1 + \frac{d}{2}\right)$$

After diffusive equilibrium is reached means that each of the gases now occupy a volume $2V$, gas A is free to roam in volume $2V$, as is gas B. Thus, the entropy increases:

$$\sigma_{\text{final}} = N_A\ln\left(\frac{n_{Q_A}2V}{N_A}\right) + N_A\left(1 + \frac{d}{2}\right) + N_B\ln\left(\frac{n_{Q_B}2V}{N_B}\right) + N_B\left(1 + \frac{d}{2}\right)$$

and so

$$\Delta\sigma = (N_A + N_B)\ln 2 = 2N\ln 2 \text{ (if } N_A = N_B = N)$$

[10] http://www.physics.umd.edu/courses/Phys404/Anlage_Spring11/hw8solv5.pdf

In general,

$$\Delta\sigma = N_A \ln\left(\frac{V_A + V_B}{V_A}\right) + N_B \ln\left(\frac{V_A + V_B}{V_A}\right)$$

Correction (mixing volumes). The second logarithm is $\ln((V_A + V_B)/V_B)$. The derivation is §133, equation (643).

For identical atoms ($A \equiv B$),

$$\sigma_A = N \ln\left(\frac{n_Q V}{N}\right) + N\left(1 + \frac{d}{2}\right) = \sigma_B$$

and so

$$\sigma_{\text{initial}} = \sigma_A + \sigma_B = 2N \ln\left(\frac{n_Q V}{N}\right) + 2N\left(1 + \frac{d}{2}\right)$$

For the final entropy after mixing, treat the system as a (whole) system of $2N$ indistinguishable particles occupying a volume of $2V$:

$$\sigma_{\text{final}} = 2N \ln\left(\frac{n_Q (2V)}{2N}\right) + 2N\left(1 + \frac{d}{2}\right)$$

and so $\Delta\sigma = 0$

From Sec. 93. Mixtures of ideal gases of Landau and Lifshitz (1980) [7], additivity of thermodynamic quantities (such as energy and entropy) holds only if interaction between various parts of body is negligible.

But for mixture of several substances, e.g. mixture of liquids, thermodynamic quantities are not equal to sums of thermodynamic quantities for individual components of the mixture.

Mixtures of ideal gases are an exception, since interaction between molecules is by definition negligible.

But for partial pressure of ith gas, $p_i = \frac{N_i \tau}{V} = \frac{N_i p}{N}$, N = total number of molecules in mixture, N_i number of molecules of ith gas.

So

$$p_j = \frac{N_j \tau}{V} = \frac{N_j}{N} p = X_j p$$

where X_j is mole fraction or particle fraction.
2008 manuscript.

135 Fermi and Bose Gases

Problems

Problem 1. Density of orbitals in one and two dimensions.

(a) Show that the density of orbitals of a free electron in one dimension is

$$\mathcal{D}_1(\epsilon) = (L/\pi)(2m/\hbar^2\epsilon)^{1/2}, \tag{655}$$

where L is the length of the line.

(b) Show that in two dimensions, for a square of area A,

$$\mathcal{D}_2(\epsilon) = Am/\pi\hbar^2 \tag{656}$$

independent of ϵ.

Solution 1. Recall, for the free electron: $H = \frac{p^2}{2m} = \frac{-\hbar^2}{2m}\nabla^2$
$\Longrightarrow \epsilon_s = \frac{\hbar^2}{2m}\frac{n_x^2\pi^2}{L^2}$ for 1 dim., $\epsilon_s = \frac{\hbar^2}{2m}\frac{(n_x^2+n_y^2)\pi^2}{L^2}$ for 2-dim.

If ϵ_F = Fermi energy, energy of the highest filled orbital,
1-dim: $N = 2n_F$. 2-dim.: $N = (2)\left(\frac{1}{4}\right)(\pi n_F^2) = \frac{\pi n_F^2}{2}$
2 factor for 2 possible spin states.

1-dim: $\epsilon_F = \frac{\hbar^2}{2m}\left(\frac{N}{2}\right)^2 \frac{\pi^2}{L^2} = \frac{1}{2m}\left(\frac{\hbar\pi N}{2}\right)^2 \left(\frac{1}{V^2}\right) = \frac{1}{2m}\left(\frac{\hbar m}{2}\right)^2 n^2 \qquad N = \sqrt{2m\epsilon_F \left(\frac{2V}{\hbar\pi}\right)^2}$

2-dim: $\epsilon_F = \frac{\hbar^2}{2m}\left(\frac{2N}{\pi}\right)\left(\frac{\pi}{L}\right)^2 = \frac{\hbar^2}{m}\frac{N}{\pi}\frac{\pi^2}{V} = \frac{\hbar^2}{m}\frac{N\pi}{V} = \frac{\hbar^2\pi}{m}n \qquad N = \frac{mV\epsilon_F}{\pi\hbar^2}$

1-dim.: $\mathcal{D}(\epsilon) = \frac{dN}{d\epsilon} = \sqrt{2m\left(\frac{2V}{\hbar\pi}\right)^2}\frac{1}{2}(\epsilon_F)^{-1/2} = \frac{N}{2\epsilon} = \left(\frac{L}{\pi}\right)\left(\frac{2m}{\hbar}^2\epsilon\right)^{1/2}$

2-dim.: $N\epsilon = Am/\pi\hbar^2$

Problem 2. Energy of relativistic Fermi gas. For electrons with an energy $\epsilon \gg mc^2$, where m is the rest mass of the electron, the energy is given by $\epsilon \simeq pc$, where p is the momentum. For electrons in a cube of volume $V = L^3$ the momentum is of the form $(\pi\hbar/L)$, multiplied by $(n_x^2 + n_y^2 + n_z^2)^{1/2}$, exactly as for the nonrelativistic limit.

(a) Show that in this extreme relativistic limit the Fermi energy of a gas of N electrons is given by

$$\epsilon_F = \hbar\pi c(3n/\pi)^{1/3}, \tag{657}$$

where $n = N/V$.

(b) Show that the total energy of the ground state of the gas is

$$U_0 = \frac{3}{4}N\epsilon_F. \tag{658}$$

The general problem is treated by F. Jüttner, Zeitschrift für Physik **47**, 542 (1928).

Solution 2.

(a) $\epsilon \simeq pc = \frac{\hbar n\pi}{L}c, \qquad \epsilon_F = \frac{\hbar n_F\pi}{L}c.$

Recall, for 3-dim.: $N = (2)\left(\frac{1}{8}\right)\left(\frac{4}{3}\pi n_F^3\right) = \frac{\pi}{3}n_F^3. \qquad n_F = \left(\frac{3N}{\pi}\right)^{1/3}.$

$\epsilon_F = \frac{\pi\hbar}{L}\left(\frac{3N}{\pi}\right)^{1/3} = \hbar\pi c\left(\frac{3n}{\pi}\right)^{1/3}$

(b)

$$U_0 = 2\sum_{n\le n_F}\epsilon_n = 2 * \frac{1}{8} * 4\pi\int_0^{n_F} dn n^2 \frac{\hbar n\pi c}{L} = \frac{\hbar\pi^2 c}{L}\int_0^{n_F} dn n^3 = \frac{\hbar\pi^2 c}{4L}n_F^4 =$$

$$= \frac{\hbar\pi^2 c}{4L}\left(\frac{\epsilon_F L}{\hbar\pi c}\right)^4 = \frac{\hbar\pi^2 c}{4L}\left(\frac{3N}{\pi}\right)\left(\frac{\epsilon_F L}{\hbar\pi c}\right) = \boxed{\frac{3}{4}N\epsilon_F}$$

Problem 3. Pressure and entropy of degenerate Fermi gas.

(a) Show that a Fermi electron gas in the ground state exerts a pressure

$$p = \frac{(3\pi^2)^{2/3}}{5}\frac{\hbar^2}{m}\left(\frac{N}{V}\right)^{5/3} \tag{659}$$

In a uniform decrease of the volume of a cube every orbital has its energy raised: The energy of an orbital is proportional to $1/L^2$ or to $1/V^{2/3}$.

(b) Find an expression for the entropy of a Fermi electron gas in the region $\tau \ll \epsilon_F$. Notice that $\sigma \to 0$ as $\tau \to 0$.

Solution 3.

(a) Recall $U_0 = \frac{3}{5}N\frac{\hbar^2}{2m}(3\pi^2 N)^{2/3}V^{-2/3}$

$$\frac{\partial U}{\partial V} = \frac{3}{5}N\frac{\hbar^2}{2m}(3\pi^2 N)^{2/3}\left(\frac{-2}{3}\right)V^{-5/3} = \frac{-1}{5}\frac{\hbar^2}{m}(3\pi^2)^{2/3}\left(\frac{N}{V}\right)^{5/3}$$

So then

$$\boxed{p = \frac{-\partial U_0}{\partial V} = \frac{1}{5}\frac{\hbar^2}{m}(3\pi^2)^{2/3}\left(\frac{N}{V}\right)^{5/3}}$$

(b) Recall that $\epsilon_F \equiv \tau_F = \frac{\hbar^2}{2m}\left(\frac{3\pi^2 N}{V}\right)^{2/3}$ and that the heat capacity of an electron gas is $C_{el} = \frac{1}{2}\pi^2 N\frac{\tau}{\tau_F} = \frac{\partial U}{\partial \tau}$, which *helps directly* with finding the entropy.

$$\sigma(\tau) - \sigma(\tau_0) = \int_{\tau_0}^{\tau} \frac{1}{\tau} dU = \int_{\tau_0}^{\tau} \frac{1}{\tau}\frac{1}{2}\pi^2 N\frac{\tau}{\tau_F} d\tau = \frac{\pi^2 N}{2\tau_F}\tau$$

Let $\sigma(\tau_0 = 0) = 0$,

$$\boxed{\sigma(\tau) = \frac{\pi^2 N}{2\tau_F}\tau}$$

Problem 4. Chemical potential versus temperature. Explain graphically why the initial curvature of μ versus τ is upward for a fermion gas in one dimension and downward in three dimensions (Figure 7.7). *Hint:* The $\mathcal{D}_1(\epsilon)$ and $\mathcal{D}_3(\epsilon)$ curves are different, where $\mathcal{D}_1$ is given in Problem 1. It will be found useful to set up the integral for N, the number of particles, and to consider from the graphs the behavior of the integrand between zero temperature and a finite temperature.

Solution 4. Recall, $N = \int_0^{\epsilon_F} d\epsilon \mathcal{D}(\epsilon)$.

1-dim:

$$N = \int_0^{\epsilon_F} d\epsilon \mathcal{D}_1(\epsilon) = \int_0^{\epsilon_F} d\epsilon \left(\frac{L}{\pi}\right)\left(\frac{2m}{\hbar^2\epsilon}\right)^{1/2} = \left(\frac{L}{\pi}\right)\frac{(2m)^{1/2}}{\hbar}\int_0^{\epsilon_F} d\epsilon \epsilon^{-1/2} =$$
$$= \left(\frac{L}{\pi}\right)\frac{(2m)^{1/2}}{\hbar}(2\epsilon^{1/2})\Big|_0^{\epsilon_F} = \left(\frac{L}{\pi}\right)\frac{(2m)^{1/2}}{\hbar}2\epsilon_F^{1/2}$$

3-dim:

$$N = \int_0^{\epsilon_F} d\epsilon \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}\epsilon^{1/2} = \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}\left(\frac{2}{3}\epsilon^{3/2}\right)\Big|_0^{\epsilon_F} = \frac{V}{3\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}\epsilon_F^{3/2}$$

Note the difference in the concavity of the $N(\epsilon)$ curves.

Solution 5.

(a) For ^3He, given $I = 1/2$, density of liquid $0.081\,g\,cm^{-3}$, we want to find v_F, ϵ_F, τ_F.

$$\epsilon_F = \left(\frac{\hbar^2}{2m}\right)(3\pi^2 n)^{2/3} =$$
$$= \frac{(6.582\times 10^{-22}\,MeV\cdot s)^2}{2\cdot 3\cdot 938\,MeV/c^2\left(\frac{1\,c}{3\times 10^{10}\,cm/s}\right)^2}\left(\left(3\pi^2\frac{0.081\,g}{cm^3}\right)\left(\frac{1\,kg}{10^3\,g}\right)\left(\frac{1\,u}{1.67\times 10^{-27}\,kg}\right)\left(\frac{1\,^3He}{3\,u}\right)\right)^{2/3} =$$
$$= 4.24\times 10^{-10}\,MeV = 4.24\times 10^{-4}\,eV$$

Now suppose we have a nonrelativistic gas. Then $\frac{1}{2}mv_F^2$ or $v_F^2 = \frac{2\epsilon_F}{m}$.

$$v_F = 1.675\times 10^4\,\frac{cm}{sec}$$

$$T_F = \frac{4.24\times 10^{-4} eV}{0.8619\times 10^{-4}\,eV/K} = 4.92\,K$$

(b)

$$C_{el} = \frac{\pi^2}{3}\mathcal{D}(\epsilon_F)\tau = \frac{\pi^2}{3}\frac{3N}{2\tau_F}\tau = \frac{\pi^2}{2}N\frac{\tau}{\tau_F} = 1.003 k_B T N$$

Solution 6. Mass-radius relationship for white dwarfs.

(a)

$$U = \int \rho(r)\phi(r) r^2 4\pi dr = -4\pi\int \rho\frac{G\frac{4}{3}\pi r^3\rho}{r} r^2 dr = 4\pi\frac{M}{\frac{4}{3}\pi R^3} G\frac{4}{3}\pi\frac{1}{5}R^5\frac{M}{\frac{4}{3}\pi R^3} = \boxed{\frac{-3GM^2}{5R}}$$

(b)

$$\epsilon_F = \left(\frac{\hbar^2}{2m}\right)(3\pi^2 n)^{2/3}$$

is the Fermi energy.

With $V = \frac{4}{3}\pi R^3$,

$$T_{tot} = N\frac{1}{2}mv^2 = N\epsilon_F = N\frac{h^2}{2m}\left(\frac{3\pi^2 N}{V}\right)^{2/3} = \frac{(3\pi^2)^{2/3}}{2}\left(\frac{\hbar^2}{m}\frac{N^{5/3}}{V^{2/3}}\right) = \frac{(3\pi^2)^{2/3}}{2}\frac{\hbar^2}{m}\frac{N^{5/3}}{\left(\frac{4}{3}\pi\right)^{2/3}R^2} =$$

$$= \frac{\left(\frac{9\pi}{4}\right)^{2/3}}{2}\frac{\hbar^2 N^{5/3}}{mR^2} \simeq \hbar^2\frac{(M/M_H)^{5/3}}{mR^2}$$

since $N = \frac{M}{M_H}$ since $M_H \ll m$.

(c)

$$\frac{\hbar^2 M^{5/3}}{mM_H^{5/3}R^2} = \frac{GM^2}{R} \quad \Longrightarrow M^{1/3}R \simeq \frac{\hbar^2/G}{mM_H^{5/3}}$$

$$\frac{(6.582\times10^{-22}\,MeV\cdot s)(1.05457\times10^{-34}\,J\cdot s)}{0.511\,MeV/c^2((1.67\times10^{-27}\,kg)\left(\frac{10^3\,g}{1\,kg}\right))^{5/3}}\left(\frac{10^3\,g}{1\,kg}\right)^2/(6.67\times10^{-11}\,\frac{m^3/s^2}{kg})\left(\frac{(3\times10^8\,m/s)^2}{1\,c^2}\right)$$

Dealing with units and dimensions,

$$J = kg\cdot\frac{m^2}{s^2}; \qquad \frac{N\cdot m^2}{kg^2} = kg\cdot\frac{m^3/s^2}{kg^2} = \frac{m^3/s^2}{kg}$$

$$\frac{(6.582\times10^{-22}\,MeV\cdot s)(1.05457\times10^{-34}\,kg\cdot\frac{m^2}{s})\left(\frac{10^2\,cm}{1\,m}\right)^2\left(\frac{10^3\,g}{1\,kg}\right)^2}{(0.511\,MeV/c^2)(1.67\times10^{-24}\,g)^{5/3}(6.67\times10^{-11}\,\frac{m^3/s^2}{kg}\times\left(\frac{10^2\,cm}{1\,m}\right)^3)}\left(\frac{(3\times10^{10}\,cm/s)^2}{1\,c^2}\right)\;10^{20}\,g^{1/3}\,cm$$

(d)

$$\rho = \frac{M}{\frac{4}{3}\pi R^3} = \frac{M}{\frac{4}{3}\pi\left(\frac{10^{20}\,g^{1/3}\,cm}{M^{1/3}}\right)^3} = \frac{M^2}{\frac{4}{3}\pi 10^{60}\,g\,cm^3} = \frac{(2\times10^{33}\,g)^2}{\frac{4}{3}\pi10^{60}\,g\,cm^3} = \frac{3\times10^{66}\,g^2}{\pi10^{60}\,g\,cm^3}$$

$$10^6\,\frac{g}{cm^3}$$

(e)

$$M^{1/3}R \simeq \frac{\hbar^2/G}{mM_H^{5/3}}\;10^{17}\,g^{1/3}\,cm \quad \Longrightarrow R = \frac{10^{17}g^{1/3}\,cm}{(2\times10^{33}\,g)^{1/3}} = 7.937\,km$$

Solution 7. Photon condensation. $N_e = 2.404V\tau^3/\pi^2\hbar^3c^3$.

The condition is that $N = N_e$:

$$N = N_e = \frac{2.404V\tau^3}{\pi^2\hbar^3c^3}$$

$$\boxed{\tau = \left(\frac{\pi^2\hbar^3c^3}{2.404}\frac{N}{V}\right)^{1/3}}$$

With a concentration of $10^{20}\,cm^{-3}$, $T = 1.7\times10^6\,K$ (the critical temperature in K below which $N_e < N$.

Solution 8. Energy, heat capacity, and entropy of degenerate boson gas.

Recall that the distribution function for bosons is

$$f(\epsilon,\tau) = \frac{1}{\exp\left[(\epsilon-\mu)/\tau\right]-1}$$

Consider N noninteracting bosons of spin zero.

$\epsilon = 0$ for ground state. Thus, $f(0,\tau) = \frac{1}{\exp\left(\frac{-\mu}{\tau}\right)-1}$. Recall that

$$N_\epsilon(\tau) = \int_0^\infty d\epsilon \mathcal{D}(\epsilon) f(\epsilon,\tau) = \int_0^\infty d\epsilon \frac{V}{4\pi^2}\left(\frac{2M}{\hbar^2}\right)^{3/2}\epsilon^{1/2}\frac{1}{\lambda^{-1}\exp\left(\frac{\epsilon}{\tau}\right)-1}$$

Recall,

$$U = 0\left(\frac{1}{\exp\left(-\mu/\tau\right)-1}\right) + \int_0^\infty \frac{\epsilon}{e^{(\epsilon-\mu)/\tau}-1}\mathcal{D}(\epsilon)d\epsilon = \int_0^\infty \frac{\epsilon}{e^{(\epsilon-\mu)/\tau}-1}\frac{V}{4\pi^2}\left(\frac{2M}{\hbar^2}\right)^{3/2}\epsilon^{1/2}d\epsilon =$$
$$= \frac{V}{4\pi^2}\left(\frac{2M}{\hbar^2}\right)^{3/2}\int_0^\infty \frac{\epsilon^{3/2}}{e^{(\epsilon-\mu)/\tau}-1}$$

$\tau < \tau_\epsilon$, so $\lambda \approx 1$, for N_0 to be sufficiently large.

$$\implies U = \frac{V}{4\pi^2}\left(\frac{2M}{\hbar^2}\right)^{3/2}\int_0^\infty \frac{\epsilon^{3/2}}{e^{\epsilon/\tau}-1}$$

Using $\begin{matrix} x = \epsilon/\tau \\ dx = d\epsilon/\tau \end{matrix}$

$$\implies U = \frac{V}{4\pi^2}\left(\frac{2M}{\hbar^2}\right)^{3/2}\tau^{5/2}\int_0^\infty \frac{x^{3/2}dx}{e^x-1} = B_0\tau^{5/2}C_0$$

It's true that $\partial_\tau(e^{\epsilon/\tau}) = e^{\epsilon/\tau}\left(\frac{-\epsilon}{\tau^2}\right)$

$$C_V = \left(\frac{5}{2}\right)\frac{V}{4\pi^2}\left(\frac{2M}{\hbar^2}\right)^{3/2}\tau^{3/2}\int_0^\infty \frac{x^{3/2}dx}{e^x-1}$$

Now $\frac{1}{\tau} = \left(\frac{\partial\sigma}{\partial U}\right)_V$.

$$\sigma(U) - \sigma(U_0) = \int \frac{1}{\tau}dU = \int \frac{VB_0\frac{5}{2}\tau^{3/2}C_0 d\tau}{\tau} = VB_0\frac{5}{2}C_0\int\tau^{1/2}d\tau =$$
$$\xrightarrow{dU = VB_0\frac{5}{2}\tau^{3/2}C_0d\tau} = VB_0\frac{5}{2}C_0\frac{2}{3}\tau^{3/2}\Big|_0^\tau$$

$$\sigma(U) = VB_0\frac{5}{3}C_0\tau^{3/2} = VB_0\frac{5}{3}C_0\frac{U^{3/5}}{(VB_0C_0)^{3/5}} = (VB_0C_0)^{2/5}\frac{5}{3}U^{3/5} =$$
$$= \left(\frac{V}{4\pi}\left(\frac{2M}{\hbar^2}\right)^{3/2}\int_0^\infty\frac{x^{3/2}dx}{e^x-1}\right)^{2/5}\frac{5}{3}U^{3/5}$$

where we had used $U = VB_0\tau^{5/2}C_0$ or $\left(\frac{U}{VB_0C_0}\right)^{3/5} = \tau^{3/2}$.

Solution 9. Boson gas in one dimension.

In one-dim.,

$$\epsilon_s = \frac{\hbar^2}{2m}\frac{\pi^2n^2}{L^2} = \frac{\hbar^2}{2m}\frac{\pi^2n^2}{V^2} \text{ or } \frac{2mV^2}{\hbar^2\pi^2}\epsilon_s = n^2$$
$$\implies n = \left(\frac{2mV^2}{\hbar^2\pi^2}\right)^{1/2}\epsilon^{1/2} \implies \frac{dn}{d\epsilon} = \left(\frac{2mV^2}{\hbar^2\pi^2}\right)^{1/2}\frac{1}{2\epsilon^{1/2}}$$

Note the difference with 3-dim.: For spinless bosons,

$$\mathcal{D}(n)dn = \frac{4\pi n^2dn}{8} = \frac{\pi}{2}\left(\frac{2mL^2}{\hbar^2\pi^2}\epsilon_s\right)\left(\frac{2mL^2}{\hbar^2\pi^2}\right)^{1/2}\frac{1}{2\epsilon^{1/2}}d\epsilon =$$
$$= \frac{\pi}{4}\left(\frac{2mL^2}{\hbar^2\pi^2}\right)^{3/2}\epsilon^{1/2}d\epsilon = \mathcal{D}(\epsilon)d\epsilon$$

$$N_\epsilon(\tau) = \int_0^\infty d\epsilon \mathcal{D}(\epsilon) f(\epsilon,\tau) = \int_0^\infty d\epsilon \left(\frac{2mV^2}{\hbar^2\pi^2}\right)^{1/2} \frac{1}{2\epsilon^{1/2}} \frac{1}{\lambda^{-1}} \exp\left(\frac{\epsilon}{\tau}\right) - 1 =$$
$$= \frac{1}{2}\left(\frac{2mV^2}{\hbar^2\pi^2}\right)^{1/2} \int_0^\infty \frac{d\epsilon}{\epsilon^{1/2}} (\lambda^{-1} \exp\left(\frac{\epsilon}{\tau}\right) - 1)$$

For $\epsilon \to 0$, $N_\epsilon(\tau) \to \infty$ which is not characteristic of a Boson.

Solution 10. Relativistic white dwarf stars. $\epsilon \simeq pc \qquad \lambda = 2\pi\hbar/p$.

Virial theorem:

$$2\langle T\rangle = -\langle U\rangle \qquad 2\langle T\rangle = k\langle U\rangle$$

$$2\left(\frac{3}{4}N\epsilon_F\right) = \frac{3}{2}N^{4/3}\hbar\pi c\left(\frac{3}{\pi}\right)^{1/3}\frac{1}{L} = -\left(-\frac{3GM^2}{5R}\right) = \frac{3GM^2}{5R}$$

$$\epsilon_F = \hbar\pi c\left(\frac{3n}{\pi}\right)^{1/3}$$

Approximating the sphere as a box,

$$L^3 = \frac{4}{3}\pi R^3$$
$$L = \left(\frac{4}{3}\right)^{1/3}\pi^{1/3}R \qquad \left(\frac{3}{4\pi}\right)^{1/3} L = R$$

$$N^{4/3} = \left(\frac{GM^2}{5R}\right)\left(\frac{2L}{\hbar\pi^{2/3}c3^{1/3}}\right) \Longrightarrow N = \left(\left(\frac{GM^2}{5R}\right)\left(\frac{2L}{\hbar\pi^{2/3}c3^{1/3}}\right)\right)^{3/4} = \left(\left(\frac{2GM^2}{5}\right)\frac{4^{1/3}}{\hbar 3^{2/3}\pi^{1/3}c}\right)^{3/4}$$

Now $M = Nm_H$, where m_H is the mass of hydrogen, so

$$1 = \left(\left(\frac{2Gm_H^2}{5}\right)\frac{4^{1/3}}{\hbar 3^{2/3}\pi^{1/3}c}\right)^{3/4} N^{1/2}$$

$$\Longrightarrow N = \left(\frac{5\hbar(3^{2/3})\pi^{1/3}c}{4^{1/3}(2Gm_H^2)}\right)^{3/2} = \left(\frac{(1.05457\times10^{-34}\,J\cdot s)(3\times10^8\,m/s)}{(6.67\times10^{-11}\,\frac{m^3}{kg\cdot s^2})(1.67\times10^{-27}\,kg)^2}\right)^{3/2}\cdot\left(\frac{5(3^{2/3})\pi^{1/3}}{4^{1/3}2}\right)^{3/2} = 2.2\times10^{58}$$

2008 manuscript.

136 Heat and Work

Energy and Entropy transfer: Definition of Heat and Work

Consider Σ, a manifold consisting of points representing thermodynamic states of a single system. For instance, for global coordinates (U, V),
$(U, V) \in \Sigma$.

Consider $W, Q \in \Omega^1(\Sigma)$, 1-forms on Σ.

Now define Q as

$$Q \equiv \tau d\sigma$$

with $\sigma = \sigma(U, V) \in C^\infty(\Sigma)$

$\tau = \tau(U, V) \in C^\infty(\Sigma)$

Recall energy conservation in this form:

$$dU = W + Q$$

Consider pure heat and, so, no work. Now $Q = \tau d\sigma$

Heat Engines: Conversion of Heat into Work

Consider curve $c : \mathbb{R} \to \Sigma$ s.t. c generates vector field $\dot{c} = \frac{\partial}{\partial \sigma}$ (U is suited for this).

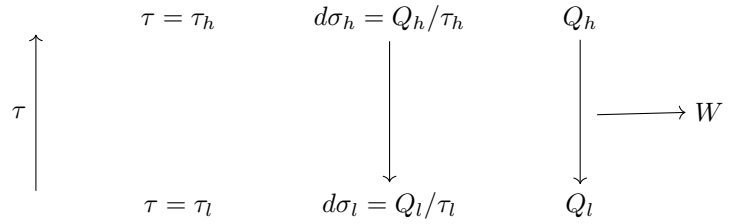
$$c(t) \in \Sigma$$

Act on this vector field $\frac{\partial}{\partial \sigma} \in \mathfrak{X}(\Sigma)$ with Q, i.e.

$$Q\left(\frac{\partial}{\partial \sigma}\right) = \tau$$

This is what's meant when it's said "reversible heat transfer accompanying 1 unit of entropy is given by temperature τ" [2].

Consider Figure 8.1 on page 229 of Kittel and Kroemer [2]. Roughly it looks like this:



But what's really going on?

Consider $Q_h = dU$, the initial heat input at high temperature (I'll show that later) τ_h.
Consider a curve $c_0 \in \Sigma$ s.t. $\dot{c}_0 = \dot{\sigma}_0 \frac{\partial}{\partial \sigma}$. Then

$$Q_h(\dot{c}_0) = dU(\dot{c}_0) = \left(\frac{\partial U}{\partial \sigma}\right)_V = \tau \dot{\sigma}_0 \equiv \tau_h \dot{\sigma}_0$$

We can integrate the 1-form $dU \in \Omega^1(\Sigma)$ for 2 reasons: mathematically, it is an exact form. Physically, we are considering a reversible process, passing through thermodynamic states of the system, starting with the system being in energy U_0 and ending up with energy U_1, and all the energy states ($\in \mathbb{R}$) in between.

$$\implies U_1 - U_0 = \int_0^1 \tau_h d\sigma$$

If this is conducted all at temperature τ_h during the whole process, then $U_1 - U_0 = \tau_h \int_0^1 d\sigma = \tau_h(\sigma_1 - \sigma_0)$. It's in this case that $d\sigma$ is an exact form and can be integrated over that curve c_0.

Legendre transforms revisited

Let's recall 2 of our favorite thermodynamic potentials, U, and Helmholtz free energy F. They are related by Legendre transformations that transform 1 coordinate into its conjugate coordinate, somewhat like how the Legendre transform transforms that Lagrangian in canonical coordinates into a Hamiltonian written with the conjugate momentum. However, I do want to point out that, for Lagrangians and Hamiltonians, the Legendre transformation is a fiber derivative between tangent bundle to the cotangent bundle on the manifold. In our current case, we want a mundane Legendre transformation between convex function to another convex function, a coordinate transformation by a C^∞ function, not a morphism between vector spaces.

Recall F. It's defined as such:
$F \equiv U - \tau\sigma$, so
$dF = dU - \tau d\sigma - \sigma d\tau = -\sigma d\tau - pdV$

Consider curve $c : \mathbb{R} \to \Sigma$

$$c(t) \in \Sigma$$

Consider 2 curves that generate vector fields:

$$\dot{c} = \dot{\tau}\frac{\partial}{\partial \tau} \text{ or } \dot{c} = \dot{V}\frac{\partial}{\partial V}$$

Now, in general, mathematically,

$$dF = +\left(\frac{\partial F}{\partial \tau}\right)_V d\tau + \left(\frac{\partial F}{\partial V}\right)_\tau dV$$

Thus,

$$\left(\frac{\partial F}{\partial \tau}\right)_V = -\sigma$$
$$\left(\frac{\partial F}{\partial V}\right)_\tau = -p$$

Maxwell relations are easily derived:

$$\frac{\partial^2 F}{\partial V \partial \tau} = \frac{\partial^2 F}{\partial \tau \partial V} \text{ so } \left(\frac{\partial \sigma}{\partial V}\right)_\tau = \left(\frac{\partial p}{\partial \tau}\right)_V$$

So-called natural coordinates for F are τ, V. So $\Sigma \ni (\tau, V)$ (i.e. after a Legendre transformation, the coordinates become (τ, V) for each thermodynamic state.

Recall U as a thermodynamic potential. Using energy conservation and how Q is defined,

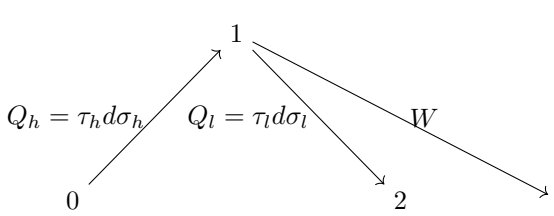
$$dU = Q + W = \tau d\sigma + pdV$$

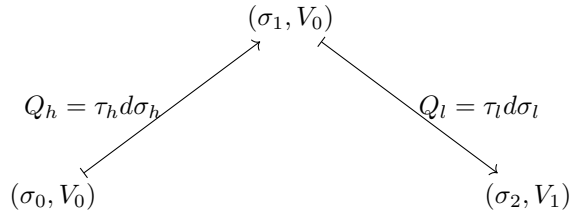
Correction (work sign). The identity used in the rest of these notes is $dU = \tau\, d\sigma - p\, dV$. The plus sign on $p\, dV$ in the display above disagrees with that identity and with the Carnot integral later in this section. See §152.3.

Natural coordinates are σ, V for U. So $\Sigma \ni (\sigma, V)$.

heat engine

ideal heat engine:

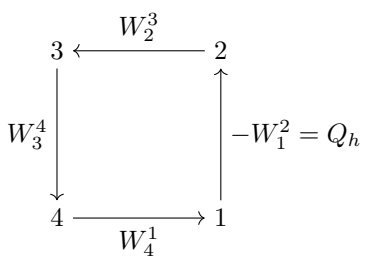


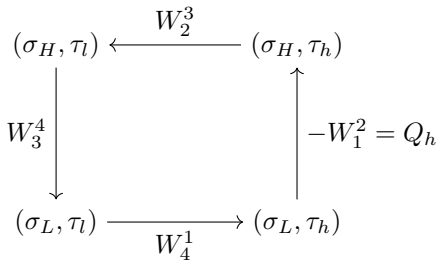


$$W + Q_l = Q_h \text{ or } W = Q_h - Q_l$$
$$W = Q_h - Q_l \le \frac{\tau_h - \tau_l}{\tau_h} Q_h = \eta_C Q_h \qquad \sigma_l = \sigma_h \text{ so } \frac{Q_h}{\tau_h} = \frac{Q_l}{\tau_l}$$

Carnot efficiency $\eta_C \equiv \frac{\tau_h - \tau_l}{\tau_h}$ is the ratio of the work generated to the heat added, in the reversible process.

Carnot cycle

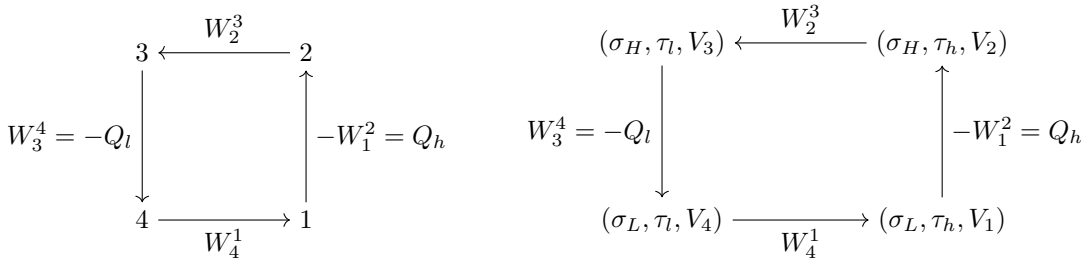




The total work is as such: $\oint dU = 0$ for 2 reasons: mathematically, the integration of an exact 1-form around a closed curve is 0, and physically, we return the system back to its original state, as this is a reversible process.

$$\oint dU = 0 = \oint \tau d\sigma - \oint pdV \Longrightarrow -\oint W = \oint \tau d\sigma = [\tau_h(\sigma_H - \sigma_L) + 0 + \tau_l(\sigma_L - \sigma_H) + 0] = (\tau_h - \tau_l)(\sigma_H - \sigma_L)$$

Example: Carnot cycle for an ideal gas



with

isothermal expansion $Q_h = -W_1^2 = \int_1^2 pdV = N\tau_h \ln\left(\frac{V_2}{V_1}\right)$

adiabatic expansion $W_2^3 = -\int_2^3 dU = U(\tau_h) - U(\tau_l) = C_V(\tau_h - \tau_l)$ $\qquad \tau_l V_3^{\gamma-1} = \tau_h V_2^{\gamma-1}$ or $\frac{V_3}{V_2} = \left(\frac{\tau_h}{\tau_l}\right)^{\frac{1}{\gamma-1}}$

isothermal compression $-Q_l = W_3^4 = \int_3^4 -pdV = N\tau_l \ln\frac{V_3}{V_4}$ $\qquad \frac{V_4}{V_1} = (\frac{\tau_h}{\tau_l})^{\frac{1}{\gamma-1}}$

adiabatic compression $W_4^1 = C_V(\tau_h - \tau_l)$

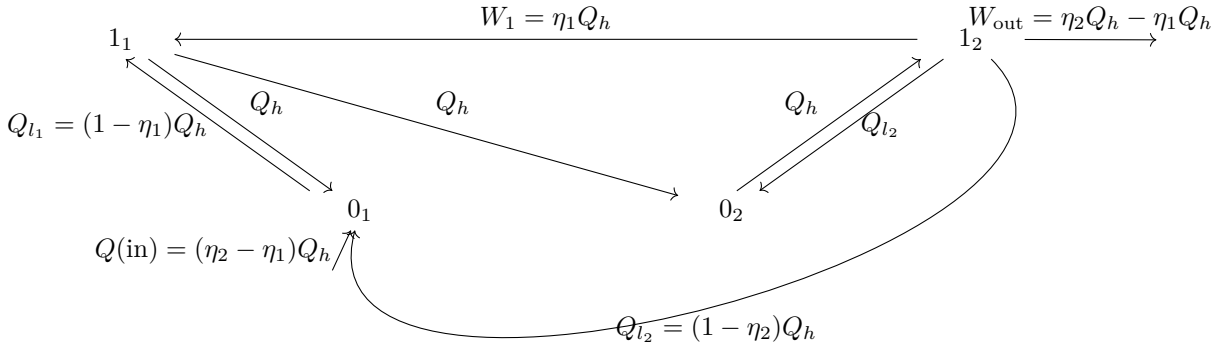
EY : 20150911 I don't have a good reason why C_V which is defined for constant V, that $C_V \equiv \left(\frac{\partial U}{\partial \tau}\right)_V$, can be used in the isentropic (i.e. adiabatic) expansion from $2 \to 3$.

Correction (heat capacity on an adiabatic leg). The subscript on $C_V = (\partial U/\partial\tau)_V$ names the derivative. On the ideal-gas surface $U = U(\tau)$, so $dU = C_V\, d\tau$ along the adiabatic leg as well. The argument is §152.1.

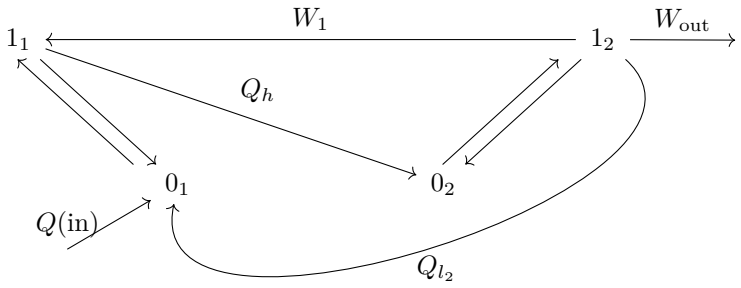
The total work done is

$$W = N(\tau_h - \tau_l) \ln\left(\frac{V_2}{V_1}\right)$$

Energy Conversion and the Second Law of Thermodynamics



Q_{l_2} waste heat



So with $Q(\text{in})$ heat in, W_{out} net work can be done. But that's a decrease in overall entropy. This violates the law of increasing entropy.

Define $H = U + pV$. $H \in C^\infty(\Sigma)$, where Σ is the manifold of equilibrium (and non-equilibrium) states of the system.

Path Dependence of Heat and Work

Mathematically, Q and W are not necessarily *exact* 1-forms. So they are path-dependent.

EY : 20150911 That Q, W are not necessarily exact 1-forms would imply that Σ has some nontrivial, interesting *topological* features.

Correction (topology). Path dependence of the reversible heat is $dQ = d\tau \wedge d\sigma \neq 0$. That failure is local on an open set of $\mathbb{R}^2$. See §152.2.

Heat and Work at Constant Temperature or Constant Pressure

isothermal work

$$dU = W + Q = W + \tau d\sigma$$
$$F = U - \tau\sigma$$
$$dF = dU - \tau d\sigma - \sigma d\tau = W - \sigma d\tau$$

If $d\tau = 0$, on an isothermal curve,
$dF = W$, W becomes an exact 1-form, with potential function F, the Helmholtz free energy.

isobaric heat and work

e.g. boiling of liquid. When liquid boils under atmospheric pressure, vapor pressure displacing atmospheric odes work against atmospheric pressure. isobaric process.

Consider this change of volume:
$dx = \frac{dV}{A}$. Now
p_{eq} = vapor pressure.

$F = p_{\text{eq}}A = p_{\text{atm}}A$ (force equilibrium)

$$W = -p_{\text{atm}}Adx = -p_{\text{atm}}dV \quad \begin{matrix}(\sigma_1, V_1)\\ \uparrow \\ (\sigma_0, V_0)\end{matrix}$$

$W = -p_{\text{atm}}dV \equiv -pdV = -d(pV)$ is part of total work done on system.

If $-d(pV) > 0$, work provided by environment and is "free".
If $-d(pV) < 0$, work delivered to environment and not extractable from system for other purposes.

$$W + d(pV) = dU - Q + d(pV) = dH - Q$$

Recall that for enthalpy $H = U + pV$,

$$dH = dU + Vdp + pdV = dU - W + Vdp = \tau d\sigma + Vdp$$

σ, p are natural coordinates of H.

$$dH - Q = W + d(pV)$$

An isobaric curve s.t. $dp = 0$,
$dH = Q + W + d(pV)$
so
$Q + W$ is an exact 1-form of $H - pV \Longrightarrow d(H - pV) = W + Q$.

2 classes of constant pressure processes:

(a)
$$W + d(pV) = 0$$
$$dH = Q$$

e.g. liquid evaporation from open vessel, because no effective work is done.
heat of evaporation is enthalpy difference between vapor phase and liquid phase

(b) constant temperature and constant pressure.

$$G = F + pV = U - \tau\sigma + pV$$
$$dG = dF + Vdp + pdV = dU - \tau d\sigma - \sigma d\tau + Vdp + pdV = Vdp - \sigma d\tau$$
$$dG = W - \sigma d\tau + d(pV) = W + d(pV) - \sigma d\tau$$

with natural variables are p, τ

at constant temperature, $W + d(pV)$ is exact 1-form, dG

Problems

Solution 1. Heat pump.

(a) For a heat pump,
input: $\sigma_h = \frac{Q_h}{\tau_h}$
output: $\sigma_l = \frac{Q_l}{\tau_l}$

Reversible condition: $\sigma_h = \sigma_l = \frac{Q_h}{\tau_h} = \frac{Q_l}{\tau_l}$ so that $Q_h = \frac{\tau_h}{\tau_l} Q_l$.

$Q_h - Q_l = Q_h - \frac{\tau_l}{\tau_h} Q_h = \frac{\tau_h - \tau_l}{\tau_h} Q_h$ net heat inputted to pump heat.

Thus,

$$\frac{W}{Q_h} = \eta_c = \frac{\tau_h - \tau_l}{\tau_h}$$

If heat pump is not reversible, $\sigma_h > \sigma_l$, so that $\frac{Q_h}{\tau_h} > \frac{Q_l}{\tau_l}$ or $\frac{\tau_l}{\tau_h} Q_h > Q_l$,

$$\frac{W}{Q_h} = \frac{Q_h - Q_l}{Q_h} < \frac{Q_h - \frac{\tau_l}{\tau_h} Q_h}{Q_h} = \eta_{c,\, ideal}$$

(b) Q_h = electricity consumed by reversible heat pump.
Carnot engine: $W = (\tau_{hh} - \tau_l)(\sigma_{hh} - \sigma_l)$, with $\sigma_{hh} = \frac{Q_{hh}}{\tau_{hh}}$, and $\sigma_l = \frac{Q_l}{\tau_l}$

Condition that electricity consumed by reversible heat pump:

$$W = (\tau_{hh} - \tau_l)\left(\frac{Q_{hh}}{\tau_{hh}} - \frac{Q_l}{\tau_l}\right) = Q_h$$

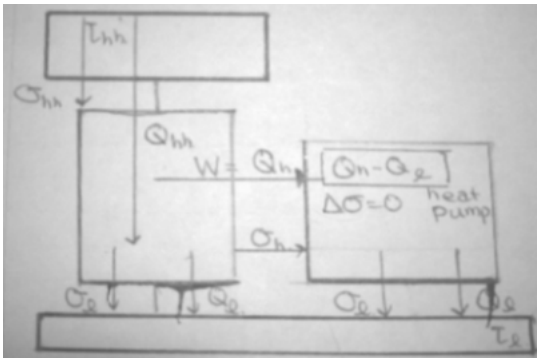
Note we let $\sigma_l = \frac{Q_l}{\tau_l}$ since both heat pump andCarnot engine are reversible.

$$\implies \left(\frac{Q_{hh}}{\tau_{hh}} - \frac{Q_h}{\tau_h}\right) = \frac{Q_h}{\tau_{hh} - \tau_l} \implies \frac{Q_{hh}}{\tau_{hh}} = Q_h\left(\frac{1}{\tau_{hh} - \tau_l} + \frac{1}{\tau_h}\right) \implies \boxed{\frac{Q_{hh}}{Q_h} = \frac{\tau_{hh}(\tau_h + \tau_{hh} - \tau_l)}{\tau_h(\tau_{hh} - \tau_l)}}$$

For $T_{hh} = 600\,K$, $T_h = 300\,K$, $T_l = 270\,K$,

$$\frac{Q_{hh}}{Q_h} = \frac{600(300 + 600 - 270)}{300(600 - 270)} = 3.82$$

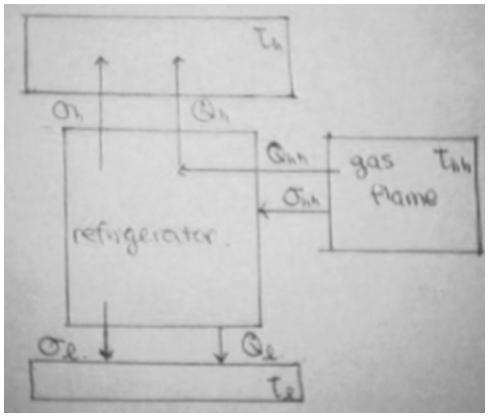
(c) See Figure (136).



Problem 8.1(c)

Solution 2. Absorption refrigerator.

(a) See Figure (136).



Problem 8.2(a)

(b) Given $\tau_{hh} > \tau_h$,
by energy conservation: $Q_{hh} + Q_l - Q_h = 0$
reversible refrigerator: $\sigma_{hh} + \sigma_l - \sigma_h = 0$,

$$\implies \frac{Q_{hh}}{\tau_{hh}} + \frac{Q_l}{\tau_l} - \frac{Q_h}{\tau_h} = 0$$

$$\frac{Q_{hh}}{\tau_{hh}}+\frac{Q_l}{\tau_l}=\frac{Q_{hh}+Q_l}{\tau_h}\text{ or }Q_{hh}\left(\frac{1}{\tau_{hh}}-\frac{1}{\tau_h}\right)=Q_l\left(\frac{1}{\tau_h}-\frac{1}{\tau_l}\right)$$

$$\frac{Q_l}{Q_{hh}}=\left(\frac{1}{\tau_{hh}}-\frac{1}{\tau_h}\right)/\left(\frac{1}{\tau_h}-\frac{1}{\tau_l}\right)=\left(\frac{\tau_h-\tau_{hh}}{\tau_h\tau_{hh}}\right)/\left(\frac{\tau_l-\tau_h}{\tau_h\tau_l}\right)=\boxed{\left(\frac{\tau_{hh}-\tau_h}{\tau_h-\tau_l}\right)\left(\frac{\tau_l}{\tau_{hh}}\right)=\frac{Q_l}{Q_{hh}}}$$

Note that $Q_l-Q_h=Q_l-(Q_{hh}+Q_l)=-Q_{hh}$; we've removed Q_{hh} heat from refrigerator's inside.

Solution 3. Photon Carnot engine. Recall, photons are relativistic: $\epsilon=pc$. Recall $p=\frac{\hbar}{i}\nabla$. $\implies \epsilon_s=pc=\hbar k_s c=\left(\frac{n_s\pi}{L}\right)\hbar c$.

Recalling that there are 2 polarization states for a photon in 3-dim. space,

$$U=(2)\left(\frac{1}{8}\right)(4\pi)\int_0^\infty n^2dn\left(\frac{n\pi}{L}\hbar c\right)e^{-\frac{n\pi}{L}\hbar c/\tau}=\frac{\pi^2\hbar c}{L}\int_0^\infty n^3dnexp\left(\frac{-\pi\hbar c}{L\tau}n\right)=$$
$$=\left(\frac{\pi^2\hbar c}{L}\right)\{n^3exp\left(\frac{-\pi\hbar c}{L\tau}n\right)\left(\frac{L\tau}{-\pi\hbar c}\right)\Big|_0^\infty-\int_0^\infty 3n^2exp\left(\frac{-\pi\hbar c}{L\tau}n\right)\left(\frac{L\tau}{-\pi\hbar c}\right)dn\}=$$
$$=\left(\frac{\pi^2\hbar c}{L}\right)\{(-1)\left(\frac{-K\tau}{\pi\hbar c}\right)3\int_0^\infty n^2exp\left(\frac{-\pi\hbar c}{L\tau}n\right)dn\}=$$
$$=\left(\frac{\pi^2\hbar c}{L}\right)\{(-1)\left(\frac{-L\tau}{\pi\hbar c}\right)3\{n^2exp\left(\frac{-\pi\hbar c}{L\tau}n\right)\left(\frac{L\tau}{-\pi\hbar c}\right)\Big|_0^\infty-\int_0^\infty 2nexp\left(\frac{-\pi\hbar c}{L\tau}n\right)\left(\frac{L\tau}{-\pi\hbar c}\right)dn=$$

$$=\left(\frac{\pi^2\hbar c}{L}\right)(-1)^2\left(\frac{-L\tau}{\pi\hbar c}\right)^2 3(2)\int_0^\infty nexp\left(\frac{-\pi\hbar c}{L\tau}n\right)\left(\frac{L\tau}{-\pi\hbar c}\right)dn=$$
$$=\left(\frac{\pi^2\hbar c}{L}\right)(-1)^3\left(\frac{-L\tau}{\pi\hbar c}\right)^3 3(2)(1)\int_0^\infty exp\left(\frac{-\pi\hbar c}{L\tau}n\right)dn=\left(\frac{\pi^2\hbar c}{L}\right)(-1)^3\left(\frac{-L\tau}{\pi\hbar c}\right)^4 3(2)1\left(\exp\left(\frac{-\pi\hbar c}{L\tau}n\right)\right)\Big|_0^\infty=$$
$$=6\left(\frac{L}{\pi^2\hbar c}\right)^3\tau^4=\boxed{6\frac{V}{(\pi^2\hbar c)^3}\tau^4=U}$$

To get the entropy, *recall*, $\left(\frac{\partial\sigma}{\partial U}\right)_V=\frac{1}{\tau}$, and using this is usually the *most direct way to obtain entropy.*

$$\implies d\sigma=\int\frac{dU}{\tau}=\int\frac{6V}{(\pi^2\hbar^2c)^3}4\tau^3\frac{d\tau}{\tau}=\frac{6V}{(\pi^2\hbar^2c)^3}\frac{1}{3}\tau^3\implies\boxed{\sigma(\tau)=\frac{8V\tau^3}{(\pi^2\hbar^2c)^3}}$$

Consider
Isothermal expansion: Helmholtz free energy F is needed.

$$F=U-\tau\sigma=\frac{6V}{(\pi^2\hbar^2c)^3}\tau^4-\tau\frac{8V\tau^3}{(\pi^2\hbar^2c)^3}=-\frac{2V\tau^4}{(\pi^2\hbar^2c)^3}$$

Then

$$p=-\left(\frac{\partial F}{\partial V}\right)_{\tau,N}=\frac{2\tau^4}{(\pi^2\hbar^2c)^3}\qquad\sigma_{12}=\frac{8\tau_h^3}{(\pi^2\hbar^2c)^3}(V_2-V_1)$$
$$W_{12}=p(V_2-V_1)=\frac{2\tau_h^4}{(\pi^2\hbar^2c)^3}(V_2-V_1)\qquad\Delta Q_{12}=\frac{\tau}{\Delta}\sigma=\frac{8\tau_h^4}{(\pi^2\hbar^2c)^3}(V_2-V_1)$$

Isentropic expansion: $\implies V_2\tau_h^3=V_3\tau_l^3$ or $V_3=V_2\left(\frac{\tau_h}{\tau_l}\right)^3$.

So for this isentropic process, $V_2\tau_h^3=V\tau^3$,

$$U=\frac{6V}{(\pi^2\hbar^2c)^3}\left(\frac{V_2\tau_h^3}{V}\right)^{4/3}=\frac{6(V_2\tau_h^3)^{4/3}}{(\pi^2\hbar^2c)^3}V^{-1/3}$$

$$p=\frac{-\partial U}{\partial V}=-\frac{6(V_2\tau_h^3)^{4/3}}{(\pi^2\hbar^2c)^3}\left(\frac{-1}{3}V^{-4/3}\right)=\frac{2(V_2\tau_h^3)^{4/3}}{(\pi^2\hbar^2c)^3}V^{-4/3}$$

$$W_{23} = \int pdV = \int \frac{2(V_2\tau_h^3)^{4/3}}{(\pi^2\hbar^2c)^3}V^{-4/3}dV = \frac{2(V_2\tau_h^3)^{4/3}}{(\pi^2\hbar^2c)^3}(-3V^{-1/3})\Big|_{V_2}^{V_3} = \frac{-6(V_2\tau_h^3)^{4/3}}{(\pi^2\hbar^2c)^3}\left(\frac{1}{V_3^{1/3}} - \frac{1}{V_2^{1/3}}\right) =$$

$$= \frac{6V_2\tau_h^4}{(\pi^2\hbar^2c)^3}\left(1 - \frac{\tau_l}{\tau_h}\right)$$

Isothermal compression: $W_{34} = \frac{2\tau_l^4}{(\pi^2\hbar^2c)^3}(V_4 - V_3) = \frac{2\tau_h^3\tau_l}{(\pi^2\hbar^2c)^3}(V_1 - V_2)$.
$\sigma_{34} = \frac{8\tau_l^3}{(\pi^2\hbar c)^3}(V_4 - V_3) = \frac{8\tau_h^3}{(\pi^2\hbar c)^3}(V_1 - V_2)$.

Isentropic compressiong: $V_4\tau_l^3 = V_1\tau_h^3$ or $V_4 = V_1\left(\frac{\tau_h}{\tau_l}\right)^3$.

$$W_{41} = \frac{6(V_4\tau_l^3)^{4/3}}{(\pi^2\hbar^2c)^3}\left(\frac{1}{V_4^{1/3}} - \frac{1}{V_1^{1/3}}\right) = \frac{-6V_1\tau_h^4}{(\pi^2\hbar^2c)^3}\left(1 - \frac{\tau_l}{\tau_h}\right)$$

$$\Delta W = \frac{2\tau_h^4}{(\pi^2\hbar^2c)^3}(V_2 - V_1) + \frac{6V_2\tau_h^4}{(\pi^2\hbar^2c)^3}\left(1 - \frac{\tau_l}{\tau_h}\right) + \frac{2\tau_h^3\tau_l}{(\pi^2\hbar^2c)^3}(V_1 - V_2) + \frac{-6V_1\tau_h^4}{(\pi^2\hbar^2c)^3}\left(1 - \frac{\tau_l}{\tau_h}\right) = \frac{8\tau_h^4(V_2 - V_1)}{(\pi^2\hbar^2c)^3}\left(1 - \frac{\tau_l}{\tau_h}\right)$$

$$Q_h = \frac{8\tau_h^4(V_2 - V_1)}{(\pi^2\hbar c)^3} \Longrightarrow \boxed{\frac{\Delta W}{Q_h} = 1 - \frac{\tau_l}{\tau_h}}$$

Solution 4. Heat engine-refrigerator cascade. Consider the heat engine as a Carnot cycle.

$$W + W_r = (\tau_h - \tau_l)\sigma_h$$

where W_r = work consumed by refrigerator.

$$\sigma_h = \frac{Q_h}{\tau_h} = \frac{Q_l}{\tau_l} = \sigma_l$$

This must be true for *any* heat engine undergoing Carnot cycle; furthermore, we can say it's the most efficient heat engine possible.

reversible refrigerator: $Q_L + W_r = Q_H$, (by E-consv.)
$\sigma_L = \sigma_H = \frac{Q_L}{\tau_L} = \frac{Q_H}{\tau_H}$, (by reversible condition)

Note, Q_l is energy transfer from heat engine to τ_l reservoir. Q_L is energy transfer from τ_l reservoir to refrigerator. $Q_L \geq Q_l$, otherwise, no cooling, no thermal energy extracted from τ_l resevoir to lower its temperature. $Q_L = Q_l$ at equilibrium; no further cooling, τ_r reached.

Note that τ_l is given as the environmental temperature. Assume refrigerator throws out Q_H heat into the environment. $\rightarrow \tau_H = \tau_l$. Since Q_L heat inputed into refrigerator from a τ_l reservoir now lowered to τ_r, $\tau_l \rightarrow \tau_r$.

$$W_r = \frac{\tau_l}{\tau_r}Q_L - Q_L = \left(\frac{\tau_l}{\tau_r} - 1\right)Q_L$$

since for a reversible refrigerator, $\sigma_L = \sigma_H = \frac{Q_L}{\tau_L} = \frac{Q_H}{\tau_H}$.

$$\Longrightarrow \frac{W}{Q_h} = \left(1 - \frac{\tau_r}{\tau_h}\right) - \left(\frac{\tau_l}{\tau_r} - 1\right)\frac{Q_L}{Q_h} = \left(1 - \frac{\tau_r}{\tau_h}\right) - \left(\frac{\tau_l}{\tau_r} - 1\right)\left(\frac{\tau_r}{\tau_h}\right) = \boxed{1 - \frac{\tau_l}{\tau_h}}$$

Combinations of reversible systems = reversible system.

Solution 5. Thermal pollution. Given $\begin{matrix} T_l = 20^\circ\, C \\ T_h = 500^\circ\, C \end{matrix}$. Consider a Carnot cycle.

$$W = (\tau_h - \tau_l)\sigma_l = (\tau_h - \tau_l)\frac{Q_l}{\tau_l} = \left(\frac{\tau_h}{\tau_l} - 1\right)Q_l = \left(\frac{500}{20} - 1\right)1500\,MW = \boxed{36000MW}$$

If improvements in hot-steam technology would permit raising T_h by $100^\circ\, C$,

$$W = \left(\frac{600}{20} - 1\right)1500\,MW = (29)(1500\,MW) = 43500\,MW$$

There was a $17.2\,\%$ increase in output.

Solution 6. Room air conditioner.

(a)

$$W = (\tau_h - \tau_l)\frac{Q_l}{\tau_l} = \left(\frac{\tau_h}{\tau_l} - 1\right) Q_l$$

$$P = \left(\frac{\tau_h}{\tau_l} - 1\right)\frac{dQ_l}{dt} = \left(\frac{\tau_h}{\tau_l} - 1\right) A(\tau_h - \tau_l) \Longrightarrow \frac{P}{A}\tau_l = (\tau_h - \tau_l)(\tau_h - \tau_l) = \tau_h^2 - 2\tau_h\tau_l + \tau_l^2$$

$$\Longrightarrow \tau_l^2 - 2\tau_h\tau_l - \frac{P}{A}\tau_l + \tau_h^2 = 0$$

$$\boxed{\tau_l = \tau_h + \frac{P}{2A} - \sqrt{(\tau_h + \frac{P}{2A})^2 - \tau_h^2}}$$

(b) For $T_l = 17^\circ\, C = 290\, K$, $T_h = 310\, K$,

$$A = \frac{P\tau_l}{(\tau_h - \tau_l)^2} = \frac{(2\, kW)(290\, K)}{(310 - 290)^2} = \frac{580 \times 10^3\, W}{400\, K} = \boxed{1450\, \frac{W}{K}}$$

Solution 7. Light bulb in a refrigerator

Carnot refrigerator draws 100 W. For any Carnot cycle,

$$W = (\tau_h - \tau_l)\frac{Q_l}{\tau_l} = \left(\frac{\tau_h}{\tau_l} - 1\right) Q_l = \left(1 - \frac{\tau_l}{\tau_h}\right) Q_h$$

Carnot refrigerator expels Q_h thermal energy to hot τ_h environment and *inputs* Q_l thermal energy from τ_l reservoir.

$$Q_l + W = Q_h$$

Work W must be drawn by Carnot refrigerator to do work. Suppose Carnot cycle part of the refrigerator must input in heat from light bulb to cool down its inside, i.e. consider Carnot refrigerator in *equilibrium with light bulb, now inputting in heat from light bulb* Q_{ext}, and drawing in work to expend out Q_h thermal energy into the environment.

$$\Longrightarrow \boxed{Q_{ext} = Q_l}$$

$\dot{W} = \dot{Q}_l$ in this case, so

$$\left(\frac{\tau_h}{\tau_l} - 1\right) Q_l - Q_l = 0 \text{ or } \left(\frac{\tau_h}{\tau_l} - 2\right) Q_l = 0$$

$$\Longrightarrow \boxed{\tau_l = \frac{\tau_h}{2}} = \frac{300\, K}{2} = 150\, K$$

Solution 8. Geothermal energy.

Given $\Delta Q_h = -MCdT_h$,

T_l lower reservoir temperature stays constant. τ_h decreasing, $d\tau_h < 0$.

$$\Delta W = (\tau_h - \tau_l)\frac{\Delta Q_h}{\tau_h} = \left(1 - \frac{\tau_l}{\tau_h}\right)(-MC)\frac{d\tau_h}{k_B}$$

$$\Longrightarrow W = -\left(\frac{MC}{k_B}\right)(\tau_h - \tau_l \ln \tau_h)|_{\tau_i}^{\tau_f} = -\left(\frac{MC}{k_B}\right)\left(\tau_l \ln\left(\frac{\tau_i}{\tau_f}\right) - (\tau_i - \tau_f)\right)$$

For $M = 10^{17}\, g$, $C = 1\, J/g \cdot K$, $T_l = 20^\circ\, C = 293\, K$, $T_i = 600^\circ\, C = 873\, K$, $T_f = 110^\circ\, C = 383\, K$

$$W = 2.486 \times 10^{19}\, J$$

Note that $10^{14}\, kWh = 10^{17}\, \frac{J}{s} \cdot h\left(\frac{3600\, sec}{1\, h}\right) = 3.6 \times 10^{20}\, J$.

Solution 9. Cooling of nonmetallic solid to $T = 0$. Recall that $C = aT^3 = \left(\frac{\partial U}{\partial T}\right)_V$. Then $dQ_l = aT_l^3 dT_l$. Now $d\tau_l < 0$ since τ_l decreasing.

For the refrigerator: $Q_l + W = Q_h$.

$$dW = (\tau_h - \tau_l)\frac{Q_l}{\tau_l} = -\left(\frac{\tau_h}{\tau_l} - 1\right)(a\tau_l^3 d\tau_l)\left(\frac{1}{k_B^4}\right) = \frac{-a}{k_B^4}\left(\frac{\tau_h}{\tau_l} - 1\right)\tau_l^3 d\tau_l = \frac{-a}{k_B^4}(\tau_h\tau_l^2 - \tau_l^3)d\tau_l$$

$$W = \frac{-a}{k_B^4}\left(\tau_h \frac{1}{3}\tau_l^3 - \frac{1}{4}\tau_l^4\right)\Big|_{\tau_h}^0 = \boxed{\frac{aT_h^3}{12k_B} = W}$$

137 Cryogenics

The 2008 heading was empty. The process written here is cooling by adiabatic demagnetization of the two-state paramagnet of §121. The energy is the expectation of the spin Hamiltonian, not a Legendre transform that moves the field term to the other side of the identity.

137.1 Canonical free energy of N independent moments

Each element has energies $-mB$ and $+mB$. The single-element partition function is

$$Z_1 = e^{mB/\tau} + e^{-mB/\tau} = 2\cosh x, \qquad x = \frac{mB}{\tau}. \tag{660}$$

The N elements are fixed on sites and are distinguishable by those sites, so there is no $1/N!$. The Helmholtz free energy is

$$F(\tau, B) = -N\tau \ln Z_1 = -N\tau \ln(2\cosh x). \tag{661}$$

Hold B fixed and differentiate in τ. Then $\partial x/\partial\tau = -x/\tau$, and

$$\begin{aligned} \frac{\partial}{\partial\tau}\big(\tau \ln(2\cosh x)\big) &= \ln(2\cosh x) + \tau \cdot \tanh x \cdot \left(-\frac{x}{\tau}\right) \\ &= \ln(2\cosh x) - x\tanh x. \end{aligned} \tag{662}$$

Therefore

$$\sigma = -\left(\frac{\partial F}{\partial \tau}\right)_B = N\big[\ln(2\cosh x) - x\tanh x\big]. \tag{663}$$

At $B = 0$, one has $x = 0$ and $\sigma = N\ln 2$, which is $\ln(2^N)$. The mean energy per element is $-\partial \ln Z_1/\partial\beta$ with $\beta = 1/\tau$:

$$U = -NmB\tanh x = -N\tau\, x\tanh x. \tag{664}$$

The total moment along the field, defined by $U = -\mathcal{M}B$, is

$$\mathcal{M} = Nm\tanh x. \tag{665}$$

Check $F = U - \tau\sigma$:

$$\begin{aligned} U - \tau\sigma &= -N\tau\, x\tanh x - N\tau\ln(2\cosh x) + N\tau\, x\tanh x \\ &= F. \end{aligned} \tag{666}$$

The τ derivative at fixed B was arranged so that $(\partial F/\partial\tau)_B = -\sigma$. The B derivative at fixed τ is

$$\left(\frac{\partial F}{\partial B}\right)_\tau = -N\tau \cdot \tanh x \cdot \frac{m}{\tau} = -\mathcal{M}, \tag{667}$$

because $\partial x/\partial B = m/\tau$. Thus

$$dF = -\sigma\, d\tau - \mathcal{M}\, dB. \tag{668}$$

From $U = F + \tau\sigma$,

$$\begin{aligned} dU &= dF + \sigma\, d\tau + \tau\, d\sigma \\ &= -\mathcal{M}\, dB + \tau\, d\sigma. \end{aligned} \tag{669}$$

Equation (669) is the thermodynamic identity for this choice of U. A convention that uses $U' = U + \mathcal{M}B$ moves the work term to $+B\, d\mathcal{M}$. That U' is not the expectation of $-mB\sum_i \epsilon_i$.

For $|x| \ll 1$, $\tanh x = x + O(x^3)$, so

$$U = -NmB\big(x + O(x^3)\big) = -\frac{Nm^2B^2}{\tau} + O(x^3)\, N\tau. \tag{670}$$

The leading term is (615), from the microcanonical Gaussian. The two counts agree in the regime where both apply.

137.2 Isothermal magnetization

Hold $\tau = \tau_i$ and increase B from 0 to B_i. Then x goes from 0 to $x_i = mB_i/\tau_i$. The entropy change of the spins is

$$\Delta\sigma = N\big[\ln(\cosh x_i) - x_i \tanh x_i\big]. \tag{671}$$

The subtraction of $N \ln 2$ produced $\ln(\cosh x_i) = \ln(2\cosh x_i) - \ln 2$. Define $f(x) = \ln(\cosh x) - x\tanh x$. Then $f(0) = 0$ and

$$f'(x) = \tanh x - \tanh x - x\,\mathrm{sech}^2 x = -x\,\mathrm{sech}^2 x. \tag{672}$$

For $x > 0$, $f'(x) < 0$, so $f(x) < 0$ and $\Delta\sigma < 0$. The heat entering the spin system along this reversible isotherm is

$$Q = \tau_i\,\Delta\sigma < 0. \tag{673}$$

The bath receives $-Q$.

137.3 Adiabatic demagnetization

After the magnetization, isolate the sample and reduce the field. Two cases use (663).

If the spins are the whole system, isolation at fixed entropy means $\sigma(x)$ is constant. Equation (663) depends on τ and B only through $x = mB/\tau$. Constant σ therefore means constant x, hence

$$\frac{\tau}{B} = \text{constant} \tag{674}$$

along that path. Reducing B reduces τ in the same ratio. Reaching $B = 0$ at this frozen entropy would require $\tau \to 0$, because $\sigma < N\ln 2$ whenever $x \neq 0$, while $\sigma(B = 0) = N\ln 2$ at every finite τ.

A lattice in the Debye regime has heat capacity $C_V = A\tau^3$ with A constant, at temperatures far below the Debye temperature. Then

$$\left(\frac{\partial\sigma_{\text{latt}}}{\partial\tau}\right)_V = \frac{C_V}{\tau} = A\tau^2, \qquad \sigma_{\text{latt}} = \frac{A}{3}\tau^3, \tag{675}$$

where the integration constant is zero because $\sigma_{\text{latt}} \to 0$ as $\tau \to 0$. Assume the spins and the lattice share one τ during the demagnetization. That is an equilibrium assumption. If the spins decouple from the lattice, (674) applies to the spins and the lattice entropy stays where it was. When they do share τ, isolation conserves the sum

$$N\big[\ln(2\cosh x) - x\tanh x\big] + \frac{A}{3}\tau^3 = \text{constant}, \qquad x = \frac{mB}{\tau}. \tag{676}$$

The constant is fixed by the state at the end of the isothermal step: x_i and τ_i. Lowering B at fixed τ would increase the spin entropy toward $N\ln 2$. With the sum fixed, that increase is paid for by a decrease of σ_{latt}, which by (675) means a lower τ. Equation (676) is the equation for the final temperature once a final field and a value of A are given. No particular A is assumed here.

The refrigerator cycle is then: magnetize in contact with the bath, isolate, demagnetize. The heat $-Q$ of (673) left the sample during the contact step. The later drop in τ is the useful cooling. Repeating the cycle requires a heat switch, because the sample has to be put back into contact with a bath at the new, lower temperature if another isothermal magnetization is to dump entropy again. The switch is a thermal path that can be opened and closed. The entropy accounting above does not model the switch; it states where the contact is required. *2015 manuscript.*

138 Thermodynamic potentials

In the 2015 manuscript these subsections stood in the section Reactions, after the Gibbs sum.

138.1 Gibbs Free Energy

cf. Section "Gibbs Free Energy" of Ch. 9 Gibbs Free Energy and Chemical Reactions from Kittel and Kroemer (1980) [2]

Recall

$$F = U - \tau\sigma$$
$$dF = dU - \tau d\sigma - \sigma d\tau = -\sigma d\tau - pdV + \mu dN$$

where the *generalized* thermodynamic identity

$$dU = Q + W + \mu dN = \tau d\sigma - pdV + \mu dN$$

was used.

Now

$$G \equiv U - \tau\sigma + pV = F + pV$$
$$dG = dU - \tau d\sigma - \sigma d\tau + pdV + Vdp = \mu dN - \sigma d\tau + Vdp$$

and so $G = G(N, \tau, p)$.

Now

$$\sigma = \frac{1}{\tau}[U + pV - G]$$
$$\implies d\sigma = \frac{-1}{\tau^2} d\tau (U + pV - G) + \frac{1}{\tau}(dU + Vdp + pdV - dG)$$

Consider a curve $\gamma : \mathbb{R} \to \Sigma$ s.t. $dU(\dot{\gamma}) = dp(\dot{\gamma}) = dV(\dot{\gamma}) = 0$ corresponding to minimized internal energy, constant pressure, and constant volume. Then

$$d\sigma(\dot{\gamma}) = 0 + 0 + 0 + -dG(\dot{\gamma}) \geq 0$$

and so $dG(\dot{\gamma}) \leq 0$, so G must be a minimum, because $d\sigma(\dot{\gamma}) \geq 0$ represents the fact that entropy must always increase for any thermodynamic process. G must move to a minimum at equilibrium, i.e. "For any irreversible change taking place entirely within $\mathcal{S}$ will increase σ and thus decrease $G_{\mathcal{S}}$" [2] for $\mathcal{S}$ signifying the system.

τ, p are **intensive quantities**; they do not change value when 2 identical systems are put together.

U, σ, V, G are linear in N; their values doubles when 2 identical systems are put together; apart from interface effects

U, σ, V, N, G are **extensive quantities**.

$$G = N\varphi(\tau, p)$$
$$\left(\frac{\partial G}{\partial N}\right)_{p,\tau} = \varphi(\tau, p) = \mu \implies G(N, \tau, p) = N\mu(\tau, p)$$

If more than 1 chemical species present,

$$\boxed{\begin{aligned} G &= \sum_j N_j \mu_j \\ \tau d\sigma &= dU + pdV - \sum_j \mu_j dN_j \\ dG &= \sum_j \mu_j dN_j - \sigma d\tau + Vdp \end{aligned}} \tag{677}$$

138.2 Summary of Thermodynamic Potentials

With Eq. 677, I will summarize the thermodynamic potentials we have, obtained through Legendre transformations:

$$Q = dU - W = dU + pdV$$

For curve or thermodynamic process in Σ, γ, s.t. $dp(\dot{\gamma}) = 0$ (thermodynamic process at *constant pressure*),

$$Q(\dot{\gamma}) = dH(\dot{\gamma}) - Vdp(\dot{\gamma}) = dH(\dot{\gamma}) - 0$$
$$\implies \int_\gamma Q(\dot{\gamma}) = \int dH(\dot{\gamma}) = H_f - H_i \qquad \text{for constant } p$$

with

$$H = U + pV$$
$$dH = dU + pdV + Vdp = Q + Vdp$$

For the thermodynamic potential

$$\begin{gathered}
dU = \tau d\sigma - pdV \\
\boxed{U = U(\sigma, V)} \\
dU = \tau d\sigma - pdV + \sum_j \mu_j dN_j \qquad U = U(\sigma, V, N_j) \\
F = U - \tau\sigma \\
dF = dU - \tau d\sigma - \sigma d\tau = -pdV - \sigma d\tau \Longrightarrow \boxed{F = F(\tau, V)} \\
dF = Q + W - d(\tau\sigma) \\
G = F + pV \\
dG = -\sigma d\tau + V dp \Longrightarrow \boxed{G = G(\tau, p)} \\
dG = (Q - pdV) + pdV + Vdp - d(\tau\sigma) = Q + Vdp - d(\tau\sigma)
\end{gathered} \tag{678}$$

Let $\Delta H := H_f - H_i$.
If $\Delta H < 0$, $\int_\gamma Q(\dot\gamma) < 0$, heat transferred to surroundings: reaction is *exothermic*.
If $\Delta H > 0$, $\int_\gamma Q(\dot\gamma) > 0$, heat transferred to system: reaction is *endothermic*. Now $dH = Q + Vdp$. Suppose $Q = C_p(\tau)d\tau$. then for constant presure, $dp(\dot\gamma) = 0$ and so

$$\int_\gamma dH = \Delta H = \int_\gamma C_p(\tau)d\tau$$

For constant volume, $dV(\dot\gamma_V) = 0$, and so

$$Q(\dot\gamma_V) \equiv Q_V = dU(\dot\gamma_V)$$

138.2.1 Relations between derivatives of thermodynamic quantities

For an adiabatic process, $Q = 0$.
For $Q = Q(\tau, V)$,

$$Q = \tau d\sigma = C_V d\tau + \left(\frac{\partial Q}{\partial V}\right)_\tau dV \tag{679}$$

and $\sigma = \sigma(\tau, V)$, so

$$\tau d\sigma = \tau \left(\frac{\partial \sigma}{\partial \tau}\right)_V d\tau + \tau \left(\frac{\partial \sigma}{\partial V}\right)_\tau dV \tag{680}$$

So by comparing terms between Eq. 679 and Eq. 680, conclude that

$$\left(\frac{\partial Q}{\partial V}\right)_\tau = \tau \left(\frac{\partial \sigma}{\partial V}\right)_\tau \tag{681}$$

For curve $c = c(t) = (\tau(t), V(t))$ s.t. $\sigma = \sigma(\tau, V) =$ constant,

$$\begin{gathered}
\dot c = \dot\tau \frac{\partial}{\partial \tau} + \dot V \frac{\partial}{\partial V} \\
Q(\dot c) = 0 = C_V \dot\tau + \dot V \left(\frac{\partial Q}{\partial V}\right)_\tau \text{ or } \dot\tau = \frac{-\dot V}{C_V}\left(\frac{\partial Q}{\partial V}\right)_\tau
\end{gathered}$$

Let $t = V$. Let $\dot\tau = \left(\frac{\partial \tau}{\partial V}\right)_\sigma$, $\dot V = 1$. So using Eq. 681,

$$\left(\frac{\partial \tau}{\partial V}\right)_\sigma = \frac{-\tau}{C_V}\left(\frac{\partial \sigma}{\partial V}\right)_\tau \tag{682}$$

Likewise, for $Q = Q(\tau, p)$, $\sigma = \sigma(\tau, p)$,

$$Q = Q(\tau, p) = C_p d\tau + \left(\frac{\partial Q}{\partial p}\right)_\tau dp = \tau d\sigma = \tau \left(\frac{\partial \sigma}{\partial \tau}\right)_p d\tau + \tau \left(\frac{\partial \sigma}{\partial p}\right)_\tau dp$$

and repeating all steps for p, but instead using V (what I call "label symmetry"),

$$\left(\frac{\partial\tau}{\partial p}\right)_\sigma = \frac{-\tau}{C_p}\left(\frac{\partial\sigma}{\partial p}\right)_\tau \tag{683}$$

Divide Eq. 683 by Eq. 682:

$$\frac{\left(\frac{\partial\tau}{\partial p}\right)_\sigma}{\left(\frac{\partial\tau}{\partial V}\right)_\sigma} = \frac{C_V}{C_p}\frac{\left(\frac{\partial\sigma}{\partial p}\right)_\tau}{\left(\frac{\partial\sigma}{\partial V}\right)_\tau} \tag{684}$$

Since $F = F(\tau, V)$,

$$dF = -pdV - \sigma d\tau$$
$$\left(\frac{\partial F}{\partial \tau}\right)_V = -\sigma$$
$$\left(\frac{\partial F}{\partial V}\right)_\tau = -p$$

and it is sufficient that $\left(\frac{\partial F}{\partial \tau}\right)_V$, $\left(\frac{\partial F}{\partial V}\right)_\tau$ are continuous on a neighborhood of (τ, V) and $\frac{\partial^2 F}{\partial\tau\partial V}$ exists and is continuous on neighborhood of (τ, V), then

$$\left(\frac{\partial^2 F}{\partial\tau\partial V}\right) = \left(\frac{\partial^2 F}{\partial V\partial\tau}\right)$$

(cf. search for Clairaut's Thm., mean value Thm., or Fubini's Thm.)

$$\implies \left(\frac{\partial\sigma}{\partial V}\right)_\tau = \left(\frac{\partial p}{\partial\tau}\right)_V \qquad \text{(from Helmholtz's free energy } F\text{)} \tag{685}$$

Similarly, with the Gibbs free energy G given in Eq. 678,
since $G = G(\tau, p)$,

$$dG = -\sigma d\tau + V dp$$

then

$$\left(\frac{\partial G}{\partial\tau}\right)_p = -\sigma$$
$$\left(\frac{\partial G}{\partial p}\right)_\tau = V$$

so

$$-\left(\frac{\partial\sigma}{\partial p}\right)_\tau = \left(\frac{\partial V}{\partial\tau}\right)_p \qquad \text{(from Gibbs free energy } G\text{)} \tag{686}$$

Use Eqns. 685, 686 in Eq. 684:

$$\frac{\left(\frac{\partial\tau}{\partial p}\right)_\sigma}{\left(\frac{\partial\tau}{\partial V}\right)_\sigma} = \frac{-C_V}{C_p}\frac{\left(\frac{\partial V}{\partial\tau}\right)_p}{\left(\frac{\partial p}{\partial\tau}\right)_V} \tag{687}$$

Consider an *adiabatic* process, such as adiabatic compression. Then $Q = \tau d\sigma = 0$. So choose curves in Σ s.t. σ is constant along the curve.

Let $\sigma(\tau, V), d\sigma = \left(\frac{\partial\sigma}{\partial\tau}\right)_V d\tau + \left(\frac{\partial\sigma}{\partial V}\right)_\tau dV$.
Let $c = c(t) = (\tau, p) = (\tau(t), p(t))$ s.t. σ constant. Then

$$d\sigma(\dot{c}) = 0 = \left(\frac{\partial\sigma}{\partial\tau}\right)_V \dot{\tau} + \left(\frac{\partial\sigma}{\partial V}\right)_\tau \dot{p}\left(\frac{\partial V}{\partial p}\right)_\sigma \implies$$
$$\left(\frac{\partial V}{\partial p}\right)_\sigma = \frac{-\dot{\tau}\left(\frac{\partial\sigma}{\partial\tau}\right)_V}{\dot{p}\left(\frac{\partial\sigma}{\partial V}\right)_\tau} \xrightarrow{t=p} \frac{-\left(\frac{\partial\tau}{\partial p}\right)_\sigma\left(\frac{\partial\sigma}{\partial\tau}\right)_V}{\left(\frac{\partial\sigma}{\partial V}\right)_\tau} \tag{688}$$

Now let $c = c(t) = (\tau, V) = (\tau(t), V(t))$ s.t. σ constant.

$$d\sigma(\dot{c}) = 0 = \left(\frac{\partial\sigma}{\partial\tau}\right)_V \dot{\tau} + \left(\frac{\partial\sigma}{\partial V}\right)_\tau \dot{V} \text{ or } \frac{\left(\frac{\partial\sigma}{\partial\tau}\right)_V}{\left(\frac{\partial\sigma}{\partial V}\right)_\tau} = \frac{-\dot{V}}{\dot{\tau}} \xrightarrow{t=V} \frac{-1}{\left(\frac{\partial\tau}{\partial V}\right)_\sigma}$$

Then plugging in the previous expression into Eq. 688, we obtain

$$\implies \left(\frac{\partial V}{\partial p}\right)_\sigma = \frac{\left(\frac{\partial\tau}{\partial p}\right)_\sigma}{\left(\frac{\partial\tau}{\partial V}\right)_\sigma} \tag{689}$$

Using label symmetry for $p, V \mapsto V, p$, then

$$\left(\frac{\partial p}{\partial V}\right)_\sigma = \frac{-\left(\frac{\partial\tau}{\partial V}\right)_\sigma \left(\frac{\partial\sigma}{\partial\tau}\right)_p}{\left(\frac{\partial\sigma}{\partial p}\right)_\tau} = \frac{\left(\frac{\partial\tau}{\partial V}\right)_\sigma}{\left(\frac{\partial\tau}{\partial p}\right)_\sigma} \text{ since}$$
$$\frac{\left(\frac{\partial\sigma}{\partial\tau}\right)_p}{\left(\frac{\partial\sigma}{\partial p}\right)_\tau} = \frac{-1}{\left(\frac{\partial\tau}{\partial p}\right)_\sigma}$$

The previous expression and its derivation from first principles (namely that $d\sigma$ is a 1-form on 2-dim. Σ and thermodynamic quantities are coordinates for Σ) leads me to strongly exhort the following: stop memorizing and using "hand-wavey" formulae that claim to transform partial derivatives "to different independent variables" such as the "Jacobians" in Landau and Lifshitz (1980) [7]), pp. 53, Ch. 11 "Thermodynamic Quantities", Sec. 16 "Relations between the Derivatives of Thermodynmaic Quantities", Eq. (IV) and related:

$$\frac{\partial(u,v)}{\partial(x,y)} = \frac{\partial(u,v)}{\partial(t,s)} \cdot \frac{\partial(t,s)}{\partial(x,y)}$$

or in Appendix 4, "Summary of Thermodynamic Relations", pp. 577 of Sabersky, et. al. (1998) [8], which claim that the "rules of differential calculus" (???) allows for some to write, for $z = z(x, y)$,

$$dz = \left(\frac{\partial z}{\partial x}\right)_y dx + \left(\frac{\partial z}{\partial y}\right)_x dy \text{ or}$$
$$\left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial z}{\partial x}\right)_y \left(\frac{\partial y}{\partial z}\right)_x = -1$$

For the latter, the "differential", dx, dy is ill-defined. It is "an outdated concept from the time of Liebniz which was used to define derivatives and integrals before limits came along.". For the former, the "Jacobian" is not general enough (at least for me), being only applicable to Euclidean spaces and is another unnecessary formula to memorize.

Instead, consider the coordinate transformation of 2-dim. manifold Σ:

$$(\tau, V) \mapsto (\tau, p)$$

Consider then that in general, $p = p(\tau, V)$, so that

$$dp = \left(\frac{\partial p}{\partial\tau}\right)_V d\tau + \left(\frac{\partial p}{\partial V}\right)_\tau dV$$

Consider curve $c \subset \Sigma$, $c = c(t) = (\tau, p) = (\tau(t), p(t))$, so that

$$\dot{c} = (\dot{\tau}, \dot{p}) = \dot{\tau}\frac{\partial}{\partial\tau} + \dot{p}\frac{\partial}{\partial p}$$

Then

$$dp(\dot{c}) = \dot{p} = \dot{\tau}\left(\frac{\partial p}{\partial\tau}\right)_V + \dot{p}\left(\frac{\partial p}{\partial V}\right)_\tau \left(\frac{\partial V}{\partial p}\right)_\tau \tag{690}$$

where we used the fact that $dV\left(\frac{\partial}{\partial\tau}\right)=0$ (τ and V are still independent coordinates for Σ) and $dV\left(\frac{\partial}{\partial p}\right)=\left(\frac{\partial V}{\partial p}\right)_\tau$ since τ is held constant for the component of $\dot{c}\in\mathfrak{X}(\Sigma)$ in p (i.e. $\frac{\partial}{\partial p}$).

Consider letting parameter $t=V$ for c. Then

$$\dot{\tau}=\left(\frac{\partial\tau}{\partial V}\right)_p,\qquad \dot{p}=\left(\frac{\partial p}{\partial V}\right)_\tau$$

Consider letting $t=p$. Then $\dot{\tau}=\left(\frac{\partial\tau}{\partial p}\right)_V$ and $\dot{p}=1$. V is help constant for $\dot{\tau}=\left(\frac{\partial\tau}{\partial p}\right)_V$ because of how $p=p(\tau,V)$ for

$$dp=\left(\frac{\partial p}{\partial\tau}\right)_V d\tau+\left(\frac{\partial p}{\partial V}\right)_\tau dV$$

so

$$\implies 1=\left(\frac{\partial\tau}{\partial p}\right)_V\left(\frac{\partial p}{\partial\tau}\right)_V+\left(\frac{\partial p}{\partial V}\right)_\tau\left(\frac{\partial V}{\partial p}\right)_\tau \tag{691}$$

Plug in Eq. 691 into Eq. 690:

$$\dot{p}=\dot{\tau}\left(\frac{\partial p}{\partial\tau}\right)_V+\dot{p}\left[1-\left(\frac{\partial\tau}{\partial V}\right)_p\left(\frac{\partial p}{\partial\tau}\right)_V\right]\implies\dot{p}\left(\frac{\partial\tau}{\partial V}\right)_p\left(\frac{\partial p}{\partial\tau}\right)_V=\dot{\tau}\left(\frac{\partial p}{\partial\tau}\right)_V\text{ or}$$

$$\boxed{\left(\frac{\partial p}{\partial V}\right)_\tau=\frac{\left(\frac{\partial\tau}{\partial V}\right)_p}{\left(\frac{\partial\tau}{\partial p}\right)_V}} \tag{692}$$

Consider the coordinate transformation $(\tau,p)\mapsto(\tau,V)$. By "label symmetry" (i.e. substitute p for V, V for p, for example),

$$\boxed{\left(\frac{\partial V}{\partial p}\right)_\tau=\frac{\left(\frac{\partial\tau}{\partial p}\right)_V}{\left(\frac{\partial\tau}{\partial V}\right)_p}} \tag{693}$$

In fact, furthermore, by label symmetry, consider $(\tau,p)\mapsto(\tau,\rho)$, $\rho\equiv$ density, then

$$\boxed{\left(\frac{\partial p}{\partial\rho}\right)_\tau=\frac{\left(\frac{\partial\tau}{\partial\rho}\right)_p}{\left(\frac{\partial\tau}{\partial p}\right)_\rho}} \tag{694}$$

Compare the previous expressions to the equation before (16.14) and (16.14) of Landau and Lifshitz (1980) [7], pp. 54,

$$\left(\frac{\partial V}{\partial P}\right)_S=\frac{\partial(V,S)}{\partial(P,S)}=\frac{\partial(V,S)/\partial(V,T)}{\partial(P,S)/\partial(P,T)}\cdot\frac{\partial(V,T)}{\partial(P,T)}=\frac{(\partial S/\partial T)_V}{(\partial S/\partial T)_P}\cdot\left(\frac{\partial V}{\partial P}\right)_T$$

$$\left(\frac{\partial V}{\partial P}\right)_S=\frac{C_V}{C_p}\left(\frac{\partial V}{\partial P}\right)_T$$

Similarly with Eq. (A-4.9a) on pp. 577 with Appendix 4 of Sabersky, et. al. (1998) [8]:

$$\left(\frac{\partial p}{\partial\rho}\right)_s=-\frac{(\partial s/\partial\rho)_p}{(\partial s/\partial p)_\rho}=-\frac{(\partial s/\partial T)_p(\partial T/\partial\rho)_p}{(\partial s/\partial p)_\rho(\partial T/\partial p)_\rho}=\frac{(\partial s/\partial T)_p}{(\partial s/\partial T)_\rho}\left(\frac{\partial p}{\partial\rho}\right)_T$$

The expression we'll need uses V:

$$V=V(\tau,p)\implies$$
$$dV=\left(\frac{\partial V}{\partial\tau}\right)_p d\tau+\left(\frac{\partial V}{\partial p}\right)_\tau dp$$

Let $c=(\tau,p)$ be a curve for when V is constant.

$$dV(\dot{c}) = 0 = \left(\frac{\partial V}{\partial \tau}\right)_p \dot{\tau} + \left(\frac{\partial V}{\partial p}\right) \tau \dot{p} \xrightarrow{t=\tau} -\left(\frac{\partial V}{\partial \tau}\right)_p = \left(\frac{\partial V}{\partial p}\right)_\tau \left(\frac{\partial p}{\partial \tau}\right)_V \text{ or}$$

$$\left(\frac{\partial V}{\partial p}\right)_\tau = -\frac{\left(\frac{\partial V}{\partial \tau}\right)_p}{\left(\frac{\partial p}{\partial \tau}\right)_V} \tag{695}$$

Using label symmetry for $V, p \mapsto p, V$, then also

$$\left(\frac{\partial p}{\partial V}\right)_\tau = -\frac{\left(\frac{\partial p}{\partial \tau}\right)_V}{\left(\frac{\partial V}{\partial \tau}\right)_p}$$

Then using both Eq. 689 and the previous equation in Eq. 687, then we get

$$\boxed{\left(\frac{\partial V}{\partial p}\right)_\sigma = \frac{C_V}{C_p}\left(\frac{\partial V}{\partial p}\right)_\tau} \tag{696}$$

This is the equation for the *adiabatic compressibility* $\left(\frac{\partial V}{\partial P}\right)_\sigma$ of a body. The inequality $C_p > C_v$ implies that adiabatic compressibility is always smaller in absolute value than isothermal compressibility. Compare this to Eq. (16.14) of Landau and Lifshitz (1980) and Eq. (A-4.10) of Appendix 4, "Summary of Thermodynamic Relations", pp. 577 of Sabersky, et. al. (1998) [8].

The previous equation can be expressed as

$$\left(\frac{\partial p}{\partial V}\right)_\sigma = \frac{C_p}{C_V}\left(\frac{\partial p}{\partial V}\right)_\tau$$

From the definition of density ρ, $\rho := \frac{MN}{V}$ or $V = \frac{MN}{\rho}$, and assuming N constant, then

$$\frac{\partial}{\partial V} = \frac{\partial \rho}{\partial V}\frac{\partial}{\partial \rho} = \frac{-MN}{V^2}\frac{\partial}{\partial \rho} = \frac{-\rho^2}{MN}\frac{\partial}{\partial \rho}$$

so that

$$\left(\frac{\partial p}{\partial \rho}\right)_\sigma = \frac{C_p}{C_V}\left(\frac{\partial p}{\partial \rho}\right)_\tau$$

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Solution 1. Thermal expansion near absolute zero

(a)

$$\left(\frac{\partial G}{\partial \tau}\right)_{N,p} = -\sigma \qquad \left(\frac{\partial G}{\partial p}\right)_\tau = V$$
$$\left(\frac{\partial^2 G}{\partial p \partial \tau}\right)_\tau = -\left(\frac{\partial \sigma}{\partial p}\right)_\tau \qquad \left(\frac{\partial^2 G}{\partial \tau \partial p}\right)_p = \left(\frac{\partial V}{\partial \tau}\right)_p \qquad \Longrightarrow \boxed{\left(\frac{\partial V}{\partial \tau}\right)_p = -\left(\frac{\partial \sigma}{\partial p}\right)_\tau}$$

$$\left(\frac{\partial G}{\partial N}\right)_p = \mu \qquad \left(\frac{\partial G}{\partial p}\right)_\tau = V$$
$$\left(\frac{\partial^2 G}{\partial p \partial N}\right)_N = \left(\frac{\partial \mu}{\partial p}\right)_N \qquad \left(\frac{\partial^2 G}{\partial N \partial p}\right)_p = \left(\frac{\partial V}{\partial N}\right)_p \qquad \Longrightarrow \boxed{\left(\frac{\partial V}{\partial N}\right)_p = \left(\frac{\partial \mu}{\partial p}\right)_N}$$

$$\left(\frac{\partial^2 G}{\partial \tau \partial N}\right)_N = \left(\frac{\partial \mu}{\partial \tau}\right)_N \qquad \left(\frac{\partial^2 G}{\partial N \partial \tau}\right)_\tau = -\left(\frac{\partial \sigma}{\partial N}\right)_\tau \qquad \Longrightarrow \boxed{\left(\frac{\partial \mu}{\partial \tau}\right)_N = -\left(\frac{\partial \sigma}{\partial N}\right)_\tau}$$

(b) $\alpha = \frac{1}{V}\left(\frac{\partial V}{\partial \tau}\right)_p = \frac{-1}{V}\left(\frac{\partial \sigma}{\partial p}\right)_\tau = 0$ as $\tau \to 0$ since $\sigma \to$ constant as $\tau \to 0$ by third law of thermodynamics.

Solution 2. Thermal ionization of hydrogen.

(a) Given $e + H^+ \rightleftarrows H$, note that $e + H^+ - H = 0$. Recall

$$\frac{[e][H^+]}{[H]} = K(\tau) = \prod_j n_{Q_j}^{\nu_j} \exp\left[-\nu_j F_j(int)/\tau\right]$$

where $n_Q = \left(\frac{M\tau}{2\pi\hbar^2}\right)^{3/2} V$.

For dissocation of H into $e^- + H^+$ choose zero of internal energy of each composite particle (here H) to concide with energy of dissociated particles (here H^+, e^-) at rest; place energy of ground state of composite particle H at $-I$, I is energy required in reaction to dissociate composite particle into its constituents and is taken to be positive, i.e. the ionization energy.

$$K(\tau) = (n_{e^-})^1 \exp\left[-F_{int}(e^-)/\tau\right] \cdot (n_{H^+})^1 \exp\left[-F_{int}(H^+)/\tau\right](n_H)^{-1} \exp\left(-(-1)(F_{int}(H)/\tau)\right)$$

Note that $n_{H^+} \simeq n_H$. Let $n_{e^-} = n_Q$. *Importantly*, note

$$\boxed{F_{int}(e^-) + F_{int}(H^+) - F_{int}(H) = I}$$

$F_{int}(H)$ is at a lower free energy than e^- and H^+.

$$\implies K(\tau) = n_Q e^{-I/\tau} \implies \boxed{\frac{[e][H^+]}{[H]} = n_Q e^{-I/\tau}}$$

(b) By charge conservation, $[e] = [H^+]$, so that

$$[e] = [H]^{1/2} n_Q^{1/2} \exp\left(-I/2\tau\right)$$

Given $[H] \simeq 10^{23}\, cm^{-3}$, $m_e = 0.511\, MeV/c^2$, $T = 5000\, K$, $I = 13.6\, eV$ ionization energy,

$$[e] = (10^{23}\, cm^{-3})^{1/2}(2.92 \times 10^{10} 1/cm^{3/2})^{1/2} \exp\left(-13.6\, eV/2k_B 5000\, K\right) = 1.3 \times 10^{15}\, cm^{-3}$$

Note that $H(exc)$ and H are just two different states of atomic hydrogen. Their concentrations must therefore be proportional to the probability of occurrence of these states, and the ratio of probabilities is the ratio of the respective Boltzmann

$$\frac{[H(exc)]}{[H]} = \frac{p(H(exc))}{p(H)}$$

If $\epsilon_{H(exc)}$ is the internal energy of the first excited state and ϵ_H is the internal energy of the ground state of atomic hydrogen, we are given that $\epsilon_{H(exc)} - \epsilon_H = \frac{3}{4}I$. **We also need to take into account the fact that the first excited electronic state of hydrogen is 4-fold degenerate i.e. one 2s-orbital and three 2p-orbitals.**[11] Therefore,

$$\frac{[H(exc)]}{[H]} = \frac{p(H(exc))}{p(H)} = \frac{4e^{-\epsilon_{H(exc)}/\tau}}{e^{-\epsilon_H/\tau}} = 4e^{-3I/4\tau}$$

$$[H(exc)] = 4[H]e^{-3I/4\tau} = \boxed{2.092 \times 10^{13}\, cm^{-3}}$$

$$\frac{[e]}{[H(exc)]} = 62$$

Solution 3. Ionization of donor impurities in semiconductors.

Consider a pentavalent impurity (donor), introduced in place of tetravalent silicon atom, acts like a hydrogen atom in free space.

[11] (from solutions to Homework 8, Ph12c, Caltech, June 6, 2008, by Prabha Mandayam, Heywood Tam)

Then recall, from Problem 3, $e + H^+ \rightleftarrows H$, and so

$$\frac{[e][H^+]}{[H]} = n_Q \exp(-I/\tau)$$

where $n_Q \equiv \left(\frac{m\tau}{2\pi\hbar^2}\right)^{3/2}$ refers to the electron.

By charge conservation, $[e] = [H^+]$.

So

$$[e] = [H]^{1/2} n_Q^{1/2} \exp(-I/2\tau)$$

Now given $[H] = 10^{17}$ donors per cm^3, $\tau = 100\,K$,

$$n_Q = \left(\frac{m^*\tau}{2\pi\hbar^2}\right)^{3/2} = \left(\frac{0.3 \cdot 0.511\,MeV/c^2 * 100\,K \cdot 0.00008617\,eV/K}{2\pi(4.136 \times 10^{-15}\,eV/s)^2}\right)^{3/2} = 1263.\text{ donors per cm}^3$$

Now $F = \frac{q_1 q_2}{r^2}$,

$$W = \int -F dr = -\int_{r_0}^{\infty} \frac{q_1 q_2}{r^2} dr = +\left(\frac{q_1 q_2}{r}\right)\Big|_{r_0}^{\infty} = \frac{-q_1 q_2}{r_0}$$

and so $I = \frac{e^2}{\epsilon r_0} = \frac{1}{\epsilon}(13.6\,eV)$ where $13.6\,eV$ is the ionization energy of an electron from a (real) hydrogen atom.

So

$$[e] = 2. \times 10^{-18}\text{ donors per cm}^3$$

Correction (donor density). The value $1263\,\text{cm}^{-3}$ for n_Q does not evaluate $(m\tau/(2\pi\hbar^2))^{3/2}$. The hydrogenic binding energy and the quadratic for n are in §140. At $m^* = 0.3\,m_e$, $\epsilon_r = 11.7$, and $T = 100\,\text{K}$, that calculation gives $n = 2.965 \times 10^{16}\,\text{cm}^{-3}$.

Solution 4. Biopolymer growth.

Recall that $G(N, p, \tau) = N\mu(p, \tau)$, since G was chosen to be an *extensive* quantity (it scales with size). For more than one chemical species $G = \sum_j N_j \mu_j$.
$dF = 0$ for equilibrium, for constant P, τ.

μ_j = chemical potential of species j, $\mu_j = (\partial G/\partial N_j)_{\tau,p}$.

Given $\sum_i \nu_j A_j$, e.g. $H_2 + Cl_2 = 2HCl$,
$dG = (\sum_j \nu_j \mu_j) d\hat{N}$ where $dN_j = \nu_j d\hat{N}$, $dG = 0 \to \sum_j \nu_j \mu_j = 0$.

Recall the mass action law derivation: assume constituents act as ideal gases; $\mu_j = \tau(\ln n_j - \ln c_j)$, n_j concentration of species j; $c_j \equiv n_{Q_j} Z_j(int)$.

$$\sum_j \nu_j \ln n_j = \sum_j \nu_j \ln c_j \Longrightarrow \sum_j \ln n_j^{\nu_j} = \sum_j \ln c_j^{\nu_j} = \ln \prod_j n_j^{\nu_j} = \ln K(\tau)$$

$$\prod_j n_j^{\nu_j} = K(\tau) \qquad \text{mass action law}$$

(a) By mass action law, $\frac{[\text{monomer}][N\text{mer}]}{[(N+1)\text{mer}]} = \frac{[1][N]}{[N+1]} = K_N$.

$$\frac{[1][1]}{[2]} = K_1 \qquad \frac{[1]^2}{[2]}\frac{[1][2]}{[3]} = \frac{[1]^3}{[3]} = K_1 K_2$$

$$\frac{[1]^{j+1}}{[j+1]}\frac{[1][j+1]}{[j+2]} = \prod_{l=1}^{j} K_l K_{j+1} = \frac{[1]^{j+2}}{[j+2]} = \prod_{l=1}^{j+1} K_l \qquad \Longrightarrow \boxed{[N+1] = [1]^{N+1}/K_1 K_2 K_3 \ldots K_N}$$

(b) Recall that $K(\tau) = \prod_j n_{Q_j}^{\nu_j} \exp\left[-\nu_j F_j(int)/\tau\right]$

$$K_N = \frac{n_Q(N) n_Q(1)}{n_Q(N+1)} \exp\left[\frac{-F_N}{\tau} - \frac{F_1}{\tau}\right] \exp\left[\frac{F_{N+1}}{\tau}\right] = \frac{n_Q(N) n_Q(1)}{n_Q(N+1)} \exp\left[\frac{-(F_N + F_1 - F_{N+1})}{\tau}\right]$$

where $n_Q(N) = \left(\frac{M_N \tau}{2\pi\hbar^2}\right)^{3/2}$ and M_N is the mass of Nmer molecules, F_N is the free energy of one Nmer molecule.

(c) Assume $N \ll 1$ so $n_Q(N) \simeq n_Q(N+1)$. Assume $[1] = 10^{20}\, cm^{-3}$. Assume $\Delta F = F_{N+1} - F_N - F_1 = 0$, meaning zero free energy change in the basic reaction step. We're given the molecular weight of the monomer to be 200.

We want $\frac{[N+1]}{[N]}$ at room temperature. Now $K_N \simeq n_Q(1) = \left(\frac{M_1 \tau}{2\pi\hbar^2}\right)^{3/2}$.

$$\frac{[1][N]}{[N+1]} = n_Q(1) = \left(\frac{M_1 \tau}{2\pi\hbar^2}\right)^{3/2} \text{ or } \frac{[N+1]}{[N]} = \frac{[1]}{n_Q(1)} = \left(\frac{2\pi\hbar^2}{M_1 \tau}\right)^{3/2} 10^{20}\, cm^{-3}$$

Note that $\left(\frac{2\pi(6.582\times10^{-22}\, MeV\cdot s)^2}{200(938 MeV/c^2)(0.8617\times10^{-4} eV/K)(298\, K)} \left(\frac{3\times10^{10}\, cm/s}{1\, c}\right)^2\right)^{3/2} = 0.3627 \times 10^{-27}\, cm.$

$$\Longrightarrow \boxed{\frac{[N+1]}{[N]} = 3.627 \times 10^{-8}}$$

(d) We want the condition

$$1 < \frac{[N+1]}{[N]} = \frac{[1]}{n_Q(1)} \exp\left(\frac{-\Delta F}{\tau}\right) \text{ or } \ln \frac{n_Q(1)}{[1]} < \frac{-\Delta F}{\tau}$$

$$\Longrightarrow \Delta F < \tau \ln \frac{[1]}{n_Q(1)} = \boxed{-0.44\, eV}$$

140 Semiconductor statistics

The 2008 heading was empty. The preserved solution of the donor-impurity problem, in the Gibbs chapter, adapts the hydrogen ionization mass action and then evaluates a number. The quantum concentration in that arithmetic is not the concentration defined by the formula next to it. This section writes the mass action, the unit-consistent n_Q, and the hydrogenic binding energy, and it solves the quadratic that the low-ionization shortcut does not justify at the corrected numbers. The preserved solution is unchanged in the Gibbs chapter; a pointer there sends the numerical claim here.

140.1 Classical electrons and holes

Use Kittel and Kroemer's orbital quantum concentration, which does not include a spin factor,

$$n_Q(m, \tau) = \left(\frac{m\tau}{2\pi\hbar^2}\right)^{3/2}. \tag{697}$$

In electron-volt units the same quantity is

$$n_Q = \left(\frac{(mc^2)\,(k_B T)}{2\pi(\hbar c)^2}\right)^{3/2}, \qquad \tau = k_B T. \tag{698}$$

The factors of c are required because mc^2 is an energy and $\hbar c$ is an energy times a length. A quotient of mc^2 and $(\hbar)^2$ with $\hbar$ in eV $\cdot$ s is not a number per volume.

For a parabolic conduction band edge at energy ϵ_c, the classical density of electrons, still without a spin factor 2, is

$$n = n_Q(m_e^*, \tau)\, \exp\big((\mu - \epsilon_c)/\tau\big). \tag{699}$$

For holes in a valence band edge at ϵ_v,

$$p = n_Q(m_h^*, \tau)\, \exp\big((\epsilon_v - \mu)/\tau\big). \tag{700}$$

The product is independent of μ:

$$np = n_Q(m_e^*, \tau)\, n_Q(m_h^*, \tau)\, \exp\big(-(\epsilon_c - \epsilon_v)/\tau\big). \tag{701}$$

In an intrinsic sample the only source of carriers is the promotion across the gap, so charge neutrality says $n = p$. Then each density is the square root of (701). A spin factor 2 in both n and p multiplies the product by 4 and each density by 2. The preserved hydrogen solution does not put that factor into n_Q, and the donor arithmetic below follows that choice. The omission is a convention to keep visible, not a claim that spin is absent.

140.2 Donors

The preserved solution treats a donor as a hydrogenic center, with the reaction written there as $e + H^+ \rightleftharpoons H$, and with the mass action

$$\frac{n\, n_+}{n_0} = n_Q(m^*, \tau)\, \exp(-I/\tau). \tag{702}$$

Here n is the conduction-electron density, n_+ the density of ionized donors, and n_0 the density of neutral donors. Charge neutrality, with no acceptors, is $n = n_+$. Number conservation is $n_0 + n_+ = N_d$, so $n_0 = N_d - n$. Equation (702) becomes

$$\frac{n^2}{N_d - n} = K, \qquad K = n_Q(m^*, \tau)\, e^{-I/\tau}. \tag{703}$$

Rearrange:

$$n^2 + Kn - KN_d = 0. \tag{704}$$

The positive root is

$$n = \frac{-K + \sqrt{K^2 + 4KN_d}}{2}. \tag{705}$$

If $n \ll N_d$, then $N_d - n$ may be replaced by N_d and $n \approx \sqrt{KN_d}$. That replacement has to be checked against the root it produces.

The preserved numerical inputs are $N_d = 10^{17}\,\mathrm{cm}^{-3}$, $T = 100\,\mathrm{K}$, and $m^*/m_e = 0.3$. The constants used to evaluate (698) are

$$\begin{aligned} k_B &= 8.617333262145 \times 10^{-5}\,\mathrm{eV/K}, \\ m_e c^2 &= 0.5109989461 \times 10^{6}\,\mathrm{eV}, \\ \hbar c &= 197.3269718\,\mathrm{eV \cdot nm}. \end{aligned} \tag{706}$$

They give

$$n_Q(0.3\, m_e,\, k_B \cdot 100\,\mathrm{K}) = 3.9677 \times 10^{17}\,\mathrm{cm}^{-3}. \tag{707}$$

The value $1263\,\mathrm{cm}^{-3}$ in the preserved solution is the result of dividing an energy by $(\mathrm{eV \cdot s})^2$ and calling the quotient a concentration. Equation (707) is the evaluation of the displayed formula.

The binding energy in the preserved solution is written $I = (13.6\,\mathrm{eV})/\epsilon$, which divides the hydrogen potential by a dielectric constant and holds the Bohr radius fixed. For a hydrogenic donor the radius scales as $\epsilon_r m_e/m^*$, and the energy scales as

$$I = (13.6\,\mathrm{eV})\, \frac{m^*/m_e}{\epsilon_r^2}. \tag{708}$$

The static dielectric constant of silicon is not in the preserved solution. The usual value $\epsilon_r = 11.7$ is used here so that (708) has a number, and it is an added assumption:

$$I = (13.6\,\mathrm{eV})\, \frac{0.3}{(11.7)^2} = 0.02980\,\mathrm{eV}. \tag{709}$$

Then $I/\tau = I/(k_B \cdot 100\,\mathrm{K}) = 3.458$, and

$$K = 1.249 \times 10^{16}\,\mathrm{cm}^{-3}. \tag{710}$$

The quadratic (705) gives

$$n = 2.965 \times 10^{16}\,\mathrm{cm}^{-3}, \qquad \frac{n}{N_d} = 0.296. \tag{711}$$

The shortcut $\sqrt{KN_d} = 3.53 \times 10^{16}\,\mathrm{cm}^{-3}$ is about twenty percent higher, and it is not consistent with $n \ll N_d$. The reported density is (711). Changing m^*/m_e or ϵ_r inside their experimental range moves n by more than this arithmetic does. The preserved value $2 \times 10^{-18}\,\mathrm{cm}^{-3}$ comes from $n_Q \sim 10^3\,\mathrm{cm}^{-3}$ together with a binding energy of order one electron-volt. Both inputs differ from (707) and (709).

A spin degeneracy $g = 2$ of the neutral donor, or a factor 2 in the conduction-band density, replaces K by $K/2$ or by $2K$. Either change is outside the mass action as written in the preserved solution, so it is not folded into (711). *2015 manuscript.*

141 Equilibrium in Reactions

cf. Section Equilibrium in Reactions for Chapter 9: Gibbs Free Energy and Chemical Reactions of Kittel and Kroemer (1980) [2]

$$\nu_1 A_1 + \nu_2 A_2 + \cdots + \nu_l A_l = 0 \text{ or } \sum_j \nu_j A_j = 0 \tag{712}$$

where $\nu_j \in \mathbb{Z}$, A_j chemical species, where we consider the finite set of chemical elements, and nonnegative integers $\mathbb{Z}^+$ for each element, signifying how many of the element in the molecule.

Now clearly,

$$dN_j = \nu_j d\widehat{N}$$

where $d\widehat{N} \equiv$ how many times reaction takes place.

Now

$$dG = \sum_j \mu_j dN_j - \sigma d\tau + V dp$$

and so for constant temperature, constant pressure,

$$dG = \sum_j \mu_j dN_j = \sum_j \mu_j \nu_j d\widehat{N}$$

then for equilibrium $dG = 0$,

$$\boxed{\sum_j \nu_j \mu_j = 0} \tag{713}$$

141.1 Equilibrium for Ideal Gases

cf. Subsection Equilibrium for Ideal Gases for Chapter 9: Gibbs Free Energy and Chemical Reactions of Kittel and Kroemer (1980) [2]

From Eq. (6.48) of Kittel and Kroemer (1980) [2],

$$\mu = \tau[\ln\left(\frac{n}{n_Q}\right) - \ln Z_{\mathrm{int}}]$$

and so

$$\mu_j = \tau(\ln n_j - \ln c_j)$$

where

$$c_j \equiv n_{Q_j} Z_j(\mathrm{int})$$

where

$$Z_j(\mathrm{int}) = \sum_{\mathrm{int}} \exp\left(\frac{-\epsilon_{\mathrm{int}}}{\tau}\right)$$

Thus

$$\Longrightarrow \sum_j \nu_j \ln n_j = \sum_j \nu_j \ln c_j$$

and so the **law of mass action** is obtained:

$$\boxed{\prod_j n_j^{\nu_j} = K(\tau) \equiv \prod_j n_{Q_j}^{\nu_j} \exp\left(-\nu_j F_j(\text{int})/\tau\right)} \tag{714}$$

where

$$F_j(\text{int}) = -\tau \ln Z_j(\text{int})$$

From Problem 6 **Rotation of diatomic molecules** of Chapter 3: Boltzmann Distribution and Helmholtz Free Energy, Kittel and Kroemer [2],
rotational energy at level $j \in \mathbb{Z}^+$, $\epsilon(j) = j(j+1)\epsilon_0$
multiplicity $g(j) = 2j+1$

$$Z_R = \sum_{j=0}^{\infty}(2j+1)\exp\left(\frac{1}{-\tau}j(j+1)\epsilon_0\right) = \frac{-\tau}{\epsilon_0}\sum_{j=0}^{\infty}\frac{d}{dj}\left[(e^{-\epsilon_0/\tau})^{j^2+j}\right]$$

For $1 \gg \frac{\epsilon_0}{\tau}$,

$$Z_R(\tau) = \frac{-\tau}{\epsilon_0}\int_0^{\infty}\frac{d}{dx}\left[(e^{-\epsilon_0/\tau})^{x^2+x}\right] = \frac{\tau}{\epsilon_0}$$

So

$$F_R = -\tau \ln Z_R = -\tau \ln\left(\frac{\tau}{\epsilon_0}\right)$$

and

$$\exp\left(-\nu_j F_j(\text{int})/\tau\right) = \exp\left(\nu_j \ln\left(\frac{\tau}{\epsilon_{0;j}}\right)\right) = \exp\left(\nu_j \ln\left(\frac{\tau}{\epsilon_{0;j}}\right)\right) = \left(\frac{\tau}{\epsilon_{0;j}}\right)^{\nu_j}$$

and so

$$K(\tau) = \prod_j (n_{Q_j})^{\nu_j}\left(\frac{\tau}{\epsilon_{0;j}}\right)^{\nu_j} = \prod_j\left[\left(\frac{M_j\tau}{2\pi\hbar^2}\right)^{d/2}\frac{\tau}{\epsilon_{0;j}}\right]^{\nu_j}$$

2015 manuscript.

142 Chemical Reactions

Topics:

- Heat of formation i.e. standard enthalpy of formation

I will follow Chapter 10, "Chemical Reactions", of Landau and Lifshitz (1980) [7].

From Sec. 101. The condition for chemical equilibrium of Landau and Lifshitz (1980) [7], for a chemical reaction,

$$\sum_i \nu_i A_i = 0 \qquad \nu_i \in \mathbb{Z}$$

Denote thermodynamic potential Φ.

In equilibrium, say at constant temperature and pressure,

$$d\Phi\left(\frac{\partial}{\partial N_i}\right) = \frac{\partial\Phi}{\partial N_1} + \frac{\partial\Phi}{\partial N_2}\frac{\partial N_2}{\partial N_1} + \frac{\partial\Phi}{\partial N_3}\frac{\partial N_3}{\partial N_1} + \cdots = 0 \text{ for}$$
$$d\Phi = \frac{\partial\Phi}{\partial N_1}dN_1 + \frac{\partial\Phi}{\partial N_2}dN_2 + \ldots$$

where $N_1, N_2 \cdots =$ number of particles of particles of various substances taking place in reaction.

It's clear that

$$dN_i = \frac{\nu_i}{\nu_1}dN_1$$

so

$$\sum_i \frac{\partial\Phi}{\partial N_i}\frac{\nu_i}{\nu_1} = 0$$

Let $\mu_i := \frac{\partial \Phi}{\partial N_i}$.

$$\implies \sum_i \nu_i \mu_i = 0 \tag{715}$$

Here, I will follow Section 3.5 Chemical potential of Le Bellac, Mortessagne, Batrouni (2004) [9].

Recall that

$$G(N, \tau, p) = N\mu(\tau, p)$$
$$G = \sum_j \mu_j N_j$$

from Eq. 677, and Kittel and Kroemer (1980) [2]. Therefore,

$$dG = \sum_j \mu_j dN_j + N_j d\mu_j = -\sigma d\tau + V dp + \sum_j \mu_j dN_j$$

and so the **Gibbs-Duhem** relation is (correctly) obtained:

$$0 = \sigma d\tau - V dp + \sum_j N_j d\mu_j \tag{716}$$

142.0.1 Conduction and Convection

In $\tau d\sigma$, let's distinguish the *conduction* component and *convection* component.

Consider system at equilibrium, thermal equilibrium with thermostat at temperature τ
kept at pressure p
able to exchange particles with reservoir at chemical potential μ

Define

$$\widehat{\sigma} := \frac{\sigma}{N} \qquad \textbf{entropy per particle}$$
$$\widehat{\epsilon} := \frac{E}{N} \qquad \textbf{energy per particle}$$

Then,

$$\boxed{\tau d\sigma = \tau d(\widehat{\sigma} N) = \tau N d\widehat{\sigma} + \tau \widehat{\sigma} dN}$$

where

$\tau N d\widehat{\sigma}$ is *entropy change due to change in entropy per particle*, i.e. *conduction term* and

$\tau \widehat{\sigma} dN$ is *entropy change due to change in number of particles* i.e. *convection term.*

So

$$dE = Q + W + \mu dN = \tau d\sigma + 0 + \mu dN = \tau N d\widehat{\sigma} + (\tau\widehat{\sigma} + \mu) dN = \tau N d\widehat{\sigma} + \widehat{h} dN$$

where

$$\widehat{h} := \frac{H}{N} = \widehat{\epsilon} + pV \text{ is } \textbf{enthalpy per particle}$$

since

$$\begin{gathered} H = E + pV \\ G = E - \tau\sigma + pV = \mu N = H - \tau\sigma \end{gathered} \implies \tau\widehat{\sigma} + \mu = \frac{H}{N} = \widehat{h}$$

Interpretation of $\widehat{h} dN$: If dN particles are transported into a given volume by convection, energy increases by $\widehat{\epsilon} dN$.
However, to return to initial volume, it's necessary to compress by $v dN =: \frac{V}{N} dN$ and add work done by pressure $pv dN$

$$\implies \widehat{h} dN = (\widehat{\epsilon} + pv) dN$$

An example of this is *Joule-Thomson expansion.*

Correction (enthalpy per particle). $\hat{h} = H/N = \hat{\epsilon} + pv$ with $v = V/N$. The display that adds pV to $\hat{\epsilon}$ mixes an energy per particle with an energy. See §152.5.

2008 manuscript.

143 Phase Transformations

Vapor Pressure Equation

Derivation of the Coexistence Curve p versus τ

Suppose $\mu_g(p_0,\tau_0)=\mu_l(p_0,\tau_0)$ and $\mu_g(p_0+dp,\tau_0+d\tau)=\mu_l(p_0+dp,\tau_0+d\tau)$

$$\implies \mu_g(p_0,\tau_0)+\left(\frac{\partial\mu_g}{\partial p}\right)_\tau dp+\left(\frac{\partial\mu_g}{\partial\tau}\right)_p d\tau+\cdots=\mu_l(p_0,\tau_0)+\left(\frac{\partial\mu_l}{\partial p}\right)_\tau dp+\left(\frac{\partial\mu_l}{\partial\tau}\right)_p d\tau+\ldots$$

$$\frac{dp}{d\tau}=\frac{\left(\frac{\partial\mu_l}{\partial\tau}\right)_p-\left(\frac{\partial\mu_g}{\partial\tau}\right)_p}{\left(\frac{\partial\mu_g}{\partial\tau}\right)_\tau-\left(\frac{\partial\mu_l}{\partial p}\right)_\tau}$$

$v\equiv V/N$

$s\equiv\sigma/N$

$$\frac{1}{N}\left(\frac{\partial G}{\partial p}\right)_{N,\tau}=v=\left(\frac{\partial\mu}{\partial p}\right)_\tau$$

$$\frac{1}{N}\left(\frac{\partial G}{\partial\tau}\right)_{N,p}=\frac{-\sigma}{N}=-s=\left(\frac{\partial\mu}{\partial\tau}\right)_p$$

s_g-s_l related directly to quantity of heat that must be added to system to transfer 1 molecule reversibly from liquid to gas, while temperature constant. (if heat isn't added to system from outside in the process, temperature will decrease when molecule transferred to gas)

$$L\equiv\tau(s_g-s_l) \tag{717}$$

latent heat of vaporization

$$\Delta v=v_g-v_l \tag{718}$$

change of volume when 1 molecule transferred from liquid to gas

$$\frac{dp}{d\tau}=\frac{L}{\tau\Delta v} \tag{719}$$

is the **Clausius-Clapeyron equation** or **vapor pressure equation**.

Assumptions:

(a) if assume $v_g\gg v_l$, then $\Delta v\cong v_g=V_g/N_g$ at $p=1$ atm (at atmospheric pressure), $v_g/v_l\approx 10^3$

(b) ideal gas law $pV_g=N_g\tau\implies\Delta v\cong\tau/p$

$$\frac{dp}{d\tau}=\frac{L}{\tau^2}p \text{ or } \frac{d}{d\tau}\ln p=\frac{L}{\tau^2}$$

$L\equiv$ latent heat per molecule

If latent heat L independent of temperature over temperature range of interest, $L=L_0$

$\int\frac{dp}{p}=L_0\int\frac{d\tau}{\tau^2}$ whence $\ln p=-L_0/\tau+\text{const}$ or $p(\tau)=p_0\exp(-L_0/\tau)$

Consider $\frac{L_0}{\tau}$, L_0, latent heat of vaporization of *1 molecule*.

Now compare it with $\frac{L_0}{RT}$,

where $R\equiv N_0k_B\equiv$ gas constant.

$N_0=6.02205\times10^{23}\frac{1}{\text{mol}}$

$k_B=1.38066\times10^{-23}\,J/K$ (usual Boltzmann constant)

$R=8.31441\,J/K\cdot\text{mol}$

So for

$$\frac{L_0}{RT}=\frac{L_0}{N_0k_BT}$$

Then that implies

$$\frac{1}{1\text{ particle}}\frac{1}{N_0}=\frac{1}{1\text{ particle }\frac{1}{\text{mol}}}$$

which implies that L_0 now is the latent heat of vaporization of a *mole (mol)* of the gas.

Latent heat and enthalpy

crossing the coexistence curve,

$$\tau d\sigma = dU + pdV - (\mu_g - \mu_l)dN$$

if liquid $\to$ gas, $\mu_l > \mu_g$ (or $0 > \mu_g - \mu_l$)

on coexistence curve, $\mu_g = \mu_l$

$$L = \tau d\sigma(\dot{\gamma}) = dU(\dot{\gamma}) + pdV(\dot{\gamma}) = (dU + pdV)(\dot{\gamma}) + 0 = (dU + pdV)(\dot{\gamma}) + Vdp(\dot{\gamma}) = dH(\dot{\gamma}) = H_g - H_l$$

where $dp(\dot{\gamma}) = 0$ since γ is a path s.t. pressure p is constant.

Recall that $H = U + pV$.

$dH = dU + pdV + Vdp$

$C_p = \tau \left(\frac{\partial \sigma}{\partial \tau}\right)_p = \tau\sigma(\tau, p)(\dot{\gamma}) = (dU(\tau, p)(\dot{\gamma}) + pdV(\tau, p)(\dot{\gamma})) = \left(\frac{\partial H}{\partial \tau}\right)_p$

Then $H = \int C_p d\tau = \int_\gamma C_p d\tau$

Example: Model system for gas-solid equilibrium (pp.285 Ch. 10 "Phase transformations", Kittel and Kroemer (1980) [2])

imagine solid of N atoms

each atom bound as harmonic oscillator of frequency ω

binding energy of each atom in ground state ϵ_0, i.e. energy of atom in ground state is $-\epsilon_0$

energy of a single oscillator $\epsilon = n\hbar\omega - \epsilon_0$, $n \in \mathbb{Z}^+$

$Z_s \equiv$ partition function of a single oscillator in solid, $\beta \equiv 1/\tau$

$$Z_s = \sum_n \exp\left[-\beta(n\hbar\omega - \epsilon_0)\right] = \epsilon^{\beta\epsilon_0} \sum_n (e^{-\beta\omega})^n = \frac{e^{\beta\epsilon_0}}{1 - e^{-\beta\omega}}$$

Free energy $F_s = U_s - \tau\sigma_s = -\tau \ln Z_s$

Gibbs free energy in the solid, per atom

$$G_s = U_s - \tau\sigma_s + pv_s = F_s + pV_s = \mu_s$$

Supposing $v_s \ll v_g$ (volume v_s per atom in solid phase much smaller than volume v_g per atom in gas phase)

$$\mu_s = F_s + pv_s \approx F_s$$

$$\lambda_s := \exp(\beta\mu_s) \simeq \exp(\beta F_s) = \exp(-\ln Z_s) = \frac{1}{Z_s} = \exp(-\beta\epsilon_0)(1 - \exp(-\beta\omega))$$

use ideal gas approximation to describe gas phase, and spin of atom to be 0. From Ch. 6,

$$\lambda_g = \frac{n}{n_Q} = \frac{p}{\tau n_Q} = \frac{p}{\tau}\left(\frac{2\pi\hbar^2}{M\tau}\right)^{3/2}$$

gas in equilibrium with solid with $\lambda_g = \lambda_s$, or

$$\frac{p}{\tau n_Q} = \lambda_g = \lambda_s = e^{-\beta\epsilon_0}(1 - e^{-\beta\omega}) \Longrightarrow p = \left(\frac{M}{2\pi\hbar^2}\right)^{3/2} \tau^{5/2} \exp(-\epsilon_0/\tau)(1 - \exp(-\hbar\omega/\tau))$$

Van Der Waals Equation of State

$$F(\text{ideal gas}) = -N\tau[\ln(n_Q/n) + 1]$$

$n := \frac{N}{V} = \frac{N}{V - Nb}$ (hardcore repulsion at short distances)

Mean Field Method

Let $\varphi(r) :=$ potential energy of interaction of 2 atoms separated by distance r.
average value of total interaction of all other atoms on atom at $r = 0$

$$\int_0^\infty dV\varphi(r)n = n\int_b^\infty dV\varphi(r) = -2na \quad \text{(2 factor is convention)}$$

excluded hardcore sphere of volume b from volume of integration.
assumed mean value of n, concentration n constant throughout volume accessible to gas molecules.
ignored increase of concentration in regions of strong attractive potential energy i.e.
mean field method meglects correlations between interacting molecules.
$\Delta F \simeq \Delta U = \frac{-1}{2}(2Nna) = -N^2a/V$
where $\frac{1}{2}N(N-1) \simeq \frac{1}{2}N^2$ (N large)

$$\implies F(\text{vdW}) = -N\tau(\ln[n_Q(V-Nb)/N]+1) - N^2a/V$$

$$p = -\left(\frac{\partial F}{\partial V}\right)_{\tau,N} = \frac{N\tau}{V-Nb} - \frac{N^2a}{V^2} \text{ or } \left(p+\frac{N^2a}{V^2}\right)(V-Nb) = N\tau$$

Critical Points for the Van der Waals Gas

$$p_c = \frac{a}{27b^2}$$
$$V_c = 3Nb$$
$$\tau_c = \frac{8a}{27b}$$
$$\widehat{p} := \frac{p}{p_c}$$
$$\widehat{V} := \frac{V}{V_c}$$
$$\widehat{\tau} := \frac{\tau}{\tau_c}$$

since

$$\left(\frac{p}{p_c}+\frac{3}{(V/V_c)^2}\right)\left(\frac{V}{V_c}-\frac{1}{3}\right) = \frac{8\tau}{3\tau_c} \qquad \frac{8\tau}{3\tau_c} = \frac{8\tau}{3\frac{8a}{27b}} = \frac{9\tau}{a/b}$$

$$\left(\frac{p}{p_c}+\frac{3}{(V/V_c)^2}\right) = \frac{1}{p_c}(p+\frac{3a/27b^2}{V^2/9N^2b^2}) = \frac{1}{p_c}(p+\frac{N^2a}{V^2}) \qquad \frac{1}{p_cV_c} = \frac{1}{\frac{a}{27b^2}3Nb} = \frac{9}{a/b}$$

$$\left(\frac{V}{V_c}-\frac{1}{3}\right) = \frac{1}{V_0}(V-Nb)$$

$$\implies \left(\widehat{p}+\frac{3}{\widehat{V}^2}\right)\left(\widehat{V}-\frac{1}{3}\right) = \frac{8}{3}\widehat{\tau} \text{ or } \widehat{p} = \frac{\frac{8}{3}\widehat{\tau}}{\widehat{V}-\frac{1}{3}} - \frac{3}{\widehat{V}^2}$$

at $\widehat{p}=1$, $\widehat{V}=1$, $\widehat{\tau}=1$, $\widehat{\tau}$ constant, $\left(\frac{\partial\widehat{p}}{\partial\widehat{V}}\right)_{\widehat{\tau}} = 0$, $\left(\frac{\partial^2\widehat{p}}{\partial\widehat{V}^2}\right)_{\widehat{\tau}} = 0$
Above τ_c, no phase separation exists.

Gibbs Free Energy of the Van der Waals Gas

$G = F + pV$

$$G(\tau,V,N) = \frac{N\tau V}{V-Nb} - \frac{2N^2a}{V} - N\tau(\ln[n_Q(V-Nb)/N]+1)$$

Recall $dG = \mu dN - \sigma d\tau + Vdp$

$$\implies \left(\frac{\partial G}{\partial N}\right)_{\tau,p} = \mu = \mu(p,\tau) \text{ or } G(N,p,\tau) = N\mu(p,\tau)$$

cf. (9.13) of Kittel and Kroemer (1980) [2]
$G(\tau, p, N)$ solved numerically.
on p-V diagram,
coexistence line: consider $dG = -\sigma d\tau + Vdp + \mu dN$
at constant τ, constant total number of particles, N,
$dG(\dot{\gamma}) = Vdp(\dot{\gamma}) = G_g - G_l = \int Vdp$
if $\int Vdp = 0$, $G_p = G_l$, $G_g(\tau, p) = G_l(\tau, p)$, so $\mu_g(\tau, p) = \mu_l(\tau, p)$ along horizontal coexistence line

Nucleation

Let $\Delta\mu = \mu_g - \mu_l$ chemical potential difference between
vapor surrounding small liquid droplet and
liquid in bulk (infinitely large drop)
surface free energy of liquid drop is positive and tends to increase free energy of liquid
for small drop radii, surface can be dominant and drop can be unstable with respect to gas
$n_l \equiv$ concentration of molecules in liquid

$$\Delta G = G_l - G_g = -\left(\frac{4\pi}{3}\right) R^3 n_l \Delta\mu + 4\pi R^2 \gamma$$

where γ surface free energy per limit area.

$$\frac{d\Delta G}{dR} = 0 = -4\pi R^2 n_l \Delta\mu + 8\pi R\gamma \text{ or } R_c = \frac{2\gamma}{n_l \Delta\mu}$$

$R_c \equiv$ critical radius for nucleation
when $R < R_c$, $\Delta G > 0$, so drop will evaporate (to gas) spontaneously
when $R > R_c$, $\Delta G < 0$, drop will tend to grow spontaneously

$$(\Delta G)_c = \frac{16\pi}{3}[\frac{\gamma^3}{n_l^2(\Delta\mu)^2}]$$

where $(\Delta G)_c = \Delta G(R = R_c)$
assume vapor behaves like ideal gas:

$$\Delta\mu = \tau \ln\left(\frac{p}{p_{\text{eq}}}\right)$$

Eq. (12a), chemical potential of the ideal gas, $\mu = \tau \ln\left(\frac{n}{n_Q}\right) = \tau \ln\left(\frac{p}{p_{\text{eq}}}\right)$
$p \equiv$ vapor pressure in gas phase
$p_{\text{eq}} \equiv$ equilibrium vapor pressure of bulk liquid ($R \to \infty$)
use $\gamma = 72\,\frac{\text{erg}}{\text{cm}^2}$, $p = 1.1 p_{\text{eq}}$, $\tau = 300\,K$ (H_2O), $R_c = \frac{2\gamma}{n_l \Delta\mu} = 1 \times 10^{-6}$ cm.

Problems

Problem 1. Entropy, energy, and enthalpy of van der Waals gas.

(a) Recall the generalized thermodynamic identity; starting with $\sigma = \sigma(U, V, N) \in C^\infty(\Sigma) = C^\infty((U, V, N))$, the identity is $dU = \tau d\sigma - pdV + \mu dN$.

By definition, $F = U - \tau\sigma$, so that

$$dF = dU - d\tau\sigma - \tau d\sigma = -pdV + \mu dN - \sigma d\tau$$

Then clearly, $\left(\frac{\partial F}{\partial \tau}\right)_{V,N} = -\sigma$.

Now

$$F(\text{vdW}) = -N\tau(\ln[n_Q(V - Nb)/N] + 1) - N^2 a/V$$

as was postulated about the hardcore repulsion inside a molecule, making V into $V - Nb$.

Remembering that quantum concentration does depend on τ: $n_Q \equiv \left(\frac{M\tau}{2\pi\hbar^2}\right)^{3/2}$

Consider

$$\left(\ln\left(\left(\frac{M\tau}{2\pi\hbar^2}\right)^{3/2}\frac{(V-Nb)}{N}\right)+1\right)=\left(\frac{3}{2}\ln\tau+\ln\left(\left(\frac{M}{2\pi\hbar^2}\right)^{3/2}\frac{(V-Nb)}{N}\right)+1\right)\xrightarrow{\frac{\partial}{\partial\tau}}\frac{3}{2\tau}$$

Thus

$$\sigma=-\left(\frac{\partial F}{\partial\tau}\right)_{V,N}=N\left(\ln\left(\frac{n_Q(V-Nb)}{N}\right)+1\right)+N\frac{3}{2}=N\left(\ln\left(\frac{n_Q(V-Nb)}{N}\right)+\frac{5}{2}\right)$$

(b) Recall that $F:=U-\tau\sigma$ or $U=F+\tau\sigma=F-\tau\left(\frac{\partial F}{\partial\tau}\right)_{V,N}=-\tau^2\left(\frac{\partial}{\partial\tau}\left(\frac{F}{\tau}\right)\right)_{V,N}$

$$U=-\tau^2\frac{3}{2\tau}(-N)+\frac{N^2a}{V}=\boxed{\frac{3N\tau}{2}+\frac{N^2a}{V}=U}$$

(c) Give to first order in the van der Waals correction terms a,b.

$$p=\frac{N\tau}{V-Nb}-\frac{N^2a}{V^2}=\frac{N\tau}{V(1-Nb/V)}-\frac{N^2a}{V^2}=\frac{N\tau}{V}(1+\frac{Nb}{V})-\frac{N^2a}{V^2}$$

where $Nb\ll V$.

Thus,

$$\boxed{H(\tau,V):=U+pV=\frac{5}{2}N\tau+\frac{N^2b\tau}{V}-\frac{2N^2a}{V}}$$

Problem 2. Calculation of dT/dp for water

Using the **Clausius-Clapeyron equation** or **vapor pressure equation**,

$$\frac{dp}{d\tau}=\frac{L}{\tau\Delta v}$$

assume $\Delta v:=v_g-v_l\approx v_g\equiv\frac{V_g}{N_g}$ (i.e. the volume of the gas is much bigger than the volume of the liquid), and $pV_g=N_g\tau$ or $\frac{V_g}{N_g}=\frac{\tau}{p}$ (i.e. water vapor acts like an ideal gas)

$$\frac{L}{\tau\Delta v}\approx\frac{L}{\tau^2p}$$

Looking up fundamental constants and unit conversions *manually* appears, to me at least, in the 21st century, to be an obsolete exercise ("an exercise in insanity?"). I would like to propose presenting the fundamental constants and unit conversions, sourced from the NIST (National Institute of Standards and Technology) website, and using Python library `pandas`, as panda dataframe objects, to make the data accessible and computable, and persisting them or databasing the values as panda dataframe objects.

Open up `thermo.py`, which relies on the `Physique` repository (I'll put it up on my `github`, `ernestyalumni`).

```
N_A = FundConst[ FundConst["Quantity"].str.contains("Avogadro")].loc[42,:]
k_BOLTZ = FundConst[ FundConst["Quantity"].str.contains("Boltzmann") ].loc[49,:]
P_frmATM = conv[ conv["Toconvertfrom"].str.contains("atm") ].loc[15,:] # 1 atm to Pascal conversion
    for pressure

...

## 2. Calculation of $dT/dp$ for water

L = 2260 # J g^{-1}
boilingwatertemp_K = KCconv.subs(T_C,100).rhs # room temperature in Kelvin
tau_boilingwater = boilingwatertemp_K * k_BOLTZ.Value
P = 1 # atm

dpdtau_1002 = L*18.0153/float(N_A.Value)/( tau_boilingwater**2 )
dtaudp_1002 = 1./ dpdtau_1002 # in J/Pascal
dTdp_1002 = dtaudp_1002 / (k_BOLTZ.Value )  # 28.4348535111262 K/atm
```

$$\boxed{\frac{dT}{dp} = 28.43 K/\text{atm}}$$

Problem 3. Heat of vaporization of ice

Given

$$P(\text{H}_2\text{O vapor})(T = -2^\circ C) = 3.88\, mm\, Hg$$
$$P(\text{H}_2\text{O vapor})(T = 0^\circ C) = 4.58\, mm\, Hg$$

Use the **Clausius-Clapeyron equation** or **vapor pressure equation** (for coexistence equilibrium).

$$\frac{dp}{d\tau} = \frac{L}{\tau \Delta v} \approx \frac{L}{\tau^2/p}$$
$$\text{with } \frac{dp}{d\tau} \approx \frac{\Delta p}{\Delta \tau} = \frac{p(\tau_2) - p(\tau_1)}{\tau_2 - \tau_1}$$

Now idealizing water vapor as a perfect ideal gas, $p = \frac{N}{V}\tau$, so p linear in τ, i.e. $p = p(\tau)$.

Note that while p is linear in τ, the given temperature values correspond to T, in K, and so one would need this form of the slope equation $y = mx + b$, where $m = \frac{N}{V}$ times a bunch of conversion factors, and one must also include a nonzero b constant, the result of choice of units (celsius versus energy units).

Thus, $L = \frac{dp}{d\tau}\frac{\tau^2}{p}$, with L being the latent heat of vaporization per particle.

Note, I treat Avogadro's number as a units conversion, $1 = N_A = 6.02205 \times 10^{23}\, \frac{\text{particles}}{\text{mol}}$. So in the end, for the desired J/mol, multiply L by N_A, LN_A.

```
L1003 = ( 4.58 - 3.88 )/(0 - (-2.) )*((KCconv.subs(T_C,1.).rhs)**2)/( (4.58-3.88)/(0 - (-2)) * ( 1.) +
    3.88 ) *
k_BOLTZ.Value*N_A.Value  # 51705.6757640485
```

$$\boxed{L = 5.17 \times 10^4\, J/\text{mol}}$$

Problem 4. Gas-solid equilibrium

(a) Let's review the quantum harmonic oscillator (qho).

Recall the Hamiltonian in d dimensions

$$\widehat{H} = \frac{\widehat{p}^2}{2m} + \frac{1}{2}m\omega^2\widehat{x}^2$$

where $\widehat{p} = -i\frac{\partial}{\partial x^i}\mathbf{e}_i$.

Suppose for the Hilbert space $\mathcal{H}$, that $\mathcal{H} = \mathcal{H}_{x^1} \otimes \mathcal{H}_{x^2} \otimes \mathcal{H}_{x^3}$,

$$|\psi\rangle \in \mathcal{H} = \mathcal{H}_{x^1} \otimes \mathcal{H}_{x^2} \otimes \mathcal{H}_{x^3} \Longrightarrow |\psi\rangle = |\psi_{x^1}\rangle \otimes |\psi_{x^2}\rangle \otimes |\psi_{x^3}\rangle \equiv |\psi_{x^1}\rangle|\psi_{x^2}\rangle|\psi_{x^3}\rangle$$

Now

$$\widehat{H}|\psi\rangle = \sum_{i=1}^{d}\left(\frac{-1}{2m}\left(\frac{\partial}{\partial x^i}\right)^2 + \frac{1}{2}m\omega^2(\widehat{x}^i)^2\right)|\psi\rangle \equiv \left(\frac{-1}{2m}\left(\frac{\partial}{\partial x^i}\right)^2 + \frac{1}{2}m\omega^2(\widehat{x}^i)^2\right)|\psi\rangle$$

The ladder operators make it much more simpler and straightforward than directly solving the differential equation. Let

$$a_i = \sqrt{\frac{m\omega}{2}}\left(\widehat{x}^i + \frac{i}{m\omega}\widehat{p}_i\right) \qquad \widehat{x}^i = \sqrt{\frac{1}{2m\omega}}(a_i + a_i^\dagger)$$
$$a_i^\dagger = \sqrt{\frac{m\omega}{2}}\left(\widehat{x}^i - \frac{i}{m\omega}\widehat{p}_i\right) \qquad \widehat{p}_i = i\sqrt{\frac{m\omega}{2}}(a_i^\dagger - a_i)$$

Now let

$$N_i = a_i^\dagger a_i = \frac{m\omega}{2}((\widehat{x}^i)^2 + \frac{(\widehat{p}_i)^2}{(m\omega)^2}) - \frac{1}{2}$$

where the uncertainty principle commutator relation was used:

$$\widehat{x}^i\widehat{p}_i - \widehat{p}_i\widehat{x}^i = [\widehat{x}^i, \widehat{p}_i] = i$$

Thus

$$\widehat{H} = (N_1 + \frac{1}{2})\omega + (N_2 + \frac{1}{2})\omega + (N_3 + \frac{1}{2})\omega$$

$$\widehat{H}|\psi\rangle = ((n_x + \frac{1}{2})\omega + (n_y + \frac{1}{2})\omega + (n_z + \frac{1}{2})\omega)|\psi\rangle$$

Consider

$$Z_s = \sum_{n_x,n_y,n_z} \exp\left[-\beta((n_x + \frac{1}{2})\omega + (n_y + \frac{1}{2})\omega + (n_z + \frac{1}{2})\omega - \epsilon_0)\right] = e^{\beta\epsilon_0}\left(\sum_n (e^{-\beta\omega})^n\right)^d = e^{\beta\epsilon_0}\left(\frac{1}{1-e^{-\beta\omega}}\right)^d$$

The derivation of $p = p(\tau)$ goes, as above (with Example: Model system for gas-solid equilibrium (pp.285 Ch. 10 "Phase transformations", Kittel and Kroemer (1980) [2])); recall

$$\lambda_g = \lambda_s\,,\ \lambda_g = \frac{n}{n_Q} = \frac{p}{\tau n_Q}$$

and so for $\tau \gg \hbar\omega$ or $1 \gg \beta\omega$,

$$p = \left(\frac{M}{2\pi\hbar^2}\right)^{3/2} \tau^{5/2} e^{-\beta\epsilon_0}(1-e^{-\beta\omega})^d \cong \left(\frac{M}{2\pi}\right)^{3/2} \frac{1}{\hbar^3}\tau^{5/2} e^{-\beta\epsilon_0}\beta^d\omega^d$$

For $d = 3$,

$$\boxed{p = \left(\frac{M}{2\pi}\right)^{3/2} \frac{\omega^3}{\tau^{1/2}} e^{-\beta\epsilon_0}}$$

(b) From above part (a),

$$\frac{dp}{d\tau}\frac{1}{p} = \frac{-1}{2}\tau^{-1} + (-\epsilon_0\left(\frac{-1}{\tau^2}\right))$$

Now using the approximation that $\Delta v \approx v_g$, and the ideal gas idealization,

$$L = \frac{dp}{d\tau}\frac{\tau^2}{p} = \epsilon_0 - \frac{\tau}{2}$$

My explanation: we must excite the atom by its (bounded) ground state energy, ϵ_0. On the other hand, thermal energy of gas, with increasing linear temperature, helps to remove atom from solid to gas.

Problem 5. Gas-solid equilibrium

(a) Now

$$F_s = -\tau \ln \frac{Z_s^{N_s}}{N_s!} = -\tau\left(N_s \ln Z_s - \ln N_s!\right) \simeq -\tau(N_s \ln Z_s - N_s \ln N_s + N_s)$$

where Stirling's approximation was used and $(1/N_s!)$ factor is for indistinguishability (from quantum mechanics).

Recall

$$Z_s = \frac{e^{\beta\epsilon_0}}{(1-e^{-\beta\omega})^d} \text{ so } \ln Z_s = \beta\epsilon_0 - d\ln(1-e^{-\beta\omega})$$

from Example: Model system for gas-solid equilibrium (pp.285 Ch. 10 "Phase transformations", Kittel and Kroemer (1980) [2]).

Then

$$F_s = -N_s\epsilon_0 + N_s\tau d\ln(1-e^{-\beta\omega}) + \tau N_s \ln N_s - N_s\tau \tag{720}$$

Recall this useful relation (it's easy to calculate σ from Helmholtz free energy F):

$$\left(\frac{\partial F}{\partial \tau}\right)_V = \sigma$$

Correction (Helmholtz derivative). Differentiating $F = U - \tau\sigma$ gives $(\partial F/\partial\tau)_V = -\sigma$. The minimization in N_g below does not use this line. The boxed N_g and p_g are unchanged. See §152.4.

and so

$$\begin{aligned}\frac{\partial F_s}{\partial\tau} &= N_s d\ln\left(1-e^{-\omega/\tau}\right) + N_s\ln N_s - N_s + N_s\tau d\left(\frac{1}{1-e^{-\beta\omega}}\right)(-e^{-\beta\omega})(-\omega)\left(\frac{-1}{\tau}\right) = \\ &= N_s d\ln\left(1-e^{-\beta\omega}\right) + N_s\ln N_s - N_s - N_s d\beta\omega\frac{e^{-\beta\omega}}{1-e^{-\beta\omega}} = \sigma\end{aligned} \tag{721}$$

Suppose $\tau \gg \omega$ or $1 \gg \beta\omega$. Then $\sigma = N_s d\ln\left(1-e^{-\beta\omega}\right) + N_s\ln N_s - N_s(1+d)$.

So take the (given) "extreme assumption that the entropy of the solid maybe neglected over the temperature range of interest." [2]

Then comparing σ from Eq. 721 and F_s from Eq. 720, then

$$F_s \simeq -N_s\epsilon_0$$

$$\boxed{F = F_s + F_g = -N_s\epsilon_0 + N\tau\left[\ln\left(\frac{N_g}{Vn_Q}\right) - 1\right]}$$

(b) Now

$$F = -N_s\epsilon_0 + N_g\tau[\ln\left(\frac{N_g}{Vn_Q}\right) - 1] = -\epsilon_0(N - N_g) + N_g\tau\left(\ln\left(\frac{N_g}{Vn_Q}\right) - 1\right)$$

Now, taking the partial derivative,

$$\frac{\partial F}{\partial N_g} = \epsilon_0 + \tau\left(\ln\left(\frac{N_g}{Vn_Q}\right) - 1\right) + N_g\tau\frac{1}{N_g}$$

Taking the minimum free energy with respect to N_g, i.e. set the partial derivative to 0,

$$\frac{\partial F}{\partial N_g} = 0 = \epsilon_0 + \tau\ln\left(\frac{N_g}{Vn_Q}\right) \text{ or } \boxed{N_g = n_Q V\exp\left(\frac{-\epsilon_0}{\tau}\right)}$$

(c) Idealize the gas with the ideal gas law.

$$pV = N_g\tau \Longrightarrow p_g = n_Q\tau\exp\left(\frac{-\epsilon_0}{\tau}\right) = \boxed{\left(\frac{M}{2\pi\hbar^2}\right)^{3/2}\tau^{5/2}\exp\left(\frac{-\epsilon_0}{\tau}\right) = p_g}$$

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144 Phase Equilibrium

144.1 Condition of coexistence; Coexistence of phases

cf. Chapter 10: Phase Transformation, earlier sections of Kittel and Kroemer (1980) [2]; Subsection 3.5.2. "Coexistence of phases" of Le Bellac, Mortessagne, Batrouni (2004) [9]

The *condition of coexistence* is

$$\mu_1(\tau, p) = \mu_2(\tau, p)$$

For example $1 = g =$ gas

$2 = l =$ liquid

Then Taylor expand:

$$\mu_1(\tau_0,p_0)=\mu_2(\tau_0,p_0)=\mu_1(\tau_0,p_0)+\left(\frac{\partial\mu_1}{\partial p}\right)_\tau dp+\left(\frac{\partial\mu_1}{\partial\tau}\right)_p d\tau+\cdots=\mu_2(\tau_0,p_0)+\left(\frac{\partial\mu_2}{\partial p}\right)_\tau dp+\left(\frac{\partial\mu_2}{\partial\tau}\right)d\tau+\ldots$$

$$\Longrightarrow\frac{dp}{d\tau}=\frac{\left(\frac{\partial\mu_1}{\partial\tau}\right)_p-\left(\frac{\partial\mu_2}{\partial\tau}\right)_p}{\left(\frac{\partial\mu_2}{\partial p}\right)_\tau-\left(\frac{\partial\mu_1}{\partial p}\right)_\tau}$$

Now

$$G=N\mu(\tau,p)$$

and so define

$$\frac{1}{N}\left(\frac{\partial G}{\partial\tau}\right)_{N,p}=\frac{-\sigma}{N}\equiv-\widehat{\sigma}=\left(\frac{\partial\mu}{\partial\tau}\right)_{N,p}$$
$$\frac{1}{N}\left(\frac{\partial G}{\partial p}\right)_{N,\tau}=\frac{V}{N}\equiv v=\left(\frac{\partial\mu}{\partial p}\right)_{N,\tau}$$

and so

$$\Longrightarrow\frac{dp}{d\tau}=\frac{\widehat{\sigma}_2-\widehat{\sigma}_1}{v_2-v_1}$$

Equation 11 in Chapter 10 of Kittel and Kroemer (1980) [2] and Eq. 3.96 of Le Bellac, Mortessagne, Batrouni (2004) [9] agree.

Now consider $\tau N d\widehat{\sigma}$ for constant τ,

$$Q=\tau N d\widehat{\sigma}\Longrightarrow\int Q=\tau N(\widehat{\sigma}_2-\widehat{\sigma}_1)\text{ or }\frac{\int Q}{N}=\tau(\widehat{\sigma}_2-\widehat{\sigma}_1)$$

which is the heat added in the transfer.

Define the **latent heat of vaporization**

$$L:=\tau(\widehat{\sigma}_2-\widehat{\sigma}_1)\qquad\text{(latent heat of vaporization)}\tag{722}$$

For the change of volume when 1 molecule is transferred from liquid to gas or 1 to 2, Δv, $\Delta v\equiv v_2-v_1$, and so

$$\left.\frac{dp}{d\tau}\right|_{\text{coexist}}=\frac{L}{\Delta v\tau}\tag{723}$$

144.2 Latent heat

p,T const., liquid $\to$ gas, e.g. $p=1$ atm

$$T=373\,K$$

latent heat of vaporization L_{vap} (by def.) amount of energy supplied by heating.

$Q=dU+W$

$\epsilon:=U/N=$ average (internal) energy per molecule

$v:=V/N=$ volume per molecule

$L_{\text{vap}}=d\epsilon+pdV$

$d\epsilon(\dot\gamma)=\epsilon_{\text{vap}}-\epsilon_{\text{liq}}$

$dv(\dot\gamma)=v_{\text{vap}}-v_{\text{liq}}>0$

if p const., $L_{\text{vap}}=d(\epsilon+pV)=dh$, $h:=H/N=\frac{U+pV}{N}$

EY : 20151031 Another way to think about it is this: recall that

$$Q=dU+W=dU+pdV=\tau d\sigma$$
$$H=U+pV\text{ so }dH=dU+pdV+Vdp=Q+Vdp$$

Consider a path $\gamma\in M$ s.t. $d\tau(\dot\gamma)=0$ (constant τ,p)

$$dp(\dot\gamma)=0$$

$$Q(\dot\gamma) = dH(\dot\gamma) - V dp(\dot\gamma) = dH(\dot\gamma) - 0 = dH(\dot\gamma) \quad Q(\dot\gamma) = \tau d\sigma(\dot\gamma)$$
$$\int Q(\dot\gamma) = \int \tau d\sigma(\dot\gamma) = \tau(\sigma_g - \sigma_l)$$

Thus

$$\frac{\int Q(\dot\gamma)}{N} = \tau(s_g - s_l) = \frac{H_g}{N} - \frac{H_l}{N}$$
$$L \equiv \tau(s_g - s_l) = \frac{\int Q(\dot\gamma)}{N} = \frac{1}{N}(H_g - H_l)$$

Latent heat versus heat capacity

Take slow, reversible process.

Now

$$C_V := \tau \left(\frac{\partial \sigma}{\partial \tau}\right)_V$$
$$C_P := \tau \left(\frac{\partial \sigma}{\partial \tau}\right)_P$$

It's stated in Kittel and Kroemer (1980), pp. 166, Equation (37), Chapter 6: Ideal Gas, Subsection "Heat capacity" [2], that

$$C_P = \tau \left(\frac{\partial \sigma}{\partial \tau}\right)_P = \left(\frac{\partial U}{\partial \tau}\right)_P + p \left(\frac{\partial V}{\partial \tau}\right)_p \tag{724}$$

Suppose $\sigma = \sigma(\tau, V)$. With $\tau d\sigma = dU + W = \tau d\sigma = dU + pdV$,

$$d\sigma = \frac{\partial \sigma}{\partial \tau} d\tau + \frac{\partial \sigma}{\partial V} dV = \frac{dU}{\tau} + \frac{p}{\tau} dV$$
$$d\sigma(\dot\gamma) = \left(\frac{\partial \sigma}{\partial \tau}\right)_V + 0 = \frac{1}{\tau} dU(\dot\gamma) = \frac{1}{\tau}\left(\frac{\partial U}{\partial \tau}\right)_V$$

Then it's clear that

$$\tau \left(\frac{\partial \sigma}{\partial \tau}\right)_V = \left(\frac{\partial U}{\partial \tau}\right)_V = C_V$$

Now suppose $\sigma = \sigma(\tau, p)$.

Consider also the enthalpy, $H = U + pV$, and so

$$dH = dU + V dp + pdV = \tau d\sigma + V dp$$

Now for $\sigma = \sigma(\tau, p)$,

$$\sigma = \sigma(\tau, p)$$
$$d\sigma = \frac{\partial \sigma}{\partial \tau} d\tau + \frac{\partial \sigma}{\partial p} dp = \frac{dH}{\tau} - \frac{V}{\tau} dp \implies d\sigma(\dot\gamma) = \left(\frac{\partial \sigma}{\partial \tau}\right)_p + 0 = \frac{1}{\tau}\left(\frac{\partial H}{\partial \tau}\right)_p - 0$$

So

$$C_p := \tau \left(\frac{\partial \sigma}{\partial \tau}\right)_p = \left(\frac{\partial H}{\partial \tau}\right)_p$$

Now

$$dH(\dot\gamma) = dU(\dot\gamma) + V dp(\dot\gamma) + pdV(\dot\gamma) = \left(\frac{\partial U}{\partial \tau}\right)_p + 0 + p\left(\frac{\partial V}{\partial \tau}\right)_p = \left(\frac{\partial H}{\partial \tau}\right)_p \text{ so}$$
$$C_p = \tau \left(\frac{\partial \sigma}{\partial \tau}\right)_p = \left(\frac{\partial U}{\partial \tau}\right)_p + p\left(\frac{\partial V}{\partial \tau}\right)_p$$

Kittel and Kroemer (1980) [2] argues that for ideal gas,

$$\left(\frac{\partial U}{\partial \tau}\right)_p = \left(\frac{\partial U}{\partial \tau}\right)_V$$

since $U = U(\tau)$.

144.3 Conditions for coexistence

Sec. 12.3 of Baierlein (1999) [1].

Recall

$$G = F + pV = U + pV - \tau\sigma = G(\tau, p, N)$$

Now

$$G = G(\tau, p, N_{\text{vap}}, N_{\text{liq}}) \Longrightarrow dG = \frac{\partial G}{\partial N_{\text{vap}}} dN_{\text{vap}} + \frac{\partial G}{\partial N_{\text{liq}}} dN_{\text{liq}} = \mu_{\text{vap}} 1 + \mu_{\text{liq}}(-1) = 0$$
$$\Longrightarrow \mu_{\text{vap}}(\tau, p) = \mu_{\text{liq}}(\tau, p)$$

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[6]

145 Properties of a Pure Substance

EY : 20151030 Is the word vapor the same as gas? vapour, gaz, gas, vapore

For coexistence equilibrium,

$$\mu_g(p_0, \tau_0) = \mu_l(p_0, \tau_0) \text{ and}$$
$$\mu_g(p_0 + dp, \tau_0 + d\tau) = \mu_l(p_0 + dp, \tau_0 + d\tau)$$

and so

$$\frac{dp}{d\tau} = \frac{L}{\tau \Delta v} \tag{725}$$

where $v \equiv \frac{V}{N}$, the so-called **vapor pressure equation** or **Clausius-Clapeyron** equation.

For (2) approximations, $\Delta v = v_g - v_l \approx v_g = \frac{V_g}{N_g}$ and idealize vapor as ideal gas, $pV = N\tau$ so $\frac{dp}{d\tau} = \frac{L}{\tau^2/p}$.

Second, if L constant,

$$p(\tau) = p_0 \exp\left(-L_0/\tau\right) \text{ or } \ln\left(\frac{p(\tau)}{p_0}\right) = \frac{-L_0}{\tau} \tag{726}$$

cf. Ch. 10 Phase Transformations, pp. 278-284, "Derivation of the Coexistence Curve, p Versus τ" of Kittel and Kroemer (1980) [2].

Eq. 726 explains the shape of the coexistence curve between solid and gas (vapor) (sublimation) and liquid and gas (vapor) (vaporization; vaporization curve).

Saturation is this $p = p(\tau)$ coexistence curve.

Isotherms, Isothermals

Recall $G = F + pV$ and so $dG = dF + Vdp + pdV = dU - \tau d\sigma - \sigma d\tau + pdV + Vdp = -\sigma d\tau + Vdp$

$F = U - \tau\sigma$ $\qquad$ $dG = -\sigma d\tau + Vdp$

So then $G = G(\tau, p)$

$G = G(\tau, p, N)$

where the latter statement is when we include particle transfer, so that

$$dG = -\sigma d\tau + Vdp + \mu dN \text{ for } \mu = \mu(\tau, p)$$

For the *ideal gas*:

$$F(\tau, V) = F = -N\tau\left(\ln\left(\frac{n_Q}{n}\right) + 1\right)$$

$$\text{where } n_Q = \left(\frac{M\tau}{2\pi\hbar^2}\right)^{3/2}$$

$$n \equiv N/V = \frac{p}{\tau}$$

$$G(\tau, p, N) = -N\tau(\ln\left(\frac{n_Q}{n}\right) + 1) + N\tau = -N\tau(\ln\left(\frac{n_Q}{n}\right)) = -N\tau\ln\left(\left(\frac{M\tau}{2\pi\hbar^2}\right)^{3/2}\frac{\tau}{p}\right)$$

$$\text{so then } \left(\frac{\partial G}{\partial N}\right)_{\tau,p} = \mu = -\tau\ln\left(\left(\frac{M\tau}{2\pi\hbar^2}\right)^{3/2}\frac{\tau}{p}\right)$$

For the *Van der Waals gas*

$$F(vdW) = -N\tau\left(\ln\left(\frac{n_Q(V - Nb)}{N}\right) + 1\right) - \frac{N^2a}{V}$$

$$p = -\left(\frac{\partial F}{\partial V}\right)_{\tau,N} = \frac{N\tau}{V - Nb} - \frac{N^2a}{V^2}$$

$$G(\tau, V, N) = \frac{N\tau V}{V - Nb} - \frac{2N^2a}{V} - N\tau(\ln\left(\frac{n_Q(V - Nb)}{N}\right) + 1)$$

Nevertheless, consider, when considering isotherms, isothermals,

$$dG = -\sigma d\tau + Vdp + \mu dN$$

Consider a path γ on constant τ, constant total number of particles N, and so

$$dG(\dot{\gamma}) = Vdp(\dot{\gamma}) = G_g - G_l = \int_\gamma Vdp$$

2015 manuscript.

146 The coexistence curve

Manuscript section title: Phase Equilibrium.

Phase Equilibrium

Phase Diagram

(Phase) Coexistence Curve

Antoine Equation (parameters)

For coexistence equilibrium, $\mu_g(p_0, \tau_0) = \mu_l(p_0, \tau_0)$ and $\mu_g(p_0 + dp, \tau_0 + d\tau) = \mu_l(p_0 + dp, \tau_0 + d\tau)$, and so $\frac{dp}{d\tau} = \frac{L}{\tau\Delta v}$ with $v \equiv \frac{V}{N}$.

Make the approximation that $\Delta v \equiv v_g - v_l \approx v_g = \frac{V_g}{N_g}$ and idealize vapor as ideal gas, $pV = N\tau$, so $\frac{dp}{d\tau} = \frac{L}{\tau^2/p}$.

If L constant,

$$p(\tau) = p_0\exp\left(-L_0/\tau\right) \text{ or } \ln\left(\frac{p(\tau)}{p_0}\right) = \frac{-L_0}{\tau} \tag{727}$$

Consider the Antoine Equation Parameters given in the NIST (National Institute of Standards and Technology) Chemistry WebBook [12]

$$\log_{10}(P) = A - (B/(T + C)) \tag{728}$$

[12] Phase change data for Oxygen

Then

$$P = 10^{A=\frac{B}{T+C}} = 10^A 10^{-\frac{B}{T+C}} = 10^A \exp\left(\frac{-B}{T+C}\ln 10\right)$$

Now $\frac{D}{T+C} = \frac{D}{T} - \frac{DC}{T(T+C)}$, and so

$$P = 10^A \exp\left(\frac{(B\ln 10)C}{T(T+C)}\right)\exp\left(\frac{-B\ln 10}{T}\right)$$

Consider how much the pressure is changed due to this C parameter. Consider P_0, P_1, defined as such:

$$P_0 = 10^A \exp\left(\frac{-B\ln 10}{T}\right) \quad P_1 = 10^A \exp\left(\frac{(B\ln 10)C}{T(T+C)}\right)\exp\left(\frac{-B\ln 10}{T}\right) \Longrightarrow \frac{P_0}{P_1} = \exp\left(\frac{(B\ln 10)C}{T(T+C)}\right)$$

So a deviation can be estimated from $1 - \frac{P_0}{P_1}$.

Open up `thermochem.py`. The *saturation curve* or *coexistence curve* for, for example, oxygen, from liquid to gas, can be reproduced. One needs to input in the Antoine parameters from the NIST website [13]

```
>>> O2coec = CoexistCurve(3.85845, 325.675, -5.667 )
>>> plot( O2coec.curveSI.rhs, (T,54.36,100.16) )

>>> N(O2coec.curveSI.rhs.subs(T,60.))
731.804072053687
>>> N(O2coec.curveSI.rhs.subs(T,70.))
6253.42680398774
>>> N(O2coec.curveSI.rhs.subs(T,75.))
14494.1195824433
```

2015 manuscript.

147 Joule-Thomson Effect

cf. Problem 1.7.6 of Le Bellac, Mortessagne, Batrouni (2004) [9], Lecture 18, Sections "Throttling Processes", "The Joule-Thomson Effect" of Groth, Phys301 Fall 1999 [14]

From Prob. 1.7.6 of Le Bellac, Mortessagne, Batrouni (2004) [9], their setup is the following:
gas initially confined in fixed volume escapes freely without exchanging heat with its environment,

- initially, gas in compartment and (T_1, V_1, p_1)
- expands into container of volume V_2 under constant pressure p_2 $(p_2 < p_1)$
- 2 compartments are thermally isolated; connected by porous plug to maintain pressure difference
- adiabatic (but not quasi-static) expansion.

$Q = 0$. So $Q = 0 = dU - W$. Then

$$\int dU - \int W = \Delta U - \int W = 0$$

Now

$$\Delta H = H_2 - H_1 = u_2 + p_2V_2 - U_1 - p_1V_1 = \Delta U + p_2V_2 - p_1V_1$$

Treat the process of gas evacuating, or going from compartment 1 to container 2 as 2 processes: $\int W = W_1 + W_2$ where
$W_1 = -p_1(0 - V_1) = p_1V_1 \equiv$ work done on gas as the gas evacuates the compartment.
$W_2 = -p_2(V_2 - 0) = -p_2V_2 =$ work done by gas as gas expands into container at constant pressure p_2.

Thus $\int W = W_1 + W_2 = p_1V_1 - p_2V_2$ and so

$$\boxed{\Delta H = 0}$$

[13] http://webbook.nist.gov/cgi/cbook.cgi?ID=C7782447&Mask=4#Thermo-Phase
[14] Edward J. Groth, 1999, Fall 1999 Physics 301, Phys301 23-Nov-1999 18-10

The enthalpy is constant for this expansion.

For an ideal gas, $H = U + pV = \frac{\gamma}{\gamma-1}N\tau$, for an ideal gas has $U = \frac{1}{\gamma-1}N\tau$. $\gamma = \frac{5}{3}$ and $\frac{\gamma}{\gamma-1} = \frac{5/3}{2/3} = 5/2$. Nevertheless, since H is constant, T is constant, and so there's no cooling of an ideal gas in a Joule-Thomson apparatus! But an ideal gas has no interaction energy, and it can be argued that one has to consider interaction energy between molecules for liquefaction (cf. Groth (1999) [10], Lecture 18 Physics 301 Fall 1999).

From the "Derivation of the Joule-Thomson (Kelvin) coefficient" section of the *Joule-Thomson effect* article in Wikipedia, recall the definition of enthalpy $H := U + pV$, and so that $dH = dU + pdV + Vdp = \tau d\sigma + Vdp$.

Let $\sigma = \sigma(\tau, p)$.

$$dH = \tau \left(\frac{\partial \sigma}{\partial \tau}\right)_p d\tau + (V + \tau \left(\frac{\partial \sigma}{\partial p}\right)_\tau)dp = C_p d\tau + (V + \tau \left(\frac{\partial \sigma}{\partial p}\right)_\tau)dp$$

since

$$Q = Q(\tau, p) = \frac{\partial Q}{\partial \tau}d\tau + \frac{\partial Q}{\partial p}dp = C_p d\tau + \frac{\partial Q}{\partial p}dp$$

and so for a thermodynamic process γ such that $dp(\dot{\gamma}) = 0$ (i.e. constant pressure), then we can say

$$\frac{C_p}{\tau}d\tau = d\sigma \xrightarrow{\int_\gamma} \int_\gamma \frac{C_p}{\tau}d\tau = \Delta\sigma \Longrightarrow \tau \left(\frac{\partial \sigma}{\partial \tau}\right)_p = C_p$$

And now

$$dG = -\sigma d\tau + Vdp = \left(\frac{\partial G}{\partial \tau}\right)_p d\tau + \left(\frac{\partial G}{\partial p}\right)_\tau dp$$

and since $\frac{\partial^2 G}{\partial \tau \partial p} = \frac{\partial^2 G}{\partial p \partial \tau}$,

$$\left(\frac{\partial \sigma}{\partial p}\right)_\tau = -\left(\frac{\partial V}{\partial \tau}\right)_p = -V\alpha$$

Then

$$dH = C_p d\tau + V(1 - \tau\alpha)dp$$

and so

$$\left(\frac{\partial \tau}{\partial p}\right)_H = \frac{V(1 - \tau\alpha)}{-C_p} \tag{729}$$

is the **Joule-Thomson coefficient**, where α is the coefficient of thermal expansion.

For an ideal gas,

$$\begin{aligned} pV &= N\tau \\ V &= \frac{N\tau}{p} \end{aligned} \Longrightarrow \frac{\partial V}{\partial \tau} = \frac{N}{p} = \frac{V}{\tau}$$

so

$$dH = C_p d\tau + (V + \tau \left(\frac{-V}{\tau}\right))dp = C_p d\tau$$

so that

$$\left(\frac{\partial \tau}{\partial p}\right)_H = 0$$

for an ideal gas.

For every real gas, every real gas has an *inversion temperature* such that

below the inversion temperature, $\left(\frac{\partial T}{\partial p}\right)_H$ is positive, and so $dp < 0$ always in an expansion, and so $dT < 0$ and the gas cools, and

above the inversion temperature, $\left(\frac{\partial T}{\partial p}\right)_H$ is negative, and so $dp < 0$ always in an expansion, and so $dT < 0$ and the gas warms.

Correction (Joule–Thomson warming). Above inversion the coefficient is negative and $dp < 0$, so $d\tau > 0$. The gas warms. The clause $dT < 0$ in the sentence has the wrong sign. See §152.6.

An inversion temperature must exist because when the molecules are close enough, there are intermolecular forces that'll make them attract each other (such as temporary configurations so a slightly (electrically) negative cloud of electrons attract slightly positively charged part of another molecule) so the interaction energy is negative.

However, it appears that this inversion temperature is not found from the Joule-Thomson coefficient expression, Eq. 729, but from, possibly, the Van der Waals expression or another model (cf. Lecture 18 of Groth (1999) [10]).

Looking at Figure 1.16 of Le Bellac, Mortessagne, Batrouni (2004) [9], for isenthalpy curves (H = const.), in (T, p) plane, T vs. p, the dashed line intersects the points on each isenthalpy curves where $\left(\frac{\partial T}{\partial p}\right)_H = 0$ and for lower pressures ($dp < 0$, gas expands), cooling occurs.

There's a maximum temperature, about $300° C$ for nitrogen, where the dashed line meets the $p = 0\,\text{atm}$ axis, meaning $\left(\frac{\partial \tau}{\partial p}\right)_H < 0 \quad \forall\, p > 0$, so no cooling occurs.
2008 manuscript.

148 Kinetic Theory

The wall-collision argument for $P = N\tau/V$ is in §123.2.

Kinetic Theory of the Ideal Gas Law

Consider molecule strike unit area of wall.
Let $v_z \equiv$ velocity component normal to plane of wall.
Suppose molecules, of mass M, reflected specularly (mirror-like) from wall,

$$\Delta p_z = -2M|v_z|$$

Let $a(v_z)dv_z$, number of molecules per unit volume with z-component of velocity between v_z and $v_z + dv_z$.

$$\int a(v_z)dv_z = \frac{N}{V} = n$$

$a(v_z)v_z dv_z$ number of molecules in $(v_z, v_z + dv_z)$ velocity range that strike unit area of wall in (per) unit time

$$\text{pressure } p = \int_0^\infty 2Mv_z a(v_z)v_z dv_z = M\int_{-\infty}^\infty v_z^2 a(v_z)dv_z = Mn\langle v_z^2\rangle$$

$\frac{1}{2}M\langle v_z^2\rangle = \frac{1}{2}\tau$ by equipartition of energy (Ch.3)

$$p = nM\langle v_z^2\rangle = n\tau = \frac{N\tau}{V}$$

Maxwell Distribution of Velocities

cf. Ch.6. distribution function of ideal gas $f(\epsilon_n) = \lambda \exp\left(\frac{-\epsilon_n}{\tau}\right)$

Recall, Ch. 6, Sec. "Classical Limit" of Kittel and Kroemer [2], **an ideal gas is defined as a system of free noninteracting particles in the classical regime**.
$f(\epsilon) \equiv$ average occupancy of an orbital at energy ϵ
$\epsilon \equiv$ energy of orbital occupied by 1 particle; not energy of system of N particles
Fermi-Dirac and Bose-Einstein distribution $f(\epsilon) = \frac{1}{\exp[(\epsilon-\mu)/\tau]\pm 1}$
In order for $f(\epsilon) \ll 1\ \forall$ orbitals, $\exp[(\epsilon - \mu)/\tau] \gg 1\,\forall\,\epsilon$.

$$\Longrightarrow f(\epsilon) \simeq \exp[(\mu - \epsilon)/\tau] = \lambda \exp(-\epsilon/\tau) \quad \lambda \equiv \exp\left(\frac{\mu}{\tau}\right)$$

$f(\epsilon)$, average occupancy of orbital of energy ϵ, is classical distribution function.

particle in a box: $\epsilon_n = \frac{1}{2M}\left(\frac{\pi n}{L}\right)^2$ (for, recall $\frac{1}{2M}\left(\frac{1}{i}\partial\right)^2\psi = \frac{-1}{2M}\partial^2\psi = E\psi$)

number of orbitals in range of quantum number $(n, n + dn)$, probability such orbital is occupied

$$(\frac{1}{2}\pi n^2 dn)f(\epsilon_n) = \frac{1}{2}\pi n^2 \lambda \exp(-\epsilon_n/\tau)dn$$

$$\frac{1}{2}Mv^2 = \frac{1}{2M}\left(\frac{\pi n}{L}\right)^2 \text{ or } n^2 = \frac{(ML)^2}{\pi^2}v^2 \text{ or } n = \frac{MLv}{\pi}$$

Consider system of N particles in volume V.

Let $NP(v)dv$ number of atoms with velocity magnitude in range dv at v

$$NP(v)dv = \frac{1}{2}\pi n^2 \lambda \exp\left(-\epsilon_n/\tau\right)\frac{dn}{dv}dv = \frac{1}{2}\pi\lambda\left(\frac{ML}{\pi}\right)^3 v^2 \exp\left(\frac{-Mv^2}{2\tau}\right)dv$$

cf. Ch. 6, $\lambda = \frac{n}{n_Q} = \frac{N}{L^3}\left(\frac{2\pi\hbar^2}{M\tau}\right)^{3/2}$

$$\frac{1}{2}\pi\frac{N}{L^3}\left(\frac{2\pi}{M\tau}\right)^{3/2}\left(\frac{ML}{\pi}\right)^3 = 4\pi N\left(\frac{N}{2\pi\tau}\right)^{3/2}$$

Correction (Maxwell prefactor). The right-hand side above is not the prefactor of the speed density in the next display. That density is normalized, as checked in §152.8.

$$\Longrightarrow P(v) = 4\pi\left(\frac{M}{2\pi\tau}\right)^{3/2} v^2 \exp\left(\frac{-Mv^2}{2\tau}\right) \tag{730}$$

$P(v)$ is **Maxwell velocity distribution**, $P(v)dv$ is probability particle has speed in dv at v

Experimental verification

velocity distribution of atoms which exit from slit of oven.

exit beam weighted in favor of atoms of high velocity at expense of those at low velocity.

weight factor is velocity component $v\cos\theta$ normal to plane of hole.

$$\int(\cos\theta)drrd\varphi r\sin\theta d\theta = (\frac{1}{3}(2\pi)R^3)\int\cos\theta\sin\theta d\theta = (\frac{2\pi}{3}R^3)\int\frac{\sin 2\theta d\theta}{2} = \frac{2\pi R^3}{3}\left(\frac{-\cos 2\theta}{4}\right)\Big|_0^{\pi/2} =$$
$$= \frac{2\pi R^3}{3}\left(\frac{1+1}{4}\right) = \frac{2\pi R^3}{3}\left(\frac{1}{2}\right)$$

Probability atom leaves hole will have velocity in $(v, v+dv)$: $P_{\text{beam}}(v)dv$

$$P_{\text{beam}}(v) \propto vP_{\text{Maxwell}} \propto v^3 \exp\left(\frac{-Mv^2}{2\tau}\right)$$

with, recall $P_{\text{Maxwell}} = 4\pi\left(\frac{M}{2\pi\tau}\right)^{3/2} v^2 \exp\left(\frac{-Mv^2}{2\tau}\right)$

P_{beam} distribution of transmission through a hole is Maxwell transmission distribution.

$$\langle v_{\text{out}}\rangle = \int\int_0^{\pi/2} v\cos\theta\sin\theta d\theta P_{\text{Maxwell}}(v)dv = 4\pi\left(\frac{M}{2\pi\tau}\right)^{3/2}\frac{1}{2}\int_0^\infty v^3\exp\left(\frac{-M}{2\tau}v^2\right)dv =$$
$$= 4\pi\left(\frac{M}{2\pi\tau}\right)^{3/2}\frac{1}{2}\left(\frac{1}{-2\alpha}\right)\left[0-\frac{1}{\alpha}\right] = 4\pi\left(\frac{M}{2\pi\tau}\right)^{3/2}\frac{1}{4}\left(\frac{4\tau^2}{M^2}\right) = \boxed{\frac{2^{1/2}}{\pi^{1/2}}\left(\frac{\tau}{M}\right)^{1/2}}$$

for (doing the integration by hand)

$$(e^{-\alpha v^2})' = -2\alpha v e^{-\alpha v^2}$$
$$(v^2 e^{-\alpha v^2})' = -2\alpha v^3 e^{-\alpha v^2} + 2ve^{-\alpha v^2}$$
$$(v^2 e^{-\alpha v^2} + \frac{e^{-\alpha v^2}}{\alpha})' = -2\alpha v^3 e^{-\alpha v^2}$$

Particle Diffusion

pp. 399 of Kittel and Kroemer [2]

Consider a system.

One end in diffusive contact with reservoir at chemical potential μ_1

Other end in diffusive contact with reservoir at chemical potential μ_2

Constant temperature τ.

If $\mu_1 > \mu_2$, particle flow through system from reservoir 1 to reservoir 2; $1 \to 2$.

$n_i \equiv$ particle concentration in i

Take

$$\mathbf{j}_n = -D\text{grad}n \tag{731}$$

which is **Fick's law**, and where $D \equiv$ particle diffusion constant or **diffusivity**.

Mean free path l. Particles freely travel over l.

Assume in a collision at z, particles come into local equilibrium at local chemical potential $\mu(z)$, local concentration $n(z)$.

At z, particle flux density in positive z direction $\quad \frac{1}{2}n(z-l_z)\bar{c}_z$

particle flux density in negative z direction $\quad -\frac{1}{2}n(z+l_z)\bar{c}_z$

Note $n(z-l_z)$ is particle concentration at $z-l_z$

$$J_n^z = \frac{1}{2}\left[n(z-l_z) - n(z+l_z)\right]\bar{c}_z = -\frac{dn}{dz}\bar{c}_z l_z$$

where $\bar{c}_z = \bar{c}\cos\theta$

$\bar{l}_z = \bar{l}\cos\theta$

$$\langle \bar{c}_z l_z \rangle = \bar{c}\bar{l}\frac{\int_{\text{hemisphere}} \cos^2\theta dS}{\int_{\text{hemisphere}} dS} = \bar{c}\bar{l}\frac{\int_0^{\pi/2} d\theta \int_0^{2\pi} d\varphi \cos^2\theta \sin\theta d\theta}{\int_0^{\pi/2} d\theta \sin\theta \int_0^{2\pi} d\varphi} = \bar{c}\bar{l}\frac{1}{3}$$

Comparing with Fick's law,

$$J_n^z = \frac{-1}{3}\bar{c}\bar{l}\frac{dn}{dz} \text{ or } \mathbf{J}_N = -\frac{1}{3}\bar{c}\bar{l}\nabla n$$

For diffusivity D is then $D = \frac{1}{3}\bar{c}\bar{l}$.

Now recall that $\bar{l}$, the *mean free path*, was derived from kinetic theory:

$$l = \frac{1}{n\pi d^2}$$

where d is the diameter of the particle.

The mean thermal velocity was derived from the Maxwell distribution:

$$\bar{c} = \left(\frac{8\tau}{M\pi}\right)^{1/2}$$

Problems

Problem 1. Mean speeds in a Maxwellian distribution.

(a) $P(v) = 4\pi\left(\frac{M}{2\pi\tau}\right)^{3/2} v^2 \exp\left(\frac{-Mv^2}{2\tau}\right)$

Now

$$\int_0^\infty dv v^4 \exp\left(-\alpha v^2\right) = \int_0^\infty v^3\left(\frac{\exp\left(-\alpha v^2\right)}{-2\alpha}\right)' = 0 - \int 3v^2\frac{\exp\left(-\alpha v^2\right)}{-2\alpha} = \int_0^\infty \frac{3}{2\alpha}v^2\exp\left(-\alpha v^2\right) =$$

$$= \frac{3}{2\alpha}\int v\left(\frac{e^{-\alpha v^2}}{-2\alpha}\right)' = \frac{3}{2\alpha}\left[0 - \int\frac{e^{-\alpha v^2}}{-2\alpha}\right] = \frac{3}{4\alpha^2}\frac{\sqrt{\pi}}{2\sqrt{\alpha}}$$

So

$$\langle v^2\rangle = \int_0^\infty v^2 P(v) = 4\pi\left(\frac{M}{2\pi\tau}\right)^{3/2}\frac{3}{4\left(\frac{M}{2\tau}\right)^2}\frac{\sqrt{\pi}}{\sqrt{\frac{M}{2\tau}}}\frac{1}{2} = \frac{3}{\frac{M}{2\tau}}\frac{1}{2} = \frac{3\tau}{M}$$

so

$$\boxed{v_{rms} = \left(\frac{3\tau}{M}\right)^{1/2}}$$

Now

$$\langle v^2\rangle = 3\langle v_x^2\rangle = \frac{3\tau}{M} \text{ so } \langle v_x^2\rangle^{1/2} = \left(\frac{\tau}{M}\right)^{1/2}$$

(b)

$$\frac{\partial P(v)}{\partial v} = \frac{2P(v)}{v} + P(v)\left(\frac{-M}{\tau}\right)v = P(v)\left[\frac{2}{v} - \frac{M}{\tau}v\right] = 0 \text{ if } v_{mp} = 0 \text{ or } v_{mp} = \boxed{\sqrt{\frac{2\tau}{M}}}$$

where mp stands for most probable.

$$v_{mp} = \sqrt{\frac{2\tau}{M}} < \sqrt{\frac{3\tau}{M}} = v_{rms}$$

(c)

$$\bar{c} = \int_0^\infty dv v P(v) =: \langle v\rangle = \int_0^\infty dv 4\pi\left(\frac{M}{2\pi\tau}\right)^{3/2} v^3 \exp\left(\frac{-Mv^2}{2\tau}\right) = 4\pi\left(\frac{M}{2\pi\tau}\right)^{3/2}\int_0^\infty dv v^2\left(\frac{e^{-\frac{Mv^2}{2\tau}}}{\frac{-M}{\tau}}\right)' =$$

$$= 4\pi\left(\frac{M}{2\pi\tau}\right)^{3/2}\left[0 - \int_0^\infty dv 2v\frac{e^{-\frac{Mv^2}{2\tau}}}{\frac{-M}{\tau}}\right] = 4\pi\left(\frac{M}{2\pi\tau}\right)^{3/2}\left[\frac{2\tau}{M}\left(\frac{e^{-\frac{Mv^2}{2\tau}}}{-\frac{M}{\tau}}\right)\Bigg|_0^\infty\right] = 4\pi\left(\frac{M}{2\pi\tau}\right)^{3/2}\frac{2\tau^2}{M^2} =$$

$$= \frac{1}{(2\pi)^{3/2}}8\pi\sqrt{\frac{\tau}{M}} = \boxed{\sqrt{\frac{8\tau}{\pi M}}}$$

Note

$$\frac{v_{rms}}{\bar{c}} = \frac{\left(\frac{3\tau}{M}\right)^{1/2}}{\left(\frac{8\tau}{\pi M}\right)^{1/2}} = \left(\frac{3\pi}{8}\right)^{1/2}$$

(d) $|v_z| = |v\cos\theta| = v|\cos\theta|$. For for a fixed θ, $v\cos\theta$ is 1 point on a sphere, whereas $v|\cos\theta|$ could be 2 pts on a sphere.

Now

$$\int_0^\pi |\cos\theta|\sin\theta d\theta = \int_0^{\pi/2}\cos\theta\sin\theta d\theta + \int_{\pi/2}^\pi -\cos\theta\sin\theta d\theta = \int_0^{\pi/2}\frac{\sin 2\theta d\theta}{2} + \int_{\pi/2}^\pi\frac{-\sin 2\theta d\theta}{2} =$$

$$= \left(\frac{\cos 2\theta d\theta}{-4}\right)\Bigg|_0^{\pi/2} + \left(\frac{\cos 2\theta}{4}\right)\Bigg|_{\pi/2}^\pi = \frac{1}{2} + \frac{1+1}{4} = 1$$

$P(v)dv$ is probability that particle has speed or velocity magnitude in $(v, v+dv)$.
It counts the event of $v\cos\theta$ and $v\cos(\pi-\theta)$, so it counts $v|\cos\theta|$ twice. $P(v)dv = \int_0^{\pi/2} d\theta 2P(v|\cos\theta|)$

$$\Longrightarrow \bar{c}_z = \langle|v_z|\rangle = \langle|v\cos\theta|\rangle = \langle v|\cos\theta|\rangle = \int_0^\pi\int_0^\infty v\frac{P(v)dv}{2}|\cos\theta|d\theta = \left(\frac{2\tau}{\pi M}\right)^{1/2}$$

149 Propagation

Heat Conduction Equation

nonrelativistic case:

Let manifold $N = \mathbb{R} \times M$, with $\dim M = n$.

Let $\mathbf{J} \in \mathfrak{X}(N) = \mathfrak{X}(\mathbb{R} \times M)$ be a vector field in N.

Let $\rho \in C^\infty(N)$ be a smooth function on N.

$\rho = \rho(t, x)$ locally

Let $J \in \Omega^1(N)$ be a 1-form on N that is isomorphic to $\mathbf{J}$ (Tangent-Cotangent isomorphism theorem), i.e.

$$J = \mathbf{J}^\flat$$
$$J_i = g_{ij}\mathbf{J}^j$$

with g_{ij} being the metric on N (not just M!)

Note that as $N = \mathbb{R} \times M$, $g_{0j} = \delta_{0j}$

The local form of $\mathbf{J}$ is the following:

$$\mathbf{J} = \rho\frac{\partial}{\partial t} + \mathbf{j}^i\frac{\partial}{\partial x^i} \qquad i = 1 \dots n$$

So

$$J = J_i dx^i = g_{ij}\mathbf{J}^j dx^i = \rho dt + j_k dx^k \qquad k = 1 \dots n$$

Thus

$$J = \rho dt + j_k dx^k \qquad k = 1 \dots n \tag{732}$$

Now do the Hodge star operator, resulting in a n-form

$$*J \in \Omega^n(N)$$

and so

$$*J = \rho * dt + j_k * dx^k = \rho \mathrm{vol}^n + i_{\mathbf{j}}\mathrm{vol}^{n+1}$$

as

$$\rho * dt = \rho\frac{\sqrt{g}}{n!}\epsilon^0{}_{i_1 \dots i_n} dx^i_1 \wedge \dots \wedge dx^{i_n} \qquad i_1 \dots i_n \in \{1 \dots n\}$$
$$j_k * dx^k = g_{kl}j^l\frac{\sqrt{g}}{(n+1)!}\epsilon^k{}_{i_1 \dots i_n} dx^{i_1} \wedge \dots \wedge dx^{i_n} = i_{\mathbf{j}}\mathrm{vol}^{n+1}$$

so thus

$$*J = \rho \mathrm{vol}^n + i_{\mathbf{j}}\mathrm{vol}^{n+1}$$

Hence

$$d * J = \frac{\partial}{\partial t}(\rho\sqrt{g})\frac{1}{\sqrt{g}}\mathrm{vol}^{n+1} + \frac{\partial}{\partial x^k}(\sqrt{g}j_k)\frac{1}{\sqrt{g}}\mathrm{vol}^{n+1} = d(\rho\mathrm{vol}^n) + di_{\mathbf{j}}\mathrm{vol}^{n+1} \tag{733}$$

Special case: $\frac{\partial\sqrt{g}}{\partial t} = 0$

$$d * J = \frac{\partial\rho}{\partial t}\mathrm{vol}^{n+1} + \left(\frac{\partial}{\partial x^k}\ln\sqrt{g}\right)j_k\mathrm{vol}^{n+1} + \frac{\partial j_k}{\partial x^k}\mathrm{vol}^{n+1} = 0$$
$$\implies \frac{\partial\rho}{\partial t} + \left(\frac{\partial}{\partial x^k}\ln\sqrt{g}\right)j_k + \frac{\partial j_k}{\partial x^k} = 0$$

Special case: if $\sqrt{g}$ constant, $\frac{\partial\rho}{\partial t} + \frac{\partial j_k}{\partial x^k} = 0$

Let $j \equiv -D\mathbf{d}\rho$ (j is a closed form on M) where $\mathbf{d}\rho = \frac{\partial\rho}{\partial x^i}dx^i$ $\quad i = 1 \dots n$, D constant

$$d * J = d\rho\mathrm{vol}^n + -Dd * \mathrm{d}\rho = 0$$

For flat metric, $\sqrt{g}$ constant,

$$\frac{\partial \rho}{\partial t} = D \frac{\partial^2 \rho}{(\partial x^k)^2}$$

For relativistic case,
Consider manifold M, with dimension $\dim M = n+1$

$$\mathbf{J} = \rho \frac{\partial}{\partial t} + \mathbf{j}^i \frac{\partial}{\partial x^i}$$

Let $J = \mathbf{J}^\flat$. $J_\mu = g_{\mu\nu}\mathbf{J}^\nu$
For special case of flat Minkowski space,

$$J = -\rho dt + j_i dx^i$$

$$\begin{aligned} *J &= -\rho * dt + j_i * dx^i \\ &\quad - \rho \frac{\sqrt{g}}{(n+1)!} \epsilon^0{}_{i_1 \dots i_n} dx^{i_1} \wedge \cdots \wedge dx^{i_n} + j_i \frac{\sqrt{g}}{(n+1)!} \epsilon^i{}_{\mu_1 \dots \mu_n} dx^{\mu_1} \wedge \cdots \wedge dx^{\mu_n} \\ d * J &= -\frac{\partial(\rho\sqrt{g})}{\partial t} \frac{1}{\sqrt{g}} \mathrm{vol}^{n+1} + \frac{\partial(j_i \sqrt{g})}{\partial x^i} \frac{1}{\sqrt{g}} \mathrm{vol}^{n+1} = 0 \\ &\implies -\frac{\partial(\rho\sqrt{g})}{\partial t} + \frac{\partial(j_i\sqrt{g})}{\partial x^i} = 0 \end{aligned}$$

Fick law (14.19) for particle flux density, $j = -D_n \mathbf{d}n$ where D_n particle diffusivity constant
n particle concentration

$$J = n dt + j$$

thermal conductivity; homogeneous medium $\widehat{C}$ heat capacity per unit volume. $j_u = -K\mathbf{d}\tau$
$J = \widehat{C}\tau dt + j_u$

$$\widehat{C}\frac{\partial \tau}{\partial t} + \frac{\partial (j_u)_k}{\partial x^k} = 0 \qquad (5) \tag{734}$$

$$\frac{\partial \tau}{\partial t} = D_\tau \frac{\partial^2 \tau}{(\partial x^k)^2} \qquad D_\tau \equiv K/\widehat{C} \qquad (6) \tag{735}$$

Propagation of Sound Waves in Gases

pressure associated with sound wave

$$\delta p = \delta p_0 \exp\left[i(kx - \omega t)\right] \qquad (27) \tag{736}$$

Suppose ideal gas:

$$pV = N\tau \text{ or } p = \rho\tau/M \qquad (28) \tag{737}$$

Consider "solid ball" or "billiard ball" particle (extended particle, not pt. particle, but no internal structure)

$$\rho = \frac{NM}{V}$$

Force on particle

$$F = \frac{dP}{dt} = \frac{d}{dt}\int \rho \mathrm{vol}^n u = \frac{M}{V}\int \mathcal{L}_{\frac{\partial}{\partial t}+\mathbf{u}}\mathbf{u} = \frac{M}{V}\int \frac{\partial u}{\partial t} + [u,u]$$

Suppose $[u,u] = 0$ (certainly for flat spaces; what about for curved spaces? $[u,u] \neq 0$? Possibly? I don't know. EY: 20150317

$$dU + pdV = \tau d\sigma \tag{738}$$

define fractional deviations s, θ

$$\begin{aligned} \rho &= \rho_0(1+s) \\ \tau &= \tau_0(1+\theta) \end{aligned} \qquad (5) \tag{739}$$

where ρ_0, τ_0 are density and temperature in absence of sound wave.
assume u, s, θ have form of traveling $\exp[i(kx - \omega t)]$

$$\omega u - \left(\frac{k\tau}{M}\right) s - \left(\frac{k\tau}{M}\right)\theta = 0 \qquad (39) \tag{740}$$

$$\omega s - ku = 0 \qquad (40) \tag{741}$$

$$\tau \widehat{C}_V \theta - ps = 0 \text{ or } \widehat{C}_V \theta - ns = 0 \qquad (41) \tag{742}$$

$$\omega u = \frac{k\tau}{M}\left(1 + \frac{n}{\widehat{C}_V}\right) s$$

$$\omega = \frac{k\tau}{M}\left(1 + \frac{n}{\widehat{C}_V}\right)\frac{k}{\omega}$$

So

$$\omega = \left(\frac{\gamma\tau}{M}\right)^{1/2} k \qquad (42) \tag{743}$$

$$\gamma = \frac{\widehat{C}_V + n}{\widehat{C}_V} = \frac{\widehat{C}_p}{\widehat{C}_V}$$

$$v_s = \frac{\partial \omega}{\partial k} = \left(\frac{\gamma\tau}{M}\right)^{1/2}$$

Problems

Problem 1. Fourier analysis of pulse

$t = 0$

$$\theta(x,0) = \delta(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} dk \exp(ikx) \qquad (58) \tag{744}$$

$$\theta(x,t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} dk \exp[i(kx - \omega t)] \qquad (59) \tag{745}$$

Given a dispersion relation at this form:

$$Dk^2 = i\omega \qquad (10) \tag{746}$$

$$\theta(x,t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} dk \exp[ikx - Dk^2 t] \qquad (60) \tag{747}$$

and so, doing the Gaussian integral,

$$\theta(x,t) = \frac{1}{\sqrt{4\pi Dt}} \exp\left(\frac{-x^2}{4Dt}\right) \qquad (14) \tag{748}$$

Problem Diffusion in two and three dimensions.

(a)

$$\frac{\partial \theta_2}{\partial t} = -\frac{\theta_2}{t} + \frac{r^2}{4}\frac{\theta_2}{Dt^2}$$

$$\frac{\partial^2 \theta_2}{(\partial x^i)^2} = -\frac{1}{2}\frac{\theta_2}{Dt} - \frac{1}{2}\frac{\theta_2}{Dt} + \frac{1}{4}\frac{\theta_2}{D^2 t^2}(x^2 + y^2)$$

$$\implies \frac{\partial \theta_2}{\partial t} = D\frac{\partial^2 \theta_2}{(\partial x^i)^2}$$

(b)

(c)

2015 manuscript.

150 Hydrodynamics

Another example: hydrodynamics; Eulerian description in fixed volume.

Define $\epsilon := \frac{E}{V} = \widehat{\epsilon} \cdot \left(\frac{N}{V}\right) \equiv \widehat{\epsilon} n$.

$$dE = \tau N d\widehat{\sigma} + \widehat{h} dN \xrightarrow{\frac{1}{V}} d\epsilon = \tau n d\widehat{\sigma} + \widehat{h} dn$$

in isentropic process, $\left(\frac{1}{2}\rho u^2 + h\right)\mathbf{u}$ is energy current; $h = \frac{H}{V}$, $\mathbf{u}$ velocity of fluid

2015 manuscript.

Part XXXIII
Nonequilibrium

151 Irreversible processes: macroscopic theory

Chapter 6 of Le Bellac, Mortessagne, Batrouni (2004) [9]

151.1 Flux, affinities, transport coefficients

Section 6.1 of Le Bellac, Mortessagne, Batrouni (2004) [9].

Assume system composed of homogeneous cells, small on macroscopic scale, but large on microscopic, labelled $(a, b, \dots)$.

Assume cells weakly interacting, so each cell independently attains local equilibrium with microscopic relaxation time τ_{micro}

Assume $\tau_{\text{micro}} \ll \tau_{\text{macro}}$ where

τ_{macro} macroscopic relaxation time, for global equilibrium

151.1.1 Affinities and transport coefficients

Suppose neighboring cells a, b have different intensive variables γ_i, an exchange of A_i will take place between them.

Recall, from the Gibbs-Duhem relation, 716,

$$N d\mu + \sigma d\tau - V dp = 0$$

For the $d\tau = 0$ isothermal case, define

$$n := \frac{N}{V} \equiv \text{ particle density} \tag{749}$$

Then

$$\begin{gathered} dp = n d\mu \\ \Longrightarrow \boxed{\left(\frac{\partial p}{\partial \mu}\right)_\tau = n} \end{gathered} \tag{750}$$

Recall

151.1.2 Dissipation and Entropy Production

cf. Subsection 6.1.5 of Le Bellac, Mortessagne, Batrouni (2004) [9].

$$\frac{dS(a)}{dt} + \sum_{b\neq a} \Phi_S(a\to b) = \frac{1}{2}\sum_{i,b\neq a} \Gamma_i(a,b)\Phi_i(a\to b) \tag{751}$$

since

$$\frac{dS(a)}{dt} = -\sum_{i,b\neq a}\frac{1}{2}(\gamma_i(a)+\gamma_i(b))\Phi_i(a\to b) + \frac{1}{2}\sum_{i,b\neq a}\Gamma_i(a,b)\Phi_i(a\to b)$$

and since

$$\Phi_S(a\to b) = \frac{1}{2}\sum_i(\gamma_i(a)+\gamma_i(b))\Phi_i(a\to b) = -\Phi_S(b\to a) \tag{752}$$

Calculate $\frac{dS_{\rm tot}}{dt}$, evolution of total entropy.

$$\frac{dS_{\rm tot}}{dt} = \frac{d}{dt}\sum_a S(a) = \sum_a \frac{dS(a)}{dt}$$

Note that

$$\Phi_S(b\to a) + \Phi_S(a\to b) = 0$$

by antisymmetry of $\Phi_S(a\to b)$.

The $\sum_a\sum_{b\neq a}$ is the summation of all the ways of making a size 2 ordered choice of 2 elements out of the possible cells $\{a,b,\dots\}$; if there are n total cells, then there are $n(n-1)$ ways of picking out 2, *in order*. Note that $n(n-1)$ *must* be even. Then

$$\sum_a\sum_{b\neq a}\Phi_S(a\to b) = \frac{1}{2}\sum_a\sum_{b\neq a}\Phi_S(a\to b)+\frac{1}{2}\sum_b\sum_{a\neq b}\Phi_S(b\to a) = \frac{1}{2}\sum_a\sum_{b\neq a}\Phi_S(a\to b)+\frac{1}{2}\sum_b\sum_{a\neq b}-\Phi_S(a\to b) = 0$$

So Φ_S doesn't contribute to $\frac{dS_{\rm tot}}{dt}$: $\implies$ *reversible exchange*

$$\frac{dS_{\rm tot}}{dt} = \frac{1}{2}\sum_a\sum_{i,b\neq a}\Gamma_i(a,b)\Phi_i(a\to b)$$

entropy production, called *dissipation*; all physical phenomena accompanied by entropy production called *dissipative*.

151.1.3 Particle Diffusion

cf. pp. 399 of Kittel and Kroemer [2]

Consider a system.

One end in diffusive contact with reservoir at chemical potential μ_1

Other end in diffusive contact with reservoir at chemical potential μ_2

Constant temperature τ.

If $\mu_1 > \mu_2$, particle flow through system from reservoir 1 to reservoir 2; $1\to 2$.

$n_i \equiv$ particle concentration in i

Take

$$\mathbf{j}_n = -D\text{grad}n \tag{753}$$

which is **Fick's law**, and where $D \equiv$ particle diffusion constant or **diffusivity**.

Mean free path l. Particles freely travel over l.

Assume in a collision at z, particles come into local equilibrium at local chemical potential $\mu(z)$, local concentration $n(z)$.

At z, particle flux density in positive z direction $\quad \frac{1}{2}n(z-l_z)\bar{c}_z$

particle flux density in negative z direction $\quad -\frac{1}{2}n(z+l_z)\bar{c}_z$

Note $n(z-l_z)$ is particle concentration at $z-l_z$

$$J_n^z = \frac{1}{2}\left[n(z-l_z)-n(z+l_z)\right]\bar{c}_z = -\frac{dn}{dz}\bar{c}_z l_z$$

where $\overline{c}_z = \overline{c}\cos\theta$

$\overline{l}_z = \overline{l}\cos\theta$

$$\langle \overline{c}_z l_z \rangle = \overline{c}\overline{l}\frac{\int_{\text{hemisphere}} \cos^2\theta dS}{\int_{\text{hemisphere}} dS} = \overline{c}\overline{l}\frac{\int_0^{\pi/2} d\theta \int_0^{2\pi} d\varphi \cos^2\theta \sin\theta d\theta}{\int_0^{\pi/2} d\theta \sin\theta \int_0^{2\pi} d\varphi} = \overline{c}\overline{l}\frac{1}{3}$$

Comparing with Fick's law,

$$J_n^z = \frac{-1}{3}\overline{c}\overline{l}\frac{dn}{dz} \text{ or } \mathbf{J}_N = -\frac{1}{3}\overline{c}\overline{l}\nabla n$$

For diffusivity D is then $D = \frac{1}{3}\overline{c}\overline{l}$.

Now recall that $\overline{l}$, the *mean free path*, was derived from kinetic theory:

$$l = \frac{1}{n\pi d^2}$$

where d is the diameter of the particle.

The mean thermal velocity was derived from the Maxwell distribution:

$$\overline{c} = \left(\frac{8\tau}{M\pi}\right)^{1/2}$$

151.1.4 Momentum conservation

Introduce *stress tensor* $\mathcal{P}_{\alpha\beta}$, $-\mathcal{P}_{\alpha\beta}$ is α component of force per unit area applied by fluid outside parallelopiped on surface $\mathcal{S}_\beta$.

151.1.5 Hydrodynamics of the perfect fluid

cf. Subsection 6.5.2 Hydrodynamics of the perfect fluid, of Le Bellac, Mortessagne, Batrouni (2004) [9].

Consider a fluid where the only internal force is pressure. Assume there's no thermal conduction. In such a fluid, dissipation is absent. The fluid is a so-called "perfect fluid." Then right hand side for these transport equations (Eq. (6.87), (6.88) of Le Bellac, Mortessagne, Batrouni (2004) [9] is 0:

$$\mathbf{j}'_E = L_{EE}\nabla\frac{1}{T}$$
$$\mathcal{P}_{\alpha\beta} - \delta_{\alpha\beta}\mathcal{P} = -\xi\delta_{\alpha\beta}(\partial_\gamma u_\gamma) - 2\eta\Delta_{\alpha\beta}$$

1. Consider the force acting on a fluid volume element $\text{vol}^n \in \Omega^n(N)$, where spatial manifold has dimension n, i.e. $\dim N = n$.

   The **total force** applied by rest of fluid in fluid enclosed in parallelpipe (generalized to V), integrate over ∂V (all forces)

   $$-\int_{\partial V} \mathcal{P}_{\alpha\beta} dx^\alpha \otimes dS^\beta = -\int_{\partial V} \mathcal{P}_{\alpha\beta} dS^\beta \otimes dx^\alpha = -\int_V d(\mathcal{P}_{\alpha beta} dS^\beta) \otimes dx^\alpha = -\int_V \frac{\partial(\mathcal{P}_{\alpha\beta}\sqrt{g})}{\partial x^\beta}\frac{1}{\sqrt{g}}\text{vol}^n \otimes dx^\alpha$$

   Now consider the "left-hand side" which describes the dynamics (or kinematics?). For *total* momentum Π

   $$\Pi = \int_V \rho u_i dx^i \otimes \text{vol}^n$$

   $$\frac{d\Pi}{dt} = \dot{\Pi} = \frac{d}{dt}\int_V \rho u_i \text{vol}^n \otimes dx^i = \int_V \frac{\partial(\rho u_i)}{\partial t}\text{vol}^n \otimes dx^i + \int_V \mathcal{L}_{\mathbf{u}}\rho u_i \text{vol}^n \otimes dx^i = \int_V \frac{\partial(\rho u_i)}{\partial t}\text{vol}^n \otimes dx^i + \int_V \mathbf{d}(\rho u_i u^j dS_j)$$

   Now

   $$\mathbf{d}(\rho u_i u^j dS_j) = \frac{\partial\rho}{\partial x^j}u^j u_i \text{vol}^n + \rho\left[\frac{\partial u_i}{\partial x^j}u^j \text{vol}^n + u_i\frac{\partial u^j}{\partial x^j}\text{vol}^n\right]$$

   where, recalling

   $$dS_j = \frac{\sqrt{g}}{(n-1)!}\epsilon_{ji_2\ldots i_n} dx^{i_2} \wedge \cdots \wedge dx^{i_n}$$

then

$$dx^k \wedge dS_j = \frac{\sqrt{g}}{(n-1)!}\epsilon_{ji_2\ldots i_n} dx^k \wedge dx^{i_2} \wedge \cdots \wedge dx^{i_n} = \frac{\sqrt{g}}{(n-1)!}\epsilon_{ji_2\ldots i_n}\epsilon^{ki_2\ldots i_n}_{12\ldots n}\frac{dx^1 \wedge \cdots \wedge dx^n}{n} = \delta^k_j \mathrm{vol}^n$$

2. Consider the *lab frame* (with unprimed notation).
Consider the frame *at rest in the fluid's frame* (with primed notation).

Note that

$$\frac{d}{dt}\int_V \mathrm{vol}^n = \int_V \mathcal{L}_{\frac{\partial}{\partial t}+\mathbf{u}}\mathrm{vol}^n = \int_V 0 + \mathbf{d}i_{\mathbf{u}}\mathrm{vol}^n + 0 = \int_{\partial V} i_{\mathbf{u}}\mathrm{vol}^n = \int_{\partial V} u^i dS_i$$

Let $s \equiv S/V \equiv \overline{\sigma} \equiv \frac{\sigma}{V}$ entropy per unit volume in fluid's frame
$h' \equiv H/V$ enthalpy per unit volume in *fluid's frame.*

Note that $s = s'$ since entropy is Galilean invariant.

Now

$$\check{s} \equiv S/N \equiv \check{\sigma} \equiv \sigma/N$$
$$\check{h}' \equiv H/N$$

Let's review Subsubsection 142.0.1, which is on pp.137 at the end of Section 3.5.1. Basic formulae of Le Bellac, Mortessagne, Batrouni (2004) [9].

151.1.6 Conduction and Convection

Consider system at equilibrium, with thermostate at temperature T, kept at pressure p, able to exchange particles, with reservoir at chemical potential μ.

For 1 kind of particle, let

$$\check{s} \equiv S/N \text{ entropy per particle}$$
$$\check{\epsilon} \equiv E/N \text{ energy per particle}$$

Thus

$$\tau d\sigma = \tau d(N\check{\sigma}) = \tau N d\check{\sigma} + \tau\check{\sigma} dN$$

where

$\tau N d\check{\sigma}$ is entropy change due to change in entropy per particle, i.e. **conduction term**
$\tau\check{\sigma} dN$ is entropy change due to change in number of particles, i.e. **convection term**

For constant volume,

$$dE = TNd\check{\sigma} + (T\check{\sigma} + \mu)dN = TNd\check{\sigma} + \check{h}dN$$

where $\check{h} = \overline{H}/N = \check{\epsilon} + Pv$ is enthalpy per particle, since, recall,

$$\overline{H}(S,P) = E + PV$$
$$\mu N = E - TS + PV = G$$

physical interpretation of $\check{h}dN$: if dN particles transported into a given volume by convection, energy increases by $\check{\epsilon}dN$.

But to return to initial volume, it's necessary to compress by vdN and so add energy to do work $PvdN$ by pressure.

Example: hydrodynamics, where 1 uses Eulerian description in fixed volume.

Define energy density $\epsilon = \frac{E}{V} = \check{\epsilon}n$.

$$d\epsilon = Tnd\check{s} + \check{h}dn$$

separates conduction and convection terms.

To reiterate, in my notation,

$$H = E + pV \xrightarrow{1/N} \check{h} = \check{\epsilon} + pV$$
$$G = F + pV = E - \tau\sigma + pV = \mu N = H - \tau\sigma \Longrightarrow \mu = \check{h} - \tau\check{\sigma}$$
$$dE = Q + W + \mu dN \Longrightarrow dE = \tau d\sigma - pdV + \mu dN$$
$$= \tau N d\check{\sigma} + \tau\check{\sigma} dN - pdV + \mu dN = \tau N d\check{\sigma} + (\tau\check{\sigma} + \mu) dN = \tau N d\check{\sigma} + \check{h} dN$$

where the process considered was for constant volume, and so $dV = 0$, and one must keep in mind that the general form of the (internal) energy is as a function of $\sigma, V, \{N_i\}$, so that

Correction (enthalpy per particle, again). The same per-particle enthalpy is $\check{\epsilon} + pv$, with $v = V/N$.

$$dE = \tau d\sigma - pdV + \sum_i \mu_i dN_i$$

152 Resolutions

Each item states the correct relation, the hypotheses, and the manuscript line it replaces. The manuscript text is still in the preceding sections.

152.1 Ideal-gas internal energy, and the adiabatic path

The 2015 notes ask why, on an adiabatic path, one may write $dU = C_V\, d\tau$ and drop $(\partial U/\partial V)_\tau\, dV$. They also write, in that question, $dU = C_V \tau$. The differential is $C_V\, d\tau$. The product $C_V \tau$ is not that differential.

Start from (595) at fixed N,

$$dU = \tau\, d\sigma - p\, dV. \tag{754}$$

Take $U = U(\tau, V)$ on the equilibrium surface, so

$$dU = \left(\frac{\partial U}{\partial \tau}\right)_V d\tau + \left(\frac{\partial U}{\partial V}\right)_\tau dV. \tag{755}$$

The definition $C_V = (\partial U/\partial\tau)_V$ turns the first coefficient into C_V. It remains to compute the second. From $F = U - \tau\sigma$ and $dF = -\sigma\, d\tau - p\, dV$,

$$\left(\frac{\partial \sigma}{\partial V}\right)_\tau = \left(\frac{\partial p}{\partial \tau}\right)_V. \tag{756}$$

Hold τ fixed in (754) and solve for the volume derivative:

$$\left(\frac{\partial U}{\partial V}\right)_\tau = \tau\left(\frac{\partial \sigma}{\partial V}\right)_\tau - p = \tau\left(\frac{\partial p}{\partial \tau}\right)_V - p. \tag{757}$$

The second equality is (756). This identity is general at fixed N. It has not used an ideal-gas law and it has not used $Q = 0$.

Now restrict to a classical ideal gas,

$$p = \frac{N\tau}{V}. \tag{758}$$

Then $(\partial p/\partial\tau)_V = N/V = p/\tau$, and (757) collapses:

$$\left(\frac{\partial U}{\partial V}\right)_\tau = \tau \cdot \frac{p}{\tau} - p = 0. \tag{759}$$

So $U = U(\tau)$ only. Equation (755) becomes

$$dU = C_V(\tau)\, d\tau \tag{760}$$

on every path on the ideal-gas surface, adiabatic or not. The term that the 2015 question could not place is zero because of (758), before $Q = 0$ is imposed.

The reversible heat is then

$$Q = dU + p\,dV = C_V\,d\tau + \frac{N\tau}{V}\,dV. \tag{761}$$

An adiabatic reversible path is $Q = 0$. If C_V is independent of τ, which is the classical gas with a temperature-independent heat capacity,

$$\frac{d\tau}{\tau} + \frac{N}{C_V}\frac{dV}{V} = 0. \tag{762}$$

Integrate from an initial state to a final state:

$$\ln\frac{\tau_f}{\tau_i} + \frac{N}{C_V}\ln\frac{V_f}{V_i} = 0, \qquad \tau\,V^{N/C_V} = \text{constant}. \tag{763}$$

The difference $C_p - C_V$ uses the same two facts, $U = U(\tau)$ and (758), and does not need the monatomic value of C_V. The enthalpy is $H = U + pV = U + N\tau$, so

$$C_p = \left(\frac{\partial H}{\partial \tau}\right)_p = \left(\frac{\partial U}{\partial \tau}\right)_p + N = C_V + N, \tag{764}$$

where $(\partial U/\partial\tau)_p = (\partial U/\partial\tau)_V$ because U does not depend on V or on p. Therefore

$$\frac{N}{C_V} = \frac{C_p - C_V}{C_V} = \gamma - 1, \qquad \gamma = \frac{C_p}{C_V}, \tag{765}$$

and $\tau V^{\gamma-1}$ is constant. For a monatomic gas in three dimensions, equipartition gives $C_V = 3N/2$ and $\gamma = 5/3$. For a diatomic gas with $C_V = 5N/2$, the same $C_p - C_V = N$ holds and $\gamma = 7/5$. The 2015 question about molecules other than monatomic ones is this split: $C_p - C_V = N$ is the ideal-gas law, while the value of C_V is the internal spectrum.

The Carnot-cycle notes ask why C_V, whose definition holds V fixed, may be used on the adiabatic leg $2 \to 3$, where V changes. The subscript V names the variable held fixed in the partial derivative that defines the state function $C_V(\tau)$. It does not name a constraint on the path. Once (759) is in force, (760) is the differential of U along the adiabatic leg and along every other leg.

152.2 Path dependence is the failure of $dQ = 0$

The 2008 notes say that a reversible process is one whose 1-forms are exact, and later that Q and W are not necessarily exact, so the state space would have to be topologically nontrivial. Those two sentences cannot stand together with $Q = \tau\,d\sigma$.

The energy U is a state function, so dU is exact. On a reversible path the heat 1-form is defined to be

$$Q = \tau\,d\sigma. \tag{766}$$

Its exterior derivative, with τ and σ treated as coordinates, or as functions whose differentials are known, is

$$dQ = d(\tau\,d\sigma) = d\tau \wedge d\sigma. \tag{767}$$

The 2-form $d\tau \wedge d\sigma$ vanishes when τ is a function of σ alone. In the (τ, V) chart it does not vanish: $\sigma = \sigma(\tau, V)$ and C_V is not the whole story of Q, because of the second term in (761). So Q fails to be closed. On an open set of $\mathbb{R}^2$, which is contractible, the Poincaré lemma says that a closed 1-form is exact. The contrapositive used here is local: a 1-form that is not closed is not exact, on a domain with no interesting topology. Path dependence of the reversible heat is $dQ \neq 0$. It is already visible for an ideal gas in the (τ, V) plane.

The work 1-form $W_{\text{by}} = p\,dV$ is likewise not closed once $p = p(\tau, V)$. Their sum $Q - W_{\text{by}} = dU$ is exact. Exactness belongs to dU, including on irreversible processes when U is evaluated between equilibrium endpoints. It is not the definition of a reversible path.

152.3 Work sign in the heat-and-work notes

One line in the heat-and-work notes writes $dU = Q + W = \tau\,d\sigma + p\,dV$. The identity in force everywhere else in that discussion is (754), and the Carnot calculation a few lines later uses $\oint dU = 0 = \oint \tau\,d\sigma - \oint p\,dV$. The plus sign on $p\,dV$ in that one line is the slip. The Carnot algebra that follows it uses the minus sign and is not changed by the slip.

152.4 Derivative of the Helmholtz free energy

From $F = U - \tau\sigma$ and (754),

$$dF = -\sigma\, d\tau - p\, dV \tag{768}$$

at fixed N, so

$$\left(\frac{\partial F}{\partial \tau}\right)_V = -\sigma. \tag{769}$$

The heat-and-work notes record (769). The gas–solid solution in the phase chapter states the opposite, $(\partial F/\partial\tau)_V = \sigma$, and then differentiates F_s. The entropy computed there is $-\sigma$ under (769). That solution then drops the solid entropy by the assumption stated in the problem, and the boxed N_g and p_g come from minimizing F in N_g:

$$F = -\epsilon_0(N - N_g) + N_g\tau\left[\ln\left(\frac{N_g}{V n_Q}\right) - 1\right],$$
$$\left(\frac{\partial F}{\partial N_g}\right) = \epsilon_0 + \tau \ln\left(\frac{N_g}{V n_Q}\right). \tag{770}$$

The τ terms $-\tau + \tau$ from the product rule cancelled. Setting the derivative to zero gives $N_g = V n_Q e^{-\epsilon_0/\tau}$, and $p_g = N_g\tau/V$ gives the boxed pressure. Those two boxed results do not use the flipped entropy identity.

152.5 Enthalpy per particle

The extensive enthalpy is $H = U + pV$. Divide by N:

$$\hat{h} = \frac{H}{N} = \hat{\epsilon} + pv, \qquad \hat{\epsilon} = \frac{U}{N}, \qquad v = \frac{V}{N}. \tag{771}$$

The sum $\hat{\epsilon} + pV$ adds an energy per particle to an energy. The 2015 definition writes that sum, and the later reminder writes it again as $\check{h} = \check{\epsilon} + pV$. The line $\hat{h}\, dN = (\hat{\epsilon} + pv)\, dN$, in the same subsection, has the volume right. Dimensions select (771). The conduction term $\tau N\, d\hat{\sigma}$ and the convection term $\hat{h}\, dN$ then add as energies. This is the comparison the 2019 note in the differential-geometry dump asked for.

152.6 Joule–Thomson sign

The 2015 derivation of the coefficient is kept. With $\alpha = V^{-1}(\partial V/\partial\tau)_p$ and $(\partial\sigma/\partial p)_\tau = -(\partial V/\partial\tau)_p$ from $dG = -\sigma\, d\tau + V\, dp$,

$$dH = C_p\, d\tau + V(1 - \tau\alpha)\, dp. \tag{772}$$

At fixed H,

$$\left(\frac{\partial \tau}{\partial p}\right)_H = -\frac{V(1 - \tau\alpha)}{C_p}. \tag{773}$$

For the ideal gas, $\alpha = 1/\tau$, so the coefficient vanishes and an isenthalpic expansion does not change τ.

Below the inversion temperature one has $\tau\alpha > 1$, so (773) is positive. An expansion has $dp < 0$, and therefore $d\tau < 0$: the gas cools. The 2015 sentence for that side agrees.

Above the inversion temperature, $\tau\alpha < 1$, so (773) is negative. The same expansion has $dp < 0$, and the product is

$$d\tau = \left(\frac{\partial \tau}{\partial p}\right)_H dp > 0. \tag{774}$$

The gas warms. The 2015 sentence for that side says both $dT < 0$ and that the gas warms. The warming conclusion matches (774). The clause $dT < 0$ does not.

The inversion curve itself, where the coefficient vanishes, is $\tau\alpha = 1$. For a van der Waals gas that locus is a curve in the (τ, p) plane. Equation (773) does not by itself produce the numerical inversion temperature; a model for $\alpha(\tau, p)$ does. The 2015 note already says that.

152.7 Rotational factor in the ideal-gas summary

In the 2015 summary the energy of translation plus rotation is written correctly as

$$\epsilon_{\mathbf{n},j} = \frac{\hbar^2}{2M}\left(\frac{\pi}{L}\right)^2 \mathbf{n}^2 + j(j+1)\epsilon_0, \tag{775}$$

and then the Boltzmann factor is written with 2π in place of $2M$ in the translational piece and with $e^{j(j+1)\epsilon_0}$ in place of the rotational factor. The factor required by (775) is

$$\exp(-\epsilon_{\mathbf{n},j}/\tau) = \exp\left(-\frac{\hbar^2\pi^2\mathbf{n}^2}{2ML^2\tau}\right)\exp\left(-\frac{j(j+1)\epsilon_0}{\tau}\right). \tag{776}$$

One line in that summary also sets the power on $M\tau/(2\pi\hbar^2)$ equal to 2. The quantum concentration in three dimensions is the power 3/2, as in (697).

The later replacement of the rotational sum by τ/ϵ_0 is the high-temperature integral, and it is consistent with the minus sign in (776). For a linear rotor, $g(j) = 2j+1$. When $\epsilon_0 \ll \tau$ the sum over j is the integral

$$\sum_{j=0}^{\infty}(2j+1)\exp\left(-\frac{j(j+1)\epsilon_0}{\tau}\right) \approx \int_0^{\infty}(2j+1)\exp\left(-\frac{j(j+1)\epsilon_0}{\tau}\right)dj. \tag{777}$$

Set $y = j(j+1)$. Then $dy = (2j+1)\,dj$, and

$$\int_0^{\infty} e^{-y\epsilon_0/\tau}\,dy = \frac{\tau}{\epsilon_0}. \tag{778}$$

The hypothesis $\epsilon_0 \ll \tau$ is what licenses the integral. The factorization $Z_1 = n_Q V\, Z_{\text{int}}$ in that summary survives; the displayed exponential between (775) and (778) is the line to replace by (776).

152.8 Maxwell speed distribution

The kinetic-theory notes contain an intermediate reduction whose right-hand side is $4\pi N(N/(2\pi\tau))^{3/2}$. That expression does not have the dimension of the prefactor of a speed distribution, and it is not the prefactor that appears in the final formula. The final formula is the Maxwell speed density

$$P(v) = 4\pi\left(\frac{M}{2\pi\tau}\right)^{3/2} v^2 \exp\left(-\frac{Mv^2}{2\tau}\right), \qquad v \geq 0. \tag{779}$$

It is normalized. Set $\alpha = M/(2\tau)$. The Gaussian integrals

$$\int_{-\infty}^{\infty} e^{-\alpha v^2}\,dv = \sqrt{\frac{\pi}{\alpha}}, \qquad \int_{-\infty}^{\infty} v^2 e^{-\alpha v^2}\,dv = \frac{1}{2}\sqrt{\frac{\pi}{\alpha^3}} \tag{780}$$

give, because $v^2 e^{-\alpha v^2}$ is even,

$$\int_0^{\infty} v^2 e^{-\alpha v^2}\,dv = \frac{1}{4}\sqrt{\frac{\pi}{\alpha^3}}. \tag{781}$$

Then

$$\begin{aligned}\int_0^{\infty} P(v)\,dv &= 4\pi\left(\frac{\alpha}{\pi}\right)^{3/2}\cdot\frac{1}{4}\sqrt{\frac{\pi}{\alpha^3}} \\ &= \pi\cdot\frac{\alpha^{3/2}}{\pi^{3/2}}\cdot\frac{\pi^{1/2}}{\alpha^{3/2}} = 1.\end{aligned} \tag{782}$$

The mean-speed solution that follows (779) in the manuscript is using this normalized density. The intermediate prefactor line is not an input to that solution.

152.9 Donor arithmetic

The preserved donor solution reports $n_Q = 1263\,\mathrm{cm}^{-3}$ and $n = 2 \times 10^{-18}\,\mathrm{cm}^{-3}$ at $T = 100\,\mathrm{K}$, $m^* = 0.3\,m_e$, and $N_d = 10^{17}\,\mathrm{cm}^{-3}$. Section 140 evaluates (698) and the hydrogenic binding energy (708) at $\epsilon_r = 11.7$, and it solves the quadratic (705). The result is $n = 2.965 \times 10^{16}\,\mathrm{cm}^{-3}$.

152.10 A numerical check that already matched

The adiabatic compression in the 2015 background uses $\gamma = 1.4$, $V_i/V_f = 15$, and $\tau_i = 293.15\,\mathrm{K}$, which is 20°C on the Celsius shift used in `Propulsion/thermo.py`. Equation (763) gives

$$\tau_f = \tau_i \left(\frac{V_i}{V_f} \right)^{\gamma - 1} = 293.15 \times 15^{0.4} = 866.017\,\mathrm{K}. \tag{783}$$

The pressure ratio at fixed N is

$$\frac{p_f}{p_i} = \frac{V_i}{V_f}\frac{\tau_f}{\tau_i} = 15^{1.4} = 44.313. \tag{784}$$

Both match the values written in that solution, 866.01 K and 44.31.

152.11 Left blank on purpose

Radiation solution 16 is absent from the 2008 file. Parts (b) and (c) of the last propagation problem are empty and the problem statement is not in the file. Those three slots were not filled with a guessed question. Entropy of mixing, which was an empty heading, is §133.

Part XXXIV
Statistical Mechanics: Ising Model

153 Ising Model

153.1 Definition of Ising Model

cf. Wikipedia, "Ising model"

Consider set of lattice sites Λ, each with set of adjacent sites (e.g. **graph**) forming d-dim. lattice.
$\forall$ lattice site $k \in \Lambda$, $\exists$ discrete variable σ_k, s.t. $\sigma_k \in \{-1, 1\}$.
spin configuration $\equiv \sigma = (\sigma_k)_{k\in\Lambda}$ is an assignment of spin value to each lattice site.

i.e.

$d = 1$, consider "line" configuration: $i \in \mathbb{Z}$, $i = 0, 1, \dots L-1$. Lattice site $k \in \Lambda = \Lambda_{d=1}$. $\forall\, k \in \Lambda$, $\exists$ bijection to its index i, $k \mapsto i$, and $\exists\, \sigma_k$ i.e.

$$\sigma : \Lambda \leftrightarrow \sigma : \mathbb{Z} \to \mathbb{Z}_2$$
$$\sigma(k) \equiv \sigma_k \leftrightarrow \sigma(i) \equiv \sigma_i \mapsto \{-1, 1\}$$

spin configuration $\sigma : \Lambda \mapsto (\sigma_k)_{k\in\Lambda} \in \{-1, 1\}^{|\Lambda|}$, where $|\Lambda| = L$.
$\forall\, k \in \Lambda$, $\exists!$ only at most 2 edges, given, for $k \mapsto i$, $i+1, i-1$, $\forall\, i = 1 \dots L-2$.

$d = 2$, "rectangle" configuration. $(i, j) \in \mathbb{Z}^2$. $i \in 0, 1, \dots L_x - 1$. Lattice site $\mathbf{k} \in \Lambda = \Lambda_{d=2}$.
$j \in 0, 1, \dots L_y - 1$
$\forall\, \mathbf{k} \in \Lambda$, $\exists$ bijection to its "grid coordinates" (i, j), $\mathbf{k} \mapsto (i, j)$, and $\exists\, \sigma_{\mathbf{k}}$ i.e. $\sigma_{\mathbf{k}} = \sigma_{ij} \in \{-1, 1\}$.
spin configuration $\sigma : \Lambda \mapsto (\sigma_{\mathbf{k}})_{\mathbf{k}\in\Lambda} \in \{-1, 1\}^{|\Lambda|}$, where $|\Lambda| \equiv |\Lambda_{d=2}| = L_x L_y$.

$\forall\, \mathbf{k} \in \Lambda$, $\exists!$ only at most 4 edges, given by $\mathbf{k} \mapsto (i, j)$, $(i \pm 1, j), (i, j \pm 1)$, $i = 1 \dots L_x - 2$.
$j = 1 \dots L_y - 2$

Note that in both cases, I haven't yet defined the boundary conditions, and leave that to be discussed thoroughly in the future (i.e. following sections).

There are $2^{|\Lambda|}$ number of configurations in any dim. d.

cf. Wikipedia, "Ising model"

153.1.1 Interaction $J_{ij} \equiv J_{\mathbf{kl}}$, Hamiltonian (energy functional)for a configuration $H(\sigma)$

$\forall$ 2 adjacent (lattice) sites, $i, j \equiv \mathbf{k}, \mathbf{l} \in \Lambda$, let there be an interaction $J_{ij} \equiv J_{\mathbf{kl}}$ i.e. $J : \Lambda^2 \to \mathbb{R}$.
$J : (\mathbf{k}, \mathbf{l}) \mapsto J_{\mathbf{kl}}$

Adjacent means $\exists$ edge $\mathbf{k} \mapsto \mathbf{l}$ (the mapping is the edge)

Suppose $\forall$ site $j \equiv \mathbf{l} \in \Lambda$, $\exists$ external magnetic field $h_j \equiv h_{\mathbf{l}}$ interacting with it.

Given (site) configuration $\sigma : \Lambda \mapsto (\sigma_{\mathbf{k}})_{\mathbf{k} \in \Lambda} \in \{-1, 1\}^{|\Lambda|}$.

$$H(\sigma) = -\sum_{\langle ij \rangle} J_{ij} \sigma_i \sigma_j - \mu \sum_j h_j \sigma_j \equiv H(\sigma(\Lambda)) = -\sum_{\langle \mathbf{kl} \rangle} J_{\mathbf{kl}} \sigma_{\mathbf{k}} \sigma_{\mathbf{l}} - \mu \sum_{\mathbf{k} \in \Lambda} h_{\mathbf{k}} \sigma_{\mathbf{k}} \tag{785}$$

where $\sum_{\langle \mathbf{kl} \rangle}$ is overall pairs of adjacent spins (every pair is counted once),

$\langle \mathbf{k}, \mathbf{l} \rangle \equiv$ sites $\mathbf{k}, \mathbf{l}$ are nearest neighbors.

Note sign in 2nd. term, $-\mu \sum_{\mathbf{k}} h_{\mathbf{k}} \sigma_{\mathbf{k}}$ should be positive because of electron's magnetic moment is antiparallel to its spin, but negative term used conventionally.

Nothing was said about boundary conditions, I propose that it can be either fixed in the summation or by setting $J_{\mathbf{kl}} = 0$.

$\forall \mathbf{k} \in \Lambda$, let $\mathbf{y} : \Lambda \to E$, with $\{\langle \mathbf{k}, \mathbf{l} \rangle\}_{\mathbf{l}}$ be set of all edges from $\mathbf{k}$
$\mathbf{y} : \mathbf{k} \mapsto \{\langle \mathbf{k}, \mathbf{l} \rangle_{\mathbf{l}}$

Then clearly $\sum_{\langle \mathbf{kl} \rangle} = \frac{1}{2} \sum_{\mathbf{k} \in \Lambda} \sum_{\{\langle \mathbf{kl} \rangle\}_{\mathbf{l}}}$.

Taking into account only interaction between adjoining dipoles, on a square lattice:

$$E(\sigma) = -J \sum_{k,l=0}^{L-1} (\sigma_{kl} \sigma_{k,l+1} + \sigma_{kl} \sigma_{k+1,l})$$

cf. Landau and Lifshitz [51]

EY : 20171223 Things to check from Hjorth-Jensen (2015) [52]:

2-dim. Ising model, with $\mathcal{B} \equiv h_j = 0$, undergoes phase transition of 2nd. order: meaning below given critical temperature T_C, there's spontaneous magnetization with $\langle \mathcal{M} \rangle \equiv \langle \mathbf{M} \rangle \neq 0$. $\langle \mathbf{B} \rangle \to 0$ at T_C with *infinite* slope, a behavior called *critical phenomena*. Critical phenomenon normally marked by 1 or more thermodynamical variables which is 0 above a critical point. In this case, $\langle \mathbf{B} \rangle \neq 0$, such a parameter normally called *order parameter*.

Critical phenomena; we still don't have a satisfactory understanding of system's properties close to the critical point, even for simplest 3-dim. systems. Even mean-field models can predict wrong physics; mean-field theory results in a 2nd.-order phase transition for 1-dim. Ising model, wherea 1-dim. Ising model doesn't predict any spontaneous magnetization at any finite temperature T.

e.g. Consider 1-dim. N-spin system. Assume periodic boundary conditions. Consider state of all spins up, with total energy $-NJ$ and magnetization N. Flip half of spins (e.g. all spins of index $i > N/2$) so 1st half of spins point upwards and last half points downwards. Energy is $-NJ + 4J$, net magnetization 0. This is an example of a possible disordered state with net magnetization 0. Change in energy is too small to stabilize disordered state (to $-NJ$).

Definition 220 (configuration probability)**. configuration probability** $P_\beta(\sigma)$ given by Boltzmann distribution:

$$P_\beta(\sigma) = \frac{\exp(-\beta H(\sigma))}{Z_\beta} = \text{ prob. of configuration } \sigma \equiv \sigma(\Lambda) \equiv (\sigma_{\mathbf{k}})_{\mathbf{k} \in \Lambda} \tag{786}$$

with the partition function as normalization constant Z_β:

$$Z_\beta = \sum_\sigma \exp -\beta H(\sigma) \tag{787}$$

cf. pp. 504 Sec. 151 Phase transitions of the second kind in a 2-dim. lattice, Landau and Lifshitz [51]

$$Z = 2^N (1-x^2)^{-N} \coprod_{p,q=0}^{L-1} \left[(1+x^2)^2 - 2x(1-x^2) \left(\cos\frac{2\pi p}{L} + \cos\frac{2\pi q}{L} \right) \right]^{1/2} \tag{788}$$

cf. (151.11) of Landau and Lifshitz [51], where $x = \tanh\theta$, $\theta = J/T \equiv J/\tau = \beta J$.

$$\begin{aligned} \Phi \equiv F = -\tau \ln Z = \\ = -\tau N \ln 2 + \tau N \ln(1-x^2) - \frac{\tau}{2} \sum_{p,q=0}^{L} \ln \left[(1+x^2)^2 - 2x(1-x^2) \left(\cos\frac{2\pi p}{L} + \cos\frac{2\pi q}{L} \right) \right] \end{aligned} \tag{789}$$

Let $\omega_1 = \dfrac{2\pi p}{L}$ with $p \to 0$ as $L \to \infty$ so $\dfrac{L d\omega_1}{2\pi} = dp$ and using $L^2 = N$.
$\omega_2 = \dfrac{2\pi q}{L}$ with $q \to 0$ as $L \to \infty$ $\quad \dfrac{L d\omega_2}{2\pi} = dq$

$$\Phi = -\tau N \ln 2 + \tau N \ln(1-x^2) - \frac{N\tau}{2(2\pi)^2} \int_0^{2\pi} \int_0^{2\pi} d\omega_1 d\omega_2 \ln\left[(1-x^2) - 2x(1-x^2)(\cos\omega_1 + \cos\omega_2) \right]$$

$F \equiv \Phi$ has singularity when $(1-x^2) - 2x(1-x^2)(\cos\omega_1 + \cos\omega_2)$ in $\ln\left[(1-x^2) - 2x(1-x^2)(\cos\omega_1 + \cos\omega_2)\right]$. $(1-x^2) - 2x(1-x^2)(\cos\omega_1 + \cos\omega_2)$ minimized when $\cos\omega_1 = \cos\omega_2 = 1$ (since $-1 < x < 1$)

$$\implies (1+x^2)^2 - 4x(1-x^2) = 1 + 2x^2 + x^4 - 4x + 4x^3 = (x^2 + 2x - 1)^2 = 0 \implies x = \frac{-2 \pm \sqrt{4 - 4(-1)}}{2} = -1 + \sqrt{2}$$

$$x = \tanh\theta = \frac{e^\theta - e^{-\theta}}{e^\theta + e^{-\theta}} = \sqrt{2} - 1 \text{ or}$$

$$\begin{aligned} e^\theta - e^{-\theta} &= \sqrt{2} e^\theta + \sqrt{2} e^{-\theta} - e^\theta - e^{-\theta} \text{ so} \\ (2-\sqrt{2}) e^\theta &= \sqrt{2} e^{-\theta} \\ e^{2\theta} &= \frac{\sqrt{2}}{2-\sqrt{2}} \left(\frac{2+\sqrt{2}}{2+\sqrt{2}} \right) \text{ or} \\ 2\theta &= \ln(1+\sqrt{2}) \end{aligned}$$

$$\frac{J}{T_c} = \frac{1}{2} \ln(1+\sqrt{2}) \text{ or}$$

$$\boxed{\tau_c = \frac{2J}{\ln(1+\sqrt{2})}} \tag{790}$$

so that $\tau_C \equiv T_C$ is where phase transition occurs.

Let $t := \tau - \tau_c$. $\theta = \frac{J}{\tau} = \frac{J}{t+\tau_C}$

Expand about minimum

EY:20171230 do this explicitly

$$\int_0^{2\pi} \int_0^{2\pi} d\omega_1 d\omega_2 \ln\left[c_1 t^2 + c_2(\omega_1^2 + \omega_2^2) \right]$$

$$F \equiv \Phi \simeq a + \frac{1}{2} b(\tau - \tau_c)^2 \ln|\tau - \tau_c|$$

$$C = \frac{\partial^2 F}{\partial \tau} \simeq -b\tau_c \ln|\tau - \tau_c|$$

with C being heat capacity.

Order parameter $\langle M \rangle \equiv \eta = \text{constant}(\tau_c - \tau)^{1/8} = \begin{cases} 0 & \text{if } \tau > \tau_c \\ \text{constant } (\tau_c - \tau)^{1/8} & \text{if } \tau < \tau_c \end{cases}$

cf. pp. 505 Sec. 151 Phase transitions of the second kind in a 2-dim. lattice, Landau and Lifshitz [51], L.Onsager 1947.

153.2 An actual calculation of a small number of spins with Ising model

Sec. 3.7 "An actual calculation" on pp. 76 of Newman and Barkema (1999) [53] goes through a simple actual Monte Carlo calculation as a test case check so to compare this exact calculation/solution to the simulation, as a test of whether the simulation/program is correct. This is done in Sec. 1.3 of Newman and Barkema (1999) [53].

However, none of these promised simple calculations were shown explicitly in Newman and Barkema (1999) [53]. I will forego this simple case.

153.3 Explicit calculation showing stencil operation on each spin on a periodic lattice grid

Consider

$$H(\sigma) = -\sum_{\langle \mathbf{kl} \rangle} J\sigma_{\mathbf{k}}\sigma_{\mathbf{l}} = -J\sum_{i=0}^{L_x-1}\sum_{j=0}^{L_y-1}\sigma_{ij}(\sigma_{i+1j}+\sigma_{ij+1}) =$$

$$= \frac{-J}{2}\left(\sum_{i=0}^{L_x-1}\sum_{j=0}^{L_y-1}\sigma_{ij}(\sigma_{i+1j}+\sigma_{ij+1}) + \sum_{i=1}^{L_x}\sum_{j=0}^{L_y-1}\sigma_{i-1j}(\sigma_{ij}+\sigma_{i-1j+1})\right) =$$

$$= \frac{-J}{2}\left(\sum_{i=0}^{L_x-1}\sum_{j=0}^{L_y-1}\sigma_{ij}(\sigma_{i+1j}+\sigma_{ij+1}) + \sum_{i=1}^{L_x}\sum_{j=0}^{L_y-1}\sigma_{i-1j}\sigma_{ij} + \sum_{i=0}^{L_x-1}\sum_{j=1}^{L_y}\sigma_{ij-1}\sigma_{ij}\right)$$

Now for each of these terms,

$$\sum_{i=1}^{L_x}\sum_{j=0}^{L_y-1}\sigma_{i-1j}\sigma_{ij} = \sum_{i=1}^{L_x}\left(\sum_{j=1}^{L_y-1}\sigma_{i-1j}\sigma_{ij}+\sigma_{i-10}\sigma_{i0}\right) = \sum_{i=1}^{L_x-1}\left(\sum_{j=1}^{L_y-1}\sigma_{i-1j}\sigma_{ij}+\sigma_{i-10}\sigma_{i0}\right) + \left(\sum_{j=1}^{L_y-1}\sigma_{L_x-1j}\sigma_{L_xj}\right) + \sigma_{L_x-10}\sigma_{L_x0}$$

$$\sum_{i=0}^{L_x-1}\sum_{j=1}^{L_y}\sigma_{ij-1}\sigma_{ij} = \sum_{j=1}^{L_y-1}\left(\sum_{i=1}^{L_x-1}\sigma_{ij-1}\sigma_{ij}+\sigma_{0j-1}\sigma_{0j}\right) + \sum_{i=1}^{L_x-1}\sigma_{iL_y-1}\sigma_{iL_y} + \sigma_{0L_y-1}\sigma_{0L_y}$$

$$\sum_{i=0}^{L_x-1}\sum_{j=0}^{L_y-1}\sigma_{ij}(\sigma_{i+1j}+\sigma_{ij+1}) = \sum_{i=0}^{L_x-1}\left(\sum_{j=1}^{L_y}\sigma_{ij}(\sigma_{i+1j}+\sigma_{ij+1}) + \sigma_{i0}(\sigma_{i+10}+\sigma_{i1})\right) =$$

$$\sum_{i=1}^{L_x-1}\left(\sum_{j=1}^{L_y-1}\sigma_{ij}(\sigma_{i+1j}+\sigma_{ij+1}) + \sigma_{i0}(\sigma_{i+10}+\sigma_{i1})\right) + \sum_{j=1}^{L_y-1}\sigma_{0j}(\sigma_{1j}+\sigma_{0j+1}) + \sigma_{00}(\sigma_{10}+\sigma_{01})$$

Apply periodic boundary conditions. Adding up all the terms above, clearly we obtain 1 term which shows the stencil operation for spins on the "interior" of the grid:

$$\sum_{i=1}^{L_x-1}\sum_{j=1}^{L_y-1}\sigma_{ij}(\sigma_{i+1j}+\sigma_{ij+1}+\sigma_{i-1j}+\sigma_{ij-1})$$

and if we apply *periodic* boundary conditions, neatly, we'll see all the lattice sites at the boundary also will have this stencil operation:

$$\sum_{i=1}^{L_x-1}\sigma_{i0}(\sigma_{i+10}+\sigma_{i1}) + \sum_{j=1}^{L_y-1}\sigma_{0j}(\sigma_{1j}+\sigma_{0j+1}) + \sigma_{00}(\sigma_{10}+\sigma_{01}) + \left(\sum_{i=1}^{L_x-1}\sigma_{iL_y-1}\sigma_{i0}\right) + \sigma_{0L_y-1}\sigma_{00} + \sum_{j=1}^{L_y-1}\sigma_{0j-1}\sigma_{0j} +$$
$$+ \sum_{j=1}^{L_y-1}\sigma_{L_x-1j}\sigma_{0j} + \sigma_{L_x-10}\sigma_{00} + \sum_{i=1}^{L_x-1}\sigma_{i-10}\sigma_{i0}$$

Now, we can obtain the following for Hamiltonian, given spin configuration σ with a lattice grid obeying periodic conditions:

$$H(\sigma) = -\frac{J}{2}\sum_{i=0}^{L_x-1}\sum_{j=0}^{L_y-1}\sigma_{ij}(\sigma_{i+1j} + \sigma i - 1j + \sigma_{ij+1} + \sigma_{ij-1}) =$$

$$= \frac{-J}{2}\left[\sum_{i=0}^{L_x-1}\left(\sum_{\substack{j=0\\ j\neq j'}}^{L_y-1}\sigma_{ij}(\sigma_{i+1j} + \sigma_{i-1j} + \sigma_{ij+1} + \sigma_{ij-1}) + \sigma_{ij'}(\sigma_{i+1j'} + \sigma_{i-1j'} + \sigma_{ij'+1} + \sigma_{ij'-1})\right) + \right. \tag{791}$$

$$\left.\sum_{\substack{j=0\\ j\neq j'}}^{L_y-1}\sigma_{i'j}(\sigma_{i'+1j} + \sigma_{i'-1j} + \sigma_{i'j+1} + \sigma_{i'j-1}) + \sigma_{i'j'}(\sigma_{i'+1j'} + \sigma_{i'-1j'} + \sigma_{i'j'+1} + \sigma_{i'j'-1})\right]$$

Consider a psin flip of $\sigma_{i'j'}$. Contribution to ΔH at stencil operation on $\sigma_{i'j'}$, at $(i'j') \in \Lambda$, is

$$\frac{-J}{2}(-\sigma_{i'j'} - \sigma_{i'j'})(\sigma_{i'+1j'} + \sigma_{i'-1j'} + \sigma_{i'j'+1} + \sigma_{i'j'-1}) = J\sigma_{i'j'}(\sigma_{i'+1j'} + \sigma_{i'-1j'} + \sigma_{i'j'+1} + \sigma_{i'j'-1})$$

Consider $\sigma_{i'j'}\sigma_{i'+1j'}$. Clearly, term $\sigma_{i-1j'}\sigma_{ij'}$ with $i = i' + 1$ only occurs once more in the summation. Thus, we can definitely conclude that for $\Delta H \equiv \Delta H(\Delta\sigma_{i'j'})$ due to a single spin-flip is

$$\Delta H(\Delta\sigma_{i'j'}) = 2J\sigma_{i'j'}(\sigma_{i'+1j'} + \sigma_{i'-1j'} + \sigma_{i'j'+1} + \sigma_{i'j'-1}) \tag{792}$$

https://www.colorado.edu/physics/phys7240/phys7240_fa12/notes/Week3.pdf Victor Gurarie, Advanced Statistical Mechanics, Fall 2012 Exact solution by transfer matrices for 2-dim. Ising model.

Part XXXV
Quantum Mechanics

154 The Wave function and the Schrödinger Equation, its probability interpretation, some postulates

cf. Ch. 2 "The Wave Function and the Schrödinger Equation" in **Quantum Mechanics** by Franz Schwabl (2007) [49].

From experimental considerations (Sec. 1.2.2, Schwabl (2007) [49]), with electron diffraction, electrons, e^-, have wavelike properties; let this wave be $\psi(\mathbf{x}, t)$.

For free e^- of momentum $\mathbf{p}$, energy $E = \frac{\mathbf{p}^2}{2m}$, in accordance with diffraction experiments, consider as free plane waves

$$\psi(\mathbf{x}, t) = C\exp\left(i(\mathbf{k}\cdot\mathbf{x} - \omega t)\right), \qquad \omega = E/\hbar = E,\ \mathbf{k} = \mathbf{p}/hbar = \mathbf{p}$$

with $\hbar = 1$

Hypothesis: wave function $\psi(\mathbf{x}, t)$ gives probability distribution

$$\rho(\mathbf{x}, t) = |\psi(\mathbf{x}, t)|^2$$

$\rho(\mathbf{x}, t)d^3x$ = probability of finding e^- at location $\mathbf{x}$ in volume element d^3x.

e.g. e^- waves $\psi_1(\mathbf{x}, t)$, $\psi_2(\mathbf{x}, t)$

If both slits open, superposition of wave functions $\psi_1(\mathbf{x}, t) + \psi_2(\mathbf{x}, t)$

Note $|\psi_1(\mathbf{x}, t) + \psi_2(\mathbf{x}, t)|^2 \neq |\psi_1(\mathbf{x}, t)|^2 + |\psi_2(\mathbf{x}, t)|^2$ if there are no interference terms.

Important remarks:

(i) Single e^- not smeared out. $\rho(\mathbf{x}, t)$ is **not** the charge distribution of e^-, but is the probability density for measuring particle at position $\mathbf{x}$ at time t.

(ii) Prob. distribution doesn't occur by interference of many simultaneously incoming e^-, but one obtains same interference pattern if each e^- enters separately, i.e. even for very low intensity source. Thus, wave function applies to every electron and describes state of single e^-.

cf. 2.2 "The Schrödinger Equation for Free Particles" in **Quantum Mechanics** by Franz Schwabl (2007) [49].
(i) 1st. order DE (differential equation); (ii) linear in ψ for linear superposition (iii) "homogeneous" $\int d^3x|\psi(\mathbf{x},t)|^2 = 1$, (iv) plane waves

$$\psi(\mathbf{x},t) = C\exp\left[i(\mathbf{p}\cdot\mathbf{x} - \frac{p^2}{2m}t)/\hbar\right] \qquad \text{plane waves}$$

Should be solutions of the equations.

From postulates (i-iv),

$$i\hbar\frac{\partial}{\partial t}\psi(\mathbf{x},t) = \frac{-\hbar^2}{2m}\nabla^2\psi(\mathbf{x},t)$$

Time-dependent Schrödinger equation for free particles.

$$\int_{-\infty}^{\infty} d^3k e^{i\mathbf{k}\cdot\mathbf{x}}e^{-k^2\alpha^2} = \prod_{j=x}^{z}\int_{-\infty}^{\infty} dk_j e^{ik_jx_j}e^{-k_j^2\alpha^2} = \prod_{j=x}^{z}\left(\sqrt{\frac{\pi}{\alpha^2}}\exp\left(\frac{-x_j^2}{4\alpha^2}\right)\right) = \left(\frac{\sqrt{\pi}}{\alpha}\right)^3\exp\left(\frac{-x^2}{4\alpha^2}\right)$$

Part XXXVI

Tensors, Tensor networks; Singular Value Decomposition, QR decomposition, Density Matrix Renormalization Group (DMRG), Matrix Product states (MPS)

155 Introductions to Tensor Networks

José Barbon (IFT-CSIC, Univ. Autonoma de Madrid) gave the https://youtu.be/nsxgAOAEgbg for the workshop "Black Holes, Quantum Information, Entanglement, and all that," (29 May-1 June, 2017, with the organizing committee of Thibault Damour (IHES), Vasily Pestun (IHES), Eliezer Rabinovici (IHES & Hebrew Univ. of Jerusalem).

In the talk,

cf. 43:13

The church of the doubled Hilbert space

Any thermal box can be obtained by tracing over a second identical copy, if appropriately entangled into a global pure state.

$$\rho_R = \mathrm{Tr}_L \sum_n C_n \Psi_n^L \otimes \Psi_n^R$$

$$(C_n)_{\text{thermal}} = \left[\frac{e^{-\beta E_n}}{\sum_m e^{-\beta E_M}} \right]^{1/2}$$

But!!

If the entanglement basis is taken to be the high-energy band of two "entangled" CFTs ...

$$|TFD\rangle \sim \sum_{E_n} e^{-\beta E_n/2} |E_n\rangle_L \otimes |E_n\rangle_R$$

neglecting the tiny e^{-S} spacings. we can approximate by continuous spectrum of fields in the background of an AdS black hole, to get ...

$$\int_E e^{-\beta E/2} |E\rangle_L \otimes |E\rangle_R$$

The HH state of the bulk fields!

cf. 46:16

SLOGAN: EPR = ER Maldacena-Susskind

Accumulating a density of entanglement of $S \gg 1$ well-separated Bell pairs within a transversal size of order $(GS)^{1/2}$ seems to generate a geometrical bridge of area GS.

cf. 49:26

Parametrizing complexity of entanglement

Pick a tensor decomposition of Hilbert space of dimension $\exp(S)$ into S factors of $O(1)$ dimension.

$$\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \cdots \otimes \mathcal{H}_S$$

A tensor of S indices gives a generic state.

cf. 50:27

The decomposition of the big tensor in small building blocks gives a notion of "complexity of entanglement"

rather simple entanglement pattern

somewhat more complex entanglement pattern

picture from M von Raamsdonk

cf. 55:10

A list of open questions & problems

- Need exactly calculable toy models of AdS/CFT along the lines of SYK model
- Give a "renormalized" definition of quantum complexity for continuum CFTs
- Can tensor networks describe bulk gravitons?
- What is the space-time meaning of quantum complexity saturation?
- Can we define approximate local observables for black hole inferiors?
- Are there obstructions related to firewalls and/or fuzzballs?

Workshop introductory overview by José Barbon for the Institut des Hautes Études Scientifiques (IHÉS) gave me the first impetus to understand tensor networks as I sought to also understand the condensates of entanglement pairs within the black hole. A Google search for introductions to tensor networks that are on arxiv ("Introduction Tensor Network arxiv") yielded Bridgeman and Chubb's course notes (bf. Bridgeman and Chubb (2017) [61]).

155.1 List of stuff I want to look at/do/study

I would like to compare/contrast the following:

- Rotman (2010) [57], Ch. 8, but starting from 8.4 Tensor Products, pp. 574
- Jeffrey Lee (2009) [60], Ch. 7 Tensors
- http://www.irisa.fr/sage/bernard/publis/SVD-Chapter06.pdf, https://math.stackexchange.com/questions/694339/parallel-algorithms-for-svd

Maldacena and Susskind (2013) [66]
Lectures on Gravity and Entanglement. Mark Van Raamsdonk [69]

- Consider as physical system AdS-Schwarzchild black hole
- CFT
  - PFL Lectures on Conformal Field Theory in $D \geq 3$ Dimensions, Rychkov (2016) [67].

Evenbly and Vidal (2011) [68], Tensor network states and geometry
Loose ends (might not be useful links)

- https://arxiv.org/pdf/1506.06958.pdf
- https://arxiv.org/pdf/1512.02532.pdf One-point Functions in AdS/dCFT from Matrix Product States

Numerical implementation strategy: 1st, CUDA cuSolver, 2nd, Numerical Recepes version, 3rd, parallel algorithm review.

155.2 Tensor operations; Tensor properties

155.2.1 rank

r = rank tensor of dim. $d_1 \times \cdots \times d_r$ is element of $\mathbb{C}^{d_1 \times \cdots \times d_r}$
Tensor product

$$[A \otimes B]_{i_1 \ldots i_r, j_1 \ldots j_s} := A_{i_1 \ldots i_r} \cdot B_{j_1 \ldots j_s} \tag{793}$$

155.2.2 Trace

Given tensor A, xth, yth indices have identical dims. $(d_x = d_y)$,
partial trace over these 2 dims. is simply joint summation over that index

$$[\mathrm{Tr}_{x,y} A]_{i_1 \ldots i_{x-1} i_{x+1} \ldots i_{y-1} i_{y+1} \ldots i_r} = \sum_{\alpha=1}^{d_x} A_{i_1 \ldots i_{x-1} \alpha i_{x+1} \ldots i_{y-1} \alpha i_{y+1} \ldots i_r} \tag{794}$$

155.2.3 Contraction

155.2.4 Group and splitting, Bridgeman and Chubb (2017) [61]

"Rank is a rather fluid concept in the study of tensor networks." Bridgeman and Chubb (2017) [61].

$\mathbb{C}^{a_1\times\cdots\times a_n}\simeq\mathbb{C}^{b_1\times\cdots\times b_m}$ isomorphic as vector spaces if $\prod_i a_i=\prod_i b_i$.

We can "group" or "split" indices to lower or raise rank of given tensor, resp.

Consider contracting 2 arbitrary tensors.

If we group together indices which are and are not involved in contraction,

"It should be noted that not only is this reduction to matrix multiplication pedagogically handy, but this is precisely the manner in which numerical tensor packages perform contraction, allowing them to leverage highly optimised matrix multiplication code." (cf. Bridgeman and Chubb (2017) [61]; check this)

"Owing to freedom in choice of basis, precise details of grouping and splitting aren't unique." (cf. Bridgeman and Chubb (2017) [61]).

1 specific choice of convention:

tensor product basis, defining basis on product space by product of respective bases.

"The canonical use of tensor product bases in quantum information allows for grouping and splitting described above to be - dealt with implicitly."

$$|0\rangle\otimes|1\rangle\equiv|0\rangle \tag{795}$$

and precisely this grouping,

$$\begin{gathered}|0\rangle\otimes|1\rangle\in\mathrm{Mat}_{\mathbb{C}}(2,2),\ \text{whilst}\\ |01\rangle\in\mathbb{C}^4\end{gathered} \tag{796}$$

Suppose rank $n+m$ tensor T, group its first n indices, last m indices together.

$$T_{I,J}:=T_{i_1\ldots i_n,j_1\ldots j_m}$$

where

$$\begin{aligned}I&:=i_1+d_1^{(i)}i_2+d_1^{(i)}d_2^{(i)}i_3+\cdots+d_1^{(i)}\ldots d_{n-1}^{(i)}i_n\\ J&:=j_1+d_1^{(j)}j_2+d_1^{(j)}d_2^{(j)}j_3+\cdots+d_1^{(j)}\ldots d_{m-1}^{(j)}j_m\end{aligned}$$

EY : 20170627 to elaborate, consider a functor `flatten` that does what's described above, in the context of category theory (and so this is the generalization):

$$\begin{gathered}\mathbb{K}^{d_1^{(i)}}\times\mathbb{K}^{d_2^{(i)}}\times\cdots\times\mathbb{K}^{d_n^{(i)}}\times\mathbb{K}^{d_1^{(j)}}\times\mathbb{K}^{d_2^{(j)}}\times\cdots\times\mathbb{K}^{d_m^{(j)}}\xrightarrow{\text{flatten}}\mathbb{K}^{\prod_{p=1}^n d_p^{(i)}}\times\mathbb{K}^{\prod_{q=1}^m d_q^{(j)}}\\ T_{i_1\ldots i_n,j_1\ldots j_m}\overset{\text{flatten}}{\longmapsto}T_{I,J}\\ \{0,1,\ldots d_1^{(i)}\}\times\{0,1,\ldots d_2^{(i)}\}\times\cdots\times\{0,1,\ldots d_n^{(i)}\}\times\{0,1,\ldots d_1^{(j)}\}\times\{0,1,\ldots d_2^{(j)}\}\times\cdots\times\{0,1,\ldots d_m^{(j)}\}\xrightarrow{\text{flatten}}\\ \xrightarrow{\text{flatten}}\{0,1,\cdots\prod_{p=1}^n d_p^{(i)}-1\}\times\{0,1,\cdots\prod_{q=1}^m d_q^{(j)}-1\}\\ (i_1,i_2,\ldots i_n,j_1,j_2\ldots j_m)\overset{\text{flatten}}{\longmapsto}(I,J):=(i_1+d_1^{(i)}i_2+\cdots+d_1^{(i)}\ldots d_{n-1}^{(i)}i_n,j_1+d_1^{(j)}j_2+\cdots+d_1^{(j)}\ldots d_{m-1}^{(j)}j_m)\end{gathered} \tag{797}$$

It doesn't make sense to call this "row-major" or "column-major" ordering generalization, because we are not dealing with only 2 indices where we can definitely say the first index indexes the "row" and the second index indexes the "column." At most, possibly, you can alternatively have this:

$$(i_1\ldots i_n,j_1\ldots j_m)\overset{\text{flatten}}{\longmapsto}(I,J):=(d_2^{(i)}\ldots d_n^{(i)}i_1+d_3^{(i)}\ldots d_n^{(i)}i_2+\cdots+i_n,d_2^{(j)}\ldots d_m^{(j)}j_1+\cdots+j_m)$$

Note that this is all 0-*based counting* (i.e. we start counting from 0 just like in C,C++,Python, etc.). If you really wanted 1-based counting, you'd have to complicate the above formulas as such:

$$(I,J):=(i_1+d_1^{(i)}(i_2-1)+\cdots+d_1^{(i)}\ldots d_{n-1}^{(i)}(i_n-1),j_1+d_1^{(j)}(j_2-1)+\cdots+d_1^{(j)}\ldots d_{m-1}^{(j)}(j_m-1))$$

Note that formulas are easily checked by pluggin in the minimum and maximum values for the indices and seeing if they make sense (e.g. plug in $(0,0,\ldots,0)$ for all indices for 0-based counting and make sure you get back $I=0$ or $J=0$).

155.3 Singular Value Decomposition

$$\begin{aligned} T_{I,J} &= \sum_{\alpha} U_{I,\alpha} S_{\alpha,\alpha} \overline{V}_{J,\alpha} \\ \mathrm{Mat}_{\mathbb{K}}(N,M) &\xrightarrow{\text{SVD}} \mathrm{Mat}_{\mathbb{K}}(N,P) \times \mathrm{Mat}_{\mathbb{K}}(P,P) \times \mathrm{Mat}_{\mathbb{K}}(M,P) \\ T_{I,J} &\xmapsto{\text{SVD}} U_{I,\alpha}, S_{\alpha,\alpha}, \overline{V}_{I,\alpha} \text{ s.t.} \\ T_{I,J} &= \sum_{\alpha} U_{I,\alpha} S_{\alpha,\alpha} \overline{V}_{J,\alpha} \\ T &= USV^{\dagger} \end{aligned} \tag{798}$$

For the higher-dimensional version of SVD,

$$\begin{aligned} \mathbb{K}^{d_1^{(i)}} \otimes \cdots \otimes \mathbb{K}^{d_N^{(i)}} \otimes \mathbb{K}^{d_1^{(j)}} \otimes \cdots \otimes \mathbb{K}^{d_M^{(j)}} &\xrightarrow{\text{flatten}} \mathrm{Mat}_{\mathbb{K}}(N,M) \xrightarrow{\text{SVD}} \mathrm{Mat}_{\mathbb{K}}(N,P) \times \mathrm{Mat}_{\mathbb{K}}(P,P) \times \mathrm{Mat}_{\mathbb{K}}(M,P) \xrightarrow{\text{splitting}} \\ &\xrightarrow{\text{splitting}} \mathbb{K}^{d_1^{(i)}} \otimes \cdots \otimes \mathbb{K}^{d_N^{(i)}} \otimes \mathbb{K}^{P} \times \mathrm{Mat}_{\mathbb{K}}(P,P) \times \mathbb{K}^{d_1^{(j)}} \otimes \cdots \otimes \mathbb{K}^{d_M^{(j)}} \otimes \mathbb{K}^{P} \\ T_{i_1 \ldots i_N, j_1 \ldots j_M} &= \sum_{\alpha} U_{i_1 \ldots i_N, \alpha} S_{\alpha,\alpha} \overline{V}_{j_1 \ldots j_M, \alpha} \end{aligned} \tag{799}$$

156 Density Matrix Renormalization Group; Matrix Product States (MPS)

156.1 Introduction; physical system (physical setup)

cf. "Density Matrix Renormalization Group/Matrix Product States" lectures by Schollwöck (2017) [64].

Recall the fundamental Hamiltonian (frequently in solid state physics), for *electrons moving in a Hamiltonian potential.*

$$H = \sum_{j=1}^{e^-} \frac{\mathbf{p}_j^2}{2m_e} + \frac{1}{2} \frac{1}{4\pi\epsilon_0} \frac{q_e^2}{|\mathbf{r}_i - \mathbf{r}_j|^2} + \sum_{j=1}^{e} V_{\text{eff}}(\mathbf{r}_j) \tag{800}$$

where $\frac{\mathbf{p}_j^2}{2m_e}$ is the kinetic energy term, $\sum_{j=1}^{e^-} V_{\text{eff}}(\mathbf{r}_j)$ is the lattice potential. The problem is in the 2nd. term, electron-electron interaction, $\frac{1}{2} \frac{1}{4\pi\epsilon_0} \frac{q_e^2}{|\mathbf{r}_i - \mathbf{r}_j|^2}$

Typical models include the following:

- Hubbard model (tight, binding-like model; basis states are not energy states but *Wannier basis* states):

$$H = -t \sum_{\langle i,j \rangle, \sigma} c_{i\sigma}^{\dagger} c_{j\sigma} + h.c. + U \sum_i n_{i\uparrow} n_{i\downarrow} \tag{801}$$

  where $\langle i, j \rangle$ denotes nearest neighbors, σ index is for all possible states, *h.c.* stands for hermitian conjugate, and
  $d \equiv$ number of states of single spin site.
  $-t \sum_{\langle i,j \rangle, \sigma} c_{i\sigma}^{\dagger} c_{j\sigma} + h.c.$ is the kinetic energy term,
  $U \sum_i n_{i\uparrow} n_{i\downarrow}$ is the Coulomb energy.

  Hilbert space for the Hubbard model is

$$\{|\emptyset\rangle, |\uparrow\rangle, |\downarrow\rangle, |\uparrow\downarrow\rangle\}^{\otimes L}, \qquad d = 4 \tag{802}$$

- Heisenberg model (large -U Hubbard at half-filling)

$$H = J \sum_{\langle i,j \rangle} \mathbf{S}_i \cdot \mathbf{S}_j = J \sum_{\langle i,j \rangle} \frac{1}{2} (S_i^+ S_j^- + S_j^+ S_i^-) + S_i^z S_j^z) \tag{803}$$

  Hilbert space $\{|\uparrow\rangle, |\downarrow\rangle\}^{\otimes L}$, $d = 2$

156.1.1 Compression of information viewpoint for solid-state Hamiltonians, quantum many-body systems

.

"emergent" macroscopic quantities, τ, p (temperature, pressure). For

$$H = J\sum_{\langle i,j\rangle}\mathbf{S}_i\cdot\mathbf{S}_j = J\sum_{\langle i,j\rangle}\frac{1}{2}(S_i^+S_j^- + S_j^+S_i^-) + S_i^zS_j^z)$$

H as classical spins: thermodynamic limit $N\to\infty$. 2 angles required to describe unit vector on unit sphere (S^3) $\Longrightarrow 2N$ degrees of freedom (linear)
quantum spins: superposition of states, thermodynamic limit: $N\to\infty$, 2^N degrees of freedom (exponential).

156.1.2 Definitions; notation and conventions

Quantum system living on L lattice sites; cf. Schollwöck (2017) [64], lattice can be in any dim., effectively most useful in 1-dim., think of the example of a 1-dim. chain of L sites.

d local states per site $\{\sigma_i\rangle$, $i\in\{1,2,\dots L\}$
e.g. spin $\frac{1}{2}$, $d=2$, $|\uparrow\rangle, |\downarrow\rangle$.
Hilbert space: $\mathcal{H}=\otimes_{i=1}^L\mathcal{H}_i$, $\mathcal{H}_i=\{|1_i\rangle,\dots|d_i\rangle\}$.
Notice, there are *exponentially many coefficients*, c's. Most general state (not necessarily 1-dim.) is

$$|\psi\rangle = \sum_{\sigma_1\dots\sigma_L} c^{\sigma_1\dots\sigma_L}|\sigma_1\dots\sigma_L\rangle \tag{804}$$

abbreviations: $\{\sigma\}=\sigma_1\dots\sigma_L$. And so we can write $c^{\{\sigma\}}$.

156.2 MPS, matrix product states

$$|\psi\rangle = \sum_{\sigma_1\dots\sigma_L} M^{\sigma_1}M^{\sigma_2}\dots M^{\sigma_L}|\sigma_1\sigma_2\dots\sigma_L\rangle \tag{805}$$

The $\sum_{\sigma_1\dots\sigma_L}$ means that all basis states participate; Schollwöck is not kicking out any states arbitrarily.

$$c^{\{\sigma\}} = M^{\sigma_1}M^{\sigma_2}\dots M^{\sigma_L}\in\mathbb{C}$$

so

$M^{\sigma_1}\in \mathrm{Mat}_{\mathbb{C}}(1,n_1)$ so to get a scalar in the product of matrices. Likewise, $M^{\sigma_L}\in\mathrm{Mat}_{\mathbb{C}}(m_L,1)$
(variational) constraint is in expansion coefficients.
$\forall$ d local basis states, $|\sigma_i\rangle\in V_i\equiv V$,dim$V=d$, let there be 1 matrix M, i.e. M^{σ_i}.
Thus, dL matrices altogether (in total).
Assume matrix size has upper limit D (a computer limitation).
Up to dLD^2 coefficients, instead of exponentially many ($c^{\{\sigma\}}$, and sum over $\{\sigma\}$).

156.2.1 Product States and MPS

Mean-filed approximation/product state misses essential quantum feature: **entanglement**.
Consider 2 spin $\frac{1}{2}$ systems: $\mathcal{H}=\mathcal{H}_1\otimes\mathcal{H}_2$, $\mathcal{H}_i=\{|\uparrow\rangle,|\downarrow\rangle\}$
General state is

$$|\psi\rangle = c^{\uparrow\uparrow}|\uparrow\uparrow\rangle + c^{\uparrow\downarrow}|\uparrow\downarrow\rangle + c^{\downarrow\uparrow}|\downarrow\uparrow\rangle + c^{\downarrow\downarrow}|\downarrow\downarrow\rangle$$

e.g. singlet state: $|\psi\rangle=\frac{1}{\sqrt{2}}|\uparrow\downarrow\rangle - \frac{1}{\sqrt{2}}|\downarrow\uparrow\rangle$.
As an exercise, show that the singlet state cannot be written as product of local coefficients, i.e.

$$c_{\uparrow\downarrow}\neq c^{\uparrow}c^{\downarrow}$$

Instead of writing products of scalars, write product of matrices, i.e. $e^{\sigma_1}\cdot e^{\sigma_2}\to M^{\sigma_1}M^{\sigma_2}$

$$M^{\uparrow 1} = \begin{bmatrix}1 & 0\end{bmatrix}\quad M^{\downarrow 1}=\begin{bmatrix}0 & 1\end{bmatrix}\qquad M^{\uparrow 2}=\begin{bmatrix}0\\ \frac{-1}{\sqrt{2}}\end{bmatrix}\quad M^{\downarrow 2}=\begin{bmatrix}\frac{1}{\sqrt{2}}\\ 0\end{bmatrix}$$

$$M^{\uparrow 1}M^{\downarrow 2} = \frac{1}{\sqrt{2}}$$
$$M^{\downarrow 1}M^{\uparrow 2} = \frac{-1}{\sqrt{2}}$$

156.2.2 AKLT model (Affleck-Kennedy-Lieb-Tasaki)

MPS is useful even for matrices of dim. 2.

156.3 General matrix product state (MPS) and SVD (Singular Value Decomposition)

cf. Schollwöck (2017) [64]

The general matrix product state (MPS) is the following:

$$|\psi\rangle = \sum_{\sigma_1 \dots \sigma_L} M^{\sigma_1} M^{\sigma_2} \dots M^{\sigma_L} |\sigma_1 \sigma_2 \dots \sigma_L\rangle \tag{806}$$

where $\sigma_i \in V_i$, $\dim V_i = d_i$ and
$M^{\sigma_1} \in \mathrm{Mat}_{\mathbb{C}}(1, D_1)$
$M^{\sigma_2} \in \mathrm{Mat}_{\mathbb{C}}(D_1, D_2)$
$\vdots$
$M^{\sigma_{L-1}} \in \mathrm{Mat}_{\mathbb{C}}(D_{L-2}, D_{L-1})$
$M^{\sigma_L} \in \mathrm{Mat}_{\mathbb{C}}(D_{L-1}, 1)$

Notice the non-unique **gauge degree of freedom**:
$\forall\, A \in \mathrm{Mat}_{\mathbb{C}}(m,n)$, then for $k = \min(m,n)$,

$$A = USV^\dagger \equiv U\Sigma V^\dagger \text{ where} \tag{807}$$

$U \in \mathrm{Mat}_{\mathbb{C}}(m,k)$, $U^\dagger U = 1$ (i.e. U consists of orthonormal columns, or k number of u's $\in \mathbb{C}^m$); if $m = k$, $UU^\dagger = 1$,
$S \in \mathrm{Mat}_{\mathbb{C}}(k,k)$ s.t. $S \in \mathrm{diag}_{\mathbb{C}}(k)$, $s_1 \geq s_2 \geq s_3 \geq \dots s_i \geq 0$, s_j's non-negative "singular values" (adjacent "singular" in name doesn't imply anything), non-vanishing $= \mathrm{rank} r \leq k$.
$V^\dagger \in \mathrm{Mat}_{\mathbb{C}}(k,n)$, $V^\dagger V = 1$, (orthonormal rows, or k number of $v \in \mathbb{C}^n$); if $k = n$, $VV^\dagger = 1$

Recall eigenvalue equation and thus so-called eigenvalue decomposition.

For $A \in \mathrm{Mat}_{\mathbb{C}}(m,m)$,

$$Au_j = \lambda_j u_j; \qquad j = 1 \dots r;\ r \equiv \text{ rank }, \quad u_j \in \mathrm{Mat}_{\mathbb{C}}(m,1)$$
$$A_{ik}u_{kj} = \lambda_j u_{ij} = u_{ik}\delta_{kj}\lambda_j \Longrightarrow AU = U\Lambda$$

with $U \in \mathrm{Mat}_{\mathbb{C}}(m,r)$, $\Lambda \in \mathrm{Mat}_{\mathbb{C}}(r,r)$.

And so

$$AA^\dagger = USV^\dagger VSU^\dagger = US^2U^\dagger \Longrightarrow (AA^\dagger)U = US^2$$
$$A^\dagger A = VSU^\dagger USV^\dagger = VS^2V^\dagger \Longrightarrow (A^\dagger A)V = VS^2$$

so if we treat U and V, matrices of left, right singular vectors, then S^2 singular value squared are eigenvalues.

Start with

$$|\psi\rangle = \sum_{\sigma_1 \dots \sigma_L} c^{\sigma_1 \dots \sigma_L} |\sigma_1 \dots \sigma_L\rangle \in V \text{ s.t. } \dim V = d^L \tag{808}$$

Note the *abuse of notation*: while $c^{\sigma_1 \dots \sigma_L} \in \mathbb{C}$ itself, also denote $c^{\sigma_1 \dots \sigma_L} \in \mathbb{C}^{d^L}$ as a shorthand for $\sum_{\sigma_1 \dots \sigma_L} c^{\sigma_1 \dots \sigma_L} |\sigma_1 \dots \sigma_L\rangle$

Reshape coefficient vector into matrix of (size) dimension $(d \times d^{L-1})$.

$$\mathbb{C}^{d^L} \xrightarrow{\text{reshape}} \mathrm{Mat}_{\mathbb{C}}(d, d^{L-1})$$
$$c^{\sigma_1 \dots \sigma_L} \xmapsto{\text{reshape}} \Psi_{\sigma_1, (\sigma_2 \dots \sigma_L)}$$

Then do SVD:

$$\Psi_{\sigma_1, (\sigma_2 \dots \sigma_L)} \overset{\text{SVD}}{=} \sum_{a_1} U_{\sigma_1 a_1} S_{a_1 a_1} V^\dagger_{a_1, \sigma_2 \dots L} = U_{\sigma_1 a_1} S_{a_1 a_1} V^\dagger_{a_1, \sigma_2 \dots \sigma_L}$$

Let's utilize commutative diagrams to summarize the reshaping and SVD operations that we've done.

$$\mathbb{C}^{d^L} = \mathrm{Mat}_{\mathbb{C}}(1, d^L) \xrightarrow{\text{reshape}} \mathrm{Mat}_{\mathbb{C}}(d, d^{L-1}) \xrightarrow{\text{SVD}} \mathrm{Mat}_{\mathbb{C}}(d, r_1) \times \mathrm{Mat}_{\mathbb{C}}(r_1, r_1) \times \mathrm{Mat}_{\mathbb{C}}(r_1, d^{L-1})$$

$$|\Psi\rangle \equiv c^{\sigma_1 \dots \sigma_L} \xmapsto{\text{reshape}} \Psi_{\sigma_1, (\sigma_2 \dots \sigma_L)} \xmapsto{\text{SVD}} \Psi_{\sigma_1, (\sigma_2 \dots \sigma_L)} \overset{\text{SVD}}{=} U_{\sigma_1 a_1} S_{a_1 a_1} V^\dagger_{a_1, \sigma_2 \dots \sigma_L}$$

where I abuse notation for the SVD operation in that, SVD maps a matrix (in this case, Ψ) into 3 matrices, that obey the equality relationship when they're multiplied together (i.e. $\Psi = USV^\dagger$).

Slice U into d row vectors, i.e. for $U \in \mathrm{Mat}_{\mathbb{C}}(d, r_1)$.

$$\begin{aligned} &\mathrm{Mat}_{\mathbb{C}}(d, r_1) \xrightarrow{\text{slice}} \mathrm{Mat}_{\mathbb{C}}(1, r_1)^d \\ &\quad U_{\sigma_1 a_1} \mapsto \{A^{\sigma_1}\} \equiv \{A^{\sigma_1}_{1, a_1}\}_{\sigma_1} \text{ s.t. } A^{\sigma_1}_{1, a_1} = U_{\sigma_1 a_1} \text{ and } |\{A^{\sigma_1}_{1, a_1}\}| = d \end{aligned}$$

Collecting all the operations, and doing the following notation rewrite,

$$c^{\sigma_1 \sigma_2 \dots \sigma_L} \mapsto \Psi_{\sigma_1 \sigma_2 \dots \sigma_L} = \sum_{a_1} A^{\sigma_1}_{1 a_1} S_{a_1 a_1} V^\dagger_{a_1, \sigma_2 \dots \sigma_L} = \sum_{a_1} A^{\sigma_1}_{1 a_1} c^{a_1 \sigma_2 \sigma_3 \dots \sigma_L}$$

where

$$c^{a_1 \sigma_2 \sigma_3 \dots \sigma_L} = S_{a_1 a_1} V^\dagger_{a_1 \sigma_2 \dots \sigma_L}$$

Do the same procedure again.

$$\mathrm{Mat}_{\mathbb{C}}(r_1, d^{L-1}) \xrightarrow{\text{reshape}} \mathrm{Mat}_{\mathbb{C}}(r_1 d, d^{L-2}) \xrightarrow{\text{SVD}} \mathrm{Mat}_{\mathbb{C}}(r_1 d, r_2) \times \mathrm{Mat}_{\mathbb{C}}(r_2, r_2) \times \mathrm{Mat}_{\mathbb{C}}(r_2, d^{L-2})$$

$$c^{a_1, \sigma_2 \sigma_3 \dots \sigma_L} \xmapsto{\text{reshape}} \Psi_{a_1 \sigma_2, (\sigma_3 \dots \sigma_L)} \xmapsto{=} \Psi_{a_1 \sigma_2, (\sigma_3 \dots \sigma_L)} \overset{\text{SVD}}{=} U_{a_1 \sigma_2, a_2} S_{a_2 a_2} V^\dagger_{a_2, \sigma_3 \dots \sigma_L}$$

Then slice U into d matrices, and then matrix multiply the S and $V^\dagger$ matrices together:

$$\mathrm{Mat}_{\mathbb{C}}(r_1 d, r_2) \times \mathrm{Mat}_{\mathbb{C}}(r_2, r_2) \times \mathrm{Mat}_{\mathbb{C}}(r_2, d^{L-2}) \xrightarrow{\text{slice and multiply}} \mathrm{Mat}_{\mathbb{C}}(r_1, r_2)^d \times \mathrm{Mat}_{\mathbb{C}}(r_2, d^{L-2})$$

$$\textstyle\sum_{a_2} U_{a_1 \sigma_2, a_2} S_{a_2 a_2} V^\dagger_{a_2, \sigma_3 \dots \sigma_L} \xmapsto{=} = \sum_{a_2} A^{\sigma_2}_{a_1 a_2} c^{a_2, a_3 \dots \sigma_L} \text{ where } A^{\sigma_2}_{a_1 a_2} = U_{a_1 \sigma_2, \sigma_3 \dots \sigma_L}$$

Thus, generalize the *ith procedure*: for $i = 1 \dots L$,
Let $r_0 = 1$.

$$\boxed{\begin{aligned} &\mathrm{Mat}_{\mathbb{C}}(r_{i-1}, d^{L-(i-1)}) \xrightarrow{\text{reshape}} \mathrm{Mat}_{\mathbb{C}}(r_{i-1} d, d^{L-i}) \xrightarrow{\text{SVD}} \mathrm{Mat}_{\mathbb{C}}(r_{i-1} d, r_i) \times \mathrm{Mat}_{\mathbb{C}}(r_i, r_i) \times \mathrm{Mat}_{\mathbb{C}}(r_i, d^{L-i}) \xrightarrow{\text{slice and multiply}} \mathrm{Mat}_{\mathbb{C}}(r_{i-1}, r_i)^d \times \mathrm{Mat}_{\mathbb{C}}(r_i, d^{L-i}) \\ &c^{a_{i-1}, \sigma_i \sigma_{i+1} \dots \sigma_L} \xmapsto{\text{reshape}} \Psi_{a_{i-1} \sigma_i, (\sigma_{i+1} \sigma_{i+2} \dots \sigma_L)} \xmapsto{=} U_{a_{i-1} \sigma_i, a_i} S_{a_i a_i} V^\dagger_{a_i, \sigma_{i+1} \dots \sigma_L} \xmapsto{=} A^{\sigma_i}_{a_{i-1}, a_i} c^{a_i, \sigma_{i+1} \dots \sigma_L} \end{aligned}} \tag{809}$$

Remember that $r_i \leq \min(r_{i-1} d, d^{L-i})$ and for $i = L$, there is no need to do a SVD, but only a reshape, and slice and multiply.
Collecting all the A matrices:

$$\boxed{\begin{aligned}
&A^{\sigma_1}_{1,a_1} \in \mathrm{Mat}_{\mathbb{C}}(1, r_1); \qquad r_1 \leq d \\
&A^{\sigma_2}_{a_1,a_2} \in \mathrm{Mat}_{\mathbb{C}}(r_1, r_2); \qquad r_2 \leq r_1 d \\
&\vdots \\
&A^{\sigma_i}_{a_{i-1},a_i} \in \mathrm{Mat}_{\mathbb{C}}(r_{i-1}, r_i); \qquad r_i \leq \min\left(r_{i-1}d, d^{L-i}\right) \\
&\vdots \\
&A^{\sigma_L}_{a_{L-1},a_L} \in \mathrm{Mat}_{\mathbb{C}}(r_{L-1}, 1); \qquad r_{L-1} \leq d
\end{aligned}} \tag{810}$$

156.3.1 Left and Right Normalization, A and B matrices, "special gauge" from normalization

Choose orthonormal basis states $\forall\, a_l$, $\forall\, l = 1, 2, \ldots L$ For

$$\begin{aligned}
|a_l\rangle &= \sum_{a_{l-1}\sigma_l} M^{\sigma_l}_{a_{l-1}a_l} |a_{l-1}\sigma_l\rangle \\
\langle a'_l| &= \sum_{a'_{l-1}\sigma'_l} \langle a'_{l-1}\sigma'_l| (M^{\sigma'_l}_{a'_{l-1}a'_l})^*
\end{aligned}$$

then,

$$\begin{aligned}
\delta_{a'_l a_l} = \langle a'_l | a_l \rangle = \sum_{a'_{l-1}\sigma'_l, a_{l-1}\sigma_l} M^{\sigma'^*_l}_{a'_{l-1}a'_l} M^{\sigma_l}_{a_{l-1}a_l} \langle a'_{l-1}\sigma'_l | a_{l-1}\sigma_l \rangle = \sum_{a_{l-1}\sigma_l} M^{\sigma_l *}_{a_{l-1}a'_l} M^{\sigma_l}_{a_{l-1}a_l} = \\
= \sum_{\sigma_l} ((M^{\sigma_l})^\dagger M^{\sigma_l})_{a'_l a_l}
\end{aligned} \tag{811}$$

Left normalization comes from a property of SVD in that $\forall U$ matrices, $U^\dagger U = 1$, and so

$$\begin{aligned}
(U^\dagger)_{a'_i k_i} U_{k_i a_i} = \delta_{a'_i a_i} = U^*_{k_i a'_i} U_{k_i a_i} = U^*_{a'_{i-1}\sigma_i, a'_i} U_{a''_{i-1}\sigma_i, a_i} = \\
= A^{\sigma_i *}_{a''_{i-1}, a'_i} A^{\sigma_i}_{a''_{i-1}, a_i} = (A^{\sigma_i})^\dagger A^{\sigma_i} = \boxed{\sum_{\sigma_i} (A^{\sigma_i})^\dagger A^{\sigma_i} = 1}
\end{aligned} \tag{812}$$

For right normalization, consider doing the operations of Eq. 809 "on the right":

$$\mathrm{Mat}_{\mathbb{C}}(d^L, 1) \xrightarrow{\text{reshape}} \mathrm{Mat}_{\mathbb{C}}(d^{L-1}, d) \xrightarrow{\text{SVD}} \mathrm{Mat}_{\mathbb{C}}(d^{L-1}, r_1) \times \mathrm{Mat}_{\mathbb{C}}(r_1, r_1) \times \mathrm{Mat}_{\mathbb{C}}(r_1, d) \xrightarrow{\text{slice and multiply}} \mathrm{Mat}_{\mathbb{C}}(d^{L-1}, r_1) \times \mathrm{Mat}_{\mathbb{C}}(r_1, 1)^d$$

$$c^{\sigma_1\sigma_2\ldots\sigma_L} \xmapsto{\text{reshape}} \Psi_{\sigma_1\ldots\sigma_{L-1},\sigma_L} \xmapsto{=} U_{\sigma_1\ldots\sigma_{L-1}a_1} S_{a_1a_1} V^\dagger_{a_1,\sigma_L} \xmapsto{=} \textstyle\sum_{\sigma_L} c^{\sigma_1\ldots\sigma_{L-1},a_1} B^{\sigma_L}_{a_1,1}$$

$$\mathrm{Mat}_{\mathbb{C}}(d^{L-1}, r_1) \xrightarrow{\text{reshape}} \mathrm{Mat}_{\mathbb{C}}(d^{L-2}, r_1 d) \xrightarrow{\text{SVD}} \mathrm{Mat}_{\mathbb{C}}(d^{L-2}, r_2) \times \mathrm{Mat}_{\mathbb{C}}(r_2, r_2) \times \mathrm{Mat}_{\mathbb{C}}(r_2, r_1 d) \xrightarrow{\text{slice and multiply}} \mathrm{Mat}_{\mathbb{C}}(d^{L-2}, r_2) \times \mathrm{Mat}_{\mathbb{C}}(r_2, r_1)^d$$

$$c^{\sigma_1\ldots\sigma_{L-1}a_1} \xmapsto{\text{reshape}} \Psi_{\sigma_1\ldots\sigma_{L-2},\sigma_{L-1}a_1} \xmapsto{=} U_{\sigma_1\ldots\sigma_{L-2},a_2} S_{a_2a_2} V^\dagger_{a_2,\sigma_{L-1}a_1} \xmapsto{=} \textstyle\sum_{\sigma_{L-1}} c^{\sigma_1\ldots\sigma_{L-2},a_2} B^{\sigma_{L-1}}_{a_2,a_1}$$

$$\vdots$$

$$\boxed{\begin{array}{c}\mathrm{Mat}_{\mathbb{C}}(d^{L-(i-1)}, r_{i-1}) \xrightarrow{\text{reshape}} \mathrm{Mat}_{\mathbb{C}}(d^{L-i}, r_{i-1}d) \xrightarrow{\text{SVD}} \mathrm{Mat}_{\mathbb{C}}(d^{L-i}, r_i) \times \mathrm{Mat}_{\mathbb{C}}(r_i, r_i) \times \mathrm{Mat}_{\mathbb{C}}(r_i, r_{i-1}d) \xrightarrow{\text{slice and multiply}} \mathrm{Mat}_{\mathbb{C}}(d^{L-i}, r_i) \times \mathrm{Mat}_{\mathbb{C}}(r_i, r_{i-1})^d \\ \\ c^{\sigma_1 \dots \sigma_{L-(i-1)} a_{i-1}} \xmapsto{\text{reshape}} \Psi_{\sigma_1 \dots \sigma_{L-i}, \sigma_{L-(i-1)} a_{i-1}} \xmapsto{=} U_{\sigma_1 \dots \sigma_{L-i}, a_i} S_{a_i a_i} V^{\dagger}_{a_i, \sigma_{L-(i-1)} a_{i-1}} \xmapsto{=} \sum_{\sigma_{L-(i-1)}} c^{\sigma_1 \dots \sigma_{L-i}, a_i} B^{\sigma_{L-(i-1)}}_{a_i, a_{i-1}} \end{array}} \tag{813}$$

Remember that $r_i \leq \min(d^{L-i}, r_{i-1}d)$ and for $i = L$, just do reshape and slice and multiply operations.
Then, finally, the **right normalization** is derived and is such:

$$\begin{aligned} V^{\dagger}V = 1 \implies \\ (V^{\dagger}V)_{a_i a_i'} = \delta_{a_i a_i'} = V^{\dagger}_{a_i, \sigma_{L-(i-1)} a_{i-1}} V_{\sigma_{L-(i-1)} a_{i-1}, a_i'} = B^{\sigma_{L-(i-1)}}_{a_i, a_{i-1}} (V^{\dagger})^{\dagger}_{\sigma_{L-(i-1)} a_{i-1}, a_i'} = \\ = B^{\sigma_{L-(i-1)}}_{a_i a_{i-1}} (V^{\dagger})^{*}_{a_i', \sigma_{L-(i-1)}, a_{i-1}} = B^{\sigma_{L-(i-1)}}_{a_i, a_{i-1}} B^{\sigma_{L-(i-1)}}_{a_i', a_{i-1}} = B^{\sigma_{L-(i-1)}}_{a_i a_{i-1}} (B^{\dagger})^{\sigma_{L-(i-1)}}_{a_{i-1} a_i'} \qquad \forall i = 1 \dots L \\ \implies \sum_{\sigma_{L-(i-1)}} B^{\sigma_{L-(i-1)}} (B^{\dagger})^{\sigma_{L-(i-1)}} = 1 \end{aligned} \tag{814}$$

cf. Sec. 4, Matrix Product States (MPS) of Schollwöck [63].
Necessarily, given matrix $M \in \mathrm{Mat}_{\mathbb{K}}(M, N)$ (notation in Bridgeman and Chubb (2017) [61] and CUDA Toolkit Documentation; I will follow the notation in Schollwöck [63] since his A,B denote specific physical meaning).
For

$$\begin{aligned} U \in \mathrm{Mat}_{\mathbb{K}}(N_A, \min(N_A, N_B)) \text{ s.t. } UU^{\dagger} = 1 \\ S \in \mathrm{Mat}_{\mathbb{K}}(\min(N_A, N_B), \min(N_A, N_B)) \end{aligned}$$

s.t. S diagonal with nonnegative $S_{aa} = s_a$, i.e. $S_{ij} = \delta_{ij} s_i$ s.t. $s_i \geq 0 \quad \forall i = 1, 2, \dots \min(N_A, N_B)$.
$r \equiv$ (Schmidt) rank of $M :=$ number of nonzero singular values.
Assume $s_1 \geq s_2 \geq \cdots \geq s_r \geq 0$.
$V^{\dagger} \in \mathrm{Mat}_{\mathbb{K}}(\min(N_A, N_B), N_B)$ s.t. $V^{\dagger}V = 1$.

$$\mathrm{Mat}_{\mathbb{K}}(N_A, N_B) \xrightarrow{\text{SVD}} U_{\mathbb{K}}(N_A, \min(N_A, N_B)) \times \mathrm{diag}_{\mathbb{K}}(\min(N_A, N_B)) \times U_{\mathbb{K}}(\min(N_A, N_B), N_B)$$

$$M \xmapsto{\text{SVD}} USV^{\dagger}$$

Optimal approximation of M (rank r by matrix M' (rank $r' < r$) property.
In Frobenius norm $\|M\|_F^2 := \sum_{i,j} |M_{ij}|^2$, induced by inner product $\langle M|N\rangle = \mathrm{tr} M^{\dagger}N$. Indeed,

$$\mathrm{tr} M^{\dagger}N = (M^{\dagger})_{ik} N_{ki} = \overline{M}_{ki} N_{ki}$$

and so for

$$M' = US'V^{\dagger}, \qquad S' = \mathrm{diag}(s_1, s_2 \dots s_{r'}, 0 \dots) \tag{815}$$

cf. Eq. (19) of Schollwöck [63], i.e. 1 sets all but 1st r' singualr values to 0.
Use singular value decomposition (SVD) to derive Schmidt decomposition of general quantum state.
$\forall$ pure state $|\psi\rangle$ on AB,

$$|\psi\rangle = \sum_{i,j} \Psi_{ij} |i\rangle_A |j\rangle_B$$

where $\{|i\rangle_A\}, \{|j\rangle_B\}$ orthonormal bases of A, B ((complex) Hilbert spaces), with dim. N_A, N_B, respectively.
Let $\Psi_{i,j} \in \mathrm{Mat}_{\mathbb{K}}(N_A, N_B)$.
Then **reduced density operators** $\widehat{\rho}_A, \widehat{\rho}_B$ are such that

$$\begin{aligned} \widehat{\rho}_A = \mathrm{tr}_B |\psi\rangle\langle\psi| \\ \widehat{\rho}_B = \mathrm{tr}_A |\psi\rangle\langle\psi| \end{aligned}$$

In matrix form,

$$\rho_A = \Psi\Psi^\dagger$$
$$\rho_B = \Psi^\dagger\Psi$$

Indeed,

$$(\rho_A)_{ij} = \Psi_{ik}\overline{\Psi}_{jk}$$
$$(\rho_B)_{ij} = \overline{\Psi}_{ki}\Psi_{kj}$$
$$|\psi\rangle\langle\psi| = \sum_{i,j}\Psi_{ij}|i\rangle_A|j\rangle_B\sum_{l,m}\overline{\Psi}_{lm}\langle l|_A\langle m|_B$$
$$\mathrm{tr}_B|\psi\rangle\langle\psi| = \sum_{i,j}\Psi_{ik}\overline{\Psi}_{jk}|i\rangle_A\rangle j|_A$$

In matrix form,

$$\rho_A = \Psi\Psi^\dagger$$
$$\rho_B = \Psi^\dagger\Psi$$

Carry out SVD on Ψ in Eq. (20) of Schollwöck [63],

$$|\psi\rangle = \sum_{i,j}\Psi_{ij}|i\rangle_A|j\rangle_B$$

$$|\psi\rangle = \sum_{ij}\Psi_{ij}|i\rangle_A|j\rangle_B = \sum_{ij}\sum_{a=1}^{\min(N_A,N_B)}U_{ia}S_{aa}\overline{V}_{ja}|i\rangle_A|j\rangle_B = \sum_{a=1}^{\min(N_A,N_B)}\sum_i U_{ia}|i\rangle_A s_a\sum_j\overline{V}_{ja}|j\rangle_B = \sum_{a=1}^{\min(N_A,N_B)}s_a|a\rangle_A|a\rangle_B$$

Due to orthogonality of $U, V^\dagger$, $\{|a\rangle_A\}, \{|a\rangle_B\}$ orthonormal, and can be extended to be orthonormal bases of A, B.

If we restrict the sum to run only over the $r \leq \min(N_A, N_B)$ positive nonzero singular values (i.e., for $\sum_{a=1}^{\min(N_A,N_B)}$, $_a > 0 \quad \forall\, a \leq r$, and so

$$|\psi\rangle = \sum_{a=1}^{r}s_a|a\rangle_A|a\rangle_B$$

$r = 1$ (classical) product states. $|\psi\rangle = s_1|1\rangle_A|1\rangle_B$.
$r > 1$ entangled (quantum) states.

Schmidt decomposition on reduced density operators for A and B:

$$\widehat{\rho}_A = \sum_{a=1}^{r}s_a^2|a\rangle_A\langle a|_A$$
$$\widehat{\rho}_B = \sum_{a=1}^{r}s_a^2|a\rangle_B\langle a|_B$$

Respective eigenvectors are left and right singular vectors.

Von Neumann entropy can be read off:

$$S_{A|B}(|\psi\rangle) = -\mathrm{tr}\widehat{\rho}_A\log_2\widehat{\rho}_A = -\sum_{a=1}^{r}s_a^2\log_2 s_a^2$$

In view of large size of Hilbert spaces, approximate $|\psi\rangle$ by some $|\widetilde{\psi}\rangle$ spanned over state spaces A, B that have dims. r' only.

Since 2-norm of $|\psi\rangle$,

$$\||\psi\rangle\|_2^2 = \sum_{ij}|\Psi_{ij}|^2 = \|\Psi\|_F^2$$

since

$$\||\psi\rangle\|_2^2 = \sum_{a=1}^{r}s_a^2 = \sum_{ij}|\Psi_{ij}|^2$$

iff $\{|i\rangle\}, \{|j\rangle\}$ orthonormal. Optimal approx. of 2-norm given by optimal approx. of Ψ by $\overline{\Psi}$ in Frobenius norm, where $\overline{\Psi}$ is matrix of rank r'.
$\overline{\Psi} = USV^\dagger$, $S' = \text{diag}(s_1, \dots s_{r'}, 0 \dots)$ from above.
$\implies$ Schmidt decomposition of approximate state

$$|\overline{\Psi}\rangle = \sum_{a=1}^{r'} s_a |a\rangle_A |a\rangle_B \tag{816}$$

cf. Eq. (27) of Schollwöck [63], where s_a must be rescaled if normalization desired.

156.4 QR decomposition

cf. 4.1.2. of Schollwöck [63].
If actual value of singular values not used explicitly, then use *QR decomposition*.
QR decomposition: $\forall M \in \text{Mat}_{\mathbb{K}}(N_A, N_B)$,

$$M = QR,\ Q \in U_{\mathbb{K}}(N_A),\ \text{i.e. } Q^\dagger Q = 1 = QQ^\dagger,\ R \in \text{Mat}_{\mathbb{K}}(N_A, N_B) \text{ s.t. upper triangular, i.e. } R_{ij} = 0 \text{ if } i > j \tag{817}$$

thin QR decomposition: assume $N_A > N_B$. Then bottom $N_A - N_B$ rows of R are 0, so

$$M = Q \begin{bmatrix} R_1 \\ 0 \end{bmatrix} = \begin{bmatrix} Q_1 & Q_2 \end{bmatrix} \begin{bmatrix} R_1 \\ 0 \end{bmatrix} = Q_1 R_1$$
$$Q_1 \in \text{Mat}_{\mathbb{K}}(N_A, N_B)$$
$$R_1 \in \text{Mat}_{\mathbb{K}}(N_B, N_B)$$

While $Q_1^\dagger Q_1 = 1$ in general $Q_1 Q_1^\dagger \neq 1$

157 Matrix Product States (MPS)

cf. Section 4.13 Decomposition of arbitrary quantum states into MPS of Schollwöck [63].
Consider lattice of L sites, d-dim. local state spaces $\{\sigma_i\}_{i=1,\dots L}$.
Most general pure quantum state on lattice (assume normalized)

$$|\psi\rangle = \sum_{\sigma_1 \dots \sigma_L} c_{\sigma_1 \dots \sigma_L} |\sigma_1 \dots \sigma_L\rangle \tag{818}$$

cf. Eq. (30) of Schollwöck [63],

157.1 Left-canonical matrix product state

cf. Schollwöck [63],
Consider the process of refactoring or "flattening", which I claim to be a functor *flatten*:

$$|\psi\rangle \in \mathcal{H} \text{ s.t. } \dim \mathcal{H} = d^L \mapsto \Psi \in \text{Mat}_{\mathbb{K}}(d, d^{L-1})$$
$$\Psi_{\sigma_1, (\sigma_2 \dots \sigma_L)} = c_{\sigma_1 \dots \sigma_L}$$
$$\xrightarrow{\text{SVD}} c_{\sigma_1 \dots \sigma_L} = \Psi_{\sigma_1, (\sigma_2 \dots \sigma_L)} = \sum_a^{r_1} U_{\sigma_1, a_1} S_{a_1, a_1} (V^\dagger)_{a_1, (\sigma_2 \dots \sigma_L)} \equiv \sum_{a_1}^{r_1} U_{\sigma_1, a_1} c_{a_1, \sigma_2 \dots \sigma_L} \tag{819}$$

i.e.

$$(\mathbb{K}^d)^L \to \text{Mat}_{\mathbb{K}}(1, r) \times \text{Mat}_{\mathbb{K}}(r_1 d, d^{L-2})$$
$$c_{\sigma_1 \dots \sigma_L} \mapsto A^{\sigma_1}_{a_1}, \Psi_{(a_1 \sigma_2), (\sigma_3 \dots \sigma_L)}$$

s.t.

$$c_{\sigma_1 \dots \sigma_L} = \sum_{a_1}^{r_1} A^{\sigma_1}_{a_1} \Psi_{(a_1 \sigma_2), (\sigma_3 \dots \sigma_L)}$$

where rank $r_1 \leq d$.

$$U \in \text{Mat}_{\mathbb{K}}(d, \min(d, r)) = \text{Mat}_{\mathbb{K}}(d, r)$$

Consider d row vectors A^{σ_1}, $A^{\sigma_1}_{a_1} = U_{\sigma_1,a_1}$.

$$c_{a_1\sigma_2\ldots\sigma_L} = \sum_{a_1}^{r_1} A^{\sigma_1}_{a_1}\Psi_{(a_1,\sigma_2),(\sigma_3\ldots\sigma_L)} \text{ with}$$
$$\Psi_{(a_1\sigma_2),(\sigma_3\ldots\sigma_L)} \in \mathrm{Mat}_{\mathbb{K}}(r_1 d, d^{L-2})$$

So from Eq. (34) of Schollwöck [63],

$$c_{\sigma_1\ldots\sigma_L} = \sum_{a_1}^{r_1}\sum_{a_2}^{r_2} A^{\sigma_1}_{a_1}U_{(a_1\sigma_2),a_2}S_{a_2,a_2}(V^\dagger)_{a_2,(\sigma_3\ldots\sigma_L)} = \sum_{a_1}^{r_1}\sum_{a_2}^{r_2} A^{\sigma_1}_{a_1}A^{\sigma_2}_{a_1,a_2}\Psi_{(a_2\sigma_3),(\sigma_4\ldots\sigma_L)} \tag{820}$$

So for

$$U \in \mathrm{Mat}_{\mathbb{K}}(d, r_1 \times r_2) \mapsto \{A^{\sigma_2}\}_{\sigma_2}, \qquad |\{A^{\sigma_2}\}_{\sigma_2}| = d, \qquad A^{\sigma_2} \in \mathrm{Mat}_{\mathbb{K}}(r_1, r_2)$$

$A^{\sigma_2}_{a_1,a_2} = U_{(a_1,\sigma_2),a_2}$ and multiplied S and $V^\dagger$,

$$SV^\dagger \mapsto \Psi \in \mathrm{Mat}_{\mathbb{K}}(r_2 d, d^{L-3}); \qquad r_2 \leq r_1 d \leq d^2$$

and so continuing the application of SVD and refactoring (what I call applying the *flatten* functor)

$$\xrightarrow{\text{SVD}} c_{\sigma_1\ldots\sigma_L} = \sum_{a_1\ldots a_{L-1}} A^{\sigma_1}_{a_1}A^{\sigma_2}_{a_1a_2}\ldots A^{\sigma_{L-1}}_{a_{L-2},a_{L-1}}A^{\sigma_L}_{a_L-1} \equiv A^{\sigma_1}A^{\sigma_2}\ldots A^{\sigma_{L-1}}A^{\sigma_L}$$

157.1.1 Matrix Product State (definition)

Definition 221 (Matrix Product State)**.**

$$|\psi\rangle = \sum_{\sigma_1\ldots\sigma_L} A^{\sigma_1}A^{\sigma_2}\ldots A^{\sigma_{L-1}}A^{\sigma_L}|\sigma_1\ldots\sigma_L\rangle \tag{821}$$

Maximally, the dims. are

$$(1\times d),(d\times d^2)\ldots(d^{L/2-1}\times d^{L/2}),(d^{L/2}\times d^{L/2-1})\ldots(d^2\times d),(d\times 1)$$

Since $\forall$ SVD, $U^\dagger U = 1$,

$$\delta_{a_l,a'_l} = \sum_{a_{l-1}a_l}(U^\dagger)_{a_l,(a_{l-1}\sigma_l)}U_{(a_{l-1}\sigma_l),a'_l} = \sum_{a_{l-1}\sigma_l}(A^{\sigma_l})^\dagger_{a_l,a_{l-1}}A^{\sigma_l}_{a_{l-1},a'_l} = \sum_{\sigma_l}((A^{\sigma_2})^\dagger A^{\sigma_l})_{a_l,a'_l}$$

or

$$\sum_{\sigma_l}(A^{\sigma_l})^\dagger A^{\sigma_l} = 1 \tag{822}$$

cf. Eq. (38) of Schollwöck [63],

If for $\{A^{\sigma_l}\}_{\sigma_l}$, $\sum_{\sigma_l}(A^{\sigma_l})^\dagger A = 1$, $\{A^{\sigma_l}\}_{\sigma_l}$ are **left-normalized**; matrix product states that consist of only left-normalized matrices are **left-canonical**.

View Density Matrix Renormalization Group (DMRG) decomposition of universe into blocks A and B, split lattice into parts A,B, where A comprises sites 1 through l and B sites $l+1$ through L.

$$|a_l\rangle_A = \sum_{\sigma_1\ldots\sigma_l}(A^{\sigma_1}A^{\sigma_2}\ldots A^{\sigma_l})_{a_l,1}|\sigma_1\ldots\sigma_l\rangle$$
$$|a_l\rangle_B = \sum_{\sigma_{l+1}\ldots\sigma_L}(A^{\sigma_{l+1}}A^{\sigma_{l+2}}\ldots A^{\sigma_L})_{a_l,1}|\sigma_{l+1}\ldots\sigma_L\rangle$$

s.t. matrix product state (MPS) is

$$|\psi\rangle = \sum_{a_l}|a_l\rangle_A|a_l\rangle_B$$

157.1.2 Summarize this procedure of constructing, from a pure state, the matrix product state (version) by successive application Singular Value Decomposition (SVD) from the Category Theory point of view

$$(\mathbb{K}^d)^L \xrightarrow{\text{SVD}} (\mathrm{Mat}_{\mathbb{K}}(1,r_1))^d \times (\mathrm{Mat}_{\mathbb{K}}(r_1,r_2))^d \times \cdots \times (\mathrm{Mat}_{\mathbb{K}}(r_{L-2},r_{L-1}))^d \times (\mathrm{Mat}_{\mathbb{K}}(r_{L-1},1))^d$$

$$c_{\sigma_1 \ldots \sigma_L} \xmapsto{\text{SVD}} c_{\sigma_1 \ldots \sigma_L} = \sum_{a_1 \ldots a_{L-1}} A^{\sigma_1}_{a_1} A^{\sigma_2}_{a_1 a_2} \ldots A^{\sigma_{L-1}}_{a_{L-2}, a_{L-1}} A^{\sigma_L}_{a_{L-1}}$$

Consider all applications of SVD to get to a matrix product state (MPS): and remember the maximal values that the r_i's can take:

$$\begin{array}{lll} r_1 \leq d & r_{L/2} \leq d^{L/2} & r_{L-2} \leq d^2 \\ r_2 \leq d^2 & r_{L/2+1} \leq d^{L/2-1} & r_{L-1} \leq d \end{array}$$

Let us explicitly note the functors (that were applied) flatten (and its inverse), and the application of SVD, explicitly:

$$(\mathbb{K}^d)^L \xrightarrow{\text{flatten}^{-1}} \mathrm{Mat}_{\mathbb{K}}(d, d^{L-1}) \xrightarrow{\text{SVD}} U_{\mathbb{K}}(d,r_1) \times \mathrm{diag}_{\mathbb{K}}(r_1) \times U_{\mathbb{K}}(r_1, d^{L-1}) \xrightarrow{\cong} (\mathrm{Mat}_{\mathbb{K}}(1,r_1))^d \times \mathrm{Mat}_{\mathbb{K}}(r_1 d, d^{L-2}) \xrightarrow{\text{flatten}} (\mathrm{Mat}_{\mathbb{K}}(1,r_1))^d \times (\mathbb{K}^{r_1}) \times (\mathbb{K}^d)^{L-1}$$

$$c_{\sigma_1 \ldots \sigma_L} \xmapsto{\text{flatten}^{-1}} c_{\sigma_1 \ldots \sigma_L} = \Psi_{\sigma_1, (\sigma_2 \ldots \sigma_L)} \xmapsto{\text{SVD}} \Psi_{\sigma_1, (\sigma_2 \ldots \sigma_L)} = \sum_{a_1}^{r_1} U_{\sigma_1 a_1} S_{a_1, a_1} (V^\dagger)_{a_1, (\sigma_2 \ldots \sigma_L)} \xmapsto{\cong} c_{a_1 \sigma_2 \ldots \sigma_L} = \sum_{a_1}^{r_1} A^{\sigma_1}_{a_1} \Psi_{(a_1, a_2), (\sigma_3 \ldots \sigma_L)} \xmapsto{\text{flatten}} c_{a_1 \sigma_2 \ldots \sigma_L} = \sum_{a_1}^{r_1} A^{\sigma_1}_{a_1} c_{a_1 \sigma_2 \ldots \sigma_L}$$

with $\cong$ in this case denoting an isomorphism (clearly).

In considering some kind of recursive algorithm, so to repeat some series of steps until a matrix product state is obtained, consider this:

$$(\mathbb{K}^d)^L \longrightarrow (\mathrm{Mat}_{\mathbb{K}}(1,r_1))^d \times \mathbb{K}^{r_1} \times (\mathbb{K}^d)^{L-1}$$

$$c_{\sigma_1 \ldots \sigma_L} \longmapsto c_{\sigma_1 \ldots \sigma_L} = \sum_{a_1}^{r_1} A^{\sigma_1}_{a_1} c_{a_1 \sigma_2 \ldots \sigma_L}$$

So in summary, to obtain matrix product states, starting from a matrix,

$$\boxed{\begin{array}{l} \mathrm{Mat}_{\mathbb{K}}(d, d^{L-1}) \longrightarrow (\mathrm{Mat}_{\mathbb{K}}(1,r_1))^d \times \mathrm{Mat}_{\mathbb{K}}(r_1 d, d^{L-2}) \longrightarrow \ldots \longrightarrow (\mathrm{Mat}_{\mathbb{K}}(1,r_1))^d \times (\mathrm{Mat}_{\mathbb{K}}(r_1,r_2))^d \times \cdots \times (\mathrm{Mat}_{\mathbb{K}}(r_{n-1},r_n))^d \times (\mathrm{Mat}_{\mathbb{K}}(r_n d, d^{L-(n+1)}))^d \\ \\ \Psi_{\sigma_1, (\sigma_2 \ldots \sigma_L)} \longmapsto \sum_{a_1}^{r_1} A^{\sigma_1}_{a_1} \Psi_{(a_1,\sigma_2),(\sigma_3 \ldots \sigma_L)} \longmapsto \cdots \longmapsto \sum_{a_1, a_2, \ldots a_n}^{r_1, r_2, \ldots r_n} A^{\sigma_1}_{a_1} A^{\sigma_2}_{a_1 a_2} \ldots A^{\sigma_n}_{a_{n-1} a_n} \Psi_{(a_n \sigma_{n+1}), (\sigma_{n+2} \ldots \sigma_L)} \end{array}} \tag{823}$$

157.2 Right-canonical matrix product state

cf. Schollwöck [63],

We can start from right in order to obtain

$$c_{\sigma_1 \ldots \sigma_L} = \Psi_{(\sigma_1 \ldots \sigma_{L-1}), \sigma_L} = \sum_{a_{L-1}} U_{(\sigma_1 \ldots \sigma_{L-1}), a_{L-1}} S_{a_{L-1}, a_{L-1}} (V^\dagger)_{a_{L-1}, \sigma_L} = \sum_{a_{L-1}} \Psi_{(\sigma_1 \ldots \sigma_{L-2}), (\sigma_{L-1} a_{L-1})} B^{\sigma_L}_{a_{L-1}} =$$

$$= \sum_{a_{L-1}, a_{L-2}} U_{(\sigma_1 \ldots \sigma_{L-2}), a_{L-2}} S_{a_{L-2}, a_{L-2}} (V^\dagger)_{a_{L-2}, (\sigma_{L-1} a_{L-1})} B^{\sigma_L}_{a_{L-1}} = \sum_{a_{L-2}, a_{L-1}} \Psi_{(\sigma_1 \ldots \sigma_{L-3}), (\sigma_{L-2} a_{L-2})} B^{\sigma_{L-1}}_{a_{L-2}, a_{L-1}} B^{\sigma_L}_{a_{L-1}} = \ldots$$

or consider

$$(\mathbb{K}^d)^L \xrightarrow{\text{flatten}^{-1}} \text{Mat}_{\mathbb{K}}(d^{L-1}, d) \xrightarrow{\text{SVD}} U_{\mathbb{K}}(d^{L-1}, r_{L-1}) \times \text{diag}_{\mathbb{K}}(r_{L-1}) \times U_{\mathbb{K}}(r_{L-1}, d) \xrightarrow{\cong} \text{Mat}_{\mathbb{K}}(d^{L-2}, dr_{L-1}) \times (\text{Mat}_{\mathbb{K}}(r_{L-1}, 1))^d \xrightarrow{\text{SVD}}$$

$$c_{\sigma_1 \dots \sigma_L} \xmapsto{\text{flatten}^{-1}} c_{\sigma_1 \dots \sigma_L} = \Psi_{(\sigma_1 \dots \sigma_{L-1}), \sigma_L} \xmapsto{\text{SVD}} c_{\sigma_1 \dots \sigma_L} = \sum_{a_{L-1}}^{r_{L-1}} U_{(\sigma_1 \dots \sigma_{L-1}), a_{L-1}} S_{a_{L-1}, a_{L-1}} (V^\dagger)_{a_{L-1}, \sigma_L} \xmapsto{\cong} \begin{matrix} U_{(\sigma_1 \dots \sigma_{L-1}), a_{L-1}} S_{a_{L-1}, a_{L-1}} = \Psi_{(\sigma_1 \dots \sigma_{L-2}), (\sigma_{L-1} a_{L-1})} \\ (V^\dagger)_{a_{L-1}, \sigma_L} = B^{\sigma_L}_{a_{L-1}} \\ c_{\sigma_1 \dots \sigma_L} = \sum_{a_{L-1}} \Psi_{(\sigma_1 \dots \sigma_{L-2}), (\sigma_{L-1}, a_{L-1})} B^{\sigma_L}_{a_{L-1}} \end{matrix} \xrightarrow{\text{SVD}}$$

$$\xrightarrow{\text{SVD}} U_{\mathbb{K}}(d^{L-2}, r_{L-2}) \times \text{diag}_{\mathbb{K}}(r_{L-2}) \times U_{\mathbb{K}}(r_{L-2}, dr_{L-1}) \times (\text{Mat}_{\mathbb{K}}(r_{L-1}, 1))^d \xrightarrow{\cong} \text{Mat}_{\mathbb{K}}(d^{L-3}, dr_{L-2}) \times (\text{Mat}_{\mathbb{K}}(r_{L-2}, r_{L-1}))^d \times (\text{Mat}_{\mathbb{K}}(r_{L-1}, 1))^d$$

$$\xmapsto{\text{SVD}} c_{\sigma_1 \dots \sigma_L} = \sum_{a_{L-1}, a_{L-2}} U_{(\sigma_1 \dots \sigma_{L-2}), a_{L-2}} S_{a_{L-2}, a_{L-2}} (V^\dagger)_{a_{L-2}, (\sigma_{L-1} a_{L-1})} B^{\sigma_L}_{a_{L-1}} \xmapsto{\cong} \begin{matrix} U_{(\sigma_1 \dots \sigma_{L-2}), a_{L-2}} S_{a_{L-2}, a_{L-2}} = \Psi_{(\sigma_1 \dots \sigma_{L-3}), (\sigma_{L-2} a_{L-2})} \\ (V^\dagger)_{a_{L-2}, (\sigma_{L-1} a_{L-1})} = B^{\sigma_{L-1}}_{a_{L-2} a_{L-1}} \\ c_{\sigma_1 \dots \sigma_L} = \sum_{a_{L-1}, a_{L-2}} \Psi_{(\sigma_1 \dots \sigma_{L-3}), (\sigma_{L-2}, a_{L-2})} B^{\sigma_{L-1}}_{a_{L-2}, a_{L-1}} B^{\sigma_L}_{a_{L-1}} \end{matrix}$$

with $\cong$ in this case denoting an isomorphism (clearly).

And so we can explicitly state the recursion step, for the purpose of writing numerical implementations/algorithms: $\forall\, l = 1, 2 \dots L$,

$$\text{Mat}_{\mathbb{K}}(d^{L-l}, dr_{L-(l-1)}) \longrightarrow \text{Mat}_{\mathbb{K}}(d^{L-(l+1)}, dr_{L-l}) \times (\text{Mat}_{\mathbb{K}}(r_{L-l}, r_{L-(l-1)}))^d$$

$$\Psi_{(\sigma_1 \dots \sigma_{L-l}), (\sigma_{L-(l-1)} a_{L-(l-1)})} \longmapsto \Psi_{(\sigma_1 \dots \sigma_{L-l}), (\sigma_{L-(l-1)} a_{L-(l-1)})} = \sum_{a_{L-l}} \Psi_{(\sigma_1 \dots \sigma_{L-(l+1)}), (\sigma_{L-l} a_{L-l})} B^{\sigma_{L-(l-1)}}_{a_{L-l}, a_{L-(l-1)}}$$

and we finally obtained, after successive applications SVD, the matrix product state:

$$(\mathbb{K}^d)^L \longrightarrow \text{Mat}_{\mathbb{K}}(d^{L-1}, d) \longrightarrow (\text{Mat}_{\mathbb{K}}(1, r_1))^d \times (\text{Mat}_{\mathbb{K}}(r_1, r_2))^d \times \dots \times (\text{Mat}_{\mathbb{K}}(r_{L-2}, r_{L-1}))^d \times (\text{Mat}_{\mathbb{K}}(r_{L-1}, 1))^d$$

$$c_{\sigma_1 \dots \sigma_L} \longmapsto \Psi_{(\sigma_1 \dots \sigma_{L-l}), \sigma_L} \longmapsto c_{\sigma_1 \dots \sigma_L} = \sum_{a_1 \dots a_{L-1}} B^{\sigma_1}_{a_1} B^{\sigma_2}_{a_1 a_2} \dots B^{\sigma_{L-1}}_{a_{L-2} a_{L-1}} B^{\sigma_L}_{a_{L-1}}$$

Since

$$V^\dagger V = 1 \tag{824}$$

, then

$$\delta_{a_l a'_l} = \sum_{\sigma_m a_m} (V^\dagger)_{a_l (\sigma_m a_m)} V_{(\sigma_m a_m) a'_l} = \sum_{\sigma_m a_m} B^{\sigma_m}_{a_l a_m} \overline{B}^{\sigma_m}_{a'_l a_m} \Longrightarrow \sum_{\sigma_m} \boxed{B^{\sigma_m} (B^{\sigma_m})^\dagger = 1} \tag{825}$$

The B-matrices that obey this condition are referred to as **right-normalized** matrices. A matrix product state (MPS) entirely consisting of a product of these right-normalized matrices is called **right-canonical**.

157.3 Matrix Product Operators (MPO)

The form of a general operator, $\widehat{O}$ is the following:

$$\widehat{O} = \sum_{\{\sigma\}} \sum_{\{\sigma'\}} c^{\sigma_1 \dots \sigma_L, \sigma'_1 \dots \sigma'_L} |\sigma_1 \dots \sigma_L\rangle \langle \sigma'_1 \dots \sigma'_L| \in \mathcal{H} \otimes \mathcal{H}^* \tag{826}$$

with $\dim \mathcal{H} = \dim \mathcal{H}^* = d^L$.

For MPO, do the same decomposition as done in Eq. 809 or in 813, but with the double index $\sigma_i \sigma'_i$ taking the role of index σ_i in MPS (i.e. do this substitution and the decomposition will proceed *exactly* as before).

$$
\begin{array}{l}
\mathrm{Mat}_{\mathbb{C}}(d^L, d^L) \\
\quad\downarrow \text{reorder} \\
\mathrm{Mat}_{\mathbb{C}}(1, (d^2)^L) \xrightarrow{\text{reshape}} \mathrm{Mat}_{\mathbb{C}}(d^2, (d^2)^{L-1}) \xrightarrow{\text{SVD}} \mathrm{Mat}_{\mathbb{C}}(d^2, r_i) \times \mathrm{Mat}_{\mathbb{C}}(r_i, r_i) \times \mathrm{Mat}_{\mathbb{C}}(r_i, (d^2)^{L-1}) \xrightarrow{\text{slice and multiply}} \mathrm{Mat}_{\mathbb{C}}(1, r_i)^d \times \mathrm{Mat}_{\mathbb{C}}(r_i, (d^2)^{L-1})
\end{array}
$$

$$
\begin{array}{l}
c^{\sigma_1 \dots \sigma_L, \sigma'_1 \dots \sigma'_L} \\
\quad\downarrow \text{reorder} \\
c^{\sigma_1 \sigma'_1 \sigma_2 \sigma'_2 \dots \sigma_L \sigma'_L} \xmapsto{\text{reshape}} \Psi_{\sigma_1 \sigma'_1, (\sigma_2 \sigma'_2, \sigma_3 \sigma'_3 \dots \sigma_L \sigma'_L)} \xmapsto{=} U_{\sigma_1 \sigma'_1, a_1} S_{a_1 a_1} V^{\dagger}_{a_1, (\sigma_2 \sigma'_2 \dots \sigma_L \sigma'_L)} \xmapsto{=} \sum_{\sigma_1 \sigma'_1} A^{\sigma_1 \sigma'_1}_{1, a_1} c^{a_1, \sigma_2 \sigma'_2 \dots \sigma_L \sigma'_L}
\end{array}
$$

$$
\boxed{
\begin{array}{l}
\mathrm{Mat}_{\mathbb{C}}(r_{i-1}, (d^2)^{L-(i-1)}) \xrightarrow{\text{reshape}} \mathrm{Mat}_{\mathbb{C}}(r_{i-1} d^2, (d^2)^{L-i}) \xrightarrow{\text{SVD}} \mathrm{Mat}_{\mathbb{C}}(r_{i-1} d^2, r_i) \times \mathrm{Mat}_{\mathbb{C}}(r_i, r_i) \times \mathrm{Mat}_{\mathbb{C}}(r_i, (d^2)^{L-i}) \xrightarrow{\text{slice and multiply}} \mathrm{Mat}_{\mathbb{C}}(r_{i-1}, r_i)^{d^2} \times \mathrm{Mat}_{\mathbb{C}}(r_i, (d^2)^{L-i}) \\
\\
c^{a_{i-1}, \sigma_i \sigma'_i \sigma_{i+1} \sigma'_{i+1} \dots \sigma_L \sigma'_L} \xmapsto{\text{reshape}} \Psi_{a_{i-1} \sigma_i \sigma'_i, (\sigma_{i+1} \sigma'_{i+1} \sigma_{i+2} \sigma'_{i+2} \dots \sigma_L \sigma'_L)} \xmapsto{=} U_{a_{i-1} \sigma_i \sigma'_i, a_i} S_{a_i a_i} V^{\dagger}_{a_i, \sigma_{i+1} \sigma'_{i+1} \dots \sigma_L \sigma'_L} \xmapsto{=} \sum_{\sigma_i \sigma'_i}^{max} A^{\sigma_i \sigma'_i}_{a_{i-1}, a_i} c^{a_i, \sigma_{i+1} \sigma'_{i+1} \dots \sigma_L \sigma'_L}
\end{array}
} \tag{827}
$$

157.3.1 Numerical implementation; both in BLAS and cuBLAS

As stated in the CUDA Toolkit Documentation v8.0 for cuSOLVER, under section 5.3.6. `cusolverDn<t>gesvd()` and Remark 1, `gesvd` "only supports" `m>=n`, for matrix you want to decompose $A \in \mathrm{Mat}_{\mathbb{K}}(m, n)$. So number of rows must be greater than or equal to number of columns. And so we can only consider right-normalized matrices in a practical implementation.

I suspect it's the same in BLAS.

Consider the very first step, $l = 1$, in a procedure to calculate the matrix product state.

$$
\mathrm{Mat}_{\mathbb{K}}(d^{L-1}, d) \xrightarrow{\text{SVD}} U_{\mathbb{K}}(d^{L-1}, r_{L-1}) \times \mathrm{diag}_{\mathbb{K}}(r_{L-1}) \times U_{\mathbb{K}}(r_{L-1}, d) \xrightarrow{\cong} \mathrm{Mat}_{\mathbb{K}}(d^{L-2}, d r_{L-1}) \times (\mathrm{Mat}_{\mathbb{K}}(r_{L-1}, 1))^d
$$

$$
\Psi_{(\sigma_1 \dots \sigma_{L-1}), \sigma_L} \xmapsto{\text{SVD}} = \sum_{a_{L-1}}^{r_{L-1}} U_{(\sigma_1 \dots \sigma_{L-1}), a_{L-1}} S_{a_{L-1}, a_{L-1}} (V^{\dagger})_{a_{L-1}, \sigma_L} \xmapsto{\cong} \begin{array}{l} U_{(\sigma_1 \dots \sigma_{L-1}), a_{L-1}} S_{a_{L-1}, a_{L-1}} = \Psi_{(\sigma_1 \dots \sigma_{L-2}), (\sigma_{L-1} a_{L-1})} \\ (V^{\dagger})_{a_{L-1}, \sigma_L} = B^{\sigma_L}_{a_{L-1}} \end{array}
$$

with $\cong$ in this case denoting an isomorphism, the *reshaping* of a matrix into different matrix size dimensions, which should be the inverse of a "flatten" functor, which I'll denote as flatten^{-1} as well (and this is this same isomorphism we're talking about).

Let's deal with the specific procedure of flatten^{-1}, how it reshapes indices in accordance with different matrix size dimensions, and with the so-called "stride" when going from, say, 2-dimensional indices to a "flattened" 1-dimensional index.

Note also as a practical numerical implementation design point, LAPACK's linear algebra BLAS library package and CUBLAS assumes *column*-major ordering.

Consider $i = 1, 2, \dots L-1$ (for site i) (or for 0-based counting, starting to count from 0, $i = 0, 1, \dots L-2$; be aware of this difference as in practical numerical implementation, in C, C++, Python, it assumes 0-based counting).

For a state space of dimension d, we can consider the specific example of $d = 2$, representing say a spin-1/2 system. Then index σ_i can be 0 or 1: $\sigma_i \in \{0, 1\}$. In general, $\sigma_i \in \{0, 1, \dots d-1\}$. I may use d or 2 in the context of the number of states (basis vectors) of the spin system (state vector space).

Consider site i. Suppose the spin system there interacts most with sites $i-1$, $i+1$, and then next sites $i-2$, $i+2$, etc. So the values at $\sigma_{i-1}, \sigma_{i+1}$, etc. are most important in calculating interactions with spin system at site i.

$$
\{0, 1\}^{L-1} \xrightarrow{(\text{flatten})^{-1}} \{0, 1, \dots 2^{L-1} - 1\}
$$

$$
(\sigma_0, \sigma_1, \dots \sigma_{L-2}) \xmapsto{(\text{flatten})^{-1}} I_{L-1} := \sigma_0 + 2\sigma_1 + \dots + 2^i \sigma_i + \dots + 2^{L-2} \sigma_{L-2} = \sum_{i=0}^{L-2} 2^i \sigma_i
$$

Then we seek this reshaping of the matrix index - assuming 0-based counting/ordering, for $l = 1$:

In this way, states of a site i are closest in memory addresses in the allocation of a 1-dim. array, on CPU or GPU memory, so that memory access operations should be efficient.

Assuming SVD doesn't change the striding, and defining the result of matrix multiplication:

$$U_{(\sigma_0,\sigma_1\dots\sigma_{L-2}),a_{L-1}}S_{a_{L-1},a_{L-1}} =: (US)_{(\sigma_0\dots\sigma_{L-2}),a_{L-1}} \in \mathrm{Mat}_{\mathbb{K}}(d^{L-1}, r_{L-1})$$

We can reshape (i.e. $(\text{flatten})^{-1}$) in such a manner:

$$\mathrm{Mat}_{\mathbb{K}}(d^{L-1}, r_{L-1}) \xrightarrow{(\text{flatten})^{-1}} \mathrm{Mat}_{\mathbb{K}}(d^{L-2}, dr_{L-1})$$

$$(US)_{(\sigma_0\dots\sigma_{L-2}),a_{L-1}} \xmapsto{(\text{flatten})^{-1}} \Psi_{(\sigma_0,\sigma_1,\dots\sigma_{L-3}),(\sigma_{L-2}a_{L-1})}$$

$$\{0,1,\dots 2^{L-1}-1\}\times\{0,1,\dots r_{L-1}-1\} \xrightarrow{(\text{flatten})^{-1}} \{0,1,\dots 2^{L-2}-1\}\times\{0,1,\dots dr_{L-1}-1\}$$

$$I_{L-1}, a_{L-1} \xmapsto{(\text{flatten})^{-1}} I_{L-1} \mod 2^{L-2}, \tfrac{I_{L-1}}{2^{L-2}} + da_{L-1}$$

Reshaping $V^\dagger$ at iteration $l = 1$ can be done as follows:

$$U_{\mathbb{K}}(r_{L-1}, d) \xrightarrow{(\text{flatten})^{-1}} (\mathrm{Mat}_{\mathbb{K}}(r_{L-1}, 1))^d$$

$$(V^\dagger)_{a_{L-1},\sigma_{L-1}} \xmapsto{(\text{flatten})^{-1}} (V^\dagger)_{a_{L-1},\sigma_{L-1}} = B^{\sigma_{L-1}}_{a_{L-1}}$$

$$\{0,1,\dots r_{L-1}-1\}\times\{0,1,\dots d-1\} \xrightarrow{(\text{flatten})^{-1}} (\{0,1,\dots r_{L-1}-1\})^d$$

$$a_{L-1}, \sigma_{L-1} \xmapsto{(\text{flatten})^{-1}} a_{L-1}$$

Let's do this same procedure, reshaping or $(\text{flatten})^{-1}$, for a general l iteration.

$$\mathrm{Mat}_{\mathbb{K}}(d^{L-l}, r_{L-l}) \xrightarrow{(\text{flatten})^{-1}} \mathrm{Mat}_{\mathbb{K}}(d^{L-(l+1)}, dr_{L-l})$$

$$(US)_{(\sigma_0\dots\sigma_{L-(l+1)}),a_{L-l}} \xmapsto{(\text{flatten})^{-1}} \Psi_{(\sigma_0,\sigma_1,\dots\sigma_{L-(l+2)}),(\sigma_{L-(l+1)}a_{L-l})}$$

$$\{0,1,\dots d^{L-l}-1\}\times\{0,1,\dots r_{L-l}-1\} \xrightarrow{(\text{flatten})^{-1}} \{0,1,\dots d^{L-(l+1)}-1\}\times\{0,1,\dots dr_{L-l}-1\}$$

$$I_{L-l}, a_{L-l} \xmapsto{(\text{flatten})^{-1}} I_{L-l} \mod d^{L-(l+1)}, \tfrac{I_{L-l}}{d^{L-(l+1)}} + da_{L-l}$$

$$U_{\mathbb{K}}(r_{L-l}, dr_{L-(l-1)}) \xrightarrow{\text{(flatten)}^{-1}} (\text{Mat}_{\mathbb{K}}(r_{L-l}, r_{L-(l-1)}))^d$$

$$(V^\dagger)_{a_{L-l},(\sigma_{L-l}a_{L-(l-1)})} \xmapsto{\text{(flatten)}^{-1}} (V^\dagger)_{a_{L-l},(\sigma_{L-l}a_{L-(l-1)})} = B^{\sigma_{L-l}}_{a_{L-l},a_{L-(l-1)}}$$

$$\{0,1,\dots r_{L-l}-1\} \times \{0,1,\dots dr_{L-(l-1)}-1\} \xrightarrow{\text{(flatten)}^{-1}} (\{0,1,\dots r_L-1\} \times \{0,1,\dots r_{L-(l-1)}-1\})^d$$

$$a_{L-l}, (\sigma_{L-l}a_{L-(l-1)}) := a_{L-l}, \sigma_{L-1} + da_{L-(l-1)} \xmapsto{\text{(flatten)}^{-1}} a_{L-l}, \tfrac{(\sigma_{L-1}a_{L-(l-1)})}{d}\,;\, \sigma_{L-l} = (\sigma_{L-l}a_{L-(l-1)}) \mod d$$

157.3.2 Numerical implementations of initial states

Something else that shouldn't be overlooked is the numerical implementation of *initial states*, the c's of a state $|\psi\rangle = \sum_{\{\sigma\}} c^\sigma |\{\sigma\}\rangle$ for a many-body quantum system. Remember what the postulates of quantum mechanics say and interpret accordingly (and correctly). While we call them "probability amplitudes", one should be careful about what physical interpretation we may (or may not!) assign them. One thing's for certain: $c \in \mathbb{C}$ and normalization of the quantum state: $|\langle\psi|\psi\rangle|^2 = 1$

Here are some setups to try:

$d = 2, L = 2$, $d^L = 2^2 = 4$.

$$\begin{bmatrix} c_{\uparrow\uparrow} & c_{\uparrow\downarrow} & c_{\downarrow\uparrow} & c_{\downarrow\downarrow} \end{bmatrix} \mapsto \begin{bmatrix} c_{\uparrow\uparrow} & c_{\uparrow\downarrow} \\ c_{\downarrow\uparrow} & c_{\downarrow\downarrow} \end{bmatrix}$$

Singlet state: $|\psi\rangle = \frac{1}{\sqrt{2}}|\uparrow\downarrow\rangle - \frac{1}{\sqrt{2}}|\downarrow\uparrow\rangle$, $\begin{bmatrix} 0 & \frac{1}{\sqrt{2}} \\ \frac{-1}{\sqrt{2}} & 0 \end{bmatrix}$

$d = 2, L = 3, d^L = 2^8 = 8$

For notational convenience, let $\uparrow \equiv 1$, $\downarrow \equiv 0$

$$\begin{bmatrix} c_{000} & c_{001} & c_{010} & c_{011} & c_{100} & \dots & c_{111} \end{bmatrix} \mapsto \begin{bmatrix} c_{000} & c_{001} & \dots & c_{011} \\ c_{100} & c_{001} & \dots & c_{111} \end{bmatrix}$$

$d = 3, L = 2$, $d^L = 3^2 = 9$

$$\begin{bmatrix} c_{-1-1} & c_{-10} & c_{-11} & \dots & c_{11} \end{bmatrix} \mapsto \begin{bmatrix} c_{-1-1} & c_{-10} & c_{-11} \\ c_{0-1} & c_{00} & c_{01} \\ c_{1-1} & c_{10} & c_{11} \end{bmatrix}$$

Part XXXVII
Conformal Field Theory; Virasoro Algebra

cf. Schottenloher (2008) [50]

158 Conformal Transformations

158.1 Semi-Riemannian manifolds (review and (key) examples)

cf. pp. 7, Ch. 1 "Conformal Transformations and Conformal Killing Fields." Schottenloher (2008) [50]

Semi-Riemannian manifold is a pair (M, g) s.t.
smooth manifold M, $\dim M = n$,
smooth tensor field g s.t. $g : a \in M \mapsto \Omega^2(T_a M)$, i.e. $\forall\, a \in M$, g assigns a a nonnegative and symmetric bilinear form on tangent space $T_a M$.

In local coordinates, $x^1 \dots x^n$ of manifold M,
given chart $\phi : U \to V$, open subset $U \subseteq M$, open subset $V \subseteq \mathbb{R}^n$,

$$\phi(a) = (x^1(a) \dots x^n(a)),\ a \in M$$

Bilinear form g_a on $T_a M$, written

$$g_a(X, Y) = g_{\mu\nu}(a) X^\mu Y^\nu$$

Tangent vectors $X = X^\mu \partial_\mu$, $Y = Y^\nu \partial_\nu \in T_a M$ basis $\partial_\mu := \frac{\partial}{\partial x^\mu}$, $\mu = 1 \dots n$ of tangent space $T_a M$, induced by chart ϕ.

By assumption, matrix

$$g_{\mu\nu}(a)$$

Nondegenerate and symmetric, $\forall\, a \in U$, i.e.

$$\det(g_{\mu\nu}(a)) \neq 0, \qquad (g_{\mu\nu}(a))^T = (g_{\mu\nu}(a))$$

Differentiating of g_a implies matrix $g_{\mu\nu}(a)$ depends differentiably on a.
That means that in its dependence on local coordinates x^j, coefficient $g_{\mu\nu} = g_{\mu\nu}(x)$ are smooth functions.
In general, $g_{\mu\nu} X^\mu X^\nu > 0$ doesn't hold $\forall\, X \neq 0$, i.e. $g_{\mu\nu}(a)$ not required to be positive-definite.
2 important subcases: [15]
Riemannian manifold: metric g positive definite, signature $n = \dim M$.
Lorentz manifold specified as semi-Riemannian manifold with $(p, q) = (n-1, 1)$ or $(p, q) = (1, n-1)$.
Metric g has signature $n - 2$ (positive convention) or $2 - n$ (negative convention).

158.1.1 Examples (of Riemannian manifolds for Conformal Field Theory)

$\mathbb{R}^{p,q} = (\mathbb{R}^{p,q}, g^{p,q})$, $p, q \in \mathbb{N}$, where

$$g^{p,q}(X, Y) := \sum_{i=1}^{p} X^i Y^i - \sum_{i=p+1}^{p+q} X^i Y^i$$

Hence

$$(g_{\mu\nu}) = \begin{pmatrix} 1_p & \\ & -1_q \end{pmatrix} = \mathrm{diag}(1 \dots 1, -1, \cdots - 1)$$

$\mathbb{R}^{1,3} = \mathbb{R}^{3,1}$, usual Minkowski space.
$\mathbb{R}^{1,1}$, 2 -dim. Minkowski space (Minkowski plane).
$\mathbb{R}^{2,0}$, Euclidean plane.
$\mathbb{S}^2 \subset \mathbb{R}^{3,0}$, compactification of $\mathbb{R}^{2,0}$, structure of Riemannian manifold on 2-sphere $\mathbb{S}^2$ induced by inclusion in $\mathbb{R}^{2,0}$
$\mathbb{S} \times \mathbb{S} \subset \mathbb{R}^{2,2}$, compactification of $\mathbb{R}^{1,1}$. More precisely,
$\mathbb{S} \times \mathbb{S} \subset \mathbb{R}^{2,0} \times \mathbb{R}^{0,2} \simeq \mathbb{R}^{2,2}$ where structure of semi-Riemannian manifold on $\mathbb{S} \times \mathbb{S}$ induced by inclusion into $\mathbb{R}^{2,2}$.

[15] https://doc.sagemath.org/html/en/reference/manifolds/sage/manifolds/differentiable/pseudo_riemannian.html

$\mathbb{S}^p \times \mathbb{S}^q \subset \mathbb{R}^{p+1,0} \times \mathbb{R}^{0,q+1} \simeq \mathbb{R}^{p+1,q+1}$ with p-sphere $\mathbb{S}^p = \{X \in \mathbb{R}^{p+1} : g^{p+1,0}(X,X) = 1\} \subset \mathbb{R}^{p+1,0}$, q-sphere $\mathbb{S}^q \subset \mathbb{R}^{0,q+1}$
yields a compactification of $\mathbb{R}^{p,q}$ for $p, q \geq 1$

Compact semi-Riemannian manifold denoted by $\mathbb{S}^{p,q}$, for $p, q \geq 0$.

Quadrics $N^{p,q}$ (of Sec. 2.1) are locally isomorphic to $\mathbb{S}^{p,q}$ from point of view of conformal geometry.

For the "*negative convention*":

$$g^{p,q}(X,Y) = -\sum_{i=0}^{p-1} X^i Y^i + \sum_{i=p}^{p+q} X^i Y^i$$

$$(g_{\mu\nu}) = \begin{pmatrix} -1_p & \\ & 1_q \end{pmatrix} = \mathrm{diag}(-1, \cdots -1, 1 \ldots 1)$$

$\mathbb{R}^{1,3}$, Minkowski space.
$\mathbb{R}^{1,1}$, 2 -dim. Minkowski space.
$\mathbb{R}^{0,2}$, Euclidean plane.
$\mathbb{S}^2 \subset \mathbb{R}^{0,3}$, compactification of $\mathbb{R}^{0,2}$
$\mathbb{S} \times \mathbb{S} \subset \mathbb{R}^{0,2} \times \mathbb{R}^{2,0} \simeq \mathbb{R}^{2,2}$
$\mathbb{S}^p \times \mathbb{S}^q \subset \mathbb{R}^{0,p+1} \times \mathbb{R}^{q+1,0} \simeq \mathbb{R}^{p+1,q+1}$ with p-sphere $\mathbb{S}^p = \{X \in \mathbb{R}^{p+1} : g^{0,p+1}(X,X) = 1\} \subset \mathbb{R}^{0,p+1}$, q-sphere $\mathbb{S}^q \subset \mathbb{R}^{q+1,0}$
yields a compactification of $\mathbb{R}^{p,q}$

Definition 222 (Conformal transformation or conformal map)**.** Let 2 semi-Riemannian manifolds(M,g), (M',g'), $\dim M = \dim M'$, let open $U \subset M$, open $V \subset M'$.

conformal transformation or **conformal map** is a smooth $\varphi : U \to V$ of maximal rank, if $\exists$ smooth $\Omega : U \to \mathbb{R}^+$ s.t.

$$\varphi^* g' = \Omega^2 g \tag{828}$$

where $\varphi * g'(X,Y) := g'(T\varphi(X), T\varphi(Y))$ and $T\varphi : TU \to TV$ denote tangent map (derivative) of φ.

$\Omega \equiv$ *conformal factor* of φ.

Locally, $y^i = \varphi^i(x)$,

$$\frac{\partial \varphi^i}{\partial x^j} = \frac{\partial y^i}{\partial x^j}$$

Then

$$X = X^k \frac{\partial}{\partial x^k} = X^k \frac{\partial y^i}{\partial x^k} \frac{\partial}{\partial y^i} = X^k \frac{\partial \varphi^i}{\partial x^k} \frac{\partial}{\partial y^k} \in TM$$

and so

$$\varphi^* g'(X,Y) = g'(T\varphi(X), T\varphi(Y)) = (g')_{ij} X^k \frac{\partial y^i}{\partial x^k} Y^l \frac{\partial y^j}{\partial x^l} = (g')_{ij} X^k \frac{\partial \varphi^i}{\partial x^k} Y^l \frac{\partial y^j}{\partial x^l}$$

$$\implies (\varphi^* g')_{kl} = (g')_{ij} \frac{\partial y^i}{\partial x^k} \frac{\partial y^j}{\partial x^l}$$

$$\implies (\varphi^* g')_{kl} = (g')_{ij} \frac{\partial \varphi^i}{\partial x^k} \frac{\partial \varphi^j}{\partial x^l} = \Omega^2 g_{kl}$$

Definition 223. extension of G by group A is (given by) an *exact sequence* of group homomorphisms.

$$1 \longrightarrow A \xrightarrow{i} E \xrightarrow{\pi} G \longrightarrow 1$$

cf. Def. 3.1 of Schottenloher (2008) [50].

Recall that an exact sequence, if $\begin{array}{l} \mathrm{im}(1 \to A) = \ker(i) \\ \mathrm{im}(i) = \ker(\pi) \\ \mathrm{im}(\pi) = \ker(G \to 1) \end{array}$

By Thm., $1 \to A \xrightarrow{i} E$ exact so i injective.
$E \xrightarrow{\pi} G \to 1$ exact so π surjective.

Extension is called **central** if A abelian and image im$i$ is in center of E, i.e. $a \in A, b \in E \implies i(a)b = bi(a)$.

158.1.2 Examples of extensions of G, and central extensions of G (which has a particular E)

- e.g. central extension has form

$$1 \longrightarrow A \xrightarrow{i} A \times G \xrightarrow{\mathrm{pr}_2} G \longrightarrow 1$$

where
$i : A \to A \times G$
$a \mapsto (a, 1)$

$$i(a)(a', g) = (a, 1)(a', g) = (aa', g) =$$
$$= (a'a, g \cdot 1) = (a', g)(a, 1) = (a', g)i(a)$$

Notice that what the *exactness* property of an exact sequence does:

$$\mathrm{pr}_2 i(a) = \mathrm{pr}_2(a, 1) = 1$$

- e.g. of a nontrivial central extension is exact sequence

$$1 \longrightarrow \mathbb{Z}/k\mathbb{Z} \longrightarrow E \times U(1) \xrightarrow{\pi} U(1) \longrightarrow 1 \tag{829}$$

with $\pi(z) = z^k \quad \forall\, k \in \mathbb{N},\ k \geq 2$, since $E = U(1)$ and $\mathbb{Z}/k\mathbb{Z}$ are not isomorphic.

Also, homomorphism $\tau : U(1) \to E$ with $\pi \circ \tau = 1_{U(1)}$, doesn't exist, since there's no global kth root.

EY : 20170926 It's that in integer division of the argument in a complex number $z \in U(1)$, and exponent multiplication by k, you go from 1 to many and many to 1, depending upon the "branch" you're mapping to for complex numbers.

For $[n] \in \mathbb{Z}/k\mathbb{Z}$,

$$[n] \overset{i}{\mapsto} \exp\left(\frac{[n]}{k} 2\pi i\right)$$

and so

$$\ker\pi = \{z | \pi(z) = 1\} \text{ so that } \ker\pi = \{z = \exp\left(\frac{i2\pi n}{k}\right)\}$$

- e.g. *Semidirect products.*
group G acting on another group H, by homomorphism

$$\tau : G \to \mathrm{Aut}(H)$$

Definition 224 (semi-direct product)**. semidirect product** group $G \ltimes H$ is set $H \times G$, with multiplication

$$(x, g) \cdot (x', g') := (x\tau(g)(x'), gg') \qquad \forall\, (x, g), (x', g') \in H \times G$$

$$1 \longrightarrow H \xrightarrow{i} G \ltimes H \xrightarrow{\pi} G \longrightarrow 1 \tag{830}$$

with

$$\begin{aligned} &i : H \to G \ltimes H \\ &i(x) = (x, 1) \end{aligned} \tag{831}$$

i group homomorphism, since

$$i(x_1 x_2) = (x_1 x_2, 1) = (x_1 \tau(1) x_2, 1) = (x_1, 1) \cdot (x_2, 1) = i(x_1) i(x_2)$$

$$\begin{aligned} &\pi : G \ltimes H \to G \\ &\pi(x, g) = g \end{aligned} \tag{832}$$

cf. http://sierra.nmsu.edu/morandi/oldwebpages/math683fall2002/GroupExtensions.pdf

Observe that

$$\pi i(x) = \pi(x, 1) = 1 \text{ so } \ker\pi = \mathrm{im} i$$

Definition 225 (Semi-direct product (2); with direct product)**. direct product** $G = HK$ if H, K subgroups of group G, s.t.

- H and K are normal in G ($gkg^{-1} \in K \quad \forall\, g \in G, \forall\, k \in K$)
- $H \cap K = \{1\}$
- $HK = G$.

semi-direct product. Relax the 1st condition (of direct products) so H still normal in G, but K need not be.

- H normal in G ($ghg^{-1} \in H, \forall\, g, \forall\, h \in H$)
- $H \cap K = \{1\}$
- $HK = G$

Connection between Def. 224 and Def. 225 for the *semidirect product*: Consider $\tau : G \to \mathrm{Aut}(H)$. Consider $G \ltimes H$ - what is the identity $1_{G \ltimes H} \equiv (1_H, 1_G)$ of this group?

$$(x, g) \cdot (1_H, 1_G) = (x\tau(g)1_H, g1_G) = (x\tau(g)1_H, g) \Longrightarrow 1_H = \tau(g^{-1})1,\ 1_G = 1$$

and so the inverse, $\forall (x, g) \in G \ltimes H$, $(x, g)^{-1} \equiv ((x^{-1}), (g^{-1}))$,

$$(x, g)(x, g)^{-1} = (x\tau(g)(x^{-1}), g(g^{-1})) = (x\tau(g)(x^{-1}), 1) \text{ (if } (g^{-1}) = g^{-1})$$

Moving along,

$$\begin{aligned} & x\tau(g)(x^{-1}) = \tau(g^{-1})1 \\ \Longrightarrow\ & (x^{-1}) = \tau(g^{-1})x^{-1}\tau(g^{-1})1 \end{aligned}$$

Checking out the H being a normal subgroup of $G \ltimes H$ condition, i.e. $H \lhd G$,

$$\begin{aligned} (x, g)(h, 1)(\tau(g^{-1})x^{-1}\tau(g^{-1}), g^{-1}) &= (x\tau(g)h, g)(\tau(g^{-1})x^{-1}\tau(g^{-1}), g^{-1}) = \\ &= (x\tau(g)h\tau(g)\tau(g^{-1})x^{-1}\tau(g^{-1}), 1) = (x\tau(g)hx^{-1}\tau(g^{-1}), 1) \end{aligned}$$

$\Longrightarrow H$ normal subgroup of $G \ltimes H \equiv H \lhd (G \ltimes H)$.

Notes on Semidirect products

- extension

$$1 \longrightarrow SL(n, \mathbb{R}) \xrightarrow{\ i\ } GL(n, \mathbb{R}) \xrightarrow{\ \det\ } \mathbb{R}^* \longrightarrow 1 \tag{833}$$

with
$GL(n, \mathbb{R}) \equiv Gl_n(\mathbb{R}) = \{A | A \in \mathrm{Mat}_{\mathbb{R}}(n, n); \det A \neq 0\}$
$\det : GL(n, \mathbb{R}) \to \mathbb{R}^* \equiv \mathbb{R} \backslash \{0\}$, det surjective homomorphism
$SL(n, \mathbb{R}) \equiv Sl_n(\mathbb{R}) = \{A | A \in \mathrm{Mat}_{\mathbb{R}}(n, n);\ \det A = 1\}$

Note that $\ker(\det) = SL(n, \mathbb{R})$.

Now

$$\mathbb{R}^* \simeq \{a1_n | a \in \mathbb{R}^*\}$$

and $\det(a1_n) = a^n$.

If n odd, and $\det(a1_n) = a^n = 1$, then $a = 1$. If n even, $a = \{-1, 1\}$.

By the second definition of a semi-direct product, Def. 225, it's required that $SL(n, \mathbb{R}) \cap \mathbb{R}^* = 1$ (i.e. the intersection is only the identity). This will only be the case if n odd.

cf. http://sierra.nmsu.edu/morandi/oldwebpages/math683fall2002/GroupExtensions.pdf

Part XXXVIII
Quantum Field Theory

159 Relativistic fields and the scalar action

A particle trajectory assigns a position to each time. A classical field assigns a value to each event. For a real scalar field the value lies in $\mathbb{R}$; for a spinor field it lies in a complex representation space. The transformation law of that space is part of the definition of the field. Quantization will replace the classical field by an operator-valued distribution; it does not turn a classical field into a one-particle wavefunction.

We begin with a real scalar field and derive its dynamics before introducing spin. Compare Srednicki, Chapters 2–3, pp. 15–30 [122]. The Lorentz representations and spinor algebra below connect with Chapters 33–36, pp. 205–231; the trace identities connect with Chapter 47, pp. 294–297. The order here follows the objects and their relations rather than the order of exercises.

159.1 The interval and the mass shell

Work on flat spacetime $\mathbb{R}^{1,3}$, with $\hbar = c = 1$. Use

$$x^\mu = (t, \mathbf{x}), \qquad \eta_{\mu\nu} = \operatorname{diag}(-1, 1, 1, 1), \qquad x_\mu = \eta_{\mu\nu} x^\nu. \tag{834}$$

Greek indices range from 0 to 3 and spatial indices from 1 to 3. Srednicki writes this metric as $g_{\mu\nu}$; our $\eta_{\mu\nu}$ has the same signature. In particular,

$$x \cdot y = \eta_{\mu\nu} x^\mu y^\nu = -x^0 y^0 + \mathbf{x} \cdot \mathbf{y}. \tag{835}$$

For a nonzero future-directed on-shell four-momentum $p^\mu = (E_{\mathbf{p}}, \mathbf{p})$,

$$p^2 = -m^2, \qquad E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2} > 0. \tag{836}$$

Here $m \geq 0$ is the mass in natural units. Before quantization it is the inverse-length parameter of the classical wave equation; the particle interpretation follows from its energy spectrum.

A Lorentz transformation preserves Eq. (835). Substituting $x'^\mu = \Lambda^\mu{}_\rho x^\rho$ and $y'^\nu = \Lambda^\nu{}_\sigma y^\sigma$ gives

$$\begin{aligned} \eta_{\mu\nu} x'^\mu y'^\nu &= \eta_{\mu\nu} \Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma x^\rho y^\sigma, \\ \eta_{\mu\nu} \Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma &= \eta_{\rho\sigma}. \end{aligned} \tag{837}$$

For $\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu + O(\omega^2)$, the terms linear in ω are

$$\eta_{\rho\sigma} + \omega_{\sigma\rho} + \omega_{\rho\sigma} = \eta_{\rho\sigma}, \qquad \omega_{\rho\sigma} = -\omega_{\sigma\rho}. \tag{838}$$

Thus the infinitesimal transformations have six independent parameters: three spatial rotations and three boosts.

A scalar field satisfies $\varphi'(x') = \varphi(x)$. The chain rule, with $x = \Lambda^{-1} x'$, gives

$$\frac{\partial \varphi'}{\partial x'^\mu}(x') = (\Lambda^{-1})^\nu{}_\mu \frac{\partial \varphi}{\partial x^\nu}(x). \tag{839}$$

The gradient is a covector. Contracting two such gradients with $\eta^{\mu\nu}$ produces a scalar by Eq. (837).

159.2 Stationary action and the field equation

For a smooth real field, take the free action

$$S[\varphi] = \int d^4x \, \mathcal{L}, \qquad \mathcal{L} = -\frac{1}{2} \eta^{\mu\nu} \frac{\partial \varphi}{\partial x^\mu} \frac{\partial \varphi}{\partial x^\nu} - \frac{1}{2} m^2 \varphi^2. \tag{840}$$

This is a choice of a local, quadratic, Lorentz-invariant dynamics. Lorentz invariance alone would also permit interactions such as φ^4; the word *free* specifies their absence. With dimensionless action, d^4x has mass dimension -4, so

$$[\mathcal{L}] = 4, \qquad [\varphi] = 1, \qquad [m] = 1, \tag{841}$$

where brackets denote mass dimension, not a commutator.

Let f be a smooth compactly supported variation and set $\varphi_\epsilon = \varphi + \epsilon f$. Differentiating at $\epsilon = 0$ and using the symmetry of $\eta^{\mu\nu}$,

$$\begin{aligned}\left.\frac{dS[\varphi_\epsilon]}{d\epsilon}\right|_{\epsilon=0} &= \int d^4x \left[-\frac{1}{2}\eta^{\mu\nu}\left(\frac{\partial f}{\partial x^\mu}\frac{\partial \varphi}{\partial x^\nu} + \frac{\partial \varphi}{\partial x^\mu}\frac{\partial f}{\partial x^\nu}\right) - m^2\varphi f\right] \\ &= \int d^4x \left[-\eta^{\mu\nu}\frac{\partial \varphi}{\partial x^\nu}\frac{\partial f}{\partial x^\mu} - m^2\varphi f\right] \\ &= \int d^4x \left[\eta^{\mu\nu}\frac{\partial^2 \varphi}{\partial x^\mu \partial x^\nu} - m^2\varphi\right] f. \end{aligned} \tag{842}$$

The last line follows by integration by parts; compact support removes the boundary integral. Since f is arbitrary, stationarity gives

$$\left(-\frac{\partial^2}{\partial t^2} + \sum_{i=1}^{3}\frac{\partial^2}{\partial (x^i)^2} - m^2\right)\varphi = 0. \tag{843}$$

Equivalently, define $\Box = \eta^{\mu\nu}\frac{\partial^2}{\partial x^\mu \partial x^\nu}$, so the equation is $(\Box - m^2)\varphi = 0$. Multiplying by -1 gives Srednicki's $(-\partial^2 + m^2)\varphi = 0$, with his $\partial^2 = \Box$.

For $\varphi(x) = Ae^{ip\cdot x}$ the derivatives are

$$\frac{\partial \varphi}{\partial x^\mu} = ip_\mu \varphi, \qquad (\Box - m^2)\varphi = (-p^2 - m^2)\varphi. \tag{844}$$

A nonzero plane wave therefore lies on the mass shell (836). Its positive-frequency phase is $p\cdot x = -E_{\mathbf{p}}t + \mathbf{p}\cdot\mathbf{x}$.

159.3 Canonical momentum and energy

Separating time and space in Eq. (840),

$$\mathcal{L} = \frac{1}{2}\left(\frac{\partial \varphi}{\partial t}\right)^2 - \frac{1}{2}|\boldsymbol{\nabla}\varphi|^2 - \frac{1}{2}m^2\varphi^2. \tag{845}$$

At fixed φ and spatial derivatives, the canonical momentum is

$$\pi = \frac{\partial \mathcal{L}}{\partial(\frac{\partial \varphi}{\partial t})} = \frac{\partial \varphi}{\partial t}. \tag{846}$$

The Legendre transform gives

$$\begin{aligned}\mathcal{H} &= \pi\frac{\partial \varphi}{\partial t} - \mathcal{L} \\ &= \pi^2 - \frac{1}{2}\pi^2 + \frac{1}{2}|\boldsymbol{\nabla}\varphi|^2 + \frac{1}{2}m^2\varphi^2 \\ &= \frac{1}{2}\left(\pi^2 + |\boldsymbol{\nabla}\varphi|^2 + m^2\varphi^2\right), \qquad H = \int d^3x\, \mathcal{H}. \end{aligned} \tag{847}$$

For fields with sufficient spatial decay, variation at fixed time and spatial integration by parts give

$$\begin{aligned}\delta H &= \int d^3x \left[\pi\,\delta\pi + \sum_{i=1}^{3}\frac{\partial \varphi}{\partial x^i}\frac{\partial \delta\varphi}{\partial x^i} + m^2\varphi\,\delta\varphi\right] \\ &= \int d^3x \left[\pi\,\delta\pi + \left(-\sum_{i=1}^{3}\frac{\partial^2 \varphi}{\partial (x^i)^2} + m^2\varphi\right)\delta\varphi\right]. \end{aligned} \tag{848}$$

Reading off the functional derivatives, Hamilton's equations are

$$\begin{aligned}\frac{\partial \varphi}{\partial t} &= \frac{\delta H}{\delta \pi} = \pi, \\ \frac{\partial \pi}{\partial t} &= -\frac{\delta H}{\delta \varphi} = \sum_{i=1}^{3}\frac{\partial^2 \varphi}{\partial (x^i)^2} - m^2\varphi. \end{aligned} \tag{849}$$

Substituting the first equation into the second recovers Eq. (843). The nonnegative energy in Eq. (847) fixes the kinetic sign for this free scalar theory.

160 From field modes to particles

160.1 Each free Fourier mode is an oscillator

For classical fields of sufficient decay write

$$\varphi(t,\mathbf{x}) = \int \frac{d^3p}{(2\pi)^3}\widetilde{\varphi}(t,\mathbf{p})e^{i\mathbf{p}\cdot\mathbf{x}}. \tag{850}$$

Reality requires $\widetilde{\varphi}(t,-\mathbf{p}) = \widetilde{\varphi}(t,\mathbf{p})^*$. The spatial second derivatives contribute $-\mathbf{p}^2$, so Eq. (843) gives

$$\frac{\partial^2\widetilde{\varphi}}{\partial t^2} + (\mathbf{p}^2+m^2)\widetilde{\varphi} = 0. \tag{851}$$

Its frequency is precisely $E_{\mathbf{p}}$ in Eq. (836). This is why oscillator quantization enters a free field theory.

For the oscillator construction take $m > 0$, so every momentum has positive frequency. If $m = 0$, the zero-momentum mode needs separate treatment. In a periodic box of volume V, write its spatially constant field as $\varphi_0(t) = q_0(t)/\sqrt{V}$. Its Lagrangian and Hamiltonian are

$$L_0 = \frac{1}{2}\left(\frac{dq_0}{dt}\right)^2, \qquad p_{q_0} = \frac{dq_0}{dt}, \qquad H_0 = \frac{1}{2}p_{q_0}^2. \tag{852}$$

This is a free particle degree of freedom, with no positive-frequency oscillator vacuum. It cannot be included by setting $E_{\mathbf{0}} = 0$ in the ladder-operator formulas.

For one oscillator, with $[q,p_q] = i$ and frequency $\omega > 0$, define

$$a = \sqrt{\frac{\omega}{2}}\,q + \frac{i}{\sqrt{2\omega}}p_q, \qquad a^\dagger = \sqrt{\frac{\omega}{2}}\,q - \frac{i}{\sqrt{2\omega}}p_q. \tag{853}$$

Direct multiplication gives

$$\begin{aligned}
[a,a^\dagger] &= -\frac{i}{2}[q,p_q] + \frac{i}{2}[p_q,q] = 1,\\
a^\dagger a &= \frac{\omega}{2}q^2 + \frac{1}{2\omega}p_q^2 + \frac{i}{2}[q,p_q] = \frac{\omega}{2}q^2 + \frac{1}{2\omega}p_q^2 - \frac{1}{2},\\
\frac{1}{2}(p_q^2+\omega^2q^2) &= \omega\left(a^\dagger a + \frac{1}{2}\right).
\end{aligned} \tag{854}$$

160.2 Equal-time commutators and normalization

Canonical quantization postulates, at equal time,

$$[\varphi(t,\mathbf{x}),\pi(t,\mathbf{y})] = i\delta^3(\mathbf{x}-\mathbf{y}), \qquad [\varphi(t,\mathbf{x}),\varphi(t,\mathbf{y})] = [\pi(t,\mathbf{x}),\pi(t,\mathbf{y})] = 0. \tag{855}$$

Linear fields and their commutators are interpreted as distributions, with field operators defined by smearing against suitable test functions. Smearing alone does not define coincident products such as $\pi(x)^2$; these require a regulator or a specified composite operator, such as a Wick product. A field at a point is not an ordinary finite-dimensional matrix.

Choose the oscillator normalization

$$[a(\mathbf{p}),a^\dagger(\mathbf{q})] = (2\pi)^3\delta^3(\mathbf{p}-\mathbf{q}), \qquad [a(\mathbf{p}),a(\mathbf{q})] = 0. \tag{856}$$

The Hermitian free field and its momentum are then

$$\varphi(x) = \int \frac{d^3p}{(2\pi)^3\sqrt{2E_{\mathbf{p}}}}\left[a(\mathbf{p})e^{ip\cdot x} + a^\dagger(\mathbf{p})e^{-ip\cdot x}\right], \tag{857}$$

$$\pi(x) = \int \frac{d^3p}{(2\pi)^3}\left[-i\sqrt{\frac{E_{\mathbf{p}}}{2}}a(\mathbf{p})e^{ip\cdot x} + i\sqrt{\frac{E_{\mathbf{p}}}{2}}a^\dagger(\mathbf{p})e^{-ip\cdot x}\right]. \tag{858}$$

The sign in the second line follows by differentiating $p\cdot x = -E_{\mathbf{p}}t + \mathbf{p}\cdot\mathbf{x}$. Using Eq. (856), the two nonzero terms at equal time are

$$\begin{aligned}[\varphi(t,\mathbf{x}),\pi(t,\mathbf{y})] &= \frac{i}{2}\int \frac{d^3p}{(2\pi)^3}\left[e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} + e^{-i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})}\right] \\ &= i\delta^3(\mathbf{x}-\mathbf{y}).\end{aligned} \tag{859}$$

The other two commutators are

$$\begin{aligned}[\varphi(t,\mathbf{x}),\varphi(t,\mathbf{y})] &= \int \frac{d^3p}{(2\pi)^3\,2E_{\mathbf{p}}}\left[e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} - e^{-i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})}\right] = 0, \\ [\pi(t,\mathbf{x}),\pi(t,\mathbf{y})] &= \int \frac{d^3p}{(2\pi)^3}\frac{E_{\mathbf{p}}}{2}\left[e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} - e^{-i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})}\right] = 0.\end{aligned} \tag{860}$$

In each integral, changing $\mathbf{p}\mapsto -\mathbf{p}$ in the second term makes it the negative of the first.

Srednicki also uses a covariant oscillator normalization. Write that oscillator as $b(\mathbf{p})$ here to distinguish it from ours:

$$\begin{aligned} b(\mathbf{p}) &= \sqrt{2E_{\mathbf{p}}}\,a(\mathbf{p}), \\ [b(\mathbf{p}), b^\dagger(\mathbf{q})] &= (2\pi)^3\,2E_{\mathbf{p}}\delta^3(\mathbf{p}-\mathbf{q}), \\ \varphi(x) &= \int \frac{d^3p}{(2\pi)^3\,2E_{\mathbf{p}}}\left[b(\mathbf{p})e^{ip\cdot x} + b^\dagger(\mathbf{p})e^{-ip\cdot x}\right].\end{aligned} \tag{861}$$

The change is a normalization of the modes, not a change of the field. For a massive shell, the measure follows by integrating out positive energy:

$$\begin{aligned}&\int \frac{d^4p}{(2\pi)^3}\,heta(p^0)\delta(p^2+m^2)F(p) \\ &\quad= \int \frac{d^3p}{(2\pi)^3}\int_0^\infty dp^0\,\frac{\delta(p^0-E_{\mathbf{p}})+\delta(p^0+E_{\mathbf{p}})}{2E_{\mathbf{p}}}F(p) = \int \frac{d^3p}{(2\pi)^3\,2E_{\mathbf{p}}}F(E_{\mathbf{p}},\mathbf{p}).\end{aligned} \tag{862}$$

The four-dimensional measure, the mass-shell constraint and the positive-energy condition are invariant under proper orthochronous Lorentz transformations. The massless formula holds away from the zero-momentum tip, with its usual distributional interpretation.

Let $a(\mathbf{p})|0\rangle = 0$. Evaluate the quadratic quantum Hamiltonian first in a finite periodic box with a momentum cutoff; sums then replace the integrals below. Continue to use $m>0$. Normal ordering places creation operators to the left of annihilation operators and fixes the free vacuum energy to zero.

To see every contribution, substitute Eqs. (857) and (858) into Eq. (847). The time derivatives contribute $(-iE_{\mathbf{p}})(-iE_{\mathbf{q}})$ to aa and $(-iE_{\mathbf{p}})(iE_{\mathbf{q}})$ to $aa^\dagger$. The spatial derivatives give $(ip^i)(iq^i)$ and $(ip^i)(-iq^i)$, respectively. Thus define

$$\begin{aligned} C_-(\mathbf{p},\mathbf{q}) &= -E_{\mathbf{p}}E_{\mathbf{q}} - \mathbf{p}\cdot\mathbf{q} + m^2, \\ C_+(\mathbf{p},\mathbf{q}) &= E_{\mathbf{p}}E_{\mathbf{q}} + \mathbf{p}\cdot\mathbf{q} + m^2.\end{aligned} \tag{863}$$

The conjugate terms have the same respective coefficients. The full substitution is

$$\begin{aligned} H &= \int \frac{d^3p\,d^3q}{(2\pi)^6}\int \frac{d^3x}{4\sqrt{E_{\mathbf{p}}E_{\mathbf{q}}}}\mathcal{K}(p,q;x), \\ \mathcal{K}(p,q;x) &= C_-(\mathbf{p},\mathbf{q})\big[a(\mathbf{p})a(\mathbf{q})e^{i(p+q)\cdot x} \\ &\qquad + a^\dagger(\mathbf{p})a^\dagger(\mathbf{q})e^{-i(p+q)\cdot x}\big] \\ &\quad + C_+(\mathbf{p},\mathbf{q})\big[a(\mathbf{p})a^\dagger(\mathbf{q})e^{i(p-q)\cdot x} \\ &\qquad + a^\dagger(\mathbf{p})a(\mathbf{q})e^{-i(p-q)\cdot x}\big].\end{aligned} \tag{864}$$

For the equal-sign terms,

$$\int d^3x\,e^{i(\mathbf{p}+\mathbf{q})\cdot\mathbf{x}} = (2\pi)^3\delta^3(\mathbf{p}+\mathbf{q}),$$

$$\frac{C_-(\mathbf{p},-\mathbf{p})}{4E_{\mathbf{p}}} = \frac{-E_{\mathbf{p}}^2+\mathbf{p}^2+m^2}{4E_{\mathbf{p}}} = 0. \tag{865}$$

For the mixed terms,

$$\int d^3x\, e^{i(\mathbf{p}-\mathbf{q})\cdot\mathbf{x}} = (2\pi)^3\delta^3(\mathbf{p}-\mathbf{q}),$$
$$\frac{C_+(\mathbf{p},\mathbf{p})}{4E_{\mathbf{p}}} = \frac{E_{\mathbf{p}}^2+\mathbf{p}^2+m^2}{4E_{\mathbf{p}}} = \frac{E_{\mathbf{p}}}{2}. \tag{866}$$

Hence the Hamiltonian and its normal-ordered form are

$$H = \frac{1}{2}\int \frac{d^3p}{(2\pi)^3} E_{\mathbf{p}} \left[a(\mathbf{p})a^\dagger(\mathbf{p}) + a^\dagger(\mathbf{p})a(\mathbf{p})\right],$$
$$:H: = \int \frac{d^3p}{(2\pi)^3} E_{\mathbf{p}} a^\dagger(\mathbf{p})a(\mathbf{p}),$$
$$[:H:, a^\dagger(\mathbf{q})] = \int \frac{d^3p}{(2\pi)^3} E_{\mathbf{p}} a^\dagger(\mathbf{p})(2\pi)^3\delta^3(\mathbf{p}-\mathbf{q}) = E_{\mathbf{q}} a^\dagger(\mathbf{q}). \tag{867}$$

The first line carries the regulated vacuum constant. Normal ordering here fixes the zero of energy in the free flat-spacetime theory; vacuum effects in interacting theories and in gravity require their own analysis. Equation (867) gives the particle interpretation: $a^\dagger(\mathbf{q})|0\rangle$ has energy $E_{\mathbf{q}}$ above the vacuum. Sharp-momentum states are delta-normalized; normalizable states are wave packets obtained by integrating them against momentum profiles.

161 Pauli matrices, Lorentz vectors and two-component spinors

161.1 The Pauli product

Take the orientation $\epsilon_{123} = 1$ and define

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{868}$$

Multiplication gives $\sigma_i^2 = I_2$ and

$$\sigma_1\sigma_2 = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} = i\sigma_3, \qquad \sigma_2\sigma_3 = i\sigma_1, \qquad \sigma_3\sigma_1 = i\sigma_2. \tag{869}$$

Each reversed product has the opposite sign. Therefore

$$\sigma_i\sigma_j = \delta_{ij}I_2 + i\epsilon_{ijk}\sigma_k, \qquad \operatorname{tr}\sigma_i = 0. \tag{870}$$

For ordinary commuting vectors $\mathbf{u}, \mathbf{v}$, this gives

$$(u^i\sigma_i)(v^j\sigma_j) = u^iv^j\delta_{ij}I_2 + iu^iv^j\epsilon_{ijk}\sigma_k$$
$$= (\mathbf{u}\cdot\mathbf{v})I_2 + i(\mathbf{u}\times\mathbf{v})^k\sigma_k. \tag{871}$$

In particular, $(\mathbf{p}\cdot\boldsymbol{\sigma})^2 = \mathbf{p}^2 I_2$.

161.2 A four-vector as a Hermitian matrix

Associate to a real four-vector p the Hermitian matrix

$$P = p^0 I_2 + p^i\sigma_i = \begin{pmatrix} p^0+p^3 & p^1-ip^2 \\ p^1+ip^2 & p^0-p^3 \end{pmatrix}. \tag{872}$$

Its determinant is the Lorentz norm with a minus sign:

$$\det P = (p^0+p^3)(p^0-p^3) - (p^1-ip^2)(p^1+ip^2)$$

$$= (p^0)^2 - (p^3)^2 - (p^1)^2 - (p^2)^2 = -p^2. \tag{873}$$

For $M \in SL(2,\mathbb{C})$ set $P' = MPM^\dagger$. Then

$$(P')^\dagger = P', \qquad \det P' = \det M \, \det P \, \overline{\det M} = \det P. \tag{874}$$

The induced real linear transformation preserves the interval. The matrices M and $-M$ have the same action on P, but opposite actions on a spinor $\lambda \in \mathbb{C}^2$, which transforms as $\lambda' = M\lambda$. For example, rotations about the third axis are represented by

$$M(\theta) = e^{-i\theta\sigma_3/2} = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}, \quad M(2\pi) = -I_2, \quad M(4\pi) = I_2. \tag{875}$$

This relates spinors to the $SU(2)$ rotations already developed in the quaternion part; the spinor itself is not a spatial vector.

161.3 The alternating form and spinor brackets

For undotted indices $a, b \in \{1,2\}$, choose

$$(\varepsilon_{ab}) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad (\varepsilon^{ab}) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \varepsilon^{ab}\varepsilon_{bc} = \delta^a{}_c. \tag{876}$$

These conventions fix lowering and raising:

$$\lambda_a = \varepsilon_{ab}\lambda^b, \qquad \lambda^a = \varepsilon^{ab}\lambda_b, \qquad (\lambda_1, \lambda_2) = (\lambda^2, -\lambda^1). \tag{877}$$

For $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$, the explicit invariance calculation is

$$M^T \varepsilon M = \begin{pmatrix} A & C \\ B & D \end{pmatrix} \begin{pmatrix} C & D \\ -A & -B \end{pmatrix} = (AD - BC)\varepsilon = \varepsilon \quad (\det M = 1). \tag{878}$$

Thus the alternating pairing of two commuting spinors is invariant:

$$\langle \lambda \, \chi \rangle := \varepsilon_{ab}\lambda^a\chi^b = \lambda^1\chi^2 - \lambda^2\chi^1 = -\langle \chi \, \lambda \rangle. \tag{879}$$

The word *commuting* matters. Classical fermion fields in a Grassmann formulation have anticommuting components; exchanging them introduces an additional sign. The spinors in the determinant identities here have commuting complex components.

The dimension of the spinor space gives the Schouten identity. In the first component,

$$\begin{aligned} &\langle u \, v \rangle w^1 + \langle v \, w \rangle u^1 + \langle w \, u \rangle v^1 \\ &= u^1 v^2 w^1 - u^2 v^1 w^1 + v^1 w^2 u^1 - v^2 w^1 u^1 + w^1 u^2 v^1 - w^2 u^1 v^1 = 0. \end{aligned} \tag{880}$$

The second component cancels independently:

$$\begin{aligned} &\langle u \, v \rangle w^2 + \langle v \, w \rangle u^2 + \langle w \, u \rangle v^2 \\ &= u^1 v^2 w^2 - u^2 v^1 w^2 + v^1 w^2 u^2 - v^2 w^1 u^2 + w^1 u^2 v^2 - w^2 u^1 v^2 = 0. \end{aligned} \tag{881}$$

Hence, without dividing by any bracket,

$$\langle u \, v \rangle w^a + \langle v \, w \rangle u^a + \langle w \, u \rangle v^a = 0. \tag{882}$$

Contracting this equation with $\varepsilon_{ab}z^b$ gives

$$\langle u \, v \rangle \langle w \, z \rangle + \langle v \, w \rangle \langle u \, z \rangle + \langle w \, u \rangle \langle v \, z \rangle = 0. \tag{883}$$

It also holds for linearly dependent spinors, where attempts to solve by dividing by a bracket would fail.

For a nonzero future-directed null momentum, Eq. (872) has eigenvalues $p^0 \pm |\mathbf{p}| = 2p^0, 0$. It is therefore a positive rank-one matrix and can be written

$$P = \lambda\lambda^\dagger, \qquad \lambda^\dagger\lambda = 2p^0. \tag{884}$$

Indeed, if e is a unit eigenvector for $2p^0$, take $\lambda = \sqrt{2p^0}\, e$. The phase replacement $\lambda \mapsto e^{i\alpha}\lambda$ leaves P unchanged. For complexified momenta, Hermitian conjugation is no longer the relevant constraint; two independent spinors replace λ and its conjugate. This is the entry to massless spinor-helicity variables; compare Srednicki, Chapter 50, pp. 308–313 [122].

162 Clifford algebra and Dirac traces

162.1 Factoring the relativistic wave operator

The scalar wave equation is second order. Seek a first-order matrix operator whose square supplies its second derivatives. Define

$$\sigma^{\mu} = (I_2, \sigma_1, \sigma_2, \sigma_3), \qquad \bar{\sigma}^{\mu} = (I_2, -\sigma_1, -\sigma_2, -\sigma_3). \tag{885}$$

By Eq. (870), the time-time, mixed and spatial cases are respectively

$$\begin{aligned} \sigma^0\bar{\sigma}^0 + \sigma^0\bar{\sigma}^0 &= 2I_2, \\ \sigma^0\bar{\sigma}^i + \sigma^i\bar{\sigma}^0 &= -\sigma_i + \sigma_i = 0, \\ \sigma^i\bar{\sigma}^j + \sigma^j\bar{\sigma}^i &= -(\sigma_i\sigma_j + \sigma_j\sigma_i) = -2\delta_{ij}I_2. \end{aligned} \tag{886}$$

Together these say

$$\sigma^{\mu}\bar{\sigma}^{\nu} + \sigma^{\nu}\bar{\sigma}^{\mu} = -2\eta^{\mu\nu}I_2. \tag{887}$$

Exchanging the bars gives the same relation. Form the block matrices

$$\gamma^{\mu} = \begin{pmatrix} 0 & \sigma^{\mu} \\ \bar{\sigma}^{\mu} & 0 \end{pmatrix}. \tag{888}$$

Block multiplication and Eq. (887) give the Clifford relation in Srednicki's convention:

$$\gamma^{\mu}\gamma^{\nu} + \gamma^{\nu}\gamma^{\mu} = -2\eta^{\mu\nu}I_4. \tag{889}$$

In particular $(\gamma^0)^2 = I_4$ and $(\gamma^i)^2 = -I_4$.

For a smooth four-component spinor ψ, commuting partial derivatives remove the antisymmetric matrix part. Explicitly,

$$\begin{aligned} \gamma^{\mu}\gamma^{\nu}\frac{\partial^2\psi}{\partial x^{\mu}\partial x^{\nu}} &= \frac{1}{2}(\gamma^{\mu}\gamma^{\nu} + \gamma^{\nu}\gamma^{\mu})\frac{\partial^2\psi}{\partial x^{\mu}\partial x^{\nu}} = -\Box\psi, \\ \left(i\gamma^{\mu}\frac{\partial}{\partial x^{\mu}} + m\right)\left(i\gamma^{\nu}\frac{\partial}{\partial x^{\nu}} - m\right)\psi &= (\Box - m^2)\psi. \end{aligned} \tag{890}$$

The cross terms cancel because m and the matrices are constant. A solution of the free Dirac equation

$$\left(i\gamma^{\mu}\frac{\partial}{\partial x^{\mu}} - m\right)\psi = 0 \tag{891}$$

therefore has Klein–Gordon components. The converse does not follow: the first-order equation also relates the spinor components.

162.2 Ordering products without losing metric signs

The defining relation can be used in either direction:

$$\gamma^{\mu}\gamma^{\nu} = -\gamma^{\nu}\gamma^{\mu} - 2\eta^{\mu\nu}I_4. \tag{892}$$

The second term is essential even when reordering a long product. Introduce $h^{\mu\nu} = -\eta^{\mu\nu}$ only when comparing the generic Clifford convention $\{\gamma^{\mu}, \gamma^{\nu}\} = 2h^{\mu\nu}I_4$. For the rest of this derivation we use η and Eq. (892) directly.

Define $\not{p} = p_{\mu}\gamma^{\mu}$. For ordinary commuting four-vectors,

$$\begin{aligned} (\not{p})^2 &= \frac{1}{2}p_{\mu}p_{\nu}(\gamma^{\mu}\gamma^{\nu} + \gamma^{\nu}\gamma^{\mu}) = -p^2 I_4, \\ \not{p}\,\not{q} + \not{q}\,\not{p} &= -2(p\cdot q)I_4. \end{aligned} \tag{893}$$

On shell, $(\not{p})^2 = m^2 I_4$. The contraction uses the physical metric η, even though the Clifford anticommutator contains $-\eta$.

162.3 Odd traces and the first even traces

We now use the four-dimensional complex Dirac representation: $\operatorname{tr} I_4 = 4$. Define

$$\gamma_5 = i\gamma^0\gamma^1\gamma^2\gamma^3 = \begin{pmatrix} -I_2 & 0 \\ 0 & I_2 \end{pmatrix}, \qquad \gamma_5^2 = I_4, \qquad \gamma_5\gamma^\mu = -\gamma^\mu\gamma_5. \tag{894}$$

For a product $G = \gamma^{\mu_1}\cdots\gamma^{\mu_n}$, cyclicity and Eq. (894) give

$$\operatorname{tr} G = \operatorname{tr}(\gamma_5 G \gamma_5) = (-1)^n \operatorname{tr}(G\gamma_5^2) = (-1)^n \operatorname{tr} G. \tag{895}$$

Thus odd traces vanish. This argument is specifically four-dimensional; a continuation in dimension requires additional assumptions, particularly for γ_5.

For two factors, trace cyclicity and Eq. (889) give

$$\begin{aligned} 2\operatorname{tr}(\gamma^\mu\gamma^\nu) &= \operatorname{tr}(\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu) = -2\eta^{\mu\nu}\operatorname{tr} I_4, \\ \operatorname{tr}(\gamma^\mu\gamma^\nu) &= -4\eta^{\mu\nu}. \end{aligned} \tag{896}$$

For four factors, move the first matrix through the remaining three, using Eq. (892) at every step:

$$\begin{aligned} \gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma &= -\gamma^\nu\gamma^\mu\gamma^\rho\gamma^\sigma - 2\eta^{\mu\nu}\gamma^\rho\gamma^\sigma \\ &= \gamma^\nu\gamma^\rho\gamma^\mu\gamma^\sigma + 2\eta^{\mu\rho}\gamma^\nu\gamma^\sigma - 2\eta^{\mu\nu}\gamma^\rho\gamma^\sigma \\ &= -\gamma^\nu\gamma^\rho\gamma^\sigma\gamma^\mu - 2\eta^{\mu\sigma}\gamma^\nu\gamma^\rho + 2\eta^{\mu\rho}\gamma^\nu\gamma^\sigma - 2\eta^{\mu\nu}\gamma^\rho\gamma^\sigma. \end{aligned} \tag{897}$$

The trace of the first term in the last line is minus the original trace. Bringing it to the left and substituting Eq. (896),

$$\begin{aligned} 2\operatorname{tr}(\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma) &= -2\eta^{\mu\sigma}(-4\eta^{\nu\rho}) + 2\eta^{\mu\rho}(-4\eta^{\nu\sigma}) - 2\eta^{\mu\nu}(-4\eta^{\rho\sigma}), \\ \operatorname{tr}(\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma) &= 4\left(\eta^{\mu\nu}\eta^{\rho\sigma} - \eta^{\mu\rho}\eta^{\nu\sigma} + \eta^{\mu\sigma}\eta^{\nu\rho}\right). \end{aligned} \tag{898}$$

For example, $\mu = \nu = 0$ in Eq. (896) gives 4, while $\mu = \nu = i$ with fixed spatial i gives -4, as required by the squares of the matrices. Contracting Eq. (898) with four covectors produces

$$\begin{aligned} &\operatorname{tr}(\not{p}\,\not{q}\,\not{r}\,\not{s}) \\ &\quad = 4\left[(p\cdot q)(r\cdot s) - (p\cdot r)(q\cdot s) + (p\cdot s)(q\cdot r)\right]. \end{aligned} \tag{899}$$

This expression is a Lorentz scalar and needs no particular matrix basis.

162.4 A contracted Clifford identity

Lower the matrix index with the same physical metric, $\gamma_\mu = \eta_{\mu\nu}\gamma^\nu$. Contracting the anticommutator,

$$\begin{aligned} 2\gamma^\mu\gamma_\mu &= \eta_{\mu\nu}(\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu) = -2\eta_{\mu\nu}\eta^{\mu\nu} I_4 = -8I_4, \\ \gamma^\mu\gamma_\mu &= -4I_4. \end{aligned} \tag{900}$$

Now move γ^ν past γ_μ and apply Eq. (900):

$$\begin{aligned} \gamma^\mu\gamma^\nu\gamma_\mu &= \gamma^\mu(-\gamma_\mu\gamma^\nu - 2\delta^\nu{}_\mu I_4) \\ &= -\gamma^\mu\gamma_\mu\gamma^\nu - 2\gamma^\nu = 4\gamma^\nu - 2\gamma^\nu = 2\gamma^\nu. \end{aligned} \tag{901}$$

The sign would change if the contracted index were lowered using $h = -\eta$ instead. A contraction is therefore specified by both the algebraic relation and the bilinear form used to lower the index.

Part XXXIX
The Standard Model

gauged symmetry group

$$\begin{array}{ll} SU_c(3)\times & SU_i(2)\times U_1(1) \\ \downarrow & \downarrow\downarrow \\ 8G_\mu^\alpha & 3W_\mu^a B_\mu \\ \alpha = 1\ldots 8 & a = 1,2,3 \end{array}$$

8 spin-1 particles $G^{\alpha}_{\mu}(x) \equiv$ gluons, c is meant to denote "color"
Gluons are strong to be massless,
strong interaction

3 spin-1 particles $W^{a}_{\mu}(x)$ associated with $SU_L(2)$, L indicates that only left-handed fermions turn out to carry this quantum number.

1 $B_{\mu}(x)$ with factor $U_Y(1)$, subscript Y to distinguish group associated with the quantum number (defined below) of weak hypercharge, from group associated with ordinary electric charge, $Q \equiv U_{\text{em}}(1)$.

4 spin-1 bosons associated with factors $SU_L(2) = U_Y(1)$ are related to physical bosons that mediate we are interactions, $W^{\pm}$ and Z^0, and familiar photon γ from QED.

Part XL
General Relativity

$\hbar = c = 1$

Classically, entropy is a measure of the phase space available to the system in question. Then for S/E, it should have an upper bound, since,
let its energy be no more than E: this amounts to a limitation on momentum space available to system's components, provided potential energy bounded from below. Bekenstein (1981) [111]

From wikipedia, "Bekenstein bound"

Suppose we have a black hole of mass M,
box of energy E outside of it of size R
radius of the black hole R_{bh} goes as GM
entropy of black hole goes as $\frac{R^2_{\text{bh}}}{G} \equiv \frac{kc^3 R^2_{\text{bh}}}{\hbar G} \sim GM^2$

If we dump the box into the black hole, mass of black hole goes up to $M + E$

entropy goes up by $\frac{kGME}{\hbar c^3} \equiv GME$
since entropy can't decrease $S \lesssim GME$
In order for the box to fit inside black hole R $\lesssim GM$
If sizes are comparable, then $S \lesssim \frac{kRE}{\hbar c} \equiv RE$

162.4.1 The case of a weakly gravitating system falling into a black hole

cf. Bekenstein (2018) [112]

Consider a weakly self-gravitating system $\mathcal{U}$ of proper mass energy E, entropy $\mathcal{S}$.
Assume spherical box of radius R enclosing $\mathcal{U}$, concentric to its center of mass of radius R.
Box dropped into Schwarzchild black hold of mass M satisfying $M \gg E$, $2M \gg R$.
Assume black hole absorbs $\mathcal{U}$, and increases its mass by E.

Black hole's entropy, originally $S_{BH} = \frac{4\pi M^2}{l_p^2}$ increases by $\Delta S_{BH} = \frac{8\pi ME}{l_p^2} + O(E^2)$

By GSL (generalized Second Law), $\Delta S_{BH}+$ radiation entropies are at least as large as $\mathcal{S}$

$$\implies S \leq \frac{8\pi ME}{l_p^2}$$

Choose $M = \frac{1}{8}R^2/E$

$$S \leq \frac{\pi R^2}{l_p^2}$$

Claim: justify neglect of energy losses to Hawking's radiance (small)
appropximate power emitted by Schwarzchild black hole by sphere with radius $2M$ radiating according to Stefan-Boltzmann law at Hawking temperature $T_H = \frac{l_p^2}{8\pi M}$

$$\dot{M} \equiv \frac{dM}{dt} \approx \frac{-Nl_p^3}{15360\pi M^2} \tag{902}$$

where $N \equiv$ effective number of radiated species (photons contribute unity to N).
If $\mathcal{U}$ dropped at Schwarzschild time $t = 0$, when its CM (center of mass) was at Schwarzschild radial coordinate $r = \alpha 2M$ ($\alpha \gg 1$), then bottom of sphere reaches horizon $r = 2M$, and infall practically finished, at time

$$t \approx 4M\left[\frac{1}{3}\alpha^{3/2} - \frac{1}{4}\ln\left(\frac{R}{4M}\right) + O(\alpha^{1/2})\right] \tag{903}$$

Part XLI
WE Heraeus International Winter School on Gravity and Light

Introduction (from EY)

The International Winter School on Gravity and Light held *central lectures* given by Dr. Frederic P. Schuller. These lectures on General Relativity and Gravity are unequivocally and undeniably, the best and most lucid and well-constructed lecture series on General Relativity and Gravity. The mathematical foundation from topology and differential geometry from which General Relativity arises from is solid, well-selected in rigor. The lectures themselves are well-thought out and clearly explained.

Even more so, the International Winter School provided accompanying Tutorial Sessions for each of the lectures. I had given up hopes in seeing this component of the learning process ever be put online so that anyone and everyone in the world could learn through the Tutorial process as well. I was afraid that nobody would understand how the Tutorial or "Office Hours" session was important for students to digest and comprehend and work out-doing exercises-the material presented in the lectures. This International Winter School gets it and shows how online education has to be done, to do it in an excellent manner, moving forward.

For anyone who is serious about learning General Relativity and Gravity, I would simply point to these video lectures and tutorials.

What I want to do is to build upon the material presented in this International Winter School. Why it's important to me, and to the students and practicing researchers out there, is that the material presented takes the student from an introduction to the research frontier. That is the stated goal of the International Winter School. I want to dig into and help contribute to the cutting edge in research and this entire program with lectures and tutorials appears to be the most direct and sensible route directly to being able to do research in General Relativity and Gravity. -EY 20150323

20200321 update: Someone collected all the lecture videos and tutorials:
http://mathswithphysics.blogspot.com/2016/07/the-we-heraeus-international-winter.html

163 Lecture 1: Topology

163.1 Lecture 1: Topological Spaces

Definition 226. Let M be a set.
A **topology** $\mathcal{O}$ is a subset $\mathcal{O} \subseteq \mathcal{P}(M)$, $\mathcal{P}(M)$ power set of M: set of all subsets of M. satisfying

(i) $\emptyset \in \mathcal{O}$, $M \in \mathcal{O}$

(ii) $U \in \mathcal{O}$, $V \in \mathcal{O} \Longrightarrow U \bigcap V \in \mathcal{O}$

(iii) $U_\alpha \in \mathcal{O}$, $\alpha \in \mathcal{A} \Longrightarrow \left(\bigcup_{\alpha\in\mathcal{A}} U_\alpha\right) \in \mathcal{O}$

$\mathcal{O}\}$ utterly useless

Definition 227. $\mathcal{O}_{\text{standard}} \subseteq \mathcal{P}(\mathbb{R}^d)$

EY : 20150524
I'll fill in the proof that $\mathcal{O}_{\text{standard}}$ is a topology.

Proof. $\emptyset \in \mathcal{O}_{\text{standard}}$
since $\forall\, p \in \emptyset$, $\exists\, r \in \mathbb{R}^+$: $\mathcal{B}_r(p) \subseteq \emptyset$ (i.e. satisfied "vacuously")

Suppose $U, V \in \mathcal{O}_{\text{standard}}$.

Let $p \in U \bigcap V$. Then $\exists\, r_1, r_2 \in \mathbb{R}^+$ s.t. $\mathcal{B}_{r_1}(p) \subseteq U$
$\mathcal{B}_{r_2}(p) \subseteq V$

Let $r = \min\{r_1, r_2\}$.
Clearly $\mathcal{B}_r(p) \subseteq U$ and $\mathcal{B}_r(p) \subseteq V$. Then $\mathcal{B}_r(p) \subseteq U \bigcap V$. So $U \bigcap V \in \mathcal{O}_{\text{standard}}$.

Suppose, $U_\alpha \in \mathcal{O}_{\text{standard}}$, $\forall\, \alpha \in \mathcal{A}$.
Let $p \in \bigcup_{\alpha\in\mathcal{A}} U_\alpha$. Then $p \in U_\alpha$ for at least 1 $\alpha \in \mathcal{A}$.
$\exists\, r_\alpha \in \mathbb{R}^+$ s.t. $\mathcal{B}_{r_\alpha}(p) \subseteq U_\alpha \subseteq \bigcup_{\alpha\in\mathcal{A}} U_\alpha$. So $\bigcup_{\alpha\in\mathcal{A}} U_\alpha \in \mathcal{O}_{\text{standard}}$ □

163.2 2. Continuous maps

163.3 3. Composition of continuous maps

163.4 4. Inheriting a topology

EY : 20150524
I'll fill in the proof that given f continuous (cont.), then the restriction of f onto a subspace S is cont. If you want a reference, check out Klaus Jänich [16, pp. 13, Ch. 1 Fundamental Concepts, Sec. Continuous Maps]
If cont. $f : M \to N$, $S \subseteq M$, then $f|_S$ cont.

Proof. Let open $V \subseteq N$, i.e. $V \in \mathcal{O}_N$ i.e. V in the topology $\mathcal{O}_N$ of N.

$$f|_S^{-1}(V) = \{m \in M|\ f|_S(m) \in V\}$$

Now $f^{-1}(V) = \{m \in M | f(m) \in V\}$.
So $f^{-1}(V) \bigcap S = f|_S^{-1}(V)$
Now f cont. So $f^{-1}(V) \in \mathcal{O}_N$.
and recall $\mathcal{O}_S| := \{U \bigcap S | U \in \mathcal{O}_M\}$.
so $f^{-1}(V) \bigcap S = f|_S^{-1}(V) \in \mathcal{O}_S$ i.e. $f|_S^{-1}(V)$ open.
$\implies f|_S$ cont. □

Topology Tutorial Sheet

filename : `main.pdf`
The WE-Heraeus International Winter School on Gravity and Light: Topology

EY : 20150524
What I won't do here is retype up the solutions presented in the Tutorial (cf. https://youtu.be/_XkhZQ-hNLs): the presenter did a very good job. If someone wants to type up the solutions and copy and paste it onto this LaTeX file, in the spirit of open-source collaboration, I would encourage this effort.
Instead, what I want to encourage is the use of as much CAS (Computer Algebra System) and symbolic and numerical computation because, first, we're in the 21st century, second, to set the stage for further applications in research. I use Python and Sage Math alot, mostly because they are open-source software (OSS) and fun to use. Also note that the structure of Sage Math modules matches closely to Category Theory.

In checking whether a set is a topology, I found it strange that there wasn't already a function in Sage Math to check each of the axioms. So I wrote my own; see my code snippet, which you can copy, paste, edit freely in the spirit of OSS here, titled `topology.sage`:

gist github ernestyalumni topology.sage
Download topology.sage

Loading `topology.sage`, after changing into (with the usual Linux terminal commands, cd, ls) by

```
sage: load(''topology.sage'')
```

Exercise 2: Topologies on a simple set.

Question Does $\mathcal{O}_1 := \ldots$ constitute a topology ...?.

Solution: Yes, since we check by typing in the following commands in Sage Math:

```
emptyset in O_1
Axiom2check(O_1) # True
Axiom3check(O_1) # True
```

Question What about $\mathcal{O}_2$ ...?.

Solution: No since the 3rd. axiom fails, as can be checked by typing in the following commands in Sage Math:

```
emptyset in O_2
Axiom2check(O_2) # True
Axiom3check(O_2) # False
```

164 Lecture 2: Topological Manifolds

Lecture 2: Manifolds

Topological spaces: $\exists$ so man that mathematicians cannot even classify them.

For spacetime physics, we may focus on topological spaces $(M, \mathcal{O})$ that can be charted, analogously to how the surface of the earth is charted in an atlas.

164.1 Topological manifolds

Definition 228. A topological space $(M, \mathcal{O})$ is called a **d-dimensional topological method** if
$\forall p \in M : \exists U \in \mathcal{O}, U \ni p : \exists x : U \subseteq M \to x(U) \subseteq \mathbb{R}^d \qquad (M, \mathcal{O}), (\mathbb{R}^d, \mathcal{O}_{\text{std}})$

(i) x **invertible** :
$$x^{-1} : x(U) \to U$$

(ii) x **continuous**

(iii) x^{-1} **continuous**

164.2 Terminology

164.3 3. Chart transition maps

Imagine 2 charts (U, x) and (V, y) with overlapping regions.

164.4 4. Manifold philosophy

Often it is desirable (or indeed the way) to define properties ("continuity") of real-world object ("$\mathbb{R} \xrightarrow{\gamma} M$") by judging suitable coordinates not on the "real-world" object itself, but on a chart-representation of that real world object.

EY's add-ons

This lecture gives me a good excuse to review Topology and Topological Manifolds from a mathematician's point of view. I find John M. Lee's **Introduction to Topological Manifolds** book good because it's elementary and thorough and it's fairly recent (2010) so it's up to date [17]. See my notes and solutions for the book; it's a file titled `LeeJM_IntroTopManifolds_sol.pdf` of which I'll try to keep the pdf and LaTeX file available for download on my ernestyalumni Google Drive (so try to search for it on Google).

Tutorial Topological manifolds

filename: `Sheet_1.2.pdf`

Exercise 4: Before the invention of the wheel.

Another one-dimensional topological manifold. Another one?

Consider set $F^1 := \{(m,n) \in \mathbb{R}^2 | m^4 + n^4 = 1\}$, equipped with subset topology $\mathcal{O}_{\text{std}}|_{F^1}$

Question $x : F^1 \to \mathbb{R}$ is what?.

Solution . EY : 20150525 The tutorial video https://youtu.be/ghfEQ3u_B6g is really good and this solution is how I'd write it, but it's really the same (I needed the practice).

$$\boxed{\begin{aligned} x : F^1 &\to \mathbb{R} \\ (m,n) &\mapsto m \end{aligned}}$$

If $m = 0$, $n^4 = 1$ so $n = \pm 1$ so it's not injective.

Let the closed n-dim. upper half-space $\mathbb{H}^n \subseteq \mathbb{R}^1$. Then

$$\begin{aligned} \mathbb{H}^n &= \{(x_1 \dots x_n) \in \mathbb{R}^n | x_n \geq 0\} \\ \text{int}\mathbb{H}^n &= \{(x_1 \dots x_n) \in \mathbb{R}^n | x_n > 0\} \\ -\mathbb{H}^n &= \{(x_1 \dots x_n) \in \mathbb{R}^n | x_n \leq 0\} \\ -\text{int}\mathbb{H}^n &= \{(x_1 \dots x_n) \in \mathbb{R}^n | x_n < 0\} \end{aligned}$$

Question This map x may be made injective by restricting its domain to either of 2 maximal open subsets of F^1. Which ones?.

Solution .

Let

$$\begin{aligned} U_+ &= F^1 \cap \text{int}\mathbb{H}^2 \\ U_- &= F^1 \cap -\text{int}\mathbb{H}^2 \end{aligned}$$

Look at

$$\begin{aligned} x^4 &= 1 - n^4 \\ \implies x &= \pm(1-n^4)^{1/4} \end{aligned}$$

Then for

$$\begin{aligned} x_+^{-1} : (-1,1) \subseteq \mathbb{R} &\to U_+ \\ m &\mapsto (m, (1-m^4)^{1/4}) \\ x_-^{-1} : (-1,1) \subseteq \mathbb{R} &\to U_- \\ m &\mapsto (m, -(1-m^4)^{1/4}) \end{aligned}$$

x_+,x_- injective (since left inverse exists).

Question Construct injective y.

Solution .

Let

$$V_+ = F^1 \cap \text{int}\mathbb{H}^1$$
$$V_- = F^1 \cap -\text{int}\mathbb{H}^1$$

Then

$$y_+ : V_+ \to (-1,1) \subseteq \mathbb{R}$$
$$(m,n) \mapsto n$$
$$y_- : V_- \to (-1,1) \subseteq \mathbb{R}$$
$$(m,n) \mapsto n$$

Question Construct inverse y^{-1}. Solution .

For

$$y_+^{-1} : (-1,1) \subseteq \mathbb{R} \to V_+$$
$$n \mapsto ((1-n^4)^{1/4}, n)$$
$$y_-^{-1} : (-1,1) \subseteq \mathbb{R} \to V_-$$
$$n \mapsto (-(1-n^4)^{1/4}, n)$$

y_+,y_- injective (since left inverse exists).

Note $(-1,0) \notin U_+, U_-$
$(1,0) \notin U_+, U_-$
and

$(0,1) \notin V_+, V_-$
$(0,-1) \notin V_+, V_-$

Question construct *transition map* $x \circ y^{-1}$.

Solution .

$$x_+ y_+^{-1} : (0,1) \subseteq \mathbb{R} \to (0,1) \subseteq \mathbb{R}$$
$$n \mapsto (1-n^4)^{1/4}$$
$$x_- y_+^{-1} : (-1,0) \subseteq \mathbb{R} \to (0,1) \subseteq \mathbb{R}$$
$$n \xrightarrow{y_+^{-1}} ((1-n^4)^{1/4}, n) \xrightarrow{x_-} (1-n^4)^{1/4}$$
$$x_+ y_-^{-1} : (0,1) \subseteq \mathbb{R} \to (-1,0) \subseteq \mathbb{R}$$
$$n \mapsto -(1-n^4)^{1/4}$$
$$x_- y_-^{-1} : (-1,0) \subseteq \mathbb{R} \to (-1,0) \subseteq \mathbb{R}$$
$$n \mapsto -(1-n^4)^{1/4}$$

Question . . . Does the collection of these domains and maps form an atlas of F^1?.

Yes, with atlas

$$\mathcal{A} = \{ \begin{matrix} (U_+, x_+) & (V_+, y_+) \\ (U_-, x_-)' & (V_-, y_-) \end{matrix} \}$$

Clearly

$$U_+ \cup U_- \cup V_+ \cup V_- = (F^1 \cap \text{int}\mathbb{H}^2) \cup (F^1 \cap -\text{int}\mathbb{H}^2) \cup (F^1 \cap \text{int}\mathbb{H}^1) \cup (F^1 \cap -\text{int}\mathbb{H}^1) = \\ = F^1 \cap \mathbb{R}^2 \backslash \{(0,0)\} = F^1$$

and (the point is that) $x_\pm, y_\pm$ are homeomorphisms of open sets of F^1 onto open sets of 1 dim. $\mathbb{R}^1$ (namely $(-1,1) \subseteq \mathbb{R}^1$), and so $\mathcal{A}$ is an atlas of F^1.

165 Lecture 3: Multilinear Algebra

Lecture 3: Multilinear Algebra (International Winter School on Gravity and Light 2015)

We will **not** equip space(time) with a vector space structure. Do you know where

$$5 \cdot \begin{matrix} \text{Paris} \\ \text{lie ?} \end{matrix} =? \qquad \text{Paris} + \text{Vienna} =?$$

Moreover, the tangent spaces T_pM (lecture 5) smooth manifolds (Lecture 4)

Beneficial to first study vector spaces abstractly for two reason

(i) for construction of T_pM one needs an intermediate vector space $C^\infty(M)$

(ii) tensor technique are most easily understood in an abstract setting.

165.1 Vector spaces

Definition 229. A vector space $(V, +, -)$ is

(i) a set V

(ii) $+ : V \times V \to V$ "addition"

(iii) $\cdot : \mathbb{R} \times V \to V$ "s-multiplication" EY : 20160317 s for "scalar"

satisfying:

CANIADDU

$$\begin{aligned} C^+ &: \quad v + w = w + v \\ A^+ &: \quad (u + v) + w = u + (v + w) \\ N^+ &: \quad \exists\, 0 \in V : \forall\, v \in V : v + 0 = v \\ I^+ &: \quad \forall\, v \in V : \exists\, (-v) \in V : v + (-v) = 0 \\ A &: \quad \lambda \cdot (\mu + v) = (\lambda \cdot \mu) \cdot v \qquad (\forall\, \lambda, \mu \in \mathbb{R}) \\ D &: \quad (\lambda + \mu) \cdot v = \lambda \cdot v + \mu \cdot v \\ D &: \quad \lambda \cdot v + \lambda \cdot w = \lambda \cdot (v + w) \\ U &: \quad 1 \cdot v = v \end{aligned}$$

Terminology

An element of a vector space is often referred to, informally as a vector.

Example

def. **set** of polynomials (fixed) degree $\mathcal{P} := \{p : (-1,+1) \to \mathbb{R} | p(x) = \sum_{n=0}^{N} p_n \cdot x^n\}$

Thought bubble: is $\square$ a vector?

$\square(x) = x^2$

No $\square \in \mathcal{P}$.

$+ : \mathcal{P} \times \mathcal{P} \to \mathcal{P}$

$(p, q) \mapsto p + q$

where $(p + q)(x) = p(x) +_{\mathbb{R}} q(x)$

$\cdot : \mathbb{R} \times \mathcal{P} \to \mathcal{P}$

$(\lambda, p) \mapsto \lambda \cdot p$

where $(\lambda \cdot p)(x) := \lambda \cdot_{\mathbb{R}} p(x)$

Thought bubble: $\square$ a vector? Yes, but who cares?

$(\mathcal{P}, +, \cdot)$ is a vector space.

$\square \in \mathcal{P}$

165.2 Linear maps

These are the structure-respecting maps between vector spaces.

EY : 20160316 out of tradition, they're called "linear" maps

Definition 230. $(V, +_V, \cdot_V)$ and $(W, +_W, \cdot_W)$ vector spaces
Then a map

$$\varphi : V \to W$$

is called **linear** if

(i) $\varphi(v +_V \widetilde{v}) = \varphi(v) +_W \varphi(\widetilde{v})$

(ii) $\varphi(\lambda \cdot_V v) = \lambda \cdot_W \varphi(v)$

Example

$\delta : \mathcal{P} \to \mathcal{P}$

:

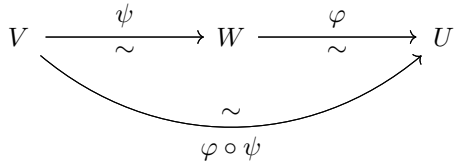
$p \mapsto \delta(p) := p'$

linear:

(i) $\delta(p + q) = (p +_{\mathcal{P}} q)' \underbrace{=}_{\text{sum rule}} p' +_{\mathcal{P}} q' = \delta(p) +_{\mathcal{P}} \delta(q)$

(ii) $\delta(\lambda p) = (\lambda p)' = \lambda \cdot p' = \lambda \cdot \delta(p)$

Notation: $\varphi : V \to W$ linear $\iff$: $\varphi : V \xrightarrow{\sim} W$



165.2.1 Example*

$\delta \circ \delta : \mathcal{P} \xrightarrow{\sim} \mathcal{P}$

165.3 Vector space of Homomorphisms

fun fact: $(V, +, \cdot)$ $(W, +, \cdot)$ vector spaces
def. $\mathrm{Hom}(V, W) := \{\varphi : V \xrightarrow{\sim} W\}$ set.

We can make this into a vector spaces.

$$\oplus : \mathrm{Hom}(V, W) \times \mathrm{Hom}(V, W) \to \mathrm{Hom}(V, W)$$
$$(\varphi, \psi) \mapsto \varphi \oplus \psi$$

where $(\varphi \otimes \psi)(v) := \varphi(v) +_W \psi(v)$

$\otimes : \ldots$ similarly.

$(\mathrm{Hom}(V, W), \oplus, \otimes)$ **is** a vector space.

165.3.1 Example*

$\mathrm{Hom}(\mathcal{P},\mathcal{P})$ is a vector space.
$\delta \in \mathrm{Hom}(\mathcal{P},\mathcal{P})$
$\delta \circ \delta \in \mathrm{Hom}(\mathcal{P},\mathcal{P})$
$\vdots$
$\underbrace{\delta \circ \cdots \circ \delta}_{M} \in \mathrm{Hom}(\mathcal{P},\mathcal{P})$

$$\implies 5 \circ \delta \oplus_{\mathrm{Hom}(\mathcal{P},\mathcal{P})} \delta \circ \delta \in \mathrm{Hom}(\mathcal{P},\mathcal{P})$$

165.4 Dual vector space

heavily used special case:
$(V,+,\cdot)$ vector space:

Definition 231.

$$V^* := \{\varphi : V \xrightarrow{\sim} \mathbb{R}\} = \mathrm{Hom}(V,\mathbb{R})$$

$\underbrace{(V^*,\oplus,\otimes)}_{\text{dual vector space (to } V)}$ is a vector space

Terminology: $\varphi \in V^*$ is called, informally, a covector.

Example

$I : \mathcal{P} \xrightarrow{\sim} \mathbb{R}$
i.e. $I \in \mathcal{P}^*$
def. $I(p) := \int_0^1 dx p(x)$
linear:

$$I(p+q) = \int_0^1 dx \underbrace{(p+q)(x)}_{p(x)+q(x)}$$
$$= \cdots = I(q) + I(p)$$

$I(\lambda p) = \lambda \cdot I(p)$
i.e. $I = \int_0^1 dx$

165.5 Tensors

Definition 232. Let $(V,+,\cdot)$ be a vector space.
An (r,s)-tensor T over V $\quad r,s \in \mathbb{N}_0$
is a multi-linear map

$$T : \underbrace{V^* \times \cdots \times V^*}_{r} \times \underbrace{V \times \cdots \times V}_{s} \xrightarrow[\vdots\}r+s]{\sim} \mathbb{R}$$

165.5.1 Example

: T $(1,1)$-tensor

$$T(\varphi+\psi,v) = T(\varphi,v) + T(\psi,v) \qquad T(\varphi,v+w) = T(\varphi,v) + T(\varphi,w)$$
$$T(\lambda\varphi,v) = \lambda \cdot T(\varphi,v) \qquad T(\varphi,\lambda\cdot v) = \lambda T(\varphi,v)$$

$$T(\varphi+\psi,v+w) =$$
$$= t(\varphi,v) + T(\varphi,w) + T(\psi,v) + t(\psi,w)$$

Excursion: Given $T : V^* \times V \xrightarrow{\sim} \mathbb{R}$

$$\phi_T : \qquad V \xrightarrow{\sim} (V*)^* \underbrace{=}_{\dim V<\infty} V$$

Define $v \mapsto \underbrace{T(\cdot, v)}_{V^* \xrightarrow{\sim} \mathbb{R}}$

Given $\phi : V \xrightarrow{\sim} V$

Construct $T_\phi : V^* \times V \xrightarrow{\sim} \mathbb{R}$

$$(\varphi, v) \mapsto \varphi(\phi(v))$$

$\Longrightarrow$ given $T : T = T_{\varphi_T}$

given $\phi : \phi = \phi_{T_\phi}$

Example

$g : P \times P \xrightarrow{\sim} \mathbb{R}$

$$(p, q) \mapsto \int_{-1}^{1} dx p(x) q(x)$$

is a $(0, 2)$-tensor over P.

Info: If $T \in \mathrm{Hom}(V, W)$

165.6 Vectors and covectors as tensors

Theorem 77. *(including proof)*
"covector" $\varphi \in V^* \Longleftrightarrow \varphi : V \xrightarrow{\sim} \mathbb{R} \Longleftrightarrow \varphi$*(0, 1)-tensor.*

Theorem 78. $v \in V \underbrace{=}_{dim V<\infty} (V^*)^* \Longleftrightarrow v : V^* \xrightarrow{\sim} \mathbb{R} \Longleftrightarrow v$ *is* $(1, 0)$*-tensor.*

165.7 Bases

Definition 233. $(V, +, \cdot)$ vector space.
A subset $B \subset V$ is called
a basis if

Thought bubble: Hamel (L.A.) EY : 20160316 Hamel basis, Linear Algebra

$$\forall\, v \in V \quad \exists \ \underline{\text{finite}} \ \underbrace{F}_{\{f_1, \ldots, f_n\}} \subset B : \exists ! \underbrace{v^1, v^2, \ldots, v^n}_{\in \mathbb{R}}, \qquad v = v^1 f_1 + \cdots + v^n f_n$$

Definition 234. If $\exists$ basis $\mathcal{B}$ with finitely many elements, say d many, then we call $d =: \dim V$
This is well-defined.

Remark: $(V, +, \cdot)$ be a finite-dim. vector space.
Having chosen a basis $e_1, \ldots, e_n$ of $(V, +, \cdot)$ we may uniquely associate
(Thought bubble: this requires a chosen basis)

$$v \mapsto (v^1, \ldots, v^n) \text{ called the components of } v \text{ w.r.t. chosen basis}$$

where: $v^1 e_1 + \cdots + v^n e + n = v$

165.8 Basis for the dual space

choose Basis $e_1, \ldots, e_n$ for V
can choose Basis $\epsilon^1, \ldots, \epsilon^n$ for V^*

However, more economical to require
once $e_1, \ldots, e_n$ on V has been chosen, that

$$\epsilon^a(e_b) = \delta^a_b$$

This uniquely determines choice of $\epsilon^1, \ldots, \epsilon^n$ from choice of $e_1, \ldots, e_n$

Definition 235. If a basis $\epsilon^1, \dots, \epsilon^n$ of V^* satisfies this, it is called the **dual basis** (of the dual space)

Example: P $(N = 3)$

e_0, e_1, e_2, e_3 basis if $\begin{matrix} e_0(x) = 1 \\ e_1(x) = x \\ e_2(x) = x^2 \\ e_3(x) = x^3 \end{matrix}$ $\{e_a(x) := x^a$

$\epsilon^0, \epsilon^1, \epsilon^2, \epsilon^3$ dual basis $\epsilon^a := \frac{1}{a!}\, \partial^a|_{x=0}$

Proof. $\epsilon^a(e_b) = \delta^a_b$ □

165.9 Components of tensors

Let T be an (r, s)-tensor on a finite-dim. vs. V. Then define the $(r+s)^{\dim V}$ many real numbers.

$$\underbrace{T^{i_1 \dots i_r}{}_{j_1 \dots j_s}}_{\in \mathbb{R}} := T(\epsilon^{i_1}, \dots, \epsilon^{i_r}, e_{j_1}, e_{j_2}, \dots, e_{j_s})$$

$i_1 \dots i_r, j_1 \dots j_s \in \{1, \dots, \dim V\}$

Thought bubble: $\underbrace{T^{i_1 \dots i_r}{}_{j_1 \dots j_s}}_{\in \mathbb{R}}$ are the components of the tensor w.r.t. chosen basis

Useful: Knowing components (and basis) one can reconstruct the entire tensor.

Example

T $(1,1)$- tensor

$$T^i{}_j := T(\epsilon^i, e_j)$$

reconstruct

$$T(\varphi, v) = T(\sum_{i=1}^{\dim V} \varphi_i \epsilon^i, \sum_{j=1}^{\dim V} v^j e_j) \qquad \varphi_i \in \mathbb{R}, v^j \in \mathbb{R}$$

$$= \sum_{i=1}^{\dim V} \sum_{j=1}^{\dim V} \varphi_i v^j \underbrace{T(\epsilon^i, e_j)}_{T^i{}_j}$$

$$=: \varphi_i v^j T^i{}_j$$

166 Lecture 4: Differentiable Manifolds

so far: top. mfd. $(M, \mathcal{O})$

$\dim M = d$

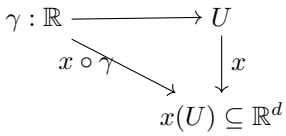
we wish to define a notion of differentiable

- curves $\mathbb{R} \to M$
- function $M \to \mathbb{R}$
- maps $M \to N$

166.1 1. Strategy

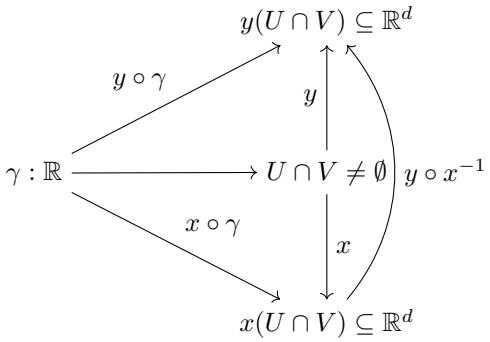
choose a chart (U, x)

$\gamma : \mathbb{R} \to M$ portion of curve in chart domain



idea. try to "lift" the undergraduate notion of differentiability of a curve on $\mathbb{R}^d$ to a notion of differentiability of a curve on M

Problem Can this be well-defined under change of chart?



$x \circ \gamma$ undergraduate differentiable ("as a map $\mathbb{R} \to \mathbb{R}^d$")

$$\underbrace{y \circ \gamma}_{\text{maybe only continuous, but not undergraduate differentiable}} = \underbrace{\overbrace{(y \circ x^{-1})}^{\mathbb{R}^d \to \mathbb{R}^d}}_{\text{continuous}} \circ \underbrace{\overbrace{(x \circ \gamma)}^{\mathbb{R} \to \mathbb{R}^d}}_{\text{undergrad differentiable}} = y \circ (x^{-1} \circ x) \circ \gamma$$

At first sight, strategy does not work out.

166.2 2. Compatible charts

In section 1, we used any imaginable charts on the top. mfd. $(M, \mathcal{O})$.

To emphasize this, we may say that we took U and V from the *maximal atlas* $\mathcal{A}$ of $(M, \mathcal{O})$.

Definition 236. Two charts (U, x) and (V, y) of a top. mfd. are called ❀-compatible if either

(a) $U \cap V = \emptyset$ or

(b) $U \cap V \neq \emptyset$

chart transition maps have undergraduate ❀ property.

EY : 20151109 e.g. since $\mathbb{R}^d \to \mathbb{R}^d$, can use undergradate ❀ property such as continuity or differentiability.

$$y \circ x^{-1} : x(U \cap V) \subseteq \mathbb{R}^d \to y(U \cap V) \subseteq \mathbb{R}^d$$
$$x \circ y^{-1} : y(U \cap V) \subseteq \mathbb{R}^d \to x(U \cap V) \subseteq \mathbb{R}^d$$

Philosophy:

Definition 237. An atlas $\mathcal{A}_❀$ is a ❀-compatible atlas if any two charts in $\mathcal{A}_❀$ are ❀-compatible.

Definition 238. A **❀-manifold** is a triple $(\underbrace{M, \mathcal{O}}_{\text{top. mfd.}}, \mathcal{A}_❀)$ $\quad \mathcal{A}_❀ \subseteq \mathcal{A}_{\text{maximal}}$

❀	undergraduate ❀	
C^0	$C^0(\mathbb{R}^d \to \mathbb{R}^d) =$	continuous maps w.r.t. $\mathcal{O}$
C^1	$C^1(\mathbb{R}^d \to \mathbb{R}^d) =$	differentiable (once) and is continuous
C^k		k-times continuously differentiable
D^k		k-times differentiable
$\vdots$		
C^∞	$C^\infty(\mathbb{R}^d \to \mathbb{R}^d)$	
$\cap\!\!\cup$		
C^ω	$\exists$ multi-dim. Taylor exp.	
$\mathbb{C}^\infty$	satisfy Cauchy-Riemann equations, pair-wise	

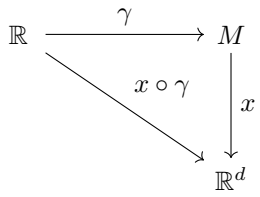
EY : 20151109 Schuller says: C^k is easy to work with because you can judge k-times cont. differentiability from existence of all partial derivatives **and** their continuity. There are examples of maps that partial derivatives exist but are not D^k, k-times differentiable.

Theorem 79 (Whitney)**.** *Any $C^{k\geq 1}$-atlas, $\mathcal{A}_{C^{k\geq 1}}$ of a topological manifold* contains *a C^∞-atlas.*

Thus we may w.l.o.g. always consider C^∞-manifolds, "smooth manifolds", unless we wish to define Taylor expandibility/complex differentiability ...

EY : 20151109 Hassler Whitney [16]

Definition 239. A smooth manifold ($\underbrace{M, \mathcal{O}}_{\text{top. mfd.}}$, $\underbrace{\mathcal{A}}_{C^\infty-\text{atlas}}$)



EY: 20151109 Schuller was explaining that the trajectory is real in M; the coordinate maps to obtain coordinates is $x \circ \gamma$

166.3 4. Diffeomorphisms

$M \xrightarrow{\phi} N$

If M, N are naked sets, the structure preserving maps are the bijections (invertible maps).

e.g. $\{1,2,3\} \to \{a,b\}$

Definition 240. $M \cong_{\text{set}} N$ (set-theoretically) isomorphic if $\exists$ bijection $\phi : M \to N$

Examples. $\mathbb{N} \cong_{\text{set}} \mathbb{Z}$

$\mathbb{N} \cong_{\text{set}} \mathbb{Q}$ (EY: 20151109 Schuller says from diagonal counting)

$\mathbb{N} \not\cong_{\text{set}} \mathbb{R}$

Now $(M, \mathcal{O}_M) \cong_{\text{top}} (N, \mathcal{O}_N)$ (topl.) isomorphic = "homeomorphic" $\exists$ bijection $\phi : M \to N$

ϕ, ϕ^{-1} are continuous.

$(V, +, \cdot) \cong_{\text{vec}} (W, +_w, \cdot_w)$ (EY: 20151109 vector space isomorphism) if

$\exists$ bijection $\phi : V \to W$ linearly

finally

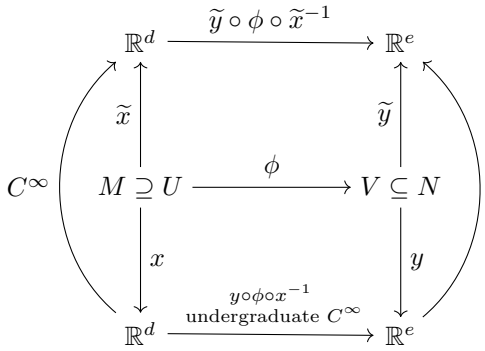
Definition 241. Two C^∞-manifolds

$(M, \mathcal{O}_M, \mathcal{A}_M)$ and $(N, \mathcal{O}_N, \mathcal{A}_N)$ are said to be **diffeomorphic** if $\exists$ bijection $\phi : M \to N$ s.t.

$$\phi : M \to N$$
$$\phi^{-1} : N \to M$$

are both C^∞-maps

[16] http://mathoverflow.net/questions/8789/can-every-manifold-be-given-an-analytic-structure



Theorem 80. *# = number of C^∞-manifolds one can make out of a given C^0-manifolds (if any) - up to diffeomorphisms.*

dimM	#	
1	*1*	*Morse-Radon theorems*
2	*1*	*Morse-Radon theorems*
3	*1*	*Morse-Radon theorems*
4	*uncountably infinitely many*	
5	*finite*	*surgery theory*
6	*finite*	*surgery theory*
⋮	*finite*	*surgery theory*

EY : 20151109 cf. http://math.stackexchange.com/questions/833766/closed-4-manifolds-with-uncountably-many-differentiable-structures
The wild world of 4-manifolds

Tutorial 4 Differentiable Manifolds

EY : 20151109 The gravity-and-light.org website, where you can download the tutorial sheets *and* the full length videos for the tutorials and lectures, are no longer there. = (

Hopefully, the YouTube video will remain: https://youtu.be/FXPdKxOq1KA?list=PLFeEvEPtX_0RQ1ys-7VIsKlBWz7RX-FaL

Exercise 1: True or false?. *These basic questions are designed to spark discussion and as a self-test.*

Tick the correct statements, but not the incorrect ones!

(a) The function $f : \mathbb{R} \to \mathbb{R}$, ...

-
-
- ..., defined by $f(x) = |x^3|$, lies in $C^3(\mathbb{R} \to \mathbb{R})$.

  EY : 20151109 **Solution 1a3.** For $f : \mathbb{R} \to \mathbb{R}$, $f(x) = |x^3| = \begin{cases} x^3 & \text{if } x \geq 0 \\ -x^3 & \text{if } x < 0 \end{cases}$

$$f'(x) = \begin{cases} 3x^2 & \text{if } x \geq 0 \\ -3x^2 & \text{if } x < 0 \end{cases}$$

$$f''(x) = \begin{cases} 6x & \text{if } x \geq 0 \\ -6x & \text{if } x < 0 \end{cases}$$

  Thus,

$$\boxed{f(x) = |x^3| \in C^1(\mathbb{R}) \text{ but } f(x) \notin C^2(\mathbb{R}) \subseteq C^3(\mathbb{R})}$$

-

•

(b)

(c)

Short Exercise 4: Undergraduate multi-dimensional analysis .

A good notation and basic results for partial differentiation.

For a map $f : \mathbb{R}^d \to \mathbb{R}$ we denote by the map $\partial_i f : \mathbb{R}^d \to \mathbb{R}$ the partial derivative with respect to the i-th entry.

Question :. Given a function

$$f : \mathbb{R}^3 \to \mathbb{R};\ (\alpha, \beta, \delta) \mapsto f(\alpha, \beta, \delta) := \alpha^3\beta^2 + \beta^2\delta + \delta$$

calculate the values of the following derivatives:
Solution :.

- $(\partial_2 f)(x, y, z) =$
- $(\partial_1 f)(\square, \circ, *) =$
- $(\partial_1\partial_2 f)(a, b, c) =$
- $(\partial_3^2 f)(299, 1222, 0) =$

EY: 20151110
For $f(\alpha, \beta, \delta) := \alpha^3\beta^2 + \beta^2\delta + \delta$, or $f(x, y, z) = x^3y^2 + y^2z + z$,

$$\begin{aligned}(\partial_2 f) &= 2(x^3y + yz)\\ (\partial_1 f) &= 3x^2y^2\\ (\partial_1\partial_2 f) &= 6x^2y\\ (\partial_3^2 f) &= 0\end{aligned}$$

and so

- $(\partial_2 f)(x, y, z) = 2(x^3y + yz)$
- $(\partial_1 f)(\square, \circ, *) = 3\square^2\circ^2$
- $(\partial_1\partial_2 f)(a, b, c) = 6a^2b$
- $(\partial_3^2 f)(299, 1222, 0) = 0$

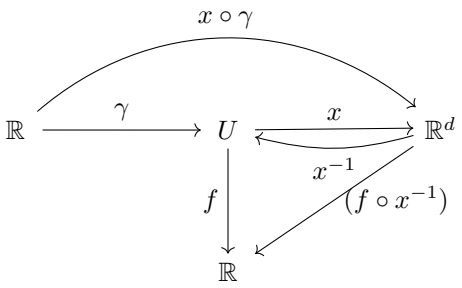
Exercise 5: Differentiability on a manifold.

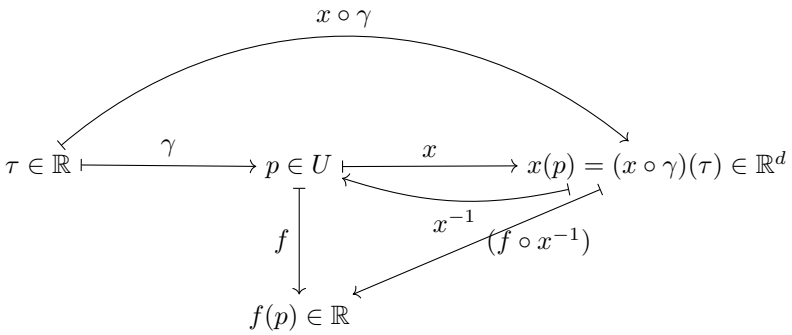
How to deal with functions and curves in a chart

Let $(M, \mathcal{O}, \mathcal{A})$ be a smooth d-dimensional manifold. Consider a chart (U, x) of the atlas $\mathcal{A}$ together with a smooth curve $\gamma : \mathbb{R} \to U$ and a smooth function $f : U \to \mathbb{R}$ on the domain U of the chart.

Question :. Draw a commutative diagram containing the chart domain, chart map, function, curveand the

respective representatives of the function and the curve in the chart.
Solution :.





Question :. Consider, for $d = 2$,

$$(x \circ \gamma)(\lambda) := (\cos(\lambda), \sin(\lambda)) \text{ and } (f \circ x^{-1})((x, y)) := x^2 + y^2$$

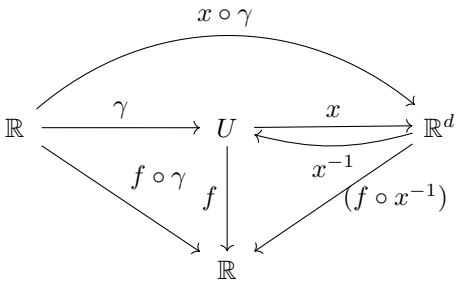
Using the chain rule, calculate

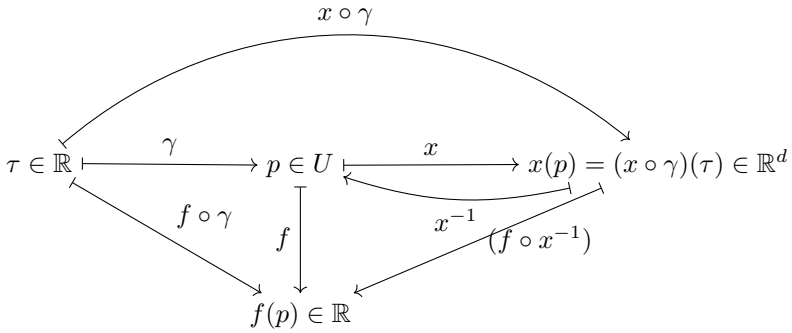
$$(f \circ \gamma)'(\lambda)$$

explicitly.

Solution :.

EY : 20151109 Indeed, the domains and codomains of this $f\gamma$ mapping makes sense, from $\mathbb{R} \to \mathbb{R}$ for





$$(f \circ \gamma)'(\lambda) = (Df) \cdot \dot{\gamma}(\lambda) = \frac{\partial f}{\partial x^j} \dot{\gamma}^j(\lambda) = 2x(-\sin\lambda) + 2y\cos\lambda = 2(-\cos\lambda\sin\lambda + \sin\lambda\cos\lambda) = 0$$

167 Lecture 5: Tangent Spaces

lead question: "what is the velocity of a curve γ point p?

167.1 Velocities

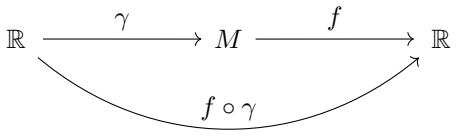
Definition 242. $(M, \mathcal{O}, \mathcal{A})$ smooth mfd.
curve $\gamma : \mathbb{R} \to M$ at least C^1.
Suppose $\gamma(\lambda_0) = p$
The **velocity** of γ p is the linear map

$$v_{\gamma,p} : C^\infty(M) \xrightarrow{\sim} \mathbb{R}$$

$C^\infty(M) := \{f : M \to \mathbb{R} | f \text{ smooth function } \}$ equipped with $(f \oplus g)(p) := f(p) + g(p)$
$(\lambda \otimes g)(p) := \lambda \cdot g(p)$

$\sim$ denotes linear map on top of $\to$.

$$f \mapsto v_{\gamma,p}(f) := (f \circ \gamma)'(\lambda_0)$$



intuition

Schuller says: children run around the world. Temperature function as temperature contour lines. You feel the temperature. You observe the rate of change of temperature as you run around. f is temperature.

past: " $\underbrace{v^i}_{\text{vector}}(\partial_i f) = (\underbrace{v^i \partial_i}_{\text{vector}})f$

167.2 Tangent vector space

Definition 243. For each point $p \in M$
def the **set** "tangent space $\neq_0 M$ p "

$$T_pM := \{v_{\gamma,p} | \gamma \text{ smooth curves } \}$$

picture:
rather M than (embedded) p T_pM EY : 20151109 see https://youtu.be/pepU_7NJSGM?t=12m38s for the picture

Observation: T_pM can be made into a vector space.

$$\begin{aligned}
\oplus :& T_pM \times T_pM \to \\
& (v_{\gamma,p} \oplus v_{\delta,p})(\underbrace{f}_{\in C^\infty(M)}) := v_{\gamma,p}(f) +_{\mathbb{R}} v_{\delta,p}(f) \\
\odot :& \mathbb{R} \times T_pM \to \mathrm{Hom}(C^\infty(\mathbb{R}), \mathbb{R}) \\
& (\alpha \odot v_{\gamma,p})(f) := \alpha \cdot_{\mathbb{R}} v_{\gamma,p}(f)
\end{aligned}$$

Remains to be shown that

(i) $\exists \sigma$ curve : $v_{\gamma,p} \oplus v_{\delta,p} = v_{\sigma,p}$

(ii) $\exists \tau$ curve : $\alpha \odot v_{\gamma,p} = v_{\tau,p}$

Claim: $\tau : \mathbb{R} \to M$ where $\mu_\alpha : \mathbb{R} \to \mathbb{R}$, does the trick.
$\mapsto \tau(\lambda) := \gamma(\alpha\lambda + \lambda_0) = (\gamma \circ \mu_\alpha)(\lambda)$ $r \mapsto \alpha \cdot r + \lambda_0$
$\tau(0) = \gamma(\lambda_0) = p$

$$\begin{aligned}
v_{\tau,p} &:= (f \circ \tau)'(0) = (f \circ \gamma \circ \mu_\alpha)'(0) \\
&= (f \circ \gamma)'(\lambda_0) \cdot \alpha = \\
&= \alpha \cdot v_{\gamma,p}
\end{aligned}$$

Now for the sum:
$v_{\gamma,p} \oplus v_{\delta,p} \stackrel{?}{=} v_{\sigma,p}$
make a choice of chart $(\underbrace{U}_{\ni p}, x)$ In cloud: ill definition alarm bells.

and define:
Claim:

$$\begin{aligned}
&\sigma : \mathbb{R} \to M \\
&\sigma(\lambda) := x^{-1}(\underbrace{(x \circ \gamma)(\lambda_0 + \lambda)}_{\mathbb{R} \to \mathbb{R}^d} + (x \circ \delta)(\lambda_1 + \lambda) - (x \circ \gamma)(\lambda_0))
\end{aligned}$$

does the trick.

Proof. Since:

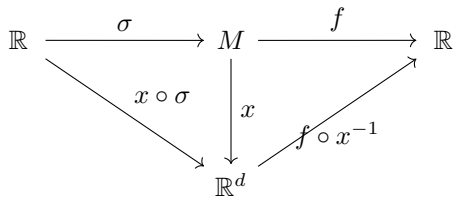
$$\begin{aligned}
\sigma_x(0) &= x^{-1}((x \circ \gamma)(\lambda_0) + (x \circ \delta)(\lambda_1) - (x \circ \gamma)(\lambda_0)) \\
&= \delta(\lambda_1) = p
\end{aligned}$$

Now:

$$\begin{aligned} v_{\sigma_x,p}(f) &:= (f\circ\sigma_x)'(0) = \\ &= (\underbrace{(f\circ x^{-1})}_{\mathbb{R}^d\to\mathbb{R}}\circ\underbrace{(x\circ\sigma_x)}_{\mathbb{R}\to\mathbb{R}^d})'(\gamma) = \underbrace{(x\circ\sigma_x)'(0)}_{(x\circ\gamma)'(\lambda_0)+(x\circ\delta)'(\lambda_1)}\cdot\left(\partial_i(f\circ x^{-1})\right)(x(\underbrace{\sigma(0)}_{p})) = \\ &= (x\circ\gamma)'(\lambda_0)(\partial_i(f\circ x^{-1}))(x(p)) + (x\circ\delta)(\lambda_1)(\partial_i(f\circ x^{-1}))(x(p)) \\ &= (f\circ\gamma)'(\lambda_0) + (f\circ\delta)'(\lambda_1) = \\ &= v_{\gamma,p}(f) + v_{\delta,p}(f) \quad \forall\, f\in C^\infty(M) \end{aligned}$$

$$\boxed{v_{\gamma,p}\oplus v_{\delta,p} = v_{\sigma,p}}$$

□



picture: (cf. https://youtu.be/pepU_7NJSGM?t=39m5s)

$$\gamma:\mathbb{R}\to M$$
$$\delta:\mathbb{R}\to M$$

$(\gamma\oplus)(\lambda) := \gamma(\lambda)+\delta(\lambda)$

EY : 20151109 Schuller says adding trajectories is chart dependent, bad. Adding velocities is good.

167.3 Components of a vector wrt a chart

Definition 244. Let $(U,x)\in\mathcal{A}_{\text{smooth}}$.

Let $\gamma:\mathbb{R}\to U$ $\gamma(0)=p$.

Calculate

$$\begin{aligned} v_{\gamma,p}(f) := (f\circ\gamma)'(0) &= (\underbrace{(f\circ x^{-1})}_{\mathbb{R}^d\to\mathbb{R}}\circ\underbrace{(x\circ\gamma)}_{\mathbb{R}\to\mathbb{R}^d})'(0) \\ &= \underbrace{(x\circ\gamma)^{i'}(0)}_{\dot\gamma^i_x(0)}\cdot\underbrace{(\partial_i(f\circ x^{-1}))(x(p))}_{=:\left(\frac{\partial f}{\partial x^i}\right)_p} \end{aligned}$$

think cloud $f:M\to\mathbb{R}$

$$= \boxed{\dot\gamma^i_x(0)\cdot\left(\frac{\partial}{\partial x^i}\right)_p} f \quad \forall\, f\in C^\infty(M)$$

$\therefore$ as a map.

$$v_{\gamma,p} \underbrace{=}_{\text{use of chart}} \underbrace{\dot\gamma^i_x(0)}_{\text{“components of the velocity } v_{\gamma,p}\text{”}} \underbrace{\left(\frac{\partial}{\partial x^i}\right)}_{\substack{\text{basis elements of the } T_pM \text{ wrt which the components need to be understood.}\\ \text{“chart induced basis of } T_pM\text{”}}}$$

Picture: https://youtu.be/pepU_7NJSGM?t=1h16s

167.4 4. Chart-induced basis

Definition 245. $(U,x) \in \mathcal{A}_{\text{smooth}}$
the $\left(\frac{\partial}{\partial x^1}\right)_p, \ldots, \left(\frac{\partial}{\partial x^d}\right)_p \in T_pU \subseteq T_pM$
constitute a **basis** of T_pU

Proof. remains: linearly independent

$$\lambda^i \left(\frac{\partial}{\partial x^i}\right)_p \stackrel{!}{=} 0$$

$$\implies \lambda^i \left(\frac{\partial}{\partial x^i}\right)_p (x^j) = \lambda^i \partial_i (\underbrace{x^j \circ x^{-1}}_{x^j \circ x^{-1} : \mathbb{R}^d \to \mathbb{R},\ (\alpha^1,\ldots,\alpha^d) \mapsto \alpha^j})(x(p)) =$$

$$= \lambda^i \delta_i{}^j = \lambda^j \qquad j = 1, \ldots, d$$

in cloud: $x^j : U \to \mathbb{R}$ differentiable

□

Corollary 16. $dim T_pM = d = dim M$

Terminology: $X \in T_pM \to \exists\, \gamma : \mathbb{R} \to M : X = v_{\gamma,p}$ and
$\exists\, \underbrace{X_1^1, \ldots, X^d}_{\in \mathbb{R}} : X = X^i \left(\frac{\partial}{\partial x^i}\right)_p$

167.5 5. Change of vector *components* under a change of chart

✘ vector does **not** change under change of chart.
Let (U,x) and (V,y) be overlapping charts and $p \in U \cap V$.
Let $X \in T_pM$

$$X^i_{(y)} \cdot \left(\frac{\partial}{\partial y^i}\right)_p \underbrace{=}_{(V,y)} X \underbrace{=}_{(U,x)} X^i_x \left(\frac{\partial}{\partial x^i}\right)_p$$

to study change of components formula:

$$\left(\frac{\partial}{\partial x^i}\right)_p f = \partial_i (f \circ x^{-1})(x(p)) =$$

$$= \partial_i (\underbrace{(f \circ y^{-1})}_{\mathbb{R}^d \to \mathbb{R}} \circ \underbrace{(y \circ x^{-1})}_{\mathbb{R}^d \to \mathbb{R}^d})(x(p))$$

$$= (\partial_i (y^i \circ x^{-1}))(x(p)) \cdot (\partial_j (f \circ y^{-1}))(y(p)) =$$

$$= \boxed{\left(\frac{\partial y^p}{\partial x^i}\right)_p \cdot \left(\frac{\partial f}{\partial y^j}\right)_p} f$$

$$\implies X^i_{(x)} \left(\frac{\partial y^j}{\partial x^i}\right)_p \left(\frac{\partial}{\partial y^j}\right)_p = X^j_{(y)} \left(\frac{\partial}{\partial y^j}\right)_p$$

$$\implies \boxed{X^j_{(y)} = \left(\frac{\partial y^j}{\partial x^i}\right)_p X^i_{(x)}}$$

167.6 6. Cotangent spaces

$T_pM = V$
trivial $(T_pM)^* := \{\varphi : T_pM \xrightarrow{\sim} \mathbb{R}\}$
Example: $f \in C^\infty(M)$

$$(df)_p : T_pM \xrightarrow{\sim} \mathbb{R}$$
$$X \mapsto (df)_p(X)$$

i.e. $\boxed{(df)_p \in T_pM^*}$

$(df)_p$ called the gradient of f $p \in M$.

Calculate components of gradient w.r.t. chart-induced basis (U, x)

$$((df)_p)_j := (df)_p \left(\left(\frac{\partial}{\partial x^j} \right)_p \right)$$
$$= \left(\frac{\partial f}{\partial x^j} \right)_p = \partial_j (f \circ x^{-1})(x(p))$$

Theorem 81. *Consider chart* $(U, x) \Longrightarrow x^i : U \to \mathbb{R}$

Claim: basis of T_p^*M

$\Longrightarrow$ *In fact: dual basis:*

$$(dx^a)_p \left(\left(\frac{\partial}{\partial x^b} \right)_p \right) = \left(\frac{\partial x^a}{\partial x^b} \right)_p = \cdots = \delta^a_b$$

167.7 7. Change of *components* of a covector under a change of chart:

$$\underbrace{T_p^*M}_{\ni \omega} \text{ with } \omega_{(y)} (dy^j)_p = \omega = \omega_{(x)i} (dx^i)_p$$

$$\Longrightarrow \boxed{\omega_{(y)i} = \frac{\partial x^j}{\partial y^i} \omega_{(x)j}}$$

Lecture 6: Fields

cf. Lecture 6: Fields (International Winter School on Gravity and Light 2015)

So far:

T_pM
$\vdots \downarrow$
T_p^*M
$\vdots \downarrow$
$\vdots$
,

now

in Thought Cloud: theory of bundles

167.8 Bundles

Definition 246. A **bundle** is a triple

$$E \xrightarrow{\pi} M$$

E smooth manifold "**total space**"
π smooth map (surjective) "projection map"
M smooth manifold "base space"

Example E = cylinder M = circle

Definition 247.

$$E \xrightarrow{T} M$$

bundle.

$p \in M$

define **fibre over** p
$:= \text{preim}_\pi(\{p\})$

Definition 248. A **section** σ of a bundle

$$\begin{array}{c} E \\ \pi \big\downarrow \big\nwarrow \sigma \\ M \end{array}$$

require $\pi \circ \sigma = \text{id}_M$

Schuller says: in quantum mechanics, Aside: $\psi : M \to \mathbb{C}$

167.9 Tangent bundle of smooth manifold

$(M, \mathcal{O}, \mathcal{A})$ smooth manifold

(a) as a **set** $TM := \dot{\bigcup}_{p \in M} T_p M$

(b) surjective $\pi : TM \to M$ the *unique* point $p \in M$, $X \in T_p M$
$X \mapsto p$

situation: $\underbrace{TM}_{\text{set}} \quad \underbrace{\xrightarrow{\pi}}_{\text{surjective map}} \quad \underbrace{M}_{\text{smooth manifold}}$

(c) Construct topology on TM that is the coarsest topology such that π (just) continuous. ("initial topology with respect to π").

$$\mathcal{O}_{TM} := \{\text{preim}_\pi(U) | \mathcal{U} \in \mathcal{O}\}$$

Show: Tutorial $\mathcal{O}_{TM}$ Schuller says this is shown in the tutorial
$(TM, \mathcal{O}_{TM})$

Construction of a C^∞-atlas on TM from the C^∞-atlas $\mathcal{A}$ on M.

$$\mathcal{A}_{TM} := \{(T\mathcal{U}, \xi_x) | (U, x) \in \mathcal{A}\}$$

where

$$\begin{aligned} \xi_x : & T\mathcal{U} \to \mathbb{R}^{2 \cdot \dim M} \\ & X \mapsto (\underbrace{(x^1 \circ \pi)(X), \ldots, (x^d \circ \pi)(X)}_{(U,x)- \text{ coords of } \pi(X) \ \ (d \text{ many })}, (dx^1)_{\pi(X)}(X), \ldots, (dx^d)_{\pi(X)}(X)) \end{aligned}$$

where $X \in T_{\pi(X)} M$
$X = X^i_{(x)} \left(\frac{\partial}{\partial x^i}\right)_{\pi(X)}$

$$\begin{aligned} (dx^j)_{\pi(X)}(X) &= (dx^j)_{\pi(X)} \left(X^i_{(x)} \left(\frac{\partial}{\partial x^i} \right)_{\pi(X)} \right) = \\ &= X^i_{(x)} \delta^j_i = X^j_{(x)} \end{aligned}$$

Write $\xi_x^{-1} : \xi_x(TU) \subseteq \mathbb{R}^{2 \dim M} \to TU$

$$(\alpha^1, \ldots, \alpha^d, \beta^1, \ldots, \beta^d) := \beta^i \left(\frac{\partial}{\partial x^i} \right)_{\underbrace{x^{-1}(\alpha^1, \ldots, \alpha^d)}_{\pi(X)}}$$

Check:

$$(\xi_y \circ \xi_x^{-1})(\alpha^1, \dots, \alpha^d, \beta^1, \dots, \beta^d) =$$
$$= \xi_y \left(\beta^i \left(\frac{\partial}{\partial x^i} \right)_{x^{-1}(\alpha^1, \dots, \alpha^d)} \right)$$
$$= \left(\dots, (y^i \circ \pi)(\beta^m \cdot \left(\frac{\partial}{\partial x^m} \right)_{x^{-1}(\alpha^1 \dots \alpha^d)}), \dots, \dots (dy^i)_{x^{-1}(\alpha^1, \dots \alpha^d)} \left(\beta^m \left(\frac{\partial}{\partial x^m} \right)_{x^{-1}(\alpha^1 \dots \alpha^d)} \right), \dots \right) =$$
$$= (\dots, (y^i \circ x^{-1})(\alpha^1, \dots, \alpha^d), \dots, \dots, \underbrace{\beta^m (dy^i)_{x^{-1}(\alpha^1, \dots, \alpha^d)} \left(\left(\frac{\partial}{\partial x^m} \right)_{x^{-1}(\alpha^1 \dots \alpha^d)} \right)}_{\beta^m \left(\frac{\partial y}{\partial x^m} \right)_{x^{-1}(\alpha^1 \dots \alpha^d)}})$$

Check transition map: $(U, x), (V, y), \quad U \cap V \neq 0$
$\left(\frac{\partial y}{\partial x^m} \right)_{x^{-1}(\alpha^1 \dots \alpha^d)} = \partial_m (y^i \circ x^{-1})(x \circ (x^{-1}(\alpha^1 \dots \alpha^d))) = \partial_m (y^i \circ x^{-1})(\alpha^1 \dots \alpha^d)$ smooth.
upshot

$$\underbrace{TM}_{\text{smooth manifold}} \quad \underbrace{\overset{\pi}{\to}}_{\text{smooth map}} \quad \underbrace{M}_{\text{smooth manifold}}$$

bundle, called the tangent bundle.

3. Vector fields

Definition 249. A **smooth vector field** χ is a *smooth* map, (where)

$$\begin{array}{ccc} TM & & \\ {\scriptstyle \pi}\downarrow & \curvearrowleft & {\scriptstyle \chi} \\ M & & \end{array}$$

Example:

$$\begin{array}{c} TS^1 \\ \downarrow {\scriptstyle \pi} \\ S^1 \end{array}$$

4. The $C^\infty(M)$-module $\Gamma(TM)$

$C^\infty(M)$-module $\leftarrow (C^\infty(M), +, \cdot)$ (satisfies) $C^+, A^+, N^+, I+$, C^+, A^+, N^+, D^+. Not a ~~field~~. A *ring*.

set $\Gamma(TM) = \{\chi \quad M \to TM | \text{ smooth section } \}$
$(\chi \oplus \widetilde{\chi})(f) := (\chi f) \underbrace{+}_{C^\infty(M)} \widetilde{\chi}(f)$
$(\underbrace{g}_{C^\infty(M)} \odot \xi)(f) := \underbrace{g}_{C^\infty(M)} \cdot \chi(f)$

$$\chi : M \to TM$$
$$p \mapsto \chi(p)$$
$$\chi f : M \to \mathbb{R}$$
$$p \mapsto \chi(p) f$$

$(\Gamma(TM), \oplus, \odot)\, C^\infty(M)$ - **module**

upshot: set of all smooth vector fields can be made into a $C^\infty(M)$-module.
Fact:

(1) ZFC $\Longrightarrow$ every vector space has a basis. (You have to have C - axiom of choice in set theory)

(2) no such result exists for modules.

This is a shame, because otherwise, we could have chosen (for any manifolds) vector fields,

$$\chi_{(1)}, \dots, \chi_{(d)} \in \Gamma(TM)$$

and would be able to write every vector field Ξ

$$\chi = \underbrace{f^i}_{\text{component functions}} \cdot \chi_{(i)}$$

Simple counterexample
Schuller says: Take a sphere, Morse Theorem, every smooth vector field must vanish at 2 pts. "mustn't choose a global basis"

However: $\dfrac{\partial}{\partial x^i} : U \xrightarrow{\text{smooth}} TU$

$$p \mapsto \left(\frac{\partial}{\partial x^i}\right)_p$$

167.10 Tensor fields

so far
$\Gamma(M)$ ="set of vector fields" $C^\infty(M)$-module
$\Gamma(T^*M)$ = "covector fields" $C^\infty(M)$-module

Definition 250. An (r, s)-tensor field T is a multi-linear map

$$T : \underbrace{\Gamma(T^*M) \times \cdots \times \Gamma(T^*M)}_{r} \times \Gamma(TM) \times \cdots \times \Gamma(TM) \xrightarrow{\sim} C^\infty(M)$$

Example: $f \in C^\infty(M)$

$$df : \Gamma(TM) \xrightarrow{\sim} C^\infty(M)$$
$$\chi \mapsto df(\chi) := \chi[f]$$

df $(0,1)$-T.F. (tensor field)
where $(\chi f)(\underbrace{p}_{\in M}) := \underbrace{\chi(p)}_{\in T_pM} f$
can check: df is $C^\infty-$linear

Lecture 7: Connections

cf. Lecture 7: Connections (International Winter School on Gravity and Light 2015)
So far: saw that a vector field X can be used to provide a directional derivative

$$\nabla_X f := Xf$$

of a function $f \in C^\infty(M)$.
Remark: from now on: consider mostly vector fields.
Notational overkill?

$$\nabla_X f = Xf = (df)(X)$$

In Thought Bubble: $\nabla_X(f \cdot g) = X(fg) = (Xf) \cdot g + fX(g)$ Product rule, because it's a derivative.
not quite:

$$X : C^\infty(M) \to C^\infty(M)$$
$$df : \Gamma(TM) \to C^\infty(M)$$
$$\nabla_X : C^\infty(M) \to C^\infty(M)$$

$$\begin{array}{ccc} \nabla_X : C^\infty(M) & \longrightarrow & C^\infty(M) \\ \vdots\downarrow & & \vdots\downarrow \end{array}$$

$$\nabla_X : \begin{array}{c} TM^p \otimes T^*M^q \text{ i.e.} \\ \binom{p}{q} \text{ tensor field} \end{array} \longrightarrow \begin{array}{c} TM^p \otimes T^*M^q \text{ i.e.} \\ \binom{p}{q} \text{ tensor field} \end{array}$$

1. Directional derivatives of tensor fields

We formulate a wish list of properties which the ∇_X acting on a tensor field should have.

In Thought Bubble: Any remaining freedom in choosing ∇, will need to be provided as additional structure beyond $(M, \mathcal{O}, \mathcal{A})$

Definition 251 (connection)**.** In Thought Bubble: linear connection, covariant derivatve, affine connection

A **connection** ∇ on a smooth manifold $(M, \mathcal{O}, \mathcal{A})$ is a map that takes a pair consisting of a vector (field) X and a (p, q)-tensor *field* T and sends them to a (p, q)-tensor (field) $\nabla_X T$ satisfying

(i) $\nabla_X f = Xf \quad \forall f \in C^\infty(M)$

(ii) $\nabla_X(T + S) = \nabla_X T + \nabla_X S$

(iii) $\nabla_X(\underbrace{T(\omega, Y)}_{\in C^\infty(M)}) = (\nabla_X T)(\omega, Y) + T(\nabla_X \omega, Y) + T(\omega, \nabla_X Y)$

In Thought Bubble:for $(1, 1)$-TF T, but analogously for any (p, q) - TF

"Leibnitz" rule.

(iv) $\nabla_{fX+Z} T = f\nabla_X T + \nabla_Z T$
$f \in C^\infty(M)$

A manifold with connection is quadruple $(M, \mathcal{O}, \mathcal{A}, \nabla)$
Remark: ∇_X is the extension of X.
∇ — " ———- of d

2. New structure on $(M, \mathcal{O}, \mathcal{A})$ required to fix ∇

Q: How much freedom do we have in choosing such a structure.
Consider X, Y vector fields

$$\nabla_X Y \underbrace{=}_{\text{In Thought Bubble:}(U,x)} \nabla_{X^i \frac{\partial}{\partial x^i}} \left(Y^m \frac{\partial}{\partial x^m} \right)$$

$$\underbrace{=}_{\text{(iii)}} X^i \left(\nabla_{\frac{\partial}{\partial x^i}} Y^m \right) \frac{\partial}{\partial x^m} + X^i Y^m \underbrace{\nabla_{\frac{\partial}{\partial x^i}} \left(\frac{\partial}{\partial x^m} \right)}$$

$$\underbrace{=}_{\text{(i)}} X^i \left(\frac{\partial}{\partial x^i} Y^m \right) \frac{\partial}{\partial x^m} + X^i Y^m \underbrace{\Gamma^q{}_{mi}(x)}_{\text{connection coefficient functions (on } M\text{) of } \nabla \text{ wrt } (U,x)} \frac{\partial}{\partial x^q}$$

$$(\underbrace{T}_{(p,q)} \otimes \underbrace{S}_{(r,s)})(\omega, \dots, X, \dots) := T(\omega, \dots, X, \dots) \underbrace{\cdot}_{C^\infty(M)} S(\dots, \dots)$$

$$\nabla_X (T \otimes S) = (\nabla_X T) \otimes S + T \otimes (\nabla_X S)$$

Definition 252 (Connection coefficient functions)**.** $(M, \mathcal{O}, \mathcal{A}, \nabla)$, $(\mathcal{U}, x) \in \mathcal{A}$.
Then the **connection coefficient functions** ("Γ"s) with respect to (wrt) (U, x) on the $(\dim(M))^3$ many functions

$$\begin{aligned} \Gamma^i{}_{jk} : & \mathcal{U} \to \mathbb{R} \\ & p \mapsto \left(dx^i \left(\nabla_{\frac{\partial}{\partial x^k}} \frac{\partial}{\partial x^j} \right) \right)(p) \end{aligned}$$

Thus:

$$(\nabla_X Y)^i = X^m \left(\frac{\partial}{\partial x^m} Y^i \right) + \Gamma^i{}_{nm} \underbrace{\cdot}_{C^\infty(M)} Y^n X^m$$

Remark: On a chart domain U, choice of the $(\dim M)^3$ functions $\Gamma^i{}_{jk}$ suffices to fix the action of ∇ on a *vector* field.

Fortunately, the same $(\dim M)^3$ functions fix the action of ∇ on any tensor field.

key observation:

$$\nabla_{\frac{\partial}{\partial x^m}} (dx^i) = \sum_{jm}^{i} dx^j$$

but now:

$$\underbrace{\nabla_{\frac{\partial}{\partial x^m}} (\underbrace{dx^i \left(\frac{\partial}{\partial x^j} \right)}_{\delta^i{}_j})}_{\| \quad \text{(iii)}} = \frac{\partial}{\partial x^m} (\delta^i{}_j) = 0$$

$$= \left(\nabla_{\frac{\partial}{\partial x^m}} dx^i \right) \left(\frac{\partial}{\partial x^j} \right) + dx^i (\underbrace{\nabla_{\frac{\partial}{\partial x^m}} \frac{\partial}{\partial x^j}}_{\Gamma^q_{jm} \frac{\partial}{\partial x^q}}) = 0$$

$$\implies \left(\nabla_{\frac{\partial}{\partial x^m}} dx^i \right) \left(\frac{\partial}{\partial x^j} \right) = -\Gamma^i_{jm}$$

Summary so far:

$$(\nabla_X Y)^i = X(Y^i) \underbrace{+}_{\text{act on } \textbf{vector} \text{ field}} \Gamma^i_{jm} Y^j X^m$$

$$(\nabla_X \omega)_i = X(\omega_i) + -\Gamma^j_{im} \omega_j X^m$$

Note that for the immediately above expression for $(\nabla_X Y)^i$, in the second term on the right hand side, Γ^i_{jm} has the last entry at the bottom, m going in the direction of X, so that it matches up with X^m. This is a good mnemonic to memorize the index positions of Γ.

similarly, by further application of Leibnitz

T a $(1,2)$-TF (tensor field)

$$(\nabla_X T)^i{}_{jk} = X(T^i_{jk}) + \Gamma^i{}_{sm} T^s_{jk} X^m - \Gamma^s{}_{jm} T^i{}_{sk} X^m - \Gamma^s{}_{km} T^i{}_{js} X^m$$

Question: If in a Euclidean space, the Γs all vanish in a (then existing) global chart.

Answer: Yes, but: What is a Euclidean space:
$(M = \mathbb{R}^n, \mathcal{O}_{\mathrm{st}}, \mathcal{A})$ smooth manifold.
Assume $(\mathbb{R}^n, \mathrm{id}_{\mathbb{R}^n}) \in \mathcal{A}$ and

$$(\Gamma^i_{(x)})_{jk} = dx^i \left((\nabla_{\underline{\mathrm{E}}})_{\frac{\partial}{\partial x^k}} \frac{\partial}{\partial x^j} \right) \overset{!}{=} 0$$

Intuition:

$\mathbb{R}^2$: $\nabla_{\mathrm{Euclidean}}$

$\mathbb{R}^2$: $\nabla_{\mathrm{Hyperbolic}}$

Definition 253. X vector field on $(M, \mathcal{O}, \mathcal{A}, \nabla)$.
Then divergence of X is the function:

$$\mathrm{div}(X) := \left(\nabla_{\frac{\partial}{\partial x^i}} X \right)^i$$

Claim: chart-independent.

3. Change of Γ's under change of chart

(U, x), $(V, y) \in \mathcal{A}$ and $U \cap V \neq \emptyset$

$$\Gamma^i_{jk}(y) := dy^i \left(\nabla_{\frac{\partial}{\partial y^k}} \frac{\partial}{\partial y^j} \right) = \frac{\partial y^i}{\partial x^q} dx^q \left(\nabla_{\frac{\partial x^p}{\partial y^k} \frac{\partial}{\partial x^p}} \frac{\partial x^s}{\partial y^j} \frac{\partial}{\partial x^s} \right)$$

Note ∇_{fX} is C^∞-linear for fX
covector dy^i is C^∞-linear in its argument

$$\begin{aligned} \Longrightarrow \Gamma^i_{jk}(y) &= \frac{\partial y^i}{\partial x^q} dx^q \left(\frac{\partial x^p}{\partial y^k} \left[\left(\nabla_{\frac{\partial}{\partial x^p}} \frac{\partial x^s}{\partial y^j} \right) \frac{\partial}{\partial x^s} + \frac{\partial x^s}{\partial y^j} \left(\nabla_{\frac{\partial}{\partial x^p}} \frac{\partial}{\partial x^s} \right) \right] \right) = \\ &= \frac{\partial y^i}{\partial x^q} \underbrace{\frac{\partial x^p}{\partial y^k} \frac{\partial}{\partial x^p}}_{\frac{\partial}{\partial y^k}} \frac{\partial x^s}{\partial y^j} \delta^q{}_s + \frac{\partial y^i}{\partial x^q} \frac{\partial x^p}{\partial y^k} \frac{\partial x^s}{\partial y^j} \Gamma^q{}_{sp}(x) \end{aligned}$$

in summary:

$$\Gamma^i_{jk}(y) = \frac{\partial y^i}{\partial x^q} \frac{\partial^2 x^q}{\partial y^j \partial y^k} + \frac{\partial y^i}{\partial x^q} \frac{\partial x^s}{\partial y^j} \frac{\partial x^p}{\partial y^k} \Gamma^q{}_{sp}(x) \tag{904}$$

Eq. (904) is the change of connection coefficient function under the change of chart $(U \cap V, x) \to (U \cap V, y)$

4. Normal Coordinates

Let $p \in M$ of $(M, \mathcal{O}, \mathcal{A}, \nabla)$

Then one can construct a chart (U, x) with $p \in U$ such that

$$\Gamma(x)^i{}_{(jk)}(p) = 0$$

at the point p. **Not** necessarily in any neighborhood.

Proof. Let (V, y) be any chart (with) $p \in V$.

Thus, in general: $\Gamma(y)^i{}_{jk} \neq 0$

Then consider a new chart (U, x) to which one transits by virtue of

$$(x \circ y^{-1})^i(\alpha^1, \ldots, \alpha^d) := \alpha^i - \frac{1}{2}\Gamma(y)^i{}_{(jk)}(p)\alpha^j\alpha^k$$

$p = x^{-1}(\alpha^1, \ldots, \alpha^d)$

$$\left(\frac{\partial x^i}{\partial y^j}\right)_p = \partial_j(x^i \circ y^{-1}) = \delta^i_j - \Gamma(y)^i{}_{mj}(p)\ \alpha^m|_{\alpha=0} = \delta^i_j$$

$$\frac{\partial x^i}{\partial y^k \partial y^j}(p) = -\Gamma(y)^i{}_{kj}(p)$$

$$\begin{aligned} \Longrightarrow \Gamma(x)^i_{jk}(p) &= \Gamma(y)^i{}_{jk}(p) - \Gamma(y)^i{}_{kj}(p) = 0 \\ &= \Gamma(y)^i_{[jk]}(p) = T(y)^i{}_{jk} \end{aligned}$$

Terminology: (U, x) is called a **normal coordinate chart** of ∇ at $\mathbf{p} \in \mathbf{M}$.

□

Tutorial 7 Connections

Exercise 1.: True or false?

(a)

- $\nabla_{fX}Y = f\nabla_X Y$ by definition so $\nabla_{fX} = f\nabla_X$ i.e. ∇_X is $C^\infty(M)$-linear in X
- $f \in C^\infty(M)$ is a $(0,0)$-tensor field. $\nabla_X f = Xf \equiv X(f)$ by definition.
- If the manifold is flat, I'm assuming that means that the manifold is globally a Euclidean space, and by definition, $\Gamma = 0$.

  $$\nabla_X Y = X^j \frac{\partial}{\partial x^j}(Y^i)\frac{\partial}{\partial x^i} + \Gamma^i_{jk}Y^k X^k \frac{\partial}{\partial x^i} = X^j \frac{\partial Y^i}{\partial x^j}\frac{\partial}{\partial x^i} + 0$$

  and similarly for any (p,q)-tensor field, i.e.

  $$\nabla_X T = X^j \frac{\partial T^{i_1 \ldots i_p}_{j_1 \ldots j_q}}{\partial x^j}$$

- $$\nabla_X f = X^j \frac{\partial f}{\partial x^j} = X \cdot \mathrm{grad}(f)$$

- $\forall\, (U, x) \in \mathcal{A}$, locally (after working out the first few cases, and doing induction, one can look up the expression for the local form; I found it in Nakahara's **Geometry, Topology and Physics**, Eq. 7.26, and it needs to be modified for the convention of order of bottom indices for Γ:

  $$\nabla_\nu t^{\lambda_1 \ldots \lambda_p}_{\mu_1 \ldots \mu_q} = \partial_\nu t^{\lambda_1 \ldots \lambda_p}_{\mu_1 \ldots \mu_q} + \Gamma^{\lambda_1}_{\kappa\nu} t^{\kappa\lambda_2 \ldots \lambda_p}_{\mu_1 \ldots \mu_q} + \cdots + \Gamma^{\lambda_p}_{\kappa\nu} t^{\lambda_1 \ldots \lambda_{p-1}\kappa}_{\mu_1 \ldots \mu_q} - \Gamma^\kappa_{\mu_1\nu} t^{\lambda_1 \ldots \lambda_p}_{\kappa\mu_2 \ldots \mu_q} - \cdots - \Gamma^\kappa_{\mu_q\nu} t^{\lambda_1 \ldots \lambda_p}_{\mu_1 \ldots \mu_{q-1}\kappa}$$

  Clearly, ∇_X is uniquely fixed $\forall\, p \in M$ by choosing each of the $(\dim M)^3$ many connection coefficient functions Γ.

(b)
- $\nabla : \mathfrak{X}(M) \to \mathfrak{X}(M)$
  $\nabla : (p,q)$-tensor field $\mapsto (p,q)$-tensor field
- By definition, ∇ satisfies the Leibniz rule.
-
-
-

Exercise 2.: **Practical rules for how ∇ acts** Torsion-free covariant derivative boils down to a connection coefficient function Γ that is symmetric in the bottom indices.

- $$\nabla_X f = X(f) = X^i \frac{\partial f}{\partial x^i}$$
- $$(\nabla_X Y)^a = X^i \frac{\partial Y^a}{\partial x^i} + \Gamma^a_{jk} Y^j X^k$$
- $$(\nabla_X \omega)_a = X^i \frac{\partial \omega_a}{\partial x^j} - \Gamma^i_{ak} \omega_i X^k$$
- $$(\nabla_m T)^a{}_{bc} = \frac{\partial}{\partial x^m}(T^a{}_{bc}) + \Gamma^a{}_{im} T^i_{bc} - \Gamma^i_{bm} T^a_{ic} - \Gamma^j_{cm} T^a_{bj}$$
- $$(\nabla_{[m} A)_{n]} = (\nabla_m A)_n - (\nabla_n A)_m = \frac{\partial A_n}{\partial x^m} - \Gamma^i_{nm} A_i - \left(\frac{\partial A_m}{\partial x^n} - \Gamma^i_{mn} A_i \right) = \frac{\partial A_m}{\partial x^m} - \frac{\partial A_m}{\partial x^n}$$
- $$(\nabla_m \omega)_{nr} = \frac{\partial \omega_{nr}}{\partial x^m} - \Gamma^i_{nm} \omega_{ir} - \Gamma^i_{rm} \omega_{ni}$$

Exercise 3.: Connection coefficients

Question .

The connection coefficient functions Γ in chart $(U \cap V, y)$ is given, in terms of chart $(U \cap V, x)$ as follows: Recall Eq. (904)

$$\Gamma^i_{jk}(y) = \frac{\partial y^i}{\partial x^q} \frac{\partial^2 x^q}{\partial y^j \partial y^k} + \frac{\partial y^i}{\partial x^q} \frac{\partial x^s}{\partial y^j} \frac{\partial x^p}{\partial y^k} \Gamma^q_{sp}(x)$$

Lecture 8: Parallel Transport & Curvature (International Winter School on Gravity and Light 2015)

Experiment

$\nabla_{v_\gamma} Y \stackrel{!}{=} 0$
(please move in this fashion)

round sphere
$(S^2, \mathcal{O}, \mathcal{A}, \nabla)$

ellipsoid
$(S^2, \mathcal{O}, \mathcal{A}, \widetilde{\nabla})$

ellisoid embedded in $\mathbb{R}^3$ with $\mathbb{R}^3$ Euclidean connection
$(\mathbb{R}^3, \nabla_E)$

1. Parallelity of vector fields

$(M, \mathcal{O}, \mathcal{A}, \nabla)$ manifold with connection.

Definition 254 (parallely transported, parallel)**.** (1) A vector field X on M is said to be **parallely transported** along a smooth curve $\gamma : \mathbb{R} \to M$
if

$$\boxed{\nabla_{v_\gamma} X = 0}$$

i.e.

$$\left(\nabla_{v_{\gamma,\gamma(\lambda)}} X\right)_{\gamma(\lambda)} = 0$$

(2) A slightly weaker condition than "parallely transported" is "parallel":

$$(\nabla_{v_{\gamma,\gamma(\lambda)}} X)_{\gamma(\lambda)} = \mu(\lambda) X_{\gamma(\lambda)}$$

for $\mu : \mathbb{R} \to \mathbb{R}$

Example: *Euclidean plane* $(\mathbb{R}^2, \mathcal{O}, \mathcal{A}, \nabla_E)$,

parallely transported X along γ, parallel X along γ, not parallel, not parallely transported.

2. Autoparallely transported curves

Thought bubble: self- is "Auto"

Child: "How do I get to the toy store." Adult: "Just follow your nose."

(Move in direction of nose means move along curve whose tangent vector is your nose. Means move in the straightest line. We are not talking about shortest line because we do not have a notion of distance.)

Definition 255. A (smooth) curve $\gamma : \mathbb{R} \to M$ is called
autoparallely transported if

$$\nabla_{v_\gamma} v_\gamma = 0 \tag{905}$$

$$\iff \left(\nabla_{v_{\gamma,\gamma(\lambda)}} v_\gamma\right)_{\gamma(\lambda)} = 0$$

Remark: Define weaker notion of autoparallel curve
Thought bubble: !

$$\nabla_{v_\gamma} v_\gamma = \mu v_\gamma$$

Example. Euclidean plane $(\mathbb{R}^2, \mathcal{O}, \mathcal{A}, \nabla_E)$

autoparallely transported (uniform straight motion, Newton's First law), autoparallel (straight motion) but not autoparallely transported,

3. Autoparallel equation

Autoparallely transported curve γ
Consider that portion of the curve that lies in U, where $(U, x) \in \mathcal{A}$.

$$\nabla_{v_\gamma} v_\gamma = 0$$

Express $\nabla_{v_\gamma} v_\gamma = 0$ in terms of chart representation.

$$(\nabla_{v_\gamma} v_\gamma) = \left(\nabla_{\dot{\gamma}^m(x)\left(\frac{\partial}{\partial x^m}\right)_{\gamma(\lambda)}} \dot{\gamma}^n(x) \frac{\partial}{\partial x^n}\right)$$

where

$$v_{\gamma,\gamma(\lambda)} = \dot{\gamma}^m(x) \cdot \left(\frac{\partial}{\partial x^m}\right)$$

$$\gamma^m(x) := x^m \circ \gamma$$

$$\Longrightarrow (\nabla_{v_\gamma} v_\gamma) =$$
$$= \underbrace{\dot\gamma^m \frac{\partial \dot\gamma^q}{\partial x^m}}_{\ddot\gamma(x)^m} \cdot \frac{\partial}{\partial x^q} + \dot\gamma^m \dot\gamma^n \Gamma^q{}_{nm} \frac{\partial}{\partial x^q}$$

20200306 EY: For further clarification,

$$\dot\gamma(t) = v(\gamma(t)) \in T_{\gamma(t)} U$$
$$\dot\gamma(t) = v^i(\gamma^1(t), \ldots, \gamma^n(t))$$
$$\frac{d}{dt}\dot\gamma^i(t) = \frac{d}{dt} v^i(\gamma^1(t), \ldots, \gamma^n(t)) = \frac{d}{dt} v^i(x^1, \ldots, x^n)$$
$$= \frac{\partial v^i}{\partial x^j}(x)\dot x^j = \dot\gamma^j \frac{\partial \dot\gamma^i}{\partial x^j}$$

in summary:

$$\boxed{\ddot\gamma^m_{(x)}(\lambda) + (\Gamma^m_{(x)})_{ab}(\gamma(\lambda))\dot\gamma^a_{(x)}(\lambda)\dot\gamma^b_{(x)}(\lambda) = 0} \tag{906}$$

chart expression of the condition that γ be autoparallely transported.

Examples

(a) Euclidean plane, $U = \mathbb{R}^2$, $x = \mathrm{id}_{\mathbb{R}}$, $\Gamma(x)^i{}_{jk} = 0$

$$\Longrightarrow \ddot\gamma(x)^m = 0 \Longrightarrow \gamma(x)^m(\lambda) = a^m\lambda + b^m a, b \in \mathbb{R}^d$$

(b) round sphere $(S^2, \mathcal{O}, \mathcal{A}, \nabla_{\mathrm{round}})$
Consider a chart:

$$x(p) = (\theta, \varphi)$$
$$\theta \in (0, \pi)$$
$$\varphi \in (0, 2\pi)$$

all other ($\Gamma^1{}_{11} = 0$) vanish.

$$\Gamma(x)^1{}_{22}(x^{-1}(\theta, p)) := -\sin\theta\cos\theta \quad \Gamma(x)^2{}_{21} = \Gamma(x)^2{}_{12} := \cot\theta$$

Sloppy notation (classical mechanics)

$$x^1(p) = \theta(p)$$
$$x^2(p) = \varphi(p)$$

autoparallel equation

$$\ddot\theta + \Gamma^1{}_{22}\dot\varphi\dot\varphi = 0$$
$$\ddot\varphi + 2\Gamma^2{}_{12}\dot\theta\dot\varphi = 0$$
$$\Longleftrightarrow \boxed{\begin{aligned}\ddot\theta - \sin(\theta)\cos(\theta)\dot\varphi\dot\varphi = 0\\ \ddot\varphi + 2\cot(\theta)\dot\theta\dot\varphi = 0\end{aligned}}$$

Solve: e.g. $\theta(\lambda) = \dfrac{\pi}{2} \quad \varphi(\lambda) = \underline{\omega}\cdot\lambda + \varphi_0$
Similarly,
∇_{potato}, $(S^2, \mathcal{O}, \mathcal{A})$

3. Torsion

Q: Can one use ∇ to define tensors on $(M, \mathcal{O}, \mathcal{A}, \nabla)$?

Definition 256. The **torsion** of a connection ∇ is the $(1,2)$-**tensor field**

$$T(\omega, X, Y) := \omega(\nabla_X Y - \nabla_Y X - [X, Y]) \tag{907}$$

(Inside a cloud)
$[X, Y]$ vector field defined by

$$[X, Y]f := X(Yf) - Y(Xf)$$

Proof. check T is C^∞-linear in each entry

$$T(f \cdot \omega, X, Y) = f \cdot \omega(\dots) = fT(\omega, X, Y)$$
$$T(\omega + \psi, X, Y) = \dots = T(\omega, X, Y) + T(\psi, X, Y)$$
$$T(\omega, fX, Y) = \omega(\nabla_{fX} Y - \nabla_Y(fX) - [fX, Y]) =$$
$$= \omega(f\nabla_X Y - (Yf)X - f\nabla_Y X - f[X, Y] + (Yf)X) = fT(\omega, X, Y)$$

since

$$[fX, Y]g = fX(Yg) - Y(fXg) =$$
$$= fX(Yg) - (Yf)(Xg) - fY(Xg) =$$
$$= f \cdot [X, Y]g - (Yf)Xg =$$
$$= (f \cdot [X, Y] - (Yf)X)g$$

$$T(\omega, X, Y) = -T(\omega, Y, X) \quad \checkmark$$

□

Definition 257. A $(M, \mathcal{O}, \mathcal{A}, \nabla)$ is called torsion-free if $T = 0$

In a chart

$$T^i{}_{ab} := T\left(dx^i, \frac{\partial}{\partial x^a}, \frac{\partial}{\partial x^b}\right) = dx^i(\dots)$$
$$= \Gamma^i{}_{ab} - \Gamma^i{}_{ba} = 2\Gamma^i{}_{[ab]}$$

From now on, in these lectures, we only use torsion-free connections.

4. Curvature

Definition 258. The **Riemann curvature** of a connection ∇ is the $(1,3)$-tensor field

$$\mathrm{Riem}(\omega, Z, X, Y) := \omega(\nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z) \tag{908}$$

Proof. do it: C^∞-linear in each slot. □

Tutorials $\mathrm{Riem}^i{}_{jab} = \dots$

Algebraic relevance of Riem

$$(\nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z) = R(\cdot, Z, X, Y) + \nabla_{[X,Y]} Z$$

In one chart (U, x)

$$(\underbrace{\nabla_a}_{=\nabla_{\frac{\partial}{\partial x^a}}} \nabla_b Z - \nabla_b \nabla_a Z)^m = \mathrm{Riem}^m{}_{nab} Z^n + \nabla_{\underbrace{\left[\frac{\partial}{\partial x^a}, \frac{\partial}{\partial x^b}\right]}_{=0}} Z$$

$$\boxed{(\nabla_a \nabla_b Z)^m - (\nabla_b \nabla_a Z)^m = \mathrm{Riem}^m{}_{nab} Z^n}$$

Geometric significance

Idea:
$[X, Y] = 0$

Figure 1: Closed Riem

If $[X, Y] \neq 0$

$$(\delta Z)^m = \cdots = R^m{}_{nab} X^a Y^b Z^n \delta s \delta t + \mathcal{O}(\delta s^2 \delta t, \delta t^2 \delta s)$$

Holonomy

Tutorial 8 Parallel transport & Curvature

Exercise 1.

Exercise 2.: Where connection coefficients appear

It was suggested in the tutorial sheets and hinted in the lecture that the following should be committed to memory.

Question : Recall the autoparallel equation for a curve γ.

(a)

$$\nabla_{v_\gamma} v_\gamma = 0$$

(b)

$$\nabla_{v_\gamma} v_\gamma = \nabla_{\dot\gamma \frac{\partial}{\partial x^\mu}} v_\gamma = \dot\gamma^\nu \nabla_{\partial_\nu} v_\gamma = \dot\gamma^\nu \left[\frac{\partial v^\mu_\gamma}{\partial x^\nu} + \Gamma^\rho_{\mu\nu} v^\mu_\gamma \right] \frac{\partial}{\partial x^\rho} = \dot\gamma^\nu \left[\frac{\partial \dot\gamma^\rho}{\partial x^\nu} + \Gamma^\rho_{\mu\nu} \dot\gamma^\mu \right] \frac{\partial}{\partial x^\rho} = 0$$

$$\implies \boxed{\ddot\gamma^\rho + \Gamma^\rho_{\mu\nu} \dot\gamma^\mu \dot\gamma^\nu}$$

as, for example, for $F(x(t))$,

$$\frac{dF(x(t))}{dt} = \dot x \frac{\partial F}{\partial x} = \frac{d}{dt} F$$

so that

$$\dot\gamma^\nu \frac{\partial v^\mu_\gamma}{\partial x^\nu} = \frac{d}{d\lambda} v^\mu_\gamma = \frac{d^2}{d\lambda^2} \gamma^\mu$$

Question : Determine the coefficients of the Riemann tensor with respect to a chart (U, x).

Recall this manifestly covariant definition

$$\mathrm{Riem}(\omega, Z, X, Y) = \omega(\nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z)$$

We want $R^i{}_{jab}$.
now

$$\nabla_X \nabla_Y Z = \nabla_X ((Y^\mu \frac{\partial}{\partial x^\mu} Z^\rho + \Gamma^\rho_{\mu\nu} Z^\mu Y^\nu) \frac{\partial}{\partial x^\rho}) = (X^\alpha \frac{\partial}{\partial x^\alpha} (Y^\mu \frac{\partial}{\partial x^\mu} Z^\rho + \Gamma^\rho_{\mu\nu} Z^\mu Y^\nu) + \Gamma^\rho_{\alpha\beta} (Y^\mu \frac{\partial}{\partial x^\mu} Z^\alpha + \Gamma^\alpha_{\mu\nu} Z^\mu Y^\nu) X^\beta) \frac{\partial}{\partial x^\rho}$$

For $X = \partial_a$, $Y = \partial_b$, $Z = \partial_j$, then the partial derivatives of the coefficients of the input vectors become zero.

$$\implies \nabla_{\partial_a}\nabla_{\partial_b}\partial_j = \frac{\partial}{\partial x^a}(\Gamma^i_{jb}) + \Gamma^i_{\alpha a}\Gamma^\alpha_{jb}$$

Now

$$[X,Y]^i = X^j\frac{\partial}{\partial x^j}Y^i - Y^j\frac{\partial X^i}{\partial x^j}$$

For coordinate vectors, $[\partial_i, \partial_j] = 0\ \forall\, i,j = 0, 1\ldots d$.

Thus

$$\boxed{R^i{}_{jab} = \frac{\partial}{\partial x^a}\Gamma^i_{jb} - \frac{\partial}{\partial x^b}\Gamma^i_{ja} + \Gamma^i_{\alpha a}\Gamma^\alpha_{jb} - \Gamma^i_{\alpha b}\Gamma^\alpha_{ja}}$$

Question :Ric$(X,Y) :=$ Riem$^m_{amb}X^aY^b$ define $(0,2)$-tensor?.

Yes, transforms as such:

EY developments

I roughly follow the spirit in Theodore Frankel's **The Geometry of Physics: An Introduction** Second Ed. 2003, Chapter 9 Covariant Differentiation and Curvature, Section 9.3b. The Covariant Differential of a Vector Field. P.S. EY : 20150320 I would like a copy of the Third Edition but I don't have the funds right now to purchase the third edition: go to my tilt crowdfunding campaign, http://ernestyalumni.tilt.com, and help with your financial support if you can or send me a message on my various channels and ernestyalumni gmail email address if you could help me get a hold of a digital or hard copy as a pro bono gift from the publisher or author. EY : 20221220 Update: I'm gainfully employed now and I've obtained a digital copy of the 3rd Ed. of Frankel! I don't have the tilt campaign, but if you'd like to donate still, PayPal me @ernestyalumni or better yet Venmo me @ernestyalumni

The spirit of the development is the following:

> "How can we express connections and curvatures in terms of forms?" -Theodore Frankel.

From Lecture 7, connection ∇ on vector field Y, in the "direction" X,

$$\nabla_{\frac{\partial}{\partial x^k}}Y = \left(\frac{\partial Y^i}{\partial x^k} + \Gamma^i_{jk}Y^j\right)\frac{\partial}{\partial x^i}$$

Make the ansatz (approche, impostazione) that the connection ∇ acts on Y, the vector field, first:

$$\nabla Y(X) = \left(X^k\frac{\partial Y^i}{\partial x^k} + \Gamma^i_{jk}Y^jX^k\right)\frac{\partial}{\partial x^i} = X^k\left(\nabla_{\frac{\partial}{\partial x^k}}Y\right)^i\frac{\partial}{\partial x^i} = (\nabla_X Y)^i\frac{\partial}{\partial x^i} = \nabla_X Y$$

Now from Lecture 7, Definition for Γ,

$$dx^i\left(\nabla_{\frac{\partial}{\partial x^k}}\frac{\partial}{\partial x^j}\right) = \Gamma^i_{jk}$$

Make this ansatz (approche, impostazine)

$$\nabla\frac{\partial}{\partial x^j} = \left(\Gamma^i_{jk}dx^k\right)\otimes\frac{\partial}{\partial x^i} \in \Omega^1(M, TM) = T^*M\otimes TM$$

where $\Omega^1(M,TM) = T^*M\otimes TM$ is the set of all TM or vector-valued 1-forms on M, with the 1-form being the following:

$$\Gamma^i_{jk}dx^k = \Gamma^i{}_j \in \Omega^1(M) \qquad \begin{matrix} i = 1\ldots\dim(M) \\ j = 1\ldots\dim(M)\end{matrix}$$

So $\Gamma^i{}_j$ is a $\dim M \times \dim M$ matrix of 1-forms (EY !!!).

Thus

$$\nabla Y = (d(Y^i) + \Gamma^i_j Y^j) \otimes \frac{\partial}{\partial x^i}$$

So the connection is a (smooth) map from TM to the set of all vector-valued 1-forms on M, $\Omega^1(M, TM)$, and then, after "eating" a vector Y, yields the "covariant derivative":

$$\nabla : TM \to \Omega^1(M, TM) = T^*M \otimes TM$$
$$\nabla : Y \mapsto \nabla Y$$
$$\nabla Y : TM \to TM$$
$$\nabla Y(X) \mapsto \nabla Y(X) = \nabla_X(Y)$$

Now

$$\left[\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}\right] f = \frac{\partial}{\partial x^i}\left(\frac{\partial}{\partial x^j}\right) - \frac{\partial}{\partial x^j}\left(\frac{\partial}{\partial x^i}\right) = 0$$

(this is okay as on $p \in (U, x)$; x-coordinates on same chart (U, x))

EY : 20150320 My question is when is this nontrivial or nonvanishing (i.e. not equal to 0).

$$[e_a, e_b] = ?$$

for a frame (e_c) and would this be the difference between a tangent bundle TM vs. a (general) vector bundle?

Wikipedia helps here. cf. wikipedia, "Connection (vector bundle)"

$$\nabla : \Gamma(E) \to \Gamma(T^*M \otimes E) = \Omega^1(M, E)$$
$$\nabla e_a = \omega^c_{ab} f^b \otimes e_c$$

$f^b \in T^*M$ (this is the dual basis for TM and, note, this is for the manifold, M

$$\nabla_{f_b} e_a = \omega^c_{ab} e_c \in E$$
$$\omega^c_a = \omega^c_{ab} f^b \in \Omega^1(M)$$

is the connection 1-form, with $a, c = 1 \ldots \dim V$. EY : 20150320 This V is a vector space living on each of the fibers of E. I know that $\Gamma(T^*M \otimes E)$ looks like it should take values in E, but it's meaning that it takes vector values of V. Correct me if I'm wrong: ernestyalumni at gmail and various social media.

Let $\sigma \in \Gamma(E)$, $\sigma = \sigma^a e_a$

$$\nabla\sigma = (d\sigma^c + \omega^c_{ab}\sigma^a f^b) \otimes e_c \text{ with}$$
$$d\sigma^c = \frac{\partial \sigma^c}{\partial x^b} f^b$$
$$\implies \nabla_X \sigma = \left(X^b \frac{\partial \sigma^c}{\partial x^b} + \omega^c_{ab}\sigma^a X^b\right) e_c = X^b \left(\frac{\partial \sigma^c}{\partial x^b} + \omega^c_{ab}\sigma^a\right) e_c$$

Lecture 9: Newtonian spacetime is curved!

Axiom 2 (Newton I:)**.** *A body on which* no *force acts moves uniformly along a straight line*

Axiom 3 (Newton II:)**.** *Deviation of a body's motion from such uniform straight motion is effected by a force, reduced by a factor of the body's reciprocal mass.*

Remark:

(1) 1st axiom - in order to be relevant - must be read as a measurement prescription for the geometry of space ...

(2) Since gravity universally acts on every particle, in a universe with at least two particles, gravity must not be considered a force if Newton I is supposed to remain applicable.

167.11 Laplace's questions

Laplace $*1749$
$\dagger 1827$

Q: "Can gravity be encoded in a curvature of space, such that its effects show if particles under the influence of (no other) force we postulated to move along straight lines in this curved space?"

Answer: No!

Proof. gravity is a force point of view

$$m\ddot{x}^\alpha(t) = F^\alpha(x(t))$$
$$m\ddot{x}^\alpha(t) = \underbrace{mf^\alpha}_{F^\alpha}(x(t))$$

$-\partial_\alpha f^\alpha = 4\pi G\rho$ (Poisson)
ρ mass density of matter

$$m\ddot{x}^\alpha(t) = \underbrace{mf^\alpha}_{F^\alpha}(x(t))$$

True?

(EY : 20150330) You know this, $F = Gm_1m_2/r^2$

Yes

weak equivalence principle

$$\ddot{x}^\alpha(t) - f^\alpha(x(t)) = 0$$

Laplace asks: Is this $(\ddot{x}(t))$ of the form

$$\ddot{x}^\alpha(t) + \Gamma^\alpha_{\beta\gamma}(x(t))\dot{x}^\beta(t)\dot{x}^\gamma(t) = 0$$

Conclusion: One cannot find Γ s such that Newton's equation takes the form of an autoparallel.

□

Question (from audience) We can evaluate the autoparallel equation pointwise?! But at each point, we can set the Gammas to zero?!

Then, one should be able to write Newton's second law in the usual form?

Prof. Schuller: you (observer) fall with the mass (i.e. accelerated reference frame) and so you transform Γ's to be 0. The problem with this is if you do the same experiment in the North pole and fall with the body. If someone else at the South Pole does the same experiment at the same time, with that same transformation (of reference frames), the effect of gravity cannot be transformed out.

In a homogeneous gravitational field, you can possibly transform away gravity, $\Gamma = 0$. But in an inhomogeneous gravitational field, no.

2. The full wisdom of Newton I

use also the information from Newton's first law that particles (no force) move uniformly

introduce the appropriate setting to talk about the difference easily

insight: in spacetime | uniform & straight motion in space | is simply **straight** motion

So let's try in spacetime:

let $x : \mathbb{R} \to \mathbb{R}^3$

be a particle's trajectory in space $\longleftrightarrow$ worldline (history) of the particle

$$X : \mathbb{R} \to \mathbb{R}^4$$
$$t \mapsto (t, x^1(t), x^2(t), x^3(t)) :=$$
$$:= (X^0(t), X^1(t), X^2(t), X^3(t))$$

That's all it takes:

Trivial rewritings:

$$\dot{X}^0 = 1$$

$$\Longrightarrow \boxed{\begin{aligned} \ddot{X}^0 &= 0 \\ \ddot{X}^\alpha - f^\alpha(X(t)) \cdot \dot{X}^0 \cdot \dot{X}^0 &= 0 \end{aligned}} \quad (\alpha = 1,2,3) \Longrightarrow \quad \begin{matrix} a = 0,1,2,3 \\ \boxed{\ddot{X}^a + \Gamma^a_{\ bc} \dot{X}^b \dot{X}^c = 0} \\ \text{antoparallel eqn in spacetime} \end{matrix}$$

Yes, choosing $\Gamma^0_{\ ab} = 0$

$$\Gamma^\alpha_{\ \beta\gamma} = 0 = \Gamma^\alpha_{\ 0\beta} = \Gamma^\alpha_{\ \beta 0}$$

only: $\boxed{\Gamma^\alpha_{\ 00} \overset{!}{=} -f^\alpha}$

Question: Is this a coordinate-choice artifact?

No, since $R^\alpha_{\ 0\beta 0} = -\frac{\partial}{\partial x^\beta} f^\alpha$ (only non-vanishing components) (tidal force tensor, $-$ the Hessian of the force component)

Ricci tensor $\Longrightarrow R_{00} = R^m_{\ 0m0} = -\partial_\alpha f^\alpha = 4\pi G\rho$

Poisson: $-\partial_\alpha f^\alpha = 4\pi G \cdot \rho$

writing: $T_{00} = \frac{1}{2}\rho$

$$\Longrightarrow \boxed{R_{00} = 8\pi G T_{00}}$$

Einstein in 1912 $\boxed{\cancel{R_{ab} = 8\pi G T_{ab}}}$

Conclusion: Laplace's idea works in spacetime

Remark

$$\Gamma^\alpha_{\ 00} = -f^\alpha$$
$$R^\alpha_{\ \beta\gamma\delta} = 0 \qquad \alpha,\beta,\gamma,\delta = 1,2,3$$
$$\boxed{R_{00} = 4\pi G\rho}$$

Q: What about transformation behavior of LHS of

$$\underbrace{\underbrace{\ddot{x}^a + \Gamma^a_{\ bc}\dot{X}^b\dot{X}^c}_{(\nabla_{v_X} v_X)^a}}_{:= a^a \text{ "acceleration vector"}} = 0$$

3. The foundations of the geometric formulation of Newton's axiom

new start

Definition 259. A **Newtonian spacetime** is a quintuple

$$(M, \mathcal{O}, \mathcal{A}, \nabla, t)$$

where $(M, \mathcal{O}, \mathcal{A})$ 4-dim. smooth manifold

$$t : M \to \mathbb{R} \text{ smooth function}$$

(i) "There is an absolute space"

$$(dt)_p \neq 0 \qquad \forall\, p \in M$$

(ii) "absolute time flows uniformly"

$$\nabla dt \underbrace{=}_{\text{space of } (0,2)\text{-tensor fields}} 0 \qquad \text{everywhere}$$

∇dt is a $(0,2)$-tensor field

(iii) add to axioms of Newtonian spacetime $\nabla = 0$ torsion free

Definition 260. absolute space at time τ

$$S_\tau := \{p \in M | t(p) = \tau\}$$

$$\xrightarrow{dt \neq 0} M = \coprod S_\tau$$

Definition 261. A vector $X \in T_pM$ is called

(a) future-directed if

$$dt(X) > 0$$

(b) spatial if

$$dt(X) = 0$$

(c) past-directed if

$$dt(X) < 0$$

picture

Newton I: The worldline of a particle under the influence of no force (gravity isn't one, anyway) is a future-directed autoparallel i.e.

$$\nabla_{v_X} v_X = 0$$
$$dt(v_X) > 0$$

and (iii) ∇ is torsion-free.

Newton II:

$$\nabla_{v_X} v_X = \frac{F}{m} \iff m \cdot \mathfrak{a} = F$$

where F is a spatial vector field:

$$dt(F) = 0$$

Convention: restrict attention to atlases $\mathcal{A}_{\text{stratefied}}$ whose charts $(\mathcal{U}, x)$ have the property

$$\begin{array}{l} x^0 : \mathcal{U} \to \mathbb{R} \\ x^1 : \mathcal{U} \to \mathbb{R} \\ \vdots \quad \vdots \\ x^3 \end{array} \qquad x^0 = t|_{\mathcal{U}} \qquad \Longrightarrow \begin{array}{l} 0 \overset{\text{"absolute time flows uniformly"}}{=} \nabla dt \\ 0 = \nabla_{\frac{\partial}{\partial x^a}} dx^0 = -\Gamma^0{}_{ba} \qquad a = 0, 1, 2, 3 \end{array}$$

Let's evaluate in a chart $(\mathcal{U}, x)$ of a stratified atlas $\mathcal{A}_{\text{sheet}}$: Newton II:

$$\nabla_{v_X} v_X = \frac{F}{m}$$

in a chart.

$$(X^0)'' + \cancel{\Gamma^0_{cd}(X^a)'(X^b)'}^{\text{stratified atlas}} = 0$$
$$(X^\alpha)'' + \Gamma^\alpha_{\gamma\delta} X^{\gamma'} X^{\delta'} + \Gamma^\alpha_{00} X^{0'} X^{0'} + 2\Gamma^\alpha_{\gamma 0} X^{\gamma'} X^{0'} = \frac{F^\alpha}{m} \qquad \alpha = 1, 2, 3$$

EY: 20160623: where the factor of 2 comes from torsion free ∇

$$\Longrightarrow (X^0)''(\lambda) = 0 \Longrightarrow X^0(\lambda) = a\lambda + b \quad \text{constants } a, b \text{ with}$$
$$X^0(\lambda) = (x^0 \circ X)(\lambda) \overset{\text{stratified}}{=} (t \circ X)(\lambda)$$

convention parametrize worldline by absolute time

$$\frac{d}{d\lambda} = a \frac{d}{dt}$$

$$a^2\ddot{X}^\alpha + a^2\Gamma^\alpha_{\gamma\delta}\dot{X}^\gamma\dot{X}^\delta + a^2\Gamma^\alpha_{00}\dot{X}^0\dot{X}^0 + 2a^2\Gamma^\alpha_{\gamma 0}\dot{X}^\gamma\dot{X}^0 = \frac{F^\alpha}{m}$$
$$\Longrightarrow \underbrace{\ddot{X}^\alpha + \Gamma^\alpha_{\gamma\delta}\dot{X}^\gamma\dot{X}^\delta + \Gamma^\alpha_{00}\dot{X}^0\dot{X}^0 + 2\Gamma^\alpha_{\gamma 0}\dot{X}^\gamma\dot{X}^0}_{\mathfrak{a}^\alpha} = \frac{1}{a^2}\frac{F^\alpha}{m}$$

Student question: How can you get the Lagrange equations in this formalism?
Schuller says: In *all* systems they take the form like this:

$$L = \underbrace{T}_{\frac{m}{2}g(v_x,v_x)} + V$$

where v_x is velocity to the spatial curves x.
Schuller says: $\frac{m}{2}\sqrt{g(v_x,v_x)}$, parameter invariance, you get straight curves; if you take $\sqrt{}$ away, $\frac{m}{2}g(v_x,v_x)$ you get straight, *uniform* curves.
Schuller says: Only in inertial systems, which is defined as:

$$\boxed{\nabla_{\frac{\partial}{\partial x^a}}\frac{\partial}{\partial x^b} = 0}$$

inertial frame, non-accelerating, non-rotating.
Schuller says: Apart from g, a 2-tensor, you need to know ω.
Schuller references 2nd. year Mechanik in German: https://www.video.uni-erlangen.de/course/id/272

168 Lecture 10: Metric Manifolds

cf. Lecture 10: Metric Manifolds (International Winter School on Gravity and Light 2015)
We establish a structure on a smooth manifold that allows one to assign vectors in each tangent space a length (and an angle between vectors in the same tangent space).
From this structure, one can then define a notion of length of a curve.
Then we can look at shortest curves.
Requiring then that the shortest curves coincide with the straightest curves (wrt ∇) will result in ∇ being determined by the metric structure.

$$g \overset{\text{straight=short}}{\underset{T=0}{\rightsquigarrow}} \nabla \rightsquigarrow \text{Riem}$$

168.1 Metrics

Definition 262. A metric g on a smooth manifold $(M,\mathcal{O},\mathcal{A})$ is a $(0,2)$-tensor field satisfying

(i) symmetry $g(X,Y) = g(Y,X) \quad \forall X,Y$ vector fields

(ii) non-degeneracy: the musical map

$$\text{“flat”}\ \flat : \Gamma(TM) \to \Gamma(T^*M)$$
$$X \mapsto \flat(X)$$

where $\flat(X)(Y) := g(X,Y)$
$\flat(X) \in \Gamma(T^*M)$
In thought bubble: $\flat(X) = g(X,\cdot)$
. . . is a C^∞-isomorphism in other words, it is invertible.

Remark: $(\flat(X))_a$ or
X_a
$(\flat(X))_a := g_{am}X^m$
Thought bubble: $\flat^{-1} = \sharp$
$\flat^{-1}(\omega)^a := g^{am}\omega_m$
$\flat^{-1}(\omega)^a := (g^{\text{“}-1\text{”}})^{am}\omega_m \Longrightarrow$ not needed. (all of this is not needed)

Definition 263. The $(2,0)$-tensor field $g^{\text{“}-1''}$ with respect to a metric g is the symmetric

$$g^{\text{“}-1''} : \Gamma(T^*M) \times \Gamma(T^*M) \to C^\infty(M)$$
$$(\omega, \sigma) \mapsto \omega(\flat^{-1}(\sigma)) \qquad \flat^{-1}(\sigma) \in \Gamma(TM))$$

chart: $g_{ab} = g_{ba}$
$(g^{-1})^{am} g_{mb} = \delta^a_b$

Example: $(S^2, \mathcal{O}, \mathcal{A})$
chart $(\mathcal{U}, x)$
$\varphi \in (0, 2\pi)$
$\theta \in (0, \pi)$
define the metric

$$g_{ij}(x^{-1}(\theta, \varphi)) = \begin{bmatrix} R^2 & 0 \\ 0 & R^2 \sin^2\theta \end{bmatrix}_{ij}$$

$R \in \mathbb{R}^+$
"the metric of the round sphere of radius R"

168.2 Signature

$$A^a{}_m v^m = \lambda v^a \qquad \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$$

Linear algebra:

$$g_{am} v^m = \lambda \cdot v^a ? \rightsquigarrow \qquad \begin{pmatrix} 1 & & & & & & & \\ & \ddots & & & & & & \\ & & 1 & & & & & \\ & & & -1 & & & & \\ & & & & \ddots & & & \\ & & & & & -1 & & \\ & & & & & & 0 & \\ & & & & & & & \ddots \\ & & & & & & & & 0 \end{pmatrix}$$

$(1,1)$ tensor has eigenvalues
$(0,2)$ has signature (p,q) (well-defined)

$$\left.\begin{array}{l}(+++)\\(++-)\\(+--)\\(---)\end{array}\right\} d+1 \text{ if } p+q = \dim V$$

Definition 264. A metric is called
Riemannian if its signature is $(+ + \cdots +)$
Lorentzian if $(+ - \cdots -)$

168.3 Length of a curve

Let γ be a smooth curve.
Then we know its velocity $v_{\gamma,\gamma(\lambda)}$ at each $\gamma(\lambda) \in M$.

Definition 265. On a Riemannian metric manifold $M, \mathcal{O}, \mathcal{A}, g)$, the **speed** of a curve at $\gamma(\lambda)$ is the number

$$(\sqrt{g(v_\gamma, v_\gamma)})_{\gamma(\lambda)} = s(\lambda)$$

F. Schuller: "I feel the need for speed." -Top Gun.
(I feel the need for speed, then I feel the need for a metric)
Aside: $[v^a] = \frac{1}{T}$
$[g_{ab}] = L^2$
$[\sqrt{g_{ab}v^a v^b}] = \sqrt{\frac{L^2}{T^2}} = \frac{L}{T}$

Definition 266. Let $\gamma : (0,1) \to M$ a smooth curve.
Then the **length of** γ is the number

$$\mathbb{R} \ni L[\gamma] := \int_0^1 d\lambda s(\lambda) = \int_0^1 d\lambda \sqrt{(g(v_\gamma, v_\gamma))_{\gamma(\lambda)}}$$

F. Schuller: "velocity is more fundamental than speed, speed is more fundamental than length"
Example: reconsider the round sphere of radius R
Consider its equator:
$\theta(\lambda) := (x^1 \circ \gamma)(\lambda) = \frac{\pi}{2}$
$\varphi(\lambda) := (x^2 \circ \gamma)(\lambda) = 2\pi\lambda^3$
$\theta'(\lambda) = 0$
$\varphi'(\lambda) = 6\pi\lambda^2$
on the same chart $g_{ij} = \begin{bmatrix} R^2 & \\ & R^2 \sin^2\theta \end{bmatrix}$
F.Schuller: do everything in this chart

$$L[\gamma] = \int_0^1 d\lambda \sqrt{g_{ij}(x^{-1}(\theta(\lambda), \varphi(\lambda)))(x^i \circ \gamma)'(\lambda)(x^j \circ \gamma)'(\lambda)} = \int_0^1 d\lambda \sqrt{R^2 \cdot 0 + R^2 \sin^2(\theta(\lambda)) 36\pi^2\lambda^4} =$$

$$= 6\pi R \int_0^1 d\lambda \lambda^2 = 6\pi R [\frac{1}{3}\lambda^3]_0^1 = 2\pi R$$

Theorem 82. $\gamma : (0,1) \to M$ *and*
$\sigma : (0,1) \to (0,1)$ *smooth bijective and increasing "reparametrization"*

$$L[\gamma] = L[\gamma \circ \sigma]$$

Proof. $\Longrightarrow$ Tutorials □

168.4 Geodesics

Definition 267. A curve $\gamma : (0,1) \to M$ is called a **geodesic** on a Riemannian manifold $(M, \mathcal{O}, \mathcal{A}, g)$ if its a *stationary* curve with respect to a length functional L.

Thought bubble: in classical mechanics, deform the curve a little, ϵ times this deformation, to first order, it agrees with $L[\gamma]$

Theorem 83. γ *geodesic iff it satisfies the Euler-Lagrange equations for the Lagrangian*

$$\mathcal{L} : TM \to \mathbb{R}$$
$$X \mapsto \sqrt{g(X,X)}$$

In a chart, the Euler Lagrange equations take the form:

$$\left(\frac{\partial \mathcal{L}}{\partial \dot{x}^m}\right)^{\cdot} - \frac{\partial \mathcal{L}}{\partial x^m} = 0$$

F.Schuller: this is a chart dependent formulation
here:

$$\mathcal{L}(\gamma^i, \dot{\gamma}^i) = \sqrt{g_{ij}(\gamma(\lambda))\dot{\gamma}^i(\lambda)\dot{\gamma}^j(\lambda)}$$

Euler-Lagrange equations:

$$\frac{\partial \mathcal{L}}{\partial \dot{\gamma}^m} = \frac{1}{\sqrt{\cdots}} g_{mj}(\gamma(\lambda))\dot{\gamma}^j(\lambda)$$

$$\left(\frac{\partial \mathcal{L}}{\partial \dot{\gamma}^m}\right)^{\cdot} = \left(\frac{1}{\sqrt{\cdots}}\right)^{\cdot} g_{mj}(\gamma(\lambda)) \cdot \dot{\gamma}^j(\lambda) + \frac{1}{\sqrt{\cdots}} \left(g_{mj}(\gamma(\lambda))\ddot{\gamma}^j(\lambda) + \dot{\gamma}^s(\partial_s g_{mj})\dot{\gamma}^j(\lambda)\right)$$

Thought bubble: reparametrize $g(\dot{\gamma}, \dot{\gamma}) = 1$ (it's a condition on my reparametrization)

By a clever choice of reparametrization $(\frac{1}{\sqrt{\cdots}})^{\cdot} = 0$

$$\frac{\partial \mathcal{L}}{\partial \gamma^m} = \frac{1}{2\sqrt{\cdots}} \partial_m g_{ij}(\gamma(\lambda))\dot{\gamma}^i(\lambda)\dot{\gamma}^j(\lambda)$$

putting this together as Euler-Lagrange equations:

$$g_{mj}\ddot{\gamma}^j + \partial_s g_{mj}\dot{\gamma}^s\dot{\gamma}^j - \frac{1}{2}\partial_m g_{ij}\dot{\gamma}^i\dot{\gamma}^j = 0$$

Multiply on both sides $(g^{-1})^{qm}$

$$\ddot{\gamma^q} + (g^{-1})^{qm}(\partial_i g_{mj} - \frac{1}{2}\partial_m g_{ij})\dot{\gamma}^i\dot{\gamma}^j = 0$$

$$\boxed{\ddot{\gamma^q} + (g^{-1})^{qm}\frac{1}{2}(\partial_i g_{mj} + \partial_j g_{mi} - \partial_m g_{ij})\dot{\gamma}^i\dot{\gamma}^j = 0}$$

geodesic equation for γ in a chart.

$$\boxed{(g^{-1})^{qm}\frac{1}{2}(\partial_i g_{mj} + \partial_j g_{mi} - \partial_m g_{ij}) =: \Gamma^q{}_{ij}(\gamma(\lambda))}$$

Thought bubble: $\left(\frac{\partial \mathcal{L}}{\partial \xi_x^{a+\dim M}}\right)^{\cdot}_{\sigma(x)} - \left(\frac{\partial \mathcal{L}}{\partial x i_x^a}\right)_{\sigma(x)} = 0$

Definition 268. "Christoffel symbol" ${}^{\text{L.C.}}\Gamma$ are the connection coefficient functions of the so-called Levi-Civita connection ${}^{\text{L.C.}}\nabla$

We usually make this choice of ∇ if g is given.
$(M, \mathcal{O}, \mathcal{A}, g) \to (M, \mathcal{O}, \mathcal{A}, g, {}^{\text{L.C.}}\nabla)$
abstract way: $\nabla g = 0$ and $T = 0$ (torsion)
$\Longrightarrow \overline{\nabla = {}^{\text{L.C.}}\nabla}$

Definition 269. (a) The Riemann-Christoffel curvature is defined by

$$R_{abcd} := g_{am}R^m_{bcd}$$

(b) Ricci: $R_{ab} = R^m_{amb}$
Thought bubble: with a metric, ${}^{\text{L.C.}}\nabla$

(c) (Ricci) scalar curvature:

$$R = g^{ab}R_{ab}$$

Thought bubble: ${}^{\text{L.C.}}\nabla$

Definition 270. Einstein curvature $(M, \mathcal{O}, \mathcal{A}, g)$

$$G_{ab} := R_{ab} - \frac{1}{2}g_{ab}R$$

Convention: $g^{ab} := (g^{\text{“}-1''})^{ab}$
F. Schuller: these indices are not being pulled up, because what would you pull them up with
(student) Question: Does the Einstein curvature yield new information?
Answer:

$$g^{ab}G_{ab} = R_{ab}g^{ab} - \frac{1}{2}g_{ab}g^{ab}R = R - \delta^a_a R = R - \frac{1}{2}\dim M\, R = (1 - \frac{d}{2})R$$

Tutorial 9: Metric manifolds

Exercise 3: Levi-Civita Connection. Suppose torsion-free $T = 0$ and metric-compatible connection $\nabla g = 0$

Question Recall $T = 0$ on a chart.

$$\boxed{\Gamma^c_{ba} = \frac{1}{2}(g^{-1})^{cm}\left(\frac{\partial g_{bm}}{\partial x^a} + \frac{\partial g_{ma}}{\partial x^b} - \frac{\partial g_{ab}}{\partial x^m}\right)}$$

or

$$\Gamma^a_{bc} = \frac{1}{2}(g^{-1})^{am}\left(\frac{\partial g_{bm}}{\partial x^c} + \frac{\partial g_{mc}}{\partial x^b} - \frac{\partial g_{bc}}{\partial x^m}\right)$$

169 Symmetry

EY : 20150321 This lecture tremendously and lucidly clarified, for me at least, what a symmetry of the Lie algebra is, and in comparing structures $(M, \mathcal{O}, \mathcal{A})$ vs. $(M, \mathcal{O}, \mathcal{A}, \nabla)$, clarified differences, and asking about differences is a good way to learn, the difference between $\mathcal{L}$ and ∇, respectively.

Feeling that the round sphere

$$(S^2, \mathcal{O}, \mathcal{A}, g^{\text{round}})$$

has rotational symmetry, while
the potato

$$(S^2, \mathcal{O}, \mathcal{A}, g^{\text{potato}})$$

does not.

169.1

169.2

Important

169.3 Flow of a complete vector field

Let $(M, \mathcal{O}, \mathcal{A})$ smooth X vector field on M

Definition 271. A curve $\gamma : I \subseteq \mathbb{R} \to M$ is called an integral curve of X
if

$$v_{\gamma,\gamma(\lambda)} = X_{\gamma(\lambda)}$$

Definition 272. A vector filed X is **complete** if all integral curves have $I = \mathbb{R}$ EY: 20150321 (i.e. domain is all of $\mathbb{R}$)

Ex. minute 48:30 EY : reall good explanation by F.P.Schuller; take a pt. out for an incomplete vector field.

Theorem 84. *compactly supported smooth vector field is complete.*

Definition 273. The flow of a complete vector field X is a 1-parameter family

$$h^X = \mathbb{R} \times M \to M$$

where $\gamma_p : \mathbb{R} \to M$ is the integral curve of X with

$\gamma(0) = p$

Then for fixed $\lambda \in \mathbb{R}$

$$h^X_\lambda : M \to M \text{ smooth}$$

picture $h^X_\lambda(S) \neq S($ if $X \neq 0)$

169.4 Lie subalgebras of the Lie algebra $(\Gamma(TM), [\cdot,\cdot])$ of vector fields

(a) $\Gamma(TM) = \{$ set of all vector fields $\}$ $\quad C^\infty(M)$-module $= \mathbb{R}$-vector space

$$\implies [X,Y] \in \Gamma(TM) \qquad [X,Y]f := X(Yf) - Y(Xf)$$

(i) $[X,Y] = -[Y,X]$

(ii) $[\lambda X + Z, Y] = \lambda[X,Y] + [Z,Y]$

(iii) $[X,[Y,Z]] + [Z,[X,Y]] + [Y,[Z,X]] = 0$

$(\Gamma(TM), [\cdot,\cdot])$ Lie algebra

(b) Let $X_1 \dots X_s$ for s (many) vector fields on M, such that

Tutorial 11 Symmetry

Exercise 1.: True or false?

(a)
-
- $\phi^* : T^*N \to T*M$ i.e. $\phi^*\nu(X) = \nu(\phi_* X)$ for smooth $\phi : M \to N$, so the pullback of a covector $\nu \in T^*N$ maps to a covector in $T*M$.
-
-
-
-

(b)

(c)

Exercise 2.: Pull-back and push-forward

Question . Let's check this locally

$$\phi^*(df)(X) = (df)(\phi_* X) = (df)(X^i \frac{\partial y^j}{\partial x^i}\frac{\partial}{\partial y^j}) = X^i \frac{\partial y^j}{\partial x^i}\frac{\partial f}{\partial y^j} \text{ where } \quad \phi_* X = X^i \frac{\partial y^j}{\partial x^i}\frac{\partial}{\partial y^j}$$

$$d(\phi^* f)(X) = d(f(\phi))(X) = \frac{\partial f}{\partial y^j}\frac{\partial y^j}{\partial x^i} dx^i(X) = X^i \frac{\partial y^j}{\partial x^i}\frac{\partial f}{\partial y^j}$$

So

$$\boxed{\phi^*(df) = d(\phi^* f)} \qquad \forall\, p \in M,\ \forall\, X \in \mathfrak{X}(M)$$

The big idea is that this is a showing of the **naturality** of the pullback ϕ^* with d, i.e. that this commutes:

$$\begin{array}{ccc} \Omega^1(M) & \xleftarrow{\phi^*} & \Omega^1(N) \\ {\scriptstyle d}\uparrow & & \uparrow{\scriptstyle d} \\ C^\infty(M) & \xleftarrow[\phi^*]{} & C^\infty(N) \end{array}$$

Question .

$$(\phi_*)^a_{\ b} := (dy^a)(\phi_*(\frac{\partial}{\partial x^b}))$$

$$\text{Let } g \in C^\infty(N)$$

$$\phi_* \left(\frac{\partial}{\partial x^b} \right) g = \frac{\partial x^b}{g} \phi(p) = \frac{\partial}{\partial x^b} g \phi x^{-1} x(p) = \frac{\partial}{\partial x^b} (g y y^{-1} \phi x^{-1})(x) =$$

$$= \frac{\partial}{\partial x^b} (g y^{-1} (y \phi x^{-1}(x(p)))) = \frac{\partial g}{\partial y}^b \Big|_y \frac{\partial y^a}{\partial x^b} \Big|_x = \frac{\partial y^a}{\partial x^b} \frac{\partial g}{\partial y^a}$$

Then

$$\phi_* \left(\frac{\partial}{\partial x^b} \right) = \frac{\partial y^a}{\partial x^b} \frac{\partial}{\partial y^a}$$

and so

$$(\phi_*)^a_{\ b} = \frac{\partial y^a}{\partial x^b}$$

Question .

Exercise 3.:Lie derivative-the pedestrian way

Question . While it is true that $\forall\, p \in S^2$, for $x(p) = (\theta, \varphi)$, and $(yix^{-1})(\theta, \varphi) = (y^1, y^2, y^3) \in \mathbb{R}^3$ and that, at this point p, $(y^1)^2/a^2 + (y^2)^2/b^2 + (y^3)^2/c^3 = 1$, this doesn't imply (EY: 20150321 I think) that, globally, it's an ellipsoid (yet). In the familiar charts given,
spherical chart $(U, x) \in \mathcal{A}$ and
$(\mathbb{R}^3, y = \mathrm{id}_{\mathbb{R}^3}) \in \mathcal{B}$
it looks like an ellipsoid, but change to another choice of charts, and it could look something very different.

Question .

Equip $(\mathbb{R}^3, \mathcal{O}_{\mathrm{st}}, \mathcal{B})$ with the Euclidean metric g, and pullback g.
Note that the pullback of the inclusion from $\mathbb{R}^3$ onto S^2 for the Euclidean metric is the following:

$$i^* g \left(\frac{\partial}{\partial \theta^i}, \frac{\partial}{\partial \theta^j} \right) = g \left(i_* \frac{\partial}{\partial \theta^i}, i_* \frac{\partial}{\partial \theta^j} \right) = g \left(\frac{\partial x^a}{\partial \theta^i} \frac{\partial}{\partial x^a}, \frac{\partial x^b}{\partial \theta^j} \frac{\partial}{\partial x^b} \right) = g_{ab} \frac{\partial x^a}{\partial \theta^i} \frac{\partial x^b}{\partial \theta^j}$$

With $g_{ab} = \delta_{ab}$, the usual Euclidean metric, this becomes the following:

$$g_{ij}^{\text{ellipsoid}} = \frac{\partial x^a}{\partial \theta^i} \frac{\partial x^a}{\partial \theta^j}$$

At this point, one should get smart (we are in the 21st century) and use some sort of CAS (Computer Algebra System). I like Sage Math (version 6.4 as of 20150322). I also like the Sage Manifolds package for Sage Math.

I like Sage Math for the following reasons:

- Open source, so it's open and freely available to anyone, which fits into my principle of making online education open and freely available to anyone, anytime
- Sage Math structures everything in terms of Category Theory and Categories and Morphisms naturally correspond to Classes and Class methods or functions in Object-Oriented Programming in Python and they've written it that way

and I like Sage Manifolds for roughly the same reasons, as manifolds are fit into a category theory framework that's written into the Python code. e.g.

```
sage: S2 = Manifold(2, 'S^2', r'\mathbb{S}^2', start_index=1) ; print S2
sage: print S2
2-dimensional manifold 'S^2'
sage: type(S2)
<class 'sage.geometry.manifolds.manifold.Manifold_with_category'>
```

With code (I've provided for convenience; you can make your own as I wrote it based upon to example of S^2 on the sagemanifolds documentation website page), load it and do the following:

cf. https://github.com/ernestyalumni/diffgeo-by-sagemnfd/blob/master/S2.sage
http://sagemanifolds.obspm.fr/examples.html

```
sage: load("S2.sage")
sage: U_ep = S2.open_subset('U_{ep}')
sage: eps.<the,phi> = U_ep.chart()
sage: a = var("a")
sage: b = var("b")
sage: c = var("c")
sage: inclus = S2.diff_mapping(R3, {(eps, cart): [ a*cos(phi)*sin(the), b*sin(phi)*sin(the),c*cos(the) ]} , name="inc",latex_name=r'\mathcal{i}')
sage: inclus.pullback(h).display()
inc_*(h) = (c^2*sin(the)^2 + (a^2*cos(phi)^2 + b^2*sin(phi)^2)*cos(the)^2) dthe*dthe - (a^2 - b^2)*cos(phi)*cos(the)*sin(phi)*sin(the) dthe*dphi
- (a^2 - b^2)*cos(phi)*cos(the)*sin(phi)*sin(the) dphi*dthe + (b^2*cos(phi)^2 + a^2*sin(phi)^2)*sin(the)^2 dphi*dphi
sage: inclus.pullback(h)[2,2].expr()
(b^2*cos(phi)^2 + a^2*sin(phi)^2)*sin(the)^2
```

A new open subset U_{ep} was declared in S^2, a new chart $(U_{\text{ep}}, (\theta, \phi))$ was declared, the constants, a, b, c, were declared, and the inclusion map given in the problem

$$y \circ \mathfrak{i} \circ x^{-1} : (\theta, \phi) \mapsto (a \cos\phi \sin\theta, b \sin\phi \sin\theta, c \cos\theta)$$

Then the pullback of the inclusion map $\rangle$ was done on the Euclidean metric h, defined earlier in the file

`S2.sage`

. Then one can access the components of this metric and do, for example,

`simplify_full(),full_simplify(), reduce_trig()`

on the expression.

In Python, I could easily do this, and give an answer quick in LaTeX:

```
sage: for i in range(1,3):
....:     for j in range(1,3):
....:         print inclus.pullback(h)[i,j].expr()
....:         latex(inclus.pullback(h)[i,j].expr() )
....:
c^2*sin(the)^2 + (a^2*cos(phi)^2 + b^2*sin(phi)^2)*cos(the)^2
```

(EY: I'll suppress the LaTeX output but this sage math function gives you LaTeX code)

and so

$$\boxed{\begin{aligned} i^* g = c^2 \sin(the)^2 + \left(a^2 \cos(\phi)^2 + b^2 \sin(\phi)^2\right) \cos(the)^2 \, d\theta \otimes d\theta + \\ -2\left(a^2 - b^2\right) \cos(\phi) \cos(the) \sin(\phi) \sin(the) \, d\theta \otimes d\phi + \\ + \left(b^2 \cos(\phi)^2 + a^2 \sin(\phi)^2\right) \sin(the)^2 \, d\phi \otimes d\phi \end{aligned}}$$

Question .

```
sage: polar_vees = eps.frame()
sage: X_1 = - sin(phi) * polar_vees[1] - cot( the ) * cos(phi) * polar_vees[2]
sage: X_2 = cos( phi ) * polar_vees[1] - cot( the ) * sin( phi) * polar_vees[2]
sage: X_3 = polar_vees[2]
sage: X_2.lie_der(X_1).display()
(cos(the)^2 - 1)/sin(the)^2 d/dphi
sage: X_3.lie_der(X_1).display()
cos(phi) d/dthe - cos(the)*sin(phi)/sin(the) d/dphi
sage: X_3.lie_der(X_2).display()
sin(phi) d/dthe + cos(phi)*cos(the)/sin(the) d/dphi
```

Indeed, one can check on a scalar field $f_{\text{eps}} \in C^\infty(S^2)$:

```
sage: f_eps = S2.scalar_field({eps: function('f', the, phi ) }, name='f' )
sage: (X_1( X_2(f_eps)) - X_2(X_1(f_eps) ) ).display()
U_{ep} --> R
(the, phi) |--> -D[1](f)(the, phi)
sage: X_2.lie_der(X_1) == -X_3
True
sage: X_3.lie_der(X_1) == X_2
True
sage: X_3.lie_der(X_2) == -X_1
True
```

$$\implies \boxed{[X_i, X_j] = -\epsilon_{ijk} X_k}$$

So $\text{span}_{\mathbb{R}}\{X_1, X_2, X_3\}$ equipped with $[\,,\,]$ constitute a Lie subalgebra on S^2 (It's closed under $[\,,\,]$

170 Integration

170.1

170.2

170.3 Volume forms

Definition 274. On a smooth manifold $(M, \mathcal{O}, \mathcal{A})$
a $(0, \dim M)$-tensor field Ω is called a volume form if

(a) Ω vanishes nowhere (i.e. $\Omega \neq 0 \; \forall\, p \in M$)

(b) totally antisymmetric

$$\Omega(\dots, \underbrace{X}_{i\text{th}}, \dots, \underbrace{Y}_{j\text{th}} \dots) = -\Omega(\dots, \underbrace{Y}_{i\text{th}}, \dots, \underbrace{X}_{j\text{th}} \dots)$$

In a chart:

$$\Omega_{i_1 \dots i_d} = \Omega_{[i_1 \dots i_d]}$$

Example $(M, \mathcal{O}, \mathcal{A}, g)$ metric manifold
construct volume form Ω from g
In any chart: (U, x)

$$\Omega_{i_1 \dots i_d} := \sqrt{\det(g_{ij}(x))}\, \epsilon_{i_1 \dots i_d}$$

where **Levi-Civita symbol** $\epsilon_{i_1 \dots i_d}$ is defined as $\epsilon_{123\dots d} = +1$

$$\epsilon_{1\dots d} = \epsilon_{[i_1 \dots i_d]}$$

Proof. (well-defined) Check: What happens under a change of charts

$$\begin{aligned}
\Omega(y)_{i_1 \dots i_d} &= \sqrt{\det(g(y)_{ij})}\, \epsilon_{i_1 \dots i_d} = \\
&= \sqrt{\det(g_{mn}(x) \frac{\partial x^m}{\partial y^i} \frac{\partial x^n}{\partial y^j})} \frac{\partial y^{m_1}}{\partial x^{i_1}} \cdots \frac{\partial y^{m_d}}{\partial x^{i_d}} \epsilon_{[m_1 \dots m_d]} = \\
&= \sqrt{|\det g_{ij}(x)|} \left| \det\left(\frac{\partial x}{\partial y}\right) \right| \det\left(\frac{\partial y}{\partial x}\right) \epsilon_{i_1 \dots i_d} = \sqrt{\det g_{ij}(x)}\, \epsilon_{i_1 \dots i_d} \text{sgn}\left(\det\left(\frac{\partial x}{\partial y}\right)\right)
\end{aligned}$$

□

EY : 20150323
Consider the following:

$$\begin{aligned}\Omega(y)(Y_{(1)}\dots Y_{(d)}) &= \Omega(y)_{i_1\dots i_d}Y^{i_1}_{(1)}\dots Y^{i_d}_{(d)} = \\ &= \sqrt{\det(g_{ij}(y))}\epsilon_{i_1\dots i_d}Y^{i_1}_{(1)}\dots Y^{i_d}_{(d)} = \\ &= \sqrt{\det(g_{mn}(x))\frac{\partial x^m}{\partial y^i}\frac{\partial x^n}{\partial y^j}}\epsilon_{i_1\dots i_d}\frac{\partial y^{i_1}}{\partial x^{m_1}}\dots\frac{\partial y^{i_d}}{\partial x^{m_d}}X^{m_1}\dots X^{m_d} = \\ &= \sqrt{\det(g_{mn}(x))\frac{\partial x^m}{\partial y^i}\frac{\partial x^n}{\partial y^j}}\det\left(\frac{\partial y}{\partial x}\right)\epsilon_{m_1\dots m_d}X^{m_1}\dots X^{m_d} = \\ &= \sqrt{\det(g_{mn}(x))}\left|\det\left(\frac{\partial x}{\partial y}\right)\right|\det\left(\frac{\partial y}{\partial x}\right)\epsilon_{m_1\dots m_d}X^{m_1}\dots X^{m_d} = \\ &= \sqrt{\det(g_{mn}(x))}\epsilon_{m_1\dots m_d}\mathrm{sgn}\left(\det\left(\frac{\partial x}{\partial y}\right)\right)X^{m_1}\dots X^{m_d} = \mathrm{sgn}(\det\left(\frac{\partial x}{\partial y}\right))\Omega_{m_1\dots m_d}(x)X^{m_1}\dots X^{m_d}\end{aligned}$$

If $\det\left(\frac{\partial y}{\partial x}\right) > 0$,

$$\Omega(y)(Y_{(1)}\dots Y_{(d)}) = \Omega(x)(X_{(1)}\dots X_{(d)})$$

This works also if Levi-Civita symbol $\epsilon_{i_1\dots i_d}$ doesn't change at all under a change of charts. (around 42:43 https://youtu.be/2XpnbvPy-Zg)

Alright, let's require,
restrict the smooth atlas $\mathcal{A}$
to a subatlas ($\mathcal{A}^\uparrow$ still an atlas)

$$\mathcal{A}^\uparrow \subseteq \mathcal{A}$$

s.t. $\forall\,(U,x),(V,y)$ have chart transition maps $y\circ x^{-1}$
$x\circ y^{-1}$

s.t. $\det\left(\frac{\partial y}{\partial x}\right) > 0$
such $\mathcal{A}^\uparrow$ called an **oriented** atlas

$$(M,\mathcal{O},\mathcal{A},g) \Longrightarrow (M,\mathcal{O},\mathcal{A}^\uparrow,g)$$

Note: associated bundles.

Note also: $\det\left(\frac{\partial y^b}{\partial x^a}\right) = \det(\partial_a(y^b x^{-1}))$ $\frac{\partial y^b}{\partial x^a}$ is an endomorphism on vector space V. $\varphi: V\to V$
$\det\varphi$ independent of choice of b
g is a $(0,2)$ tensor field, not endomorphism (not independent of choice of basis) $\sqrt{|\det(g_{ij}(y))|}$

Definition 275. Ω be a volume form on $(M,\mathcal{O},\mathcal{A}^\uparrow)$ and consider chart (U,x)

Definition 276. $\omega_{(X)} := \Omega_{i_1\dots i_d}\epsilon^{i_1\dots i_d}$ same way $\epsilon^{12\dots d} = +1$
$\epsilon^{[\dots]}$

one can show

$$\boxed{\omega_{(y)} = \det\left(\frac{\partial x}{\partial y}\right)\omega_{(x)}}$$ scalar density

170.4 Integration on one chart domain U

Definition 277.

$$\boxed{\int_U f :\overset{(U,y)}{=} \int_{y(U)} d^d\beta \omega_{(y)}(y^{-1}(\beta)) f_{(y)}(\beta)} \tag{909}$$

Proof. : Check that it's (well-defined), how it changes under change of charts

$$\int_U f :\overset{(U,y)}{=} \int_{y(U)} d^d\beta \omega_{(y)}(y^{-1}(\beta)) f_{(y)}(\beta) = \underset{(U,y)}{=} \int_{x(U)} \int d^d\alpha \left|\det\left(\frac{\partial y}{\partial x}\right)\right| f_{(x)}(\alpha)\omega_{(x)}(x^{-1}(\alpha)\det\left(\frac{\partial x}{\partial y}\right) =$$
$$= \int_{x(U)} d^d\alpha \omega_{(x)}(x^{-1}(x)) f_{(x)}(\alpha)$$

□

On an oriented metric manifold $(M, \mathcal{O}, \mathcal{A}^\uparrow, g)$

$$\int_U f := \int_{x(U)} d^d\alpha \underbrace{\sqrt{\det(g_{ij}(x))(x^{-1}(\alpha))}}_{\sqrt{g}} f_{(x)}(\alpha)$$

170.5 Integration on the entire manifold

171 Lecture 13: Relativistic spacetime

Recall, from Lecture 9, the definition of Newtonian spacetime

$$(M, \mathcal{O}, \mathcal{A}, \nabla, t) \qquad \begin{array}{l} \nabla \text{ torsion free} \\ t \in C^\infty(M) \\ dt \neq 0 \\ \nabla dt = 0 \quad \text{(uniform time)} \end{array}$$

and the definition of relativistic spacetime (before Lecture)

$$(M, \mathcal{O}, \mathcal{A}^\uparrow, \nabla, g, T) \qquad \begin{array}{l} \nabla \text{ torsion-free} \\ g \text{ Lorentzian metric}(+---) \\ T \text{ time-orientation} \end{array}$$

171.1 Time orientation

Definition 278. $(M, \mathcal{O}, \mathcal{A}^\uparrow, g)$ a Lorentzian manifold. Then a time-orientation is given by a vector field T that

(i) does **not** vanish anywhere

(ii) $g(T, T) > 0$

Newtonian vs. relativistic
Newtonian
X was called future-directed if

$$dt(X) > 0$$

$\forall\, p \in M$, take half plane, half space of T_pM
also stratified atlas so make planes of constant t straight
relativistic
half cone $\forall\, p, q \in M$, half-cone $\subseteq T_pM$

This definition of spacetime

Question
I see how the cone structure arises from the new metric. I don't understand however, how the T, the time orientation, comes in

Answer
$(M, \mathcal{O}, \mathcal{A}, g)$ $g \overset{(}{\leftarrow} +---)$
requiring $g(X, X) > 0$, select cones
T chooses which cone

This definition of spacetime has been made to enable the following physical postulates:

(P1) The worldline γ of a massive particle satisfies

(i) $g_{\gamma(\lambda)}(v_{\gamma,\gamma(lambda)}, v_{\gamma,\gamma(\lambda)}) > 0$
(ii) $g_{\gamma(\lambda)}(T, v_{\gamma,\gamma(\lambda)}) > 0$

(P2) Worldlines of massless particles satisfy

(i) $g_{\gamma(\lambda)}(v_{\gamma,\gamma(\lambda)}, v_{\gamma,\gamma(\lambda)}) = 0$
(ii) $g_{\gamma(\lambda)}(T, v_{\gamma,\gamma(\lambda)}) > 0$

picture: spacetime:

Answer (to a question) T is a smooth vector field, T determines future vs. past, "general relativity: we have such a time orientation; smoothness makes it less arbitrary than it seems" -FSchuller,
Claim: 9/10 of a metric are determined by the cone
spacetime determined by distribution, only one-tenth error

171.2 Observers

$(M, \mathcal{O}, \mathcal{A}^{\uparrow}, \nabla, g, T)$

Definition 279. An observer is a worldline γ with

$$g(v_\gamma, v_\gamma) > 0$$
$$g(T, v_\gamma) > 0$$

together with a choice of basis

$$v_{\gamma,\gamma(\lambda)} \equiv e_0(\lambda), e_1(\lambda), e_2(\lambda), e_3(\lambda)$$

of each $T_{\gamma(\lambda)}M$ where the observer worldline passes, if $g(e_a(\lambda), e_b(\lambda)) = \eta_{ab} = \begin{bmatrix} 1 & & & \\ & -1 & & \\ & & -1 & \\ & & & -1 \end{bmatrix}_{ab}$

precise: observer = smooth curve in the frame bundle LM over M

171.2.1 Two physical postulates

(P3) A **clock** carried by a specific observer (γ, e) will measure a **time**

$$\tau := \int_{\lambda_0}^{\lambda_1} d\lambda \sqrt{g_{\gamma(\lambda)}(v_{\gamma,\gamma(\lambda)}, v_{\gamma,\gamma(\lambda)})}$$

between the two "events"

$\gamma(\lambda_0)$ "start the clock"

and

$\gamma(\lambda_1)$ "stop the clock"

Compare with Newtonian spacetime:

$$t(p) = 7$$

Thought bubble: proper time/eigentime τ

Application/Example.
$$M = \mathbb{R}^4$$
$$\mathcal{O} = \mathcal{O}_{\text{st}}$$
$$\mathcal{A} \ni (\mathbb{R}^4, \text{id}_{\mathbb{R}^4})$$
$$g : g_{(x)ij} = \eta_{ij} \quad ; \qquad T^i_{(x)} = (1,0,0,0)^i$$
$$\implies \Gamma^i_{(x)\ jk} = 0 \text{ everywhere}$$

$\implies (M, \mathcal{O}, \mathcal{A}^\uparrow, g, T, \nabla)$ Riemm $= 0$
$\implies$ spacetime is flat

This situation is called special relativity.

Consider two observers:
$$\gamma : (0,1) \to M$$
$$\gamma^i_{(x)} = (\lambda, 0, 0, 0)^i$$
$$\delta : (0,1) \to M$$
$$\alpha \in (0,1) : \delta^i_{(x)} = \begin{cases} (\lambda, \alpha\lambda, 0, 0)^i & \lambda \leq \frac{1}{2} \\ (\lambda, (1-\lambda)\alpha, 0, 0)^i & \lambda > \frac{1}{2} \end{cases}$$

let's calculate:
$$\tau_\gamma := \int_0^1 \sqrt{g_{(x)ij}\dot{\gamma}^i_{(x)}\dot{\gamma}^j_{(x)}} = \int_0^1 d\lambda 1 = 1$$
$$\tau_\delta := \int_0^{1/2} d\lambda\sqrt{1-\alpha^2} + \int_{1/2}^1 \sqrt{1^2 - (-\alpha)^2} = \int_0^1 \sqrt{1-\alpha^2} = \sqrt{1-\alpha^2}$$

Note: piecewise integration

Taking the clock postulate (P3) seriously, one better come up with a realistic clock design that supports the postulate. idea.

2 little mirrors

(P4) Postulate

Let (γ, e) be an observer, and
δ be a *massive* particle worldline that is parametrized s.t. $g(v_\gamma, v_\gamma) = 1$ (for parametrization/normalization convenience)

Suppose the observer and the particle *meet* somewhere (in spacetime)
$$\delta(\tau_2) = p = \gamma(\tau_1)$$

This observer measures the 3-velocity (spatial velocity) of this particle as
$$v_\delta : \epsilon^\alpha(v_{\delta,\delta(\tau_2)})e_\alpha \qquad \alpha = 1,2,3 \tag{910}$$

where $\epsilon^0, \boxed{\epsilon^1, \epsilon^2, \epsilon^3}$ is the unique dual basis of $e_0, \boxed{e_1, e_2, e_3}$

EY:20150407
There might be a major correction to Eq. (910) from the Tutorial 14 : Relativistic spacetime, matter, and Gravitation, see the second exercise, Exercise 2, third question:
$$v := \frac{\epsilon^\alpha(v_\delta)}{\epsilon^0(v_\delta)}e_\alpha \tag{911}$$

Consequence: An observer (γ, e) will extract quantities measurable in his laboratory from objective spacetime quantities always like that.

Ex: F Faraday $(0,2)$-tensor of electromagnetism:

$$F(e_a, e_b) = F_{ab} = \begin{bmatrix} 0 & E_1 & E_2 & E_3 \\ -E_1 & 0 & B_3 & -B_2 \\ -E_2 & -B_3 & 0 & B_1 \\ -E_3 & B_2 & -B_1 & 0 \end{bmatrix}$$

observer frame e_a, e_b
$E_\alpha := F(e_0, e_\alpha)$
$B^\gamma := F(e_\alpha, e_\rho)\epsilon^{\alpha\beta\gamma}$ where $\epsilon^{123} = +1$ totally antisymmetric

171.3 Role of the Lorentz transformations

Lorentz transformations emerge as follows:
Let (γ, e) and $(\widetilde{\gamma}, \widetilde{e})$ be observers with $\gamma(\tau_1) = \widetilde{\gamma}(\tau_2)$
(for simplicity $\gamma(0) = \widetilde{\gamma}(0)$
Now

$$\begin{aligned} e_0, \ldots, e_1 \qquad & \text{at } \tau = 0 \\ \text{and } \widetilde{e}_0, \ldots, \widetilde{e}_1 \qquad & \text{at } \tau = 0 \end{aligned}$$

both bases for the same $T_{\gamma(0)}M$
Thus: $\widetilde{e}_a = \Lambda^b{}_a e_b \qquad \Lambda \in GL(4)$
Now:

$$\begin{aligned} \eta_{ab} = g(\widetilde{e}_a, \widetilde{e}_b) = g(\Lambda^m{}_a e_m, \Lambda^n{}_b e_n) = \\ = \Lambda^m{}_a \Lambda^n{}_b \underbrace{g(e_m, e_n)}_{\eta_{mn}} \end{aligned}$$

i.e. $\Lambda \in O(1,3)$
Result: Lorentz transformations relate the *frames* of *any two observers* at the same point.
"$\widetilde{x}^\mu - \Lambda^\mu{}_\nu x^\nu$" is utter nonsense

Tutorial

I didn't see a tutorial video for this lecture, but I saw that the Tutorial sheet number 14 had the relevant topics. Go there.

172 Lecture 14: Matter

two types of matter
point matter
field matter
point matter
massive point particle
more of a phenomenological importance
field matter
electromagnetic field
more fundamental from the GR point of view
both classical matter types

172.1 Point matter

Our postulates (P1) and (P2) already constrain the possible particle worldlines.
But what is their precise law of motion, possibly in the presence of "forces",

(a) without external forces

$$S_{\text{massive}}[\gamma] := m \int d\lambda \sqrt{g_{\gamma(\lambda)}(v_{\gamma,\gamma(\lambda)}, v_{\gamma,\gamma(\lambda)})}$$

with:

$$g_{\gamma(\lambda)}(T_{\gamma(\lambda)}, v_{\gamma,\gamma(\lambda)}) > 0$$

dynamical law Euler-Lagrange equation

similarly

$$S_{\text{massless}}[\gamma, \mu] = \int d\lambda \mu g(v_{\gamma,\gamma(\lambda)}, v_{\gamma,\gamma(\lambda)})$$

$$\begin{array}{ll} \delta_\mu & g(v_{\gamma,\gamma(\lambda)}, v_{\gamma,\gamma(\lambda)}) = 0 \\ \delta_\gamma & \text{e.o.m.} \end{array}$$

Reason for describing equations of motion by actions is that composite systems have an action that is the sum of the actions of the parts of that system, possibly including "interaction terms."

Example.

$$S[\gamma] + S[\delta] + S_{\text{int}}[\gamma, \delta]$$

(b) presence of external forces
or rather presence of fields to which a particle "couples"

Example

$$S[\gamma; A] = \int d\lambda m \sqrt{g_{\gamma(\lambda)}(v_{\gamma,\gamma(\lambda)}, v_{\gamma,\gamma(\lambda)})} + qA(v_{\gamma,\gamma(\lambda)})$$

where A is a **covector field** on M. A fixed (e.g. the electromagnetic potential)

Consider Euler-Lagrange eqns. $L_{\text{int}} = qA_{(x)}\dot{\gamma}^m_{(x)}$

$$m(\nabla_{v_\gamma} v_\gamma)_a + \underbrace{\left(\frac{\partial \dot{L}_{\text{int}}}{\partial \dot{\cdot}^m_{(x)}}\right)\gamma - \frac{\partial L_{\text{int}}}{\partial \gamma^m_{(x)}}}_{*} = 0 \Longrightarrow \boxed{m(\nabla_{v_\gamma} v_\gamma)^a = \underbrace{-qF^a_{\ m}\dot{\gamma}^m}_{\text{Lorentz force on a charged particle in an electromagnetic field}}}$$

$$\frac{\partial L}{\partial \dot{\gamma}^a} = qA_{(x)a}, \qquad \left(\frac{\partial \dot{L}}{\partial \dot{\cdot}^m}\right)\gamma = q \cdot \frac{\partial}{\partial x^m}(A_{(x)m}) \cdot \dot{\gamma}^m_{(x)}$$

$$\frac{\partial L}{\partial \gamma^a} = q \cdot \frac{\partial}{\partial x^a}(A_{(x)m})\dot{\gamma}^m$$

$$* = q\left(\frac{\partial A_a}{\partial x^m} - \frac{\partial A_m}{\partial x^a}\right)\dot{\gamma}^m_{(x)} \quad = q \cdot F_{(x)am}\dot{\gamma}^m_{(x)}$$

$F \leftarrow$ Faraday

$$S[\gamma] = \int (m\sqrt{g(v_\gamma, v_\gamma)} + qA(v_\gamma))d\lambda$$

172.2 Field matter

Definition 280. Classical (non-quantum) field matter is any tensor field on spacetime where equations of motion derive from an action.

Example:

$$S_{\text{Maxwell}}[A] = \frac{1}{4}\int_M d^4x\sqrt{-g}F_{ab}F_{cd}g^{ac}g^{bd}$$

A $(0,1)$-tensor field
= thought cloud: for simplicity one chart covers all of M

– for $\sqrt{-g}$ $(+---)$

$F_{ab} := 2\partial_{[a}A_{b]} = 2(\nabla_{[a}A)_{b]}$
Euler-Lagrange equations for fields

$$0 = \frac{\partial \mathcal{L}}{\partial A_m} - \frac{\partial}{\partial x^s}\left(\frac{\partial \mathcal{L}}{\partial \partial_s A_m}\right) + \frac{\partial}{\partial x^s}\frac{\partial}{\partial x^t}\frac{\partial^2 \mathcal{L}}{\partial \partial_t \partial_s A_m}$$

Example ...

$$(\nabla_{\frac{\partial}{\partial x^m}} F)^{ma} = j^a$$

inhomogeneous Maxwell
thought bubble $j = qv_\gamma$

$$\partial_{[a}F_{b]} - ()$$

homogeneous Maxwell
Other example well-liked by textbooks

$$S_{\text{Klein-Gordon}}[\phi] := \int_M d^4x\sqrt{-g}[g^{ab}(\partial_a\phi)(\partial_b\phi) - m^2\phi^2]$$

ϕ $(0,0)$-tensor field

172.3 Energy-momentum tensor of matter fields

At some point, we want to write down an action for the metric tensor field itself.
But then, this action $S_{\text{grav}}[g]$ will be added to any $S_{\text{matter}}[A, \phi, \dots]$ in order to describe the total system.

$$S_{\text{total}}[g, A] = S_{\text{grav}}[g] + S_{\text{Maxwell}}[A, g]$$

$$\begin{aligned} \delta A \quad &:\Longrightarrow \text{ Maxwell's equations} \\ \delta g_{ab} \quad &: \boxed{\frac{1}{16\pi G}G^{ab}} + (-2T^{ab}) = 0 \end{aligned}$$

G Newton's constant

$$G^{ab} = 8\pi G_N T^{ab}$$

Definition 281. $S_{\text{matter}}[\Phi, g]$ is a matter action, the **so-called energy-momentum tensor** is

$$T^{ab} := \frac{-2}{\sqrt{-g}}\left(\frac{\partial \mathcal{L}_{\text{matter}}}{\partial g_{ab}} - \partial_s \frac{\partial \mathcal{L}_{\text{matter}}}{\partial \partial_s g_{ab}} + \dots\right)$$

– of $\frac{-2}{\sqrt{g}}$ is Schrödinger minus (EY : 20150408 F.Schuller's joke? but wise)
choose all sign conventions s.t.

$$T(\epsilon^0, \epsilon^0) > 0$$

Example: For S_{Maxwell}:

$$T_{ab} = F_{am}F_{bn}g^{mn} - \frac{1}{4}F_{mn}F^{mn}g_{ab}$$

$T_{ab} \equiv T_{\text{Maxwell}ab}$

$$\begin{aligned} T(e_0, e_0) &= \underline{E}^2 + \underline{B}^2 \\ T(e_0, e_\alpha) &= (E \times B)_\alpha \end{aligned}$$

Fact: One often does not specify the fundamental action for some matter, but one is rather satisfied to assume certain properties / forms of

$$T_{ab}$$

Example Cosmology: (homogeneous & isotropic)
perfect fluid

of pressure p and density ρ modelled by

$$T^{ab} = (\rho + p)u^a u^b - pg^{ab}$$

radiative fluid
What is a fluid of photons:

$$T^{ab}_{\text{Maxwell}} g_{ab} = 0$$

observe: $T^{ab}_{\text{p.f.}} g_{ab} \stackrel{!}{=} 0$

$$= (\rho + p)u^a u^b g_{ab} - p \underbrace{g^{ab} g_{ab}}_{4}$$

$$\leftrightarrow \rho_p 04p = 0$$

$$\rho = 3p$$

$p = \frac{1}{3}\rho$

Reconvene at 3 pm? (EY : 20150409 I sent a Facebook (FB) message to the International Winter School on Gravity and Light: there was no missing video; it continues on Lecture 15 immediately)

Tutorial 14: Relativistic Spacetime, Matter and Gravitation

Exercise 2: Lorentz force law.

Question electromagnetic potential.

173 Lecture 15: Einstein gravity

Recall that in Newtonian spacetime, we were able to reformulate the Poisson law $\Delta\phi = 4\pi G_N \rho$ in terms of the Newtonian spacetime curvature as

$$R_{00} = 4\pi G_N \rho$$

R_{00} with respect to ∇_{Newton}

G_N = Newtonian gravitational constant

This prompted Einstein to postulate $<$ 1915 that the relativistic field equations for the Lorentzian metric g of (relativistic) spacetime

$$R_{ab} = 8\pi G_N T_{ab} \not/$$

However, this equation suffers from a problem

LHS: $(\nabla_a R)^{ab} \neq 0$
generically

RHS:

$$(\nabla_a T)^{ab} = 0$$

thought bubble: = formulated from an action

Einstein tried to argue this problem away.

Nevertheless, the equations cannot be upheld.

173.1 Hilbert

Hilbert was a specialist for variational principles.

To find the appropriate left hand side of the gravitational field equations, Hibert suggested to start from an action

$$S_{\text{Hilbert}}[g] = \int_M \sqrt{-g} R_{ab} g^{ab}$$

thought bubble = "simplest action"

aim: varying this w.r.t. metric g_{ab} will result in some tensor

$$G^{ab} = 0$$

173.2 Variation of S_{Hilbert}

$$0 \stackrel{!}{=} \underbrace{\delta}_{g_i} S_{\text{Hilbert}}[g] = \int_M [\underbrace{\delta\sqrt{-g} g^{ab} R_{ab}}_{1} + \underbrace{\sqrt{-g}\delta g^{ab} R_{ab}}_{2} + \underbrace{\sqrt{-g} g^{ab}\delta R_{ab}}_{3}]$$

$$\text{and } 1: \ \delta\sqrt{-g} = \frac{-(\det g) g^{mn}\delta g_{mn}}{2\sqrt{-g}} = \frac{1}{2}\sqrt{-g} g^{mn}\delta g_{mn}$$

thought bubble

$$\delta\det(g) = \det(g) g^{mn}\delta g_{mn}$$

e.g. from

$$\det(g) = \exp \operatorname{trln} g$$

ad 2: $g^{ab} g_{bc} = \delta^a_c$

$$\Longrightarrow (\delta g^{ab}) g_{bc} + g^{ab}(\delta g_{bc}) = 0$$
$$\Longrightarrow \delta g^{ab} = -g^{am} g^{bn}\delta g_{mn}$$

ad 3:

$$\Delta R_{ab} \underbrace{=}_{\text{normal coords at point}} \delta\partial_b \Gamma^m_{\ am} - \delta\partial_m \Gamma^m_{\ ab} + \Gamma\Gamma - \Gamma\Gamma =$$
$$= \partial_b \delta\Gamma^m_{\ am} - \partial_m \delta\Gamma^m_{\ ab} =$$
$$= \nabla_b (\delta\Gamma)^m_{\ am} - \nabla_m (\delta\Gamma)^m_{\ ab}$$
$$\Longrightarrow \sqrt{-g} g^{ab}\delta R_{ab} = \sqrt{-g}$$

"if you formulate the variation properly, you'll see the variation δ commute with ∂_b" EY : 20150408 I think one uses the integration at the bounds, integration by parts trick

$\Gamma^i_{(x)\ jk} - \widetilde{\Gamma}^i_{(x)\ jk}$ are the components of a $(1,2)$-tensor.

Notation: $(\nabla_b A)^i_{\ g} =: A^i_{\ j;b}$

$$\Longrightarrow \sqrt{-g} g^{ab}\delta R_{ab}$$
$$\underbrace{=}_{\nabla g = 0} \sqrt{-g}(g^{ab}\delta\Gamma^m_{\ am})_{;b} - \sqrt{-g}(g^{ab}\delta\Gamma^m_{\ ab})_{;m} = \sqrt{-g} A^b_{\ ;b} - \sqrt{-g} B^m_{\ ,m}$$

Question: Why is the difference of coefficients a tensor?

Answer:

$$\Gamma^i_{(y)\ jk} = \frac{\partial y^i}{\partial x^m}\frac{\partial x^m}{\partial y^j}\frac{\partial x^q}{\partial y^k}\Gamma^m_{(x)\ \ ,nq} + \frac{\partial y^i}{\partial x^m}\frac{\partial^2 x^m}{\partial y^j \partial y^k}$$

Collecting terms, one obtains

$$0 \stackrel{!}{=} \delta S_{\text{Hilbert}} = \int_M [\frac{1}{2}\sqrt{-g} g^{mn}\delta g_{mn} g^{ab} R_{ab} - \sqrt{-g} g^{am} g^{bn}\delta g_{mn} R_{ab} + \underbrace{(\sqrt{-g} A^a)_{,a}}_{\text{surface}} - \underbrace{(\sqrt{-g} B^b)_{,b}}_{\text{surface term}}]$$

$$= \int_M \sqrt{-g}\delta \underbrace{g_{mn}}_{\text{arbitrary variation}} [\frac{1}{2} g^{mn} R - R^{mn}] \Longrightarrow G^{mn} = R^{mn} - \frac{1}{2} g^{mn} R$$

Hence Hilbert, from this "mathematical" argument, concluded that one may take

$$\boxed{R_{ab} - \frac{1}{2} g_{ab} R = 8\pi G_N T_{ab}}$$

Einstein equations

$$S_{E-H}[g] = \int_M \sqrt{-g} R$$

173.3 3. Solution of the $\nabla_a T^{ab} = 0$ issue

One can show ($\rightarrow$ Tutorials) that the Einstein curvature

$$G_{ab} = R_{ab} - \frac{1}{2} g_{ab} R$$

satisfy the so-called contracted differential Bianchi identity

$$(\nabla_a G)^{ab} = 0$$

173.4 Variants of the field equations

(a) a simple rewriting:

$$R_{ab} - \frac{1}{2} g_{ab} R = 8\pi G_N T_{ab} = T_{ab}$$

$G_N = \frac{1}{8\pi}$

Contract on both sides g^{ab}

$$\begin{aligned} R_{ab} - \frac{1}{2} g_{ab} R &= T_{ab} || g^{ab} \\ R - 2R = T &:= T_{ab} g^{ab} \\ \Longrightarrow R &= -T \\ \Longrightarrow R_{ab} + \frac{1}{2} g_{ab} T &= T_{ab} \\ \Longleftrightarrow R_{ab} = (T_{ab} - \frac{1}{2} T g_{ab}) &=: \widehat{T}_{ab} \end{aligned}$$

$$\boxed{R_{ab} = \widehat{T}_{ab}}$$

(b)

$$S_{E-H}[g] := \int_M \sqrt{-g}(R + 2\Lambda)$$

thought bubble: Λ cosmological constant

History:

1915: $\Lambda < 0$ (Einstein) in order to get a non-expanding universe

>1915: $\Lambda = 0$ Hubble

today $\Lambda > 0$ to account for an accelerated expansion

$\Lambda \neq 0$ can be interpreted as a contribution

$-\frac{1}{2}\Lambda g$ to the energy-momentum

"dark energy"

Question: surface terms scalar?

Answer: for a careful treatment of the surface terms which we discarded, see, e.g. E. Poisson, "A relativist's toolkit" C.U.P. "excellent book"

Question: What is a constant on a manifold?

Answer: $\int \sqrt{-g}\Lambda = \Lambda \int \sqrt{-g}1$

[back to dark energy]

[Weinberg, QCD, calculated]
idea: 1 could arise as the vacuum energy of the standard model fields

$\Lambda_{\text{calculated}} = 10^{120} \times \Lambda_{\text{obs}}$

"worst prediction of physics"

Tutorials: check that

- Schwarzscheld metric (1916)
- FRW metric
- pp-wave metric
- Reisner-Nordstrom

$\Longrightarrow$ are solutions to Einstein's equations

in high school
$m\ddot{x} + m\omega^2 x^2 = 0$
$x(t) = \cos(\omega t)$
ET: [elementary tutorials]
study motion of particles & observers in Schwarzscheld S.T.
Satellite: Marcus C. Werner
Gravitational lensing
odd number of pictures Morse theory (EY:20150408 Morse Theory !!!)
Domenico Giulini
Hamiltonian form Canonical Formulations
Key to Quantum Gravity

Tutorial 13 Schwarzschild Spacetime

EY : 20150408 I'm not sure which tutorial follows which lecture at this point.

The tutorial video is excellent itself. Here, I want to encourage the use of CAS to do calculations. There are many out there. Again, I'm partial to the Sage Manifolds package for Sage Math which are both open-source and based on Python. I'll use that here.

Exercise 1. Geodesics in a Schwarzschild spacetime

Question Write down the Lagrangian.

Load "Schwarzschild.sage" in Sage Math, which will always be available freely here https://github.com/ernestyalumni/diffgeo-by-sagemnfd/blob/master/Schwarzschild.sage:

```
sage: load("Schwarzschild.sage")
4-dimensional manifold 'M'
open subset 'U_sph' of the 4-dimensional manifold 'M'
Levi-Civita connection 'nabla_g' associated with the Lorentzian metric 'g' on the 4-dimensional manifold 'M'
```

and so on.

Look at the code and I had defined the Lagrangian to be

`L`

. To get out the coefficients of L of the components of the tangent vectors to the curve, i.e. t', r', θ', ϕ', denoted

`tp,rp,thp,php`

in my .sage file, do the following:

```
sage: L.expr().coefficients(tp)[1][0].factor().full_simplify()
(2*G_N*M_0 - r)/r
sage: L.expr().coefficients(rp)[1][0].factor().full_simplify()
-r/(2*G_N*M_0 - r)
sage: L.expr().coefficients(php)[1][0].factor().full_simplify()
r^2
sage: L.expr().coefficients(thp)[1][0].factor().full_simplify()
r^2*sin(th)^2
```

Question There are 4 Euler-Lagrange equations for this Lagrangian. Derive the one with respect to the function $t(\lambda)$!.

```
sage: L.expr().diff(t)
0
```

This confirms that $\frac{\partial L}{\partial t} = 0$

For $\frac{d}{d\lambda}\frac{\partial L}{\partial t'}$, then one needs to consider this particular workaround for Sage Math (computer technicality). One takes derivatives with respect to declared variables (declared with var) and then substitute in functions that are dependent upon λ, and then take the derivative with respect to the parameter λ. This does that:

```
sage: L.expr().diff( thp ).factor().subs( r == gamma1 ).subs( thp == gamma3.diff( tau ) ).subs( th == gamma3 ).diff(tau)\
....: .factor()
2*(2*cos(gamma3(tau))*gamma1(tau)*D[0](gamma3)(tau)^2 + 2*sin(gamma3(tau))*D[0](gamma1)(tau)*D[0](gamma3)(tau)
+ gamma1(tau)*sin(gamma3(tau))*D[0, 0](gamma3)(tau))*gamma1(tau)*sin(gamma3(tau))
```

Question Show that the Lie derivative of g with respect to the vector fields $K_t := \frac{\partial}{\partial t}$.

The first line defines the vector field by accessing the frame defined on a chart with spherical coordinates and getting the time vector. The second line is the Lie derivative of g with respect to this vector field.

```
sage: K_t = espher[0]
sage: g.lie_der(K_t).display() # 0, as desired
0
```

EY : 20150410 My question is this: $\forall\, X \in \Gamma(TM)$ i.e. X is a vector field on M, or, specifically, a section of the tangent bundle, then does

$$\mathcal{L}_X g = 0$$

instantly mean that X is a symmetry for (M, g)? $\mathcal{L}_X g$ is interpreted geometrically as how g changes along the flow generated by X, and if it equals 0, then g doesn't change.

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176 Canonical Formulation of GR I

Dynamical and Hamiltonian formulation of General Relativity.

Purpose

(1) formulate and solve initial-value problems

(2) integrate Einstein's Equations by numerical codes

(3) characterize degrees of freedom

(4) characterize isolated systems, associated symmetry groups and conserved quantities, like Energy/Mass, Momenta (linear and angular), Poincaré charges

(5) starting point for "canonical quantization" program.

How. We will rewrite Einstein's Eq. in form of a *constrained Hamiltonian system.*

$$\underbrace{R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R}_{G_{\mu\nu}} + \underbrace{\Lambda}_{\text{kosm.}const.}\, g_{\mu\nu} = \underbrace{k}_{\frac{8\pi G}{c^4}} T_{\mu\nu}$$

$(-+++)$

$$T^{\mu\nu} = \begin{pmatrix} W & \frac{1}{c}S^m \\ cg^m & \mathbf{t}^{mn} \end{pmatrix}$$

W = Energy density (1 component)
g^m = Momentum density, (3 components)
S^m = Energy current-density (3 components)
$\mathbf{t}^{mn}$ = Momentum current-density (6 components)

$$T^{\mu\nu} = T^{\nu\mu} \Longrightarrow S^m = c^2 g^m$$

10 independent komp. (components)

Phys. dim. $[T^{\mu\nu}] = \frac{J}{m^3}$

$[G^{\mu\nu}] = \frac{1}{m^2}$

$$k = \frac{\text{curvature}}{\text{Energy } \cdot \text{ density}}$$

$[k] = \frac{1}{m^2} / \frac{J}{m^3}$, ${}^2k$ $\quad \dfrac{\text{Curvature}}{\text{mass density}} = \left(\dfrac{1}{1.5\,\text{AU}}\right)^2$ / Density of water

$$= \left(\frac{1}{10\,km}\right)^2 \text{ / Nuclear density in core of neutron star } \simeq 5 \cdot 10^{17}\, kg/m^3$$

If "Ein" for Einstein Tensor, $G_{\mu\nu} = \text{Ein}\left(\frac{\partial}{\partial x^\mu}, \frac{\partial}{\partial x^\nu}\right)$

$$\text{Ein}(v,w) = \frac{1}{4}[\text{Ein}(v+w, v+w) - \text{Ein}(v-w, v-w)$$

$$\text{Ein}(w,w) = -g(w,w)\sum_{\perp w} \text{Sec}$$

where $\perp w$ is take the sum over any triple of mutually perp. 2-planes in $\perp w$

$\text{Sec}(\text{Span}\{v,w\}) = \frac{\text{Riem}(v,w,v,w)}{[g(v,w)]^2 - g(v,v)g(w,w)}$

"sectional curvature"

Identity: $\nabla_\mu G^{\mu\nu} = 0$ (follows from twice-contracted II. Bianchi Identity

$\sum_{\lambda\mu\nu \text{ cycl}} \nabla_\lambda R_{\alpha\beta\mu\nu} = 0$)

$$\underbrace{\partial_0 G^{0\nu}}_{\text{contains at most 1st time der.}} + \underbrace{\partial_k G^{k\nu} + \Gamma G + \Gamma G \equiv 0}_{\text{contains at most 2nd. time derivatives}}$$

$\Longrightarrow$ 4 out of 10 Einstein Eq. do not evolve the fields but rather constrain the initial data. The space-space components (6 Eqns.) are the evolution Eqns.

10 Einstein Eq. - 4 constraints (underdetermined elliptic type)

\ - 6 evolution equations (undetermined hyperbolic type)

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180 Lecture 22: Black Holes

Only depends on Lectures 1-15, so does lecture on "Wednesday"

Schwarzschild solution also vacuum solution (from tutorial EY : oh no, must do tutorial)

Study the Schwarzschild as a vacuum solution of the Einstein equation:

$m = G_N M$ where M is the "mass"

$$g = \left(1 - \frac{2m}{r}\right) dt \otimes dt - \frac{1}{1 - \frac{2m}{r}} dr \otimes dr - r^2 (d\theta \otimes d\theta + \sin^2\theta d\varphi \otimes d\varphi$$

in the so-called Schwarzschild coordinates

t	r	θ	φ
$(-\infty, \infty)$	$(0, \infty)$	$(0, \pi)$	$(0, 2\pi)$

What staring at this metric for a while, two questions naturally pose themselves:

(i) What exactly happens $r = 2m$?

t	r	θ	φ
$(-\infty, \infty)$	$(0, 2m) \cup (2m, \infty)$	$(0, \pi)$	$(0, 2\pi)$

(ii) Is there anything (in the real world) beyond $t \to -\infty$?
$t \to +\infty$

idea: Map of Linz, blown up

Insight into these two issues is afforded by stopping to stare.

Look at *geodesic* of g, instead.

180.1 Radial null geodesics

null - $g(v_\gamma, v_\gamma) = 0$

Consider null geodesic in "Schd"

$$S[\gamma] = \int d\lambda \left[\left(1 - \frac{2m}{r}\right) \dot{t}^2 - \left(1 - \frac{2m}{r}\right)^{-1} \dot{r}^2 - r^2 (\dot{\theta}^2 + \sin^2\theta \dot{\varphi}^2) \right]$$

with $[\dots] = 0$

and one has, in particular, the t-eqn. of motion:

$$\left(\left(1 - \frac{2m}{r}\right) \dot{t} \right)^{\cdot} = 0$$

$\Longrightarrow$

$$\boxed{\left(1 - \frac{2m}{r}\right) \dot{t} = k} = \text{ const.}$$

Consider radial null geodesics

$\theta \overset{!}{=} \text{ const.}$ $\quad \varphi = \text{ const.}$

From $\square$ and $\square$

$$\Longrightarrow \dot{r}^2 = k^2 \leftrightarrow \dot{r} = \pm k$$

$$\implies r(\lambda) = \pm k \cdot \lambda$$

Hence, we may consider

$$\widetilde{t}(r) := t(\pm k\lambda)$$

Case A: $\oplus$

$$\frac{d\widetilde{t}}{dr} = \frac{\dot{\widetilde{t}}}{\dot{r}} = \frac{k}{\left(1 - \frac{2m}{r}\right) k} = \frac{r}{r - 2m}$$

$$\implies \widetilde{t}_+(r) = r + 2m \ln |r - 2m|$$

(**outgoing** null geodesics)

Case b. $\pm$ (Circle around $-$, consider $-$):

$$\widetilde{t}_-(r) = -r - 2m \ln |r - 2m|$$

(**ingoing** null geodesics)

Picture

180.2 Eddington-Finkelstein

Brilliantly simple idea:

change (on the domain of the Schwarzschild coordinates) to different coordinates, s.t. in those new coordinates,

ingoing null geodesics appear as straight lines, of slope -1

This is achieved by

$$\bar{t}(t, r, \theta, \varphi) := t + 2m \ln |r - 2m|$$

Recall: ingoing null geodesic has

$$\widetilde{t}(r) = -(r + 2m \ln |r - 2m|) \qquad (Schdcoords)$$

$$\iff \bar{t} - 2m \ln |r - 2m| = -r - 2m \ln |r - 2m| + \text{ const.}$$

$$\therefore \bar{t} = -r + \text{ const.}$$

(Picture)

outgoing null geodesics

$$\bar{t} = r + 4m \ln |r - 2m| + \text{ const.}$$

Consider the new chart (V, g) while (U, x) was the Schd chart.

$$\underbrace{U}_{\text{Schd}} \bigcup \{ \text{ horizon } \} = V$$

"chart image of the horizon"

Now calculate the *Schd metric* g w.r.t. Eddington-Finkelstein coords.

$$\bar{t}(t, r, \theta, \varphi) = t + 2m \ln |r - 2m|$$
$$\bar{r}(t, r, \theta, \varphi) = r$$
$$\bar{\theta}(t, r, \theta, \varphi) = \theta$$
$$\bar{\varphi}(t, r, \theta, \varphi) = \varphi$$

EY : 20150422 I would suggest that after seeing this, one would calculate the metric by your favorite CAS. I like the Sage Manifolds package for Sage Math.

Schwarzschild_BH.sage on github

Schwarzschild_BH.sage on Patreon

Schwarzschild_BH.sage on Google Drive

```
sage: load(''Schwarzschild_BH.sage'')
4-dimensional manifold 'M'
expr = expr.simplify_radical()
Levi-Civita connection 'nabla_g' associated with the Lorentzian metric 'g' on the 4-dimensional
    manifold 'M'
Launched png viewer for Graphics object consisting of 4 graphics primitives
```

Then calculate the Schwarzschild metric g but in Eddington-Finkelstein coordinates. Keep in mind to calculate the set of coordinates that uses $\bar{t}$, not $\widetilde{t}$:

```
sage: gI.display()
gI = (2*m - r)/r dt*dt - r/(2*m - r) dr*dr + r^2 dth*dth + r^2*sin(th)^2 dph*dph
sage: gI.display( X_EF_I_null.frame())
gI = (2*m - r)/r dtbar*dtbar + 2*m/r dtbar*dr + 2*m/r dr*dtbar + (2*m + r)/r dr*dr + r^2 dth*dth + r
    ^2*sin(th)^2 dph*dph
```

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